

LUFTHANSA GROUP

Prospectus

**for the public offering
in the Federal Republic of Germany**

of

597,742,822 newly issued ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*) from a combined capital increase utilizing authorized capital against (i) contribution in cash with indirect subscription rights (*mittelbare Bezugsrechte*) to existing shareholders, except for the German Economic Stabilization Fund (*Wirtschaftsstabilisierungsfonds*), and (ii) contribution in kind by the German Economic Stabilization Fund (*Wirtschaftsstabilisierungsfonds*) with direct subscription rights, resolved by the Executive Board (*Vorstand*) on September 19, 2021, with approval of the Supervisory Board (*Aufsichtsrat*) on the same date

and

for the admission to trading

on the regulated market segment (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with simultaneous admission to the sub-segment thereof with additional post-admission obligations (Prime Standard)

of

597,742,822 newly issued ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*)

– each such share representing a notional share in the share capital of €2.56 and carrying full dividend rights as from January 1, 2021 –

of

Deutsche Lufthansa Aktiengesellschaft
Cologne, Germany

Subscription price: €3.58

International Securities Identification Number (ISIN): DE0008232125
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Joint Global Coordinators

BofA Securities

Deutsche Bank

Goldman Sachs

J.P. Morgan

Joint Bookrunners

Barclays

BNP PARIBAS

COMMERZBANK

HSBC

UniCredit Bank AG

**Crédit Agricole Corporate
and Investment Bank**

DZ BANK

**Landesbank
Baden-Württemberg**

SMBC Nikko

Société Générale

September 20, 2021

The validity of this Prospectus is expected to expire at the end of the day on October 11, 2021. The closing of the offer period is expected to occur on October 5, 2021, and the time when trading on a regulated market begins is expected to occur on October 11, 2021. The obligation to supplement this Prospectus in the event of any significant new factors, material mistakes or material inaccuracies no longer applies when this Prospectus is no longer valid.

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I. SUMMARY

A. Introduction and Warnings

This prospectus (the “**Prospectus**”) relates to the offer and admission to trading of 597,742,822 newly issued ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*) (the “**New Shares**” and each, a “**New Share**”) of Deutsche Lufthansa Aktiengesellschaft, a German stock corporation (*Aktiengesellschaft*) with its corporate seat in Cologne, Germany and its registered business address at Venloer Straße 151-153, 50672 Cologne, Germany, Legal Entity Identifier (“**LEI**”) 529900PH63HYJ86ASW55 (the “**Company**” and, together with its consolidated subsidiaries, the “**Lufthansa Group**,” “**we**,” “**us**” and “**our**”). The International Securities Identification Number (“**ISIN**”) of the Company’s shares is DE0008232125. The New Shares will be offered by BofA Securities Europe SA, 51 rue La Boétie, 75008 Paris, France, LEI 549300FH0WJAPEHTIQ77 (“**BofA Securities**”); Deutsche Bank Aktiengesellschaft, Taunusanlage 12, 60325 Frankfurt am Main, Germany, LEI 7LTWFZYICNSX8D621K86 (“**Deutsche Bank**”); Goldman Sachs Bank Europe SE, Marienturm, Taunusanlage 9-10, 60329 Frankfurt am Main, Germany, LEI 8IBZUGJ7JPLH368JE346 (“**Goldman Sachs**”); J.P. Morgan AG, Taunustor 1, 60310 Frankfurt am Main, Germany, LEI 549300ZK53CNGEEI6A29 (“**J.P. Morgan**” and, together with BofA Securities, Deutsche Bank and Goldman Sachs, the “**Joint Global Coordinators**”); Barclays Bank Ireland PLC, One Molesworth Street, Dublin 2, D02 RF29, Ireland, LEI 2G5BKIC2CB69PRJH1W31 (“**Barclays**”); BNP PARIBAS SA, 16, Boulevard des Italiens, 75009 Paris, France, LEI R0MUWSFPU8MPRO8K5P83 (“**BNP PARIBAS**”); COMMERZBANK Aktiengesellschaft, Kaiserstraße 16 (Kaiserplatz), 60311 Frankfurt am Main, Germany, LEI 851WYGNLUQLFZBSYGB56 (“**COMMERZBANK**”); HSBC Trinkaus & Burkhardt AG, Königsallee 21/23, 40212 Düsseldorf, Germany, LEI JUNTA405OW8OY5GN4DX16 (“**HSBC**”); UniCredit Bank AG, Arabellastraße 12, 81925 Munich, Germany, LEI 2ZCNRR8UK83OBTEK2170 (“**UniCredit Bank**”); Crédit Agricole Corporate and Investment Bank, 12 Place des Etats-Unis, CS 70052, 92547 Montrouge Cedex, France, LEI 1VUV7VQFKUOQJSJ21A208 (“**Crédit Agricole Corporate and Investment Bank**”); DZ BANK AG Deutsche Zentral-Genossenschaftsbank, Frankfurt am Main, Platz der Republik, 60325 Frankfurt am Main, Germany, LEI 529900HNOAA1KXQJUQ27 (“**DZ BANK**”); Landesbank Baden-Württemberg, Am Hauptbahnhof 2, 70173 Stuttgart, Germany, LEI B81CK4ESI35472RHJ606 (“**Landesbank Baden-Württemberg**”); SMBC Nikko Capital Markets Europe GmbH, Neue Mainzer Straße 52-58, 60311 Frankfurt am Main, Germany, LEI: 5493008794GM6YH5G766 (“**SMBC Nikko**”) and Société Générale S.A., 29, boulevard Haussmann, 75009 Paris, France, LEI O2RNE8IBXP4R0TD8PU41 (“**Société Générale**” and, together with Barclays, BNP PARIBAS, COMMERZBANK, HSBC, UniCredit Bank, Crédit Agricole Corporate and Investment Bank, DZ BANK, Landesbank Baden-Württemberg and SMBC Nikko, the “**Joint Bookrunners**” and, together with the Joint Global Coordinators, the “**Underwriters**”).

This Prospectus is dated September 20, 2021, and has been approved by the German Federal Financial Supervisory Authority (*Bundesanstalt für Finanzdienstleistungsaufsicht*) (“**BaFin**”) as competent authority under Regulation (EU) 2017/1129 of the European Parliament and of the Council of June 14, 2017, on the prospectus to be published when securities are offered to the public or admitted to trading on a regulated market, and repealing Directive 2003/71/EC (the “**Prospectus Regulation**”) on that date. BaFin can be contacted at Marie-Curie-Str. 24-28, 60439 Frankfurt am Main, Germany, by telephone +49 228 4108-0 or via its website: <http://www.bafin.de>.

*This summary should be read as an introduction to this Prospectus. Any decision to invest in the New Shares (together with the existing shares of the Company, the “**Shares**”) should be based on a consideration of this Prospectus as a whole by the investor. Investors could lose all or part of their invested capital. Where a claim relating to the information contained in a prospectus is brought before a court, the plaintiff investor might, under national law, have to bear the costs of translating this Prospectus before the legal proceedings are initiated. Civil liability attaches only to those persons who have tabled the summary including any translation thereof, but only where the summary is misleading, inaccurate or inconsistent, when read together with the other parts of this Prospectus, or where it does not provide, when read together with the other parts of this Prospectus, key information in order to aid investors when considering whether to invest in the New Shares.*

B. Key Information on the Issuer

B.1. Who is the issuer of the securities?

The legal and commercial name of the Company is “Deutsche Lufthansa Aktiengesellschaft.” The Company is incorporated as a stock corporation (*Aktiengesellschaft*) under the laws of Germany. The Company’s registered business address is at Venloer Straße 151-153, 50672 Cologne, Germany. The Company is registered with the commercial register (*Handelsregister*) of the local court (*Amtsgericht*) of Cologne, Germany, under number HRB 2168. The Company’s LEI is 529900PH63HYJ86ASW55.

Principal activities. We are the leading airline group in Europe and one of the largest airline groups globally by total revenue (*source*: Statista, Leading Airline Groups in Europe in 2020, Company Information). The airlines that form the core of the Lufthansa Group have their home markets in Germany, Switzerland, Austria and Belgium. As an airline group, our core business is the operation of domestic, short-haul and long-haul scheduled passenger and cargo flights within a worldwide network, as well as the provision of logistical and technical services related to flight operations. In addition, we provide aircraft maintenance, repair and overhaul services and catering. We generated total revenue of €5,771 million and €13,589 million in the six-month period ended June 30, 2021, and the financial year ended December 31, 2020, respectively, compared to €8,335 million and €36,424 million in the six-month period ended June 30, 2020, and the financial year ended December 31, 2019, respectively.

Major shareholders. To the Company’s knowledge, based on the Company’s shareholder register (*Aktienregister*) or the shareholding notifications it received, the German Economic Stabilization Fund (*Wirtschaftsstabilisierungsfonds*) (“**WSF**”) holds 15.94% of the share capital of the Company, The Goldman Sachs Group, Inc. holds 0.08% of the share capital of the Company and instruments pursuant to Section 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*) relating to 6.80% of the share capital of the Company and Société Générale holds 0.74% of the share capital of the Company and instruments pursuant to Section 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*) relating to 4.28% of the share capital of the Company.

Controlling Shareholders. The Company is not controlled by any of its direct or indirect shareholders.

Key managing directors. The Company's executive board (*Vorstand*) (the “**Executive Board**”) consists of six members: Carsten Spohr (Chief Executive Officer), Remco Steenbergen (Chief Financial Officer), Christina Foerster (Chief Customer Officer), Harry Hohmeister (Chief Commercial Officer), Dr. Detlef Kayser (Chief Operations Officer) and Dr. Michael Niggemann (Chief HR & Legal Officer).

Statutory Auditors. Ernst & Young GmbH Wirtschaftsprüfungsgesellschaft, with its registered seat in Stuttgart, Germany, acting through its Frankfurt office at Mergenthalerallee 3-5, 65760 Eschborn, Germany (“**EY**”) was appointed as statutory auditor of the Company's consolidated financial statements and annual financial statements, each for the financial year ended December 31, 2020. As an independent auditor, EY has audited the Company's consolidated financial statements and annual financial statements, each for the financial year ended December 31, 2020. The Company's statutory auditor for the financial years ended December 31, 2019, and December 31, 2018, was PricewaterhouseCoopers GmbH Wirtschaftsprüfungsgesellschaft with its registered seat in Frankfurt am Main, Germany, acting through its Düsseldorf office, Moskauer Straße 19, 40227 Düsseldorf, Germany (“**PwC**”). As an independent auditor, PwC has audited the Company's consolidated financial statements for the financial years ended December 31, 2019, and December 31, 2018.

B.2. What is the key financial information regarding the issuer?

The consolidated financial statements of the Lufthansa Group as of and for the financial years ended December 31, 2020, 2019, and 2018 (the “**Audited Consolidated Financial Statements**”) were prepared in accordance with International Financial Reporting Standards as adopted by the European Union (“**IFRS**”) and the additional requirements of German commercial law pursuant to Section 315e para. 1 of the German Commercial Code (*Handelsgesetzbuch*) and have been audited in accordance with Section 317 of the German Commercial Code (*Handelsgesetzbuch*) and in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the German Institute of Public Auditors (*Institut der Wirtschaftsprüfer*) and Regulation (EU) No 537/2014 of the European Parliament and of the Council of April 16, 2014, on specific requirements regarding statutory audit of public-interest entities, by EY (for the financial year 2020), who issued an unqualified independent auditor's report thereon, and PwC (for the financial years 2019 and 2018), who issued unqualified auditor's reports thereon. The unaudited condensed consolidated interim financial statements of the Company as of and for the six-month period ended June 30, 2021 (the “**Unaudited Condensed Consolidated Interim Financial Statements**” and, together with the Audited Consolidated Financial Statements, the “**Consolidated Financial Statements**”), were prepared in accordance with IFRS applicable to interim financial reporting.

The financial information presented in this summary for the financial year 2019 was taken from the comparative prior-year figures contained in the audited consolidated financial statements for the financial year ended December 31, 2020, to the extent such information was adjusted in the audited consolidated financial statements for the financial year ended December 31, 2020. The financial information presented in this summary for the financial year 2018 was taken from the comparative prior-year figures contained in the audited consolidated financial statements for the financial year ended December 31, 2019, unless indicated otherwise.

In this summary, where financial information is labeled “audited” in tables, this information was taken from the Audited Consolidated Financial Statements. The label “unaudited” is used in tables in this summary to indicate financial information that was taken from the Unaudited Condensed Consolidated Interim Financial Statements, or from our accounting records or internal management reporting systems or has been calculated based on figures from the abovementioned sources.

Selected Key Financial Information from the Consolidated Income Statements

	For the six-month period ended June 30,		For the financial year ended December 31,		
	2021	2020	2020	2019	2018 ⁽¹⁾
	(unaudited)		(audited)		
	(in € million, unless otherwise indicated)				
Total revenue.....	5,771	8,335	13,589	36,424	35,542
Profit/loss from operating activities.....	(2,050)	(3,288)	(7,089)	1,689	2,800
Profit/loss before income taxes.....	(2,234)	(4,419)	(8,631)	1,860	2,784
Profit/loss after income taxes	(1,813)	(3,627)	(6,766)	1,245	2,196
Net profit/loss attributable to the Company's shareholders.....	(1,805)	(3,617)	(6,725)	1,213	2,163
Basic/diluted earnings per share in €	(3.02)	(7.56)	(12.51)	2.55	4.58

(1) The figures for the fiscal year ended December 31, 2018, were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.

Selected Key Financial Information from the Consolidated Statements of Financial Position

	As of June 30,	As of December 31,		
	2021	2020	2019	2018
	(unaudited)	(audited)		
		(in € million)		
Total assets	40,838	39,484	42,659	38,213
Shareholder's equity	3,145	1,387	10,256	9,573

Selected Key Financial Information from the Consolidated Cash Flow Statements

	For the six-month period ended June 30,		For the financial year ended December 31,		
	2021	2020	2020	2019	2018
	(unaudited)		(in € million) (audited)		
Net cash from/used in operating activities ⁽¹⁾	18	363	(2,328)	4,030	4,109
Net cash from/used in investing and cash management activities	(1,362)	(1,083)	(2,342)	(3,867)	(3,269)
Net cash from/used in financing activities.....	1,572	534	5,076	(161)	(626)

(1) Referred to as cash flow from operating activities in the Unaudited Condensed Consolidated Interim Financial Statements.

Key Performance Indicators and Alternative Performance Measures

	For the six-month period ended June 30,		For the financial year ended December 31,		
	2021	2020	2020	2019	2018 ⁽¹⁾
	(unaudited, unless otherwise indicated)		(in € million, unless otherwise indicated)		
Adjusted EBITDA ⁽²⁾	(970)	(1,578)	(2,890)	4,718	5,016
Adjusted EBIT ⁽²⁾	(2,095)	(2,899)	(5,451)(*)	2,026(*)	2,836(*)
Adjusted EBIT Margin (in %) ⁽³⁾	(36.3)	(34.8)	(40.1)	5.6	8.0

(*) Audited.

(1) The figures for the fiscal year ended December 31, 2018, were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.

(2) The reconciliations of EBIT to Adjusted EBIT and of Adjusted EBIT to Adjusted EBITDA on group level are shown in the table below.

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018 ^(a)
	(unaudited)		(audited, unless otherwise indicated)		
			(in € million)		
Total revenue.....	5,771	8,335	13,589	36,424	35,542
Changes in inventories and work performed by entity and capitalized	49	158	175	685	531
Other operating income	655	797	1,897	1,889	1,818
Cost of materials and services	(3,204)	(5,127)	(8,453)	(19,827)	(18,367)
Staff costs	(2,907)	(3,620)	(6,436)	(9,121)	(8,811)
Depreciation, amortization and impairment	(1,135)	(1,783)	(4,389)	(2,776)	(2,205)
Other operating expenses	(1,279)	(2,048)	(3,472)	(5,585)	(5,708)
Profit/loss from operating activities.....	(2,050)	(3,288)	(7,089)	1,689	2,800
Result of equity investments accounted for using the equity method...	(71)	(184)	(276)	88	114
Result of other equity investments	7	4	12	80	60
EBIT	(2,114)	(3,468)	(7,353)	1,857	2,974
Impairment losses/gains	6	552	1,870	139	9
Effects from pension provisions	(3)	8	31	10	(113)
Results of disposal of assets	16	9	1	20	(34)
Adjusted EBIT	(2,095)	(2,899)	(5,451)	2,026	2,836
Depreciation and amortization.....	1,125	1,321	2,561	2,692	2,180
Adjusted EBITDA^(b)	(970)	(1,578)	(2,890)	4,718	5,016

(a) The figures for the fiscal year ended December 31, 2018, were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.

(b) Unaudited.

(3) Defined as Adjusted EBIT divided by total revenue.

B.3. What are the key risks that are specific to the issuer?

An investment in the Company's shares is subject to a number of risks, some of which are presented in this section and under section "C.3. What are the key risks attached to the securities?" of this summary. If any of these risks were to materialize, investors could lose all or part of their investments. The following risks are key risks specific to the Company:

- The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected the demand for our services and the conditions under which we operate. We generally expect these developments to continue, but the magnitude of the impact of COVID-19 will depend on various factors beyond our control.

- Our business activities and operations are sensitive to changes in entry regulations in relation to our destinations, including mandatory quarantines and travel bans, and a loss of take-off and landing slots, including in connection with and as a result of COVID-19, which may ultimately intensify the decrease in ticket sales and the increase in reimbursement claims.
- We are exposed to adverse developments in relation to our customer stream resulting from our various joint venture arrangements on certain, in particular international, routes and our membership in the “Star Alliance.”
- We face increasing susceptibility for the demand of our business as a result of increasing competition from alternative means of transportation, including rail travel, as well as alternatives to business travel, and due to changes in the applicable legal framework or consumer behavior, in particular in relation to climate change and generally increased environmental consciousness. Against this background, we may also be required to reduce certain short-haul flights or discontinue such flights entirely.
- We are exposed to adverse developments in relation to the sourcing of jet fuel, in particular its availability and price, the applicability of tax exemption regulations for jet fuel and our related fuel hedging activities with respect to price fluctuations, because we may be unable to source jet fuel in sufficient volumes and at acceptable prices and/or we may be unable to sufficiently hedge against fluctuations in prices for jet fuel.
- We are dependent on our ability to obtain sufficient funding to continue our business activities and, as a result of COVID-19, are exposed to significantly increased uncertainties in relation thereto. In the event that we are unable to secure sufficient funding, we may be unable to maintain adequate liquidity.
- We are subject to significant limitations in our corporate and financial flexibility under the state aid received and financial stabilization measures obtained following the outbreak of COVID-19 and could be required to partially or fully repay such state aid and/or stabilization measures in the event of non-compliance with such limitations.
- The highly competitive airline industry as a whole is susceptible to the effects of adverse economic developments, which may lead to lower demand for flights and overcapacity in the market and, as a result, reduced fares and profitability.
- We may be exposed to legal sanctions and penalties as we may not be able to prevent or detect violations of legal compliance and other economic and administrative regulations, including violations of trade and economic sanctions, anti-bribery and anti-corruption laws and regulations as our business regularly involves contact with representatives of national governments and other officials, who may be in the focus of anti-bribery and anti-corruption laws and regulations.
- We face significant labor law implications as the working conditions and circumstances, primarily due to the effects of COVID-19, have made it necessary for us to apply reductions in pay, implement pay freezes, short-time work and lay-offs, and we ultimately may not be able to retain or attract qualified staff. Additionally, we are and could become subject to claims in connection with labor agreements and may, as a result, be required to make additional payments to our employees or increase or form provisions.

C. Key Information on the Securities

C.1. *What are the main features of the securities?*

The Shares are ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*), each such share representing a notional share in the share capital of the Company of €2.56. The ISIN of the New Shares is DE0008232125. The Shares are denominated in euro. As of the date of this Prospectus, the Company has 597,742,822 shares outstanding. All existing Shares are fully paid in. After the Offering (as defined below), the Company will have 1,195,485,644 shares outstanding. Each Share carries one vote at the Company’s general meeting (*Hauptversammlung*). There are no restrictions on voting rights. The Shares participate in any liquidation proceeds in proportion to their notional share in the share capital of the Company. The Shares rank equally with all other shares of their class in the event of insolvency. They rank junior to all our liabilities. The German Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) stipulates that the Shares must be registered shares with restricted transferability, such that the transfer of such shares requires the consent of the Company. All shares of the Company, including the New Shares, will therefore be issued as registered shares with no-par value and with restricted transferability. Following the exercise of subscription rights, a shareholder is generally required to record its identity and nationality in the Company’s share register. In the event of a corporation or other legal entity, nationality is determined by the country of registration.

The New Shares carry full dividend rights as from January 1, 2021, and for all subsequent financial years. Shareholders participate in any dividend paid in proportion to their notional share in the share capital of the Company. The Lufthansa Group’s dividend policy is to distribute to shareholders 20% to 40% of its net profit, adjusted for non-recurring gains and losses. However, the Company is obligated not to pay dividends to shareholders as long as any amounts under the WSF stabilization financings remain outstanding and for as long as the WSF is a shareholder of the Company. We are not entitled to distribute any dividends, including for the fiscal year ending December 31, 2021, for as long as the WSF remains a shareholder of the Company. The WSF has agreed to sell all of its Company’s shares by not later than December 31, 2023, provided that the Silent Participation I (as defined below), including interest and any additional payments, and the silent participation II in the aggregate amount of €1.0 billion (the “**Silent Participation II**”), including interest, have been fully repaid and subject to achieving a certain minimum sales price of (i) the market price or (ii) €2.56 per Company’s share plus 12% p.a. calculated for the period between acquisition of the Company’s shares by the WSF and the divestment, whichever is higher (the “**Minimum Sales Price**”). The Minimum Sales Price is subject to adjustments in the event of certain corporate actions. However, if the WSF subscribes and receives New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I, the WSF shall sell all of its Company’s shares within 18 months following full repayment of the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, and subject to achieving the Minimum Sales Price, which is being adjusted to reflect the weighted average of the purchase prices paid by the WSF in relation to existing shares of the Company and the New Shares. However, such 18-months period commences only after the six-months lock-up period in relation to existing shares of the Company and the New Shares, which the WSF agreed to, has expired, if the repayment of the Silent Participation I and Silent Participation II occurs prior to expiration of such lock-up. Following expiration of the applicable period and provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have been repaid in full, the Company shall have the right to request that the WSF sells all of its Company’s shares to investors designated by the Company at the Minimum Sales Price.

C.2. *Where will the securities be traded?*

The Company, together with Deutsche Bank, who is acting as listing agent, will apply for admission to trading of the New Shares under the symbol “LHA” on the regulated market segment (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with simultaneous admission to the sub-segment thereof with additional post-admission obligations (Prime Standard).

C.3. *What are the key risks attached to the securities?*

The following risks are key risks specific to the Company’s shares:

- The terms of the framework agreement regarding granting stabilization measures entered into between the WSF and the Company and certain other stabilization measures place significant restrictions on us. Additionally, the interests of the WSF may not necessarily coincide with the interests of other shareholders.
- No assurance can be given that a trading market will develop for the subscription rights and, if a market does develop, the subscription rights may be subject to greater volatility than the Shares.
- Failure by a shareholder of the Company to exercise allocated subscription rights during the applicable exercise period will result in a dilution of such shareholder’s percentage ownership of the Shares.

D. Key Information on the Offer of Securities to the Public and Admission to Trading on a Regulated Market

D.1. *Under which conditions and timetable can I invest in this security?*

The Offering. The offering relates to 597,742,822 newly issued ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*) of the Company from a combined capital increase utilizing authorized capital against (i) contribution in cash with indirect subscription rights (*mittelbare Bezugsrechte*) to existing shareholders, except for the WSF, pursuant to Section 186 para. 5 of the German Stock Corporation Act (*Aktiengesetz*), and (ii) contribution in kind by the WSF with direct subscription rights, if and to the extent that the WSF participates in the capital increase, resolved by the Executive Board on September 19, 2021, with approval of the supervisory board of the Company (*Aufsichtsrat*) (the “**Supervisory Board**”) on the same date, to increase the share capital from €1,530,221,624.32 by €1,530,221,624.32 to €3,060,443,248.64 by issuing 597,742,822 New Shares. The New Shares will be offered to existing shareholders at a subscription ratio of 1:1, i.e., one New Share may be acquired at the Subscription Price (as defined below) for one existing share of the Company (the “**Subscription Offer**”). Subscription rights which are not exercised during the Subscription Period (as defined below) will lapse and be of no value. No compensation will be payable for subscription rights that are not exercised.

The Subscription Offer will include (i) a public offering in Germany, (ii) private placements in certain jurisdictions outside Germany and the United States of America (“**United States**”) in offshore transactions in reliance on Regulation S under the U.S. Securities Act of 1933, as amended (the “**Securities Act**”), and (iii) private placements in the United States pursuant to an exemption from registration under the Securities Act to qualified institutional buyers as defined in Rule 144A under the Securities Act. The New Shares and the subscription rights have not been and will not be registered under the Securities Act, or the securities laws of any other jurisdiction of the United States, its territories and possessions, any state of the United States or the District of Columbia, and may not be offered, sold or otherwise transferred within the United States, except pursuant to an exemption from, or in a transaction not subject to, the registration requirements of the Securities Act and in compliance with any applicable securities laws of any state or other jurisdiction in the United States.

The WSF previously held an interest of 20.00% of the share capital of the Company. On August 16, 2021, the WSF announced the sell-down of up to 5% of the share capital of the Company and, as of September 17, 2021, held an interest of 15.94% of the share capital of the Company. The WSF might sell existing shares of the Company before or during the subscription period (but only to the extent that such sales will not result in a shareholding of the WSF of less than 15% of the share capital of the Company) and, additionally, subscription rights relating to the New Shares that the WSF is entitled to subscribe for during the subscription rights trading from and including September 22, 2021, up to and including September 30, 2021 (until the respective closing auction for subscription rights) on the regulated market (Xetra and Xetra Frankfurt Specialist) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) and in off-market transactions. Furthermore, the WSF might exercise any of its (remaining) subscription rights to subscribe for New Shares in connection with the Offering against contribution in kind of a part of its repayment claim under the Silent Participation I. However, based on guidance provided by the European Commission, the amount of such part of such repayment claim shall not exceed in total the proceeds that the WSF has received in connection with the sell-down and any sale of its subscription rights.

On September 13, 2021, funds and accounts under the management of BlackRock, Inc. held in total 3.40% of the share capital of the Company, i.e., a shareholding of 1.42% of the share capital of the Company and instruments pursuant to Section 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*) relating to 1.98% of the share capital of the Company. Additionally, on September 19, 2021, certain funds and accounts under the management of BlackRock, Inc., which are predominantly different from such funds and accounts holding 3.40% of the share capital of the Company, entered into a sub-underwriting letter with the Underwriters (the “**Sub-Underwriting Letter**”) and committed to purchase New Shares with an aggregate value of up to €300 million that were neither subscribed during the Subscription Period (as defined below) nor placed in the Rump Placement (as defined below) and, on September 17, 2021, committed to exercise any subscription rights in relation to the Company’s shares held by them. The Sub-Underwriting Letter is subject to customary closing conditions.

Each member of the Executive Board has committed to participate in the Offering (as defined below) by fully exercising all subscription rights received in relation to the Company’s shares.

Any New Shares that are not subscribed for in the Subscription Offer or by the WSF (the “**Rump Shares**”) will be offered by the Underwriters in a rump placement by way of private placements (i) to selected qualified investors in certain jurisdictions outside the United States in offshore transactions in reliance on Regulation S under the Securities Act, and (ii) in the United States to qualified institutional buyers in reliance on Rule 144A under the Securities Act (the “**Rump Placement**” and, together with the Subscription Offer, the “**Offering**”).

Timetable. The following is a summary expected timetable for the Offering:

September 20, 2021	• Approval of this Prospectus by BaFin • Publication of this Prospectus on the Company's website
September 21, 2021	• Publication of the Subscription Offer in the German Federal Gazette (<i>Bundesanzeiger</i>)
September 22, 2021	• Commencement of Subscription Period (as defined below) • Commencement of subscription rights trading • "Ex rights" trading of the Company's shares • Credit of the subscription rights by the depositary banks to the depositary accounts of the Company's shareholders expected based on their holdings in the Shares as of September 21, 2021, in the evening, under consideration of open stock-exchange transactions (entitlement date)
September 30, 2021	• End of subscription rights trading
October 5, 2021	• End of Subscription Period (as defined below) • Last day for payment of the Subscription Price (as defined below)
October 6, 2021	• Announcement of the results of the Subscription Offer on the Company's website • Rump Placement • Announcement of the final results of the Offering on the Company's website
October 7, 2021	• Registration of the implementation of the capital increase from authorized capital with the commercial register (<i>Handelsregister</i>) expected • Admission of the New Shares to trading on the regulated market segment (<i>regulierter Markt</i>) of the Frankfurt Stock Exchange (<i>Frankfurter Wertpapierbörse</i>) with simultaneous admission to the sub-segment thereof with additional post-admission obligations (Prime Standard) • Publication of the admission in the German Federal Gazette (<i>Bundesanzeiger</i>) and on the website of the Frankfurt Stock Exchange (<i>Frankfurter Wertpapierbörse</i>) (http://www.deutsche-boerse.com)
October 11, 2021	• Commencement of trading in the New Shares by inclusion in the Company's current stock quotation on the regulated market segment (<i>regulierter Markt</i>) of the Frankfurt Stock Exchange (<i>Frankfurter Wertpapierbörse</i>) and the sub-segment thereof with additional post-admission obligations (Prime Standard) • Book-entry delivery of the New Shares subscribed for during the Subscription Period (as defined below) • Book-entry delivery of the Rump Shares placed in the Rump Placement

Terms of the Offering. The subscription price per New Share is €3.58 (the "**Subscription Price**"). The Subscription Price must be paid at the latest on October 5, 2021. Subscription rights must be exercised during the period from and including September 22, 2021, up to and including October 5, 2021 (the "**Subscription Period**"). Instructions by investors regarding the exercise of subscription rights have to be addressed to their respective depositary banks. Investors are recommended to follow the instructions by their depositary banks.

Trading in Subscription Rights. In connection with the Subscription Offer, the subscription rights (ISIN DE000A3E5B58/WKN A3E5B5) for the New Shares will be traded on the regulated market (*regulierter Markt*) (Xetra and Xetra Frankfurt Specialist) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) during the period from September 22, 2021, up to and including September 30, 2021 (until the respective closing auction for subscription rights). Neither the Company nor the subscription agent will apply for admission of the subscription rights to trading on any other stock exchange.

Dilution. Shareholders who fully exercise their subscription rights for the New Shares will maintain their percentage of ownership of the Company's share capital and voting rights following the Offering. To the extent that shareholders do not exercise any of their subscription rights, based on 597,742,822 Shares outstanding immediately prior to publication of the Subscription Offer and assuming that 597,742,822 New Shares will be issued, each shareholder's percentage ownership in the Company's share capital and its voting rights would decrease by 50.0% per existing Share.

Expenses. On the basis of the Subscription Price and issuance of 597,742,822 New Shares, the Company expects to incur commissions, fees and other expenses related to the Offering in the amount of approximately €72.3 million. No expenses will be charged to investors by the Company or the Underwriters. Investors may, however, have to bear customary transaction and handling fees charged by their account-keeping financial institutions in connection with the subscription of the New Shares as well as for the sale and purchase of subscription rights.

Admission to Trading. The application for admission to trading of 597,742,822 New Shares will be filed on or around September 21, 2021. The commencement of trading in the New Shares is expected to occur on or about October 11, 2021, by inclusion of each of the New Shares on the regulated market segment (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with simultaneous admission to the sub-segment thereof with additional post-admission obligations (Prime Standard).

D.2. Who is the offeror and the person asking for admission to trading?

Offerors. In addition to the Company, the New Shares are being offered by BofA Securities, Deutsche Bank, Goldman Sachs, J.P. Morgan, Barclays, BNP PARIBAS, COMMERZBANK, HSBC, UniCredit Bank, Crédit Agricole Corporate and Investment Bank, DZ BANK, Landesbank Baden-Württemberg, SMBC Nikko and Société Générale.

Admission to Trading. The Company, together with Deutsche Bank, who is acting as listing agent, will apply for admission to trading of 597,742,822 New Shares on the regulated market segment (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with simultaneous application for the sub-segment thereof with additional post-admission obligation (Prime Standard).

D.3. *Why is this prospectus being produced?*

Reasons for the Offering and the Listing and Use of Proceeds. As of the date of this Prospectus, the Company has drawn an amount of €1,500 million under the silent participation I (*Stille Einlage I*) provided by the WSF (the “**Silent Participation I**”). The Company will use the net proceeds from the Offering in the amount of approximately €2,068 million to repay the drawn and repayable amounts of the Silent Participation I. The net proceeds from the Offering and, correspondingly, the amount repayable under the Silent Participation I will decrease, if and to the extent that the WSF subscribes for New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I. Additionally, the Company intends to use the remaining net proceeds from the Offering to strengthen its equity base as well as its liquidity position, to increase the Lufthansa Group’s flexibility to respond to its evolving financing requirements.

Proceeds. The net proceeds to the Company from the Offering result from the gross proceeds less the underwriting commissions and other expenses described below. On the basis of a subscription price of €3.58 per New Share and issuance of 597,742,822 New Shares, the Company is seeking to raise gross proceeds of approximately €2,140 million in this Offering. The overall commissions to be paid by the Company to the Underwriters are expected to amount to approximately €58.8 million. Other issue costs to be incurred by the Company are expected to amount to approximately €13.5 million. Investors will not be charged with expenses by the Company or the Underwriters. On this basis, the Company expects net proceeds from this Offering of approximately €2,068 million. However, the net proceeds from the Offering and, correspondingly, the amount repayable under the Silent Participation I will decrease, if and to the extent that the WSF subscribes for New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I.

Underwriting. The Company and the Underwriters entered into an underwriting agreement dated September 19, 2021 (the “**Underwriting Agreement**”), which provides for a firm underwriting of the New Shares not sold in the Offering by the Underwriters or subscribed for by the WSF. The Underwriters may offer or sell such New Shares, which they are required to subscribe for as underwriters otherwise than in connection with the Offering (including by way of a coordinated sell-down). The Joint Global Coordinators (on behalf of themselves and the other Underwriters) may also arrange sub-underwriting for some, all or none of the New Shares. The Offering is subject to, among other things, registration of the implementation of the capital increase in the commercial register of the Company, which is expected to occur on or about October 7, 2021.

Most Material Conflicts of Interest Pertaining to the Offering. Each of the Underwriters, including Société Générale holding 0.74% of the share capital of the Company and instruments pursuant to Section 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*) relating to 4.28% of the share capital of the Company, and their respective affiliates may have engaged in transactions with, and provided various commercial banking, investment banking, financial advisory transactions and services in the ordinary course of their business with the Company and/or its affiliates for which they would have received customary fees and commissions. Each of the Underwriters and their respective affiliates may provide such services to the Company and/or its affiliates in the future.

All of our Executive Board members as well as all of our Supervisory Board members, except for Dr. Thomas Enders, Jürgen Jennerke, Dr. Michael Kerkloh, Harald Krüger, Miriam Sapiro, Ilja Schulz, Britta Seeger, Dr. Astrid Stange, and Angela Titzrath, directly or indirectly, hold Shares, stock option rights or phantom shares of the Company. As a result, they have a financial interest in the development of the Company’s share price, which constitutes a potential conflict of interest.

Since the Company will utilize the net proceeds from the Offering to repay amounts drawn under the Silent Participation I, the WSF has a financial interest in Offering.

II. GERMAN TRANSLATION OF THE SUMMARY – ZUSAMMENFASSUNG

A. Einleitung mit Warnhinweisen

Dieser Prospekt (der „**Prospekt**“) bezieht sich auf das Angebot und die Zulassung zum Handel von 597.742.822 neu ausgegebenen vinkulierten Namensaktien (die „**Neuen Aktien**“ und jede für sich, eine „**Neue Aktie**“) der Deutsche Lufthansa Aktiengesellschaft, einer deutschen Aktiengesellschaft mit Sitz in Köln, Deutschland, und eingetragene Geschäftsanschrift Venloer Straße 151-153, 50672 Köln, Deutschland, Rechtsträgerkennung (*Legal Entity Identifier*) („**LEI**“) 529900PH63HYJ86ASW55 (die „**Gesellschaft**“ und, gemeinsam mit ihren konsolidierten Tochtergesellschaften, die „**Lufthansa Gruppe**“, „**wir**“, „**uns**“ und „**unsere**“). Die internationale Wertpapier-Identifikationsnummer (*International Securities Identification Number*) („**ISIN**“) der Aktien der Gesellschaft lautet DE0008232125. Die Neuen Aktien werden von BofA Securities Europe SA, 51 rue La Boétie, 75008 Paris, Frankreich, LEI 549300FH0WJAPEHTIQ77 („**BofA Securities**“), Deutsche Bank Aktiengesellschaft, Taunusanlage 12, 60325 Frankfurt am Main, Deutschland, LEI 7LTWFZYICNSX8D621K86 („**Deutsche Bank**“), Goldman Sachs Bank Europe SE, Marienturm, Taunusanlage 9-10, 60329 Frankfurt am Main, Deutschland, LEI 8IBZUGJ7JPLH368JE346 („**Goldman Sachs**“), J.P. Morgan AG, Taunustor 1, 60310 Frankfurt am Main, Deutschland, LEI 549300ZK53CNGEEI6A29 („**J.P. Morgan**“ und, zusammen mit BofA Securities, Deutsche Bank und Goldman Sachs, die „**Joint Global Coordinators**“), Barclays Bank Ireland PLC, One Molesworth Street, Dublin 2, D02 RF29, Irland, LEI 2G5BKIC2CB69PRJH1W31 („**Barclays**“), BNP PARIBAS SA, 16 Boulevard des Italiens, 75009 Paris, Frankreich, LEI R0MUWSFPU8MPRO8K5P83 („**BNP PARIBAS**“), COMMERZBANK Aktiengesellschaft, Kaiserstraße 16 (Kaiserplatz), 60311 Frankfurt am Main, Deutschland, LEI 851WYGNLUQLFZBSYGB56 („**COMMERZBANK**“), HSBC Trinkaus & Burkhardt AG, Königsallee 21/23, 40212 Düsseldorf, Deutschland, LEI JUNT405OW8OY5GN4DX16 („**HSBC**“), UniCredit Bank AG, Arabellastraße 12, 81925 München, Deutschland, LEI 2ZCNRR8UK83OBTEK2170 („**UniCredit Bank**“), Crédit Agricole Corporate and Investment Bank, 12 Place des Etats-Unis, CS 70052, 92547 Montrouge Cedex, Frankreich, LEI 1VUV7VQFKUOQJSJ21A208 („**Crédit Agricole Corporate and Investment Bank**“), DZ BANK AG Deutsche Zentral-Genossenschaftsbank, Frankfurt am Main, Platz der Republik, 60325 Frankfurt am Main, Deutschland, LEI 529900HNOAA1KXQJUQ27 („**DZ BANK**“), Landesbank Baden-Württemberg, Am Hauptbahnhof 2, 70173 Stuttgart, Deutschland, LEI B81CK4ESI35472RHJ606 („**Landesbank Baden-Württemberg**“), SMBC Nikko Capital Markets Europe GmbH, Neue Mainzer Straße 52-58, 60311 Frankfurt am Main, Deutschland, LEI 5493008794GM6YH5G766 („**SMBC Nikko**“) und Société Générale S.A., 29, boulevard Haussmann, 75009 Paris, Frankreich, LEI O2RNE8IBXP4R0TD8PU41 („**Société Générale**“ und, zusammen mit Barclays, BNP PARIBAS, COMMERZBANK, HSBC, UniCredit Bank, Crédit Agricole Corporate and Investment Bank, DZ BANK, Landesbank Baden-Württemberg und SMBC Nikko, die „**Joint Bookrunners**“ und, zusammen mit den Joint Global Coordinators, die „**Underwriters**“) angeboten.

Dieser Prospekt datiert vom 20. September 2021 und wurde von der Bundesanstalt für Finanzdienstleistungsaufsicht („**BaFin**“) als zuständige Behörde gemäß der Verordnung (EU) 2017/1129 des Europäischen Parlaments und des Rates vom 14. Juni 2017, unter Aufhebung der Richtlinie 2003/71/EG, betreffend den Prospekt, der beim öffentlichen Angebot von Wertpapieren oder bei deren Zulassung zum Handel an einem geregelten Markt zu veröffentlichen ist (die „**Prospektverordnung**“) an diesem Tag gebilligt. Die BaFin ist erreichbar unter Marie-Curie-Str. 24-28, 60439 Frankfurt am Main, Deutschland, telefonisch unter +49 228 4108-0 oder über ihre Website: <http://www.bafin.de>.

*Diese Zusammenfassung sollte als Einleitung zu diesem Prospekt gelesen werden. Jede Entscheidung, in die Neuen Aktien (zusammen mit den bereits existierenden Aktien der Gesellschaft, die „**Aktien**“) zu investieren, sollte auf Grundlage einer Prüfung dieses Prospekts als Ganzes durch den Anleger erfolgen. Anleger könnten ihr investiertes Kapital ganz oder teilweise verlieren. Wird ein Anspruch in Bezug auf die in einem Prospekt enthaltenen Informationen vor Gericht geltend gemacht, könnte der klagende Anleger nach nationalem Recht die Kosten für die Übersetzung dieses Prospekts vor Einleitung des Gerichtsverfahrens zu tragen haben. Zivilrechtlich haften nur diejenigen Personen, die die Zusammenfassung einschließlich etwaiger Übersetzungen vorgelegt und übermittelt haben, und dies auch nur für den Fall, dass die Zusammenfassung, wenn sie zusammen mit den anderen Teilen des Prospekts gelesen wird, irreführend, unrichtig oder widersprüchlich ist oder dass sie, wenn sie zusammen mit den anderen Teilen dieses Prospekts gelesen wird, nicht die Basisinformationen vermittelt, die in Bezug auf Anlagen in die betreffenden Wertpapiere für die Anleger eine Entscheidungshilfe darstellen würden.*

B. Basisinformationen über die Emittentin

B.1. Wer ist die Emittentin der Wertpapiere?

Der rechtliche und kommerzielle Name der Gesellschaft lautet „Deutsche Lufthansa Aktiengesellschaft“. Die Gesellschaft ist eine Aktiengesellschaft nach deutschem Recht. Die Gesellschaft hat ihre eingetragene Geschäftsanschrift in Venloer Straße 151-153, 50672 Köln, Deutschland. Die Gesellschaft ist im Handelsregister des Amtsgerichts Köln, Deutschland, unter der Nummer HRB 2168 eingetragen. Die LEI der Gesellschaft lautet 529900PH63HYJ86ASW55.

Haupttätigkeiten. Wir sind die führende Airline-Gruppe in Europa und eine der größten Airline-Gruppen weltweit nach Gesamtumsatz (*Quelle:* Statista, Leading Airline Groups in Europe in 2020, Gesellschaftsdaten). Die Fluggesellschaften, die den Kern der Lufthansa Gruppe bilden, haben ihre Heimatmärkte in Deutschland, der Schweiz, Österreich und Belgien. Als Airline-Gruppe ist unser Kerngeschäft der Betrieb von Inlands-, Kurz- und Langstrecken-Linienflügen im Passagier- und Frachtverkehr in einem weltweiten Netzwerk, sowie die Erbringung von logistischen und technischen Dienstleistungen rund um den Flugbetrieb. Darüber hinaus bieten wir Flugzeugwartung, Reparatur- und Instandsetzungsleistungen sowie Catering an. Im Sechsmonatszeitraum zum 30. Juni 2021 bzw. im Geschäftsjahr zum 31. Dezember 2020 erzielten wir Umsatzerlöse in Höhe von €5.771 Millionen und €13.589 Millionen, verglichen mit €8.335 Millionen bzw. €36.424 Millionen im Sechsmonatszeitraum zum 30. Juni 2020 und im Geschäftsjahr zum 31. Dezember 2019.

Hauptanteilseigner. Nach Kenntnis der Gesellschaft, basierend auf dem Aktienregister der Gesellschaft oder der ihr zugegangenen Stimmrechtsmitteilungen, sind der Wirtschaftsstabilisierungsfonds („**WSF**“) mit 15,94 % am Grundkapital der Gesellschaft beteiligt, The Goldman Sachs Group, Inc. mit 0,08 % am Grundkapital der Gesellschaft beteiligt und hält Instrumente gemäß § 38

Wertpapierhandelsgesetz bezüglich 6,80 % des Grundkapitals der Gesellschaft und Société Générale mit 0,74 % am Grundkapital der Gesellschaft beteiligt und hält Instrumente gemäß § 38 Wertpapierhandelsgesetz bezüglich 4,28 % des Grundkapitals der Gesellschaft.

Beherrschende Anteilseigner. Die Gesellschaft wird durch keinen ihrer unmittelbaren oder mittelbaren Anteilseigner beherrscht.

Hauptgeschäftsführer. Der Vorstand der Gesellschaft (der „**Vorstand**“) besteht aus sechs Mitgliedern: Carsten Spohr (Chief Executive Officer), Remco Steenberg (Chief Financial Officer), Christina Foerster (Chief Customer Officer), Harry Hohmeister (Chief Commercial Officer), Dr. Detlef Kayser (Chief Operations Officer) und Dr. Michael Niggemann (Chief HR & Legal Officer).

Abschlussprüfer. Ernst & Young GmbH Wirtschaftsprüfungsgesellschaft mit Sitz in Stuttgart, Deutschland, handelnd durch ihre Frankfurter Niederlassung, Mergenthalerallee 3-5, 65760 Eschborn, Deutschland („**EY**“) wurde zum Abschlussprüfer des Konzernabschlusses und des Jahresabschlusses der Gesellschaft, jeweils für das Geschäftsjahr zum 31. Dezember 2020, bestellt. Als unabhängiger Abschlussprüfer hat EY den Konzernabschluss und den Jahresabschluss der Gesellschaft jeweils für das Geschäftsjahr zum 31. Dezember 2020 geprüft. Der Abschlussprüfer der Gesellschaft für die Geschäftsjahre zum 31. Dezember 2019 und 31. Dezember 2018 war PricewaterhouseCoopers GmbH Wirtschaftsprüfungsgesellschaft mit Sitz in Frankfurt am Main, Deutschland, handelnd durch ihre Niederlassung in Düsseldorf, Moskauer Straße 19, 40227 Düsseldorf, Deutschland („**PwC**“). PwC hat als unabhängiger Wirtschaftsprüfer die Konzernabschlüsse der Gesellschaft für die Geschäftsjahre zum 31. Dezember 2019 und 31. Dezember 2018 geprüft.

B.2. Welches sind die wesentlichen Finanzinformationen über die Emittentin?

Die Konzernabschlüsse der Lufthansa Gruppe zum 31. Dezember 2020, 2019 und 2018 (die „**Geprüften Konzernabschlüsse**“) wurden in Übereinstimmung mit den International Financial Reporting Standards, wie sie in der Europäischen Union anzuwenden sind („**IFRS**“), und den ergänzend nach § 315e Abs. 1 HGB anzuwendenden handelsrechtlichen Vorschriften aufgestellt und gemäß § 317 HGB und unter Beachtung der vom Institut der Wirtschaftsprüfer festgestellten deutschen Grundsätze ordnungsmäßiger Abschlussprüfung sowie der Verordnung (EU) Nr. 537/2014 des Europäischen Parlaments und des Rates vom 16. April 2014 über spezifische Anforderungen an die Abschlussprüfung bei Unternehmen von öffentlichem Interesse, von EY (für das Geschäftsjahr 2020) geprüft und mit einem uneingeschränkten unabhängigen Bestätigungsvermerk versehen und von PwC (für die Geschäftsjahre 2019 und 2018) geprüft und mit einem uneingeschränkten Bestätigungsvermerk versehen. Der ungeprüfte verkürzte Konzernzwischenabschluss der Gesellschaft zum und für den Sechsmontatszeitraum zum 30. Juni 2021 (der „**Ungeprüfte Verkürzte Konzernzwischenabschluss**“ und zusammen mit den Geprüften Konzernabschlüssen, die „**Konzernabschlüsse**“) wurde in Übereinstimmung mit den für die Zwischenberichterstattung geltenden IFRS Regelungen erstellt.

Die in dieser Zusammenfassung dargestellten Finanzinformationen für das Geschäftsjahr 2019 wurden aus den im geprüften Konzernabschluss 2020 enthaltenen Vergleichszahlen des Vorjahres entnommen, sofern diese Finanzinformationen im geprüften Konzernabschluss 2020 angepasst wurden. Die in dieser Zusammenfassung dargestellten Finanzinformationen für das Geschäftsjahr 2018 wurden aus den im geprüften Konzernabschluss 2019 enthaltenen Vergleichszahlen des Vorjahres entnommen, sofern nicht anders angegeben.

In dieser Zusammenfassung wurden die Finanzinformationen, die in den Tabellen mit „geprüft“ gekennzeichnet sind, den Geprüften Konzernabschlüssen entnommen. Die Bezeichnung „ungeprüft“ wird in Tabellen in dieser Zusammenfassung verwendet, um Finanzinformationen zu kennzeichnen, die dem Ungeprüften Verkürzten Konzernzwischenabschluss oder unseren Buchhaltungsunterlagen oder internen Management-Berichtssystemen entnommen wurden oder die auf der Grundlage von Zahlen aus den oben genannten Quellen berechnet wurden.

Ausgewählte wesentliche Finanzinformationen aus der Konzerngewinn- und Verlustrechnung

	Für den zum 30. Juni endenden Sechsmonatszeitraum		Für das zum 31. Dezember endende Geschäftsjahr		
	2021	2020	2020	2019	2018 ⁽¹⁾
	(ungeprüft)		(geprüft)		
	(in Millionen €, soweit nicht anders angegeben)				
Umsatzerlöse	5.771	8.335	13.589	36.424	35.542
Ergebnis der betrieblichen Tätigkeit.....	(2.050)	(3.288)	(7.089)	1.689	2.800
Ergebnis vor Ertragsteuern	(2.234)	(4.419)	(8.631)	1.860	2.784
Ergebnis nach Ertragsteuern.....	(1.813)	(3.627)	(6.766)	1.245	2.196
Auf Aktionäre der Gesellschaft entfallendes Konzernergebnis	(1.805)	(3.617)	(6.725)	1.213	2.163
„Unverwässertes“/„Verwässertes“ Ergebnis pro Aktie in €	(3,02)	(7,56)	(12,51)	2,55	4,58

(1) Die Zahlen für das Geschäftsjahr zum 31. Dezember 2018 wurden rückwirkend angepasst, um die Re-Klassifizierung von Kompensationszahlungen für Fluganullierungen und -verspätungen zu berücksichtigen.

Ausgewählte wesentliche Finanzinformationen aus der Konzernbilanz

	<u>Zum 30. Juni</u>	<u>Zum 31. Dezember</u>		
	<u>2021</u>	<u>2020</u>	<u>2019</u>	<u>2018</u>
	(ungeprüft)	(geprüft)		
		(in Millionen €)		
Bilanzsumme.....	40.838	39.484	42.659	38.213
Eigenkapital.....	3.145	1.387	10.256	9.573

Ausgewählte wesentliche Finanzinformationen aus der Konzern-Kapitalflussrechnung

	Für den zum 30. Juni endenden Sechsmonatszeitraum		Für das zum 31. Dezember endende Geschäftsjahr		
	2021	2020	2020	2019	2018
	(ungeprüft)		(geprüft)		
	(in Millionen €)				
Nettozu-/ -abflüsse aus operativer Tätigkeit ⁽¹⁾	18	363	(2.328)	4.030	4.109
Nettozu-/ -abflüsse aus der Investitionstätigkeit und Geldanlagen...	(1.362)	(1.083)	(2.342)	(3.867)	(3.269)
Nettozu-/ -abflüsse aus der Finanzierungstätigkeit.....	1.572	534	5.076	(161)	(626)

(1) Wird im ungeprüften verkürzten Konzernzwischenabschluss als operativer Cashflow bezeichnet.

Wesentliche Leistungskennzahlen und alternative Leistungskennzahlen

	Für den zum 30. Juni endenden Sechsmonatszeitraum		Für das zum 31. Dezember endende Geschäftsjahr		
	2021	2020	2020	2019	2018 ⁽¹⁾
	(ungeprüft, soweit nicht anders angegeben)		(in Millionen €, soweit nicht anders angegeben)		
Adjusted EBITDA ⁽²⁾	(970)	(1.578)	(2.890)	4.718	5.016
Adjusted EBIT ⁽²⁾	(2.095)	(2.899)	(5.451) ^(*)	2.026 ^(*)	2.836 ^(*)
Adjusted EBIT Marge (in %) ⁽³⁾	(36,3)	(34,8)	(40,1)	5,6	8,0

(*) Geprüft.

(1) Die Zahlen für das Geschäftsjahr zum 31. Dezember 2018 wurden rückwirkend angepasst um die Re-Klassifizierung von Kompensationszahlungen für Fluganullierungen und –verspätungen zu berücksichtigen.

(2) Die Überleitungen vom EBIT zum bereinigten EBIT (*Adjusted EBIT*) und vom bereinigten EBIT (*Adjusted EBIT*) zum bereinigten EBITDA (*Adjusted EBITDA*) auf Konzernebene sind in der nachfolgenden Tabelle dargestellt.

	Für den zum 30. Juni endenden Sechsmonatszeitraum		Für das zum 31. Dezember endende Geschäftsjahr		
	2021	2020	2020	2019	2018 ^(a)
	(ungeprüft)		(geprüft, soweit nicht anders angegeben)		
	(in Millionen €)				
Umsatzerlöse	5.771	8.335	13.589	36.424	35.542
Bestandsveränderungen und andere aktivierte Eigenleistungen	49	158	175	685	531
Sonstige betriebliche Erträge	655	797	1.897	1.889	1.818
Materialaufwand	(3.204)	(5.127)	(8.453)	(19.827)	(18.367)
Personalaufwand	(2.907)	(3.620)	(6.436)	(9.121)	(8.811)
Abschreibungen	(1.135)	(1.783)	(4.389)	(2.776)	(2.205)
Sonstige betriebliche Aufwendungen	(1.279)	(2.048)	(3.472)	(5.585)	(5.708)
Ergebnis der betrieblichen Tätigkeit	(2.050)	(3.288)	(7.089)	1.689	2.800
Ergebnis aus at equity bewerteten Beteiligungen	(71)	(184)	(276)	88	114
Übriges Beteiligungsergebnis	7	4	12	80	60
EBIT	(2.114)	(3.468)	(7.353)	1.857	2.974
Außerplanmäßige Abschreibung/Zuschreibung	6	552	1.870	139	9
Pensionssondereffekte	(3)	8	31	10	(113)
Ergebnis Anlagenabgang	16	9	1	20	(34)
Adjusted EBIT	(2.095)	(2.899)	(5.451)	2.026	2.836
Planmäßige Abschreibungen	1.125	1.321	2.561	2.692	2.180
Adjusted EBITDA^(b)	(970)	(1.578)	(2.890)	4.718	5.016

(a) Die Zahlen für das Geschäftsjahr zum 31. Dezember 2018 wurden rückwirkend angepasst, um die Re-Klassifizierung von Kompensationszahlungen für Fluganullierungen und –verspätungen zu berücksichtigen.

(b) Ungeprüft.

(3) Definiert als Adjusted EBIT geteilt durch die Umsatzerlöse.

B.3. Welches sind die zentralen Risiken, die für die Emittentin spezifisch sind?

Eine Investition in die Aktien der Gesellschaft ist mit einer Reihe von Risiken verbunden, von denen einige in diesem Abschnitt und im Abschnitt „C.3. Welche wesentlichen Risiken sind mit den Wertpapieren verbunden?“ dieser Zusammenfassung dargestellt sind. Sollte sich eines dieser Risiken verwirklichen, könnten Anleger ihre Investition ganz oder teilweise verlieren. Bei den folgenden Risiken handelt es sich um unternehmensspezifische Schlüsselrisiken:

- Der Ausbruch der COVID-19-Pandemie und ihre Auswirkungen sowie COVID-19 Mutationen und der Ausbruch ähnlicher Krankheiten und ihrer Auswirkungen auf die Märkte, in denen wir tätig sind, sowie die damit verbundenen makroökonomischen Entwicklungen haben die Nachfrage nach unseren Dienstleistungen und die Bedingungen, unter denen wir tätig sind, wesentlich nachteilig beeinträchtigt. Wir gehen grundsätzlich davon aus, dass dies auch weiterhin der Fall sein wird, wobei das Ausmaß der Auswirkungen von COVID-19 von verschiedenen Faktoren abhängt, die außerhalb unserer Kontrolle liegen.
- Unsere Geschäftstätigkeit und Betriebsabläufe reagieren empfindlich auf Änderungen der Einreisebestimmungen unserer Reiseziele, einschließlich obligatorischer Quarantänen und Reiseverbote, sowie dem Verlust von Start- und Landeslots, auch im Zusammenhang mit und als Folge von COVID-19, was letztlich den Rückgang von Ticketverkäufen und den Anstieg der Erstattungsansprüche verstärken kann.
- Wir sind nachteiligen Entwicklungen hinsichtlich unseres Kundenstroms ausgesetzt, der aus unseren Beteiligungen an verschiedenen Joint-Ventures auf bestimmten, insbesondere internationalen Strecken sowie unserer Mitgliedschaft in der „Star Alliance“ resultiert.
- Wir sind einer zunehmenden Anfälligkeit der Nachfrage nach unserem Geschäft ausgesetzt, die sich aus dem zunehmenden Wettbewerb mit alternativen Transportmitteln, einschließlich Bahnreisen, sowie aus Alternativen zu Geschäftsreisen und aus Veränderungen im Verbraucherverhalten oder der anwendbaren rechtlichen Regeln, insbesondere in Bezug auf den Klimawandel und das allgemein gestiegene Umweltbewusstsein, ergeben. Vor diesem Hintergrund könnte es auch erforderlich sein, dass wir bestimmte Kurzstreckenflüge reduzieren oder vollständig einstellen.
- Wir sind nachteiligen Entwicklungen im Zusammenhang mit der Beschaffung von Flugzeugtreibstoff ausgesetzt, insbesondere hinsichtlich der Verfügbarkeit und des Preises, der Anwendbarkeit von Steuerbefreiungsregelungen und unserer Absicherungsgeschäfte für Preisschwankungen von Flugzeugtreibstoff, weil es uns nicht möglich sein könnte, Flugzeugtreibstoff in ausreichender Menge und zu akzeptablen Preisen zu beziehen und/oder wir könnten nicht in der Lage sein, uns gegen Preisschwankungen für Flugzeugtreibstoff ausreichend abzusichern.
- Wir sind von unserer Fähigkeit abhängig, ausreichende Finanzmittel zu erhalten, um unsere Geschäftsaktivitäten fortzusetzen, und sind aufgrund von COVID-19 diesbezüglich erheblich größeren Unsicherheiten ausgesetzt. Falls wir nicht in der Lage sind, ausreichende Finanzmittel zu erhalten, sind wir möglicherweise nicht in der Lage, eine angemessene Liquidität aufrechtzuerhalten.
- Wir unterliegen erheblichen Einschränkungen unserer unternehmerischen und finanziellen Flexibilität nach Maßgabe der staatlichen Beihilfen und finanziellen Stabilisierungsmaßnahmen, die wir nach Ausbruch von COVID-19 erhalten haben, und könnten verpflichtet sein, diese staatlichen Beihilfen und/oder Stabilisierungsmaßnahmen teilweise oder vollständig zurückzuzahlen, sofern wir die jeweiligen Einschränkungen nach Maßgabe der staatlichen Beihilfen und/oder finanziellen Stabilisierungsmaßnahmen nicht einhalten.
- Die wettbewerbsintensive Airline-Branche ist insgesamt anfällig für die Auswirkungen ungünstiger wirtschaftlicher Entwicklungen, die zu einer geringeren Nachfrage nach Flügen, Überkapazitäten im Markt und infolgedessen zu niedrigeren Flugpreisen und geringerer Rentabilität führen können.
- Wir könnten rechtlichen Sanktionen und Strafen ausgesetzt sein, da wir möglicherweise nicht in der Lage sind, Verstöße gegen gesetzliche Compliance- und andere wirtschaftliche und administrative Vorschriften, einschließlich Verstößen gegen Handels- und Wirtschaftssanktionen, Anti-Bestechungs- und Anti-Korruptionsgesetze und -vorschriften, zu verhindern oder aufzudecken, da unser Geschäft regelmäßig den Kontakt mit Vertretern nationaler Regierungen und anderen Beamten beinhaltet, die möglicherweise im Fokus von Anti-Bestechungs- und Anti-Korruptionsgesetzen und -vorschriften stehen.
- Wir sind mit erheblichen arbeitsrechtlichen Auswirkungen konfrontiert, da die Arbeitsbedingungen und -umstände, vor allem aufgrund der Auswirkungen von COVID-19, es erforderlich machen, dass wir Gehaltskürzungen und Gehaltsstopps vornehmen, Kurzarbeit einführen und Entlassungen durchsetzen, und wir letztlich möglicherweise nicht in der Lage sind, qualifiziertes Personal zu halten oder anzuwerben. Außerdem sind wir in Rechtsstreitigkeiten mit unseren Mitarbeitern involviert bzw. können zukünftig in diese Rechtsstreitigkeiten involviert sein, auf Grund derer wir letztendlich verpflichtet sein könnten, Zahlungen an unsere Mitarbeiter zu leisten oder Rückstellungen zu erhöhen oder zu bilden.

C. Basisinformationen über die Wertpapiere

C.1. *Welches sind die wichtigsten Merkmale der Wertpapiere?*

Bei den Aktien handelt es sich um vinkulierte Namensaktien mit einem rechnerischen Anteil am Grundkapital der Gesellschaft von €2,56 je Aktie. Die ISIN der Neuen Aktien lautet DE0008232125. Die Aktien lauten auf Euro. Zum Datum dieses Prospekts hat die Gesellschaft 597.742.822 Aktien im Umlauf. Alle bestehenden Aktien sind voll eingezahlt. Nach dem Angebot (wie unten definiert) wird die Gesellschaft 1.195.485.644 im Umlauf befindliche Aktien haben. Jede Aktie gewährt eine Stimme in der Hauptversammlung der Gesellschaft. Es gibt keine Beschränkungen der Stimmrechte. Die Aktien nehmen an einem etwaigen Liquidationserlös im Verhältnis zu ihrem rechnerischen Anteil am Grundkapital der Gesellschaft teil. Die Aktien sind im Falle einer Insolvenz gleichrangig mit allen anderen Aktien ihrer Klasse. Sie sind gegenüber allen unseren Verbindlichkeiten nachrangig. Das Luftverkehrsnachweissicherungsgesetz schreibt vor, dass die Aktien vinkulierte Namensaktien sein müssen, so dass die Übertragung der Aktien der Zustimmung der Gesellschaft bedarf. Alle Aktien der Gesellschaft, einschließlich der Neuen Aktien, werden daher als vinkulierte Namensaktien ohne Nennwert und mit beschränkter Übertragbarkeit ausgegeben. Nach der Ausübung von Bezugsrechten ist ein Aktionär verpflichtet, seine Identität und Nationalität in das Aktienregister der Gesellschaft einzutragen. Im Falle einer Gesellschaft oder einer anderen juristischen Person richtet sich die Staatsangehörigkeit nach dem Land des Sitzes dieser juristischen Person.

Die Neuen Aktien sind ab dem 1. Januar 2021 und für alle folgenden Geschäftsjahre voll dividendenberechtigt. Die Aktionäre nehmen an einer etwaigen Dividendenausschüttung im Verhältnis ihres rechnerischen Anteils am Grundkapital der Gesellschaft teil. Die Dividendenpolitik der Lufthansa Gruppe sieht vor, 20 % bis 40 % des um nicht wiederkehrende Gewinne und Verluste bereinigten Konzerngewinns an die Aktionäre auszuschütten. Die Gesellschaft ist jedoch verpflichtet, keine Dividenden an die Aktionäre auszuschütten, solange Beträge aus den Stabilisierungsfinanzierungen des WSF ausstehen und solange der WSF Aktionär der Gesellschaft ist. Wir sind nicht berechtigt Dividenden auszuschütten, auch nicht für das am 31. Dezember 2021 endende Geschäftsjahr, solange der WSF Aktionär der Gesellschaft ist. Der WSF hat sich verpflichtet, alle Aktien der Gesellschaft bis spätestens zum

31. Dezember 2023 zu veräußern, sofern die Stille Einlage I (wie unten definiert), einschließlich Zinsen und etwaiger zusätzlicher Zahlungen, und die Stille Einlage II in Höhe von insgesamt €1,0 Milliarden (die „**Stille Einlage II**“), einschließlich Zinsen, vollständig zurückgezahlt sind und ein bestimmter Mindestverkaufspreis von (i) dem Börsenkurs oder (ii) €2,56 je Aktie der Gesellschaft zuzüglich 12 % pro Jahr, berechnet für den Zeitraum zwischen dem Erwerb der Aktien der Gesellschaft durch den WSF und der Veräußerung, erzielt wird, je nachdem, welcher Wert höher ist (der „**Mindestverkaufspreis**“). Der Mindestverkaufspreis wird im Falle bestimmter Unternehmensmaßnahmen angepasst. Zeichnet und erhält der WSF jedoch Neue Aktien gegen Einbringung eines Teils seines Rückzahlungsanspruchs aus der Stillen Einlage I, ist er verpflichtet, alle Aktien der Gesellschaft innerhalb von 18 Monaten nach vollständiger Rückzahlung der Stillen Einlage I, einschließlich Zinsen und etwaiger zusätzlicher Zahlungen, und der Stillen Einlage II, einschließlich Zinsen, und vorbehaltlich des Erreichens des Mindestverkaufspreises, der angepasst wird, um den gewichteten Durchschnitt der vom WSF gezahlten Kaufpreise für die bestehenden Aktien der Gesellschaft und die Neuen Aktien zu berücksichtigen, zu veräußern. Diese Frist von 18 Monaten beginnt jedoch erst nach Ablauf einer mit dem WSF vereinbarten sechsmonatigen Haltefrist in Bezug auf die bestehenden Aktien der Gesellschaft und die Neue Aktien, soweit die Stille Beteiligung I und die Stille Beteiligung II vor Ablauf der Haltefrist zurückgezahlt wird. Nach Ablauf der maßgeblichen Frist und unter der Voraussetzung, dass die Stille Einlage I, einschließlich Zinsen und etwaiger zusätzlicher Zahlungen, und die Stille Einlage II, einschließlich Zinsen, vollständig zurückgezahlt worden sind, hat die Gesellschaft das Recht zu verlangen, dass der WSF alle von ihm gehaltenen Aktien an der Gesellschaft an von der Gesellschaft benannte Investoren zum Mindestverkaufspreis verkauft.

C.2. Wo werden die Wertpapiere gehandelt?

Die Gesellschaft wird gemeinsam mit der Deutschen Bank, die als Zulassungsantragssteller fungiert, die Zulassung zum Handel der Neuen Aktien unter dem Symbol „LHA“ am regulierten Markt der Frankfurter Wertpapierbörse mit gleichzeitiger Zulassung zum Teilbereich des regulierten Marktes mit weiteren Zulassungspflichten (*Prime Standard*) beantragen.

C.3. Welches sind die zentralen Risiken, die für die Wertpapiere spezifisch sind?

Die folgenden Risiken sind wesentliche Risiken, die sich auf die Aktien der Gesellschaft beziehen:

- Die Bedingungen des Rahmenvertrags zur Gewährung von Stabilisierungsmaßnahmen zwischen dem WSF und der Gesellschaft und bestimmte andere Stabilisierungsmaßnahmen legen uns erhebliche Beschränkungen auf. Darüber hinaus müssen die Interessen des WSF nicht unbedingt mit den Interessen anderer Aktionäre übereinstimmen.
- Es kann keine Zusicherung gegeben werden, dass sich ein Markt für die Bezugsrechte entwickeln wird und, falls ein solcher Markt sich entwickeln würde, könnten die Bezugsrechte einer größeren Volatilität als die Aktien unterliegen.
- Übt ein Aktionär der Gesellschaft zugewiesene Bezugsrechte nicht während des maßgeblichen Ausübungszeitraums aus, führt dies zu einer Verwässerung des prozentualen Anteilsbesitzes dieses Aktionärs an den Aktien.

D. Basisinformationen über das öffentliche Angebot von Wertpapieren und die Zulassung zum Handel an einem geregelten Markt

D.1. Zu welchen Konditionen und nach welchem Zeitplan kann ich in dieses Wertpapier investieren?

Angebotsumfang. Das Angebot betrifft 597.742.822 neu ausgegebene vinkulierte Namensaktien der Gesellschaft aus einer kombinierten Kapitalerhöhung unter Ausnutzung des genehmigten Kapitals gegen (i) Bareinlagen mit mittelbaren Bezugsrechten der Altaktionäre mit Ausnahme des WSF gemäß § 186 Abs. 5 AktG und (ii) Sacheinlagen des WSF mit unmittelbarem Bezugsrecht, wenn und soweit der WSF an der Kapitalerhöhung teilnimmt, durch Beschluss des Vorstands vom 19. September 2021, mit Zustimmung des Aufsichtsrats der Gesellschaft (der „**Aufsichtsrat**“) vom gleichen Tag, das Grundkapital von €1.530.221.624,32 um €1.530.221.624,32 auf €3.060.443.248,64 durch Ausgabe von 597.742.822 Neuen Aktien zu erhöhen. Die Neuen Aktien werden den Altaktionären in einem Bezugsverhältnis von 1:1 angeboten, d.h., eine Neue Aktie kann zum Bezugspreis (wie unten definiert) für eine bestehende Aktie der Gesellschaft bezogen werden (das „**Bezugsangebot**“). Bezugsrechte, die während der Bezugsfrist (wie unten definiert) nicht ausgeübt werden, werden verfallen und wertlos sein. Für nicht ausgeübte Bezugsrechte ist keine Entschädigung zu zahlen.

Das Bezugsangebot umfasst (i) ein öffentliches Angebot in Deutschland, (ii) Privatplatzierungen in bestimmten Jurisdiktionen außerhalb Deutschlands und der Vereinigten Staaten von Amerika („**Vereinigte Staaten**“) im Rahmen von sog. Offshore-Transaktionen gemäß Regulation S des U.S. Securities Act von 1933 in seiner jeweils gültigen Fassung (der „**Securities Act**“) und (iii) Privatplatzierungen in den Vereinigten Staaten gemäß einer Ausnahme von der Registrierungspflicht nach dem Securities Act an qualifizierte institutionelle Käufer im Sinne von Rule 144A des Securities Act. Die Neuen Aktien und die Bezugsrechte wurden und werden nicht gemäß dem Securities Act oder den Wertpapiergesetzen in einer anderen Jurisdiktion der Vereinigten Staaten, ihrer Territorien und Besitztümer, eines Bundesstaates der Vereinigten Staaten oder des *District of Columbia* registriert und dürfen nicht innerhalb der Vereinigten Staaten angeboten, verkauft oder anderweitig übertragen werden, es sei denn, dies geschieht gemäß einer Befreiung von den Registrierungserfordernissen des Securities Act oder im Rahmen einer Transaktion, die nicht den Registrierungserfordernissen des Securities Act unterliegt, und in Übereinstimmung mit den geltenden Wertpapiergesetzen eines Bundesstaates oder einer anderen Jurisdiktion der Vereinigten Staaten.

Der WSF war zuvor mit einem Anteil von 20,00 % am Grundkapital der Gesellschaft beteiligt. Am 16. August 2021 kündigte der WSF den Verkauf von bis zu 5 % des Grundkapitals der Gesellschaft an und hielt am 17. September 2021 einen Anteil von 15,94 % am Grundkapital der Gesellschaft. Der WSF wird möglicherweise bestehende Aktien der Gesellschaft vor oder während der Bezugsfrist verkaufen (jedoch nur insoweit, als diese Verkäufe nicht dazu führen, dass der WSF weniger als 15 % des Grundkapitals der Gesellschaft hält) und zusätzlich, Bezugsrechte für die Neuen Aktien, zu deren Zeichnung er berechtigt ist, im Bezugsrechtshandel im Zeitraum vom einschließlich 22. September 2021 bis einschließlich 30. September 2021 (bis zur jeweiligen Schlussauktion für Bezugsrechte) im regulierten Markt (Xetra und Xetra Frankfurt Spezialist) der Frankfurter Wertpapierbörse und in außerbörslichen Geschäften veräußern. Darüber hinaus könnte der WSF möglicherweise (verbleibende) Bezugsrechte ausüben, um Neue Aktien im Zusammenhang mit dem Angebot gegen Sacheinlage eines Teils seines Rückzahlungsanspruchs aus der Stillen Einlage I zu zeichnen. Nach entsprechender Abstimmung mit der Europäischen Kommission würde der Betrag dieses Teils des Rückzahlungsanspruches

jedoch nicht den Betrag der Erlöse übersteigen, die der WSF durch den Aktien- und einen etwaigen Bezugsrechtsverkauf insgesamt erhalten hat.

Am 13. September 2021 hielten von BlackRock, Inc. verwaltete Investmentfonds insgesamt 3,40 % des Grundkapitals der Gesellschaft, d. h. eine Beteiligung von 1,42 % am Grundkapital der Gesellschaft und Instrumente gemäß § 38 des Wertpapierhandelsgesetzes bezüglich 1,98 % des Grundkapitals der Gesellschaft. Darüber hinaus haben bestimmte Investmentfonds unter der Verwaltung von BlackRock, Inc., die sich überwiegend von den Investmentfonds unterscheiden, die 3,40 % des Grundkapitals der Gesellschaft halten, am 19. September 2021 eine Übernahmeerklärung mit den Underwriter (die „**Übernahmeerklärung**“) abgeschlossen und sich darin verpflichtet, Neue Aktien im Gesamtwert von bis zu €300 Millionen zu erwerben, die weder während der Bezugsfrist (wie unten definiert) bezogen wurden, noch im Rahmen der Platzierung nicht bezogener Aktien (*Rump Placement*) (wie unten definiert) erworben wurden, sowie sich am 17. September 2021 verpflichtet, etwaige Bezugsrechte in Bezug auf die von ihnen gehaltenen Aktien der Gesellschaft auszuüben. Die Übernahmeerklärung unterliegt den üblichen Vollzugsbedingungen.

Jedes Mitglied des Vorstands hat sich bereit erklärt, an dem Angebot (wie unten definiert) teilzunehmen und Bezugsrechte für die jeweils gehaltenen Aktien vollständig auszuüben.

Sämtliche Neuen Aktien, die nicht im Rahmen des Bezugsangebots oder vom WSF gezeichnet werden (die „**Restaktien**“), werden von den Underwritern im Wege von Privatplatzierungen (i) an ausgewählte qualifizierte Anleger in bestimmten Jurisdiktionen außerhalb der Vereinigten Staaten in Offshore-Transaktionen unter Berufung auf Regulation S des Securities Act und (ii) in den Vereinigten Staaten an qualifizierte institutionelle Käufer unter Berufung auf Rule 144A des Securities Act angeboten (die „**Platzierung nicht bezogener Aktien (Rump Placement)**“) und zusammen mit dem Bezugsangebot, das „**Angebot**“).

Zeitplan. Die folgende Tabelle zeigt eine Zusammenfassung des Zeitplans für das Angebot:

20. September 2021	• Billigung dieses Prospekts durch die BaFin • Veröffentlichung dieses Prospekts auf der Website der Gesellschaft
21. September 2021	• Veröffentlichung des Bezugsangebots im Bundesanzeiger
22. September 2021	• Beginn der Bezugsfrist (wie unten definiert) • Beginn des Bezugsrechtshandels • „Ex-Bezugsrecht“-Handel der Aktien der Gesellschaft • Gutschrift der Bezugsrechte durch die Depotbanken auf die Depotkonten der Aktionäre der Gesellschaft erwartet, auf Grundlage der Bestände vom 21. September 2021, abends, unter Berücksichtigung der offenen Börsengeschäfte (Berechtigungs-Tag)
30. September 2021	• Ende des Bezugsrechtshandels
5. Oktober 2021	• Ende der Bezugsfrist (wie unten definiert) • Letzter Tag für die Zahlung des Bezugspreises (wie unten definiert)
6. Oktober 2021	• Bekanntgabe des Ergebnisses des Bezugsangebots auf der Website der Gesellschaft • Platzierung nicht bezogener Aktien (<i>Rump Placement</i>) • Bekanntgabe der Ergebnisse des Angebots auf der Website der Gesellschaft
7. Oktober 2021	• Eintragung der Durchführung der Kapitalerhöhung aus genehmigtem Kapital in das Handelsregister erwartet • Zulassung der Neuen Aktien zum Handel im regulierten Markt an der Frankfurter Wertpapierbörse mit gleichzeitiger Zulassung zum Teilbereich des regulierten Marktes mit weiteren Zulassungsfolgepflichten (<i>Prime Standard</i>) • Veröffentlichung der Zulassung im Bundesanzeiger und auf der Internetseite der Frankfurter Wertpapierbörse (http://www.deutsche-boerse.com)
11. Oktober 2021	• Aufnahme des Handels der Neuen Aktien durch Einbeziehung in die laufende Notierung der Gesellschaft im regulierten Markt an der Frankfurter Wertpapierbörse und dessen Teilbereich mit weiteren Zulassungsfolgepflichten (<i>Prime Standard</i>) • Buchmäßige Lieferung der während der Bezugsfrist (wie unten definiert) bezogenen Neuen Aktien • Buchmäßige Lieferung der im Rahmen der Platzierung nicht bezogener Aktien (<i>Rump Placement</i>) platzierten Aktien

Konditionen des Angebots. Der Bezugspreis je Neuer Aktie beträgt €3,58 (der „**Bezugspreis**“). Der Bezugspreis ist spätestens am 5. Oktober 2021 zu zahlen. Bezugsrechte müssen im Zeitraum vom 22. September 2021 bis einschließlich 5. Oktober 2021 (die „**Bezugsfrist**“) ausgeübt werden. Anweisungen von Anlegern bezüglich der Ausübung von Bezugsrechten sind an ihre jeweilige Depotbank zu richten. Den Anlegern wird empfohlen, die Anweisungen ihrer Depotbanken zu befolgen.

Handel von Bezugsrechten. Im Zusammenhang mit dem Bezugsangebot werden die Bezugsrechte (ISIN DE000A3E5B58/WKN A3E5B5) für die Neuen Aktien in der Zeit vom 22. September 2021 bis einschließlich 30. September 2021 (bis zur jeweiligen Schlussauktion für Bezugsrechte) im regulierten Markt (Xetra und Xetra Frankfurt Spezialist) der Frankfurter Wertpapierbörse gehandelt. Weder die Gesellschaft noch die Bezugsstelle werden die Zulassung der Bezugsrechte zum Handel an einer anderen Börse beantragen.

Verwässerung. Aktionäre, die ihre Bezugsrechte für die Neuen Aktien vollständig ausüben, behalten auch nach dem Angebot ihren prozentualen Anteil am Grundkapital und an den Stimmrechten der Gesellschaft. Soweit Aktionäre ihre Bezugsrechte, basierend auf 597.742.822 unmittelbar vor der Veröffentlichung des Bezugsangebots ausgegebenen Aktien und unter der Annahme, dass 597.742.822 Neue Aktien ausgegeben werden, nicht ausüben, würde sich der prozentuale Anteil eines jeden Aktionärs am Grundkapital der Gesellschaft und an den Stimmrechten um 50,0 % je bestehender Aktie verringern.

Gesamtkosten. Auf Grundlage des Bezugspreises und der Ausgabe von 597.742.822 Neuen Aktien erwartet die Gesellschaft, dass ihr Provisionskosten, Gebühren und sonstige Aufwendungen im Zusammenhang mit dem Angebot in Höhe von ca. €72,3 Millionen entstehen werden. Anlegern werden weder von der Gesellschaft noch von den Underwritern Kosten in Rechnung gestellt. Anlegern können jedoch übliche Transaktions- und Bearbeitungsgebühren entstehen, die von ihren kontoführenden Finanzinstituten im Zusammenhang mit dem Bezug der Neuen Aktien sowie für den Verkauf und Kauf von Bezugsrechten erhoben werden.

Zulassung zum Handel. Der Antrag auf Zulassung zum Handel von 597.742.822 Neuen Aktien wird am oder um den 21. September 2021 gestellt. Die Aufnahme des Handels mit den Neuen Aktien wird voraussichtlich am oder um den 11. Oktober 2021 durch Einbeziehung der Neuen Aktien in die laufende Notierung der Gesellschaft am regulierten Markt der Frankfurter Wertpapierbörse mit gleichzeitiger Zulassung zum Teilbereich des regulierten Marktes mit weiteren Zulassungsfolgepflichten (*Prime Standard*) erfolgen.

D.2. Wer ist der Anbieter und die die Zulassung zum Handel beantragende Person?

Anbieter. Neben der Gesellschaft werden die Neuen Aktien von der BofA Securities, der Deutsche Bank, der Goldman Sachs, der J.P. Morgan, der Barclays, der BNP PARIBAS, der COMMERZBANK, der HSBC, der UniCredit Bank, der Crédit Agricole Corporate and Investment Bank, der DZ BANK, der Landesbank Baden-Württemberg, der SMBC Nikko und der Société Générale angeboten.

Zulassung zum Handel. Die Gesellschaft wird, zusammen mit der Deutschen Bank, die als Zulassungsantragssteller fungiert, die Zulassung von 597.742.822 Neuen Aktien zum Handel im regulierten Markt an der Frankfurter Wertpapierbörse mit gleichzeitiger Beantragung für den Teilbereich des regulierten Marktes mit weiteren Zulassungsfolgepflichten (*Prime Standard*) beantragen.

D.3. Weshalb wird der Prospekt erstellt?

Gründe für das Angebot und die Zulassung zum Handel und Verwendung des Erlöses. Zum Datum dieses Prospekts hat die Gesellschaft einen Betrag in Höhe von €1.500 Millionen unter der vom WSF zur Verfügung gestellten stillen Einlage I (die „**Stille Einlage I**“) in Anspruch genommen. Die Gesellschaft wird den Nettoerlös aus dem Angebot in Höhe von etwa €2.068 Millionen zur Rückzahlung der gezogenen und rückzahlbaren Beträge der Stillen Einlage I verwenden. Der Nettoerlös aus dem Angebot und damit der Rückzahlungsbetrag aus der Stillen Einlage I wird sich verringern, wenn und soweit der WSF Neue Aktien gegen Sacheinlage eines Teils seines Rückzahlungsanspruchs aus der Stillen Einlage I zeichnet. Darüber hinaus beabsichtigt die Gesellschaft, den verbleibenden Nettoerlös aus dem Angebot sowohl zur Stärkung der Eigenkapitalbasis als auch der Liquidität zu verwenden, um damit die Flexibilität der Lufthansa Gruppe zu erhöhen und auf sich entwickelnde Finanzierungsanforderungen reagieren zu können.

Erlöse. Die der Gesellschaft aus dem Angebot zufließenden Nettoerlöse resultieren aus den Bruttoerlösen abzüglich Übernahmeprovisionen und anderen im Folgenden beschriebenen Kosten. Auf Basis eines Bezugspreises von €3,58 je Neue Aktie und Ausgabe von 597.742.822 Neuen Aktien strebt die Gesellschaft durch das Angebot Bruttoerlöse in Höhe von ca. €2.140 Millionen an. Die gesamten von der Gesellschaft an die Underwriter zu zahlenden Provisionen betragen voraussichtlich ca. €58,8 Millionen. Weitere auf die Gesellschaft entfallenden Kosten des Angebots werden voraussichtlich ca. €13,5 Millionen betragen. Den Anlegern werden weder von der Gesellschaft noch von den Underwritern Kosten in Rechnung gestellt. Auf dieser Grundlage erwartet die Gesellschaft Nettoerlöse aus dem Angebot in Höhe von ca. €2.068 Millionen. Der Nettoerlös aus dem Angebot und dementsprechend der Rückzahlungsbetrag aus der Stillen Einlage I verringert sich jedoch, wenn und soweit der WSF Neue Aktien gegen Sacheinlage eines Teils seines Rückzahlungsanspruchs aus der Stillen Einlage I zeichnet.

Übernahmevertrag. Die Gesellschaft und die Underwriter haben einen Übernahmevertrag vom 19. September 2021 (der „**Übernahmevertrag**“) abgeschlossen, der eine verbindliche Übernahme der Neuen Aktien vorsieht, die nicht im Rahmen des Angebots von den Underwritern verkauft oder vom WSF gezeichnet werden. Die Underwriter können solche Neue Aktien, die sie als Underwriter zu zeichnen haben, anderweitig als im Zusammenhang mit dem Angebot anbieten oder verkaufen (auch im Wege eines koordinierten Verkaufs). Die Joint Global Coordinators (in ihrem Namen und dem der Underwriter) können auch eine Zeichnung für einige, alle oder keine der Neuen Aktien veranlassen. Das Angebot steht u. a. unter dem Vorbehalt der Eintragung der Durchführung der Kapitalerhöhung in das Handelsregister, die voraussichtlich am oder um den 7. Oktober 2021 erfolgen wird.

Wesentliche Interessenkonflikte im Zusammenhang mit dem Angebot. Jeder der Underwriter, einschließlich der Société Générale die mit 0,74 % am Grundkapital beteiligt ist und Instrumente gemäß § 38 Wertpapierhandelsgesetz bezüglich 4,28 % des Grundkapitals der Gesellschaft hält, und ihre jeweiligen verbundenen Unternehmen haben möglicherweise im Rahmen ihrer gewöhnlichen Geschäftstätigkeit mit der Gesellschaft und/oder ihren verbundenen Unternehmen Transaktionen durchgeführt und verschiedene Transaktionen und Dienstleistungen im Bereich Commercial Banking, Investment Banking und Finanzberatung erbracht, für die sie marktübliche Gebühren und Provisionen erhalten hätten. Jeder Underwriter und seine jeweiligen verbundenen Unternehmen können solche Dienstleistungen für die Gesellschaft und/oder ihre verbundenen Unternehmen in Zukunft erbringen.

Alle Mitglieder unseres Vorstands sowie unseres Aufsichtsrats, mit Ausnahme von Dr. Thomas Enders, Jürgen Jennerke, Dr. Michael Kerkloh, Harald Krüger, Miriam Sapiro, Ilja Schulz, Britta Seeger, Dr. Astrid Stange und Angela Titzrath, halten unmittelbar oder mittelbar Aktien, Aktienoptionsrechte oder Phantomaktien der Gesellschaft. Dadurch haben sie ein finanzielles Interesse an der Entwicklung des Aktienkurses der Gesellschaft, was einen potenziellen Interessenkonflikt darstellt.

Da die Gesellschaft den Nettoerlös aus dem Angebot zur Rückzahlung von Beträgen verwendet, die unter der Stillen Einlage I in Anspruch genommen wurden, hat der WSF ein finanzielles Interesse am Angebot.

1. RISK FACTORS

*Prospective investors should carefully consider the risk factors set out below, together with other information contained in this prospectus (the “**Prospectus**”), before investing in the newly issued ordinary registered no-par value shares with restricted transferability (vinkulierte Namensaktien) (the “**New Shares**” and each, a “**New Share**”) of Deutsche Lufthansa Aktiengesellschaft (the “**Company**” and, together with its consolidated subsidiaries, the “**Lufthansa Group**,” “**we**,” “**us**” or “**our**,”). In each category, the most material risks, in the assessment undertaken by the Company, taking into account the expected magnitude of their negative impact on the Company and the probability of their occurrence, are set out first, with at least the two most material risk factors mentioned at the beginning of each category.*

1.1 Risks Related to Our Business Activities and Operations

1.1.1 The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.

Pandemics, epidemics (see “1.3.5 The airline industry is particularly vulnerable to the effects of epidemics and natural disasters including extreme weather conditions.”), outbreaks of infectious diseases or any other serious public health concerns, such as the outbreak of SARS-CoV-2 first identified in December 2019 and its associated disease, as well as any mutations thereof (“**COVID-19**”), together with any measures aimed at mitigating a further expansion thereof, including restrictions on travel, imposition of quarantines, prolonged closures of workplaces, or curfews or other social distancing measures, have had and in the future will have a material adverse effect on the global economy, international financial markets in general and, in particular, on the markets and segments in which we operate as well as on our liquidity.

In addition to the severe shock caused to the global economy in general, measures related to COVID-19 had a particularly severe effect on the aviation industry, due to their specific negative impact on passengers’ ability and willingness to travel by air and may have generally changed customer behavior.

Such measures include, among others, travel bans and social distancing measures as well as cancellations of business conferences, live entertainment and sports events. This has resulted in a sudden and material decrease in the demand for our services, as well as a significant increase in cancelled bookings, since the outbreak and global spread of the COVID-19 pandemic in early 2020. Among others, the impact of the COVID-19 pandemic on our business is evidenced by the significant decrease in passengers we transported from 145,299 thousand during the fiscal year ended December 31, 2019 (“**Fiscal Year 2019**”), to 36,354 thousand during the fiscal year ended December 31, 2020 (“**Fiscal Year 2020**”). During the six-month period ended June 30, 2021 (“**6M 2021**”), we transported 10,022 thousand passengers (compared to 23,475 thousand passengers transported during the six-month period ended June 30, 2020 (“**6M 2020**”)).

This, in turn, continues to have a material adverse effect on our operational and financial performance since the beginning of 2020. The impact of the COVID-19 pandemic on our operational and financial performance is further evidenced by the significant decrease in our total revenue from €36,424 million during the Fiscal Year 2019 to €13,589 million during the Fiscal Year 2020. During the 6M 2021, we generated a total revenue of €5,771 million (6M 2020: €8,335 million).

The COVID-19 pandemic may also have long-term negative effects on air travel demand due to potential changes in travelers’ perception of the air travel experience and a general reluctance to travel by air. We believe that the demand for air travel bookings in the short-term is particularly affected by the uncertainty concerning the scope and duration of travel restrictions. Additional restrictions or other measures to dissuade people from air travel may also be introduced in the future as a result of the COVID-19 pandemic, prolonging the negative effects of the COVID-19 pandemic on our business. Health precautions such as mandatory disease testing, the temperature screening of passengers, limiting passengers’ movement within the cabin during flights, the mandatory requirements to wear protective face masks or other measures introduced at airports or on flights may make the air travel experience more burdensome for passengers. This may result in lower demand even after the improvement of macroeconomic factors that would normally

support consumer spending, and consequently increase the demand for air travel. The COVID-19 pandemic could also result in a decreased demand for air travel due to the increasing adoption of virtual and tele- and video-conferencing tools that replace business travel (see “1.1.4 *We face increasing susceptibility for the demand of our business as a result of increasing competition from alternative means of transportation, including rail travel, as well as alternatives to business travel, and due to changes in the applicable legal framework or consumer behavior, in particular in relation to climate change and generally increased environmental consciousness.*”), or due to the perceived uncertainty relating to the current pandemic or other similar health threats in the future.

The consequences of COVID-19 on our business depend on a number of factors, including the duration and spread of the respective outbreak(s), the progress of vaccinations as well as the timing, suitability and effectiveness of measures imposed by governments and regional and local authorities (in particular regarding local or regional lockdowns and the imposition of quarantine requirements for individuals, including travelers), the availability of resources, including human, material, infrastructure and financial (e.g., governmental stimulus packages and/or measures introduced by central banks) required to implement effective responses to the respective situation at the international, national and regional level as well as the level of civil compliance with such measures. Moreover, our business is directly and indirectly dependent on measures taken by central banks, such as stimulus or monetary relief measures, as well as cutback measures, and the response of consumers and market participants to such measures. There can be no assurance that such measures, or a combination thereof, are effective means to combat such outbreak(s) or the consequences resulting therefrom, which have already resulted in and may cause a further increase of significant liquidity risk (see “1.2.3 *We may be unable to maintain adequate liquidity.*”), credit risk (see “1.2.1 *We are dependent on our ability to obtain sufficient funding to continue our business activities but, as a result of COVID-19, are exposed to significant uncertainties in relation thereto.*”) and operational risk for us and, ultimately, have had and may further have material adverse effects on our financial condition and results of operations. For example, our equity ratio (unaudited), defined as shareholders’ equity divided by total assets (the “**Equity Ratio**”), decreased from 24.0% as of December 31, 2019, to 3.5% as of December 31, 2020. As of June 30, 2021, the Equity Ratio was 7.7% (compared to 14.3% as of June 30, 2020).

The impact of COVID-19 remained difficult to predict in the past and continue to remain difficult to predict. There can be no assurance that a recovery in passenger traffic, such as during the summer of 2020, will occur again in the future. For example, the International Air Transport Association (the “**IATA**”) reported a decrease of the global passenger load factor during April 2020 by 47% compared to 2019, followed by a reported decline of 58% in July 2020 (source: IATA, Air Passenger Market Analysis, April 2020 and IATA, Air Passenger Market Analysis, July 2020). For the year 2020, IATA forecasted an overall decline of 55% in global passenger numbers compared to 2019 (source: IATA, Press Release: Recovery Delayed as International Travel Remains Locked Down). However, IATA ultimately reported a decrease of the total number of revenue passenger-kilometers (“**RPK**”) sold worldwide by 66% in 2020 (source: IATA, Air Passenger Market Analysis, December 2020). European Airlines posted a 70% decline in RPKs in 2020 (source: IATA, Air Passenger Market Analysis, December 2020). According to the Federal Association of the German Aviation Industry (“**BDL**”), sales for airlines in Germany fell overall by 76% in 2020 compared to 2019 (source: BDL, Bericht zur Lage der Branche 2020). Industry-wide RPKs in June and July 2021 were 60% and 53% below industry-wide RPKs during June and July 2019, respectively (source: IATA, Air Passenger Market Analysis, June 2021 and IATA, Air Passenger Market Analysis, July 2021). In relation to passenger traffic (in terms of RPK), IATA reported a global decrease of 67% in the 6M 2021 compared to the 6M 2019 (source: IATA, Air Passenger Market Analysis, June 2021). IATA is predicting that airlines worldwide will generate USD 458 billion in revenue in 2021, which represents an increase of 23% to 2020 and a decrease of 45% compared to 2019 (source: IATA, Airline Industry Economic Performance, April 2021 and IATA, Industry Statistics, April 2021). In 2021, global passenger traffic (in terms of RPK) is expected to decrease by 57% compared to 2019, resulting in cumulative net losses of USD 48 billion (source: IATA, Airline Industry Economic Performance, April 2021).

Any prolongation or increase of the aforementioned travel bans and related restrictive measures would reduce consumers’ ability to travel and ultimately materially adversely affect our financial condition and results of operations and may lead to an increased liquidity risk (see “1.2.3 *We may be unable to maintain adequate liquidity.*”).

Additionally, we have been and may continue to be required to renegotiate certain contractual arrangements in light of the COVID-19 pandemic and its potential effects on the markets in which we operate as well as related macroeconomic developments, and there is no assurance that we will be able to enter into contractual arrangements on commercially reasonable conditions. Furthermore, counterparties under certain contractual arrangements to which we are a party may fail to perform or may be unable to meet their respective contractual obligations due to the COVID-19 pandemic. Our counterparties may also seek to establish an exclusion of liability or to terminate a contractual arrangement under force majeure clauses or crisis clauses or allege a failure to meet contractual obligations.

1.1.2 Our business activities and operations are sensitive to changes in entry regulations, travel bans and a loss of take-off and landing slots, including in connection with and as a result of COVID-19, which may ultimately intensify the decrease in ticket sales and the increase in reimbursement claims.

In connection with the outbreak of COVID-19, many countries had issued and still have issued current entry regulations and travel bans. Entry regulations, including mandatory quarantines, and travel bans varied and still vary widely from country to country and are subject to constant change. Such entry bans continue to materially affect the flight capacity we offer and our customers' booking behavior and, ultimately, our revenue development.

Many countries, including the United States of America (the **"United States"**), imposed a travel ban for individuals from the European Union. This led to a significant decline in bookings and to flight cancellations across all of Lufthansa German Airlines, SWISS, Austrian Airlines, Brussels Airlines and the other airlines (including Eurowings Discover) reported under our 'Network Airlines' business segment (collectively, the **"Network Airlines"**) as well as Eurowings and Eurowings Europe (collectively, **"Eurowings"**).

In connection with COVID-19, we decided to significantly reduce flight capacity and offered only a reduced short-haul program and certain long-haul routes during the first peak of the COVID-19 pandemic in 2020. For example, our capacity measured in available seat-kilometers (**"ASK"**) decreased to 4% and 3% in April and May 2020, respectively, compared to April and May 2019. The vast majority of the 760 aircraft we operated as of June 30, 2020, were temporarily parked during the first peak of the COVID-19 pandemic. Additionally, we suspended the operations of Austrian Airlines and Brussels Airlines in their entirety for three months. As of June 30, 2021 a total of approximately 200 aircraft (out of 734 as of June 30, 2021) from all Lufthansa Group airlines had been temporarily parked.

Travel bans, restrictions and entry regulations have led and may further lead to a significant decline in bookings and result in flight cancellations across the Network Airlines and Eurowings. Among others, the impact of the resulting flight cancellations is evidenced by the significant decrease of ASK from 358,803 million during the Fiscal Year 2019 to 109,828 million during the Fiscal Year 2020. During the 6M 2021, ASK amounted to 44,171 million (6M 2020: 68,604 million). As a consequence, entry regulations and travel bans have and will continue to significantly affect our ability to run our business activities and operations and, ultimately, negatively affect our financial and business performance and simultaneously increase liquidity risk (see **"1.2.3 We may be unable to maintain adequate liquidity."**).

There can be no assurance as to whether or when these entry regulations and travel bans will be eased or lifted or that, following any easement in the future, such entry regulations and travel bans will not be re-enforced in the future. Additionally, borders may often be closed (and, in some cases, reopened) at very short notice and for periods that are initially unspecified. Any further continuation, intensification or re-enforcement of entry regulations or travel bans may hinder our ability to resume or operate flights to any affected country as scheduled, if at all.

Moreover, as a result of flight cancellations by the Network Airlines and Eurowings in connection with imposed entry regulations and travel bans and the accompanying deterioration of customer demand, the Network Airlines and Eurowings may lose take-off and landing slots that we are required to operate with scheduled services. According to Article 10 of Council Regulation (EEC) No 95/93 of January 18, 1993 on common rules for the allocation of slots at community airports (the **"Slot Allocation Regulation"**), air carriers, such as the Network Airlines and Eurowings, are subject to the **"80/20 rule"** (the **"80/20 Rule"**). Under the 80/20 Rule, air carriers must use at least 80% of their slot series at congested and therefore

coordinated European airports or may lose them to a competitor in the following scheduling period. Due to the spread of COVID-19 and ensuing entry regulations for crew and passengers, air carriers are generally not able to fulfill the requirement of the 80/20 Rule. Following the request by IATA and Airports Council International Europe, the European Commission had suspended the 80/20 Rule during the COVID-19 crisis to allow airlines to reduce frequencies until the end of March 2021. For the summer season 2021, further slot relief was provided in the EU (*source*: European Commission, Aviation: slot relief enacted). 50% of the slot series could be returned early to be protected and the minimum usage requirement was lowered from 80% to 50% for the remaining slot series. Some other countries extended the suspension of the 80/20 Rule (e.g., United States, United Kingdom, Japan), whereas others have made a decision or implemented slot relief broadly comparable to the EU for the summer season 2021. For the winter season 2021, further slot relief was provided in the EU (*source*: European Commission, Aviation: slot relief amended and extended), but limited to a lowered minimum usage requirement of 50% for all slot series. Most other countries either extended the suspension of the 80/20 Rule (e.g., China, Turkey, Brazil) or implemented slot relief broadly comparable to the EU for the summer season 2021. As there can be no assurance whether the competent authorities will approve any such or similar suspensions in the future, the Network Airlines and Eurowings may be unable to retain a significant portion of the take-off and landing slots in the coming flight seasons which they require to maintain their operations and which we require to ultimately maintain our business activities after the end of March 2022. Additionally, following the (re-)application of such slot allocation procedures, our Network Airlines and Eurowings may not be able to ramp up their operations in time, or at all, and may ultimately lose slots at congested airports, whether individually or entirely. Any loss of slots could jeopardize network structures that have been established over decades and that are the base of Lufthansa Group's flight operations. The Lufthansa Group's profitability could be negatively affected especially if these slots are taken over by competitors that start to operate on the same routes offering a competitive pricing.

1.1.3 We are exposed to adverse developments in relation to our customer stream resulting from our various joint venture arrangements on certain, in particular international, routes and our membership in the "Star Alliance."

We are party to numerous interline, codeshare (*i.e.*, marketing agreements pursuant to which one airline operates a flight and whereby other airlines can, in their own name, also sell their own tickets for such flights) and joint business agreements under which we market and cross-sell ticket and cargo capacities to offer customers more travel options and seamless transfers. In addition, many of the international flights we operate include passengers transported through cooperation agreements, predominantly with other Star Alliance member airlines. We derive significant benefits, including enhanced international market access, additional passengers and increased brand recognition, from our membership in the Star Alliance and from co-operations with other partner airlines.

The frameworks and requirements in relation to our joint venture, codeshare offerings as well as interlining- and inter-carrier agreements are increasing in their complexity. We are required to continuously review and re-negotiate revenue-sharing agreements and compliance as well as coordinate between our partner carriers. Ultimately, our financial condition and results of operations could be negatively affected, including by an increasing exposure to a disproportionate interest in the gains or losses generated under any of the aforementioned or similar types of agreements and any failure to effectively manage the increased complexity thereof.

We are a founding member of the Star Alliance, the world's largest airline alliance in terms of market share (measured by ASK) (*source*: Statista, Leading Airline Alliances in 2018, August 2019), and participate in various cooperative and commercial joint venture arrangements, including with United Airlines and Air Canada on routes between Europe and North America, and with All Nippon Airways, Singapore Airlines and Air China on routes between Europe and Japan as well as Singapore and China, respectively. The majority of joint venture agreements to which we are a party provide for the sharing of profits and losses on these flights. Therefore, our results of operations, cash flows and financial condition could be materially affected by negative effects associated with such profit and loss sharing agreements, including if a partner airline generates lower revenue or has a higher cost base than we do, or if a partner airline terminates a cooperation agreement that is profitable for us.

The Star Alliance may not be successful in competing with other airlines or airline alliances in the future. Other airlines or alliances acting in the context of joint venture agreements could newly establish or reinforce existing market positions through mergers, reorganizations or otherwise. Ultimately, certain advantages that we derive from our membership in the Star Alliance could be reduced or eliminated completely. The Star Alliance brings together 26 member airlines as of December 31, 2020. Its combined fleet of over 5,000 aircraft served more than 1,300 destinations in 195 countries worldwide in the Fiscal Year 2019. In the Fiscal Year 2018, the Star Alliance was the largest global airline alliance in terms of market share (measured by ASK) (*source*: Statista, Leading Airline Alliances in 2018, August 2019). Accordingly, any significant adverse developments affecting the Star Alliance, such as one or more principal members leaving the Star Alliance, whether voluntarily or as a result of bankruptcy proceedings or a consolidation with a member of a competing alliance, including in light of COVID-19, any loss of market share or appeal to passengers as a result of changes to its membership composition, particularly in the United States, China and India, could have a material adverse effect on our network and our offering of destinations through feeder flights (flights to carry passengers between non-hub airports and our hub airports in connection with long-distance flights) and connecting flights. In particular, we expect that any such event would also increase customers' tendency to choose an airline other than ours that offer(s) the benefits associated with the membership of another airline alliance or desired destinations.

Additionally, we rely on positive brand recognition to attract customers. Our brands could be harmed by the actions of one or more members of the Star Alliance or partner(s) under joint venture or similar agreements. Any damage to our reputation, brand image or brand name through either a single event or series of events involving, or due to perceptions (such as overall quality) concerning, our position as a member of the Star Alliance could materially adversely affect our ability to market our services and attract and retain customers.

1.1.4 We face increasing susceptibility for the demand of our business as a result of increasing competition from alternative means of transportation, including rail travel, as well as alternatives to business travel, and due to changes in the applicable legal framework or consumer behavior, in particular in relation to climate change and generally increased environmental consciousness.

In densely populated countries and regions in which we operate, such as Germany and Central Europe, we face increasing competition from alternative means of transportation, in particular high-speed trains. With the railway network as a whole and an increase in frequencies, routes that have traditionally been served by us have become more competitive. Among others, passengers who book rail-and-fly tickets, *i.e.*, tickets which include a train journey replacing a typically domestic or neighboring-country flight segment and which are also sold by the Network Airlines, may choose to increasingly rely on train travel within Germany or neighboring countries and, ultimately, contribute to a decrease in demand for shorter-distance domestic or neighboring-country flights. Our efforts to increase a train-and-plane ticket mix, as evidenced by our recent extension of the rail-and-fly offerings within Germany, may accelerate such decrease. Any such decrease in passenger numbers as a result of increased high-speed rail transport service offerings could adversely affect the demand for our business activities.

In addition, Germany and other European countries are generally supportive of a further expansion of rail transport, particularly in the high-speed sector. For example, the German government has launched an extensive investment program, pursuant to which €12.2 billion were scheduled to be invested into, among others, the modernization of 1,800 kilometers of railway to increase availability and connections within Germany in 2020 (*source*: Deutsche Bahn, Investitionen 2020). We are exposed to risks in connection with a shift of customer preferences and political support on national and European levels, including as a result of public opinion. Ultimately, any shift of interest could result in or be driven by direct or indirect subsidies for rail travel, coach transport services or other direct or indirect measures aimed at achieving a reduction of short-distance air travel (for example, due to changes in tax or environmental regulations), and could have a material adverse effect on our business and result of operations.

We estimate that 29% of our passengers were business travelers in the Fiscal Year 2019. Accordingly, a significant amount of our revenue is generated by and we are dependent on business travelers. We may lose a significant share of our business travel customers, including as a result of the further expansion of rail transport (as outlined above). We also face increasing competition from alternatives to business travel, in

particular from methods of electronic communication, such as tele- and video-conferencing, as these technologies continue to develop and become more widely used. The aforementioned technologies may in the future substitute a substantial part or even the majority of meetings previously held in-person. As a result of the ongoing COVID-19 pandemic, video conferences currently replace almost all in-person meetings in their entirety, including in an effort to comply with social distancing requirements and to cope with entry restrictions as well as travel bans. Among others, the impact of the COVID-19 pandemic on our business is evidenced by the significant decrease in passengers we transported from 145,299 thousand during the Fiscal Year 2019 to 36,354 thousand during the Fiscal Year 2020. During the 6M 2021, we transported 10,022 thousand passengers (6M 2020: 23,475 thousand passengers transported). The continued implementation of tele- and video-conferencing in a business context, including as a result of cost- and time-savings, would likely materially change consumer behavior and ultimately result in a material decrease in flight bookings and demand for air travel.

Moreover, air transportation may be perceived as one of the most environmentally unfriendly methods of travel and transport, in particular in terms of carbon emissions. We generated 63.0%, 66.8% and 77.2% of our total revenue from traffic revenue during the 6M 2021 and each of the Fiscal Years 2020 and 2019, respectively.

The demand for our business activities is highly susceptible to flight shaming, which includes an increasing public and political sensitivity on environmental issues, in particular in relation to carbon emissions and climate change in general in the context of air transportation, particularly domestic travel. Air travel is increasingly perceived as a driver for climate change. As a result of significant carbon emissions associated with the airline industry, public discussion tends to express a view that air travel should be reduced and, in particular in relation to domestic flights, be replaced by alternative, more environmentally conscious methods of transportation. Ultimately, any changes of the applicable legal framework or increased sensitivity in relation to environmental issues could result in a decrease of demand for air travel and, therefore, revenue generated from air transportation. Against this background, we may also be required to reduce certain short-haul flights or discontinue such flights entirely.

1.1.5 We are exposed to adverse developments in relation to the sourcing of jet fuel, in particular its availability and price, the applicability of tax exemption regulations for jet fuel and our related fuel hedging activities with respect to price fluctuations, because we may be unable to source jet fuel in sufficient volumes and at acceptable prices and/or we may be unable to sufficiently hedge against fluctuations in prices for jet fuel.

During the 6M 2021, we generated €3,637 million, or 63.0%, of our total revenue, from traffic revenue compared to €5,641 million, or 67.7%, during the 6M 2020 (Fiscal Year 2020: €9,078 million, or 66.8%; Fiscal Year 2019: €28,136 million, or 77.2%). Accordingly, we are significantly dependent on our ability to source sufficient volumes of jet fuel at acceptable prices and, moreover, to successfully hedge against fluctuations in jet fuel prices. Due to the highly competitive nature of the airline industry, we may not be able to pass increases in jet fuel prices on to our customers, and any increases in jet fuel costs or disruptions in fuel supplies could have a material adverse effect on our operations and, in consequence, our cash flows, financial condition and results of operations.

Jet fuel prices and availability have historically fluctuated widely, and are expected to continue to fluctuate in the future as a result of the level of economic activity, the rate of economic growth, political events, weather conditions, refinery outages or maintenance and the coordinated pricing decisions of producer cartels, such as the Organization of Petroleum Exporting Countries (*OPEC*), as well as speculative trading activity regarding the price of oil.

For example, the oil price declined and was very volatile over the course of 2020 with prices ranging between USD 19.33/barrel and USD 68.91/barrel. As of December 31, 2020, a barrel of Brent Crude was at USD 51.80 (year-end 2019: USD 66.00/barrel). The jet fuel crack, *i.e.*, the price difference between crude oil and kerosene, moved between USD -8.32/barrel and USD 16.36/barrel in 2020. On average over the year 2020, it traded at USD 2.56/barrel and thus 84% lower than in the previous year 2019.

Our operational and financial planning measures include a hedging policy. We use standard market instruments for fuel hedging and hedge fuel price risks mostly with option structures.

The volatility of the jet fuel market is evidenced by, among others, a comparison of our jet fuel exposure hedging activities over recent years. Prior to the outbreak of the COVID-19 pandemic, in 2019, we had hedged 73% of the forecasted fuel requirement for the Fiscal Year 2020 with fuel price derivatives designated under hedge accounting and we incurred fuel expenses (aircraft fuel and lubricants) of €6,715 million for the Fiscal Year 2019. In contrast, in 2020, we had hedged 53% of the forecasted fuel requirement for the fiscal year ended December 31, 2021 (“**Fiscal Year 2021**”), with fuel price derivatives designated under hedge accounting and we incurred fuel expenses (aircraft fuel and lubricants) of €1,875 million for the Fiscal Year 2020.

Due to general economic uncertainties, there can be no assurance that our hedging measures in place for the Fiscal Year 2021 were, are or will be sufficient and will not ultimately negatively affect our financial position. Our hedging forecasts are commonly based on the expected fuel consumption, existing hedges designated under hedge accounting, Brent crude oil forward prices and EUR/USD forward exchange rates. As of June 30, 2021, the decline in flight traffic due to COVID-19 meant that fuel prices and foreign currencies were still “over-hedged,” meaning hedging relationships previously designated under hedge accounting rules had to be terminated early and are still not matured. The corresponding hedging instruments are accounted for as profit or loss as standalone derivatives until their due date. The realized result of hedging relationships terminated in the 6M 2021 resulted in an income of €41 million for fuel hedging (Fiscal Year 2020: expense of €203 million).

Due to the significant decline in business activity in the Fiscal Year 2020, there were “overhedges” on fuel hedges and, to a lesser extent, on currencies due to a lack of underlying transactions. These overhedges are recognized in the income statement and negatively affected our financial result by a total of €752 million in the Fiscal Year 2020. Given that we did not enter into new hedging arrangements in relation to our fuel exposure following the outbreak of the COVID-19 pandemic in 2020, we believe that, subject to the accuracy of our underlying forecasts, the effect from fuel overhedging could be less significant in the Fiscal Year 2021. The hedges in place for the Fiscal Year 2021 correspond to a coverage ratio of 36% compared to the fuel consumption for the Fiscal Year 2019. Going forward, we aim to hedge around 65% of our fuel exposure for our airlines, a decrease of around 20% compared to our previous hedging strategy. Our hedging remains options-based and continues to be built on a 24-months’ time horizon and associated assumptions and estimates. Moreover, we have adjusted our hedging structure in an effort to reduce the financial risk of over-hedging, but this in turn, is resulting in an increased exposure to negative effects of fuel price fluctuations. Hedging transactions are generally concluded on the basis of crude oil prices, which can deviate from kerosene prices. However, hedging instruments do not fully protect us against short-term or long-term price increases, as we generally only hedge against specific margins of fluctuation and specific time periods.

In the event that we do not adequately or successfully hedge our fuel price exposure, a significant change in fuel prices could significantly affect our results of operations. Additionally and to the extent that we are unable to pass on jet fuel cost increases to customers or, where such increases occur subsequent to the fixing of a selling price to a customer, we do not hedge this exposure effectively, an increase in costs will have a negative impact on our results of operations.

Moreover, regulations aimed at repealing tax exemptions on kerosene and proposed regulation introducing a gradually increasing mandatory minimum share of sustainable aviation fuels at airports located in the European Union from January 1, 2025, may be enacted at the level of the European Union or in Germany. The elimination of such tax exemption and the proposed mandatory minimum share of sustainable aviation fuels requirement might result in disadvantages compared to our competitors operating outside Europe and could lead to a substantial increase in our jet fuel costs.

1.1.6 Our profitability depends on our ability to accurately estimate capacity development, in particular with a view to commitments in binding aircraft orders which could ultimately prove less profitable than originally anticipated.

Our transport capacity, both in terms of passengers and cargo, is a significant driver for our results of operations and, ultimately, profitability. Typically, a significant portion of our capital expenditures relate to primary investments in down payments and final payments for aircraft, aircraft components, and aircraft and engine overhauls. During the Fiscal Years 2020 and 2019, these amounted to €983 million and €2,998

million, respectively. Due to the long delivery lead-time, aircraft orders are based on long-term estimates and forecasts which ultimately may prove to be inaccurate or incorrect and, moreover, subject to significant general uncertainties. As of June 30, 2021, our order book comprised 178 aircraft for delivery by 2029 with an order commitment of approx. €14 billion. More specifically, we have placed firm orders for 20 Boeing 777, 25 Boeing 787, 28 Airbus A350 for the long-haul fleet across the Network Airlines and one Boeing 777F for Lufthansa Cargo. In relation to our airlines' short-haul fleet, we have placed firm orders for 104 aircraft from the Airbus A320-family. Moreover, cumulatively, we have 74 order options in relation to new aircraft as of June 30, 2021. However, capacity development predictions underlying these orders may turn out to be incorrect. Following a positive outlook for air passenger transport in 2020 (*source*: IATA, Airline Industry Economic Performance, December 2019), as a result of the COVID-19 pandemic, a significant portion of our passenger flight services in 2020 were temporarily suspended, leading to a decrease of 69.4% in our capacity measured in ASK. As of the date of this Prospectus, a significant number of our scheduled long-haul routes continue to be operated at a reduced frequency and capacity. There can be no assurance that the capacity development estimates and predictions underlying the anticipated demand for air transportation by the time of the respective aircraft deliveries were based on an accurate or thorough pattern of assumptions or facts.

As part of our "ReNew" restructuring program, we intend to significantly reduce our fleet through the postponement of new deliveries, the termination of lease agreements, long-term storage and sale. We have, between January 1, 2020, and June 30, 2021, terminated leases for 18 aircraft and sold 39 aircraft (six Boeing MD-11Fs, five Boeing 747s, two Boeing 767, one Airbus A321, seven Airbus A320s, four Airbus A319s, three Bombardier CRJ900 and eleven Bombardier Q Series aircraft). In addition, a total of approximately 200 aircraft (out of 734 aircraft as of June 30, 2021) from all Lufthansa Group airlines had been temporarily parked, as of June 30, 2021. Moreover, we have scheduled approximately 100 aircraft for long-term storage and/or retirement, which includes among others our entire Airbus A380 fleet and is primarily aimed at adjusting the number of flights and the capacity of our fleet to the current projections for the market and demand, which is, as a result of the COVID-19 crisis, expected to remain below previous years' levels in the longer term. However, due to existing contractual obligations, during the Fiscal Year 2020, we took delivery of 18 newly built aircraft, two of which were Boeing 777Fs, two of which were Airbus A350-900s, three of which were Airbus A321neos and eleven of which were Airbus A320neos, as well as four used Airbus A320neos. In the 6M 2021, we took delivery of six newly built aircraft, one of which was an Airbus A320neo, four of which were Airbus A321neo and one of which was an Airbus A220. It is estimated that the demand for passenger air travel will only partially recover from the effects of COVID-19 in the near future and not achieve levels comparable to those of 2019 prior to 2024 (*source*: IATA, Press Release Deep Losses Continue Into 2021). Following discussions with aircraft manufacturers, aircraft delivery schedules have been adjusted in order to align these with decreased financial resources available to us and, in particular, the expected decrease in demand for 2022 and beyond due to COVID-19. As of the date of this Prospectus, we expect to receive 16 aircraft during 2021. The effects of COVID-19 on the demand for passenger air transport remain very difficult to predict and we, as a result of our early aircraft retirement on the one hand and delay in aircraft delivery on the other hand, are and will continue to be exposed to the risks of adverse effects caused by structural over- or under-capacity and a negative effect on our ability to operate our core business.

Simultaneously, any of the aircraft which we ordered or may order in the future could prove to be less profitable than expected, including due to our inability to realize the utilization or revenue targets that we originally envisaged to achieve in connection with these aircraft investments, such as the Airbus A380. Moreover, if any of the aircraft we ordered exceed estimated operating costs or do not meet envisaged technical performance levels or environmental standards, we would be unable to utilize this aircraft as planned or at all.

In addition, any delays in the deliveries of the aircraft we ordered and a delay in or cancellation of their commercial introduction, may result in the forfeiture of market demand and, in turn, negative impact on our revenue generation. Moreover, any newly-introduced aircraft which we have ordered and that have not yet entered the commercial use phase may be subject to delays and grounding orders by the relevant authorities or manufactures.

1.1.7 We may be unable to implement additional cost-saving measures as well as operational and business improvement programs or to ultimately achieve any cost savings anticipated thereunder, which may adversely affect our financial position and business outlook.

We maintain a consistent focus on the improvement of our operations and cost efficiency and revenue quality through relevant projects, initiatives and programs aimed at such improvements, particularly following the outbreak of the COVID-19 pandemic. For example, in response to the effects of COVID-19, we announced our restructuring program “ReNew” in 2020. “ReNew” guides and coordinates our restructuring activities on the basis of four modules, ReStructure, ReOrg, ReFocus and RePay: (i) the combination of the decentralized restructuring programs for our airlines and Aviation Services business segment, including the reduction of our operating fleet and workforce (*ReStructure*) (see “1.5.1 We face significant labor law implications as the working conditions and circumstances, primarily due to the effects of COVID-19, have made it necessary for us to apply reductions in pay, implement pay freezes, short-time work and lay-offs, and we ultimately may not be able to retain or attract qualified staff.”), (ii) the implementation of an efficient organizational structure and the organizational separation of Lufthansa German Airlines functions from Lufthansa Group functions (*ReOrg*), (iii) increasingly focus on the operations of our core airline business, including resuming the sales process for the remaining non-European business of LSG Lufthansa Service Holding AG and its subsidiaries (together, the “**LSG Group**”) and the review of disposals of other non-core companies (e.g., companies not involved in our core airline business) (*ReFocus*), and (iv) the refinancing and repayment of all funding received from government (*RePay*) (see “1.2.2 We are subject to significant limitations in our financial flexibility as a result of the state aid received and financial stabilization measures obtained from governments in response to the outbreak of COVID-19 and face risks associated with triggering events of default under these state aid and stabilization measures.”).

As part of our “ReNew” restructuring program, we intend to structurally adjust our workforce to the market decline caused by COVID-19. As of June 30, 2021, the number of total employees (by headcount) decreased by 22,000, or 17%, of the total workforce, including 5,000 employees (by headcount) in Germany compared to December 31, 2019 (excluding a workforce reduction by approximately 8,600 employees (headcount) in connection with the divestment of LSG Group’s European business). We implemented short-term measures in 2020 to reduce staff costs and to counteract the effects of the pandemic. In addition, crisis agreements were concluded for the majority of the other groups of our employees. On August 5, 2021, Lufthansa Group has communicated that it already implemented measures to reduce personnel costs by €1.1 billion. Taking into account additional measures such as, in particular, savings of still to be negotiated amended tariff agreements as well as further staff reduction, we expect to achieve a total of €1.8 billion structural annual personnel cost savings from 2024 onwards.

We have entered into multiple groundhandling arrangements relating to various services provided to airport users, including ground administration and supervision (e.g., handling, storage and administration of unit load devices), passenger handling (any kind of assistance to arriving, departing, transfer or transit passengers, including checking tickets and travel documents, registering baggage and carrying it to the sorting area) or freight and mail handling. There is no assurance that providers of groundhandling services will comply with the terms of the respective groundhandling arrangements at all times. Providers of groundhandling services could exhibit unsatisfactory performance or may stop to deliver groundhandling services entirely. In this event, we may be unable to maintain our flight schedules as planned or at all in respect of such airports with unsatisfactory groundhandling service levels. Additionally, we are renegotiating the contractual terms of certain supply arrangements and groundhandling arrangements from time to time and we may not be able to enter into new arrangements at the same or similarly favorable commercial terms.

If the implementation of the “ReNew” program, or any other project or initiative that we implement in an effort to restructure our business, increase cost efficiency or achieve profitability, as well as any related change management measures, is not successful and the targeted cost savings and other improvements cannot be realized, our results of operations could be adversely affected. Even if we achieve the expected benefits, they may not be achieved within the anticipated timeframe. The cost savings anticipated by us are based on estimates and assumptions made by us that are inherently uncertain, although considered reasonable by us, and may be subject to significant business, economic and competitive uncertainties and contingencies, all of which are difficult to predict and a number of which are beyond our control. As a result, there can be

no assurance that such restructuring measures as well as related change management measures, cost savings and operating improvements will be achieved.

In addition to flight cancellations as a result of COVID-19, any changes in the political and geopolitical conditions could require us to cancel additional flights or services temporarily or indefinitely, including as a result of trade restrictions.

1.1.8 We may be adversely affected by aircraft accidents and similar disasters, design defects, operational malfunctions and significant losses from safety incidents.

As an airline, we risk suffering adverse effects and significant losses if an aircraft is involved in an incident, an accident or is lost, including as a result of design defects, operational malfunctions, human error or wrongdoing or similar safety incidents. Incidents and wreckages may be caused by any number of factors, whether individually or in the aggregate and including, but not limited to, human error or wrongdoing (including terrorist attacks and acts of sabotage), technical, human or design failures, operational malfunctions and deferred maintenance as well as similar events. The occurrence of any incident in relation to our fleet, resulting in an accident or the grounding of such aircraft or type of aircraft would adversely affect our revenue, operations, reputation and customer confidence. The risk pertaining to such problems and accidents exposes us to the incurrence of unexpected costs in the form of passenger claims, repair and replacement costs and losses connected to any public perception in relation to the safety or reliability of our fleet, which could ultimately cause air travelers to be reluctant to use our aircraft and, hence, ultimately poses a material risk to our ability to generate revenue. For example, on March 24, 2015, an Airbus A320 of our subsidiary Germanwings operating under the call sign 4U9525 crashed north-west of Nice in France, which resulted in the loss of 150 lives.

In particular, there can be absolutely no assurance that the insurance coverage available to or obtained by us would be adequate to cover costs or losses resulting from the occurrence of such an event, if available at all. We may be obliged to bear substantial losses in particular regarding damage claims for the loss of lives, personal injuries or other material or immaterial damages itself, irrespective of our insurance coverage. This could, for example, be the case if any of our insurers was unwilling or unable to pay out contractual compensations or if losses were to occur that exceed insurance coverage.

Any incidents involving aircraft and comparable disasters, whether in relation to us, the Star Alliance or other joint ventures to which we are a party, could tarnish our reputation and, thereby, ultimately result in an indefinite decline in demand for our services. Moreover, due to the heightened public sensitivity and media attention related to major incidents commonly reported internationally and the effects on our customers' confidence in us and the aircraft operated by us, we could be exposed to additional material risks from a general decline in demand for our flights and products and ultimately incur significant losses.

1.1.9 We are dependent on the resilience and uncompromised operation of our reservation, data processing, technology and management systems as well as those operated by third parties and are exposed to risks related to the poor performance, the failure of and any unauthorized access to these systems, including as a result of data security breaches.

Our business operations are dependent on the suitability, reliability, resilience and durability of our technology platforms, systems and processes, including automated systems and third-party infrastructure that we source. In order for our operations to work efficiently, our information technology ("IT") systems, including our communications and logistics systems, must be able to accommodate a high volume of traffic, maintain secure information, dispatch flights and deliver flight information, issue electronic tickets and ultimately process critical financial and non-financial information in a timely manner. Additionally, we are outsourcing the operation of certain IT systems to third parties and are dependent on the performance of such third parties. Any poor performance by a third party will expose us to the risk that we will not be able to operate our IT systems effectively or at all. Furthermore, we are dependent on the availability of passenger service systems used by travel agents and sales partners, including through third parties, such as those indirectly linked to us through the Star Alliance.

We face both significant external cyber-attack threats and internal risks to our data, IT and software systems. Besides in-place mitigation measures, our data and systems may be vulnerable to theft, payment fraud, loss,

damage and interruption due to unauthorized access, security breaches, cyber-attacks, computer viruses, power loss or other disruptive events. A security breach could have a negative impact on customer confidence in our systems and negatively impact our reputation. Cyber-attacks could also force us to temporarily or for a longer period ground aircraft or shut down our operations, in part or entirely.

We have experienced and are exposed to an increasing risk that individuals or groups (including, in particular, criminals, foreign governments, competitors and new entrants to the market) attempt to disrupt the availability, confidentiality, integrity and resilience of our IT systems or IT systems we rely on and which have access to and store our customers data, such as within the Star Alliance program. This, in turn, could disrupt key operations, including our passenger service systems, make it difficult to recover critical services, damage assets and compromise the integrity and security of both corporate and customer data.

For example, in December 2019, the integrity of the Miles & More program software was partially compromised due to an erroneous update, as a result of which personal data of some customers became visible to other customers. Moreover, in February 2020, due to a software malfunction, the booking software of Eurowings was compromised. As a result, third parties were able to obtain personal data of customers and were in a position to change or cancel existing bookings. In addition, in February 2021, the integrity of a software provider that processes certain data in connection with our involvement in the Star Alliance was compromised and third parties were able to obtain personal data of customers, in particular data in relation to Miles & More memberships.

Any inability to detect or efficiently counteract such incidents, if at all, may result in fraud, the extent of which is not quantifiable in advance, additional expenses and compensation claims brought against us. Moreover, data security and the protection of personal data has become a topic of heightened sensitivity, including as a result of evolving general data protection laws and regulations (see “*1.1.10 We are exposed to risks associated with breaches of GDPR compliance, which we particularly face in connection with cyber security breaches.*”). Ultimately, we face reputational risks as a result of data theft and failure of any of our IT systems.

1.1.10 We are exposed to risks associated with breaches of GDPR compliance, which we particularly face in connection with cyber security breaches.

As part of our operations, we retain a significant amount of personal data we receive from our customers, any of which may be subject to certain regulatory data privacy protection in the European Union and elsewhere. We collect, store and process personal data, including credit card numbers, booking data and other personal information as well as data in connection with our Miles & More Program, any of which is sensitive. With respect to the European Union, the General Data Protection Regulation (EU) 2016/679 of the European Parliament and of the Council of April 27, 2016, on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (the “**GDPR**”), which has applied to all European Union member states since May 25, 2018, has increased the financial and reputational implications we face following any breach of our IT systems or those of third-party suppliers, with competent authorities being in a position to impose significant fines and to demand remedial steps. Any security breaches may impact GDPR requirements (see “*1.1.9 We are dependent on the resilience and uncompromised operation of our reservation, data processing, technology and management systems as well as those operated by third parties and are exposed to risks related to the poor performance, the failure of and any unauthorized access to these systems, including as a result of data security breaches.*”).

We, as well as third parties with whom we contractually engage for the provision of related services, collect, store and process the personal data of our customers throughout the world. We and any of our business associates may experience a breach under the GDPR and/or other similar laws (the “**Data Protection Laws**”) when there has been impermissible use, access, or disclosure of unsecured personal data, a breach under the Data Protection Laws when we or any of our business associates neglect to implement the required administrative, technical and physical safeguards of its electronic systems and devices, or a data breach that results in impermissible use, access or disclosure of personal identifying information of its employees and customers. The GDPR imposes substantial worldwide obligations on the processing of personal data. These laws continue to develop globally and differ from jurisdiction to jurisdiction, which increases the complexity and costs of our global data protection and security compliance programs.

If personal data were processed, stored or made available to third parties in violation of the applicable laws and regulations, including under the GDPR, or if a third party were to misappropriate such information, our reputation could be harmed, we could become subject of regulatory action, proceedings and/or fines and customers could bring legal claims against us. Under the GDPR alone, national authorities are in a position to levy significant levels of fines for non-compliance of up to €20 million or 4% of our annual global turnover.

Any inability to detect or efficiently counteract such incidents, if at all, may result in fraud, the extent of which is not quantifiable in advance, additional expenses and compensation claims brought against us. Moreover, data security and the protection of personal data has become a topic of heightened sensitivity, including as a result of the GDPR. Ultimately, we face reputational risks as a result of data theft and failure of any of our IT systems.

1.1.11 We are dependent on the strength of our brands and the brands we are associated with and our reputation is exposed to risks related to any damage thereof.

Our reputation and brand names have contributed significantly to our market position. Accordingly, we are dependent on the strength of our brands and those associated with our business.

Our ability to attract bookings depends on the public recognition of our brands including, in particular, of the Network Airlines, Eurowings and Miles & More. Our brands are, and will continue to be, vulnerable to adverse market and customer perception. Reliability, including on-time performance, is an integral element of the brands and of each customer's experience. Unreliable operational performance, an inability to react to customer expectations as a result of routine and ongoing disruption could negatively affect customer satisfaction and perception of our brands.

For example, following the cancellation of, during certain periods, almost all of our flights during the first half of 2020 in response to the COVID-19-related travel bans and entry restrictions, we faced significant delays in processing refund claims and addressing customer service tickets. If we and, in particular, any of our Network Airlines or Eurowings, are unable to provide, on a consistent basis, the level of services associated with our brands, including in relation to booking and in-flight experiences, or if we are unable to maintain the current or implement additional cleanliness and hygiene standards in response to COVID-19, the reputation of our brands could be adversely affected.

Moreover, the reputation of any of our brands could be adversely affected if we, or any corporation or association with which we are associated, fail or fails to maintain ethical, social or environmental sustainability practices, or if customers negatively perceive our strategy or approach to environmental, social and governance issues, including, in particular, greenhouse gas emissions, employee and union relations and general corporate responsibility (see “1.1.4 We face increasing susceptibility for the demand of our business as a result of increasing competition from alternative means of transportation, including rail travel, as well as alternatives to business travel, and due to changes in the applicable legal framework or consumer behavior, in particular in relation to climate change and generally increased environmental consciousness.”). The value of our brands could also be negatively affected by external factors outside of our control, including changes in applicable regulations related to the aviation industry and resulting changes in the perception of our brands.

1.1.12 The airline industry is highly competitive, and we face intense competition from low-cost airlines, state-owned or subsidized airlines or other companies that operate in any of our other business segments, all of which may further increase competitive and financial pressure in the future.

Already prior to the outbreak of COVID-19, we operated in a highly competitive market and were in intense competition with a large number of airlines and air carriers across the majority of markets in which we operate. More specifically, Lufthansa Group is facing competition in the Network carrier sector (including from U.S. carriers, such as American Airlines and Delta Airlines, from Asian carriers, such as Air China and Cathay Pacific, carriers from Turkey and the Middle East, such as Turkish Airlines, Emirates, Qatar Airways and Etihad Airways) as well as in the low-cost point-to-point segment (such as easyJet, Ryanair and WizzAir). Among others, changes in customer behavior and the emergence and rapid growth of low-cost

airlines in a variety of our markets have intensified the price and cost-competitive nature of the airline industry.

In particular, we continue to face heavy competition from low-cost and ultra-low-cost airlines. We believe that the cost at which we produce our services, including, among others, as a result of labor-related cost impact, is higher compared to the cost base of low-cost and ultra-low-cost airlines. Accordingly, our competitors may be in a position to offer flights at significantly lower prices than we can or at prices below cost in order to capture or secure market share. The further growth of low-cost airlines and ultra-low-cost airlines could impair our growth, result in a decrease of our profitability or may lead to a decline in market share. This competition is expected to further increase as various low cost and ultra-low-cost-airlines have announced their intention to enter into a price competition. Until 2018, through our Network Airlines business segment, we focused on providing all-in fares, meaning that, in the majority of our booking classes, on-board food and beverages, carry-on item(s) and seat reservations were included. In contrast, we believe that the majority of European low-cost airlines and ultra-low-cost airlines aim to attract customers by, among others, offering “flight only” fares, meaning that the price of a ticket would typically only include the transportation, limited or no carry-on luggage and frequently not include seat reservations or on-board services, to attract customers by very low initial ticket prices. There can be no assurance that we will be able to successfully adapt our fare structure in the future or to ultimately win customers over our competitors.

In addition, economic uncertainty in specific markets or on specific routes may cause competitors to progressively transfer their capacity to markets and routes that we also serve, resulting in increased competition in these markets and on these routes. During the 6M 2021, we generated 39%, 24%, 10% and 14% of our external traffic revenue of €1,628 million in our Network Airlines business segment in Europe, America, Asia/Pacific and Middle East/Africa, respectively. During the Fiscal Year 2020, we generated 39%, 25%, 13% and 10% of our external traffic revenue of €5,488 million in our Network Airlines business segment in Europe, America, Asia/Pacific and Middle East/Africa, respectively. During the Fiscal Year 2019, we generated 41%, 33%, 19% and 8% of our external traffic revenue of €23,055 million in our Network Airlines business segment in Europe, America, Asia/Pacific and Middle East/Africa, respectively. However, competitors of the Network Airlines include airlines serving larger catchment areas than we do. These airlines may have greater financial resources and lower cost structures than we do, particularly with regard to point-to-point flights within Continental Europe and flights to Asia and North America.

Some of the airlines with which we compete are wholly or partially owned by governments (aside from any government involvement in the context of stabilization measures, see “1.2.1 *We are dependent on our ability to obtain sufficient funding to continue our business activities but, as a result of COVID-19, are exposed to significant uncertainties in relation thereto.*”). Governmental shareholdings have in the past, and could continue to in the future, given these airlines access to larger and less expensive sources of funding (including state subsidies) which may, in particular during times of crisis, such as COVID-19, entail a material competitive advantage. If governments were to provide one or more of our competitors with unilateral subsidies or other forms of government assistance, including the buildup of extensive infrastructure, this could result in market distortions and materially weaken our competitive position. The expansion of state-owned airlines continues to be a competitive threat to us. Certain airlines that are operating in relatively small home markets continue to ensure utilization of their extensive capacity by transit passengers transported through their hubs, which they primarily achieve by offering very low fares for travel to and from Europe and other large travel markets or by forming co-operations with or by acquiring interests in other, in particular European, airlines. While we operate the majority of our business based on the framework of EU regulations (including the principles of an economic competition between market participants under comparable legal prerequisites (e.g., labor law, the regulation/absence of state aid, merger control, infrastructural access)), some of our global (state-owned) competitors focus on unique aspects of business (e.g., infrastructural development, deviation of business and creation of touristic traffic streams), rather than a commercially successful operation. We do not believe to be in a position to adequately counter-act such global market imbalances in the foreseeable future, if at all.

In connection with the ongoing COVID-19 pandemic, some of the airlines with which we compete may be able to benefit from protection under insolvency laws in their respective home countries. Simultaneously, the airline industry may undergo a market consolidation and any of our competitors may be in a position to participate in mergers and acquisitions activities and, in particular, acquire commercially beneficial

companies while we are currently restricted from engaging in acquisition activities (see “1.2.2 *We are subject to significant limitations in our financial flexibility as a result of the state aid received and financial stabilization measures obtained from governments in response to the outbreak of COVID-19 and face risks associated with triggering events of default under these state aid and stabilization measures.*”).

In some cases, we are exposed to strong competitive pressure in our maintenance, repair and overhaul (“MRO”) and Catering business segments, which intensified due to the outbreak of COVID-19 and the related decline in customer demand and multiple restrictions impacting the ordinary course of business. During the 6M 2021, our MRO and Catering business segments generated total revenue of €1,717 million and €447 million, respectively. During the Fiscal Year 2020, our MRO and Catering business segments served more than 800 and 300 customers and generated total revenue of €3,747 million and €1,305 million, respectively. During the Fiscal Year 2019, our MRO and Catering business segments served more than 850 and 300 customers and generated total revenue of €6,572 million and €3,360 million, respectively. We face significant risks in relation to the loss of customers to other suppliers within these business segments, including as a result of a de-consolidation and increase price competition. Risks arise, for example, in cases where long-term contracts between our companies with their customers, especially in the MRO and Catering business segments are not renewed, which would lead to sustained deterioration in the income situation of the affected companies.

The MRO business segment is exposed to the financial risk posed by a demanding competitive situation, which is, in particular, due to original equipment manufacturers (“OEMs”), who restrict access to intellectual property.

1.1.13 Any increase in fees related to air traffic control, airport, passenger and take-off and landing, as well as the costs we incur to ensure air traffic security, may result in additional costs and ultimately increase our operating expenses.

Airport, passenger and landing fees, as well as security charges and initiatives represent a significant share of our operating costs. However, air traffic control, airport, passenger and take-off and landing fees, as well as security charges are costs that are not subject to individual negotiations but, for example in Germany, are subject to the approval of regional or national aviation authorities. We may be and partly already are subject to increased fees and charges. Any future increases of fees and charges may be significant and there is no assurance that we will be able to recharge such significantly increased fees and charges to our customers, which could negatively affect our margins. In particular, airport infrastructure providers as well as air navigation service providers may likely increase fees and charges to recover losses incurred as a result of the outbreak of COVID-19, particularly if such providers are generally financed by fees and charges. Currently, consultations are ongoing in the course of setting the revised air traffic control charges in the Eurocontrol area, which will lead to additional costs from 2022 onwards of a medium three-digit-million Euro amount. Additionally, we are exposed to the risk that new fees and charges could be implemented for an unforeseeable number of other reasons, including if airport, noise, emission or landing charges and fees were to be levied based on environmental criteria such as aircraft noise or emission levels, if airlines were forced to assume additional security responsibilities or not equally in the jurisdictions in which we operate.

If a terrorist attack or safety incident were to occur within the aviation industry and, in particular, at an airport, security regulations worldwide could be further tightened. In addition, security charges or other costs arising from additional security measures at airports in any of the markets in which we operate, including in the United States, could increase further. However, the degree to which increases in fees related to air traffic control, airport, passenger and take-off and landing as well as costs incurred in connection with air traffic security cannot be predicted.

1.1.14 Our business activities are dependent on the availability of airspace, air traffic controllers, airport slots, services provided by airports and other third parties, ultimately, as suitable airport infrastructures are a prerequisite for our business activities, we are highly susceptible to changes and shortfalls thereof.

The capacity and amount of airspace and airport slots available for the use of the civil airline industry is limited and any further increase in air traffic density could adversely affect our ability to offer our services. Increases in air traffic, especially at high-density hubs, including Frankfurt, Germany, and Munich,

Germany, may lead to shortages of available slots and increasing cost competition. During the 6M 2021, 57% of our flights took off and landed at our hub airports in Frankfurt, Germany, and Munich, Germany (Fiscal Year 2020: 54%; Fiscal Year 2019: 51%).

Although we believe that we, as one of the largest airlines operating from the airports in Frankfurt, Germany and Munich, Germany, may benefit from regional strength and legacy commitments, there can be no assurance that we will be in a position to retain our take-off and landing slots at either airport to the extent we so require to efficiently operate our business, if at all (see “*1.1.2 Our business activities and operations are sensitive to changes in entry regulations, travel bans and a loss of take-off and landing slots, including in connection with and as a result of COVID-19, which may ultimately intensify the decrease in ticket sales and the increase in reimbursement claims.*”). Additionally, any changes to the offering at or an increase in competition for slots at the airports in Frankfurt, Germany, and Munich, Germany, and at any of the other coordinated airports which we frequent, including Zurich, Switzerland, and Vienna, Austria, may affect our ability to maintain or increase the flight services we offer.

In response to the outbreak of COVID-19 and its impact on the German economy and, in particular the adversely affected financial situation of businesses across all sectors, the German government decided to establish an economic stabilization fund (*Wirtschaftsstabilisierungsfonds*) (the “**WSF**”). On May 25, 2020, the WSF agreed to grant certain stabilization measures to the Company (see “*1.2.2 We are subject to significant limitations in our financial flexibility as a result of the state aid received and financial stabilization measures obtained from governments in response to the outbreak of COVID-19 and face risks associated with triggering events of default under these state aid and stabilization measures.*”). As such stabilization measures by the WSF constitute a state investment into a publicly owned company, the European Commission’s approval of the stabilization package was a prerequisite. As part of the approval process, the European Commission imposed certain measures on us to remedy antitrust considerations. These measures included, among others, the requirement for us to transfer 24 take-off and landing slots per day to a competitor that was unable to obtain take-off and landing slots assigned during the ordinary slot allocation procedure from the competent authorities (whereas, following a period of 1.5 years, these competitors may include Ryanair and Condor), assigned to a base of up to four aircraft at each of the airports in Frankfurt, Germany and Munich, Germany, thereby allowing reasonable aircraft rotation for short-haul flights. We are required to uphold the offer to transfer take-off and landing slots for six IATA seasons (*i.e.*, a period of three years in total) following full restoration of the Slot Allocation Regulation, which is, as of the date of this Prospectus, not expected to commence prior to the end of the IATA northern-hemisphere winter season 2021.

We are the leading airline group in Europe and one of the largest airline groups globally by total revenue (*source*: Statista, Leading Airline Groups in Europe in 2020, Company Information). As a result, we may be subject to antitrust proceedings and risks associated with our dominant market position in Germany and within the European Union. Among others, we could be required, under current or future regulations, to transfer commercially significant take-off and landing slots to competitors, be prohibited from obtaining additional take-off and landing slots, which would otherwise require us to expand our business activities and operations in the future, and maintain feeder flight arrangements.

The legal basis for the allocation of take-off and landing slots, noise-related operational restrictions and ground handling within the European Union have been under review since 2011. Regulation (EU) No 598/2014 of the European Parliament and of the Council of April 16, 2014, on the establishment of rules and procedures with regard to the introduction of noise-related operating restrictions at airports located in the European Union within a Balanced Approach has been in force since 2014. A European regulation on ground handling procedures was withdrawn in 2015. We believe that political sensitivities have caused legislative procedures on take-off and landing slots to come to a halt. The European Commission is conducting a study to update the latest available data in relation to airline slots from 2011 and is expected, on the basis of these results, to decide on how to proceed with these legislative procedures. However, the timing for any legislative initiatives remains unclear. Ultimately, there can be no assurance whether European regulations will be implemented or that any regulations would not adversely affect our ability to conduct our business. Furthermore, the regulations regarding ground handling at European airports may, in the future, be revised, the scope, timing or effects of which on our business or the airline industry as a whole cannot be predicted.

In addition to the general availability of airspace, air traffic controllers and take-off and landing slots, we are dependent on the provision of services by third parties, such as providers of ground handling services (including aircraft fueling and baggage handling), general airport services and the availability of the requisite airport infrastructure. Furthermore, other than in relation to third parties for the provision of operational services, we are also dependent on general third-party service providers and suppliers. These provide, for example, distribution systems (such as “Global Distribution Systems” or “Computer Reservation Systems”), which serve as an interface between the various indirect booking channels, such as agencies, and our inventory system booking and ticketing platforms, general IT services, insurances, communication providers and energy suppliers. If any of these third-party services were temporarily or permanently unavailable, including as a result of strikes or cancellation of contracts, or were only available on commercially unreasonable terms, our operating performance could be adversely affected.

The number of flights performed by the global airline industry increased from 23.8 million in 2004 to 38.9 million in 2019 (*source*: Statista, Global Flights Performed) and, ultimately, contributed to the emergence of airspace-related bottlenecks. These bottlenecks, caused by, among others, a lack of remaining available airspace capacity, in the fragmented European air traffic control system continue to pose a significant problem for the European airline industry, including ourselves. Bottlenecks result in considerable delays to air traffic, unscheduled and financially significant detours, holding periods, increased fuel consumption and emissions. Ultimately, bottlenecks have a negative impact on the earnings of all European airlines, including our own, as well as on the environment and continue to jeopardize growth in air traffic. In 2009, the European Union introduced a revised concept of European-wide air traffic management, the Single European Skies (“SES II”). The SES II includes legislative framework measures aimed at capitalizing on the achievements of a previous single-sky concept and increasing the overall performance of the air traffic management system in Europe. The measures include an increased integration of air security organizations within the framework of “functional airspace blocks,” the development of a uniform air traffic management system across Europe and the improved utilization of slots. However, there can be no assurance that the SES II measures will be implemented successfully, ease the existing bottlenecks or not include measures aimed at an equal distribution of airspace to our disadvantage and, ultimately, adversely affect our competitive position and ability to perform our operations.

1.1.15 Demand for our MRO and other non-carrier services are highly susceptible to the demand for air transportation and adverse economic developments.

In addition to passenger and freight transport services, which we conduct under our Network Airlines, Eurowings and Logistics business segments, we generate a significant share of our revenue from the provision of services not directly related to air transport services, namely under our MRO and Catering business segments. During the 6M 2021, business operations from our MRO business segment contributed external revenue of €1,361 million, or 23.6%, to our total revenue (Fiscal Year 2020: €2,724 million, or 20.0%; Fiscal Year 2019: €4,378 million, or 12.0%). Likewise, during the same period, business operations from our Catering business segment contributed external revenue of €435 million, or 7.5%, to our total revenue (Fiscal Year 2020: €1,064 million, or 7.8%; Fiscal Year 2019: €2,623 million, or 7.2%).

In addition to servicing the 734 aircraft in our fleet as of June 30, 2021, we provide MRO services to many customers worldwide, including other airlines, OEMs, aircraft leasing companies and operators of VIP jets through Lufthansa Technik Aktiengesellschaft (“**Lufthansa Technik**”) at 35 maintenance plants. Any mid- or long-term decline in demand for air transportation, whether in relation to airliners, cargo or VIP jets will result in a decline of the use of aircraft and, ultimately, adversely affect demand for our MRO services. Commonly, aircraft owners and operators use an economic decline, such as occurred during the first half of 2020 as a result of the spread of COVID-19, to ground and service their aircraft, resulting in a short-term increase in demand for our MRO services. If, however, as a result of a mid- or long-term downturn in economic developments or demand for aircraft transportation, fewer aircraft are required, demand and the price we can charge for aircraft MRO services may deteriorate. Among others, the COVID-19 pandemic resulted in a significant decrease in total revenue (before consolidation) generated in our MRO business segment, from €6,572 million during the Fiscal Year 2019 to €3,747 million during the Fiscal Year 2020. During the 6M 2021, our MRO business segment generated €1,717 million of total revenue (before consolidation) (compared to €2,280 million during the 6M 2020).

Additionally, the susceptibility of the airline industry to adverse economic developments can lead to a further increase in price pressure along other parts of our value chain, including the Catering business segment and the prices we can charge for catering and related services. Among others, the COVID-19 pandemic resulted in a significant decrease in total revenue (before consolidation) generated in our Catering business segment, from €3,360 million during the Fiscal Year 2019 to €1,305 million during the Fiscal Year 2020. During the 6M 2021, our Catering business segment generated total revenue (before consolidation) of €447 million (compared to €814 million during the 6M 2020).

Ultimately, the demand and financial performance of our MRO and other non-carrier services are directly linked to the demand for air transportation and are adversely affected by a decline thereof as well as negative economic developments. Accordingly, the entirety of our revenue generating services is susceptible to a similar risk profile and a decrease in demand for our air transportation services may exacerbate and adversely affect the financial performance of each of our MRO and Catering business segments. Additionally, the COVID-19 pandemic may also have structural long-term effects on the MRO business because we are accelerating the modernization of our fleet requiring less maintenance, repair and overhaul, which ultimately results in a decreased demand for MRO services.

1.1.16 Our revenue and profits are susceptible to seasonal fluctuations.

Demand for our services by passengers, in particular leisure travelers, varies over the course of the year, which causes our quarterly results to fluctuate significantly. During the winter months, our revenue is typically lower than in the rest of the year, which is generally reflected in lower operating results in the first and fourth quarters. Our passenger numbers are traditionally highest in September and October of each year. As a result of quarterly fluctuations, the level of our aircraft utilization and profitability fluctuates during the year. For example, in the Fiscal Year 2019, our profit/loss from operating activities plus result of equity investments accounted for using the equity method and result of other equity investments (“**EBIT**”), adjusted for impairment losses/gains, effects from pension provisions and results of disposal of assets (“**Adjusted EBIT**”) amounted to a loss of €336 million in the first quarter, a profit of €754 million in the second quarter, a profit of €1,297 million in the third quarter and a profit of €311 million in the fourth quarter. Due to the effects of the COVID-19 pandemic and travel uncertainty, customers are booking flights with less lead time, which has a negative impact on cash flow. The COVID-19 pandemic has significantly disrupted our historical seasonal pattern. For example, in contrast to 2019, in the Fiscal Year 2020, our Adjusted EBIT amounted to a loss of €1,220 million in the first quarter, a loss of €1,679 million in the second quarter, a loss of €1,262 million in the third quarter and a loss of €1,290 million in the fourth quarter.

1.1.17 We are dependent on our ability to adapt our workforce to a changing demand situation and any shift in international regulations.

Our business operations depend on the availability of a sufficient number of staff, including in relation to air passenger transport, where we are required to staff and operate flights with varying numbers of minimum on-board personnel as required by international regulations, depending on the number of seats offered. Accordingly, staff costs, which during the 6M 2021 amounted to €2,907 million, contribute significantly to costs related to our operations (Fiscal Year 2020: €6,436 million; Fiscal Year 2019: €9,121 million). We may be unable to adequately adapt our workforce to the changes in the domestic and international economic environments that may be required as a result of changes in entry regulations or a loss of slots. Changes in various domestic and international economic environments require corresponding increases or decreases of our workforce across all of our business activities and markets in which we operate. We have and will continue to take into consideration labor law measures in the context of and as a response to the effects of the COVID-19 pandemic including, but not limited to, early retirements, re-allocation of human resources and termination benefits. Furthermore, we entered into various crisis agreements with collective bargaining partners across the Network Airlines and Eurowings over the course of the Fiscal Year 2020 (see “1.1.1 The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.”). We are dependent on our ability to adapt the size of our workforce, which includes personnel for operating aircraft (*i.e.*, flight staff) and personnel for ground operations (*i.e.*, ground-staff), to correspond to volatile economic situations or highly competitive

situations and markets (e.g., due “low-cost carriers” or “ultra-low cost carriers”). As of June 30, 2021, we had 108,072 employees (by headcount) (compared to 129,356, 110,065 and 138,353 employees (by headcount) as of June 30, 2020, December 31, 2020, and December 31, 2019, respectively). As of the same date, 66,379 of those employees were ground staff (compared to 83,609, 66,070 and 91,642 ground staff as of June 30, 2020, December 31, 2020, and December 31, 2019, respectively). Despite a partial restructuring of our employee base during 2020 in connection with the COVID-19 pandemic, we may in the future be unable to quickly or accurately adapt the size of our workforce to changing economic conditions and to keep a highly motivated workforce during times of crisis. Additionally, during the COVID-19 pandemic, a significant number of employees left our workforce at their own request, as a result of which we may have difficulties or ultimately may not be able to find adequately talented substitutes, including against the background of our financial performance and outlook.

1.1.18 We are exposed to adverse developments of our joint ventures, divestments, acquisitions, and other investments.

We are exposed to a variety of risks associated to previous and future joint venture agreements, divestments, acquisitions and investments. The commercial success of any of these measures, in each case including, in relation to a financial upside potential for us, the achievement of synergies and the reduction of costs, depends on a number of materially uncertain factors.

We are and will in the future be engaged in numerous joint ventures. For example, Lufthansa Cargo Aktiengesellschaft (“**Lufthansa Cargo**”) and Deutsche Post Beteiligungen Holding GmbH agreed on a 50-50 joint venture, Aerologic GmbH (“**AeroLogic**”), which holds long-term flight services agreements with Lufthansa Cargo and DHL International GmbH (“**DHL**”). The flight services agreements provide that AeroLogic will sell the capacity of all of its aircraft currently in service to Lufthansa Cargo and DHL, in accordance with an agreed share of capacity and use. Additionally, Lufthansa Technik participates in numerous joint ventures, including with MTU Aero Engines AG and Rolls Royce plc. There can be no assurance that joint ventures we have entered into or may enter into in the future yield any synergies, profits or similar favorable business development for us.

Likewise, any previous or future decisions to divest any of our subsidiaries or material business operations, such as the sale of LSG Group’s European business operations in 2019/2020 or the potential divestment of the remaining LSG Group’s business activities, including those that were or are profitable for us, could yield a lower-than-adequate return, if at all, and subject us to legal proceedings. For example, the share and asset purchase agreement relating to the sale of LSG Group’s European business operations provides for representations and warranties, certain indemnification provisions as well as other provisions that could ultimately require us to assume certain liabilities or otherwise indemnify the purchaser for certain liabilities in relation to LSG Group’s European business operations, including pension liabilities, pensionable income, financing costs, litigation, administrative and insolvency proceedings as well as tax. Additionally, as a result of the sale of LSG Group’s European business operations and the loss of our ability to exercise instruction rights we previously held as a shareholder, we may be exposed to shortages in relation to our services offering, among others on flights which we operate and in lounges which we offer to our guests. Moreover, we are continuously exploring options in relation to a partial or full divestiture of Lufthansa Technik and have mandated advisors to evaluate such options, which, if such divestment were to occur, could result in procurement shortages and ultimately negatively affect the availability of MRO services we require for the availability and operation of our own fleet. Disposal and partnering options for all or part of Lufthansa AirPlus Servicekarten GmbH (“**AirPlus**”) are also under review, taking into account the necessary capital expenditure, synergies and value creation potential.

The aforementioned factors and other developments not known at the time of the entering into any joint venture agreement, any divestment, any acquisition and any other investment may not yield the benefits anticipated, be associated with higher-than-expected costs and result in litigation against us, any of which could hamper our business operations and tie up significant management and staff capacity.

1.1.19 We are exposed to adverse developments relating to the occurrence of unforeseen events, in particular natural and man-made disasters that may not be covered by our insurance policies, and changes in the political, geopolitical and economic conditions.

Our ability to manage our business activities, in particular in relation to our Network Airlines, Eurowings, Logistics and MRO business segments, with the required level of insurance coverage against the risk of losses from both natural and man-made disasters is dependent on, among others, the availability of insurance policies and their economic terms. Our insurance provides liability coverage for passengers, mail, cargo, product liability, third-party liability and hull damage. Since insurance companies continue to be reluctant in providing coverage, it is difficult for us and other European airlines to effectively insure against terrorist attacks, pandemics, such as the COVID-19 pandemic, and certain natural disasters, such as volcanic eruptions, and official administrative grounding orders for aircraft in connection with such disasters. We believe that, going forward, insurers may explicitly exclude, among others, risks related to pandemics from their coverage which could, ultimately, increase our insurance-related risk exposure. Specialized insurers now offer only limited hull insurance policies for these risks that cover damage to aircraft. Such policies generally stipulate a number of conditions under which the insurers may terminate policies. In addition, the policies must be renewed at regular intervals (see “1.3.5 The airline industry is particularly vulnerable to the effects of epidemics and natural disasters including extreme weather conditions.”).

Our third-party liability insurance for war and allied perils covers damages caused to third parties by our aircraft due to war and allied perils, including terrorist attacks. Insurance companies may stop providing coverage under such comprehensive or third-party liability insurance policies at commercially acceptable terms or may suspend such insurance entirely. Further, terrorist attacks, acts of sabotage and other disasters, especially if they occur during air travel or are directed against aircraft, could result in insurance coverage for air traffic risks becoming even more expensive, or in certain risks becoming insurable only to a limited degree or becoming completely uninsurable, or could result in upper limits being established for insurable losses.

We are significantly exposed to adverse developments of the political, geopolitical and economic landscape, as any such development can adversely affect demand and conditions for air travel between different countries. This is particularly relevant in relation to changes in Europe, where we conduct the majority of our business by number of flights and generated 61.0%, 64.5% and 66.1% of our traffic revenue by area of operations (sector) during the 6M 2021 and each of the Fiscal Years 2020 and 2019, respectively. However, volatility in North America and Asia/Pacific, where we generated 31.4%, 29.2% and 27.7% of our traffic revenue by area of operations (sector) during the 6M 2021 and each of the Fiscal Years 2020 and 2019, respectively, may also negatively affect demand for our business.

Recent examples for changes in political conditions include, for example, political instabilities in Belarus and Hong Kong, domestic political volatility within the United States, and uncertainties relating to the exit of the United Kingdom from the European Union (the “**Brexit**”). Although the Brexit has formally taken place with effect from January 1, 2021, heightened business and consumer uncertainty remains in relation to the mid- to long-term effects on business, consumer behavior travel and life in the United Kingdom and the European Union. There can be no assurance that the shape of future aviation regulations in the United Kingdom and between the European Union and the United Kingdom will allow us to continue to operate into and out of the United Kingdom.

In addition to flight cancellations as a result of COVID-19, any changes in the political and geopolitical conditions could require us to cancel additional flights or services temporarily or indefinitely, including as a result of trade restrictions.

Ultimately, there can be no assurance that scenarios and consequences which occurred in connection with COVID-19 or are related thereto are sufficiently covered by our insurance policies, if at all. Additionally, insurance premiums may substantially increase as a consequence of occurrences such as the COVID-19 pandemic and changes in the international political landscape.

1.2 Risks Related to Our Financial Situation

1.2.1 *We are dependent on our ability to obtain sufficient funding to continue our business activities but, as a result of COVID-19, are exposed to significant uncertainties in relation thereto.*

We require significant funds to operate our business such as for the acquisition of additional aircraft, the replacement of existing aged aircraft, the refinancing of existing obligations as they fall due and, as a result of COVID-19, to finance our general business activities which, during the 6M 2021 resulted in an Adjusted EBIT of a loss of €2,095 million and an Adjusted EBIT margin (defined as Adjusted EBIT divided by total revenue, “**Adjusted EBIT margin**”) of (36.3)% (Fiscal Year 2020: loss of €5,451 million, Adjusted EBIT margin of (40.1)%; Fiscal Year 2019: €2,026 million, Adjusted EBIT margin of 5.6%). Ultimately, our operations resulted in an average monthly operating cash drain (defined as Adjusted EBITDA adjusted for significant non-cash items, state aid grants and the redemption portion of IFRS 16 leasing expenses) of €217 million during the 6M 2021, resulting in an aggregate operating cash drain of €1,299 million as of June 30, 2021. We are not expecting any operating cash drain in the third quarter of 2021.

As of June 30, 2021, our order book comprised 178 aircraft for delivery by 2029 with an order commitment of approx. €14 billion. More specifically, we have placed firm orders for 20 Boeing 777, 25 Boeing 787, 28 Airbus A350 for the long-haul fleet across the Network Airlines and one Boeing 777F for Lufthansa Cargo. In relation to our airlines’ short-haul fleet, we have placed firm orders for 104 aircraft from the Airbus A320-family. Moreover, cumulatively, we have 74 order options in relation to new aircraft as of June 30, 2021. If we are unable to secure financing for the acquisition of new aircraft, we could be unable to renew our aircraft fleet and, ultimately, be required to operate an older fleet of aircraft which are typically less energy- and cost-efficient and less attractive to our customers.

We have significant financial borrowings and may be unable to partially or fully repay such borrowings as they fall due. As of June 30, 2021, we have contractual obligations in an amount of €2,538 million, €7,126 million and €6,305 million that become due in less than a year, in one to five years and in more than five years, respectively. As of June 30, 2021, the Equity Ratio amounted to 7.7%. The amount of debt we incur directly affects our financial condition and liquidity. Any inability to maintain our liquidity could expose us to risks related to the delay of default of payments under our indebtedness. Additionally, our assumptions and expectations regarding our ability to satisfy our current and non-current financial liabilities which, as of June 30, 2021, amount to €15,571 million (including bonds, commercial paper to banks, borrower’s note loans, credit lines, state-guaranteed loans, aircraft financing, other borrowings and leasing liabilities), are based on forecasts of our cash flows and liquidity needs for the short- and mid-term future. Due to the number of underlying assumptions and individual uncertainties in relation thereto, in particular the impact of COVID-19, we may be exposed to a significantly lower cash flow than expected, forecasted or required to operate our business.

Our ability to obtain funding may be affected by lenders who, as a result of COVID-19, are exposed to increasing risks of default and to borrowers whose credit levels may change. Among others, default risk assessments are based on ratings related to our financial position. In 2020, our senior unsecured debt was downgraded and received a “non-investment grade” rating from S&P Global Ratings Europe Limited (“**Standard & Poor’s**”) (BB-) with a “negative” outlook, downgraded and received a “non-investment grade” rating from Moody’s Deutschland GmbH (“**Moody’s**”) (Ba2) with “negative” outlook and downgraded and received an “investment grade” rating from Scope Ratings GmbH (“**Scope**”) (BBB-) with a “negative” outlook (see “1.2.4 Any rating downgrade could impair our ability to raise financing.”). Each of Standard & Poor’s, Moody’s and Scope has a registered seat in the European Union and has been declared to be registered in accordance with Regulation (EC) No. 1060/2009 of the European Parliament and of the Council of September 16, 2009 on rating agencies by the European Securities and Markets Authority.

Any adverse change in general market and financing conditions may negatively affect the investment and credit appetite and ultimately make it more difficult for us to obtain financing. Our inability to obtain or borrow sufficient funding at economically reasonable terms could make it impossible for us to meet our financing needs and materially increase our liquidity risk (see “1.2.3 We may be unable to maintain adequate liquidity.”).

1.2.2 We are subject to significant limitations in our financial flexibility as a result of the state aid received and financial stabilization measures obtained from governments in response to the outbreak of COVID-19 and face risks associated with triggering events of default under these state aid and stabilization measures.

As a result of the outbreak of the COVID-19 pandemic our operations resulted in an average monthly operating cash drain (defined as Adjusted EBITDA adjusted for significant non-cash items, state aid grants and the redemption portion of IFRS 16 leasing expenses) of €217 million during the 6M 2021, resulting in an aggregate operating cash drain of €1,299 million as of June 30, 2021. We are not expecting any operating cash drain in the third quarter of 2021. In an effort to maintain our operations and preserve financial flexibility, we entered into various contractual agreements with governments and government-owned banks in order to receive financial stabilization measures and state aid.

In response to the outbreak of COVID-19 and its impact on the German economy and, in particular the adversely affected financial situation of businesses across all sectors, the German government decided to establish the WSF. On May 25, 2020, the economic stabilization fund committee resolved to grant stabilization measures to the Company (consisting of silent participation contributions and the subscription to new shares) in the aggregate amount of up to €6 billion (the “**WSF Stabilization Measures**”). The WSF Stabilization Measures are governed by a framework agreement based on the respective EU Commission’s state aid decision regarding granting stabilization measures (*Rahmenvertrag zur Gewährung von Stabilisierungsmassnahmen*) and certain ancillary agreements entered into between the WSF and the Company on June 29, 2020, as amended from time to time (the “**Framework Agreement**”).

The Company has initially issued a silent participation I (*Stille Einlage I*) to the WSF in the amount of €4.7 billion, which was subsequently reduced to an amount of €4.5 billion, that is accounted for as equity according to IFRS to the extent drawn by the Company and provides for a loss participation and coupon deferral right (the “**Silent Participation I**”). The Silent Participation I can only be drawn until December 31, 2021, in not more than six tranches. The Silent Participation I, which has no fixed maturity, includes interest rate step ups from 4% p.a. (in 2020 and 2021) to up to 9.5% p.a. from 2027 onwards. As of the date of this Prospectus, the Company has drawn an amount of €1.5 billion under the Silent Participation I.

Additionally, the WSF has acquired the silent participation II-A (*Stille Einlage II-A*) (the “**Silent Participation II-A**”) and the silent participation II-B (*Stille Einlage II-B*) (the “**Silent Participation II-B**”) and, together with the Silent Participation II-A, the “**Silent Participation II**” and, together with the Silent Participation I, the “**Silent Participations**”) in the aggregate amount of €1.0 billion, which are accounted for as liability and not as equity according to IFRS, do not provide for loss participation and coupon deferral rights. The Silent Participation II has a six-year term and set forth interest rate step-ups from 4% p.a. (in 2020 and 2021) to 8% p.a. (in 2025 and 2026) and 9.5% p.a. from 2027 onwards.

In addition to the financings received under the WSF Stabilization Measures, we have also received state aid in other countries which including the United States and Italy, in which certain significant subsidiaries operate.

In Switzerland, under the Swiss Stabilization Package, a syndicate of Swiss banks provided a credit facility (85% state-guaranteed) in the amount of CHF 1.5 billion to SWISS and Edelweiss on August 20, 2020, as amended on November 12, 2020 (the “**Swiss Loan**”). The repayment of the Swiss Loan is scheduled for 2023, but can be extended to 2025 subject to the exercise of two one-year extension options. It does not include an equity portion. The Swiss Loan is secured, including by a pledge over the shares in SWISS, Edelweiss and its parent company AirTrust Ltd. that are ultimately held by the Company. Furthermore, the Company provided a subordinated loan in the amount of CHF 500 million in connection with the Swiss Loan.

In Austria, the Austrian Federal Government provided state aid in an aggregate amount of €450 million, comprising a credit facility (90% state-guaranteed) in the amount of €300 million (the “**Austrian Stabilization Loan**”), of which an amount of €30 million was repaid on July 15, 2021 and the Austrian Stabilization Loan was reduced accordingly, and a non-repayable subsidy grant (*Katastrophenbeihilfe*) in the amount of €150 million (the “**Austrian Subsidy Grant**”) to Austrian Airlines on August 28, 2020, which has been fully utilized on October 15, 2020. The Austrian Stabilization Loan carries an interest rate of 1%

p.a. and provides for a repayment in instalments, with the last tranche due for repayment on December 31, 2025. The Austrian Stabilization Loan is additionally secured, including by a pledge over the shares in Austrian Airlines as well as a lien on 38 certain aircraft owned by Austrian Airlines. The Austrian Subsidy Grant is not repayable provided that certain conditions are met, including that the net loss in the Fiscal Year 2020 exceeds €150 million and that such loss has been audited. Furthermore, the Company provided an equity contribution in the amount of €150 million in connection with the Austrian Stabilization Loan.

In Belgium, the Belgian Government provided state aid in the amount of €290 million, comprising a credit facility in the aggregate amount of €287 million (the “**Belgian Stabilization Loan**”) and profit participation certificates in the amount of €3 million (the “**Belgian Profit Participation Certificates**”) to SN Airholding SA/NV, for on-lending to Brussels Airlines, on September 2, 2020. Any amounts available for drawdown under the Belgian Stabilization Loan can only be drawn until December 31, 2021. The Belgian Stabilization Loan has a term of six years, carries an IBOR based interest rate plus a margin between 50 and 200 basis points and is due for repayment on July 31, 2026, at the latest, whereas the Belgian Profit Participation Certificates have been issued for an indefinite term. The Belgian Stabilization Loan is secured by pledges in SN Airholding SA/NV owned by the Company.

We face risks related to the breach of covenants under state aid packages we have been granted, and may trigger other events of default, which may ultimately result in a loss of shares in the companies SWISS (including Edelweiss), Austrian Airlines and Brussel Airlines. The state aid granted to SWISS (including Edelweiss) includes financial covenants regarding minimum liquidity, minimum EBITDA and minimum equity. The Austrian state aid includes financial covenants regarding, among others, maximum net debt, minimum EBITDA, maximum net debt/EBITDA and minimum equity. Further, an event of default as defined in each of the state aid packages may lead to a mandatory repayment of those state aids and a loss of securities pledged in connection therewith. Additionally, the interest rate applicable in connection with our state aid packages may increase as a result of a rating downgrade.

The Lufthansa Group is currently subject to, among others, the following behavioral commitments according to the Framework Agreement based on the decision of the European Commission dated June 25, 2020 regarding the approval of WSF stabilization measures (the “**EC Decision**”): M&A ban, remuneration cap and bonus ban, dividend ban and coupon/share buy-back ban as well as cross-subsidization ban. The Company has to abide to the restrictions and limitations in connection with these commitments and the financial stabilization measures that the Company has received and may continue to receive. For example, on May 19, 2021, we decided to suspend any coupon payments in relation to our €500,000,000 4.382% hybrid bond due 2075 and issued on August 12, 2015 (the “**2015 Hybrid Bond**”). The terms and conditions of the 2015 Hybrid Bond permit the suspension of coupon payments and such coupon payments shall remain suspended as long as the Silent Participation I and/or the Silent Participation II is drawn and/or the WSF is a shareholder of the Company. This decision was based on the European Commission’s view that a coupon payment under the 2015 Hybrid Bond would constitute a violation of the Temporary Framework for state aid measures to support the economy in the current COVID-19 outbreak adopted on March 19, 2020, as amended (the “**EU Temporary Framework**”), and the conditions of the EC Decision. As a result of our decision to suspend the coupon payments, Standard & Poor’s announced on May 25, 2021, that it lowered its rating of the 2015 Hybrid Bond to (CC). Additionally, this rating may be further lowered to (D) in the event that the coupon payments remain suspended at the next coupon payment date in February 2022. For further details on the Framework Agreement, see “*1.6.1 The terms of the Framework Agreement and certain other stabilization measures place significant restrictions on us. Additionally, the interests of the WSF may not necessarily coincide with the interests of other shareholders.*”

Any non-compliance with the commitments and the terms and conditions set out in the Framework Agreement, the EU Temporary Framework or the EC Decision could result in significant fines, penalties or other measures to sanction such non-compliance and may ultimately require us to repay state aid. As of the date of this Prospectus, the European Commission and the Company are discussing the preliminary assessment of the European Commission regarding potential violations of these commitments. The European Commission initially considered that the Company has violated certain commitments regarding the remuneration and dividend ban. However, the Company and the German government are still providing further information regarding those preliminary assessments. Additionally, the European Commission has requested that the Company provides further information to assess other measures taken or intended by the

Company with respect to their compliance with the M&A ban and the cross-subsidization ban. Provided that the European Commission maintains its current assessment, it may initiate a formal investigation procedure. If the European Commission were to uphold its assessment in such a procedure, it may order a partial recovery of the state aid granted to the Company and/or other measures or, ultimately, an annulment of the EC Decision and an early repayment of outstanding stabilization measures which could have negative effects on our business, financial positions and results of operation. Following such formal investigation, the Company could challenge any decision in court, but there is no assurance that such challenge will ultimately be successful.

Any future funding measures are subject to a consideration in line with market conditions, any material development of which could result in additional costs for us. Moreover, it cannot be excluded that we might incur higher funding costs when refinancing any funding instruments previously covered by governmental or other financial support programs and there can be no assurance that the credit, money and capital markets are accessible to us when these instruments mature. Consequences of any risks associated thereto may significantly affect our available cash flows and our financial performance.

1.2.3 We may be unable to maintain adequate liquidity.

As of June 30, 2021, we had available liquidity of €11.1 billion, including €6.7 billion cash at hand (equals sum of cash and cash equivalents plus securities, each as shown in the Unaudited Condensed Consolidated Interim Financial Statements), €0.5 billion free credit lines, €0.9 billion undrawn state aid (excluding Germany) and €3.0 billion undrawn under the Silent Participation I, which are available for drawdown until December 31, 2021. In the future, we may be unable to maintain, an adequate minimum liquidity to operate our business.

We have a significant amount of financial leverage and substantial non-cancelable commitments for capital expenditures, including the acquisition of new aircraft and related spare engines. As a result of a significant adverse effect of the COVID-19 pandemic on our business operations since the outbreak in December 2019, our business activities did not yield positive operating cash flows until the end of the second quarter of 2021, which materially negatively affected our liquidity. Our operations resulted in an average monthly operating cash drain (defined as Adjusted EBITDA adjusted for significant non-cash items, state aid grants and the redemption portion of IFRS 16 leasing expenses) of €217 million during the 6M 2021, resulting in an aggregate operating cash drain of €1,299 million as of June 30, 2021. We are not expecting any operating cash drain in the third quarter of 2021. Additionally, we have recorded contract liabilities from unused flight documents in the amount of €3,089 million as of June 30, 2021. However, in the event that air travel would significantly decrease or entirely come to a halt as a result of the further development of the COVID-19 pandemic, we could be exposed to a substantial cash outflow as customers could seek refunds for such unused flight documents. We have also recorded deferred import-turnover tax (*Einfuhrumsatzsteuer*) in the amount of €898 million as of June 30, 2021 due to the conclusion of deferral arrangements with the competent tax authorities in light of the COVID-19 pandemic.

While our cash flows from operations and available capital, including the proceeds from financing transactions, have been sufficient to meet these obligations and commitments to date, our liquidity could be negatively impacted by a number of risks, including, but not limited to, volatility in the price of fuel, adverse economic conditions, including the COVID-19 pandemic, rating downgrades, disruptions in the global capital markets and catastrophic external events or the failure in whole or in part of our measures and programs we aimed at improving earnings and reducing costs. Furthermore, we may have received certain subsidies in connection with growth arrangements entered into with certain airport operators that we could be obliged to fully or partially pay back. Under such arrangements, we are obliged to station a certain number of aircraft at an airport to expand its capacity and have in turn received subsidies therefore. Due to COVID-19, we may not have been able to station the required number of aircraft at an airport or may decide to discontinue stationing the relevant aircraft entirely given a decreased demand for air travel. In addition, if we were required to provide collateral under our existing hedging agreements, including as a result of a downgrade in our rating or market changes, our ability to access funds could be negatively affected and ultimately lead to reduced liquidity.

Moreover, our liquidity may be adversely affected by a discontinuation of state-paid work unemployment benefits, which would in turn ultimately have to be borne and financed by us. For example, in the 6M 2021, we were granted government support in the amount of €609 million in Germany, Austria and Switzerland in connection with short-time working. However, there is no assurance that we have been eligible for short-time work funds that we have received and/or complied with the requirements for short-time work at all times or that we will be eligible to receive short-time work funds going forward. In the event that we have not been fully eligible in relation to the short-time work funds that we have received or if we have not complied with the requirements for short-time work at all times, we may be required to repay short-time work funds, partially or in full.

If our liquidity is constrained due to any of these factors or otherwise, or if we fail to timely pay our debts or large invoices, comply with other material provisions of our contractual obligations, including, for example, covenants in relation to our credit card processing, we could become subject to a variety of adverse consequences, including the acceleration of debt, providing reserves under credit card processing agreements, the withholding of credit card sale proceeds by our credit card service providers and the exercise of other remedies by our creditors that could result in a material adverse effect on our cash flows, financial condition and results of operations.

Furthermore, constrained liquidity may limit our ability to withstand competitive pressure and limit our flexibility in responding to changing business and economic conditions, including increased competition and demand for new services, placing us at a disadvantage when compared to competitors.

Due to consequences arising in the context of COVID-19, we face increasing risks in the context of holding available sufficient liquidity (see “1.1.1 *The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.*”).

1.2.4 Any rating downgrade could impair our ability to raise financing.

As of the date of this Prospectus, we hold the following corporate ratings: Standard & Poor’s (BB-) with a “negative outlook”, Moody’s (Ba2) with a “negative” outlook” and Scope (BBB-) with a “negative” outlook. Additionally, our senior unsecured debt holds a “non-investment grade” rating from Standard & Poor’s (BB-) with a “negative” outlook, a “non-investment grade” rating from Moody’s (Ba2) with “negative” outlook and an “investment grade” rating from Scope (BBB-) with a “negative” outlook. Additionally, Standard & Poor’s announced on May 25, 2021, that it lowered the rating of our 2015 Hybrid Bond to (CC) as a result of our decision to suspend the coupon payments and may further lower the rating to (D) in the event that coupon payments remain suspended at the next coupon payment date of the 2015 Hybrid Bond in February 2022. There is no assurance that our credit rating could not be further downgraded at any time by Standard & Poor’s, Moody’s or Scope, in particular, if our operating cash flows or other financial indicators, including financial debt or pension obligations, fall short of or exceed targets and expectations or due to the sell-down of the WSF resulting in a shareholding of 15.94% of the share capital of the Company as of September 17, 2021. Downgrades of any of our ratings may have a material adverse effect on the cost and availability of financing or refinancing opportunities and could significantly adversely affect the willingness of our business partners to conduct business with us. For example, if our credit ratings were downgraded further, we may face significant difficulties in finding counterparties for our hedging transactions or may be required to fully cash collateralize these transactions with a corresponding negative effect on our liquidity. In addition, the downgrading of our credit ratings could also force us to accept termination clauses that would be triggered in the event of renewed negative rating actions by certain credit rating agencies in future contracts that we enter into.

Moreover, if our ratings are further downgraded, we may only be able to meet our financing and refinancing requirements on significantly less-favorable terms, for example at higher interest rates or with additional collateral requirements or not at all. This also applies to current funding instruments, including state stabilization measures that we have received such as the Swiss Loan, which provide for interest step-ups (based on a rating grid) in case of rating events or similar provisions with negative funding impact in the

form of direct or indirect increased costs of funding for us. Furthermore, downgrades in the credit ratings assigned to us may also lead to negative impact concerning the participation in governmental stimulus packages and/or measures introduced by central banks (see “1.1.1 The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.”).

1.2.5 Our profit forecast and other profitability, revenue or cost-saving targets could differ materially from our actual results of operations.

As of the date of this Prospectus, depending on the extent of the recovery in the airline business and the progress made with certain restructuring measures, the operating loss for the Lufthansa Group, as measured by Adjusted EBIT is expected to be lower for the Fiscal Year 2021 than in the previous year. The Adjusted EBIT margin is therefore also expected to be less negative than in the previous year.

Our profit forecast is based on a variety of factors beyond our control, as well as related assumptions, which include, but are not limited to: (i) uncertainties relating to COVID-19, (ii) the general economic and political development in the European Economic Area, (iii) the stability of the legal and regulatory framework, (iv) general developments in the European and global airline industries, (v) certain fuel cost fluctuations, and (vi) other unforeseen events.

The Company’s executive board (the “**Executive Board**”) has based the profit forecast as well as the other profitability and revenue targets on a number of assumptions, which are inherently subject to significant business, operational, economic and other risks, many of which are beyond our control. Accordingly, any such assumptions may change or may not materialize to the extent envisaged, if at all. In the event that one or more of the assumptions underlying the profit forecast as well as the other profitability and revenue targets should prove to be incorrect or inadequate, our actual results of operations for the respective future period(s) may differ materially from those outlined above. Accordingly, investors are urged to not place undue reliance on our profit forecast, the underlying assumption or any development of the Company’s business as a result thereof.

1.2.6 Fluctuations in currency exchange rates can have material adverse effects on our cash flows, financial condition and results of operations.

Given the international nature of our business, we generate a substantial portion of our revenue, and incur a substantial portion of our operating expenses, in foreign currencies, especially the U.S. dollar, Chinese renminbi, the Swiss franc, British pound sterling, the Japanese yen and the Indian rupee. In addition, we hold interests in a series of companies, especially in the Network Airlines, Catering and MRO business segments, that are incorporated outside the Eurozone and whose cash flows are therefore generated in currencies other than the Euro. Therefore, fluctuations in exchange rates between the Euro and non-Euro currencies will affect the translation of the financial results of our consolidated non-Euro subsidiaries into Euro and will also affect the value of any contributions that our business segments generate. Exchange rate changes may also affect our consolidated balance sheet. As of June 30, 2021, we have outstanding foreign exchange derivatives with a nominal volume of €240 million which are not designated as hedging. Changes in the Euro values of our consolidated assets and liabilities resulting from exchange rate movements may cause us to record foreign currency gains and losses. In addition, most of our consolidated debt and capital expenditure commitments are denominated in Euro or U.S. dollars. Therefore, our ability to use cash received in a currency other than the Euro or U.S. dollars to service our debt or capital expenditure commitments could be adversely affected by changes in exchange rates against the U.S. dollar or Euro. Further, all of our aviation fuel expenses are denominated in U.S. dollars.

We are subject to significant exposure to currency fluctuations, including the exchange rate of the Euro and the U.S. dollar, and have implemented a policy of hedging such currency risks. However, there is no assurance that our existing currency hedging arrangements fully protect us against currency fluctuations or may otherwise reduce or negate the benefit we are able to derive from positive changes in exchange rates. If our hedging policy proves unsuccessful, it could have a material adverse effect on our business and financial position.

1.2.7 *Fluctuations in interest rates could have material adverse effects on our financial condition and results of operations.*

As of June 30, 2021, our financial liabilities amounted to €15,571 million. A significant amount of our financial liabilities as of June 30, 2021, are either at floating rates from the outset or are swapped into floating rates using derivatives. Floating interest rates tend to fluctuate based on general economic conditions, prevailing market interest rates and the supply of, as well as demand for, credit. To the extent floating interest rates increase, our interest expenses will increase, in the case of which we may have difficulties making interest payments and funding our other fixed costs, and our cash flow available for general corporate purposes may ultimately be adversely affected.

1.2.8 *Our pension obligations could substantially increase and exceed the provisions we have recognized for these obligations in our accounts.*

We have significant pension obligations towards our employees. Our pension obligations comprise both defined-benefit and defined-contribution plans and include both obligations to make current payments and entitlements to future pension payments. Obligations under defined-benefit pension plans for our employees relate mostly to pension obligations in Germany, Switzerland, Austria and the United States. Various commitments have been made to different groups of employees. The defined-benefit pension obligations are, in a relevant part, covered by pension funds or insurance and partly by provisions. The amount of these provisions is based on certain actuarial assumptions, including discount factors, demographic trends, pension trends, future salary trends and expected returns on plan assets. If actual results, particularly in relation to discount factors, were to deviate from these assumptions, or if actuarial assumptions change, there could be a substantial increase in pension obligations and a resulting increase in the provisions for pensions on our balance sheet. We bear the risk that the value of plan assets will decrease, including with respect to pension obligations whose coverage has been outsourced to external funds, since our employees will in all cases have direct subsidiary claims against us.

Additionally, under IAS 19 “Employee Benefits,” which is applicable since January 1, 2013, actuarial gains and losses are to be recognized directly in “other comprehensive income,” taking deferred taxes into account. Changes in the discount rate used to measure defined-benefit pension obligations and fluctuations in the market value of plan assets for funded pension plans, can in particular result in considerable and unpredictable fluctuations in the balance sheet, as well as shifts between equity and liabilities applying the revised IAS 19. With regard to pension fund assets, we are exposed to general financial market risks of below-average portfolio performance as well as to the risk of errors in the choice of investments. All the above factors could have material adverse effects on our net assets, as well as our financial and earnings position. For further details, see “1.4.2 *We are and could in the future be subject to claims in connection with legal disputes, administrative proceedings and investigations, which could adversely affect our reputation and financial position.*”

As of June 30, 2021, the present value of our defined benefit obligations was €25,077 million. As of June 30, 2021, the fair value of external plan assets was €17,470 million for pension obligations. As of June 30, 2021, our pension provisions amounted to €7,607 million (as of December 31, 2020: €9,531 million). The discount rate used to calculate these obligations in Germany was 1.2% (as of December 31, 2020: 0.8%).

1.2.9 *We are exposed to the risk of payment default by our contractual partners.*

We generate a significant portion of our revenue from a comparatively small number of customers in our Logistics, MRO and Catering business segments. For example, the ten largest revenue-generating customers (excluding companies within the Lufthansa Group) accounted for 32% and 72% of external revenue generated in the MRO and in the Catering business segments during the 6M 2021, respectively (Fiscal Year 2020: 33% and 66%, respectively; Fiscal Year 2019: 30% and 59%, excluding customers of LSG Group’s European business operations, respectively). The share of external revenue generated in the Logistics segment attributable to the top ten customers (excluding companies within the Lufthansa Group) was 42% during the 6M 2021 (Fiscal Year 2020: 40%; Fiscal Year 2019: 39%).

For certain categories of transactions, we have a limited number of counterparties. For example, we enter into transactions to hedge currency, fuel price, and other risks with a limited number of banks and financial

institutions. In addition, we insure our fleet of aircraft and their operations with companies such as Delvag Luftfahrtversicherungs AG, one of our subsidiaries, which, in turn, obtains insurance from only a limited number of re-insurers. Moreover, we regularly pay deposits on aircraft orders to aircraft manufacturers. In some business segments, including the MRO business segment, the advance performance of services to customers is significant.

As a result of our contractual partner structure, total receivables in respect of individual customers, re-insurers and banks or financial institutions sometimes accrue in considerable amounts. There is a risk that these receivables could be uncollectible in whole or in part if contractual partners fail to pay or experience a temporary inability to pay or become insolvent and it is expected that such risks will increase due to the current situation around COVID-19 and the economic downturn in this context (see “1.1.1 *The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.*”). In addition, a reduction or elimination of demand for our services by a key customer due to insolvency could lead to a decrease in revenue. We have been and may, in the future, be unable to recover considerable amounts owed to us by contractual partners that are subject to insolvency, restructuring or liquidation proceedings, including due to the COVID-19 pandemic, which could adversely affect our cash flows, financial condition and results of operations.

1.2.10 We are required to make estimates, assumptions and judgments, applying substantial discretion, when preparing our consolidated financial statements, which could prove to be materially incomplete or inaccurate, requiring us to record additional impairments.

The preparation of our consolidated financial statements in accordance with IFRS requires making estimates, assumptions and judgments that ultimately affect our results or financial statements. Some of these estimates, assumptions and judgments are critical due to the high degree of uncertainty of the relevant parameters at the time they are used. The discretion and optionality available when making such decisions could, if such estimates, assumptions and judgments would turn out to be incomplete or inaccurate, materially impact our results or financial statements. This applies, in particular, to the following items: (i) initial consolidation and the recognition of goodwill, (ii) calculation of defined benefit obligations, (iii) net pension provisions, (iv) deferred tax assets, (v) financial assets, and (vi) impairment losses on intangible assets and property, plant and equipment.

The outbreak of the COVID-19 pandemic and the steps taken globally to contain the pandemic have had a significant impact on our business operations and accordingly on the estimates, assumptions and judgments that we make when preparing our financial statements. Management assesses against the backdrop of the COVID-19 pandemic, whether future taxable profits will be available against which deductible temporary differences and unused tax losses can probably be utilized. While the planning period generally is between three and five years, the Company also assessed certain positive and negative indicators, such as the restructuring and re-dimensioning of our business or expected challenges resulting from climate change, and assuming a utilization period of more than 10 years for tax losses carry-forward (including the temporary braking effects as a result of the German minimum profit taxation) and particularly for deductible temporary differences on pension provisions subject to their duration. This prolonged planning period is susceptible to increased uncertainty and, therefore, the estimates, assumptions and judgments made to determine the relevant indicators affecting the value of our deferred tax assets and/or the planning period could prove to be inaccurate.

Additionally, the determination of the amount of any impairments under our accounting principles and policies with respect to tangible assets is based upon our periodic evaluation and assessment of known and inherent risks associated with the respective asset class. We test property, plant and equipment for impairment if events or changes in circumstances indicate that the carrying amount of an asset may not be recoverable. In addition to depreciation, impairment losses are also recognized at the end of each reporting period if the asset's recoverable amount or the recoverable amount of the asset's cash-generating unit has fallen below its carrying amount.

In this regard, the Company was made aware that, following an individual allegation, a preliminary inquiry for the clarification of facts was commenced by the German Auditor Supervisory Authority (*Abschlussprüferaufsichtsstelle* - APAS). Subject to the preliminary inquiry is, among others, the Company's accounting for deferred tax assets and the accounting for the aircraft fleet in its statutory and consolidated financial statements for the fiscal year 2020 in light of the COVID-19 pandemic. While the Company, without being an addressee of the preliminary inquiry, is of the view that its estimates, assumptions and judgments for the accounting of its deferred tax items and aircraft impairment are reasonable and it has accurately accounted based on these estimates, assumptions and judgments, there is no assurance that a third party will come to the same conclusion and that our estimates, assumptions and judgments that were made could not be made differently. As of June 30, 2021, the carrying amount of our deferred tax assets amounted to €4,725 million, of our aircraft and reserve engines to €15,516 million and of our property, plant and other equipment to €3,493 million. Accordingly, if our estimates, assumptions and judgments prove to be inaccurate, the carrying amount could have to be impaired.

1.3 Risks Related to Our Industry

1.3.1 The highly competitive airline industry as a whole is susceptible to the effects of adverse economic developments, which may lead to lower demand for flights and overcapacity in the market and, as a result, reduced fares and profitability.

Economic downturns generally lead to an overall decline in flight demand, both in the passenger and cargo flight segments. Furthermore, effects resulting from the COVID-19 pandemic have already led to such decline in flight demands.

In particular, our Network Airlines business segment generates higher margins on ticket sales to first class and business class passengers than on ticket sales to premium economy class and economy class passengers. Economic downturns generally lead to lower demand for these high-margin tickets and effects resulting from the COVID-19 pandemic have already led to such lower demand as corporate and business clients were and continue to refrain from travelling due to global travel restrictions and travel bans. Furthermore, due to the uncertainties resulting from the COVID-19 pandemic, airlines offer more flexible fare conditions across all booking classes and it remains uncertain when, or if at all, airlines will be able to return to pre-COVID-19 fare conditions. In addition, economic downturns resulting from the COVID-19 pandemic or resulting from other circumstances and effects cause leisure travelers, which are also materially affected by global travel restrictions and travel bans, to book flights increasingly in less expensive, and, therefore, less profitable fare classes and to take advantage of deals offered by low-cost airlines. As spending on leisure travel is largely discretionary, this spending also is likely to be significantly reduced or cut during economic downturns.

The lower number of passengers in economic downturns leads to excess capacity in the passenger airline industry, which results in increased competitive price pressure.

There is a high level of uncertainty regarding the global economic outlook. Significant downside risks originate from consequences of political uncertainty, including an increasing trend towards populism and the current situation and consequences of COVID-19 (see “1.1.1 *The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.*”). The ongoing trade dispute between the United States and China could result in a further weakening of the world economy and increase uncertainty regarding the global economic outlook. Separately, a downturn in any of the world's major economies, such as China, the United Kingdom, the United States, Germany or any other market in which we operate could also adversely affect the world economy.

The nature of our cost structure makes it difficult for us to respond flexibly to the potential adverse effects of economic downturns and shocks. Flight operations have a high percentage of fixed costs, including those related to fuel, labor costs, aircraft depreciation or lease rates, air traffic control fees and take-off and landing fees. Regardless of the number of passengers, these costs remain constant so that the marginal cost for each additional passenger or additional unit of cargo is low. Airline revenue, however, results primarily from the fares or freight rates paid so that the marginal revenue from each additional passenger or unit of cargo is

relatively high. Any decline in passenger numbers or freight volume will lead to a decline in revenue without a proportionate drop in costs because such fixed costs generally cannot be reduced on short notice, and some of these costs cannot be reduced by any meaningful amount or at all. To increase profitability during economic downturns, we have in the past reduced the number of flights we offer. However, reducing flight frequency through the cancellation of flights may decrease the demand for our services, as we may no longer offer the necessary minimum flight frequency to our customers.

1.3.2 An economic downturn may negatively affect spending on airline tickets and airfreight transport.

In addition to the current situation in the context of COVID-19 (see “*1.1.1 The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.*”), national economies, and the global economy as a whole, are currently still experiencing ongoing uncertainty, which results in a period of economic slowdown or even downturn. If domestic economies that are particularly important to our business, such as Germany, Switzerland, Austria, Belgium, the United States or China and/or the global economy undergo a prolonged period of uncertainty or a significant downturn, such as the downturns that a number of countries have experienced as a result of the European sovereign debt crisis, or if our customers believe such a period of uncertainty or a downturn will reemerge for a sustained period, our customers may reduce their air travel and airfreight spending and air travel and airfreight budgets. Such reductions could lead to a further drop in the profitability of, or even losses in, our Network Airlines, our most important business segment, and our Logistics business segment.

Further, the number of new aircraft ordered by competitors and the current growth prospects mean that overcapacities are expected to persist in both the passenger and airfreight markets in the future. This could increase pressure on our average yields. Our ability to remain competitive under these conditions depends primarily on how flexible we are in reducing costs and adjusting capacities and how fast we can react to changes in demand. We cannot guarantee that measures taken to optimize income by adjusting our prices and capacity to the demand situation will successfully remedy the adverse effects of an economic downturn.

1.3.3 Consolidation in the aviation market could lead to increased competition for us and playing an active role in the market consolidation could lead to write downs and future losses.

The European aviation market underwent, and is continuing to undergo, noticeable consolidation in the past years and is highly volatile. This volatility has increased as a result of uncertainties in relation to the COVID-19 pandemic. Ultimately, there can be no assurance whether the recent consolidation activities increase, including as a result of reduced valuations, or decrease rapidly, for example as a result of the increase in state-supported airlines (see “*1.1.1 The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.*”). In particular, due to the spread of COVID-19 and also due to the insolvency filings of, among others, Alitalia (as well as its nationalization), Air Berlin, Germania, Thomas Cook and Norwegian, shifts in market shares are likely to occur and lead to or, as the case may be, occurred and led to new market entries and stronger market positions of airlines which were previously not as active in our main markets. Although we aim at playing an active role in these consolidations, it cannot be excluded that we may not be able to obtain assets such as aircraft as well as crew to increase or even keep our market share, including as a result of the restrictions to participate in mergers and acquisitions activities in connection with the WSF Stabilization Measures (see “*1.2.2 We are subject to significant limitations in our financial flexibility as a result of the state aid received and financial stabilization measures obtained from governments in response to the outbreak of COVID-19 and face risks associated with triggering events of default under these state aid and stabilization measures.*”). Further new aircraft and routes may not be as profitable as estimated which could lead to material losses. Our competitive position might be weakened in the consolidation process following airline insolvencies.

1.3.4 Terrorist attacks, political uprisings, armed conflicts and their consequences could have a material adverse effect on the industry in which we operate as a whole.

Terrorist attacks, political uprising and armed conflicts worldwide continue to have significant negative effects on the international airline and tourism industry, including us. Moreover, the threat posed by terrorist attacks, including cyber-attacks, sabotage, bioterrorism and new forms of terrorism, as well as war, civil war and riots, has a growing adverse effect on the global business and political environment. This uncertainty is likely to continue for the foreseeable future and would likely intensify dramatically in the event of further attacks, particularly if they were targeted against civil aviation, business centers or tourist destinations. The occurrence of such events could result in, for example, a sharp decline in the demand for air travel and other services offered by our various business segments. Significant cost cutting as a result of such a decline by major airlines that are our customers could have negative consequences for us, including in relation to each of our MRO and Catering business segments. We could incur higher costs and reduced revenue as a result of additional security precautions, whether undertaken voluntarily or in accordance with regulatory requirements. In the event of armed conflicts, we could be restricted to or decide not to fly to certain destinations or over certain areas, which would curtail our flight operations and route planning. All of the aforementioned factors and additional consequences of terrorist attacks or armed conflicts could have a material adverse effect on our cash flows, financial condition and results of operations.

1.3.5 The airline industry is particularly vulnerable to the effects of epidemics and natural disasters including extreme weather conditions.

Natural disasters, including extreme weather conditions, or epidemics of regional or global proportions could restrict our operations or result in substantial reductions in, and cancellations of, bookings and cause overall demand for our services to drop (see “1.1.19 We are exposed to adverse developments relating to the occurrence of unforeseen events, in particular natural and man-made disasters that may not be covered by our insurance policies, and changes in the political, geopolitical and economic conditions.”). This relates to the passenger business in particular but also to our Logistics, Catering and MRO business segments.

Activity from volcanoes or other natural or man-made disasters are likely to materially and adversely affect our passenger and cargo volumes. In response to future natural or man-made disasters, regulatory authorities may impose operating restrictions at airports, such as landing and take-off curfews, mandatory flight paths, runway restrictions and limits on the number of average daily departures. These restrictions may limit our ability to provide services at such airports and may cause us to incur additional costs, which could result in a material adverse effect on our cash flows, financial condition and results of operations.

Health epidemics and outbreaks of contagious diseases, including but not limited to the aforementioned, could materially and adversely affect the sustainability of our business. The spread of an epidemic, such as during the worldwide spread of COVID-19 since the beginning of 2020, can result in the permanent closure of many businesses and the application of comprehensive quarantine measures to prevent the transmission and contain the contagion. In addition, governments have issued and could issue further bans on air travel during epidemics (with regard to COVID-19 see “1.1.1 The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.”). Any prolonged recurrence of diseases or the prolonged onset of COVID-19 or other adverse public health developments in Germany or elsewhere may have a material adverse effect on our business operations, with materially adverse effects, especially but not limited to its flight network and flights scheduled.

Additionally, the occurrence of viruses, pandemics or epidemics could cause severe illness and loss of the Company’s management and key employees, as well as temporary closure of its offices, subsidiaries, plants and comparable adverse effects. Such losses would severely disrupt our business operations.

1.4 Legal, Regulatory and Tax Risks

1.4.1 *We may be exposed to legal sanctions and penalties as we may not be able to prevent or detect violations of legal compliance and other economic and administrative regulations.*

As a globally operating airline with a substantial market presence particularly in Europe, Asia and North America, and due to our integration in Star Alliance, the world's largest airline alliance in terms of market share (measured by ASK) (*source*: Statista, Leading Airline Alliances in 2018, August 2019), we are exposed to the risk that antitrust authorities or courts could restrict or entirely prohibit inorganic expansion. Antitrust issues could also arise in relation to future partnerships and alliances, or the strengthening of existing ones. In such cases, there would be an examination by the antitrust authority of each country in which the cooperative arrangement would have effects and the antitrust laws and regulations of such countries would apply.

To enable the Lufthansa Group to improve its liquidity position and meet its general capital requirements, the Lufthansa Group received state aid and government-backed financial stabilization measures, including in Germany, Switzerland, Austria and Belgium. Any non-compliance with the commitments and the terms and conditions set out in the Framework Agreement, the EU Temporary Framework or the EC Decision could result in significant fines, penalties or other measures to sanction such non-compliance and may ultimately require us to repay state aid. As of the date of this Prospectus, the European Commission and the Company are discussing the preliminary assessment of the European Commission regarding potential violations of these commitments. The European Commission initially considered that the Company has violated certain commitments regarding the remuneration and dividend ban. However, the Company and the German government are still providing further information regarding those preliminary assessments. Additionally, the European Commission has requested that the Company provides further information to assess other measures taken or intended by the Company with respect to their compliance with the M&A, acquisition ban and the cross-subsidization ban. Provided that the European Commission maintains its current assessment, it may initiate a formal investigation procedure. If the European Commission were to uphold its assessment in such a procedure, it may order a partial recovery of the state aid granted to the Company and/or other measures or, ultimately, an annulment of the EC Decision and an early repayment of outstanding stabilization measures which could have negative effects on our business, financial positions and results of operation. The Company could challenge any decision following such a formal investigation in court, but there is no assurance that such challenge will ultimately be successful. For further details on the Framework Agreement, see “1.2.2 *We are subject to significant limitations in our financial flexibility as a result of the state aid received and financial stabilization measures obtained from governments in response to the outbreak of COVID-19 and face risks associated with triggering events of default under these state aid and stabilization measures.*” and “1.6.1 *The terms of the Framework Agreement and certain other stabilization measures place significant restrictions on us. Additionally, the interests of the WSF may not necessarily coincide with the interests of other shareholders.*”

There is also a risk that, if we are in violation of other economic or administrative regulations, including permit and reporting obligations, we could be prohibited from pursuing certain business activities and could receive penalties or fines or have other conditions or obligations imposed on it.

We are subject to trade and economic sanctions, anti-bribery and anti-corruption laws and regulations in the jurisdictions in which we operate. These laws and regulations evolve over time and it could be difficult for us to predict the interpretation, implementation or enforcement of such laws and regulations as well as related governmental policies with respect to our business. Additionally, our business involves contact with representatives of national governments and other officials, who may be in the focus of anti-bribery and anti-corruption laws and regulations. While we continuously review our existing policies and procedures to ensure compliance with applicable laws and regulations to address any detected and reasonably foreseeable instances of non-compliance, there can be no assurance that these policies and procedures will be followed at all times or that our internal controls will effectively prevent violations of such laws and regulations by our employees, consultants, agents or partners. Any failure or alleged failure to comply with such laws and regulations may expose us to penalties, including criminal fines, imprisonment, civil fines, disgorgement of profits, injunctions and debarment of government contracts (and termination of existing contracts), as well as other remedial measures. In addition, any potential allegation or actual violation could result in adverse media coverage and may result in reputational harm.

Since the introduction of the EU Market Abuse Regulation (“**Market Abuse Regulation**” or “**MAR**”) in 2016, there is an increased risk of substantial penalties for delayed or omitted ad-hoc notifications, infringements of the insider trading prohibition, the prohibition of market manipulation or other organizational failures by us or our employees and managers. Furthermore, we face the risk of significant fines and cost increases resulting from regulations, such as the European Market Infrastructure Regulation and the Markets in Financial Instruments Directive, as amended (“**MiFID II**”). These regulations impose stricter rules on capital market, trading and market infrastructure compliance. The increase of regulation leads to increased costs for internal compliance measures and potentially to the increase of hedging costs for us. Similarly, stricter capital requirements for banks, including due to the changes to global bank capital requirements under Basel IV standards, could increase our financing costs.

Examinations by authorities to determine potential violations of antitrust regulations or other economic or administrative regulations, including Data Protection Laws (see “1.1.10 *We are exposed to risks associated with breaches of GDPR compliance, which we particularly face in connection with cyber security breaches.*”), even unsubstantiated suspicions of such violations, could have substantial negative effects on our results of operations. Actual violations of these regulations could lead to significant fines and/or claims for damages by injured parties. By way of example, we have been subject to investigations instituted by various antitrust authorities, charging that we, together with several other air cargo carriers coordinated surcharges (such as fuel and security surcharges) and agreed to eliminate or prevent discounting of surcharges. While these proceedings have in the meantime been settled or are in the process of being appealed, we are currently subject to private civil actions for damages by customers in relation to price fixing charges in the air cargo sector. Decisions rendered against us might result in significant fines or burdensome conditions and obligations (for example, the surrender of slots without compensation).

Any decision rendered or procedure initiated against us or our officers by an authority or court, including the prohibition of a cooperative arrangement or a merger or the imposition of penalties, large fines or burdensome conditions and compliance obligations, as well as violations of data protection regulations, could have material adverse effects on our cash flows, financial condition and results of operations.

1.4.2 *We are and could in the future be subject to claims in connection with legal disputes, administrative proceedings and investigations, which could adversely affect our reputation and financial position.*

In the ordinary course of our business, we are involved in a number of lawsuits in Germany and in other countries, both as plaintiff and as defendant. A large number of these lawsuits involve, among others, the Network Airlines business segment and relate to personal injuries alleged to have been suffered on flights, service disruptions (including flight delays and lost or damaged luggage), disputes and litigations regarding intellectual property, including a trademark dispute regarding the tail of our aircraft, as well as allegations of inadequate information regarding visas and other conditions of entry. There are currently several proceedings pending against us in connection with anticompetitive arrangements by Lufthansa Cargo, and we are facing antitrust investigations and claims for damages by customers. See “1.4.1 *We may be exposed to legal sanctions and penalties as we may not be able to prevent or detect violations of legal compliance and other economic and administrative regulations.*” We may not be able to accurately predict the outcome of pending or threatened legal proceedings. This is particularly true of lawsuits under common law, for example in the United States or the United Kingdom, which may result in significant damages being awarded, specifically in the event of successful class actions.

In addition, we are party to a number of lawsuits relating to labor and employment law matters, including in relation to the training of our pilots and the financing thereof, our collective bargaining agreements as well as the transfer of undertakings. Among others, following the merger of Tyrolean Airways with Austrian Airlines in 2012, pilots of Austrian Airlines filed lawsuits regarding procedures related to seniority rankings.

We are also engaged in several legal proceedings with flight attendants claiming the continuous applicability of former collective agreements with the union “ver.di” and with the union “UFO” which arguably have been superseded by more recent collective agreements with the union “UFO.” The legal proceedings mainly relate to claims regarding early retirement schemes (*Übergangsversorgung*) and company pension schemes (*Betriebsrente*) that had been transitioned from a defined benefits system into a defined contribution system under a collective agreement with the union “UFO” in July 2016, resulting in a decrease of accruals for

pension liabilities of €650 million in the fiscal year 2016. Certain employees claim that the collective agreements do not apply to them in respect of the transition of their early retirement and company pension scheme from the defined benefits system into defined contributions system because they consider the previous collective agreements with the union “ver.di” to remain applicable. In recent decisions at the end of April 2021, the German Federal Labor Court confirmed this view and ruled that, due to the interpretation of the reference clause to collective agreements included in the respective employment agreements, employees may request the application of previous collective agreements, which would result in the applicability of the defined benefits system in place prior to July 2016. However, following such request and as a result of the application of such collective agreements, the previous remuneration tariff agreement would equally apply, which does not provide for dynamic remuneration increases. We have recently received the reasoning of the German Federal Labor Court and are determining its potential impact on the Lufthansa Group. At this stage, neither the number of claims that may be asserted nor the merits of any potential claims can be reliably assessed. Accordingly, potential revaluation effects from our pension liabilities can neither be quantified at this stage. In the event that we would be required to increase our pension liabilities from revaluation effects as a result of this ruling, a potential revaluation could amount to a low three-digit million amount.

Additionally, several legal proceedings are pending against us, in particular, in connection with the outbreak of COVID-19. Among others, we are facing claims by flight attendants that are members of the ver.di workers union regarding the non-applicability of the collective agreement regarding COVID-19-related employment adjustments and in relation to the application of certain of its provisions. If the claims were to be partially or fully successful, any ver.di flight attendant could claim that the savings measures pursuant to the collective agreement are not applicable and request payment of the withheld amounts. Additionally, we are facing similar lawsuits by our ground staff regarding the cancellation of the Christmas bonus and holiday pay (*Urlaubsgeld*) in connection with the collective agreements with ver.di and demanding payment of a 13th monthly salary based on their individual employment agreement. Moreover, Ryanair DAC initiated an action for annulment in relation to the state aid we received and the financial stabilization measures we obtained from various governments. If their claims were partially or fully successful, we could be obliged to, among others, repay the aforementioned funds in their entirety.

Additionally, we are exposed to additional risks in connection with the loss of the aircraft on Germanwings flight 4U9525 on March 24, 2015, when the aircraft crashed on its way from Barcelona, Spain, to Dusseldorf, Germany, in the French Alps, killing all 150 people on board. The data recovered from the Airbus A320's two black boxes have led the investigators to believe that the aircraft was intentionally sent into descent by the co-pilot. This finding was confirmed by the French Bureau of Enquiry and Analysis for Civil Aviation Safety (*BEA*) in its final report issued in March 2016. Germanwings (whose passenger flight operations have been terminated) is compensating the families of the passengers according to the applicable provisions of the relevant jurisdictions. Although a significant number of families have signed full and final settlement agreements, some other families have opted to sue Germanwings or us in various jurisdictions in order to obtain higher compensation. As of the date of this Prospectus, court proceedings remain pending in Spain, Germany and Venezuela, although a recent decision of the Higher Regional Court (*Oberlandesgericht*) of Hamm, which is not yet binding, was in favor of the Company. However, any of such litigation bears risks and could potentially result in large awards of compensatory damages. Though we believe that we have adequate insurances in place there is no absolute certainty that we will be fully indemnified by our insurers. In addition, the incident itself, any protracted legal disputes with the bereaved families or any other kind of criminal or administrative proceeding or investigations could harm our reputation.

Furthermore, it cannot be excluded that award miles or other features of our Miles & More Program could be qualified as e-money, including within the meaning of the German Payment Services Supervision Act (*Zahlungsdiensteaufsichtsgesetz*) and, as a consequence thereof or otherwise, would have to be retransferred into money at any time on request. In the event of a classification as e-money, we would be required to obtain a license from BaFin. There is no assurance that BaFin will grant such license. In this event, we may not be able to continue to operate our Miles & More Program as previously or at all.

1.4.3 *We are subject to compensation claims for flight delays and cancellations, including under the European Air Passengers Regulation, and may, including for circumstances beyond our control, be required to make significant compensatory payments to passengers.*

A number of jurisdictions have implemented provision on passenger rights, obliging airlines to provide assistance and care, as well as rerouting or reimbursement to passengers in cases of flight disruptions or denied boarding. In addition, airlines have to compensate passengers in certain cases.

Among others, we may continuously be exposed to a significant number of claims for compensation for passengers under Regulation (EC) No 261/2004 of the European Parliament and of the Council of February 11, 2004 establishing common rules on compensation and assistance to passengers in the event of denied boarding and of cancellation or long delay of flights, and repealing Regulation (EEC) No 295/91 (the “**Air Passengers Regulation**”) in the context of flight delays and cancellations we encounter. Although the European Commission has indicated that it would label COVID-19 as an extraordinary circumstance (within the meaning of the Air Passengers Regulation), as a result of which we would not be obliged to compensate passengers in certain cases, we are exposed to compensation claims for flights cancelled for operational reasons beyond the immediate effects of COVID-19, whereas such determination ultimately lies with the competent courts. Under the Air Passengers Regulation, such compensation is between €250 and €600 depending on the length of the flight and the causes of the disruption. Simultaneously, we are exposed to risks associated with court rulings, including those to which we are not a party, which may, on a national or supra-national level, broaden the scope or definition of causes of disruption within the meaning of the Air Passengers Regulation. For example, in 2018, the European Court of Justice ruled that a spontaneous absence of a significant part of the flight crew staff (in the context of a restructuring announcement and a subsequent spike in sick leaves of flight crew staff) does not fall within the meaning of ‘extraordinary circumstances’ under the Air Passengers Regulation and, accordingly, claims are payable thereunder. Accordingly, the European Court of Justice also ruled, in 2021, that a strike organized by a trade union of the staff of an air carrier does not fall within the meaning of ‘extraordinary circumstance’ under the Air Passengers Regulation.

In 2013, the European Commission published a proposal for the revision of the Air Passengers Regulation, aiming at defining the rights of passengers more clearly and finding a fair burden sharing in cases of force majeure, such as the volcanic ash crisis (e.g., caused by the outbreak of the volcano Eyjafjallajökull in Iceland in 2010) and airport closures due to adverse weather conditions. The revision has not been finalized yet and the discussions between European Council, European Parliament and European Commission are still ongoing. New passenger rights regulation might increase potential compensation payments which could have material adverse effects on our cash flows, financial condition and results of operations.

Additionally, we are exposed to a substantial number of refund claims from customers whose flights were cancelled or who, for future bookings, redeem flight vouchers issued as a result of flight cancellations in connection with COVID-19. In the Fiscal Year 2020, we refunded tickets in an amount of €3,851 million. Any increase in cash outflow (for refund claims) and decrease in cash inflow (as vouchers redeemed do not generate cash) had and will in the future have a material adverse effect on our financial condition and, ultimately, increase our liquidity risk (see “1.2.3 *We may be unable to maintain adequate liquidity.*”). In addition, we are defending class-action lawsuits in the United States and Canada related to the delays in providing passengers refunds for cancelled flights as a result of the COVID-19 pandemic. We also entered into a settlement with the German consumer organization Baden-Wuerttemberg (*Verbraucherzentrale Baden-Württemberg e.V.*, “**VZBW**”) and committed to processing ticket refund requests within seven days in accordance with the Air Passengers Regulation. In the event that we do not process ticket refund requests within such deadline, whether negligently or intentionally, VZBW may seek action, including legal proceedings, which could result in additional costs, expenses and administrative fines.

1.4.4 *Future changes in tax and social security laws and changes that have already taken place, the effect of which on our tax obligations cannot be predicted and will significantly depend on future developments, could result in an increase of tax or social security burden for us.*

It is possible that changes in applicable tax or social security laws in Germany, at the EU level or in third countries we are operating in, may increase our tax or social security burden. For example, we may

experience a material adverse effect if the tax exemption applying to kerosene is repealed. In addition, we are subject to higher regulatory taxes in relation to, in particular, governmental initiatives to reduced carbon emissions. See “1.5.4 *We are subject to additional costs as a result of our requirement to comply with increasing regulatory measures aimed at restricting the emission of greenhouse gases and related models of emission rights trading.*” Further, changes to the German Foreign Tax Act (*Außensteuergesetz*) may result in an increase of our tax burden. Correspondingly, changes in labor laws concerning employee leasing or external workforce may result in additional social security payments.

We have taken out numerous loans in connection with our business activities. These borrowings require interest and principal payments. For corporate income tax purposes, the deduction of interest on loans may be restricted by Section 4h of the German Income Tax Act (*Einkommensteuergesetz*) in conjunction with Section 8a of the German Corporate Income Tax Act (*Körperschaftsteuergesetz*) (the “interest barrier”). According to this provision, interest expenses of a business can generally be taken into account in a tax-reducing manner in the amount of the interest income of the same business year. If the balance of interest expenses and interest income is negative, the deductibility of the interest balance is generally limited to 30% of the taxable profit before interest income, interest expenses and depreciation. The applicability of the interest barrier, therefore, depends on the earnings we achieve; these earnings fluctuate and therefore cannot be predicted with any certainty. If we are increasingly affected by the applicability of these regulations in the future, this would result in a higher tax burden and would in turn have adverse effects on our cash flows, financial condition and results of operations.

1.4.5 *We are subject to regular external tax audits and proceedings which may require us to make additional payments.*

We are involved in regular external tax audits and, from time to time, tax proceedings in Germany and the other jurisdictions in which we operate. Some external tax audits and tax proceedings involve complex tax matters and are subject to substantial uncertainty and could give rise to imposition of material additional tax payments. The German external tax audits have mainly related to impairment losses recognized before 2008 on shareholder loans to various domestic and foreign companies, individual leasing structures, the treatment of income from special investment funds, acquisition costs of SWISS, the classification and valuation of spare parts and material at Lufthansa Technik and various provisions. As of the date of this Prospectus, our most recent external tax audit covered the fiscal years up to and including 2012 and was completed in February 2019. The main prospective additional tax payments for the periods 2006-2009 and 2010-2012 are already paid. The external tax audit for the years 2013 to 2015 started in 2017 and is still ongoing. The provisions we have recorded for findings in relation to these tax audits as of the date of this Prospectus may ultimately not be sufficient to cover tax claims that may be enforced against us as a result of such proceedings. No provisions have been recorded for pending matters under discussion in the tax audit that we believe are more likely than not to be decided in our favor. However, these matters could result in substantial additional tax and interest payments in an amount of up to €200 million. Additionally, we are continuously engaged in legal and administrative tax proceedings regarding our tax positions in Germany and certain other jurisdictions in which we operate.

As a global business, we are particularly exposed to tax risks regarding transfer pricing rules applicable to cross-borders business transactions and relationships in Germany and elsewhere. There is no assurance that the German tax authorities or one or more foreign tax authorities may not challenge the transfer pricing principles implemented by us and, for example, initiate a tax audit. This may result in penalties and/or additional tax payments. Furthermore, transfer pricing risks may increase due to the tax authorities’ interpretation of the so-called arm’s length principle which evolves from time to time.

Because of varying legal interpretations and applications of law by the tax authorities regarding tax matters or underlying circumstances, there is a possibility that the tax authorities will demand additional tax payments on the occasion of current or future external tax audits. Additional tax payments or demands for additional tax payments could have an adverse effect on our cash flows and financial condition even if those payments are subsequently contested in court.

1.4.6 We are exposed to risks in connection with our investments in airport infrastructure, including liability risks.

Together with Air France, Japan Airlines and Korean Air, we are a stakeholder in the operator of Terminal 1 at John F. Kennedy International Airport in New York and are jointly liable, together with the operator, for a minimum lease payment to the appropriate Port Authority in respect of Terminal 1. In their use agreements with the operator, the airlines further gave their unrestricted, unconditional and irrevocable undertaking and guarantee to pay all payment obligations of the operator. If any of the parties fails to pay amounts due under a use agreement, the use agreements furthermore provide that the amounts in question will be paid by all of the parties that are not in arrears in proportion to their respective interest in the operator. If the total rental proceeds generated by the operator of Terminal 1 from these airlines and any other air carriers and lessees at Terminal 1 on the basis of sub-lease agreements were to fall short of the minimum lease payments contractually agreed with the Port Authority, for example, because of rent losses, delayed payments, vacancies or other delays in leasing, the four guarantor airlines would be jointly liable for the shortfall. Accordingly, we, together with the other three airlines, would be responsible for the deficit resulting from such loss of rental income. If any of the other three airlines defaults, the non-defaulting airlines would be held liable for the defaulting airline's stake. If all other three airlines were to default on their respective obligations under the joint liability, we would have sole liability.

We also hold an indirect general partner's interest of 40% in Terminal 2 Gesellschaft mbH & Co oHG ("T2"), which constructed and operates the Terminal 2 passenger terminal at Munich Airport. We are jointly and severally liable to the full extent for all liabilities and payment obligations of this partnership. T2 has obtained borrowings in the amount of €2,107 million, including a syndicated credit facility in the total amount of €1,107 million to finance the construction of Terminal 2, which was drawn in several tranches with a gradual repayment schedule with the last tranche being repaid in the year 2033 and a further syndicated credit facility in the total amount of €725 million for the construction of a satellite to Terminal 2. If the Munich Airport partnership were no longer able to meet its obligations, including due to a decline in flights departing from or arriving at Munich Airport, we will be obligated to cover any shortfalls together with our other partner in accordance with our shareholding in the partnership pursuant to Section 302 of the German Stock Corporation Act (*Aktiengesetz*), if resolved by the shareholders with at least a 2/3-majority. However, if our partner was to default on its obligation, we would be obligated to cover the entire shortfall due to our joint liability (*gesamtschuldnerische Haftung*).

We are also exposed to liability risks arising from our interests in companies involved in fueling, fuel storage and hydrant systems operation and handling companies at airports we use in Germany, the United States and Canada. The investing activities of these companies are financed mostly by external capital. The interest and principal payments these companies make for their credit liabilities, as well as the operating costs of the companies, must be generated from the charges for fueling and handling services, as well as for the use of the respective fuel depot or hydrant system. If the income generated falls short of the interest payments, principal payments and operating costs (for example, because of payment delays, defaults or non-use of the services offered by the companies), the respective partners are liable for the difference and, if loans are called, for the liabilities of the respective company under such loans. If one or several of these companies were no longer able to meet their obligations, the shortfall would have to be borne by us alone or together with other shareholders. In the latter case, we would also be exposed to the risk that the remaining shareholders might be unable to pay.

1.4.7 National and international conflicts, in particular in regions in which we operate, and sanction mechanisms affecting our business activities could negatively affect feasibility of and demand for air travel.

As a globally operating organization, we are highly dependent on smooth and uninterrupted cross-border and cross-cultural operation conditions which are currently under severe regulation, *inter alia*, due to the current situation around COVID-19 (see "1.1.1 The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate."). Furthermore, local as well as international conflicts might negatively impact our business directly

and indirectly. Political tension between the United States, China, Iran and North Korea as well as Russia and the conflicts in Syria, Turkey, Ukraine and Belarus could have a major impact on the global economy and therefore also on the aviation industry worldwide. Any conflict, especially with regard to regions where we provide relevant services or on which we depend to be provided with services might result in a material adverse effect on our operations and business as a whole.

The occurrence of any of the following could have a material adverse effect on our cash flows, financial condition and results of operations:

- difficulties in staffing and managing international operations;
- potentially adverse tax consequences and governmental fees, including as a result of inconsistent or unforeseeable interpretations of tax laws;
- costs of complying with various regulatory regimes;
- restrictions and costs relating to compliance with different legal standards and enforcement mechanisms, as well as limitation on transfers of capital;
- imposition of quotas relating to the composition of the employee base;
- volatile political situations, economic instability and related impact on foreign currency exchange rates;
- local permissions and access to local infrastructure;
- different terms and payment delays of accounts receivable in the countries in which we operate; and
- changes in foreign and domestic laws and policies that govern operations of foreign-based companies.

1.5 Environmental, Social and Governance Risks

1.5.1 We face significant labor law implications as the working conditions and circumstances, primarily due to the effects of COVID-19, have made it necessary for us to apply reductions in pay, implement pay freezes, short-time work and lay-offs, and we ultimately may not be able to retain or attract qualified staff.

As part of our “ReNew” restructuring program, we intend to structurally adjust our workforce to the market decline caused by COVID-19. As of June 30, 2021, the number of total employees (by headcount) decreased by 22,000, or 17%, of the total workforce, including 5,000 employees (by headcount) in Germany compared to December 31, 2019 (excluding a workforce reduction by approximately 8,600 employees (headcount) in connection with the divestment of LSG Group’s European business). We implemented short-term measures in 2020 to reduce staff costs and to counteract the effects of the pandemic. In addition, crisis agreements were concluded for the majority of the other groups of our employees. On August 5, 2021, Lufthansa Group has communicated that it already implemented measures to reduce personnel costs by €1.1 billion. Taking into account additional measures such as, in particular, savings of still to be negotiated amended tariff agreements as well as further staff reduction, we expect to achieve a total of €1.8 billion structural annual personnel cost savings from 2024 onwards.

Due to the outbreak of COVID-19, the measures imposed by authorities and our severely reduced operations, a significant number of our employees were forced to cease to work but with continued payment entitlement. Governments and public authorities in several countries where we operate introduced new schemes or improved existing schemes to mitigate the employer’s financial situation, e.g., reductions in pay by introducing short-time work, temporary layoffs, enforced leave or longer-term or permanent home office employment. Applying cost-saving schemes requires us to have specific agreements with unions, works councils and/or employees. The magnitude of cost savings depends on the final conditions agreed by collective or individual agreements with affected employees. Among others, in March and April 2020, the competent German employment agency (*Agentur für Arbeit*) agreed at the first time on arrangements for short-time work (*Kurzarbeit*) for some of our staff for a period until, as of the date of this Prospectus, December 31, 2021. As a result, the German government pays 60% to 87% of the foregone net salaries, but

capped at the social security assessment ceiling, of the affected staff. On the basis of these arrangements, we supplement the short-time working allowance for affected employees up to 90% of the foregone net salary. The short-time work arrangements are further supplemented with additional personnel measures, including, among others, the reduction of overtime work, unpaid leaves or early retirements (including related one-off payments). In addition, we have applied for short-time work funds in, among others, Austria, Belgium and Switzerland. More specifically, in the 6M 2021, we were granted government support in the amount of €609 million in Germany, Austria and Switzerland in connection with short-time working. However, there is no assurance that we have been eligible for short-time work funds that we have received and complied with the requirements for short-time work at all times or that we will be eligible to receive short-time work funds going forward. In the event that we have not been fully eligible in relation to the short-time work funds that we have received or if we have not complied with the requirements for short-time work at all times, we may be required to repay short-time work funds, partially or in full.

The continued low demand for air travel, ongoing travel restrictions for many countries and the reduced number of aircraft under operation can make it necessary for us to further downsize our staff significantly or to achieve adequate personnel cost savings. We require both cooperation and agreements with unions and works councils and/or employees to achieve cost savings from downsizing or other cost saving measures required to meet our targets. Downsizing measures can result in the need for us to pay severance payments or other benefits / costs for terminations.

In addition to the above-mentioned schemes, certain training courses and new hires have been cancelled. There exists a risk that the current working conditions and circumstances, including applying reductions in pay, pay freezes, short-time work, dismissals or similar measures lead to increased fluctuation and loss of qualified staff, in particular of qualified executives.

1.5.2 As a significant portion of our employees is member of a union, we are dependent on good relations with our employees and their unions and, consequently, any union disputes, employee strikes or slowdowns and other labor-related disruptions expose us to risks related to our ability to operate our business and may adversely impair our financial performance.

Staff costs are one of our largest operating expenses. In the 6M 2021, staff costs amounted to €2,907 million and represented 34% of our total operating expenses (€8,525 million). In the Fiscal Years 2020 and 2019, staff costs amounted to €6,436 million (or 28% of our total operating expenses (€22,750 million)) and €9,121 million (or 24% of our total operating expenses (€37,309 million)), respectively.

Our employees have traditionally been represented by unions. As of June 30, 2021, 84% of our around 62,000 German employees are tariff employees, who are covered by collective bargaining agreements that are customary for the industry or are members of unions, including the cabin crew union “UFO,” the pilot’s union “Vereinigung Cockpit” and the workers’ union “ver.di.” Correspondingly, we face the risk of complicated or even failed negotiations with unions due to increased activism of each union seeking to obtain the best terms, which may lead to complicated and extended negotiations and have in the past, for example, in relation to ver.di led to extended negotiations. Due to the numerous collective agreements and different interests within and between the unions, there is also a risk that we may not be able to agree with all unions on necessary possible cost reductions and efficiency improvements. These risks are increased due to the current measures we take in the context of COVID-19 (see “1.1.1 The outbreak of the COVID-19 pandemic and its effects, as well as mutations of COVID-19 and the outbreak of similar diseases or effects thereof, on the markets in which we operate as well as related macroeconomic developments have materially adversely affected and will continue to materially adversely affect the demand for our services and the conditions under which we operate.”). If we are unable to negotiate collective agreements with all unions on commercially reasonable terms, it could have material adverse effects on our cash flows, financial condition and results of operations through workplace actions such as strikes, slowdowns, or other actions designed to disrupt the normal operations. Additionally, we are aiming at total annual cost savings of €3.5 billion by 2024, which we may not be able to achieve if we are unable to negotiate collective agreements on commercially reasonable terms. Furthermore, our operational performance could be materially adversely affected in the event of workplace actions such as strikes, slowdowns, or other actions by our employees resulting in disruptions of our operations. Additionally, we are aiming to achieve structural annual personnel cost savings of €1.8 billion by 2024, which we may not be able to achieve in this event. If we are unable to

negotiate collective agreements with the employees' union representatives on commercially reasonable terms, it could have material adverse effects on our cash flows, financial condition and results of operations.

Our success depends to a large extent on the services of our key executives and qualified personnel. Since competition for executives is fierce, there is no certainty that in the future we will be able to retain the required key executives and qualified personnel and to hire new ones. The loss of one or more members of the Executive Board or of other key personnel, as well as the difficulties presented by having to hire new qualified executives, including as a result of a COVID-19, deadly infections or other sources of deaths (e.g., accidents), could impair our abilities in relation to crisis management and future growth strategies and could ultimately have material adverse effects on our cash flows, financial condition and results of operations. These risks are likely to increase due to the current measures taken by us in the context of COVID-19.

1.5.3 We may be exposed to liability risks relating to potential environmental damage.

In our operations, especially in the passenger business as well as in the Logistics, MRO and Catering business segments, inadvertent environmental damage might occur in the form of leaks of harmful or hazardous substances, particularly kerosene or other oil products, that could contaminate real estate owned by us or third parties, or pollute waterways or groundwater. This is particularly applicable with regard to the facilities operated by Lufthansa Technik, where hazardous substances are stored, processed and discharged, as well as the other facilities and storage areas used by us. For example, we have in the past experienced groundwater contaminations at certain facilities in Hamburg, Germany, and Frankfurt, Germany. The event of such contamination or pollution could result not only in possible fines or other public law sanctions, but also in considerable costs for removal, restoration and disposal, as well as further liability risks. Facilities owned or operated by us might also contain asbestos contamination, which would require us to refurbish such facilities. Additionally, there is no assurance that we are in compliance with all applicable fire protection regulations in relation to our building permits. In the event of non-compliance with such fire protection regulations, we may be unable to maintain such building permits. Environmental regulations could be tightened, which could lead to considerable costs or have other negative effects on our operations. Public knowledge of such environmental damage caused by us could also damage our reputation significantly.

1.5.4 We are subject to additional costs as a result of our requirement to comply with increasing regulatory measures aimed at restricting the emission of greenhouse gases and related models of emission rights trading.

The increasing significance of and focus on sustainability may lead to increasing restrictions for airlines with regard to pollution, greenhouse gas emissions, fuel composition and quality, noise and other environmental laws and regulations.

Pursuant to the United Nations Framework Convention on Climate Change and the Kyoto Protocol, the signatory states have undertaken to control and reduce the emission of greenhouse gases. In order to meet its obligations under international law, in 2003 the European Union introduced a model for the restriction of greenhouse gases and the trade in emission certificates that applies to certain industries.

In the event that any such regulation would be specific to a region or country in which we operate, we would be exposed to a risk of competitive disadvantage. We may be subject to additional costs due to changes in "cap and trade"-schemes such as the European Emissions Trading Scheme (the "EU ETS"). The EU included air traffic as part of the EU ETS in 2012. However, the European institutions decided to exempt all flights between countries in the European Economic Area and between third countries from the EU ETS until 2023 (the so-called "stop-the-clock").

On July 14, 2021, the European Commission adopted a set of proposals to revise and update legislation aimed at aligning current laws with the EU's greenhouse gas emission reduction target of at least 55% below 1990 levels by 2030, by, among other, extending the application of emissions trading to new sectors and tightening the existing EU ETS (the so-called "Fit for 55 Package"). Notably, as part of the Fit for 55 Package, the European Commission is proposing a regulation aiming at the mandatory introduction and gradual ramp-up of sustainable aviation fuels at airports located in the European Union, as well as proposing to phase out free emission allowances for aviation by 2027, reducing the total number of allowances in line with the new 55% target and aligning emission treatment with CORSIA (as defined below). The proposals

also include a revision of the Energy Taxation Directive (2003/96/EC) of 27 October 2003 by introducing minimum tax rates applicable to aviation fuels used on flights within the European Union. The proposed tax rates on aviation fuel are intended to increase gradually over a ten year period to reach an EU-wide minimum rate. Sustainable and alternative aviation fuels will benefit from a zero minimum tax rate during this transition phase. In the event that such proposals were to be enacted only in Germany or Europe any such proposed legislative changes may result in competitive disadvantage compared to our competitors operating primarily outside of Europe.

In addition to the EU ETS and Swiss Emissions Trading Scheme, some national governments have implemented country-specific air traffic taxes with the pretext of targeting the carbon emissions or other environmental impact caused by flights. For example, the German Air Traffic Tax Act (*Luftverkehrsteuergesetz*) (the “**LuftVStG**”), imposes a travel tax on all bookings from September 1, 2010, and charged for passengers departing from a German airport with a commercial airline to a domestic or international destination from January 1, 2011. Since January 2012, the tax has been slightly reduced and, as of the date of this Prospectus, amounts to €12.88, €32.62 or €58.73 per passenger depending on the flight’s destination.

The LuftVStG provides for a reduction mechanism in section 11 para. 2 LuftVStG which shall be offset against the German government’s revenue from aviation-related EU ETS trading. Pursuant to this mechanism, the tax rates can be lowered on a percentage basis at the beginning of each calendar year by statutory order. The percentage reduction is calculated on the basis of the ratio of the German governments previous year’s revenue from the EU ETS trading up to an amount of €1 billion. This mechanism takes into account that aviation has been included in emissions trading since 2012, and that, since then, the German government has received proceeds from the auctioning of certificates. According to the explanatory memorandum to the bill, the reference figure for calculating the reduction of €1 billion is based on the originally estimated target revenue from the aviation tax. There was no ecological earmarking of this tax, *e.g.*, to promote innovations in more energy-efficient and low-noise aircraft. Various studies estimate that Germany lost between 1.4 and 5 million passengers in 2011, which decided to switch to nearby non-German/EU airports due to the duty.

Austria, France, Sweden and the United Kingdom, among others, have also introduced similar air travel taxes. The introduction of these air travel taxes has made it necessary for us to pass on the resulting cost increase to passengers. Nevertheless, we may not be able to pass on current and future air travel tax and carbon emission permit costs in their entirety to our customers via ticket prices and freight charges.

In 2016, the assembly of the International Civil Aviation Organization (“**ICAO**”) concluded a historic agreement on market-based measures aimed at stabilizing airline emissions at a baseline-level (*i.e.*, Carbon Offsetting and Reduction Scheme for International Aviation (“**CORSIA**”)). Since January 1, 2021, airlines, including ours, are obligated to compensate their emissions exceeding the target by purchasing emission reduction units. As of the date of this Prospectus, the EU ETS and the Swiss emission trading scheme (“**CH ETS**”) apply to flights between airports located in the European Economic Area (“**EEA**”) as well as Switzerland, whereas the United Kingdom emission trading scheme applies to flights from the United Kingdom to the EEA and Switzerland. In addition, we expect that flights from countries in the EEA and Switzerland to the United Kingdom will be regulated under the EU ETS and the CH ETS, respectively, with effect as of January 1, 2021. Otherwise, flights between countries in the EEA or Switzerland and third countries are exempted. The costs of emissions trading may increase in the coming years due to changes to these current emission trading schemes and/or future implementation of the CORSIA agreement.

We believe that the airline industry as a whole is expected to face stricter regulations on emissions and noise, with increased regulations resulting in additional costs, including, in particular, tax-like payments. From time to time, there have been discussions at the level of the European Union and within member states of the European Union about whether existing tax exemptions for jet fuel should be reviewed; and there can be no assurance that the current tax exemptions for jet fuel will be upheld. The elimination of these exemptions would lead to a substantial increase in our jet fuel costs.

Additionally, any noise control regulations to be implemented in the future may limit our operations and/or cause additional costs. Noise control regulations typically concentrate on the level of noise and its

environmental impact, especially in areas close to airports. Permits may, for example, include restrictions on night flights to meet the imposed noise level requirements. Any such restrictions on night flights could particularly affect our scheduling and operations at our airport hubs in Frankfurt, Germany, and Munich, Germany, which also serve as our home hubs; any additional restrictions may have a material adverse effect on our ability to conduct our business.

Moreover, we may be unable to adapt our fleet or our operations to any new environmental regulations in a timely or appropriate manner, if at all. Such failures may require us to make significant investments and could potentially result in sanctions imposed by authorities for non-compliance. In addition to the direct financial impact, there is a risk that insufficient measures to comply with environmental regulations and best practices, or to meet customer expectations for sustainability, may give rise to negative publicity, which may in turn have a material adverse effect on our brand and lead to a loss of reputation.

This may result in a shift in consumer attitudes that is more negative towards air travel in general, or cause consumers to view our competitors more favorably. Such shifts in consumer attitude or demand and negative publicity could reduce our sales significantly and ultimately have a significant effect on our financial condition.

Additionally, German and European regulators are discussing intensely the introduction of a fuel, carbon or passenger tax and other instruments (e.g., a ban for short-haul flights) to reduce aviation's carbon emission. National taxes, levies, and bans will impact our business and hub operation as customers might seek for alternative connections and cheaper prices and it will increase operational cost. The same effect will happen if taxes and levies were raised on a European level.

1.6 Risks Related to the Shares and the Offering

1.6.1 The terms of the Framework Agreement and certain other stabilization measures place significant restrictions on us. Additionally, the interests of the WSF may not necessarily coincide with the interests of other shareholders.

In response to the outbreak of COVID-19 and its impact on the German economy and, in particular, the adversely affected financial situation of businesses across all sectors, the WSF and the Company entered into the Framework Agreement and the WSF became a shareholder of the Company. By resolution of an extraordinary general meeting (*außerordentliche Hauptversammlung*) of the Company on June 25, 2020, the Company's shareholders (the "**Shareholders**") approved the issuance of new shares as part of the WSF Stabilization Measures. Subsequently, 119.5 million Shares were issued to the WSF at a price of €2.56 per share, which corresponded to a shareholding of 20.00% of the share capital of the Company at the time. The WSF has agreed to sell all of its Company's shares by not later than December 31, 2023, provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have been fully repaid and subject to achieving a certain minimum sales price of (i) the market price or (ii) €2.56 per Company's share plus 12% p.a. calculated for the period between acquisition of the Company's shares by the WSF and the divestment, whichever is higher (the "**Minimum Sales Price**"). The Minimum Sales Price is subject to adjustments in the event of certain corporate actions, including capital reductions and share splits. However, if the WSF subscribes and receives New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I, the WSF shall sell all of its Company's shares within 18 months following full repayment of the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, and subject to achieving the Minimum Sales Price, which is being adjusted to reflect the weighted average of the purchase prices paid by the WSF in relation to existing shares of the Company and the New Shares. However, such 18-months period commences only after the six-months lock-up period in relation to existing shares of the Company and the New Shares, which the WSF agreed to, has expired, if the repayment of the Silent Participation I and Silent Participation II occurs prior to expiration of such lock-up. Following expiration of the applicable period and provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have been repaid in full, the Company shall have the right to request that the WSF sells all of its Company's shares to investors designated by the Company at the Minimum Sales Price. Key terms of the Framework Agreement include provisions pursuant to which:

- In the event of a takeover of the Company, the WSF may increase its shareholding in the Company by, fully or partially, converting the Silent Participation II-A into equity against subscribing up to 39,849,522 Company's shares.
- In the event of non-payment by the Company of interest in relation to the Silent Participation I for the years until 2023, the WSF can convert a certain portion of the Silent Participation II-B into equity corresponding to a shareholding of not more than 5% of the Company's share capital at the earliest from 2024 and in the event of a further non-payment by the Company of interest in relation to the Silent Participation I for the years 2024 and 2025, the WSF can convert an additional portion of the Silent Participation II-B into equity corresponding to a further shareholding of not more than 5% of the Company's share capital at the earliest from 2026. Such additional conversion right, however, only applies to the extent that the WSF has not previously increased its shareholding by converting the Silent Participation II-A.
- In the event that the WSF subscribes and receives New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I and provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have not been fully repaid by March 31, 2022, the Company shall propose to the General Meeting to resolve on a new contingent capital (*bedingtes Kapital*) and a new authorized capital (*genehmigtes Kapital*) to be utilized in connection with a conversion of the Silent Participation II and/or contribution of amounts drawn under the Silent Participation I but not repaid for the purpose of anti-dilution protection.
- The Company shall not pay any dividends to its shareholders or repurchase shares or other financial instruments or make non-obligatory coupon payments or non-obligatory distributions to parties other than the WSF before the WSF Stabilization Measures are fully terminated.
- The Company and the WSF have agreed that the WSF may only approve a rights issue of the Company if the exercise of all of its subscription rights does not result in a total subscription price to be paid by the WSF in excess of €1 billion. In the event of a rights issue, the WSF also shall be entitled to contribute the Silent Participation I or II, in full or in part, in kind in lieu of a contribution in cash. Furthermore, the WSF may receive the option to purchase any shares not subscribed for during the subscription period at a discount of 5% to the subscription price as permitted under the German Economic Stabilization Acceleration Act (*Wirtschaftsstabilisierungsgesetz*).
- Restrictions apply to the remuneration of the members of the Executive Board, the Supervisory Board and the first and second management level below board level of the Company.
- Two seats on the Supervisory Board shall be filled in agreement with the German government, one of whom shall become a member of the Company's audit committee.
- Except in the event of a takeover, the WSF has undertaken not to exercise its voting rights at the General Meeting in connection with the resolutions customary for such meeting.
- Transactions of significant importance such as the creation or utilization of authorized capital, high-volume M&A and financing transactions require prior approval by the WSF.
- The Company has agreed not to cross-subsidize any consolidated subsidiaries that were in difficulties within the meaning of Regulation (EU) No 651/2014 prior to December 31, 2019.
- The Company agreed to timely implement the commitment that the Federal Government of Germany made to the European Commission regarding the transfer of take-off and landing rights to the extent legally permissible, including stationing up to four aircraft at each of the airports of Frankfurt, Germany, and Munich, Germany, and to transfer in each case up to 24 take-off and landing rights (slots) to a competitor, *i.e.*, three take-off and three landing rights per aircraft per calendar day.

Subject to certain exceptions, including a takeover, the WSF has undertaken not to exercise its voting rights at the Company's general meetings (*Hauptversammlungen*) in connection with the resolutions customary for such meetings, including the appointment of the Company's auditor, the appointment of members to the Supervisory Board and the approval of the Company's financial statements. However, the size of its shareholding may give the WSF a de facto blocking minority with respect to important resolutions to be

passed by the Company's shareholders' meetings, in particular following a potential increase of its shareholding in the Company to 25% plus one share. There is no assurance that the interests of the WSF will coincide with those of other Shareholders.

In light of the Stabilization Measures and subsequent corporate actions implemented under the German Economic Stabilization Acceleration Act (*Wirtschaftsstabilisierungsgesetz*), Shareholders may take actions that could result in a delay or disruption of the implementation of the Offering. Any such delay or failure to implement the offering of the New Shares (the "**Offering**") as well as legal disputes in relation thereto could, in particular, limit our ability to repay stabilization measures as planned or at all.

1.6.2 No assurance can be given that a trading market will develop for the subscription rights and, if a market does develop, the subscription rights may be subject to greater volatility than the Shares.

Trading in the subscription rights for the New Shares on the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) is expected to begin on September 22, 2021, and will end on September 30, 2021 (with the closing auction for subscription rights). There will be no application for the subscription rights to be traded on any other exchange. There can be no assurance that an active trading market in the subscription rights will develop during that period and, if a market does develop, there can be no assurance regarding the nature of such trading market. If an active trading market does not develop or is not substantial, the liquidity and trading price of the subscription rights could be adversely affected. The trading price of the subscription rights depends on a variety of factors, including but not limited to the performance of the price of the Shares, but may also be subject to significantly greater price fluctuations than the Shares. Additionally, the WSF might sell subscription rights relating to the New Shares that the WSF is entitled to subscribe for during the subscription rights trading from and including September 22, 2021, up to and including September 30, 2021 (until the respective closing auction for subscription rights) on the regulated market (Xetra and Xetra Frankfurt Specialist) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) and in off-market transactions. Any such sale could result in significant price fluctuations and a significant decrease of the trading price of the subscription rights.

1.6.3 Failure by a shareholder of the Company to exercise subscription rights allocated to it during the subscription period will result in such subscription rights becoming worthless, which will ultimately result in a dilution of such shareholder's percentage ownership of the Shares.

The Offering enables the Company to raise capital in a manner that gives the Shareholders, to the extent eligible, the opportunity to subscribe for the New Shares *pro rata* to their shareholding at the record date, subject to applicable securities laws. To the extent that the Shareholders do not exercise their subscription rights during the subscription period, such subscription rights will become worthless and their proportionate ownership and voting interest in the Company will decrease. If Shareholders elect to sell their subscription rights, the consideration they receive, if any, may not be sufficient to fully compensate them for the dilution of their percentage ownership of the New Shares that may be issued in connection with the Offering. The New Shares will be offered to the Shareholders at a subscription ratio of 1:1, whereby one New Share may be acquired at the Subscription Price for one existing share of the Company.

1.6.4 If the Offering is not consummated, including as a result of the termination of the Underwriting Agreement, the subscription rights would become null and void. Additionally, the size of the actual capital increase may be lower than contemplated.

BofA Securities Europe SA, 51 rue La Boétie, 75008 Paris, France, LEI 549300FH0WJAPEHTIQ77 ("**BofA Securities**"), Deutsche Bank Aktiengesellschaft, Taunusanlage 12, 60325 Frankfurt am Main, Germany, LEI 7LTWFZYICNSX8D621K86 ("**Deutsche Bank**"), Goldman Sachs Bank Europe SE, Marienturm, Taunusanlage 9-10, 60329 Frankfurt am Main, Germany, LEI 8IBZUGJ7JPLH368JE346 ("**Goldman Sachs**"), J.P. Morgan AG, Taunustor 1, 60310 Frankfurt am Main, Germany, LEI 549300ZK53CNGEEI6A29 ("**J.P. Morgan**" and, together with BofA Securities, Deutsche Bank and Goldman Sachs, the "**Joint Global Coordinators**"), Barclays Bank Ireland PLC, One Molesworth Street, Dublin 2, D02 RF29, Ireland, LEI 2G5BKIC2CB69PRJH1W31 ("**Barclays**"), BNP PARIBAS SA, 16, Boulevard des Italiens, 75009 Paris, France, LEI R0MUWSFPU8MPRO8K5P83 ("**BNP PARIBAS**"), COMMERZBANK Aktiengesellschaft, Kaiserstraße 16 (Kaiserplatz), 60311 Frankfurt am Main, Germany,

LEI 851WYGNLUQLFZBSYGB56 (“**COMMERZBANK**”), HSBC Trinkaus & Burkhardt AG, Königsallee 21/23, 40212 Düsseldorf, Germany, LEI JUNTA405OW8OY5GN4DX16 (“**HSBC**”), UniCredit Bank AG, Arabellastraße 12, 81925 Munich, Germany, LEI 2ZCNRR8UK83OBTEK2170 (“**UniCredit Bank**”), Crédit Agricole Corporate and Investment Bank, 12 Place des Etats-Unis, CS 70052, 92547 Montrouge Cedex, France, LEI 1VUV7VQFKUOQSJ21A208 (“**Crédit Agricole Corporate and Investment Bank**”), DZ BANK AG Deutsche Zentral-Genossenschaftsbank, Frankfurt am Main, Platz der Republik, 60325 Frankfurt am Main, Germany, LEI 529900HNOAA1KXQJUQ27 (“**DZ BANK**”), Landesbank Baden-Württemberg, Am Hauptbahnhof 2, 70173 Stuttgart, Germany, LEI B81CK4ESI35472RHJ606 (“**Landesbank Baden-Württemberg**”), SMBC Nikko Capital Markets Europe GmbH, Neue Mainzer Straße 52-58, 60311 Frankfurt am Main, Germany, LEI: 5493008794GM6YH5G766 (“**SMBC Nikko**”) and Société Générale S.A., 29, boulevard Haussmann, 75009 Paris, France, LEI O2RNE8IBXP4R0TD8PU41 (“**Société Générale**” and, together with Barclays, BNP PARIBAS, COMMERZBANK, HSBC, UniCredit Bank, Crédit Agricole Corporate and Investment Bank, DZ BANK, Landesbank Baden-Württemberg and SMBC Nikko, the “**Joint Bookrunners**” and, together with the Joint Global Coordinators, the “**Underwriters**”) have agreed, subject to certain conditions, to subscribe for the New Shares and to offer such shares in a public offering to the Shareholders for subscription. The subscription to the New Shares will be made on the basis of an underwriting agreement entered into between the Company and the Joint Global Coordinators on September 19, 2021 (the “**Underwriting Agreement**”). The Offering may be terminated under certain circumstances. These include the withdrawal by the Underwriters from the Underwriting Agreement in accordance with its terms. If the Offering is terminated prior to registration of the implementation of the capital increase with the commercial register (*Handelsregister*), the Offering will not take place and subscription rights may lapse and be of no value. Under these circumstances, investors may not be entitled to the delivery of New Shares, and the capital increase may not be consummated as contemplated hereby. Investors who purchased subscription rights could accordingly suffer a loss. No prior trading in the subscription rights will be unwound and no compensation will be paid for the subscription rights. In addition, any investors engaging in short selling transactions bear the risk of being unable to meet their obligation to deliver New Shares. The agents brokering the subscription rights transactions will not reverse such short selling transactions.

1.6.5 The market price of the New Shares may fluctuate and may decline below the Subscription Price.

The market price of the New Shares may fluctuate and may decline below the subscription price of €3.58 per New Share (the “**Subscription Price**”). Moreover, given that the market price of the subscription rights depends to a significant degree on the price of the Company’s existing shares (the “**Shares**”), a decline in the trading price of the Shares would also be expected to negatively affect the trading price of the subscription rights. The price and the trading volume of the Shares is determined by supply and demand, which is influenced by a variety of factors.

The Company cannot assure holders of subscription rights that the market price of the Shares will not decline below the Subscription Price after such holders elect to exercise their subscription rights. For example, existing shareholders may divest Shares during the subscription period, which could result in a decline of the market price of the Shares. In the event that the market price of the Shares declines below the Subscription Price after such holders elect to exercise their subscription rights, such holders have committed to buy the New Shares at a price above the prevailing market price, and such holders will suffer an immediate unrealized loss as a result thereof.

In the event that holders of subscription rights do not take up their subscription rights during the subscription period, the Underwriters will seek to procure purchasers for all of the New Shares not subscribed for in the subscription offer or by the WSF (the “**Rump Shares**”) in private placements to eligible or qualified investors in certain jurisdictions at a price at least equivalent to the Subscription Price (the “**Rump Placement**”). Any Rump Shares not sold in connection with the Rump Placement are being purchased by the Underwriters in accordance with the Underwriting Agreement, which provides for a firm underwriting of the New Shares not sold in the Offering by the Underwriters or subscribed for by the WSF. The Underwriters may offer or sell such New Shares, which they are required to subscribe for as underwriters otherwise than in connection with the Offering (including by way of a coordinated sell-down), which could result in the market price of the New Shares to decline significantly.

Additionally, the market price of the New Shares may fluctuate significantly as a result of various factors, many of which are beyond the Company's control, including, but not limited to:

- any subsequent divestment by the WSF of Shares held by it;
- volatility in the market as a whole or investor perception of the Company's industry and competitors, in particular volatility related to the COVID-19 pandemic;
- market expectations for the Company's financial performance;
- actual or anticipated fluctuations in the Company's business, results of operations and financial condition;
- changes in the estimates of the Company's results of operations by securities analysts;
- investor perception of the impact of the Offering on the Company and its shareholders;
- potential or actual sales of blocks of shares in the market or short selling of shares, including any sales by the Underwriters (as defined below) of shares purchased for their own accounts pursuant to the Underwriting Agreement (as defined below) or in connection with the enforcement of security under margin loans made to investors in the Offering;
- announcements by the Company or its competitors of significant contracts;
- acquisitions, strategic partnerships, joint ventures, capital commitments or new business activities;
- additions or departures of key personnel;
- any shortfall in revenue or net income or any increase in losses from levels expected by securities analysts;
- future issues or sales of shares;
- new government regulation;
- general, economic, financial and political conditions; and
- any other risk in relation to the Company and its industry set out above.

The market price of the New Shares may be adversely affected by most of the preceding or other factors regardless of the Company's actual results of operations and financial condition. Publicly traded securities, including the Shares, from time to time experience significant price and volume fluctuations that may be unrelated to the performance of the companies that have issued them. The Company cannot make any predictions about the market price of the Shares. No assurance can be given that the market price of the New Shares will not decline below the Subscription Price or their market price at the commencement of the Offering.

1.6.6 The ability of the Company to distribute dividends is dependent on a variety of factors. The previous dividend distributions are no indication for future dividend distributions. We are prohibited from paying any dividends under the Framework Agreement. Additionally, the dividend policy of the Company may change in the future.

The General Meeting will decide on matters relating to the payment of future dividends. These decisions will be based on our particular situation at the time. The Company's ability to pay dividends depends upon, among other things, our results of operations, financing and investment requirements, as well as the availability of distributable profit (*Bilanzgewinn*). Certain reserves must be established by law and must be deducted when calculating the distributable profit (*Bilanzgewinn*). In addition, the Framework Agreement entered into in connection with the WSF Stabilization Measures contains, and future debt financing arrangements could contain, covenants which impose restrictions on our business and joint ventures and, in particular, our ability to distribute dividends. Following termination of the Framework Agreement we may continue to be subject to covenants which limit the Company's ability to pay dividends under certain circumstances under future debt financing arrangements. Any of these factors, individually or in combination, could restrict the Company's ability to pay dividends.

There can be no certainty that future dividends can be distributed in line with previous dividend distributions or with the dividend policy of the Company applicable at the given time, or any dividend at all. Given that the Shares and the New Shares are, and any dividends to be paid in respect of them will be, denominated in Euro, an investment in the Shares and the New Shares by an investor whose principal currency is not the Euro exposes the investor to an additional foreign currency exchange rate risk and any such investor is subject to adverse movements to its local currency against the Euro.

1.6.7 Future offerings of debt or equity securities may adversely affect the market price of the Shares, and future capital measures could lead to a dilution of existing shareholdings.

We may require further capital in the future to finance our business operations and planned growth, to strengthen our balance sheet or to fulfill regulatory requirements. Therefore, we may seek to raise capital through offerings of debt securities (possibly including convertible debt securities) or additional equity securities of the Company. An issuance of additional equity securities, through a share capital increase against cash or contributions in kind, or of securities with a right to convert into equity, such as convertible bonds or warrant bonds, could adversely affect the market price of the Shares and would dilute the economic and voting interests of the Shareholders if made without granting subscription rights to the Shareholders. Subject to certain conditions, we may, for example, limit or cancel the subscription rights of the Shareholders in the event of a capital increase through contributions in kind. Even if the Shareholders were granted subscription rights, investors in certain jurisdictions (particularly in the United States) may not be able to acquire or exercise any subscription rights due to local laws.

Because the timing and nature of any future offering would depend on market conditions, it is not possible to predict or estimate the amount, timing, or nature of future offerings. In addition, the acquisition of other companies or investments in companies in exchange for newly issued Shares, as well as the exercise of stock options by our employees in the context of possible future stock option programs or the issuance of shares to employees in the context of possible future employee stock participation programs, could lead to a dilution of the economic and voting interests of existing shareholders. Furthermore, a proposal to the Company's general meeting to take any of the abovementioned measures, with dilutive effects on the existing shareholdings, or any other announcement thereof, could adversely affect the market price of the Shares.

1.6.8 Overseas shareholders may not and holders of American Depositary Receipts will not be able to acquire New Shares in the context of the Offering.

Securities laws of certain jurisdictions may restrict the Company's ability to allow participation by certain Shareholders in the Offering. In particular, and subject to certain exceptions, Shareholders who are located in the United States may not be permitted to exercise their entitlements under the Offering unless an exemption from the registration requirements is available under the United States Securities Act of 1933, as amended (the "**Securities Act**"). The New Shares and the subscription rights have not been and will not be registered under the Securities Act, or the securities laws of any other jurisdiction of the United States, its territories and possessions (including Puerto Rico, the U.S. Virgin Islands, Guam, American Samoa, Wake Island and the Northern Mariana Islands), any state of the United States or the District of Columbia, and may not be offered, sold or otherwise transferred within the United States, except pursuant to an exemption from, or in a transaction not subject to, the registration requirements of the Securities Act and in compliance with any applicable securities laws of any state or other jurisdiction in the United States. In addition, holders of American Depositary Receipts ("**ADRs**") will not be permitted to receive or exercise any rights related to the ADRs respectively held or subscribe for New Shares on the basis of any ADRs respectively held. Securities laws of certain other jurisdictions may restrict the Company's ability to allow participation by Shareholders in such jurisdictions in any future issue of shares carried out by the Company. Shareholders who have a registered address in or who are incorporated, registered, resident or located in countries other than Germany are encouraged to consult their own professional advisers as to whether they require any governmental or other consents or need to observe any other formalities to enable them to take up, sell or exercise their subscription rights, including to ultimately acquire any New Shares. Any Shareholder who is not entitled to participate in the Offering carried out by the Company will suffer dilution.

1.6.9 The ability of Shareholders to bring actions or enforce judgments against the Company or members of the Executive Board or Supervisory Board may be limited.

The ability of shareholders to bring an action against the Company may be limited. The Company is a stock corporation (*Aktiengesellschaft*) incorporated under the laws of Germany. The rights of shareholders are governed by German law and by the articles of association of a company (*Satzung*). These rights differ from the rights of shareholders in other jurisdictions. It may be difficult for a shareholder to prevail in a claim against the Company or to enforce liabilities predicated upon the laws of jurisdictions other than Germany. A Shareholder may not be able to enforce a judgment against some or all of the members of the Executive Board or of the Company's supervisory board (the "**Supervisory Board**"). It may not be possible for a shareholder to effect service of process upon members of the Executive Board or Supervisory Board within the shareholder's country of residence, or to enforce against members of the Executive Board or Supervisory Board judgments of courts of such shareholder's country of residence based on civil liabilities under that country's securities laws. There can be no assurance that a Shareholder will be able to enforce any judgment in civil and commercial matters or any judgments against the members of the Executive Board or Supervisory Board who are residents of countries other than those in which the judgment is made. In addition, German and other courts may not impose civil liability on members of the Executive Board or Supervisory Board in any original action based solely on foreign securities laws brought against the Company or members of the Executive Board or Supervisory Board in a competent court in Germany or other countries.

The members of the Executive Board and Supervisory Board may be non-residents of the United States, and a significant portion of the assets of the Lufthansa Group and those of the Company's directors and executive officers may be located outside the United States. As a result, a shareholder may be unable to enforce judgments obtained in U.S. courts against them. Moreover, in light of recent decisions of the Supreme Court of the United States, actions of the Company may not be subject to the civil liability provisions of U.S. federal securities laws.

2. GENERAL INFORMATION

2.1 Responsibility Statement

The following persons assume responsibility for the contents of this prospectus (the “**Prospectus**”) pursuant to section 8 of the German Securities Prospectus Act (*Wertpapierprospektgesetz*) in conjunction with Article 11 of Regulation (EU) 2017/1129 of the European Parliament and of the Council of June 14, 2017 on the prospectus to be published when securities are offered to the public or admitted to trading on a regulated market and repealing Directive 2003/71/EC, as amended (the “**Prospectus Regulation**”), and declare that the information contained in this Prospectus is, to the best of their knowledge, in accordance with the facts, and that this Prospectus makes no omission likely to affect its import:

- Deutsche Lufthansa Aktiengesellschaft (the “**Company**” and, together with its consolidated subsidiaries, the “**Lufthansa Group**,” “**we**,” “**our**” and “**us**”), with its registered offices at Venloer Straße 151-153, 50672 Cologne, Federal Republic of Germany (“**Germany**”), registered with the commercial register (*Handelsregister*) of the local court (*Amtsgericht*) of Cologne, Germany, under HRB 2168, Legal Entity Identifier (“**LEI**”) 529900PH63HYJ86ASW55;
- BofA Securities Europe SA, 51 rue La Boétie, 75008 Paris, France, LEI 549300FH0WJAPEHTIQ77 (“**BofA Securities**”);
- Deutsche Bank Aktiengesellschaft, Taunusanlage 12, 60325 Frankfurt am Main, Germany, LEI 7LTWFZYICNSX8D621K86 (“**Deutsche Bank**”);
- Goldman Sachs Bank Europe SE, Marienturm, Taunusanlage 9-10, 60329 Frankfurt am Main, Germany, LEI 8IBZUGJ7JPLH368JE346 (“**Goldman Sachs**”);
- J.P. Morgan AG, Taunustor 1, 60310 Frankfurt am Main, Germany, LEI 549300ZK53CNGEEI6A29 (“**J.P. Morgan**” and, together with BofA Securities, Deutsche Bank and Goldman Sachs, the “**Joint Global Coordinators**”);
- Barclays Bank Ireland PLC, One Molesworth Street, Dublin 2, D02 RF29, Ireland, LEI 2G5BKIC2CB69PRJH1W31 (“**Barclays**”);
- BNP PARIBAS SA, 16, Boulevard des Italiens, 75009 Paris, France, LEI R0MUWSFPU8MPRO8K5P83 (“**BNP PARIBAS**”);
- COMMERZBANK Aktiengesellschaft, Kaiserstraße 16 (Kaiserplatz), 60311 Frankfurt am Main, Germany, LEI 851WYGNLUQLFZBSYGB56 (“**COMMERZBANK**”);
- HSBC Trinkaus & Burkhardt AG, Königsallee 21/23, 40212 Düsseldorf, Germany, LEI JUNT405OW8OY5GN4DX16 (“**HSBC**”);
- UniCredit Bank AG, Arabellastraße 12, 81925 Munich, Germany, LEI 2ZCNR8UK83OBTEK2170 (“**UniCredit Bank**”);
- Crédit Agricole Corporate and Investment Bank, 12 Place des Etats-Unis, CS 70052, 92547 Montrouge Cedex, France, LEI 1VUV7VQFKUOQSJ21A208 (“**Crédit Agricole Corporate and Investment Bank**”);
- DZ BANK AG Deutsche Zentral-Genossenschaftsbank, Frankfurt am Main, Platz der Republik, 60325 Frankfurt am Main, Germany, LEI 529900HNOAA1KXQJUQ27 (“**DZ BANK**”);
- Landesbank Baden-Württemberg, Am Hauptbahnhof 2, 70173 Stuttgart, Germany, LEI B81CK4ESI35472RHJ606 (“**Landesbank Baden-Württemberg**”);
- SMBC Nikko Capital Markets Europe GmbH, Neue Mainzer Straße 52-58, 60311 Frankfurt am Main, Germany, LEI: 5493008794GM6YH5G766 (“**SMBC Nikko**”); and
- Société Générale S.A., 29, boulevard Haussmann, 75009 Paris, France, LEI O2RNE8IBXP4R0TD8PU41 (“**Société Générale**” and, together with Barclays, BNP PARIBAS, COMMERZBANK, HSBC, UniCredit Bank, Crédit Agricole Corporate and Investment

Bank, DZ BANK, Landesbank Baden-Württemberg and SMBC Nikko, the “**Joint Bookrunners**” and, together with the Joint Global Coordinators, the “**Underwriters**”).

2.2 General Disclaimers

Where a claim relating to the information contained in a prospectus is brought before a court, the plaintiff investor might, under the respective national legislation of the relevant member state of the European Economic Area (“**EEA**”), have to bear the costs of translating this Prospectus before the legal proceedings are initiated.

Neither the Company nor the Underwriters are required by law to update this Prospectus subsequent to the date hereof, except in accordance with Article 23 of the Prospectus Regulation, which stipulates that every significant new factor, material mistake, or material inaccuracy relating to the information included in a prospectus which may affect the assessment of the securities and which arises or is noted between the time when this prospectus is approved and the closing of the offer period or the time when trading on a regulated market begins, whichever occurs later, shall be mentioned in a supplement to this Prospectus without undue delay.

The validity of this Prospectus is expected to expire at the end of the day on October 11, 2021. The closing of the offer period is expected to occur on October 5, 2021, and the time when trading on a regulated market begins is expected to occur on October 11, 2021. The obligation to supplement this Prospectus in the event of any significant new factors, material mistakes or material inaccuracies no longer applies when this Prospectus is no longer valid.

2.3 Competent Authority Approval

This Prospectus constitutes a prospectus for the purposes of Article 3 of the Prospectus Regulation and has been approved by the German Federal Financial Supervisory Authority (*Bundesanstalt für Finanzdienstleistungsaufsicht*) (“**BaFin**”), as the competent authority under the Prospectus Regulation. BaFin only approves this Prospectus as meeting the standards of completeness, comprehensibility and consistency imposed by the Prospectus Regulation and such approval should not be considered as an endorsement of the Company or the quality of the Company’s shares (the “**Shares**”). Investors should make their own assessment as to the suitability of investing in the Shares. BaFin can be contacted at Marie-Curie-Str. 24-28, 60439 Frankfurt am Main, Germany, by telephone +49 228 4108-0 or via its website: <http://www.bafin.de>.

2.4 Subject-Matter of this Prospectus

This Prospectus relates to the offer of 597,742,822 newly issued ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*) of the Company (the “**New Shares**” and each, a “**New Share**”), each such share representing a notional share in the share capital of €2.56 in the share capital and carrying full dividend rights as from January 1, 2021, from the capital increase against contribution in cash from authorized capital resolved by the Company’s executive board (*Vorstand*) (the “**Executive Board**”) on September 19, 2021, with approval of the Company’s supervisory board (*Aufsichtsrat*) (the “**Supervisory Board**”) on the same date.

For purposes of the admission to trading on the regulated market segment (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with simultaneous application for the sub-segment thereof with additional post-admission obligation (Prime Standard), this Prospectus relates to 597,742,822 New Shares.

This Prospectus does not constitute an offer to sell, or a solicitation of an offer to buy, any shares offered by any person in any jurisdiction in which it is unlawful for such person to make such an offer or solicitation. For further information on certain selling restrictions with respect to the New Shares, see “3.8 Selling Restriction Notices.” For further information on the subject matter of this Prospectus, see “3.1 Subject-Matter of the Offering.”

2.5 Forward-Looking Statements

This Prospectus contains certain forward-looking statements. A forward-looking statement is any statement that does not relate to historical facts or events or to facts or events as of the date of this Prospectus. This applies, in particular, to statements in this Prospectus containing information on our future earnings capacity, plans and expectations regarding our business, growth, and profitability, as well as the general economic and legal conditions to which we are exposed. The terms “aim,” “anticipate,” “expect,” “intend,” “plan,” “predict,” “project,” and “target” are an indication of forward-looking statements.

The forward-looking statements contained in this Prospectus are based on our current estimates and assessments. They are based on assumptions and subject to risks, uncertainties, and other factors, the occurrence or non-occurrence of which could cause actual circumstances – including with regard to our business, results of operations, financial position, cash flows, and prospects – to differ materially from or fail to meet the expectations expressed or implied in the forward-looking statements. Even if our future results meet the expectations expressed herein, they may not be indicative of the results of any succeeding periods.

Our business is also subject to a number of risks and uncertainties that could cause a forward-looking statement, estimate, or prediction in this Prospectus to become inaccurate. Accordingly, investors are strongly advised to consider this Prospectus as a whole and particularly ensure that they have read the following sections of this Prospectus, which include more detailed descriptions of factors that might influence our business performance and the markets in which we operate: “*1 Risk Factors*,” “*9 Management’s Discussion and Analysis of Financial Condition and Results of Operations*,” “*11 Markets and Competition*,” and “*23 Recent Developments and Outlook*.”

In light of these assumptions, as well as risks, uncertainties, and other factors, it is also possible that the future events mentioned in this Prospectus may not occur or may differ materially from actual events. In addition, the forward-looking estimates and forecasts reproduced in this Prospectus from third-party sources could prove to be inaccurate. For more information on third-party sources, see “*2.6 Information from Third Parties*.” Actual results, performance, or events may differ materially from those in such statements due to, among other reasons:

- adverse developments of general economic conditions in the markets in which we operate, including changes due to the impact of the COVID-19 pandemic;
- reduced demand for air travel and our maintenance, repair and overhaul (“**MRO**”) services;
- competition, in particular from low-cost, state-owned or subsidized airlines;
- changes in laws, regulations and governmental policies and compliance with such laws, regulations and governmental policies;
- inaccurate capacity development forecasts, including aircraft orders;
- fluctuations in currency exchange and interest rates;
- interruptions of the Lufthansa Group’s information technology systems;
- litigation, administrative proceedings and investigations; and
- deteriorating relations with our employees and their workers unions as well as inability to retain key employees of the Lufthansa Group.

However, the forward-looking statements contained in this Prospectus speak only as of the date of this Prospectus. Investors are advised that neither we nor the Underwriters assume any obligation and do not intend, except as required by law, to publicly release any updates or revisions to these forward-looking statements to reflect any change in our expectations with regard thereto or any change in events, conditions, or circumstances on which any such statement is based or to adjust them in line with future events or developments.

2.6 Information from Third Parties

This Prospectus contains industry- and customer-related data, as well as calculations sourced from industry reports published by third parties, market research reports, publicly available information and commercial publications of third parties. These publications generally state that the information they contain has originated from sources assumed to be reliable but that the accuracy and completeness of such information is not guaranteed and that the calculations contained therein are based on assumptions. These sources may not fully reflect the potential impact of COVID-19 due to, among other things, uncertainties surrounding its further development. In view of the potential effects of this pandemic on the economy, society and markets in which we operate, all current forecasts are made only with a considerably higher degree of uncertainty. This applies particularly in the context of links and interrelations between the global financial markets, economies and political decisions, which each individually may have an influence on the economic and political development, and when combined are currently impossible to assess with any certainty *ex ante*.

Irrespective of the assumption of responsibility for the contents of this Prospectus by the Company and the Underwriters (see “2.1 Responsibility Statement”), neither the Company nor the Underwriters have independently verified such information, or make any representation, or give any warranty as to the accuracy or completeness of such information.

Where information in this Prospectus has been sourced from a third party, the Company confirms that this information has been accurately reproduced and that, as far as the Company is aware and able to ascertain from information published by such third party, no facts have been omitted which would render the reproduced information inaccurate or misleading. Nevertheless, prospective investors are advised to consider this information with caution.

In preparing this Prospectus, the following sources of third-party information were used:

- Air Cargo, “Air Cargo News Awards 2020,” <https://aircargonewsawards.net/live/en/page/2020-winners> (“**Air Cargo Awards 2020**”);
- BDL, “Bericht zur Lage der Branche,” <https://www.bdl.aero/wp-content/uploads/2021/01/Jahreszahlen-2020.pdf> (“**BDL, Bericht zur Lage der Branche 2020**”);
- Halldale Group, “Civil Full Flight Simulator Census” (“**CAT SIM Census**”);
- Deutsche Bahn, “2020: 12,2 Milliarden Euro Investitionen für ein besseres Netz,” https://www.deutschebahn.com/de/presse/pressestart_zentrales_uebersicht/2020-12-2-Milliarden-Euro-Investitionen-fuer-ein-besseres-Netz-attraktive-Bahnhoefe-und-kundenfreundlicheres-Bauen-4844644 (“**Deutsche Bahn, Investitionen 2020**”);
- European Commission, Press Release, “Aviation: slot relief enacted,” https://ec.europa.eu/transport/modes/air/news/2021-02-16-aviation-slot-relief-enacted_en#:~:text=Aviation%3A%20slot%20relief%20enacted%2016%2F02%2F2021%20Yesterday%2C%20the%20EU,is%20no%20longer%20justified%20for%20the%20next%20season (“**European Commission, Aviation: slot relief enacted**”);
- European Commission, Press Release, “Aviation: slot relief rules for airlines amended and extended,” https://ec.europa.eu/transport/modes/air/news/2021-07-26-slot-relief-rules_en (“**European Commission, Aviation: slot relief amended and extended**”);
- Mathieu, E., Ritchie, H., Ortiz-Ospina, E. et al., A global database of COVID-19 vaccinations. Nat Hum Behav (2021), Our World in Data, <https://ourworldindata.org/covid-vaccinations> (“**Our World in Data**”);
- Global Insight, “World Review” (“**Global Insight, World Review, January 2021**”);
- IATA, “Air Cargo Market Analysis July 2021,” <https://www.iata.org/en/iata-repository/publications/economic-reports/air-freight-monthly-analysis---july-2021/> (“**IATA, Air Cargo Market Analysis, July 2021**”);

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This Prospectus also contains estimates of market and other data and information derived from such data that cannot be obtained from publications by market research institutes or from other independent sources. Such information is partly based on our own market observations, the evaluation of industry information (from conferences, sector events etc.), and internal assessments. We believe that our estimates of market and other data and the information derived from such data assist investors in gaining a better understanding of the industry in which we operate and our position therein. Our own estimates have not been checked or verified externally. We nevertheless believe that our own market observations are reliable. They may however differ from estimates made by our competitors or from current and future studies conducted by market research institutes or other independent sources.

Information contained on any website mentioned in this Prospectus, including our own website, is not incorporated by reference in this Prospectus and is not part of this Prospectus.

2.7 Documents Available for Inspection

For the period during which this Prospectus is valid, copies of the following documents will be available for inspection at the Company’s business address at Deutsche Lufthansa Aktiengesellschaft, Investor Relations, FRA CW, Lufthansa Aviation Center, Airportring, 60546 Frankfurt am Main, Germany and on the Company’s website (<http://www.lufthansagroup.com>):

- the Company’s articles of association (the “**Articles of Association**”);
- the Company’s unaudited condensed consolidated interim financial statements prepared in accordance with International Financial Reporting Standards as adopted by the European Union (“**IFRS**”) applicable to interim financial reporting as of and for the six-month period ended June 30, 2021 (the “**Unaudited Condensed Consolidated Interim Financial Statements**”);
- the Company’s audited consolidated financial statements prepared in accordance with IFRS and the additional requirements of German commercial law pursuant to section 315e para. 1 of the German Commercial Code (*Handelsgesetzbuch*) as of and for the financial year ended December 31, 2020, (the “**Audited Consolidated Financial Statements 2020**”);
- the Company’s audited consolidated financial statements prepared in accordance with IFRS and the additional requirements of German commercial law pursuant to section 315e para. 1 of the German Commercial Code (*Handelsgesetzbuch*) as of and for the financial year ended December 31, 2019, (the “**Audited Consolidated Financial Statements 2019**”);

- the Company’s audited consolidated financial statements prepared in accordance with IFRS and the additional requirements of German commercial law pursuant to section 315e para. 1 of the German Commercial Code (*Handelsgesetzbuch*) as of and for the financial year ended December 31, 2018, (the “**Audited Consolidated Financial Statements 2018**” and, together with the Audited Consolidated Financial Statements 2020 and the Audited Consolidated Financial Statements 2019, the “**Audited Consolidated Financial Statements**” and, together with the Unaudited Condensed Consolidated Interim Financial Statements, the “**Consolidated Financial Statements**”); and
- the Company’s audited financial statements prepared in accordance with the German Commercial Code (*Handelsgesetzbuch*) as of and for the financial year ended December 31, 2020 (the “**Audited Annual Financial Statements**”).

2.8 Presentation of Financial and Other Information

2.8.1 General

Unless otherwise indicated, financial information contained in this Prospectus was prepared in accordance with IFRS.

The fiscal years ended December 31, 2020, December 31, 2019, and December 31, 2018, are also referred to in this Prospectus as the “**Fiscal Year 2020**,” the “**Fiscal Year 2019**” and the “**Fiscal Year 2018**,” respectively. The six-month periods ended June 30, 2021, and June 30, 2020, are referred to in this Prospectus as the “**6M 2021**” and the “**6M 2020**,” respectively. The current fiscal year, which will end on December 31, 2021, is referred to as the “**Fiscal Year 2021**.”

The financial information presented in this Prospectus for the Fiscal Year 2019 was taken from the comparative prior-year figures contained in the Audited Consolidated Financial Statements 2020, unless indicated otherwise. The financial information presented in this Prospectus for the Fiscal Year 2018 was taken from the comparative prior-year figures contained in the Audited Consolidated Financial Statements 2019, unless indicated otherwise.

Where financial information in the tables in this Prospectus is presented as “audited,” this information was taken from the Audited Consolidated Financial Statements and the Audited Annual Financial Statements. Financial information in this Prospectus presented as “unaudited” in the tables in this Prospectus is not taken from the Audited Consolidated Financial Statements or the Audited Annual Financial Statements but either from the Unaudited Condensed Consolidated Interim Financial Statements or our accounting records or internal management reporting systems or was calculated based on figures from the above-mentioned sources.

The Audited Consolidated Financial Statements were prepared in accordance with IFRS and the additional requirements of German commercial law pursuant to section 315e para. 1 of the German Commercial Code (*Handelsgesetzbuch*) and were audited in accordance with section 317 of the German Commercial Code (*Handelsgesetzbuch*) and in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the Institute of Public Auditors in Germany (*Institut der Wirtschaftsprüfer*) (“**IDW**”) and Regulation (EU) No 537/2014 of the European Parliament and of the Council of April 16, 2014, on specific requirements regarding statutory audit of public-interest entities. The German-language version of the Audited Consolidated Financial Statements 2020 was audited by Ernst & Young GmbH Wirtschaftsprüfungsgesellschaft, with its registered seat in Stuttgart, Germany, acting through its Frankfurt office at Mergenthalerallee 3-5, 65760 Eschborn, Germany (“**EY**”), which issued an unqualified independent German-language auditor’s report thereon, and the Audited Consolidated Financial Statements 2019 and Audited Consolidated Financial Statements 2018 were audited by PricewaterhouseCoopers GmbH Wirtschaftsprüfungsgesellschaft with its registered seat in Frankfurt am Main, Germany, acting through its Düsseldorf office at Moskau Str. 19, 40227 Düsseldorf, Germany (“**PwC**”), which issued unqualified German-language auditor’s reports thereon, English-language translations are included in section “21 Financial Information” beginning on page F-1, respectively. The Audited Annual Financial Statements were prepared in accordance with the requirements of German commercial law applicable to business corporations and the German-language version of the Audited Annual Financial Statements was audited in accordance with section 317 of the German Commercial Code (*Handelsgesetzbuch*) and in compliance with

German Generally Accepted Standards for Financial Statement Audits promulgated by IDW, by EY, who issued an unqualified German-language independent auditor's report thereon, an English-language translation thereof is included in section "21 Financial Information" beginning on page F-1. The Unaudited Condensed Consolidated Interim Financial Statements were prepared in accordance with IFRS applicable to interim financial reporting. An English-language translation thereof is included in section "21 Financial Information" beginning on page F-1.

2.8.2 Currency

The amounts set forth in this Prospectus in "€," "euro" or "EUR" refer to the single currency of the participating member states in the third stage of the European Economic Union pursuant to the Treaty Establishing the European Community. The amounts in "\$," "U.S. dollar" or "USD" refer to the legal currency of the United States. Fluctuations in the currency exchange rates between the euro and the U.S. dollar or other currencies will affect the U.S. dollar or other currency amounts received by owners of the Shares on conversion of dividends, if any, paid in euro on the Shares. The Lufthansa Group's principal functional currency is the euro, and the Consolidated Financial Statements were prepared in euro.

2.8.3 Figures and Technical Terms

Financial information presented in parentheses in this Prospectus denote a negative amount. Certain numerical data, financial information and market data in this Prospectus was rounded in accordance with the rules of commercial rounding. Unless indicated otherwise, all financial data presented in the text and tables in this section of this Prospectus is shown in millions of euros (€ million), commercially rounded to the nearest million. Variances, percentage changes and ratios in the text and tables of this Prospectus are calculated based on the underlying numbers as presented in this Prospectus, *i.e.*, after rounding of such underlying numbers, and then commercially rounded to a whole percentage or to one digit after the decimal point. In some instances, such rounded figures and percentages may not add up to 100% or to the totals or subtotals contained in this Prospectus. Furthermore, totals and subtotals in tables may differ slightly from unrounded figures contained in this Prospectus due to rounding in accordance with commercial rounding. A dash ("—") signifies that the relevant figure is not available or equal to zero, while a zero ("0") or nil signifies that the relevant figure was rounded to zero.

A glossary of certain technical and financial terms and abbreviations used in this Prospectus is provided at the end of this Prospectus under the heading "22 Glossary."

2.9 Alternative Performance Measures

In accordance with the European Securities and Markets Authority ("ESMA") Guidelines on alternative performance measures of October 5, 2015 (the "ESMA Guidelines"), the following sections set out information related to certain financial measures of the Lufthansa Group that are not defined by IFRS and which the Lufthansa Group regards as alternative performance measures ("APMs") within the meaning of the ESMA Guidelines.

2.9.1 Overview

This Prospectus contains the following APMs: (i) EBIT, (ii) Adjusted EBIT, (iii) Adjusted EBITDA, (iv) Adjusted EBIT Margin, and (v) Adjusted ROCE. For more information on these APMs, including a reconciliation, see "9.5 Alternative Performance Measures and Other Operating Metrics."

We present these APMs and operating metrics as (i) supplemental information because they are used by our management to measure operating performance or as an auxiliary control profitability parameter, including in presentations to our management, and as a basis for strategic planning and forecasting, and (ii) they represent measures that we believe are widely used by certain investors, securities analysts, and other parties as supplemental measures of operating and financial performance. These APMs and operating metrics may enhance management's and investors' understanding of our financial performance by excluding items that are not classified as part of our ongoing operations such as non-recurring income and expenses and non-cash expenses from purchase price allocations.

These APMs are not defined by IFRS or any other accepted accounting principles, and such items should not be considered as an alternative to the historical financial results or other indicators of our results of operations and financial position based on IFRS measures. In particular, they should not be considered as alternatives to the Lufthansa Group's net income/loss as an indicator of the Lufthansa Group's performance and profitability, or as alternatives to cash flow from operating activities as an indicator of its financial strength. We derive the APMs used in this Prospectus from our internal accounting records or internal management reporting systems or from calculations based on figures from the Consolidated Financial Statements. The APMs, as defined by us, have not been audited (except for Adjusted EBIT) and may not be comparable to similarly titled measures as presented by other companies due to differences in the way our APMs are calculated. Even though the APMs are used by management to assess ongoing operating performance and indebtedness, and though these types of measures are commonly used by investors, they have important limitations as analytical tools, and you should not consider them in isolation or as substitutes for analysis of our results, cash flows or assets and liabilities as reported under IFRS.

2.9.2 Definitions

The table below presents a definition of each of the APMs and the relevance of its use for the Lufthansa Group:

APM	Definition	Relevance of its use
EBIT	EBIT (earnings before interest and tax) is profit/loss from operating activities plus result of equity investments accounted for using the equity method and result of other equity investments.	EBIT is a measure of performance that is adjusted to include equity investments and represent the capacity of the company to earn its costs of capital.
Adjusted EBIT	Adjusted EBIT is EBIT adjusted for impairment losses/gains, effects from pension provisions and results of disposal of assets.	Adjusted EBIT is the key figure for operational management and analysis of our financial performance and excludes effects from transactions that are not expected to occur on a regular recurring basis.
Adjusted EBITDA	Adjusted EBITDA is Adjusted EBIT plus depreciation and amortization, which excludes impairments.	Adjusted EBITDA is a measure of performance that is adjusted for company-specific impairments and assets held for sale.
Adjusted EBIT Margin	Adjusted EBIT Margin is Adjusted EBIT divided by total revenue.	See Adjusted EBIT.
Adjusted ROCE	Adjusted ROCE (return on capital employed) is Adjusted EBIT plus interest on liquidity less 25% taxes divided by the average capital employed, whereby the capital employed results from total assets adjusted for non-operating items (deferred taxes, derivative financial instruments) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).	Adjusted ROCE is a measure of the profitability and efficiency of the capital employed. Capital employed is the working capital that is required to achieve our operational objectives. Adjusted ROCE is only presented for the full fiscal year.

3. THE OFFERING

3.1 Subject-Matter of the Offering

The Prospectus relates to the Offering (as defined below) of 597,742,822 newly issued ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*) of the Company (the “**New Shares**” and each, a “**New Share**”), each such share with a notional value of €2.56 in the share capital of the Company and full dividend rights as from January 1, 2021.

The New Shares will result from a combined capital increase against (i) contribution in cash with indirect subscription rights (*mittelbare Bezugsrechte*) to existing shareholders, except for the WSF, pursuant to Section 186 para. 5 of the German Stock Corporation Act (*Aktiengesetz*), and (ii) contribution in kind by the German Economic Stabilization Fund (*Wirtschaftsstabilisierungsfonds*) (the “**WSF**”) with direct subscription rights, if and to the extent that the WSF participates in the capital increase. Exercising the authorization pursuant to section 4 para. 8 of the Articles of Association (authorized capital created by the Company’s annual general meeting (the “**General Meeting**”) on May 4, 2021, in accordance with Section 7b of the German Economic Stabilization Acceleration Act (*Wirtschaftsstabilisierungsgesetz*)), resolved by the executive board of the Company (*Vorstand*) (the “**Executive Board**”) on September 19, 2021, with approval of the supervisory board of the Company (*Aufsichtsrat*) (the “**Supervisory Board**”) on the same date, to increase the share capital from €1,530,221,624.32 by €1,530,221,624.32 to €3,060,443,248.64 by issuing 597,742,822 New Shares. The New Shares will be offered to existing shareholders at a subscription ratio of 1:1, *i.e.*, one New Share may be acquired at the Subscription Price (as defined below) for one existing share of the Company (the “**Subscription Offer**”). Subscription rights which are not exercised during the period from and including September 22, 2021, up to and including October 5, 2021 (the “**Subscription Period**”) will lapse and be of no value. No compensation will be payable for subscription rights that are not exercised.

The Subscription Offer will include (i) a public offering in Germany, (ii) private placements in certain jurisdictions outside Germany and the United States of America (“**United States**”) in offshore transactions in reliance on Regulation S under the U.S. Securities Act of 1933, as amended (the “**Securities Act**”), and (iii) private placements in the United States pursuant to an exemption from registration under the Securities Act to qualified institutional buyers as defined in Rule 144A under the Securities Act. The New Shares and the subscription rights have not been and will not be registered under the Securities Act, or the securities laws of any other jurisdiction of the United States, its territories and possessions, any state of the United States or the District of Columbia, and may not be offered, sold or otherwise transferred within the United States, except pursuant to an exemption from, or in a transaction not subject to, the registration requirements of the Securities Act and in compliance with any applicable securities laws of any state or other jurisdiction in the United States.

The WSF previously held an interest of 20.00% of the share capital of the Company. On August 16, 2021, the WSF announced the sell-down of up to 5% of the share capital of the Company and, as of September 17, 2021, held an interest of 15.94% of the share capital of the Company. The WSF might sell existing shares of the Company before or during the subscription period (but only to the extent that such sales will not result in a shareholding of the WSF of less than 15% of the share capital of the Company) and, additionally, subscription rights relating to the New Shares that the WSF is entitled to subscribe for during the subscription rights trading from and including September 22, 2021, up to and including September 30, 2021 (until the respective closing auction for subscription rights) on the regulated market (Xetra and Xetra Frankfurt Specialist) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) and in off-market transactions. Furthermore, the WSF might exercise any of its (remaining) subscription rights to subscribe for New Shares in connection with the Offering against contribution in kind of a part of its repayment claim under the Silent Participation I. However, based on guidance provided by the European Commission, the amount of such part of such repayment claim shall not exceed in total the proceeds that the WSF has received in connection with the sell-down and any sale of its subscription rights.

On September 13, 2021, funds and accounts under the management of BlackRock, Inc. held in total 3.40% of the share capital of the Company, *i.e.*, a shareholding of 1.42% of the share capital of the Company and instruments pursuant to Section 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*) relating to 1.98% of the share capital of the Company. Additionally, on September 19, 2021, certain funds and

accounts under the management of BlackRock, Inc., which are predominantly different from such funds and accounts holding 3.40% of the share capital of the Company, entered into a sub-underwriting letter with the Underwriters (the “**Sub-Underwriting Letter**”) and committed to purchase New Shares with an aggregate value of up to €300 million that were neither subscribed during the Subscription Period nor placed in the Rump Placement (as defined below) and, on September 17, 2021, committed to exercise any subscription rights in relation to the Company’s shares held by them. The Sub-Underwriting Letter is subject to customary closing conditions, including (i) the admission of the New Shares, including the Rump Shares (as defined below), to trading on the regulated market (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) and, simultaneously, to the sub-segment thereof with additional post-admission obligations (Prime Standard) and (ii) the closing of the Offering occurs.

Each member of the Executive Board has committed to participate in the Offering (as defined below) by fully exercising all subscription rights received in relation to the Company’s shares.

Any New Shares that are not subscribed for in the Subscription Offer or by the WSF (the “**Rump Shares**”) will be offered for sale by the Underwriters at a price at least equivalent to the subscription price (i) to qualified investors in private placements outside the United States in offshore transactions in reliance on Regulation S under the Securities Act, and (ii) in the United States to qualified institutional buyers in reliance on Rule 144A under the Securities Act (the “**Rump Placement**” and, together with the Subscription Offer, the “**Offering**”).

The Company and the Underwriters entered into an underwriting agreement dated September 19, 2021 (the “**Underwriting Agreement**”), which provides for a firm underwriting of the New Shares not sold in the Offering by the Underwriters or subscribed for by the WSF. The Underwriters may offer or sell such New Shares, which they are required to subscribe for as underwriters otherwise than in connection with the Offering (including by way of a coordinated sell-down). The Joint Global Coordinators (on behalf of themselves and the other Underwriters) may also arrange sub-underwriting for some, all or none of the New Shares. The Offering is subject to, among other things, registration of the implementation of the capital increase in the commercial register of the Company, which is expected to occur on or about October 7, 2021.

3.2 Scheduled Timetable

The scheduled timetable, which remains subject to change, for the Offering is as follows:

September 20, 2021	• Approval of this Prospectus by BaFin • Publication of this Prospectus on the Company’s website
September 21, 2021	• Publication of the Subscription Offer in the German Federal Gazette (<i>Bundesanzeiger</i>)
September 22, 2021	• Commencement of the Subscription Period • Commencement of subscription rights trading • “Ex rights” trading of the Company’s shares • Credit of the subscription rights by the depositary banks to the depositary accounts of the Company’s shareholders expected based on their holdings in the Shares as of September 21, 2021, in the evening, under consideration of open stock-exchange transactions (entitlement date)
September 30, 2021	• End of subscription rights trading
October 5, 2021	• End of Subscription Period • Last day for payment of the Subscription Price (as defined below)
October 6, 2021	• Announcement of the results of the Subscription Offer on the Company’s website • Rump Placement • Announcement of the final results of the Offering on the Company’s website
October 7, 2021	• Registration of the implementation of the capital increase from authorized capital with the commercial register (<i>Handelsregister</i>) expected • Admission of the New Shares to trading on the regulated market segment (<i>regulierter Markt</i>) of the Frankfurt Stock Exchange (<i>Frankfurter Wertpapierbörse</i>) with simultaneous admission to the sub-segment thereof with additional post-admission obligations (Prime Standard) • Publication of the admission in the German Federal Gazette (<i>Bundesanzeiger</i>) and on the website of the Frankfurt Stock Exchange (<i>Frankfurter Wertpapierbörse</i>) (http://www.deutsche-boerse.com)

- October 11, 2021 • Commencement of trading in the New Shares by inclusion in the Company's current stock quotation on the regulated market segment (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) and the sub-segment thereof with additional post-admission obligations (Prime Standard)
- Book-entry delivery of the New Shares subscribed for during the Subscription Period
 - Book-entry delivery of the Rump Shares placed in the Rump Placement

This Prospectus will be published on the internet website of the Deutsche Lufthansa Aktiengesellschaft (<https://www.lufthansagroup.com/capital-increase>). Printed copies of this Prospectus will be available for distribution free of charge at and upon inquiry with Deutsche Lufthansa Aktiengesellschaft, Investor Relations, FRA CW, Lufthansa Aviation Center, Airportring, 60546 Frankfurt am Main, Germany, during regular business hours.

3.3 Subscription Offer

The following is an English-language convenience translation of the Subscription Offer. The German-language version will be published in the German Federal Gazette (*Bundesanzeiger*) on or around September 21, 2021.

“DEUTSCHE LUFTHANSA AKTIENGESELLSCHAFT

Cologne, Germany
ISIN DE0008232125/WKN 823212

Subscription Offer

On September 19, 2021, the executive board of Deutsche Lufthansa Aktiengesellschaft (the “**Company**”) resolved, with the approval of the Company's supervisory board on the same date, to exercise the authorization pursuant to section 4 para. 8 of the Company's articles of association (authorized capital created by the Company's general meeting on May 4, 2021, in accordance with Section 7b of the German Economic Stabilization Acceleration Act (*Wirtschaftsstabilisierungsgesetz*)) to increase the share capital from €1,530,221,624.32 by €1,530,221,624.32 to €3,060,443,248.64 by issuing 597,742,822 new ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*) (the “**New Shares**” and each, a “**New Share**”). The New Shares will result from a combined capital increase against (i) contribution in cash with indirect subscription rights (*mittelbare Bezugsrechte*) to existing shareholders, except for the German Economic Stabilization Fund (*Wirtschaftsstabilisierungsfonds*) (“**WSF**”), pursuant to Section 186 para. 5 of the German Stock Corporation Act (*Aktiengesetz*), and (ii) contribution in kind by the WSF with direct subscription rights, if and to the extent that the WSF participates in the capital increase. The New Shares will be issued at an issue price of €2.56 per New Share (the “**Issue Price**”) and with full dividend rights as from January 1, 2021.

BofA Securities Europe SA (“**BofA Securities**”), Deutsche Bank Aktiengesellschaft (“**Deutsche Bank**”), Goldman Sachs Bank Europe SE (“**Goldman Sachs**”) and J.P. Morgan AG (“**J.P. Morgan**”) and, together with BofA Securities, Deutsche Bank and Goldman Sachs, the “**Joint Global Coordinators**”), Barclays Bank Ireland PLC (“**Barclays**”), BNP PARIBAS SA (“**BNP PARIBAS**”), COMMERZBANK Aktiengesellschaft (“**COMMERZBANK**”), HSBC Trinkaus & Burkhardt AG (“**HSBC**”), UniCredit Bank AG (“**UniCredit Bank**”), Crédit Agricole Corporate and Investment Bank (“**Crédit Agricole Corporate and Investment Bank**”), DZ BANK AG Deutsche Zentral-Genossenschaftsbank, Frankfurt am Main (“**DZ BANK**”), Landesbank Baden-Württemberg (“**Landesbank Baden-Württemberg**”), SMBC Nikko Capital Markets Europe GmbH (“**SMBC Nikko**”) and Société Générale S.A. (“**Société Générale**”) and, together Barclays, BNP PARIBAS, COMMERZBANK, HSBC, UniCredit Bank, Crédit Agricole Corporate and Investment Bank, DZ BANK, Landesbank Baden-Württemberg and SMBC Nikko, the “**Joint Bookrunners**” and, together with the Joint Global Coordinators, the “**Underwriters**”) have agreed, pursuant to an underwriting agreement dated September 19, 2021 (the “**Underwriting Agreement**”), to subscribe the New Shares and acquire and offer the New Shares to the Company's existing shareholders during the subscription period for indirect subscription at the subscription ratio and at the subscription price per New Share (the “**Subscription Offer**”), except for the new shares directly subscribed by the WSF, if and to the extent that the WSF participates in the capital increase. The Underwriting Agreement provides for a firm underwriting of the New Shares not sold in the offering by the Underwriters or subscribed for by the WSF.

The WSF previously held an interest of 20.00% of the share capital of the Company. On August 16, 2021, the WSF announced the sell-down of up to 5% of the share capital of the Company and, as of September 17, 2021, held an interest of 15.94% of the share capital of the Company. The WSF might sell existing shares of the Company before or during the subscription period (but only to the extent that such sales will not result in a shareholding of the WSF of less than 15% of the share capital of the Company) and, additionally, subscription rights relating to the New Shares that the WSF is entitled to subscribe for during the subscription rights trading from and including September 22, 2021, up to and including September 30, 2021 (until the respective closing auction for subscription rights) on the regulated market (Xetra and Xetra Frankfurt Specialist) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) and in off-market transactions. Furthermore, the WSF might exercise any of its (remaining) subscription rights to subscribe for New Shares in connection with the Offering against contribution in kind of a part of its repayment claim under the Silent Participation I. However, based on guidance provided by the European Commission, the amount of such part of such repayment claim shall not exceed in total the proceeds that the WSF has received in connection with the sell-down and any sale of its subscription rights.

Each member of the Executive Board has committed to participate in the Offering (as defined below) by fully exercising all subscription rights received in relation to the Company's shares.

The registration of the implementation of the capital increase in the commercial register (*Handelsregister*) of the local court (*Amtsgericht*) of Cologne is expected to occur on or about October 7, 2021.

On September 22, 2021, the depository banks are expected to credit to the depository accounts of the Company's shareholders the subscription rights (ISIN DE000A3E5B58/WKN A3E5B5) attributable to the existing shares of the Company on the basis of the holdings as of September 21, 2021, in the evening, under consideration of open stock-exchange transactions (entitlement date). On September 24, 2021, Clearstream Banking Aktiengesellschaft, Mergenthalerallee 61, 65760 Eschborn, Federal Republic of Germany ("Germany"), will automatically credit the subscription rights to the custodian banks on the basis of the holdings as of September 23, 2021 (in the evening) (Record Date).

We kindly request our shareholders to exercise their subscription rights to the New Shares during the period

from and including September 22, 2021, up to and including October 5, 2021

(the "Subscription Period")

through their respective depository bank at the subscription agent, Deutsche Bank Aktiengesellschaft (the "Subscription Agent"), during regular banking hours. Investors are recommended to follow the respective instructions by their custodian banks. Subscription rights that are not exercised during this period will lapse and be of no value. No compensation will be awarded for subscription rights that will not be exercised.

The Subscription Agent has its registered office in Frankfurt am Main, Germany.

In accordance with the subscription ratio of 1:1, one existing share of the Company entitles the holder to subscribe for one New Share at the subscription price per New Share. Shareholders can subscribe for one share or multiples thereof. The notice of the exercise of subscription rights is binding upon its receipt by the Subscription Agent and cannot be altered afterwards. The exercise of the subscription rights is, however, conditional upon the registration of the implementation of the capital increase in the commercial register and subject to the other limitations set forth below under "*Important Notice*."

Subscription Price

The subscription price per New Share is €3.58 (the "**Subscription Price**"). The Subscription Price must be paid at the latest on October 5, 2021.

Trading in Subscription Rights

In connection with the Subscription Offer of the New Shares, the subscription rights (ISIN DE000A3E5B58/WKN A3E5B5) for the New Shares will be traded during the period from and including September 22, 2021, up to and including September 30, 2021 (until the respective closing auction for subscription rights) on the regulated market (Xetra and Xetra Frankfurt Specialist) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*). Neither the Company nor the Subscription Agent has applied for admission of the subscription rights to trading on any other stock exchange nor does it intend to do so.

The market price of the subscription rights depends, inter alia, on the development of the price of the Company's shares but it may deviate substantially from the price of the Company's shares. No compensation will be paid for subscription rights not exercised. Upon expiration of the subscription period, subscription rights not exercised will lapse and be of no value. The purchase of one subscription right enables the exercise of the subscription right for the purchase of one New Share, *i.e.*, one New Share may be purchased for one subscription right.

The price of the subscription rights is determined continuously during the ordinary times of trading. On September 30, 2021, the subscription rights trading on Xetra will end with a closing auction starting not before 11:45 a.m. CEST and on Xetra Frankfurt Specialist with an independent special auction starting at 12:00 (noon) CEST.

The market price for the subscription rights depends, among other things, on the development of the share price of the Company but may fluctuate more strongly than the share price.

Share Certificates and Delivery of the New Shares

The New Shares (ISIN DE0008232125/WKN 823212) will be issued as registered shares with no-par value and with restricted transferability (*vinkulierte Namensaktien*). The New Shares will be represented by one global share certificate, which will be held in custody with Clearstream Banking Aktiengesellschaft.

According to the Company's articles of association, the rights of shareholders to receive definitive share certificates for their shares shall be excluded. Likewise, the right of shareholders to dividend coupons and renewal coupons shall be excluded.

Unless the Subscription Period is extended or the Subscription Offer is cancelled, the New Shares subscribed for in the Subscription Offer are expected to be made available to the collective securities custody as co-ownership interests in the global share certificate on October 11, 2021. In the same way, the New Shares acquired in the Rump Placement (as defined below) are expected to be made available on October 11, 2021. The New Shares hold the same rights as all other shares of the Company and are not vested with any additional rights or benefits.

German Aviation Compliance Documentation Act (Luftverkehrsnachweissicherungsgesetz)

The German Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) stipulates that the shares of the Company must be registered shares with restricted transferability (*vinkulierte Namensaktien*), such that the transfer of such shares requires the consent of the Company. All shares of the Company are, and the New Shares will be, registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*) and may only be acquired and transferred with the Company's consent. The Company may only withhold its consent if the registration of the acquisition or transfer would jeopardize the Company's licenses, rights and prerogatives under aviation law and agreements. Only persons registered in the share register are considered shareholders of the Company. The Company's consent is, therefore, not required in order for shareholders already registered in the share register to exercise their subscription rights. However, investors who acquire subscription rights by means of subscription rights trading and who subsequently exercise these rights will be subject to this consent requirement. Following the exercise of subscription rights, a shareholder is generally required to record its identity and nationality in the Company's share register. In the event of a corporation or other legal entity, nationality is determined by the country of registration.

Commissions Charged by Custodian Banks

The custodian banks may charge a customary commission in connection with the subscription of the New Shares as well as for the sale and purchase of subscription rights.

Placement of Unsubscribed Shares (Rump Placement)

Any New Shares that are not subscribed for in the Subscription Offer or by the WSF (the “**Rump Shares**”) will be offered for sale by the Underwriters at a price at least equivalent to the Subscription Price (i) to qualified investors in private placements outside the United States of America (the “**United States**”) in offshore transactions in reliance on Regulation S under the U.S. Securities Act of 1933, as amended (the “**Securities Act**”), and (ii) in the United States of America (“**United States**”) to qualified institutional buyers in reliance on Rule 144A under the Securities Act (the “**Rump Placement**” and, together with the Subscription Offer, the “**Offering**”).

Admission to Trading and Inclusion of the New Shares

The admission of the New Shares to trading on the regulated market (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with simultaneous admission to the sub-segment of the regulated market with additional post-admission obligations (Prime Standard) is expected to be granted on October 7, 2021. The New Shares are expected to be included in the existing quotation for the Company’s listed shares on the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) on October 11, 2021.

Selling Restrictions

The New Shares and the subscription rights will only be publicly offered in Germany. The New Shares and the subscription rights have not been and will not be registered under the Securities Act or with the securities regulatory authority of any individual state or other jurisdiction of the United States. The New Shares and the subscription rights may neither be offered, sold, exercised, pledged, transferred nor delivered directly or indirectly to or within the United States, except pursuant to an exemption from the registration requirements of the Securities Act and the securities laws of the respective individual states of the United States apply and in accordance with further applicable laws of the United States.

The acceptance of this offer outside Germany may be subject to restrictions. Persons who intend to accept this offer outside Germany are requested to inform themselves with regard to and comply with the restrictions that exist outside Germany.

Important Notices

Existing shareholders and new investors are advised to carefully read the Prospectus and to take particular note of the risks described in the section “*I Risk Factors*” starting at page 1 of the Prospectus and to consider such information when making their decision, before making a decision to exercise, acquire or sell any subscription rights or to acquire any shares. The Prospectus is available on the Company’s website (<https://www.lufthansagroup.com/capital-increase>). The Underwriters are entitled to terminate the Underwriting Agreement or decide together with the Company to extend the Subscription Period or to postpone the implementation of the Subscription Offer under certain conditions. These conditions include, in particular, material adverse changes in the business or financial condition, prospects, shareholders’ equity or results of operations of the Company and/or its subsidiaries, material restrictions on stock exchange trading or banking activities, the outbreak or escalation of hostilities or war, or the occurrence of acts of terrorism or other calamity or crisis which have a material adverse impact on the financial markets in Germany, the United Kingdom or the United States. The Underwriters are also relieved from their obligations under the Underwriting Agreement if the implementation of the capital increase has not been registered by October 15, 2021, 11:59 p.m. CEST with the commercial register (*Handelsregister*) of the local court (*Amtsgericht*) of Cologne and if the Company and the Underwriters fail to agree on a later deadline.

In case the Underwriters withdraw from the Underwriting Agreement prior to the registration of the capital increase with the commercial register (*Handelsregister*), subscription rights may expire without compensation. In such a case, a reverse transaction relating to the trading of subscription rights will not take place and investors who purchased subscription rights could suffer a loss. In addition, if at the time of the termination any sales of New Shares have already been made, the seller of the relevant shares bears the risk

of not being able to meet the delivery obligation by delivering New Shares. In case the Underwriters terminate the Underwriting Agreement after the capital increase has been registered with the commercial register (*Handelsregister*), shareholders and purchasers of subscription rights who have exercised their subscription rights may acquire New Shares at the Subscription Price. In case the Underwriters withdraw from the Underwriting Agreement after the Subscription Offer has been completed, which is also possible following delivery, settlement and the listing of the New Shares subscribed for in the offering, such withdrawal would only apply to New Shares that were not subscribed for.

Publication of the Prospectus

In connection with the Subscription Offer, a securities prospectus of the Company dated September 20, 2021 (the “**Prospectus**”) has been published on the Company’s website (<https://www.lufthansagroup.com/capital-increase>). Printed copies of this Prospectus will be available for distribution free of charge at and upon inquiry with Deutsche Lufthansa Aktiengesellschaft, Investor Relations, FRA CW, Lufthansa Aviation Center, Airportring, 60546 Frankfurt am Main, Germany, during regular business hours.

Cologne, September 2021

DEUTSCHE LUFTHANSA AKTIENGESELLSCHAFT

The Executive Board”

3.4 Sale of Subscription Rights and Subscription Rights Remaining Unexercised

The subscription rights are freely transferable. To the extent provided for in the terms and conditions of the custodian banks, these banks will use their best efforts to dispose of the subscription rights at the best possible price, unless the party entitled to the subscription issues an instruction regarding the exercise of its subscription rights. Subscription rights remaining unexercised will lapse and be of no value.

3.5 Underwriting Agreement

Pursuant to the Underwriting Agreement dated September 19, 2021, the Underwriters have agreed, under certain conditions, to subscribe the New Shares and acquire and offer the New Shares to the Company’s existing shareholders during the subscription period for indirect subscription at the subscription ratio and at the subscription price per New Share. Furthermore, the number of New Shares to be ultimately subscribed by the Underwriters will decrease if and to the extent that the WSF subscribes for New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I.

The following table presents the legal names and registered addresses of the Underwriters:

	Address	Number of New Shares	Percentage of New Shares
BofA Securities Europe SA	51 rue La Boétie 75008 Paris France	89,661,424	15.0%
Deutsche Bank Aktiengesellschaft	Taunusanlage 12 60325 Frankfurt am Main Germany	89,661,424	15.0%
Goldman Sachs Bank Europe SE	Marienturm, Taunusanlage 9-10 60329 Frankfurt am Main Germany	89,661,424	15.0%
J.P. Morgan AG	Taunustor 1 60310 Frankfurt am Main Germany	89,661,424	15.0%
Barclays Bank Ireland PLC	One Molesworth Street Dublin 2, D02 RF29 Ireland	35,864,570	6.0%

	Address	Number of New Shares	Percentage of New Shares
BNP PARIBAS SA	16, Boulevard des Italiens 75009 Paris France	35,864,569	6.0%
COMMERZBANK Aktiengesellschaft	Kaiserstraße 16 (Kaiserplatz) 60311 Frankfurt am Main Germany	35,864,569	6.0%
HSBC Trinkaus & Burkhardt AG	Königsallee 21/23 40212 Düsseldorf Germany	35,864,569	6.0%
UniCredit Bank AG	Arabellastraße 12 81925 Munich Germany	35,864,569	6.0%
Crédit Agricole Corporate and Investment Bank	12 Place des Etats-Unis CS 70052 92547 Montrouge Cedex France	11,954,856	2.0%
DZ BANK AG Deutsche Zentral- Genossenschaftsbank, Frankfurt am Main	Platz der Republik 60325 Frankfurt am Main Germany	11,954,856	2.0%
Landesbank Baden-Württemberg	Am Hauptbahnhof 2 70173 Stuttgart Germany	11,954,856	2.0%
SMBC Nikko Capital Markets Europe GmbH	Neue Mainzer Straße 52-58 60311 Frankfurt am Main Germany	11,954,856	2.0%
Société Générale S.A.	29, boulevard Haussmann 75009 Paris France	11,954,856	2.0%
Total		597,742,822	100.0%

Assuming gross proceeds of approximately €2,140 million from the sale of the New Shares, the total costs related to the Offering, including commissions for the Underwriters, are expected to be approximately €72.3 million and the net proceeds from the Offering €2,068 million. In addition, the Company has agreed to indemnify the Underwriters against certain liabilities.

The Underwriting Agreement also provides that the obligations of the Underwriters are subject to the satisfaction of certain conditions, including, for example, the receipt of a customary confirmation and legal opinions satisfactory to the requirements of the Underwriters. The Company has further agreed in the Underwriting Agreement to indemnify the Underwriters against certain liabilities, including liabilities under applicable securities laws that may arise in connection with the Offering.

The Underwriters are entitled to terminate the Underwriting Agreement or decide together with the Company to extend the Subscription Period or to postpone the implementation of the Subscription Offer under certain conditions. These conditions include, in particular, material adverse changes in the business or financial condition, prospects, shareholders' equity or results of operations of the Company and/or its subsidiaries, material restrictions on stock exchange trading or banking activities, the outbreak or escalation of hostilities or war, or the occurrence of acts of terrorism or other calamity or crisis which have a material adverse impact on the financial markets in Germany, the United Kingdom or the United States. The Underwriters are also relieved from their obligations under the Underwriting Agreement if the implementation of the capital increase has not been registered by October 15, 2021, 11:59 p.m. CEST, with the commercial register (*Handelsregister*) of the local court (*Amtsgericht*) of Cologne and if the Company and the Underwriters fail to agree on a later deadline.

Any New Shares not subscribed in the Subscription Offer or by the WSF will be offered for sale by the Underwriters at a price at least equivalent to the Subscription Price in the Rump Placement.

3.6 Lock-Up Agreement

In the Underwriting Agreement, the Company has undertaken that it will not, without the prior written consent of the Joint Global Coordinators, which consent may not be unreasonably withheld, during the period ending 180 days after the date of the book-entry delivery of the New Shares and the Rump Shares, which is expected to take place on October 11, 2021:

- carry out any capital increases or issue any convertible bonds or other securities which are convertible, exercisable into, or otherwise give the right to subscribe for or acquire its ordinary shares, whether directly or indirectly; or
- enter into any swap or other agreement, arrangement or transaction that transfers, confers or allots, in whole or in part, directly or indirectly, any of the economic consequences of the ownership of its shares (whether any such swap, agreement, arrangement or transaction is to be settled by delivery of ordinary shares, cash or otherwise).

The Company may however (a) issue or sell Shares or other securities to employees and members of the executive bodies of the Company or its subsidiaries under employee and management compensation or bonus arrangements, (b) issue shares to bondholders in connection with the convertible bond due 2025 (for further details, see “12.16.6(e) Hybrid Bond and Convertible Bond”), (c) issue shares to the WSF to fulfill its obligations under the Framework Agreement and (d) undertake any corporate action for purposes of entering into joint ventures, other forms of co-operations or acquisitions, provided that the respective other party assumes towards the Underwriters to comply with customary restrictions on the disposal of shares for the then remaining period of the 180-day period after the book-entry delivery of the New Shares and the Rump Shares.

Additionally, the WSF has undertaken to the Company, if and to the extent that it subscribes and receives New Shares, that during the period of six-months after the date of the admission of the New Shares to trading, not to offer, pledge, sell, otherwise dispose of, or to contract to offer, pledge or sell, or otherwise dispose of, directly or indirectly, any New Shares or existing shares of the Company, including any securities of the Company that are substantially similar to the New Shares or existing shares of the Company, including but not limited to any securities that are convertible into or exchangeable for, or that represent the right to receive, New Shares or existing shares of the Company or any such substantially similar securities or do anything with the same economic effect as any of the foregoing, without the prior written consent of the Joint Global Coordinators that shall not be given without the prior written consent of the Company, which shall not be unreasonably withheld by the Company (the “**WSF Lock-Up**”). The WSF Lock-Up shall in each case not apply to sales of New Shares or existing shares of the Company to purchasers who themselves agree to the aforementioned lock-up.

3.7 Material Interests, Including Conflicts of Interest

In connection with the Offering and admission of the New Shares to trading on the regulated market segment (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with simultaneous admission to the sub-segment thereof with additional post-admission obligations (Prime Standard), the Underwriters have a contractual relationship with the Company. The Underwriters advise the Company on the transaction and coordinate the structuring and execution of the transaction. In addition, Deutsche Bank is appointed to act as Subscription Agent, settlement agent and as listing agent. Upon execution of the transaction, the Underwriters will receive fees. As a result, the Underwriters have a financial interest in the successful completion of the Offering.

Each of the Underwriters, including Société Générale holding 0.74% of the share capital of the Company and instruments pursuant to Section 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*) relating to 4.28% of the share capital of the Company, and their respective affiliates may have engaged in transactions with, and provided various commercial banking, investment banking, financial advisory transactions and services in the ordinary course of their business with the Company and/or its affiliates for which they would have received customary fees and commissions.

Furthermore, in connection with the Offering, each of the Underwriters and any of their respective affiliates may take up a portion of the New Shares in the Offering as a principal position and in that capacity may

retain, purchase, or sell for their own account such New Shares or related investments and may offer or sell such New Shares or other investments otherwise than in connection with the Offering. In addition, the Underwriters or companies affiliated with the Underwriters have engaged in securities trading and brokerage activities, as well as providing investment banking, asset management, financing and financial advisory services and other commercial and investment banking products and services to a wide range of corporations and individuals. They may from time to time enter into business relationships with companies of Lufthansa Group or perform services on their behalf as part of their normal course of business including such relating to lending and asset-backed securities transactions. In the ordinary course of the Underwriters' trading, brokerage, asset management, and financing activities, the Underwriters may at any time deal as principal or agent for more than one party in, or hold long or short positions, and may trade or otherwise effect transactions, for its own account or the accounts of customers, in debt or equity securities or senior loans of the Company, its affiliates or other entities that may be involved in or connected with the transactions contemplated hereby, including in connection with the Offering. Accordingly, references in this document to the New Shares being issued, offered, subscribed, acquired, placed or otherwise dealt in should be read as including any issue, offer, subscription, acquisition, placing or dealing by each of the Underwriters and any of their affiliates acting as investors for their own accounts. Accordingly, the Underwriters and companies affiliated with the Underwriters may in the future face conflicts of interests with shareholders in the Company. In addition, certain of the Underwriters or their affiliates may enter into financing arrangements (including swaps, contracts for difference and margin loans) with investors in connection with which such Underwriters (or their affiliates) may from time to time acquire, hold or dispose of Shares of the Company. Except as required by applicable law or regulation, the Underwriters do not propose to make any public disclosure in relation to such transactions.

Because of these business relationships between the Underwriters, or any of their respective affiliates, and the Company, the interests of the Underwriters may not be aligned with those of the Company or the Company's other shareholders, which constitutes a potential conflict of interest.

All of our Executive Board members as well as all of our Supervisory Board members, except for Dr. Thomas Enders, Jürgen Jennerke, Dr. Michael Kerkloh, Harald Krüger, Miriam Sapiro, Ilja Schulz, Britta Seeger, Dr. Astrid Stange and Angela Titzrath, directly or indirectly, hold Shares, stock option rights or phantom shares of the Company (see "*18.2.3 Compensation*" and "*18.3.4 Compensation*"). As a result, they have a financial interest in the development of the Company's share price, which constitutes a potential conflict of interest. As to further implications of the Offering with respect to the compensation of the members of the Executive Board, see "*18.2.3(c) Remuneration Limited for the Duration of Stabilization Package*."

Since the Company will utilize the net proceeds from the Offering to repay amounts drawn under the Silent Participation I, the WSF has a financial interest in Offering.

Other than the interests described above, there are no material interests, in particular no material conflicts of interests, with respect to the Offering.

3.8 Selling Restriction Notices

The distribution of this Prospectus and the sale of the New Shares and subscription rights for the New Shares may be restricted by law in certain jurisdictions. The New Shares and subscription rights for the New Shares will be offered to the existing shareholders of the Company and holders of subscription rights by way of public offering solely in Germany. The acceptance of the Subscription Offer outside Germany may be subject to restrictions. The New Shares and subscription rights for the New Shares may not be offered or sold, directly or indirectly, and neither this Prospectus nor any other offering material or advertisement in connection with the New Shares and subscription rights for the New Shares may be distributed or published in or from any country or jurisdiction other than Germany, except in compliance with any applicable rules and regulations of such country or jurisdiction. Persons who intend to accept the Subscription Offer outside Germany are requested to inform themselves of or seek professional advice for and comply with the restrictions that exist outside Germany. It is the responsibility of any person who receives a copy of this Prospectus to inform themselves about and observe any laws and restrictions, including, but not limited to, those set out below. Failure to comply with these restrictions may constitute a violation of securities laws.

No action has been or will be taken by the Company or the Underwriters to permit a public offering of any New Shares and subscription rights for the New Shares or the possession or distribution of this Prospectus in any country or jurisdiction other than Germany where action for such purposes may be required. This Prospectus has been approved solely by BaFin as competent authority under the Prospectus Regulation, see “2.1 Responsibility Statement.”

The New Shares and the subscription rights for the New Shares have not been, and will not be, registered under the Securities Act or the securities laws of any state or other jurisdiction of the United States. They may not be offered, sold, otherwise transferred or delivered, directly or indirectly, within or into the United States its territories and possessions (including Puerto Rico, the U.S. Virgin Islands, Guam, American Samoa, Wake Island and the Northern Mariana Islands), any state of the United States or the District of Columbia (together, the “**United States**”), except pursuant to an exemption from the registration and reporting requirements of the Securities Act and in compliance with any applicable securities laws of any state or other jurisdiction in the United States and all other applicable provisions of U.S. law. Accordingly, the New Shares and subscription rights for the New Shares are being offered and sold only (a) in the United States pursuant to an exemption from registration under the Securities Act, to persons reasonably believed to be qualified institutional buyers pursuant to Rule 144A and (b) outside the United States in an offshore transaction in reliance on Regulation S. Thus, pursuant to the Underwriting Agreement, each of the Underwriters, severally and not jointly, has covenanted to and agreed with the Company that neither it nor any of its affiliates (as defined in Rule 405 under the Securities Act), nor any third party acting on its or their behalf, with regard to the subscription rights or the New Shares:

- (i) directly or through any agent, has or have offered, solicited offers to buy or sold, or will offer, solicit offers to buy or sell these (y) by any form of “general solicitation” or “general advertising” in the United States, each within the meaning of Regulation D under the Securities Act, or in any manner involving a public offering within the meaning of Section 4(a)(2) of the Securities Act, or (z) to any persons in the United States, except to such persons whom they reasonably believe to be “qualified institutional buyers” (within the meaning of Rule 144A) for its own account or for the account of another qualified institutional buyer in transactions meeting the requirements of Rule 144A; or
- (ii) has or have engaged or will engage in any “directed selling efforts” in the United States within the meaning of Regulation S under the Securities Act.

The Company does not intend to register the Offering or any portion thereof in the United States or to conduct a public offering of the New Shares or subscription rights for the New Shares in the United States.

Until 40 days after commencement of the Offering, the offer, sale, purchase or transfer of the subscription rights or New Shares within the United States by a dealer (whether or not participating in the Offering) may violate the registration requirements of the Securities Act if such offer or sale is made otherwise than in accordance with Rule 144A or another exemption from, or in a transaction not subject to, the registration requirements of the Securities Act. The subscription rights and the New Shares have not been approved or rejected by any federal or state securities commission or regulatory authority of the United States. Furthermore, none of these authorities have confirmed or approved the terms of the Offering, the subscription rights, the New Shares or the accuracy or completeness of this Prospectus.

The subscription rights will not be distributed by the Company to any shareholder of the Company that is located or resident in Canada, except in limited circumstances after such shareholder has established its eligibility to receive the subscription rights to the Company’s satisfaction. The New Shares may be sold in certain provinces of Canada pursuant to an exemption from the prospectus requirements of Canadian securities laws.

Sales in the United Kingdom are also subject to restrictions. Each of the Underwriters has, severally and not jointly, covenanted to and agreed with the Company that it or any of its affiliates or any person on its or their behalf:

- (i) has only communicated or caused to be communicated and will only communicate or cause to be communicated any invitation or inducement to engage in investment activity within the meaning of Section 21 of the United Kingdom Financial Services and Markets Act 2000 (“**FSMA**”) received by it

in connection with the issue or sale of any New Shares in circumstances in which Section 21 para. 1 of the FSMA does not apply to the Company; and

- (ii) has complied and will comply with all applicable provisions of the FSMA with respect to anything done by it in relation to the subscription rights and the New Shares in, from, or otherwise involving the United Kingdom.

Each of the Underwriters has, severally and not jointly, represented and warranted to the Company in the Underwriting Agreement that it has not publicly offered and will not publicly offer the subscription rights or the New Shares to existing shareholders of the Company or holders of subscription right to the New Shares in any member state of the European Economic Area or the United Kingdom (each, a “**Relevant State**”) bound by the Prospectus Regulation (other than the offer in Germany) from the date of application of the Prospectus Regulation unless:

- (i) a prospectus for the New Shares has been approved by BaFin and published and notified in accordance with the Prospectus Regulation;
- (ii) the Offering is addressed solely to “qualified investors” (within the meaning of the Prospectus Regulation);
- (iii) the Offering is made to fewer than 150 natural or legal persons (other than a “qualified investor” within the meaning of (ii) above) in the respective Relevant State;
- (iv) for the existence of any other circumstances falling within Article 1(3) or Article 1(4) of the Prospectus Regulation; or
- (v) for the existence of any other circumstance falling within Article 3(2) of the Prospectus Regulation, as a result of which the publication of a prospectus by the Company is not required in the respective Relevant State;

and provided that, in each of (ii) to (v) above, no such offer of New Shares or subscription rights shall result in the requirement by the Company or the Underwriters to publish a prospectus under the Prospectus Regulation, and whereas, for the purposes of this Offering, any limitation arising as a result of or in connection with the requirement for the publication of a prospectus shall apply mutatis mutandis to the requirement to publish a supplement to prospectus pursuant to Article 23 of the Prospectus Regulation.

In addition, each of the Underwriters has, severally and not jointly, represented and warranted to the Company in the Underwriting Agreement that:

- (i) it and none of its affiliates has taken or will take any action in any jurisdiction other than in Germany that would constitute an offer to the public of any subscription rights or New Shares; and
- (ii) it will comply in all material respects with such laws and regulations that are applicable to the offering in any jurisdictions in which it offers, sells or delivers subscription rights or New Shares.

3.9 Identification of Target Market, Information to Distributors

Each distributor is responsible for undertaking its own target market assessment with respect to the Company’s shares and subscription rights and determining appropriate distribution channels. Solely for the purposes of the product governance requirements contained within: (a) Directive 2014/65/EU of May 15, 2014 on markets in financial instruments, as amended (“**MiFID II**”); (b) Articles 9 and 10 of Commission Delegated Directive (EU) 2017/593 supplementing MiFID II; and (c) local implementing measures (together, the “**MiFID II Product Governance Requirements**”), and disclaiming all and any liability, whether arising in tort, contract or otherwise, which any “manufacturers” (for the purposes of the MiFID II Product Governance Requirements) may otherwise have with respect thereto, the Company’s shares have been subject to a product approval process by the Underwriters. As a result, it has been determined that such Shares are: (i) compatible with an end target market of retail investors and investors who meet the criteria of professional clients and eligible counterparties, each as defined in MiFID II; and (ii) eligible for distribution through all distribution channels as are permitted by MiFID II (the “**Target Market Assessment**”). Notwithstanding the Target Market Assessment, any person subsequently offering, selling or recommending the Company’s shares (a “distributor” for the purpose of the MiFID II Product Governance Requirements) should note that: the value of the subscription rights and of the price of the Company’s shares may decline

and investors could lose all or part of their investment; the Company's shares offer no guaranteed income and no capital protection; and an investment in the Company's shares is compatible only with investors who do not need a guaranteed income or capital protection, who (either alone or in conjunction with an appropriate financial or other adviser) are capable of evaluating the merits and risks of such an investment and who have sufficient resources to be able to bear any losses that may result therefrom. The Target Market Assessment is without prejudice to the requirements of any contractual, legal or regulatory selling restrictions in relation to any subsequent offering or sale of the Company's shares and does not constitute (i) an assessment of suitability or appropriateness for the purposes of MiFID II or (ii) a recommendation to any investor or group of investors to invest in, purchase, or take any other action whatsoever with respect to, the Company's shares.

In addition, other selling restrictions may apply in other jurisdictions in relation to any subsequent offering or sale of the Company's shares. Potential investors or distributors should make their own assessment on the suitability, appropriateness and lawfulness of any such resale or offering and as to their own investment decision and, if appropriate, seek professional advice for the relevant jurisdiction or jurisdictions.

4. THE STABILIZATION PACKAGE

As a result of the outbreak of COVID-19, the Lufthansa Group experienced unprecedented demands on its liquidity and financial position resulting in an average monthly operating cash drain (defined as Adjusted EBITDA adjusted for significant non-cash items, state aid grants and the redemption portion of IFRS 16 leasing expenses) of €260 million in the Fiscal Year 2020, resulting in an aggregate operating cash drain of €3,119 million in the Fiscal Year 2020. Furthermore, Lufthansa Group's operations resulted in an average monthly operating cash drain of €217 million during the 6M 2021, resulting in an aggregate operating cash drain of €1,299 million as of June 30, 2021. We are not expecting any operating cash drain in the third quarter of 2021.

To enable the Lufthansa Group to improve its liquidity position and meet its general capital requirements, the Lufthansa Group received various government-backed financial stabilization measures in Germany regarding the Company (the **“German Stabilization Package”**), in Switzerland regarding SWISS and Edelweiss (the **“Swiss Stabilization Package”**), in Austria regarding Austrian Airlines (the **“Austrian Stabilization Package”**) and in Belgium regarding Brussels Airlines (the **“Belgian Stabilization Package”**) in the aggregate amount of up to €9 billion (collectively, the **“Stabilization Package”**).

4.1 German Stabilization Package

The German Stabilization Package comprises the following measures:

- The Company entered into a syndicated credit facility with, among others, Deutsche Bank Luxembourg S.A. as agent and as security agent, dated July 1, 2020, in the amount of up to €3 billion with the participation of, among other lenders, the German government-owned development bank (*Kreditanstalt für Wiederaufbau*) (**“KfW”**) under the KfW Program 855 “Direct Participation for Syndicated Financing” (the **“KfW Financing”**). The KfW Financing had a term of three years with bullet repayment and an initial interest rate of EURIBOR (zero-floored) plus a margin of 3.00% p.a., subject to adjustments based on the Company's credit rating. The KfW Financing was secured by pledges and/or charges over the shares of and/or partnership interests in special purpose entities in Malta, Ireland and Austria owning aircraft. The KfW Financing was initially drawn down in an amount of €1 billion. Following the issuance of a €1.6 billion bond in 2021, the amounts previously drawn under the KfW Financing were repaid and the pledges and charges were released on March 19, 2021, following which it was terminated.
- Silent participations (*Stille Einlagen*) issued by the Company to the Economic Stabilization Fund of the Federal Republic of Germany (*Wirtschaftsstabilisierungsfonds*) (**“WSF”**) in an aggregate amount of approximately €5.5 billion includes:
 - i. The silent participation I (*Stille Einlage I*) in the amount of €4.5 billion, which is accounted for as equity according to IFRS to the extent drawn by the Company, provides for a loss participation and coupon deferral right (the **“Silent Participation I”**). The Silent Participation I can only be drawn until December 31, 2021, in not more than six tranches. The Silent Participation I does not have a fixed term and set forth interest rate step-ups from 4% p.a. (in 2020 and 2021) to up to 9.5% p.a. from 2027 onwards.
 - ii. The silent participation II-A (*Stille Einlage II-A*) (the **“Silent Participation II-A”**) and the silent participation II-B (*Stille Einlage II-B*) (the **“Silent Participation II-B”**) and, together with the Silent Participation II-A, the **“Silent Participation II”** and, together with the Silent Participation I, the **“Silent Participations”**) in the aggregate amount of €1.0 billion, which are accounted for as liability and not as equity according to IFRS, do not provide for loss participation and coupon deferral rights. The Silent Participation II has a six-year term and sets forth interest rate step-ups from 4% p.a. (in 2020 and 2021) to 8% p.a. (in 2025 and 2026) and to 9.5% p.a. from 2027 onwards.
- As of the date of this Prospectus, the Silent Participation I has been utilized in the amount of €1.5 billion and the Silent Participation II has been fully utilized in the amount of €1.0 billion.
- Certain stabilization measures provided under the Belgian Stabilization Package (*i.e.*, the Belgian Profit Participation Certificates (as defined below)) and the Austrian Stabilization Package (*i.e.*, the

Austrian Subsidy Grant (as defined below)) as well as certain other stabilization measures, including certain other non-domestic equity-linked stabilization measures, are credited against the maximum amount available under the Silent Participation I. Accordingly, the maximum amount available under the Silent Participation I of initially €4.7 billion was reduced to €4.5 billion. Following completion of the Offering, but only if the WSF subscribes and receives New Shares, the maximum amount available for drawdown under the Silent Participation I is being further reduced in the amount of the gross proceeds from the Offering.

- Furthermore, the Company issued 119.5 million new shares to the WSF at a subscription price of €2.56 per share in connection with a capital increase against cash (the “**WSF Capital Increase**”), resulting in total subscription amount of €306.0 million. The WSF Capital Increase was approved by the General Meeting on June 25, 2020, and was registered in the Company’s commercial register (*Handelsregister*) on June 26, 2020.
- The WSF has agreed to sell all of its Company’s shares by not later than December 31, 2023, provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have been fully repaid and subject to achieving a minimum sales price of (i) the market price or (ii) €2.56 per Company’s share plus 12% p.a. calculated for the period between acquisition of the Company’s shares by the WSF and the divestment, whichever is higher (the “**Minimum Sales Price**”). The Minimum Sales Price is subject to adjustments in the event of certain corporate actions, including capital reductions and share splits. Following expiration of this period and provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have been repaid in full, the Company has the right to request that the WSF sells all of its Company’s shares to investors designated by the Company at the Minimum Sales Price.
- However, if the WSF subscribes and receives New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I, the WSF shall sell all of its Company’s shares within 18 months following full repayment of the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, and subject to achieving the Minimum Sales Price, which is being adjusted to reflect the weighted average of the purchase prices paid by the WSF in relation to existing shares of the Company and the New Shares. However, such 18-months period commences only after the WSF Lock-Up has expired, if the repayment of the Silent Participation I and Silent Participation II occurs prior to expiration of the WSF Lock-Up. Following expiration of this period and provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have been repaid in full, the Company shall have the right to request that the WSF sells all of its Company’s shares to investors designated by the Company at the Minimum Sales Price.

The Silent Participation I, the Silent Participation II and the WSF Capital Increase (together, the “**WSF Stabilization Measures**”) are governed by a framework agreement based on the respective EU Commission’s state aid decision regarding granting stabilization measures (*Rahmenvertrag zur Gewährung von Stabilisierungsmaßnahmen*) and certain ancillary agreements entered into between the WSF and the Company on June 29, 2020, as amended from time to time (the “**Framework Agreement**”).

Key terms of the Framework Agreement include provisions pursuant to which:

- In the event of a takeover of the Company, the WSF may increase its shareholding in the Company by, fully or partially, converting the Silent Participation II-A into equity against subscribing 39,849,522 Company’s shares.
- In the event of non-payment by the Company of interest in relation to the Silent Participation I for the years until 2023, the WSF can convert a certain portion of the Silent Participation II-B into equity corresponding to a shareholding of not more than 5% of the Company’s share capital at the earliest from 2024 and in the event of a further non-payment by the Company of interest in relation to the Silent Participation I for the years 2024 and 2025, the WSF can convert an additional portion of the Silent Participation II-B into equity corresponding to a further shareholding of not more than 5% of the Company’s share capital at the earliest from 2026. Such additional conversion right,

however, only applies to the extent that the WSF has not previously increased its shareholding by converting the Silent Participation II-A.

- In the event that the WSF subscribes and receives New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I and provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have not been fully repaid by March 31, 2022, the Company shall propose to the General Meeting to resolve on a new contingent capital (*bedingtes Kapital*) and a new authorized capital (*genehmigtes Kapital*) to be utilized in connection with a conversion of the Silent Participation II and/or contribution of amounts drawn under the Silent Participation I but not repaid for the purpose of anti-dilution protection.
- The Company shall not pay any dividends to its shareholders or repurchase shares or other financial instruments or make non-obligatory coupon payments or non-obligatory distributions to parties other than the WSF before the WSF Stabilization Measures are fully terminated.
- The Company and the WSF have agreed that the WSF may approve a rights issue of the Company if the exercise of all of its subscription rights does not result in a total subscription price to be paid by the WSF in excess of €1 billion. In the event of a rights issue, the WSF also shall be entitled to contribute the Silent Participation I or II, in full or in part, in kind in lieu of a contribution in cash. Furthermore, the WSF may receive the option to purchase any shares not subscribed for during the subscription period at a discount of 5% to the subscription price as permitted under the German Economic Stabilization Acceleration Act (*Wirtschaftsstabilisierungsgesetz*).
- Restrictions apply to the remuneration of the members of the Executive Board, the Supervisory Board and the first and second management level below board level of the Company. For further details, see “12.16.5 State Aid / Stabilization Package” and “18.2.3(c) Remuneration Limited for the Duration of Stabilization Package.”
- Two seats on the Supervisory Board shall be filled in agreement with the German government, one of whom shall become a member of the Company’s audit committee. For further details, see “18.3.1 Overview.”
- Except in the event of a takeover, the WSF has undertaken not to exercise its voting rights at the General Meeting in connection with the resolutions customary for such meeting.
- Transactions of significant importance such as the creation or utilization of authorized capital, high-volume M&A and financing transactions require prior approval by the WSF.
- The Company has agreed not to cross-subsidize any consolidated subsidiaries that were in difficulties within the meaning of Regulation (EU) No 651/2014 prior to December 31, 2019.
- The Company agreed to timely implement the commitment that the Federal Government of Germany made to the European Commission regarding the transfer of take-off and landing rights to the extent legally permissible. For further details, see “4.5 Approval of the Stabilization Package.”

4.2 Swiss Stabilization Package

Under the Swiss Stabilization Package, a syndicate of Swiss banks provided a credit facility (85% state-guaranteed) in the amount of CHF 1.5 billion to SWISS and Edelweiss on August 20, 2020, as amended on from time to time (the “**Swiss Loan**”). The repayment of the Swiss Loan is scheduled for 2023, but can be extended to 2025 subject to the exercise of two one-year extension options. It does not include an equity portion. The Swiss Loan is secured, including by a pledge over the shares in SWISS, Edelweiss and its parent company AirTrust Ltd. that are ultimately held by the Company. Furthermore, the Company provided a subordinated loan in the amount of CHF 500 million in connection with the Swiss Loan.

As of the date of this Prospectus, an amount of CHF 490 million has been drawn down under the Swiss Loan. For a more detailed description of the key terms underlying the Swiss Loan, see “12.16.5 State Aid / Stabilization Package.”

4.3 Austrian Stabilization Package

The Austrian Federal Government provided state aid in an aggregate amount of €450 million, comprising a credit facility (90% state-guaranteed) in the amount of €300 million (the “**Austrian Stabilization Loan**”), of which an amount of €30 million was repaid on July 15, 2021 and the Austrian Stabilization Loan was reduced accordingly, and a non-repayable subsidy grant (*Katastrophenbeihilfe*) in the amount of €150 million (the “**Austrian Subsidy Grant**”) to Austrian Airlines on June 25, 2020, which has been fully utilized on October 15, 2020. The Austrian Stabilization Loan carries an interest rate of 1% p.a. and provides for a repayment in instalments, with the last tranche due for repayment on December 31, 2025. The Austrian Stabilization Loan is additionally secured, including by a pledge over the shares in Austrian Airlines as well as a lien on 38 certain aircraft owned by Austrian Airlines. The Austrian Subsidy Grant is not repayable provided that certain conditions are met, including that the net loss in the Fiscal Year 2020 exceeds €150 million and that such loss has been audited. Furthermore, the Company provided an equity contribution in the amount of €150 million in connection with the Austrian Stabilization Loan.

As of the date of this Prospectus, the Austrian Stabilization Loan has been fully drawn down in the amount of €270 million. Additionally, Austrian Airlines has received the Austrian Subsidy Grant in the Fiscal Year 2020 in full, which is credited against the Silent Participation I. For a more detailed description of the key terms of the Austrian Stabilization Package, see “12.16.5 State Aid / Stabilization Package.”

4.4 Belgian Stabilization Package

The Belgian Government provided state aid in the amount of €290 million, comprising a credit facility in the aggregate amount of €287 million (the “**Belgian Stabilization Loan**”) and profit participation certificates in the amount of €3 million (the “**Belgian Profit Participation Certificates**”) to SN Airholding SA/NV, for on-lending to Brussels Airline, on September 2, 2020. Any amounts available for drawdown under the Belgian Stabilization Loan can only be drawn until December 31, 2021. The Belgian Stabilization Loan has a term of six years, carried an IBOR based interest rate plus a margin between 50 and 200 basis points and is due for repayment on July 31, 2026, at the latest, whereas the Belgian Profit Participation Certificates have been issued for an indefinite term. The Belgian Stabilization Loan is secured by pledges in SN Airholding SA/NV owned by the Company.

In addition, the Company agreed to an equity contribution in the amount of €170 million in connection with the Belgian Stabilization Loan.

As of the date of this Prospectus, the Belgian Stabilization Loan has been drawn down in an amount of €250 million and the Belgian Participation Certificates were issued, which is credited against the Silent Participation I. For a more detailed description of the key terms of the Belgian Stabilization Package, see “12.16.5 State Aid / Stabilization Package.”

4.5 Approval of the Stabilization Package

The German Stabilization Package was approved by the European Commission on June 25, 2020. The Federal Government of Germany made certain commitments in connection with the approval of the German Stabilization Package, including that the Company will station up to four aircraft at each of the airports of Frankfurt, Germany, and Munich, Germany, and that it will transfer in each case up to 24 take-off and landing rights (slots) to a competitor, *i.e.*, three take-off and three landing rights per aircraft per calendar day. In addition, the European Commission approved the Austrian Stabilization Package on July 6, 2020, and the Belgian Stabilization Package on September 2, 2020. The Swiss Competition Commission approved the Swiss Stabilization Package on May 20, 2020.

4.6 State Aid in the United States

In the United States, certain subsidiaries of LSG Lufthansa Service Holding AG located in the United States received state aid under the Coronavirus Aid, Relief and Economic Security Act, as amended from time to time (“**CARES**”), and the Consolidated Appropriations Act, partially as loans and partially as grants and an amount of USD 463 million remains outstanding as of the date of this Prospectus. In addition, certain subsidiaries of Lufthansa Technik Aktiengesellschaft (“**Lufthansa Technik**”) located in the United States

received state aid under CARES, partially as loans and partially as grants, and an amount of USD 46 million remains outstanding as of the date of this Prospectus.

4.7 Additional Financing Measures

The Company implemented further financing measures to strengthen its liquidity and financial position, including:

- In November 2020, the Company issued a convertible bond with an aggregate amount of €600 million. For further details regarding the convertible bond, see “12.16.6(e) *Hybrid Bond and Convertible Bond*.”
- In November 2020, the Company also issued a corporate bond in the amount of €1 billion under its debt issuance program. For further details regarding the debt issuance program, see “12.16.6(a) *Debt Issuance Program*.”
- Furthermore, the Company concluded or extended several short-term borrower’s note loans (*kurzfristige Schuldscheindarlehen*) in an aggregate amount of €750 million, issued money market paper (*Geldmarktpapiere*) in the amount of €350 million and issued commercial papers in the amount of €385 million, all of which remained outstanding in full as of June 30, 2021. For further details regarding these financing measures, see “12.16.6(b) *Multi-Currency Commercial Paper Program*.”
- The Company entered into a various financing instruments, including sale and lease back financings, secured loans and secured promissory notes (*Schuldscheindarlehen*), raising an aggregate amount of around €500 million. Several of these financing instruments were secured by aircraft of the Lufthansa Group. For further details, see “12.16.6(c) *Unsecured Promissory Notes (Schuldscheindarlehen)*” and “12.16.6(d) *Aircraft-Secured Financing*.”
- In February and in July 2021, the Company issued a dual-tranche corporate bond in the amount of €1.6 billion and €1.0 billion, respectively, under its debt issuance program. For further details regarding the debt issuance program, see “12.16.6(a) *Debt Issuance Program*.”

5. REASONS FOR THE OFFERING, USE OF PROCEEDS

5.1 Proceeds and Costs of the Offering

The net proceeds to the Company from the Offering result from the gross proceeds less the underwriting commissions and other expenses described below. On the basis of a subscription price of €3.58 per New Share and issuance of 597,742,822 New Shares, the Company is seeking to raise gross proceeds of approximately €2,140 million in this Offering. The overall commissions to be paid by the Company to the Underwriters are expected to amount to approximately €58.8 million. Other issue costs to be incurred by the Company are expected to amount to approximately €13.5 million. Investors will not be charged with expenses by the Company or the Underwriters (regarding potential charges by custodian banks, see “3.3 Subscription Offer”). On this basis, the Company expects net proceeds from this Offering of approximately €2,068 million. However, the net proceeds from the Offering and, correspondingly, the amount repayable under the Silent Participation I will decrease, if and to the extent that the WSF subscribes for New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I.

5.2 Reasons for the Offering and Use of Proceeds

As of the date of this Prospectus, the Company has drawn an amount of €1,500 million under the Silent Participation I. The Company will use the net proceeds from the Offering in the amount of approximately €2,068 million to repay the drawn and repayable amounts of the Silent Participation I. The amount repayable under the Silent Participation I and, at the same time, the net proceeds from the Offering will decrease, if and to the extent that the WSF subscribes for New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I. Additionally, the Company intends to use the remaining net proceeds from the Offering to strengthen its equity base as well as its liquidity position, to increase the Lufthansa Group’s flexibility to respond to its evolving financing requirements. For further details, see “8 Capitalization and Indebtedness” and regarding the requirement to use the net proceeds from the Offering primarily to repay state aid made available to the Company by the WSF, see “17.5.3 Authorized Capital C.”

6. DILUTION

The rights of the shareholders to subscribe for the New Shares from the capital increase ensure that each shareholder exercising its subscription rights will continue to hold its original, nearly unchanged percentage share in the share capital of the Company. Assuming the completion of the Offering and the repayment of amounts drawn under the Silent Participation I, the shareholders' percentage ownership in the Company's share capital and its voting rights will be diluted by 50.0% if such shareholder does not exercise any of its subscription rights.

The net book value of the shareholders' equity of the Company as of June 30, 2021, is calculated as the difference of "*Total assets*" less "*Non-current provisions and liabilities*" and "*Current provisions and liabilities*" each as presented as line items in the Unaudited Condensed Consolidated Interim Financial Statements, and was €3,145 million and therefore €5.26 per Share, calculated on the basis of 597,742,822 issued Shares as of June 30, 2021.

Based on the foregoing, and assuming a full implementation of the capital increase from €1,530,221,624.32 by €1,530,221,624.32 to €3,060,443,248.64 by issuing 597,742,822 New Shares against contribution in cash, as well as assuming a Subscription Price of €3.58 per New Share, and following the deduction of the commissions, fees and other expenses related to the Offering in the amount of approximately €72.3 million, the net book value of the shareholders' equity of the Company would have been €3,713 million or €3.11 per Share as of June 30, 2021 (calculated on the basis of the number of 1,195,485,644 Shares issued after the implementation of the capital increase in connection with the Offering).

This corresponds to a dilution in the net book value of the shareholders' equity of the Company by €(2.16) or (41.0)% per Share for the existing shareholders, as the adjusted net book value of the shareholders' equity of the Company per share exceeds the prior net book value of the shareholders' equity by this amount or this percentage. For purchasers of New Shares at the Subscription Price (of €3.58), this entails a theoretical loss of €0.47 or 13.3% per Share against the net book value of the shareholders' equity of the Company.

7. DIVIDEND POLICY

7.1 General Provisions Relating to Profit Allocation and Dividend Payments

Shareholders have a share in the Company's profits determined in proportion to their interest in the Company's share capital. The participation of new shares in the profits may be determined in a different manner.

Distributions of dividends on shares of a stock corporation, such as the Company, are for a given financial year generally resolved by the General Meeting in the subsequent financial year under the German Stock Corporation Act (*Aktiengesetz*). The Executive Board and the Supervisory Board submit a proposal for the distribution of dividends to the General Meeting generally held within the first eight months of the subsequent financial year. The General Meeting then adopts a resolution on such distribution with simple majority of the votes cast without being bound by such proposal.

Under German law, dividends can only be resolved upon and distributed if the unconsolidated financial statements of the Company show distributable profit (*Bilanzgewinn*). Compared to the Company's consolidated financial statements, which are prepared in accordance with IFRS and the additional requirements of German commercial law pursuant to Section 315e of the German Commercial Code (*Handelsgesetzbuch*), the unconsolidated financial statements of the Company are prepared in accordance with the German Commercial Code (*Handelsgesetzbuch*). These accounting principles differ from IFRS in material respects. The unconsolidated financial statements of the Company are prepared by the Executive Board and approved by the Supervisory Board, unless the Executive Board and the Supervisory Board refer the approval to the General Meeting.

When determining the distributable profit (*Bilanzgewinn*), the net profit or loss for the year (*Jahresüberschuss/-fehlbetrag*) is adjusted for profits or losses carried forward from the previous financial year (*Gewinn-/ Verlustvorträge*), as well as for withdrawals from and transfers to reserves. Certain reserves are required to be set up by law and amounts mandatorily allocated to these reserves in the given financial year must be deducted when calculating the distributable profit (*Bilanzgewinn*). Subject to certain statutory restrictions, the General Meeting is entitled to transfer additional amounts to the reserves or carry them forward. The Executive Board must prepare unconsolidated financial statements and a management report for the previous financial year by the statutory deadline and present these to the Supervisory Board and the auditors immediately after preparation. At the same time, the Executive Board must present to the Supervisory Board a proposal for the allocation of the Company's distributable profit (*Bilanzgewinn*) pursuant to Section 170 para. 2 of the German Stock Corporation Act (*Aktiengesetz*). The Supervisory Board must review the unconsolidated financial statements, the Executive Board's management report and the proposal for the allocation of the distributable profit (*Bilanzgewinn*) and report to the General Meeting in writing on the results. If the Executive Board and the Supervisory Board approve the unconsolidated financial statements, they may, pursuant to Section 58 para. 2 of the German Stock Corporation Act (*Aktiengesetz*), transfer 50% of the profit for the financial year (*Jahresüberschuss*) remaining after deducting any transfers to statutory reserves and any losses carried forward to non-statutory reserves. The portion transferred to non-statutory reserves may not exceed 50% of the net income to the extent the non-statutory reserves exceed, or would exceed following such transfer, 50% of the registered share capital of the Company. Pursuant to the Articles of Association and subject to applicable statutory law, the General Meeting may resolve to pay dividends in kind (*Sachdividende*) in accordance with Section 58 para. 5 of the German Stock Corporation Act (*Aktiengesetz*) in addition to or in lieu of a cash distribution.

Besides requirements of mandatory German law, the Company's ability to distribute dividends may be limited by the availability of distributable funds.

Generally, withholding tax (*Kapitalertragsteuer*) is withheld from dividends paid. For more information on the taxation of dividends, see "20.1 Taxation of Dividends."

Any dividends not claimed become time-barred within three years pursuant to the general statute of limitations. Once the statute of limitations applies, the right to receive the relevant dividend payments passes to the Company.

We publish information about any dividends that the General Meeting resolves and about the paying agents we engage in the German Federal Gazette (*Bundesanzeiger*). We also publish them on the Company's website (<http://www.lufthansagroup.com>).

7.2 Dividend Policy and Historical Dividends

The Company's dividend policy is to distribute to shareholders 20% to 40% of its net profit, adjusted for non-recurring gains and losses. However, the outbreak of COVID-19 prompted the General Meeting in 2020 to suspend a dividend payment for the Fiscal Year 2019. To respond to the impact of the COVID-19 pandemic on our business, we have requested and received certain stabilization financings from the WSF (for further details, see "*4 The Stabilization Package*"). As a consequence, we are obligated not to pay any dividends to shareholders as long as any amounts under these stabilization measures remain outstanding, *i.e.*, particularly until the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, are repaid in full and the WSF sold all of the Company's shares held by it. The WSF has agreed to sell all of its Company's shares by not later than December 31, 2023, provided that the Silent Participation I (as defined below), including interest and any additional payments, and the Silent Participation II, including interest, have been fully repaid and subject to achieving the Minimum Sales Price. However, if the WSF subscribes and receives New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I, the WSF shall sell all of its Company's shares within 18 months following full repayment of the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, and subject to achieving the Minimum Sales Price, which is being adjusted to reflect the weighted average of the purchase prices paid by the WSF in relation to existing shares of the Company and the New Shares. However, such 18-months period commences only after the WSF Lock-Up has expired, if the repayment of the Silent Participation I and Silent Participation II occurs prior to expiration of the WSF Lock-Up. Following expiration of the applicable period and provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have been repaid in full, the Company shall have the right to request that the WSF sells all of its Company's shares to investors designated by the Company at the Minimum Sales Price. Following the repayment of all stabilization measures, the Company will be able to pay dividends to its shareholders in accordance with its dividend policy. Until then, the Company will use its free cash flow primarily for repaying its financial liabilities, including the stabilization measures. Against this background, the Shareholders are expected to once again be in a position to participate in the Company's success.

The table below shows the net income attributable to shareholders of the Company and the corresponding earnings per share (basic and diluted) of the Lufthansa Group for the Fiscal Years 2020, 2019 and 2018:

	For the year ended December 31,		
	2020	2019	2018
	(audited, unless otherwise indicated)		
Net profit/loss attributable to the Company's shareholders (in € million).....	(6,725)	1,213	2,163
Weighted average number of the Company's shares (in million)	537.6	475.7	472.6
Earnings per share (in €).....	(12.51)	2.55	4.58
Total dividend pay-out in the following financial year for the financial year indicated (in € million) (unaudited).....	—	—	380
Dividend paid per share in the following financial year for the financial year indicated (in €) (unaudited)	—	—	0.80

8. CAPITALIZATION AND INDEBTEDNESS

Investors should read this section in conjunction with “9 Management’s Discussion And Analysis Of Financial Condition And Results Of Operations” and the Consolidated Financial Statements, including the notes thereto, which are included in the financial section of this Prospectus.

8.1 Capitalization

The following table sets forth the Company’s consolidated capitalization as of June 30, 2021, taken or derived from the Unaudited Condensed Consolidated Interim Financial Statements or the Lufthansa Group’s accounting records or internal management reporting systems, (i) on an actual basis, and (ii) as adjusted for the completion of the Offering and the repayment of amounts drawn under the Silent Participation I, and assuming no tax effects. The amount repayable under the Silent Participation I and, at the same time, the net proceeds from the Offering will decrease, if and to the extent that the WSF subscribes for New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I. For details regarding the proceeds from the Offering and their intended use, see “5 Reasons for the Offering, Use of Proceeds.”

	As of June 30, 2021	Adjusted for the Offering and the repayment of the Silent Participation I As of June 30, 2021
	(in € million) ^(*) (unaudited)	
Total current debt⁽¹⁾	15,322	15,322
<i>thereof guaranteed</i>	—	—
<i>thereof secured⁽²⁾</i>	1,287	1,287
<i>thereof unguaranteed/unsecured</i>	14,035	14,035
Total non-current debt⁽³⁾	22,371	22,371
<i>thereof guaranteed</i>	—	—
<i>thereof secured⁽²⁾</i>	7,248	7,248
<i>thereof unguaranteed/unsecured</i>	15,123	15,123
Shareholders’ equity	3,145	3,713
<i>thereof share capital⁽⁴⁾</i>	1,530	3,060
<i>thereof legal reserves⁽⁵⁾</i>	378	988
<i>thereof other reserves⁽⁶⁾</i>	1,237	(335)
Total	40,838	41,406

(*) Columns may not add up due to rounding.

(1) Referred to as “current provisions and liabilities” in the Unaudited Condensed Consolidated Interim Financial Statements.

(2) Reflects the portion of total current debt and total non-current debt, as applicable, which was primarily secured by way of liens on aircraft and pledges over shares in, among others, Austrian Airlines, SN Airholding SA/NV and SWISS as of June 30, 2021. For further details on such pledges, see “4 The Stabilization Package.”

(3) Referred to as “non-current provisions and liabilities” in the Unaudited Condensed Consolidated Interim Financial Statements.

(4) Referred to as “issued capital” in the Unaudited Condensed Consolidated Interim Financial Statements. The increase of €1,530 million in share capital in the adjusted column reflects the issuance of 597,742,822 New Shares multiplied with the notional share in the share capital of €2.56 per New Share.

(5) Referred to as “capital reserve” in the in the Unaudited Condensed Consolidated Interim Financial Statements. The increase of €610 million in legal reserves in the adjusted column reflects the difference between the assumed gross proceeds of €2,140 million from the sale of the New Shares and the increase of €1,530 million in issued capital resulting from the assumed gross proceeds of €2,140 million from the sale of the New Shares already allocated to share capital.

(6) Comprises “silent participation of the Economic Stabilization Fund,” “retained earnings,” “other neutral reserves,” “net profit/loss” and “minority interests,” each as referred to in the Unaudited Condensed Consolidated Interim Financial Statements. The decrease of €1,572 million in other reserves in the adjusted column reflects the assumed repayment of Silent Participation I amounting to €1,500 million and the subtraction of the total costs related to the Offering of €72 million.

8.2 Indebtedness

The following table sets forth the Company's consolidated indebtedness as of June 30, 2021, taken or derived from the Unaudited Condensed Consolidated Interim Financial Statements or the Lufthansa Group's accounting records or internal management reporting systems, (i) on an actual basis, and (ii) as adjusted for the completion of the Offering and the repayment of amounts drawn under the Silent Participation I, and assuming no tax effects. The amount repayable under the Silent Participation I and, at the same time, the net proceeds from the Offering will decrease, if and to the extent that the WSF subscribes for New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I. For details regarding the proceeds from the Offering and their intended use, see "5 Reasons for the Offering, Use of Proceeds."

	As of June 30, 2021	Adjusted for the Offering and the repayment of the Silent Participation I As of June 30, 2021
	(in € million) ^(*) (unaudited)	
A. Cash ⁽¹⁾	2,063	2,631
B. Cash equivalents	—	—
C. Other current financial assets ⁽²⁾	4,603	4,603
D. Liquidity (A)+(B)+(C)	6,666	7,234
E. Current financial debt ⁽³⁾	6,023	6,023
F. Current portion of non-current financial debt	—	—
G. Current financial indebtedness (E + F)	6,023	6,023
H. Net current financial indebtedness (G - D)	(643)	(1,211)
I. Non-current financial debt (excluding current portion and debt instruments) ⁽⁴⁾	89	89
J. Debt Instruments ⁽⁵⁾	13,266	13,266
K. Non-current trade and other payables	—	—
L. Non-current financial indebtedness (I)+(J)+(K)...	13,355	13,355
M. Total financial indebtedness (H)+(L)	12,712	12,144

(*) Columns may not add up due to rounding.

- (1) Referred to as "cash and cash equivalents" in the Unaudited Condensed Consolidated Interim Financial Statements, which includes fixed-term deposits in the amount of €21 million with a term of four to twelve months. The increase of €568 million in cash in the adjusted column reflects the difference between assumed net proceeds of €2,068 million from the sale of the New Shares and the reduction in cash due to the assumed repayment of Silent Participation I amounting to €1,500 million.
- (2) Referred to as "securities" in the Unaudited Condensed Consolidated Interim Financial Statements.
- (3) Referred to as "borrowings" (current) and "trade payables and other financial liabilities" in the Unaudited Condensed Consolidated Interim Financial Statements.
- (4) Referred to as "other financial liabilities" (non-current) in the Unaudited Condensed Consolidated Interim Financial Statements.
- (5) Referred to as "borrowings" (non-current) in the Unaudited Condensed Consolidated Interim Financial Statements.

As of June 30, 2021, current financial debt included current liabilities related to leases in the amount of €431 million and debt instruments included non-current liabilities related to leases in the amount of €2,030 million.

8.3 Indirect and Contingent Indebtedness

As of June 30, 2021, our indirect and contingent indebtedness amounted to €914 million (presented as contingent liabilities in the Unaudited Condensed Consolidated Interim Financial Statements) and comprised (i) €693 million of contingent liabilities from guarantees, bills of exchange and cheque guarantees; (ii) €205 million of contingent liabilities from guarantees from warranty contracts, and (iii) €16 million of contingent liabilities from providing collateral for third-party liabilities. For additional information, see "9.8.2 Financial and Contingent Liabilities."

8.4 Working Capital Statement

The Company is of the opinion that Lufthansa Group is in a position to meet the payment obligations that become due within at least the next twelve months from the date of this Prospectus, regardless of the completion of the Offering.

9. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

You should read the following discussion together with our Consolidated Financial Statements, including the related notes, that you will find beginning on page F-1. For a description of our Consolidated Financial Statements, see "2.8 Presentation of Financial and Other Information."

*The financial data contained in the following tables is extracted or derived from the audited consolidated financial statements of the Company as of and for the fiscal year ended December 31, 2020 (the "**Fiscal Year 2020**"), the audited consolidated financial statements of the Company as of and for the fiscal year ended December 31, 2019 (the "**Fiscal Year 2019**") and the audited consolidated financial statements of the Company as of and for the fiscal year ended December 31, 2018 (the "**Fiscal Year 2018**") as well as the unaudited condensed consolidated interim financial statements of the Company as of and for the six-month period ended June 30, 2021 (the "**6M 2021**"), including comparative figures as of and for the six-month period ended June 30, 2020 (the "**6M 2020**").*

Where financial data in the following tables is labeled "audited," this means that it has been extracted from the audited financial statements mentioned above. The label "unaudited" is used in the following tables to indicate financial data that has not been taken from the audited financial statements mentioned above, but either from the unaudited condensed consolidated interim financial statements mentioned above or from accounting records or internal management reporting systems or was calculated based on figures from the above-mentioned sources. All of the financial data presented in the text and tables below are shown in millions of euro (in € million), except as otherwise stated. In order to ensure that figures given in the text and the tables sum up to the totals given, the numbers are commercially rounded to the nearest whole number or in some cases to such number that facilitates the summing up. The percentage changes that are stated in the text and the tables have been commercially rounded to one decimal point unless stated otherwise. Financial data presented in parentheses denotes the negative of such number presented. In respect of financial data set out in this Prospectus, a dash ("–") signifies that the relevant figure is not available, while a zero ("0.0") signifies that the relevant figure is available, but has been rounded to zero.

Our historical results are not necessarily indicative of the results that should be expected in the future. The following discussion contains forward-looking statements that reflect our plans, estimates and beliefs. Our actual results may differ materially from those discussed in these forward-looking statements. Factors that could cause or contribute to such differences include those discussed below and elsewhere in this prospectus, including under "1 Risk Factors" and "2.5 Forward-Looking Statements."

9.1 Overview

We are the leading airline group in Europe and one of the largest airline groups globally by total revenue (*source*: Statista, Leading Airline Groups in Europe in 2020, Company Information). The airlines that form the core of the Lufthansa Group have their home markets in Germany, Switzerland, Austria and Belgium. As an airline group, our core business is the operation of domestic, short-haul and long-haul scheduled passenger and cargo flights within a worldwide network, as well as the provision of logistical and technical services related to flight operations. In addition, we provide aircraft maintenance, repair and overhaul ("**MRO**") services and catering.

We divide our business into six business segments: (i) Network Airlines; (ii) Eurowings; (iii) Logistics; (iv) MRO; (v) Catering; and (vi) Additional Businesses and Group Functions.

Network Airlines	Mainly comprises Lufthansa German Airlines (including Eurowings Discover, regional partners and Miles & More), SWISS, Austrian Airlines, Brussels Airlines and Eurowings Discover.
Eurowings	Mainly comprises Eurowings and Eurowings Europe, providing point-to-point short-haul flights in Europe to price-sensitive and service-oriented customers.

Logistics	Mainly comprises Lufthansa Cargo Aktiengesellschaft (“ Lufthansa Cargo ”), and the Jettainer group, the time:matters group and the equity investment of 50% in the cargo airline Aerologic GmbH (“ AeroLogic ”).
MRO.....	Mainly comprises the Lufthansa Technik group, consisting of 35 plants offering technical aviation services worldwide and direct and indirect stakes in 62 companies.
Catering	Mainly comprises LSG Lufthansa Service Holding AG and its subsidiaries (together, the “ LSG Group ”) and, <i>inter alia</i> , the brand ‘LSG Sky Chefs.’ In December 2020, the sale of the European business operations of LSG Group was closed. The planned sale of the international operations is currently under review.
Additional Businesses and Group Functions	Mainly comprises the service and financial companies, particularly Lufthansa AirPlus Servicekarten GmbH (“ AirPlus ”), Lufthansa Aviation Training (“ LAT ”) and Lufthansa Systems, as well as the group functions of the Lufthansa Group.

We generated total revenue of €5,771 million and €13,589 million in the 6M 2021 and the Fiscal Year 2020, respectively, compared to €8,335 million and €36,424 million in the 6M 2020 and the Fiscal Year 2019, respectively, impacted significantly by the outbreak of SARS-CoV-2 first identified in December 2019 and its related disease (“**COVID-19**”).

The global spread of COVID-19 and the ensuing travel restrictions had a significantly negative impact on the demand for air travel. Passenger airline traffic temporarily came to an almost complete standstill worldwide. Air traffic on Asian routes was first affected in February 2020 after the outbreak of the virus in China. This trend progressed rapidly, with air traffic in Europe, the Middle East and North America increasingly impacted. While the summer months saw a slight recovery, with the number of infections rising significantly again, passenger numbers fell significantly again from September 2020 onwards. In June 2021, we offered 33,823 flights, compared to 11,291 and 105,220 in June 2020 and 2019, respectively.

The combination of new and re-imposed government travel restrictions and other counter-measures significantly limited our ability to resume operations. Notwithstanding the immediate effects of the COVID-19 pandemic, we believe air travel, in particular in the tourism sector, has strong long-term prospects.

9.2 Key Factors Affecting Our Results of Operations

Our results of operations have been, and will continue to be, affected by a number of events and actions, some of which are beyond our control. We believe that the specific factors discussed below have affected our results in the periods for which financial information is presented in this Prospectus and will continue to affect our results in the future.

9.2.1 Impact of COVID-19

The global spread of COVID-19 has created significant macroeconomic uncertainty, volatility and disruption. In response, many governments have implemented policies intended to stop or slow the further spread of the disease, such as travel restrictions and general restricted movement guidelines and lockdowns, and these measures may remain in place for a significant amount of time. The global spread of COVID-19 and the ensuing travel restrictions had a significant and unprecedented negative impact on the demand for air travel. Air traffic on Asian routes was first affected in February 2020 after the outbreak of the virus in China. This trend progressed rapidly in March 2020, with air traffic in Europe, the Middle East and North America being increasingly impacted. Global passenger traffic then came to a nearly complete standstill in April and May 2020. During this period, the connections on offer were mainly limited to return flights and domestic services, in addition to a few international connections.

The June to August 2020 summer months saw a slight recovery as a result of the gradual lifting of travel restrictions. However, from September 2020 onwards, the number of infections rose significantly again, and as a consequence, passenger numbers also decreased significantly again. This trend persisted for the rest of the Fiscal Year 2020 after widespread travel restrictions were re-established in Europe as well as other countries. The airlines reacted to this development by once again significantly reducing the capacity on offer.

Substantial travel restrictions were maintained in the first quarter of 2021. Due to, among others, increasing vaccination rates in regions such as North America and Europe, travel restrictions were gradually lifted in the summer of 2021, especially within the European Union.

The total number of revenue passenger-kilometers (“**RPK**”) sold worldwide decreased by 66% in the year 2020 according to figures published by the International Air Transport Association (“**IATA**”) (2019: increase of 4%) (*source*: IATA, Air Passenger Market Analysis, December 2020 and IATA, Annual Review 2020). Airlines from the Middle East reported the largest decline in RPK at 72% (*source*: IATA, Air Passenger Market Analysis, December 2020). In the same period, airlines from Europe posted a 70% decline in RPK (*source*: IATA, Air Passenger Market Analysis, December 2020). According to the Federal Association of the German Aviation Industry (“**BDL**”), sales for airlines in Germany fell overall by 76% in 2020 compared to 2019 (*source*: BDL, Bericht zur Lage der Branche 2020). Average yields in global passenger traffic fell by 8.7% in 2020 (2019: decline by 3.7%) (*source*: IATA, Industry Statistics, April 2021). During the 6M 2021, the global passenger business improved slightly due to increasing vaccination rates, decreasing infection rates and the gradual lifting of travel restrictions resulting in an increase of RPKs by 193% compared to 6M 2020 (*source*: IATA, Air Passenger Market Analysis, June 2021).

For the Fiscal Year 2020, our capacity (measured by available seat-kilometers (“**ASK**”)) was 69% below the level of our capacity for the Fiscal Year 2019. Additionally, our capacity amounted to 21% in the first quarter of 2021 compared to the first quarter of 2019 and to 29% in the second quarter of 2021 compared to and the second quarter in 2019. To mitigate the effects of the COVID-19 pandemic, we initiated extensive measures over the course of the Fiscal Year 2020 to reduce costs and investment spending and to maintain liquidity. Among other things, a large part of our fleet was temporarily decommissioned and short-time working was introduced for a majority of the employees. Crisis agreements reached with the trade unions further reduced staff costs. Other key cost-cutting measures included terminating leases where the lessor provides the aircraft and crew (the “**Wet Leases**”), postponing maintenance and servicing work on the fleet that was not safety-relevant, renegotiating supply contracts, reducing marketing activities and deferring projects that are not crucial for business. See “9.2.7 “*ReNew*” Restructuring Program” below.

The global market for airfreight declined less significantly than passenger transport in the year 2020. Following the global spread of COVID-19, demand for airfreight initially declined and reached its low point in April 2020 (*source*: IATA, Air Cargo Market Analysis, April 2020). Starting in May 2020, market demand improved significantly again (*source*: IATA, Air Cargo Market Analysis, May 2020). However, despite an increase in yields, the freight capacity on offer did not recover to the same degree, as a large proportion of the belly capacities of passenger aircraft were not available for airfreight due to the greatly reduced passenger flows. According to IATA, global airfreight volumes in 2020 fell by 11% (2019: decline by 3%) (*source*: IATA, Air Cargo Market Analysis, December 2020 and Air Freight Market Analysis, December 2019). Our Logistics business segment, however, recorded a significant increase in yields driven by general capacity reductions and thereby, to a large extent, benefited from the increase in demand for cargo transport as a result of the COVID-19 pandemic. This positive trend continued in the 6M 2021. Revenue tonne-kilometres increased worldwide by 24% compared to the first six-month of 2020, thereby exceeding the pre-COVID-19 levels by 8% (*source*: IATA, Air Cargo Market Analysis, June 2021).

In addition, our MRO and catering business segments were also impacted by the COVID-19 pandemic. Demand for MRO services fell significantly in the Fiscal Year 2020 as the market environment suffered from a significant decline in capacity utilization, especially in engine maintenance and in the components business. With the crisis-related decline in passenger numbers in air travel, particularly in the long-haul business, demand for in-flight catering services also decreased. This was compounded by service restrictions related to hygiene concepts that minimize contact between crews and passengers. While both business segments continued to be impacted by the COVID-19 pandemic in the 6M 2021, the gradual recovery of the passenger business resulted in a slowly increasing demand for MRO and catering services.

9.2.2 Developments in the European and Global Airline Industries

The airline passenger and air freight industries are highly sensitive to general economic conditions. Because a substantial portion of business and leisure travel is discretionary, the airline industry tends to experience

adverse financial results during economic downturns. Changing economic cycles have an impact on national income, unemployment rates as well as consumption and are thereby affecting touristic supply and demand. During a recession, consumers are more likely to reduce spending for non-essential products such as leisure travel, leading suppliers to lower their prices, which in turn has a negative impact on profitability.

The dynamics between supply and demand are not only influenced by macro-economic factors, but also by market-specific conditions, such as the competitive forces. The European airline market is a particular example of a highly competitive market, which is – especially through the entrance of low-cost carriers such as easyJet, Ryanair and WizzAir – characterized by overcapacities and thus high pressure on margins, even during economic upward cycles.

See also “*11 Markets and Competition*” for more information on the European and global airline industries.

9.2.3 Foreign Currency Translation and Exchange Rate Effects

As an international airline group, we generate income in numerous non-euro currencies, whereas we prepare our consolidated financial statements in euro. In preparing these statements, we translate the book values of assets, liabilities, income and expenses that are incurred in other currencies into euro amounts.

Consequently, our results have been subject to currency translation risks from fluctuations of non-euro currencies against the euro during the period under review. In addition, our results have been affected by foreign currency exchange transaction gains and losses. For example, in the Fiscal Year 2020, the average exchange rate against both the U.S. dollar and the Chinese renminbi was 2% higher than in the Fiscal Year 2019. During the same period, the euro also gained 1% against the pound sterling and the euro remained almost unchanged against the Japanese yen, whereas it fell by 4% year-on-year against the Swiss franc.

We have a foreign currency exposure surplus in most non-euro currencies in which we earn income, which means that revenues denominated in these currencies exceed the costs denominated in these currencies that were incurred in the relevant period. For U.S. dollars, the Lufthansa Group is in a net payer position as regards currency risks from its operating business, since fuel payments are U.S. dollar-denominated. We hedge part or all of our expected exposure to fluctuations in currency exchange rates. For a description of our foreign currency hedging policy, see “*9.13 Quantitative and Qualitative Disclosures about Market Risks.*”

9.2.4 Fuel Cost Fluctuations

Jet fuel cost is a significant variable that has substantially affected our results of operations in the period under review. Jet fuel prices have historically fluctuated widely, and are likely to do so in the future as a result of the level of economic activity, the rate of economic growth, political events, weather conditions, refinery outages or maintenance and the coordinated pricing decisions of producer cartels, such as the Organization of Petroleum Exporting Countries, as well as speculative trading activity regarding the price of oil.

The oil price declined and was very volatile over the course of 2020 with prices ranging between USD 19.33/barrel and USD 68.91/barrel. As of December 31, 2020, a barrel of Brent Crude was at USD 51.80 (year-end 2019: USD 66.00/barrel). The jet fuel crack, *i.e.*, the price difference between crude oil and kerosene, moved between USD -8.32/barrel and USD 16.36/barrel in 2020. On average over the year 2020, it traded at USD 2.56/barrel and thus 84% lower than in the previous year 2019.

We hedge a significant proportion of these costs in advance by using financial derivatives. To the extent that we do not adequately or successfully hedge our fuel price exposure, a significant change in fuel prices could significantly affect our results of operations. To the extent that we are unable to pass on jet fuel cost increases to customers or, where such increases occur subsequent to the fixing of a selling price to a customer, we do not hedge this exposure effectively, an increase in costs will have a negative impact on our results of operations. Fuel price hedging was suspended over large periods of 2020, and, following the outbreak of the COVID-19 pandemic in March 2020 and its impact on the predictability of fuel demand, no further fuel price hedges were entered into. As a result of the capacity adjustments, the fuel hedging concluded was in part not matched by any underlying transactions.

We continuously analyze commodity price risks. Based on such analysis, the Executive Board, in March 2021, considered the fuel price risk in the current business environment as significant and chose to re-initiate the hedging program with a reduced target hedge ratio of 65%. As a result of the increase of fuel prices during the fourth quarter of the Fiscal Year 2020 and the 6M 2021, the market value of existing hedges is no longer negative and banks have re-initiated their trading lines for fuel derivatives.

9.2.5 High Fixed Costs

The airline industry is characterized by high levels of fixed costs. A significant portion of our air transport expenses, including certain staff costs, depreciation, maintenance, overhaul, aviation fuel expenses, aircraft handling fees and traffic control charges, finance costs and operating lease payments vary minimally based on passenger and cargo load factors. Thus, changes in our operating expenses may not correspond and have not corresponded to changes in our revenue, which may have and has had a material effect on our results of operations. In addition, a significant portion of our expenses, including depreciation, finance costs, operating lease payments and some staff costs, do not vary based on the number of flights flown or aircraft utilization rates. Therefore, our results of operations have been and may be materially affected by relatively minor changes in aircraft utilization rates. This high operating leverage further exacerbates revenue fluctuations prevailing in the airline industry generally.

9.2.6 Seasonality

Historically, we have earned most of our profits in the European summer months. In fiscal years unaffected by major disruptions, such as the COVID-19 pandemic, the level of demand for our services has fluctuated over the course of the year, causing our results to fluctuate accordingly.

Demand has historically been highest in the summer season from May through October and lowest in the winter season from November through April (except for the days around Christmas, New Year and Easter). At the same time, we incur a significant proportion of our expenses more evenly throughout the year. Our profitability therefore fluctuates during the year, with the majority of profits generated in the summer season. As a result of this seasonality, disruptions to our business operations that occur during the summer season could have particularly adverse effects. Cash flows are similarly seasonal as cash inflows tend to be higher in the summer. Due to the effects of the COVID-19 pandemic and travel uncertainty, customers are booking with less lead time, which has an impact on cash flow.

The COVID-19 pandemic has significantly disrupted our historical seasonal pattern. As of the date of this Prospectus, we can neither predict whether this historical pattern will return after the effects of the pandemic have subsided nor, if it does, when our financial results will resume this pattern.

9.2.7 “ReNew” Restructuring Program

We believe that successfully coping with the COVID-19 pandemic is a prerequisite for our strategic development. Accordingly, we have implemented a comprehensive restructuring program titled “ReNew,” aimed at adapting to the new market conditions. ReNew combines the restructuring programs at our airlines and services companies. On the basis of this fundamental restructuring path, we intend to re-introduce and improve profitability and cash flows, refinance and repay funding received in connection with the Stabilization Package in the near- to medium term whilst preserving our goal of financial stability, and to ultimately achieve long-term strategic objectives.

We divide the “ReNew” program into four modules: (i) ReStructure, (ii) ReOrg, (iii) ReFocus and (iv) RePay.

- **ReStructure** combines the decentralized restructuring programs for the airlines and aviation services. Its overarching goal is to catch up with the leading competitors in the respective markets in terms of profitability. The main focus is on the implementation of a much smaller and more efficient production structure in all of our group companies. This entails a significant reduction of the fleet. In the Fiscal Year 2020, we signed crisis agreements with all the main trade unions of the Lufthansa Group in Germany and initiated measures to streamline our organizational structure. As of June 30, 2021, the number of total employees (by headcount) decreased by 22,000, or 17%, of the total workforce, including 5,000 employees (by headcount) in Germany compared to December 31, 2019

(excluding a workforce reduction by approximately 8,600 employees (headcount) in connection with the divestment of LSG Group's European business). We implemented short-term measures in 2020 to reduce staff costs and to counteract the effects of the pandemic. In addition, crisis agreements were concluded for the majority of the other groups of our employees. On August 5, 2021, Lufthansa Group has communicated that it already implemented measures to reduce personnel costs by €1.1 billion. Taking into account additional measures such as, in particular, savings of still to be negotiated amended tariff agreements as well as further staff reduction, we expect to achieve a total of €1.8 billion structural annual personnel cost savings from 2024 onwards.

- **ReOrg** serves to implement a more efficient organizational structure, including the separation of Lufthansa German Airlines functions from Lufthansa Group functions as well as restructuring the group functions and matrix process organization. This will entail, among other things, the reduction of around 1,000 administrative full-time employees in group functions and the reduction of management positions across the Lufthansa Group by at least 20% compared with the period before the crisis in connection with ReStructure. As a part of this module, the Executive Board and the management boards of certain of the Company's subsidiaries were downsized.
- **ReFocus** aims to focus the Lufthansa Group even more on its core airline business, so that the portfolio of operating segments is made up predominantly of synergistic units. This means that the sales process for the remaining non-European business of the LSG Group will be resumed. Disposals of other non-airline companies are also under review. In December 2020, we closed the sale of the European business of the LSG Group. See "9.6.3 Sale of LSG Group's European Business."
- **RePay** aims to plan, coordinate and implement the refinancing and prompt repayment of all the funding from the government support packages and existing financial liabilities on the best possible terms. Suitable financial instruments for this purpose are being evaluated. In addition, suitable measures are to be taken to strengthen the balance sheet of the Lufthansa Group and optimize its gearing in order to obtain an investment grade rating again.

9.3 Consolidated Income Statement Items

The following is a description of certain line items in our consolidated income statement.

Traffic revenue comprises revenue generated solely from flight operations for passengers and cargo as well as related ancillary services.

Other revenue comprises revenue generated mainly in our business segments MRO, Catering and Additional Businesses and Group Functions.

Changes in inventories and work performed by entity and capitalized comprises the increase or decrease in finished goods and work in progress as well as other internally produced and capitalized assets, which mainly relates to aircraft and engine overhauls.

Other operating income comprises foreign exchange gains, income from the reversal of provisions and accruals, subsidies, compensation received for damages, reversal of write-downs on receivables, rental income, income from the reversal of impairment losses on fixed assets, services provided by the Lufthansa Group, income from the disposal of non-current available-for-sale financial assets, income from the disposal of non-current assets, income from staff secondment, commissions income, income from operating lease aircraft and miscellaneous other operating income.

Cost of materials and services comprises (i) costs of raw materials, in particular aircraft fuel lubricants, consumables and supplies and purchased goods, and (ii) costs of purchased services, including fees and charges, external MRO services, charter expenses, external IT services, in-flight services, flight irregularities and other services.

Staff costs comprises wages and salaries, social security contributions and expenses for pension plans and other employee benefits.

Depreciation, amortization and impairment comprises the depreciation of aircraft and other tangible assets and amortization of intangible assets as well as the impairment of goodwill, other intangible assets, aircraft and reserve engines, other tangibles assets, right-of-use assets and financial assets.

Other operating expenses comprises rental and maintenance expenses, foreign exchange losses, staff-related expenses, write-downs on receivables, auditing, consulting and legal expenses, other services, advertising and sales promotions, expenses for computerized distribution systems, sales commission paid to agencies, communications costs, commissions for credit cards, other taxes, insurance premiums for flight operations, losses on disposal of non-current assets and miscellaneous other operating expenses.

Result of equity investments accounted for using the equity method comprises the result of joint ventures and associated companies accounted for using the equity method, but does not include any impairment losses on investments accounted for using the equity method.

Result of other equity investments comprises dividends from other joint ventures and other associated companies, income from profit transfer agreements, expenses from loss transfer agreements and dividends from other equity investments.

Interest income comprises income from other securities and non-current financial loans as well as other interest and similar income.

Interest expenses comprises interest expenses on pension obligations, other provisions and other similar expenses.

Other financial items comprises the result of fair value hedges (the change in time value of hedged transactions), the result of fair value hedges (change in time value of hedging instruments), the ineffective portion of derivatives used as cash flow hedges, the result of derivatives held for trading classified as at fair value through profit or loss, the result of measuring securities classified as at fair value through profit or loss and exchange rates effects from financial liabilities.

Income taxes comprises current and deferred income taxes.

9.4 Business Segments

We divide our business into six business segments: (i) Network Airlines; (ii) Eurowings; (iii) Logistics; (iv) MRO; (v) Catering; and (vi) Additional Businesses and Group Functions.

9.4.1 Network Airlines

The Network Airlines business segment mainly comprises Lufthansa German Airlines (regional partners and Miles & More), SWISS, Austrian Airlines, Brussels Airlines and Eurowings Discover. Lufthansa German Airlines is the biggest German airline in the segment, with hubs in Frankfurt and Munich. Lufthansa CityLine and Air Dolomiti regional airlines are also part of Lufthansa German Airlines. Additionally, the business segment includes the operations of Eurowings Discover and Germanwings, whereas Germanwings' passenger flight operations have been discontinued in 2020.

9.4.2 Eurowings

The Eurowings business segment mainly comprises Eurowings and Eurowings Europe, providing point-to-point short-haul flights in Europe to price-sensitive and service-oriented customers.

The structure of our business segments was changed at the start of the Fiscal Year 2020. Brussels Airlines and the Eurowings long-haul business have since been managed by the Network Airlines group and have therefore been allocated to the Network Airlines business segment.

9.4.3 Logistics

In addition to Lufthansa Cargo, the Logistics business segment includes the Jettainer group, the time:matters group, which specializes in particularly urgent consignments, and the equity investment of 50% in the cargo airline AeroLogic. Lufthansa Cargo also has equity investments in various handling companies.

9.4.4 MRO

The MRO business segment mainly comprises Lufthansa Technik, consisting of 35 plants offering technical aviation services worldwide and direct and indirect stakes in 62 companies. As of June 30, 2021, the Lufthansa Technik group served more than 800 customers worldwide, including airlines, original equipment manufacturers (“OEMs”), aircraft leasing companies and operators of VIP jets.

9.4.5 Catering

The Catering business segment is made up of the LSG Group, which consists of more than 150 companies with catering facilities and numerous partnerships serving more than 135 airports in 44 countries as of June 30, 2021. The business segment includes, *inter alia*, the brands ‘LSG Sky Chefs’ and ‘Retail inMotion.’

In December 2020, the Lufthansa Group sold the European catering business of the LSG Group (including the divisions Spiriant, Ringeltaube and Lounge as well as the convenience retail brand Evertaste). The LSG Group continues to be active in the European market with the ‘Retail inMotion’ brand and its pan-European customer contracts.

9.4.6 Additional Businesses and Group Functions

The Additional Businesses and Group Functions business segment comprises the service and financial companies, particularly AirPlus, Lufthansa Aviation Training and Lufthansa Systems, as well as the group functions of the Lufthansa Group. The Additional Business and Group Functions business segment is not a reporting segment according to IFRS 8 (Operating Segments). Business activities not allocated to a reportable segment are included under this business segment.

See “12 Business” for additional information on our business segments.

9.5 Alternative Performance Measures and Other Operating Metrics

9.5.1 Alternative Performance Measures

We present certain financial measures that are not required by, or not presented in accordance with, IFRS or the German Commercial Code (*Handelsgesetzbuch*) or any other generally accepted accounting principle as alternative performance measures in accordance with the Commission Delegated Regulation (EU) 2016/301 and the guidelines on alternative performance measures published by the European Securities and Markets Authority on October 5, 2015. In this Prospectus, we present APMs as (i) supplemental information because they are used by the Executive Board to measure operating performance, including in presentations to the Executive Board or the Supervisory Board, and as a basis for strategic planning and forecasting, and (ii) they represent measures that we believe are widely used by certain investors, securities analysts and other parties as supplemental measures of operating and financial performance. These financial measures may enhance investors’ understanding of our financial performance by excluding items that are not classified as part of our ongoing operations.

However, the APMs as used in this Prospectus are not defined by IFRS or any other accepted accounting principles, and investors should not consider such items as an alternative to the historical financial results or other indicators of our performance, assets or liabilities based on IFRS measures. In particular, they should not be considered as alternatives to our net profit/loss as an indicator of our profitability, or as alternatives to cash flows from operating activities as an indicator of our financial strength. We derive the APMs used in this Prospectus from our internal accounting records or internal management reporting systems or from calculations based on figures from the Consolidated Financial Statements. The APMs, as defined by us, have not been audited (except for Adjusted EBIT) and may not be comparable to similarly titled measures as presented by other companies due to differences in the way our APMs are calculated. Even though the APMs are used by the Executive Board to assess ongoing operating performance and indebtedness, and though these types of measures are commonly used by investors, they have important limitations as analytical tools and investors should not consider them in isolation, or as substitutes for, the analysis of our results of operations, financial position and cash flows as reported under IFRS.

We have defined the APMs as follows:

- “**EBIT**” is defined as profit/loss from operating activities plus result of equity investments accounted for using the equity method and result of other equity investments.
- “**Adjusted EBIT**” is defined as EBIT adjusted for impairment losses/gains, effects from pension provisions and results of disposal of assets.
- “**Adjusted EBITDA**” is defined as Adjusted EBIT plus depreciation and amortization. Depreciation and amortization which excludes impairments.
- “**Adjusted EBIT Margin**” is defined as Adjusted EBIT divided by total revenue.
- “**Adjusted ROCE**” is defined as Adjusted EBIT plus interest on liquidity less 25% taxes divided by the average capital employed. The capital employed results from total assets adjusted for non-operating items (deferred taxes, derivative financial instruments) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

The following table provides an overview of the APMs for the Lufthansa Group and by business segment for the periods presented:

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019 ⁽¹⁾	2018 ⁽¹⁾
	(unaudited) (in € million, unless otherwise indicated)		(unaudited, unless otherwise indicated) (in € million, unless otherwise indicated)		
Total Lufthansa Group⁽²⁾					
Adjusted EBITDA ⁽³⁾	(970)	(1,578)	(2,890)	4,718	5,016
Adjusted EBIT	(2,095)	(2,899)	(5,451) ^(*)	2,026 ^(*)	2,836 ^(*)
Adjusted EBIT Margin in %	(36.3)	(34.8)	(40.1)	5.6	8.0
Adjusted ROCE in % ⁽⁴⁾	—	—	(16.7)	6.6	10.6
Network Airlines⁽²⁾					
Adjusted EBITDA ⁽³⁾	(1,664)	(1,460)	(2,860)	3,703	3,926
Adjusted EBIT	(2,450)	(2,416)	(4,674) ^(*)	1,776 ^(*)	2,429 ^(*)
Adjusted EBIT Margin in %	(107.1)	(53.3)	(70.6)	7.1	10.8
Eurowings					
Adjusted EBITDA ⁽³⁾	(151)	(254)	(504)	111	141
Adjusted EBIT	(252)	(358)	(703) ^(*)	(122) ^(*)	(231) ^(*)
Adjusted EBIT Margin in %	(159.5)	(95.0)	(117.6)	(5.3)	(5.6)
Logistics					
Adjusted EBITDA ⁽³⁾	710	355	950	161	372
Adjusted EBIT	640	277	772 ^(*)	1 ^(*)	268 ^(*)
Adjusted EBIT Margin in %	38.3	21.0	28.0	0.0	9.9
MRO					
Adjusted EBITDA ⁽³⁾	191	(22)	(186)	654	572
Adjusted EBIT	102	(122)	(383) ^(*)	463 ^(*)	446 ^(*)
Adjusted EBIT Margin in %	5.9	(5.4)	(10.2)	7.0	7.3
Catering					
Adjusted EBITDA ⁽³⁾	57	(134)	(167)	247	181
Adjusted EBIT	17	(195)	(284) ^(*)	128 ^(*)	115 ^(*)
Adjusted EBIT Margin in %	3.8	(24.0)	(21.8)	3.8	3.6
Additional Business and Group Functions					
Adjusted EBITDA ⁽³⁾	(99)	(64)	(190)	(120)	(157)
Adjusted EBIT	(158)	(122)	(314) ^(*)	(227) ^(*)	(209) ^(*)

(*) Audited.

- (1) The figures for the Fiscal Year 2019 were adjusted retrospectively to account for the changes in the segment allocation of Brussels Airlines, Germanwings, the long-haul operation of Eurowings and the transfer of maintenance activities from the MRO segment to the Network Airlines Segment. The figures for the Fiscal Year 2018 were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.
- (2) A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects. In addition, a translation of the figures from Lufthansa German Airlines, SWISS and Austrian Airlines to the Network Airlines business is also only possible when considering consolidation effects. As of January 1, 2020, Brussels Airlines is considered as a part of the Network Airlines business segment.
- (3) The reconciliations of EBIT to Adjusted EBIT and of Adjusted EBIT to Adjusted EBITDA on group level as well as the reconciliations of Adjusted EBIT to EBIT and of Adjusted EBIT to Adjusted EBITDA of the business segments as presented in the segment reporting of the Company's consolidated financial statements are shown in the table below.

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019 ^(a)	2018 ^(a)
	(unaudited) (in € million, unless otherwise indicated)		(audited, unless otherwise indicated) (in € million, unless otherwise indicated)		
Total Lufthansa Group^(b)					
Total revenue	5,771	8,335	13,589	36,424	35,542
Changes in inventories and work performed by entity and capitalized	49	158	175	685	531
Other operating income.....	655	797	1,897	1,889	1,818
Cost of materials and services.....	(3,204)	(5,127)	(8,453)	(19,827)	(18,367)
Staff costs.....	(2,907)	(3,620)	(6,436)	(9,121)	(8,811)
Depreciation, amortization and impairment.....	(1,135)	(1,783)	(4,389)	(2,776)	(2,205)
Other operating expenses	(1,279)	(2,048)	(3,472)	(5,585)	(5,708)
Profit/loss from operating activities	(2,050)	(3,288)	(7,089)	1,689	2,800
Result of equity investments accounted for using the equity method.....	(71)	(184)	(276)	88	114
Result of other equity investments.....	7	4	12	80	60
EBIT.....	(2,114)	(3,468)	(7,353)	1,857	2,974
Impairment losses/gains.....	6	552	1,870	139	9
Effects from pension provisions	(3)	8	31	10	(113)
Results of disposal of assets.....	16	9	1	20	(34)
Adjusted EBIT	(2,095)	(2,899)	(5,451)	2,026	2,836
Adjusted EBIT Margin (in %)^(c).....	(36.3)	(34.8)	(40.1)	5.6	8.0
Depreciation and amortization.....	1,125	1,321	2,561	2,692	2,180
Adjusted EBITDA^(c)	(970)	(1,578)	(2,890)	4,718	5,016
Network Airlines^(b)					
Total revenue	2,288	4,531	6,622	25,078	22,549
Other operating income.....	346	451	994	1,014	699
Total operating income^(d).....	2,634	4,982	7,616	26,092	23,248
Operating expenses	5,062	(7,377)	(12,258)	(24,362)	(20,854)
Results of equity investments ^(e)	(22)	(21)	(32)	46	35
Adjusted EBIT	(2,450)	(2,416)	(4,674)	1,776	2,429
Impairment losses/gains.....	(8)	(268)	(1,398)	(26)	1
Effects from pension provisions	3	—	(27)	(3)	110
Results of disposal of assets.....	4	(2)	—	(23)	9
EBIT.....	(2,451)	(2,686)	(6,099)	1,724	2,549
Adjusted EBIT Margin (in %)^(c).....	(107.1)	(53.3)	(70.6)	7.1	10.8
Adjusted EBIT	(2,450)	(2,416)	(4,674)	1,776	2,429
Depreciation and amortization.....	786	956	1,814	1,927	1,497
Adjusted EBITDA^(c)	(1,664)	(1,460)	(2,860)	3,703	3,926
Eurowings					
Total revenue	158	377	598	2,311	4,098
Other operating income.....	40	75	147	176	290

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019 ^(a)	2018 ^(a)
	(unaudited) (in € million, unless otherwise indicated)		(audited, unless otherwise indicated) (in € million, unless otherwise indicated)		
Total operating income^(d)	198	452	745	2,487	4,388
Operating expenses	409	(746)	(1,346)	(2,633)	(4,643)
Results of equity investments ^(e)	(41)	(64)	(102)	24	24
Adjusted EBIT	(252)	(358)	(703)	(122)	(231)
Impairment losses/gains.....	(2)	(73)	(97)	—	—
Effects from pension provisions	—	—	—	—	—
Results of disposal of assets.....	3	(1)	(2)	(4)	—
EBIT	(251)	(432)	(802)	(126)	(231)
Adjusted EBIT Margin (in %)^(c)	(159.5)	(95.0)	(117.6)	(5.3)	(5.6)
Adjusted EBIT	(252)	(358)	(703)	(122)	(231)
Depreciation and amortization	101	104	199	233	372
Adjusted EBITDA^(c)	(151)	(254)	(504)	111	141
Logistics					
Total revenue	1,671	1,320	2,757	2,478	2,713
Other operating income.....	32	31	69	103	57
Total operating income^(d)	1,703	1,351	2,826	2,581	2,770
Operating expenses	1,073	(1,089)	(2,082)	(2,621)	(2,538)
Results of equity investments ^(e)	10	15	28	41	36
Adjusted EBIT	640	277	772	1	268
Impairment losses/gains.....	1	(19)	(55)	(38)	(4)
Effects from pension provisions	—	—	—	(1)	—
Results of disposal of assets.....	2	—	—	5	(1)
EBIT	643	258	717	(33)	263
Adjusted EBIT Margin (in %)^(c)	38.3	21.0	28.0	0.0	9.9
Adjusted EBIT	640	277	772	1	268
Depreciation and amortization	70	78	178	160	104
Adjusted EBITDA^(c)	710	355	950	161	372
MRO					
Total revenue	1,717	2,280	3,747	6,572	6,105
Other operating income.....	174	184	437	256	270
Total operating income^(d)	1,891	2,464	4,184	6,828	6,375
Operating expenses	1,779	(2,554)	(4,502)	(6,425)	(5,936)
Results of equity investments ^(e)	(10)	(32)	(65)	60	7
Adjusted EBIT	102	(122)	(383)	463	446
Impairment losses/gains.....	1	(65)	(108)	12	3
Effects from pension provisions	—	(2)	(3)	(2)	—
Results of disposal of assets.....	(2)	(4)	(14)	(1)	(4)
EBIT	101	(193)	(508)	472	445
Adjusted EBIT Margin (in %)^(c)	5.9	(5.4)	(10.2)	7.0	7.3
Adjusted EBIT	102	(122)	(383)	463	446
Depreciation and amortization	89	(100)	197	191	126
Adjusted EBITDA^(c)	191	(22)	(186)	654	572
Catering					
Total revenue	447	814	1,305	3,360	3,217
Other operating income.....	139	33	143	95	77
Total operating income^(d)	586	847	1,448	3,455	3,294
Operating expenses	561	(1,029)	(1,698)	(3,355)	(3,208)
Results of equity investments ^(e)	(8)	(13)	(34)	28	29
Adjusted EBIT	17	(195)	(284)	128	115
Impairment losses/gains.....	1	(111)	(201)	(42)	(5)

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019 ^(a)	2018 ^(a)
	(unaudited) (in € million, unless otherwise indicated)		(audited, unless otherwise indicated) (in € million, unless otherwise indicated)		
Effects from pension provisions	—	—	1	(1)	—
Results of disposal of assets.....	(23)	—	53	13	—
EBIT.....	(5)	(306)	(431)	98	110
Adjusted EBIT Margin (in %)^(c).....	3.8	(24.0)	(21.8)	3.8	3.6
Adjusted EBIT	17	(195)	(284)	128	115
Depreciation and amortization	40	61	117	119	66
Adjusted EBITDA^(c).....	57	(134)	(167)	247	181
Additional Businesses and Group Functions					
Total revenue	215	249	470	793	783
Other operating income.....	1,023	920	1,571	1,939	1,879
Total operating income^(d).....	1,238	1,169	2,041	2,732	2,662
Operating expenses	1,402	(1,288)	(2,361)	(2,971)	(2,914)
Results of equity investments ^(e)	6	(3)	6	12	43
Adjusted EBIT	(158)	(122)	(314)	(227)	(209)
Impairment losses/gains.....	—	1	(43)	7	(5)
Effects from pension provisions	1	(6)	(2)	(4)	2
Results of disposal of assets.....	(2)	(2)	(4)	3	30
EBIT	(159)	(129)	(363)	(221)	(182)
Adjusted EBIT	(158)	(122)	(314)	(227)	(209)
Depreciation and amortization	59	58	124	107	52
Adjusted EBITDA^(c).....	(99)	(64)	(190)	(120)	(157)

(a) The figures for the Fiscal Year 2019 were adjusted retrospectively to account for the changes in the segment allocation of Brussels Airlines, Germanwings, the long-haul operation of Eurowings and the transfer of maintenance activities from the MRO segment to the Network Airlines Segment. The figures for the Fiscal Year 2018 were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.

(b) A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects. In addition, a translation of the figures from Lufthansa German Airlines, SWISS and Austrian Airlines to the Network Airlines business is also only possible when considering consolidation effects. As of January 1, 2020, Brussels Airlines is considered as a part of the Network Airlines business segment.

(c) Unaudited.

(d) Labelled as “operating income” in the Unaudited Condensed Consolidated Interim Financial Statements.

(e) Does not include any impairment losses on investments accounted for using the equity method.

(4) The following table shows the reconciliations of Adjusted ROCE on a group level.

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018
	(unaudited) (in € million)		(unaudited, unless otherwise indicated) (in € million, unless otherwise indicated)		
Total Lufthansa Group					
Adjusted EBIT.....	(2,095)	(2,899)	(5,451)^(*)	2,026^(*)	2,836^(*)
Interest on liquidity	—	—	84	79	68
Taxes ^(a)	—	—	1,817	(484)	(761)
Cost of Capital ^(b)	—	—	(1,011)	(1,007)	(860)
EACC^(c).....	—	—	(6,463)	445	1,422
Adjusted ROCE^(d) (in %).....	—	—	(16.7)	6.6	10.6

(*) Audited.

- (a) Assumption of 25% of EBIT plus interest on liquidity.
- (b) The average return required on the capital employed multiplied by average capital employed. The return on capital is calculated using the weighted average return for both debt and equity.
- (c) Earnings after cost of capital. Calculated as EBIT plus interest on liquidity minus 25% taxes on EBIT and cost of capital.
- (d) Calculated by dividing the sum of Adjusted EBIT plus interest on liquidity minus 25% taxes on Adjusted EBIT and average capital employed.

9.5.2 Other Operating Metrics

The following table provides an overview of the other operating metrics for the Lufthansa Group and by business segment for the periods presented. We believe that these metrics are important performance indicators for the Lufthansa Group.

	For the six-month period ended June 30,		For the year ended December 31,		
	2021 (unaudited)	2020 (unaudited)	2020 (unaudited)	2019 ⁽¹⁾ (unaudited)	2018 ⁽¹⁾ (unaudited)
Total Lufthansa Group⁽²⁾					
Passengers (in millions).....	10.0	23.4	36.4	145.3	141.9
Passenger load factor (in %) ⁽³⁾	48.9	72.2	63.2	82.6	81.5
Available seat-kilometers (in millions) ⁽⁴⁾	44,171	68,604	109,828	358,803	349,391
Revenue seat-kilometers (in millions) ⁽⁴⁾	21,616	49,512	69,462	296,217	284,639
Available cargo ton-kilometers (in millions) ⁽⁴⁾	5,381	5,464	10,591	17,379	16,349
Revenue cargo ton-kilometers (in millions) ⁽⁴⁾	4,074	3,595	7,373	10,664	10,896
Cargo load factor (in %) ⁽⁵⁾	75.7	65.8	69.6	61.4	66.6
Number of flights.....	120,435	229,934	390,900	1,187,728	1,163,565
Network Airlines⁽²⁾					
Passengers (in thousands).....	8,564	19,220	29,119	118,328	103,639
Passenger load factor (in %) ⁽³⁾	47.7	71.6	62.2	82.6	81.5
Available seat-kilometers (in millions) ⁽⁴⁾	41,314	63,173	99,568	326,420	284,642
Revenue seat-kilometers (in millions) ⁽⁴⁾	19,721	45,230	61,961	269,578	232,030
Number of flights.....	100,603	185,366	310,023	945,621	840,945
Unit revenue (in € cent) ⁽⁶⁾	5.9	7.5	7.2	7.8	8.1 ⁽⁷⁾
Unit cost excluding fuel (in € cent) ⁽⁶⁾	10.5	9.5	10.3	5.5	5.6 ⁽⁷⁾
Eurowings					
Passengers (in thousands).....	1,458	4,255	7,235	26,971	38,297
Passenger load factor (in %) ⁽³⁾	66.3	78.9	73.1	82.3	81.3
Available seat-kilometers (in millions) ⁽⁴⁾	2,857	5,431	10,260	32,383	64,748
Revenue seat-kilometers (in millions) ⁽⁴⁾	1,895	4,283	7,501	26,639	52,609
Number of flights.....	15,854	40,571	71,829	232,461	322,620
Unit revenue (in € cent) ⁽⁶⁾	6.1	7.9	6.5	7.5	6.8 ⁽⁷⁾
Unit cost excluding fuel (in € cent) ⁽⁶⁾	12.2	11.6	11.0	6.1	5.7 ⁽⁷⁾
Logistics					
Cargo load factor (in %) ⁽⁵⁾	74.3	65.9	69.1	61.3	65.9
Available cargo ton-kilometers (in millions) ⁽⁴⁾	4,683	4,738	9,350	14,507	13,555
Revenue cargo ton-kilometers (in millions) ⁽⁴⁾	3,481	3,122	6,461	8,899	8,934

- (1) The figures for the Fiscal Year 2019 were adjusted retrospectively to account for the changes in the segment allocation of Brussels Airlines, Germanwings, the long-haul operation of Eurowings and the transfer of maintenance activities from the MRO segment to the Network Airlines Segment. The figures for the Fiscal Year 2018 were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.
- (2) A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects. In addition, a translation of the figures from Lufthansa German Airlines, SWISS and Austrian Airlines to the Network Airlines business is also only possible when considering consolidation effects. As of January 1, 2020, Brussels Airlines is considered as a part of the Network Airlines business segment.
- (3) Measure of capacity utilization in per cent. The passenger load factor refers to passenger transportation.

- (4) Standard output units for air transport. An available seat-kilometer (ASK) denotes one offered seat flown for one kilometer. Revenue seat-kilometer corresponds to revenue passenger-kilometer (RPK), which denotes one paying passenger transported for one kilometer. A ton-kilometer offered (TKO) denotes the offered capacity equivalent of one ton of load (passengers and/or cargo) for one kilometer; a revenue ton-kilometer (RTK) denotes one ton of load (passengers and/or cargo) transported one kilometer.
- (5) Measure of capacity utilization in per cent. The cargo load factor expresses the ratio of capacity sold to available capacity. The cargo load factor refers to freight transport or total traffic.
- (6) Key performance indicator for air transport. Unit costs (CASK) denote the operating expenses divided by offered seat-kilometers. Unit revenue (RASK) denotes the revenue divided by offered seat-kilometers.
- (7) Without effects from reclassification of compensation payments for flight delays.

9.6 Factors Affecting the Comparability of Our Financial Results

Various events can limit the comparability of our financial results from period to period. During the period under review, these events related primarily to the COVID-19 pandemic, IFRS 16 and the sale of LSG Group's European business.

9.6.1 COVID-19 Pandemic

The effects of the COVID-19 pandemic limit the comparability of the financial statements presented in this Prospectus with each other, as well as the comparability of our historical financial statements with future financial statements. Our financial statements for the Fiscal Year 2020 and 6M 2021 were strongly influenced by factors and events that do not pertain to the other financial statements we present in this Prospectus. Overall, our total revenue fell by 62.7% in the Fiscal Year 2020 compared to the Fiscal Year 2019 and by 30.8% in the 6M 2021 compared to the 6M 2020 due to the COVID-19 pandemic.

9.6.2 IFRS 16

We have applied IFRS 16 (Leases) since January 1, 2019, using the modified retrospective approach. The comparative figures for the Fiscal Year 2018 were therefore not adjusted. In particular, our net assets, financial and earnings position have been affected by IFRS 16 (Leases). Our payment obligations from contracts previously classified as operating leases are now discounted using the incremental borrowing rate and recognized as lease liabilities. Right-of-use assets are recognized for the same amount.

See "General remarks" in the notes to our Audited Consolidated Financial Statements 2019 for more detailed information about the effects of IFRS 16 on our lease liabilities and on the financial position of the Lufthansa Group.

9.6.3 Sale of LSG Group's European Business

On December 2, 2020, we closed the sale of the European business of the LSG Group to gategroup Holding AG ("gategroup"). The entities sold accounted for revenue of €366 million from January 1, 2020, until their disposal on December 2, 2020. Approximately 7,500 employees left the Lufthansa Group when the disposal was completed. In addition, the disposal of the LSG Group's European business resulted in a decrease in cash and cash equivalents (proceeds from the disposal of consolidated equity investments less cash outflows) of €218 million.

9.7 Results of Operations

The following table provides an overview of our results of operations for the periods presented:

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018 ⁽¹⁾
	(unaudited) (in € million)		(audited) (in € million)		
Traffic revenue	3,637	5,641	9,078	28,136	27,801
Other revenue	2,134	2,694	4,511	8,288	7,741
Total revenue	5,771	8,335	13,589	36,424	35,542

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018 ⁽¹⁾
	(unaudited) (in € million)		(audited) (in € million)		
Changes in inventories and work performed by entity and capitalized.....	49	158	175	685	531
Other operating income.....	655	797	1,897	1,889	1,818
Cost of materials and services.....	(3,204)	(5,127)	(8,453)	(19,827)	(18,367)
Staff costs.....	(2,907)	(3,620)	(6,436)	(9,121)	(8,811)
Depreciation, amortization and impairment.....	(1,135)	(1,783)	(4,389)	(2,776)	(2,205)
Other operating expenses.....	(1,279)	(2,048)	(3,472)	(5,585)	(5,708)
Profit/loss from operating activities.....	(2,050)	(3,288)	(7,089)	1,689	2,800
Result of equity investments accounted for using the equity method.....	(71)	(184)	(276)	88	114
Result of other equity investments.....	7	4	12	80	60
Interest income.....	(2)	33	83	79	68
Interest expenses.....	(211)	(195)	(417)	(394)	(212)
Other financial items.....	93	(789)	(944)	318	(46)
Financial result.....	(184)	(1,131)	(1,542)	171	(16)
Profit/loss before income taxes.....	(2,234)	(4,419)	(8,631)	1,860	2,784
Income taxes.....	421	792	1,865	(615)	(588)
Profit/loss after income taxes.....	(1,813)	(3,627)	(6,766)	1,245	2,196

(1) The figures for prior periods were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.

9.7.1 Comparison of the Six-Month Periods Ended June 30, 2021, and June 30, 2020

(a) Total Revenue

Total revenue decreased by €2,564 million, or 31%, from €8,335 million in the 6M 2020 to €5,771 million in the 6M 2021 primarily due to the effects of the COVID-19 pandemic and the related restrictions applying to the Lufthansa Group, resulting in a decrease of our traffic revenue, which was primarily caused by reduced capacity and number of flights due to lower passenger demand as well as a decrease in our other revenue primarily caused by reduced flying hours across the industry resulting in a significant adverse impact on customer demand for MRO and Catering services.

The following table sets forth a breakdown of our total revenue by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects.

	For the six-month period ended June 30,		% Change
	2021	2020	
	(unaudited) (in € million)		
Network Airlines.....	2,288	4,531	(50)
Eurowings.....	158	377	(58)
Logistics.....	1,671	1,320	27
MRO.....	1,717	2,280	(25)
Catering.....	447	814	(45)
Additional Businesses and Group Functions.....	215	249	(14)

Total revenue in our Network Airlines business segment decreased by €2,243 million, or 50%, from €4,531 million in the 6M 2020 to €2,288 million in the 6M 2021 primarily due to a decrease in traffic revenue which

was caused by fewer capacity, reduced number of flights and lower sales due to travel restrictions and reduced customer demand in light of the COVID-19 pandemic.

Total revenue in our Eurowings business segment decreased by €219 million, or 58%, from €377 million in the 6M 2020 to €158 million in the 6M 2021 primarily due to reduced traffic revenue which was caused by travel restrictions and the resulting decrease in customer demand in light of the COVID-19 pandemic.

Total revenue in our Logistics business segment increased by €351 million, or 27%, from €1,320 million in the 6M 2020 to €1,671 million in the 6M 2021 primarily due to strong demand for airfreight capacity which was offset by reduced belly capacities on passenger aircraft, however resulting in higher yields than in the previous year.

Total revenue in our MRO business segment decreased by €563 million, or 25%, from €2,280 million in the 6M 2020 to €1,717 million in the 6M 2021 primarily due to reduced flying hours across the industry as well as the retirement and decommissioning of aircraft, which had an adverse impact on customer demand for MRO services.

Total revenue in our Catering business segment decreased by €367 million, or 45%, from €814 million in the 6M 2020 to €447 million in the 6M 2021 primarily due to reduced number of flights, particularly long-haul flights, across the industry, resulting in a decrease of customer demand for catering services.

Total revenue in our Additional Businesses and Group Functions business segment decreased by €34 million, or 14%, from €249 million in the 6M 2020 to €215 million in the 6M 2021 primarily due to travel restrictions in light of the COVID-19 pandemic resulting in a decrease of customer demand for services such as aviation training and travel cost management.

The following table provides a breakdown of our total revenue by traffic revenue and other revenue for the periods presented:

	For the six-month period ended June 30,		% Change
	2021	2020	
	(unaudited) (in € million)		
Traffic revenue	3,637	5,641	(36)
Other revenue	2,134	2,694	(21)
Total Revenue	5,771	8,335	(31)

(i) *Traffic Revenue*

Traffic revenue decreased by €2,004 million, or 36%, from €5,641 million in the 6M 2020 to €3,637 million in the 6M 2021 primarily due to the effects of the COVID-19 pandemic and the accompanying travel restrictions resulting in a decrease of capacity and number of flights due to reduced customer demand both in the Network Airlines business segment as well as in the Eurowings business segment.

Most of our traffic revenue by area of operations was generated in Europe (measured by the original location of the sale). In particular, traffic revenue by area of operations for the Lufthansa German Airlines decreased from €2,486 million in the 6M 2020 to €1,040 million in the 6M 2021. Traffic revenue by area of operations for the other airlines SWISS, Austrian Airlines and Brussels Airlines decreased from €1,044 million, €294 million and €235 million in 6M 2020 to €566 million, €159 million and €118 million in the 6M 2021, respectively. In Europe, the traffic revenue by area of operations decreased from €3,685 million in the 6M 2020 to €2,220 million in the 6M 2021. In the 6M 2021, the traffic revenue by area of operations amounted to €445 million (6M 2020: €786 million) in North America, €106 million (6M 2020: €133 million) in Central- and South America, €698 million (6M 2020: €839 million) in Asia/Pacific, €72 million (6M 2020: €96 million) in the Middle East and €96 million (6M 2020: €102 million) in Africa.

(ii) *Other Revenue*

Other revenue decreased by €560 million, or 21%, from €2,694 million in the 6M 2020 to €2,134 million in the 6M 2021 primarily due to the COVID-19 pandemic and the resulting decrease in the number of flights and customer demand for MRO and Catering services.

(b) *Changes in Inventories and Work Performed by Entity and Capitalized*

Changes in inventories and work performed by entity and capitalized decreased by €109 million, or 69%, from €158 million in the 6M 2020 to €49 million in the 6M 2021 primarily due to the COVID-19 pandemic and the resulting reduction in maintenance events that can be capitalized.

(c) *Other Operating Income*

Other operating income decreased by €142 million, or 18%, from €797 million in the 6M 2020 to €655 million in the 6M 2021 primarily due to reduced foreign exchange rate gains and reduced income from release of provisions partially offset by an increase in government grants.

(d) *Cost of Materials and Services*

Cost of materials and services decreased by €1,923 million, or 38%, from €5,127 million in the 6M 2020 to €3,204 million in the 6M 2021 primarily due to a decrease in the number of flights and reduced capacity across the Group airlines in light of the COVID-19 pandemic resulting in lower expenses for fuel, other raw materials, consumables and supplies, fees and charges as well as external MRO services.

The following table sets forth a breakdown of our cost of materials and services by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects.

	For the six-month period ended June 30,		% Change
	2021	2020	
	(unaudited) (in € million)		
Network Airlines	1,886	3,277	(42)
Eurowings.....	157	427	(63)
Logistics	734	702	5
MRO	902	1,346	(33)
Catering.....	153	345	(56)
Additional Businesses and Group Functions	90	118	(24)

Cost of materials and services in our Network Airlines business segment decreased by €1,391 million, or 42%, from €3,277 million in the 6M 2020 to €1,886 million in the 6M 2021 primarily due to a decrease in the number of flights in light of the COVID-19 pandemic resulting in lower expenses for fuel, other raw materials, consumables and supplies, fees and charges as well as external MRO services.

Cost of materials and services in our Eurowings business segment decreased by €270 million, or 63%, from €427 million in the 6M 2020 to €157 million in the 6M 2021 primarily due to a decrease in customer demand and reduced capacity in light of the COVID-19 pandemic resulting in lower expenses for fuel, other raw materials, consumables and supplies, fees and charges as well as for external wet leases.

Cost of materials and services in our Logistics business segment increased by €32 million, or 5%, from €702 million in the 6M 2020 to €734 million in the 6M 2021 primarily due to higher fuel costs and higher belly expenses paid to Group Companies, partly offset by lower expenses for MRO services and the lack of depreciation related to capitalized engine overhauls.

Cost of materials and services in our MRO business segment decreased by €444 million, or 33%, from €1,346 million in the 6M 2020 to €902 million in the 6M 2021 primarily due to reduced demand for MRO services resulting in a decrease in expenses for raw materials, consumables and supplies and a decline in costs for external maintenance services.

Cost of materials and services in our Catering business segment decreased by €192 million, or 56%, from €345 million in the 6M 2020 to €153 million in the 6M 2021 primarily due to reduced demand for Catering services resulting in a decrease in expenses for raw materials, consumables and supplies as well as expenses for external services.

Cost of materials and services in our Additional Businesses and Group Functions business segment decreased by €28 million, or 24%, from €118 million in the 6M 2020 to €90 million in the 6M 2021 primarily due to decrease in IT costs as well as expenses for other services supplied.

(e) *Staff Costs*

Staff costs decreased by €713 million, or 20%, from €3,620 million in the 6M 2020 to €2,907 million in the 6M 2021 primarily due to a decrease in the average number of employees as well as short-time working and the related state aids such as the reimbursement of wage-replacement benefits and social security contributions. In addition, crisis agreements concluded in 2020 in most of the Group's companies, containing measures such as suspension of pay increase, reduction in the income supplement for short-time working pay and in employer contributions to the pension schemes, continue to be in place.

The following table sets forth a breakdown of our staff costs (without past service costs/settlement) by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects.

	For the six-month period ended June 30,		
	2021	2020	% Change
	(unaudited) (in € million)		
Network Airlines	1,486	1,880	(21)
Eurowings.....	78	93	(16)
Logistics	180	188	(4)
MRO	556	648	(14)
Catering.....	285	446	(36)
Additional Businesses and Group Functions	327	358	(9)

Staff costs in our Network Airlines business segment decreased by €394 million, or 21%, from €1,880 million in the 6M 2020 to €1,486 million in the 6M 2021 primarily due to a decrease of the number of our employees as well as short-time working and the related state aids.

Staff costs in our Eurowings business segment decreased by €15 million, or 16%, from €93 million in the 6M 2020 to €78 million in the 6M 2021 primarily due to a decrease of the number of our employees as well as short-time working and the related state aids.

Staff costs in our Logistics business segment decreased by €8 million, or 4%, from €188 million in the 6M 2020 to €180 million in the 6M 2021 primarily due to a decrease of the number of our employees as well as short-time working and the related state aids.

Staff costs in our MRO business segment decreased by €92 million, or 14%, from €648 million in the 6M 2020 to €556 million in the 6M 2021 primarily due to a decrease of the number of our employees as well as short-time working and the related state aids.

Staff costs in our Catering business segment decreased by €161 million, or 36%, from €446 million in the 6M 2020 to €285 million in the 6M 2021 primarily due to a decrease of the number of our employees as well as short-time working and the related state aids.

Staff costs in our Additional Businesses and Group Functions business segment decreased by €31 million, or 9%, from €358 million in the 6M 2020 to €327 million in the 6M 2021 primarily due to a decrease of the number of our employees as well as short-time working and the related state aids.

(f) Depreciation, Amortization and impairment

Depreciation, amortization and impairment decreased by €648 million, or 36%, from €1,783 million in the 6M 2020 to €1,135 million in the 6M 2021 primarily due to impairment losses recognized in the previous year as well as lower investing activities than in the previous year.

The following table sets forth a breakdown of our depreciation and amortization (without impairment losses) by business segment for the periods indicated.

	For the six-month period ended June 30,		% Change
	2021 (unaudited) (in € million)	2020	
Network Airlines	786	956	(18)
Eurowings.....	101	104	(3)
Logistics	70	78	(10)
MRO	89	100	(11)
Catering.....	40	61	(34)
Additional Businesses and Group Functions.....	59	58	2

Depreciation and amortization in our Network Airlines business segment decreased by €170 million, or 18%, from €956 million in the 6M 2020 to €786 million in the 6M 2021 primarily due to lower depreciation of aircraft and engine overhaul following the recognition of impairment losses in the previous year as well as the deferral of aircraft deliveries resulting in lower investing activities.

Depreciation and amortization in our Eurowings business segment decreased by €3 million, or 3%, from €104 million in the 6M 2020 to €101 million in the 6M 2021 primarily due to the termination of lease arrangements resulting in lower depreciation of right-of-use assets for aircraft, which were partially offset by increased depreciation on capitalized engine overhaul events.

Depreciation and amortization in our Logistics business segment decreased by €8 million, or 10%, from €78 million in the 6M 2020 to €70 million in the 6M 2021 primarily due to lower depreciation of intangible assets and engine overhaul.

Depreciation and amortization in our MRO business segment decreased by €11 million, or 11%, from €100 million in the 6M 2020 to €89 million in the 6M 2021 primarily due to lower depreciation of right-of-use assets for aircraft.

Depreciation and amortization in our Catering business segment decreased by €21 million, or 34%, from €61 million in the 6M 2020 to €40 million in the 6M 2021 primarily due to a lower depreciation on other fixed assets resulting from decreased non-current assets.

Depreciation and amortization in our Additional Businesses and Group Functions business segment increased by €1 million, or 2%, from €58 million in the 6M 2020 to €59 million in the 6M 2021 primarily due to an increase of the depreciation of right-of-use assets.

(g) *Other Operating Expenses*

Other operating expenses decreased by €769 million, or 38%, from €2,048 million in the 6M 2020 to €1,279 million in the 6M 2021 primarily due to lower sales and marketing expenses, a decline in other costs directly linked to business activities and lower write-downs on receivables.

The following table sets forth a breakdown of our other operating expenses (without book losses) by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects.

	For the six-month period ended June 30,		
	2021	2020	% Change
	(unaudited) (in € million)		
Network Airlines	904	1,264	(28)
Eurowings.....	73	122	(40)
Logistics	89	121	(26)
MRO	232	460	(50)
Catering.....	83	177	(53)
Additional Businesses and Group Functions	926	754	23

Other operating expenses in our Network Airlines business segment decreased by €360 million, or 28%, from €1,264 million in the 6M 2020 to €904 million in the 6M 2021 primarily due to a decrease in other expenses, other personnel and travel costs, sales provisions and marketing expenses.

Other operating expenses in our Eurowings business segment decreased by €49 million, or 40%, from €122 million in the 6M 2020 to €73 million in the 6M 2021 primarily due to decreased value adjustments on receivables and other current assets.

Other operating expenses in our Logistics business segment decreased by €32 million, or 26%, from €121 million in the 6M 2020 to €89 million in the 6M 2021 primarily due to a decrease in exchange rate losses, other personnel and travel costs as well as decreased value adjustments on other current assets.

Other operating expenses in our MRO business segment decreased by €228 million, or 50%, from €460 million in the 6M 2020 to €232 million in the 6M 2021 primarily due to decreased value adjustments on receivables and decreased other personnel and travel costs and rent and maintenance expenses.

Other operating expenses in our Catering business segment decreased by €94 million, or 53%, from €177 million in the 6M 2020 to €83 million in the 6M 2021 primarily due to a decrease in rent and maintenance expenses, other personnel and travel costs and other expenses, primarily related to the divestment of LSG Group's European business.

Other operating expenses in our Additional Businesses and Group Functions business segment increased by €172 million, or 23%, from €754 million in the 6M 2020 to €926 million in the 6M 2021 primarily due to an increase in exchange rate losses, which was partially offset by reduced consulting costs and other personnel and travel costs.

(h) *Profit/Loss from Operating Activities*

Loss from operating activities decreased by €1,238 million, or 38%, from €(3,288) million in the 6M 2020 to €(2,050) million in the 6M 2021 due to the reasons stated above.

(i) *Result of Equity Investments Accounted for Using the Equity Method*

Result of equity investments accounted for using the equity method increased by €113 million, or 61%, from a loss of €184 million in the 6M 2020 to a loss of €71 million in the 6M 2021 primarily due to decreased recognizable losses of joint ventures in the MRO and Eurowings segment.

(j) *Result of Other Equity Investments*

Result of other equity investments increased by €3 million, or 75%, from €4 million in the 6M 2020 to €7 million in the 6M 2021 primarily due to increased income from profit absorption agreements, which was partially offset by decreased dividend payments received from joint ventures and other associated companies.

(k) *Interest Income*

Interest income decreased by €35 million, or 106%, from €33 million in the 6M 2020 to €(2) million in the 6M 2021 primarily due to the loss of interest income resulting from certain cash deposits as well negative interest income.

(l) *Interest Expenses*

Interest expenses increased by €16 million, or 8%, from €195 million in the 6M 2020 to €211 million in the 6M 2021 primarily due to an increase in borrowings resulting in higher interest payments on financial liability.

(m) *Other Financial Items*

Other financial items increased by €882 million, or 112%, from €(789) million in the 6M 2020 to €93 million in the 6M 2021 primarily due to positive changes in the market value of debt instruments recognized in profit and loss in the 6M 2021 compared to valuation losses on fuel hedges in the 6M 2020, which were recognized in the financial result, because fuel consumption was lower as a result of the COVID-19 pandemic.

(n) *Financial Result*

Financial result increased by €947 million, or 84%, from a loss of €1,131 million in the 6M 2020 to a loss of €184 million in the 6M 2021 due to the reasons stated above.

(o) *Profit/Loss Before Income Taxes*

Profit/loss before income taxes increased by €2,185 million, or 49%, from a loss of €4,419 million in the 6M 2020 to a loss of €2,234 million in the 6M 2021 due to the reasons stated above.

(p) *Income Taxes*

Income taxes decreased by €371 million, or 47%, from an income of €792 million in the 6M 2020 to an income of €421 million in the 6M 2021 primarily due to the recognition of lower deferred tax assets for tax losses.

(q) *Profit/Loss After Income Taxes*

Profit/loss after income taxes decreased by €1,814 million, or 50%, from a loss of €3,627 million in the 6M 2020 to a loss of €1,813 million in the 6M 2021 due to the reasons stated above.

9.7.2 Comparison of the Fiscal Years Ended December 31, 2020, and December 31, 2019

(a) Total Revenue

Total revenue decreased by €22,835 million, or 62.7%, from €36,424 million in the Fiscal Year 2019 to €13,589 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting global travel restrictions.

The following table sets forth a breakdown of our total revenue by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects.

	For the Fiscal Year ended December 31,		% Change
	2020	2019	
	(audited) (in € million)		
Network Airlines	6,622	25,078	(73.6)
Eurowings.....	598	2,311	(74.1)
Logistics	2,757	2,478	11.3
MRO	3,747	6,572	(43.0)
Catering.....	1,305	3,360	(61.2)
Additional Businesses and Group Functions.....	470	793	(40.7)

Total revenue in our Network Airlines business segment decreased by €18,456 million, or 73.6%, from €25,078 million in the Fiscal Year 2019 to €6,622 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting global travel restrictions.

Total revenue in our Eurowings business segment decreased by €1,713 million, or 74.1%, from €2,311 million in the Fiscal Year 2019 to €598 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting global travel restrictions.

Total revenue in our Logistics business segment increased by €279 million, or 11.3%, from €2,478 million in the Fiscal Year 2019 to €2,757 million in the Fiscal Year 2020 primarily due to higher average prices, which offset the decreased belly capacities as a result of the COVID-19 pandemic.

Total revenue in our MRO business segment decreased by €2,825 million, or 43.0%, from €6,572 million in the Fiscal Year 2019 to €3,747 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic resulting in lower revenues from maintenance and overhaul events.

Total revenue in our Catering business segment decreased by €2,055 million, or 61.2%, from €3,360 million in the Fiscal Year 2019 to €1,305 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic resulting in lower revenue from flight security concepts, in-flight service equipment, transport/warehouse logistics and lounge operations.

Total revenue in our Additional Businesses and Group Functions business segment decreased by €323 million, or 40.7%, from €793 million in the Fiscal Year 2019 to €470 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic resulting in lower revenue from training and reductions in travel management.

The following table provides a breakdown of our total revenue by traffic revenue and other revenue for the periods presented.

	For the Fiscal Year ended December 31,		% Change
	2020	2019	
	(audited) (in € million)		
Traffic revenue	9,078	28,136	(67.7)
Other revenue	4,511	8,288	(45.6)
Total Revenue	13,589	36,424	(62.7)

(i) *Traffic Revenue*

Traffic revenue decreased by €19,058 million, or 67.7%, from €28,136 million in the Fiscal Year 2019 to €9,078 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting global travel restrictions. Passenger numbers decreased by 75% in the Fiscal Year 2020 compared to the Fiscal Year 2019 while sales (revenue passenger-kilometers) decreased by 77% and capacity (available passenger-kilometers) decreased by 69% during the same period. The passenger load factor fell by 19.4 percentage points to 63.2%.

Our cargo business was particularly affected by the loss of belly capacities on passenger aircraft. Capacity (available cargo ton-kilometers) decreased by 39% and sales (revenue cargo ton-kilometers) decreased by 31% from the Fiscal Year 2019 to the Fiscal Year 2020. However, the cargo load factor of 69.6% was 8.2 percentage points higher in the Fiscal Year 2020 compared to the Fiscal Year 2019, supported by the market-wide decline in available capacity.

Most of our traffic revenue by sector was generated in Europe (measured by the original location of the sale). In particular, traffic revenue by sector for the Lufthansa German Airlines decreased from €15,170 million in the Fiscal Year 2019 to €3,465 million in the Fiscal Year 2020. Traffic revenue by sector for the other airlines SWISS, Austrian Airlines and Brussels Airlines decreased from €5,010 million, €1,941 million and €1,386 million in the Fiscal Year 2019 to €1,635 million, €415 million and €384 million in the Fiscal Year 2020, respectively. In Europe, the traffic revenue by sector decreased from €18,586 million in the Fiscal Year 2019 to €5,852 million in the Fiscal Year 2020. In the Fiscal Year 2020, the traffic revenue by sector amounted to €1,113 million (Fiscal Year 2019: €4,707 million) in North America, €222 million (Fiscal Year 2019: €615 million) in Latin America, €1,541 million (Fiscal Year 2019: €3,073 million) in Asia/Pacific, €166 million (Fiscal Year 2019: €621 million) in the Middle East and €184 million (Fiscal Year 2019: €534 million) in Africa.

(ii) *Other Revenue*

Other revenue decreased by €3,777 million, or 45.6%, from €8,288 million in the Fiscal Year 2019 to €4,511 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting declining revenue in the MRO and Catering business segment, in particular.

(b) *Changes in Inventories and Work Performed by Entity and Capitalized*

Changes in inventories and work performed by entity and capitalized decreased by €510 million, or 74.5%, from €685 million in the Fiscal Year 2019 to €175 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting global travel restrictions, which led to parts of the fleet being decommissioned for the longer term in the Fiscal Year 2020. As a result, we recorded a decrease in other own work capitalized, as fewer maintenance inspections took place than in the Fiscal Year 2019.

(c) *Other Operating Income*

Other operating income largely remained unchanged with an increase by €8 million, or 0.4%, from €1,889 million in the Fiscal Year 2019 to €1,897 million in the Fiscal Year 2020 primarily due to the government subsidies in an amount of €261 million received as well as income from the reversal of provisions and accruals which were partially offset by a decrease in foreign exchange gains from €761 million in the Fiscal Year 2019 to €677 million in the Fiscal Year 2020.

(d) *Cost of Materials and Services*

Cost of materials and services decreased by €11,374 million, or 57.4%, from €19,827 million in the Fiscal Year 2019 to €8,453 million in the Fiscal Year 2020 primarily due to the volume-related decline, particularly for fuel, fees and charges and other measures to reduce costs in light of the COVID-19 pandemic.

The following table sets forth a breakdown of our cost of materials and services by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects.

	For the Fiscal Year ended December 31,		% Change
	2020	2019	
	(audited) (in € million)		
Network Airlines	5,101	13,843	(63.2)
Eurowings.....	699	1,861	(62.4)
Logistics	1,300	1,778	(26.9)
MRO	2,372	3,902	(39.2)
Catering.....	545	1,441	(62.2)
Additional Businesses and Group Functions.....	203	287	(29.3)

Cost of materials and services in our Network Airlines business segment decreased by €8,742 million, or 63.2%, from €13,843 million in the Fiscal Year 2019 to €5,101 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting decrease in fuel costs, fees and MRO services.

Cost of materials and services in our Eurowings business segment decreased by €1,162 million, or 62.4%, from €1,861 million in the Fiscal Year 2019 to €699 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting decrease in fuel costs, fees and charter expenses.

Cost of materials and services in our Logistics business segment decreased by €478 million, or 26.9%, from €1,778 million in the Fiscal Year 2019 to €1,300 million in the Fiscal Year 2020 primarily due to a decrease in fuel costs as a result of the lower average fuel prices, the modernization of the fleet as well as a decline in charter expenses.

Cost of materials and services in our MRO business segment decreased by €1,530 million, or 39.2%, from €3,902 million in the Fiscal Year 2019 to €2,372 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting decrease in costs of raw materials, consumables and supplies as well as a decline in costs for external services. In particular, costs of external MRO services decreased as the number of engine maintenance events was significantly lower than in the Fiscal Year 2019 due to the COVID-19 pandemic.

Cost of materials and services in our Catering business segment decreased by €896 million, or 62.2%, from €1,441 million in the Fiscal Year 2019 to €545 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic resulting in a decrease of products and services supplied.

Cost of materials and services in our Additional Businesses and Group Functions business segment decreased by €84 million, or 29.3%, from €287 million in the Fiscal Year 2019 to €203 million in the Fiscal

Year 2020 primarily due to the COVID-19 pandemic resulting in a decrease of products and services supplied.

(e) *Staff Costs*

Staff costs decreased by €2,685 million, or 29.4%, from €9,121 million in the Fiscal Year 2019 to €6,436 million in the Fiscal Year 2020 primarily due to a reduction in our headcount and savings as a result of the use of short-time working to preserve jobs, salary waivers and the reduction of variable salary components, all of which were measured to counter the effects of the COVID-19 pandemic. In the Fiscal Year 2020, income of €1,020 million was received in reimbursement of wage-replacement benefits and social security contributions paid in the context of short-time working in Germany, Austria and Switzerland.

The following table sets forth a breakdown of our staff costs (without past service costs/settlement) by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects and the aforementioned adjustments.

	For the Fiscal Year ended December 31,		% Change
	2020	2019	
	(audited) (in € million)		
Network Airlines	3,340	4,801	(30.4)
Eurowings.....	176	259	(32.0)
Logistics	377	406	(7.1)
MRO	1,113	1,448	(23.1)
Catering.....	742	1,290	(42.5)
Additional Businesses and Group Functions.....	660	913	(27.7)

Staff costs in our Network Airlines business segment decreased by €1,461 million, or 30.4%, from €4,801 million in the Fiscal Year 2019 to €3,340 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting introduction of short-time working and cost cuts based on crisis agreements with various trade unions.

Staff costs in our Eurowings business segment decreased by €83 million, or 32.0%, from €259 million in the Fiscal Year 2019 to €176 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting introduction of short-time working, the decrease in the average number of employees and the restructuring program in general.

Staff costs in our Logistics business segment decreased by €29 million, or 7.1%, from €406 million in the Fiscal Year 2019 to €377 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting introduction of short-time working, the decrease in the average number of employees as well as the initial effects of our ProFlex cost-cutting program, which also includes automated planning and control at hubs and gateways as well as simplified cargo net storage, automated embargo check and mobile digital handling improvement.

Staff costs in our MRO business segment decreased by €335 million, or 23.1%, from €1,448 million in the Fiscal Year 2019 to €1,113 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting introduction of short-time working.

Staff costs in our Catering business segment decreased by €548 million, or 42.5%, from €1,290 million in the Fiscal Year 2019 to €742 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting lower average number of employees.

Staff costs in our Additional Businesses and Group Functions business segment decreased by €253 million, or 27.7%, from €913 million in the Fiscal Year 2019 to €660 million in the Fiscal Year 2020 primarily due to

the COVID-19 pandemic and the resulting introduction of short-time work and cost cuts based on crises agreements with trade unions.

(f) Depreciation, Amortization and impairment

Depreciation, amortization and impairment increased by €1,613 million, or 58.1%, from €2,776 million in the Fiscal Year 2019 to €4,389 million in the Fiscal Year 2020 primarily due to impairments of aircraft and reserve engines in an amount of €1,461 million. The adjustments include depreciation and amortization of €1,475 million on aircraft and right-of-use assets for a total of 115 aircraft that were transferred to long-term storage mode or which are no longer planned to return to service and aircraft which are held for sale.

The following table sets forth a breakdown of our depreciation and amortization (without impairment losses) by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects and the aforementioned adjustments.

	For the Fiscal Year ended December 31,		% Change
	2020 (audited) (in € million)	2019	
Network Airlines	1,814	1,927	(5.9)
Eurowings.....	199	233	(14.6)
Logistics	178	160	11.3
MRO	197	191	3.1
Catering.....	117	119	(1.7)
Additional Businesses and Group Functions.....	124	107	15.9

Depreciation and amortization in our Network Airlines business segment decreased by €113 million, or 5.9%, from €1,927 million in the Fiscal Year 2019 to €1,814 million in the Fiscal Year 2020 primarily due to lower depreciation of aircraft.

Depreciation and amortization in our Eurowings business segment decreased by €34 million, or 14.6%, from €233 million in the Fiscal Year 2019 to €199 million in the Fiscal Year 2020 primarily due to lower expenses for right-of-use assets for leased aircraft.

Depreciation and amortization in our Logistics business segment increased by €18 million, or 11.3%, from €160 million in the Fiscal Year 2019 to €178 million in the Fiscal Year 2020 primarily due to the reduction of the useful life of our IT-systems.

Depreciation and amortization in our MRO business segment increased by €6 million, or 3.1%, from €191 million in the Fiscal Year 2019 to €197 million in the Fiscal Year 2020 primarily due to higher depreciation of other fixed assets.

Depreciation and amortization in our Catering business segment decreased by €2 million, or 1.7%, from €119 million in the Fiscal Year 2019 to €117 million in the Fiscal Year 2020 primarily due to lower depreciation on other fixed assets.

Depreciation and amortization in our Additional Businesses and Group Functions business segment increased by €17 million, or 15.9%, from €107 million in the Fiscal Year 2019 to €124 million in the Fiscal Year 2020 primarily due to additional depreciation on rights of use for leased properties.

(g) Other Operating Expenses

Other operating expenses decreased by €2,113 million, or 37.8%, from €5,585 million in the Fiscal Year 2019 to €3,472 million in the Fiscal Year 2020 primarily due to lower staff-related expenses, a decline in

other costs directly linked to business activities, and reduced marketing expenses, which were partially offset by increased impairment losses on receivables.

The following table sets forth a breakdown of our other operating expenses (without book losses) by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects and the aforementioned adjustments.

	For the Fiscal Year ended December 31,		% Change
	2020	2019	
	(audited) (in € million)		
Network Airlines	2,003	3,791	(47.2)
Eurowings.....	272	280	(2.9)
Logistics	227	277	(18.1)
MRO	820	884	(7.2)
Catering.....	294	505	(41.8)
Additional Businesses and Group Functions.....	1,374	1,664	(17.4)

Other operating expenses in our Network Airlines business segment decreased by €1,788 million, or 47.2%, from €3,791 million in the Fiscal Year 2019 to €2,003 million in the Fiscal Year 2020 primarily due to reductions in travel expenses and in costs for distribution services, marketing and sales-related IT-systems.

Other operating expenses in our Eurowings business segment decreased by €8 million, or 2.9%, from €280 million in the Fiscal Year 2019 to €272 million in the Fiscal Year 2020 primarily due to lower sales, marketing and distribution costs due to reduced capacity as well as lower legal and advisory costs, partially offset by the write-downs on receivables following the insolvency of our lease partner Luftfahrtgesellschaft Walter (“LGW”) and restructuring costs for Germanwings.

Other operating expenses in our Logistics business segment decreased by €50 million, or 18.1%, from €277 million in the Fiscal Year 2019 to €227 million in the Fiscal Year 2020 primarily due to lower travel, project and consulting expenses.

Other operating expenses in our MRO business segment decreased by €64 million, or 7.2%, from €884 million in the Fiscal Year 2019 to €820 million in the Fiscal Year 2020 primarily due to reduced costs for third-party personnel and travel costs, which were partially offset by several debt provisions.

Other operating expenses in our Catering business segment decreased by €211 million, or 41.8%, from €505 million in the Fiscal Year 2019 to €294 million in the Fiscal Year 2020 primarily due to a reduction in costs for third-party personnel and leases, as well as related maintenance costs.

Other operating expenses in our Additional Businesses and Group Functions business segment decreased by €290 million, or 17.4%, from €1,664 million in the Fiscal Year 2019 to €1,374 million in the Fiscal Year 2020 primarily due to a decline in expenses for third-party personnel, consulting and marketing, as well as a reduction in exchange rate losses.

(h) Profit/Loss from Operating Activities

Profit/loss from operating activities decreased by €8,778 million from €1,689 million in the Fiscal Year 2019 to a loss of €7,089 million in the Fiscal Year 2020 due to the reasons stated above.

(i) Result of Equity Investments Accounted for Using the Equity Method

Result of equity investments accounted for using the equity method decreased by €364 million from €88 million in the Fiscal Year 2019 to a loss of €276 million in the Fiscal Year 2020 primarily due to the

COVID-19 pandemic resulting in a decrease in the result of joint ventures and associated companies accounted for using the equity method. The result mainly relates to losses incurred by Günes Ekspres Havacilik A.S. (“**SunExpress**”), and Terminal 2 mbH & Co oHG. In addition, the book value of the two joint venture companies within the MRO segmented had to be impaired due to a revision of the business plans of these entities.

(j) *Result of Other Equity Investments*

Result of other equity investments decreased by €68 million, or 85%, from €80 million in the Fiscal Year 2019 to €12 million in the Fiscal Year 2020 primarily due to a decrease in dividends from other joint venture and other equity investments as well as a decrease in income from profit transfer agreements and an increase in expenses from loss transfer agreements for entities that are not consolidated.

(k) *Interest Income*

Interest income increased by €4 million, or 5.1%, from €79 million in the Fiscal Year 2019 to €83 million in the Fiscal Year 2020 primarily due to an increase in other interest income which was partially offset by a decrease in income from other securities and non-current financial loans.

(l) *Interest Expenses*

Interest expenses increased by €23 million, or 5.8%, from a loss of €394 million in the Fiscal Year 2019 to a loss of €417 million in the Fiscal Year 2020 primarily due to an increase in interest expenses in connection with borrowings, which outweighed the absence of interest expenses incurred in the Fiscal Year 2019 in connection with a supplementary tax payment for previous years. The increase was partially offset by lower expenses for the compounding of pension provisions, since the base rate was 0.6% lower on average between December 31, 2018, and December 31, 2019.

(m) *Other Financial Items*

Other financial items decreased by €1,262 million from €318 million in the Fiscal Year 2019 to a loss of €944 million in the Fiscal Year 2020 primarily due to the COVID-19 pandemic and the resulting decline in business activity which caused “overhedges” on fuel hedges and, to a lesser extent, on currencies due to a lack of underlying transactions.

(n) *Financial Result*

Financial result decreased by €1,713 million from €171 million in the Fiscal Year 2019 to a loss of €1,542 million in the Fiscal Year 2020 due to the reasons stated above.

(o) *Profit/Loss Before Income Taxes*

Profit/loss before income taxes decreased by €10,491 million, or 564.0%, from a profit of €1,860 million in the Fiscal Year 2019 to a loss of €8,631 million in the Fiscal Year 2020 due to the reasons stated above.

(p) *Income Taxes*

Income taxes increased by €2,480 million from a tax expense of €615 million in the Fiscal Year 2019 to tax income of €1,865 million in the Fiscal Year 2020 primarily due to the recognition of deferred tax assets for the losses incurred.

(q) *Profit/Loss After Income Taxes*

Profit/loss after income taxes decreased by €8,011 million from a profit of €1,245 million in the Fiscal Year 2019 to a loss of €6,766 million in the Fiscal Year 2020 due to the reasons stated above.

9.7.3 Comparison of the Fiscal Years Ended December 31, 2019, and December 31, 2018

In this section, reference is made to the figures published for the Fiscal Year 2019, which do not reflect the changes in segmentation carried out in the Fiscal Year 2020. See “9.4 Business Segments.”

(a) Total Revenue

Total revenue increased by €882 million, or 2.5%, from €35,542 million in the Fiscal Year 2018 to €36,424 million in the Fiscal Year 2019 primarily due to higher sales in the passenger business and positive exchange rate effects, which were partially offset by the negative price trend.

The following table sets forth a breakdown of our total revenue by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects.

	For the Fiscal Year ended December 31,		% Change
	2019	2018	
	(audited) (in € million)		
Network Airlines	23,106	22,549	2.5
Eurowings.....	4,123	4,098	0.6
Logistics	2,478	2,713	(8.7)
MRO	6,921	6,105	13.4
Catering.....	3,360	3,217	4.4
Additional Business and Group Functions	793	783	1.3

Total revenue in our Network Airlines business segment increased by €557 million, or 2.5%, from €22,549 million in the Fiscal Year 2018 to €23,106 million in the Fiscal Year 2019 primarily due an increase in passenger numbers and exchange rates.

Total revenue in our Eurowings business segment largely remained stable with a slight increase by €25 million, or 0.6%, from €4,098 million in the Fiscal Year 2018 to €4,123 million in the Fiscal Year 2019.

Total revenue in our Logistics business segment decreased by €235 million, or 8.7%, from €2,713 million in the Fiscal Year 2018 to €2,478 million in the Fiscal Year 2019 primarily due to the difficult conditions in the airfreight industry and the steep fall in yields as a result.

Total revenue in our MRO business segment increased by €816 million, or 13.4%, from 6,105 million in the Fiscal Year 2018 to €6,921 million in the Fiscal Year 2019 primarily due to more favorable exchange rates and an increase in demand for components and engine overhaul events.

Total revenue in our Catering business segment increased by €143 million, or 4.4%, from €3,217 million in the Fiscal Year 2018 to €3,360 million in the Fiscal Year 2019 primarily due to new business, price increases and more favorable exchange rates.

Total revenue in our Additional Businesses and Group Functions business segment increased by €10 million, or 1.3%, from €783 million in the Fiscal Year 2018 to €793 million in the Fiscal Year 2019 primarily due to increased IT-services and training revenues.

The following table provides a breakdown of our total revenue by traffic revenue and other revenue for the periods presented.

	For the Fiscal Year ended December 31,		% Change
	2019	2018	
	(audited) (in € million)		
Traffic revenue	28,136	27,801	1.2
Other revenue	8,288	7,741	7.1
Total Revenue	36,424	35,542	2.5

(i) *Traffic Revenue*

Traffic revenue increased by €335 million, or 1.2%, from €27,801 million in the Fiscal Year 2018 to €28,136 million in the Fiscal Year 2019 primarily due to higher sales in the passenger business and positive exchange rate effects which were partially offset by the negative price trend.

Most of our traffic revenue by sector was generated in Europe (measured by the original location of the sale). In particular, traffic revenue by sector for the Lufthansa German Airlines increased from €14,496 million in the Fiscal Year 2018 to €14,875 million in the Fiscal Year 2019. Traffic revenue by sector for the other airlines SWISS and Austrian Airlines amounted to €4,749 million and €1,956 million in the Fiscal Year 2018 compared to €5,010 million and €1,941 million in the Fiscal Year 2019, respectively. In Europe, the traffic revenue by sector decreased from €18,785 million in the Fiscal Year 2018 to €18,586 million in the Fiscal Year 2019. In the Fiscal Year 2019, the traffic revenue by sector amounted to €4,707 million (Fiscal Year 2018: €4,160 million) in North America, €615 million (Fiscal Year 2018: €669 million) in Latin America, €3,073 million (Fiscal Year 2018: €3,123 million) in Asia/Pacific, €621 million (Fiscal Year 2018: €590 million) in the Middle East and €534 million (Fiscal Year 2018: €474 million) in Africa.

(ii) *Other Revenue*

Other revenue increased by €547 million, or 7.1%, from €7,741 million in the Fiscal Year 2018 to €8,288 million in the Fiscal Year 2019 primarily due to revenue increases in the MRO and the Catering business segments.

(b) *Changes in Inventories and Work Performed by Entity and Capitalized*

Changes in inventories and work performed by entity and capitalized increased by €154 million, or 29.0%, from €531 million in the Fiscal Year 2018 to €685 million in the Fiscal Year 2019 primarily due to the larger volume of engine overhaul events that qualify for capitalization.

(c) *Other Operating Income*

Other operating income increased by €71 million, or 3.9%, from €1,818 million in the Fiscal Year 2018 to €1,889 million in the Fiscal Year 2019 primarily due to non-periodic income, such as the release of provisions and credit notes for prior years.

(d) *Cost of Materials and Services*

Cost of materials and services increased by €1,460 million, or 7.9%, from €18,367 million in the Fiscal Year 2018 to €19,827 million in the Fiscal Year 2019 primarily due to an increase in fuel costs and growth-related additional costs for raw materials, consumables and supplies in the MRO business.

The following table sets forth a breakdown of our cost of materials and services by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects.

	For the Fiscal Year ended December 31,		% Change
	2019	2018	
	(audited) (in € million)		
Network Airlines	12,799	11,714	9.3
Eurowings.....	3,005	3,042	(1.2)
Logistics	1,778	1,753	1.4
MRO	3,927	3,376	16.3
Catering.....	1,441	1,385	4.0
Additional Businesses and Group Functions.....	287	246	16.7

Cost of materials and services in our Network Airlines business segment increased by €1,085 million, or 9.3%, from €11,714 million in the Fiscal Year 2018 to €12,799 million in the Fiscal Year 2019 primarily due to an increase in fuel, maintenance and charter costs and costs for emission certificates.

Cost of materials and services in our Eurowings business segment decreased by €37 million, or 1.2%, from €3,042 million in the Fiscal Year 2018 to €3,005 million in the Fiscal Year 2019 primarily due to lower MRO expenses following the retirement of older, higher-maintenance aircraft and the sale of LGW, reduced lease expenses resulting from the initial application of IFRS 16 as well as the introduction of a new catering concept which compensated the increased fuel costs.

Cost of materials and services in our Logistics business segment increased by €25 million, or 1.4%, from €1,753 million in the Fiscal Year 2018 to €1,778 million in the Fiscal Year 2019 primarily due to increase charter expenses relating to additional AeroLogic aircraft and higher belly expenses paid to Lufthansa Group companies, including the takeover of the belly capacities of Brussels Airlines. This increase was partially offset by volume-related lower fuel costs.

Cost of materials and services in our MRO business segment increased by €551 million, or 16.3%, from €3,376 million in the Fiscal Year 2018 to €3,927 million in the Fiscal Year 2019 primarily due to growth-related additional costs for raw materials, consumables and supplies for the engine and component business as well as higher purchased services.

Cost of materials and services in our Catering business segment increased by €56 million, or 4.0%, from €1,385 million in the Fiscal Year 2018 to €1,441 million in the Fiscal Year 2019 primarily due to unfavorable exchange rate movements, additional volumes and price increases.

Cost of materials and services in our Additional Businesses and Group Functions business segment increased by €41 million, or 16.7%, from €246 million in the Fiscal Year 2018 to €287 million in the Fiscal Year 2019 primarily due to increased IT costs and costs for other services supplied.

(e) *Staff Costs*

Staff costs increased by €310 million, or 3.5%, from €8,811 million in the Fiscal Year 2018 to €9,121 million in the Fiscal Year 2019 primarily due to the larger workforce.

The following table sets forth a breakdown of our staff costs (without past service costs/settlement) by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects and the aforementioned adjustments.

	For the Fiscal Year ended December 31,		% Change
	2019	2018	
	(audited) (in € million)		
Network Airlines	4,210	4,159	1.2
Eurowings.....	628	619	1.5
Logistics	406	420	(3.3)
MRO	1,671	1,552	7.7
Catering.....	1,290	1,218	5.9
Additional Businesses and Group Functions.....	913	963	(5.2)

Staff costs in our Network Airlines business segment increased by €51 million, or 1.2%, from €4,159 million in the Fiscal Year 2018 to €4,210 million in the Fiscal Year 2019 primarily due to an increase in our workforce.

Staff costs in our Eurowings business segment increased by €9 million, or 1.5%, from €619 million in the Fiscal Year 2018 to €628 million in the Fiscal Year 2019 primarily due to the expansion of the Brussels Airlines long-haul business which was partially offset by increased crew productivity.

Staff costs in our Logistics business segment decreased by €14 million, or 3.3%, from €420 million in the Fiscal Year 2018 to €406 million in the Fiscal Year 2019 primarily due to reduced expenses for bonus accruals.

Staff costs in our MRO business segment increased by €119 million, or 7.7%, from €1,552 million in the Fiscal Year 2018 to €1,671 million in the Fiscal Year 2019 primarily due to an increase in our workforce.

Staff costs in our Catering business segment increased by €72 million, or 5.9%, from €1,218 million in the Fiscal Year 2018 to €1,290 million in the Fiscal Year 2019 primarily due to an increase in our workforce.

Staff costs in our Additional Businesses and Group Functions business segment decreased by €50 million, or 5.2%, from €963 million in the Fiscal Year 2018 to €913 million in the Fiscal Year 2019 primarily due to reduced expenses for bonus and severance provisions.

(f) Depreciation, Amortization and impairment

Depreciation, amortization and impairment increased by €571 million, or 25.9%, from €2,205 million in the Fiscal Year 2018 to €2,776 million in the Fiscal Year 2019 primarily due to an increase in depreciation and amortization resulting from the first time application of IFRS 16 and the write-down of several aircraft and the capitalized development expenses for a digital distribution platform of the entire Lufthansa Group. In addition, impairment losses increased primarily due to four Boeing MD-11Fs being held for sale as well as write-downs of other receivables and discontinued IT projects.

The following table sets forth a breakdown of our depreciation and amortization (without impairment losses) by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects and the aforementioned adjustments.

	For the Fiscal Year ended December 31,		% Change
	2019	2018	
	(audited) (in € million)		
Network Airlines	1,678	1,497	12.1
Eurowings.....	465	372	25.0
Logistics	160	104	53.8

	For the Fiscal Year ended December 31,		% Change
	2019	2018	
	(audited) (in € million)		
MRO	207	126	64.3
Catering.....	119	66	80.3
Additional Businesses and Group Functions.....	107	52	105.8

Depreciation and amortization in our Network Airlines business segment increased by €181 million, or 12.1%, from €1,497 million in the Fiscal Year 2018 to €1,678 million in the Fiscal Year 2019 primarily due to first time application of IFRS 16.

Depreciation and amortization in our Eurowings business segment increased by €93 million, or 25.0%, from €372 million in the Fiscal Year 2018 to €465 million in the Fiscal Year 2019 primarily due to IFRS 16 effects in connection with leased aircraft.

Depreciation and amortization in our Logistics business segment increased by €56 million, or 53.8%, from €104 million in the Fiscal Year 2018 to €160 million in the Fiscal Year 2019 primarily due to IFRS 16 effects relating to the depreciation of right-of-use assets from leases and increased depreciation resulting from additional aircraft.

Depreciation and amortization in our MRO business segment increased by €81 million, or 64.3%, from €126 million in the Fiscal Year 2018 to €207 million in the Fiscal Year 2019 primarily due to the IFRS 16 effects relating to the recognition of right-of-use assets from leases.

Depreciation and amortization in our Catering business segment increased by €53 million, or 80.3%, from €66 million in the Fiscal Year 2018 to €119 million in the Fiscal Year 2019 primarily due to IFRS 16 effects relating to the depreciation of right-of-use assets from leases.

Depreciation and amortization in our Additional Businesses and Group Functions business segment increased by €55 million, or 105.8%, from €52 million in the Fiscal Year 2018 to €107 million in the Fiscal Year 2019 primarily due to IFRS 16 effects relating to the depreciation of right-of-use assets from leases.

(g) Other Operating Expenses

Other operating expenses decreased by €123 million, or 2.2%, from €5,708 million in the Fiscal Year 2018 to €5,585 million in the Fiscal Year 2019 primarily due to the shift of lease expenses from other operating expenses to depreciation in connection with the initial application of IFRS 16 and lower expenses from exchange rate effects.

The following table sets forth a breakdown of our other operating expenses (without book losses) by business segment for the periods indicated. A translation of the figures from the business segments to the Lufthansa Group is only possible when considering consolidation effects and the aforementioned adjustments.

	For the Fiscal Year ended December 31,		% Change
	2019	2018	
	(audited) (in € million)		
Network Airlines	3,445	3,484	(1.1)
Eurowings.....	557	610	(8.7)
Logistics	277	261	6.1
MRO	943	882	6.9
Catering.....	505	539	(6.3)
Additional Businesses and Group Functions.....	1,664	1,653	0.7

Other operating expenses in our Network Airlines business segment decreased by €39 million, or 1.1%, from €3,484 million in the Fiscal Year 2018 to €3,445 million in the Fiscal Year 2019 primarily due to shift of lease expenses from other operating expenses to depreciation in connection with the initial application of IFRS 16 and reduced exchange rate expenses.

Other operating expenses in our Eurowings business segment decreased by €53 million, or 8.7%, from €610 million in the Fiscal Year 2018 to €557 million in the Fiscal Year 2019 primarily due to the reduction in marketing expenses as part of the turnaround program and lower crew travel costs.

Other operating expenses in our Logistics business segment increased by €16 million, or 6.1%, from €261 million in the Fiscal Year 2018 to €277 million in the Fiscal Year 2019 primarily due to exchange rate effects and increased consulting and legal expenses, which overcompensated the shift of lease expenses from other operating expenses to depreciation in connection with the initial application of IFRS 16.

Other operating expenses in our MRO business segment increased by €61 million, or 6.9%, from €882 million in the Fiscal Year 2018 to €943 million in the Fiscal Year 2019 primarily due to increased travel and third party expenses and value adjustments on receivables.

Other operating expenses in our Catering business segment decreased by €34 million, or 6.3%, from €539 million in the Fiscal Year 2018 to €505 million in the Fiscal Year 2019 primarily due to the shift of lease expenses from other operating expenses to depreciation in connection with the initial IFRS 16 application.

Other operating expenses in our Additional Businesses and Group Functions business segment increased by €11 million, or 0.7%, from €1,653 million in the Fiscal Year 2018 to €1,664 million in the Fiscal Year 2019 primarily due to increased miscellaneous expenses and consulting and legal expenses, which overcompensated the shift of lease expenses from other operating expenses to depreciation in connection with the initial application of IFRS 16.

(h) Profit/Loss from Operating Activities

Profit/loss from operating activities decreased by €1,111 million, or 39.7%, from €2,800 million in the Fiscal Year 2018 to €1,689 million in the Fiscal Year 2019 due to the reasons stated above.

(i) Result of Equity Investments Accounted for Using the Equity Method

Result of equity investments accounted for using the equity method decreased by €26 million, or 22.8%, from €114 million in the Fiscal Year 2018 to €88 million in the Fiscal Year 2019 primarily due to an impairment of the shares of the joint venture Alpha Holding Ltd. in the Catering segment, which was partially offset by a one-off disposal gain for an associate in the MRO segment.

(j) Result of Other Equity Investments

Result of other equity investments increased by €20 million, or 33.3%, from €60 million in the Fiscal Year 2018 to €80 million in the Fiscal Year 2019 primarily due to an increase in dividends from other equity investments and in results recognized in connection with profit and loss transfer agreements between the Company and non-consolidated entities.

(k) Interest Income

Interest income increased by €11 million, or 16.2%, from €68 million in the Fiscal Year 2018 to €79 million in the Fiscal Year 2019 primarily due to an increase in income from other securities and non-current financial loans as well as an increase in income from other securities and non-current financial loans.

(l) *Interest Expenses*

Interest expenses increased by €182 million, or 85.8%, from €212 million in the Fiscal Year 2018 to €394 million in the Fiscal Year 2019 primarily due to interest expenses in connection with a supplementary tax payment in the Fiscal Year 2019 in connection with a tax ruling on non-deductibility of partial write-downs on cross-border loans as well as interest expenses for leases recognized for the first time as a result of the adoption of IFRS 16 as of January 1, 2019.

(m) *Other Financial Items*

Other financial items increased by €364 million from a loss of €46 million in the Fiscal Year 2018 to €318 million in the Fiscal Year 2019 primarily due to a gain resulting from the early termination of long-term currency hedge transactions in connection with a conversion of non-binding purchase orders for 14 Boeing 777 aircraft into purchase options.

(n) *Financial Result*

Financial result increased by €187 million from a loss of €16 million in the Fiscal Year 2018 to €171 million in the Fiscal Year 2019 due to the reasons stated above.

(o) *Profit/Loss Before Income Taxes*

Profit/loss before income taxes decreased by €924 million, or 33.2%, from €2,784 million in the Fiscal Year 2018 to €1,860 million in the Fiscal Year 2019 due to the reasons stated above.

(p) *Income Taxes*

Income taxes increased by €27 million, or 4.6%, from a tax expense of €588 million in the Fiscal Year 2018 to a tax expense of €615 million in the Fiscal Year 2019 primarily due to an increase in deferred income taxes which was partially offset by a decrease in current income taxes. Current income tax did not reduce in line with the profit before income tax, because of additional tax expenses for prior years resulting from a tax ruling on the non-deductibility of partial write-downs on cross-border loans.

(q) *Profit/Loss After Income Taxes*

Profit/loss after income taxes decreased €951 million, or 43.3%, from €2,196 million in the Fiscal Year 2018 to €1,245 million in the Fiscal Year 2019 due to the reasons stated above.

9.8 Liquidity and Capital Resources

Historically, our primary sources of liquidity have been cash flows from operating activities, capital markets proceeds (bonds and promissory notes) and aircraft financing. During the Fiscal Year 2020, in light of the COVID-19 pandemic, our liquidity significantly declined primarily due to the severely restricted flight operations. This decline was partially offset by positive changes in working capital, lower tax payments and tax deferrals. During the Fiscal Year 2020, we received significant liquidity in the form of €3.3 billion stabilization measures in Germany, Switzerland, Austria and Belgium as well as the issuance of long-term debt such as a €1.0 billion bond, a €0.6 billion convertible bond and several aircraft-secured financings, cumulatively in excess of €500 million. During the Fiscal Year 2020, our Adjusted Free Cash Flow amounted to negative €3,669 million. During the 6M 2021, we issued a dual-tranche bond with an aggregate volume of €1.6 billion, promissory notes (*Schuldscheindarlehen*) of €350 million and several aircraft-secured financings with an aggregate volume of €672 million. For an overview of our material financing agreements, see “12.16.6 Additional Sources of Funding.” Over the near to medium term, we expect that our primary sources of liquidity will also include, in addition to the proceeds we receive from the Offering, proceeds mainly from corporate bonds, promissory notes and aircraft financing.

9.8.1 Cash flows

The following table provides an overview of the Lufthansa Group's cash flows for the periods presented:

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018
	(unaudited) (in € million)		(audited) (in € million)		
Cash and cash equivalents at the beginning of the period.....	1,804	1,431	1,431	1,434	1,218
Net profit/loss before income taxes	(2,234)	(4,419)	(8,631)	1,860	2,784
Depreciation, amortization and impairment losses on non-current assets (net of reversals)	1,134	1,782	4,393	2,738	2,178
Depreciation, amortization and impairment losses on current assets (net of reversals).....	(20)	49	159	99	23
Net proceeds on disposal of non-current assets	19	9	9	20	(34)
Result of equity investments	64	180	264	(168)	(174)
Net interest	213	162	334	315	144
Income tax payments/reimbursements	(72)	112	81	(1,009)	(670)
Significant non-cash-relevant expenses/income	(160)	385	212	(134)	(276)
Change in trade working capital	650	1,434	(683)	490	410
Change in other assets/shareholders' equity and liabilities	424	669	1,534	(181)	(276)
Net cash from/used in operating activities⁽¹⁾.....	18	363	(2,328)	4,030	4,109
Capital expenditure for property, plant and equipment and intangible assets	(604)	(883)	(1,249)	(3,486)	(3,709)
Capital expenditure for financial investments	(8)	(14)	(24)	(73)	(48)
Additions/loss to repairable spare parts for aircraft.....	70	86	276	(231)	(388)
Proceeds from disposal of non-consolidated equity investments	—	(1)	3	1	2
Proceeds from disposal of consolidated equity investments ⁽²⁾	—	—	(218)	3	4
Cash outflows for acquisitions/capital increases of/at non-consolidated equity investments.....	(7)	(5)	(39)	(107)	(48)
Cash outflows for acquisitions of consolidated equity investments	—	—	—	—	(12)
Proceeds from disposal of intangible assets, property, plant and equipment and other financial investments	99	84	166	132	146
Interest income	—	47	66	70	51
Dividends received	7	13	57	243	143
Net cash from/used in investing activities.....	(443)	(673)	(962)	(3,448)	(3,859)
Purchase of securities/fund investments.....	(2,851)	(5,450)	(10,678)	(4,989)	(3,289)
Disposal of securities/fund investments	1,932	5,040	9,298	4,570	3,879
Net cash from/used in investing and cash management activities	(1,362)	(1,083)	(2,342)	(3,867)	(3,269)
Capital increase/Silent Participation I	1,500	—	306	—	—
Transactions by non-controlling interests.....	—	—	—	1	1
Non-current borrowing.....	2,864	1,784	6,738	3,843	987
Repayment of non-current borrowing	(2,672)	(1,126)	(1,724)	(3,413)	(1,196)
Dividends paid.....	—	(17)	(21)	(414)	(349)
Interest paid	(120)	(107)	(223)	(178)	(69)
Net cash from/used in financing activities	1,572	534	5,076	(161)	(626)
Net increase/decrease in cash and cash equivalents	228	(186)	406	2	214
Changes due to currency translation differences	10	(8)	(33)	(5)	2

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018
	(unaudited) (in € million)		(audited) (in € million)		
Cash and cash equivalents at the end of the period.....	2,042	1,237	1,804	1,431	1,434
Less cash and cash equivalents of companies held for sale at the end of the period	—	26	—	16	—
Cash and cash equivalents of companies not classified as held for sale at the end of the period⁽³⁾ ...	2,042	1,211	1,804	1,415	1,434
Securities	4,603	2,448	3,654	1,970	1,735
Liquidity	6,645	3,659	5,458	3,385	3,169
Net increase/decrease in liquidity	1,187	274	2,073	216	(600)

(1) Referred to as cash flow from operating activities in the Unaudited Condensed Consolidated Interim Financial Statements.

(2) Less cash outflows.

(3) The difference between the bank balances and cash-in-hand shown in the statement of financial position for the Fiscal Year 2020 comes from fixed-term deposits of €2 million with terms of four to twelve months (6M 2021: €21 million; 6M 2020: €0 million; Fiscal Year 2019: €0 million; Fiscal Year 2018: €66 million).

(a) Comparison of the Six-Month Periods Ended June 30, 2021, and June 30, 2020

(i) Net cash from/used in Operating Activities

Net cash from/used in operating activities decreased by 95% from €363 million net cash from operating activities in the 6M 2020 to €18 million net cash used in operating activities in the 6M 2021 primarily due to lower non-cash earnings components and higher cash flow from operating activities recognized directly in equity in the previous year. In addition, usual seasonal cash surpluses from ticket sales compared to tickets used decreased due to lower demand for flights, exacerbated by the payment of ticket refunds to customers.

(ii) Net cash from/used in Investing Activities

Net cash used in investing activities decreased by 34% from €673 million in the 6M 2020 to €443 million in the 6M 2021 primarily due to the postponement of planned aircraft deliveries resulting in lower capital expenditure.

(iii) Net cash from/used in Financing Activities

Net cash from in financing activities increased by 194% from €534 million in the 6M 2020 to €1,572 million in the 6M 2021 primarily due to increased financial liabilities, including bonds, borrower's note loans, Japanese operating leases and state-guaranteed loans as well as the drawdown under the Silent Participation I, which was partially offset by financial liabilities that were repaid, including the KfW Financing, borrower's note loans and short-term borrowings as well as aircraft financings and other lease obligations.

(b) Comparison of the Fiscal Years Ended December 31, 2020, and December 31, 2019

(i) Net cash from/used in Operating Activities

Net cash from/used in operating activities decreased by 157.8% from €4,030 million net cash from operating activities in the Fiscal Year 2019 to €2,328 million net cash used in operating activities in the Fiscal Year 2020 primarily due to lower pre-tax earnings. The decrease was lower than for pre-tax earnings due to the non-cash components of earnings. These included, in particular, impairment losses of €1,828 million on non-current assets (Fiscal Year 2019: €84 million) and valuation effects from financial derivatives of €334 million.

(ii) *Net cash from/used in Investing Activities*

Net cash from/used in investing activities decreased by 72.1% from €3,448 million in the Fiscal Year 2019 to €962 million in the Fiscal Year 2020 primarily due to the decrease in capital expenditures for property, plant and equipment and intangible assets as well as capital expenditures for financial investments which was partially offset by the reduction in additions/loss in repairable spare parts for aircraft. In addition, the disposal of the LSG Group's European business resulted in a decrease in cash while dividend income also decreased.

(iii) *Net cash from/used in Financing Activities*

Net cash from/used in financing activities increased from €161 million net cash used in financing activities in the Fiscal Year 2019 to €5,076 million net cash from financing activities in the Fiscal Year 2020 primarily due to an increase in non-current borrowing in connection with the government assisted stabilization efforts and the issuance of capital markets debt as well as the capital increase of €306 million which took effect on July 2, 2020.

(c) *Comparison of the Fiscal Years Ended December 31, 2019, and December 31, 2018*

(i) *Net cash from/used in Operating Activities*

Net cash from/used in operating activities decreased by 1.9% from €4,109 million in net cash from operating activities in the Fiscal Year 2018 to €4,030 million in net cash from operating activities in the Fiscal Year 2019 primarily due to lower pre-tax earnings and higher tax payments.

(ii) *Net cash from/used in Investing Activities*

Net cash from/used in investing activities decreased by 10.7% from €3,859 million in net cash used in investing activities in the Fiscal Year 2018 to €3,448 million in net cash used in investing activities in the Fiscal Year 2019 primarily due to a decrease in capital expenditures for property, plant and equipment and intangible assets and additions/loss to repairable spare parts for aircraft.

(iii) *Net cash from/used in Financing Activities*

Net cash from/used in financing activities decreased by 74.3% from €626 million in net cash used in financing activities in the Fiscal Year 2018 to €161 million in net cash used in financing activities in the Fiscal Year 2019 primarily due to a borrowing switch from net redemption of financial debt to a net increase of financial debt, which was partially offset by increased interest payments and increased dividends paid.

9.8.2 *Financial and Contingent Liabilities*

(a) *Financial Liabilities*

Our principal financial liabilities comprise bonds, bank loans and lease liabilities. The following table sets forth our financial liabilities (sum of current and non-current borrowings as labelled in the statements of financial position) outstanding as of the dates indicated below:

	As of June 30, 2021 (unaudited) (in € million)	As of December 31, 2020 2019 2018 (audited) (in € million)		
Bonds	4,238	2,907	1,094	1,007
Liabilities to banks	3,300	4,938	2,110	1,957
Leasing liabilities	2,461	2,637	2,872	596
Other loans	5,572	4,886	3,954	3,125
Financial liabilities	15,571	15,368	10,030	6,685

For further information on our main financings refer to “12.16 Material Agreements.”

(b) *Contingent Liabilities*

As of June 30, 2021, we had contingent liabilities totaling €914 million (as of December 31, 2020: €872 million) from (i) guarantees, bills of exchange and cheque guarantees, (ii) warranty contracts, and (iii) providing collateral for third-party liabilities. Contingent liabilities from warranty contracts and collateral agreements also relate to contingent liabilities towards creditors of joint ventures. Of the total, €712 million (as of December 31, 2020: €692 million) related to joint and several guarantees and warranties, which are matched by compensatory claims against the other co-debtors.

9.9 Off-Balance Sheet Arrangements

We do not have any off-balance sheet arrangements that we regard as material.

9.10 Contractual Obligations

Our business is very capital intensive, requiring significant funds for the acquisition of assets. In the past, we have primarily acquired aircraft through outright purchase, which are partly used for structured finance arrangements, but also entered into leases. We also enter into long-term lease commitments with airports to ensure access to terminal, cargo, maintenance and other facilities.

The following table sets forth our contractual obligations (on an undiscounted cash flow basis) as of June 30, 2021.

	Less than 1 year	1-5 years	More than 5 years	Total
		(unaudited) (in € million)		
Bonds.....	—	2,350	1,850	4,200
Liabilities to banks	1,613	1,574	110	3,297
Leasing liabilities	496	1,255	1,288	3,039
Other loans	333	1,469	3,692	5,493
Total contractual obligations.....	2,441	6,648	6,940	16,029

9.11 Capital Expenditures

In response to the COVID-19 pandemic, we reduced our capital expenditures, primarily by restructuring payment plans for capital expenditure on aircraft. Our capital expenditures primarily related to aircraft, aircraft components and aircraft and engine overhauls. Capital expenditures for other items of property, plant and equipment primarily related to technical equipment and machinery as well as operating and office equipment while capital expenditures for intangible assets primarily related to licenses and software. Financial investments (not including equity investments) mainly related to cash outflows for loans granted to affiliated, non-consolidated companies.

The following table provides an overview of our capital expenditures for the periods presented:

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018
	(unaudited)		(unaudited)		
	(in € million)		(in € million)		
Aircraft, aircraft components, and aircraft and engine overhauls.....	541	722	983	2,998	3,303
Other items of property, plant and equipment	41	119	186	332	297
Intangible assets.....	22	43	80	156	109
Financial investments (excluding equity investments)...	8	14	24	73	48
Total.....	612	897	1,273	3,559	3,757

As of the date of this Prospectus, we have made commitments in respect of capital expenditures of €14 billion, which primarily relate to the acquisition of new aircraft. We expect to make capital expenditures in a total amount of approximately €1.5 billion in the fiscal year 2021. As these are budgeted amounts, there can be no assurance that our capital expenditures will not deviate from the amounts set forth above. Apart from the capital expenditures mentioned above, we have not resolved on any significant investments for the fiscal year 2021 or beyond.

9.12 Information on Our Consolidated Statement of Financial Position

9.12.1 Assets

The following table below sets forth a breakdown of our assets as of June 30, 2021 and December 31, 2020, 2019 and 2018.

	As of June 30, 2021 (unaudited) (in € million)	As of December 31, 2020 2019 2018 (audited) (in € million)		
Intangible assets with an indefinite useful life ⁽¹⁾	1,165	1,169	1,395	1,381
Other intangible assets.....	442	469	547	512
Aircraft and reserve engines.....	15,516	15,842	18,349	16,776
Repairable spare parts for aircraft	1,777	1,823	2,270	2,133
Property, plant and other equipment.....	3,493	3,671	4,041	2,221
Investments accounted for using the equity method....	353	403	672	650
Other equity investments.....	252	252	256	246
Non-current securities	38	54	53	41
Loans and receivables	443	440	469	512
Derivative financial instruments.....	491	363	906	828
Deferred charges and prepaid expenses.....	82	91	116	118
Effective income tax receivables.....	36	34	32	10
Deferred tax assets.....	4,725	4,833	2,268	2,131
Non-current assets.....	28,813	29,444	31,374	27,559
Inventories.....	669	726	980	968
Contract assets.....	180	142	277	234
Trade receivables and other receivables.....	3,510	2,843	5,417	5,576
Derivative financial instruments.....	390	260	459	357
Deferred charges and prepaid expenses.....	242	193	245	217
Effective income tax receivables.....	288	282	153	58
Securities	4,603	3,654	1,970	1,735
Cash and cash equivalents	2,063	1,806	1,415	1,500
Assets held for sale.....	80	134	369	9
Current assets	12,025	10,040	11,285	10,654
Total assets	40,838	39,484	42,659	38,213

(1) Including goodwill.

(a) Comparison of June 30, 2021, and December 31, 2020

(i) Non-Current Assets

Non-current assets decreased by 2% from €29,444 million as of December 31, 2020, to €28,813 million as of June 30, 2021, primarily due to impairment losses on aircraft and reserve engines, which were partly offset by down payments on aircraft orders and the addition of five new Airbus A320 aircraft and one Airbus A220 aircraft; deferred tax assets also declined due to the tax effects of the revaluation of pension provisions recognized directly in equity, which resulted in lower obligations due to the change in interest rates.

(ii) *Current Assets*

Current assets increased by 20% from €10,040 million as of December 31, 2020, to €12,025 million as of June 30, 2021, primarily due to an increase in cash including current securities following the drawdown of the Silent Participation I, as well as an increase in current trade and other receivables.

(b) *Comparison of December 31, 2020, and December 31, 2019*

(i) *Non-Current Assets*

Non-current assets decreased by 6.2% from €31,374 million as of December 31, 2019, to €29,444 million as of December 31, 2020, primarily due to write-downs on non-current assets and the decline in the market values of hedging instruments, which were partially offset by the increase in deferred income tax assets.

(ii) *Current Assets*

Current assets decreased by 11.0% from €11,285 million as of December 31, 2019, to €10,040 million as of December 31, 2020, primarily due to a decrease in current trade and other receivables as well as inventories resulting from the effects of the COVID-19 pandemic. This decrease was partially offset by the increase in cash and cash equivalents, including current securities, resulting from net borrowings and cash inflows recognized directly in equity.

Assets held for sale of €134 million mainly related to 40 aircraft held for sale. In the Fiscal Year 2019, the assets of the disposal group of the companies of LSG Group's European business represented the majority of the assets held for sale. These assets were derecognized with the completion of the disposal in December 2020.

(c) *Comparison of December 31, 2019, and December 31, 2018*

(i) *Non-Current Assets*

Non-current assets increased by 13.8% from €27,559 million as of December 31, 2018, to €31,374 million as of December 31, 2019, primarily due to the IFRS 16 effect of €2,364 million. In addition, investments in aircraft and reserve engines (without right-of-use assets) and advance payments on them led to a net increase of 7% in these assets. Deferred tax assets also increased due to interest rate-related measurement gains/losses on pension obligations.

(ii) *Current Assets*

Current assets increased by 5.9% from €10,654 million as of December 31, 2018, to €11,285 million as of December 31, 2019, primarily due to the increase of €360 million to €369 million in assets held for sale in the Fiscal Year 2019 compared to €9 million in the Fiscal Year 2018. In addition, we increased our securities held as a strategic liquidity reserve. The increase was partially offset by a decrease in receivables mainly related to the decline in the freight business and the reclassification of receivables in the LSG Group's European business.

9.12.2 *Shareholders' Equity*

The following table below sets forth a breakdown of our shareholders' equity as of June 30, 2021, and December 31, 2020, 2019 and 2018.

	As of June 30, 2021	As of December 31,		
	(unaudited) (in € million)	2020	2019	2018
		(audited) (in € million)		
Issued capital.....	1,530	1,530	1,224	1,217
Capital reserve	378	378	378	343
Silent Participation of the Economic Stabilization	1,500	—	—	—

	As of June 30, 2021 (unaudited) (in € million)	As of December 31, 2020 2019 2018 (audited) (in € million)		
Fund				
Retained earnings.....	(210)	4,868	5,617	4,555
Other neutral reserves	1,720	1,296	1,715	1,185
Net profit/loss	(1,805)	(6,725)	1,213	2,163
Equity attributable to shareholders of Deutsche Lufthansa Aktiengesellschaft.....	3,113	1,347	10,147	9,463
Non-controlling interests ⁽¹⁾	32	40	109	110
Shareholders' equity	3,145	1,387	10,256	9,573

(1) Referred to as minority interests in the Unaudited Condensed Consolidated Interim Financial Statements.

(a) Comparison of June 30, 2021, and December 31, 2020

Shareholders' equity increased by 127% from €1,387 million as of December 31, 2020, to €3,145 million as of June 30, 2021, primarily due to the Silent Participation I classified within equity and positive valuation effects recognized directly in equity for pensions and derivatives, which offset the net loss for the period.

(b) Comparison of December 31, 2020, and December 31, 2019

Shareholders' equity decreased by 86.5% from €10,256 million as of December 31, 2019, to €1,387 million as of December 31, 2020, primarily due to the net loss and the negative valuation effects for pensions and derivatives recognized directly in equity. The decrease was slightly offset by the capital increase of €306 million which took effect on July 2, 2020. The new shares were subscribed by the WSF.

(c) Comparison of December 31, 2019, and December 31, 2018

Shareholders' equity increased by 7.1% from €9,573 million as of December 31, 2018, to €10,256 million as of December 31, 2019, primarily due to the positive after-tax result and the increase of €396 million in the market value of derivatives. The increase was partially offset by measurement losses recognized directly in equity and deferred taxes of €763 million as well as dividend payments to the shareholders of the Company of €380 million.

9.12.3 Provisions and Liabilities

The following table below sets forth a breakdown of our provisions and liabilities as of June 30, 2021, and December 31, 2020, 2019 and 2018.

	As of June 30, 2021 (unaudited) (in € million)	As of December 31, 2020 2019 2018 (audited) (in € million)		
Pension provisions.....	7,607	9,531	6,659	5,865
Other provisions	593	558	490	537
Borrowings.....	13,266	12,252	8,396	5,008
Contract liabilities	35	36	25	22
Other financial liabilities	89	86	76	137
Advance payments received, deferred income and other non-financial liabilities.....	33	33	32	51
Derivative financial instruments.....	247	457	128	222
Deferred tax liabilities	501	485	611	583

	As of June 30, 2021 (unaudited) (in € million)	As of December 31, 2020 2019 2018 (audited) (in € million)		
Non-current provisions and liabilities	22,371	23,438	16,417	12,425
Other provisions	866	831	794	894
Borrowings	2,305	3,116	1,634	1,677
Trade payables and other financial liabilities	3,718	3,321	5,351	5,720
(Contract) Liabilities from unused flight documents	3,089	2,064	4,071	3,969
Other contract liabilities	2,793	2,977	2,675	2,391
Advance payments received, deferred income and other non-financial liabilities	1,640	1,295	382	388
Derivative financial instruments	252	366	137	393
Effective income tax obligations	659	689	402	783
Liabilities in connection with assets held for sale	—	—	540	—
Current provisions and liabilities	15,332	14,659	15,986	16,215

(a) *Comparison of June 30, 2021, and December 31, 2020*

(i) *Non-Current Provisions and Liabilities*

Non-current provisions and liabilities decreased by 5% from €23,438 million as of December 31, 2020, to €22,371 million as of June 30, 2021, primarily due to a decrease in pension liabilities, which was largely caused by an increase in the interest rate of 0.4 percentage points to 1.2% which was used to discount pension obligations and the positive performance of plan assets. This effect was partially offset by an increase in borrowings.

(ii) *Current Provisions and Liabilities*

Current provisions and liabilities increased by 5% from €14,659 million as of December 31, 2020, to €15,322 million as of June 30, 2021, primarily due to higher (contract) liabilities from unused flight documents.

(b) *Comparison of December 31, 2020, and December 31, 2019*

(i) *Non-Current Provisions and Liabilities*

Non-current provisions and liabilities increased by 42.8% from €16,417 million as of December 31, 2019, to €23,438 million as of December 31, 2020, primarily due to the increase in borrowings and pension liabilities.

Non-current borrowings increased from €8,396 million as of December 31, 2019, to €12,252 million as of December 31, 2020, primarily due to the utilization of loans from the state stabilization packages (including fees) in the amount of €2,935 million as well as other borrowings in the amount of €3,773 million. Pension provisions increased from €6,659 million as of December 31, 2019, to €9,531 million as of December 31, 2020, primarily due to the decrease in the interest rate by 0.6 percentage points to 0.8% which was used to discount pension obligations in Germany and Austria, and the negative performance of plan assets.

(ii) *Current Provisions and Liabilities*

Current provisions and liabilities decreased by 8.3% from €15,986 million as of December 31, 2019, to €14,659 million as of December 31, 2020, primarily due to the decrease in liabilities from unused flight tickets and trade payables, which were partially offset by the increase in current borrowings and derivative financial instruments. Liabilities in connection with assets held for sale in the Fiscal Year 2019 related to the disposal of LSG Group's European business, which was sold in December 2020.

(c) *Comparison of December 31, 2019, and December 31, 2018*

(i) *Non-Current Provisions and Liabilities*

Non-current provisions and liabilities increased by 32.1% from €12,425 million as of December 31, 2018, to €16,417 million as of December 31, 2019, primarily due to the increase in borrowings and pension liabilities.

Non-current borrowings increased from €5,008 million as of December 31, 2018, to €8,396 million as of December 31, 2019, primarily due to the effect of the first-time application of IFRS 16 and an increase in lease liabilities. Pension liabilities increased from €5,865 million as of December 31, 2018, to €6,659 million as of December 31, 2019, primarily due to the decrease in the interest rate by 0.6 percentage points to 1.4% which was used to discount pension obligations in Germany and Austria, and which was partially offset by the positive performance of plan assets.

(ii) *Current Provisions and Liabilities*

Current provisions and liabilities decreased by 1.4% from €16,215 million as of December 31, 2018, to €15,986 million as of December 31, 2019, primarily due to a decrease in trade payables and other financial liabilities, driven by lower deferrals for performance-related remuneration and the reclassification of debt of the LSG Group's European business. Effective income tax obligations also decreased due to payments made for prior-year periods and negative market values of derivatives due to higher crude oil prices.

9.13 Quantitative and Qualitative Disclosures about Market Risks

The following discussion of risk serves a purpose different to that of the section "*1 Risk Factors.*" It is intended to discuss certain market risks that we face in our operations and the measures we take to mitigate those risks. As discussed below, the primary market risks to which we are exposed are foreign exchange risk, interest rate risk, fuel price risk, liquidity risk and credit risk. For further information on market risks that we face in our operations, including a sensitivity analysis, see Note 45 to our Audited Consolidated Financial Statements 2020, included in this Prospectus beginning on page F-1.

9.13.1 Foreign Exchange Risk

We are exposed to foreign exchange risk from fluctuations in exchange rates between the euro and other currencies. For U.S. dollars, we are in a net payer position with regard to currency risks from our operating business, since fuel payments are denominated in U.S. dollar. The main risks relate to the Chinese renminbi, the Swiss franc, British pound sterling, the Japanese yen and the Indian rupee. Depending on market liquidity, we hedge currency risks from projected operational exposure gradually over a period of 24 months by means of futures contracts, which are accounted for as cash flow hedges. We reduced the volume of hedging following the outbreak of COVID-19.

We hedge 50% of currency risks from capital expenditure on aircraft when we sign a contract. We review and increase the hedging level where necessary, if, over the lifetime of the contract, the exchange rate goes significantly above or below that used to calculate the investment. In the last 24 months before payment, we increase the hedging level in half-yearly steps of 10%, reaching 90% by the end. These investment hedges are therefore also accounted for as cash flow hedges. Our capital expenditures on aircraft are denominated in U.S. dollars and are hedged in euros or in Swiss francs, depending on the functional currency of the Lufthansa Group company making the purchase.

9.13.2 Interest Rate Risk

We are exposed to interest rate risk from floating rate and derivative financial instruments. In the event that we seek to pay interest on our financial liabilities in euros at floating rates, we arrange interest rate swaps for interest-bearing, fixed-rate financial debt and leasing liabilities. Financial liabilities, if hedged, are hedged using "plain vanilla" interest rate swaps or cross-currency interest rate swaps. Depending on the interest rate structure of the exposure to be hedged, hedges are either classified as fair value hedges or cash flow hedges. The interest rate risk is monitored constantly; strategic interest rate hedges are used as needed in response to different market situations. Depending on the counterparties and the instruments used, cash collateral for interest rate swaps is either deposited with or received from counterparties.

9.13.3 Fuel Price Risk

We are exposed to market price risk from the purchase of fuel for our aircraft. In 2020, fuel costs accounted for 9.0% of our operating expenses (€20,846 million as shown in our segment reporting) (2019: 18.1% of our operating expenses; €37,124 million as shown in our segment reporting). Significant changes in fuel prices can therefore have a significant effect on our result. Fuel price risk is limited by the use of crude oil hedges. The hedging level and the time horizon depend on the risk profile, which is derived from the business model of a group company. As a rule, we hedge up to 5% of exposure monthly for up to 24 months by spread options and other combinations of hedges.

We continuously analyze commodity price risks. Based on such analysis, the Executive Board, in March 2021, considered the fuel price risk in the current business environment as significant and chose to re-initiate the hedging program with a reduced target hedge ratio of 65%. As a result of the increase of fuel prices during the fourth quarter of the Fiscal Year 2020 and the 6M 2021, the market value of existing hedges is no longer negative and banks have re-initiated their trading lines for fuel derivatives.

9.13.4 Liquidity Risk

We are exposed to liquidity risk from our financial instruments. We have established a monthly rolling liquidity plan differentiated by currency with a planning horizon of 24 months. As of June 30, 2021, we had unused credit lines in the amount of €510 million (December 31, 2019: €774 million). In addition to our unused credit lines, we have certain undrawn amounts available to us under state aid programs, as of the date of this Prospectus. However, any amounts available for drawdown under the Silent Participation I or the Belgian Stabilization Loan can only be drawn until December 31, 2021. For additional information and an overview for utilized state aids, see “12.16.5 State Aid / Stabilization Package.”

We have taken specific and general measures to safeguard and manage our liquidity to avoid any future potential liquidity restrictions that could result from the COVID-19 pandemic. They include the implementation of monitoring on the basis of detailed, rolling short-term cash plans in order to manage liquidity effectively and track the current financing requirement closely. Specific liquidity risks resulting from reimbursements for cancelled flights were also analyzed and managed. In addition, liquidity management in connection with current orders for goods and services was also optimized, greater transparency was achieved across the Lufthansa Group, including an early warning system and an escalation process for outstanding receivables, additional approval steps were required to place orders and monitoring concerning short-time working requirements was introduced.

9.13.5 Credit Risk

We are exposed to credit risk from our financial instruments. The sale of passenger travel and freight documents mostly takes place via agencies. These agencies are mostly connected to national clearing systems for billing passenger and freight sales. The credit rating of the agents is reviewed by the responsible clearing systems. Due to the broad diversification, credit risk for the agencies is relatively low worldwide. Nonetheless, credit terms for agents in some markets were tightened significantly in cooperation with IATA in order to reduce credit risks even further.

Receivables and liabilities between airlines are offset through bilateral arrangements or via the IATA clearing house, insofar as the contracts underlying services do not explicitly specify otherwise. Systematic settlement of weekly receivables and liability balances significantly reduces the default risk. Fidelity guarantee insurance also covers partial risks within a certain range. Service contracts occasionally require collateral for miscellaneous transactions. All other contractual relationships are subject to credit rules, which, depending on the type and volume of the contract involved, require collateral, credit ratings/references or historical data from prior dealings, particularly payment history, in order to avoid defaults. Credit risks from our MRO business segment are monitored and managed with a separate credit risk management system. It comprises the calculation, authorization and monitoring of customer specific credit limits and the daily monitoring of payments received and receivables past due.

Counterparty risks in connection with credit card companies are monitored closely and incoming payments are reviewed daily. To reduce risks even further, we examine, through a permanent analysis process, whether

to further tighten credit terms for some settlement partners. In addition to the monitoring of receivables at the Company or segment level, there is also counterparty monitoring at the Lufthansa Group level, with individually assigned limits, in order to identify the accumulation of portfolio risks across the entire Lufthansa Group and take appropriate action where necessary. The maximum credit risk for financial assets from the potential insolvency of customers is their carrying amount.

9.14 Critical Accounting Policies

The preparation of our consolidated financial statements in accordance with IFRS requires management to use estimates and make assumptions and judgments that affect our results or financial statements. Some of these estimates, assumptions and judgments are critical to us due to the high degree of uncertainty of the relevant parameters at the time they are used or due to various alternatives available to management in making decisions that would lead to significantly different results reflected on our financial statements.

The outbreak of the COVID-19 pandemic and the steps taken worldwide to contain the virus have had a significant impact on our business operations. The uncertainties resulting from the crisis are vital for the general assessment of specific accounting judgments and estimates. The following is a summary of our critical accounting policies. For more information about our critical accounting policies, see “*Summary of significant accounting policies and valuation methods*” in the notes to our Audited Consolidated Financial Statements 2020, included in this Prospectus beginning on page F-1.

9.14.1 Passenger Transport and Ancillary Services

We sell flight tickets and related ancillary services primarily via agents, our own websites or other airlines in the course of interlining. The payments are received by us via credit card billing companies, agents or other airlines generally before the corresponding service is provided. Receivables from the sale of flight tickets and related ancillary services are only amounts payable by credit card billing companies, agents or other airlines.

We initially recognize all ticket sales as liabilities from unused flight documents. These are presented as contract liabilities in accordance with IFRS 15. Depending on the terms of the selected fare, the contract liabilities reflect a range of possibilities for refunding services that have not yet been provided. Liabilities include both the deferred income for future flights and ancillary services that are recognized as revenue when the flight documents are used, and the liabilities for award miles credited to the passenger when the flight documents are used. We allocate the transaction price to all of the performance obligations identified on the flight ticket on the basis of their individual transaction prices. The individual transaction prices for flight segments are determined using the IATA procedure, which allocates the total price payable to individual flight segments using what is known as a prorate calculation. Amounts calculated in this way meet the IFRS 15 definition of a relative individual transaction price. The individual transaction prices for ancillary services that are not included in the fare are directly observable prices within the meaning of IFRS 15. Prior to the outbreak of the COVID-19 pandemic, on average, it took 2.5 months for a flight coupon to be realized.

This does not apply to fiscal years with severe flight restrictions, such as the Fiscal Year 2020, when the use of flight documents fluctuated from month to month and altogether the large number of flight cancellations meant that customers could not use their tickets as planned. Additional cancellation and rebooking options were introduced in connection with the restrictions and flight cancellations caused by the COVID-19 pandemic.

We reduce liabilities from unused flight documents and recognize revenue for each flight segment (including the related ancillary revenue) when the respective document is used. For tickets that cover more than one flight segment, we identify each flight segment as a distinct performance obligation, since each flight segment is independent and can be distinguished in the context of the contract.

Interlining means that the passenger is carried by another airline for one (or more) flight segments. Only the commission paid by the other carrier is recognized as revenue for these flight segments, since we act solely as an agent in terms of these performance obligations. If passengers with tickets sold by other airlines are carried partly or fully by us, we show the *pro rata* ticket income received from the other airlines less the commission retained by the ticketing airline as revenue.

Generally speaking, we do not expect to receive any amount if a flight document is not used (or does not expect the amount to be material) and so for this reason do not anticipate the possibility that documents for a flight segment will not be used. The expected amount if flight documents are not used is only recognized as revenue if the probability that the passengers (in accordance with the portfolios) exercise their remaining rights is low, and no later than when the expiry of flight documents is certain and known.

IFRS 15 requires that income from the expiry of miles is recognized in parallel with revenue from the performance obligations that do not expire. Other than in relation to miles from status customers and from sponsored credit cards, a period of three years is therefore assumed for revenue recognition, and the revenue from miles expected to expire is recognized on a straight-line basis over this time. The effects of the COVID-19 pandemic on air travel have changed the recognition parameters, partly because there were fewer opportunities to fly and partly because customers' redemption patterns changed. The average percentage decline in redemption volumes was used as an indicator for adjusting the *pro rata* recognition volume.

Revenue for award miles is recognized at the point in time or over the time at which the goods and services purchased with the award miles are transferred.

9.14.2 Initial Consolidation and Goodwill

The initial consolidation of the Lufthansa Group companies takes place using the acquisition method. This involves measuring the fair value of the assets, liabilities and contingent liabilities identified, in accordance with the provisions of IFRS 3, of the company acquired at the acquisition date, and allocating the acquisition costs to them. The proportion of fair value of assets and liabilities not acquired is shown under non-controlling interests.

The ancillary acquisition costs are recognized as expenses in the periods in which they occur. Any excess of cost over the value of equity acquired is capitalized as goodwill. If the value of the acquirer's interest in the shareholders' equity exceeds the purchase price paid by the acquiring company, the difference is recognized immediately in profit or loss.

Differences from non-controlling interests acquired after control has been gained are set off directly against equity. Goodwill is not amortized, but is tested annually for impairment. The impairment tests applied to goodwill are carried out using established discounted cash flow methods. This is done on the basis of expected future cash flows from the latest business plan, which are extrapolated on the basis of long-term revenue growth rates and assumptions with regard to margin development and are discounted for the capital costs of the business unit. Tests are performed at the cash generating unit (CGU) level. For the individual premises on which impairment tests were based in the Fiscal Year 2020.

Additional impairment tests are also applied during the course of the year if events give reason to believe that goodwill could be permanently impaired. Once an impairment loss has been recognized on goodwill, it is not reversed in subsequent periods. Notwithstanding the principles described above, Lufthansa Group companies that have no material impact on the Lufthansa Group's net assets, financial and earnings position are not consolidated, but rather recognized in the consolidated financial statements at cost less any impairments.

9.14.3 Other Intangible Assets (Except Goodwill)

Acquired intangible assets are shown at cost, while internally generated intangible assets from which we expect to derive future benefit and that can be measured reliably are capitalized at cost of production and amortized regularly using the straight-line method over an estimated useful life. The cost of production includes all costs directly attributable to the production process, including borrowing costs as required under IAS 23, as well as appropriate portions of production-related overhead. Intangible assets with indefinite useful lives (mainly brands and purchased, re-sellable take-off and landing rights) are not amortized, but rather subjected to a regular annual impairment test, as is goodwill.

9.14.4 Pension Provisions

The pension provisions for defined benefit plans correspond to the present value of the defined benefit obligations ("DBO") on the reporting date less the fair value of plan assets, if necessary taking the rules on the maximum surplus of plan assets over the obligation (asset ceiling) into account. The DBO is calculated

annually by independent actuaries using the projected unit credit method prescribed in IAS 19 for defined benefit pension plans. The measurement of pension provisions within the statement of financial position is based on a number of actuarial assumptions.

Capital account plans are measured using the market value of the assets assigned to the individual capital accounts as of the reporting date, whereby the present value of the minimum benefit payable when the beneficiary becomes entitled to the benefit is compared with the amount of contributions already paid in, measured using the assumptions for the benefit plans. Additional risk premiums that the employer contributes to insure against early entitlements are included in current service expense. They include, in particular, assumptions about long-term salary and pension trends as well as average life expectancy.

The assumptions about salary and pension trends are based on developments observed in the past and take into account national interest and inflation rates and labor market trends. Estimates of average life expectancy are based on recognized biometric calculation formulas. The interest rate used to discount the individual future payment obligations is based on the return from investment grade corporate bonds in the same currency and with a similar term to maturity. The discount rate is determined by reference to high-quality corporate bonds with an issue volume of at least €100 million and an AA rating from at least one of the rating agencies Moody's Investor Service, Fitch Ratings or Standard & Poor's Rating Services.

Actuarial gains and losses arising from the regular adjustment of actuarial assumptions are recognized directly in equity in the period in which they arise, taking deferred taxes into account. Also presented without effect on profit and loss are differences between the interest income at the beginning of the period calculated on plan assets based on the interest rate used to discount the pension obligations and the earnings from plan assets actually recorded at the end of the period. The actuarial gains and losses and any difference between the forecast result and the actual result from plan assets form part of the remeasurement.

Past service costs are recognized immediately in profit or loss. Payments to pension providers for defined contribution retirement benefit commitments for which the pension provider or the beneficiary assumes the financial risks are recognized in staff costs as they fall due.

9.14.5 Government Grants

Government grants are recognized at fair value when there is reasonable assurance that the grant will be received and that we will fulfill all conditions attached to such grants. Government grants include, for example, income subsidies or social security subsidies for short-time working pay. If the supplement for short-time working pay is a personal benefit for the employee, the corresponding payments are transitory items. Another example is government loans at below-market rates, where the interest rate advantage is allocated *pro rata temporis* over the term of the loan. Government grants for the acquisition of property, plant and equipment are included in other liabilities as deferred income and recognized in other operating income on a straight-line basis over the estimated useful life of the corresponding asset.

Non-monetary grants are only recognized in the income statement when the necessary eligibility criteria have been fulfilled. Until then, the corresponding amounts must also be shown under deferred income.

9.14.6 Deferred Tax Assets

In accordance with IAS 12, deferred taxes are recognized for temporary differences between the statements of financial position with regard to tax of individual companies and the consolidated financial statements and for tax losses carry-forwards. As of December 31, 2020, the deferred tax assets were (i) tax loss carry-forwards in an amount of €1.3 billion and (ii) deductible temporary differences in the amount (prior to netting with deferred tax liabilities) of €4.1 billion of which €3.4 billion related to pensions. Deferred tax assets are to be recorded on the basis whether future taxable profit will be available against which deductible temporary differences and unused tax losses can probably (probability > 50%) be utilized.

Tax loss carry-forwards are recognized to the extent that the tax losses are likely to be used in the future. Company earnings forecasts are used to determine whether deferred tax assets are usable or not, i.e., whether they have a value that can be realized. The accounting for deferred tax assets requires management's forward-looking assessments, which are subject to inherent risks of estimation inaccuracies. The planning period used to assess this probability is determined by the individual Lufthansa Group company according to

the specific circumstances and lies generally between three and five years unless there is convincing evidence of possible prolonged use beyond the general horizon of the official Lufthansa Group planning.

This was the case as of December 31, 2020 due to the high losses caused by the COVID-19 pandemic, especially in Germany, resulting in the application of longer periods. As of December 31, 2020, the Company used positive and negative indicators to assess the probability of sufficient taxable income to recognize the deferred tax assets. For example, positive indicators included (i) the COVID-19 pandemic as a one-off item, most likely a non-recurring event, (ii) a loss that did not result from the business core, (iii) efficiency programs of the business units to increase profitability (crises of the last 20 years have shown the crisis resistance and adaptability of the Group), (iv) the planning assumptions available at the time of the financial statements which showed a return to profitability in 2022 (IFRS) and 2023 (tax planning) and a complete utilization of the tax loss carry-forward until 2031, (v) strong profit history (no loss history) before the COVID-19 pandemic; (vi) fundamentally positive future perspectives of the industry derived from external analyzes (among others, IATA statistics), (vii) budget plans of the last five years which showed a resilient robustness (actual values average approximately 20% above budget forecasts), and (viii) we are part of the critical infrastructure. Negative indicators included (i) volatile results in the past, (ii) challenges / risks from climate change, (iii) challenges / risks from changing travel behavior and (iv) loss expectation in the near future. For further details, see *“1.2.10 We are required to make estimates, assumptions and judgments, applying substantial discretion, when preparing our consolidated financial statements, which could prove to be materially incomplete or inaccurate, requiring us to record additional impairments.”*

9.14.7 Impairment Loss on aircraft

We test property, plant and equipment for impairment if events or changes in circumstances indicate that the carrying amount of an asset or the carrying amount of the asset's cash-generating unit may not be recoverable. In addition to depreciation and amortization, impairment losses are also recognized on the balance sheet date if the asset's recoverable amount has fallen below its carrying amount. The recoverable amount of an asset or a cash generating unit is determined as the higher of an asset's/cash generating unit's fair value less costs to sell and the present value of the estimated net future cash flows from continued use of the asset or the cash generating unit (value in use). We derive the fair value less costs to sell from recently observed market transactions – insofar as they are available – or in the case of aircraft, from general external information on current market prices.

Aircrafts are recognized at acquisition cost less scheduled depreciation, but at no more than their value in use. In determining the value in use of aircraft, a distinction is being made between aircraft that are permanently retired or released for sale and aircraft that are currently in use or for which future use is planned. Aircraft that are permanently retired or released for sale are valued individually and recognized at their realizable value (which is equivalent to their value in use).

The value in use of an aircraft currently in use or planned to be used again in the future cannot be determined individually, as no separate cash inflows can be attributed to an individual aircraft. Cash inflows from operating activities can only be determined at the level of the respective business unit (i.e., cash generating unit in accordance with IAS 36 (“CGU”). For this reason, the impairment of the aircraft to be used in the future is required to be reviewed as part of the impairment test for the respective business unit. In this respect, the carrying amount of the CGU (including the carrying amount of the aircrafts) is compared with the discounted expected cash flows of the CGU. If the discounted future cash flows exceed the carrying amount of the CGU, the assets are considered to be recoverable and no impairment loss is recognized.

9.15 Recently Adopted Accounting Principles

The following new standards, or amendments to existing standards, published by the IASB and already endorsed, had to be applied for the first time in the audited consolidated financial statements as of and for the Fiscal Year 2020.

- Conceptual Framework: Amendments to references to the conceptual framework in IFRS,
- Amendments to IAS 1 and IAS 8: Definition of materiality,
- Amendments to IFRS 9, IAS 39 and IFRS 7: Interest rate benchmark reform,

- Amendments to IFRS 3: Definition of a business, and
- Amendments to IFRS 16: COVID-19 related rent concessions.

The first-time application from January 1, 2020, or June 1, 2020, as applicable, of the mandatory accounting standards had no or no material effect on the presentation of our net assets, financial and earnings position or on earnings per share.

9.16 Information from the Annual Financial Statements of Deutsche Lufthansa Aktiengesellschaft According to HGB for the Fiscal Year Ended December 31, 2020

Some information from the audited annual financial statements (the “**Audited Annual Financial Statements**”) of the Company prepared in accordance with the German Commercial Code (*Handelsgesetzbuch*) (“**HGB**”) as of and for the Fiscal Year 2020 is presented below. The Audited Annual Financial Statements are reproduced on pages F-21 *et seqq.* in section “21 Financial Information” of this Prospectus.

In the Fiscal Year 2020, Deutsche Lufthansa Aktiengesellschaft recorded a net loss of €780 million compared to a net income of €595 million in the Fiscal Year 2019. As was the case for the consolidated results of the Lufthansa Group for the Fiscal Year 2020, the decrease primarily related to the COVID-19 pandemic and only partially offset by the recognition of previously unrealized book gains from the release of hidden reserves in the shares of AirTrust AG.

No dividend for the Fiscal Year 2020 was proposed to the General Meeting, among other reasons due to the restrictions on dividends under various state aid measures that we have received or that we expect to receive as part of the Stabilization Package.

As of December 31, 2020, the total assets of Deutsche Lufthansa Aktiengesellschaft were €32,272 million.

10. PROFIT FORECAST

This forecast of Deutsche Lufthansa Aktiengesellschaft (the “**Company**”) and its consolidated subsidiaries (the “**Lufthansa Group**,” “**we**,” “**us**” or “**our**”) for our Adjusted EBIT and Adjusted EBIT Margin for the fiscal year ending December 31, 2021 (the “**Fiscal Year 2021**”), together with the explanatory notes, is referred to collectively as the “**Profit Forecast**.” This Profit Forecast is not a representation of facts and should not be interpreted as such by investors.

This Profit Forecast is based on assumptions made by the Company’s executive board (*Vorstand*) (the “**Executive Board**”) with respect to the development and preparation of our Adjusted EBIT and Adjusted EBIT Margin for the Fiscal Year 2021 as set out below. These assumptions relate to factors that are beyond our control and to factors that can be influenced by us. While the Executive Board believes that these assumptions were reasonable on the date on which this Profit Forecast was prepared, they may subsequently prove to have been unjustified, incomplete or incorrect. If any of these assumptions prove to have been unjustified, incomplete or incorrect, our actual Adjusted EBIT and Adjusted EBIT Margin for the Fiscal Year 2021 could materially deviate from this Profit Forecast. Accordingly, prospective investors should treat this information with caution and should not place undue reliance on this Profit Forecast.

10.1 Definition of Adjusted EBIT and Adjusted EBIT Margin

We use Adjusted EBIT and Adjusted EBIT Margin as alternative performance measures and believe Adjusted EBIT and Adjusted EBIT Margin to be indicative of our operating performance. We understand that these measures are broadly used by analysts and investors in assessing our operating performance.

However, the alternative performance measures Adjusted EBIT and Adjusted EBIT Margin as used in this Profit Forecast are not defined by the International Financial Reporting Standards as adopted in the European Union (“**IFRS**”) or any other accepted accounting principles, and prospective investors should not consider such items as an alternative to the historical financial results or other indicators of our performance, assets or liabilities based on IFRS measures. In particular, they should not be considered as alternatives to our net profit/loss as an indicator of our profitability, or as alternatives to cash flows from operating activities as an indicator of our financial strength. The alternative performance measures Adjusted EBIT and Adjusted EBIT Margin, as defined by us, may not be comparable to similarly titled measures as presented by other companies due to differences in the way our alternative performance measures are calculated. Even though the alternative performance measures are used by the Executive Board to assess ongoing operating performance, and though these types of measures are commonly used by investors, they have important limitations as analytical tools and prospective investors should not consider them in isolation, or as substitutes for, the analysis of our results of operations, financial position and cash flows as reported under IFRS.

For purposes of this Profit Forecast, our Adjusted EBIT is defined as the profit/loss from operating activities plus result of equity investments accounted for using the equity method and result of other equity investments (“**EBIT**”), adjusted for impairment losses/gains, effects from pension provisions and results of disposal of assets. The following table provides a reconciliation of total revenue for the period to our Adjusted EBIT for purposes of this Profit Forecast:

Reconciliation	Line item
	Traffic revenue
+/-	Other revenue
+	Changes in inventories and work performed by entity and capitalized
+	Other operating income
-	Cost of materials and services
-	Staff costs
-	Depreciation, amortization and impairment
-	Other operating expenses
+/-	Result of equity investments accounted for using the equity method
+/-	Result of other equity investment
=	EBIT

Reconciliation	Line item
+/-	Impairment losses/gains
-/+	Effects from pension provisions
-/+	Results of disposal of assets
=	Adjusted EBIT

In addition, our Adjusted EBIT Margin is defined as our Adjusted EBIT divided by total revenue for purposes of this Profit Forecast.

10.2 Profit Forecast for the Fiscal Year 2021

The Company expects the operating loss as measured by Adjusted EBIT of the Lufthansa Group for the Fiscal Year 2021 to be less negative than Adjusted EBIT in the fiscal year ending December 31, 2020 (the “**Fiscal Year 2020**”), which amounted to an operating loss of €5,451 million. Furthermore, the Company expects the Adjusted EBIT Margin of the Lufthansa Group for the Fiscal Year 2021 to be less negative than in the Fiscal Year 2020, which amounted to negative 40.1%.

10.3 Explanatory Notes to this Profit Forecast

10.3.1 Basis of Preparation

This Profit Forecast was prepared in accordance with the principles of the Institute of Public Auditors in Germany (*Institut der Wirtschaftsprüfer in Deutschland e. V.*) (“**IDW**”) set forth in IDW Accounting Practice Statement: Preparation of Forecasts and Estimates in Accordance with the Specific Requirements of the Regulation on Prospectuses (*IDW AcPS AAB 2.003*) (*Rechnungslegungshinweis: Erstellung von Gewinnprognosen und -schätzungen nach den besonderen Anforderungen der Prospektverordnung (IDW RH HFA 2.003)*).

While Adjusted EBIT and Adjusted EBIT Margin are not IFRS measures of operating income, operating performance or liquidity, this Profit Forecast was prepared on the basis of the IFRS. With respect to the accounting policies applied, reference is made to the notes to the audited consolidated annual financial statements of the Company for the Fiscal Year 2020 prepared in accordance with IFRS.

This Profit Forecast has been compiled and prepared on a basis which is both (i) comparable with the historical financial information of the Lufthansa Group included in this Prospectus, and (ii) consistent with the Company’s accounting policies.

This Profit Forecast has been prepared solely for the inclusion in this Prospectus and represents our best estimate as of the date of this Profit Forecast (September 17, 2021). In preparing this Profit Forecast, based on our assessment, we have considered significant factors to take into account the operational and financial performance for this Profit Forecast. The expected development of these factors is based on assumptions made by the Executive Board which are set forth below.

10.3.2 Factors Beyond Our Control and Related Assumptions

This Profit Forecast is subject to factors beyond our control. These factors, and our assumptions taken about their impact, are described below:

(a) COVID-19

For purposes of this Profit Forecast, we estimate that the spread and persistence of the COVID-19 pandemic continues to have a negative impact on our operating performance. We have no control over factors relating to the COVID-19 pandemic, and our ability to foresee their future effects is limited and uncertain. These factors include:

- the severity, extent and duration of the global pandemic and its impact on the demand for air travel and on consumer spending more broadly;

- actions taken by national, state and local governments to contain COVID-19 or mitigate its impact, including travel restrictions, quarantine requirements and general restricted movement guidelines and lockdowns, required closures of nonessential businesses and aid and economic stimulus efforts;
- the duration and extent of short-time employment, including governmental assistance in relation thereto, as a result of COVID-19;
- the speed and extent of the recovery across the broader travel ecosystem;
- the duration, timing and severity of the impact on customer spending, including any economic recession resulting from the pandemic;
- the development and regulatory approval of vaccines against COVID-19 as well as the manufacture, distribution and administering of vaccines to a sufficiently large percentage of the population in our source markets and destinations;
- the acceptance of negative testing for COVID-19 as entry requirement to other countries in light of the lack of vaccinations; and
- the sufficiency, as determined by the various relevant governmental authorities, of the hygienic measures that we began developing and implementing in the summer of 2020.

The COVID-19 pandemic and the measures implemented to contain it, particularly travel restrictions and bans, together with the ensuing economic effects, significantly reduced and continues to reduce discretionary spending for air travel and related activities and services. This reduction in spending has affected and will continue to affect us directly and adversely, because our financial results and prospects are almost entirely dependent on the sale of travel and travel-related services. For purposes of this Profit Forecast, the Company has assumed that the pandemic will continue to have a significant negative impact on our business and results of operations. However, the Company has assumed that the increased availability of effective vaccines will drive customer demand in the Fiscal Year 2021 resulting in increasing demand for air travel. The recovery of air travel started to slowly gain momentum in the course of the summer of 2021, mainly driven by touristic travel to key summer destinations, but we do not expect that leisure air travel will reach pre-crisis levels in the Fiscal Year 2021. The recovery on long-haul routes and in the business travel segment is expected to be slower. Furthermore, customers are expected to continue to book at very short notice. Accordingly, the Company expects that the capacity of the Network Airlines and Eurowings (measured by ASK) will increase in the Fiscal Year 2021 to around 40% of the Fiscal Year 2019 and pre-COVID-19 levels. In order to respond to shifting demands and ensure a high degree of operational agility, we are temporarily parking a certain number of aircraft and maintain short-time working arrangements to be able to adjust our flight capacities swiftly.

(b) Economic and Political Development in the European Economic Area

Global growth in passenger demand (measured in RPKs) historically tracked growth of global gross domestic product (“GDP”) but, for example, during the period from 2003 to 2006 significantly outpaced GDP (*source*: IATA, Annual Review 2020). We generally estimate that the GDP in Europe will increase by 5.2% in the Fiscal Year 2021 compared to the Fiscal Year 2020 (*source*: Global Insight, World Review, August 2021) and therefore expect a positive impact on our Adjusted EBIT and Adjusted EBIT Margin for the Fiscal Year 2021. We also assume that the political environment in those countries most relevant to the Lufthansa Group remains stable and therefore will not have a particular impact on our Profit Forecast.

(c) Legal and Regulatory Framework

The airline industry is heavily regulated. Areas of our business that are substantially affected by law and regulation include:

- airlines’ requirement to hold operating licenses;
- compliance with security regulations;
- the availability of take-off and landing slots for airlines;
- air travel passenger rights;

- environmental regulations affecting airlines;
- sanctions imposed by national and international authorities, such as the United States; and
- rules and requirements, and in particular financial undertakings, imposed by the International Air Transport Association (the “IATA”), as a prerequisite for selling flight tickets of airlines that are IATA members.

Applicable regulations could be extended to include further environmental, consumer protection or other areas of regulation. In addition, the various regulatory regimes to which we are subject may conflict, so that compliance with the regulatory requirements in one jurisdiction may create regulatory conflicts in another. For purposes of this Profit Forecast, the Company has assumed that (except as described above under “10.3.2(a) COVID-19”) political, legislative and other regulatory measures will not materially impact its total revenue or Adjusted EBIT for the Fiscal Year 2021.

(d) Developments in the European and Global Airline Industries

The airline passenger and air freight industries are highly sensitive to general economic conditions. Because a substantial portion of business and leisure travel is discretionary, the airline industry tends to experience adverse financial results during economic downturns. Changing economic cycles have an impact on national income, unemployment rates as well as consumption and are thereby affecting particularly touristic supply and demand. During a recession, consumers are more likely to reduce spending for non-essential products such as leisure travel, leading suppliers to lower their prices, which in turn has a negative impact on profitability.

The dynamics between supply and demand are not only influenced by macro-economic factors, but also by market-specific conditions, such as the competitive forces. The European airline market is a particular example of a highly competitive market, which is – especially through the entrance of low-cost carriers such as easyJet, Ryanair and WizzAir – characterized by overcapacities and thus high pressure on margins, even during economic upward cycles. These overcapacities are expected to persist during and immediately after the COVID-19 pandemic due to stabilization measures that certain carriers, including Air France-KLM and IAG (including British Airways, Iberia, Aer Lingus and Vueling), received.

For purposes of this Profit Forecast, the Company has assumed that, except as described above under “10.3.2(a) COVID-19,” developments in the European and global airline industries will not materially impact its total revenue or Adjusted EBIT for the Fiscal Year 2021.

(e) Fuel Cost Fluctuations

Jet fuel prices have historically fluctuated widely, and are likely to do so in the future as a result of the level of economic activity, the rate of economic growth, political events, weather conditions, refinery outages or maintenance and the coordinated pricing decisions of producer cartels, such as the Organization of Petroleum Exporting Countries (OPEC), as well as speculative trading activity regarding the price of oil.

For purposes of this Profit Forecast, the Company has assumed an increase of average jet fuel prices for the Fiscal Year 2021 compared to the Fiscal Year 2020.

(f) Unforeseen Events

For purposes of this Profit Forecast, we assume that no material unforeseen events will occur that could result in material or lasting constraints on the ongoing operations of the entities comprising the Lufthansa Group, such as but not limited to force majeure, including natural disasters (e.g., fires, floods, hurricanes, storms and earthquakes), war or terrorist attacks, extraordinary macroeconomic events or cyber-attacks, maintenance outages, power or equipment failure, social unrest, work stoppages and public health concerns, except for the spread and persistence of the COVID-19 pandemic.

10.3.3 Factors That Can be Influenced by Us and Related Assumptions

In addition to the factors that are beyond our control, this Profit Forecast is subject to factors that can be influenced by us. These factors and our assumptions regarding their impact are described below.

(a) *Total Revenue*

For purposes of this Profit Forecast, the Company has assumed that air travel will gradually increase in the course of the Fiscal Year 2021 and as a result our total revenue is expected to be higher than in the Fiscal Year 2020. However, the Company expects that air travel and, therefore, our operating performance will return to pre-COVID-19 levels only after the Fiscal Year 2021. The recovery of air travel started to slowly gain momentum in the course of the summer 2021, mainly driven by touristic travel to key summer destinations, but we do not expect that leisure air travel will reach pre-crisis levels in the Fiscal Year 2021. The recovery on long-haul routes and in the business travel segment is expected to be slower.

(b) *Other Operating Income*

For purposes of this Profit Forecast, we estimate that other operating income will be less positive compared to the Fiscal Year 2020, since we assume lower income from reversals of provisions and lower income from currency gains in the Fiscal Year 2021 compared to the Fiscal Year 2020.

(c) *Our Restructuring Measures*

We have implemented our “ReNew” restructuring program to adapt to the new market conditions and further decrease cost of materials and services and staff costs. ReNew combines the decentralized restructuring programs for the airlines and aviation services with a focus on implementing a much smaller and more efficient production structure in all of our group companies. To achieve this objective, we signed crisis agreements with all the main trade unions of the Lufthansa Group in Germany and initiated measures to streamline our organizational structure in the Fiscal Year 2020. We also intend to structurally adjust our workforce to the market decline caused by COVID-19. As of June 30, 2021, we have reduced our workforce to 108,072 employees (headcount), compared to 110,065 employees (headcount) as of December 31, 2020 and 129,356 employees (headcount) as of June 30, 2020. The reduction of workforce from June 30, 2020 to June 30, 2021, includes an additional workforce reduction by approximately 8,100 employees (headcount) in connection with the divestment of LSG Group’s European business. In addition, we assume the ongoing financial support from short-time work and the effect of certain agreements, which we concluded with various trade unions for key parts of the Group’s workforce in light of the COVID-19 pandemic. As a result, the Company has assumed for purposes of this Profit Forecast that staff costs will decrease in the Fiscal Year 2021 compared to the Fiscal Year 2020 due to this reduction of our workforce in connection with our “ReNew” restructuring program.

Our “ReNew” program already affected our business, results of operations or financial condition in the Fiscal Year 2020 and we expect that it will continue to affect our business, results of operations or financial condition in the future. For the Fiscal Year 2021, we have assumed the following measures will have been implemented and, therefore, will impact the Company’s Adjusted EBIT: the implementation of a hiring freeze, the reduction of management positions to implement a more efficient organizational structure, the adoption of short-time hours (*Kurzarbeit*) across the Group, the postponement of non-safety relevant aircraft and engine maintenance, the renegotiation of key supplier contracts, the postponement or cancellation of business projects. Additionally, we intend to significantly reduce our fleet. As a result, we expect to achieve cost savings in the Fiscal Year 2021 resulting from the “ReNew” program that exceed the cost savings generated in the Fiscal Year 2020.

10.3.4 Other Explanatory Notes

This Profit Forecast does not take into account any extraordinary events, results from non-recurring activities and extraordinary tax expenses within the meaning of IDW Accounting Practice Statement 2.003 (AcPS AAB 2.003), except where explicitly stated otherwise in the explanatory notes above.

As this Profit Forecast relates to a period that has not yet ended and is based on several assumptions regarding uncertain events and actions, it inherently involves considerable uncertainties. As a result of such uncertainties, our Adjusted EBIT and Adjusted EBIT Margin for the Fiscal Year 2021 may differ from the respective forecast of Adjusted EBIT and Adjusted EBIT Margin for the Fiscal Year 2021, even substantially.

This Profit Forecast was prepared on September 17, 2021.

11. MARKETS AND COMPETITION

Most of the projections and other information set forth in this section have been derived from external sources. We believe that such industry publications and forecasts are reliable but we have not independently verified them and cannot guarantee their accuracy or completeness. The projections and forward-looking statements in this section are not guarantees of future performance and actual events and circumstances could differ materially from current expectations. Numerous factors could cause or contribute to such differences. See “1 Risk Factors” and “2.5 Forward-Looking Statements.”

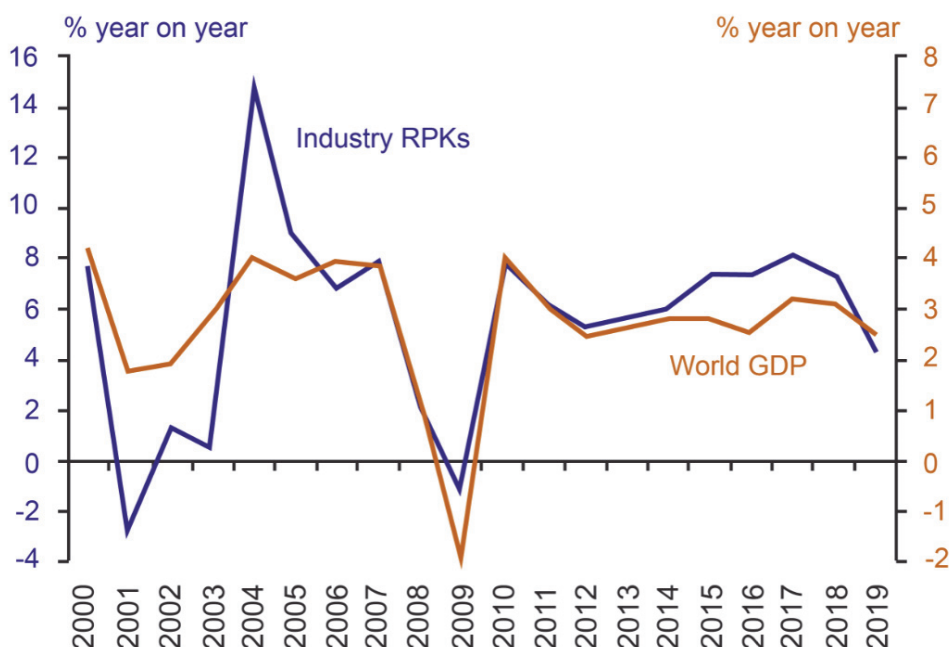
11.1 Global Passenger Airline Industry

11.1.1 Overview

The global passenger aviation market increased in terms of revenue from USD 721 billion in 2015 to USD 838 billion in 2019, corresponding to an increase of 16.2% (*source: IATA, Industry Statistics, April 2021*). Additionally, the global passenger aviation market has seen demand (revenue passenger-kilometers (“**RPKs**”)) grow at a compound annual growth rate (“**CAGR**”) of 6.8% while capacity (available seat-kilometer (“**ASKs**”)) has grown at a CAGR of 6.1% between 2015 and 2019 (*source: IATA, Industry Statistics, April 2021*).

We believe that RPK and ASK are one of the most frequently used metrics for operational data analysis within the airline industry. The RPK metric sets forth the number of kilometers travelled by paying passengers and is calculated as the number of revenue passengers multiplied by the total distance travelled. We utilize RPK as an indicator of, among others, demand in a given market, the requirement to make capacity adjustments and, simultaneously, to improve efficiency by utilizing existing capacity. The ASK metric sets forth the total flight passenger capacity in kilometers and is calculated by multiplying the total number of seats available for scheduled passengers and the total number of kilometers in which those seats were flown. We utilize ASK as an indicator of, among others, available passenger capacity (to generate revenue), load factor as well as revenue and cost of production.

Global growth in passenger demand (measured in RPKs) historically tracked growth of global gross domestic product (“**GDP**”) but, for example, during the period from 2003 to 2006 significantly outpaced GDP (*source: IATA, Annual Review 2020*).

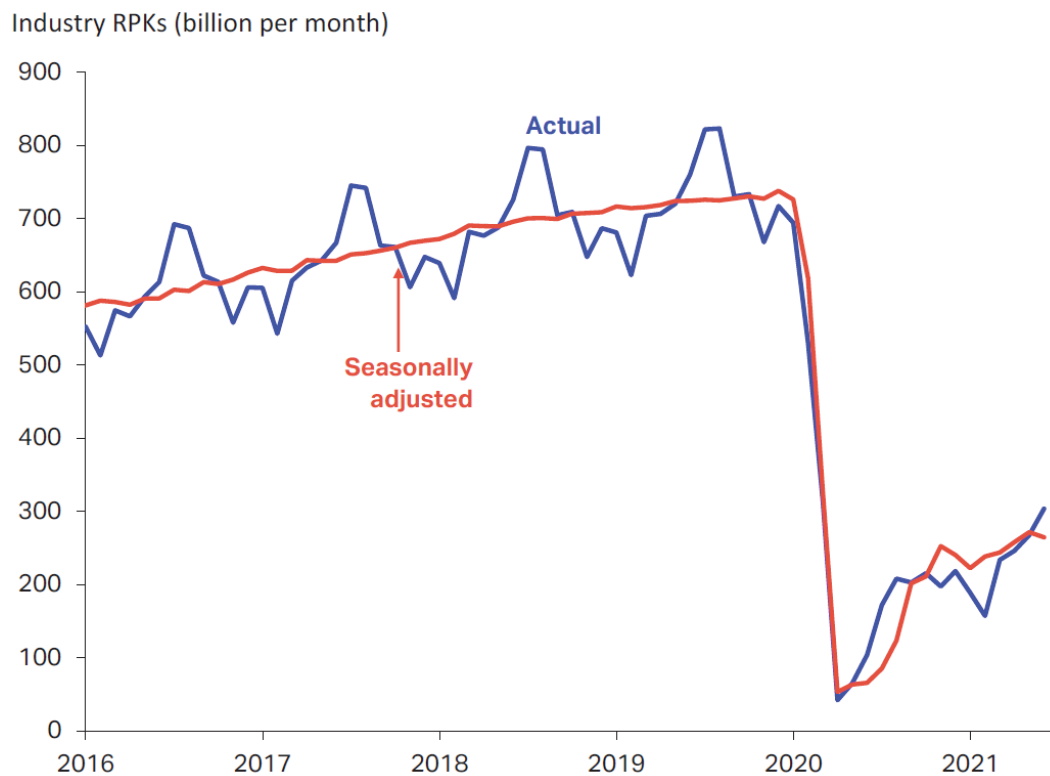


Source: IATA, Annual Review 2020.

From March 2020, the global spread of COVID-19 and the ensuing travel restrictions had a significant negative impact on the demand for air travel. Air traffic on Asian routes was first affected after the outbreak of the virus in China in February 2020. This trend progressed rapidly in March 2020, with air traffic in Europe, the Middle East and North America increasingly impacted. Global passenger traffic then came to a nearly complete standstill in April and May 2020. During this period, the connections on offer were mainly limited to repatriation flights and domestic services, plus a few international connections. The June to August 2020 summer months saw a slight recovery as a result of the gradual lifting of travel restrictions. However, after the end of the summer travel season and with the number of infections rising significantly again, passenger numbers declined again from September 2020 onwards. This situation remained largely unchanged for the rest of the year after widespread travel restrictions were re-established in Europe as well. The total number of RPKs worldwide declined by 66% in 2020 according to figures published by the International Air Transport Association (“**IATA**”) (2019: increase of 4%) (*source: IATA, Air Passenger Market Analysis, December 2020 and IATA, Annual Review 2020*). Airlines from the Middle East reported the largest decline in RPKs, at 72%. In the same period, airlines from Europe posted a 70% decline in RPK (*source: IATA, Air Passenger Market Analysis, December 2020*).

The following chart sets forth the industry-wide RPKs actual and seasonally adjusted from April 2017 to April 2021. We believe that industry-wide RPKs generally followed the underlying market demand.

The following chart sets forth the industry-wide RPKs actual and seasonally adjusted from 2016 to June 2021. We believe that industry-wide RPKs generally followed the underlying market demand.



Source: IATA, Air Passenger Market Analysis, June 2021.

According to the Federal Association of the German Aviation Industry (“**BDL**”), sales for airlines in Germany fell overall by 76% in 2020 compared to 2019 (*source: BDL, Bericht zur Lage der Branche 2020*) in 2020. Average yields in global passenger traffic fell by 8.7% in 2020 (2019: decline by 3.7%) (*source: IATA, Industry Statistics, April 2021*).

The following table sets forth the sales performance (by RPK) in the passenger airline industry for 2020.

Region	Passenger kilometers in % compared with 2019
Europe	(70)
North America.....	(65)
Latin America	(62)
Asia/Pacific	(62)
Middle East	(72)
Africa	(69)
Global	(66)

Source: IATA, Air Passenger Market Analysis, December 2020.

The global airline industry saw a significant decline in earnings with a net loss of USD 126 billion for 2020 (2019: net profit of USD 26 billion) due to the strong negative development of demand in passenger traffic. On a regional basis, the highest net losses of USD 35 billion were in the North America region (2019: net profit of USD 17 billion). European airlines also generated a net loss of USD 35 billion (2019: net profit of USD 7 billion) (source: IATA, Industry Statistics, April 2021).

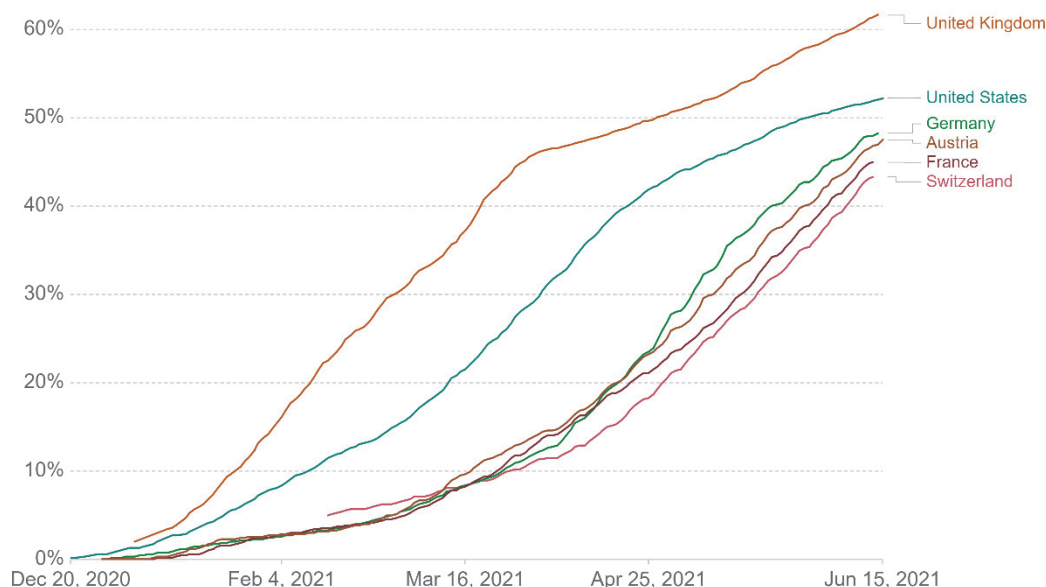
The following table sets forth the earnings development in the airline industry for 2020 and 2019.

Region	2020	2019
	(in USD billion)	
Europe	(35)	7
North America.....	(35)	17
Latin America.....	(12)	(1)
Asia/Pacific	(35)	5
Middle East	(8)	(2)
Africa	(2)	0
Industry	(126)	(26)

Source: IATA, Industry Statistics, April 2021.

Many airlines were dependent on government support in this unprecedented crisis situation. In addition to general support, such as short-time working, the individual states also adopted specific rescue measures for airlines. Nonetheless, some airlines had to file for insolvency in 2020. The continuing challenging situation could still lead to further market consolidation, including as a result of governmental funding and state aid granted to economically fragile airlines.

Since the end of 2020, countries around the world have begun to vaccinate their population with various vaccines against COVID-19. As of September 10, 2021, more than 61% of the total population of Germany, more than 64% of the total population of the UK and more than 52% of the total population of the United States have been fully vaccinated against COVID-19 and more than a further 4% of the total population of Germany, more than a further 6% of the total population of the UK and more than a further 9% of the total population of the United States have received at least the first vaccine dose (source: Our World in Data).



The recovery in air traffic is highly dependent on government travel restrictions. As vaccination roll-out progresses over the course of the rest of the Fiscal Year 2021, governments around the world are expected to gradually lift travel restrictions, which is expected to be a major catalyst for air travel demand recovery.

11.1.2 Airline Business Models

The four main categories of passenger airlines are: (i) full-service network carriers, (ii) low-cost point-to-point carriers, (iii) regional carriers and (iv) charter airlines/touristic carriers. The Lufthansa Group serves its customers within the business models full-service network airlines and low-cost point-to-point carriers with a strong footprint in the touristic segment (including, for example, through the Eurowings business), which we aim to further increase in the future.

Full-service network carriers are generally located in primary airports, from which they operate a hub-based network. These carriers tend to have greater complexity in their business models (including, for example, as a result of significant fluctuations in demand) but typically generate higher yields through the products and services that are part of that business model. Full-service carriers account for the majority of ASKs in the airline industry.

Low-cost point-to-point carriers tend to compete in the segment of the market that is the most price sensitive. By focusing on removing the complexity associated with full-service network models by operating a point-to-point network with a standardized fleet, they are able to reduce their cost of production, which in turn results in lower ticket prices relative to full-service carriers. These business models have tended to operate in short-haul markets, mostly in North America, Europe and Asia Pacific. Carriers such as JetStar and Air Asia also operate low-cost mid-haul business models in Asia Pacific. The expansion of the low-cost long-haul business model in Europe seems to have lost momentum in the most recent past.

Regional carriers operate smaller-capacity aircraft, and tend to focus on providing passenger air service to secondary and tertiary catchments without sufficient demand to fill larger gauge aircraft. In addition, these carriers provide feeder services on behalf of full-service network carriers.

Charter airlines/touristic carriers mostly operate flights where passengers originate from capacity allotments (for example in connection with tour operators) on regular (*i.e.*, scheduled) flights or individual bookings for remaining seat capacities on the aforementioned regular flights, which are primarily leisure based.

11.2 Our Home Markets

The airlines form the core of the Lufthansa Group, with their comprehensive networks in their home markets of Germany, Switzerland, Austria and Belgium. We hold a competitive position in each of our home

markets. In particular, we are one of the leading passenger airlines in our home markets, Germany, Switzerland, Austria and Belgium with market shares (by number of passengers) of 35%, 34%, 46%, and 26% in 2019, respectively (*source*: IATA, Direct Data Solutions).

As outlined above, the global spread of COVID-19 has led to an unprecedented decrease in the demand for both domestic and international air travel and the shape of recovery remains uncertain. However, the lifting of travel restrictions has resulted in an immediate increase in bookings. Given the country-by-country and regional specific impact of the COVID-19 pandemic, there is no certainty that the relative size of the principal markets in which we compete, including the relative size of our home markets, will remain the same as it was prior to the COVID-19 pandemic.

11.3 Overview of the Aviation Cargo Market

The global market for airfreight declined less significantly than passenger transport in 2020. Following the global spread of COVID-19, demand for airfreight initially declined and reached its low point in April 2020 (*source*: IATA, Air Cargo Market Analysis, April 2020). This development was driven by the significant decline in world trade and the collapse of global supply chains. Countering this trend, especially at the beginning of the COVID-19 pandemic, was the increased need for transport of medical and protective equipment. Starting in May 2020, market demand improved significantly again (*source*: IATA, Air Cargo Market Analysis, May 2020). According to IATA, global airfreight volumes in 2020 fell by 11% (2019: decline by 3%) (*source*: IATA, Air Cargo Market Analysis, December 2020 and IATA, Air Freight Market Analysis, December 2019).

The following table sets forth the volume performance in the aviation cargo industry for 2020.

Region	Cargo ton-kilometers in % compared with 2019
Europe.....	(16)
North America	1
Latin America	(21)
Asia/Pacific.....	(15)
Middle East.....	(10)
Africa.....	1
Global.....	(11)

Source: IATA, Air Cargo Market Analysis, December 2020.

Cargo airlines from Latin America reported the fastest decline at 21% compared with 2019. European providers saw a decline of 16% compared with 2019 (*source*: IATA, Air Cargo Market Analysis, December 2020). In 2021, industry-wide cargo ton-kilometers (“CTKs”) increased to pre-COVID-19 pandemic levels for the first time since the crisis started (+1.1% compared to January 2019).

The below chart shows the CTK levels, actual and seasonally adjusted from 2016 to July 2021.

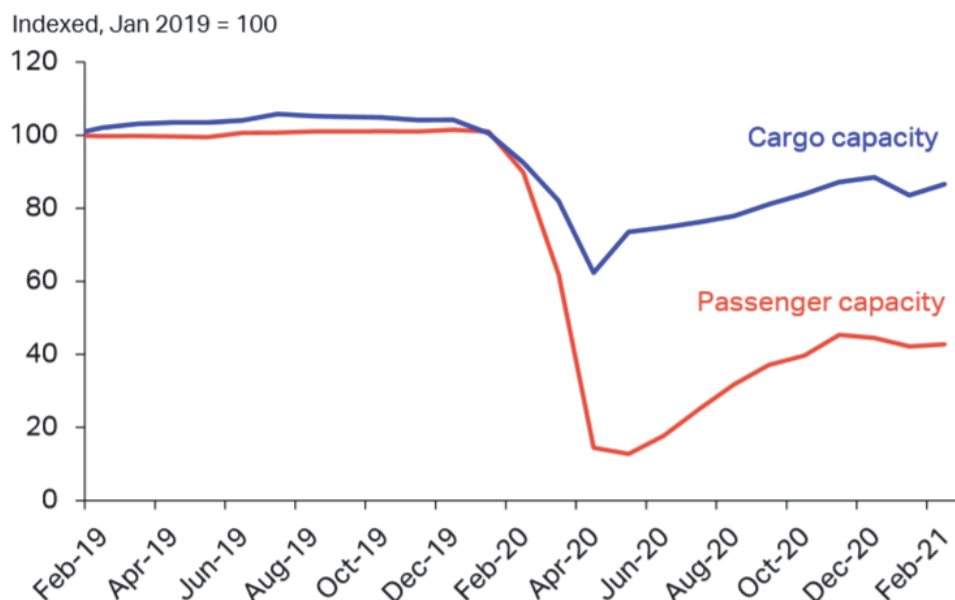


Source: IATA, Air Cargo Market Analysis, July 2021.

The rebound in CTK from May 2020 onwards was mainly due to more robust conditions in the manufacturing sector despite the slowed recovery in air passenger volumes. In addition to an expected increase in consumer spending, including, to a large extent, as a result of e-commerce, the distribution for vaccines is expected to be one of the drivers for air cargo volumes in the near term (*source*: IATA, Air Cargo Market Analysis, January 2021). In January 2021, all regions demonstrated cargo demand improvements, in particular North America (+5.9% month-on-month) and Latin America (+6.9% month-on-month), while international air cargo volumes increased by 1.2% in January 2021 compared to the same period in 2020 (*source*: IATA, Air Cargo Market Analysis, January 2021). This increase corresponded to pre-crisis levels. In February 2021, global industry-wide CTKs continued to recover slightly, rising by 2.3% from January 2021. Posting a robust 44.2% increase compared to February 2019, Africa was a significant contributing factor. International CTKs flown by cargo carriers from North America increased by 0.9% (from January 2021) and 17.4% compared to February 2019 (*source*: IATA, Air Cargo Market Analysis, February 2021).

However, the freight capacity on offer did not recover to the same degree, as a large proportion of the belly capacities of passenger aircraft were not available for airfreight due to the greatly reduced passenger flows. This imbalance between rising demand and significantly reduced capacity caused average yields in global airfreight traffic to go up by 40.0% in 2020 (2019: decline by 8.2%) (*source*: IATA, Industry Statistics, April 2021). Volumes also fell as a result of the significantly lower capacity. Capacity in terms of industry-wide available cargo ton-kilometers (“**ACTKs**”) also decreased again at the beginning of 2021 due to the increase in COVID-19 cases and the resulting re-instated heightened travel restrictions, again reducing the belly capacities of passenger aircraft.

The chart below shows the ACTK levels between February 2019 and February 2021, compared to the development of ASK over the same period.



Source: IATA, Air Cargo Market Analysis, February 2021.

The industry-wide cargo load factor reached 54.4% in July 2021 (corresponding to an increase of 9.5% compared to July 2020), a record high for any month of July in the history of IATA reporting (started in 1990), with Asia Pacific posting the highest cargo load factor at 65.4%, followed by European carriers with a load factor of 59.8% (source: IATA, Air Cargo Market Analysis, July 2021).

11.4 Overview of MRO and Catering Market

11.4.1 MRO

Demand for aircraft MRO services also fell significantly in 2020. The market environment suffered from a significant decline in capacity utilization, especially in engine maintenance and in the components business. Analyses provided by consultancy firm ICF International Inc. indicate that the MRO market declined by around 33% during the Fiscal Year 2020. Additionally, MRO market volumes fell by 36% in the Europe/Middle East/Africa (“EMEA”) region, by 34% in Asia and by 27% in the Americas during the Fiscal Year 2020. Digitalization in the MRO industry has affected repair processes and business models. These developments in combination with limited flight operations and increasing surplus availability continue to put pressure on prices for MRO services.

In recent years, OEMs, in particular, extended their after-market activities and new business models with aircraft and engine lessors increasingly seeking an after-market exposure. Whether this trend continues or might be reversed by the COVID-19 pandemic cannot be anticipated as of the date of this Prospectus.

11.4.2 Catering

The economic impact of the COVID-19 pandemic has also drastically changed the airline catering industry. With the crisis-related decline in passenger numbers in air travel, particularly in the long-haul business, demand for in-flight catering services also decreased temporarily. This was compounded by service restrictions related to hygiene concepts that minimize contact between crews and passengers. However, the North American airline catering market is slowly recovering. Additionally, the COVID-19 pandemic has subsequently accelerated the digitalization of pre- and post-flight and in-flight services. The pre-order and in-flight retail trend continued to increase, especially for short-haul traffic, economy class and leisure travel.

The LSG Group’s market environment is characterized by continuous cost pressures and rigorous competition. Many airlines are adapting new service models, in particular onboard retail programs, or hybrid concepts to achieve higher digitalization as well as cost efficiencies. As part of our “ReNew” restructuring program, we intend to also dispose of the remaining non-European business of the LSG group as soon as market conditions support such disposal. See “9.2.7 “ReNew” Restructuring Program.”

11.5 Competition

Lufthansa Group is facing competition in the Network carrier sector (including from U.S. carriers, such as American Airlines and Delta Airlines, from Asian carriers, such as Air China and Cathay Pacific, carriers from Turkey and the Middle East, such as Turkish Airlines, Emirates, Qatar Airways and Etihad Airways) as well as in the low-cost point-to-point segment (such as easyJet, Ryanair and WizzAir).

Our competitors are affected in similar ways as we are from the impact of COVID-19 on air travel demand.

11.5.1 Network Airlines Business Segment

In Europe, our network airlines' main competitors are the large network carrier groups Air France-KLM and IAG (including British Airways, Iberia, Aer Lingus and Vueling), in addition to a multitude of small national airlines. Our network airlines' primary competitors on routes from Germany to other European cities are generally the national airlines that operate in these countries. In addition, in Europe our network airlines are facing competition from low-cost airlines such as easyJet, Ryanair and WizzAir. Yields for the Network Airlines business segment have been put under pressure in the European short-haul market in previous years due to price pressure caused by carriers willing to accept significant losses to expand their market share, including in our home markets Germany, Austria and Belgium.

11.5.2 Eurowings Business Segment

The competition in the Eurowings business segment primarily consists of value- or low cost airlines operating point-to-point routes in the European short-haul market, primarily connecting medium- and large-sized airports outside of the major hubs. Competitors include easyJet, Ryanair, WizzAir and Vueling.

11.5.3 Logistics Business Segment

Competitors of Lufthansa Cargo are other airlines with significant cargo capacities on their long-haul passenger aircraft (such as Air France and IAG), airlines with a mix of cargo and passenger aircraft (such as Qatar Airways, Emirates and Etihad Airways) and airlines only operating cargo aircraft (such as Cargolux).

11.5.4 MRO Business Segment

Competitors of Lufthansa Technik are in general OEMs servicing the after-market and complementing their product-offering (such as Airbus, Boeing, General Electric and MTU Aero Engines), airline affiliates (such as Air France-KLM and IAG), independent and globally active MRO contractors offering a broad spectrum of services (such as ST Aerospace and HAECO), and small, regionally and technologically focused suppliers.

11.5.5 Catering Business Segment

LSG Group's competition consists of a few international and a growing number of local or regional companies. The biggest market player is gategroup (including Servair, acquired in 2018, and LSG Group's European business acquired in 2020), followed by the LSG Group, dnata, SATS, Do&Co and newrest, which, with the exception of the LSG Group's decrease as a result of the divestment of its European business, are all growing continuously. The past several years have also seen some companies from the logistics and other industries entering the airline catering market with a different value-chain setup, product portfolio and aggressive pricing. They, similar to the LSG Group and other traditional catering providers, offer services specifically geared towards the onboard retail programs of low-cost carriers and less complex onboard services on short-haul flights. They also target the economy class service where freshness and culinary excellence are not a priority in relation to the food selection.

12. BUSINESS

12.1 Overview

We are the leading airline group in Europe and one of the largest airline groups globally by total revenue (*source*: Statista, Leading Airline Groups in Europe in 2020, Company Information). The airlines that form the core of the Lufthansa Group have their home markets in Germany, Switzerland, Austria and Belgium. As an airline group, our core business is the operation of domestic, short-haul and long-haul scheduled passenger and cargo flights within a worldwide network, as well as the provision of logistical and technical services related to flight operations. In addition, we provide aircraft maintenance, repair and overhaul (“MRO”) services and catering.

We divide our business into six business segments: (i) Network Airlines; (ii) Eurowings; (iii) Logistics; (iv) MRO; (v) Catering; and (vi) Additional Businesses and Group Functions.

Network Airlines	Mainly comprises Lufthansa German Airlines (including regional partners and Miles & More), SWISS, Austrian Airlines, Brussels Airlines and Eurowings Discover.
Eurowings	Mainly comprises Eurowings and Eurowings Europe, providing point-to-point short-haul flights in Europe to price-sensitive and service-oriented customers.
Logistics	Mainly comprises Lufthansa Cargo and the Jettainer group, the time:matters group and the equity investment of 50% in the cargo airline AeroLogic.
MRO	Mainly comprises the Lufthansa Technik group, consisting of 35 plants offering technical aviation services worldwide and direct and indirect stakes in 62 companies.
Catering	Mainly comprises the LSG Group and, <i>inter alia</i> , the brand ‘LSG Sky Chefs.’ In December 2020, the sale of the European business operations of LSG Group was closed. The planned sale of the international operations is currently under review.
Additional Businesses and Group Functions	Mainly comprises the service and financial companies, particularly AirPlus, Lufthansa Aviation Training and Lufthansa Systems, as well as the group functions of the Lufthansa Group.

We generated total revenue of €5,771 million and €13,589 million in the 6M 2021 and the Fiscal Year 2020, respectively, compared to €8,335 million and €36,424 million in the 6M 2020 and the Fiscal Year 2019, respectively, impacted significantly by the outbreak of COVID-19.

The global spread of COVID-19 and the ensuing travel restrictions had a significantly negative impact on the demand for air travel. Passenger airline traffic temporarily came to an almost complete standstill worldwide. Air traffic on Asian routes was first affected in February 2020 after the outbreak of the virus in China. This trend progressed rapidly, with air traffic in Europe, the Middle East and North America increasingly impacted. While the summer months 2020 saw a slight recovery, with the number of infections rising significantly again, passenger numbers fell significantly again from September 2020 onwards. In June 2021, we offered 11,291 flights, compared to 22,291 and 105,200 in June 2020 and 2019, respectively.

The combination of new and re-imposed government travel restrictions and other counter-measures significantly limited our ability to resume operations. Notwithstanding the immediate effects of the COVID-19 pandemic, we believe air travel, in particular in the tourism sector, has strong long-term prospects.

12.2 Our Strengths

12.2.1 Leading Airline in Europe

We are the leading airline group in Europe and one of the largest airline groups globally by total revenue (*source*: Statista, Leading Airline Groups in Europe in 2020, Company Information) as well as one of

Europe's leading freight airlines providing comprehensive airport-to-airport airfreight services (*source*: Air Cargo Awards 2020) and are particularly capitalizing on our strong foothold in Germany as a leading global economy. In particular, we are one of the leading passenger airlines in our home markets, Germany, Switzerland, Austria and Belgium with market shares (by number of passengers) of 35%, 34%, 46%, and 26% in 2019, respectively (*source*: IATA, Direct Data Solutions).

The Network Airlines and Eurowings as well as our cargo business form the core of the Lufthansa Group. This multi-brand airline business model allows us to capitalize on our market share and leading position in our key markets. Through our differentiated portfolio of Network Airlines and Eurowings, we offer a comprehensive route network through our hubs and point-to-point connections to serve all relevant market segments and geographic markets with an attractive offering both for premium customers and for more price-sensitive travelers, covering both the leisure and corporate travel segments. We believe our brands are among the strongest in their respective hubs and the numerous awards we have received confirm this. For example, Lufthansa German Airlines is the first airline and the only airline in Europe to receive a five-star rating from Skytrax. Lufthansa German Airlines was also voted Europe's Leading Airline 2020 at the World Travel Awards and SWISS took first place in the category Europe's Leading Airline – First Class 2020. SWISS was voted Best Airline for Business Travellers in Europe for the year 2020 by the magazine Business Traveller.

Our Network Airlines offer their customers a premium experience with high-quality products and services. With their wide range of destinations and high frequencies in their global hubs, Lufthansa German Airlines and SWISS offer extensive connectivity to their customers across key markets. Austrian Airlines and Brussels Airlines connect their home markets with Europe and the world through their national hubs. The multi-hub strategy offers a comprehensive route network generating maximum choice and flexibility for our customers. Commercial joint ventures and numerous codeshare agreements across our global networks add further destinations to our global route network. We enjoy flexibility of production and the ability to respond rapidly to changes in market dynamics. We believe our strong product offering and strong positioning generates industry-leading yields.

With Eurowings, the Lufthansa Group has as competitive offering in the structural growth segment of European direct traffic. Eurowings addresses both price-sensitive and service-oriented customers by offering customer-tailored offers through low-cost basic fares and a selection of add-on service options that can be booked flexibly. We believe that Eurowings is among the market leaders in its core home bases of Dusseldorf, Hamburg, Cologne and Stuttgart. We expect that the recovery of air travel, which started to slowly gain momentum in the course of the summer of 2021, mainly driven by touristic travel to key summer destinations, will allow us to capitalize on Eurowings' market position.

We believe that leadership on a city/market/route level is a key driver of overall and relative performance. We are therefore continually looking to strengthen and protect our leadership positions.

12.2.2 Strategic Global Partnerships Further Leveraging Strong Network and Geographic Position

We believe that extended partnerships strengthen our network, competitive position and revenue quality due to a joint capacity management, joint product strategy, mutual market access, and last but not least revenue sharing. In 1996, Lufthansa German Airlines signed the commercial transatlantic joint venture A++ with the Star Alliance carriers United Airlines and Air Canada to foster growth and strengthen their presence in the North Atlantic market. SWISS and Austrian Airlines effectively joined the agreement in 2011.

Furthermore, we also have extended joint ventures in Asia, which we believe will be instrumental to a swift recovery from the COVID-19 pandemic as the Asian travel market is expected to outperform on growth. Lufthansa German Airlines and All Nippon Airways implemented J+, the first commercial joint venture between Europe and Japan, in 2011. The C+ Joint Venture between Air China, Lufthansa German Airlines, SWISS and Austrian Airlines, establishing a commercial joint venture among the companies for the Network Airlines has been concluded in September 2016. The S+ Joint Venture between Lufthansa German Airlines, SWISS, Austrian Airlines and Singapore Airlines established a commercial joint venture in November 2015 to operate key routes between Singapore and Europe, manifesting a wide-ranging partnership in operating key routes between Singapore and Europe on a contractual joint venture basis. The agreement also significantly expands codeshare ties and deepens commercial cooperation. For further information regarding our commercial joint ventures, see “12.7.1 Joint Ventures.”

In addition, we are the largest European member of the Star Alliance, the world's leading aviation alliance in terms of market share (measured by ASK) (*source*: Statista, Leading Airline Alliances in 2018, August 2019). The Star Alliance network served approximately 195 countries with more than 19,000 daily departure prior to the outbreak of COVID-19 (*source*: Star Alliance Facts & Figures, October 2019). The mutual recognition of frequent flyer miles among Star Alliance member airlines enables us to attract new customers from other member airlines. We believe that the strength of the Star Alliance route network and the consolidation of the frequent flyer programs in addition to our commercial joint ventures and numerous codeshare agreements across our global networks provide additional benefits to our customers. By cooperating with other Star Alliance member airlines, we are also able to achieve cost synergies, for example, by sharing ground facilities and through joint purchasing activities and revenue synergies by providing network feed for onward journeys.

12.2.3 Strongly Positioned for the Post-COVID-19 Recovery

The COVID-19 pandemic has had an unprecedented impact on the aviation industry. The travel restrictions imposed in reaction to the pandemic and the global economic situation pose fundamental challenges for the industry and the Lufthansa Group. It is estimated that demand will not regain its pre-crisis level any earlier than 2024 (*source*: IATA, Press Release Deep Losses Continue Into 2021). Nevertheless, we believe once COVID-19 subsides and travel restrictions start to be relaxed globally, the airline industry, in particular the tourism sector, will likely start to recover from the effects of the pandemic.

We have tackled the short-term effects of the pandemic by quickly adjusting available capacity to meet reduced demand, cutting costs, taking decisive steps to safeguard liquidity, and ensuring the Company's solvency by raising new funds, particularly in the form of government stabilization measures. See "*4 The Stabilization Package*."

Ultimately, we were also able to quickly respond to the effects of the COVID-19 pandemic with additional cost-cutting measures, including temporarily decommissioning our fleet, introduction of short-time work for the majority of our employees as well as terminating Wet Leases, postponing maintenance and servicing work on the fleet that is not safety-relevant, renegotiating supply contracts, reducing marketing activities and deferring projects that are not crucial for our business. In addition, we are implementing our "ReNew" restructuring program, which encompasses the consolidation of the individual restructuring programs undertaken at the airlines and service companies into one program at the Lufthansa Group level. See "*9.2.7 'ReNew' Restructuring Program*" and "*12.3 Our Strategy*."

We are anticipating the long-term effects of the crisis by adapting to a permanently smaller market with a comprehensive restructuring program and adjusting our strategy to make the best possible use of the market opportunities resulting from the changes brought by the crisis. Above all, this means focusing even more closely on individual customer needs, expanding our position in the tourism and short-haul segments and differentiating the various group airlines within the framework of our multi-airline model, while simultaneously increasing synergies and reducing complexity to further drive efficiency. We believe that we are well positioned to capitalize on these market developments. For example, we were able to almost double our Adjusted EBIT margin and our Adjusted ROCE between 2012 and 2019. We are set-up for maximum flexibility with regards to customer and route mix for the upcoming years. We have always serviced a large share of leisure and 'visiting friends and relatives' traffic and plan to adjust the composition of our fleet and the configuration of our aircraft and cabins flexibly, for example by expanding the premium offering, to cater to possible changes in the mix of customer demand, including a potentially slower recovery of demand in the business travel segment. Correspondingly, we seek to increase the relative share of our premium economy seats compared to the share of business and first class seats to offset the slower recovery of the business travel segment. With our ability to respond resolutely and quickly to changing market conditions, we firmly believe that we are very well positioned for the post-COVID-19 recovery phase.

12.2.4 Sustainability and Innovation Leadership

We believe we are a sustainability and innovation leader in the global aviation landscape. Sustainable management across environmental, employee and social aspects was and continues to be an integral pillar of our strategy. We aim to reduce emissions by 50% by 2030. In the Fiscal Year 2020, for example, 34 fuel-saving projects were under way across the Lufthansa Group. These projects comprised activities relating to

performance and procedures, weight reduction, flight route optimization and technical developments. Through these projects, we were able to permanently avoid 52.6 thousand tons of CO₂ emissions in the Fiscal Year 2020 alone. In 2020, the quantity of kerosene saved amounted to approximately 16.7 thousand tons – this is equivalent to approximately 196 return flights between Munich and New York with an Airbus A350-900 aircraft. We are engaged in a variety of projects to promote availability of sustainable aviation fuels. Alternative fuels produced from used cooking oil and cooking fats from agricultural production are generally available in small quantities. In recent years, we have been intensively involved in researching, testing and using sustainable aviation fuels and, in 2011, we pioneered the world's first long-term testing of bio kerosene in regular flight operations. Additionally, in 2002, the Company was the first airline globally to sign the United Nations Global Compact, an initiative to adopt sustainable and socially responsible policies, and to report on their implementation.

In addition, we intend to include sustainability aspects as a vital factor in the design of all of our products and services. Our platform Compensaid offers our passengers and those from other airlines attractive and personalized ways of offsetting their carbon emissions via various sales channels. In our product and service development, we systematically apply the principles “reduce – recycle – reuse – replace” to replace disposable plastic products with environmentally friendly products where possible and to reduce plastic packaging overall.

Digitalization has been at the forefront of our customer proposition, touching all aspects of the customer journey. Digitalization drives enhanced operational excellence through harmonization of our standards, processes, data and key performance measures – improving steering and planning throughout the Lufthansa Group. Digitalization is a key factor of developments in revenue management, customer loyalty and satisfaction. For instance, with One ID our customers will be able to establish a single profile, utilizing connected permissions across all our carriers and digital touchpoints, including Miles & More, thus improving their experience through better data usage and platform compatibility. Conversely, One ID will allow us to increase access to our customers, accelerate direct distribution and generate cost savings due to the integration of our platforms. We intend to make One ID available to our customers in the course of 2022. We have also won multiple awards supporting our innovation leadership in this area. For example, Lufthansa German Airlines won the German Innovation Prize in the category Future Digital Inflight Experience for its new in-flight experience platform, creating seamless, consistent and contextualized touchpoints along the entire travel chain. The Lufthansa Innovation Hub won the Best Digital Lab Award for the third time in 2020. This was largely due to its development of Compensaid. Eurowings Digital won the Best Digital Lab Award 2020 from the business magazine Capital, acknowledging the development of smart services with which Eurowings accompanied its passengers throughout their journey. Additionally, Star Alliance completed its development of a cross-system biometric identity and identification platform known as Star Alliance Biometrics in November 2020. Customers in the Miles & More frequent flyer program thus enjoy contact-free access to the security checks and boarding gates without removing their face masks, which also represents an important contribution to health and safety during the COVID-19 pandemic.

12.2.5 Experienced Management Team

We have a strong and experienced management team, with members from diverse backgrounds, both from within the aviation industry and other sectors. Our management team incorporates best practices throughout the organization and has extensive experience with strong track records of driving value and delivering attractive shareholder returns. We believe that the diversity of executive backgrounds and experience allows the management team to contribute a balanced mix of perspectives on how to deal with the current COVID-19 situation, adapt to the post-crisis travel landscape and integrate best practices from outside the aviation sector into areas such as technology and digitalization. The strength and experience of the management team facilitates an agile response to sudden changes in passenger demand and preferences, such as the impact of the COVID-19 pandemic.

12.3 Our Strategy

12.3.1 Focus on our Core Airline Business

The Lufthansa Group is a portfolio of strong, market leading businesses across the aviation value chain. Our strategic goal to focus the Lufthansa Group on its core airline business was set already before the COVID-19

crisis. As part of our “ReNew” restructuring program, this process has been reconfirmed as a main aspect of our strategy. Following the successful sale of the European business of the LSG group, disposals of other non-airline companies offering only limited synergies with our core business are also under review and may follow in case their full value can be realized. For example, disposal and partnering options for all or part of Lufthansa Technik as well as the disposal of AirPlus and the non-European business of the LSG group are under review, taking into account the necessary capital expenditures, synergies and value creation potential, and we have mandated advisors with respect to Lufthansa Technik to evaluate such options.

In addition, we are implementing an enhanced matrix steering initiative that combines certain core airline functions into the clusters “Corporate Function,” “Competence Center” and “Core Airline Matrix Functions.” The objective of this initiative is to increase efficiency and performance, resulting in a leaner organization with streamlined board structures. Additionally, management decisions shall, in principle, be taken on the lowest possible level and only be centralized in the event that a certain value or synergies for the Lufthansa Group can be established. We plan to increase synergies around the core airline business of the Lufthansa Group. Key drivers include joint production in the cargo business, for example, by which Lufthansa Cargo transports almost half of its freight in the belly capacities of the Lufthansa Group’s passenger aircraft, or revenue synergies between the airlines and our loyalty program Miles & More in the loyalty business. Other key synergy areas relate to aircraft purchasing and maintenance, staff training, marketing and the appropriate usage of our digital platforms and customer data.

In addition, we still believe there is considerable scope for consolidation within the European airline sector, which continues to be significantly more fragmented than the North American market. In the mid to long-term, we believe we are well positioned to be a key driver of this consolidation, leveraging our existing strength and platform to capitalize on synergistic opportunities as they arise.

12.3.2 Integrating Sustainability to Reinforce our Environmentally and Socially Responsible Organization

We aspire to lead the airline industry with high standards of corporate and social responsibility. We therefore continuously build on our environmental commitment, are dedicated to socially responsible actions and are committed to rigorous standards of responsible and fair treatment of employees. We target to cut our net carbon emissions in half by 2030 compared to 2019 and support the objective of making aviation carbon neutral by 2050. As a result, we continue to invest in fuel-efficient aircraft, accompanied by steps to increase the operational efficiency of the airlines, utilizing carbon offsets and the further expansion of intermodal traffic in cooperation with Deutsche Bahn, Austrian Railways and various bus operators. We also work with policymakers and partners in industry, technology and research to support the industrialization and use of sustainable fuels. We have been closely involved with the research, testing and use of sustainable aviation fuels and have been a key driver of the development of new technologies through multiple partnerships.

The ongoing optimization of our fleet has been accelerated as a result of the COVID-19 pandemic. Key targets are the systematic renewal of the fleet to cut fuel consumption, reduce carbon emissions and the simplification of the long-haul fleet by storing, retiring and phasing out older, less efficient aircraft, such as the Airbus A340-600 and Airbus A380. Phasing out large aircraft also increases operating flexibility. Accordingly, as part of the “ReNew” restructuring program, the Lufthansa Group intends to significantly reduce its fleet.

Regarding the responsibility for our employees, we put great emphasis on offering our staff an attractive working environment with transparent structures, efficient processes and a wide range of voluntary social benefits. This is vital for retaining committed employees who can tackle future challenges and for ensuring that we remain competitive and successful. In terms of social responsibility, we promote greater equality of opportunity for disadvantaged people worldwide via the financial and material support for educational establishments and training provided by helpalliance, our own aid organization.

12.3.3 Return to Value Creation by Restoring our Profitability and Generating Strong Cash Flows

Sustainable value creation is a key objective of our financial strategy. We apply a value-based management system focused on maximizing capital returns. Our “ReNew” restructuring program aims at aligning the business model and cost structures with changes in the market environment resulting from the COVID-19

crisis. It is designed to right-size our business and ensure that the cost structures reflect the lower market volumes that we expect to persist for several years. The necessary measures are being identified and implemented in all our business segments, so that we can return to a positive operating result once demand normalizes.

In addition, we have a structured approach to optimizing our use of capital through strict working capital management and a targeted, return-focused allocation of capital. Our aim is to increase structural profitability by reducing unit costs, simplifying processes and structures and improving productivity to ultimately improve our operating cash flow. Working capital measures focus on strict receivables management, improving or extending the payment targets for suppliers and optimizing inventories. Capital expenditure or investment measures will extend to, but are not limited to, demand-led fleet investment at those airlines which offer the highest returns and investment in innovative technologies that allow for greater operational and financial efficiency.

12.3.4 Restore Financial Stability

The refinancing and repayment of governmental stabilization measures we received is an integral element of the future financial strategy aimed at restoring financial stability. In particular, we strive to (i) reduce our net debt, (ii) return to investment grade rating, (iii) ensure a sufficient level of liquidity at all times and (iv) minimize our financial risks. We believe that the combination of these measures will ensure we have the stability and financial flexibility to support the businesses recovery in the short-term, and to position the Lufthansa Group to re-establish a sustainable growth trajectory and shareholder returns in the mid to long-term.

Reduce our State Aid and Debt. We aim to repay the state aid that we have received as quickly as possible and also reduce our overall level of debt after the COVID-19 crisis. We believe that cash inflows from the operating business and proceeds from the disposal of non-core business segments will contribute to reducing our net indebtedness. Net indebtedness (calculated as the sum of current and non-current borrowings (€15,571 million) and other borrowings (€25 million) less cash and cash equivalents (€2,063 million) and securities (€4,603 million)) was €8,930 million as of June 30, 2021. We aim to use a mix of different instruments for borrowing, in particular aircraft financing, bonds and promissory notes (*Schuldscheindarlehen*). By optimizing the funding mix, we aim to reduce financing costs, maintain a balanced maturity profile and diversify our portfolio of creditors.

Return to Investment Grade Rating. Investment grade ratings for our debt ensure low funding costs and grant financial flexibility. The conditions for an investment grade rating are a return to a positive operating result, lower net debt and a stronger equity base. Gearing, measured by the ratio of adjusted net debt (including pension provisions) to Adjusted EBITDA, is targeted to be reduced to a figure of less than 3.5x.

Ensure sufficient liquidity level. Following the outbreak of the COVID-19 pandemic, capital markets were temporarily inaccessible for the Lufthansa Group, resulting in the need of government stabilization measures to maintain solvency. To ensure that we have sufficient liquidity for future crises, we have decided to increase the targeted minimum liquidity from around €2.3 billion before the crisis to approximately €6-8 billion. Additionally, we seek to achieve a positive adjusted free cash flow in 2022 also driven by the expected increase in capacity compared to the Fiscal Year 2021, cost reduction measures and active working capital management.

Minimize our Financial Risks. We aim to ensure our financial stability by means of integrated risk management. For example, hedging fuel, exchange rate and interest rate risks is geared towards minimizing our short-term financial risks. The hedges smooth price fluctuations by means of rule-based processes. The significant fall in capacity in 2020 meant that fuel was overhedged. As a result, we adjusted our fuel hedging strategy by lowering the hedge ratio based on our ability to pass on fuel price increases to customers and by increasing protection against sudden and parallel declines in consumption and prices.

We will continuously analyze potential additional measures targeted to maintain financial stability during a crisis. This may be particularly challenging knowing that, as evidenced by the COVID-19 pandemic, we may find ourselves in a position unable to achieve protection against significant levels of loss, such as a global pandemic.

12.4 History

The Company was founded in Cologne, Germany, in 1953 under the name “Aktiengesellschaft für Luftverkehrsbedarf” and has been operating since 1954 as “Deutsche Lufthansa Aktiengesellschaft.” We started operating flights within Germany on April 1, 1955 and later expanded our flight operations abroad.

Parallel to the development of a European route network, flights to destinations in North America and then to South America and the Far East were also added in 1955 and 1956, respectively. In 1958, it carried one million passengers for the first time. In 1959, the Company operated its first scheduled flight to Asia.

In 1966, the Shares were first traded on the German stock exchanges. In 1997, the German government sold its remaining interest in the Company, thereby completing its full privatization. The Shares were listed on the Frankfurt Stock Exchange on October 13, 1997. Our organically grown MRO and cargo businesses were integrated into the Lufthansa Group in 1994 as Lufthansa Technik and Lufthansa Cargo. In 2006, we acquired and subsequently integrated SWISS and, furthermore, in 2009, we acquired and integrated Austrian Airlines as well as Germanwings, which predominantly operates under the commercial name Eurowings since November 2015.

12.5 Our Business Segments

We divide our business into six business segments: (i) Network Airlines; (ii) Eurowings; (iii) Logistics, (iv) MRO; (v) Catering; and (vi) Additional Businesses and Group Functions.

12.5.1 Network Airlines Business Segment

We provide a comprehensive route network through our multi-hub strategy combined with a high level of flexibility. As of June 30, 2021, the Network Airlines business segment comprised Lufthansa German Airlines (including regional partners and Miles & More), SWISS, Austrian Airlines, Brussels Airlines and Eurowings Discover as well as Germanwings, whose passenger flight operations have been discontinued in 2020. Lufthansa German Airlines is the biggest German airline, with hubs in Frankfurt, Germany and Munich, Germany. SWISS is, together with its affiliate Edelweiss Air, based in Zurich, Switzerland, while Austrian Airlines is based in Vienna, Austria. Brussels Airlines’ flights depart from and to Brussels, Belgium. The Lufthansa CityLine and Air Dolomiti regional airlines are also part of Lufthansa German Airlines, which primarily operate within Germany and into and out of Italy, respectively. In order to exploit opportunities in the long-distance leisure travel market, we envisage to expand our travel offering aimed at private travelers from our hubs in Frankfurt, Germany, and Munich, Germany. Eurowings Discover will carry out certain routes, including flights from Frankfurt, Germany to Punta Cana (Dominican Republic), Mombasa (Kenya) and Zanzibar (Tanzania). Ultimately, passengers of Eurowings Discover will be in a position to benefit from the broad network and the established ground processes of Lufthansa German Airlines. Additionally, we engage in a variety of commercial joint ventures (see “12.7.1 Joint Ventures” for additional information).

As a result of the COVID-19 pandemic, the Network Airlines have reduced their flight capacity significantly and temporarily decommissioned a large part of the fleet. At times, the Network Airlines only operated a minimum flight program. Austrian Airlines and Brussels Airlines completely suspended flight operations between March 18 and 21, 2020, respectively, and June 14, 2020. For the 6M 2021, capacity (measured by ASK) decreased by 35% compared to the 6M 2020 and the total revenue of our Network Airlines business segment for the 6M 2021 decreased by 50% year-over-year. All Network Airlines carried out numerous repatriation flights on behalf of their governments to fly travelers home from all over the world, sometimes even while their regular flight operations were suspended. Passenger aircraft were also used for cargo flights in order to transport urgently needed medical equipment.

In the 6M 2021, our Network Airlines business segment included 286 destinations in 102 countries, which we served via the global hubs in Frankfurt, Germany, Munich, Germany, and Zurich, Switzerland, as well as the national hubs in Brussels, Belgium, and Vienna, Austria (Fiscal Year 2020: 328 destinations in 109 countries, Fiscal Year 2019: 357 destinations in 109 countries, and Fiscal Year 2018: 371 destinations in 111 countries).

The table below shows certain operating metrics by company for the Network Airlines business segment for the periods indicated.

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019 ⁽¹⁾	2018 ⁽¹⁾
	(unaudited)		(unaudited)		
Network Airlines⁽²⁾					
Passengers (in thousands).....	8,564	19,220	29,119	118,328	103,639
Number of flights	100,603	185,366	310,023	945,621	840,945
Available seat-kilometers (in millions) ⁽³⁾	41,314	63,173	99,568	326,420	284,642
Revenue seat-kilometers (in millions) ⁽³⁾	19,721	45,230	61,961	269,578	232,030
Passenger load factor (in %) ⁽⁴⁾	47.7	71.6	62.2	82.6	81.5
Unit revenue (in € cent) ⁽⁵⁾	5.9	7.5	7.2	7.8	8.1 ⁽⁶⁾
Unit cost excluding fuel (in € cent) ⁽⁵⁾	10.5	9.5	10.3	5.5	5.6 ⁽⁶⁾
Lufthansa German Airlines⁽⁷⁾					
Passengers (in thousands).....	5,432	12,071	17,996	72,470	69,886
Number of Flights	62,348	115,990	192,216	565,703	554,615
Available seat-kilometers (in millions) ⁽³⁾	27,283	41,350	64,480	212,948	196,789
Revenue seat-kilometers (in millions) ⁽³⁾	13,164	29,586	40,064	175,762	160,074
Passenger load factor (in %) ⁽⁴⁾	48.2	71.6	62.1	82.5	81.3
SWISS⁽⁸⁾					
Passengers (in thousands).....	1,343	3,602	5,677	21,531	20,432
Flights.....	15,680	32,239	54,835	167,119	159,533
Available seat-kilometers (in millions) ⁽³⁾	8,159	13,406	21,489	63,321	60,519
Revenue seat-kilometers (in millions) ⁽³⁾	3,214	9,747	13,069	53,116	50,283
Passenger load factor (in %) ⁽⁴⁾	39.4	72.7	60.8	83.9	83.1
Austrian Airlines					
Passengers (in thousands).....	1,112	1,986	3,114	14,613	13,934
Flights (in numbers)	16,286	23,635	42,310	139,230	135,293
Available seat-kilometers (in millions) ⁽³⁾	2,906	4,598	7,127	28,508	27,703
Revenue seat-kilometers (in millions) ⁽³⁾	1,543	3,131	4,412	23,043	21,965
Passenger load factor (in %) ⁽⁴⁾	53.1	68.1	61.9	80.8	79.3
Brussels Airlines					
Passengers (in thousands).....	676	1,590	2,362	10,285	-
Flights (in numbers)	6,295	14,114	21,280	81,540	-
Available seat-kilometers (in millions) ⁽³⁾	2,966	3,834	6,487	21,994	-
Revenue seat-kilometers (in millions) ⁽³⁾	1,801	2,776	4,428	17,929	-
Passenger load factor (in %) ⁽⁴⁾	60.7	72.4	68.3	81.5	-

- (1) The figures for the Fiscal Year 2019 were adjusted retrospectively to account for the changes in the segment allocation of Brussels Airlines, Germanwings, the long-haul operation of Eurowings and the transfer of maintenance activities from the MRO segment to the Network Airlines Segment. The figures for the Fiscal Year 2018 were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.
- (2) A translation of the figures from Lufthansa German Airlines, SWISS and Austrian Airlines to the Network Airlines business is only possible when considering consolidation effects. As of January 1, 2020, Brussels Airlines is considered as a part of the Network Airlines business segment.
- (3) Standard output units for air transport. An available seat-kilometer (ASK) denotes one offered seat flown for one kilometer. Revenue seat-kilometer corresponds to revenue passenger-kilometer (RPK), which denotes one paying passenger transported for one kilometer.
- (4) Measure of capacity utilization in per cent. The passenger load factor refers to passenger transportation.
- (5) Key performance indicator for air transport. Unit costs (CASK) denote the operating expenses divided by offered seat kilometers. Unit revenue (RASK) denotes the revenue divided by offered seat-kilometers.
- (6) Without effects from reclassification of compensation payments for flight delays.
- (7) Including regional partners.
- (8) Including Edelweiss Air.

As a result of the COVID-19 pandemic, we aim to scale back and to continuously modernize our fleet. Among other things, this is expected to entail the long-term storage and/or retirement of long-haul aircraft that are older or less efficient than the rest of the fleet. This comprises the entire Airbus A380 and Airbus A340-600 sub-fleets, which are expected to be stored or decommissioned in the short- to medium-term. At the same time, we believe to be in a position to deploy modern aircraft models which could ultimately reduce costs, fuel consumption and CO₂ and noise emission levels.

In order to give customers maximum flexibility during the COVID-19 pandemic, we offer rebooking options and guarantee that our passengers will have a return flight on all European routes irrespective of the fare booked. Passengers have also the option to purchase additional COVID-19 insurance services. We aim to guarantee the highest safety and health precautions for our passengers. In order to comply with governmental orders, we imposed mandatory wearing of face masks throughout the flight. Furthermore, Lufthansa German Airlines and Austrian Airlines carried out a pilot project to use rapid antigen tests on all passengers on selected flights between Munich and Hamburg and between Vienna and Hamburg. Boarding passes were only activated and access to the gate provided once a negative result was obtained. The aforementioned actions are some of the results of our strategic 'New Premium' vision which is aimed at optimizing our products, such as by offering our customers an option to book free adjacent seats on European continental routes in the Economy class, and services along the full range of the travel chain by improving health and safety while, simultaneously, making flight bookings more flexible. Among others, we currently offer flexible rebookings for all fares on short, medium and long-haul flights.

12.5.2 Eurowings Business Segment

Through our Eurowings business segment, we offer an innovative point-to-point traffic for price-sensitive and value-oriented customers in the structural growth segment of European point-to-point traffic. Our flight schedules include, among others, a focus on tourism-relevant routes, as a result of which we maintain significant relationships with tour operators.

As of June 30, 2021, the Eurowings business segment comprised Eurowings and Eurowings Europe, as well as SunExpress, in which we hold a 50% equity investment and of which a subsidiary, SunExpress Germany, has discontinued its passenger flight operations in 2020. The Eurowings concept is based on high productivity, the central management of different flight operations and on a scalable company structure that enables the integration of new partners with a variety of cooperation models, both in terms of pricing as well as packaged travel deals. Eurowings Digital won the Best Digital Lab Award 2020 from the business magazine Capital for its development of innovative and smart services that accompany passengers throughout their journey.

For the 6M 2021, the Eurowings business segment included 97 destinations in 28 countries (Fiscal Year 2020: 149 destinations in 42 countries; Fiscal Year 2019: 168 destinations in 44 countries, and Fiscal Year 2018: 171 destinations in 46 countries).

The table below shows certain operating metrics for the Eurowings business segment for the periods indicated.

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018 ⁽¹⁾
	(unaudited)		(unaudited)		
Passengers (in thousands).....	1,458	4,255	7,235	26,971	38,297
Number of flights	15,854	40,571	71,829	232,461	322,620
Available seat-kilometers (in millions) ⁽²⁾	2,857	5,431	10,260	32,383	64,748
Revenue seat-kilometers (in millions) ⁽²⁾	1,895	4,283	7,501	26,639	52,609
Passenger load factor (in %) ⁽³⁾	66.3	78.9	73.1	82.3	81.3

(1) The figures for the Fiscal Year 2018 were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.

- (2) Standard output units for air transport. An available seat-kilometer (ASK) denotes one offered seat flown for one kilometer. Revenue seat-kilometer corresponds to revenue passenger-kilometer (RPK), which denotes one paying passenger transported for one kilometer.
- (3) Measure of capacity utilization in per cent. The passenger load factor refers to passenger transportation.

By reducing staff costs, decommissioning of older aircraft and making the Airbus A320 our aircraft standard, we aim to achieve savings in operating costs and project budgets. In addition, we also aim to guarantee highest safety, service and health measures for our passengers while offering them flexible and customer-friendly rebooking options. This included the option to book a free middle seat before the start of their journey.

In 2019, Eurowings successfully implemented a substantial restructuring program aimed at streamlining its cost base and achieving operational synergies. In order to prepare the Eurowings division for the post-crisis period and in an effort to restore financial stability and profitability, we intensified our restructuring program by applying various measures. On the one hand, we focus on further increasing the productivity of aircraft and crew as well as the reduction of overhead costs. Additionally, our strategy includes increasing ancillary revenues (seat reservations, free middle seats, etc.) and the expansion of leisure and ‘visiting friends and relatives’ traffic. Moreover, we envisage to further expand Eurowings Europe, including by opening new operations bases across the EU.

12.5.3 Logistics Business Segment

We are one of Europe’s leading freight airlines providing comprehensive airport-to-airport airfreight services (*source*: Air Cargo Awards 2020). As of June 30, 2021, the Logistics business segment included Lufthansa Cargo, the airfreight container management specialist Jettainer group, the time:matters group, which specializes in particularly urgent consignments, our subsidiary Heyworld, which specializes in tailored solutions for the e-commerce sector, and the equity investment in the cargo airline AeroLogic GmbH (“**AeroLogic**”). Moreover, Lufthansa Cargo has equity investments in various handling companies and smaller companies involved in aspects of digitalizing the sector. Additionally, we engage in a variety of commercial joint ventures, including in relation to our offering of logistics solutions (see “12.7.1 Joint Ventures” for additional information).

We offer standard and express freight as well as highly specialized products. These include the transport of live animals, valuable cargo, post and dangerous goods, as well as meeting growing market demand for the carriage of temperature-sensitive goods. We have a specialized infrastructure at the Frankfurt airport to handle these goods, including the Animal Lounge and the Lufthansa Cargo Cool Center.

The table below shows certain financial information for the Logistics business segment for the periods indicated.

	For the six-month period ended June 30,		For the year ended December 31,		
	2021	2020	2020	2019	2018 ⁽¹⁾
	(unaudited)		(unaudited)		
Cargo load factor (in %) ⁽²⁾	74.3	65.9	69.1	61.3	65.9
Available cargo ton-kilometers (in millions) ⁽³⁾	4,683	4,738	9,350	14,507	13,555
Revenue cargo ton-kilometers (in millions) ⁽²⁾	3,481	3,122	6,461	8,899	8,934

- (1) The figures for the Fiscal Year 2018 were adjusted retrospectively to reflect the restated compensation payments for flight cancellations and delays.
- (2) Measure of capacity utilization in per cent. The cargo load factor expresses the ratio of capacity sold to available capacity. The cargo load factor refers to freight transport or total traffic.
- (3) Standard output units for air transport. An offered ton-kilometer (TKO) denotes the offered capacity equivalent of one ton of load (passengers and/or cargo) for one kilometer; a revenue ton-kilometer (RTK) denotes one ton of load (passengers and/or cargo) transported one kilometer.

Normally, around half of the freight volume at Lufthansa Cargo is transported in the belly capacities of passenger aircraft that are operated by Lufthansa German Airlines, Brussels Airlines, Austrian Airlines, Eurowings long-haul routes and SunExpress. The other half is transported by our freighter fleet, which consists of thirteen Boeing 777F (including four at AeroLogic) and two Boeing MD-11F as of June 30, 2021.

A part of the belly capacities carried through our Network Airlines was not available in the Fiscal Year 2020 or in the 6M 2021. As a result of the COVID-19 pandemic, capacity fell by 36% and 1% in the Fiscal Year 2020 and the 6M 2021, respectively, compared to the previous periods. However, we were able to offset the decreased capacity due to reduced belly capacities by means of flexible network planning, the temporary use of converted passenger aircraft, deferral of the retirement of the MD11-fleet and our Proflex cost-cutting program, which includes automated planning and control at hubs and gateways as well as simplified cargo net storage, automated embargo check and mobile digital handling improvement, resulting in structural improvements to the cost base.

Furthermore, we expanded our infrastructure for the transport of temperature-sensitive pharmaceutical products and opened pharma hubs in Chicago and Munich in addition to 30 Center of Excellence for Independent Validators in Pharmaceutical Logistics (“CEIV”) certified stations worldwide. We further opened the Frankfurt pharma hub, which is the largest of its kind in Europe and employs specially trained staff. We believe that we operate the world’s largest airline pharmaceutical network. For the transport of COVID-19 vaccines, we specifically developed a premium service – COVID-19 Temp Premium – which is in operation since January 11, 2021. It provides a high level of comprehensive and personalized customer service along the travel chain, including seamless monitoring of vaccine shipments throughout the entire process and a 24/7 hotline. We believe that Lufthansa Cargo is ideally prepared for the rapid intercontinental distribution of COVID-19 vaccines and other highly sensitive pharmaceutical products.

12.5.4 Maintenance, Repair and Overhaul Services Business Segment

We believe Lufthansa Technik is one of the world’s leading providers of MRO services for civilian commercial aircraft. Lufthansa Technik is certified around the globe as a maintenance, design and production organization (within the International Civil Aviation Organization (the “ICAO”) and EU Aviation Safety Agency (the “EASA”) regulations to Part 145, Part 21/J and Part 21/G, respectively). Headquartered in Hamburg, Germany, and with its services firmly embedded into our global network, as of June 30, 2021, Lufthansa Technik comprises 35 plants globally, through which it offers technical aviation services worldwide. In addition, Lufthansa Technik holds direct as well as indirect stakes in 62 companies as of June 30, 2021. The portfolio covers a variety of products and service combinations, ranging from the repair of individual components to consultancy and digital services to the fully integrated supply for and of our fleet. As of June 30, 2021, our service portfolio encompasses eight divisions: (i) aircraft maintenance, (ii) aircraft overhaul, (iii) engine maintenance, (iv) component maintenance, (v) aircraft systems, (vi) development and manufacture of cabin products, (vii) digital solutions, and (viii) initial equipment and servicing of VIP aircraft. However, we have streamlined organizational structures and aligned our service portfolio towards five divisions. From July 1, 2021 onwards, our service portfolio encompasses five divisions: (i) aircraft maintenance and overhaul, (ii) engine maintenance and production, (iii) component maintenance, (iv) digital solutions technical flight operations, and (v) modification and special aircraft.

In the Fiscal Year 2020, Lufthansa Technik group serviced 4,529 aircraft under exclusive contracts from more than 800 customers worldwide. We also won 16 new customers and signed 515 contracts with a volume of €2.3 billion during the same period.

As a result of the COVID-19 pandemic, total revenue of our MRO business segment decreased by 43% for the Fiscal Year 2020 compared to the Fiscal Year 2019 and 25% for the 6M 2021 compared to the 6M 2020. In order to safeguard continued viability, we implemented comprehensive cost-cutting measures such as the introduction of short-time working, recruitment freezes, extensive reductions in operating costs and the cancellation or postponement of investment projects.

We believe that the COVID-19 pandemic is resulting in an increase in demand for digital MRO services such as crisis products, as a result of which physical maintenance events can be organized more reliably and cost-efficiently in the future. Analyses provided by consultancy firm ICF International Inc. indicate that the MRO will recover to its 2019 level by 2024. One of our proprietary digital MRO services includes the integrated digital platform AVIATAR. AVIATAR supports customers in real time with the management of complex fleet operations and helps to diagnose errors in individual components, thereby supporting the digital transformation of airline operations. United Airlines joined the platform with more than 600 aircraft towards the end of the Fiscal Year 2020. Furthermore, as part of Lufthansa Technik’s flexible response to the

COVID-19 pandemic, it is expanding its product offering to cover crisis products including, among others, the facilitated conversion of passenger aircraft into freighters.

12.5.5 Catering Business Segment

We offer, through the LSG Group, a broad product, concepts and services portfolio related to in-flight service. In particular, the range of service includes several catering activities, in-flight sales and entertainment, in-flight service equipment, the associated logistics, consultancy services and the operation of lounges.

We believe that the LSG Group, headquartered in Neu-Isenburg, Germany, is one of the leading global airlines caterers. As of June 30, 2021, we operate with two well-established and independent expert brands: LSG Sky Chefs and Retail inMotion. LSG Sky Chefs is a global catering specialist with a particular focus on hygiene and quality standards for airlines as well as for the home-delivery and retail markets. Retail inMotion specializes in on-board retail, product development and technology solutions. In the Fiscal Year 2020, the LSG Group generated revenues of €945 million and its approximately 12,500 employees produced more 240 million meals for over 190 active airline customers globally.

On December 2, 2020, we closed the sale of the European business of the LSG Group, which generated approximately 30% of the LSG Group's revenue in the Fiscal Year 2019 and comprises a team of approximately 7,500 employees, to gategroup. In addition to the European catering facilities, the transaction included the sale of the lounge business, the European activities of the convenience-retail specialist Evertaste, the LSG Group's equipment business SPIRIANT and the Germany-based Ringeltaube retail shops. The transaction also includes a long-term catering contract with Lufthansa German Airlines for its operations in Germany, including its hubs in Frankfurt, Germany, and Munich, Germany and a long-term catering contract with SWISS for its operations in Switzerland, including its hub in Zurich, Switzerland. In order to ensure and facilitate a successful transfer of the catering business, we retained a minority interest in the two aforementioned facilities (see "12.16.4 Lufthansa Hub Catering Contract (LHCC)"). Divestment and partnering options for all or part of the international business of the LSG Group are also under review, taking into account the necessary capital expenditure, synergies and value creation potential.

As the strongest revenue driver in the LSG Group, LSG Sky Chefs offers classical catering for airlines, as well as lounge management. After the sale of its European business, the LSG Group continued to, as of June 30, 2021, serve more than 190 customers from its 134 facilities in 44 countries.

Shortly after the outbreak of the COVID-19 pandemic, the LSG Group counteracted the significant revenue losses incurred as a result thereof with a strict cost management emphasis and a strategic realignment process. Key components of the cost-cutting and restructuring package included global staff outplacements. During the Fiscal Year 2020 and as a result of the drastic adverse business impact of the COVID-19 pandemic, the LSG Group was forced to discontinue relationships with approximately 52% of its workforce and approximately 30% of individuals in management positions.

Going forward, the LSG Group has re-positioned itself to focus on five revenue streams: (i) classical catering services, (ii) in-flight retail sales, (iii) end-to-end IT and platform solutions, (iv) convenience retail, and (v) home delivery. Furthermore, LSG Sky Chefs was able to renew or sign new key catering contracts in all regions during the Fiscal Year 2020, including with American Airlines, Atlas Air, Azul, British Airways, Delta Airlines, Emirates, Etihad, Hawaiian Airlines, United Airlines and Westjet, as well as five-year contracts with Frontier Airlines, which operates in North America. The LSG Group has decided to remain active in the European market with Retail inMotion and its pan-European customer contracts. Moreover, during the Fiscal Year 2020, it secured long-term contracts in the convenience-retail business across all of the regions in which it operates, and, as a part of its strategic realignment, launched additional projects in the home-delivery segment. Further to its dedication to translate its expertise into new markets, the LSG Group has committed to contribute to sustainability, among others by the application of six sustainable development goals (SDGs), as recommended by the United Nations.

12.5.6 Additional Businesses and Group Functions

The Additional Businesses and Group Functions business segment comprises the service and financial companies, particularly AirPlus, Lufthansa Aviation Training and Lufthansa Systems, as well as the group functions of the Lufthansa Group.

We believe that AirPlus is a leading international provider in the global market for payment and billing services in terms of market share. Under the AirPlus International brand, we offer solutions in over 60 countries worldwide. The company served more than 48,000 corporate customers in the Fiscal Year 2020. Whereas our business travel management unit suffered from lower volumes in the 6M 2021 and the Fiscal Year 2020 as a result of global restrictions to contain the COVID-19 pandemic, the strategic expansion of activities in the corporate payment sector demonstrated significant success. The new AirPlus Virtual Cards for Procurement reported a positive performance and were already in use by companies in eleven countries as of June 30, 2021. Divestment and partnering options for all or part of AirPlus are also under review, taking into account the necessary capital expenditure, synergies and value creation potential.

LAT is one of the leading flight training companies by number of flight simulators operated (*source: CAT SIM Census*), providing vocational and professional training for cockpit and cabin crew at twelve training centers. LAT's customer portfolio includes approximately 250 national and international airlines as of June 30, 2021. As a result of the effects of the COVID-19 pandemic, Lufthansa Aviation Training is undergoing a restructuring process in an effort to align its pilot flight school to adequately service a decreasing demand for pilots in the future. Lufthansa Aviation Training implemented the Future One EFA Pilot School project to maintain in-house training facilities while realizing a sustainable flight school strategy which reflects the increasing volatility in the demand for pilots.

Lufthansa Systems offers a range of solutions and advisory services relating to improving the efficiency and differentiating all areas of an airline, and to optimizing passengers' entire travel experience. Lufthansa Systems' customer base includes more than 350 airlines around the world as of June 30, 2021. Despite the difficult market environment caused by the COVID-19 pandemic, Lufthansa Systems was able to maintain its strong customer base throughout the international airline IT market by offering flexible solutions to its customers. This enabled it to win additional new customers and bring numerous projects to a successful close in the Fiscal Year 2020.

12.6 Operations

12.6.1 Fleet

Our fleet comprises aircraft of almost every size and uses state-of-the-art technology. As of June 30, 2021, our fleet included 734 aircraft, of which we owned 88% and leased 12%. The average age of the aircraft in the fleet was 12.6 years (December 31, 2020: 12.5 years; December 31, 2019: 12.1 years and December 31, 2018: 11.9 years).

Between January 1, 2018, and June 30, 2021, a total of 105 aircraft were added to the fleet, while 99 aircraft left the fleet. Fleet additions included 77 newly built aircraft (four Boeing 777Fs, four Boeing 777-300s, ten Airbus A350-900s, 12 Airbus A321neos, 23 Airbus A320neos, nine Airbus A320ceos and 15 Airbus A220s), 15 used aircraft (mainly Airbus A330s and Airbus A320ceos) and 13 leased aircraft. In contrast, 66 aircraft were sold between January 1, 2018, and June 30, 2021 (ten Boeing MD-11Fs, five Boeing 747s, two Boeing 767, seven Airbus A340-600s, one Airbus A340-300, one Airbus A321ceo, 18 Airbus A320ceos, four Airbus A319s, three Bombardier CRJ900, 14 Bombardier Q Series and one Fokker 100 aircraft) and leases were terminated for 33 aircraft. Additionally, we did not renew dry leases expiring in the Fiscal Year 2020 and the 6M 2021 and we terminated dry leases during the same period.

The table below sets forth the Lufthansa Group's fleet owned and leased as of June 30, 2021 and planned additions from July 1, 2021, to 2029, including options but excluding disposals. This table does not indicate whether the Company or one of its group companies holds legal title to the aircraft, nor does it indicate whether the aircraft are actually in operation.

Manufacturer and Model	Operator						Group fleet	Thereof leases ⁽¹⁾	Planned additions from July 1, 2021 to 2029 ⁽²⁾	Additional options
	LH ^(*)	LX ^(*)	OS ^(*)	SN ^(*)	EW ^(*)	LCAG				
Airbus A220.....	–	30	–	–	–	–	30	–	–	30
Airbus A319.....	42	–	7	18	34	–	101	29	–	–
Airbus A320.....	100	32	29	16	56	–	233	36	68	–
Airbus A321.....	73	11	6	–	4	–	94	2	36	–
Airbus A330.....	26 ⁽³⁾	16	–	8	–	–	50	8	–	–
Airbus A340.....	34	9	–	–	–	–	43	–	–	–
Airbus A350.....	17	–	–	–	–	–	17	1	28	–
Airbus A380.....	14	–	–	–	–	–	14	–	–	–
Boeing 747.....	27	–	–	–	–	–	27	–	–	–
Boeing 767.....	–	–	4	–	–	–	4	–	–	–
Boeing 777.....	–	12	6	–	–	–	18	2	20	24
Boeing 787.....	–	–	–	–	–	–	–	–	25	20
Boeing 777F.....	–	–	–	–	–	13 ⁽⁴⁾	13	4	1	–
Boeing MD-11F.....	–	–	–	–	–	2	2	–	–	–
Bombardier CRJ.....	32	–	–	–	–	–	32	–	–	–
Bombardier Q Series.....	–	–	4	–	9	–	13	9	–	–
Embraer.....	26	–	17	–	–	–	43	–	–	–
Total aircraft	391	110	73	42	103	15	734	91	178	74

(*) LH refers to Lufthansa German Airlines (including regional partners); LX refers to SWISS including Edelweiss; OS refers to Austrian Airlines; EW refers to Eurowings including Germanwings; SN refers to Brussels Airlines; and LCAG refers to Lufthansa Cargo.

(1) From January 1, 2019, operating leases are recognized as finance leases pursuant to IFRS 16 and are therefore consolidated under “Leases.”

(2) Excluding disposals.

(3) Partly operated by Brussels Airlines.

(4) Partly operated by AeroLogic, of which two aircraft are attributed *pro rata*.

Due to the COVID-19 pandemic, we expect the market to stay smaller in the longer term and are therefore reducing our operating fleet to adjust the number of flights and the capacity of our fleet to the new market conditions. This fleet reduction will be achieved primarily through the retirement of certain older aircraft types. For instance, as of the date of this Prospectus, the entire sub-fleets of Airbus A340-600s (17 aircraft) and Airbus A380s (14 aircraft) at Lufthansa German Airlines are scheduled for long-term storage and/or retirement in the short- to medium-term. As of June 30, 2021, six Airbus A380s have been sold and are scheduled to be handed over to their respective buyers in 2022 and 2023. The eight Airbus A380s remaining following the agreed sale of six aircraft in 2022 and 2023 have been fully decommissioned for several years. Lufthansa Cargo’s two remaining Boeing MD-11F freighters are also scheduled to be sold.

As of June 30, 2021 a total of approximately 200 aircraft (out of 734 aircraft as of June 30, 2021) from all Lufthansa Group airlines had been temporarily parked. This involved temporarily parking the aircraft at various airports in Europe and abroad. The locations were selected with a view to keeping costs as low as possible while maintaining technical capacity. This will ensure the prompt reintegration of the aircraft into the operational fleet, if necessary, in order to maintain flexibility.

As part of the fleet strategy, we aim to continuously scale back the number of aircraft types in operation across the Lufthansa Group to minimize complexity. As a result of the COVID-19 pandemic, we postponed a significant number of firm orders for aircraft, allowing us to extend our previously planned capital expenditures in accordance with adjusted delivery schedules for aircraft. Capital expenditures are expected to slightly increase in the Fiscal Year 2021 to around €1.5 billion compared to the Fiscal Year 2020 and is expected to reach €2.5 billion by 2023 primarily due to the adjusted delivery schedules. As of June 30, 2021, our order book comprised 178 aircraft for delivery by 2029 with an order commitment of approx. €14 billion. More specifically, we have placed firm orders for 20 Boeing 777, 25 Boeing 787, 28 Airbus A350 for the long-haul fleet across the Network Airlines and one Boeing 777F for Lufthansa Cargo. In relation to our airlines’ short-haul fleet, we have placed firm orders for 104 aircraft from the Airbus A320. Moreover, cumulatively, we have 74 order options in relation to new aircraft as of June 30, 2021. In addition to one Airbus A320neo, four Airbus A321neo and one Airbus A220 delivered to us in 6M 2021, we expect to take delivery of up to ten additional aircraft (six aircraft of the Airbus A320neo family, one Airbus A350, one Boeing B789 and two Boeing B777F) during the Fiscal Year 2021.

The order placed in the Fiscal Year 2019 for 20 new Airbus A350-900s and 20 new Boeing 787-9s continues to be the foundation for a fundamental modernization and optimization of the long-haul fleet. In the short term, we aim to deploy the most efficient aircraft types to serve the current route network, in particular new Boeing 777-900ERs as well as Airbus A350-900s and Airbus A320neo family aircraft. We plan to reduce the number of long-haul aircraft operated by the Lufthansa Group, by no later than the middle of the decade, from 13 to seven. The complete retirement of the Boeing 747-400s, Boeing 777-200s, Airbus A340-600s, Airbus A340-300s, Airbus A330-200s, Boeing 767-300s and Boeing MD-11F is expected to be offset by the introduction of the new Boeing 787-9s. We expect significant cost savings from these measures, especially in the areas of crew training, maintenance and operations.

We intend to maintain a high share of owned aircraft but at the same time intend to increase the number of leasing aircraft in our fleet in order to respond flexibly to fluctuations in demand and adjust capacity at short notice. Depreciated aircraft owned by the Lufthansa Group can at short notice remain in service for longer, which may result in reversals of impairments, or be decommissioned before their planned phase-out.

12.6.2 Route Network

In the 6M 2021, we served 299 destinations in 104 countries (Fiscal Year 2020: 347 destinations in 110 countries, Fiscal Year 2019: 381 destinations in 111 countries and Fiscal Year 2018: 397 destinations in 114 countries). A part of our multi-hub strategy, we offer our customers a wide range of flights via our global hubs in Frankfurt, Germany, Munich, Germany, Zurich, Switzerland, as well as our national hubs in Brussels, Belgium, and Vienna, Austria. Our route network is complemented by the route networks of our alliance and joint venture partners, which offer extensive transfer connections.

The global spread of COVID-19 had a severe impact on our route network during the course of the Fiscal Year 2020 and the 6M 2021. The resulting expansion of travel restrictions led to a drastic decrease in the number of connections offered, particularly at the beginning of the COVID-19 pandemic and during the winter season 2020. During this time, we fulfilled the social responsibility and maintained a minimum flight program throughout. In order to optimize the consolidation of the greatly reduced passenger flows, traffic was primarily routed via the Frankfurt, Germany, and Zurich, Switzerland, hubs, and the number of connecting flights from the wider network was reduced to a small number of flights per day. In addition, we have operated hundreds of repatriation flights to fly travelers home from around the world. Passenger aircraft were also used on some route exclusively to transport airfreights, primarily medical products.

The number of flights on offer increased again over the summer of 2020 with a pronounced recovery in demand for European short and medium-haul tourist destinations in particular. Flights to many long-haul destinations were also resumed. With the close of the summer travel season, the resurgence of infections and the resulting increase in travel restrictions, the number of flights again declined from the end of the third quarter of 2020, although not to the same extent as of the beginning of the second quarter. To protect our liquidity, we only operated flights with a positive cash contribution.

After the COVID-19 pandemic, we aim to rebuild our route network from the hubs and to increase the share of touristic passengers, which continues to gain in importance. We expect the tourist routes to have the fastest recovery.

12.7 Alliances and Cooperation Arrangements

12.7.1 Joint Ventures

We believe that a strong joint venture network drives leading connectivity and global scale. In 1996, Lufthansa German Airlines signed the commercial transatlantic joint venture A++ with the Star Alliance carriers United Airlines and Air Canada to foster growth and strengthen their presence in the North Atlantic market. SWISS and Austrian Airlines effectively joined the agreement in 2011.

Furthermore, we also have extended joint ventures in Asia, which we believe will be instrumental to a swift recovery from the COVID-19 pandemic as the Asian travel market is expected to outperform on growth. Lufthansa German Airlines and All Nippon Airways implemented J+, the first commercial joint venture between Europe and Japan, in 2011. The C+ Joint Venture between Air China, Lufthansa German Airlines, SWISS and Austrian Airlines, establishing a commercial joint venture among the companies for the Network

Airlines has been concluded in September 2016. The S+ Joint Venture between Lufthansa German Airlines, SWISS, Austrian Airlines and Singapore Airlines established a commercial joint venture in November 2015 to operate key routes between Singapore and Europe. The agreement also significantly expands codeshare ties and deepens commercial cooperation.

We also participate in commercial joint ventures in our Logistics and MRO business segments. Lufthansa Cargo and Deutsche Post Beteiligungen Holding GmbH agreed on a 50-50 joint venture, known as AeroLogic, which holds long-term flight services agreements with Lufthansa Cargo and DHL International GmbH (“DHL”). The flight services agreements provide that AeroLogic will sell the capacity of all of its aircraft currently in service to Lufthansa Cargo and DHL, in accordance with an agreed share of capacity and use. Additionally, Lufthansa Cargo has been operating a commercial joint venture at Shanghai Pudong airport together with Shanghai Airport (Group) Co., Ltd. and Shanghai Jin Hai Jet Air International Forwarding Co., Ltd since 1999, which provides process service for cargo and mail.

Additionally, Lufthansa Technik benefits from participation in a variety of joint ventures, including with MTU Aero Engines AG and Rolls Royce plc, which are aimed at capitalizing on joint expertise, market access, intellectual property rights and talent utilization to broaden the scope of services on new aircraft/engine types and maintain competitive cost structures.

We believe that these extended partnerships strengthen our network, competitive position, and revenue quality due to a joint capacity management, joint product strategy, mutual market access, and last but not least revenue sharing.

12.7.2 Star Alliance

Alliance and other cooperation agreements are vital for us in order to provide a global network of services for our passengers. We are one of the founding members of the Star Alliance in 1997, which is the cornerstone of our global partner and network strategy, offering our customers reliable travel products and services worldwide. The Star Alliance brings together 26 member airlines as of December 31, 2020. Its combined fleet of over 5,000 aircraft served more than 1,300 destinations in 195 countries worldwide in the Fiscal Year 2019. In the Fiscal Year 2018, the Star Alliance was the largest global airline alliance in terms of market share (measured by ASK) (*source*: Statista, Leading Airline Alliances in 2018, August 2019). Through our membership in the Star Alliance, we have been able to offer our passengers access to a global network of destinations and convenient transfers. The co-operation has also made possible one travelling experience in which the traveler can earn and use bonus points and access lounges and other time saving services. In the event of flight cancellations, we and the other members of Star Alliance have an agreement to cater to shared customers, including through rebooking to the final destination with the next available Star Alliance flight.

In 2020, Star Alliance completed its development of a cross-system biometric identity and identification platform known as Star Alliance Biometrics. Customers in the Miles & More frequent flyer program thus enjoy contact-free access to the security checks and boarding gates without removing their face masks, which also represents an important contribution to health and safety during the COVID-19 pandemic.

12.7.3 Miles & More Program

We use various bonus miles programs with the aim of ensuring long-term customer loyalty. Participants in the Miles & More Program, which is the biggest bonus miles program within the Lufthansa Group in terms of customers with approximately 35 million customers and which we further believe to be the largest customer loyalty program in the European airline industry, can collect and redeem bonus miles for flights with the airlines in the Lufthansa Group as well as with more than 300 partners (including other airlines, hotels, global car hire companies, financial and insurance providers, telecommunications companies, retailers, automobile clubs). The program is developed continuously with a strong focus on customers’ interests. This entails adapting the structure of the program, for instance by giving customers the option to link their Miles & More account to One ID, and making the Miles & More partnerships broader and deeper in order to make the program even more attractive.

12.8 Employees

As of June 30, 2021, we had 108,072 employees (by headcount).

The following table provides an overview of our employees (by headcount) per business segment for the dates presented:

	As of June 30,		As of December 31,		
	2021	2020	2020	2019	2018
	(unaudited)		(unaudited)		
Network Airlines	55,508	59,953	57,363	60,913	59,738
Eurowings.....	3,227	3,219	3,088	3,432	4,090
Logistics	4,216	4,452	4,373	4,539	4,505
MRO.....	21,467	23,927	22,745	23,855	21,799
Catering	15,288	28,130	13,227	35,636	35,443
Additional Businesses and Group Functions.....	8,366	9,675	9,269	9,978	9,959
Total.....	108,072	129,356	110,065	138,353	135,534

We believe that we have good working relationships with our employees, evidenced by the average tenure of our employees of approximately 13.9 years. Our employees have traditionally been represented by unions. As of June 30, 2021, 84% of our around 62,000 German employees are tariff employees, which are covered by collective bargaining agreements that are customary for the industry or are members of labor unions, which include the cabin crew union “UFO,” pilot’s union “Vereinigung Cockpit” and workers’ union “ver.di.”

As part of our “ReNew” restructuring program, we intend to structurally adjust our workforce to the market decline caused by COVID-19. As of June 30, 2021, the number of total employees (by headcount) decreased by 22,000, or 17%, of the total workforce, including 5,000 employees (by headcount) in Germany compared to December 31, 2019 (excluding a workforce reduction by approximately 8,600 employees (headcount) in connection with the divestment of LSG Group’s European business). We implemented short-term measures in 2020 to reduce staff costs and to counteract the effects of the pandemic. In addition, crisis agreements were concluded for the majority of the other groups of our employees. On August 5, 2021, Lufthansa Group has communicated that it already implemented measures to reduce personnel costs by €1.1 billion. Taking into account additional measures such as, in particular, savings of still to be negotiated amended tariff agreements as well as further staff reduction, we expect to achieve a total of €1.8 billion structural annual personnel cost savings from 2024 onwards.

We offer pension benefits in many countries in which we operate. Based on the local situation and local laws, we have implemented several pension plans worldwide and/or contribute to local statutory plans. The majority of our pension obligations are attributable to our German companies and regulated by contractual bargaining agreements. For certain members of the management, we also offer individual pension contracts, depending on the position and years of service. These commitments are recorded in our financial statements and two-thirds of the liabilities are covered by external funds. All our external funding complies with local minimum funding regulations.

12.9 Insurance

We are insured under the Lufthansa Aviation Insurance Group’s policy together with 50 mostly European aircraft operators. This policy provides liability coverage for passengers, mail, cargo, product legal liability and third-party legal liability and coverage for hull damages. We renew most of these policies annually. Since insurance companies continue to be very reluctant in providing coverage for hull damages to aircraft caused by weapons of mass destruction, it is no longer possible for us and other European airlines to effectively insure against such threats. Since hull insurance is not a condition to operating an airline, however, there is no risk that aircraft will be grounded for this reason.

We do not purchase insurance coverage for financial business interruption losses caused by certain natural disasters that do not damage the aircraft or that are caused by pandemics, such as the COVID-19 pandemic,

as the insurance market does not provide adequate coverage with respect to such losses. Our insurance policies also cover claims resulting out of the operation of our passenger and cargo aircraft due to war or allied perils, including terrorist attacks.

We also maintain various other insurance policies to cover a number of other risks related to our business, such as director and officer liability cover, crew related loss of license cover and general liability cover. We believe that the types and amounts of insurance coverage we currently maintain are in line with customary practice in the industry and are adequate for the conduct of our business.

12.10 Sales and Marketing

In the Fiscal Year 2020, there was a significant increase in sales through our direct sales channels, in particular through our website, resulting in low administration costs per ticket sold as well as giving us the tool to conduct active revenue management by increasing load factors through active use of our website for flights where there is a large number of seats available. This increase was a direct result of the COVID-19 pandemic which has prompted our customers to look for tickets online, especially during the long lockdown periods. A substantially higher share of our tickets were sold through our own sales channels with the remaining ones sold via our business-to-business partners, travel agencies and online booking portals. We plan to increase the share of tickets sold through direct channels to 75% of the tickets sold by 2024 (Fiscal Year 2020: approximately 65%; Fiscal Year 2019: approximately 50%).

12.11 Facilities

We own or lease a number of administrative and commercial buildings. As of June 30, 2021, the carrying amount of our real estate properties was approximately €1 billion, of which €123 million represented land and €877 million represented buildings. Our operational headquarters are located in Frankfurt, Germany.

As of June 30, 2021, our real estate portfolio comprised a total of approximately 3.92 million square meters, of which approximately 1.88 million square meters were leased. In addition, we, together with Air France, Korean Air and Japan Airlines, are a partner in Terminal One Group Association, L.P. (“**TOGA**”), organized in 1994 as a limited partnership under the laws of the State of New York to lease, finance, construct, maintain and operate the passenger terminal facility ‘Terminal One’ at John F. Kennedy International Airport in New York. Furthermore, we hold an indirect general partner’s interest of 40% in Terminal 2 Gesellschaft mbH & Co oHG (“**T2**”), which operates the Terminal 2 passenger terminal at Munich Airport. Through Lufthansa Cargo, we further operate pharmaceutical hubs in Frankfurt, Germany, Munich, Germany, and Chicago, United States, which have a temperature control capacity ranging from -20°C to +25°C (-4°F to 77°F).

There are no major encumbrances on properties we own, and we are not aware of any encumbrances on properties we lease that could materially affect our business.

12.12 Environmental, Social and Governance

Sustainable and responsible entrepreneurial practice is an integral part of our corporate strategy. We aim to reduce the carbon footprint, to support regulators towards a carbon free economy, to provide digital solutions to link customer and sustainability, and to foster gender equality. To achieve this, we strive to comply with all applicable rules and regulations with our best efforts in aviation operation as well as in the daily working environment to minimize any adverse impact on the environment.

For the last 25 years, we have published our sustainability report “Balance,” which, in the Fiscal Year 2020, was incorporated into our Sustainability Fact Sheet. As of January 1, 2020, a new board resort “Customer, IT & Corporate Responsibility” was created. Specific measures to decrease our carbon footprint include, *inter alia*, the continuous investment in modern fleet to reduce CO₂ emission and noise pollution. We aim to reduce emissions by 50% by 2030. Since 2019, we voluntarily compensate 100% of the CO₂ emissions incurred by our employees on all business-related travel. In addition, we offer our customers the option to voluntarily compensate the CO₂ emissions caused by their flights and thus to make a personal contribution to climate protection. Since 2007, Lufthansa German Airlines and SWISS have cooperated with the climate protection foundation myclimate for this purpose and, since 2019, the aforementioned compensations (including compensations for CO₂ emissions incurred by our employees on all business-related travel) can be

made through the purchase of low-carbon sustainable aviation fuel on the Compensaid platform (as described below) or, in relation to business travels of our employees, through a contribution towards certified projects by myclimate, such as for the conservation of moors. We also plan to be CO₂-neutral for mobility for all ground transportation services, as well as use green electricity for all Lufthansa Group buildings in the DACH (*i.e.*, Germany, Austria and Switzerland) region by 2030.

We underscored the importance of diversity and equal opportunities over 20 years ago by creating the management function “Change Management and Diversity” and promoting equal career opportunities for men and women. The help alliance gGmbH is our central corporate citizenship pillar. The charitable aid organization acts as a catalyst for greater social engagement, combining proven projects initiated by employees with the power and network of the Lufthansa Group. The Lufthansa Group Code of Conduct, which has been established in 2017 and internally and externally communicated in 2018, contains the principles and guidelines that are binding for all bodies, executives and employees in daily business life. The code also serves as a guide aimed at promoting social governance and contribution for and within our business partners.

The aviation industry has not yet yielded an electric or hydrogen engine for large aircraft. We believe that the core problem relates to the lower energy density of batteries and the complexity of the storage of large quantities of hydrogen. Therefore, we are of the opinion that sustainable aviation fuels are a decisive technological key to advancing the future of flying towards a CO₂-neutral endeavor. We estimate that, compared to fossil fuels, sustainable aviation fuels could reduce CO₂ emissions by up to 80%.

We are engaged in a variety of projects to promote availability of sustainable aviation fuels. Alternative fuels produced from used cooking oil and cooking fats from agricultural production are generally available in small quantities. In recent years, we have been intensively involved in researching, testing and using sustainable aviation fuels and, in 2011, we pioneered the world’s first long-term testing of bio kerosene in regular flight operations. The project was accompanied by detailed emission measurements and research into the production processes and availability of biomass. We believe that we successfully demonstrated the feasibility of sustainable aviation fuels for flight operations, in particular as cost- and resource-intensive changes to the infrastructure would generally not be required.

By launching our Compensaid platform, the Lufthansa Innovation Hub has developed a forward-looking customer solution for the use of sustainable aviation fuels. Passengers can replace conventional kerosene with sustainable aviation fuels, regardless of the choice of airline, and thus fly almost “CO₂-neutral,” by means of voluntary financial compensation.

As an international airline group, social responsibility is an important topic for us and this is particularly true in times of COVID-19. For instance, as of February 2021, Lufthansa Cargo supports the World Health Organization’s (*WHO*) and UNICEF’s COVID-19 vaccines global access program titled ‘COVAX’ and the logistical facilities. Moreover, in connection with the COVID-19 pandemic, we have devoted significant resources to internationally transport millions of surgical masks and, in cooperation with the German government, conducted more than 430 repatriation flights to retrieve approximately 90,000 individuals, including 34,000 German and European travelers, back to their home countries between March 13, 2020, and April 22, 2020.

12.13 Information Technology

Our information technology (“IT”) system and logistical processes are key operational and management assets for our business. The ability to process information accurately and quickly is fundamental to our position in the aviation industry, which is characterized by constant movement of thousands of individual items across a global network. Our most material IT systems are the computerized airline passenger service system, the flight operations system, the websites, the telecommunications systems, applications for the passenger service domain, mobile applications for customers and employees and other automated systems.

Within our IT organization, various IT units oversee our IT demand and supply landscape, which are incorporated by our Group-wide ‘process domain’ information management. The responsibility for business segment-specific processes and applications are tailored to the requirements of and remain within our respective business segments (Logistics, MRO, Catering and Eurowings). In addition to this allocation,

certain individualized applications tailored to support specific local process requirements are allocated to the responsibilities of the individual Network Airlines, while common cross-sectional tasks are centralized on the Lufthansa Group level. This enables us to address the individual demands of the respective business segments and, through dedicated contact personnel, independently advance process transformation to ultimately, through standardization, realize additional synergies.

Through Lufthansa Industry Solutions, which we consider to be a driving force in digitalization and which we believe to be a market leader in artificial intelligence, and Lufthansa Systems, which we believe is an industry leader in the international airline IT market, we benefit from extensive in-house enterprise expertise. Lufthansa Systems generates additional revenue through its offering of system integration services and IT consultancy to more than 350 airline customers globally. Simultaneously, in order to maintain cost flexibility, we procure and license a variety of IT systems and IT infrastructure services from third parties.

In addition, as part of our operations, we retain personal information received from our customers, which is subject to certain regulatory data privacy protection in the EU and elsewhere. See *“1.1.9 We are dependent on the resilience and uncompromised operation of our reservation, data processing, technology and management systems as well as those operated by third parties and are exposed to risks related to the poor performance, the failure of and any unauthorized access to these systems, including as a result of data security breaches.”* and *“1.4.1 We may be exposed to legal sanctions and penalties as we may not be able to prevent or detect violations of legal compliance and other economic and administrative regulations.”* A fundamental requirement for online commerce, including sales of tickets online, is the secure transmission of confidential information. We are committed to preserving the confidentiality, integrity and availability of applicable physical and electronic information assets throughout the Lufthansa Group by using implemented controls, procedures and selected vendors. Threats and vulnerabilities associated with business applications and systems and networks are managed by scanning for technical vulnerabilities and other weaknesses, performing continuous security event monitoring, acting on threat intelligence and protecting information against targeted cyber-attacks.

12.14 Intellectual Property

We own various intellectual property (“IP”) rights in our business segments. Our main IP assets consist of the “Lufthansa” trademark and related trademarks, colors and logos in the field of aviation, which Lufthansa acquired in 1951. We believe that many of our brands, such as the Lufthansa, Eurowings, SWISS, Austrian Airlines, Brussels Airline and Miles & More brands, enjoy high rates of customer recognition and brand loyalty in their markets. Additionally, our IP base comprises a range of IP rights owned by Lufthansa Technik from which our fleet operations, our maintenance and third parties equally benefit.

We own all trademarks of our Lufthansa master brand concept and other relevant trademarks for our business. Our brand, with the crane and the colors blue and yellow, has been in usage for nearly 100 years and stands for our aspiration to ensure a consistent customer experience. In 2018, we modernized our Lufthansa brand by using a new paint scheme for the aircraft and redesigning the crane. We believe that this rebranding strategy has enhanced and strengthened our competitive position, expanded our customer base and made our expenditure to preserve and enhance consumer awareness of the Lufthansa brand more efficient, while staying loyal to our original brand.

We protect our major brands in the manner we believe appropriate to best protect and advance our business interests in each of our markets, including extending our trademarks and defending them against infringement. We also monitor “domain grabbing,” the unauthorized registration of our trademark and internet domain names. From time to time we use third-party services to monitor the internet for activity relating to our major trademarks.

We believe we are at the forefront of aviation innovation. Through Lufthansa Technik, we research and develop IP in-house. This enables us to make continuous process improvement, increase efficiency and enhance safety standards for our line maintenance services and aircraft operation in general. Lufthansa Technik allows us to benefit from an exclusive range of innovative products, such as rescue tools, communication systems as well as maintenance tools and procedures. Further, we closely collaborate with manufacturers of aircraft and aircraft equipment in order to develop bespoke solutions tailored to the needs of their customers. We have registered and obtained an extensive range of national and international patents

and licenses to cover our IP and their design and are continuously seeking to secure further patents on our developments, such as through the development of the AVIATAR platform. As of June 30, 2021, we held more than 450 patents and had more than 400 patent applications filed globally. Ultimately, we consider our intellectual property base a competitive advantage and maintain stringent control over the filing and use of our IP rights, including through international patent monitoring systems.

We experience occasional trademark oppositions, similarities with existing trademarks in local markets, domain grabbing and IP infringement. Given the size of our IP, trademark and internet domain name portfolio, we consider these minor incidents as within the ordinary course of business. Other than these incidents, there have been no material violations, disputes or litigation in relation to any of our intellectual property rights, including our master brands, in recent years.

12.15 Legal and Tax Proceedings

From time to time companies of the Lufthansa Group are involved in legal disputes and administrative proceedings as part of their ordinary business activities.

Save as described below, there are currently no, and we have not been involved in any, governmental, legal or arbitration proceedings during the period of the last twelve months, against or affecting the Company or any of its subsidiaries, nor is the Company aware of any pending or threatened proceedings, which (in either case) may have or have had in the recent past significant effects on the financial position or profitability of the Company or the Lufthansa Group.

12.15.1 Cargo Cartel Public Investigations

The European Commission and various national antitrust authorities conducted global investigations of air carriers in connection with allegations of anticompetitive collusive practices in the term between 1999 and 2006 relating to freight rates and premiums in the air cargo sector. We cooperated fully with the antitrust authorities in these proceedings and availed ourselves of the leniency policy for cooperating witnesses in the relevant jurisdictions. In doing so, we avoided any fines imposed by the European Commission, which amounted to about €800 million for the other cartel participants. Above that, we benefited from leniency programs regarding antitrust proceedings of the authorities of, *inter alia*, the United States, Australia, Canada, Switzerland and Korea in relation to the assessment of fines amounting to about €1.2 billion. Despite receiving full immunity, we have appealed the European Commission's decision to the European Court of Justice ("ECJ") strictly on legal grounds. This appeal did not affect the European Commission's grant of full immunity.

In some jurisdictions, cargo customers, freight forwarders and shippers in particular have filed objections to, or opted out of the settlements reached with the antitrust authorities or have filed civil actions for damages against the cartel members, including us. We were able to obtain dismissal of civil class actions in the United States, Australia and Canada by settling with the plaintiffs. In Germany and the United Kingdom, we settled all pending litigations in 2019. In the Netherlands, South Korea, Norway and Israel, we, together with other cargo carriers, are currently subject to, or have been joined in, class actions and other civil actions for damages. In other proceedings, particularly in the Netherlands, parties have filed claims amounting to up to three-digit million figures.

At present, there can be no assurance as to the likely outcome of these actions. There can also be no assurance that additional civil actions will not be filed against us in relation to allegations of anticompetitive collusive practices in the air cargo sector.

12.15.2 Tax Proceedings

We are engaged in tax proceedings with the tax office in Cologne-Altstadt at the German Federal Fiscal Court (*Bundesfinanzhof*) regarding several tax matters, in particular, the deductibility of write-downs on cross-border inter-company loans granted by us originating in the years between 2001 and 2005. In our financial statements as of and for the first half year of 2019, we showed an additional tax expense for our tax risk of €340 million relating to this open tax matter in Germany, which has been paid by us. Past judgments by the competent fiscal court and the German Federal Fiscal Court (*Bundesfinanzhof*) had originally affirmed our position in principal. However, the German Federal Fiscal Court (*Bundesfinanzhof*) has recently repealed

the recently established case law in relation to a comparable case. After a hearing at the German Federal Fiscal Court (*Bundesfinanzhof*) in June 2019, the German Federal Fiscal Court (*Bundesfinanzhof*) decided to refer our case back to the local fiscal court (*Finanzgericht*) in Cologne. Another tax proceeding at the local fiscal court (*Finanzgericht*) in Cologne in relation to the years 2006 to 2009 is pending. The proceeding relates to acquisition costs of a foreign subsidiary and the tax exemption of certain dividends. While we believe that we have valid arguments supporting our position in this case, it is not possible to determine the final outcome of the case at this stage. We have paid the taxes claimed by the competent tax office following its initial assessment, as a result of which we have not recorded any provisions. However, the outcome of the aforementioned proceedings may negatively affect us, including as a result of a revised application of German tax laws.

In addition to Germany, we are, from time to time, involved in legal and administrative tax proceedings in other jurisdictions in which we have operations. See “1.4.5 *We are subject to regular external tax audits and proceedings which may require us to make additional payments.*”

12.16 Material Agreements

12.16.1 AeroLogic

AeroLogic is a 50-50 joint venture between Lufthansa Cargo and Deutsche Post Beteiligungen Holding GmbH, holding long term flight services agreements with Lufthansa Cargo and DHL. These flight services agreements provide that AeroLogic will sell the capacity of all of its aircraft currently in service to Lufthansa Cargo and DHL, in accordance with an agreed share of capacity and use. Pursuant to the flight services agreements, Lufthansa Cargo and DHL must compensate AeroLogic for these flight services. These compensation payments are to be calculated in such a way that AeroLogic’s costs are covered. Additionally, each joint venture partner agreed in the joint venture agreement to provide two shareholder loans in the respective amounts of €1 million and €14 million, subject to certain terms and conditions.

12.16.2 Terminal One at John F. Kennedy International Airport in New York

We, together with Air France, Korean Air and Japan Airlines, are limited partners in TOGA, organized in 1994 as a limited partnership under the laws of the State of New York to lease, finance, construct, maintain and operate the passenger terminal facility “Terminal One” at John F. Kennedy International Airport in New York. In this respect, we have entered into a use and lease agreement with TOGA for the use of the Terminal One facilities (the “**Facility Use and Lease Agreement**”), pursuant to which we have leased premises in Terminal One for joint or exclusive use, received certain rights to use the Terminal One and consented irrevocably, without restriction or reservation, to pay our share of all of TOGA’s payment obligations and to assume certain obligations, including the debt service of bonds issued by TOGA to finance the development of Terminal One in the initial amount of approx. USD 167 million. We understand that Air France, Korean Air and Japan Airlines have each entered into respective use and lease agreements at substantially similar terms. In the event that we fail to pay amounts due under the Facility Use and Lease Agreement, it provides that the amounts due will have to be paid by all of the carrier-partners who are not in arrears in proportion to their respective charges based on volume of use.

12.16.3 Terminal 2 at Munich Airport

We hold an indirect general partner’s interest of 40% in T2. The other general partner (holding a direct interest of 60%) is Flughafen München GmbH, which is jointly owned by the Free State of Bavaria, the Federal Republic of Germany and the City of Munich. The Terminal 2 passenger terminal at Munich Airport was built by T2 and is now operated by T2. We are jointly and severally liable to the full extent for all liabilities and payment obligations of this partnership (for further details, see “1.4.6 *We are exposed to risks in connection with our investments in airport infrastructure, including liability risks.*”). T2 has obtained borrowings in the amount of €2,107 million, including a syndicated credit facility in the total amount of €1,107 million to finance the construction of Terminal 2, which was drawn in several tranches with a gradual repayment schedule with the last tranche being repaid in the year 2033 and a further syndicated credit facility in the total amount of €725 million for the construction of a satellite to Terminal 2. This satellite started operations in 2016.

12.16.4 Lufthansa Hub Catering Contract (LHCC)

As part of the sale of the LSG Group's European business to gategroup, which closed on December 2, 2020, we concluded a long-term catering contract (the "LHCC") with gategroup to cater Lufthansa German Airlines flights from airports in Germany, including its Frankfurt and Munich hubs as well as another long-term catering contract with Gate Gourmet Switzerland GmbH to cater SWISS flights from airports in Switzerland, including its hub in Zurich, Switzerland. The LHCC is a modular contract and covers food production, catering logistics and other catering services. During an initial transition phase, business disruptions and quality issues might occur post-closing which shall be mitigated by a joint venture structure of us and gategroup governing the future catering activities in Frankfurt and Munich. Further on, joint implementation teams will be formed and joint contingency plans will be in place.

12.16.5 State Aid / Stabilization Package

Due to the COVID-19 pandemic and its impact on the financial situation of the airline industry, we required additional funding and have received local state aid in Germany, Switzerland, Austria and Belgium as well as some other countries. The WSF agreed with and granted to the Company the WSF Stabilization Measures, totaling approximately €6 billion. The WSF Stabilization Measures are governed by the Framework Agreement as well as certain ancillary agreements and consist of the WSF Capital Increase, the Silent Participation I and the Silent Participation II.

The WSF Capital Increase consisted of the Company issuing 119.5 million new shares to the WSF at a subscription price of €2.56 per share in connection with a capital increase against cash, resulting in total subscription amount of €306.0 million. The WSF Capital Increase was approved by the General Meeting on June 25, 2020, and was registered in the Company's commercial register (*Handelsregister*) on June 26, 2020.

The Silent Participation I is composed of the WSF's silent participation in the Company in the amount of €4.7 billion, which was subsequently reduced to an amount of €4.5 billion, that is accounted for as equity according to IFRS to the extent drawn by the Company and provides for a loss participation and coupon deferral right. The Silent Participation I can only be drawn until December 31, 2021, in not more than six tranches. The coupon steps up from 4% p.a. in 2020 / 2021 to 9.5% p.a. from 2027 onwards and has no maturity. The Silent Participation II is composed of two tranches with conversion features, the Silent Participation II-A and the Silent Participation II-B. Due to a six-year maturity with an extension option until full repayment of the Silent Participation I, the Silent Participation II is not accounted for as equity according to IFRS. The coupon steps up from 4% p.a. in 2020 / 2021 to 8% p.a. in 2025 and 2026 and to 9.5% from 2027 onwards. In addition thereto, 119.5 million shares were issued at a price of €2.56 per share. The WSF has agreed to sell all of its Company's shares by not later than December 31, 2023, provided that the Silent Participation I (as defined below), including interest and any additional payments, and the Silent Participation II, including interest, have been fully repaid and subject to achieving the Minimum Sales Price. However, if the WSF subscribes and receives New Shares against contribution in kind of a part of its repayment claim under the Silent Participation I, the WSF shall sell all of its Company's shares within 18 months following full repayment of the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, and subject to achieving the Minimum Sales Price, which is being adjusted to reflect the weighted average of the purchase prices paid by the WSF in relation to existing shares of the Company and the New Shares. However, such 18-months period commences only after the WSF Lock-Up has expired, if the repayment of the Silent Participation I and Silent Participation II occurs prior to expiration of the WSF Lock-Up. Following expiration of the applicable period and provided that the Silent Participation I, including interest and any additional payments, and the Silent Participation II, including interest, have been repaid in full, the Company shall have the right to request that the WSF sells all of its Company's shares to investors designated by the Company at the Minimum Sales Price.

SWISS and Edelweiss have been granted the Swiss Stabilization Package, a partly state-guaranteed loan in an amount of CHF 1,500,000,000. The state aid does not include any equity portion. The loan is repayable in fixed installments between 2023 and 2025, with two extension options of one year each. The initial interest rate was 2.60% p.a. and has subsequently risen to 3.95%, whereas further step-ups may occur subject to the rating of the Company. The Swiss Loan is secured, including by a pledge over the shares in SWISS,

Edelweiss and its parent company AirTrust Ltd. that are ultimately held by the Company. Furthermore, the Company provided a subordinated loan in the amount of CHF 500 million in connection with the Swiss Loan.

Austrian Airlines has received the Austrian Stabilization Package, a state aid in the form of a loan and subsidy (“*Katastrophenbeihilfe*”) in a total amount of €450,000,000 comprising a state-guaranteed loan of €300,000,000 and a non-repayable grant of €150,000,000. The interest of the loan is 1% p.a. with a graduated repayment schedule with a last tranche to be repaid on December 31, 2025. The loan is further secured by, amongst others, a pledge in the shares of Austrian Airlines as well as certain aircraft owned by Austrian Airlines. On July 15, 2021, Austrian Airlines repaid an amount of €30 million under such loan following which it was reduced accordingly.

Brussels Airlines has been granted the Belgian Stabilization Package, a state aid in form of a loan and an equity portion in a total amount of €290,000,000 comprising a loan of €287,100,000 and profit share certificates of €2,900,000. Any amounts available for drawdown under such loan can only be drawn until December 31, 2021.

As of June 30, 2021, we had received the following state aid under the Stabilization Package:

Entity	Country	State Aid	Total State Aid	State Aid Provided	State Aid Available
Deutsche Lufthansa Aktiengesellschaft	Germany	(i) €306 million WSF Capital Increase; (ii) €4,541 million ⁽¹⁾ Silent Participation I; and (iii) €1,000 million Silent Participation II	€5,847 million ⁽¹⁾	€2,806 million	€3,041 million ⁽¹⁾
Austrian Airlines	Austria	(i) €300 million loan; and (ii) €150 million subsidy (<i>Katastrophenbeihilfe</i>) (90% guaranteed by the Austrian government)	€450 million	€450 million	—
Brussels Airlines	Belgium	(i) €287.1 million loan; and (ii) €2.9 million profit share certificates	€290 million	€253 million	€37 million
SWISS and Edelweiss	Switzerland	CHF1,500 million loan (85% guaranteed by the Swiss government)	CHF1,500 million	CHF550 million	CHF950 million

(1) Excluding the KfW Financing which was repaid in full and terminated on March 12, 2021, and certain other non-domestic stabilization measures that are credited against the maximum amount available under the Silent Participation I.

For further details on our state aid, see “4 The Stabilization Package.”

12.16.6 Additional Sources of Funding

(a) Debt Issuance Program

The Company entered into a dealer agreement which has been amended and restated on July 1, 2021, with certain financial institutions as dealers to establish a debt issuance program. Under such debt issuance program, the Company may, from time to time, issue notes with an aggregate principal amount of up to €10,000,000,000. The notes under the program will be issued on a senior unsecured basis and bear interest at a fixed or floating rate. The terms and conditions of the notes are set out in a base prospectus approved by the Luxembourg *Commission de Surveillance du Secteur Financier* and include negative pledge provisions and events of default. The notes are subject to completion by the final terms pertaining to a series of notes.

As of the date of this Prospectus, the following notes were outstanding under the debt issuance program:

ISIN	Nominal Amount	Coupon	Maturity Date
XS2049726990	€500,000,000	0.25% p.a.	September 6, 2024
XS2265369657	€1,000,000,000	3.00% p.a.	May 29, 2026
XS2296201424	€750,000,000	2.875% p.a.	February 11, 2025
XS2296203123	€850,000,000	3.75% p.a.	February 11, 2028
XS2363244513	€500,000,000	2.00% p.a.	July 14, 2024
XS2363235107	€500,000,000	3.50% p.a.	July 14, 2029

(b) *Multi-Currency Commercial Paper Program*

We hold a STEP certified Multi-Currency Commercial Paper Program with a volume of up to €1,000,000,000, of which, as of June 30, 2021, no amount was drawn.

(c) *Unsecured Promissory Notes (Schuldscheindarlehen)*

The Company regularly enters into unsecured promissory notes (*Schuldscheindarlehen*). During the 6M 2021, we entered into unsecured promissory notes (*Schuldscheindarlehen*) in the amount of €350 million. As of June 30, 2021, the total outstanding amount of unsecured promissory notes was €1,874 million.

(d) *Aircraft-Secured Financing*

Additionally, we regularly use aircraft-secured financing, mainly by Japanese operating leases with a call option. During the 6M 2021, we entered into several aircraft-secured financings with an aggregate volume of €672 million. As of June 30, 2021, the total outstanding amount of aircraft-secured financings, which are secured by a substantial number of our aircraft, was €3,753 million.

(e) *Hybrid Bond and Convertible Bond*

On August 12, 2015, the Company issued a €500,000,000 hybrid bond which as of the date of this Prospectus bears a coupon of 4.382% p.a. and matures on August 12, 2075 (the “**2015 Hybrid Bond**”). As of June 30, 2021, the full amount was outstanding under the 2015 Hybrid Bond and the next call date is scheduled for February 12, 2026. The fixed rate resets every five-years according to the Euro swap rate plus margin. The next interest rate reset is scheduled to occur on February 12, 2026, with a margin of 5.033% in addition to the applicable market interest rate. On May 19, 2021, we decided to suspend any coupon payments in relation to the 2015 Hybrid Bond in accordance with its terms and conditions. The coupon payments shall remain suspended as long as the Silent Participation I and/or the Silent Participation II is drawn and/or the WSF is a shareholder of the Company. As a result of our decision to suspend the coupon payments, Standard & Poor’s announced on May 25, 2021, that it lowered its rating of the 2015 Hybrid Bond to (CC). Additionally, this rating may be further lowered to (D) in the event that the coupon payments remain suspended at the next coupon payment date in February 2022. For further details, see “1.2.2 We are subject to significant limitations in our financial flexibility as a result of the state aid received and financial stabilization measures obtained from governments in response to the outbreak of COVID-19 and face risks associated with triggering events of default under these state aid and stabilization measures.”

On November 10, 2020, the Company issued a €600,000,000 convertible bond which bears interest at a rate of 2.00% p.a. and matures on November 17, 2025. As of June 30, 2021, the full amount was outstanding under the convertible bond.

13. REGULATORY ENVIRONMENT

The regulation of the civil aviation industry is based on international conventions and agreements as well as fundamental treaties, regulations and directives harmonizing the legislation of EU member states, as well as national legislation. These are enforced through supervision by both international and domestic authorities.

Any alleged or actual non-compliance with currently applicable regulation concerning our operations, new regulatory measures or amendments or the imposition of additional administrative charges or costs may result in additional profitability-reducing expenses and the need to divert managerial efforts and attention from operational matters, which may have a material adverse effect on our business, financial condition, results of operations and future prospects.

The following summary is a general description of the applicable legal regime and is not intended to provide, and cannot be, an exhaustive account of all laws and regulations which apply to the business of the Lufthansa Group.

13.1 Operating License and Air Operator Certificate

Regulation of the civil aviation industry in the EU has been substantially harmonized through EU directives and regulations, which has diminished the significance of national rules of EU member states and reduced differences between them. One of the most significant regulations concerning the aviation industry in the EU is Regulation (EC) No 1008/2008 on common rules for the operation of air services in the Community (the “**Air Services Regulation**”). Under the Air Services Regulation, no natural or legal person or official body established in the EU is permitted to carry by air passengers, mail and/or cargo for remuneration and/or hire unless it has been granted the appropriate operating license (*Betriebsgenehmigung*) (each, an “**Operating License**”). The Air Services Regulation clarifies the criteria for granting EU Operating Licenses as well as their duration, including also requirements for the financial condition, ownership and control, and supervision applicable to air carriers. If a competent licensing authority is no longer satisfied that the air carrier can meet its current and potential obligations for a twelve-month period, it is, in principle, required to suspend or revoke the Operating License. The EU has amended the Air Services Regulation during the COVID-19 pandemic (see below “*13.4 Aspects*”).

One prerequisite for awarding an Operating License is that EU member states and/or nationals of EU member states must own more than 50% of an air carrier and effectively control it, whether directly or indirectly through one or more intermediate undertakings (Article 4 lit. f of the Air Services Regulation).

Under the European Economic Area (the “**EEA**”) Agreement as well as the EU-Switzerland Air Transport Agreement, the European Free Trade Association (the “**EFTA**”) countries (Iceland, Liechtenstein, Norway and Switzerland) are equated with EU member states.

Therefore, in order to comply with the Air Services Regulation, a German air carrier must be owned and controlled by EU or EFTA member states, their nationals or companies and must be able to demonstrate such ownership and control in order to maintain its current portfolio of air traffic rights and aviation licenses.

Another prerequisite for the award of an Operating License is the air carrier’s possession of an Air Operator Certificate (*Luftverkehrsbetreiberzeugnis*) (the “**AOC**”) issued in accordance with Regulation (EU) 2018/1139 (Article 4 lit. b and Art. 6 para (1) of the Air Services Regulation). An AOC is a certificate delivered to an undertaking confirming that the operator has the professional ability and organization to ensure the safety of operations specified in the certificate, as provided in the relevant provisions of European regulations or national law. AOCs can be issued by the competent authorities of the member states or the EASA, as further described in Regulation (EU) 2018/1139.

The EASA is responsible for safety and environmental protection in the civil aviation industry at the EU level. The EASA’s tasks include, among others, facilitating common regulation and certification, drafting technical rules for aviation, approving aircraft and their components as well as companies manufacturing and maintaining aircraft and their components, supervision of aviation safety and related support to EU member states, promotion of European and global safety standards as well as cooperation with international stakeholders to improve safety in Europe.

13.2 The International Regulatory Framework

13.2.1 The “Chicago System”

The regulatory system for international air transport is based upon the principles established by the Chicago Convention on International Civil Aviation of 1944 (the “**Chicago Convention**”) and is referred to as the “Chicago System.” The Chicago Convention, which has been signed by almost 200 states, including Germany in 1956, lays down the general principle that each state has sovereignty over its air space and has the right to control the operation of scheduled international air services over its territory. As a consequence, international air transport rights are based primarily on traffic rights (*Verkehrsrechte*) (i.e., the Freedoms of Air) granted by individual states to other states in bilateral air service agreements (*Luftverkehrsabkommen*, “ASA”). States, in turn, grant the rights they have received in an ASA to their national air carriers by way of designation.

In addition to indirect bilateral air traffic rights, some air traffic rights have been directly granted in multilateral agreements. The Chicago Convention liberalized air traffic for non-scheduled commercial flights, including cargo flights, over, and landings in, the territories of signatory states. Non-scheduled flights are subject to the restrictions of individual states. Air carriers generally obtain traffic rights for themselves for non-scheduled flights from the relevant foreign states. Only a few bilateral ASAs to which Germany is a party address non-scheduled services.

Concerning scheduled flights (*Linienflüge*), each signatory state of the International Air Services Transit Agreement of 1944 (the “**IATA**”), which has been signed by over 100 states, including Germany in 1956, grants air carriers based in signatory states the privilege to fly across its territory without landing as well as the privilege to land for non-traffic purposes. Such rights, however, may only be exercised between signatory states. Most notably, the Russian Federation is not a signatory to the IATA.

The Chicago Convention also established the ICAO, a specialized agency of the United Nations. The ICAO has developed Standards and Recommended Practices (the “**SARPs**”) for a wide range of activities. SARPs are published by the ICAO in the form of Annexes to the Chicago Convention. The enacted SARPs include aircraft operations, personnel licensing, security, aircraft accident and incident investigations, navigation services, aerodromes and environmental protection. Nearly 200 countries are members of the ICAO, including Germany, which has adopted most of the SARPs recommended by the ICAO.

The ICAO verifies compliance with SARPs through audits of state oversight systems. As of the date of this Prospectus, there are two audit programs, namely the Universal Safety Oversight Audit Programme (USOAP) as well as the Universal Security Audit Programme (USAP).

13.2.2 Bilateral Air Service Agreements

As the international regulatory framework is based mainly on bilateral air traffic rights, Germany is currently a party to bilateral ASAs with more than 120 states. These agreements govern the designation of airlines and airports for the operation of specified routes, airline capacity and fare-approval procedures. One of the focal points of ASAs is the mutual granting of air traffic rights within the meaning of Art. 6 Chicago Convention. In addition, ASAs generally contain provisions on equal treatment in the levying of charges, exemptions from customs duties and levies, the remittance of income, the transmission of operating data and statistics, the commercial activities of air carriers in the contracting state, aviation security, entry and control of travel documents, and dispute settlement.

Pursuant to these agreements, the contracting state grants the German government air traffic rights, such as the right to fly over, and to land in, the territory of the contracting state. The German government grants its rights under these bilateral agreements to German air carriers by way of designations. In addition, bilateral agreements typically give the designated airlines the right to operate scheduled passenger and air-freight services on certain routes between the contracting states.

States have traditionally granted their rights under bilateral agreements to only one carrier, the state’s flag carrier, or to a small number of airlines. In former times, Lufthansa German Airlines was the only carrier that the German government had designated to exercise air traffic rights under bilateral agreements. Over time,

the German government has designated other German airlines rights, which has increased competition on certain routes.

Bilateral ASAs to which Germany is a party set forth certain ownership requirements. Traditionally, bilateral agreements provide that Germany may only designate airlines that are substantially held, which is generally interpreted to mean majority held, and controlled, by German nationals and/or companies. Some bilateral ASAs, however, do not require a specific ownership structure; others are not referring to German nationality but to the nationality of an EU member state. Most bilateral agreements also require that the airline receiving the designation be able to demonstrate to the foreign state, upon request, that it meets the ownership requirements for a designation. If an airline fails to do this, the foreign state may prohibit the airline from using its designation under the agreement.

The Company, as an air carrier and listed stock corporation with its registered office in the Federal Republic of Germany, must know the identity and nationality of its shareholders to preserve its Operating Licenses and international air traffic rights. The German Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) provides that a listed stock corporation with its registered office in the Federal Republic of Germany that operates an air carrier within the meaning of Regulation (EC) 1008/2008 (repealing and replacing Regulation (EEC) No. 2407/92) may register the identity and nationality of its shareholders in its shareholder register (*Aktienregister der Gesellschaft*) and gives the air carrier recourse to certain measures and intervention rights to ensure that its shareholders do not jeopardize its licenses under aviation laws and ASAs. The German Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) also provides that a German air carrier may only issue restricted, registered shares. Pursuant to section 5 of our Articles of Association, the Executive Board may withhold its consent to the transfer of shares only if the registration of the transferee could jeopardize the licenses, rights and prerogatives under aviation laws and ASAs. See “17.3 Representation, Transferability.”

As of June 30, 2021, the Company had a shareholder structure broken down by nationality, according to which 83.9% was attributable to Germany, 7.0% to the United States, 1.8% to Ireland, 1.1% to Luxembourg, 1.1% to Great Britain and 5.1% to other countries. The requirements of the German Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) for the maintenance of air traffic control authority are therefore met.

13.2.3 Open Sky Agreements

In the 1990s, some EU member states, including Germany, liberalized air traffic with the United States by entering into open sky agreements. For example, the U.S.–German Open Sky Agreement liberalized the number of airlines that may be designated beneficiaries under the agreement, the number of weekly flights and the number of airport destinations. The agreement also provided a liberal tariff system.

In November 2002, the ECJ ruled that open sky agreements of individual EU member states violate the EC Treaty. The effect of this decision has not been to render these agreements invalid but to require that each EU member state treats air carriers from other EU member states not differently than home country air carriers. Because of this jurisprudence, Regulation (EC) No 847/2004 on the negotiation and implementation of ASAs between EU member states and third countries was adopted. This regulation sets out principles designed to ensure an adequate exchange of information within the EU, so that EU member states, in their bilateral relations with third countries in the area of air service, do not risk infringing EU law. EU member states may also delegate their powers to the European Commission to negotiate agreements with third countries on behalf of all the EU member states.

The EU concluded its own open sky agreement, the EU-US-Air Transport Agreement (the “EU-U.S. ATA”) with the United States, which has finally entered into force on June 29, 2020, after it had provisionally been applied since March 30, 2008. With Decision 2007/339/EC and Decision (EU) 2020/1110, the EU-U.S. ATA was signed by the European Community and approved by the EU.

Under the EU-U.S. ATA, U.S.- and EU-based airlines can fly on all air routes between the EU and the United States. Although U.S. airlines can fly intra-European flights as well as some routes outside Europe, EU airlines are not allowed to fly domestic flights in the U.S. Ownership of U.S. airlines remains restricted.

The EU has also entered into open sky agreements with Canada and Morocco and is leading negotiations with Turkey.

The EU's ultimate objective is to create a transatlantic open aviation area, *i.e.*, a single air transport market between the EU and the U.S. with free flows of investment and no restrictions on air services, including access to the domestic markets of both parties.

Regulation (EC) No 549/2004 of the European Parliament and of the Council of March 10, 2004 laying down the framework for the creation of the single European sky and three technical regulations on the provision of air navigation services (Regulation (EC) No 550/2004, No 551/2004 and No 552/2004) were adopted in 2004. They form the first 'Single European Sky' legislation package (the "**SES-I**") designed, in particular, to improve and reinforce safety and to reorganize the airspace on the basis of traffic instead of national borders. The SES-I was revised and extended in 2009 with Regulation (EC) No 1070/2009 aimed at increasing the overall performance of the air traffic management system in Europe (the "**SES-II**").

The SES-II changed the SES-I focus from capacity to performance in general. Its goal is to increase the economic, financial and environmental performance of the air navigation services in Europe. In particular, an EU-wide performance scheme was introduced; a refocus of the functional airspace blocks to be not just about airspace but service provision in general took place, and a network manager to co-ordinate certain actions at network level was established. Furthermore, it extended the competences of the EASA.

On this basis, the European Commission adopted and implemented extensive and comprehensive implementing legislation; this framework also includes more than 20 Implementing Rules and Community Specifications adopted by the European Commission in view of ensuring the interoperability of technologies and systems as well as the performance and charges regulation. The charges are regulated in regulatory periods and the current third regulatory period runs from 2020-2024. Due to the COVID-19 pandemic, derogations to Commission Implementing Regulation (EU) 2019/317 have been introduced by Commission Implementing Regulation (EU) 2020/1627 of November 3, 2020, in order to mitigate some of the financial impact of the COVID-19 crisis, leading to revised performance targets, plans and charges for the third regulatory period.

In September 2020, the European Commission has published a draft proposal on "SES II+," the revision of the existing SES-II regulatory framework. The proposal includes, among others, economic regulation by independent authorities, an increasingly harmonized and centralized European network management as well as unbundling and liberalization of certain services. The legislative procedure is ongoing as of the date of this Prospectus.

13.2.4 Slot Allocation

In order to use airport facilities, air carriers need take-off and landing rights (*i.e.*, slots). Slots are specific allocated time windows for using the runway and other facilities to unload their arrival traffic and to pick up their departure traffic at designated airports. Slots are particularly scarce at heavily frequented or congested airports. We must have access to a sufficient number of slots in order to maintain and expand our business.

Slot allocation at EU major airports, including fifteen German airports, is governed by Regulation (EC) No 95/93 (the "**Slots Regulation**"). Corresponding national rules are provided in sections 27a and 27b of the German Air Traffic Act (*Luftverkehrsgesetz*, the "**LuftVG**").

Airports are designated as either coordinated or schedules-facilitated airports. At coordinated airports, it is necessary for an air carrier to have a slot allocated by a coordinator for each arriving or departing flight. A schedules-facilitated airport is an airport where there is potential for congestion during some periods of the day, week or year, which is amenable to resolution by voluntary co-operation between air carriers. At schedules-facilitated airports, a schedules facilitator has been appointed to facilitate the operations of air carriers which are operating services or intending to operate services at a specific airport.

Under the Slots Regulation, an airport coordinator allocates slots for each scheduling period at the respective coordinated airport. Under Article 8 (2) read in conjunction with Article 10 (2) EU Slots Regulation, air carriers must use at least 80% of a slot series allocated to them, or otherwise lose historical precedence for these slots in the next equivalent scheduling period (summer / winter season) (the "**80/20 Rule**"). If the

number of applications exceeds the number of available slots, priority is given to the carriers that held the corresponding slots in the previous scheduling period and used such slots at least 80% of the time (“grandfathering”). On the other hand, if a carrier has failed to meet the usage threshold, it may lose the relevant slot, which may be allocated to a slot pool for assignment to other carriers. However, the Slots Regulation also provides that if an air carrier has failed to use a slot for exceptional reasons (namely because of unforeseen and unavoidable circumstances outside the air carrier’s control), the air carrier may be entitled to retain the slot (Article 10 (4) Slots Regulation). The EU has amended the Slots Regulation to suspend the application for the 80/20 Rule during the COVID-19 crisis (see below “13.4 Aspects Related to COVID-19”).

Slots are generally not transferable except under limited circumstances that are spelled out in Article 8a of the Slots Regulation.

13.2.5 Groundhandling

Since 1997, the provision of groundhandling services (*Bodenabfertigungsdienste*) in the EU is covered by Directive 96/67/EC.

Within the meaning of this directive, groundhandling means certain services provided to airport users at airports such as ground administration and supervision (e.g., handling, storage and administration of unit load devices), passenger handling (any kind of assistance to arriving, departing, transfer or transit passengers, including checking tickets and travel documents, registering baggage and carrying it to the sorting area) or freight and mail handling.

The directive opened up groundhandling services to competition. Prior to this, monopolies were the norm for groundhandling services at EU airports and many airlines complained about prices and quality standards. For certain services such as baggage handling, ramp handling, passenger transport and freight services, Directive 96/67/EC only requires to open the market to not fewer than two providers but also allows EU member states under certain circumstances to assign these services to only one provider. Germany has implemented Directive 96/67/EC and limited the number of providers for those services to the required minimum of two providers at the majority of our relevant airports. Directive 96/67/EC is implemented into German law through section 19c of the LuftVG and in detail in the ordinance on groundhandling services at airports (*Verordnung über Bodenabfertigungsdienste auf Flugplätzen*) (the “**BADV**”).

13.2.6 Airport Charges

Airport operators currently charge fees for incoming and outgoing flights based on a number of criteria in order to finance the relevant airport infrastructure and services.

Airport charges are regulated at the EU level through Directive 2009/12/EC. Directive 2009/12/EC, *inter alia*, prohibits the discrimination of airport users through airport charges, and sets out a compulsory consultation and approval or appeal procedure. The airport managing body is obliged to consult the airport users, their representatives or associations of airport users on a regular basis. Adaptions of charges must be either made in agreement between the airport managing body and the airport users or, in the event of a disagreement over a decision on airport charges taken by the airport managing body, determined by independent supervisory authorities. According to the directive, the EU member states shall ensure that each time consultations between the airport managing body and airport users or the representatives or associations of airport users take place, the respective airport managing body provides transparency on the system of airport charges, the level of airport charges, the underlying costs and, as appropriate, the quality of service provided.

Directive 2009/12/EC is implemented into German law through section 19b of the LuftVG. Under this provision, the managing body of a commercial airport must adopt a charges regulation indicating the charges to be paid for the use of facilities and services related to lighting, take-off, landing and parking of aircraft, as well as handling of passengers and cargo. In order for revised charges to become effective, the competent authority must approve the charges regulation. Approval shall only be granted if the charges are appropriate, objective, transparent and non-discriminatory. Additionally, at airports with more than 5 million passengers annually, approval shall only be granted if the charges are based on actual cost, are reasonable and efficiency orientated. However, the approving authority (*Genehmigungsbehörde*) may waive this review if the airport

operator submits a written agreement with the airport users on the fee schedule and provided there is no violation of state aid law. According to a judgement of the ECJ and the German Federal Administrative Court (*Bundesverwaltungsgericht*) in 2019 and 2020, the approval of the charges scheme is subject to appeal not only by the airport but also the airlines as airport users. Thus, airport users may directly challenge the approval of the airport charges regulation by the independent supervisory authority and do not have to bring an action against the airport managing body before a civil court, claiming that the charge set in the airport charges regulation, which they are required to pay, is not fair. The decisions also stated that the approval of the charges scheme generally has a binding effect between airport and the airlines to ensure non-discrimination.

13.2.7 Insurance, Liability and Passenger Rights

Regulation (EC) No 785/2004 establishes minimum insurance requirements for air carriers and aircraft operators in respect of passengers, baggage, cargo and third parties. In respect of the carriage of mail, the insurance requirements are those set out in Regulation (EEC) No 2407/92 and in the national laws of the EU member states. Compliance with the insurance requirements specified in Article 11 of the Air Services Regulation and in Regulation (EC) No 785/2004 are prerequisites for obtaining and keeping the Operating License of EU air carriers. We have obtained the insurance coverage needed to meet our obligations, particularly our minimum coverage obligations, under EU and German law as well as international air transport treaties. Our insurance coverage includes the following liability risks, among others: passenger contractual liability, third party liability and hull damage. See “12.9 Insurance.” Liability issues are governed both by international agreements (for example, Article 17 et seq. of the Warsaw Convention and the Montreal Convention) and by EU and national legislation.

Regulation (EC) No 2027/97, as amended, sets out air carrier liability in respect of the carriage of passengers and their baggage by air. According to this regulation, the liability of a Community air carrier in respect of passengers and their baggage is governed by all provisions of the Convention for the Unification of Certain Rules Relating to International Air Carriage by Air signed at Montreal on May 28, 1999 (the “**Montreal Convention**”). Under this regulation, air carriers are liable to, among others, compensate passengers for the destruction, delay or loss of, or damages to their carriage and baggage, up to a certain amount. Up to an amount of 100,000 Special Drawing Rights (as defined by the International Monetary Fund, “**SDR**”), the claims for compensation for the possible death or injury of passengers cannot be contested by the air carrier. To the extent that they exceed the amount of 100,000 SDRs, the air carrier is not liable if it proves that such damage was not due to the negligence or other wrongful act or omission of the carrier or its servants or agents or that such damage was solely due to the negligence or other wrongful act or omission of a third party.

The EU has further adopted Regulation (EC) No 261/2004 (the “**Air Passengers Regulation**”) concerning passengers who are denied boarding on flights for which they have a valid ticket. This Air Passengers Regulation also sets out fixed amounts of compensation payable to passengers whose flights are delayed or cancelled, except in cases where the airline proves that the cancellation was due to exceptional circumstances, such as weather conditions, delays by air traffic control or security and safety issues. The compensation amounts set out in the regulation are €250, €400 or €600 for each passenger, depending on the distance of the flight and whether the flight is an intra-Community or another flight. The ECJ has extended these obligations to delays of more than three hours. The Air Passengers Regulation also sets out obligations concerning the care and assistance of passengers, which become applicable as a result of cancellations or delays of flights.

13.2.8 Safety in Civil Aviation

Regulation (EC) No 300/2008 on rules in the field of civil aviation security confirms common rules at the EU level for the protection of civil aviation from unlawful interferences that jeopardize the security of civil aviation.

According to this regulation, air carriers shall draw up, apply and maintain an air carrier security program which complies both with this regulation as well as the civil aviation security program of the EU member state from which it provides services. Council Regulation (EEC) No 3922/91, as amended, imposes restrictions on maximum total duty time for cockpit and cabin crew members and stipulates defined stopping

times and rest periods. In addition, Commission Regulation (EU) No 965/2012, as amended, lays out more detailed technical security and safety specifications on, *inter alia*, the planning of flight duty periods in order to ensure that members of the crew remain sufficiently alert. The annexes of the Commission Regulation also include technical specifications for, *inter alia*, safety specifications, supervision, approval and maintenance of aircraft, as well as the transport of dangerous goods.

13.2.9 Competition Law

Airlines operating in the EU must, in addition to the observation of national competition laws of the countries in which they operate, observe EU competition rules (Articles 101 and 102 of the Treaty on the Functioning of the European Union (the “TFEU”)) and EU rules governing the provision of public subsidies (Articles 107 to 109 TFEU).

Articles 101 and 102 TFEU are the most substantial instruments to counteract measures restricting competition. Article 101(1) TFEU prohibits all agreements between undertakings, decisions by associations of undertakings and concerted practices which restrict or distort competition. Article 102 TFEU prohibits the abuse of a dominant market position.

An agreement which would normally restrict competition as set out in Article 101(1) TFEU may not be unlawful in all cases. Provided that certain requirements are met, exemptions can generally be granted in accordance with Article 101(3) TFEU to certain categories of agreements or concerted practices in the aviation industry. In addition to individual exemptions, it is also possible to apply more generally applicable block exemptions. As of the date of this Prospectus, there are no block exemptions in force that would apply particularly to the air transport sector. However, the European Commission has adopted general block exemptions for certain types of horizontal and vertical agreements that also apply to the air transport sector.

According to Article 107(1) TFEU, aid granted through resources of an EU member state which distorts competition by favoring certain undertakings or certain fields of business are generally prohibited. However, under Article 107(2) and (3) TFEU, certain forms of aid may be considered permissible, *e.g.*, aid to facilitate the development of certain economic activities or of certain economic areas, where such aid does not adversely affect trading conditions to an extent contrary to the common interest of the EU. According to Article 108(3) TFEU, an EU member state must inform the European Commission of any plans to grant or alter aid. The EU member state concerned may not put its proposed measures into effect until the European Commission has approved the state aid. The European Commission may put its approval under certain conditions, which the aid beneficiary must meet. If the state aid has been granted unlawfully without prior approval or the conditions have not been met, there is a risk that the EU member state has to recover the aid from the beneficiary.

During the COVID-19 pandemic, the European Commission has adopted a “Temporary Framework” (the “TF”) explaining under which conditions it will approve aid measures to support the economy. Such subsidies are especially important for undertakings active in sectors particularly affected by the pandemic, including the air transport sector.

We have received state aid by Germany, Austria and Belgium approved by the European Commission under the TF. See “4 The Stabilization Package.”

13.2.10 Data Protection

Like other EU companies, we are subject to increasing regulation relating to data protection. For example, Regulation (EU) 2016/679 of April 27, 2016 on the protection of natural persons (General Data Protection Regulation) (the “GDPR”). The GDPR went into effect in May 2018, implementing more stringent requirements in relation to companies’ use of personal data relating to individuals (“data subjects”). Under the GDPR, the expanded definition of personal data include information such as name, identification number, email address, location data, online identifiers such as Internet protocol addresses and cookie identifiers, or any other type of information that can identify a living individual. The GDPR imposes a number of new requirements, which include: (i) providing expanded information about how data subjects’ personal data is or will be used; (ii) carrying out data protection impact assessments for operations which present specific risks to individuals due to the nature or scope of the processing operation; (iii) an obligation

to appoint data protection officers in certain circumstances; (iv) new rights for individuals to be “forgotten” and rights to data portability, as well as enhanced current rights; (v) the principal of accountability and demonstrating compliance through policies, procedures, training and audit; (vi) profiling restrictions; and (vii) a new mandatory data breach reporting regime. Moreover, we are subject to data protection laws and regulations introduced by other countries in which we operate, including Brazil and, with respect to the State of California, the United States. While their respective data protection laws are, to a certain extent, built on the GDPR framework, they include various further provisions with which we are obliged to comply.

13.3 The German Regulatory Framework

The Air Services Regulation has partly replaced section 20 LuftVG in respect of the granting of Operating Licenses for intra-EFTA air traffic. Air carriers must nevertheless make an application to the German Air Traffic Authority and the only grounds for denial of the application are contained in the Air Services Regulation and section 20 (2) LuftVG.

Scheduled routes to destinations outside of the EU require a license under section 21 LuftVG. Route licenses are issued for a single flight schedule period and are generally renewed for the next flight schedule period. The rights of German carriers to fly to destinations outside the EU are governed by bilateral or multilateral ASAs.

Under German law, German aircraft may only operate if they have been admitted to traffic with the award of an airworthiness certificate (*Lufttüchtigkeitszeugnis*) and are listed in the German registry of aircraft. The prerequisites for admitting aircraft to traffic are codified in the German Air Traffic Licensing Ordinance (*Luftverkehrs-Zulassungs-Ordnung*). Aircraft that are not admitted to traffic and registered in Germany may still operate in the German airspace without the additional permits or registrations in Germany if the operating air carrier holds an Operating License under the Air Services Regulation or if the aircraft is registered and certified for traffic in another EU member state and has certificate of airworthiness in accordance with the Regulation (EU) 2018/1139 (repealing and replacing Regulation (EC) No. 216/2008) (section 2(7) LuftVG).

According to Article 13 of Regulation (EC) No. 300/2008 and section 9 of the German Air Security Act (*Luftsicherheitsgesetz*), an air carrier is required to demonstrate specific security measures as set out in and in compliance with a security program approved by the German air traffic authority.

13.4 Aspects Related to COVID-19

Due to the outbreak of COVID-19, many countries have issued entry regulations and travel bans. Entry regulations and travel bans vary widely from country to country and are subject to constant change. Currently, it cannot be foreseen at what time these entry regulations and travel bans will be revoked or whether revoked entry regulations and travel bans are re-instated. Entry regulations and travel bans stop us from resuming or continuing flights into affected countries and/or regions. This has led and may further lead to a significant decline in bookings and to ongoing flight cancellations.

We are subject to claims for compensation for passengers under the Air Passenger Rights Regulation in the context of flight cancellations which fall under the scope of the Air Passengers Regulation.

Article 5 (3) of the Air Passengers Regulation waives the right to compensation on condition that the cancellation in question ‘is caused’ by extraordinary circumstances, which could not have been avoided even if all reasonable measures had been taken. The European Commission considers that, where public authorities take measures intended to contain the COVID-19 pandemic, such measures are by their nature and origin not inherent in the normal exercise of the activity of carriers and are outside their actual control (Commission notice 2020/C 89 I/01). This condition should be considered fulfilled, inter alia, (i) where public authorities either prohibit certain flights or ban the movement of persons in a manner that excludes, de facto, the flight in question to be operated or (ii) where the flight cancellation occurs in circumstances where the corresponding movement of persons is not entirely prohibited, but limited to persons benefitting from derogations (for example nationals or residents of the state concerned). According to the European Commission, where the airline decides to cancel a flight and shows that this decision was justified on, for

example, grounds of protecting the health of the crew, such cancellation should also be considered as ‘caused’ by extraordinary circumstances.

Due to decrease in demand for air travel as a result of the outbreak of the COVID-19 pandemic, airlines are not able to fulfill the requirement of the 80/20 Rule (see “13.2.4 Slot Allocation”). On March 30, 2020, the EU adopted Regulation (EU) 2020/459 amending the Slots Regulation to provide airlines with relief from the 80/20 Rule. On February 16, 2021, the EU has enacted a further alleviation from the 80/20 Rule with Regulation (EU) 2021/250. Air carriers were granted the possibility to return 50% of their slot series prior to the start of the 2021 summer season. However, they had to use at least 50% of the remaining slots to maintain such slots for upcoming scheduling periods. The European Commission is entitled to adopt delegated acts for one year to cover the following two seasons (until February 21, 2022). With these acts, the European Commission may change the minimum utilization rate to between 30% and 70%, but there will be no possibility to return slot series at the beginning of a season. This grants some flexibility to adapt to different air traffic levels based on traffic data and forecasts and other indicators. The aim of the full waiver of the 80/20 Rule and the pursuant alleviation is to protect the financial health of air carriers and avoid the negative environmental impact of empty or largely-empty flights operated only for the purpose of maintaining underlying airport slots. On July 23, 2021, the European Commission proposed to amend the Slot Allocation Regulation in order to extend the slot relief rules adopted in February 2021. Reportedly, slot relief may be extended to the upcoming winter scheduling season, running from October 31, 2021 to March 27, 2022. According to the proposal, airlines would only have to use 50% of a slot series, and not at least 80%, to obtain historic rights in those slots. Furthermore, the use requirement shall not apply at all when state-imposed measures severely impede passengers’ ability to travel on routes for which the slots were intended (the so-called ‘justified non-use of slots’ exception).

The European Commission’s approval of the stabilization package, granted by the WSF on May 25, 2020, to us, was conditioned, inter alia, on our divestment of up to 24 slots/day to operate a base of four aircraft (summer and winter season) to allow for a “reasonable aircraft rotation.” Our offer to divest slots has to be upheld for six IATA seasons after full restoration of the EU Slots Regulation.

On May 25, 2020, Regulation (EU) 2020/696 was adopted and temporarily amends the Air Services Regulation by (i) suspending, until the end of 2020, the obligation for member states to suspend or revoke the Operating License of any air carrier if the member state is no longer convinced that such carrier will fulfill its financial obligations for the next year (such exemption has subsequently been prolonged until end of 2021 by Delegated Regulation (EU) 2020/2115), (ii) providing that, until the end of 2020, member states may temporarily keep justified and proportionate emergency measures in place for a period beyond 14 days, limited to the duration of public health risks clearly linked to the pandemic, (iii) allowing for authorizations for suppliers of groundhandling services at EU airports that expire in the period May 28, 2020, to December 31, 2021, to be prolonged until December 31, 2022, and (iv) introducing, until the end of 2020, an urgent procedure for the selection of suppliers of groundhandling services during the crisis, so that contracts can be awarded more efficiently.

14. ENVIRONMENTAL MATTERS

The most essential environmental regulations concerning the operation of aircraft relate to greenhouse gas emissions and noise emissions.

14.1 Greenhouse Gas Emissions

Under the United Nations Framework Convention on Climate Change (the “UNFCCC”) and the Kyoto Protocol which was superseded by the Paris Agreement, the contracting parties entered into obligations to control and reduce emissions of greenhouse gases. To comply with its obligations under public international law, the EU introduced a scheme to limit greenhouse gas emissions and for trading of allowances applicable to certain industrial installations (the “EU Emissions Trading System” or the “EU ETS”) through the adoption of Directive 2003/87/EC (the “EU ETS Directive”).

Following the adoption of the Paris Agreement, the ICAO adopted the Carbon Offsetting and Reduction Scheme for International Aviation (the “CORSIA”) in order to monitor, report and offset annual CO₂ emissions from international civil aviation that exceed 2020 levels. Due to the outbreak of the COVID-19 pandemic, the ICAO council determined that the value of the 2019 emissions should also be applied for 2020 emission levels in an effort to avoid inappropriate economic burden on the aviation industry. The CORSIA is an “offsetting scheme” under which total emissions may also increase, but under which the emissions must be compensated by offsets. To compensate for CO₂ emissions above 2020 levels in international aviation and to achieve carbon-neutral growth over time, emitters will have to purchase emissions units. In terms of timing, the CORSIA is divided into three phases, including the pilot phase in 2021–2023, phase one in 2024–2026 and phase two in 2027–2035. The CORSIA makes aviation the first industry to establish a global market-based emissions system.

In June 2018, the Council of the ICAO adopted the First Edition of Annex 16 — Environmental Protection, Volume IV — the CORSIA which became applicable on January 1, 2019, which is also referred to as International Standards and Recommended Practices (the “SARPs”). The SARPs set the monitoring, reporting and verification rules as well as the rules for deletion of certificates, *e.g.*, offsets. To implement these rules, the EU has enacted a series of Regulations such as the Regulation (EU) 2017/2392 which amended the EU ETS Directive in 2017. One of the purposes of that Regulation was to prepare for the implementation of CORSIA from 2021, to lay down requirements under EU law for the monitoring, reporting and verification for the purpose of CORSIA, and for the reporting on and the review of the implementation of the CORSIA. Apart from Regulation (EU) 2017/2392, the Commission Implementing Regulations (EU) 2018/2066, (EU) 2018/2067 and (EU) 2019/1603 have been passed.

Participation of the ICAO member states in the CORSIA is voluntary during the pilot phase and first phase (2021–2026), becoming mandatory for all member states as of 2027 unless there are exceptions (like for Least Developed Countries). In June 2020, the EU member states have decided that they will participate voluntarily. Since May 2020, an aeroplane operator needs to submit a copy of the verified emissions report together with the associated verification report to the state to which the aeroplane operator is attributed. Starting in August 2020, the state has to submit to the ICAO the state’s emissions report for the aggregated 2019 CO₂ emissions of the aeroplane operators attributed to the state.

By the Directive 2008/101/EC, which entered into force in 2009, aviation activities have been included in the EU ETS. Since 2012, the EU ETS applies to all flights departing from or arriving in EU and the EEA states. Since 2020, the EU ETS applies also to flights departing from or arriving in Switzerland. The EU ETS is an emission trading system that requires aircraft operators operating flights between, departing from and arriving in EU member states and the EEA EFTA states as well as Switzerland to report their emissions from such flights and account for those emissions by surrendering allowances. The EU has already announced its intention to gradually reduce the number of allowances available under the EU ETS. If allowances are reduced, operating costs will increase even for airlines that hold their emissions stable, as the credits threshold will be reached earlier and additional allowances may need to be purchased.

The revision of the EU ETS Directive in 2017 translates the understanding, spelled out in Article 28b, that CORSIA would be taken into account by way of an amendment to the scheme governing the EU ETS, subject to an assessment of the key features of CORSIA, including in terms of participation and that this

should be achieved in a manner consistent with the EU climate objective for 2030. Although the aim is to reconcile the supervision and monitoring obligations under EU emissions trading and the CORSIA, diverging requirements are still possible.

On June 30, 2021, the European Union adopted Regulation (EU) 2021/1119, also known as the “European Climate Law,” aiming at reducing emissions to net zero at the latest by 2050 for the EU as a whole. This binding target covers all sectors and all greenhouse gases, and requires EU member states to take ‘necessary measures’ to meet the target of climate neutrality by 2050. The European Climate Law includes measures to keep track of progress and adjusts the actions accordingly, based on existing systems such as the governance process for member states’ national energy and climate plans, regular reports by the European Environment Agency, and the latest scientific evidence on climate change and its impact. Moreover, the European Climate Law describes a domestic reduction of net greenhouse gas emissions (emissions after deduction of removals) by at least 55% compared to 1990 levels by 2030.

Under the “European Green Deal” initiative, the European Commission presented in September 2020 an impact-assessed plan to increase the EU’s greenhouse gas emission reduction target of at least 55% below 1990 levels by 2030, as agreed in the European Climate Law. The European Commission has published an inception impact assessment and conducted a public consultation on the revision of the system from November 2020 to February 2021. On July 14, 2021, as part of the European Green Deal, the European Commission adopted a set of proposals to revise and update EU legislation aimed at aligning current laws with the EU’s greenhouse gas emission reduction target of at least 55% below 1990 levels by 2030, by, inter alia, extending the application of emissions trading to new sectors and tightening the existing EU ETS (the so-called “**Fit for 55 Package**”). To boost the supply and demand for sustainable aviation fuels in the EU and, in turn, reduce aviation’s environmental footprint and enable it to help achieve the EU’s climate targets, the Fit for 55 Package includes a proposal for a ‘Regulation on ensuring a level playing field for sustainable air transport’ (2021/0205 (COD), the “**Proposal Regulation**”).

Under the Proposal Regulation, the overarching framework set out by Directive (EU) 2018/2001 on the promotion of the use of energy from renewable sources shall be complemented with set of rules specifically applicable to air transport ensuring a harmonized set of rules on the uptake and supply of sustainable aviation fuels with a minimum share of sustainable aviation fuels becoming mandatory from January 1, 2025 and gradually increasing from 2% in 2025 to 5% in 2030, and to a minimum volume share of 63% of sustainable aviation fuels in 2050.

According to the Proposal Regulation, to avoid exacerbation of unsustainable practices of fuel tankering as a consequence of aviation fuel costs increases due to introduction and ramp-up of sustainable aviation fuels at airports located in the European Union, aircraft operators shall be required to refuel prior to departures from a given airport located in the European Union, with the amount of fuel uplifted prior to departures from such airport to be commensurate with on average 90% of the amount of fuel necessary to operate the flights departing from that airport, taking into account the necessary compliance with fuel safety rules; thereby ensuring equal conditions for operations in the EU applying equally to all operators, and, hence, ensuring that airlines operating in the EU can compete on the basis of equal opportunity and that all air carriers operating to and from the EU benefit of a level playing field. Applying to aircraft operators, airports located in the European Union, and to aviation fuel suppliers (all as defined by the Proposed Regulation, so as to cover the largest possible share of commercial air traffic operated from airports located on EU territory without imposing undue burden on small airports and aircraft operators accountable for a very low number of departures from airports located on EU territory), (i) aviation fuel suppliers shall ensure that all aviation fuel made available to aircraft operators at each airport located in the European Union contains a minimum share of sustainable aviation fuels and annually report on their supply of aviation fuel, including sustainable aviation fuels, (ii) airports located in the European Union shall take necessary measures to facilitate the access of aircraft operators to aviation fuels containing shares of sustainable aviation fuels and shall provide the infrastructure necessary for the delivery, storage and uplifting of such fuel, and (iii) aircraft operators shall report yearly to the Agency on their purchases of sustainable aviation fuels, the characteristics of this fuel, and on their actual aviation fuel uplift per airport located in the European Union, so as to prove that no fuel tinkering was performed.

Notably, the European Commission is also proposing to phase out free emission allowances for aviation in the European Emission Trading System and aligning emission treatment with CORSIA. The proposals also include a revision of the Energy Taxation Directive (2003/96/EC) of 27 October 2003 by introducing minimum tax rates applicable to aviation fuels used on flights within the European Union. The proposed tax rates on aviation fuel are intended to increase gradually over a ten year period to reach an EU-wide minimum rate. Sustainable and alternative aviation fuels will benefit from a zero minimum tax rate during this transition period.

14.2 Noise Emissions

Directive 89/629 on the limitation of noise emission from civil subsonic jet aeroplanes aims to limit noise caused by air traffic. The directive obliges EU member states to ensure that civil subsonic jet aeroplanes registered in their territory comply with Part II, Chapter 3, Volume 1 of Annex 16 to the Chicago Convention, excluding aircraft which have been registered before November 1, 1990 and with exemptions for certain overseas departments. In February 2016, the Committee on Aviation Environmental Protection (the “CAEP”) of the ICAO further approved amendments to Annex 16 Volume I “Aircraft Noise” and Volume II “Aircraft Engine Emissions” of the Chicago Convention, which have meanwhile entered into force. These restrictions may cause curtailment of service or increases in operating costs and could limit the ability to expand operations at affected airports. Directive 2002/49/EC sets a general framework for the assessment and management of noise. The directive aims at harmonizing noise indicators and noise maps, as well as encouraging states to conduct detailed evaluations of noise exposure. In addition, Regulation (EU) No 598/2014 further limits noise at and around airports. For airports with more than 50,000 civil aircraft movements per calendar year, the regulation establishes rules and procedures with regard to the introduction of noise related operating restrictions at EU airports. Airport charges are in many cases graded in accordance with the noise emission standards of aircraft.

14.3 Operation of Sites

The operation of sites is subject to international, national and, in some cases, local environmental regulation standards. With regard to operations in Germany, the German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz*) provides for a liability for harmful soil changes and contamination by, *inter alia*, any person that causes contamination, owns or leases respective property, irrespective of fault. Under certain circumstances liability also attaches to formerly owned properties. Furthermore, authorities have discretion in deciding which of the aforementioned persons may be required to undertake investigations with regard to potential harmful soil changes and contaminations. Generally, the statutory remediation obligation is not limited in time and there is no statutory cap on associated costs. If harmful soil changes or contaminations affect water bodies or ground water, the liability includes water-related remedies as well. A comparable obligation applies under the German Water Act (*Wasserhaushaltsgesetz*) to persons who cause significant adverse effects on water bodies.

15. SHAREHOLDER STRUCTURE

The Company's share capital as of the date of this Prospectus amounts to €1,530,221,624.32, divided into 597,742,822 ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*).

The German Securities Trading Act (*Wertpapierhandelsgesetz*) requires holders of shares of an issuer admitted to trading on a regulated market (*regulierter Markt*) in Germany and to which voting rights are attached or instruments which relate to such shares to notify the respective issuer and the BaFin without undue delay of the level of their holdings if they reach, exceed or fall below certain thresholds. The initial threshold triggering a notification requirement is 3.0% of the voting rights in the event of holdings in shares and 5.0% of the voting rights in the event of instruments or cumulative holdings in shares and instruments (see "17.15 Disclosure Requirements for Holdings of Shares and Other Instruments"). As of the date of this Prospectus, the following shareholders have notified us of their notifiable holdings in the Company:

Shareholders	Major Holdings ⁽¹⁾			
	Direct	Indirect	Instruments ⁽⁴⁾	Total
	Shareholdings ⁽²⁾	Shareholdings ⁽³⁾		
	(in %)			
WSF ⁽⁵⁾	15.94	—	—	15.94
The Goldman Sachs Group, Inc. ⁽⁶⁾	—	0.08	6.80	6.88
Société Générale ⁽⁷⁾	0.72	0.02	4.28	5.02

- (1) The percentage of voting rights has been calculated based on the Company's total number of voting rights (as published pursuant to Section 41 of the German Securities Trading Act (*Wertpapierhandelsgesetz*)) on the date of the respective shareholding notification or as notified otherwise.
- (2) Direct shareholdings pursuant to Section 33 of the German Securities Trading Act (*Wertpapierhandelsgesetz*).
- (3) Indirect shareholdings pursuant to Section 33, 34 of the German Securities Trading Act (*Wertpapierhandelsgesetz*).
- (4) Directly and indirectly held instruments pursuant to Section 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*).
- (5) Shareholding of the WSF, as registered in the Company's share register (*Aktienregister*) on September 17, 2021.
- (6) Shareholding of The Goldman Sachs Group, Inc., as notified to the Company on June 1, 2021.
- (7) Shareholding of Société Générale, as notified to the Company on July 6, 2021.

16. GENERAL INFORMATION ON THE COMPANY AND THE GROUP

16.1 Formation, Incorporation, Registered Corporate Seat, Commercial Name, LEI

The Company is a German stock corporation (*Aktiengesellschaft*), incorporated and operating under German law, with its registered corporate seat in Cologne and registered with the commercial register (*Handelsregister*) of the local court (*Amtsgericht*) in Cologne under registration number HRB 2168. It was incorporated on January 6, 1953 as “Aktiengesellschaft für Luftverkehrsbedarf” and, in 1954, changed its corporate name to “Deutsche Lufthansa Aktiengesellschaft.” The Company’s legal and commercial name is “Deutsche Lufthansa Aktiengesellschaft.” The Company’s Legal Entity Identifier (*LEI*) is 529900PH63HYJ86ASW55.

16.2 History of the Company

Two years after the Allies dissolved the first Lufthansa (founded in 1926) in 1951, the “Aktiengesellschaft für Luftverkehrsbedarf” (Luftag) with headquarters in Cologne was founded on January 6, 1953. On August 6, 1954, Luftag bought the name, the trademark – the crane – and the colors – blue and yellow – from the first Lufthansa, which was in liquidation at the time, and has since then called itself “Deutsche Lufthansa Aktiengesellschaft.” On April 1, 1955, the first two Lufthansa airplanes commenced the Company’s scheduled air services. The Company developed a European route network and in parallel commenced flights to destinations in America, Africa and the Far East. In 1959, the Company operated its first scheduled flight to Asia.

In 1960, the Company transferred its long-distance operations from Hamburg to Frankfurt am Main and continued to expand its cargo business. This expansion was followed by a decade of crises, including the oil crises of 1973 and 1979, which significantly increased the prices for kerosene. Simultaneously, it created a new understanding of how to manage scarce resources and drove the development of fuel-efficient and quieter jet engines. In the following decades, the airplane developed into a means of mass transportation. The Company reacted by redesigning their route network with faster connections and fewer stopovers.

In the second half of the 1990s, the Lufthansa Group faced significant changes. On the one hand, in 1995 Lufthansa Technik, Lufthansa Cargo and Lufthansa Systems were transformed into independent companies of the aviation group and on the other hand, in 1997, Lufthansa was privatized. Both developments were meant to increase the Lufthansa Group’s competitiveness and contributed to its long-term strategy of developing into the world-wide leading provider of air travel and air travel contiguous services.

16.3 Business Address, Term, Fiscal Year and Corporate Purpose

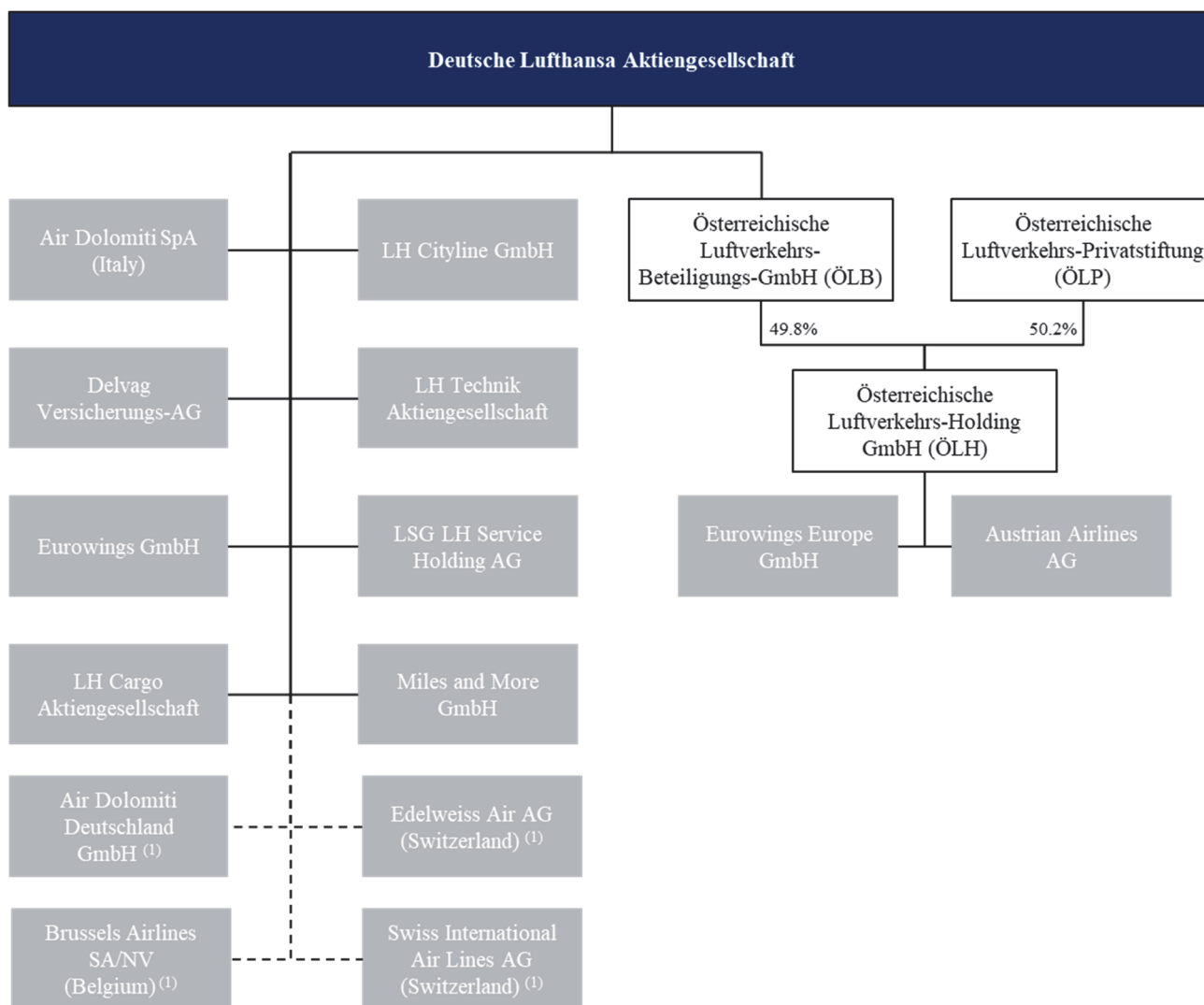
The Company’s registered office is at Venloer Straße 151-153, 50672 Cologne and its head office is located at Lufthansa Aviation Center, Airportring, 60546 Frankfurt am Main, Germany; its telephone number is: +49 (0)69 - 696 0. The Company is established for an indefinite period of time. The fiscal year of the Company is the calendar year.

Pursuant to section 2 para. 1 of the Articles of Association, the corporate purpose of the Company is national and international air traffic and the operation of all commercial activities and facilities connected directly or indirectly with and relating to civil aviation and its promotion. For the furtherance of its business purpose, the Company shall be entitled to establish domestic and foreign branches and agencies, to acquire participating interests in other domestic and foreign enterprises, to acquire outright or set up such enterprises and to conclude all manner of business contracts, including pooling agreements. It can devolve its activities completely or partially to such business ventures (section 2 para. 2 of the Articles of Association).

16.4 Group Structure

The Company is the main operating and holding company within the Lufthansa Group.

The following chart provides an overview (in simplified form) of the shareholdings of the Company as of the date of this Prospectus:



(1) Indirect holding of the Company.

16.5 Significant Subsidiaries

The following table provides an overview of the significant (in relation to the Lufthansa Group's equity or net income or from an operating perspective) direct and indirect subsidiaries of the Company as of the date of this Prospectus.

Legal Name	Registered Seat	Direct or Indirect Interest as of June 30, 2021
Austrian Airlines AG	Vienna, Austria	100.0%
Lufthansa Cargo Aktiengesellschaft	Frankfurt am Main, Germany	100.0%
Lufthansa Technik Aktiengesellschaft...	Hamburg, Germany	100.0%
SWISS International Air Lines AG	Zurich, Switzerland	100.0%

16.6 Auditors

For the Fiscal Year 2019 and the Fiscal Year 2018, the Company had appointed PricewaterhouseCoopers GmbH Wirtschaftsprüfungsgesellschaft with its registered seat in Frankfurt am Main, Germany, acting through its Düsseldorf office at Moskau Str. 19, 40227 Düsseldorf, Germany (“**PwC**”) as the auditor of the Audited Consolidated Financial Statements 2019 and the Audited Consolidated Financial Statements 2018, each of which was prepared in accordance with IFRS and included additional disclosure under requirements of German commercial law pursuant to section 315e para. 1 of the German Commercial Code (*Handelsgesetzbuch*). In each case, PwC conducted its audits of the German-language versions of these financial statements in accordance with EU Audit Regulation No. 537/2014, Section 317 of the German Commercial Code (*Handelsgesetzbuch*) and German generally accepted standards for the audit of financial statements promulgated by the Institute of Public Auditors in Germany (*Institut der Wirtschaftsprüfer e.V.*) and in supplementary compliance with the International Standards on Auditing (*ISAs*) and issued unqualified auditor’s reports in each case. PwC is a member of the German Chamber of Auditors (*Wirtschaftsprüferkammer*), Rauchstraße 26, 10787 Berlin, Germany and a member of the Institute of Public Auditors in Germany (*Institut der Wirtschaftsprüfer e.V.*).

For the Fiscal Year 2020, the Company changed its auditor in order to comply with the requirement for public interest entities (*Unternehmen von öffentlichem Interesse*) to rotate its auditor (*externe Prüferrotation*). As a result, for the Fiscal Year 2020, we appointed the independent auditors of Ernst & Young GmbH Wirtschaftsprüfungsgesellschaft, with its registered seat in Stuttgart, Germany, acting through its office at Mergenthalerallee 3-5, 65760 Eschborn, Germany (“**EY**”) as the auditor of the German-language versions of the Audited Consolidated Financial Statements 2020 and the Audited Annual Financial Statements. The Audited Consolidated Financial Statements 2020 were prepared in accordance with IFRS and included additional disclosure under requirements of German commercial law pursuant to section 315e para. 1 of the German Commercial Code (*Handelsgesetzbuch*). The Audited Annual Financial Statements were prepared in accordance with German law and the Articles of Association. In each case, EY conducted its audits of the German-language versions of these financial statements in accordance with EU Audit Regulation No. 537/2014, section 317 of the German Commercial Code (*Handelsgesetzbuch*) and German generally accepted standards for the audit of financial statements promulgated by the Institute of Public Auditors in Germany (*Institut der Wirtschaftsprüfer e.V.*) and issued unqualified German-language auditor’s reports in each case. EY is a member of the German Chamber of Auditors (*Wirtschaftsprüferkammer*), Rauchstraße 26, 10787 Berlin, Germany and a member of the Institute of Public Auditors in Germany (*Institut der Wirtschaftsprüfer e.V.*).

16.7 Notifications, Paying Agent

Pursuant to the Articles of Association, the Company’s notices are published in the German Federal Gazette (*Bundesanzeiger*), unless where another mandatory form of notice is required by law. Notices to our shareholders may also be communicated by data transmission. Notices regarding the Shares are also published in the German Federal Gazette (*Bundesanzeiger*).

Notices in connection with the approval of this Prospectus or any supplements thereto will be published in accordance with the Prospectus Regulation, in the manner of publication provided for in this Prospectus, that is, through publication on the internet website of the Deutsche Lufthansa Aktiengesellschaft (<https://www.lufthansagroup.com/capital-increase>).

COMMERZBANK is the paying agent for the Company’s Shares. The mailing address of the paying agent is: COMMERZBANK Aktiengesellschaft, Kaiserstraße 16 (Kaiserplatz), 60311 Frankfurt am Main, Germany.

17. DESCRIPTION OF SHARE CAPITAL OF THE COMPANY

17.1 Current Share Capital; Shares

As of the date of this Prospectus, the share capital of the Company amounts to €1,530,221,624.32 and is divided into 597,742,822 ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*) and with each of the Company's share representing a notional share in the share capital of €2.56. The share capital is fully paid up. The Shares were created pursuant to the laws of Germany. As of the date of this Prospectus, the Shares are admitted to trading on the regulated market segment (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with simultaneous application for the sub-segment thereof with additional post-admission obligation (Prime Standard). For further details, see "2.4 Subject-Matter of this Prospectus."

17.2 Form, Voting Rights

All Shares are ordinary registered no-par value shares with restricted transferability (*vinkulierte Namensaktien*). Each Share entitles the shareholder to one vote at the General Meeting of the Company. There are no restrictions on voting rights. Voting rights are the same for all the Company's shareholders, *i.e.*, none of the shareholders have different voting rights. Voting rights, however, cannot be exercised until the respective capital contribution is paid in full. The Company's subsidiaries do not hold shares in the Company. No Shares are held by other parties on behalf or for the account of the Company or any of its subsidiaries by other parties.

17.3 Representation, Transferability

The Company is entitled to issue share certificates representing individual shares or a global share certificate representing several shares. According to the Articles of Association, the shareholders are not entitled to individual certification of their Shares.

The provisions of the German Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) require that the Company may only issue registered shares with restricted transferability (*vinkulierte Namensaktien*) and that the transfer of shares is subject to the Company's consent. Accordingly, all shares of the Company (including the New Shares) are issued as restricted shares with limited transferability. The Executive Board is responsible for issuing the consent to transfer shares. According to section 5 para. 1 of the Articles of Association, such consent may be withheld only if the registration of the new shareholders could jeopardize the Company's licenses, rights and prerogatives under applicable aviation laws and agreements. This could occur if, as a result of a transfer of shares, German citizens or German companies no longer hold a majority of the Company's share capital or no longer control the Company.

Each shareholder's identity and nationality must be recorded in the Company's share register. The nationality of a corporation or other legal entity is determined by the entity's registered office. Entities who individually or together with other entities hold 3% or more of the voting rights in the Company must inform the Company whether they are, directly or indirectly, under foreign ownership or control, and if so, they must describe the identity of such owner or controlling entity.

17.4 Development of the Share Capital

The share capital of the Company has developed as follows between January 1, 2018, and the date of this Prospectus:

Date	Event	Subsequently Subscribed Share Capital	Number of Shares Subsequently Outstanding
June 2018	Increase of the Company's share capital by an amount of €6,130,027.52 by issuing 2,394,542 new shares in connection with a share dividend	€1,212,554,716.16	473,654,186
October 2018	Increase of the Company's share capital by an amount of €3,984,750.08 by issuing 1,556,543 new shares to the employees	€1,216,539,466.24	475,210,729

Date	Event	Subsequently Subscribed Share Capital	Number of Shares Subsequently Outstanding
October 2019	Increase of the Company's share capital by an amount of €7,637,831.68 by issuing 2,983,528 new shares to the employees	€1,224,177,297.92	478,194,257
July 2020	Increase of the Company's share capital by an amount of €306,044,326.40 by issuing 119,548,565 new shares to the WSF	€1,530,221,624.32	597,742,822
September 2021	Increase of the Company's share capital by an amount of €1,530,221,624.32 by issuing 597,742,822 New Shares	€3,060,443,248.64	1,195,485,644

17.5 Authorized Capital

17.5.1 Authorized Capital A

As of the date of this Prospectus, pursuant to section 4 para. 2 of the Articles of Association, by way of a resolution passed at the General Meeting on May 7, 2019, amended by way of resolution passed at the General Meeting on May 5, 2020, the Executive Board is authorized, with the consent of the Supervisory Board, to increase the share capital of the Company once or several times until the end of May 6, 2024, by up to €450,000,000 through the issue of new no-par registered value shares for cash or contributions in kind (Authorized Capital A). In principle, the shareholders are to be granted a subscription right. The shareholders may also be granted a subscription right indirectly pursuant to section 186 para. 5 of the German Stock Corporation Act (*Aktiengesetz*).

- a) The Executive Board is authorized, in the case of a capital increase for cash contributions, with the consent of the Supervisory Board, to exclude shareholders' subscription rights if the offering amount is not significantly below the market price, and the shares issued with subscription rights excluded, pursuant to section 186 para. 3 sentence 4 of the German Stock Corporation Act (*Aktiengesetz*), do not exceed 10% of the share capital (10% limit) at the time of the authorization taking effect or, if this amount is lower, at the time of its exercise. If during the term of the Authorized Capital A before it is utilized, other authorizations to issue or sell shares or to issue rights that enable or oblige the purchase of shares of the Company are exercised and for which subscription rights are excluded in direct or analogous application of section 186 para. 3 sentence 4 of the German Stock Corporation Act (*Aktiengesetz*), this is subject to the 10% limit stated in the above sentence.
- b) Wherever it is necessary to grant holders or creditors of warrant or conversion rights under bonds with warrants attached or convertible bonds that were or are issued by the Company or its Group companies a subscription right to new shares on a scale that would be due to them after exercise of their warrant or conversion rights and/or the meeting of conversion obligations, the Executive Board is authorized to exclude the subscription rights with the consent of the Supervisory Board.
- c) In capital increases in return for contributions in kind, specifically for the purpose of acquiring companies, business units, interests in companies or other assets or claims to the acquisition of assets, including receivables from the Company or its Group companies, or for the purpose of mergers of companies, the Executive Board is authorized to exclude the subscription rights with the consent of the Supervisory Board.
- d) The Executive Board is authorized, with the consent of the Supervisory Board, to exclude the subscription rights in order to pay a scrip dividend whereby shareholders are offered the alternative of contributing their claim to a dividend as an (either complete or partial) contribution in kind to the Company in return for being granted new shares under the Authorized Capital A.
- e) In capital increases in return for contributions in kind for the purpose of acquiring remuneration and other receivables of the members of the Executive Board as well as members of the Supervisory Board against the Company, the Executive Board is authorized to exclude the subscription rights of the shareholders with the consent of the Supervisory Board, if the subscription price is not significantly below the market price.

If the Executive Board does not make use of the aforementioned authorizations to exclude subscription rights, the Executive Board may, with the consent of the Supervisory Board, exclude the subscription rights of shareholders for fractional amounts only.

The sum of the shares issued in return for cash or contributions in kind with subscription rights of the shareholders excluded may not during the term of the Authorized Capital A exceed 10% of the share capital (10% limit) at the time of the authorization taking effect or – if this value is lower – at the time of its exercise. If during the term of the Authorized Capital A until it is utilized, other authorizations are exercised to issue or sell shares in the Company or to issue rights that enable or oblige the purchase of shares of the Company for which subscription rights are excluded, this is subject to the 10% limit stated in the above sentence.

The Executive Board is authorized, with the consent of the Supervisory Board, to determine the further details of the share rights and the conditions for the issue of shares. The Supervisory Board is authorized to adapt section 4 para. 2 of the Articles of Association in accordance with the utilization of the Authorized Capital A in each case or upon expiry of the authorization's term.

17.5.2 Authorized Capital B

As of the date of this Prospectus, pursuant to section 4 para. 3 of the Articles of Association, by way of a resolution passed at the General Meeting on May 7, 2019, the Executive Board is authorized with the consent of the Supervisory Board, to increase the share capital of the Company in one or more stages until the end of May 6, 2024, by up to €22,362,168.32, through the issue of new no-par value registered shares for cash contributions (Authorized Capital B). The new shares will be offered for purchase solely to employees of the Company and of associated companies, including in connection with our employee stock option programs. The subscription rights of shareholders shall be excluded. The Executive Board is authorized, with the consent of the Supervisory Board, to determine the further details of the utilization of the Authorized Capital B, specifically the conditions for the issue of the new no-par value registered shares, the issue amount and the further details of shareholder rights, and the implementation of capital increases. The profit participation of the new no-par value registered shares may be determined otherwise than set forth in section 60 para. 2 of the German Stock Corporation Act (*Aktiengesetz*). The Supervisory Board is authorized to adapt section 4 para. 3 sentences 1-6 of the Articles of Association in accordance with the utilization of Authorized Capital B in each case or upon expiry of the authorization's term.

17.5.3 Authorized Capital C

As of the date of this Prospectus, pursuant to section 4 para. 8 of the Articles of Association, by way of a resolution passed at the General Meeting on May 4, 2021, the Executive Board is authorized with the approval of the Supervisory Board to increase until the expiry of May 3, 2026, the share capital of the Company in connection with the agreed recapitalization pursuant to section 22 of the German Stabilization Fund Act (*Stabilisierungsfondsgesetz*) by up to €5,500,000,000.00 through the issuance, once or several times, of up to 2,148,437,500 new no-par value registered shares against contribution in cash or in kind (i) to use the net issue proceeds primarily to repay the capital made available to the Company by the WSF or (ii) for other purposes specified in section 7f of the Economic Stabilization Acceleration Act (*Wirtschaftsstabilisierungsbeschleunigungsgesetz*) (Authorized Capital C). The shareholders are to be granted subscription rights. The subscription right may also be granted to the shareholders indirectly in accordance with section 186 para. 5 of the German Stock Corporation Act (*Aktiengesetz*). The WSF is entitled to subscribe to the new registered no-par value registered shares to which it is entitled upon exercise of its subscription rights in accordance with the subscription ratio against granting of a contribution in kind by way of full or partial contribution of the Silent Participation I and/or Silent Participation II (including the claims to coupons and any additional remuneration).

The Executive Board is authorized to determine the further content of the rights conveyed by the shares and the conditions of the share issue with the approval of the Supervisory Board. This also includes the authorization, pursuant to section 7f para. 1 no. 1 in conjunction with section 7 para. 3a of the Economic Stabilization Acceleration Act (*Wirtschaftsstabilisierungsbeschleunigungsgesetz*), to offer any unsubscribed shares to the WSF for purchase after the expiration of the subscription period at the subscription price less 5% if the WSF has exercised in advance the subscription rights to which it is entitled in connection with the

corresponding capital increase. The Supervisory Board is authorized to make adjustments to the wording of section 4 of the Articles of Association in accordance with the utilization of Authorized Capital C in each case or upon expiry of the authorization's term.

17.6 Conditional Capital

17.6.1 Conditional Capital pursuant to section 4 para. 4 of the Articles of Association

By way of a resolution passed at the General Meeting on May 5, 2020, the Executive Board was authorized until May 4, 2025, subject to the approval by the Supervisory Board, to issue convertible bonds, bonds with warrants or participating bonds – or combination thereof – up to a total nominal amount of €1.5 billion (Authorization 2020). With the consent of the Supervisory Board, the Executive Board made partial use of this authorization in November 2020 by issuing convertible bonds with a total nominal amount of €600,000,000.00. For the granting of shares to the holders or creditors of above mentioned bonds, the Company's share capital is conditionally increased by up to €122,417,728 by issuing up to 47,819,425 new no-par value registered shares. The Authorization 2020 was cancelled by the General Meeting on May 4, 2021, to the extent it has not yet been utilized. The Conditional Capital pursuant to section 4 para. 4 of the Articles of Association is required to secure the conversion rights from the convertible bonds already issued on the basis of the Authorization 2020. As of the date of this Prospectus, convertible bonds in a nominal amount of €600,000,000 issued under Authorization 2020 remain outstanding.

17.6.2 Conditional Capital 2020/II

As of the date of this Prospectus, pursuant to section 4 para. 5 of the Articles of Association, on the basis of the resolution adopted by the extraordinary General Meeting of June 25, 2020, the Company's share capital is conditionally increased by up to €102,014,776.32, divided into up to 39,849,522 no-par value registered shares (Conditional Capital 2020/II). The conditional capital increase is intended for the granting of shares upon the exercise of conversion rights granted to the WSF, which was established under the German Stabilization Fund Act (*Stabilisierungsfondsgesetz*) as silent partner of the Company in respect of the Silent Participation II-A in accordance with the resolution of the extraordinary General Meeting of June 25, 2020, in case of the occurrence of a "Takeover Event." A "Takeover Event" is deemed to have occurred in the event of publication of the decision to make a takeover offer within the meaning of Section 10 of the German Securities Acquisition and Takeover Act (*Wertpapiererwerbs- und Übernahmegesetz*) or in the event of an attainment of control within the meaning of section 35 in conjunction with section 29 of the German Securities Acquisition and Takeover Act (*Wertpapiererwerbs- und Übernahmegesetz*). The new shares are issued at an issue price of €2.56 per share. The conditional capital increase is only implemented to the extent that the WSF (or a Third Party after the assignment of Silent Participation II-A) exercises the conversion right. The Executive Board is authorized, with the consent of the Supervisory Board, to determine further details of the conditional capital increase and its implementation.

17.6.3 Conditional Capital 2020/III

As of the date of this Prospectus, pursuant to section 4 para. 6 of the Articles of Association, on the basis of the resolution adopted by the extraordinary General Meeting of June 25, 2020, the Company's share capital is conditionally increased by up to €897,985,223.68, divided into up to 350,775,478 no-par value registered shares (Conditional Capital 2020/III). The conditional capital increase is intended for the granting of shares upon the exercise of conversion rights granted to the WSF, which was established under the German Stabilization Fund Act (*Stabilisierungsfondsgesetz*) as silent partner of the Company in respect of the Silent Participation II-B in accordance with the resolution of the extraordinary General Meeting of June 25, 2020, for the purpose of "Dilution Protection" and/or "Coupon Protection" (both terms are defined in section 4 para. 6 of the Articles of Association). The new shares will be issued, if issued upon exercise of the conversion right for the purpose of "Dilution Protection" (as defined in section 4 para. 6 of the Articles of Association), at the current stock exchange price at the time of conversion less 10%, if issued upon exercise of the conversion right for the purpose of "Coupon Protection" (as defined in section 4 para. 6 of the Articles of Association) at the current stock exchange price at the time of conversion less 5.25%. The conditional capital increase is only implemented to the extent that the WSF exercises the conversion right. If Silent Participation II-B is assigned, the conversion rights lapse. The Executive Board is authorized, with the

consent of the Supervisory Board, to determine further details of the conditional capital and its implementation.

17.6.4 Conditional Capital 2021

As of the date of this Prospectus, the Executive Board is authorized until May 3, 2026, subject to the approval by the Supervisory Board, to issue bearer or convertible bonds, option bonds, profit participation rights and/or participating bonds – or combination thereof – up to a total nominal amount of €1.5 billion and to grant the holders or creditors of such bonds conversion or option rights to new registered no-par value registered shares of the Company with a *pro rata* amount of the Company's share capital of up to a total of €153,022,161.92. For the granting of shares to the holders or creditors of above mentioned bonds, pursuant to section 4 para. 7 of the Articles of Association, by way of a resolution passed at the General Meeting on May 4, 2021, the Company's share capital is conditionally increased by up to €153,022,161.92 through the issue of up to 59,774,282 new no-par value registered shares (Conditional Capital 2021). The conditional capital increase is implemented only to the extent that the holders or creditors of conversion or option rights or the parties obliged to conversion under issued convertible bonds, option bonds, profit participation rights or participating bonds (or a combination of these instruments) issued by the Company or companies of the Lufthansa Group until May 3, 2026, based on the authorization resolution of the General Meeting of May 4, 2021, exercise their conversion or option rights or the holders or creditors of issued bonds with a conversion obligation to conversion fulfil their obligation to convert, or to the extent that the Company exercises an option to grant, in whole or in part, shares of the Company in lieu of payment of the cash amount due, and to the extent cash compensation is not granted or treasury shares are not used to satisfy these obligations. The new shares participate in the profits from the beginning of the fiscal year in which they are created through the exercise of conversion or option rights, through the fulfilment of conversion obligations or through the exercise of rights to offer shares. The Executive Board is authorized to stipulate the further details of the implementation of the conditional capital increase.

17.7 Authorization to Implement Measures pursuant to the Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*)

17.7.1 Authorization for a Capital Increase pursuant to section 4 para. 2 of the Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*)

In the event that the circumstances described in section 4 para. 3 of the Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) occur (*i.e.*, the share register or a notification pursuant to sections 33 et seqq. of the German Securities Trading Act (*Wertpapierhandelsgesetz*) shows that (i) 45% or more of the total voting rights, or (ii) a controlling shareholding within the meaning of section 17 of the German Stock Corporation Act (*Aktiengesetz*) is held by shareholders whose shareholdings prevent the fulfilment of the requirements for the maintenance of air traffic rights), the Executive Board is authorized, after obtaining the consent of the Supervisory Board, to increase the Company's share capital by issuing new no-par value registered shares against a contribution in cash and to rule that existing shareholders have no automatic right to subscribe to the new no-par value registered shares. The issue price for the new no-par value registered shares shall be fixed in agreement with the Supervisory Board but must not fall short of the current stock market price by a material margin. A capital increase pursuant to this provision must not exceed 10% of the Company's share capital at the time the measure is taken.

17.7.2 Divestiture Requirement

In the event that the circumstances described in section 5 para. 2 of the Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) occur, the Executive Board is authorized, after obtaining the consent of the Supervisory Board, to demand shareholders to sell all or part of the shares held by them and to furnish evidence to the Company that they have done so without undue delay. Shareholders will be called upon to take such action to the extent necessary to ensure that the Company meets the requirements for retaining its licenses, rights and prerogatives under aviation laws and agreements and in the order stipulated in section 5 para. 3 of the Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*). They shall be given an appropriate period of time to comply with this demand and shall be concurrently warned that their failure to comply may incur the legal consequence set out in section 5 para. 7 of the Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*), under which their

entitlement to own the shares can be rescinded. The demand to the shareholders, together with the warning that they may forfeit their ownership of the shares as a possible legal consequence of their failure to comply, must allow a compliance period of at least four weeks. Instead of a public announcement, an individual call addressed to the correspondent shareholders may be made; in this case, a compliance period of at least two weeks from the date of receipt of the individual call must be granted. Pursuant to a grandfathering provision, the divestiture requirement only applies to shares acquired after the Aviation Compliance Documentation Act came into effect on June 14, 1997.

17.8 Acquisition and Disposal of Treasury Shares

On the basis of the resolution adopted by the General Meeting of May 7, 2019, the Executive Board is authorized until May 6, 2024, with the consent of the Supervisory Board, to purchase up to 10% of the nominal capital existing at the time of the authorization taking effect or, if this amount is lower, at the time of its exercise, in its own shares.

The Company is entitled to repurchase shares and to sell repurchased shares in those cases defined in section 71 of the German Stock Corporation Act (*Aktiengesetz*). The authorization can be used, among other things, to issue shares for the settlement of employee share programs and to expand the financing alternatives in the event that another company or an equity stake in a company is acquired.

In this regard, the shares purchased on the basis of this authorization, together with other Company shares which the Company has already purchased and still owns or which have to be allocated to it under sections 71 *et seq.* of the German Stock Corporation Act (*Aktiengesetz*) must at no point in time amount to more than 10% of the share capital concerned of the Company.

However, under the Framework Agreement, the Company is generally required to not repurchase any Company shares, other than shares held by the WSF, until all stabilization measures provided by the WSF to the Company under the Framework Agreement have been repaid in full. This restriction does in particular not apply to the purchase of shares to fulfill contractual obligations under our employee stock option program.

17.9 General Provisions Governing the Liquidation of the Company

Apart from a liquidation as a result of insolvency proceedings, the Company may be liquidated only with a vote of 75.0% or more of the share capital represented at the general meeting at which such vote is taken. Pursuant to the German Stock Corporation Act (*Aktiengesetz*), in the event of the Company's liquidation, any assets remaining after all of the Company's liabilities have been settled will be distributed *pro rata* among its shareholders. The German Stock Corporation Act (*Aktiengesetz*) provides certain protections for creditors which must be observed in the event of liquidation.

17.10 General Provisions Governing Share Capital Increases and Decreases

The German Stock Corporation Act (*Aktiengesetz*) provides that the share capital of a stock corporation may be increased by a resolution of the general meeting. Such resolution must generally be adopted by a majority of at least 75% of the share capital represented at the general meeting at which such vote is taken unless the stock corporation's articles of association provide for a different majority. The Articles of Association provide in section 18 para. 1 that resolutions of the General Meeting are adopted by a simple majority of the votes cast unless stipulated otherwise by the Articles of Association or by mandatory law. Where the German Stock Corporation Act (*Aktiengesetz*) requires that a majority of the nominal capital represented at the General Meeting must be cast in order to carry the vote, a simple majority of the nominal capital represented shall suffice as long as that is legally permissible.

In addition, shareholders may resolve to issue authorized capital by a vote of 75% of the share capital represented at the passing of the resolution authorizing the Executive Board to issue Shares, up to a specific amount within a period not exceeding five years. The nominal amount of such issuance may not exceed 50% of the share capital in existence at the time of the authorization, that is, at the time the authorized capital is entered into the commercial register (*Handelsregister*). Such limit does not apply with respect to authorized capitals resolved in accordance with the German Economic Stabilization Acceleration Act

(*Wirtschaftsstabilisierungsgesetz*) such as the Authorized Capital C. For further details, see “17.5.3 Authorized Capital C.”

Additionally, shareholders may resolve to create contingent capital for the purpose of issuing Shares (i) to holders of convertible bonds or other securities convertible into Shares, (ii) as consideration in connection with a merger with another company or (iii) to executives and employees. A resolution to create contingent capital must be adopted by at least 75% of the share capital represented at the passing of the resolution. The nominal amount of the contingent capital created for the purpose of share issues to executives and employees may not exceed 10%, a contingent capital created for any other purpose may not exceed 50% of the nominal share capital in existence at the time such resolution is passed. The creation of contingent capital beyond this threshold is permitted only for the purpose of enabling the company to make an exchange in the event of its impending insolvency or for the purpose of averting over indebtedness.

A resolution to reduce the share capital must be adopted by at least 75% of the share capital represented at the passing of the resolution.

17.11 General Provisions on Subscription Rights

According to the German Stock Corporation Act (*Aktiengesetz*), every shareholder is generally entitled to subscription rights to any new shares issued in connection with a capital increase, including convertible bonds, bonds with warrants, profit-sharing rights or income bonds. Such subscription rights are freely transferable and may generally be traded on German stock exchanges within a specified period. However, as the Company operates as a German aviation company, the German Aviation Compliance Documentation Act (*Luftverkehrsnachweissicherungsgesetz*) is applicable, which stipulates that the shares of the Company must be registered shares with restricted transferability, such that the transfer of such shares requires the consent of the Company. All shares of the Company, including the New Shares, will therefore be issued as registered shares with no-par value and with restricted transferability (*vinkulierte Namensaktien*). Following the exercise of subscription rights, a shareholder is generally required to record its identity and nationality in the Company’s share register. In the event of a corporation or other legal entity, nationality is determined by the country of registration.

The general meeting may pass a resolution excluding subscription rights if at least 75% of the share capital represented adopts the resolution. To exclude subscription rights, the Executive Board must also make a report available to the shareholders justifying the exclusion and demonstrating that the Company’s interest in excluding the subscription rights outweighs the shareholders’ interest in keeping them. The exclusion of subscription rights upon the issuance of new Shares is permitted, in particular, if the Company increases the share capital against cash contributions, the amount of the capital increase does not exceed 10% of the existing share capital and the issue price of the new Shares is not significantly lower than the stock exchange price of the Company’s existing shares.

17.12 General Rules on Allocation of Profits and Dividend Payments

The distribution of dividends on the Company’s shares for a given fiscal year is resolved by the General Meeting of the subsequent fiscal year following a proposal by the Executive Board and the Supervisory Board.

Under the German Stock Corporation Act (*Aktiengesetz*), a resolution regarding dividends and any distribution thereof must be based on a balance sheet profit, recorded in the Company’s unconsolidated financial statements. When determining the balance sheet profit available for distribution, net income/loss must be adjusted to account for profit/loss carry-forwards of the previous years, as well as release of or allocation to reserves. Certain reserves are required by law and the respective allocations must be deducted when calculating the amount of balance sheet profit available for distribution. For further details, see “7 Dividend Policy.”

17.13 Exclusion of Minority Shareholders

Under the rules of sections 327a *et seqq.* of the German Stock Corporation Act (*Aktiengesetz*) regarding the “squeeze-out” of minority shareholders, the general meeting of a stock corporation may resolve upon request by a shareholder who holds at least 95% of the share capital (majority shareholder) that the shares held by

the remaining minority shareholders shall be transferred to the majority shareholder against payment of an adequate cash compensation. The cash compensation to be paid to the minority shareholders must reflect the situation of the company at the time the resolution is adopted by the general meeting. For the purpose of calculating the adequacy of the cash compensation, the enterprise value of the Company will be determined by applying the discounted future earnings method (*Ertragswertmethode*). The minority shareholders may file for valuation proceedings (*Spruchverfahren*) in the course of which the appropriateness of the cash compensation is reviewed.

Additionally, a majority shareholder holding at least 90% of a stock corporation's share capital can require the general meeting to resolve that the minority shareholders must sell their shares to the majority shareholder against the payment of an adequate cash compensation in connection with a merger pursuant to the German Transformation Act (*Umwandlungsgesetz*). The squeeze-out under the German Transformation Act is similar to the squeeze-out under the German Stock Corporation Act (*Aktiengesetz*) described above, including the minority shareholders' right to file for a review of the appropriateness of the cash compensation.

Furthermore, a bidder that holds at least 95% of the voting share capital of the target company following a takeover offer or a mandatory tender offer may apply to the Regional Court (*Landgericht*) of Frankfurt am Main to effect the transfer to such bidder of the remaining voting shares against payment of an adequate compensation within a period of three months following the expiration of the acceptance period. No resolution of the general meeting is required. The consideration granted under the takeover offer or the mandatory tender offer is considered as adequate compensation if the bidder, based on such offer, has acquired at least 90% of the share capital in connection with the offer. Furthermore, following a takeover offer or mandatory tender offer, the shareholders of the target company who did not accept such offer may accept the offer within three months after the expiration of the acceptance period (*i.e.*, sell-out) in accordance with section 39c of the German Securities Acquisition and Takeover Act (*Wertpapiererwerbs- und Übernahmegesetz*), provided the bidder is entitled to file an application for the transfer of the remaining voting shares pursuant to section 39a of the German Securities Acquisition and Takeover Act (*Wertpapiererwerbs- und Übernahmegesetz*).

Pursuant to the provisions of section 319 *et seq.* of the German Stock Corporation Act (*Aktiengesetz*) regarding the integration (*Eingliederung*) of a subsidiary, the general meeting of a stock corporation may resolve the integration into another company provided that the future principal company (*Hauptgesellschaft*) is a German stock corporation and holds at least 95% of the shares of the company to be integrated. The shareholders of the integrated company are entitled to adequate compensation, which is generally to be granted in the form of shares of the principal company. The amount of compensation is to be determined by the merger value ratio (*Verschmelzungswertrelation*) between the companies, *i.e.*, the exchange ratio which would have to be considered adequate in the event of a merger of the two companies.

17.14 Mandatory Takeover Bids

Pursuant to the German Securities Acquisition and Takeover Act (*Wertpapiererwerbs- und Übernahmegesetz*), every person whose share of voting rights reaches or exceeds 30% of the voting rights of the Company (*i.e.*, who has acquired control over the Company) must, without undue delay and within seven calendar days at the latest, publish this fact, including the percentage of its voting rights, unless BaFin has granted an exemption from this obligation. Subsequently, an offer document in relation to a mandatory tender offer to all shareholders of the Company must be submitted to BaFin, in principle, within four weeks thereafter. The mandatory tender offer shall be published after approval by BaFin. In connection with the mandatory tender offer, such person is required to offer an adequate consideration to all shareholders of the Company for each share of the Company. The minimum consideration to be offered for each share of the Company must at least be equal to the higher of (i) the weighted average share price during the last three months prior to the publication of the acquisition of control over the Company, and (ii) the highest consideration provided or agreed by such person or any person acting jointly pursuant to the German Securities Acquisition and Takeover Act (*Wertpapiererwerbs- und Übernahmegesetz*) or their subsidiaries for the purchase of shares in the Company within the last six-months prior to the publication of the mandatory tender offer. If a person fails to give notice of reaching or exceeding the 30% threshold or fails to submit or publish a mandatory tender offer, shareholder rights (including voting rights and, in certain cases,

the right to collect dividends and liquidation proceeds) are suspended for the duration of non-compliance under certain circumstances. In addition, a fine may be imposed.

17.15 Disclosure Requirements for Holdings of Shares and Other Instruments

The Company, as a listed company, and its shareholders are subject to the provisions of the German Securities Trading Act (*Wertpapierhandelsgesetz*) governing disclosure requirements for shareholdings.

Section 33 of the German Securities Trading Act (*Wertpapierhandelsgesetz*) requires that anyone who acquires, sells or whose shareholding in any other way reaches, exceeds or falls below 3%, 5%, 10%, 15%, 20%, 25%, 30%, 50% or 75% of the voting rights in an issuer whose home country is Germany and whose shares are admitted to trading on an organized market must immediately, and no later than within four trading days of such occurrence, notify the issuer and at the same time BaFin. The notice period commences as soon as the person obliged to notify (*Meldepflichtiger*) knows, or, under the circumstances of the case should know, that his or her voting rights reach, exceed or fall below the abovementioned thresholds, and no later than two trading days after reaching, exceeding or falling below the threshold. Only in the case where the voting rights reach, exceed or fall below the thresholds as a result of an event affecting all voting rights, then the notice period may commence at a later time. The notification requirement is set off by the establishment of an obligation to transfer such ownership immediately (*ohne zeitliche Verzögerung*).

Notice must be given using a standard form annexed to the German Securities Trading Reporting Regulation (*Wertpapierhandelsanzeigeverordnung*). It must include the address of the individual or entity, the share of voting rights held and the date of reaching, exceeding or falling below the respective threshold, and must be issued via a mandatory standard form. The Company must publish such notices immediately but no later than within three trading days after their receipt, via media outlets or outlets where it can be assumed that the notice will be disseminated in the EU and the non-EU parties to the agreement on the EEA (*Medienbündel*). The Company must also transmit the notice to BaFin and to the German Business Register (*Unternehmensregister*) for storage.

For purposes of the notification requirements, the German Securities Trading Act (*Wertpapierhandelsgesetz*) contains various rules that require the attribution (*Zurechnung*) of voting rights of certain persons associated with the shareholder or acting in concert with the shareholder. For example, shares held by a subsidiary (as defined in section 35 of the German Securities Trading Act (*Wertpapierhandelsgesetz*)) are attributed to the parent company; similarly, shares held by a third company for the account of another company are attributed to the latter. Furthermore, any kind of cooperation among shareholders that is intended to effect a permanent and material change in the business strategy of the Company can result in an attribution of voting rights. This means that the cooperation does not necessarily have to concern the exercise of voting rights specifically; coordination in individual cases, however, will not trigger the attribution of voting rights.

Pursuant to section 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*), similar obligations to notify the Company and BaFin for reaching, exceeding or falling below the abovementioned thresholds (other than the 3% threshold) apply to direct and indirect holders of certain instruments other than the Shares. This applies to instruments that grant upon maturity an unconditional right to acquire Shares of the Company, a discretionary right to acquire such shares, or instruments that refer to such shares and have a similar economic effect to the aforementioned instruments. Notifiable instruments include, *inter alia*, transferable securities, options, futures contracts, swaps, forward rate agreements and contracts for difference. The number of voting rights relevant for the notification requirement will generally be calculated by reference to the full nominal amount of shares underlying the instrument except where the instrument provides exclusively for a cash settlement. Details for such calculations are laid down in the Commission Delegated Regulation (EU) 2015/761 of December 17, 2014.

Shares or instruments held for trading by a securities services company are not taken into account for determining the notification obligation if it is ensured that the voting rights held by them are not exercised, and that they amount to no more than 5% of the voting shares, or do not grant the right to purchase more than 5% of the voting shares.

Notifiable holdings pursuant to sections 33 and 38 of the German Securities Trading Act (*Wertpapierhandelsgesetz*) must be aggregated, leading to a notification obligation for total holdings with

regard to the aforementioned thresholds (other than the 3% threshold) pursuant to section 39 of the German Securities Trading Act (*Wertpapierhandelsgesetz*).

Furthermore, a person obliged to notify (*Meldepflichtiger*) who reaches or exceeds the threshold of 10% of the voting rights, or a higher threshold, is obligated to notify the issuer within 20 trading days of the objective being pursued through the acquisition of voting rights, as well as the source of the funds used for the purchase pursuant to section 43 of the German Securities Trading Act (*Wertpapierhandelsgesetz*). Changes in those objectives must also be reported within 20 trading days. An issuer may stipulate in its articles of association that the aforementioned disclosure requirement does not apply.

In case that the disclosure requirements are not met, shareholder rights (including voting rights and, in certain cases, the right to collect dividends and liquidation proceeds) are – subject to certain exceptions – suspended for the duration of non-compliance. If the failure to comply with the disclosure requirements specifically relates to the share of voting rights and is the result of a willful or grossly negligent conduct, the suspension period is extended by six-months after the person obliged to notify (*Meldepflichtiger*) files the required notification. In addition, a fine may be imposed if a required notification is not at all, incorrectly or incompletely made, or not made in the right manner or in a timely fashion. BaFin also has the right to publish decisions on sanctions and measures relating to violations of the disclosure obligations and persons responsible for such violations.

17.16 Disclosure of Transactions of Persons Discharging Managerial Responsibilities

Pursuant to Article 19 of Regulation (EU) No 596/2014 of April 16, 2014, as amended (“**Market Abuse Regulation**” or “**MAR**”), persons discharging managerial responsibilities (“**Executives**”) shall notify the Company and BaFin of transactions conducted on their own account relating to shares or debt instruments of the Company or to derivatives or other financial instruments linked thereto (*i.e.*, managers’ transactions), subject to certain exceptions. The same applies to persons closely associated with Executives who must notify the Company and BaFin if they enter into such transactions. Transactions that must be notified also include, among others, pledging or lending of financial instruments, transactions undertaken by any person professionally arranging or executing transactions on behalf of an Executive or a closely associated person, including where discretion is exercised, as well as transactions made under a life insurance policy. The notification requirement applies to any subsequent transaction once a total amount of €20,000 has been reached within a calendar year. Notification shall be made promptly and no later than three business days after the date of the transaction.

For the purposes of the Market Abuse Regulation, Executive means a person within the Company who is a member of the administrative, management or supervisory body of the company or a senior executive who is not such member but who has regular access to inside information relating directly or indirectly to the Company and who has power to take managerial decisions affecting the future developments and business prospects of the Company. A person closely associated with an Executive means certain family members, namely a spouse, a registered civil partner (*eingetragener Lebenspartner*), a dependent child as well as a relative who has shared the same household for at least one year on the date of the transaction concerned. A person closely associated also includes a legal person, trust or partnership, the managerial responsibilities of which are discharged by an Executive of the Company or by a family member of his or hers. Finally, the term includes a legal person, trust or partnership which is directly or indirectly controlled by an Executive of the Company (or by one of its family members) or which is set up for the benefit of such a person, or the economic interests of which are substantially equivalent to those of such a person.

The Company shall ensure that information of which it is notified is made public within two business days of receipt of such a notification. The issuer shall use such media as may reasonably be relied upon for the effective dissemination of information to the public throughout the Union. Furthermore, according to the German Securities Trading Act (*Wertpapierhandelsgesetz*), the Company shall without undue delay transmit the information to the German Business Register (*Unternehmensregister*) and notify BaFin. Non-compliance with the notification requirements may result in a fine.

17.17 Post-Admission Disclosure Requirements

Due to the admission to trading of the Company's shares, the Company is subject to the legal disclosure requirements for German stock corporations with shares listed on a public exchange. These disclosure requirements include, among others, periodic financial reporting and other required disclosures according to the German Securities Trading Act (*Wertpapierhandelsgesetz*) as well as disclosure requirements under the Market Abuse Regulation. The Company will also be obliged under the Listing Rules of the Frankfurt Stock Exchange (*Börsenordnung für die Frankfurter Wertpapierbörse*), as amended from time to time, to publish quarterly statements, as the Company's shares are to be listed on the sub-segment of the regulated market (*regulierter Markt*) of the Frankfurt Stock Exchange (*Frankfurter Wertpapierbörse*) with additional post-admission obligations (Prime Standard).

Pursuant to Article 17 of the Market Abuse Regulation, the Company shall inform the public as soon as possible of inside information (as defined below) which directly concerns the Company. In such case the Company shall also, prior to informing the public, inform BaFin and the management of the trading venues and facilities (*Geschäftsführungen der Handelsplätze*) where financial instruments of the Company have been admitted to trading or been included in such trading, and, after publication, without undue delay transmit the information to the German Business Register (*Unternehmensregister*).

Inside information comprises, among others, any information of a precise nature, which has not been made public, relating, directly or indirectly, to one or more issuers or to one or more financial instruments, and which, if it were made public, would be likely to have a significant effect on the prices of those financial instruments or on the price of related derivative financial instruments.

The Company may, on its own responsibility, delay disclosure of inside information if (i) immediate disclosure is likely to prejudice the legitimate interests of the Company, (ii) delay of disclosure is not likely to mislead the public, and (iii) the Company is able to ensure that the inside information will remain confidential. In such case, the Company shall also inform BaFin that disclosure of the information was delayed and shall provide a written explanation of how the conditions set out in the preceding sentence were met, immediately after the information is disclosed to the public. Where disclosure of inside information has been delayed and the confidentiality of that inside information is no longer ensured, the Company shall disclose such inside information to the public as soon as possible.

17.18 EU Short Selling Regulation

Under Regulation (EU) No 236/2012 of the European Parliament and of the Council of March 14, 2012, on short selling and certain aspects of credit default swaps (the "**EU Short Selling Regulation**"), short sales of shares (*i.e.*, sales of shares that the seller does not own, with the intention of acquiring shares of the same class at a later point in time in order to be able to deliver the shares to the buyer), are permitted only under certain conditions. Significant net short positions in shares must be reported to BaFin and, if a certain threshold is exceeded, they must also be publicly disclosed. The reporting and publication obligations are set forth in detail in the German Regulation on Net-Short Positions (*Netto-Leerverkaufspositionsverordnung*) of December 17, 2012, as amended. Net short positions are calculated by netting the long and short positions held by a natural or legal person in the issued capital of the company concerned. The details are set forth in the EU Short Selling Regulation and the regulations adopted by the European Commission implementing it. In certain situations, described in greater detail in the EU Short Selling Regulation, BaFin is permitted to restrict short selling and comparable transactions.

18. GOVERNING BODIES

18.1 Overview

The Company's governing bodies are the Executive Board, the Supervisory Board and the Company's general meeting (*Hauptversammlung*). Their powers and responsibilities are laid down in the German Stock Corporation Act (*Aktiengesetz*) and the articles of association (*Satzung*) of the Company (the "**Articles of Association**"), as well as in the respective rules of procedure (*Geschäftsordnungen*) of the Executive Board and the Supervisory Board.

The Executive Board is responsible for the management of the Company's business; the Supervisory Board supervises and advises the Executive Board and appoints its members. The two boards are separate, and no individual may simultaneously be a member of both boards.

Each member of the Executive Board and Supervisory Board owes a duty of loyalty, duty of legality and duty of care to the Company. In discharging these duties, each member of these bodies must consider a broad spectrum of interests, particularly those of the Company and its shareholders, employees and creditors. In addition, the Executive Board must also take into consideration the shareholders' rights to equal treatment and equal access to information. If members of the Executive Board or Supervisory Board breach their duties, they may be solely or jointly and severally liable with the other members of the Executive Board or the Supervisory Board to the Company for any damages the Company has incurred.

Under German law, as a rule shareholders have no right to directly assert claims against members of the Executive Board or Supervisory Board if they believe that such members have violated their duties to the Company (*i.e.*, in principle only the Company has the right to enforce such claims against the members of the Executive Board or Supervisory Board). With respect to claims against members of the Executive Board, the Company is represented by the Supervisory Board, and with respect to claims against members of the Supervisory Board, the Company is represented by the Executive Board. The Federal Court of Justice (*Bundesgerichtshof*) has ruled that the Supervisory Board is generally required to assert claims against members of the Executive Board, if it is likely that such claims can be pursued and enforced successfully, unless significant interests of the Company conflict with the pursuit of such claims and outweigh the interests of the Company asserting such claims against members of the Executive Board.

If either the Supervisory Board or the Executive Board decides not to pursue claims of the Company against members of the respective other governing body for violations of their duties, such claims must nevertheless be asserted if the General Meeting adopts a resolution to this effect with a simple majority of the votes validly cast. The General Meeting may also appoint a special representative (*besonderer Vertreter*) to assert such claims. Shareholders whose aggregate shareholdings amount to 10% of the Company's share capital or a *pro rata* share of €1 million in the Company's share capital may also motion for the competent court to appoint such a special representative. If there are facts that justify the suspicion that the Company was harmed by dishonesty or a gross violation of laws or the Articles of Association, shareholders whose aggregate shareholdings amount to 1% of the Company's share capital or a *pro rata* share of €100,000.00 of the Company's share capital may under certain conditions assert claims of the Company against members of the Executive Board or Supervisory Board in their own names. Yet such claims become inadmissible once the Company itself files a suit to assert such claims.

In addition, the General Meeting may appoint special auditors (*Sonderprüfer*) to audit actions, particularly management transactions, by a simple majority of the votes validly cast. If the General Meeting rejects a motion to appoint special auditors, the competent court appoints such special auditors upon a motion by shareholders whose aggregate shareholdings amount to 1% of the Company's share capital or a *pro rata* share of €100,000.00 of the Company's share capital, if there are facts that justify the suspicion that the relevant occurrence involved acts of dishonesty or gross violations of the law or the Articles of Association. If the General Meeting has resolved to appoint special auditors, the competent court appoints different special auditors upon a motion by shareholders whose aggregate shareholdings amount to 1% of the Company's share capital or a *pro rata* share of €100,000.00 of the Company's share capital, if such appointment appears necessary due to reasons concerning the original special auditors.

Via the shareholders' forum of the German Federal Gazette (*Bundesanzeiger*), which is also accessible via the website of the German Company Register (*Unternehmensregister*), shareholders and shareholder associations may solicit other shareholders to file a motion, jointly or by proxy, for the appointment of special auditors, for the appointment of a special representative, the convening of a General Meeting, or the exercise of voting rights in a General Meeting.

The Company may only waive or settle claims for damages against members of the Executive Board or the Supervisory Board, if at least three years have elapsed since such claims arose and if the General Meeting has consented to such waiver or settlement by a simple majority vote, provided that a minority of the shareholders whose aggregate shareholdings amount to at least 10% of the Company's share capital does not object to such resolution in the minutes of the General Meeting.

Under German law, neither individual shareholders nor other persons may use their influence on the Company to cause a member of the Executive Board or the Supervisory Board to act in a manner that would be detrimental to the Company. Any person who uses its influence over the Company to cause a member of the Executive Board or the Supervisory Board, an authorized representative (*Prokurist*) or an authorized agent (*Handlungsbevollmächtigter*) to act to the detriment of the Company or its shareholders may be liable to compensate the Company and the affected shareholders for the resulting losses. In particular, a controlling shareholder may not use its influence to cause the Company to act contrary to its own interests, unless (i) the Company and the controlling shareholder enter into a domination agreement (*Beherrschungsvertrag*) or (ii) the controlling shareholder compensates the Company for any disadvantages resulting from its influence. Moreover, the members of the Executive Board and the Supervisory Board are solely or jointly and severally liable in addition to the person using its influence if such members acted in breach of their duty of care towards the Company.

Under the German Stock Corporation Act (*Aktiengesetz*), the German Co-Determination Act (*Mitbestimmungsgesetz*) of 1976, as amended, and the Company's Articles of Association, the Supervisory Board consists of 20 members. The principal function of the Supervisory Board is to oversee the work of the Executive Board and to appoint its members. The Supervisory Board oversees the Company's business policy, corporate planning and strategy. It also approves the annual budget as well as the unconsolidated financial statements of the Company and the consolidated financial statements of the Lufthansa Group. The Supervisory Board may not make management decisions, but the Executive Board's rules of procedure (*Geschäftsordnung*) drawn up by the Supervisory Board stipulate that certain specified types of business activities require its prior consent.

18.2 Executive Board

18.2.1 Overview

The Executive Board is responsible for managing the business of the Company in accordance with the German Stock Corporation Act (*Aktiengesetz*) and the Articles of Association and its rules of procedure. It also represents the Company in its dealings with third parties and in court. According to the Articles of Association, the Executive Board shall consist of at least two members. The Supervisory Board determines the number of and appoints the members of the Executive Board. Members of the Executive Board are appointed for a maximum term of five years and are eligible for reappointment after the completion of their term in office.

The Company shall be legally represented by two members of the Executive Board acting together, or by one member of the Executive Board jointly with an authorized representative (*Prokurist*). Otherwise, the Company shall be represented by authorized representatives (*Prokuristen*) or other duly authorized signatories as further specified by the Executive Board.

The Executive Board must report regularly to the Supervisory Board, particularly on proposed business policy and strategy, on profitability and on the current business of the Company, as well as on any exceptional matters that may arise from time to time. If not otherwise required by law, the Executive Board makes decisions by a simple majority of the votes cast. In the event of a tied vote, the chairman shall have the casting vote.

Under certain circumstances, such as a serious breach of duty or a vote of no confidence by the shareholders at the General Meeting, a member of the Executive Board may be removed by the Supervisory Board prior to expiration of his/her term. A member of the Executive Board may not deal with, or vote on, matters relating to proposals, arrangements or contracts between himself/herself and the Company.

18.2.2 Current Composition of the Executive Board

The current Executive Board consists of six members. The members of the Executive Board are jointly responsible for the overall management of the Company and inform each other of all significant activities and transactions. The entire Executive Board decides on all matters of a fundamental or significant financial importance in accordance with applicable law, the Articles of Association, or the Executive Board's rules of procedure (*Geschäftsordnung*).

The following table sets forth the members of the Executive Board, their ages, responsibilities and the years in which their current terms expire, as of the date of this Prospectus.

Name	Age	Member since	Appointed until	Current Responsibility
Carsten Spohr	54	January 1, 2011	December 31, 2023	Chairman and Chief Executive Officer
Christina Foerster	49	January 1, 2020	December 31, 2022	Chief Customer Officer
Harry Hohmeister	57	July 1, 2013	June 30, 2024	Chief Commercial Officer
Dr. Detlef Kayser	56	January 1, 2019	December 31, 2024	Chief Operations Officer
Dr. Michael Niggemann	47	January 1, 2020	December 31, 2022	Chief Human Resources and Legal Officer
Remco Steenbergen	53	January 1, 2021	December 31, 2023	Chief Financial Officer

The following description provides summaries of the *curricula vitae* of the members of the Executive Board and indicates their principal activities outside the Lufthansa Group to the extent those activities are significant with respect to the Company and as of the date of this Prospectus.

Carsten Spohr was born in Wanne-Eickel, Germany, on December 16, 1966 and has been Chairman of the Executive Board and Chief Executive Officer of the Company since May 1, 2014. He holds a degree in industrial engineering from the University of Karlsruhe and obtained a commercial pilot's license at the Lufthansa Verkehrsfliegerschule in Bremen and Phoenix. Afterwards, he completed a trainee program at Deutsche Aerospace AG in Munich. Prior to joining the Executive Board, Carsten Spohr served in various positions with increasing responsibilities within the Lufthansa Group. In 2000, Carsten Spohr was appointed head of the alliance management of the Company. In this function, he was responsible for the management and coordination of the Company's worldwide cooperation partners, including the Star Alliance and regional partners. In 2003, he also assumed responsibility for the passenger strategy and the passenger shareholdings of the Company. In October 2004, Carsten Spohr was appointed to the executive board of Lufthansa Passenger Airlines. His responsibilities included the hub management, cabin crew and personnel management of Lufthansa Passenger Airlines. With effect from January 15, 2007, Carsten Spohr was appointed chairman of the executive board of Lufthansa Cargo. Carsten Spohr holds the Lufthansa Captain's license for the Airbus A320 family aircraft types. From January 1, 2011, onwards, Carsten Spohr was a member of the Executive Board of Deutsche Lufthansa AG and chairman of the Lufthansa Passage Executive Board. On May 1, 2014, Carsten Spohr took over as chairman of the Executive Board.

Alongside his office as chairman of the Executive Board and Chief Executive Officer, Carsten Spohr is member of the supervisory board of Münchener Rückversicherungs-Gesellschaft AG (Munich RE). In addition, Carsten Spohr was a member of the supervisory board of thyssenkrupp AG until September 2019.

Besides these appointments, Carsten Spohr has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Christina Foerster was born in Wiesbaden, Germany, on September 5, 1971. As Chief Customer Officer she is responsible for the business areas customer, IT and corporate responsibility. After joining Brussels Airline as chief commercial officer in 2016, she took over the position of chief executive officer at Brussels Airlines in April 2018. Before, Christina Foerster held several positions within the Company, where she started her career in the airline industry in 2002 as project manager corporate. Afterwards she has been general manager

product development from 2005 until 2011 before being appointed vice president Network & Fleet Development in 2011. In 2014, Christina Foerster was appointed senior vice president Network, Group & Alliance Development before becoming overall process owner of the Network & Partner Management for the Lufthansa Group. Christina Foerster started her career as a consultant at The Boston Consulting Group in 1999.

Christina Foerster has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Harry Hohmeister was born in Delmenhorst, Germany, on April 19, 1964. He holds a diploma in commercial air transport and joined the Lufthansa Group in 1985. He was initially responsible for the planning and development of the fleet and the flight planning for Europe. Afterwards, Harry Hohmeister joined Thomas Cook Airlines, and held various executive management functions in Germany, Belgium and the UK for over five years. He then joined the Management Board of SWISS in 2005. After serving in various positions with increasing responsibilities within the Lufthansa Group, he has been a member of the Executive Board since July 2013 and was responsible as Chief Commercial Officer Network Airlines for the commercial management of Premium Network Airlines and the management of the hubs of these airlines until December 31, 2019. Since January 1, 2020, he has been in charge of the Executive Board department “Commercial Passenger Airlines” and is responsible for the coordination of network planning, revenue management, dispatch and sales for all passenger airlines in the Lufthansa Group.

Alongside his office as member of the Executive Board and Chief Commercial Officer, Harry Hohmeister is a non-executive member of the board of directors of SunExpress. In addition, Harry Hohmeister was member of the supervisory board of Aircraft Maintenance and Engineering Corporation (AMECO) until May 2021, a member of the board of governors of IATA until June 2019, where he chaired the audit committee until 2018, and a member of the board of directors of SN Airholding SA/NV until January 2017.

Besides these appointments, Harry Hohmeister has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Dr. Detlef Kayser was born in Kiel, Germany, on March 28, 1965. Detlef Kayser, who holds a doctorate in aerospace engineering, joined the Lufthansa Group in January 2016 and was responsible for group-wide strategy development as well as for commercial and technical fleet management of the Lufthansa Group. Besides this, he was accountable for group-wide organizational development and processes orientation as well as corporate governance and corporate investment management as executive vice president strategy and fleet. Before that, Detlef Kayser worked for McKinsey in a variety of positions, most recently as senior director in the company’s Hamburg office, heading the global “Operations Practice.” He advised clients from industry and public administration in over 100 successfully implemented projects, including several for the Lufthansa Group. As chief operating officer, he is responsible for all operational processes as well as fleet and infrastructure management within the Lufthansa Group. Since summer 2020, he is also responsible for the group wide restructuring program “ReNew” which was introduced due to the COVID-19 crisis.

Alongside his office as member of the Executive Board and Chief Operations Officer, Detlef Kayser was a member of the supervisory board of Aerodata AG until March 31, 2020, and a member of the board of directors of SunExpress, Turkey, until March 31, 2020.

Besides these appointments, Detlef Kayser has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Dr. Michael Niggemann was born in Dortmund, Germany, on April 2, 1974. From 2017 until his appointment as member of the Executive Board, Michael Niggemann was chief financial officer at SWISS in Zurich. Prior to this, he held several management positions within the legal department of the Lufthansa Group from 2007 onwards, most recently as Lufthansa Group General Counsel & Chief Compliance Officer. Before joining the Company, Michael Niggemann was an in-house legal counsel at HOCHTIEF AG. He started his career as a lawyer in 2004. He is responsible for “Human Resources and Legal Affairs” and also assumes the function of labor director.

Alongside his office as member of the Executive Board and Chief Human Resources and Legal Officer, Michael Niggemann is member of the management board of DLP Deutsche Luftverkehrs-Privatstiftung and member of the management board of ÖAP Österreichische Aviation Development Privatstiftung that is currently being dissolved (*abgewickelt*).

Besides these appointments, Michael Niggemann has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Remco Steenbergen was born in Amstelveen, Netherlands, on January 17, 1968. Until December 31, 2020, he was chief financial officer of Barry Callebaut Group based in Zurich, Switzerland. Prior to this, he worked at Philips and KPMG. Throughout his career, Remco Steenbergen has held a wide range of global leadership and financial management roles at numerous companies in the Netherlands, the United Kingdom, Taiwan, Belgium, Ireland, the United States and Switzerland. He holds a Master's degree in Business Administration from the Institute for Management and Development (IMD) in Lausanne, Switzerland and a postdoctoral degree in accounting from the Erasmus University in Rotterdam, the Netherlands. He is responsible for the finance division, which includes the areas of controlling and risk management, corporate finance, accounting and balance sheets, taxes, purchasing as well as mergers & acquisitions and investor relations.

Other than the ones listed above, Remco Steenbergen has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

18.2.3 Compensation

(a) Principles of the Executive Board's Compensation System

The remuneration system for the members of the Executive Board contributes to the promotion of the Company's corporate and financial strategy by providing incentives for sustainable and value-oriented management and taking into account the interests of all stakeholder groups. The members of the Executive Board shall be incentivized to achieve the Lufthansa Group's strategy and to ensure its sustainable and long-term positive development. Against this background, the Supervisory Board takes into account in particular the following essential principles when making decisions on the remuneration system and on the structure and the level of the remuneration of the members of the Executive Board:

- Implementing the corporate strategy;
- Aligning performance and remuneration (Pay for Performance);
- Joint and individual performance of the Executive Board members;
- Interests of shareholders and other stakeholders;
- Sustainability;
- Appropriateness of the remuneration; and
- Transparency.

The Supervisory Board as a whole is responsible for the structure of the remuneration system for Executive Board members and for defining the individual remuneration. The Steering Committee supports the Supervisory Board, monitors the appropriate design of the remuneration system and prepares the Supervisory Board's resolutions (see "*18.3.3(b) Steering Committee*"). The appropriateness of the remuneration of the Executive Board is evaluated on the basis of a comparison of the target and maximum remuneration of the companies listed in the German DAX stock index. For this horizontal market comparison, the Supervisory Board takes into account in particular the market position of the Company on the basis of the key performance indicators revenue, employees and market capitalization. Furthermore, the Supervisory Board examines the remuneration of both senior management and the workforce as a whole, based on the German Group companies within the Company's tariff association. For this purpose, the senior management group was defined by the Supervisory Board as a group of executives at the three management levels below the Executive Board of the Company. The other staff consists of the non-tariff employees below the

management levels and the tariff employees on the ground, in the cockpit and in the cabin. Furthermore, the Supervisory Board, based on a proposal of the Steering Committee, may temporarily deviate from the remuneration system in exceptional circumstances. In the event of material changes to the remuneration system, but at least every four years, the remuneration system is presented at the General Meeting for approval. The current remuneration system was approved by the General Meeting on May 5, 2020, in anticipation of the compliance with the German Act Implementing the Shareholders' Directive II (*Gesetz zur Umsetzung der zweiten Aktionärsrechterichtlinie (ARUG II)*) and Section 120a para. 1 sentence 1 of the German Stock Corporation Act (*Aktiengesetz*).

(b) Components of the Executive Board remuneration system

The remuneration of the Executive Board is made up of fixed, non-performance-related and performance-related components. In addition to the base salary, the fixed remuneration includes fringe and retirement benefits. The variable remuneration comprises a one-year (annual bonus) and a multi-year component. In the target remuneration, the share of the multi-year variable remuneration exceeds the share of the one-year variable remuneration. In certain cases, the Supervisory Board has the option of withholding the one-year and multi-year variable remuneration or of claiming back remuneration already paid (clawback).

The three main components of the remuneration system for the members of the Executive Board are the base salary, the one-year variable remuneration (annual bonus) and the multi-year variable remuneration (the "LTI").

(i) Non-Performance-Related Remuneration Components

Each member of the Executive Board receives a fixed base salary. This is paid in twelve equal monthly instalments. Each member of the Executive Board also receives certain benefits subject to the limits of the defined maximum remuneration. These include, in particular, the provision of a company car, allowances for insurance and flight allowances for private air travel. In addition, the members of the Executive Board receive retirement benefits based on a contribution-based system. During the term of employment, a fixed amount is annually credited to a personal pension account of each member of the Executive Board.

(ii) Performance-Related Remuneration Components

The performance-related remuneration of the members of the Executive Board is designed to support the Lufthansa Group's short- and long-term strategy. For this reason, it consists of a one-year variable remuneration (annual bonus) and a LTI, thus ensuring the sustainable and long-term development of the Company. One-year and multi-year variable remuneration differ with respect to the performance period and the performance criteria. Payment is subject to achieving certain financial and non-financial performance criteria. When determining such criteria, the Supervisory Board ensures that these are clearly measurable and aligned with the corporate strategy. In this context, Environment, Social and Governance criteria are also taken into account. The performance criteria are derived from the Company's strategic targets and operational management. They are aimed at increasing profitability and efficient management. For this reason, the key performance indicators for Lufthansa Group form the basis for determining the performance criteria for the variable remuneration. To achieve a balanced risk/opportunity profile, the Supervisory Board also considers the development of the Company's share price.

The Supervisory Board follows a clear "pay for performance" approach and ensures that the objectives are challenging and ambitious. If the targets are not met, the variable remuneration can fall to zero. If the targets are clearly exceeded, the target achievement is limited to 200%. Therefore, the target achievement for the financial and non-financial targets in both the annual bonus and the LTI ranges between 0% and 200%.

The one-year variable remuneration rewards the contribution to the operational implementation of the corporate strategy during a fiscal year. On this basis, 85% of the annual bonus is based on financial (*i.e.*, Adjusted EBIT margin and Adjusted Return on Capital Employed, defined as Adjusted EBIT plus interest on liquidity less 25% taxes divided by the average capital employed). The capital employed results from total assets adjusted for non-operating items (deferred taxes, derivative financial instruments) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents) (Adjusted ROCE) and 15% on non-financial overall and individual business and sustainability targets ("business and

sustainability targets”). In addition, the Supervisory Board has the option to apply an individual performance factor ranging between 0.8 and 1.2 as part of the assessment of the performance of each member of the Executive Board. Based on the weighted target achievement for the financial and non-financial targets multiplied by the individual performance factor defined for each member of the Executive Board, the Supervisory Board determines the annual bonus to be paid to each member of the Executive Board for the fiscal year in question. The annual bonus is paid out after the consolidated financial statements for the respective fiscal year have been approved. If the target is exceeded, the one-year variable remuneration is limited to a maximum of 200% of the target bonus (cap). The Supervisory Board reserves the right to make the payment in shares of the Company instead of cash. The performance criteria, target values and target achievement are reported ex post in the remuneration report for the respective fiscal year.

The LTI aims towards a long-term and sustainable corporate development and to align the interests of the members of the Executive Board and the interests of the shareholders. On this basis, 85% of the multi-year variable remuneration of the Executive Board is based on financial and 15% on non-financial performance criteria. In this regard, the Total Shareholder Return (the “TSR”) of the Company, *i.e.*, the return on Company’s shares taking into account notionally reinvested dividends, is measured by comparison with the DAX-30 companies. On the other hand, the average return on investment in the form of the Adjusted ROCE over the 4-year performance period is compared to a strategic target set by the Supervisory Board prior to the grant. In addition, non-financial strategic and sustainability targets are applied. For the strategic and sustainability targets, the Supervisory Board defines key topics for the respective performance period. At the beginning of the performance period, the members of the Executive Board are granted conditional virtual shares. The number of shares is determined by dividing the target amount of the LTI by the average share price of the Company over the first 60 trading days after the start of the respective performance period. At the end of the performance period, the number of conditionally granted shares is multiplied by the total target achievement of the financial and non-financial performance criteria. The resulting final number of shares is multiplied by the average share price of the Company’s shares over 60 trading days before the end of the performance period plus dividends paid during the performance period and, once the consolidated financial statements have been adopted, will be paid out in cash for the last year of the respective performance period. If the target is exceeded, the multi-year variable remuneration is limited to a maximum of 200% of the target amount (cap). The Supervisory Board reserves the right to make the payment in shares of the Company instead of cash. The performance criteria and target values of each grant are reported in detail in the remuneration report for the year of the grant.

(iii) Maximum Remuneration

The Supervisory Board has set uniform maximum amounts for the performance related remuneration components for all members of the Executive Board. For both, the one-year and multi-year variable remuneration, these are uniformly 200% of the target amount for all members of the Executive Board. In addition, the Supervisory Board has set a maximum remuneration for the respective members of the Executive Board for the sum of the expended remuneration amounts for a fiscal year (including fringe and retirement benefits). This amounts to €9.5 million for the chairman of the Executive Board and €5.0 million for an ordinary member of the Executive Board. If the remuneration for a fiscal year exceeds this limit, the variable remuneration will be reduced accordingly.

(iv) Share Ownership Guidelines

The Company’s share ownership guidelines (the “SOG”) have been an integral part of the remuneration system for the Executive Board since 2019. They oblige the Chief Executive Officer to acquire shares in an amount that equals twice his basic salary and the other members of the Executive Board members are required to acquire shares in an amount that equals a one year’s basic salary. Such Company’s shares shall be held at least for the term of office. From the end of the term of office, up to 25% of the shares may be sold in each calendar year.

The minimum number of Company’s shares to be purchased by each Executive Board member is determined at the beginning of their term and is based on the average share price over 125 trading days prior to the start of their service contract. Shares are to be acquired over a four-year period. Existing shareholdings are included in the calculation. In connection with the restrictions on Executive Board remuneration for the

duration of the Stabilization Package, the Supervisory Board has decided to suspend the four-year acquisition period for as long as the stabilization measures are in place. It will be resumed as soon as variable remuneration is awarded again.

(v) *Compliance and Performance Clawback*

In the event of an intentional or grossly negligent breach of statutory obligations or internal policies (compliance penalty or clawback), or if variable remuneration components dependent on achieving certain targets are paid on the basis of false data (performance clawback), the Supervisory Board has the right to withhold or demand repayment of short-term and long-term variable remuneration. Enforcement of the withholding or repayment claim is at the professional discretion of the Supervisory Board.

(vi) *End-of-service benefits*

The remuneration system also stipulates the amount of remuneration in the event of a premature termination of the Executive Board employment contract. Depending on the reason for termination, the following provisions apply to the remuneration when leaving office:

If a contract is terminated early for reasons other than good cause or a change of control, the Company will not remunerate more than the value of outstanding entitlements for the remainder of the contract, as recommended by the German Corporate Governance Code, whereby these payments may not exceed annual remuneration for two years (severance cap). The cap on severance payments is determined by the annual remuneration, which is made up of basic salary and the target amounts of one-year and long-term variable remuneration; in-kind benefits and ancillary benefits are not considered. This means the maximum severance pay for an ordinary Executive Board member is €2,560,000 p.a., or €4,864,000 for the Chief Executive Officer.

As a rule, the Executive Board members are subject to a one-year non-competition clause after leaving the Executive Board. The Company pays the Executive Board member compensation of half their annual salary for the duration of the post-contractual non-compete clause. The Company has the option of waiving compliance with the post-contractual non-compete clause up to the end of the service contract with the effect that it is no longer obliged to pay compensation six months after the waiver is granted. As of the date of this Prospectus, only the service contracts with Harry Hohmeister, Remco Steenbergen and Detlef Kayser, which were signed after the German Corporate Governance Code as amended on December 16, 2019, came into effect, as well as future appointments and reappointments, require severance pay to be offset against the non-compete compensation.

If the service contract with a member of the Executive Board is terminated in connection with a change of control at the Company, the Executive Board member is entitled to a payment equivalent to the remuneration outstanding for the remainder of the contract. The amount of payment may not exceed 150% of the contractual cap on severance pay. In line with the recommendation of the German Corporate Governance Code, the cap on severance pay also applies to a change of control for the service contracts with Harry Hohmeister, Remco Steenbergen and Detlef Kayser as well as for future appointments and reappointments.

(c) *Remuneration Limited for the Duration of Stabilization Package*

The Stabilization Package sets forth restrictions on the remuneration of the Executive Board. Subject to contractual claims against the Company arising prior to June 21, 2020, no bonuses or other variable or similar remuneration components may be awarded to Executive Board members for the duration of the stabilization measures under the Stabilization Package. The same applies to special payments in the form of share packages, gratuities and other forms of compensation in addition to the fixed salary, other discretionary payments by the Company and severance payments not required by law.

Simultaneously, no Executive Board member may receive a basic salary (inclusive of any payments for work on executive or supervisory boards within the Lufthansa Group) which is higher than their basic salary as of December 31, 2019, until at least 75% of the total of the WSF Stabilization Measures (including interest and any additional payments) have been satisfied, repaid or redeemed, sold or otherwise settled by means of a capital contribution or otherwise. Correspondingly, the upper limit for new Executive Board members is set

by the lowest fixed salary of an Executive Board member with an equivalent position as of December 31, 2019.

(d) *Remuneration of the Executive Board in 2020*

The Executive Board was reduced from seven to six members when Ulrik Svensson stepped down in April 2020. After Thorsten Dirks stepped down in June 2020, the Executive Board intermittently consisted of five members. A new Chief Financial Officer, Remco Steenberg, was appointed with effect from January 1, 2021. Since January 2021, the Executive Board consists of six members. Mr. Steenberg agreed with the remuneration restrictions during the stabilization measures.

The following tables set forth the remuneration awarded and due (paid) to each individual member of the Executive Board in the Fiscal Year 2020. Section 162 of the German Stock Corporation Act (*Aktiengesetz*) defines as awarded and due (paid) remuneration that fell due in the reporting period and have already been paid to the individual Executive Board member or which are outstanding but not yet paid.

(i) *Awarded and Granted Benefits (Allocations) 2020*

Name	Basic salary ⁽⁴⁾	Ancillary benefits	Long-term variable remuneration			Severance Payments	Total remuneration
			One-year variable remuneration 2019	Three-year variable remuneration	Option program (4 years) LH performance		
				Deferral 2017	2016 ^(*)		
(in € thousand)							
Carsten Spohr	1,471	19	576	864	900	—	3,830
Thorsten Dirks ⁽¹⁾	387	11	226 ⁽⁵⁾	360	—	3,500	4,484
Christina Foerster	774	1	—	—	—	—	775
Harry Hohmeister	774	15	303	540	600	—	2,232
Detlef Kayser.....	774	6	272 ⁽⁶⁾	—	—	—	1,052
Michael Niggemann .	774	1	—	—	—	—	775
Ulrik Svensson ⁽²⁾	272	5	303	540	—	—	1,120
Bettina Volkens ⁽³⁾	—	3	303	540	600	—	1,446
Total	3,756	61	1,983	2,844	2,100	3,500	15,713

(*) Subject to the performance of the Company's share price, Carsten Spohr and Harry Hohmeister may receive payouts from granted options for the fiscal year 2017 and the fiscal year 2018 under the LH Performance program, which members of the Executive Board were obliged to take part in up to and including the fiscal year 2018.

(1) Until June 30, 2020.

(2) Until April 30, 2020.

(3) Until December 31, 2019.

(4) Including voluntary waiver of 20% of basic salary for the period from April to September 2020.

(5) Remuneration of €77,000 paid for the Fiscal Year 2019 was offset against this.

(6) Remuneration of €68,000 paid for the Fiscal Year 2019 was offset against this.

(ii) *Contractually agreed benefits for the Fiscal Year 2020*

Contractually agreed benefits (as defined in the model table previously included in the German Corporate Governance Code) are presented in the following table at their value at the time of the agreement (corresponding to a target achievement of 100%) for the respective fiscal year.

Name	Basic salary ⁽⁴⁾	Ancillary benefits	One-year variable remuneration ⁽⁵⁾ (in € thousand)	Long-term variable remuneration	Service cost	Total remuneration
Carsten Spohr	1,471	19	1,140	2,090	925	5,645
Thorsten Dirks ⁽¹⁾	387	11	—	—	251	649
Christina Foerster	774	1	600	1,100	450	2,925
Harry Hohmeister	774	15	600	1,100	483	2,972
Detlef Kayser	774	6	600	1,100	460	2,940
Michael Niggemann	774	1	600	1,100	450	2,925
Ulrik Svensson ⁽²⁾	272	5	200	367	160	1,004
Bettina Volkens ⁽³⁾	—	—	—	—	678	678
Total	3,756	58	3,740	8,947	3,857	19,738

(1) Until June 30, 2020.

(2) Until April 30, 2020.

(3) Until December 31, 2019.

(4) Including voluntary waiver of 20% of basic salary for the period from April to September 2020.

(5) Except for Ulrik Svensson, all members of the Executive Board waived their right to the payment of the one-year variable remuneration for the Fiscal Year 2020.

(iii) Retirement Benefits

The members of the Executive Board receive retirement benefit commitments based on a defined contribution plan. As of the Fiscal Year 2019, a fixed annual amount is made available to every Executive Board member for the duration of their employment – €855,000 for the Chief Executive Officer and €450,000 for ordinary members – as a contribution to their retirement benefit account.

The investment guidelines are based on the investment concept for the Lufthansa Pension Trust, which also applies to staff members of the Company.

Retirement benefits are paid when the beneficiary reaches the retirement age of 60 years (if they are no longer an Executive Board member) or in the event of disability or death. If the service contract ends before retirement age is reached, the beneficiaries or their surviving dependents acquire a retirement benefit credit equivalent to the balance of the pension account at the time. The Company guarantees the sum of contributions paid.

A supplementary risk capital sum will be added to the pension credit in the event of a claim for a disability pension or a pension for surviving dependents. This sum will consist of the average contributions paid into the pension account over the past three years multiplied by the number of full years by which the claimant is short of the age of 60 from the time a disability pension entitlement arises.

In general, the pension credit is paid out in ten instalments. On application by the Executive Board member or his/her surviving dependents, a payment as a lump sum or in fewer than ten instalments may also be made, subject to approval by the Company. The retirement benefit credit received until December 31, 2018, by Carsten Spohr and Harry Hohmeister may also be paid as an annuity, on application and with the approval of the Company.

Under his contract as a pilot, which is currently not active, Carsten Spohr is entitled to a transitional pension in accordance with the wage agreement “Transitional pensions for cockpit staff.” If Carsten Spohr leaves the Executive Board before the age of 60 years and resumes his employment as a pilot, he is entitled to draw a “Transitional pension for cockpit staff at Lufthansa” once he turns 60 years or on request once he turns 55 years, in accordance with the provisions of the wage agreement. This additional benefit is paid if certain conditions of eligibility are met and provides for a monthly pension of up to 60% of the last modified salary until the beneficiary reaches the age of 63 years.

The total amount of pension entitlements acquired by active and former Executive Board members in 2020 is €4.0 million (previous year: €3.1 million) according to the German Commercial Code (*Handelsgesetzbuch*)

and €3.9 million (previous year: €3.3 million) under IFRS was recognized in staff costs (service cost). The individual current service cost and present values of pension entitlements are as follows for 2020:

Name	HGB	Settlement amount of pension obligations as of December 31, 2020 (in € thousand)	IFRS	Present value of pension obligations as of December 31, 2020
	Current Service costs for 2020		Current service costs for 2020	
Carsten Spohr	797	8,018	925	8,024
Thorsten Dirks ⁽¹⁾	200	1,770	251	1,770
Christina Foerster	651	654	450	666
Harry Hohmeister	427	3,572	483	3,573
Detlef Kayser	434	1,304	460	1,305
Michael Niggemann	680	684	450	701
Ulrik Svensson ⁽²⁾	142	1,890	160	1,890
Bettina Volkens ⁽³⁾	651	3,768	698	3,768
Total.....	3,982	21,660	3,877	21,697

(1) Until June 30, 2020.

(2) Until April 30, 2020.

(3) Until December 31, 2019.

(iv) *Shareholdings of current Executive Board members*

The following table sets forth the respective shareholdings of the current members of the Executive Board.

Name	Number of Lufthansa shares required to be purchased	Shareholdings as of December 31, 2020	Shareholdings as of the date of this Prospectus
Carsten Spohr	155,969	130,000	130,000
Christina Foerster	56,126	2,426	2,426
Harry Hohmeister	41,044	76,048	76,048
Dr. Detlef Kayser	41,044	2,320	2,320
Dr. Michael Niggemann	56,126	21,270	21,270
Remco Steenbergen	99,113	25,000	50,000
Total.....	449,422	257,064	282,064

The shares bought in accordance with the SOG are to be held until the end of the service contract with the Executive Board member. Thereafter, Executive Board members may sell 25% of their SOG shares per year.

(v) *Remuneration of Executive Board members who stepped down in 2020*

After Ulrik Svensson stepped down from the Executive Board for health reasons, his service contract was terminated early by resolution of the Supervisory Board on April 21, 2020, and a corresponding severance agreement on April 30, 2020. Ulrik Svensson waived his right to a severance payment. Furthermore, the Company waived the post-contractual non-compete clause and is not obliged to pay compensation. Ulrik Svensson's claims to one-year and long-term variable remuneration for the proportion attributable to the Fiscal Year 2020 are not affected. He also retains his rights to the long-term variable remuneration agreed in 2019 as a member of the Executive Board during the performance period to the extent that the conditions for a payment are met at the end of the performance period. Ulrik Svensson also retains his rights to the deferred portion of variable remuneration for 2018. Payments under the ongoing Lufthansa-performance program will be made *pro rata* for the period in which Ulrik Svensson was still an Executive Board member. The Company will make an annual contribution of €450,000 *pro rata* for 2020, *i.e.*, until April 30, 2020, (€150,000) – for retirement benefits to Ulrik Svensson's pension account. Existing internal rules for former Executive Board members apply to Ulrik Svensson's use of concessionary private travel arrangements from the time of his departure.

The employment contract with Thorsten Dirks was terminated prematurely with effect from June 30, 2020, in accordance with a Supervisory Board resolution of June 26, 2020, and the corresponding severance agreement. In accordance with the severance payment agreement, Thorsten Dirks will receive a severance payment of €3.5 million. The severance payment is less than the contractually agreed cap of two times annual remuneration (€5.1 million) and is also less than he would have earned over the remaining 2.5 years of the service contract. The Company waived the post-contractual non-compete clause and is not obliged to pay compensation. Thorsten Dirks is not entitled to the short-term and long-term variable remuneration for 2020 or to the long-term variable remuneration for 2019. However, Thorsten Dirks does retain his rights to the deferred portion of variable remuneration for 2018. He has deferred the payment until further notice. For the purposes of the ongoing Lufthansa-Performance program, Thorsten Dirks will be treated as if he had fulfilled his employment contract until the original end date of April 30, 2023. Furthermore, the Company will make an annual contribution of €450,000 – *pro rata* for 2020, *i.e.*, until June 30, 2020 (€225,000) – for retirement benefits to Thorsten Dirks' pension account. Existing internal rules for former Executive Board members apply to Thorsten Dirks' use of concessionary private travel arrangements from the time of his departure.

18.3 Supervisory Board

18.3.1 Overview

The Company's Supervisory Board consists of 20 members. The Company's shareholders elect ten members of the Supervisory Board at the General Meeting. Pursuant to the German Co-Determination Act (*Mitbestimmungsgesetz*) of 1976, as amended, the Company's employees elect the remaining ten members.

The Supervisory Board shall be composed of at least 30% women and at least 30% men, but separately by the shareholder representatives and the employee representatives on the Supervisory Board. As a result, at least three seats must be held by women and at least three seats by men among the shareholder and employee representatives. In the Fiscal Year 2020, the Supervisory Board complied with this requirement with three women and seven men as shareholder representatives and four women and six men as employee representatives.

In accordance with the Framework Agreement, the Supervisory Board shall also include two members to be proposed by the chairman of the Supervisory Board after consultation with the Nomination Committee of the Supervisory Board and appointed by the WSF, one of whom to also become a member of the Audit Committee. Accordingly, Michael Kerkloh and Angela Titzrath were appointed to the Supervisory Board and Michael Kerkloh was also appointed as a member of the Audit Committee.

In addition, the Supervisory Board has adopted a comprehensive framework for its composition, which includes requirements under the German Corporate Governance Code and the diversity concept pursuant to section 289f of the German Commercial Code (*Handelsgesetzbuch*).

The Supervisory Board endeavors to ensure that its members together possess the necessary expertise, skills and professional experience to properly perform their duties. It strives particularly to ensure that the members of the Supervisory Board possess expertise, skills and professional experience in the following areas: management and leadership of international companies, a business understanding with regard to the Company's main areas of activity, marketing, distribution and customer, politics and association work, science, finance, controlling/risk management, human resources, governance/compliance and sustainability and digitization. The Supervisory Board has also resolved to pursue diversity in its composition, for instance with regard to age, gender, education and professional background. With respect to the international business alignment of the Company, the Supervisory Board strives to ensure at all times that several of its members have international business experience or an international background in other respects. Further objectives concerning the composition of the Supervisory Board are that different age groups are suitably represented on the Supervisory Board and that, absent special circumstances, a member should not hold office beyond the end of the next General Meeting following his or her 72nd birthday, in accordance with the German Corporate Governance Code. With a view to avoiding potential conflicts of interest and taking into account the ownership structure of the Company and the number of independent Supervisory Board members, the Supervisory Board has set itself the goal that more than eight of the shareholder representatives shall be independent.

Any member of the Supervisory Board elected by the shareholders at the General Meeting may be removed by a vote of at least three quarters of the votes cast by the shareholders in such meeting. Any member elected by the employees may be removed by a majority of three quarters of the votes cast by the employees.

In the event that the term of office of the chairman of the Supervisory Board will end following a General Meeting, but in any case at least every five years, the Supervisory Board shall elect a chairman and a deputy chairman from among its members.

Unless otherwise required by law or by the Articles of Association, resolutions of the Supervisory Board are passed by a simple majority of the votes cast. According to the Articles of Association, in the event of a tied vote, a second vote is held. Should the second vote result in a tied vote as well, the chairman of the Supervisory Board shall have a casting vote. In order to constitute a quorum, at least half of the total members of the Supervisory Board must participate in the voting.

18.3.2 Members of the Supervisory Board

The persons set forth below are the members of the Company's supervisory board as of the date of this Prospectus.

Name	Age	Member since	Appointed until ⁽¹⁾	Position	Principal Activity
Dr. Karl-Ludwig Kley	69	May 7, 2013	2023	Chairman	Chairman of the supervisory board of E.ON SE and the Company
Christine Behle	52	May 7, 2013	2023	Deputy Chairwoman ⁽²⁾	Deputy Chairwoman of the national executive board of the trade union ver.di
Alexander Behrens	47	May 8, 2018	2023	Member ⁽²⁾	Flight attendant, Member of the Board of UFO e.V. trade union
Jörg Cebulla	54	November 8, 2015	2023	Member ⁽²⁾	Flight captain
Erich Clementi	63	May 5, 2020	2025	Member	Deputy chairman of the supervisory board of E.ON SE
Dr. Thomas Enders	62	May 5, 2020	2025	Member	Member in various supervisory boards
Jürgen Jennerke	57	December 8, 2020	2023	Member ⁽²⁾	Cargo handler / Works Council member on leave of absence – ver.di section
Dr. Michael Kerkloh	67	September 2, 2020	2024 ⁽³⁾	Member	Former chairman of the executive board of Flughafen München GmbH
Carsten Knobel	52	January 9, 2018	2023	Member	Chairman of the executive board and CEO Henkel AG & Co. KGaA
Dr. Holger Benjamin Koch	44	May 8, 2018	2023	Member ⁽²⁾	Senior Director Airport/Industry Charges & Commercial Provider Management
Harald Krüger	55	May 5, 2020	2025	Member	Member of the supervisory board of Deutsche Telekom AG
Birgit Rohleder	60	May 8, 2018	2023	Member ⁽²⁾	Teamlead IT Application Management Airport Services
Miriam Sapiro	60	October 24, 2017	2023	Member	Managing Director and Vice Chairman (Public Affairs), Sard Verbinen & Co.
Ilja Schulz	53	May 8, 2018	2023	Member ⁽²⁾	Pilot
Britta Seeger	51	May 4, 2021	2024	Member	Member of the executive

Name	Age	Member since	Appointed until ⁽¹⁾	Position	Principal Activity
Birgit Spineux	56	January 1, 2021	2023	Member ⁽²⁾	board of Daimler AG Purser / staff representative Lufthansa cabin on leave of absence
Dr. Astrid Stange.....	55	May 5, 2020	2025	Member	Group Chief Operating Officer of AXA SA
Olivia Stelz.....	50	May 8, 2018	2023	Member ⁽²⁾	Purser
Angela Titzrath.....	55	September 2, 2020	2025 ⁽³⁾	Member	CEO Hamburger Hafen und Logistik AG
Klaus Winkler	47	May 8, 2018	2023	Member ⁽²⁾	Engine mechanic

(1) Ending upon conclusion of the General Meeting at which the Supervisory Board is discharged of its duties.

(2) Employee representative according to the German Co-determination Act (*Mitbestimmungsgesetz*).

(3) Designated by the WSF.

The following description provides summaries of the *curricula vitae* of the current members of the Supervisory Board and indicates their principal activities outside the Lufthansa Group to the extent those activities are significant with respect to the Company.

Dr. Karl-Ludwig Kley was born on June 11, 1951 in Munich, Germany. After his law studies at Ludwig Maximilian University of Munich and working as a trainee lawyer in Hamburg and Johannesburg (South Africa), he obtained a doctorate in law (Dr. iur.) from the Ludwig Maximilian University in Munich, 1986. Karl-Ludwig Kley started his career 1982 at Bayer AG. He initially worked in corporate finance and then as an assistant to the chairman of the executive board. After a period abroad as chief financial officer in Japan, he held various senior positions in the pharma business segment. From 1991 to 1994, he worked in sales and then until 1997 as head of the pharma business segment of the Italian Bayer subsidiary. He was then head of finance and investor relations at Bayer's headquarters. From 1998 to 2006, Karl-Ludwig Kley was a member of the Executive Board. From 2007 to 2016, Karl-Ludwig Kley served as chairman of the board of directors of Merck KGaA.

Alongside his office as chairman of the Supervisory Board, Karl-Ludwig Kley is the chairman of the supervisory board of E.ON SE and the chairman of the board of trustees of the Fritz-Thyssen-Foundation. In addition, Karl-Ludwig Kley was the deputy chairman of the supervisory board of BMW AG until May 2021. Until March 3, 2018, he was a member of the board of directors of Verizon Communications Inc., United States.

Besides these appointments, Karl-Ludwig Kley has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Christine Behle was born on July 12, 1968 in Wuppertal, Germany. She holds a degree in public administration and joined the trade union ÖTV as youth education consultant in 1995. From 1999 to 2005 she served as trade union secretary for ver.di in the Wuppertal district. In 2006 Christine Behle was appointed union state director for transport. She is a member of the ver.di federal board and head of the transportation department since 2011. In September 2019 she became deputy chair of the national executive board of ver.di and state director for the departments transport, social security, state and federal states, communities and special services. From 2018 to 2020, Christine Behle was a member of the supervisory board of Dortmunder Stadtwerke Holding GmbH and Dortmunder Stadtwerke AG. From 2015 to 2019, she was a member of the supervisory board of Bochum-Gelsenkirchener Straßenbahnen AG, as well as of Bochum-Gelsenkirchener Bahngesellschaft mbH from 2018 to 2019. Christine Behle was also member of the supervisory board of Hapag-Lloyd AG from 2016 to 2019.

Alongside her office as member of the Supervisory Board, Christine Behle is member of the supervisory boards of Bremer Lagerhaus-Gesellschaft-Aktiengesellschaft, where she is deputy chairwoman, and Autobahn GmbH.

Besides these appointments, Christine Behle has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Alexander Behrens was born on March 5, 1974 in Berlin, Germany. He studied at Hochschule der Künste Berlin and joined the Lufthansa Group as a flight attendant in 1995. After becoming a member of the collective bargaining commission in 2005, he was appointed chairman of group representatives of flight attendants at the Company. Since 2014, he is the chairman of general representative council for flight personnel of the Company. From April 2016 until May 2019, he served as chairman of the board of UFO e.V.

Alexander Behrens has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Jörg Cebulla was born on June 29, 1966 in Essen, Germany. He attended the commercial flight pilot program at Lufthansa Verkehrsfliegerschule in Bremen, Germany. He is a captain at the Company since 2000 and held the chair of general representative council for flight personnel at the Company from 2005 to 2009. From 2009 until 2012, he served as an executive board member of the Vereinigung Cockpit pilots' union and is a member of the investment committee of Deutsche Lufthansa Private Pension since 2004. From 2008 until 2013, he was already a member of the Supervisory Board.

Alongside his office as member of the Supervisory Board, Jörg Cebulla is member of the supervisory board of Albatros Versicherungsdienste GmbH and a member of the supervisory board of Sparda-Bank Hessen eG until June 2026.

Besides these appointments, Jörg Cebulla has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Erich Clementi was born on December 5, 1958 in Bozen, Italy. He obtained a masters' degree in business administration at Leopold Franzens University, Innsbruck in 1982. During his time at IBM between 1984 and 2019, he took over responsibility in various roles and countries in the area of sales, strategy, product and business services. In 2009, he became vice president corporate strategy, in 2011, senior vice president IBM Global Technology Services. From 2015 to 2017, he served as senior vice president of global markets and as chairman of IBM Europe, and from 2017 to 2019 as senior vice president of global integrated accounts and as chairman of IBM Europe.

Alongside his office as member of the Supervisory Board, Erich Clementi is deputy chairman of the supervisory board of E.ON SE.

Besides these appointments, Erich Clementi has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Dr. Thomas Enders was born on December 21, 1958 in Neuschlade, Germany. From 1978 to 1983 he studied economics, politics and history at the Rheinische Friedrich-Wilhelms-University in Bonn and the University of California in Los Angeles. He obtained a doctorate degree (Dr. phil.) from the Rheinische Friedrich-Wilhelms-University in Bonn. He worked as a member of the planning staff in the Federal Ministry of Defense from 1989 until 1991. Afterwards, from 1991 to 1999, he held various positions with increasing responsibilities, including head of corporate development, at MBB/DASA. From 2000 to 2005, he worked in various managerial positions at EADS. From 2005 until 2019, he served as chief executive officer of EADS and Airbus.

Alongside his office as member of the Supervisory Board, Thomas Enders is member of the supervisory board of Knorr-Bremse AG, a member and the chairman of the board of directors of Lilium N.V. and a non-executive member of the board of directors of Linde plc.

Besides these appointments, Thomas Enders has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Jürgen Jennerke was born on April 20, 1964 in Mainz, Germany. He joined the Lufthansa Group in 1982 as a cargo handler at Lufthansa Cargo. Since 1995, he is a works council member on leave of absence. From

1995 to 2009, he served as deputy chairman of the works council of Lufthansa Cargo and deputy chairman of the general works council Lufthansa Cargo. Since 2010, he is the chairman of the general works council of Lufthansa Cargo. From 2014 to 2019, he served as chairman of the works council of the Lufthansa Group. Since 2020, Jürgen Jennerke is the chairman of the works council of Lufthansa Cargo.

Alongside his office as member of the Supervisory Board, Jürgen Jennerke is deputy chairman of the supervisory board of Lufthansa Cargo.

Jürgen Jennerke is not and has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Dr. Michael Kerkloh was born on July 5, 1953 in Ahlen, Germany. He studied economics at the Georg-August-University Göttingen, Goethe University Frankfurt and London School of Economics between 1973 and 1979. After working as a research assistant at the Goethe University Frankfurt, he obtained a doctorate degree (Dr. rer. pol.). He started his career at Norman Rentrop publishing company in Bonn in 1986. From 1987 until 1994, he worked at Flughafen FrankfurtMain AG and changed to Flughafen Hamburg GmbH in 1995, where he served as an executive director. In 2002, he became chairman of the executive board and personnel labor director of Flughafen München GmbH. He held this position until 2019. From 2017 to 2020, Michael Kerkloh was a non-executive member of the board of directors of Oman Aviation Group.

Besides these appointments, Michael Kerkloh has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Carsten Knobel was born on January 11, 1969 in Marburg, Germany. From 1990 until 1995, he studied business and technical chemistry at Technical University Berlin and attended the Executive Education Program at Harvard Business School in 2010. Carsten Knobel joined Henkel AG & Co. KGaA in 1995 as assistant of the executive director for research and development. From 1998 until 2012, he served in various leading positions at Henkel in the sectors controlling, brand management, mergers and acquisitions and business development. From 2012 until December 31, 2019, Carsten Knobel was member of the executive board and chief financial officer of Henkel AG & Co. KGaA and responsible for finance, purchasing, integrated business solutions and ONE!Global Supply Chain. Since January 1, 2020, he is chairman of the management board and chief executive officer of Henkel AG & Co. KGaA.

Besides these appointments, Carsten Knobel has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Dr. Holger Benjamin Koch was born on October 8, 1976 in Bonn, Germany. From 1995 to 1996, he studied foreign languages and music at Wagner College, New York (USA). Holger Benjamin Koch then studied business administration at the University of Cologne and the University of St. Gallen from 1996 to 2000. He obtained a doctorate degree (Dr. rer. pol.) from the University of Cologne in 2006. He joined the Lufthansa Group in 2000 at Lufthansa Consulting GmbH and worked particularly in the areas of airports, finance and processes. In 2006, he moved to the Company and worked initially as a consultant in capacity and network management Frankfurt, then in group strategy and finally in shared services. From 2011 onwards, he worked in several managerial positions with increasing responsibilities, including head of corporate and business service Europe. Since 2014, he is a member of the speakers' committees of senior managers.

Holger Benjamin Koch is not and has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Harald Krüger was born on October 13, 1965 in Freiburg, Germany. He studied mechanical engineering at the Technical University of Braunschweig and the Rheinisch-Westfälische Hochschule (RWTH) Aachen. He obtains a degree of Mechanical Engineering (*Diplomingenieur*) of the Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen. Harald Krüger started his career as research assistant at the Institute for Dynamics of Flight Systems at the German Aerospace Centre (DLR). After joining BMW AG in Munich in 1992, he held several managerial positions with increasing responsibilities, including head of technical integration. Harald Krüger became a member of the executive board of BMW AG in 2008 and served until 2015 with different responsibilities. From 2015 until August 2019, Harald Krüger was chairman of the executive board of BMW AG.

Alongside his office as member of the Supervisory Board, Harald Krüger is member of the supervisory board of Deutsche Telekom AG and managing director of KC&C GmbH and KC&C Beteiligungs GmbH.

Besides these appointments, Harald Krüger has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Birgit Rohleder was born on September 21, 1960 in Hagen, Germany. From 1980 until 1985, she studied business administration (specializing in passenger transport and IT) in Heilbronn, Germany. She joined the Lufthansa Group in 1990 working in information management with various positions in IT infrastructure at sites worldwide. Since 2002, she is member of the works council of the Company. Since 2006, she acts as team/group head and since 2014, Birgit Rohleder is chairwoman of the interest group for non-payscale employees.

Birgit Rohleder is not and has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Miriam Sapiro was born on August 23, 1960 in New York, USA. She holds a Bachelor of Arts from Williams College, a Juris Doctor of New York University, School of Law and is a Rotary Fellow at Oxford University. She began her career in various roles at the Department of State as Attorney-Advisor and on the Secretary of State's Policy Planning Staff where she served from 1988 until 1997. From 1997 until 1999, she served as director for European affairs at the National Security Council and special assistant to the president. From 2000 to 2009, she worked in the private sector, running international affairs for a technology company and starting a consulting firm, Summit Strategies International. In 2009, she returned to the U.S. Government and served, upon confirmation by the Senate as Ambassador, as Deputy U.S. Trade Representative and as Acting U.S. Trade Representative. In 2014, she joined the Brookings Institute as a Non-Resident Senior Fellow before returning to the private sector and serving as Principal at Summit Strategies International, and as a Partner at Finsbury and Head of its Washington D.C. office until February 2018. Since March 2018, she serves as managing director at Sard Verbinen & Co.

Alongside her office as member of the Supervisory Board, Miriam Sapiro is member of the board of directors of Project HOPE (the Finance and the Resource Development & Communications Committees), USA.

Besides this appointment, Miriam Sapiro has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Ilja Schulz was born on October 28, 1967 in Rüsselsheim, Germany. He attended the pilot training at Lufthansa Verkehrsfliegerschule in Bremen, Germany from 1989 until 1991. After starting as first officer in 1991 at Deutsche Lufthansa and Lufthansa Cargo, he became a captain in 2016. Between 2009 and 2012, he served as chief financial officer at the Vereinigung Cockpit trade union. In 2012, he took over as president of Vereinigung Cockpit trade union and held this position until 2018.

Ilja Schulz is not and has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Britta Seeger was born on September 25, 1969 in Bonn, Germany. She studied economics at Berufsakademie Stuttgart. In 1989 she joined Mercedes-Benz AG and held various positions in the retail and marketing sector. Between 2000 and 2016 Britta Seeger held various managerial positions at Daimler AG and Mercedes-Benz including director service operations & service sales, director sales & marketing parts Daimler AG, director Daimler Trucks and Mercedes-Benz Korea and Managing Director of Mercedes-Benz Türk A.S. with overall responsibility for all truck and bus activities for sales & production in Turkey. Since 2017, Britta Seeger is member of the executive board of Daimler AG and is responsible for the Mercedes-Benz Cars Sales division and member of the executive board of Mercedes-Benz AG since 2019.

Alongside her office as member of the Supervisory Board, Britta Seeger is member of the supervisory boards of Daimler Mobility AG, Mercedes-AMG GmbH and in the supervisory bodies of Mercedes-Benz (China) Ltd., Mercedes-Benz South Africa Ltd., Mercedes-Benz Formula E Ltd., Beijing Mercedes-Benz Sales Services Co. Ltd., Lei Shing Hong Auto International Ltd., smart Automobile Co. Ltd and Laureus World Sports Awards Ltd.

Besides these appointments, Britta Seeger has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Birgit Spineux was born on June 5, 1965 in Flensburg, Germany. She studied Business Administration at the University of Applied Sciences Wiesbaden from 1994 until 1999 and holds a degree in International Business Administration. She joined the Lufthansa Group in 1988 as a flight attendant. Since 2005, she is a staff representative purser and since 2017 staff representative Lufthansa cabin on leave of absence.

Birgit Spineux is not and has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Dr. Astrid Stange was born on December 27, 1965 in Witten, Germany. She studied economics at the Ruhr Universität Bochum and obtained a doctorate at the Technical University of Braunschweig (Dr. rer. pol.) in 1993. She started her career as an assistant to the chief financial officer Bücher/Buchclubs DACH/CEE, Bertelsmann Buch AG in 1993. From 1995 to 1998, she served as head of direct marketing services DACH Region, Bertelsmann Buchclub Germany. After moving to The Boston Consulting Group in 1998, she worked there in various positions, including senior partner and managing director. From 2014 until 2017, Astrid Stange held the position of chief officer strategy, human resources, organization and customer management at AXA Germany. Since 2017 Astrid Stange is member of the management committee of AXA SA, where she is the group chief operating officer, and since 2018, she is the chief executive officer of AXA Group Operations SAS.

Alongside her office as member of the supervisory board, Astrid Stange is member of the supervisory board of GIE AXA and chairman of the management board at AXA Group Operations SAS.

Besides these appointments, Astrid Stange has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Olivia Stelz was born on March 20, 1971 in Bad Nauheim, Germany. Olivia Stelz joined Lufthansa Express as a flight attendant in 1996. In 2007, she was appointed as purser. Since 2012, Olivia Stelz is a member of the Company's employee representative council and UFO collective bargaining commission.

Olivia Stelz is not and has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Angela Titzrath was born on April 30, 1966 in Essen, Germany. From 1986 until 1991, she studied economics and roman philology at Ruhr University Bochum, Perugia (Italy) and Coimbra (Portugal). She started her professional career as head of operational and strategic controlling at Mercedes-Benz, Rome. From 1996 until 1999, she served as chief executive officer Mercedes-Benz Credit of Canada, Toronto. Between 1999 and 2016, Angela Titzrath held various managerial positions at different companies including, vice president executive management development, Daimler AG, member of the management board, director of sales, buses division, Daimler AG and member of the executive board, personnel and labor director, Deutsche Post AG. Since 2016, Angela Titzrath is member of the executive board of Hamburger Hafen und Logistik AG and since 2017 its chief executive officer. From 2015 to 2018, Angela Titzrath was member of the supervisory board of Axa Konzern AG.

Alongside her office as member of the Supervisory Board, Angela Titzrath is member of the supervisory board of Evonik Industries AG, Talanx AG and Metrans a.s.

Besides these appointments, Angela Titzrath has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

Klaus Winkler was born on December 29, 1973 in Hamburg, Germany. He joined the Lufthansa Group in 1995 as an automotive mechanic and moved to Lufthansa Technik in 1998 as an aircraft engine mechanic. In 2014, he was elected to the works council. Klaus Winkler is an active member in the trade union and ver.di and at an international level in the European Transport Workers' Federation (*Europäische Transportarbeiter-Föderation*) and the International Transport Workers' Federation (*Internationale Transportarbeiter-Föderation*).

Klaus Winkler is not and has not been a member of any administrative, management or supervisory body of any other company or partnership outside the Lufthansa Group within the last five years.

18.3.3 Supervisory Board Committees

The Supervisory Board is entitled to appoint committees from its members and to draw up rules of procedure (*Geschäftsordnung*) defining their composition, duties and powers. The Articles of Association stipulate that the Supervisory Board must form a mediation committee (*Vermittlungsausschuss*) pursuant to Section 27 para. 3 of the German Co-Determination Act (*Mitbestimmungsgesetz*). Furthermore, the Supervisory Board has established three committees provided for by its rules of procedure: the Steering Committee (*Präsidium*), the Audit Committee (*Prüfungsausschuss*) and the Nomination Committee (*Nominierungsausschuss*).

(a) Mediation Committee

The purpose of the Mediation Committee is to make proposals if, when an appointment to the Executive Board is made or revoked, a majority of at least two-thirds of the votes of the Supervisory Board is not reached. The Mediation Committee consists of the chairman of the Supervisory Board, the deputy chairwoman, one member elected by the majority of the votes cast by the Supervisory Board members representing the shareholders and one member elected by the majority of the votes cast by the Supervisory Board members representing the employees.

As of the date of this Prospectus, the members of the Mediation Committee are:

Name	Position
Dr. Karl-Ludwig Kley	Chairman
Christine Behle	Deputy Chairwoman
Dr. Thomas Enders.....	Member
Ilja Schulz.....	Member

(b) Steering Committee

The Supervisory Board has formed the Steering Committee from among its members made up of an equal numbers of shareholder and employee representatives, consisting of the chairman of the Supervisory Board and the deputy chairwoman as well as two other Supervisory Board members to be elected by the Supervisory Board. The Steering Committee assists in the preparation of the Supervisory Board meetings, makes recommendations to the Supervisory Board on the appointment of Executive Board members, nominating the chairman of the Executive Board, setting the total remuneration for individual Executive Board members including salary and benefits and any reductions thereof, and on determining objectives and deadlines for the ratio of female Executive Board members. The Steering Committee is responsible for all other personnel matters of Executive Board members and authorized representatives (*Prokuristen*) not reserved for the entire Supervisory Board, contracts with members of the Supervisory Board and for granting of loans to members of the Supervisory Board. The Steering Committee is also responsible for approving related party transactions in accordance with section 111a *et seq.* of the German Stock Corporation Act (*Aktiengesetz*). The Steering Committee shall, in principal, determine the emergence of inside information and the publication or delay of ad-hoc notification that has arisen in or becomes known to the Supervisory Board. Furthermore, the Steering Committee approves certain other personnel matters requiring the Supervisory Board approval in accordance with the rules of procedure (*Geschäftsordnung*) for the Executive Board. In the event of a tied vote, the chairman of the Supervisory Board has the casting vote.

As of the date of this Prospectus, the members of the Steering Committee are:

Name	Position
Dr. Karl-Ludwig Kley	Chairman
Christine Behle	Deputy Chairwoman
Dr. Thomas Enders.....	Member
Ilja Schulz.....	Member

(c) *Audit Committee*

The Supervisory Board has formed an Audit Committee from among its members, consisting of six members, made up of equal numbers of shareholder and employee representatives. The Supervisory Board elects the chairman of the Audit Committee, who nominates a deputy chairman. The task of the Audit Committee is to monitor in particular the accounting, the accounting process and the non-financial reporting on corporate social responsibility, the risk management, the internal control system and the compliance management system as well as the external auditing, and to discuss the necessary independence of the external auditors, the appointment of external auditors, the focus of audits and the remuneration agreement, and to make a recommendation to the Supervisory Board, particularly in respect of the auditors to put forward for election at the General Meeting and on approval of the annual and consolidated financial statements. The Audit Committee also discusses the quarterly interim reports with the Executive Board before they are published.

As of the date of this Prospectus, the members of the Audit Committee are:

Name	Position
Harald Krüger.....	Chairman
Alexander Behrens	Member
Jörg Cebulla.....	Member
Dr. Michael Kerkloh.....	Member
Carsten Knobel	Member
Klaus Winkler.....	Member

(d) *Nomination Committee*

The Supervisory Board has formed a Nomination Committee from among the shareholder representatives, consisting of the chairman of the Supervisory Board and two further members. The Nomination Committee's task is to propose suitable candidates to the Supervisory Board to recommend for election at the General Meeting.

As of the date of this Prospectus, the members of the Nomination Committee are:

Name	Position
Dr. Karl-Ludwig Kley	Chairman
Thomas Enders	Member
Harald Krüger.....	Member

18.3.4 Compensation

(a) *Principles of Compensation System*

The chairman of the Supervisory Board receives remuneration of €240,000 and the deputy chairwoman receives remuneration of €120,000 for each fiscal year. The other members of the Supervisory Board receive remuneration of €80,000 for each fiscal year. The chairman of the Audit Committee shall receive an additional €60,000 and other members of the Audit Committee an additional €30,000 for each fiscal year. Chairmen of other committees shall receive an additional €40,000 and other members of other committees shall receive an additional €20,000 for each fiscal year. Remuneration for committee membership will only be paid, if the relevant committee has met at least once in the fiscal year.

In addition, the members of the Supervisory Board shall receive reimbursement of their expenses (in particular travel expenses) and an attendance fee of €500 for each participation in physical meetings. The Company also refunds the premium for group accident insurance and the turnover tax on their remuneration. In addition, any employer's social security contributions arising in accordance with foreign laws relating to the work of the Supervisory Board are paid or reimbursed to the member of the Supervisory Board.

If members of the Supervisory Board withdraw from the Supervisory Board in the course of a fiscal year or from an activity in one of its committees, any remaining remuneration payment shall be made on a *pro rata* basis.

Fixed remuneration and remuneration for committee work are due at the end of each fiscal year, attendance fees are due in principle at the end of each meeting. Supervisory Board remuneration and attendance fees for the Fiscal Year 2020 were paid in January 2021.

(b) Amount of Supervisory Board Remuneration

Expenses for fixed remuneration and work on Supervisory Board committees amounted to €1,887,000 in the Fiscal Year 2020 (Fiscal Year 2019: €2,170,000). The following table shows the amounts paid to the individual Supervisory Board members for the Fiscal Year 2020. Other remuneration, mainly attendance fees, amounted to €23,000 (previous year: €62,000). The Supervisory Board members were also paid €16,000 for work on supervisory boards of Group companies (previous year: €2,000). All Supervisory Board members active in 2020 voluntarily waived 25% of their remuneration for the months of April to September 2020, as well as 25% of their attendance fees for the meeting in March and all further meetings from April to September 2020.

The following table shows the remuneration of the members of the Supervisory Board for 2020.

Name	Fixed remuneration	Remuneration for committee work (in € thousand)	Total Supervisory Board remuneration
Dr. Karl-Ludwig Kley	210	53	263
Christine Behle	105	18	123
Alexander Behrens	70	26	96
Jörg Cebulla.....	70	26	96
Erich Clementi ⁽¹⁾	44	–	44
Dr. Thomas Enders ⁽²⁾	44	11	55
Herbert Hainer ⁽³⁾	26	13	39
Christian Hirsch ⁽⁴⁾	70	–	70
Jürgen Jennerke ⁽⁵⁾	5	–	5
Dr. Michael Kerkloh ⁽⁶⁾	25	8	33
Carsten Knobel.....	70	26	96
Dr. Holger Benjamin Koch..	70	–	70
Martin Koehler ⁽⁷⁾	45	11	56
Harald Krüger ⁽⁸⁾	44	–	44
Martina Merz ⁽⁹⁾	26	–	26
Michael Nilles ⁽¹⁰⁾	26	–	26
Monika Ribar ⁽¹¹⁾	45	17	62
Birgit Rohleder.....	70	–	70
Miriam Sapiro	70	–	70
Ilja Schulz.....	70	17	87
Dr. Astrid Stange ⁽¹²⁾	44	–	44
Olivia Stelz.....	70	–	70
Stephan Sturm ⁽¹³⁾	70	53	123
Angela Titzrath ⁽¹⁴⁾	25	–	25
Christina Weber ⁽¹⁵⁾	74	24	98
Klaus Winkler	70	–	70
Matthias Wissmann ⁽¹⁶⁾	26	–	26
Total	1,584	303	1,887

(1) Since May 5, 2020.

(2) Since May 5, 2020.

(3) Until May 5, 2020.

(4) Until December 31, 2020.

(5) Since December 8, 2020.

(6) Since September 2, 2020.

(7) Until August 31, 2020.

(8) Since May 5, 2020.

(9) Until May 5, 2020.

(10) Until May 5, 2020.

(11) Until August 31, 2020.

- (12) Since May 5, 2020.
- (13) Until May 4, 2021.
- (14) Since September 2, 2020.
- (15) Until December 2, 2020.
- (16) Until May 5, 2020.

(c) *Shareholdings of current Supervisory Board members*

The following table sets forth the respective shareholdings of the current members of the Supervisory Board as of the date of this Prospectus:

Name	Shareholdings
Dr. Karl-Ludwig Kley	8,397
Christine Behle	10
Alexander Behrens	204
Jörg Cebulla.....	1,000
Erich Clementi.....	5,000
Dr. Thomas Enders.....	–
Jürgen Jennerke	–
Dr. Michael Kerkloh.....	–
Carsten Knobel	360
Dr. Holger Benjamin Koch.....	1,876
Harald Krüger.....	–
Birgit Rohleder.....	3,020
Miriam Sapiro	–
Ilja Schulz.....	–
Britta Seeger.....	–
Birgit Spineux	2,005
Dr. Astrid Stange.....	–
Olivia Stelz.....	856
Angela Titzrath.....	–
Klaus Winkler	190
Total.....	22,918

18.4 Certain Information Regarding the Members of the Executive Board and Supervisory Board

During the last five years, no member of the Executive Board or Supervisory Board has been convicted of any fraudulent offense. In addition, no member of either board has been publicly incriminated or sanctioned by statutory or regulatory authorities (including professional associations) or, acting in the capacity of a member of a management or supervisory entity or as founder of an issuer, been associated with any bankruptcies and/or insolvencies, receiverships or liquidations. No member of the Executive Board or Supervisory Board has ever been deemed by a court to be unfit for membership in a management or supervisory entity of a company or to be unfit to exercise management duties for or manage the business of an issuer during the past five years.

There are no conflicts of interest or potential conflicts of interest between the duties of members of the Executive Board and duties of members of the Supervisory Board *vis-à-vis* the Company and their private interests.

No member of the Executive Board or the Supervisory Board has entered into a service agreement with a company of the Lufthansa Group that provides for special benefits, such as severance pay, at the end of the business relationship (other than pensions or compensation in the case of an early termination of the service agreement, which is determined on the basis of the remaining term of the agreement and the contractually agreed compensation). The members of the Executive Board are not bound by restrictive covenants and may therefore engage in competing activities following the end of their office.

There are no family relationships between the members of the Executive Board and those of the Supervisory Board, either among themselves or in relation to the members of the other body.

18.5 General Meeting

18.5.1 Convening of the General Meeting

Pursuant to section 175 para. 1 sentence 2 of the German Stock Corporation Act (*Aktiengesetz*), the General Meeting generally takes place within the first eight months of each fiscal year and, pursuant to the Articles of Association, has to be held at the Company's registered office or in another German city with more than 100,000 residents.

The General Meeting shall be convened by the Executive Board or the Supervisory Board. Notice must be issued in the German Federal Gazette (*Bundesanzeiger*) at least 30 days before the day of the General Meeting. The day of the meeting and the day of the publication of the convocation in the German Federal Gazette (*Bundesanzeiger*) are not taken into account when calculating this 30-day period. This period is to be extended by the days of the registration period. As a rule, shareholders must register for the General Meeting no later than six days prior to the General Meeting. The invitation to the General Meeting may provide for a shorter period to be measured in days.

For further details on virtual General Meetings, see "18.5.5 Virtual General Meeting."

18.5.2 Shareholders' Participation Rights in the General Meeting

Shareholders are entitled to participate in the General Meeting and to exercise their voting rights if they are entered in the Company's share register on the day of the General Meeting and have registered with the Company prior to the General Meeting in due time.

The shareholders can be represented at the General Meeting and voting rights may be exercised by proxy. The granting of the proxy, its revocation and the evidence of authorization to be provided to the Company must be submitted in writing (*Textform*), unless the convening notice provides for a less strict form. Details on the granting of proxy, its revocation and the evidence to be provided to the Company are provided together with the convening notice for the General Meeting. The Executive Board may allow shareholders to cast their votes in writing or by electronic communication without attending the General Meeting (absentee vote) and may determine the scope and the procedure of the exercising of rights in such way. The Executive Board may also provide that shareholders may participate in the General Meeting without being present in person at the place of the General Meeting or being represented, and may exercise all or specific shareholders' rights, in full or in part, by electronic communication (online participation).

Each Share entitles the shareholder who holds it to one vote at the General Meeting. The General Meeting adopts resolutions concerning, for example:

- appointment of members of the Supervisory Board (representatives of the shareholders);
- use of annual profit;
- approval of the actions taken by the members of the Executive Board and of the Supervisory Board and granting of discharge;
- appointment of the auditor;
- measures resulting in the procurement of capital or in the reduction of capital; and
- amendments to the Articles of Association.

18.5.3 Conduct of the General Meeting

The General Meeting shall be chaired by the chairman of the Supervisory Board or, if he is prevented from doing so, by the deputy chairman of the Supervisory Board. The members of the Supervisory Board may under certain circumstances determine a different member of the Supervisory Board or a third party as chairman of the General Meeting.

The chairman of the General Meeting shall steer the General Meeting, determine the order of the items on the agenda and the manner of voting. The chairman of the General Meeting may reasonably limit the time allowed for shareholders' right to ask questions and their right to speak. The chairman of the General

Meeting is authorized to permit complete or partial video or audio broadcasts of the General Meeting in a manner to be detailed by him.

18.5.4 Resolutions of the General Meeting

Resolutions at the General Meeting shall be passed by a simple majority of the votes cast, unless statutory law or the Articles of Association stipulate otherwise. Under German company law, resolutions of fundamental importance require the approval of at least three quarters of the share capital represented at the vote. Resolutions of fundamental importance (*grundlegende Bedeutung*) include those relating to:

- changes in the corporate purpose;
- capital increases;
- amendments to the Articles of Association;
- reductions in capital;
- creation of authorized or conditional capital;
- transformations pursuant to the German Transformation Act (*Umwandlungsgesetz*), including mergers, divisions, transfer of assets (*Vermögensübertragung*) and changes in the legal form;
- sale of all or substantially all of the Company's assets pursuant to Section 179a of the German Stock Corporation Act (*Aktiengesetz*);
- conclusion of enterprise agreements (*Unternehmensverträge*), such as domination and profit and loss transfer agreements; and
- the dissolution of the Company.

18.5.5 Virtual General Meeting

In response to the COVID-19 pandemic and pursuant to the German Act on measures in Corporate, Cooperative, Association, Foundation and Residential Property Law to Combat the Effects of the COVID-19 Pandemic (*Gesetz über Maßnahmen im Gesellschafts-, Genossenschafts-, Vereins-, Stiftungs- und Wohnungseigentumsrecht zur Bekämpfung der Auswirkungen der COVID-19-Pandemie*) as amended by the German Act on the Further Shortening of the Residual Debt Exemption Procedure and on the Adjustment of Pandemic-Related Provisions in Corporate, Cooperative, Association and Foundation Law and in Tenancy and Lease Law (*Gesetz zur weiteren Verkürzung des Restschuldbefreiungsverfahrens und zur Anpassung pandemiebedingter Vorschriften im Gesellschafts-, Genossenschafts-, Vereins- und Stiftungsrecht sowie im Miet- und Pachtrecht*) (the “COVID-19 Act”), the Executive Board may decide, with the approval of the Supervisory Board, to hold General Meetings on or before August 31, 2022, as virtual General Meetings without physical attendance of the shareholders or their representatives, provided that:

- the entire General Meeting is broadcast via audio and video transmission;
- shareholders may exercise their voting rights via electronic communication (absentee voting or electronic participation) and authorizing proxy representatives;
- shareholders have the right to ask questions by means of electronic communication; and
- shareholders who have exercised their voting rights are offered the opportunity to object to a resolution of the general meeting without the requirement to attend in person at the general meeting.

The Executive Board, with the consent of the Supervisory Board, shall decide how to answer questions in its due and free discretion; it may also specify that questions must be submitted by electronic communication no later than one day before the General Meeting. Motions or election proposals by shareholders which are to be made available pursuant to section 126 or section 127 of the German Stock Corporation Act (*Aktiengesetz*) shall be deemed to have been made at the meeting if the shareholder making the motion or submitting the election proposal is duly authorized and registered for the General Meeting. Under the COVID-19 Act, the Executive Board, with the consent of the Supervisory Board, may shorten certain periods in connection with the convocation of, registration and providing evidence of shareholding for General Meetings held on or

before August 31, 2022. In particular, the General Meeting may be convened as late as on the 21st day prior to the day of the General Meeting.

18.6 Corporate Governance

The German Corporate Governance Code as resolved by the Commission (*Regierungskommission Deutscher Corporate Governance Kodex*) on December 16, 2019 (entered into force upon publication in the Official Gazette (*Bundesgesetzblatt*) on March 20, 2020 (the “Code”), provides recommendations (“should provisions”) and suggestions (“can provisions”) for the management and supervision of German companies listed on a stock exchange. The Code incorporates nationally and internationally recognized standards of good and responsible corporate governance. The purpose of the Code is to make the German system of corporate governance and supervision transparent for investors. The Code includes recommendations and suggestions for management and supervision with regard to shareholders’, general meetings, management and supervisory boards, transparency, accounting and auditing.

There is no obligation to comply with the recommendations or suggestions of the Code. However, the German Stock Corporation Act (*Aktiengesetz*) requires that the executive board and supervisory board of a German listed company declares on a yearly basis, either that the recommendations have been or will be applied, or which recommendations have not been or will not be applied and explain why the Executive Board and the supervisory board do not/will not apply the recommendations that have not been or will not be applied. This declaration is to be made permanently accessible to shareholders. However, deviations from the suggestions contained in the Code need not be disclosed. The declaration of compliance must, however, be publicly available on the Company’s website at all times.

The Executive Board and Supervisory Board issued the last declaration of conformity regarding the Code in accordance with section 161 of the German Stock Corporation Act (*Aktiengesetz*) on September 21, 2020, as follows:

“In accordance with Section 161 of the German Stock Corporation Act (Aktiengesetz), the Executive Board and the Supervisory Board of Deutsche Lufthansa AG declare that since the last declaration of compliance, the recommendations of the German Corporate Governance Code as amended have been complied with the following exceptions and will continue to be complied with in future with the following exceptions:

According to Recommendation C.5 of the Code, Supervisory Board members being an Executive Board member of a listed company shall not hold more than two Supervisory Board mandates in non-group listed companies or similar bodies. Mrs. Titzrath, who has been appointed by the Wirtschaftsstabilisierungsfonds according to the Framework agreement between Deutsche Lufthansa AG and the Wirtschaftsstabilisierungsfonds, and was appointed to the Supervisory Board by court order as of 2 September 2020, currently holds a total of three such Supervisory Board mandates. Mrs. Titzrath has declared to Deutsche Lufthansa AG that she has sufficient time to perform her Supervisory Board duties and that she can perform these duties with due regularity and care.

In addition, restrictions on the implementation of the recommendations on the remuneration of the Executive Board in Section G of the Code result from the significant restrictions on Board remuneration within the Framework Agreement between Deutsche Lufthansa AG and the Wirtschaftsstabilisierungsfonds. In particular, the members of the Executive Board may not receive bonuses or other variable or comparable compensation components during the agreed stabilization measures. Subject to the contractual restrictions on the compensation of the Executive Board, the Supervisory Board will review its appropriateness at regular intervals in view of its competence under stock corporation law.”

As of the date of this Prospectus, the Company complies, and following the Offering, intends to comply, with the recommendations of the Code, except as declared in the declaration of conformity regarding the Code issued on September 21, 2020.

19. CERTAIN RELATIONSHIPS AND RELATED-PARTY TRANSACTIONS

19.1 Overview

In accordance with IAS 24 (Related Party Disclosures), transactions with individuals or entities that control or are controlled by the Lufthansa Group must be disclosed unless they were already included as consolidated entities in the Audited Consolidated Financial Statements. In addition, there is an obligation to disclose transactions with associates and persons who have significant influence over the financial and operating policies of the Lufthansa Group, including close family members and intermediate entities. Significant influence can mean a shareholding in the Company of 20% or more, a position on the Executive Board or the Supervisory Board or a key management position.

Set forth below are transactions with related parties in the Fiscal Year 2020, the Fiscal Year 2019 and the Fiscal Year 2018 as well as for the Fiscal Year 2021 up to the date of this Prospectus. Business relationships between the Company and other companies of the Lufthansa Group are not included.

19.2 Transactions with Shareholders, Joint Ventures, Associates and Their Affiliates

The Lufthansa Group had business transactions with related parties in the Fiscal Years 2020, 2019 and 2018 and in the Fiscal Year 2021 up to and including the date of this Prospectus. All transactions were performed substantially on the same terms as transactions of a similar nature with third party counterparts. All transactions with related parties were thus, in the Company's view, carried out in accordance with the arm's length principle. Further information, including quantitative amounts, of related party transactions are contained in the notes to the Company's Audited Consolidated Financial Statements as of and for the Fiscal Years 2020, 2019 and 2018 and the notes to the Company's Unaudited Condensed Consolidated Interim Financial Statements as of and for the 6M 2021, each of which are included in section "21 Financial Information" of this Prospectus starting on page F-1.

19.2.1 Relationships between the Lufthansa Group and its Shareholders

According to the Company's shareholder's register (*Aktienregister*) on September 17, 2021, the WSF held of 15.94% in the share capital of the Company. The WSF can exercise significant influence and represents a related party of the Company. Additionally, the German Stabilization Package entails significant transactions with the WSF.

19.2.2 Relationships with Joint Ventures, Associates and Their Affiliates

The Lufthansa Group and certain non-consolidated subsidiaries have concluded numerous billing agreements, partly governing the joint use of services. In these cases, the administrative services provided are charged as cost allocations.

The Lufthansa Group's cash management is centralized, and, in this respect, the Lufthansa Group performs a 'banking function' vis-à-vis certain non-consolidated companies. Non-consolidated companies included in the Lufthansa Group's cash management invest their available cash with the Lufthansa Group or borrow funds from the Lufthansa Group and carry out their derivative hedging transactions with the Lufthansa Group. All transactions take place at market conditions. Due to geographical proximity in many cases, a large number of subletting contracts exists between the Lufthansa Group and related parties. In these cases, the Lufthansa Group usually charges the rental costs and incidental expenses incurred to the companies in question on a *pro rata* basis.

19.3 Relationships with Members of the Executive Board and Supervisory Board

The Company defines related persons as the members of the Executive Board the Supervisory Board and employees in key management positions in the Company as well as their close family members. The Company compensates the members of the Executive Board and the Supervisory Board for their services (see "18.2.3 Compensation" and "18.3.4 Compensation").

20. TAXATION IN GERMANY

The tax laws of any jurisdiction with authority to impose taxes on the Company's shareholders and the tax laws of the Company's country of incorporation (*i.e.*, Germany) may have an impact on the income received from the shares.

The following section presents a number of key German taxation principles which generally are or can be relevant to the acquisition, holding or transfer of shares or subscription rights both by a shareholder (an individual, a partnership or a corporation) that has a tax domicile in Germany (that is, whose place of residence, habitual abode, registered office or place of management is in Germany) and by a shareholder without a tax domicile in Germany. The information is not comprehensive or exhaustive and does not constitute a definitive explanation of all possible aspects of taxation that could be relevant for shareholders. The information is based on the tax law in force in Germany as of the date of this Prospectus (and its interpretation by administrative directives and courts) as well as typical provisions of double taxation treaties that Germany has concluded with other countries. Tax law can change — sometimes retrospectively. Moreover, it cannot be ruled out that the German tax authorities or courts may consider an alternative assessment to be correct that differs from the one described in this section.

This section cannot serve as a substitute for tailored tax advice to individual shareholders. Shareholders are therefore advised to consult their tax advisors regarding the tax implications of the acquisition, holding or transfer of shares or subscription rights and regarding the procedures to be followed to achieve a possible reimbursement of German withholding tax (*Kapitalertragsteuer*). Only such advisors are in a position to take the specific tax-relevant circumstances of individual shareholders into due account.

Shareholders are in particular subject to taxation in connection with the holding of shares (see below “20.1 Taxation of Dividends”), the sale or disposal of shares or subscription rights (see “20.2 Taxation of Capital Gains”) and the gratuitous transfer of shares or subscription rights (see “20.5 Inheritance and Gift Tax”).

20.1 Taxation of Dividends

20.1.1 No Taxation in Case of Payments from a Tax-Recognized Contribution Account

In the future, the Company may pay dividends out of a tax-recognized contribution account (*steuerliches Einlagekonto*). To the extent the Company pays dividends from such tax-recognized contribution account, such dividends are not subject to withholding tax, personal income tax or corporate income tax, as the case may be (including the solidarity surcharge (*Solidaritätszuschlag*) and church tax (*Kirchensteuer*), if applicable). Any dividends paid out of a tax-recognized contribution account would, however, lower the acquisition costs of the shares, which may result in a higher amount of taxable capital gains upon the shareholder's sale of the shares. Special rules apply to the extent that dividends from the tax-recognized contribution account exceed the then lowered acquisition costs of the shares (the details are outlined below).

20.1.2 Withholding Tax

Dividends distributed by the Company that are not paid out of the tax-recognized contribution account (*steuerliches Einlagekonto*) are generally subject to a withholding tax at a rate of 25% plus a solidarity surcharge of 5.5% thereon (*i.e.*, a total of 26.375%) and church tax, if applicable. The basis for the withholding tax is the dividend resolved for distribution by the general meeting.

If the shares are admitted to be held in a collective safe custody (*Sammelverwahrung*) with a German central securities depository (*Wertpapiersammelbank*) pursuant to Section 5 of the German Act on Securities Accounts (*Depotgesetz*) and are deposited with such central securities depository for collective safe custody in Germany, the Company is generally not responsible for withholding the tax at the source. Instead, the tax is withheld for the account of the shareholder and remitted to the competent tax office by one of the following entities (each of the following entities a “**Dividend Paying Agent**”): the German bank or German financial services institution (*deutsches Kredit- oder Finanzdienstleistungsinstitut*) or German securities institution (*deutsches Wertpapierinstitut*) (including German branches of foreign banks or financial service institutions) which holds the shares in custody or that administers the shares and that disburses or credits the dividends to the shareholder or that disburses the dividends to a foreign entity, or by the central securities

depository (*Wertpapiersammelbank*) if it disburses the dividends to a foreign entity. If the shares are not held in a collective safe custody with a German central securities depository (*Wertpapiersammelbank*) or if the shares are held in a collective safe custody (*girosammelverwahrt*) by the central securities depository (*Wertpapiersammelbank*) and treated as shares being held separately (*abgesetzte Bestände*), then the Company is responsible for withholding the tax at source and the Company consequently has to withhold and remit the tax to the competent tax office.

The withholding tax must generally be deducted regardless of whether and to what extent the dividend is exempt from taxation at the level of the shareholder, whether the shareholder must report the dividend in a (corporate) income tax return and whether the shareholder is a person residing in Germany or in a foreign country.

In the case of dividends distributed to a company within the meaning of Art. 2 of the Council Directive No. 2011/96/EU of 30 November 2011 (the “**EU Parent Subsidiary Directive**”) domiciled in another Member State of the European Union, an exemption from or a refund of withholding tax may be granted by the Federal Central Tax Office (*Bundeszentralamt für Steuern*) upon request of the dividend’s creditor if further prerequisites are satisfied. This also applies to dividends distributed to a permanent establishment located in another Member State of the European Union of such a parent company or of a parent company tax resident in Germany if the participation in the Company is part of the respective permanent establishment’s business assets. The key prerequisite for the application of the EU Parent Subsidiary Directive is that the shareholder has held a direct participation in the share capital of the Company of at least 10% for at least twelve months. If there is a holding of at least 10% of the Company’s registered share capital and the shares held in collective safe custody by Clearstream are treated as shares that are being held separately (*abgesetzte Bestände*), the German tax authorities will not object if the main paying agent (*Hauptzahlstelle*) of the Company disburses dividends without deducting withholding tax, assuming a valid exemption certificate (*Freistellungsbescheinigung*) and proof that the relevant shares have been held separately are presented.

The withholding tax rate on dividends to other foreign tax resident shareholders is generally reduced in accordance with any applicable double taxation treaty if Germany has concluded such double taxation treaty with the country of residence of the shareholder and if the shareholder does not hold his shares either as part of the assets of a permanent establishment or a fixed place of business in Germany or as business assets for which a permanent representative has been appointed in Germany. The reduction of the withholding tax, if applicable, is generally granted in such a manner that the difference between the total amount withheld, including the solidarity surcharge, and the tax liability determined on the basis of the tax rate set forth in the applicable double taxation treaty (generally 15%) is refunded upon request and subject to certain requirements by the Federal Central Tax Office (*Bundeszentralamt für Steuern*). Depending on the applicable double taxation treaty, a reduced withholding tax rate may be applicable in the tax withholding process, if the shareholder has applied for an exemption certificate (*Freistellungsbescheinigung*) from the Federal Central Tax Office (*Bundeszentralamt für Steuern*).

In the case of dividends received by corporations which are not tax resident in Germany, generally two-fifths of the withholding tax deducted and remitted can be refunded upon request and subject to certain requirements but without the need to fulfill all prerequisites required for such refund under the EU Parent Subsidiary Directive or under a double taxation treaty.

In order to receive a refund pursuant to a double taxation treaty or the aforementioned option for foreign corporations, the shareholder has to submit a completed form for refund (available at the Federal Central Tax Office (*Bundeszentralamt für Steuern*) (<http://www.bzst.de>) as well as at the German embassies and consulates) together with a withholding tax certificate (*Kapitalertragsteuerbescheinigung*) issued by the institution that withheld the tax.

The exemption from withholding tax in accordance with the EU Parent Subsidiary Directive and the aforementioned options for a refund of the withholding tax depend on whether certain additional prerequisites (in particular applicable substance requirements) are fulfilled.

The aforementioned reductions of (or exemptions from) withholding tax are restricted if (i) the applicable double taxation treaty provides for a tax reduction resulting in an applicable tax rate of less than 15%, and

(ii) the shareholder is not a corporation that directly holds at least 10% in the equity/capital of the Company and is subject to tax on its income and profits in its state of residence without being exempt.

In this case as well as in the case of crediting withholding tax regarding dividends from shares held as business assets and according to the German tax authorities only in certain cases as private assets, the reduction of (or exemption from) withholding tax is subject to the following three cumulative prerequisites (together the “**Minimum Risk Test**”): (i) the shareholder must qualify as beneficial owner of the shares in the Company for a minimum holding period of 45 consecutive days occurring within a period of 45 days prior and 45 days after the due date of the dividends, (ii) the shareholder has to bear at least 70% of the change in value risk related to the shares in the Company during the minimum holding period without being directly or indirectly hedged, and (iii) the shareholder must not be required to fully or largely compensate directly or indirectly the dividends to third parties. Absent the fulfillment of all three prerequisites,

- in the case of a shareholder tax resident in Germany, three fifths of the withholding tax imposed on the dividends must not be credited against the shareholder’s (corporate) income tax liability, but may, upon application, be deducted from the shareholder’s tax base for the relevant assessment period. A shareholder that has received gross dividends without any deduction of withholding tax due to a tax exemption without qualifying for a full tax credit or that has already received a refund has to notify the competent local tax office accordingly and has to make a payment in the amount of the omitted withholding tax deduction (*i.e.*, 15% of the relevant dividends). The special rules on the restriction of withholding tax credit do not apply to a shareholder whose overall dividend earnings within an assessment period do not exceed €20,000 or that has been the beneficial owner of the shares in the Company for at least one uninterrupted year upon receipt of the dividends.
- in the case of a shareholder without a tax residence in Germany who has applied for a full or partial refund of the withholding tax pursuant to a double taxation treaty, no refund is available. However, this restriction does not apply to a shareholder (i) that directly holds at least 10% of the shares in the Company and that is subject to (corporate) income tax in the country of its tax residence without any exemptions, or (ii) that has been the beneficial owner of the shares in the Company for at least one uninterrupted year upon receipt of the dividends, or (iii) if the applicable tax rate pursuant to the applicable double taxation treaty is at least 15%.

A Dividend Paying Agent will not deduct withholding tax if the shareholder provides such Dividend Paying Agent with an application for exemption (*Freistellungsauftrag*) to the extent such shareholder’s capital income does not exceed the annual savers’ allowance (*Sparerpauschbetrag*) of €801.00 (€1,602.00 for married couples and registered partners filing jointly). Furthermore, no withholding tax will be levied if the shareholder provides the Dividend Paying Agent with a non-assessment certificate (*Nichtveranlagungsbescheinigung*) to be applied for with the competent tax office.

The Dividend Paying Agent, which keeps or administrates the shares and pays or credits the capital income, is required to create pots for offsetting losses (*Verlustverrechnungstöpfe*) to allow for negative capital income to be set off against current and future positive capital income. A set-off of negative capital income at one Dividend Paying Agent against positive capital income at another Dividend Paying Agent is only possible in the course of the income tax assessment at the level of the respective shareholder. In such case, the relevant shareholder has to apply for a certificate confirming the amount of losses not offset with the Dividend Paying Agent where the pot for offsetting losses exists. The application is irrevocable and must reach the Dividend Paying Agent until December 15 of the respective year, as otherwise the losses will be carried forward by the respective Dividend Paying Agent to the following year.

20.1.3 Taxation of Dividends Derived by German Tax Resident Shareholders

Taxation of dividend income for German tax resident shareholders who hold their shares as private assets. Dividends that are derived by a German tax resident shareholder who holds the shares as private assets constitute income from capital investment, which is subject to a special, flat income tax rate of 25% plus a solidarity surcharge of 5.5% thereon (*i.e.*, a total of 26.375%, plus church tax, if applicable). The income tax liability for the dividends is generally satisfied through the deduction of the withholding tax (*i.e.*, flat tax regime (*Abgeltungsteuer*)). Except for an annual lump-sum allowance for savers (*Sparer-Pauschbetrag*) in the amount of €801 (€1,602 for married couples and registered partners filing jointly) income-related

expenses (*Werbungskosten*) may not be deducted from income from capital investment. The shareholder may request his/her total income from capital investments (including the dividends) to be subject to taxation at the individual, progressive income tax rate together with his other taxable income rather than the flat tax rate for capital investments, if this results in his tax liability being lower. The tax base is the gross income less the lump-sum allowance for savers in the amount of €801 (€1,602 for married couples and registered partners filing jointly) with no deduction for income-related expenses to generate the income. In such a case, the withholding tax can either be credited against the income tax liability of the shareholder or refunded in the amount of the excess, subject to the rules on the restriction of withholding tax credit as described below. Dividend income can only be offset by losses from capital income, except for losses generated by the disposal of shares.

Exceptions from the final flat tax regime apply upon application to shareholders that hold at least 25% of the shares in the Company and shareholders that hold at least 1% of the shares in the Company and through professional work for the Company are able to exercise significant entrepreneurial influence on the business activities of the Company. In that case, the dividends are taxed under the partial income taxation method (see *Taxation of dividend income for German tax resident shareholders who hold their shares as business assets*).

If applicable, church tax generally has to be withheld based on an automatic data access procedure, unless the shareholder has filed a blocking notice (*Sperrvermerk*) with the Federal Central Tax Office. If church tax is withheld and remitted to the tax authority as part of the withholding tax deduction, the church tax on the dividends is also deemed to be discharged when it is deducted. The church tax withheld cannot be deducted in the tax assessment as a special expense (*Sonderausgabe*). 26.375% of the church tax withheld on the dividends is, however, deducted from the withholding tax (including the solidarity surcharge) withheld. Where church tax is not levied by way of withholding, the shareholder owing church tax is required to declare his/her dividends in his/her income tax return and then the church tax is imposed by means of an income tax assessment.

Contrary to the above, dividends that are funded from the Company's tax-recognized contribution account (*steuerliches Einlagekonto*) and are paid to a German tax resident shareholder who holds the shares as private assets, do not form part of the shareholder's taxable income, but reduce the acquisition costs for such shares. If the dividend payment funded from the Company's tax-recognized contribution account (*steuerliches Einlagekonto*) exceeds the shareholder's acquisition costs, the German tax authorities take the view that negative acquisition costs will arise which may result in a higher capital gain in case of a disposal of the shares. This will not apply if (i) the shareholder or, in the event of a gratuitous transfer, its legal predecessor, or, if the shares have been gratuitously transferred several times in succession, one of his legal predecessors at any point during the five years preceding the disposal directly or indirectly held at least 1% of the share capital of the Company (a "**Qualified Participation**"), and (ii) the dividend payment funded from the Company's tax-recognized contribution account (*steuerliches Einlagekonto*) exceeds the acquisition costs of the shares. In case of a Qualified Participation, a dividend payment funded from the Company's tax-recognized contribution account (*steuerliches Einlagekonto*) is considered a sale of the shares and is taxable as a capital gain if and to the extent the dividend payment funded from the Company's tax-recognized contribution account (*steuerliches Einlagekonto*) exceeds the acquisition costs of the shares. In this case, the taxation corresponds to the taxation of capital gains of shareholders maintaining a Qualified Participation (see "**20.2 Taxation of Capital Gains**").

Taxation of dividend income for German tax resident shareholders who hold their shares as business assets. The flat tax does not apply to dividends paid on shares held by a German tax resident shareholder as business assets. The taxation depends on whether the shareholder is a corporation, an individual entrepreneur or a partnership (co-entrepreneurship). Subject to the Minimum Risk Test, the withholding tax (including solidarity surcharge and church tax, if applicable) withheld and remitted can either be credited against the income tax or corporate income tax liability and the solidarity surcharge (and church tax, if applicable) of the shareholder or refunded in the amount of the excess.

Dividends that are funded from the Company's tax-recognized contribution account (*steuerliches Einlagekonto*) and paid to a German tax resident shareholder who holds the shares as business assets are generally fully tax-exempt in the hands of such shareholder. At the same time such dividend payments lead to a corresponding reduction of the acquisition costs/book value for the relevant shares. To the extent the

dividend payments funded from the Company's tax-recognized contribution account (*steuerliches Einlagekonto*) exceed the acquisition costs/book value of the shares, a taxable capital gain should occur. The taxation of such gain corresponds to the taxation of shareholders whose shares are held as business assets (see "20.2 Taxation of Capital Gains"). As regards the application of the 95% exemption in case of a corporation, this is, however, not undisputed.

Corporations. If the German tax resident shareholder is a corporation, effectively 95% of the dividends, subject to certain exceptions for enterprises in the financial and insurance sectors (see "20.3 Special Rules for Credit Institutions, Financial Services Institutions, Financial Enterprises as well as Life Insurance and Health Insurance Companies and Pension Funds") are generally exempt from corporate income tax and the solidarity surcharge. No minimum holding period needs to be observed. 5% of the dividends are deemed non-deductible business expenses, and are therefore subject to corporate income tax (plus solidarity surcharge) at a tax rate totaling 15.825%. However, if the corporation directly holds less than 10% of the share capital of the Company at the beginning of the calendar year, the full amount of any dividends received is taken into account when determining the corporation's taxable income for corporate income tax purposes. The acquisition of a shareholding of at least 10% is deemed to have occurred at the beginning of the calendar year. Aside from this, business expenses actually incurred and directly connected to the dividends may be deducted. The dividends are subject to trade tax in the full amount (after deduction of the business expenses economically connected to them) unless the corporation held a participation of at least 15% of the share capital of the Company at the beginning of the relevant tax assessment period (trade tax participation exemption). In this latter case, the dividends are not subject to trade tax. However, trade tax is levied on the amount deemed to be non-deductible business expenses (*i.e.*, in the amount of 5% of the dividend). Depending on the assessment rate set by the municipality in which the shareholder maintains its operations or permanent establishments the trade tax is normally imposed at an effective rate of approximately 7% to 18.55% of the trade income (*Gewerbeertrag*).

Individual entrepreneurs. If the German tax resident shareholder is an individual entrepreneur who holds the shares as business assets, only 60% of the dividends are subject to the progressive personal income tax plus solidarity surcharge at a tax rate totaling up to approximately 47.5% and, if applicable, church tax (partial income taxation method; *Teileinkünfteverfahren*). However, for church tax, if applicable, the partial income taxation method does not apply. Only 60% of the business expenses economically connected to the dividends are deductible for tax purposes. If the shares are held as business assets in a German permanent establishment of a commercial business of the shareholder, the dividend income (after deduction of the business expenses economically connected to it) are, in addition to personal income tax, also subject to trade tax in the full amount unless the shareholder held a participation of at least 15% of the share capital of the Company at the beginning of the relevant tax assessment period (trade tax participation exemption). In this latter case, the net amount of the dividends, (*i.e.*, after deduction of the business expenses directly connected to them) is exempt from trade tax. In general, the trade tax may be credited, completely or partially, against the personal income tax of the shareholder in accordance with a lump-sum tax credit method, depending on the assessment rate set by the local municipality and the personal tax situation.

Partnerships. If the German tax resident shareholder is a partnership being engaged or deemed to be engaged in a business (co-entrepreneurship), the personal income tax or corporate income tax is not charged at the level of the partnership, but at the level of the respective partner. The taxation of each partner depends on whether the partner is a corporation or an individual. If the partner is a corporation, dividends forming part of the partner's profit share are taxed in accordance with the principles applicable to corporations, *i.e.*, effectively 95% of the dividends are tax exempt, but dividends from a direct shareholding of less than 10% of the Company's share capital are fully subject to taxation (see above under "Corporations"). Indirect shareholdings via a partnership are attributed to the partners on a pro-rata basis and are deemed to be direct shareholdings. If the partner is an individual, the taxation is based on the principles applicable to individual entrepreneurs, *i.e.*, the partial income taxation method applies (see "Individual entrepreneurs") to the dividends included in the individual partner's profit share. Upon application and subject to further prerequisites, an individual being partner can have his/her personal income tax rate lowered for earnings (except for certain capital gain income) not withdrawn from the partnership.

In addition, the dividends are subject to trade tax at the level of the partnership if the shares are held as business assets of a permanent establishment of a commercial business of the partnership in Germany, and

this generally in the full amount. If the partner in the partnership is an individual, the trade tax on his/her profit share which is paid by the partnership may generally be credited, completely or partially, against his/her personal income tax, depending on the assessment rate set by the local municipality and the personal tax circumstances.

If the partnership held a participation of at least 15% of the share capital of the Company at the beginning of the relevant tax assessment period, the dividends, after deduction of the business expenses economically connected thereto, should generally not be subject to trade tax. However, in this case, trade tax should be levied on 5% of the dividends to the extent they are attributable to the profit share of such corporate partners to whom at least 10% of the shares in the Company are attributable on a look-through basis, since such portion of the dividends should be deemed to be non-deductible business expenses. The remaining portion of the dividend income attributable to other than such specific corporate partners (which includes individual partners and should, under a literal reading of the law, also include corporate partners to whom, on a look-through basis, only participations in the share capital of the Company of less than 10% are attributable) should (after the deduction of business expenses economically related thereto) not be subject to trade tax. Due to a lack of case law and administrative guidance, the application of the rules for the taxation of portfolio participations is, however, unclear. Consequently, shareholders are strongly recommended to consult their own tax advisors.

20.1.4 Shareholders Domiciled in Foreign Countries.

Shareholders (individuals and corporations) who are not tax resident in Germany and hold their shares as business assets of a permanent establishment or a fixed place of business in Germany, or as business assets for which a permanent representative has been appointed in Germany, are subject to the taxation in Germany in respect of their dividend income. The situation described above for shareholders tax resident in Germany who hold their shares as business assets applies accordingly (see section “*20.1.3 Taxation of Dividends Derived by German Tax Resident Shareholders*”). The withholding tax deducted and remitted to the tax authorities (including solidarity surcharge) is – subject to the Minimum Risk Test – either credited against the personal income tax or corporate income tax liability or refunded in the amount of an excess of such.

In all other situations, the German tax liability is satisfied for the dividends with the deduction of withholding tax. A withholding tax exemption or refund is only granted in the situations described above under section “*20.1.2 Withholding Tax.*”

Dividend payments that are funded from the Company’s tax-recognized contribution account (*steuerliches Einlagekonto*) are generally not taxable in Germany.

20.2 Taxation of Capital Gains

20.2.1 Taxation of Capital Gains for German Tax Resident Shareholders

Taxation of capital gains for German tax resident shareholders who hold their shares or subscription rights as private assets. Capital gains from the sale or disposal of shares acquired by a German tax resident shareholder after December 31, 2008, that are held as private assets are generally subject to personal income tax in Germany – irrespective of any holding period – as income from capital investment at a special, flat tax rate for income from capital investment of 25% (plus a solidarity surcharge of 5.5%, i.e., a total of 26.375%, and church tax, if applicable). This also applies to gains from the sale or disposal of subscription rights granted for such shares. By contrast, gains from the sale of shares that were acquired by the shareholder prior to January 1, 2009, and gains from the sale of subscription rights that were granted for such shares are not taxable. If the shareholder acquired shares before January 1, 2009, as well as on or after January 1, 2009, and if these shares are kept in the same custodial account, it will be deemed that those shares that were acquired first are sold first.

The amount of the taxable capital gain from the sale is the difference between (a) the proceeds from the sale or disposal and (b) the cost of acquisition of the shares or subscription rights and the expenses directly related to the sale. The acquisition costs of subscription rights granted by the Company are deemed to be €0. Under certain conditions, payments from the Company’s tax-recognized contribution account (*steuerliches Einlagekonto*) reduce the acquisition costs of the shares and may lead to negative acquisition costs, which

can increase capital gains if such shares are held as private assets and do not qualify as a Qualified Participation.

The only deduction available from the overall income from capital investment is the annual lump-sum allowance for savers (*Sparer-Pauschbetrag*) in the amount of €801 (€1,602 for married couples and registered partners filing jointly). Income-related expenses (*Werbungskosten*) may not be deducted from capital gains. Losses from the sale or disposal of shares may only be offset against capital gains arising from the sale of the shares in stock corporations (the Company or other stock corporations) during the same assessment period or in subsequent assessment periods. Losses from the sale of subscription rights may only be offset against positive income from capital investment but insofar without restrictions (*i.e.*, including such from the sale or disposal of shares in stock corporations). In case of a derecognition or transfer of worthless shares or other capital assets, the utilization of losses is further restricted and can only be offset for up to €20,000.00 per calendar year.

According to the German Federal Ministry of Finance (*Bundesministerium der Finanzen*), the exercise of subscription rights is not equivalent to a sale. Shares acquired by exercising subscription rights are considered to be acquired at the price of subscription and at the time of the exercise.

If shares or subscription rights are held in custody with or administered by a German bank, German financial services institution, German securities institution (including German branches of foreign institutions) or such an institution sells or disposes the shares or subscription rights and disburses or credits the proceeds from the sale or disposal (a “**German Disbursing Agent**”), such a German Disbursing Agent withholds withholding tax on the capital at a rate of 26.375% (including solidarity surcharge plus church tax, if applicable) and remits it to the tax authority (final flat tax regime, *Abgeltungsteuer*). If the shares are only held in custody or administered by the respective German Disbursing Agent continuously after acquisition, the amount of taxes withheld is generally based on the difference between the proceeds from the sale, after deducting expenses directly related to the sale, and the amount paid to acquire such shares. The withholding tax rate of 25% plus the current 5.5% solidarity surcharge thereon and any church tax, if applicable, will, however, be applied to 30% of the gross sales proceeds if the shares were not administered by the same custodian bank since acquisition and the original cost of the shares cannot be verified or such verification is not admissible. In this case, the shareholder is entitled to, and in case the actual gain is higher than 30% of the gross proceeds, required to, verify the original costs of the shares in his/her annual tax return.

If applicable, church tax generally has to be withheld based on an automatic data access procedure, unless the shareholder has filed a blocking notice (*Sperrvermerk*) with the Federal Central Tax Office. If church tax is withheld and remitted to the tax authority as part of the withholding tax deduction, then the church tax on the capital gain is also deemed to be discharged when it is deducted. The church tax withheld cannot be deducted in the tax assessment as a special expense (*Sonderausgabe*). 26.375% of the church tax withheld on the capital gain is, however, deducted from the withholding tax (including the solidarity surcharge) withheld. Where church tax is not levied by way of withholding, it is determined by means of an income tax assessment.

If the withholding tax on the capital gains or, if applicable, the church tax on the capital gains has not been withheld by a German Disbursing Agent, the German tax resident shareholder is required to report the capital gains in his/her income tax return. The income tax, the solidarity surcharge and, if applicable, the church tax on the capital gains are then collected by way of assessment.

The shareholder may request his/her total income from capital investments to be subject to taxation at the individual, progressive income tax rate together with his/her other taxable income rather than the flat tax rate for capital investments, if this results in his/her tax liability being lower. The tax base is the gross income less the lump-sum allowance for savers in the amount of €801 (€1,602 for married couples and registered partners filing jointly) with no deduction for income-related expenses to generate the income. In such a case, withholding tax, solidarity surcharge and church tax, if applicable, can either be credited against the individual, progressive income tax or refunded in the amount of the excess.

A gain from the sale or disposal of shares or subscription rights is not subject to the final flat tax regime, but rather the individual, progressive income tax if the German tax resident shareholder holds a Qualified Participation. In this case, the partial income taxation method (*Teileinkünfteverfahren*) applies to gains from

the sale or disposal of shares, *i.e.*, only 60% of the proceeds from the sale or disposal of shares are subject to taxation and only 60% of a loss from the sale or disposal of shares and any expenses economically related to the sale or disposal of the shares are tax deductible. In case of a Qualified Participation, the partial income taxation method should also apply to capital gains or losses associated with subscription rights. Unlike under the flat tax regime, the acquisition costs of subscription rights are calculated as a fraction of the original acquisition costs of the underlying shares which is split off from the shares and attributed to the subscription rights (aggregate value method). Upon exercise of a subscription right, its acquisition costs increase the acquisition costs of the newly acquired shares. Withholding tax (including solidarity surcharge and church tax, if applicable) is also deducted by a German Disbursing Agent in the case of a Qualified Participation, but this does not have the effect of a settlement of the shareholder's tax liability. The shareholder is therefore required to report the capital gain in his/her income tax return. Upon the shareholder's assessment, the withholding tax withheld and remitted (including solidarity surcharge and church tax, if applicable) can either be credited against the personal income tax liability or refunded in the amount of the excess. The exercise of subscription rights should not be considered equivalent to a sale in the case of a Qualified Participation.

Withholding tax will not be withheld by a German Disbursing Agent if the shareholder provides such German Disbursing Agent with an application for exemption (*Freistellungsauftrag*), to the extent such shareholder's capital income does not exceed the annual savers' allowance (*Sparerpauschbetrag*) of €801.00 (€1,602.00 for jointly filing individuals). Furthermore, no withholding tax will be levied if the shareholder provides the German Disbursing Agent with a non-assessment certificate (*Nichtveranlagungsbescheinigung*) to be applied for with the competent tax office.

Taxation of capital gains for German tax resident shareholders who hold their shares as business assets. Capital gains from the sale or disposal of shares or subscription rights held by German tax resident shareholders as business assets are not subject to the flat tax regime. The taxation of capital gains depends on whether the shareholder is a corporation, an individual or a partnership (co-entrepreneurship). Capital gains derived by enterprises in the financial and insurance sectors or pension funds are subject to the special rules described below (see section "20.3 Special Rules for Credit Institutions, Financial Services Institutions, Financial Enterprises as well as Life Insurance and Health Insurance Companies and Pension Funds"). Dividend payments that are funded from the Company's tax-recognized contribution account (*steuerliches Einlagekonto*) reduce the original acquisition costs/book value. This may give rise to a higher taxable capital gain in case of a sale or disposal of shares. If the dividend payments funded from the Company's tax-recognized contribution account exceed the original acquisition costs/book value for tax purposes, a taxable capital gain may arise.

Corporations. If the German tax resident shareholder is a corporation, effectively 95% of the gains from the sale or disposal of shares, irrespective of the amount of the participation and irrespective of any holding period, are generally exempt from corporate income tax (including solidarity surcharge) and trade tax. 5% of the gains are deemed to be non-deductible business expenses and, therefore, are subject to corporate income tax (plus solidarity surcharge) at a tax rate totaling 15.825% and trade tax (depending on the trade assessment rate set by the local municipality, generally between 7 and 18.55%). Capital losses and other reductions in profit in connection with the shares sold may, generally, not be deducted as business expenses.

By contrast, the full amount of the gains from the sale or disposal of subscription rights is subject to corporate income tax (plus solidarity surcharge) and trade tax. Capital losses and other reductions in profit in connection with the subscription rights should be tax-deductible, subject to general restrictions. The exercise of subscription rights should not be considered equivalent to a sale.

Individual entrepreneurs. If the German tax resident shareholder is an individual entrepreneur who holds the shares as business assets, 60% of the proceeds from the sale or disposal of the shares are subject to progressive personal income tax plus solidarity surcharge at a total tax rate of up to approximately 47.5% and, if applicable, church tax (partial income taxation method). However, for church tax, if applicable, the partial income taxation method does not apply. Only 60% of any capital losses and expenses economically connected to the sale or disposal can be deducted for tax purposes. If the shares are attributable to a permanent establishment of a commercial business of the shareholder in Germany, 60% of the capital gains from the sale or disposal of the shares are additionally subject to trade tax. The partial income taxation

method also applies to gains or losses from the sale or disposal of subscription rights that are held by the individual entrepreneur as business assets. The exercise of subscription rights should not be considered equivalent to a sale.

In general, the trade tax may be credited, completely or partially, against the personal income tax of the shareholder in accordance with a lump-sum tax credit method, depending on the assessment rate set by the local municipality and the individual tax circumstances.

Partnerships. If the German tax resident shareholder is a partnership being engaged or deemed to be engaged in a business (co-entrepreneurship), the personal income tax or corporate income tax is not charged at the level of the partnership, but at the level of each partner. The taxation of each partner depends on whether the partner is a corporation or an individual. If the partner is a corporation, the capital gains from the sale or disposal of shares and subscription rights included in the partner's profit share are subject to taxation in accordance with the principles applicable to corporations (see above "Corporations"). Capital gains included in the profit share of an individual partner are accordingly subject to the principles applicable to individual entrepreneurs (*i.e.*, the partial income taxation method), see above "Individual entrepreneurs"). In addition, the capital gains are subject to trade tax at the level of the partnership if the shares are attributable to a permanent establishment of a commercial business of the partnership in Germany, and this generally in the amount of 60% to the extent they are attributable to the profit share of an individual partner and generally in the amount of 5% to the extent they are attributable to the profit share of a corporate partner. Capital losses or other reductions in profit in connection with the shares sold are not taken into account for purposes of trade tax to the extent they are attributable to a partner that is a corporation, and subject to general restrictions only 60% of these losses or expenses are taken into account to the extent they are attributable to a partner who is an individual. Capital gains and losses realized from the sale or disposal of a subscription right are fully taken into account for purposes of trade tax within the scope of general restrictions to the extent they are attributable to the profit share of a corporate partner. By contrast, if the partner is an individual, the capital gains from the sale or disposal of subscription rights included in the profit share are arguably only subject to trade tax at a rate of 60%; accordingly losses and reductions of profits related to the sale or disposal of subscription rights should in this case only be deductible at a rate of 60% subject to general restrictions. The exercise of subscription rights held as business assets should not be treated as a sale or disposal of such subscription rights.

In general, if the partner of the partnership is an individual, the trade tax paid by the partnership and attributable to his/her profit share is completely or partially credited against the shareholder's personal income tax in accordance with a lump-sum tax credit method, depending on the assessment rate set by the local municipality and the individual tax circumstances.

Withholding tax. In the case of a German Disbursing Agent, capital gains from the sale or disposal of shares or subscription rights held as business assets are generally subject to withholding tax just as in the case of a shareholder who holds the shares or subscription rights as private assets (see section "20.2.1 Taxation of Capital Gains for German Tax Resident Shareholders"). The German Disbursing Agent may, however, refrain from levying withholding tax if (i) the shareholder is a corporation tax-resident in Germany, or (ii) the shareholder holds the shares or subscription rights as assets of a business in Germany and declares this to the Disbursing Agent in the officially required pre-printed form and certain further prerequisites are fulfilled. If a German Disbursing Agent nonetheless withholds tax on capital gains, the tax withheld and remitted (including solidarity surcharge and church tax, if applicable) can either be credited against the personal income tax or corporate income tax liability or refunded in the amount of the excess.

20.2.2 Shareholders Domiciled in Foreign Countries

Capital gains derived by non-German tax resident shareholders are subject to German taxation only if the selling or disposing shareholder holds a Qualified Participation in the Company or holds the shares or subscription rights as business assets of a German permanent establishment or fixed place of business or as business assets for which a permanent representative is appointed in Germany.

In the case of a Qualified Participation (as defined in section "20.1.3 Taxation of Dividends Derived by German Tax Resident Shareholders"), 5% of the gains from the sale or disposal of shares and 100% of the gains from the sale or disposal of subscription rights are generally subject to corporate income tax plus

solidarity surcharge if the shareholder is a corporation. Pursuant to a decision of the German Federal Fiscal Court (*Bundesfinanzhof*) dated May 31, 2017 (Federal Tax Gazette (*Bundessteuerblatt*), part II of 2018, p. 144), in case of a Qualified Participation, the capital gain on the disposal of shares or subscription rights is not subject to German taxation if the shareholder is a corporation which is not tax resident in Germany and neither maintains a permanent establishment nor has appointed a permanent representative in Germany. The German tax authorities have adopted this view.

If the shareholder is an individual and holds a Qualified Participation as private asset, only 60% of the gains from the sale or disposal of the shares are subject to the individual, progressive personal income tax plus solidarity surcharge (partial income taxation method). Arguably, the partial income taxation method also applies to gains from the sale or disposal of subscription rights by an individual. Where a German Disbursing Agent is involved, withholding tax on capital gains is generally levied at a rate of 25%, plus 5.5% solidarity surcharge thereon. If, however, (i) the shares are not held through a permanent establishment or fixed place of business or as business assets for which a permanent representative is appointed in Germany, and (ii) a German Disbursing Agent is involved, then in the opinion of the German tax authorities, the German Disbursing Agent is generally not required to levy withholding tax and solidarity surcharge. In the case of a Qualified Participation, the capital gains must be declared in a tax return and are taxed by way of a tax assessment, subject to an exemption under a double taxation treaty or under domestic law. Most double taxation treaties provide for an exemption from German taxation of the capital gain and assign the right to tax to the shareholder's country of residence subject to certain exemptions in case of the sale or disposal of shares in a land rich company.

In the case of capital gains or losses from the sale or disposal of shares or subscription rights held as business assets of a permanent establishment in Germany or a fixed place of business, or as business assets for which a permanent representative has been appointed in Germany, the description above for German tax resident shareholders who hold their shares as business assets applies accordingly (see section "20.1.3 Taxation of Dividends Derived by German Tax Resident Shareholders"). The German Disbursing Agent may only refrain from levying withholding tax if the shareholder declares to the German Disbursing Agent in an officially prescribed form that the shares are held as assets of a German business, and certain further prerequisites are fulfilled.

20.3 Special Rules for Credit Institutions, Financial Services Institutions, Financial Enterprises as well as Life Insurance and Health Insurance Companies and Pension Funds

If credit institutions (*Kreditinstitute*), securities institutions (*Wertpapierinstitute*) and financial services institutions (*Finanzdienstleistungsinstitute*) hold or sell shares or subscription rights, which are allocable to their trading portfolio (*Handelsbestand*) within the meaning of Section 340e para. 3 of the German Commercial Code (*Handelsgesetzbuch*), then neither the exemption from corporate income tax (95%) nor the partial-income method will apply to dividends or capital gains and capital losses arising from the sale or disposal of shares or subscription rights, *i.e.*, dividends and capital gains are fully subject to corporate income tax and correspondingly capital losses can be offset in full for tax purposes. The same applies for financial companies within the meaning of the German Banking Act (*Kreditwesengesetz*) if credit institutions, securities institutions or financial services institutions hold, directly or indirectly, a participation of more than 50% in such financial company and if the shares or subscription rights have to be recorded in the current assets (*Umlaufvermögen*) of the financial company at the time of initial recording. The dividends may be exempt from trade tax if a participation of at least 15% is held in the Company's share capital at the beginning of the relevant tax assessment period. Similarly, the 95% exemption for corporate income tax does not apply to dividends paid with respect to, or capital gains and losses arising from the sale or disposal of shares or subscription rights which are allocable to the capital investments (*Kapitalanlagen*) of life insurance and health insurance companies and pension funds; for these shareholders an exemption from trade tax in case of a participation of at least 15% in the Company's share capital is also not available.

However, an exemption to the foregoing, and thus a 95% effective tax exemption, applies to dividends obtained by the aforementioned companies, to which the EU Parent Subsidiary Directive applies. In addition, applicable double taxation treaties might provide further relief from German tax, subject to certain prerequisites, *e.g.*, substance requirements and holding periods, being met.

20.4 Amendments to the Solidarity Surcharge

As of the assessment period 2021, the solidarity surcharge has been partially abolished for certain individuals and reduced for certain other individuals, depending on certain income thresholds. The solidarity surcharge continues to apply for capital investment income and therefore to withholding tax (including the flat tax regime) as well as to corporate income tax. Should the individual income tax burden for an individual shareholder is lower than 25%, the shareholder can apply for his/her capital investment income being assessed at his/her individual tariff-based income tax rate in which case solidarity surcharge would be refunded.

20.5 Inheritance and Gift Tax

The transfer of shares or subscription rights to another person upon death or by way of a gift is generally subject to German inheritance tax or gift tax if:

- i. the decedent, the person making the gift, the heir, the person receiving the gift or the other person acquiring the assets has at the time of the transfer of the assets his/her domicile or ordinary residence (*gewöhnlicher Aufenthalt*), place of management or registered office in Germany or is a German citizen who has not permanently resided in a foreign country for longer than five consecutive years (this term is extended to ten years for German expatriates with residence in the United States) prior to the transfer without having a German residence, or
- ii. the shares or subscription rights belong to business assets of the decedent or the person making the gift for which a permanent establishment was maintained in Germany or for which a permanent representative was appointed, or
- iii. the decedent or the person making the gift, either himself or together with other persons related to him, held a direct or indirect participation of at least 10% of the share capital of the Company at the time of the transfer.

The few German double taxation treaties on inheritance tax and gift tax presently in force usually provide that German inheritance tax or gift tax can only be charged in the case of (i) above and also with certain restrictions in case of (ii). Special rules apply to certain German citizens living outside Germany and former German citizens.

20.6 Other Taxes

No German capital transfer taxes, value-added taxes, stamp taxes or similar taxes apply to the acquisition, sale or disposal or other form of transferring shares or subscription rights. However, an entrepreneur can opt to pay value-added tax on the sale or disposal of shares or subscription rights, despite being generally exempt from value added tax, if the shares or subscription rights are sold to another entrepreneur for the entrepreneur's business. Wealth tax (*Vermögensteuer*) is presently not levied in Germany.

On February 14, 2013, the European Commission published a proposal (the “**Commission's Proposal**”) for a directive for a common Financial Transaction Tax (“**FTT**”) in relation to which the negotiations between the member states of the European Union which originally intended to introduce the FTT have failed. Belgium, Germany, Greece, Spain, France, Italy, Austria, Portugal, Slovenia and Slovakia (the “**Participating Member States**”) have resumed negotiations regarding the introduction of the FTT. Additional member states of the European Union may decide to participate. The scope of such tax and the timing of its implementation remain unclear. However, according to the latest proposal brought up by the Portuguese presidency of the European Council in February 2021, the FTT could be based on a combination of the existing French and Italian financial transaction tax regimes. A financial transaction tax based on such regimes would cover transactions of issued shares of companies seated in one of the participating states with a high market-cap value and certain equity derivative transactions. Prospective holders of the shares or subscription rights are advised to monitor future developments closely and to seek their own professional tax advice in relation to the consequences of the FTT.

21. FINANCIAL INFORMATION

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***Unaudited Condensed Consolidated Interim Financial Statements of Deutsche Lufthansa
Aktiengesellschaft as of and for the six-month period ended June 30, 2021 (IFRS)***

Consolidated income statement

January - June 2021

CONSOLIDATED INCOME STATEMENT

in €m	Jan - Jun 2021	Jan - Jun 2020	Apr - Jun 2021	Apr - Jun 2020
Traffic revenue	3,637	5,641	2,095	1,102
Other revenue	2,134	2,694	1,116	792
Total revenue	5,771	8,335	3,211	1,894
Changes in inventories and work performed by entity and capitalised	49	158	25	13
Other operating income ¹⁾	655	797	339	353
Cost of materials and services	-3,204	-5,127	-1,792	-1,084
Staff costs	-2,907	-3,620	-1,517	-1,472
Depreciation, amortisation and impairment ²⁾	-1,135	-1,783	-566	-659
Other operating expenses ³⁾	-1,279	-2,048	-666	-745
Profit/loss from operating activities	-2,050	-3,288	-966	-1,700
Result of equity investments accounted for using the equity method	-71	-184	-19	-149
Result of other equity investments	7	4	6	3
Interest income	-2	33	–	-3
Interest expenses	-211	-195	-95	-103
Other financial items	93	-789	153	209
Financial result	-184	-1,131	45	-43
Profit/loss before income taxes	-2,234	-4,419	-921	-1,743
Income taxes	421	792	162	239
Profit/loss after income taxes	-1,813	-3,627	-759	-1,504
Profit/loss attributable to non-controlling interests	8	10	3	11
Net profit/loss attributable to shareholders of Deutsche Lufthansa AG	-1,805	-3,617	-756	-1,493
Basic/diluted earnings per share in €	-3.02	-7.56	-1.26	-3.12

¹⁾ The total amount includes EUR 46m (previous year: EUR 17m) from the reversal of write-downs and allowances on receivables.

²⁾ The total amount includes EUR 3m (previous year: EUR 1m) for write-downs on non-current receivables.

³⁾ The total amount includes EUR 33m (previous year: EUR 169m) for the recognition of loss allowances on current receivables.

Consolidated statement of comprehensive income

January - June 2021

STATEMENT OF COMPREHENSIVE INCOME

in €m	Jan - Jun 2021	Jan - Jun 2020	Apr - Juni 2021	Apr - Juni 2020
Profit/loss after income taxes	-1,813	-3,627	-759	-1,504
Other comprehensive income				
Other comprehensive income with subsequent reclassification to the income statement				
Differences from currency translation	30	27	12	-34
Subsequent measurement of financial assets at fair value without effect on profit and loss	-13	-14	-14	4
Subsequent measurement of hedges - cash flow hedge reserve	553	-280	156	75
Subsequent measurement of hedges - costs of hedges	60	5	17	-128
Other comprehensive income from investments accounted for using the equity method	2	2	1	1
Other expenses and income recognised directly in equity	-1	-2	—	-1
Income taxes on items in other comprehensive income	-134	59	-33	18
	497	-203	139	-65
Other comprehensive income without subsequent reclassification to the income statement				
Revaluation of defined-benefit pension plans	2,109	-672	311	-345
Subsequent measurement of financial assets at fair value	2	0	2	3
Other expenses and income recognised directly in equity	—	-4	—	-2
Income taxes on items in other comprehensive income	-462	-50	-32	144
	1,649	-726	281	-200
Other comprehensive income after income taxes	2,146	-929	420	-265
Total comprehensive income	333	-4,556	-339	-1,769
Comprehensive income attributable to minority interests	8	14	4	13
Comprehensive income attributable to shareholders of Deutsche Lufthansa AG	341	-4,542	-335	-1,756

Consolidated statement of financial position as of June June 2021

CONSOLIDATED BALANCE SHEET - ASSETS

in €m	30.06.2021	31.12.2020	30.06.2020
Intangible assets with an indefinite useful life ¹⁾	1,165	1,169	1,242
Other intangible assets	442	469	538
Aircraft and reserve engines	15,516	15,842	17,693
Repairable spare parts for aircraft	1,777	1,823	2,135
Property, plant and other equipment ²⁾	3,493	3,671	3,928
Investments accounted for using the equity method	353	403	506
Other equity investments	252	252	257
Non-current securities	38	54	54
Loans and receivables	443	440	414
Derivative financial instruments	491	363	787
Deferred charges and prepaid expenses	82	91	100
Effective income tax receivables	36	34	33
Deferred tax assets	4,725	4,833	3,074
Non-current assets	28,813	29,444	30,761
Inventories	669	726	920
Contract assets	180	142	209
Trade receivables and other receivables	3,510	2,843	3,273
Derivative financial instruments	390	260	311
Deferred charges and prepaid expenses	242	193	278
Effective income tax receivables	288	282	92
Securities	4,603	3,654	2,448
Cash and cash equivalents	2,063	1,806	1,211
Assets held for sale	80	134	384
Current assets	12,025	10,040	9,126
Total assets	40,838	39,484	39,887

¹⁾ Including Goodwill.

²⁾ These include investment property of EUR 30 million (previous year: EUR 0 million).

CONSOLIDATED BALANCE SHEET - SHAREHOLDERS' EQUITY AND LIABILITIES

in €m	30.06.2021	31.12.2020	30.06.2020
Issued Capital	1,530	1,530	1,224
Capital reserve	378	378	378
Silent participation of the Economic Stabilization Fund	1,500	–	–
Retained earnings	-210	4,868	6,108
Other neutral reserves	1,720	1,296	1,531
Net profit/loss	-1,805	-6,725	-3,617
Equity attributable to shareholders of Deutsche Lufthansa AG	3,113	1,347	5,624
Minority interests	32	40	78
Shareholders' equity	3,145	1,387	5,702
Pension provisions	7,607	9,531	7,422
Other provisions	593	558	557
Borrowings	13,266	12,252	8,131
Contract liabilities	35	36	23
Other financial liabilities	89	86	88 ¹⁾
Advance payments received, deferred income and other non-financial liabilities	33	33	160 ¹⁾
Derivative financial instruments	247	457	284
Deferred tax liabilities	501	485	603
Non-current provisions and liabilities	22,371	23,438	17,268
Other provisions	866	831	675
Borrowings	2,305	3,116	2,819
Trade payables and other financial liabilities	3,718	3,321	3,978 ¹⁾
Contract liabilities from unused flight documents	3,089	2,064	4,499
Other contract liabilities	2,793	2,977	2,654
Advance payments received, deferred income and other non-financial liabilities	1,640	1,295	707 ¹⁾
Derivative financial instruments	252	366	625
Effective income tax obligations	659	689	469
Liabilities in connection with assets held for sale	–	–	491
Current provisions and liabilities	15,322	14,659	16,917
Total shareholders' equity and liabilities	40,838	39,484	39,887

¹⁾ Comparative figure for previous year has been restated.

Consolidated statement of changes in shareholders' equity as of 30 June 2021

CONSOLIDATED STATEMENT OF CHANGES IN SHAREHOLDERS' EQUITY

in €m	Issued capital	Capital reserve	Silent participation I	Fair value measurement of financial instruments	Currency differences	Revaluation reserve (due to business combinations)	Other neutral reserves	Total other neutral reserves	Retained earnings	Net profit/loss	Equity attributable to shareholders of Deutsche Lufthansa AG	Minority interests	Total shareholders' equity
As of 01.01.2020	1,224	378	–	624	503	236	352	1,715	5,617	1,213	10,147	109	10,256
Capital increases/reductions	–	–	–	–	–	–	–	–	–	–	–	–	–
Reclassifications	–	–	–	–	–	–	–	–	1,213	-1,213	–	–	–
Dividends to Lufthansa shareholders/minority interests	–	–	–	–	–	–	–	–	–	–	–	-17	-17
Transaction with minority interests	–	–	–	–	–	–	–	–	–	–	–	–	–
Consolidated net profit/loss attributable to Lufthansa shareholders/minorities	–	–	–	–	–	–	–	–	–	-3,617	-3,617	-10	-3,627
Other expenses and income recognised directly in equity	–	–	–	-230	27	–	–	-203	-722	–	-925	-4	-929
Hedging results reclassified from non-financial assets to acquisition costs	–	–	–	19	–	–	–	19	–	–	19	–	19
As of 30.06.2020	1,224	378	–	413	530	236	352	1,531	6,108	-3,617	5,624	78	5,702
As of 01.01.2021	1,530	378	–	305	396	236	359	1,296	4,868	-6,725	1,347	40	1,387
Capital increases/reductions	–	–	1,500	–	–	–	–	–	–	–	1,500	–	1,500
Reclassifications	–	–	–	–	–	–	–	–	-6,725	6,725	–	–	–
Dividends to Lufthansa shareholders/minority interests	–	–	–	–	–	–	–	–	–	–	–	–	–
Transaction with minority interests	–	–	–	–	–	–	–	–	–	–	–	–	–
Consolidated net profit/loss attributable to Lufthansa shareholders/minorities	–	–	–	–	–	–	–	–	–	-1,805	-1,805	-8	-1,813
Other expenses and income recognised directly in equity	–	–	–	468	30	–	1	499	1,647	–	2,146	–	2,146
Hedging results reclassified from non-financial assets to acquisition costs	–	–	–	-75	–	–	–	-75	–	–	-75	–	-75
As of 30.06.2021	1,530	378	1,500	698	426	236	360	1,720	-210	-1,805	3,113	32	3,145

Consolidated cash flow statement

January - June 2021

CONSOLIDATED CASH FLOW STATEMENT

in €m	Jan - Jun 2021	Jan - Jun 2020	Apr - Jun 2021	Apr - Jun 2020
Cash and cash equivalents at start of period¹⁾	1,804	1,431	1,461	1,853
Net profit/loss before income taxes	-2,234	-4,419	-921	-1,743
Depreciation, amortisation and impairment losses on non-current assets (net of reversals)	1,134	1,782	565	659
Depreciation, amortisation and impairment losses on current assets (net of reversals)	-20	49	-10	27
Net proceeds on disposal of non-current assets	19	9	24	3
Result of equity investments	64	180	13	146
Net interest	213	162	95	106
Income tax payments/reimbursements	-72	112	-56	91
Significant non-cash-relevant expenses/income	-160	385	-218	-626
Change in trade working capital	650	1,434	1,039	-437
Change in other assets/shareholders' equity and liabilities	424	669	253	770
Cash flow from operating activities	18	363	784	-1,004
Capital expenditure for property, plant and equipment and intangible assets	-604	-883	-457	-120
Capital expenditure for financial investments	-8	-14	-2	-7
Additions/loss to repairable spare parts of aircraft	70	86	40	58
Proceeds from disposal of non-consolidated equity investments	-	-1	-	-1
Proceeds from disposal of consolidated equity investments	-	-	-	-
Cash outflows for acquisitions/capital increase of/at non-consolidated equity investments	-7	-5	-4	-2
Cash outflows for acquisitions of consolidated equity investments	-	-	-	-
Proceeds from disposal of intangible assets, property, plant and equipment and other financial investments	99	84	59	28
Interest income	-	47	2	3
Dividends received	7	13	6	10
Net cash from/used in investing activities	-443	-673	-356	-31
Purchase of securities/fund investments	-2,851	-5,450	-1,847	-2,232
Disposal of securities/fund investments	1,932	5,040	550	3,199
Net cash from/used in investing and cash management activities	-1,362	-1,083	-1,653	936
Capital increase/ Silent Participation I	1,500	-	1,500	-
Transactions by non-controlling interests	-	-	-	-
Non-current borrowing	2,864	1,784	461	284
Repayment of non-current borrowing	-2,672	-1,126	-445	-745
Dividends paid	-	-17	-	-17
Interest paid	-120	-107	-65	-64
Net cash from/used in financing activities	1,572	534	1,451	-542
Net increase/decrease in cash and cash equivalents	228	-186	582	-610
Changes due to currency translation differences	10	-8	-1	-6
Cash and cash equivalents 30 Jun²⁾	2,042	1,237	2,042	1,237
Less cash and cash equivalents of companies held for sale as of 30 Jun	-	26	-	26
Cash and cash equivalents of companies not classified as held for sale as of 30 Jun²⁾	2,042	1,211	2,042	1,211
Securities	4,603	2,448	4,603	2,448
Liquidity	6,645	3,659	6,645	3,659
Net increase/decrease in liquidity	1,187	274	1,916	-1,480

¹⁾ Amount as of 01/01/2020 includes EUR 16 m, which were included in assets held for sale as of 12/31/2019.

²⁾ The difference between the bank balance and cash-in-hand shown in the statement of financial position comes from fixed-term deposits of EUR 21m with terms of four to twelve months (previous year: EUR 0m).

1 Applied standards, changes in the group of consolidated companies and accounting principles

The consolidated financial statements of Deutsche Lufthansa AG and its subsidiaries have been prepared in accordance with the International Financial Reporting Standards (IFRS) issued by the International Accounting Standards Board (IASB), as applicable in the European Union (EU), taking account of interpretations by the IFRS Interpretations Committee (IFRIC). This interim report as of 30 June 2021 has been prepared in condensed form in accordance with IAS 34.

In preparing the interim financial statements, the standards and interpretations applicable as of 1 January 2021 have been applied. The interim financial statements as of 30 June 2021 have been prepared using the same accounting policies as those on which the preceding consolidated financial statements as of 31 December 2020 were based. The standards and interpretations mandatory from 1 January 2021 onwards had no effect on the Group's net assets, financial and earnings position, and no restatements resulting from new standards were necessary.

2 Going concern and presentation of funding measures to stabilize the economic situation

Since spring 2020, the business activities of the Lufthansa Group companies have been severely impacted by the effects of the coronavirus pandemic and the related far-reaching tightening of international travel restrictions and quarantine regulations worldwide. Only the freight business was able to significantly increase both revenue and earnings contributions.

The strongly reduced volume of business continues to affect liquidity. However, a balanced operating cash flow figure was achieved in the first half of the current financial year. This was due to increased cash flows from ticket sales, especially in the second quarter. Liquidity at Deutsche Lufthansa AG and its domestic and foreign subsidiaries was initially secured in 2020 thanks to the government aid provided as part of the stabilisation packages. From the second half of 2020 onwards, the Group was also repeatedly able to raise funds successfully on capital markets itself, some of which were used to repay the stabilisation funding drawn down to date.

The stabilisation measures were applied for, negotiated and approved in Germany, Switzerland, Austria, Belgium and the USA last year.

The framework agreement concluded in Germany between the Lufthansa Group, the Economic Stabilisation Fund (WSF) and the KfW has an overall financing framework of up to EUR 9.0bn. Funding agreed in Switzerland, Austria and Belgium is to be offset against this. The funds provided by the WSF totalling EUR 6.0bn included a 20% stake in the share capital, as well as a silent participation in Deutsche Lufthansa AG and an earnings

subsidy for Austrian Airlines. As well as the issuance of shares and the full payment made for Silent Participation II in the amount of EUR 1.0bn in the previous year, an amount of EUR 1.5bn was drawn down from Silent Participation I in the second quarter of the current financial year. While taking into account the equity capital measures of around EUR 0.2bn which have been approved outside Germany, a residual amount of EUR 3.0bn is available from Silent Participation I.

The framework agreement with the Economic Stabilisation Fund provides for extensive information and auditing rights for the Economic Stabilisation Fund and obligations for the Lufthansa Group including regarding the suspension of dividend payments, limitations on management compensation, a commitment not to make equity investments, waiver of up to 24 slots at both the Frankfurt and Munich airports and pursuit of a sustainable corporate policy. Compliance with the obligations across the Lufthansa Group is crucial and it may in some cases be demanding in the current situation since some of the obligations call for interpretation, taking into account the understanding of the EU Commission. Subject to the full repayment of the silent participations by the Company and a minimum sale price of EUR 2.56 per share plus an annual interest of 12%, the WSF undertakes to sell its shareholding in full at the market price by 31 December 2023 or after this point in time when the conditions have been fulfilled.

In addition, within the scope of the stabilisation package in Germany, KfW agreed to provide loan capital with a volume of EUR 3.0bn. After taking into consideration the loan commitments made outside Germany and following the repayment in February 2021 of the KfW credit facility which had been drawn down on 31 December 2020 with a volume of EUR 1.0bn, state-guaranteed loans were no longer available in Germany as of 30 June 2021. The repayment of the KfW credit facility increases the Lufthansa Group's financial flexibility, because the facility required shares in the leasing vehicles that own significant parts of the Lufthansa Group fleet to be pledged as collateral. Now that this no longer applies, the Lufthansa Group can use the aircraft again itself for aircraft financing. Other restrictions on the financing of subsidiaries by Deutsche Lufthansa AG were also lifted at the same time.

As well as the issuance of a EUR 1.6bn bond with two tranches within the scope of the EMTN programme, the other refinancing measures implemented in the first half of 2021 included aircraft financing. A further bond was issued after the reporting date.

CHF 550m of the state-guaranteed credit lines agreed in Switzerland as part of the stabilisation measures had been used as of the reporting date. CHF 950m is thus still available out of the total line.

The EUR 300m loan facility resulting from the aid measures agreed in Austria had been fully used as of the reporting date.

As of 30 June 2021, a total amount of EUR 250m had been drawn down from the EUR 287m credit facility included in the stabilisation package agreed with the Belgian government.

Additional funds were made available in the USA in 2021 through CARES Acts I-III. The LSG group and the Lufthansa Technik group have received commitments of a further USD 284m (of which USD 192m as a grant). The approved funds thus total USD 528m (of which USD 345m as a grant). Of the approved funds, USD 54m has not yet been paid out as of the reporting date. In 2021, USD 148m was recognised in the income statement for these grants.

The Italian government also provided funds to compensate for losses incurred due to the coronavirus pandemic. Air Dolomiti subsequently applied for a grant of EUR 16m in early 2021, which was paid out in April.

As of 30 June 2021, Deutsche Lufthansa AG had centrally available liquidity of EUR 5.4bn. A further EUR 3.0bn was available from the WSF stabilisation package (Silent Participation I). Decentralised bank and cash balances came to a further EUR 1.3bn, and a total of EUR 0.9bn has not yet been used from the state funding agreed in Switzerland and Belgium. Free credit lines of EUR 0.5bn are still available as of the reporting date. Altogether, the Lufthansa Group's available liquidity therefore comes to EUR 11.1bn.

Since there is still great uncertainty about travel opportunities and customer behaviour, the Lufthansa Group regularly updates its rolling liquidity planning to reflect the changing parameters for its forecast course of business. Its performance in the current and subsequent financial year will largely depend on the number of new infections, particularly in the context of new virus mutations as well

as the speed and scope of vaccination programmes. In addition, the nature of travel restrictions going forward will play an important role in the recovery of international travel. Within the Company, further progress was made with the ReNew programme, and management remains confident that the implementation will be successful.

State aid will still be needed for the current 2021 financial year, in the form of short-time working pay and the reimbursement of social security contributions. This aid is expected to expire by the end of the year.

Taking into account the corporate planning – which assumes a volume of business of 40% and 80% of the 2019 level in 2021 and 2022 respectively – and the resulting liquidity planning, the existing and potential funding measures and the uncertainties about the future course of business, the Executive Board of the Company considers the Group's liquidity to be secure for the next eighteen months. These interim financial statements have therefore been prepared on a going concern basis.

3 Notes to the income statement, statement of financial position, cash flow statement and segment reporting

The outbreak of the coronavirus pandemic and the steps taken worldwide to contain the virus have continued to have a massive impact on the Group's business operations in 2021. This is reflected throughout the Lufthansa Group's interim report.

In the previous year, international travel restrictions and quarantine regulations only became more strict worldwide in the spring of 2020 (March), which had a strong impact on the air traffic of the Lufthansa Group companies. As a result, the comparability of income and expenses in the two periods is limited.

TOTAL REVENUE

TRAFFIC REVENUE BY AREA OF OPERATIONS

	2021	Europe ¹⁾	North- america ¹⁾	Central- and South America ¹⁾	Asia/ Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
Network Airlines	1,883	1,242	276	47	198	54	66
Lufthansa German Airlines	1,040						
SWISS ²⁾	566						
Austrian Airlines	159						
Brussels	118						
Eurowings ²⁾	159	158	1	–	–	–	–
Logistics	1,595	820	168	59	500	18	30
Total	3,637						

¹⁾ Traffic revenue is allocated to the original location of sale.

²⁾ Disclosure of traffic revenue, including belly revenue; this is reported in the segment reporting in the reconciliation column.

TRAFFIC REVENUE BY AREA OF OPERATIONS

	2020	Europe ¹⁾	North- america ¹⁾	Central- and South America ¹⁾	Asia/ Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
Network Airlines	4,059	2,794	651	90	357	86	81
Lufthansa German Airlines	2,486						
SWISS ²⁾	1,044						
Austrian Airlines	294						
Brussels Airlines	235						
Eurowings ²⁾	363	358	3	1	1	–	–
Logistics	1,219	533	132	42	481	10	21
Total	5,641						

¹⁾ Traffic revenue is allocated to the original location of sale.

²⁾ Disclosure of traffic revenue, including belly revenue; this is reported in the segment reporting in the reconciliation column.

OTHER OPERATING REVENUE BY AREA OF OPERATIONS

	2021	Europe ¹⁾	North-America ¹⁾	Central and South America ¹⁾	Asia/Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
MRO	1,361	590	398	33	244	68	28
MRO services	1,156						
Other operating revenue	205						
Catering	435	25	331	24	34	6	15
Catering services	375						
Revenue from in-flight sales	25						
Other services	35						
Network Airlines	133	118	5	–	5	4	1
Eurowings	–	–	–	–	–	–	–
Logistics	63	37	23	–	1	2	–
Additional Businesses and Group Functions	142	95	12	5	19	8	3
IT services	79						
Travel management	34						
Other	29						
Total	2,134						

¹⁾ Other operating revenue is allocated according to the original location of sale.

OTHER OPERATING REVENUE BY AREA OF OPERATIONS

	2020	Europe ¹⁾	North-America ¹⁾	Central and South America ¹⁾	Asia/Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
MRO	1,606	691	404	70	305	81	55
MRO services	1,352						
Other operating revenue	254						
Catering	651	100	390	35	94	18	14
Catering services	554						
Revenue from in-flight sales	31						
Other services	66						
Network Airlines ²⁾	182	154	10	1	12	3	2
Eurowings ²⁾	3	3	–	–	–	–	–
Logistics	86	63	18	–	2	3	–
Additional Businesses and Group Functions	166	116	16	6	18	7	3
IT services	86						
Travel management	51						
Other	29						
Total	2,694						

¹⁾ Other operating revenue is allocated according to the original location of sale.

AIRCRAFT AND RESERVE ENGINES

The Lufthansa Group provided eight aircraft as collateral for new loans of EUR 673m taken out in the current financial year by way of aircraft financing models.

Repayment of the KfW loan in February 2021 triggered the release of shares in various leasing vehicles in Malta and Austria, which owned a total of 323 aircraft with a carrying amount of EUR 4,432m as of year-end 2020.

DEFERRED TAXES

Deferred taxes have been capitalised in full for the losses and deferred tax assets incurred in Germany and Switzerland in particular during the financial year. As the losses were triggered by an exogenous shock with a temporary impact and the Company expects to be able to use the deferred tax assets when it generates sufficient positive tax results in the foreseeable future, they are expected to continue to be recoverable in full. Tax loss

carry-forwards are not subject to any restrictions regarding the period of time in which they can be used in Germany.

ASSETS CLASSIFIED AS HELD FOR SALE

Assets with a carrying amount of EUR 80m were held for sale as of 30 June 2021. This item includes 23 aircraft held for sale with a carrying amount of EUR 79m: two Boeing MD11s, one Boeing B767, four Airbus A321s, eight Airbus A320s, four Bombardier CRJ9s and four Dash 8-400s.

SHAREHOLDERS' EQUITY

In the period under review, EUR 1.5bn was drawn down from the total volume of EUR 4.5bn provided through Silent Participation I granted by the WSF. This is reportable as equity due to the contractual provisions with regard to the indefinite term and since Lufthansa has sole discretion over servicing of the coupon and repayment. A further amount of EUR 3.0bn thus remains available from Silent Participation I in the period up to 31 December 2021.

PENSION PROVISIONS

The discount rate used to calculate obligations in Germany was 1.2%. As of 31 December 2020, the rate was 0.8%. A discount rate of 0.35% was used for the pension obligations in Switzerland (31 December 2020: 0.1%). The decline in pension provisions is largely due to the increase in the discount rate and a recovery in the market value of the plan assets. The agreement reached with the collective bargaining partners to cope with the crisis made it possible to reduce the service cost compared with the previous year, despite the interest rate-related increase it contains.

CONTRACT LIABILITIES FROM UNUSED FLIGHT DOCUMENTS

Contract liabilities from unused flight documents came to EUR 3,089m as of 30 June 2021. There are no material payments outstanding in relation to claims for refunds.

CHANGES IN ESTIMATES

On the basis of current corporate forecasts, the management of Deutsche Lufthansa AG does not consider that the long-term business prospects have changed fundamentally, even taking into account the ongoing uncertainty regarding the duration of travel restrictions and the level of future air travel once the current crisis is over. There have therefore been no material changes in estimates in this respect.

OTHER GOVERNMENT AID MEASURES

Total state subsidies of EUR 757m had been received as of 30 June 2021. They are primarily attributable to the reimbursement of wage-replacement benefits and social security contributions paid in the context of short-time working in Germany, Austria and Switzerland. This includes EUR 164m in subsidies for social security contributions, which are classified as support measures. Another EUR 148m in non-specific subsidies was reported

under other operating income. This includes grants of USD 148m received as part of the CARES Act in the USA for LSG and Lufthansa Technik companies. These were disbursed once the necessary evidence had been provided. Another EUR 16m relates to government grants for Air Dolomiti.

Loans on below-market terms were granted by the Belgian government as part of the stabilisation measures. The interest rate subsidy they contain of EUR 3m for the current financial year is netted against interest expense.

In addition to the amounts granted to the companies in connection with short-time working, employees also received direct state support in the form of salary-replacement benefits.

By way of support in the face of the crisis, German state institutions deferred payment of taxes and other levies. These consist mainly of import VAT. The amount of deferred import VAT came to EUR 898m as of 30 June 2021. This is due to be paid in instalments up to the end of 2021.

4 Seasonality

The Group's business activities are normally exposed to seasonal effects via the Network Airlines and Eurowings segments in particular. As such, revenue in the first and fourth quarters is generally lower, since people travel less, while higher revenue and operating profits are normally earned in the second and third quarters.

However, due to the impact of the coronavirus pandemic the volume of business plummeted overall and currently no longer shows any signs of seasonal effects.

5 Contingencies and events after the reporting period

CONTINGENT LIABILITIES

in €m	30.06.2021	31.12.2020
From guarantees, bills of exchange and cheque guarantees	693	664
From warranty contracts	205	192
From providing collateral for third-parties liabilities	16	16
	914	872

Provisions for other contingent liabilities were not made because it was not sufficiently probable that they would be necessary. The potential financial effect of these provisions on the result would have been EUR 68m in total (as of 31 December 2020: EUR 61m).

As well as information and auditing rights for the Economic Stabilisation Fund, the framework agreement with the Economic Stabilisation Fund provides for extensive obligations for the Lufthansa Group including the suspension of dividend payments, a commitment not to make

equity investments and a ban on cross-subsidising companies which were already in difficulty within the meaning of EU Regulation No. 651/2014 on 31 December 2019. In respect of the above-mentioned obligations, risks may arise due to a difference of interpretation between the Company and the European Commission. Lufthansa and the European Commission are continuing to exchange information (including relevant documents) in order to fully clarify these matters. However, at the present time it is impossible to reliably predict the outcome of these discussions. Significant financial risks for the Company due to ultimately determined violations of agreed obligations therefore cannot be ruled out.

As of 30 June 2021, the tax risks for which no provisions had been recognised came to some EUR 200m (as of 31 December 2020: EUR 200m).

At the end of June 2021, there were order commitments of EUR 14.4bn for capital expenditure on property, plant and equipment, including repairable spare parts, and for intangible assets. As of 31 December 2020, the order commitments came to EUR 13.0bn. This change is mainly due to the order of five Airbus A350s and five Boeing B787s, which was offset by the deduction from residual commitments on account of the aircraft additions.

EVENTS AFTER THE REPORTING PERIOD

The Lufthansa Group again successfully issued a bond for a total volume of EUR 1.0bn on 7 July 2021 and thus further strengthened its liquidity. The bond issue, with a denomination of EUR 100,000, was placed in two tranches, each with a volume of EUR 500m; the tranche with a term of three years pays interest of 2.0% p.a., while the second tranche with a term of eight years pays interest of 3.5% p.a.

On 14 July 2021, the European Commission presented its "Fit for 55" legislative package, comprising a total of twelve legislative procedures. The European Commission's proposals include a faster reduction in the number

of certificates issued in the aviation sector and discontinuing the free allocation of certificates by 2027; in addition, a kerosene tax is to be gradually introduced, along with an obligation to use an increasingly higher level of sustainable aviation fuels.

On 15 July 2021, Austrian Airlines repaid EUR 30m from the syndicated loan taken out as part of the government stabilisation measures totalling EUR 300m ahead of schedule. An additional USD 47m in subsidies were disbursed to the LSG group in the USA under the CARES Act.

On 23 July 2021, the European Commission made the decision to reduce the required slot use rate from 80% to 50% in the 2021/22 winter flight plan; this means that airlines have to use 50% of each of their slot series at slot-regulated airports so as not to lose these slot series in subsequent periods.

6 Financial instruments and financial liabilities

FINANCIAL INSTRUMENTS

The following tables show financial assets and liabilities held at fair value by level in the fair value hierarchy. The levels are defined as follows:

Level 1: Financial instruments traded on active markets, the quoted prices for which are taken for measurement unchanged.

Level 2: Measurement is made by means of valuation methods with parameters derived directly or indirectly from observable market data.

Level 3: Measurement is made by means of valuation methods with parameters not based exclusively on observable market data.

As of 30 June 2021, the fair value hierarchy for assets and liabilities held at fair value was as follows:

FAIR VALUE HIERARCHY OF ASSETS AS OF 30.06.2021

in €m	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit and loss	4,603	7	–	4,610
Financial derivatives classified as held for trading	–	7	–	7
Securities	4,603	–	–	4,603
Derivative financial instruments which are an effective part of a hedging relationship	–	874	–	874
Financial assets at fair value through other comprehensive income	13	13	–	26
Equity instruments	13	13	–	26
Debt instruments	–	–	–	–
Total assets	4,616	894	–	5,510

FAIR VALUE HIERARCHY OF LIABILITIES AS OF 30.06.2021

in €m	Level 1	Level 2	Level 2	Total
Financial liabilities at fair value through profit or loss	–	-661	–	-661
Derivative financial instruments at fair value through profit or loss	–	-34	–	-34
Derivative financial instruments which are an effective part of a hedging relationship	–	-465	–	-465
Total liabilities	–	-1,160	–	-1,160

CO₂ emissions certificates valued at EUR 67m were sold and simultaneously repurchased on the market in what are known as “repo” agreements so that economic ownership of the certificates is maintained. EUR 114m was also repaid under similar expiring repo agreements.

As of 31 December 2020, the fair value hierarchy for assets and liabilities held at fair value was as follows:

FAIR VALUE HIERARCHY OF ASSETS AS OF 31.12.2020

in €m	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit and loss	3,654	2	–	3,656
Financial derivatives classified as held for trading	–	2	–	2
Securities	3,654	–	–	3,654
Derivative financial instruments which are an effective part of a hedging relationship	–	620	–	620
Financial assets at fair value through other comprehensive income	11	12	–	23
Equity instruments	11	12	–	23
Debt instruments	–	–	–	–
Total assets	3,665	634	–	4,299

FAIR VALUE HIERARCHY OF LIABILITIES AS OF 31.12.2020

in €m	Level 1	Level 2	Level 2	Total
Financial liabilities at fair value through profit or loss	–	-712	–	-712
Derivative financial instruments at fair value through profit or loss	–	-85	–	-85
Derivative financial instruments which are an effective part of a hedging relationship	–	-738	–	-738
Total liabilities	–	-1,535	–	-1,535

The fair values of interest rate derivatives correspond to their respective market values, which are measured using appropriate mathematical methods, such as discounting expected future cash flows. Discounting takes market standard interest rates and the residual term of the respective instruments into account. Forward currency transactions and swaps are individually discounted to the reporting date based on their respective futures rates and the appropriate interest rate curve. The market prices of currency options and the options used to hedge fuel prices are determined using acknowledged option pricing models.

The fair values of debt instruments also correspond to their respective market values, which are measured using appropriate mathematical methods, such as discounting expected future cash flows. Discounting takes market standard interest rates and the residual term of the respective instruments into account.

The carrying amount for cash, trade receivables, other receivables, trade payables and other liabilities is assumed to be a realistic estimate of fair value.

FINANCIAL LIABILITIES

The following table shows the carrying amounts and market values for individual classes of financial liabilities. Market values for bonds are equal to the listed prices. The market values for other types of financial liability have been calculated using the applicable interest rates for the remaining term to maturity and repayment structures at the reporting date based on available market information (Bloomberg).

Information regarding the loan funds received under the state stabilisation measures can be found in Note 2.

FINANCIAL LIABILITIES

in €m	30.06.2021		31.12.2020	
	Carrying amount	Market value	Carrying amount	Market value
Bonds	4,238	4,292	2,707	2,643
Commercial Paper to banks	–	–	200	199
Borrower's note loans	1,945	1,975	1,900	1,894
Credit lines	250	252	763	740
State-guaranteed loans	2,184	2,380	2,907	3,147
Aircraft financing	4,267	4,037	3,603	3,590
Other borrowings	546	545	651	705
Leasing liabilities	2,141	–	2,637	–
Total	15,571	13,481	15,368	12,918

7 Earnings per share

Earnings per share

		30.06.2021	30.06.2020
Basic/diluted earnings per share	€	– 3.02	– 7.56
Consolidated net profit/loss	€m	– 1,805	– 3,617
Weighted average number of shares		597,742,822	478,194,257

8 Issued capital

SHARE CAPITAL

Deutsche Lufthansa AG's issued capital totals EUR 1,530,221,624.32. It is divided into 597,742,822 registered shares with transfer restrictions, with each share representing EUR 2.56 of issued capital.

AUTHORISED CAPITAL

A resolution passed at the Annual General Meeting on 7 May 2019 authorised the Executive Board until 6 May 2024, subject to approval by the Supervisory Board, to increase the Company's issued capital by up to EUR 450,000,000 by issuing new registered shares on one or more occasions for payment in cash or in kind (Authorised Capital A). In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board.

A resolution passed at the Annual General Meeting on 7 May 2019 authorised the Executive Board until 6 May 2024, subject to approval by the Supervisory Board, to increase the issued capital by EUR 30,000,000 by issuing new registered shares to employees (Authorised Capital B) for payment in cash. Existing shareholders' subscription rights are excluded. As of 30 June 2021, the issued capital was increased under this authorisation by a total of EUR 7,637,831.68, so that Authorised Capital B still amounted to EUR 22,362,168.32 as of the reporting date.

A resolution passed at the Annual General Meeting on 4 May 2021 authorised the Executive Board until 3 May 2026, subject to approval by the Supervisory Board, to increase the Company's issued capital by up to EUR 5,500,000,000.00 by issuing new registered shares on one or more occasions for payment in cash or in kind, in order to use the net issue proceeds largely to repay the capital provided to Deutsche Lufthansa AG by the Economic Stabilisation Fund or for other purposes mentioned in Section 7f of the German Economic Stabilisation Act (WStBG) (Authorised Capital C). Existing shareholders are to be granted subscription rights. The Economic Stabilisation Fund is entitled to subscribe for the new registered shares to which it is entitled as of its exercise of its subscription rights in accordance with the subscription ratio, in return for payment in kind through the contribution of Silent Participation I and/or II either in whole or in part (including the rights to the coupons and any additional payment). The Executive Board is authorised to prescribe the further contents of the share rights and the terms and conditions of the issuance of shares with the consent of the Supervisory Board.

The Executive Board is authorised, in the event of the fulfilment of the requirements stipulated in Section 4 Paragraph 3 of the German Aviation Compliance Documentation Act (LuftNaSiG) and with the consent of the Supervisory Board, to increase the issued capital by up to 10% by issuing new shares in return for payment in cash and without subscription rights for existing shareholders. The issue price for the new shares must be determined subject to the agreement of the Supervisory Board and may not be significantly lower than the market price. The authorisation may only be made use of insofar as this is necessary in order to achieve the non-applicability of the conditions stipulated in Section 4 Paragraph 3 LuftNaSiG.

The Executive Board is authorised, according to Section 5 Paragraph 2 LuftNaSiG and subject to the approval of the Supervisory Board, to require shareholders to sell some or all of their shares and to provide the Company with proof of this sale without delay insofar as this is necessary for compliance with the requirements for the maintenance of air traffic rights and in the sequence prescribed in Section 5 Paragraph 3 LuftNaSiG, subject to an appropriate time limit and while indicating the legal consequence which would otherwise be possible of the loss of their shares in accordance with Section 5 Paragraph 7 LuftNaSiG.

CONTINGENT CAPITAL

A resolution of the Annual General Meeting on 5 May 2020 increased the Company's contingent capital by up to EUR 122,417,728. The contingent capital increase serves to provide shares to the holders or creditors of conversion and/or option rights from convertible bonds that may be issued by the Company or its Group companies until 4 May 2025. In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board.

A resolution of the extraordinary general meeting on 25 June 2020 increased the contingent capital of Deutsche Lufthansa AG by up to EUR 102,014,776.32. The contingent capital increase serves to provide shares for the exercise of conversion rights granted to the Economic Stabilisation Fund created by the Stabilisation Fund Act as a silent shareholder of the Company for Silent Participation II-A at a strike rate of EUR 2.56 per share by resolution of the extraordinary general meeting on 25 June 2020. The rights can be exercised if a decision is published to make a takeover offer pursuant to Section 10 of the German Securities Acquisition and Takeover Act (WpÜG) or if control is acquired pursuant to Sections 35 and 29 WpÜG. The buyer can exercise the conversion rights at any time if Silent Participation II-A is sold to a private purchaser.

A resolution of the extraordinary general meeting on 25 June 2020 increased the contingent capital of Deutsche Lufthansa AG by up to EUR 897,985,223.68. The contingent capital increase serves to provide up to 350,775,478 shares for the exercise of conversion rights granted to the Economic Stabilisation Fund created by the Stabilisation Fund Act as a silent shareholder for antidilution and/or coupon protection for Silent Participation II-B by resolution of the extraordinary general meeting on 25 June 2020. If the conversion right is exercised to protect against dilution, the new shares will be issued at the current market price on the conversion date, less 10%. If the conversion right is exercised to protect the coupon, the

shares are issued at the current market price on the conversion date, less 5.25%. The conversion rights expire if Silent Participation II-B is assigned to a third party.

A resolution of the Annual General Meeting on 4 May 2021 increased the Company's contingent capital by up to EUR 153,022,161.92. The contingent capital increase serves to provide shares to the holders or creditors of conversion and/or option rights from convertible bonds that may be issued by the Company or its Group companies until 3 May 2026. In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board.

AUTHORISATION TO PURCHASE TREASURY SHARES

A resolution passed at the Annual General Meeting held on 7 May 2019 authorised the Executive Board pursuant to Section 71 Paragraph 1 No. 8 of the German Stock Corporation Act (AktG) to purchase treasury shares until 6 May 2024. The authorisation is limited to 10% of current issued capital, which can be purchased on the stock exchange or by a public purchase offer to all shareholders. The authorisation states that the Executive Board can use the shares, in particular, for the purposes defined in the resolution passed at the Annual General Meeting. According to the resolution of the Annual General Meeting held on 7 May 2019, the Executive Board is also authorised to purchase treasury shares by means of derivatives and to conclude corresponding derivative transactions.

9 Segment reporting

Segmentation has not been changed compared with the financial statements as of 31 December 2020.

SEGMENT INFORMATION FOR THE REPORTING SEGMENTS Jan - Jun 2021

	Network Airlines	Eurowings	Logistics	MRO	Catering	Total reportable operating segments	Additional Business and Group Functions	Reconciliation	Group
in €m									
External revenue	2,016	159	1,658	1,361	435	5,629	142	–	5,771
of which traffic revenue	1,628	156	1,595	–	–	3,379	–	258	3,637
Inter-segment revenue	272	-1	13	356	12	652	73	-725	–
Total revenue	2,288	158	1,671	1,717	447	6,281	215	-725	5,771
Other operating income	346	40	32	174	139	731	1,023	-1,066	688
Operating income	2,634	198	1,703	1,891	586	7,012	1,238	-1,791	6,459
Operating expenses	5,062	409	1,073	1,779	561	8,884	1,402	-1,796	8,490
of which cost of materials	1,886	157	734	902	153	3,832	90	-718	3,204
of which staff cost	1,486	78	180	556	285	2,585	327	-2	2,910
of which depreciation and amortisation	786	101	70	89	40	1,086	59	-20	1,125
of which other operating expenses	904	73	89	232	83	1,381	926	-1,056	1,251
Result of equity investments	-22	-41	10	-10	-8	-71	6	1	-64
of which result of investments accounted for using the equity method	-19	-41	9	-11	-9	-71	–	–	-71
Adjusted EBIT¹⁾	-2,450	-252	640	102	17	-1,943	-158	6	-2,095
Reconciliation items	-1	1	3	-1	-22	-20	-1	2	-19
Impairment losses/gains	-8	-2	1	1	1	-7	–	1	-6
Effects from pension provisions	3	–	–	–	–	3	1	-1	3
Result of disposal of assets	4	3	2	-2	-23	-16	-2	2	-16
EBIT	-2,451	-251	643	101	-5	-1,963	-159	8	-2,114
Other financial result									-120
Profit/loss before income taxes									-2,234
Capital employed ²⁾	10,859	1,095	2,195	3,221	915	18,285	4,653	-204	22,734
of which from investments accounted for using the equity method	–	36	60	176	80	352	1	–	353
Segment capital expenditure	528	248	28	39	8	851	20	-252	619
of which from investments accounted for using the equity method	–	–	–	7	–	7	–	–	7
Number of employees at the end of period	55,508	3,227	4,216	21,467	15,288	99,706	8,366	–	108,072

¹⁾ For detailed reconciliation from EBIT to Adjusted EBIT see table "reconciliation of results", p. 9, in the interim management report.

²⁾ The capital employed results from total assets adjusted for non-operating items, (deferred taxes, positive market values, derivatives) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

SEGMENT INFORMATION FOR THE REPORTING SEGMENTS Jan - Jun 2020

	Network Airlines	Eurowings	Logistics	MRO	Catering	Total reportable operating segments Segments	Additional Busi- nesses and Group Functions	Recon- ciliation	Group
in €m									
External revenue	4,241	366	1,305	1,606	651	8,169	166	–	8,335
of which traffic revenue	3,858	361	1,219	–	–	5,438	–	203	5,641
Inter-segment revenue	290	11	15	674	163	1,153	83	-1,236	–
Total revenue	4,531	377	1,320	2,280	814	9,322	249	-1,236	8,335
Other operating income	451	75	31	184	33	774	920	-742	952
Operating income	4,982	452	1,351	2,464	847	10,096	1,169	-1,978	9,287
Operating expenses	7,377	746	1,089	2,554	1,029	12,795	1,288	-2,014	12,069
of which cost of materials	3,277	427	702	1,346	345	6,097	118	-1,088	5,127
of which staff cost	1,880	93	188	648	446	3,255	358	-1	3,612
of which depreciation and amortisation	956	104	78	100	61	1,299	58	-36	1,321
of which other operating expenses	1,264	122	121	460	177	2,144	754	-889	2,009
Result of equity investments	-21	-64	15	-32	-13	-115	-3	1	-117
of which result of investments ac- counted for using the equity method	-19	-64	7	-33	-13	-122	–	1	-121
Adjusted EBIT¹⁾	-2,416	-358	277	-122	-195	-2,814	-122	37	-2,899
Reconciliation items	-270	-74	-19	-71	-111	-545	-7	-17	-569
Impairment losses/gains	-268	-73	-19	-65	-111	-536	1	-17	-552
Effects from pension provisions	–	–	–	-2	–	-2	-6	–	-8
Result of disposal of assets	-2	-1	–	-4	–	-7	-2	–	-9
EBIT	-2,686	-432	258	-193	-306	-3,359	-129	20	-3,468
Other financial result									-951
Profit/loss before income taxes									-4,419
Capital employed ²⁾	11,721	1,191	2,167	5,110	1,300	21,489	1,548	-571	22,466
of which from investments ac- counted for using the equity method	35	115	56	181	115	502	5	-1	506
Segment capital expenditure	664	50	89	71	21	895	32	-25	902
of which from investments ac- counted for using the equity method	–	–	–	4	–	4	–	–	4
Number of employees at the end of pe- riod	59,953	3,219	4,452	23,927	28,130	119,681	9,675	–	129,356

¹⁾ For detailed reconciliation from Adjusted EBIT to EBIT see table reconciliation of results, p. 9, in the interim management report.

²⁾ The capital employed results from total assets adjusted for non-operating items (deferred taxes, positive market values, derivatives), less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

EXTERNAL REVENUE BY REGION Jan - Jun 2021

	Europe	thereof Germany	North America	thereof USA	Central and South America	Asia/ Pacific	Middle East	Africa	Group
in €m									
Traffic revenue ¹⁾	2,220	1,154	445	416	106	698	72	96	3,637
Other operating revenue	865	366	769	688	62	303	88	47	2,134
Total revenue	3,085	1,520	1,214	1,104	168	1,001	160	143	5,771

¹⁾ Allocated according to the original location of sale.

EXTERNAL REVENUE BY REGION Jan - Jun 2020

in €m	Europe	thereof Germany	North America	thereof USA	Central and South America	Asia/ Pacific	Middle East	Africa	Group
Traffic revenue ¹⁾	3,685	1,676	786	714	133	839	96	102	5,641
Other operating revenue	1,127	394	838	671	112	431	112	74	2,694
Total revenue	4,812	2,070	1,624	1,385	245	1,270	208	176	8,335

¹⁾ Allocated according to the original location of sale.

10 Related party disclosures

As stated in [Note 50](#) to the consolidated financial statements 2020 (Annual Report 2020, p. 225 ff.), the segments in the Lufthansa Group render numerous services to related parties within the scope of their ordinary business activities and also receive services from them. These extensive supply and service relationships take place unchanged on the basis of market prices. There have been no significant changes in comparison with the reporting date. The contractual relationships with the group of related parties described in the [Remuneration Report 2020](#) (Annual Report 2020) (p. 251 ff.) and in the consolidated financial statements 2020 in [Note 51](#) (Annual Report 2020, p. 228) also still exist unchanged, but are not of material significance for the Group.

11 Published standards that have not yet been applied

Amendments of accounting standards which have been approved by the IASB as of the date of publication of this report and are applicable for financial years beginning after 1 January 2021 have no effect on the presentation of the net assets, financial and earnings position. Further information on the amendments resolved as of the preparation of the Annual Report is provided in the consolidated financial statements 2020 in [Note 3](#) "New international accounting standards in accordance with IFRS and interpretations" to the consolidated financial statements 2020 (Annual Report 2020, p. 143 ff.).

***Audited Consolidated Financial Statements of Deutsche Lufthansa Aktiengesellschaft as of
and for the financial year ended December 31, 2020 (IFRS)***

Consolidated income statement for the financial year 2020

T056 CONSOLIDATED INCOME STATEMENT

in €m	Notes	2020	2019
Traffic revenue	4	9,078	28,136
Other revenue	5	4,511	8,288
Total revenue		13,589	36,424
Changes in inventories and work performed by entity and capitalised	6	175	685
Other operating income ¹⁾	7	1,897	1,889
Cost of materials and services	8	-8,453	-19,827
Staff costs	9	-6,436	-9,121
Depreciation, amortisation and impairment ²⁾	10	-4,389	-2,776
Other operating expenses ³⁾	11	-3,472	-5,585
Profit/loss from operating activities		-7,089	1,689
Result of equity investments accounted for using the equity method	12	-276	88
Result of other equity investments	12	12	80
Interest income	13	83	79
Interest expenses	13	-417	-394
Other financial items	14	-944	318
Financial result		-1,542	171
Profit/loss before income taxes		-8,631	1,860
Income taxes	15	1,865	-615
Profit/loss after income taxes		-6,766	1,245
Profit/loss attributable to non-controlling interests		41	-32
Net profit/loss attributable to shareholders of Deutsche Lufthansa AG		-6,725	1,213
Basic/diluted earnings per share in €	16	-12.51	2.55

¹⁾ This includes EUR 36m (previous year: EUR 46m) from the reversal of write-downs on non-current receivables.

²⁾ This includes EUR 19m (previous year: EUR 23m) for the recognition of write-downs on current receivables.

³⁾ This includes EUR 251m (previous year: EUR 120m) for the recognition of loss allowances on receivables.

Statement of comprehensive income for the financial year 2020

T057 STATEMENT OF COMPREHENSIVE INCOME

in €m	2020	2019
Profit/loss after income taxes	-6,766	1,245
Other comprehensive income		
Other comprehensive income with subsequent reclassification to the income statement		
Differences from currency translation	-107	115
Subsequent measurement of financial assets and liabilities at fair value without effect on profit and loss	-29	10
Subsequent measurement of hedges – cash flow hedge reserve	-655	258
Subsequent measurement of hedges – costs of hedging	143	476
Other comprehensive income from investments accounted for using the equity method	4	13
Other expenses and income recognised directly in equity	-4	15
Income taxes on items in other comprehensive income	125	-182
	-523	705
Other comprehensive income without subsequent reclassification to the income statement		
Revaluation of defined-benefit pension plans	-2,584	-1,108
Other comprehensive income from investments accounted for using the equity method	-	-
Subsequent measurement of financial assets at fair value	1	7
Other expenses and income recognised directly in equity	-	-1
Income taxes on items in other comprehensive income	622	345
	-1,961	-757
Other comprehensive income after income taxes	-2,484	-52
Total comprehensive income	-9,250	1,193
Comprehensive income attributable to non-controlling interests	48	-31
Comprehensive income attributable to shareholders of Deutsche Lufthansa AG	-9,202	1,162

Further details on the statement of comprehensive income can be found in [Note 34, Reserves, p. 179f.](#)

Consolidated statement of financial position

as of 31 December 2020

T058 CONSOLIDATED STATEMENT OF FINANCIAL POSITION – ASSETS

in €m	Notes	31 Dec 2020	31 Dec 2019
Intangible assets with an indefinite useful life ¹⁾	17	1,169	1,395
Other intangible assets	18	469	547
Aircraft and reserve engines	19 22	15,842	18,349
Repairable spare parts for aircraft	20	1,823	2,270
Property, plant and other equipment	21 22	3,671	4,041
Investments accounted for using the equity method	23	403	672
Other equity investments	24 45	252	256
Non-current securities	24 45	54	53
Loans and receivables	25 45	440	469
Derivative financial instruments	45	363	906
Deferred charges and prepaid expenses	29	91	116
Effective income tax receivables		34	32
Deferred tax assets	15	4,833	2,268
Non-current assets		29,444	31,374
Inventories	26	726	980
Contract assets	27	142	277
Trade receivables and other receivables	28 45	2,843	5,417
Derivative financial instruments	45	260	459
Deferred charges and prepaid expenses	29	193	245
Effective income tax receivables		282	153
Securities	30 45	3,654	1,970
Cash and cash equivalents	31 45	1,806	1,415
Assets held for sale	32	134	369
Current assets		10,040	11,285
Total assets		39,484	42,659

¹⁾ Including Goodwill.

T059 CONSOLIDATED STATEMENT OF FINANCIAL POSITION – SHAREHOLDERS' EQUITY AND LIABILITIES

in €m	Notes	31 Dec 2020	31 Dec 2019
Issued capital	33	1,530	1,224
Capital reserve	34	378	378
Retained earnings	34	4,868	5,617
Other neutral reserves	34	1,296	1,715
Net profit/loss		-6,725	1,213
Equity attributable to shareholders of Deutsche Lufthansa AG		1,347	10,147
Non-controlling interests		40	109
Shareholders' equity		1,387	10,256
Pension provisions	35	9,531	6,659
Other provisions	36	558	490
Borrowings	37 45	12,252	8,396
Contract liabilities	38	36	25
Other financial liabilities		86	76
Advance payments received, deferred income and other non-financial liabilities	39	33	32
Derivative financial instruments	45	457	128
Deferred tax liabilities	15	485	611
Non-current provisions and liabilities		23,438	16,417
Other provisions	36	831	794
Borrowings	37 45	3,116	1,634
Trade payables and other financial liabilities	41 45	3,321	5,351
Liabilities from unused flight documents	40	2,064	4,071
Other contract liabilities	40	2,977	2,675
Advance payments received, deferred income and other non-financial liabilities	42	1,295	382
Derivative financial instruments	45	366	137
Effective income tax obligations		689	402
Liabilities in connection with assets held for sale	32	-	540
Current provisions and liabilities		14,659	15,986
Total shareholders' equity and liabilities		39,484	42,659

Consolidated statement of changes in shareholders' equity as of 31 December 2020

T060 CONSOLIDATED STATEMENT OF CHANGES IN SHAREHOLDERS' EQUITY

	Issued capital	Capital reserve	Fair value measurement of financial instruments	Currency differences	Revaluation reserve (due to business combinations)	Other neutral reserves	Total other neutral reserves	Retained earnings	Net profit/loss	Equity attributable to shareholders of Deutsche Lufthansa AG	Minority interests	Total shareholders' equity
in €m												
As of 31 Dec 2018	1,217	343	237	388	236	324	1,185	4,588	2,163	9,496	110	9,606
Capital increases/reductions	7	35	-	-	-	-	-	-	-	42	1	43
Reclassifications	-	-	-9	-	-	-	-9	1,792	-1,783	-	-	-
Dividends	-	-	-	-	-	-	-	-	-380	-380	-33	-413
Transactions with non-controlling interests	-	-	-	-	-	-	-	-	-	-	-	-
Earnings after taxes	-	-	-	-	-	-	-	-	1,213	1,213	32	1,245
Other expenses and income recognised directly in equity	-	-	569	115	-	28	712	-763	-	-51	-1	-52
Hedging results reclassified from non-financial assets to acquisition costs	-	-	-173	-	-	-	-173	-	-	-173	-	-173
As of 31 Dec 2019	1,224	378	624	503	236	352	1,715	5,617	1,213	10,147	109	10,256
Capital increases/reductions	306	-	-	-	-	-	-	-	-	306	-	306
Reclassifications	-	-	-	-	-	-	-	1,213	-1,213	-	-	-
Dividends	-	-	-	-	-	-	-	-	-	-	-21	-21
Transactions with non-controlling interests	-	-	-	-	-	-	-	-	-	-	-	-
Earnings after taxes	-	-	-	-	-	-	-	-	-6,725	-6,725	-41	-6,766
Other expenses and income recognised directly in equity	-	-	-415	-107	-	7	-515	-1,962	-	-2,477	-7	-2,484
Hedging results reclassified from non-financial assets to acquisition costs	-	-	96	-	-	-	96	-	-	96	-	96
As of 31 Dec 2020	1,530	378	305	396	236	359	1,296	4,868	-6,725	1,347	40	1,387

Consolidated cash flow statement for the financial year 2020

T061 CONSOLIDATED CASH FLOW STATEMENT

in €m	Notes	2020	2019
Cash and cash equivalents 1 Jan		1,431	1,434
Net profit/loss before income taxes		-8,631	1,860
Depreciation, amortisation and impairment losses on non-current assets (net of reversals)		4,393	2,738
Depreciation, amortisation and impairment losses on current assets (net of reversals)		159	99
Net proceeds on disposal of non-current assets	7 11	9	20
Result of equity investments	12	264	-168
Net interest	13	334	315
Income tax payments/reimbursements		81	-1,009
Significant non-cash-relevant expenses/income		212	-134
Change in trade working capital		-683	490
Change in other assets/shareholders' equity and liabilities		1,534	-181
Net cash from/used in operating activities		-2,328	4,030
Capital expenditure for property, plant and equipment and intangible assets	17 18 19 21 22	-1,249	-3,486
Capital expenditure for financial investments	24 25	-24	-73
Additions/loss to repairable spare parts for aircraft	20	276	-231
Proceeds from disposal of non-consolidated equity investments		3	1
Proceeds from disposal of consolidated equity investments ¹⁾	49	-218	3
Cash outflows for acquisitions/capital increases of/at non-consolidated equity investments	23 24 45	-39	-107
Cash outflows for acquisitions of consolidated equity investments	49	-	-
Proceeds from disposal of intangible assets, property, plant and equipment and other financial investments		166	132
Interest income		66	70
Dividends received		57	243
Net cash from/used in investing activities		-962	-3,448
Purchase of securities/fund investments		-10,678	-4,989
Disposal of securities/fund investments		9,298	4,570
Net cash from/used in investing and cash management activities		-2,342	-3,867
Capital increase	33 34	306	-
Transactions by non-controlling interests		-	1
Non-current borrowing		6,738	3,843
Repayment of non-current borrowing		-1,724	-3,413
Dividends paid		-21	-414
Interest paid		-223	-178
Net cash from/used in financing activities		5,076	-161
Net increase/decrease in cash and cash equivalents		406	2
Changes due to currency translation differences		-33	-5
Cash and cash equivalents as of 31 Dec	31	1,804	1,431
Less cash and cash equivalents of companies held for sale as of 31 Dec		-	16
Cash and cash equivalents of companies not classified as held for sale as of 31 Dec²⁾		1,804	1,415
Securities	30	3,654	1,970
Liquidity		5,458	3,385
Net increase/decrease in liquidity		2,073	216

¹⁾ Less cash outflows.

²⁾ The difference between the bank balances and cash-in-hand shown in the statement of financial position comes from fixed-term deposits of EUR 2m with terms of four to twelve months (previous year: EUR 0m).

Further details can be found in the section [Notes to the consolidated cash flow statement, p. 203f.](#)

Notes to the consolidated financial statements

Deutsche Lufthansa AG 2020

GENERAL REMARKS

1 Company information

The Lufthansa Group is a global aviation group whose subsidiaries and equity investments were organised into five operating segments in the financial year 2020: Network Airlines, Eurowings, Logistics, MRO and Catering.

Deutsche Lufthansa AG has its headquarters in Cologne, Germany, and is filed in the Commercial Register of Cologne District Court under HRB 2168.

The declaration on the German Corporate Governance Code required by Section 161 of the German Stock Corporation Act (AktG) was issued and made available to shareholders on the internet at www.lufthansagroup.com/declaration-of-compliance.

The consolidated financial statements of Deutsche Lufthansa AG, Cologne, and its subsidiaries have been prepared in accordance with the International Financial Reporting Standards (IFRS) issued by the International Accounting Standards Board (IASB), taking account of interpretations by the IFRS Interpretations Committee as applicable in the European Union (EU).

The commercial law provisions of Section 315e Paragraph 1 of the German Commercial Code (HGB) have also been applied. All IFRSs issued by the IASB in effect at the time that these financial statements were prepared and applied by Deutsche Lufthansa AG have been adopted by the European Commission for application in the EU. The consolidated financial statements of Deutsche Lufthansa AG are prepared in millions of euros. Its financial year is the calendar year.

The accounting policies applied in the previous year have been retained. The amendments to IFRS 9, IAS 39, IFRS 7, IFRS 4 and IFRS 16, "Interest Rate Benchmark Reform – Phase 2", were applied early in financial year 2020.

The Executive Board of Deutsche Lufthansa AG prepared the consolidated financial statements for 2020 on 26 February 2021. These consolidated financial statements were examined and approved for publication by the Supervisory Board of Deutsche Lufthansa AG in its meeting on 3 March 2021.

2 Going concern

The business of the Lufthansa Group companies were impacted significantly by the consequences of the coronavirus pandemic. International travel restrictions and quarantine rules that began in winter 2020 (in China) and spring 2020 elsewhere were tightened worldwide over the course of the year, bringing flight traffic at the Lufthansa Group companies almost to a complete standstill at times. Only the cargo business was able to continue successfully, with a sharp decline in the use of belly capacities. Around 700 Group aircraft were grounded temporarily in the course of the year. The slight recovery in summer 2020, when available capacity was around 25% of the previous year's, was slowed by the end of the year due to the renewed sharp increase in the number of infections. Payments for existing, invariable cost components and the high level of refunds for cancelled flights had an impact on liquidity. Liquidity at Deutsche Lufthansa AG and its domestic and foreign subsidiaries was secured in the middle of the financial year in an initial step by the government support measures as part of the stabilisation packages (see below).

Over the remainder of the year, the companies initiated and partly implemented wide-ranging steps to safeguard short-term liquidity and initiate a medium-term return to a profitable new normal once the pandemic comes to an end. As part of the Group-wide programme ReNew, which is made up of ReStructure, ReFocus, ReOrg and RePay, measures were taken to streamline structures and reduce the number of full-time employees by around 27,000, as well as to downsize the operating fleet by around 150 aircraft. Until further notice, some four-engined wide-bodied models (Airbus A340-600 and A380 and Boeing 747-400) are to be put into deep storage or decommissioned entirely. Previously planned investment spending, mainly in aircraft, was postponed to a later date whenever possible and appropriate. The plan is to take delivery of no more than 80 new aircraft by 2023. In addition to state aid in the form of short-time working pay and the reimbursement of social security contributions, crisis agreements extending into 2022 were reached with the collective bargaining partners to safeguard liquidity. Agreements to protect liquidity were also signed with other system partners. The Lufthansa Group's liquidity management also profited from tax deferrals in the reporting year.

In addition to the government stabilisation packages, the Lufthansa Group successfully implemented its own financing activities in the financial year, generating proceeds of EUR 3.8bn for the Lufthansa Group. Short-term credit lines, commercial paper and money-market securities were supplemented on favourable terms by a euro bond for EUR 1.0bn, a convertible bond for EUR 0.6bn and loans secured by promissory notes and aircraft for around EUR 0.7bn. They serve both to ensure ongoing liquidity and to refinance funding arrangements that expire in 2020 and 2021. At year-end, Deutsche Lufthansa AG had centrally available liquidity of EUR 4.3bn. A further EUR 4.6bn was available from the WSF stabilisation package. Decentralised bank and cash balances came to a further EUR 0.9bn, and a total of EUR 1.1bn has not yet been used from the state funding agreed in Switzerland and Belgium.

The current corporate forecast expect an end to the crisis situation only from 2024 onwards, whereby the Lufthansa Group assumes that even then only 90 % of the pre-crisis level will be reached. This means the planning is in line with sector forecasts and the most recent IATA study from February 2021, which in a positive scenario also predicts a return to pre-crisis levels in 2024 (and in a negative scenario only in 2025). The restructuring measures initiated shall then lead to a profitable growth path again. Risks remain for the ramp-up phase from 2021 until 2024. The duration of the crisis will mainly depend on when travel restrictions, particularly in North America and other important traffic regions, are loosened or lifted. The effects of potential testing options, progress with international vaccination programmes and the risks of new mutations in the COVID-19 virus will be crucial in the months ahead. On the other hand, how customers behave once travel restrictions are lifted represents a risk for the further course of business, especially for Corporate Business.

Deutsche Lufthansa AG has prepared a rolling liquidity forecast until the end of 2022 on the basis of the uncertainty mentioned above concerning travel restrictions and customers' travel patterns. The main parameters, "available capacity", "load factors" and "yield" were set over the course of the planning process in agreement with the Group companies. Specifically, they assume that average capacity should reach just under 50% of its pre-crisis level in 2021, increasing to around 80% in 2022. For 2021, the assumption is that state aid will still be needed, in the form of short-time working pay and the reimbursement of social security contributions. The parameters were reviewed against publicly available expert opinions (for example IATA, Eurocontrol) to ensure they were reasonable. On this basis, the earnings and cost forecasts

for the individual units and the Group as a whole were determined using top-down definition of parameters and bottom-up detailed planning. The resulting cash flows were adjusted for balance-sheet-related payments (for example ticket refunds), planned capital expenditure and inflows such as the planned financing measures to calculate changes in liquidity over time. The analysis comprises liquidity for the Group as a whole and for Deutsche Lufthansa AG, as well as the other important subsidiaries.

In terms of financing activities, two further bonds with a maturity of four and seven years were issued for a total of EUR 1.6bn in February 2021. They partly secure the refinancing of all liabilities falling due in 2021 and also enable the early repayment of the KfW credit facility of EUR 1.0bn. This means that certain restrictions on the Group's own financing arrangements no longer apply, such as the hypothecation of Group aircraft leasing companies and the ringfencing conditions on providing financial support to foreign subsidiaries.

In addition to the detailed forecast described above, an additional negative scenario was prepared as a top-down stress test, with more pessimistic assumptions for the level of capacity and changes over time of around 40% of pre-crisis levels, as well as for load factors and pricing. Even in this scenario, liquidity is secured for Deutsche Lufthansa AG and the Group for the planning period.

Taking into account the corporate planning, the liquidity planning, the existing and potential funding measures as presented and the existing uncertainties about the future course of business, the Executive Board of the Company considers the Group's liquidity to be secure for the eighteen months after the close of the financial year. The consolidated financial statements have therefore been prepared on a going concern basis.

The government stabilisation measures described below, which were provided in the reporting year and are still available for financing purposes, are the essential basis for this statement.

After approval by the Federal Republic of Germany's Economic Stabilisation Fund (WSF) of the stabilisation package for Deutsche Lufthansa AG on 25 May 2020 and the approval of the shareholders of Deutsche Lufthansa AG at the extraordinary general meeting held on 25 June 2020 regarding the capital measures agreed with the Economic Stabilisation Fund, the corresponding agreements with the Economic Stabilisation Fund were signed on 29 June 2020 and with KfW and a banking consortium on 1 July 2020.

The support package has a total volume of up to EUR 9bn and comprises a 20% direct equity investment by WSF in Deutsche Lufthansa AG from a capital increase of around EUR 0.3bn, a silent participation which can be recognised as equity worth EUR 4.7bn (Silent Participation I), a silent participation reported under borrowed capital worth EUR 1bn (Silent Participation II) and a credit facility of EUR 3bn provided by the KfW and a banking consortium. Funding commitments from other governments are offset against the support package from WSF. These commitments came to EUR 2.1bn as of 31 December 2020, so that the total volume of state aid provided by the German government is EUR 6.9bn.

Certain parts of Silent Participation II can be converted into shares in the event of a takeover, a dilution of the WSF stake or if no coupon payments are made on Silent Participation I. Annual coupon payments with an increasing rate of interest over the term have been agreed for the silent participations. Silent Participation I has no fixed term. Silent Participation II has a term of six years and can be extended annually thereafter, as long as Silent Participation I has not been repaid and the shareholding has not come to an end.

The capital increase agreed in the package took effect when it was entered in the commercial register on 2 July 2020. The subscription price was EUR 2.56 per share, so the cash contribution came to around EUR 0.3bn. Silent Participation II was paid out in full by the WSF at the same time.

None of Silent Participation I had been utilised as of 31 December 2020. Funding of EUR 153m agreed in Belgium and Austria is offset against Silent Participation I; EUR 4.5bn is still available from Silent Participation I.

The framework agreement with the Economic Stabilisation Fund provides for extensive information and auditing rights for the Economic Stabilisation Fund and obligations for the Lufthansa Group including regarding the suspension of dividend payments, limitations on management compensation, a commitment not to make equity investments, waiver of up to 24 slots at both the Frankfurt and Munich airports and pursuit of a sustainable corporate policy. Subject to the full repayment of the silent participations by the Company and a minimum sale price of EUR 2.56 per share plus an annual interest of 12%, the WSF undertakes to sell its shareholding in full at the market price by 31 December 2023 or after this point in time when the conditions have been fulfilled.

The credit line, originally of up to EUR 3bn, from the KfW and the banking consortium will pay interest at a floating rate dependent on the rating and has a term of three years. A total of EUR 1.0bn was used from the KfW credit facility as of the reporting date. (On Events after the reporting period

➤ **Note 46, p. 220ff.)** Funding involving state aid for the subsidiaries in Belgium, Austria and Switzerland, EUR 2.0bn is offset against the credit facility from the KfW and the banking consortium; the total volume of this credit line comes to EUR 1.0bn. The credit facility is secured by shares in various entities in Malta and Austria that own a total of 323 aircraft. The loan agreement contains covenants and obligations for the Lufthansa Group, including a requirement to hold a minimum amount of liquidity. The credit facility may not be used to finance SWISS, Edelweiss, Austrian Airlines or Brussels Airlines. At the same time, the state aid from Belgium, Austria and Switzerland is provided on condition that the funding must remain in bank accounts belonging to the Group companies in these countries or may only be used to support these companies.

The Swiss federal government guaranteed credit lines of CHF 1.5bn (approximately EUR 1.4bn) for the two Swiss airlines in the Lufthansa Group, SWISS and Edelweiss, on 29 April 2020. The facility is provided by a consortium of Swiss banks. It will pay interest at a floating rate depending on the rating and has a term of five years, with two one-year extension options. It is 85% guaranteed by the Swiss federal government and secured by shares in AirTrust AG, Swiss International Air Lines AG and Edelweiss Air AG that are held directly by Deutsche Lufthansa AG. The corresponding loan agreement was signed on 20 August 2020. It stipulates that the loans may only be used for SWISS and Edelweiss and that no dividend payments may be made by the borrowers for as long as they make use of the credit facility. A number of bank accounts were pledged as collateral in this context, but may still be used to handle payments as long as the obligations under the framework agreement are met. Furthermore, the companies are obliged to abide by covenants relating to minimum liquidity, minimum equity and future minimum earnings. As of 31 December 2020, CHF 475m of the credit facility had been used. Deutsche Lufthansa AG made its contribution in the form of a subordinated loan for CHF 500m.

The Austrian federal government, Deutsche Lufthansa AG and Austrian Airlines reached an agreement on 8 June 2020 on the key aspects of a coronavirus aid package worth EUR 600m for Austrian Airlines. The measures provide for equity capital of EUR 300m to be injected into Austrian Airlines, an amount to be split 50/50 by Deutsche Lufthansa AG and the Austrian government, as well as debt financing of EUR 300m. Deutsche Lufthansa AG's contribution of EUR 150m was made by way of a shareholder contribution by Lufthansa to Austrian Airlines via the existing holding structure. The Austrian state's contribution, also of EUR 150m, was structured in a contract on 28 August 2020 as a subordinated loan, repayment of which will be waived if Austrian Airlines demonstrates that it has incurred a loss of at least EUR 150m

as a result of the coronavirus pandemic. Since evidence had already been provided, the amount was recognised in profit and loss as a grant once the funds were received. As regards the debt component, a credit line of EUR 300m was agreed with a consortium of Austrian banks on 25 June 2020, of which the Austrian state has guaranteed 90% in the context of the special Austrian coronavirus legislation. The shares in Austrian Airlines AG, 38 aircraft and shares in an aircraft holding company that owns these aircraft were pledged to the banking consortium as additional collateral for these credit lines. In addition, it was agreed that interest and fees are to be paid a year in advance and that future income from aircraft sales and insurance payments are to be paid to accounts held as collateral. Finally, the contract prohibits Austrian Airlines AG from paying a dividend during the term of the agreement and obliges the company to maintain defined financing ratios and a certain amount of equity. The credit line runs until the end of 2025, is subject to a fixed interest rate and had been used in full as of 31 December 2020.

A stabilisation agreement was signed with the Belgian government on 1 September 2020 for a total of EUR 290m, made up of a credit facility of EUR 287m for SN Airholding SA/NV and a participation certificate for EUR 3m to be issued by Brussels Airlines SA/NV. In connection with this agreement, the Lufthansa Group pledged its shares and participation rights to the Belgian state agency Société fédérale de participations et d'investissement. The agreements also prohibit dividend payments by Brussels Airlines SA/NV, set a cap on management remuneration and provide for coordinated activities to develop the range of flights available from Brussels Airport. The participation certificate and the loan will pay interest at a floating rate with annual increases. As of 31 December 2020, the participation certificate was fully paid in and EUR 130m of the loan had been utilised. The government stabilisation measures were supplemented by an equity injection of EUR 170m from Deutsche Lufthansa AG.

The Coronavirus Aid, Relief and Economic Security Act (CARES Act), which is designed to mitigate the negative economic impact of the COVID-19 pandemic, came into force in the US on 27 March 2020. On 3 April 2020, the LSG group submitted applications to the US Department of the Treasury for the Payroll Support Program feature of the legislation for three companies in an amount totalling USD 307m, with the Lufthansa Technik group submitting applications worth USD 50m for four companies. In the third quarter of 2020, the LSG group was awarded a loan of USD 79m and grants of USD 135m, which were disbursed in the period. The

Lufthansa Technik group was awarded and received a loan of USD 8m and a grant of USD 22m. The support is tied to a commitment to maintain a defined level of employment.

Additional funding was provided in the USA in early 2021 under the CARES Act I along with further funding from the subsequently passed CARES Act II. The LSG group intends to apply for further funding of USD 143m (of which USD 95m would be a grant). The Italian government also provided funds to compensate for losses incurred due the COVID-19 pandemic. Air Dolomiti also applied for a grant of EUR 16m in early 2021.

3 New international accounting standards in accordance with IFRS and interpretations and summary of the significant accounting policies and valuation methods

INTERNATIONAL FINANCIAL REPORTING STANDARDS (IFRS) AND INTERPRETATIONS (IFRIC) TO BE APPLIED FOR THE FIRST TIME IN THE FINANCIAL YEAR AND AMENDMENTS TO STANDARDS AND INTERPRETATIONS

T062 IFRS-PRONOUNCEMENT (APPLICABLE FROM FINANCIAL YEAR 2020)

Changes to conceptual framework
Amendments to IAS 1 and IAS 8, Definition of materiality
Amendments to IFRS 9, IAS 39 and IFRS 7, Interest Rate Benchmark Reform
Amendments to IFRS 3, Definition of a Business
Amendments to IFRS 16, Covid-19-related Rent Concessions

The first-time application from 1 January 2020 or 1 June 2020 of the mandatory accounting standards had no or no material effect on the presentation of the net assets, financial and earnings position or on earnings per share.

Amendments to IFRS 9, IAS 39 and IFRS 7, Interest Rate Benchmark Reform. The amendments aim to reduce the potential impact of reforms to the interbank offered rates (IBORs) on companies' financial reporting. The Lufthansa Group has applied this pronouncement in advance since the last financial year. First-time adoption had no effect on the presentation of the net assets, the financial and earnings position or on earnings per share, but did require additional disclosures in the notes. ➤ **Note 45, p. 205ff.**

Amendments to IFRS 16, Covid-19-related Rent Concessions. As a result of the coronavirus pandemic, lessees were granted lease concessions. The IASB published an amendment to IFRS 16 in May 2020 that includes an optional exemption for lessees. As a practical expedient, the lessee can decide to exempt themselves from assessing whether

a COVID-19-related rent concession is a lease modification. Lessees that apply the exemption are to account for COVID-19-related rent concessions as they would do under IFRS 16 if they were not lease modifications. The Lufthansa Group has decided not to apply the optional exemption for lessees.

PUBLISHED INTERNATIONAL FINANCIAL REPORTING STANDARDS (IFRS) AND INTERPRETATIONS (IFRIC) NOT YET APPLIED/APPLICABLE AND AMENDMENTS TO STANDARDS AND INTERPRETATIONS

The following standards and amendments have already been adopted by the European Union but are only mandatory for financial statements after 31 December 2020:

T063 IFRS-PRONOUNCEMENT (ADOPTED BY THE EU)

	Mandatory application for financial years beginning on or after
Amendments to IFRS 9, IAS 39, IFRS 7, IFRS 4 and IFRS 16, Classification of financial liabilities – Phase 2	1 Jan 2021

Amendments to IFRS 9, IAS 39, IFRS 7, IFRS 4 and IFRS 16, Interest Rate Benchmark Reform – Phase 2. In August 2020 the IASB adopted amendments to the Interest Rate Benchmark Reform (IBOR reform) – Phase 2. The amendments provide for temporary exemptions if a reference interest rate (IBOR – Interbank Offered Rate) is replaced by an alternative, virtually risk-free rate (RFR) and this has an impact on financial reporting.

As a practical expedient, the amendments provide for contract amendments or changes in cash flows required as a direct result of the reform to be accounted for as changes to a variable interest rate, which corresponds to a change in the market rate. This expedient reflects the practical consideration that the transition from a reference interest rate to a risk-free reference interest rate takes place on the same economic basis, without any transfer of value having taken place. All other changes made at the same time are measured, for example a change in the credit spread or the maturity date. Material changes result in the derecognition of the instrument. If they are not material, the updated effective interest rate is used to determine the new carrying amount of the financial instrument, whereby all gains or losses are recognised in the income statement. A similar practical expedient is to be introduced for changes to leases (IFRS 16) resulting from the Interest Rate Benchmark Reform. In addition, the changes permit hedge accounting to be maintained if the changes are only due to the Interest Rate Benchmark Reform.

Permitted changes include changes from the transition to a risk-free rate in the hedged item, the hedging instrument and the hedged risk. IFRS 9 provides for income and expenses to be recognised in the income statement like effects from the ineffective portion of the hedge. Exemptions were also made in the event of separately identifiable risk components. The amendments to IFRS 7, Financial instruments, also require additional disclosures. The amendments are applicable retroactively for financial years beginning on or after 1 January 2021. Early application is allowed. There is no need to adjust figures for prior periods.

The amendments include practical expedients for the Lufthansa Group, particularly for hedging relationships designated as hedge accounting. Hedging relationships may be maintained where there is still uncertainty concerning an alternative interest rate, which is mainly the case for Lufthansa with USD LIBOR. The extent of exposure to the amendments of IFRS 7, Financial instruments, is explained in an additional disclosure. The Lufthansa Group applies the amendments early following their endorsement in January 2021, so that it can continue hedge accounting for its existing hedging relationships. First-time adoption has no effect on the presentation of the net assets, the financial and earnings position or on earnings per share. ➤ **Note 45, p. 205ff.**

The IASB and the IFRS Interpretations Committee have adopted other standards and interpretations whose application is not mandatory for the financial year 2020:

T064 IFRS-PRONOUNCEMENT (NOT YET ENDORSED BY THE EU)

	Anwendungspflicht für Geschäftsjahre beginnend am oder nach
IFRS 17, Insurance Contracts	1 Jan 2023
Amendments to IFRS 3, Reference to the conceptual framework	1 Jan 2022
Amendments to IAS 16, Generation of revenue before an asset's intended use	1 Jan 2022
Amendments to IAS 37, Onerous Contracts – Cost of Fulfilling a Contract	1 Jan 2022
Annual improvement process (2018 – 2020), Amendments to IFRS 1, IFRS 9 and IAS 41	1 Jan 2022
Amendments to IAS 1, Classification of Liabilities as Current or Non-current.	1 Jan 2023

Currently, the new or amended IFRS pronouncements listed in the table which are no longer discussed below are not considered to have a material effect on the presentation of the net assets, financial and earnings position.

Amendments to IAS 37, Onerous Contracts – Cost of Fulfilling a Contract. The IASB published amendments to IAS 37, Provisions, Contingent Liabilities and Contingent Assets, Onerous Contracts – Cost of Fulfilling a Contract, in May 2020 to clarify that the costs of fulfilling a contract include all directly attributable costs. They include the additional costs of fulfilling a contract such as direct costs of labour or materials and the inclusion of other costs that relate directly to fulfilling contracts. General and administrative expenses do not relate directly to the contract and so are not costs of fulfilling a contract, unless the contract specifically provides for them to be charged on to the customer. The amendments are applicable for financial years beginning on or after 1 January 2022. The Group will apply these amendments to contracts in which not all obligations have been fulfilled at the beginning of the financial year in which the amendments are applied for the first time. We do not expect any material effects.

Amendments to IAS 1, Classification of Liabilities as Current or Non-current. In January 2020 the IASB published amendments to IAS 1, Presentation of Financial Statements, to clarify the guidance for classifying liabilities as current or non-current. The amendments clarify that the classification of liabilities depends on the rights in existence at the end of the reporting period. Classification is unaffected by the expectations of management or any events. What is meant by the “settlement” of a liability in IAS 1 is also clarified. The amendments are applicable retroactively for financial years beginning on or after 1 January 2023. The Group is presently reviewing the effects the amendments will have on its current accounting practice.

Apart from the amendments to IFRS 9, IAS 39, IFRS 7, IFRS 4 and IFRS 16, Interest Rate Benchmark Reform – Phase 2, the Lufthansa Group has not applied any of the aforementioned new or amended standards in advance. If the effective dates of the standards and interpretations mentioned above fall within the year, they are applied as of 1 January of the following financial year. This is subject to the endorsement of the standards by the EU.

SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES AND VALUATION METHODS

The companies included in the consolidated financial statements use uniform accounting policies to prepare their financial statements.

The application of the accounting policies prescribed by IFRS and IFRIC requires making a large number of estimates and assumptions with regard to the future that may, naturally, not coincide with actual future conditions. All of these estimates and assumptions are, however, reviewed continuously and are based either on past experience and/or expectations

of future events that seem reasonable in the circumstances on the basis of sound business judgement. Estimates and assumptions that are of material importance in determining the carrying amounts for assets and liabilities are explained in the following description of the accounting policies applied to material items in the statement of financial position.

The outbreak of the coronavirus pandemic and the steps taken worldwide to contain the virus have had a massive impact on the Group's business operations. This is reflected throughout the Lufthansa Group's consolidated financial statements. The uncertainties resulting from the crisis are vital for the general assessment of the Company's status as a going concern, but also for specific accounting judgements and estimates. It is therefore of fundamental importance how long and at what intensity the COVID-related restrictions to air travel and the constraints for the economy as a whole will continue, and over which period a recovery will take place. In addition, it is hard to foresee what the new normal level will be for air transport in general and the Lufthansa Group in particular. The main assumptions and estimates were therefore based on the Group's liquidity and profit forecasts (➤ **Note 2, p. 140ff.**). Critical accounting areas that will be affected most severely by the ongoing uncertainty about the further course of the pandemic are:

- Carrying amounts of goodwill (➤ **Note 17, p. 165f.**) and equity investments (➤ **Note 23, p. 174f.**), which depend to a large degree on the speed of the recovery and the level of post-crisis business.
- Carrying amount of the aircraft (➤ **Note 19, p. 170**), which are affected by uncertainties regarding future capacities and so the extent to which the fleet can be used.
- Repairable spare parts (➤ **Note 20, p. 171**) and inventories (➤ **Note 26, p. 176f.**) require measurement assumptions about the extent to which lasting overcapacities will result from reducing particular fleet models or from a general long-term decline in business in the MRO segment.
- Since the assumption is that the earnings position will only recover slowly and first result in sustainably positive taxable profits once the pandemic is over, the measurement of carrying amount for deferred tax assets (➤ **Note 15, p. 163f.**), especially the tax loss carry-forwards created in the financial year, was based on their potential use over a longer time frame.
- Trade receivables (➤ **Note 28, p. 177f.**), particularly in the MRO segment, were subject to a greater default risk due to the airlines' difficult financial situation. Estimates had to be made here about the probability of future defaults, for which credit scores based on past experience were only of limited use.

- For financial instruments (➤ **Note 45, p. 205ff.**), in order to judge effective hedging relationships and so the necessity of recognising gains and losses through profit or loss, assumptions were made about the volume of future items to be hedged (particularly kerosene consumption).
- Accounting for obligations under customer loyalty programmes and unused flight documents (➤ **Note 40, p. 196**) was subject to greater uncertainty about how customers would redeem miles or use tickets, which will also depend on when and to what extent air travel is possible again. These estimates play a role in the measurement of miles accounts and forecasts of when miles and tickets are likely to expire.

The fundamental valuation method applied in the consolidated financial statements is the acquisition cost principle. Where IFRSs stipulate that other methods of measurement should be applied, these are used instead, and are referred to specifically in the following comments on measuring assets and liabilities.

Amendments to accounting policies as a result of revised and new standards are applied retrospectively unless provided otherwise for a specific standard. The income statement for the previous year and the opening statement of financial position for the comparable period are adjusted as if the new accounting policies had always been applied.

Recognition of income and expenses

Revenue and other operating income are recognised when the service has been provided.

Passenger transport and ancillary services

The Lufthansa Group sells flight tickets and related ancillary services primarily via agents, its own websites or other airlines in the case of interlining. The payments are received by the Lufthansa Group via credit card billing companies, agents or other airlines, generally before the corresponding service is provided. Receivables from the sale of flight tickets and related ancillary services are only amounts payable by credit card billing companies, agents or other airlines.

The Lufthansa Group initially recognises all ticket sales as liabilities from unused flight documents. These are presented as contract liabilities in accordance with IFRS 15. Depending on the terms of the selected fare, the contract liabilities reflect a range of possibilities for refunding services that have not yet been provided. Liabilities include both the deferred income for future flights and ancillary services that are recognised as revenue when the flight documents are used, and the

liabilities for award miles credited to the passenger when the flight documents are used. The Lufthansa Group allocates the transaction price to all of the performance obligations identified on the flight ticket on the basis of their individual transaction prices. The individual transaction prices for flight segments are determined using the IATA procedure, which allocates the total price payable to individual flight segments using what is known as a prorate calculation. Amounts calculated in this way meet the IFRS 15 definition of a relative individual transaction price. The individual transaction prices for ancillary services that are not included in the fare are directly observable prices within the meaning of IFRS 15. On average, it takes 2.5 months for a flight coupon to be realised. This does not apply to financial years with severe flight restrictions such as 2020, when the use of flight documents fluctuated from month to month and altogether the large number of flight cancellations meant that customers could not use their tickets as planned. Additional cancellation and rebooking options were introduced in connection with the restrictions and flight cancellations caused by the coronavirus pandemic.

The Lufthansa Group reduces liabilities from unused flight documents and recognises revenue for each flight segment (including the related ancillary revenue) when the respective document is used. For tickets that cover more than one flight segment, the Lufthansa Group identifies each flight segment as a distinct performance obligation, since each flight segment is independent and can be distinguished in the context of the contract.

Interlining means that the passenger is carried by another airline for one (or more) flight segments. Only the commission paid by the other carrier is recognised as revenue for these flight segments, since the Lufthansa Group acts solely as an agent in terms of these performance obligations. If passengers with tickets sold by other airlines are carried partly or fully by the Lufthansa Group, the Lufthansa Group shows the pro rata ticket income received from the other airlines less the commission retained by the ticketing airline as revenue.

Generally speaking, the Lufthansa Group does not expect to receive any amount if a flight document is not used (or does not expect the amount to be material) and so for this reason does not anticipate the possibility that documents for a flight segment will not be used. The expected amount if flight documents are not used is only recognised as revenue if the probability that the passengers (in accordance with the portfolios they are assigned to) exercise their remaining rights is low, and no later than when the expiry of flight documents is certain and known.

IFRS 15 requires that income from the expiry of miles is recognised in parallel with revenue from the performance obligations that do not expire. A period of three years is therefore assumed for revenue recognition, and the revenue from miles expected to expire is recognised on a straight-line basis over this time. The effects of the coronavirus pandemic on air travel have changed the recognition parameters, partly because there were fewer opportunities to fly and partly because customers' redemption patterns changed. The average percentage decline in redemption volumes was used as an indicator for adjusting the pro rata recognition volume.

Revenue for award miles is recognised at the point in time or over the time at which the goods and services purchased with the award miles are transferred.

Logistics

Lufthansa Cargo markets the freight capacities of passenger aircraft at Lufthansa German Airlines, Austrian Airlines, Eurowings and Brussels Airlines and operates a fleet of cargo aircraft. In addition to income from standard cargo services, Lufthansa Cargo generates part of its revenue from ancillary services that are closely connected to the freight service.

In its cargo business, the Lufthansa Group has identified the entire freight service as a distinct performance obligation. The end customer receives the benefit of the transport service and uses the service at the same time as this performance obligation is fulfilled with each transport segment. In this case, the customer takes control of the company's output while the carrier provides its service. The customer receives the benefit of the service as each transport segment is fulfilled. The corresponding cargo revenue is therefore recognised at the prorata value when the documents for each individual freight segment are used.

Lufthansa Cargo typically receives the consideration for performing its service once the transport has been carried out.

MRO

The main distinct performance obligations in the MRO segment are the provision of maintenance and aircraft and engine overhaul services, which are recognised over time since the condition of IFRS 15.35 (b) is generally met. These performance obligations involve estimating the proportion of the total contract already completed and the profit on the whole contract, so that an input-orientated measurement of the percentage of completion can be made. Contract assets and contract liabilities are therefore both recognised.

Access to Lufthansa Technik's pool of spare parts and components is another key performance obligation, which is satisfied either over time or at a point in time, depending on the contract model agreed.

In some cases, the contracts in the MRO segment make it necessary not to recognise distinct services as individual performance obligations but rather as a series, as described in IFRS 15.22 (b). Furthermore, some of the contracts include standby obligations that require the recognition of revenue over time. This is particularly the case when remuneration is paid in the form of a fixed rate per hour of flying time. For such contracts, the percentage of completion is primarily measured on the basis of the hours invoiced monthly to the customer. Revenue from component supply contracts is realised taking into consideration the margin shown in the business plans, which are updated annually.

A significant portion of the contracts in the MRO business segment run for several years and so have price adjustment clauses, which are only considered in the transaction price when the event that triggers a price adjustment (a wage increase, for example) has occurred.

Catering

The LSG group offers products and services related to in-flight service. These include catering, in-flight sales and entertainment, in-flight service equipment and the associated logistics as well as consultancy services and the operation of lounges.

Airline catering is the main business of the LSG group as far as revenue is concerned. Taking the business model and the value chain for airline catering into account, the preparation of meals and the logistics related to this catering have been identified as distinct performance obligations. The performance obligation to prepare meals is generally fulfilled when the meals are delivered to the customers. The catering logistics performance obligation is fulfilled over the time between the transport of the meals to the airport and the disposal of the waste, depending on the services ordered by the customer. For performance obligations over time, the percentage of completion is measured on an output basis in accordance with IFRS 15.B15 in conjunction with IFRS 15.B16.

Billing and payment in the Catering segment generally take place one to two months after the performance obligation has been fulfilled. This gives rise to trade receivables, but no significant contract liabilities or contract assets from catering contracts.

Variable consideration (e.g. volume discounts) must be taken into account when determining the transaction price in the catering business. The majority of the variable consideration is estimated using the expected value method on the basis of historic data and current developments. The LSG group updates the estimated transaction price at the end of each reporting period and accounts for the resulting changes in accordance with IFRS 15.87–90.

Further disclosures on the Lufthansa Group's revenue from contracts with customers can be found in [Notes 4 and 5, p. 157ff.](#)

Operating expenses are recognised when the product or service is used or the expense arises. Provisions for warranties are generally accounted for when the corresponding revenue is recognised, while provisions for onerous contracts are generally set up when they are identified.

Interest income and expenses are accrued in the appropriate period. Dividends from shareholdings not accounted for using the equity method are recognised when a legal claim to them arises.

Initial consolidation and goodwill

The initial consolidation of Group companies takes place using the purchase method. This involves measuring the fair value of the assets, liabilities and contingent liabilities identified, in accordance with the provisions of IFRS 3, of the company acquired at the acquisition date, and allocating the acquisition costs to them. The proportion of fair value of assets and liabilities not acquired is shown under non-controlling interests. The ancillary acquisition costs are recognised as expenses in the periods in which they occur.

Any excess of cost over the value of equity acquired is capitalised as goodwill. If the value of the acquirer's interest in the shareholders' equity exceeds the purchase price paid by the acquiring company, the difference is recognised immediately in profit or loss.

Differences from non-controlling interests acquired after control has been gained are set off directly against equity.

Goodwill is not amortised, but is tested annually for impairment. The impairment tests applied to goodwill are carried out using established discounted cash flow methods. This is done on the basis of expected future cash flows from the latest business plan, which are extrapolated on the basis of long-term revenue growth rates and assumptions with regard to margin development and are discounted for the capital costs of the business unit. Tests are performed at the cash

generating unit (CGU) level. For the individual premises on which impairment tests were based in the financial year 2020, see [Note 17, p. 165ff.](#)

Additional impairment tests are also applied during the course of the year if events give reason to believe that goodwill could be permanently impaired.

Once an impairment loss has been recognised on goodwill, it is not reversed in subsequent periods.

Notwithstanding the principles described above, Group companies that have no material impact on the Lufthansa Group's net assets, financial and earnings position are not consolidated, but rather recognised in the consolidated financial statements at cost less any impairments.

Currency translation and consolidation methods

The financial statements of the foreign Group companies are prepared in the relevant functional currency and translated into euros before consolidation. The functional currency is mainly the currency of the country in which the company concerned is located. Occasionally, the functional currency differs from the national currency. Assets and liabilities are translated at the middle rates on the balance sheet date. Income statements are translated at the average exchange rates for the year. Any translation differences are recognised directly in equity without effect on profit and loss and are only recognised in profit or loss when control is lost or the equity investment is disposed of.

Goodwill from capital consolidation of foreign subsidiaries prior to 2005 is carried at historical cost net of amortisation accumulated by the end of 2004. Goodwill acquired after 2005 is held in the functional currency of the purchased company and translated at the middle rates on the reporting date.

Transaction differences, however, are recognised in profit or loss. These differences arise in the financial statements of consolidated companies from the measurement of assets and liabilities denominated in a currency other than the company's functional currency. Exchange rate differences here are included in revenue (exchange rate gains and losses on trade receivables) and in other operating income (other exchange rate gains) or other operating expenses (other exchange rate losses).

Translation differences relating to items whose fair value changes are recognised in equity are also recognised in equity without effect on profit and loss.

The most important exchange rates used in the consolidated financial statements have developed in relation to the euro as follows:

T065 EXCHANGE RATES

	2020		2019	
	Balance sheet exchange rate	Income statement average rate	Balance sheet exchange rate	Income statement average rate
AUD	0.62497	0.60232	0.62443	0.62094
CAD	0.63740	0.65478	0.68299	0.67379
CHF	0.92165	0.93484	0.92137	0.89864
CNY	0.12475	0.12701	0.12767	0.12955
GBP	1.10660	1.12600	1.17019	1.14121
HKD	0.10497	0.11357	0.11456	0.11410
INR	0.01112	0.01188	0.01251	0.01271
JPY	0.00788	0.00825	0.00819	0.00820
KRW	0.00075	0.00074	0.00077	0.00077
NOK	0.09514	0.09271	0.10137	0.10171
PLN	0.21851	0.22425	0.23505	0.23209
SEK	0.09927	0.09515	0.09564	0.09444
USD	0.81377	0.88089	0.89198	0.89432

The provisions of IAS 29, Financial Accounting in Hyper-inflationary Economies, were applied to one consolidated company in Argentina. Gains and losses from adjusting for inflation on the carrying amounts of non-monetary assets and liabilities and the income statement were immaterial and were recognised in other operating income.

The effects of intra-Group transactions are completely eliminated in the course of consolidation. Receivables and liabilities between consolidated companies are offset against one another and intra-Group provisions are reversed through profit or loss. Intra-Group profits and losses in non-current assets and inventories are eliminated – mostly in connection with the internal resale of aircraft and maintenance events. Intra-Group income is set off against the corresponding expenses. Tax accruals and deferrals are made as required by IAS 12 for temporary differences arising from consolidation.

Other intangible assets (except goodwill)

Acquired intangible assets are shown at cost, while internally generated intangible assets from which the Lufthansa Group expects to derive future benefit and that can be measured reliably are capitalised at cost of production and amortised regularly using the straight-line method over an estimated useful life. The cost of production includes all costs directly attributable to the production process, including borrowing costs as required under IAS 23, as well as appropriate portions of production-related overhead.

Intangible assets with indefinite useful lives (mainly brands and purchased, resellable take-off and landing rights) are not amortised, but rather subjected to a regular annual impairment test, as is goodwill.

Property, plant and equipment

Tangible assets used in business operations for longer than one year are valued at cost less regular straight-line depreciation. The cost of production includes all costs directly attributable to the manufacturing process as well as appropriate portions of production-related overhead. Borrowing costs in close connection with the financing of the purchase or production of a qualifying asset are also capitalised.

Key components of property, plant and equipment that have different useful lives are recognised and depreciated separately. Seats and in-flight entertainment systems installed in commercial aircraft are recognised separately. If costs are incurred in connection with regular extensive maintenance work (e.g. overhauling aircraft and major engine overhauls), these costs are recognised as a separate component insofar as they meet the criteria for recognition. The useful lives and remaining carrying amounts of assets are reviewed regularly and adjusted as necessary in line with the forecast.

The following useful lives and residual carrying amounts are applied throughout the Group:

T066 USEFUL LIVES OF PROPERTY, PLANT AND EQUIPMENT

Property, plant and equipment	Useful life
Buildings	45 years
New commercial aircraft and reserve engines	20 years to a residual value of 5%
Separable aircraft components	4 to 6 years
Technical equipment and machinery	8 to 20 years
Other equipment, operating and office equipment	3 to 20 years

Buildings, fixtures and fittings on rented premises are depreciated according to the terms of the leases or over a shorter useful life.

Assets acquired second-hand are depreciated over their expected remaining useful life.

When assets are sold or scrapped, the difference between the net proceeds and the net carrying amount of the assets is recognised as a gain or loss in the other operating income or expenses, respectively.

In addition to the impairment tests for goodwill, slots and brands, individual items of property, plant and equipment and intangible assets are also tested for impairment if they are no longer intended for future use, either because they are damaged, retired or due to be sold. In this case, the assets are measured individually in line with the applicable standard (full write-down to scrap value, or disposal proceeds less costs to sell). The lowest level at which assets can form a CGU is a production facility, to the extent that separate product lines or customer (groups) can be assigned to it. When aircraft are held for service in the Lufthansa Group fleet and there is no immediate intention to sell them, they are combined with the assets of the respective operating unit for the purposes of impairment testing.

Impairment losses on intangible assets and property, plant and equipment

The Lufthansa Group tests intangible assets and property, plant and equipment for impairment if events or changes in circumstances indicate that the carrying amount of an asset may not be recoverable. In addition to depreciation and amortisation, impairment losses are also recognised on the balance sheet date if the asset's recoverable amount has fallen below its carrying amount. The recoverable amount is determined as the higher of an asset's fair value less costs to sell and the present value of the estimated net future cash flows from continued use of the asset (value in use).

Fair value less costs to sell is derived from recently observed market transactions – insofar as they are available – or in the case of aircraft, from general external information on current market prices.

If it is impossible to forecast expected cash flows for an individual asset, the cash flows for the next larger asset unit are estimated, discounted at a rate reflecting the risk involved, and the recoverable amount allocated to the individual assets in proportion to their respective carrying amounts.

If the reason for an impairment loss recognised in previous years should cease to exist in whole or in part in subsequent periods, the impairment loss is reversed up to the amount of the asset's amortised cost.

Repairable spare parts for aircraft

Initial supply of spare parts for aircraft that can be reused after repair are classified as non-current assets.

The MRO segment accounts for most of the Group's repairable spare parts. They are replaced and repaired on an ongoing basis to carry out customer orders and for the Group's own purposes and are held in stock to support the Group's long-term business. A valuation-relevant subdivision essentially distinguishes between replacement components for aircraft ("pool material"), which is provided continuously for customer orders and which is measured at production/acquisition cost less depreciation, and spare parts that are exchanged and repaired on an ongoing basis for overhaul orders ("non-pool material"), which is measured at the lower of production/acquisition cost and net realisable value. The starting point for the depreciated carrying amounts is the rolling average price of the materials. Pool material is depreciated over 5 to 20 years, depending on the expected useful life of the corresponding aircraft model. Valuation allowances for non-pool materials reflect their expected future marketability. All depreciation and impairment is recognised within the cost of materials and services, since this best reflects the business model.

Leases

The Lufthansa Group is a lessee for certain assets, particularly property and aircraft. In terms of property, the Group mainly leases airport infrastructure, including hangars, parking and handling spaces, lounges, and offices. Other office buildings, production and warehouse space are also leased. In addition, the Group uses aircraft and other operating and office equipment on the basis of leases. In accordance with IFRS 16, the Lufthansa Group assesses at contract inception whether a contract is, or contains, a lease. That is, if the contract conveys the right to control the use of an identified asset for a period of time in exchange for consideration.

Right-of-use assets are measured at cost less accumulated depreciation and impairment losses and adjusted for any change in the measurement of the lease liability. The cost of right-of-use assets includes the amount of lease liabilities recognised, initial direct costs incurred, and lease payments made at or before the commencement date less any lease incentives received.

The Lufthansa Group has opted not to apply IFRS 16 to intangible assets. Payments under leases with a term of no more than twelve months and leases in which the leased asset is of low value, will be recognised in profit or loss at the payment date in line with this option.

For contracts that include non-lease components alongside lease components, these components are separated.

Right-of-use assets are depreciated on a straight-line basis over the shorter of the lease term or the estimated useful life of the leased item.

If ownership of the leased asset passes to the Lufthansa Group at the end of the lease term or is included in the costs of exercising a purchase option, depreciation is calculated for the expected useful life of the leased asset.

Right-of-use assets are also tested for impairment.

At the commencement date of the lease, Lufthansa Group recognises **lease liabilities** measured at the present value of the lease payments to be made over the term of the lease. The lease payments include fixed payments less any lease incentives owed, variable lease payments that depend on an index or a rate, and any amounts that are expected to be paid in the context of residual value guarantees. Lease payments also include the exercise price of a purchase option or penalties for early termination if the exercise of the purchase or termination option by the lessee is reasonably certain.

The Lufthansa Group has several leases that include renewal and termination options, particularly for properties. Judgment is used when determining the probability that the option to renew or terminate the lease will be exercised. When determining lease terms, all the facts and circumstances are reasonably certain that offer an economic incentive to exercise renewal options or not to exercise termination options. After the commencement date of the lease, the Lufthansa Group remeasures the lease liability if a significant event occurs or if circumstances change.

Variable lease payments that do not depend on an index or a reference rate are recognised as expenses in the period in which the event or condition triggering the payment occurs.

Lease payments are discounted at the interest rate implicit in the lease, if it can be determined. Otherwise they are discounted at the incremental borrowing rate. Reference interest rates based on congruent, risk-free rates in major countries and currencies were used to calculate the incremental borrowing rate. A credit risk premium was added to the respective reference rates.

As the lessor in an operating lease, the Lufthansa Group presents the leased item as an asset at amortised cost in property, plant and equipment. Lease payments received in the period are shown as other operating income. The Lufthansa Group leases some of its properties and engines to other entities. There are currently no finance leases at the Lufthansa Group.

Equity investments accounted for using the equity method

Equity investments accounted for using the equity method are capitalised at cost at the time of acquisition.

In subsequent periods, the carrying amounts are either increased or reduced annually by changes in the shareholders' equity of the associated company or joint venture that is held by the Lufthansa Group. The principles of purchase price allocation that apply to full consolidation are applied accordingly to the initial measurement of any difference between the acquisition cost of the investment and the pro rata share of shareholders' equity of the company in question. An impairment test is only carried out in subsequent periods if there are indications of a potential impairment in the entire investment valuation.

Financial instruments

Financial assets are classified within the Lufthansa Group in accordance with IFRS 9 as "at amortised cost", "at fair value through profit or loss", "at fair value through other comprehensive income (with and without recycling)" and "derivative financial instruments as an effective part of a hedging relationship".

The category "**at amortised cost**" consists of financial assets that are debt instruments and are intended to be held to maturity on the basis of the company's business model. Furthermore, these instruments have fixed payment terms and meet the criteria for cash flow characteristics, i.e. contractual payments of principal and interest. For the Lufthansa Group, this item particularly includes loans and receivables, cash-in-hand and bank balances. They are classified as non-current or current assets according to their remaining maturity.

The category "**at fair value through profit or loss**" comprises debt instruments for which the business model is neither to hold nor to sell them, or which do not pass the cash flow characteristics test. This is generally not the case for the Lufthansa Group. Equity instruments are also allocated to this category as a rule, so the Lufthansa Group generally recognises shares and equity investments that are financial instruments in this category. Derivatives are also classified in this category if they do not meet the criteria for hedge accounting.

Debt instruments are classified as **“at fair value through other comprehensive income (with recycling)”** when the business model is to both hold and sell these instruments and they pass the cash flow characteristics test. For the Lufthansa Group, this applies to securities representing debt instruments.

An option can be exercised to classify specific equity instruments as **“at fair value through other comprehensive income (without recycling)”**. The Lufthansa Group exercises this option for individual share positions.

The Lufthansa Group uses derivatives for hedging, which are classified as **“derivative financial instruments as an effective part of a hedging relationship”** if all the requirements for hedge accounting are satisfied.

Financial instruments are recognised on the settlement date, i.e. on the date that they are created or transferred. Financial assets are capitalised at fair value plus transaction costs. Unrealised gains and losses are recognised directly in equity, taking deferred taxes into account. Long-term low or non-interest-bearing loans are recognised at net present value using the effective interest method. Subsequent measurement of the financial instrument depends on the classification, either at amortised cost using the effective interest method, or at fair value, through profit or loss or in equity without effect on profit and loss.

Receivables denominated in foreign currencies are measured at the balance sheet date rate.

The fair value of securities is determined by the price quoted on an active market. For unlisted fixed-interest securities, the fair value is determined from the difference between effective and market interest rate at the valuation date.

If there are doubts as to the recoverability of receivables, then impairment losses are recognised and these receivables are recognised at the lower recoverable amount. Subsequent reversals (write-backs) are recognised in profit or loss. IFRS 9 requires that when a receivable is recognised for the first time, an expected loss is provided for that reflects the credit risk of the receivable before a default event occurs. An external credit risk exists for the Lufthansa Group, especially in its portfolio of trade receivables, for which an expected credit loss is recognised.

Derivative financial instruments are measured at fair value on the basis of published market prices. If there is no quoted price on an active market, other appropriate valuation methods are applied. Appropriate valuation methods take all factors into account that independent, knowledgeable market participants would consider in arriving at a price and that constitute recognised, established economic models for calculating the price of financial instruments.

In accordance with its internal guidelines, the Lufthansa Group uses derivative financial instruments to hedge interest rate and exchange rate risks and to hedge fuel price risks. This is based on the hedging policy defined by the Executive Board and monitored by a committee. 🚩 **Note 45, p. 205ff.**

Interest rate swaps and interest rate/currency swaps are used to manage interest rate risks. Interest rate/currency swaps also hedge exchange rate risks arising from borrowing in foreign currencies.

Fuel price hedging takes the form of spread options and other hedging combinations, primarily for crude oil. To a limited extent, hedging may also be undertaken for other products, such as jet fuel or gas oil.

Hedging transactions are used to secure either fair values (fair value hedge) or future cash flows (cash flow hedge).

To the extent that the financial instruments used qualify as effective cash flow hedging instruments within the scope of a hedging relationship, in accordance with the provisions of IFRS 9, the fluctuations in market value will not affect the result for the period during the term of the derivative. They are recognised without effect on profit or loss in the corresponding reserves. If the hedged cash flow is an investment, the result of the hedging transaction that has previously been recognised in equity is set off against the cost of the capital expenditure at the time the underlying transaction matures. In all other cases, the cumulative gain or loss previously stated in equity is included in net profit or loss for the period on maturity of the hedged cash flow.

In the case of effective hedging of fair values that are designated as a fair value hedge, the changes in the market value of the hedged asset or the hedged debt and those of the financial instrument will balance out in the income statement.

Derivatives that do not meet the criteria for hedge accounting are presented in the category “at fair value through profit or loss”. Changes in fair value are then recognised directly in the income statement. For the Lufthansa Group, this generally occurs when the exposure or item being hedged cannot be measured reliably or the exposure ceases to exist prematurely over the course of the hedge.

Embedded derivatives – to the extent that they should, but cannot, be separated from the financial host contract – are also considered with these as trading transactions for measurement purposes. Changes in market value are also recognised directly as profit or loss in the income statement. Both types must be classified as financial assets stated “at fair value through profit or loss”.

It is the Lufthansa Group’s hedging policy (➤ **Note 45, p. 205ff.**) only to acquire effective derivatives for the purpose of hedging interest rate, exchange rate and fuel price risks.

Initial recognition of **financial guarantees** given to third parties is at fair value. Thereafter, financial guarantees are either measured in the category “at fair value through profit or loss” or at the higher of the originally recognised amount, less any cumulative amortisation through profit or loss in line with IFRS 15, and the value of the contractual obligation measured in line with IAS 37.

Emissions certificates

CO₂ emissions certificates are recognised as intangible assets and presented under other receivables. Rights, both those purchased and those allocated free of charge, are measured at cost and not amortised.

The Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), which was concluded with the International Civil Aviation Organization (ICAO) in October 2016, seeks to offset growth-related CO₂ emissions in international air traffic using climate protection projects from 2021 onwards. In June 2020, the ICAO adopted a resolution confirming that carbon emissions in 2019 are the sole benchmark for determining relevant industry growth in the course of the CORSIA pilot phase. On the basis of this regulatory change, the Lufthansa Group assumes that it will not have to make any carbon offsets under CORSIA for the year 2021. Whether and how many CORSIA offsets will be required in future depends primarily on how fast air traffic recovers between the states participating in the CORSIA pilot phase.

Contract assets and receivables

Contract assets represent contractual claims to receive payments from customers where the contractual performance obligations have already been fulfilled but no unconditional payment claim has yet been incurred. Receivables are recognised if the right to receive consideration is no longer subject to conditions. This is generally the case when the Group is contractually entitled to send the customer an invoice. Contract assets mainly relate to construction or service contracts for MRO and IT services. Valuation allowances are made on the respective gross amounts of expected payment defaults.

Inventories

The item “Inventories” comprises non-repairable spare parts and assets used in production or the provision of services (raw materials, consumables and supplies), purchased merchandise, finished and unfinished goods and advance payments for them. They are measured at cost, determined on the basis of average prices, or at production costs. The cost of production includes all costs directly attributable to the production process, including borrowing costs as required under IAS 23, as well as appropriate portions of production-related overheads at normal productivity rates. Measurement on the balance sheet date is at the lower of cost and net realisable value. Net realisable value is defined as the estimated selling price less the estimated cost of completion and the estimated costs necessary to make the sale. If there are indicators for future inability to pay, corresponding valuation allowances are made.

Assets held for sale

Individual, formerly non-current assets or groups of assets that are expected to be sold within the next twelve months are measured at the lower of their carrying amount at the time they are reclassified and fair value less costs to sell. Fair value less costs to sell is derived from recent market transactions, if available.

Property, plant and equipment and intangible assets are no longer depreciated or amortised and affiliated companies accounted for using the equity method are no longer accounted for in this way once they are classified as held for sale or distribution. While the impairment charge from the last measurement before reclassification is recognised as an impairment loss, all subsequent changes in the measurement of current assets held for sale, e.g. due to exchange rate movements, are shown in other operating expenses or income.

Cash and cash equivalents

Cash and cash equivalents comprise cash-in-hand, cheques received and credit balances at banks. Cash equivalents are financial investments that can be liquidated at short notice. At the time of purchase or investment, they have a maturity of three months or less.

Pension provisions

The pension provisions for defined benefit plans correspond to the present value of the defined benefit obligations (DBO) on the reporting date less the fair value of plan assets, if necessary taking the rules on the maximum surplus of plan assets over the obligation (asset ceiling) into account.

The DBO is calculated annually by independent actuaries using the projected unit credit method prescribed in IAS 19 for defined benefit pension plans. The measurement of pension provisions within the statement of financial position is based on a number of actuarial assumptions.

Capital account plans are measured using the market value of the assets assigned to the individual capital accounts as of the reporting date, whereby the present value of the minimum benefit payable when the beneficiary becomes entitled to the benefit is compared with the amount of contributions already paid in, measured using the assumptions for the benefit plans. Additional risk premiums that the employer contributes to insure against early entitlements are included in current service expense.

They include, in particular, assumptions about long-term salary and pension trends as well as average life expectancy. The assumptions about salary and pension trends are based on developments observed in the past and take into account national interest and inflation rates and labour market trends. Estimates of average life expectancy are based on recognised biometric calculation formulas.

The interest rate used to discount the individual future payment obligations is based on the return from investment grade corporate bonds in the same currency and with a similar maturity. The discount rate is determined by reference to high-quality corporate bonds with an issue volume of at least EUR 100m and an AA rating from at least one of the rating agencies Moody's Investor Service, Fitch Ratings or Standard & Poor's Rating Services.

Actuarial gains and losses arising from the regular adjustment of actuarial assumptions are recognised directly in equity in the period in which they arise, taking deferred taxes into account. Also presented without effect on profit and loss are differences between the interest income at the beginning of the period calculated on plan assets based on the interest rate used to discount the pension obligations and the earnings from plan assets actually recorded at the end of the period. The actuarial gains and losses and any difference between the forecast result and the actual result from plan assets form part of the remeasurement.

Past service costs are recognised immediately in profit or loss.

Payments to pension providers for defined contribution retirement commitments for which the pension provider or the beneficiary assumes the financial risks are recognised in staff costs as they fall due.

Other provisions

Other provisions are recognised for present legal and constructive obligations to third parties arising from past events that will probably give rise to a future outflow of resources provided that a reliable estimate can be made of the amount of the obligations as of the reporting date.

The amount of the provision is determined by the best estimate of the amount required to settle the present obligation. Past experience, current cost and price information as well as estimates from internal and external experts are used to determine the amount of provisions.

The management regularly analyses the current information on legal risks and makes provisions for probable obligations. These provisions cover estimated payments to the claimant, court and procedural costs, the costs of lawyers and of any out-of-court settlement. Internal and external lawyers assist with the estimate. When deciding on the necessity of a provision for litigation, the management takes into account the probability of an unfavourable outcome and the chance of making a sufficiently accurate estimate of the amount of the obligation. The commencement of legal proceedings, the formal assertion of a claim against the Group or the disclosure of certain litigation in the Notes does not automatically mean that a provision was made for the risk concerned. A ruling in court proceedings, a decision by a public authority or an out-of-court settlement may cause the Group to incur expenses for which no provision was made because the amount could not be reliably determined or for which the provision made and the insurance coverage is not sufficient.

Provisions for restructuring and severance payments are only recognised when the Group has a constructive obligation. In the case of restructuring, a constructive obligation exists if there is a formal restructuring plan that includes the affected business unit or the affected part of a business unit, the location and number of employees affected, the detailed estimate of the associated costs and the time schedule. In addition, the key points of the plan must have been communicated to the employees concerned. The restructuring provisions only include expenses directly attributable to the restructuring measures that are necessary for the restructuring and are not related to the future operating business. This includes, for example, expenses for severance payments to employees.

Provisions for obligations that are not expected to lead to an outflow of resources in the following year are recognised to the amount of the present value of the expected outflow, taking foreseeable price rises into account.

The assigned value of provisions is reviewed on each balance sheet date. Provisions in foreign currencies are translated at reporting date rates.

If no provision could be recognised because one of the stated criteria was not fulfilled, the corresponding obligations are shown as contingent liabilities and discussed in the relevant section.

Liabilities

Trade and other payables are initially recognised at fair value. Fair value is approximately equivalent to the carrying amount.

Measurement in subsequent periods is at amortised cost using the effective interest rate method.

Liabilities in foreign currencies are measured with the exchange rate at the balance sheet date.

Obligations from share programmes are measured at fair value as cash-settled share-based payment transactions in accordance with IFRS 2. Fair value is derived using a Monte Carlo simulation.

The liability is recognised on the basis of the resulting fair value, taking the remaining term of the programme into account. Changes are recognised as staff costs in profit or loss.

Details of the assumptions used for the model and the structure of the share programmes can be found in [Note 39, p. 192ff.](#)

Contract liabilities

A contract liability is an obligation on the part of the Group towards a customer to provide goods or services for which the customer has already performed an obligation, e.g. by making an advance payment. Contractual liabilities are recognised as revenue as soon as the Group fulfils its contractual obligations. The Group's contract liabilities consist of liabilities from unused flight documents, unredeemed miles from customer loyalty programmes, construction contracts and other contract liabilities.

Until they are used, sold flight documents are recognised as an **obligation from unused flight documents**. Coupons that are unlikely to be used any more are recognised pro rata temporis as traffic revenue in the income statement at their estimated value. The estimate is based on historical statistical data. Due to the coronavirus pandemic and the associated increased number of tickets for cancelled and rebookable flights, the parameters for the use of expired flight coupons were adjusted according to current estimates.

The Lufthansa Group uses various bonus miles programmes with the aim of ensuring long-term **customer loyalty**. Participants in the Miles & More programme, which is the biggest bonus miles programme in the Lufthansa Group, can collect and redeem bonus miles for flights with the airlines in the Lufthansa Group as well as with numerous partners (including other airlines, hotels, global car hire companies, financial and insurance providers, telecommunications companies, retailers, automobile clubs, etc.). Miles expire three years after they are collected, in accordance with the terms of membership, unless they are protected by frequent flyer status or credit card use.

Observable past redemption patterns are used to measure the premium claims that are collected on flights with the airlines in the Lufthansa Group. Miles that are expected to be used for flights with airlines of the Lufthansa Group are measured based on the average price of the premium flight or upgrade for the average number of miles used. The price is calculated on the basis of past redemption patterns, weighted for the various geographic regions and booking classes. This is then corrected to allow for the reduced flexibility of premium flights and the award miles granted for normal flights. Miles that are expected to be redeemed for other

bonuses are measured at the average price for these bonuses and the average number of miles redeemed. The prices for additional miles are recalculated every year and applied to all additions in that year. Consumption of miles is measured using the average rate for total miles at the beginning of the year (same as previous year).

Premium points collected from other partners are measured at the amounts paid by these partners in relation to the average number of miles collected and redeemed.

The calculation method for the legal and economic expiry rate entails calculating the expiry rate from the values observed in prior years, increased or decreased as necessary by reference to past trends or future enhancements to the programme. Due to the significant decline in redemption behaviour in the wake of the coronavirus pandemic, the parameter adjustments that would have resulted from the data of the current financial year were largely not taken into account for valuation purposes.

Government grants

Government grants are recognised at fair value when there is reasonable certainty that the grant will be received and that the Group will fulfil all conditions attached to such grants. Government grants include, for example, income subsidies or social security contributions for short-time allowances. If short-time allowance is a personal benefit for the employee, the respective payments are transitory items. Another example is government loans at below-market rates, where the interest rate advantage is allocated pro rata temporis over the term of the loan.

Government grants for the acquisition of property, plant and equipment are included in other liabilities as deferred income and recognised in other operating income on a straight-line basis over the estimated useful life of the corresponding asset. Non-monetary grants are only recognised in the income statement when the necessary eligibility criteria have been fulfilled. Until then, the corresponding amounts must also be shown under deferred income.

Tax liabilities

Obligations towards tax authorities that are uncertain with regard to their occurrence, probability and amount are recorded as tax liabilities on the basis of best estimates or expectations. Any contingent liabilities existing in this context are addressed separately as needed.

Deferred tax items

In accordance with IAS 12, deferred taxes are recognised for all temporary differences between the statements of financial position with regard to tax of individual companies and the consolidated financial statements. Tax loss carry-forwards are recognised to the extent that the deferred tax assets are likely to be used in the future. Company earnings forecasts and specific, realisable tax strategies are used to determine whether deferred tax assets are usable or not, i.e. whether they have a value that can be realised. The planning period used to assess this probability is determined by the individual Group company according to the specific circumstances and lies generally between three and five years unless there is convincing evidence of possible prolonged use beyond the general horizon of the official Group planning. This was the case in the financial year due to the high losses caused by the shock of the coronavirus pandemic, especially in Germany, resulting in the application of longer useful lives.

Effective income taxes

The Lufthansa Group is liable for income taxes in various countries. Material assumptions are necessary to calculate the income tax liabilities. For certain transactions and calculations, the final taxation cannot be assessed definitively in the course of normal business. The amount of the liability that may arise from future determinations of expected tax inspection is based on estimates of whether additional income taxes will be owed, and if so, at which amount. The assumptions underlying the estimates are reviewed on an ongoing basis and adjusted as necessary. Nevertheless, different tax payments may occur in the period in which taxation is definitively assessed.

NOTES TO THE CONSOLIDATED INCOME STATEMENT

4 Traffic revenue

In the consolidated income statement, the Lufthansa Group attributes revenue to the segments Network Airlines, Eurowings, Logistics, MRO, Catering, and Additional Businesses and Group Functions.

Traffic is under significant strain as a result of the worldwide travel restrictions due to the coronavirus pandemic. Passenger numbers fell by 75% in the 2020 financial year compared with the previous year.

The following table provides a breakdown of traffic revenue according to the different business models for the financial year that ended on 31 December 2020. The table following that shows the previous year's figures:

T067 TRAFFIC REVENUE BY SECTOR 2020

	Total	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia/Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
Network Airlines	5,899	4,066	859	131	556	144	143
Lufthansa German Airlines	3,465						
SWISS ²⁾	1,635						
Austrian Airlines ²⁾	415						
Brussels Airlines	384						
Eurowings ²⁾	583	576	4	1	2	–	–
Logistics	2,596	1,210	250	90	983	22	41
Total	9,078						

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Disclosure of traffic revenue, including belly revenue; this is reported in the segment reporting in the reconciliation column.

T067 TRAFFIC REVENUE BY SECTOR 2019

	Total ³⁾	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia/Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
Network Airlines ^{2),3)}	23,507	15,193	4,407	513	2,348	586	460
Lufthansa German Airlines ³⁾	15,170						
SWISS ²⁾	5,010						
Austrian Airlines	1,941						
Brussels Airlines	1,386						
Eurowings ^{2),3)}	2,311	2,240	35	8	22	4	2
Logistics	2,318	1,153	265	94	703	31	72
Total	28,136						

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Disclosure of traffic revenue, including belly revenue; this is reported in the segment reporting in the reconciliation column.

³⁾ Previous year's figures have been adjusted due to reclassification of Brussels Airlines, Germanwings GmbH and the long-haul business of Eurowings to Network Airlines.

Traffic revenue of EUR 9,078m (previous year: EUR 28,136m) includes freight and mail revenue of EUR 3,011m (previous year: EUR 2,774m). Of the total, EUR 2,596m (previous year: EUR 2,318m) relate to the Logistics segment. Other freight and mail revenue of EUR 415m (previous year: EUR 456m) mainly comes from marketing belly capacities on passenger flights by SWISS, Austrian Airlines and Eurowings.

5 Other revenue

The following table provides a breakdown of other revenue by category (type of service) and geography for 2020:

T068 OTHER OPERATING REVENUE BY AREA OF OPERATIONS 2020

	Total	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia/Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
MRO	2,724	1,175	734	90	513	133	79
MRO services	2,272						
Other operating revenue	452						
Catering	1,064	165	662	53	132	22	30
Catering services	900						
Revenue from in-flight sales	50						
Other services	114						
Network Airlines	268	233	10	2	14	5	4
Eurowings	6	6	-	-	-	-	-
Logistics	137	90	37	2	3	5	-
Additional Businesses and Group Functions	312	206	31	13	42	14	6
IT services	167						
Travel management	91						
Other	54						
Total	4,511						

¹⁾ Traffic revenue is allocated according to the original location of sale.

T068 OTHER OPERATING REVENUE BY AREA OF OPERATIONS 2019

	Total	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia/Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
MRO	4,378	1,973	1,054	282	793	173	103
MRO services	3,765						
Other operating revenue	613						
Catering	2,623	502	1,377	164	468	75	37
Catering services	2,210						
Revenue from in-flight sales	169						
Other services	244						
Network Airlines ²⁾	604	496	42	1	47	8	10
Eurowings ²⁾	10	10	-	-	-	-	-
Logistics	119	68	41	-	3	7	-
Additional Businesses and Group Functions	554	388	43	16	83	18	6
IT services	188						
Travel management	279						
Other	87						
Total	8,288						

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Adjustment due to changed allocation of Brussels Airlines, Germanwings GmbH und the long-haul business of Eurowings to Network Airlines.

The decline in other operating revenue was also recorded in the MRO and Catering business segments and in Additional Businesses and Group Functions as a result of the crisis. The Catering segment's European business was sold in December 2020 ➔ **Note 7, p. 159f.**

MRO services make up the majority of external revenue in the MRO segment. Other revenue in the MRO segment from the sale of material and hiring out material and engines, as well as logistics services, are classified as other services.

The revenue listed under catering services originates exclusively in the Catering segment. The Catering segment also generates revenue from other services, particularly in the areas of flight security concepts, in-flight service equipment, transport/warehouse logistics and lounge operations.

Other revenue also includes revenue from customer contracts that are fulfilled over a given period. These are mainly MRO and IT services.

6 Changes in inventories and work performed by entity and capitalised

T069 CHANGES IN INVENTORIES AND WORK PERFORMED BY ENTITY AND CAPITALISED

in €m	2020	2019
Increase/decrease in finished goods and work in progress	-6	-9
Other internally produced and capitalised assets	181	694
	175	685

Other own work capitalised relates almost exclusively to aircraft and engine overhauls.

Due to the coronavirus pandemic and the resulting global restrictions, parts of the fleet were retired into longterm storage in the 2020 financial year. This led to a decrease in other own work capitalised, as fewer maintenance events took place than in previous years.

7 Other operating income

T070 OTHER OPERATING INCOME

in €m	2020	2019
Foreign exchange gains	677	761
Income from the reversal of provisions and accruals	338	321
Subsidies	261	-
Compensation received for damages	45	42
Reversal of write-downs on receivables	36	27
Rental income	36	43
Income from the reversal of impairment losses on fixed assets	30	38
Services provided by the Group	24	26
Income from the disposal of non-current available-for-sale financial assets	21	2
Income from the disposal of non-current assets	17	18
Income from staff secondment	15	20
Commission income	10	14
Income from operating lease aircraft	10	49
Miscellaneous other operating income	377	528
	1,897	1,889

Foreign exchange gains (excluding financial liabilities) mainly include gains from differences between the average rate for the month on the transaction date and on the payment date, along with foreign exchange gains from measurement at the closing date rate. Income from exchange rate hedging is also recognised here. Foreign exchange losses from these transactions are reported under other operating expenses. [Note 11, p. 161f](#). The foreign currency effects of borrowing are recognised in other financial items, in the context of the net results of exchange rate hedging relationships for borrowing.

Income from the reversal of provisions and accruals relates to a number of provisions and accruals recognised in previous years that have not been fully used. In contrast, expenses from insufficient provisions recognised in previous years are recognised together with the primary expense item to which they relate.

Income from government grants mainly includes unrestricted grants paid directly to the companies affected. Government support measures for Austrian Airlines accounted for EUR 150m of this amount. LSG and Lufthansa Technik companies received a further USD 104m under the CARES Act in the USA. The grants mentioned above were received after appropriate documentation was provided.

The Lufthansa Group recognised lease income of EUR 46m in 2020 (2019: EUR 92m). The main reason for the decline was the discontinuation of operations by two lessees.

The following table shows the contractual lease payments:

T071 CONTRACTUAL LEASE PAYMENTS (LESSOR)

in €m	31 Dec 2020	31 Dec 2019
to 1 year	20	79
more than 1 year to 2 years	16	50
more than 2 years to 3 years	14	35
more than 3 years to 4 years	11	28
more than 4 years to 5 years	9	8
more than 5 years	40	22

Income of EUR 30m was recognised from write-backs on assets in 2020 (previous year: EUR 38m). Of these reversals, EUR 21m relate to the assets of LSG's European business in the Catering segment, which were recognised as assets held for sale in the previous year. The sale of the companies on 2 December 2020 resulted in a further positive effect of EUR 20m, which is reported under income from the disposal of non-current financial assets. A further EUR 8m in write-backs relates to two Boeing 767s and five Dash 8-400s from the Network Airlines segment that are held for sale.

Income from the disposal of property, plant and equipment also includes EUR 5m from aircraft sold (previous year: EUR 6m).

Other operating income includes items not attributable to any of the aforementioned categories.

8 Cost of materials and services

T072 COST OF MATERIALS AND SERVICES

in €m	2020	2019
Aircraft fuel and lubricants	1,875	6,715
Other raw materials, consumables and supplies	2,104	3,589
Purchased goods	226	512
Total cost of raw materials, consumables and supplies and of purchased goods	4,205	10,816
Fees and charges	1,796	4,523
External MRO services	1,139	1,911
Charter expenses	469	814
External IT services	318	452
In-flight services	137	409
Flight irregularities	66	199
Other services	323	703
Total cost of purchased services	4,248	9,011
	8,453	19,827

Changes in depreciation and valuation allowances on repairable spare parts are also reported within expenses for other raw materials, consumables and supplies. In the 2020 financial year, impairments and valuation allowances totalling EUR 158m were incurred as a result of the crisis.

Expenses for flight irregularities includes accommodation and meals in the case of delays, for instance, or payments for damaged luggage.

9 Staff costs

T073 STAFF COSTS

in €m	2020	2019
Wages and salaries	5,091	7,452
Social security contributions	687	1,024
Expenses for pension plans and other employee benefits	658	645
	6,436	9,121

The decline in staff costs is due on the one hand to a reduction in the headcount at the Lufthansa Group and on the other to savings such as the use of short-time working

to preserve jobs, salary waivers and the reduction of variable salary components. In 2020, income of EUR 1,020m was received in reimbursement of wage-replacement benefits and social security contributions paid in the context of short-time working in Germany, Austria and Switzerland. This includes EUR 293m in subsidies for social security contributions, which are classified as support measures.

Expenses for pension plans principally consist of additions to the pension provisions. ➔ **Note 35, p. 180ff.**

T074 EMPLOYEES

	Average for the year 2020	Average for the year 2019	As of 31 Dec 2020	As of 31 Dec 2019
Ground staff	79,878	91,241	66,070	91,642
Flight staff	44,050	45,305	42,675	45,281
Trainees	1,279	1,238	1,320	1,430
	125,207	137,784	110,065	138,353

The annual average is calculated pro rata temporis from the time companies are consolidated or deconsolidated.

10 Depreciation, amortisation and impairment

The individual line items show the breakdown of depreciation, amortisation and impairment charges between intangible assets, aircraft and other property, plant and equipment. Total depreciation, amortisation and impairment came to EUR 4,389m (previous year: EUR 2,776m).

T075 DEPRECIATION, AMORTISATION AND IMPAIRMENT

in €m	2020	2019
Amortisation of other intangible assets	134	102
Depreciation of aircraft	1,923	2,084
Depreciation of other tangible assets	504	506
Total amortisation/depreciation	2,561	2,692
Impairment of goodwill	222	-
Impairment of other intangible assets	15	24
Impairment of aircraft and reserve engines	1,461	36
Impairment of other tangible assets	51	1
Impairment of right-of-use assets	60	-
Impairment of financial assets	19	23
Total impairment	1,828	84
Total depreciation, amortisation and impairment	4,389	2,776

The decrease in depreciation and amortisation stems principally from the reduction of the active fleet.

Impairment losses of EUR 1,828m were recognised in the financial year 2020. Of this, EUR 1,402m relates to the Network Airlines segment, EUR 97m to Eurowings, EUR 54m to Logistics, EUR 45m to MRO, EUR 201m to Catering and EUR 28m to Additional Businesses and Group Functions and adjustments at Group level.

The adjustments include depreciation and amortisation of EUR 1,475m on aircraft and right-of-use assets for a total of 115 aircraft that were transferred to long-term parking mode or which are no longer planned to return to service and aircraft which are held for sale. At Lufthansa German Airlines they comprise 14 Airbus A380s, five Boeing 747s, 17 Airbus A340s, two right-of-use assets for Airbus A330s, eleven Airbus A320s and 14 Airbus A319s, some of which are leased. At Austrian Airlines, three Boeing 767s and 13 Dash 8-400s were affected. Brussels Airlines wrote down right-of-use assets for two Airbus A330s and eight Airbus A319s. Five Airbus A321s as well as right-of-use assets for 15 Dash8-400s and one Airbus A319 were affected at Eurowings. Five Boeing MD-11s were impaired in Cargo operations because they were held for sale. In addition, the depreciation and amortisation of aircraft in the MRO area included depreciation of EUR 40m on replacement engines for aircraft models that are unlikely to be used in the future.

The above amount includes impairment losses of EUR 154m which were recognised for aircraft reclassified as held for sale. The Network Airlines segment accounted for EUR 84m, Logistics for EUR 52m and Eurowings for EUR 18m.

Goodwill was written down by a total of EUR 222m in the course of impairment testing for cash-generating units. They mainly related to LSG USA, LSG Korea and Eurowings.

➤ **Note 17, p. 165ff.**

Property, plant and equipment in the catering segment as well as in the flight training operations was also impaired in the amount of EUR 31m due to the deteriorating profitability outlook. Other impairment losses included EUR 17m for high-loaders, leasehold improvements and kitchen equipment at the catering facilities in the USA, EUR 19m for financial assets and EUR 15m for IT products that are no longer in use.

A write-down of EUR 62m was recorded in the result from equity investments for the shares in the joint ventures XEOS and Spairliners (MRO segment), which is accounted for using the equity method, and of EUR 4m on Diners Club Spain (other companies). These are also related to the deterioration in the business outlook of the companies.

Other operating expenses included further impairment losses of EUR 5m on aircraft already reclassified as held for sale. These consist of three Dash 8-400s, one Boeing 767 and one Boeing MD-11 (previous year: EUR 50m on the assets of the European catering facility held for sale).

In the previous year, impairment losses mainly related to aircraft held for sale (EUR 38m), IT projects (EUR 24m) and other non-current receivables (EUR 22m).

11 Other operating expenses

T076 OTHER OPERATING EXPENSES

in €m	2020	2019
Rental and maintenance expenses	603	742
Foreign exchange losses	596	648
Staff-related expenses	543	1,201
Write-downs on receivables	251	120
Auditing, consulting and legal expenses	203	312
Other services	146	144
Advertising and sales promotions	128	384
Expenses for computerised distribution systems	126	506
Sales commission paid to agencies	97	368
Communications costs	68	85
Commissions for credit cards	58	232
Other taxes	51	84
Insurance premiums for flight operations	42	61
Losses on disposal of non-current assets	39	39
Miscellaneous other operating expenses	521	659
	3,472	5,585

Staff-related expenses also include travel and training costs for Group employees and the costs of agency staff.

Foreign exchange losses (excluding financial liabilities) mainly consist of losses from differences between the monthly average rates on the transaction date and on the payment date, expenses from exchange rate hedges and translation losses from measurement at the exchange rate on the balance sheet date. ➤ **Note 7, p. 159.** The foreign currency effects of borrowing are recognised in other financial items, in the context of the net results of exchange rate hedging relationships for borrowing.

The increase in valuation allowances on receivables was due to the crisis and, at EUR 197m (previous year: EUR 113m), mainly related to customer receivables at direct risk of default. In addition, impairments of EUR 54m (previous year: EUR 7m) were recognised with regard to existing general default risks. The MRO business segment was particularly impacted by this, with EUR 134m (previous year: EUR 51m).

Write-downs of EUR 5m were recognised in other operating expenses in connection with aircraft held for sale (previous year: EUR 50m write-downs on the assets of the European catering facility held for sale).

12 Result from equity investments

T077 RESULT FROM EQUITY INVESTMENTS

in €m	2020	2019
Result of joint ventures accounted for using the equity method	-236	14
Result of associated companies accounted for using the equity method	-40	74
Result of equity investments accounted for using the equity method	-276	88
Dividends from other joint ventures	4	12
Dividends from other associated companies	2	3
Income from profit transfer agreements	26	35
Expenses from loss transfer agreements	-35	-19
Dividends from other equity investments	15	49
Result of other equity investments	12	80
	-264	168

Write-downs on shares are also presented in the result of joint ventures accounted for using the equity method (EUR 66m; previous year: EUR 43m). ➤ **Note 10, p. 160f.**

Income and expenses from profit and loss transfer agreements include apportionments of taxes.

13 Net interest

T078 NET INTEREST

in €m	2020	2019
Income from other securities and non-current financial loans	3	8
Other interest and similar income	80	71
Interest income	83	79
Interest expenses on pensions obligations	-97	-119
Interest expenses on other provisions	-13	3
Interest and other similar expenses	-307	-278
Interest expenses	-417	-394
	-334	-315

Net interest comprises interest income and expenses – calculated using the effective interest method in accordance with IFRS 9 – from financial assets and liabilities not classified as at fair value through profit or loss.

Net interest declined year-on-year by EUR 19m. This is mainly due to interest expenses in connection with borrowing, which outweighed the absence of interest expenses incurred the previous year in connection with a supplementary tax payment. These were offset by lower expenses for the compounding of pension provisions, since the base rate was 0.6% lower on average between 31 December 2018 and 31 December 2019.

14 Other financial items

T079 OTHER FINANCIAL ITEMS

in €m	2020	2019
Result of fair value hedges – change in time value of hedged transactions	60	-90
Result of fair value hedges – change in time value of hedging instruments	-41	89
Ineffective portion of derivatives used as cash flow hedges	-144	-18
Result of derivatives held for trading classified as at fair value through profit or loss	-790	359
Result of measuring securities classified as at fair value through profit or loss	-57	-17
Exchange rates effects from financial liabilities	28	-5
	-944	318

Due to the significant decline in business activity in the 2020 financial year, there were “overhedges” on fuel hedges and, to a lesser extent, on currencies due to a lack of underlying transactions. These overhedges are recognised in the income statement and they reduced the financial result by a total of EUR 752m. In addition, there was a higher level of hedge ineffectiveness, particularly due to the restriking of currency hedges with positive market values as well as the postponement of investment spending, which resulted in a total expense of EUR 125m. In addition, expenses of EUR 96m related to the market valuation of the convertible bond issued in the financial year.

In the previous year, the release of cash flow hedges for investment spending no longer expected in the future had a positive effect on other financial items, with earnings of EUR 402m.

15 Income taxes

In the 2020 financial year, tax income of EUR 1,865m (previous year: tax expense of EUR 615m) was incurred due to the loss situation, which breaks down as follows:

T080 INCOME TAXES

in €m	2020	2019
Current income taxes	80	547
Deferred income taxes	-1,945	68
	-1,865	615

Current income taxes include corporation tax, solidarity surcharge, trade tax and other income taxes paid outside Germany totalling EUR 4m for 2020 (previous year: EUR 382m). Other tax expenses of EUR 76m related to prior years (previous year: EUR 165m). The tax rates used to calculate deferred taxes abroad in the 2020 financial year were unchanged from the previous year at 3.5% to 35.0%. For measuring deferred taxes, the relevant taxation rules in force or adopted at the balance sheet date are used. Deferred taxes includes EUR 649m (previous year: EUR 6m) in tax income from temporary differences.

The following table reconciles expected and effective tax expenses. Expected tax expense is calculated by multiplying profit before income taxes by a tax rate of 25% (previous year: 25%). This is calculated as the average estimated value for the tax group of the Group parent company and is made up of a tax rate of 15.825% (previous year: 15.825%) for corporation tax/solidarity surcharge and 9.175% for trade tax (previous year: 9.175%). The portion of trade tax related to the foreign air transport operations is deducted when calculating the tax rate for trade tax, particularly in the case of the Group parent company with its head office in Germany.

For the first time, individualised tax rates of between 15.825% and 31% were used in the financial year to determine the deferred taxes on temporary differences attributable to the individual companies of the tax group of the parent company

instead of a uniform estimated value of 25% for the entire tax group. With respect to the surpluses of deferred tax assets capitalised in previous years, this resulted in an increase of EUR 109m in total, of which EUR 51m was recognised directly in equity due to its initial recognition.

T081 TAX RECONCILIATION

in €m	2020		2019	
	Basis of assessment	Tax expenses	Basis of assessment	Tax expenses
Expected income tax expenses	-8,631	-2,158	1,860	465
Tax free gains/losses	-122	31	-	-1
Non-deductible costs	488	122	-	59
Non-taxable income	278	-70	-	-84
Non-taxable income from equity investments	-226	57	-	-51
Difference between local taxes and the deferred tax rates of the parent company as well as effects of changes in tax rates	-	-132	-	-5
Taxes from other periods ¹⁾	-	44	-	193
Effects from use or deferred tax assets not recognised	-	241	-	39
Recognised income tax expenses	-	-1,865	-	615

¹⁾ Deferred tax expenses of EUR -32m for other periods (previous year: EUR 28m tax revenue) and effective tax expenses of EUR 76m for other periods (previous year: EUR 165m).

Deferred tax liabilities of EUR 47m (previous year: EUR 48m) were not recognised on temporary differences in connection with shares in subsidiaries, as the temporary differences are not expected to reverse in the foreseeable future.

Deferred tax assets and liabilities in 2020 and 2019 were allocable to the following items in the statement of financial position:

T082 DEFERRED TAX ASSETS AND LIABILITIES

in €m	31 Dec 2020		31 Dec 2019	
	Assets	Liabilities	Assets	Liabilities
Tax loss carry-forwards and tax credits	1,341	-	45	-
Pension provisions	3,424	-	2,574	-
Intangible assets, property, plant and equipment	-	1,034	-	1,112
Non-current financial assets	-	24	-	24
Fair value measurement of financial instruments	113	-	-	235
Provisions for contingent losses	15	-	6	-
Receivables/liabilities/other provisions	215	-	163	-
Inventories	313	-	242	-
Assets held for sale	-	15	-	-
Other	-	-	-	2
Offset amounts	-588	-588	-762	-762
	4,833	485	2,268	611

In addition to a deferred tax asset on temporary differences of EUR 3,243m (previous year: EUR 57m), a deferred tax receivable on losses of EUR 1,335m (previous year: EUR 28m) was recognised for companies incurring on losses incurred a net tax loss in the reporting year or in the previous year, because tax and earnings planning indicates that there is a sufficient probability that the tax receivable will be realised.

EUR 1,200m of the capitalised loss carry-forwards related to the tax group of Deutsche Lufthansa AG. The full amount of the loss carry-forward arising in the financial year was recognised. The basis for this approach was long-term tax planning based on current corporate planning. Both external forecasts, e.g. by the industry association IATA, and internal planning currently assume that the current loss situation is an exogenous shock which will be overcome in the next few years and which does not fundamentally call into question the sustainable profitability of the industry or the company. Deutsche Lufthansa AG has shown in the past that positive tax results can be achieved over long-term periods, and the Company's planning indicates a return to sustained positive tax results by 2024 and for subsequent years. Based on these external and internal indicators, as well as the fact that under current German law the tax loss carry-forward is not restricted in time, the Company assumes that there is a high probability of sufficiently positive tax results in the future to be able to fully utilise the deferred tax assets. This means that the period of use is subject to uncertainty, but the Company does not consider the full use as such to be subject to uncertainty. An additional EUR 87m related to Swiss

International Air Lines, which, on the basis of current earnings planning, considers the use of the loss carry-forwards to be sufficiently secure for the period of their legally limited useful life.

In addition to recognised deferred tax assets from tax loss carry-forwards and tax credits, further tax loss carry-forwards and temporary differences totalling EUR 3,324m (previous year: EUR 3,050m) exist for which no deferred tax assets could be recognised.

The usage restrictions for non-capitalised tax loss carry-forwards and the resulting deferred taxes are distributed as follows:

T083 LIMITS ON THE USE OF NON-CAPITALISED LOSS CARRY-FORWARDS

in €m	Non-capitalised loss carry-forwards	Deferred taxes
Usable		
until 2024	328	74
until 2025	89	23
until 2026	106	25
until 2027	70	18
until 2028	70	17
until 2029	70	18
2030 and beyond	2,443	606
Total	3,176	781

16 Earnings per share

Basic/diluted earnings per share are calculated by dividing consolidated net profit by the weighted average number of shares in circulation during the financial year. To calculate the average number of shares, the shares bought back and reissued for the employee share programmes are included pro rata temporis.

The convertible bond issued in 2020 has not yet had an impact on earnings per share. The shares that could potentially be issued were not taken into account when determining diluted earnings, because they would have improved them.

T084 EARNINGS PER SHARE

		2020	2019
Basic/diluted earnings per share	€	-12.51	2.55
Consolidated net profit/loss	€m	-6,725	1,213
Weighted average number of shares		537,591,908	475,730,992

As the parent company of the Group, Deutsche Lufthansa AG reported a net loss according to HGB of EUR 780m for the 2020 financial year.

In 2020, the distributable profit for the 2019 financial year was transferred in full to retained earnings.

NOTES TO THE CONSOLIDATED STATEMENT OF FINANCIAL POSITION

Assets

17 Goodwill and intangible assets with an indefinite useful life

T085 GOODWILL AND INTANGIBLE ASSETS WITH AN INDEFINITE USEFUL LIFE

in €m	Goodwill from consolidation	Intangible assets with an indefinite useful life	Total
Cost as of 1 Jan 2019	1,050	648	1,698
Accumulated impairment losses	-314	-3	-317
Carrying amount 1 Jan 2019	736	645	1,381
Currency translation differences	-	14	14
Additions due to changes in consolidation	-	-	-
Additions	-	-	-
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Reclassifications to assets held for sale	-	-	-
Impairment losses	-	-	-
Reversal of impairment losses	-	-	-
Carrying amount 31 Dec 2019	736	659	1,395
Cost as of 1 Jan 2020	1,044	663	1,707
Accumulated impairment losses	-308	-4	-312
Carrying amount 1 Jan 2020	736	659	1,395
Currency translation differences	-3	-	-3
Additions due to changes in consolidation	-	-	-
Additions	-	-	-
Reclassifications	-	-	-
Disposals due to changes in consolidation	-1	-	-1
Disposals	-	-	-
Reclassifications to assets held for sale	-	-	-
Impairment losses	-222	-	-222
Reversal of impairment losses	-	-	-
Carrying amount 31 Dec 2020	510	659	1,169
Cost as of 31 Dec 2020	1,038	662	1,700
Accumulated impairment losses	-528	-3	-531

All goodwill and intangible assets with an indefinite useful life were subjected to a regular impairment test in 2020 as required by IAS 36. Furthermore, there is the obligation to perform an impairment test if there is an indication of impairment. For impairment testing following an indication of impairment ➤ Note 17, p. 165f.

Acquired brands and slots have an indefinite useful life due to their lasting legal and economic significance. The tests were performed at the level of the smallest cash generating unit (CGU) on the basis of fair value less costs to sell or value in use.

The following table provides an overview of the goodwill tested and the assumptions made in the respective impairment tests regarding the smallest possible cash-generating unit (CGU) in each case.

T086 IMPAIRMENT TESTS OF GOODWILL 2020

Name of the CGU	Lufthansa German Airlines	Brussels Airlines	Eurowings	LSG Sky Chefs USA group	LSG Sky Chefs Korea	Other ¹⁾
Segment	Network Airlines	Brussels Airlines	Eurowings	Catering	Catering	Catering/Service and financial companies
Carrying amount of goodwill (31 Dec)	€ 252m	€ 44m	–	€ 177m	–	€ 36m
Impairment losses	–	–	€ 57m	€ 100m	€ 57m	€ 8m
Duration of planning period	4 years	4 years	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	1.0%	1.0%	1.0%	2.2%	2.0%	1.0% to 3.1%
Discount rate	5.5% ²⁾	5.5% ²⁾	5.5% ²⁾	5.7% ²⁾	5.5% ²⁾	5.0% to 9.6% ²⁾

¹⁾ Goodwill of less than EUR 25m carrying amount in any individual instance.

²⁾ After-tax rate.

The assumptions on revenue growth used for the impairment tests are based on approved internal budgets and external industry-specific sources (for example IATA) in the planning period. Discounts were also applied based on the planning of the individual businesses, which take into account the corresponding planning uncertainties due to the crisis situation. Within the scope of the Group planning, the discounts were set at approximately 18% of the Adjusted EBIT figure on a long-term basis and were allocated to the units on a pro rata basis during the impairment tests. In general, a slow market recovery and with it the recovery of the earnings situation is assumed by the end of 2024, although losses are still assumed for the years 2021 and 2022. Even in the long term, capacities are not expected to return to pre-crisis levels. The duration of the ramp-up phase is an important risk factor. The margins used are based on past experience or were developed on the basis of cost-cutting measures initiated on the basis of the plans for the ReNew and ReStructure programmes. Long-term investment rates are based on past experience and take account of the replacement of any means of production envisaged during the planning period as well as current changes in fleet planning. Costs of the central functions were charged to the individual units based on their use of these functions. The tests also covered the right-of-use assets recognised in line with IFRS 16.

The weighted average cost of capital is calculated using market data to derive leverage ratios, beta factors and borrowing costs from a peer group that is reviewed annually. A market risk premium of 7.5% was used as a basis (previous year: 7.5%). Regional risks are taken into account by applying appropriate risk premiums.

With the onset of the coronavirus pandemic, the occurrence of a “triggering event” meant that interim impairment tests were carried out for all material business entities at the level of the cash-generating units during the first quarter. Compared with year-end 2019, the tests were performed with a discount rate that was 1 percentage point higher and 1 percentage point lower growth from the end of the planning period. This was based on the preliminary updates of the Company’s planning, with an additional discount applied to the earnings figures in the terminal value for the existing high uncertainties at the beginning of the crisis and the preliminary nature of the planning. The result was an impairment loss at the entities Eurowings and LSG USA. Goodwill of EUR 57m at Eurowings was written off in full and goodwill at LSG USA was impaired by EUR 100m.

At the end of the year, impairment tests were carried out at the level of the cash-generating units based on the assumptions described in the table.

The LSG Korea unit, the flight training operations and other smaller catering companies had to be written down. Goodwill of EUR 57m at LSG Korea as well as the goodwill of the flight training operations and various catering companies totalling EUR 8m were written off in full. Property, plant and equipment at the flight training operations were also impaired in the amount of EUR 22m.

An impairment test was carried out for LSG Sky Chefs Nigeria and LSG Sky Chefs Kenya as of 31 December 2020, which resulted in impairment losses of EUR 9m on property, plant and equipment.

Assuming sustained revenue growth by the CGUs as described in the table, the recoverable amounts for the units without impairment indicators at the end of the year would exceed the carrying amount by a significant figure, even if the growth assumptions were reduced by one percentage point in each case. Worsening the scenarios by one percentage point in each case, in terms of planned EBITDA margins or the discount rates used for the impairment tests, would also not reduce the recoverable amounts below the respective carrying amounts for these CGUs. A reduction in the EBITDA

margin of more than 0.6 percentage points only leads to a deficit for the LSG North America business entity. The deterioration in the parameters mentioned would also not result in any significant additional impairment needs for those units for which the impairment tests led to write-downs at the end of the year. The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation.

The following table shows the assumptions used for the previous year's impairment tests.

T086 IMPAIRMENT TESTS OF GOODWILL 2019

Name of the CGU	Lufthansa German Airlines	Brussels Airlines	Eurowings	LSG Sky Chefs USA group	LSG Sky Chefs Korea	Other ¹⁾
Segment	Network Airlines	Brussels Airlines	Eurowings	Catering	Catering	Catering/Service and financial companies
Carrying amount of goodwill (31 Dec)	€ 252m	€ 44m	€ 57m	€ 277m	€ 60m	€ 45m
Impairment losses	-	-	-	-	-	-
Duration of planning period	4 years	4 years	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	2.0%	2.0%	2.0%	1.9%	2.6%	1.7% to 3.5%
Discount rate	3.9% ²⁾	3.9% ²⁾	3.9% ²⁾	4.6% ³⁾	4.1% ³⁾	3.9% ²⁾ to 7.7% ³⁾

¹⁾ Goodwill of less than EUR 25m carrying amount in any individual instance.

²⁾ After-tax rate.

³⁾ Pre-tax rate.

The earnings situation of Austrian Airlines would also have triggered an impairment test if the value of the assets attributable to it had not already been confirmed by the impairment testing of the slots described below.

The intangible assets with indefinite useful lives consist of slots purchased as part of company acquisitions (insofar as they are tradeable) and brand names acquired. The slots were included in the impairment test for the smallest cash-generating unit (CGU) to which the respective slots are allocated.

The following table shows the assumptions made for the impairment testing.

T087 IMPAIRMENT TESTS OF SLOTS 2020

Group company	SWISS	Austrian Airlines
Carrying amount for slots (31 Dec)	€ 135m	€ 23m
Impairment losses	-	-
Duration of planning period	4 years	4 years
Revenue growth p.a. after end of planning period	1.0%	1.0%
Discount rate	5.5% ¹⁾	5.5% ¹⁾

¹⁾ After-tax rate.

Based on sustainable revenue growth according to assumptions described in the table, the recoverable amounts significantly exceed the carrying amounts. Even if the assumptions for revenue growth and margins were to be reduced or the discount rate were to be increased by one percentage point in each case, the recoverable amounts would exceed the carrying amounts.

The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation.

The following table shows the assumptions used for the previous year's impairment tests.

T087 IMPAIRMENT TESTS OF SLOTS 2019

Group company	SWISS	Austrian Airlines
Carrying amount for slots (31 Dec)	€ 135m	€ 23m
Impairment losses	-	-
Duration of planning period	4 years	4 years
Revenue growth p.a. after end of planning period	2.0%	2.0%
Discount rate	3.9% ¹⁾	3.9% ¹⁾

¹⁾ After-tax rate.

The slots purchased by Deutsche Lufthansa AG are allocated at historical cost to the Lufthansa German Airlines unit at a carrying amount of EUR 76m and to the Eurowings unit with a carrying amount of EUR 36m. Both carrying amounts as of 31 December 2020, which were unchanged compared with the previous year, were subjected to impairment testing with identical assumptions to those used for the goodwill of the CGUs Lufthansa German Airlines and Eurowings.

The regular impairment test for the brands acquired was carried out on the basis of the revenue generated from each brand.

The following additional assumptions were used in the impairment test for the acquired brands:

T088 IMPAIRMENT TESTS OF BRANDS 2020

Group company	SWISS	Austrian Airlines	Brussels Airlines	Other
Carrying amount for brand (31 Dec)	€ 238m	€ 107m	€ 37m	€ 7m
Impairment losses	–	–	–	–
Duration of planning period	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	1.0%	1.0%	1.0%	1.0% to 2.2%
Savings in hypothetical leasing payments before taxes (royalty rate)	0.63%	0.35%	0.20%	0.23% to 0.50%
Discount rate	5.5% ¹⁾	5.5% ¹⁾	5.5% ¹⁾	5.5% to 5.7% ¹⁾

¹⁾ After-tax rate.

Assuming sustained brand-related revenue growth at the end of the planning period, as described in the table, the recoverable amounts for the brands exceed their carrying amounts, significantly in some cases. If the assumptions for sustained brand-related revenue growth were to be reduced by 0.5 percentage points or the discount rate were to be increased by 0.4 percentage points, the carrying amounts of the Austrian Airlines and Brussels Airlines brands would each fall short of the recoverable amounts. With corresponding sensitivity analyses, there was no impairment charge within the other brands in 2020.

The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation.

The assumptions used for the previous year's impairment tests can be derived from the following table.

T088 IMPAIRMENT TESTS OF BRANDS 2019

Group company	SWISS	Austrian Airlines	Brussels Airlines	Other
Carrying amount for brand (31 Dec)	€ 238m	€ 107m	€ 37m	€ 7m
Impairment losses	–	–	–	–
Duration of planning period	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	2.0%	2.0%	1.0%	0.0% to 2.0%
Savings in hypothetical leasing payments before taxes (royalty rate)	0.63%	0.35%	0.20%	0.23% to 0.50%
Discount rate	3.9% ¹⁾	3.9% ¹⁾	3.9% ¹⁾	3.9% to 4.4% ¹⁾

¹⁾ After-tax rate.

18 Other intangible assets

T089 OTHER INTANGIBLE ASSETS

	Concessions, industrial property rights and similar rights and licences to such rights and assets	Internally developed software	Advance payments	Total
in €m				
Cost as of 1 Jan 2019	1,289	150	152	1,591
Accumulated amortisation	-949	-115	-15	-1,079
Carrying amount as of 1 Jan 2019	340	35	137	512
Currency translation differences	4	-1	-	3
Additions due to changes in consolidation	-	-	-	-
Additions	52	3	101	156
Reclassifications	43	11	-41	13
Disposals due to changes in consolidation	-	-	-	-
Disposals	-7	-	-8	-15
Reclassifications to assets held for sale	-1	-	-	-1
Amortisation	-99	-9	-13	-121
Reversal of impairment losses	-	-	-	-
Carrying amount as of 31 Dec 2019	332	39	176	547
Cost as of 1 Jan 2020	1,324	163	205	1,692
Accumulated amortisation	-992	-124	-29	-1,145
Carrying amount as of 1 Jan 2020	332	39	176	547
Currency translation differences	-1	-	-	-1
Additions due to changes in consolidation	-	-	-	-
Additions	18	3	59	80
Reclassifications	80	12	-88	4
Disposals due to changes in consolidation	-	-	-	-
Disposals	-8	-	-4	-12
Reclassifications to assets held for sale	-	-	-	-
Amortisation	-126	-21	-2	-149
Reversal of impairment losses	-	-	-	-
Carrying amount as of 31 Dec 2020	295	33	141	469
Cost as of 31 Dec 2020	1,393	178	157	1,728
Accumulated amortisation	-1,098	-145	-16	-1,259

Non-capitalised research and development expenses for intangible assets of EUR 25m (previous year: EUR 49m) were incurred. Fixed orders have been placed for intangible assets worth EUR 13m (previous year: EUR 14m), but they are not yet at the Lufthansa Group's economic disposal.

19 Aircraft and reserve engines including right-of-use assets

T090 AIRCRAFT AND RESERVE ENGINES INCLUDING RIGHT-OF-USE ASSETS

in €m	Aircraft and reserve engines	Advance payments for aircraft and reserve engines	Total
Cost as of 1 Jan 2019	31,609	1,690	33,299
Accumulated amortisation	-16,122	-	-16,122
Carrying amount as of 1 Jan 2019	15,487	1,690	17,177
Currency translation differences	111	10	121
Additions due to changes in consolidation	-	-	-
Additions	2,149	1,098	3,247
Reclassifications	691	-691	-
Disposals due to changes in consolidation	-	-	-
Disposals	-69	-5	-74
Reclassifications to assets held for sale	-20	-	-20
Depreciation	-2,121	-	-2,121
Reversal of impairment losses	19	-	19
Carrying amount as of 31 Dec 2019	16,247	2,102	18,349
Cost as of 1 Jan 2020	32,945	2,102	35,047
Accumulated amortisation	-16,698	-	-16,698
Carrying amount as of 1 Jan 2020	16,247	2,102	18,349
Currency translation differences	-33	-	-33
Additions due to changes in consolidation	-	-	-
Additions	1,004	258	1,262
Reclassifications	602	-602	-
Disposals due to changes in consolidation	-	-	-
Disposals	-119	-27	-146
Reclassifications to assets held for sale	-149	-	-149
Depreciation	-3,441	-	-3,441
Reversal of impairment losses	-	-	-
Carrying amount as of 31 Dec 2020	14,111	1,731	15,842
Cost as of 31 Dec 2020	32,025	1,731	33,756
Accumulated amortisation	-17,914	-	-17,914

In the aircraft item, aircraft with a carrying amount of EUR 7,048m (previous year: EUR 2,449m) serve as collateral for existing aircraft financing. This collateral is composed as described below:

The item includes 79 aircraft with a carrying amount of EUR 1,988m (previous year: 80 aircraft with a carrying amount of EUR 2,549m), which have mostly been sold to and leased back from foreign leasing companies with the aim of obtaining favourable financing conditions. The leasing companies were fully consolidated as structured entities. The Lufthansa Group is entitled to buy the aircraft back at a fixed price and at a given point in time. Another four aircraft (previous year: none) with a carrying amount of EUR 416m (previous year: EUR 0m) were pledged as collateral under loan agreements.

Furthermore, rights of lien were granted on various shares in leasing companies in Malta and Austria in connection with the state funding for Deutsche Lufthansa AG to secure the KfW loan; these companies own a total of 323 aircraft with a carrying amount of EUR 4,432m. Austrian Airlines in Austria has granted liens on 38 aircraft with a book value of EUR 212m to a banking consortium as part of the stabilisation measures.

In the reporting year, borrowing costs of EUR 19m were capitalised (previous year: EUR 24m). The financing rate used was 0.9% (previous year: 1.3%).

The additions relate to the procurement of new aircraft and engines as well as the capitalisation of engine maintenance events. This item also includes right-of-use assets for aircraft and reserve engines amounting to EUR 261m (previous year: EUR 227m).

As a result of the nearly complete shutdown of passenger flight operations, which could only be reversed in stages, the operation plans for all components of the fleet were revised. As part of these restructuring plans, charter contracts and leases were terminated and company-owned aircraft were either released for sale, permanently retired or put into deep storage. The aircraft concerned and right-of-use assets for aircraft and reserve engines were subjected to an impairment test insofar as a resumption of operations is considered unlikely on the basis of current planning. On the basis of expected realisation prices or due to the lack of additional values in use of the leases, this led to write-downs totalling EUR 1,515m (previous year: EUR 36m) for a total of 115 aircraft and other reserve engines ➤ **Note 10, p. 160ff.**

Order commitments for aircraft and reserve engines amount to EUR 12.9bn (previous year: EUR 14.3bn). We also refer to the comments on fleet orders in the management report (➤ **Principles of the Group, Fleet and route network, p. 25ff.**).

20 Repairable spare parts

T091 NOTES ON REPAIRABLE SPARE PARTS FOR AIRCRAFT

in €m	2020			2019		
	Gross acquisition cost	Accumulated depreciation	Net carrying amount	Gross acquisition cost	Accumulated depreciation	Net carrying amount
Pool material	2,283	972	1,311	2,390	938	1,452
Non-pool material	1,055	543	512	1,246	428	818
Total	3,338	1,515	1,823	3,636	1,366	2,270

The additions for the year (netted against disposals) accordingly amounted to EUR -107m (previous year: EUR +139m) for pool material and EUR -191m (previous year: EUR +100m) for non-pool material in the financial year; the net change in depreciation recognised in profit and loss was EUR 34m (previous year: EUR 60m) and EUR 115m (previous year: EUR 42m). Of the depreciation and amortisation expense

for the financial year, EUR 158m related to impairments in connection with materials that are expected to become obsolete as a result of the adjusted outlook for business development and spare parts for aircraft types that are not expected to be supplied with materials even after the end of the coronavirus crisis.

21 Property, plant and equipment including right-of-use assets

T092 PROPERTY, PLANT AND OTHER EQUIPMENT INCLUDING RIGHT-OF-USE ASSETS

in €m	Land and buildings	Technical equipment and machinery	Other equipment, operating and office equipment	Advance payments and plant under construction	Total
Cost as of 1 Jan 2019	4,248	1,315	1,528	142	7,233
Accumulated depreciation	-1,467	-946	-1,036	-13	-3,462
Carrying amount as of 1 Jan 2019	2,781	369	492	129	3,771
Currency translation differences	21	4	2	3	30
Additions due to changes in consolidation	36	-	-	-	36
Additions	692	42	151	102	987
Reclassifications	50	15	18	-98	-15
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-50	-1	-8	-3	-62
Reclassifications to assets held for sale	-112	-22	-56	-9	-199
Depreciation	-327	-58	-122	-	-507
Reversal of impairment losses	-	-	-	-	-
Carrying amount as of 31 Dec 2019	3,091	349	477	124	4,041
Cost as of 1 Jan 2020	4,714	1,245	1,470	124	7,553
Accumulated depreciation	-1,623	-896	-993	-	-3,512
Carrying amount as of 1 Jan 2020	3,091	349	477	124	4,041
Currency translation differences	-38	-6	-11	-4	-59
Additions due to changes in consolidation	-	-	-	-	-
Additions	228	34	72	61	395
Reclassifications	25	17	18	-65	-5
Disposals due to changes in consolidation	-1	-	-	-	-1
Disposals	-98	-1	-20	-19	-138
Reclassifications to assets held for sale	-11	-3	7	8	1
Depreciation	-363	-79	-120	-2	-564
Reversal of impairment losses	1	-	-	-	1
Carrying amount as of 31 Dec 2020	2,834	311	423	103	3,671
Cost as of 31 Dec 2020	4,703	1,256	1,453	105	7,517
Accumulated depreciation	-1,869	-945	-1,030	-2	-3,846

There are no charges over land and property (previous year: EUR 4m). Pre-emption rights are registered for land held at EUR 178m (previous year: EUR 185m). Other property, plant and equipment carried at EUR 4m (previous year: EUR 4m) serves as collateral for existing financing arrangements.

The following items of property, plant and equipment have been ordered, but are not yet at the Lufthansa Group's economic disposal:

T093 ORDERS OF PROPERTY, PLANT AND EQUIPMENT AS OF THE REPORTING DATE

in €m	31 Dec 2020	31 Dec 2019
Land and buildings	11	20
Technical equipment and vehicles	19	37
Operating and office equipment	26	61
	56	118

22 Leases

The following table shows the carrying amounts of the recognised right-of-use assets and the changes during the reporting period.

T094 RIGHT-OF-USE ASSETS

in €m	Aircraft and reserve engines	Land and buildings	Technical equipment and machinery	Other intangible assets and technical equipment	Total
Cost as of 1 Jan 2019	1,456	1,749	1	28	3,234
Accumulated depreciation	-477	-128	-1	-6	-612
Carrying amount as of 1 Jan 2019	979	1,621	-	22	2,622
Currency translation differences	4	14	-	-	18
Additions due to changes in consolidation	-	-	-	-	-
Additions	227	642	-	13	882
Reclassifications	-1	-6	-	-	-7
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-8	-45	-	-1	-54
Reclassifications to assets held for sale	-	-30	-	-1	-31
Depreciation	-264	-244	-	-12	-520
Reversal of impairment losses	-	-	-	-	-
Carrying amount as of 31 Dec 2019	937	1,952	-	21	2,910
Cost as of 1 Jan 2020	1,588	2,225	-	38	3,851
Accumulated depreciation	-651	-273	-	-17	-941
Carrying amount as of 1 Jan 2020	937	1,952	-	21	2,910
Currency translation differences	-	-21	-	-	-21
Additions due to changes in consolidation	-	-	-	-	-
Additions	261	205	-	6	472
Reclassifications	-3	-1	-	-	-4
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-24	-97	-	-1	-122
Reclassifications to assets held for sale	-	-2	-	-1	-3
Depreciation	-311	-264	-	-10	-585
Reversal of impairment losses	-	1	-	-	1
Carrying amount as of 31 Dec 2020	860	1,773	-	15	2,648
Cost as of 31 Dec 2020	1,652	2,273	-	37	3,962
Accumulated depreciation	-792	-500	-	-22	-1,314

The Lufthansa Group mainly leases property, particularly at airports, as well as aircraft and other operating and office equipment. Leases may include renewal and termination options. The terms of the leases are negotiated individually and cover a wide range of different areas. Longer-term leases exist particularly for property. There is a remaining lease term of up to 35 years for land and buildings (previous year: up to 36 years) as of the reporting date.

The average remaining term of the aircraft leases as of 31 December 2020 was three years (previous year: four years). The additions to rights-of-use assets for aircraft in 2020 relate to six aircraft.

The leases existing under IFRS 16 affected the consolidated income statement as follows:

T095 LEASE EXPENSES RECOGNISED IN PROFIT OR LOSS

in €m	2020	2019
Amortisation of right-of-use assets	585	520
Interest expenses for lease liabilities	69	72
Expenses for short-term leases	50	193
Expenses for low-value leases	120	120
Variable lease payments	80	126

Some of the Lufthansa Group's leases for properties and aircraft include renewal options and variable lease payments. They are used to obtain the greatest possible flexibility in terms of capacities. They have not been taken into account in various cases when measuring the lease liabilities, because it is not reasonably certain that they will be exercised. Potential future lease payments for periods after the exercise date of the renewal options are summarised in the following table:

T096 DISCLOSURES ON RENEWAL OPTIONS AND VARIABLE LEASE PAYMENTS 2020

in €m	Recognised lease liability (discounted)	Potential future lease payments not included in lease liabilities (undiscounted payments)		Total
	31 Dec 2020	Payable 2021 - 2025	Payable after 2025	
Aircraft	461	306	42	348
Property	1,812	71	435	506
Total	2,273	377	477	854

T096 DISCLOSURES ON RENEWAL OPTIONS AND VARIABLE LEASE PAYMENTS 2019

in €m	Recognised lease liability (discounted)	Potential future lease payments not included in lease liabilities (undiscounted payments)		Total
	31 Dec 2019	Payable 2020 - 2024	Payable after 2024	
Aircraft	486	296	34	330
Property	1,929	55	396	451
Total	2,415	351	430	781

Where termination options existed for individual leases, their exercise was considered unlikely, so that additional lease payments were already taken into account in the corresponding lease liability.

Amounts included in the cash flow statement are shown in the following table:

T097 CASH OUTFLOWS FOR LEASES

in €m	2020	2019
Lease expenses from short-term and low-value leases and variable lease payments not included in the measurement of lease liabilities	250	439
Repayment of the redemption portion of the lease liability	480	499
Interest payments	67	69
Total	797	1,007

Lease payments are shown as cash flows from financing activities unless they are lease payments included in the measurement of lease liabilities, which are shown as cash flow from operating activities.

The maturity analysis of lease liabilities is shown under borrowings, [Note 37, p. 191](#).

Information about operating leases in which the Lufthansa Group is lessor can be found in [Note 7, p. 159f](#).

23 Equity investments accounted for using the equity method

T098 EQUITY INVESTMENTS ACCOUNTED FOR USING THE EQUITY METHOD

in €m	Investments in joint ventures	Investments in associated companies	Total
Cost as of 1 Jan 2019	391	263	654
Accumulated impairment losses	-	-4	-4
Carrying amount as of 1 Jan 2019	391	259	650
Currency translation differences	3	9	12
Additions due to changes in consolidation	-	-	-
Additions	77	-	77
Changes with and without an effect on profit and loss	70	73	143
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Dividends paid	-62	-101	-163
Reclassifications to assets held for sale	-	-4	-4
Impairment losses	-43	-	-43
Reversal of impairment losses	-	-	-
Carrying amount as of 31 Dec 2019	436	236	672
Cost as of 1 Jan 2020	479	241	720
Accumulated impairment losses	-43	-5	-48
Carrying amount as of 1 Jan 2020	436	236	672
Currency translation differences	-11	-19	-30
Additions due to changes in consolidation	-	-	-
Additions	31	-	31
Changes with and without an effect on profit and loss	-167	-33	-200
Reclassifications	39	-	39
Disposals due to changes in consolidation	-	-	-
Disposals	2	-	2
Dividends paid	-32	-13	-45
Reclassifications to assets held for sale	-	-	-
Impairment losses	-66	-	-66
Reversal of impairment losses	-	-	-
Carrying amount as of 31 Dec 2020	232	171	403
Cost as of 31 Dec 2020	340	176	516
Accumulated impairment losses	-108	-5	-113

Reclassifications of interests in joint ventures relate to the right to profits of Terminal 2 Gesellschaft mbH & Co. OHG for 2019, which was reported under current receivables in the previous year and whose collection was partially waived in 2020.

Due to the deterioration in the business outlook in connection with the coronavirus crisis and its further impact on the market situation, impairment tests were carried out for the joint ventures XEOS and Spairliners in the MRO segment as well as at Diners Club Spain (Service & Finance companies), resulting in write-downs totalling EUR 66m. In the previous year, write-downs totalled EUR 43m and related to a Catering company. For the write-down on shares in joint ventures in 2020, see [Note 10, p. 160f.](#)

Individual interests in companies accounted for using the equity method

The following tables show condensed earnings and balance sheet data for the individual material joint ventures accounted for using the equity method.

T099 BALANCE SHEET DATA GÜNES EKSPRES HAVACILIK ANONIM SİRKETİ (SUNEXPRESS), ANTALYA, TURKEY

in €m	31 Dec 2020	31 Dec 2019
Current assets	558	594
of which cash and cash equivalents	310	317
Non-current assets	845	1,204
Current liabilities	738	769
Non-current liabilities	559	715
Current financial liabilities (except trade and other payables and provisions)	490	312
Non-current financial liabilities (except trade and other payables and provisions)	442	583
Shareholders' equity	106	314
Share of equity	53	157
Other	24	21
Carrying amount	77	178

T100 INCOME STATEMENT DATA GÜNES EKSPRES HAVACILIK ANONIM SİRKETİ (SUNEXPRESS), ANTALYA, TURKEY

in €m	2020	2019
Revenue	493	1,432
Depreciation and amortisation	103	148
Interest income	4	6
Interest expenses	27	25
Income tax expense or income	19	14
Profit or loss from continuing operations	-158	48
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	-14	9
Total comprehensive income	-172	57
Share of profit or loss from continuing operations	-102	24
Share of comprehensive income	-102	31
Dividends received	-	-

The item "Other" in the reconciliation with the carrying amount for SunExpress primarily includes the difference from the first-time consolidation of the company.

Due to the deterioration in the Company's business situation caused by the coronavirus, an impairment test was also carried out for SunExpress, whose German subsidiary SunExpress Deutschland will cease business operations. This was based on the Company's business plans, which show a crisis experience comparable to that of the Group companies in the passenger business. A discount factor of 7.0%, growth in sales of the perpetual annuity of 1.0% and a contingency discount on the sustainable result of 40% were assumed for the measurement. Even in the event of a further deterioration in the parameters by 2 percentage points or 10 percentage points, respectively, there would be no indication of an impairment of the carrying amounts.

T101 BALANCE SHEET DATA TERMINAL 2 GESELLSCHAFT MBH & CO. OHG, MUNICH AIRPORT, GERMANY

in €m	31 Dec 2020	31 Dec 2019
Current assets	15	46
of which cash and cash equivalents	-	-
Non-current assets	1,265	1,323
Current liabilities	209	400
Non-current liabilities	1,030	908
Current financial liabilities (except trade and other payables and provisions)	137	256
Non-current financial liabilities (except trade and other payables and provisions)	1,002	880
Shareholders' equity	41	61
Share of equity	17	25
Other	-	-
Carrying amount	17	25

T102 INCOME STATEMENT DATA TERMINAL 2 GESELLSCHAFT MBH & CO. OHG, MUNICH AIRPORT, GERMANY

in €m	2020	2019
Revenue	123	357
Depreciation and amortisation	83	84
Interest income	-	-
Interest expenses	31	36
Income tax expense or income	-	10
Profit or loss from continuing operations	-97	78
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	11	16
Total comprehensive income	-86	94
Share of profit or loss from continuing operations	-39	31
Share of comprehensive income	-34	38
Dividends received	12	38

The following table contains summarised aggregated data from the income statements and carrying amounts for the individually immaterial joint ventures accounted for using the equity method.

T103 INCOME STATEMENTS DATA AND CARRYING AMOUNTS OF JOINT VENTURES ACCOUNTED FOR USING THE EQUITY METHOD

in €m	2020	2019
Profit or loss from continuing operations	-30	1
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	-	-
Total comprehensive income	-30	1
Depreciation and amortisation	-66	-43
Carrying amount	138	233

The following table shows condensed earnings data and carrying amounts for the individual immaterial associated companies accounted for using the equity method.

T104 INCOME STATEMENTS DATA AND CARRYING AMOUNTS OF ASSOCIATED COMPANIES ACCOUNTED FOR USING THE EQUITY METHOD

in €m	2020	2019
Profit or loss from continuing operations	-40	74
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	3	-1
Total comprehensive income	-37	73
Carrying amount	171	236

24 Other equity investments and non-current securities

T105 OTHER EQUITY INVESTMENTS AND NON-CURRENT SECURITIES

in €m	31 Dec 2020	31 Dec 2019
Investments in affiliated companies	218	223
Investments	34	33
Other investments	252	256
Non-current securities	54	53

Shares in related parties include shares in affiliated companies, joint ventures and associates that are not consolidated for reasons of materiality. These shares are carried at amortised cost. Disclosures on the equity investments and non-current securities can be found in [Note 45, p. 205ff.](#)

25 Non-current loans and receivables

T106 NON-CURRENT LOANS AND RECEIVABLES

in €m	31 Dec 2020	31 Dec 2019
Loans to and receivables from affiliated companies	104	104
Loans to and receivables from other equity investments	-	-
Other loans and receivables	169	241
Emissions certificates	167	124
	440	469

Non-current loans and receivables are carried at amortised cost.

CO₂ emissions certificates valued at EUR 275m were sold and simultaneously repurchased on the market on a forward basis in what are known as repo agreements so that economic ownership of the certificates is maintained. Appropriate other liabilities were recognised in the amount of the consideration received. For the impairment test for emissions certificates, we refer to the disclosures on the cash-generating units (CGUs) Lufthansa German Airlines, SWISS, Austrian Airlines, Eurowings and Brussels Airlines in [Note 17, p. 165ff.](#)

Other receivables include expected reimbursements for obligations for which provisions have been made amounting to EUR 1m (previous year: EUR 2m). Of the non-current receivables, EUR 1m (previous year: EUR 1m) serve as collateral for liabilities.

26 Inventories

T107 INVENTORIES

in €m	31 Dec 2020	31 Dec 2019
Raw materials, consumables and supplies	650	888
Finished goods and work in progress	75	88
Advance payments	1	4
	726	980

No inventories have been pledged as collateral for loans. EUR 571m of the inventories (previous year: EUR 782m) consist of non-repairable spare parts for aircraft.

The gross value of written-down inventories as of 31 December 2020 was EUR 938m (previous year: EUR 998m). Inventories with a carrying amount of EUR 542m (previous year: EUR 716m) are held at net realisable value. Write-downs to net realisable value of EUR 276m were made at the beginning

of the financial year (previous year: EUR 259m). New impairment losses of EUR 129m were recognised in the reporting year (previous year: EUR 41m), most of which relate to the MRO segment. Write-downs of EUR 8m made in the previous year were reversed (previous year: EUR 18m).

27 Contract assets

The Lufthansa Group recognised the following contract assets in 2020:

T108 CONTRACT ASSETS

in €m	31 Dec 2020	31 Dec 2019
Contract assets from MRO and IT services	146	278
Impairment of contract assets	-4	-1
Total contract assets	142	277

28 Trade receivables and other receivables

T109 TRADE RECEIVABLES AND OTHER RECEIVABLES

in €m	31 Dec 2020	31 Dec 2019
Trade receivables		
Trade receivables from affiliated companies	60	58
Trade receivables from other equity investments	1	2
Trade receivables from third parties	1,516	3,802
	1,577	3,862
Other receivables		
Receivables from affiliated companies	54	89
Receivables from other equity investments	1	1
Other receivables	1,161	1,370
Emissions certificates	50	95
	1,266	1,555
Total	2,843	5,417

For the impairment test for emissions certificates, we refer to the disclosures on the cash-generating units (CGU) Lufthansa German Airlines, SWISS, Austrian Airlines, Eurowings and Brussels Airlines in [Note 17, p. 165ff.](#)

As in the previous year, there is no collateral for trade receivables. Other receivables include expected reimbursements for obligations for which provisions have been made amounting to EUR 1m (previous year: EUR 1m).

For disclosures on impairment losses, credit risks and term structures, we refer to [Note 45, p. 205ff.](#)

Other receivables include claims of EUR 140m (previous year: EUR 149m) against insurers in connection with the accident involving the Germanwings aircraft on 24 March 2015. As of the reporting date, these receivables are offset by provisions of EUR 132m for outstanding obligations relating to this accident (previous year: EUR 135m).

Other receivables of EUR 2m (previous year: EUR 0m) serve to secure negative market values of derivatives.

29 Deferred charges and prepaid expenses

Deferred charges and prepaid expenses consist of various services paid for in advance for subsequent periods.

30 Current securities

Current securities are fixed income securities, participation certificates, shares and investments in money market funds.

31 Cash and cash equivalents

This item includes fixed-term deposits in the amount of EUR 2m (previous year: EUR 0m) with terms of four to twelve months. As of 31 December 2020, bank balances in the amount of EUR 12m are pledged to lending banks. Other bank balances of EUR 70m are pledged subject to a condition precedent.

Bank balances in foreign currencies are translated at the exchange rate on the balance sheet date.

32 Assets held for sale

Assets held for sale include 40 aircraft with a carrying amount of EUR 130m. These assets consist of three Boeing 767s, two Boeing 747s, 16 Airbus A320/321s, eight Dash 8-400s, six Bombardier CRJ9s and five Boeing MD-11 freighters. A further EUR 3m relates to one building.

In the previous year, EUR 358m related to the assets in the disposal group "European activities of the LSG group".

**T110 ASSETS AND LIABILITIES IN THE DISPOSAL GROUP
"EUROPEAN BUSINESS OPERATIONS OF THE LSG GROUP"
AFTER CONSOLIDATION AS OF 31 DECEMBER 2019
AND OTHER RECEIVABLES**

in €m	31 Dec 2019
Assets	
Intangible assets	1
Property, plant and other equipment	151
Other non-current assets	13
Deferred tax assets	31
Inventories	80
Other current assets	82
Total	358
Shareholders' equity and liabilities	
Pension provisions	289
Other non-current provisions	16
Other non-current liabilities	32
Other current provisions	11
Trade payables and other non-financial liabilities	175
Other current liabilities	17
Total	540

The closing of the sale to gategroup on 2 December 2020 marked the disposal of the related assets and liabilities. In addition, three cargo aircraft and one building were available for sale.

Shareholders' equity and liabilities

33 Issued capital

SHARE CAPITAL

By resolution of the extraordinary general meeting on 25 June 2020, the issued capital of Deutsche Lufthansa AG was increased by EUR 306,044,326.40 to EUR 1,530,221,624.32 by issuing 119,548,565 registered shares excluding shareholders' subscription rights. The shares were subscribed by the Economic Stabilisation Fund at the nominal amount of EUR 2.56 per share. The capital increase took effect on 2 July 2020 when it was entered in the commercial register. Issued capital is divided into 597,742,822 registered shares, with each share representing EUR 2.56 of issued capital.

AUTHORISED CAPITAL

A resolution passed at the Annual General Meeting on 7 May 2019 authorised the Executive Board until 6 May 2024, subject to approval by the Supervisory Board, to increase the Company's issued capital by up to EUR 450,000,000 by issuing new registered shares on one or more occasions for

payment in cash or in kind (Authorised Capital A). In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board.

A resolution passed at the Annual General Meeting on 7 May 2019 authorised the Executive Board until 6 May 2024, subject to approval by the Supervisory Board, to increase the issued capital by up to EUR 30,000,000 by issuing new registered shares to employees (Authorised Capital B) for payment in cash. Existing shareholders' subscription rights are excluded. As of 31 December 2020, shares with a nominal amount of EUR 7,637,832.68 were issued under this authorization, so that Authorised Capital B still amounted to EUR 22,362,168.32 as of the reporting date.

CONTINGENT CAPITAL

A resolution of the Annual General Meeting on 5 May 2020 increased the company's contingent capital by up to EUR 122,417,728. The contingent capital increase serves to issue up to 47,819,425 shares to the holders or creditors of conversion and/or option rights from convertible bonds with a total nominal value of up to EUR 1.5bn that may be issued by the Company or its Group companies until 4 May 2025 in accordance with the resolution of the Annual General Meeting on 5 May 2020. In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board.

A resolution of the extraordinary general meeting on 25 June 2020 increased the contingent capital of Deutsche Lufthansa AG by up to EUR 102,014,776.32. The contingent capital increase serves to provide up to 39,849,522 shares for the exercise of conversion rights granted to the Economic Stabilisation Fund created by the Stabilisation Fund Act as a silent shareholder for Silent Participation II-A at a strike rate of EUR 2.56 per share by resolution of the extraordinary general meeting on 25 June 2020. The rights can be exercised if a decision is published to make a takeover offer pursuant to Section 10 Securities Acquisition and Takeover Act (WpÜG) or if control is acquired pursuant to Sections 35 and 29 WpÜG. The buyer can exercise the conversion rights at any time if the Silent Participation II-A is sold to a private purchaser.

A resolution of the extraordinary general meeting on 25 June 2020 increased the contingent capital of Deutsche Lufthansa AG by up to EUR 897,985,223.68. The contingent capital increase serves to provide up to 350,775,478 shares for the exercise of conversion rights granted to the Economic Stabilisation Fund created by the Stabilisation Fund Act as a silent shareholder for antidilution and/or coupon protection for Silent Participation II-B by resolution of the extraordinary general meeting on 25 June 2020. If the conversion right is exercised to protect against dilution, the new shares will be issued at the current market price on the conversion date, less 10%. If the conversion right is exercised to protect the

coupon, they are issued at the current market price on the conversion date, less 5.25%. The conversion rights expire if Silent Participation II-B is assigned to a third party.

AUTHORISATION TO PURCHASE TREASURY SHARES

A resolution passed at the Annual General Meeting held on 7 May 2019 authorised the Executive Board pursuant to Section 71 Paragraph 1 No. 8 Stock Corporation Act (AktG) to purchase treasury shares until 6 May 2024. The authorisation is limited to 10% of current issued capital, which can be purchased on the stock exchange or by a public purchase offer to all shareholders. The authorisation states that the Executive Board can use the shares, in particular, for the purposes defined in the resolution passed at the Annual General Meeting. According to the resolution of the Annual General Meeting held on 7 May 2019, the Executive Board is also authorised to purchase treasury shares by means of derivatives and to conclude corresponding derivative transactions.

At various points in 2020, Deutsche Lufthansa AG bought back 1,299,165 of its own shares at an average price of EUR 7.27. This corresponds to 0.22 % or EUR 3,325,862.40 of share capital.

The shares purchased were used as follows:

- 1,296,504 shares were transferred to the employees of Deutsche Lufthansa AG and to 33 other affiliated companies and equity investments as part of the profit-sharing for 2019, at a share price of EUR 7.56.
- 517 shares were transferred for employees of Deutsche Lufthansa AG and another affiliated company as part of the profit-sharing for 2018, at a share price of EUR 14.18.
- 2,144 shares were resold at a price of EUR 11.14.

On the balance sheet date, treasury shares were no longer held.

CAPITAL MANAGEMENT

After overcoming the coronavirus pandemic, the Lufthansa Group continues to aim for a sustainable equity ratio of 25% in order to ensure long-term financial flexibility and stability as a basis for the Group's growth targets. As of 31 December 2020 and 2019, equity and total assets were as follows:

T111 EQUITY AND LIABILITIES

in €m	31 Dec 2020	31 Dec 2019
Shareholders' equity	1,387	10,256
In % of total assets	3.5	24.0
Liabilities	38,097	32,403
In % of total assets	96.5	76.0
Total capital	39,484	42,659

In the financial year 2020, the equity ratio declined by 20.5 percentage points compared with the figure for the previous year to 3.5%.

Deutsche Lufthansa AG's Articles of Association do not stipulate any capital requirements.

34 Reserves

Capital reserves only include the share premium paid on capital increases and a convertible bond that was redeemed in full in previous years. The legal reserve contained in retained earnings of Deutsche Lufthansa AG is unchanged at EUR 26m; other reserves consist of other retained earnings.

The following table shows changes in other neutral reserves in 2020:

T112 NOTES ON OTHER COMPREHENSIVE INCOME

in €m	2020	2019
Other comprehensive income after income taxes		
Currency translation differences		
Profit/loss for the period	-107	115
Reclassification adjustments recognised in profit or loss	-	-
Subsequent measurement of financial assets at fair value (with recycling)	-29	10
Subsequent measurement of financial assets at fair value (without recycling)	1	7
Profit/loss for the period	-	-
Reclassification adjustments recognised in profit or loss	-	-
Subsequent measurement of hedges - cash flow hedge reserve	-655	258
Subsequent measurement of hedges - costs of hedging	143	476
Profit/loss for the period	-1,119	1,278
Reclassification adjustments recognised in profit or loss	607	-544
Other comprehensive income from investments accounted for using the equity method		
Profit/loss for the period - reclassifiable	4	13
Profit/loss for the period - non-reclassifiable	-	-
Revaluation of defined-benefit pension plans	-2,584	-1,108
Other expenses and income recognised directly in equity (with recycling)	-4	15
Other expenses and income recognised directly in equity (without recycling)	-	-1
Income taxes on items in other comprehensive income	747	163
Other comprehensive income after income taxes	-2,484	-52

T113 NOTE ON INCOME TAXES RECOGNISED FOR OTHER COMPREHENSIVE INCOME

in €m	2020			2019		
	Amount before income taxes	Tax expenses/ income	Amount after income taxes	Amount before income taxes	Tax expenses/ income	Amount after income taxes
Currency translation differences	-107	-	-107	115	-	115
Subsequent measurement of financial assets at fair value (with recycling)	-29	-7	-36	10	-	10
Subsequent measurement of financial assets at fair value (without recycling)	1	-	1	7	-	7
Subsequent measurement of hedges – cash flow hedge reserve	-655	166	-489	258	-64	194
Subsequent measurement of hedges – costs of hedging	143	-34	109	476	-118	358
Other comprehensive income from investments accounted for using the equity method – reclassifiable	4	-	4	13	-	13
Revaluation of defined-benefit pension plans	-2,584	622	-1,962	-1,108	345	-763
Other expenses and income recognised directly in equity (with recycling)	-4	-	-4	15	-	15
Other expenses and income recognised directly in equity (without recycling)	-	-	-	-1	-	-1
Other comprehensive income	-3,231	747	-2,484	-215	163	-52

The overall change in equity is shown in [T060 Consolidated statement of changes in shareholders' equity](#), p. 138.

35 Pension provisions

The Lufthansa Group's pension obligations comprise both defined benefit and defined contribution plans and include both obligations to make current payments and entitlements to future pension payments.

Obligations under defined benefit pension plans for employees of the Lufthansa Group related mostly to pension obligations in Germany, Switzerland, Austria and the USA. Various commitments have been made to different groups of employees.

For the employees in Germany and for expatriate staff who joined the Company before 1995, the supplementary pension scheme for state employees (VBL) was retained as the Company's pension scheme after the Lufthansa Group's privatisation. Employees who joined after 1994 received a retirement benefit commitment based on an average salary plan, which provided for pension units to be granted annually, with the size of these components depending on the employee's age and salary. In 2003, the VBL pension scheme was changed to the average salary plan applicable to the employees recruited after 1995. Between 2015 and 2017, the existing domestic retirement benefit commitments to the individual groups of employees have been converted successively into defined contribution schemes.

The Lufthansa collective agreement on benefits for ground staff established a new company retirement benefit plan in the form of a defined-contribution benefit commitment for the ground staff in Germany, in particular those at Deutsche Lufthansa AG, Lufthansa Cargo AG, the Lufthansa Technik group and the LSG group. For employees recruited before 1 January 2016, the entitlements vested up until 31 December 2015 are maintained. For service periods starting from 1 January 2016, employees can reach the same level of benefits by making contributions from their own pocket. For employees recruited from 1 January 2016, the contributions to the new model will be invested on the capital market. When the employee reaches retirement age, the entire account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum and while guaranteeing the contributions that were originally made.

On 17 March 2017, the "Lufthansa Pension Cabin" wage agreement was signed with the trade union UFO for cabin crew at Deutsche Lufthansa AG (DLH). It replaces the agreements on the "Lufthansa Company Pension" for cabin crew and the "Lufthansa Transitional Benefit for Cabin Crew".

For employees recruited up to 5 July 2016, the pension entitlements vested up until 30 June 2016 are maintained. For service periods from 1 July 2016, these employees receive employer contributions to the Company pension scheme depending on their eligible gross salary. An initial contribution to the transitional benefit scheme was calculated for the staff concerned as of 30 June 2016 on the basis of parameters and valuation methods defined by the collective bargaining partners. This initial transitional benefit contribution will replace all existing claims by the employees concerned under the collective agreement on “Transitional Benefit for Cabin Crew” and will be switched to a contribution commitment with a minimum guaranteed payment. All employees are free to make their own contributions on a voluntary basis. Contributions from both employer and employee, as well as the initial transitional benefit contribution, are invested on the capital markets with a capital guarantee. When the employee reaches retirement age, the available account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum.

On 21 December 2017, the “Lufthansa Pension Cockpit” wage agreement for cockpit staff was signed with the Vereinigung Cockpit pilots’ union. At the same time, a new “Transitional Benefit Cockpit” wage agreement was signed.

For employees recruited before 1 January 2017, the pension entitlements vested up until 31 December 2016 are maintained. For service periods from 1 January 2017, the employees receive employer contributions to the Company pension scheme depending on their eligible gross salary. All employees are free to make their own contributions on a voluntary basis. The capital is invested on capital markets with a capital guarantee and the guaranteed interest rate offered by the life insurance companies (currently 0.9% per annum) as an additional commitment. When the employee reaches retirement age, the available account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum.

In the new Company retirement benefit scheme for ground, cabin and cockpit staff, the obligations from the capital market-oriented components are recognised at the time value of the corresponding assets, insofar as the assets exceed the minimum guaranteed amount. Plan assets and benefit obligations are presented on a net basis. The employer contributions constitute service expense.

Cockpit staff are still additionally entitled to a transitional pension arrangement covering the period from the end of their active in-flight service until the beginning of their statutory/ Company pension plans. Benefits depend on the number of years of service and the final salary before retirement (final salary plans). Pension entitlements continue to accrue while transitional benefits are being received.

Generally speaking, the previous rules for transitional benefits continue to apply, whereby in the latest version the benefits have been extended to cover cockpit staff at Germanwings who were recruited before 1 January 2005. The collective retirement age for pilots increases gradually from 58 to 60 by 2021. In connection with the forecast long-term staff surplus among cockpit crew, the Company has proposed a temporary suspension of the clause relating to the collective retirement age. A corresponding agreement has not yet been reached on a change to the wage settlement as part of an overall agreement. This results in an increase of EUR 41m in the transitional benefit obligation in the reporting year.

Temporary reductions in employer contributions were agreed for the collective defined benefit plans for cockpit and cabin crew in 2020. These reductions will continue to apply until 2022.

Defined-benefit Company pension schemes and transitional pension arrangements for Germany are funded by plan assets, while amounts that have not yet been transferred are covered by pension provisions.

There are no minimum funding requirements in Germany. Contributions for German defined benefit plans were suspended from the second quarter of 2020 onwards. In addition, reimbursements for pension payments made were withdrawn from the plan assets for the plans of Deutsche Lufthansa AG, Lufthansa Cargo AG and Lufthansa Technik AG for the first time.

Pension obligations in Switzerland are largely based on statutory obligations. The retirement benefits are funded via pension funds known as collective foundations. In addition to retirement benefits, the plans cover invalidity and dependent persons’ benefits. Beneficiaries can choose between an annuity and a lump sum payment. The retirement age for the plans lies between 58 and 63 years. Contributions to the pension funds are made by employers and employees, whereby the Company contributions must be at least equal to the employee contributions defined in the terms of the plan. Contributions are deducted from the qualifying salary according to a sliding scale. If there is a deficit of plan assets, employer and employee contributions can be increased, a lower return can be determined or other steps permissible by law can be taken. The decision is taken by the trustees of the pension fund concerned. The trustees’ strategies for making good a deficit are based on the report by a pension fund expert and must be presented to the regulatory authority. The approval of the authority is not required, however.

In the context of the coronavirus crisis, SWISS has for the first time offered employees voluntary early retirement packages from the age of 58 until the end of 2023. This is intended to implement the socially responsible reduction of the surplus staff capacity in all groups of employees.

The pension obligations for employees of Austrian Airlines AG are mostly on a defined contribution basis and have been outsourced to a pension fund. They consist of retirement, invalidity and dependant persons' benefits.

Obligations under defined benefit plans at Austrian Airlines AG relate to former directors and Executive Board members and others already receiving their pensions. Obligations under defined benefit plans for ground staff are now contribution-free and are determined by converting plan assets into an annuity. There are no defined benefit plans but only defined contribution pension obligations for active pilots, flight attendants and members of the top management level.

The defined benefit pension plans at LSG Sky Chefs in the USA are largely closed to new entrants and no further benefits are being granted to beneficiaries still in service. Benefit payments are based on average salary and the years of service acquired before the plan was closed or frozen. The retirement age is 65. Pension payments are funded externally. Retirement benefits have been switched to defined contribution plans.

Other staff abroad are also entitled to minor retirement benefits and in some cases to medical care based mainly on length of service and salary earned. As a rule, benefits are financed by means of external funds.

Contributions for defined contribution retirement benefit commitments came to EUR 409m in 2020 (previous year: EUR 499m). These mainly consist of contributions to statutory pension funds, but also include contributions under collective bargaining agreements or voluntary contributions to other pension funds.

In the financial year 2004, work began on building up plan assets to fund and safeguard future pension payments. The aim was to fund the pension obligations under existing plans in Germany in full. Contractual trust arrangements (CTAs) in the form of a mutual two-stage trusteeship were set up for this purpose.

The main trustee is Lufthansa Pension Trust e.V., a separate legal entity subject to German regulations. Deutsche Lufthansa AG and the trustees/other trustors agree on contributions and, if such a contribution is determined, make a payment to Lufthansa Pension Trust e.V. Deutsche Lufthansa AG and its subsidiaries Lufthansa Technik AG and Lufthansa Cargo AG are parties to the contractual trust arrangement. The trust assets have largely been held by a Maltese corporate vehicle since 2007. The Investment Board of Lufthansa Malta Pension Holding decides on the fund's asset allocation. The asset management itself is delegated to fund management companies, who invest the assets in accordance with the general investment principles defined by the Investment Board.

Assets to fund pension obligations for other German subsidiaries have been invested with Deutsche Treuinvest Stiftung with the same investment strategy.

The assets to fund pension obligations in the new Lufthansa Pension Ground, Lufthansa Pension Cabin and Lufthansa Pension Cockpit capital market-based benefits system were transferred to an external trustee, Deutsche Treuinvest Stiftung, as part of a contractual trust arrangement. Capital is invested in what are known as age group funds, whose investment strategy is based on a life cycle model. As employees get older, less and less is invested in asset classes with a higher risk-return profile and a greater percentage in more conservative asset classes. The Company has set up an Investment Committee that is responsible for defining and monitoring the investment strategy, e.g. how the age group funds are composed and how the asset allocation changes over time.

EUR 85m was contributed to plan assets for employees in Germany in 2020 (previous year: EUR 250m). No further contributions have been made to Deutsche Lufthansa AG since the beginning of the crisis in March.

Amounts shown in the statement of financial position for defined benefit commitments are made up as follows:

T114 DEFINED-BENEFIT RETIREMENT COMMITMENTS

	31 Dec 2020				31 Dec 2019			
	Defined-benefit obligations (DBO)	Fair value of plan assets	Effect of asset ceiling	Net carrying amount for defined-benefit obligations	Defined-benefit obligations (DBO)	Fair value of plan assets	Effect of asset ceiling	Net carrying amount for defined-benefit obligations
in €m								
Retirement benefits Germany	19,506	-12,165	-	7,341	18,227	-13,135	-	5,092
Transitional benefits Germany	1,528	-337	-	1,191	1,462	-441	-	1,021
Switzerland	4,188	-3,606	-	582	3,840	-3,434	2	408
Austria	444	-180	-	264	450	-183	-	267
USA	342	-306	-	36	369	-327	-	42
Other countries	520	-443	-	77	517	-440	-	77
Carrying amounts	26,528	-17,037	-	9,491	24,865	-17,960	2	6,907
of which pension provisions	-	-	-	9,531	-	-	-	6,658
of which other assets	-	-	-	40	-	-	-	40
of which shown under liabilities for disposal	-	-	-	-	-	-	-	289

Reconciliation between the funding status and the amounts shown in the consolidated balance sheet is as follows:

T115 RECONCILIATION FUNDING STATUS

in €m	2020	2019
Present value of funded pension obligations	25,984	24,374
Plan assets	-17,037	-17,960
Funding status (net)	8,947	6,414
Present value of unfunded pension obligations	544	491
Adjustment for asset ceiling	-	2
Carrying amounts	9,491	6,907
of which pension provisions	9,531	6,658
of which other assets	40	40
of which shown under liabilities for disposal	-	289

During the reporting period, the present value of defined benefit pension obligations changed as follows:

T116 CHANGE IN PRESENT VALUE OF PENSION OBLIGATIONS

in €m	2020	2019
Balance on 1 Jan	24,865	21,412
on which shown under liabilities for disposal	-755	-
Current service costs	502	514
Interest expenses	319	418
Past service cost/effects of curtailments	32	11
Effects of settlements	-1	-
Revaluations		
Actuarial gains/losses from changes in demographic assumptions	24	-64
Actuarial gains/losses from changes in financial assumptions	2,433	2,461
Experience adjustments	-308	281
Currency translation differences	-51	154
Changes in the group of consolidated companies	-772	-
Plan contributions - employees	96	116
Pension payments	-602	-449
Settlement payments	-10	-
Other ¹⁾ /reclassifications	1	11
As of 31 Dec	26,528	24,865
of which shown under liabilities for disposal	-	-755

¹⁾ The amounts are partly for benefit obligations which were measured in accordance with IAS 19 for the first time.

The changes in the group of consolidated companies relate to the pension obligations disposed of with the sale of the LSG group's European business.

Actuarial gains/losses from changes in financial assumptions include losses due to the renewed fall in the discount rate for Germany, Austria and Switzerland compared with the previous year. Adjustments to obligations regarding capital market-based pension plans, which are due to changes in exchange rates, are shown in adjustments based on past experience.

The following table provides a detailed reconciliation of changes in the fair value of plan assets:

T117 CHANGE IN FAIR VALUE OF PLAN ASSETS

in €m	2020	2019
Balance on 1 Jan	17,960	15,576
on which shown under liabilities for disposal	-466	-
Interest income	222	299
Revaluations		
Income from plan assets, without amounts included in interest	-443	1,571
Currency translation differences	-46	139
Changes in the group of consolidated companies	-431	-
Plan contributions – employers	235	394
Plan contributions – employees	61	111
Pension payments	-510	-138
Settlement payments	-10	-
Administrative costs related to obligations	-3	-3
Other ¹⁾ /reclassifications	2	11
As of 31 Dec	17,037	17,960
of which shown under liabilities for disposal	-	-466

¹⁾ The amounts are partly for benefit obligations which were measured in accordance with IAS 19 for the first time.

The outflows of plan assets due to pension payments of EUR 510m included, among other things, the financing of pensions in Germany from the CTA assets for the first time in the amount of EUR 288m.

In the financial years 2020 and 2019, pension provisions developed as follows:

T118 PENSION PROVISIONS

in €m	2020	2019
Balance on 1 Jan	6,947	5,865
on which shown under liabilities for disposal	-289	-
Currency translation differences carried forward	-5	15
Changes in the group of consolidated companies	-341	-
Pensions payments	-92	-311
Current service costs	502	514
Interest expenses	319	418
Interest income	-222	-299
Effects of amendments incl. curtailments, settlements and administrative costs	34	14
Revaluations		
Actuarial gains/losses and experience adjustments	2,149	2,678
Income from plan assets, without amounts included in interest	443	-1,571
Net effect of adjustments for asset ceiling	-3	2
Plan contributions/reclassifications	-200	-378
As of 31 Dec	9,531	6,947
of which shown under liabilities for disposal	-	-289

Expenses and income for defined-benefit plans are made up as follows:

T119 EXPENSES AND INCOME FOR DEFINED-BENEFIT PENSION PLANS

in €m	2020	2019
Current service costs	502	514
Past service cost/effects of curtailments	32	11
Income from settlements	-1	-
Accrued interest on projected pension obligations	319	418
Interest income on plan assets	-222	-299
Administrative costs related to obligations	3	3
Balance of expenses and income recognised in the income statement	633	647
Income from plan assets, without amounts included in interest	443	-1,571
Actuarial gains and losses	2,149	2,678
Net effect of adjustment for asset ceiling	-3	2
Other comprehensive income	2,589	1,109
	3,222	1,756

Interest expenses on pension provisions and interest income on plan assets are shown in the financial result. Current service expense and past service expense are recognised in staff costs.

Past service expenses incurred in the reporting year stem mainly from forecast future expenses in connection with measurement effects from the temporary suspension proposed by the Company of the clause defining the average retirement age for pilots in the context of transitional benefits at Deutsche Lufthansa AG. A corresponding accord to change the respective collective labor agreement has not been reached yet and will be part of an overall agreement.

A loss of EUR 221m was generated from plan assets in the financial year 2020. This amount is made up of the interest income recognised in the income statement and the revaluation component for plan assets. Capital appreciation of EUR 1,870m was recognised in the previous year.

There were no significant effects from the asset ceiling defined in IAS 19.64.

The main actuarial assumptions used to calculate pension obligations and the corresponding plan assets are shown below:

T120 MAIN ACTUARIAL ASSUMPTIONS FOR GERMAN COMPANIES

in %	31 Dec 2020	31 Dec 2019
Interest rate		
Retirement benefits	0.8	1.4
Transitional benefits	0.8	1.4
Salary increase		
Retirement benefits	2.5	2.5
Transitional benefits	2.5	2.5
Pension increase		
Retirement benefits	1.0	1.0
Transitional benefits	1.0	1.0

The Heubeck Actuarial Tables 2018 G were used in the biometric calculations for the German companies in the Group.

T121 MAIN ACTUARIAL ASSUMPTIONS FOR FOREIGN COMPANIES

in %	31 Dec 2020	31 Dec 2019
Interest rates		
Austria	0.8	1.4
Switzerland	0.1	0.3
USA	2.3	3.2
Salary increase		
Austria	1.6	1.8
Switzerland	1.5	1.5
USA	-	-
Pension increase		
Austria	1.6	2.0
Switzerland	0.0	0.0
USA	-	-

The BVG 2015 generation tables are used for the biometric calculations for Switzerland. Country-specific mortality tables are used in the other countries.

The following table shows how the present value of defined benefit obligations would have been affected by changes in the relevant actuarial assumptions for the main pension plans described above:

T122 CHANGE IN ACTUARIAL ASSUMPTIONS, AS OF 2020

	Effect on the defined-benefit contribution as of 31 Dec 2020 in €m	Change in %
Present value of the obligation ¹⁾	26,528	-
Interest rate		
Increase by 0.5 percentage points	24,257	-8.6
Decrease by 0.5 percentage points	29,241	+10.2
Salary trend		
Increase by 0.5 percentage points	26,668	+0.5
Decrease by 0.5 percentage points	26,396	-0.5
Pension trend		
Increase by 0.5 percentage points	26,825	+1.1
Decrease by 0.5 percentage points	26,483	-0.2

¹⁾ Present value of the obligation using the assumptions shown in the "Actuarial assumptions" tables.

T122 CHANGE IN ACTUARIAL ASSUMPTIONS, AS OF 2019

	Effect on the defined-benefit contribution as of 31 Dec 2019 in €m	Change in %
Present value of the obligation ¹⁾	24,865	–
Interest rate		
Increase by 0.5 percentage points	22,862	–8.1
Decrease by 0.5 percentage points	27,201	+9.4
Salary trend		
Increase by 0.5 percentage points	25,022	+0.6
Decrease by 0.5 percentage points	24,717	–0.6
Pension trend		
Increase by 0.5 percentage points	25,140	+1.1
Decrease by 0.5 percentage points	24,606	–1.0

¹⁾ Present value of the obligation using the assumptions shown in the "Actuarial assumptions" tables.

A reduction of 10% in the mortality rates used to calculate the pension obligations increases the life expectancy of the beneficiaries by a given amount depending on their individual ages. It roughly corresponds to an increase of one year in the life expectancy of a male employee who is 55 years old today. A 10% reduction in the mortality rate would therefore increase the present value of the main benefit obligations in Germany and Switzerland by EUR 189m as of 31 December 2020 (previous year: EUR 176m).

The sensitivity analysis examines changes in one assumption and leaves the other assumptions unchanged compared with the original calculation. The effects of any correlation between the individual assumptions are therefore not taken into account.

Plan assets for funded defined benefit pension obligations consist mainly of fixed-income securities, equities and cash and cash equivalents. They do not include financial instruments issued by companies in the Group nor properties used by Group companies.

Plan assets serve solely to meet the defined benefit obligations. Funding these benefit obligations with assets provides security for future payments. In some countries, this takes place on the basis of statutory regulations, while in others (Germany, for example), this takes place on a voluntary basis.

The Lufthansa Group aims to completely cover its German pension obligations by means of cover assets and positive capital market returns in the medium to long term. Additional capital is transferred to the trustees for this purpose. Investment performance plays a crucial role in meeting this target.

The investment strategy for the capital market-based pension plans is initially defined by the Company and is regularly reviewed in the course of an allocation study. Where necessary, it is adjusted by the Investment Committee to reflect changes in capital market requirements. This may result in changes to the investment strategy for amounts that have already been invested.

The Lufthansa Group manages and monitors the financial risks that arise from outsourcing the defined benefit pension obligations. There was no change in the risk management and monitoring processes compared with the previous year. Derivative financial instruments are used, especially to manage foreign exchange risks.

The allocation of the funds to asset classes (e.g. equities) for the defined benefit plans is carried out on the basis of asset-liability matching (ALM) studies performed by the Lufthansa Group. The ALM study is conducted every three years with an external adviser in order to review the funding strategy on a regular basis and to make adjustments as necessary. The results of the study should indicate what combination of investments (annuities, equities, etc.) can be used to cover the long-term pension obligations. Step one of this process is for the actuary to draft a long-term forecast charting how the pension obligations will develop.

In addition to this, target figures are needed for the relative return and relative risk as regards coverage of the obligations. Last but not least, a risk budget must also be defined.

A simulation is used to test all permissible investment allocations for their future compliance with these objectives. Those which do not fulfil the criteria are eliminated. Preference is given to allocations that are return-oriented yet conservative and that have a high probability of achieving the investment target.

The results of the ALM study show whether there will be strategic shifts in the existing allocation. Alternative investments (e.g. property, private equity, infrastructure) are currently being further developed.

Plan assets are made up as follows:

T123 COMPOSITION OF PLAN ASSETS

	31 Dec 2020				31 Dec 2019			
	Listed price in an active market in €m	No listed price in an active market in €m	Total in €m	in %	Listed price in an active market in €m	No listed price in an active market in €m	Total in €m	in %
Equities			5,626	33.0			5,189	28.9
Europe	2,152	–			3,299	–		
Other	3,470	4			1,890	–		
Fixed-income securities			6,060	35.6			6,111	34.0
Government bonds	3,159	8			3,053	–		
Corporate bonds	2,890	3			3,058	–		
Share funds	153	–	153	0.9	847	–	847	4.7
Fixed-income funds	723	–	723	4.2	1,482	–	1,482	8.3
Mixed funds ¹⁾	77	–	77	0.5	247	–	247	1.4
Money market investments	1,298	–	1,298	7.6	1,039	–	1,039	5.8
Property			1,225	7.2			1,015	5.7
Direct investments	–	7			–	7		
Indirect investments	863	355			744	264		
Insurance contracts	–	151	151	0.9	–	133	133	0.7
Bank balances	345	184	529	3.1	334	191	525	2.9
Other investments ²⁾	207	988	1,195	7.0	288	1,084	1,372	7.6
Total	15,337	1,700	17,037	100.0	16,281	1,679	17,960	100.0

¹⁾ Includes equities and interest-bearing securities.

²⁾ Other investments include, in particular, alternative investments such as hedge funds, commodities and private equity funds.

In addition to various actuarial risks such as interest rate risk, life expectancy risk and the risk of salary increases, the pension plans expose the Group primarily to financial risks in connection with plan assets.

The return on plan assets is assumed at the beginning of the period to be the discount rate, which is determined on the basis of investment grade corporate bonds. For the old pension plans, if the actual return on plan assets is less than the discount rates applied, the net obligation from the pension plan goes up. With the new capital market-based pension plans, a gross obligation is recognised for the time value of the corresponding plan assets, taking the minimum guaranteed amount into account.

The share price risk that arises from the proportion of plan assets invested in equities is considered to be reasonable. The risk of default by bond issuers is limited, because investments are only made in investment grade bonds.

The amount of the net obligation under the old pension plans depends to a large extent on the rates of interest, whereby the current low-interest environment results in a relatively high net obligation. If yields on corporate bonds continue to decline, this would lead to a further increase in defined benefit obligations, which could probably only be partly offset by positive developments in the market values of the corporate bonds held in plan assets.

Based on current knowledge, it can be assumed that only EUR 178m will be transferred to pension plans in the 2021 financial year due to the crisis situation (previous year: EUR 808m). The transfers are made up of planned allocations and benefit payments that are not covered by equivalent reimbursements from plan assets. The weighted duration of pension obligations was 20 years as of 31 December 2020 (previous year: 18 years).

Over the next ten years, the following pension payments are forecast for the defined benefit commitments in existence as of the reporting date:

T124 FORECAST MATURITIES OF UNDISCOUNTED PENSION PAYMENTS, AS OF 2020

in €m	Forecast pension payments 31 Dec 2020
2021	524
2022	689
2023	626
2024	605
2025	627
2026 – 2030	3,699

T124 FORECAST MATURITIES OF UNDISCOUNTED PENSION PAYMENTS, AS OF 2019

in €m	Forecast pension payments 31 Dec 2019
2020	574
2021	596
2022	612
2023	637
2024	656
2025 – 2029	3,828

36 Other provisions

Other provisions disclosed in the statement of financial position as non-current and current other provisions are made up as follows:

T125 NON-CURRENT AND CURRENT OTHER PROVISIONS

in €m	31 Dec 2020			31 Dec 2019		
	Total	Non-current	Current	Total	Non-current	Current
Obligations under partial retirement contracts	32	32	–	20	18	2
Other staff costs	209	169	40	202	159	43
Obligation to return emissions certificates	50	–	50	95	–	95
Onerous contracts	88	38	50	49	20	29
Environmental restoration	31	28	3	30	27	3
Legal proceedings	65	12	53	77	11	66
Restructuring/severance payments	221	47	174	171	79	92
Maintenance of lease aircraft	323	162	161	293	139	154
Warranties	47	–	47	46	–	46
Other provisions	323	70	253	301	37	264
Total	1,389	558	831	1,284	490	794

The obligations from partial retirement agreements result from collective bargaining agreements in Germany. Obligations were measured in 2020 using an interest rate of 0.00% (previous year: 0.01%). The interest effects shown in the following table on the development of other provisions resulted from the changes in the market value of the plan assets.

A total of EUR 133m (previous year: EUR 132m) has been transferred to an external trust fund as insolvency insurance for employer's performance arrears as part of partial retirement agreements under which the employee at first works full-time for less pay and then retires early on the same reduced pay. These assets, which fulfil the requirements for plan assets and therefore reduce the gross amount of obligations accordingly, are measured at market value on the balance sheet date.

The funding status for provisions for obligations to employees under partial retirement agreements is as follows:

T126 FUNDING STATUS

in €m	2020	2019
Present value of funded obligations under partial retirement agreements	149	146
External plan assets	-130	-141
	19	5
of which other provisions	32	20
of which other assets	13	15

Provisions for other staff costs mainly relate to staff anniversary bonuses and other current obligations.

A provision for the obligation to submit CO₂ emissions certificates to the relevant authorities is recognised for an amount equivalent to the carrying amount of the capitalised CO₂ certificates. If the obligation is not fully covered by available certificates, the outstanding amount of the provision is measured using the market price of the emissions certificates as of the reporting date.

Expected losses from onerous contracts result from ongoing obligations or other contractual relationships in which performance and consideration are out of balance.

Provisions for environmental restoration are based on surveyors' findings and the assumption that all contamination is removed within ten years without any further legal requirements.

Provisions for ongoing legal proceedings were based on an assessment of the likely outcome of the proceedings.

The provisions for restructuring and severance payments are based on existing termination agreements or proposed contract terminations which the Lufthansa Group can no longer avoid.

The provisions for the overhaul of leased aircraft mainly relate to obligations for the maintenance, overhaul and repair of aircraft.

Other provisions of EUR 132m (previous year: EUR 135m) relate to outstanding obligations in connection with the accident involving the Germanwings aircraft on 24 March 2015.

Changes in groups of individual provisions in 2020 were as follows:

T127 CHANGES IN OTHER PROVISIONS 2020

in €m	Obligations under partial retirement contracts	Other staff costs	Obligation to return emissions certificates	Expected losses from onerous contracts	Environmental restoration	Legal proceedings
As of 1 Jan 2020	20	202	95	49	30	77
Changes in the group of consolidated companies	-	-	-	-	-	-
Currency translation differences	-	-2	-	-	-	-1
Utilisation	-52	-42	-64	-11	-2	-4
Increase/addition	58	49	67	61	3	19
Interest added back	11	2	-	-	-	-
Reversal	-1	-3	-48	-11	-	-22
Transfers	-4	3	-	-	-	-4
As of 31 Dec 2020	32	209	50	88	31	65

T127 CHANGES IN OTHER PROVISIONS 2020 (continued)

in €m	Restructuring/ severance payments	Maintenance of lease aircraft	Warranties	Other provisions	Total
As of 1 Jan 2020	171	293	46	301	1,284
Changes in the group of consolidated companies	-	-	-	-	-
Currency translation differences	-1	-10	-	-2	-16
Utilisation	-68	-89	-13	-62	-407
Increase/addition	124	132	21	97	631
Interest added back	-	-	-	-	13
Reversal	-2	-2	-7	-9	-105
Transfers	-3	-1	-	-2	-11
As of 31 Dec 2020	221	323	47	323	1,389

Changes in groups of individual provisions in the previous year were as follows:

T127 CHANGES IN OTHER PROVISIONS 2019

in €m	Obligations under partial retirement contracts	Other staff costs	Obligation to return emissions certificates	Expected losses from onerous contracts	Environmental restoration	Legal proceedings
As of 1 Jan 2019	31	196	63	64	27	85
Changes in the group of consolidated companies	-	-	-	-	-	-
Currency translation differences	-	-	-	-	-	-
Utilisation	-57	-30	-73	-11	-2	-14
Increase/addition	54	42	104	16	5	28
Interest added back	-7	2	-	-	-	-
Reversal	-	-1	-	-19	-	-22
Transfers	-1	-7	1	-1	-	-
As of 31 Dec 2019	20	202	95	49	30	77

T127 CHANGES IN OTHER PROVISIONS 2019 (continued)

in €m	Restructuring/severance payments	Maintenance of lease aircraft	Warranties	Other provisions	Total
As of 1 Jan 2019	210	281	49	425	1,431
Changes in the group of consolidated companies	-	-	-	-	-
Currency translation differences	-	4	-	-	4
Utilisation	-77	-95	-10	-85	-454
Increase/addition	45	127	14	84	519
Interest added back	-	-	-	1	-4
Reversal	-6	-24	-6	-34	-112
Transfers	-1	-	-1	-90	-100
As of 31 Dec 2019	171	293	46	301	1,284

The following cash outflows are estimated for the non-current portion of the other groups of provisions:

T128 CASH OUTFLOWS FOR NON-CURRENT PROVISIONS, AS OF 2020

in €m	2022	2023	2024	2025 and thereafter
Expected losses from onerous contracts	13	6	6	12
Environmental restoration	3	3	3	19
Restructuring/severance payments	29	6	1	11
Maintenance of aircraft on leases	25	6	8	35
Other provisions	17	9	14	48

At the end of 2019, the corresponding cash outflows were estimated as follows:

T128 CASH OUTFLOWS FOR NON-CURRENT PROVISIONS, AS OF 2019

in €m	2021	2022	2023	2024 and thereafter
Expected losses from onerous contracts	11	3	-	7
Environmental restoration	3	3	3	18
Restructuring/severance payments	54	23	9	5
Maintenance of aircraft on leases	87	16	12	29
Other provisions	15	6	4	44

37 Financial liabilities

Financial liabilities consist of a non-current portion with a residual term of more than one year and a current portion of less than one year, which is shown under current liabilities. The following table shows the total amount of borrowings:

T129 FINANCIAL LIABILITIES 31 DEC 2020

in €m	Total	Non-current	Current
Bonds	2,907	2,707	200
Liabilities to banks	4,938	2,867	2,071
Leasing liabilities	2,637	2,183	454
Other loans	4,886	4,495	391
	15,368	12,252	3,116

T129 FINANCIAL LIABILITIES 31 DEC 2019

in €m	Total	Non-current	Current
Bonds	1,094	996	98
Liabilities to banks	2,110	1,453	657
Leasing liabilities	2,872	2,384	488
Other loans	3,954	3,563	391
	10,030	8,396	1,634

Lufthansa pursues an interest rate risk strategy of converting fixed-interest financial liabilities in all currencies into variable-interest financial liabilities in euros by means of interest rate derivatives.

The outstanding bonds comprise with a fixed redemption amount and one convertible bond. The bonds are denominated exclusively in EUR and were issued at a fixed interest rate of between 0.25% and 5.125%. They are set to mature between February 2021 and May 2026. The convertible bond was issued with a nominal volume of EUR 600m. Unless previously converted, the bond will be redeemed at its nominal value on 17 November 2025. Investors also have the option of converting the bond into new and/or existing registered shares of Deutsche Lufthansa AG. The initial conversion price was set at EUR 12.96.

Of the liabilities to banks, EUR 1,712m related to state-guaranteed loans secured by shares, company shares and aircraft of various consolidated companies (Note 2, p. 140ff., and Note 19, p. 170). In addition, EUR 248m was attributable to loans from banks secured by three aircraft, which were concluded in 2020. Liabilities to banks in the amount of EUR 2m were collateralised in the previous year.

Under other loans, EUR 3,378m (previous year: EUR 3,524m) was attributable to structured leasing companies and other aircraft financing models (Note 19, p. 170). This amount was collateralised with the respective financed aircraft. A total of four additional aircraft financing arrangements were concluded in 2020. EUR 133m related to a loan and profit-sharing rights as part of the Belgian state aid. Interest on the loan was determined to be below market rates with an total amount of EUR 8m. The loan was collateralised with shares and profit-sharing rights of the Belgian SN Airholding SA/NV.

Other loans also include Silent Participation II of the Economic Stabilisation Fund (EUR 1,000m) and loans to the LSG companies under the US CARES Act (EUR 65m).

In both the 2020 financial year and the 2019 financial year, all payment obligations and requirements from the loan agreements described have been fulfilled.

The Lufthansa Group's lease liabilities have the following term structure: The disclosures are based on contractual, undiscounted payments.

T130 MATURITY ANALYSIS OF LEASE LIABILITIES

in €m	31 Dec 2020	31 Dec 2019
1st quarter	133	146
Up to 1 year ¹⁾	376	392
1 – 5 years	1,309	1,350
Later	1,409	1,461

¹⁾ Without payments in 1st quarter.

38 Non-current contract liabilities

T131 NON-CURRENT CONTRACT LIABILITIES

in €m	31 Dec 2020	31 Dec 2019
Non-current contract liabilities	36	25
	36	25

Non-current contract liabilities consist of long-term deferrals for construction contracts where the payments received exceed the performance to date.

39 Non-current advance payments received, deferred income and other non-financial liabilities

T132 NON-CURRENT ADVANCE PAYMENTS RECEIVED, DEFERRED INCOME AND OTHER NON-FINANCIAL LIABILITIES

in €m	31 Dec 2020	31 Dec 2019
Advance payments received	8	7
Deferred income	14	13
Other non-financial liabilities	11	12
	33	32

Deferred income includes EUR 6m (previous year: EUR 7m) for government grants and subsidies for capital expenditure, which are realised over the useful life of the assets in the following years.

Other non-financial liabilities include obligations under share-based remuneration agreements for Executive Board members, managers and non-payscale employees. As part of the share-based remuneration agreements, the Lufthansa Group and other consolidated and non-consolidated Group companies participating in the programme offer a 50% discount on employee investment in Lufthansa shares to Executive Board members, managers and non-payscale employees. The option packages granted in 2017, 2018, 2019 and 2020 consist of an outperformance option and a performance option. At the end of the programme, the participants receive a cash payment if the conditions are met.

The outperformance option is linked to the performance of the Lufthansa share compared with a notional index of shares in European competitors. When the beneficiary exercises the outperformance option, they receive a cash payment for every percentage point of outperformance, with a hurdle rate of 1%. The cash payment is capped at an outperformance of more than 20%.

The performance option is linked to the absolute performance of the Lufthansa share. The amount of the payment depends on meeting defined performance targets, whereby both a hurdle rate and a cap apply.

The programmes are scheduled to run for four years. The performance and the outperformance in all programmes are calculated on the principle of total shareholder return. The shares invested in personally may not be sold until the option is exercised.

T133 OUTPERFORMANCE OPTION – OVERVIEW

	€ per outperformance level	Maximum per tranche in €
Board member	1,000 per percentage point from 1%	20,000
Managers	400 per percentage point from 1%	8,000
Non-payscale staff	200 per 5 percentage points from 1%	1,000

T134 PERFORMANCE OPTION – PERFORMANCE TARGETS

Performance option per year	Hurdle rate	Cap
2017	23%	35%
2018	22%	33%
2019	22%	33%
2020	20%	35%

T135 PERFORMANCE OPTION – PAYMENT AMOUNTS

	€ per performance level	Maximum per tranche in €
Board member	10,000 + 1,000 per performance unit	20,000
Managers	4,000 + 500 per performance unit	8,000
Non-payscale staff	500 + 100 per performance unit	1,000

A different system of variable remuneration has applied to the Executive Board since 2019. Under this system, one of the financial targets for the share-based remuneration compares the performance of the Lufthansa share, expressed as the total shareholder return (TSR), to other DAX companies. The performance period is still four years. To calculate the TSR in the performance period, the arithmetic mean closing prices in XETRA trading at Deutsche Börse for shares in Deutsche Lufthansa AG and its peer group over the 30 days immediately preceding the beginning of the performance period are compared with the prices over the 30 days immediately preceding the end of the performance period.

The performance of all the companies in the DAX index at the beginning and end of the period is ranked and the relative position of Deutsche Lufthansa AG is determined by its achieved percentile. Performance against the target and the amount of the payment in the 2019 programme depend on the percentile position of Deutsche Lufthansa AG. Target achievement is adjusted on a linear basis for positions between the 25th and 50th percentile and between the 50th and 75th percentile.

T136 RELATIVE TSR – PERFORMANCE AND PAYMENT PROGRAMME 2019

Position of Lufthansa share compared with peer group	Target achievement	Payment in € for Executive Board members	Payment in € for Executive Board Chairman
≤ 25th percentile	0%	–	–
≥ 50th percentile	100%	467,500	888,250
≥ 75th percentile	200%	935,000	1,776,500

As a result of the regulations in the stabilisation agreement with the WSF, the 2020 management programme was only offered with a contribution of shares. The 50% discount applies only to a portion of the potential tranches to be agreed and will only be granted at the end of the programme. The payments from the 2020 Executive Board and management programme are subject to the completion of the stabilisation measures.

Starting in 2020, the TSR approach for the Executive Board programme has been adjusted so that 85% of performance is now measured by financial parameters and 15% by sustainability parameters. The financial targets are the relative total shareholder return (TSR) of the Lufthansa share compared with the DAX and the average adjusted return on capital employed (Adjusted ROCE) during the performance period, both in equal parts (42.5% each). For the TSR component, the 60 trading days immediately preceding the beginning of the performance period and the 60 trading days immediately preceding the end of the performance period are used in the performance period. The positioning of the share in relation to the peer group remains identical for performance against the target. Performance against the target for average Adjusted ROCE is based on a comparison of the average Adjusted ROCE for the four-year performance period against a strategic target set in the grant year, which has a lower weighted average cost of capital (WACC). The sustainability parameters are set by the Supervisory Board for each performance period. For both of the non-market-dependent performance targets, Adjusted ROCE and sustainability parameters, an expected performance against each target of 100% was assumed for the purposes of the valuation. At the beginning of each performance period, a further innovation is the allocation of a number of virtual shares, which is calculated by dividing the individual target amount of the long-term variable remuneration by the average price of Deutsche Lufthansa AG shares during the 60 trading days immediately following the beginning of each performance period. The payment is calculated by multiplying the degree of target achievement for this performance target by the number of virtual shares at the beginning of the performance period and the average price of Deutsche Lufthansa AG shares during the 60 trading days immediately preceding the end of the last year of each performance period.

Over the financial years 2020 and 2019, the number of options changed as follows:

T137 CHANGE IN NUMBER OF OPTIONS

	2020			2019	
	Number of options	Number of virtual shares in € thousands	Cash settlement in € thousands	Number of options	Cash settlement in € thousands
Outstanding on 1 Jan	22,313	–	–	20,546	–
Issued	2,243	571,106	–	7,420	–
Expired or unused	1,017	–	–	1,110	–
Exercised	3,957	–	17,062	4,543	31,726
Outstanding on 31 Dec	19,582	571,106	–	22,313	–

The performance and outperformance option of the share programme 2016, which has now expired, resulted in a payment of EUR 17m (previous year: EUR 32m). No new shares were issued for the share programme 2020 in the financial year due to the changed structure. As a result, no staff costs arose from the issue of new shares in the financial year. Participants in the programme hold 3,126,854 shares as of the reporting date (previous year: 3,594,679 shares).

The fair values of the option rights in the share programmes still running were calculated using Monte Carlo simulations. This involves simulating the future returns of the shares in the comparative index and of Deutsche Lufthansa AG and calculating the value of the option rights as the forecast amount of a dividend.

The following fair values were measured in total:

T138 FAIR VALUE OF OPTION RIGHTS AS OF 31 DEC 2020

	Number of options/ virtual shares	Fair value per option in €	Proportional vested benefit	Total fair value in €
Board member				
Options 2017	181	2,940	0.74	395,259
Options 2018	181	3,224	0.51	295,390
Options 2019	6	25,067	0.44	66,117
Virtuelle Aktien 2020	571,106	3.19	0.23	419,020
Managers				
Options 2017	2,172	1,176	0.79	2,022,132
Options 2018	2,386	1,301	0.54	1,681,434
Options 2019	2,922	3,212	0.29	2,737,427
Options 2020	2,243	5,326	0.02	248,880
Non-payscale staff				
Options 2017	2,291	158	0.79	286,566
Options 2018	3,154	168	0.54	287,014
Options 2019	4,046	400	0.29	472,033
Options 2020	-	-	-	-
Total options and virtual shares	590,688			8,911,272
of which options	19,582			
of which virtual shares	571,106			

T138 FAIR VALUE OF OPTION RIGHTS AS OF 31 DEC 2019

	Number of options	Fair value per option in €	Proportional vested benefit	Total fair value in €
Board member				
Options 2016	135	31,582	0.70	2,980,551
Options 2017	181	5,376	0.58	559,576
Options 2018	165	5,770	0.36	342,594
Options 2019	6	281,633	0.25	422,450
Managers				
Options 2016	2,062	12,628	0.79	20,614,158
Options 2017	2,173	2,148	0.54	2,528,286
Options 2018	2,525	2,354	0.29	1,733,623
Options 2019	3,071	5,479	0.04	701,084
Non-payscale staff				
Options 2016	2,470	1,579	0.79	3,087,603
Options 2017	2,301	274	0.54	341,507
Options 2018	3,164	286	0.29	263,930
Options 2019	4,060	659	0.04	111,481
Total	22,313			33,686,841

Staff fluctuation of 5.35% is again assumed when accounting for the liability resulting from the valuation of option rights, so that the recognised liability is less than their calculated time value. An additional provision of EUR 2.7m was made for a discount on the purchase price that is only payable at the end of the management programme 2020. This one-time provision and the measurement of option rights therefore resulted in a total provision of EUR 11.0m as of the reporting date (previous year: EUR 32.3m), of which EUR 9m (previous year: EUR 7m) is shown under non-current liabilities. The payment of EUR 17m in the financial year on the basis of option rights that have now expired reduced the previously recognised provisions, so that the change in option rights in the reporting year decreased staff costs by EUR 4.2m.

The weighted average share prices at the calculation date (excluding the Executive Board TSR programme) were used in the Monte Carlo simulation. As stated in the terms of the programme, these are 50-day averages for the shares of Deutsche Lufthansa AG and the competitors included in the comparative index. The volatilities and correlations used are forecasts for a specific date and maturity on the basis of current market estimates.

Swap rates were used as the interest rate for the remaining term of the outperformance option in each case. The maximum term of the programmes was used for measurement purposes.

Time values for the Executive Board TSR programme were also measured using a Monte Carlo simulation based on historical and current market data for the relevant peer group of DAX companies. Forecast volatilities are based on the historic TSR data. The share prices for the past four years were used to calculate historical volatility. The measurement for the programme 2019 took into account a remaining term of 25 months and a risk-free interest rate of -0.77%, for the programme 2020 a remaining term of 37 months and a risk-free interest rate of -0.79%.

The parameters used by the external service provider for the notional airline peer group index are shown in the following table:

T139 REFERENCE PRICE

		Options 2017	Options 2018	Options 2019	Options 2020
Lufthansa	EUR	23.00	21.18	14.84	8.39
Air France-KLM	EUR	13.12	8.56	9.90	3.54
IAG	GBP	612.70	646.28	480.37	182.21
Ryanair	EUR	17.25	12.82	11.00	13.31
easyJet	GBP	1,240.72	1,340.31	1,131.63	622.96
Norwegian	NOK	218.45	236.70	37.49	-
WIZZair	GBP	2,991.52	2,858.46	3,667.66	3,701.91
Turkish	TRY	-	-	-	10.71

T140 PROJECTED VOLATILITIES

in % for:	Options 2017 as of 31 Dec 2020	Options 2017 as of 31 Dec 2019	Options 2018 as of 31 Dec 2020	Options 2018 as of 31 Dec 2019	Options 2019 as of 31 Dec 2020	Options 2019 as of 31 Dec 2019	Options 2020 as of 31 Dec 2020	Options 2020 as of 31 Dec 2019
Lufthansa	64.13	31.05	50.16	29.94	44.85	31.40	41.23	-
Air France-KLM	68.98	38.66	56.19	38.07	50.83	37.73	47.74	-
IAG	99.59	27.42	73.90	26.76	61.77	32.51	54.96	-
Ryanair	56.54	34.01	47.08	31.16	42.83	32.59	39.04	-
easyJet	86.64	34.95	67.31	33.35	57.57	36.06	52.02	-
Norwegian	141.14	81.10	119.94	70.93	105.57	64.75	-	-
WIZZair	67.61	32.96	51.89	32.14	47.38	33.70	43.74	-
Turkish	-	-	-	-	-	-	40.88	-
Risk-free interest rate	Options 2017: -0.72% for euro zone (previous year: -0.66%) -0.02% for UK (previous year: 0.55%) 0.31% for Norway (previous year: 1.25%)				Options 2019: -0.79% for euro zone (previous year: -0.64%) -0.02% for UK (previous year: 0.48%) 0.31% for Norway (previous year: 1.24%)			
	Options 2018: -0.77% for euro zone (previous year: -0.66%) -0.02% for UK (previous year: 0.50%) 0.31% for Norway (previous year: 1.25%)				Options 2020: -0.80% for euro zone 0.00% for UK 15.00% für Turkey			
Fluctuation	5.35% (previous year: 4.9%)				5.35% (previous year: 4.9%)			

40 Current contract liabilities

The Lufthansa Group recognised the following contract liabilities:

T141 CONTRACT LIABILITIES

in €m	31 Dec 2020	31 Dec 2019
Contract liabilities from unused flight documents	2,064	4,071
Liabilities from customer loyalty programmes	2,220	2,200
Liabilities from MRO and IT services	297	196
Miscellaneous contract liabilities	460	279
Other contract liabilities	2,977	2,675
Liabilities from contracts with customers	5,041	6,746
Revenue recognised in the reporting period		
Revenue recognised that was included in the contract liability balance at the beginning of the period		
Revenue from unused flight documents	2,873	3,768
Revenue from customer loyalty programmes	210	576
Revenue from MRO and IT services	148	76
Other	226	50
Total	3,457	4,470

The decline in contract liabilities for unused flight documents is caused by the temporary nearly complete cancellation of flights since the beginning of March 2020 and lower ticket sales in 2020, and partly from higher refunds in connection with cancellations by passengers or airlines. EUR 1,045m of the contract liabilities as at 31 December 2019 could not be realised and were refunded to customers. Total ticket refunds of EUR 3,851m were made in connection with flights cancelled as a result of the pandemic. Only a small amount of expiry is currently assumed with regard to potentially unused tickets relating to flights during the crisis period. Consequently, no material revenue realisation was recognised for breakage for this period as part of the estimation assumptions in the measurement of the obligation.

Liabilities under customer loyalty programmes as of 31 December 2020 included 228 billion miles from bonus miles programmes (previous year: 225 billion miles). As a rule, the

miles that are expected to expire are recognised pro rata over the general validity period of three years. The pro rata revenue realisations were reduced in accordance with the declining total redemptions in the financial year due to the severely restricted redemption options for air travel in connection with the coronavirus pandemic when measuring the scope of the obligation. This is in line with the assumption that the proportion of expiring miles will not fundamentally increase due to the effects of the pandemic, but will merely shift to later periods.

The unsatisfied performance obligation under existing long-term service contracts came to EUR 6.1bn in total, assuming that the services are performed as agreed, of which EUR 1.1bn relate to the next twelve months. These essentially consist of maintenance contracts in the MRO segment for the long-term maintenance and overhaul of airline sub-fleets. To calculate the outstanding performance obligations, the number of maintenance events derived from the respective flight plans and agreed in the contracts are taken into account, along with the expected revenue and fixed prices for certain services (VIP and cabin modifications). Around 61% of performance obligations beyond twelve months are expected to have been fulfilled by 2026.

As in the previous year, no revenue was recognised in 2020 for performance obligations fulfilled in prior financial years.

In line with the simplification rules of IFRS 15 disclosures are not made on performance obligations as of 31 December 2020 or 31 December 2019 that have a forecast original term of one year or less. The option of rebooking flights means that there may be a period of time between the conclusion of the contract and the provision of the service that exceeds one year, although this cannot be foreseen when the contract is concluded. Due to the advance booking period of a maximum of one year and short-term rebooking possibilities, the Lufthansa Group assumes that the application of the simplification rule is justified. Award miles can be redeemed for at least three years, but may also be redeemed at short notice and for this reason are also reported as current.

The Lufthansa Group applies the simplification rule defined in IFRS 15.94, which allows contract initiation costs to be expensed if the amortisation period otherwise to be taken into account would be twelve months or less.

41 Trade payables and other current financial liabilities

T142 TRADE PAYABLES AND OTHER CURRENT FINANCIAL LIABILITIES

in €m	31 Dec 2020	31 Dec 2019
Trade payables		
Trade payables to affiliated companies	55	42
Trade payables to other equity investments	-	2
Trade payables to third parties	2,032	3,678
	2,087	3,722
Other liabilities		
Liabilities to banks	14	17
Other liabilities to affiliated companies	316	260
Other liabilities to equity investments	-	-
Liabilities from equity investments	61	356
Other financial liabilities	843	996
	1,234	1,629
Total	3,321	5,351

The carrying amount of these liabilities corresponds to their fair value. Other liabilities of EUR 38m (previous year: EUR 109m) serve as collateral for positive market values of derivatives.

42 Current advance payments received, deferred income and other non-financial liabilities

T143 CURRENT ADVANCE PAYMENTS RECEIVED, DEFERRED INCOME AND OTHER NON-FINANCIAL LIABILITIES

in €m	31 Dec 2020	31 Dec 2019
Advance payments received	21	17
Deferred income	88	59
Other non-financial liabilities	1,186	306
	1,295	382

As of 31 December 2020, deferred income includes a balance of EUR 43m from grants received by catering and MRO companies in the USA under the CARES Act totalling USD 157m. Deferred income and prepaid expenses are written back in accordance with the use of the funds received.

The increase in other non-financial liabilities is mainly due to tax liabilities. To support affected companies during the crisis in Germany and Austria, tax and levy deferments were granted by state institutions in the amount of EUR 918m. These are mainly import VAT and air traffic tax. Other non-financial liabilities also include the current portion of obligations under share-based remuneration agreements measured at fair value (➤ Note 39, p. 192ff.). In addition, EUR 193m (previous year: EUR 279m) relates to deferred amounts for outstanding holiday allowance and overtime.

NOTES TO THE SEGMENT REPORTING

43 Notes to the reportable segments and segment data

NOTES TO THE REPORTABLE SEGMENTS

As of 31 December 2020, the Lufthansa Group operates in five reporting segments, which make up its Group activities. The segments are defined in line with the internal reporting and management structure.

The airline activities were combined in their respective reporting segments based on the similarity between the economic characteristics of the individual airlines, such as network and sales structures, as well as customers and services. The Network Airlines segment comprises Lufthansa German Airlines, SWISS, Austrian Airlines, Brussels Airlines and Germanwings. Further information about the individual airlines can be found in the Group management report [starting on p. 51](#).

The Eurowings segment comprises the airlines under the Eurowings brand and the equity investment in SunExpress.

The Logistics segment comprises the scheduled airfreight activities of the Lufthansa Cargo group. Lufthansa Cargo is Europe's leading cargo airline.

The MRO segment is a leading global provider of maintenance, repair and overhaul services for civil and commercial aircraft and is represented by Lufthansa Technik group.

The Catering segment, represented by the LSG Lufthansa Service/Sky Chefs group, is a leading airline caterer and also offers catering services in the retail segment as well as service and logistics services.

Business activities not allocated to a reportable segment are presented in the "Additional Businesses and Group Functions" column of the segment reporting along with the income and expenses of central Group functions. They include income and expenses of Lufthansa Commercial Holding GmbH, Lufthansa AirPlus group, the Lufthansa Systems group, the Lufthansa Aviation Training group and other Group companies.

NOTES TO SEGMENT DATA AND INTERNAL MANAGEMENT

The accounting policies of the reportable segments are the same as those described in [Note 3, p. 143ff](#).

Segmentation has been changed compared with the financial statements as of 31 December 2019. Brussels Airlines, Germanwings and the Eurowings long-haul business are managed by the Network Airlines group as of the start of financial year 2020 and have therefore been allocated to the Network Airlines segment. As of 1 January 2020, the line maintenance business of Lufthansa Technik was transferred to Deutsche Lufthansa AG, which will carry out the work itself from this point on, and is now part of the Network Airlines segment. The figures for the previous year have been adjusted accordingly. The segmentation changes increased previous-year revenue for the Network Airlines segment by EUR 1,972m and reduced its Adjusted EBIT by EUR 29m. In the Eurowings segment, the previous year's revenue fell by EUR 1,812m, while Adjusted EBIT increased by EUR 44m. The figures for the previous year in the MRO segment for revenue and Adjusted EBIT fell by EUR -349m and by EUR -30m, respectively.

The Lufthansa Group measures the performance of its segments using both segment result indicators: EBIT and Adjusted EBIT. EBIT is made up of the IFRS operating result and the result from equity investments. Adjusted EBIT is obtained by correcting EBIT for gains and losses on the disposal of assets, write-downs and write-backs and earnings attributable to other periods in connection with pension obligations (plan adjustments and plan settlements).

Sales and revenue between reportable segments are based on arm's length prices. Administrative services are charged as cost allocations.

For information on external traffic revenue [Note 4, p. 157](#).

Capital employed largely comprises segment assets, adjusted for derivative financial instruments, and deferred tax items less non-interest-bearing debt.

The result of the equity valuation for the segment's equity investments is part of its segment result. However, from a Group perspective, it is not attributed to the operating result, but rather to the financial result.

T144 SEGMENT INFORMATION FOR THE 2020 REPORTING SEGMENTS

	Network Airlines	Eurowings	Logistics	MRO	Catering	Total reportable operating segments	Additional Businesses and Group Functions	Reconciliation		Group
in €m								Not allocated	Consoli- dation	
External revenue	6,167	589	2,733	2,724	1,064	13,277	312	-	-	13,589
of which traffic revenue	5,488	579	2,596	-	-	8,663	-	415	-	9,078
Inter-segment revenue	455	9	24	1,023	241	1,752	158	-	-1,910	-
Total revenue	6,622	598	2,757	3,747	1,305	15,029	470	-	-1,910	13,589
Other operating income	994	147	69	437	143	1,790	1,571	-	-1,356	2,005
Total operating income	7,616	745	2,826	4,184	1,448	16,819	2,041	-	-3,266	15,594
Operating expenses	12,258	1,346	2,082	4,502	1,698	21,886	2,361	-	-3,401	20,846
of which cost of materials and services	5,101	699	1,300	2,372	545	10,017	203	-	-1,767	8,453
of which staff costs	3,340	176	377	1,113	742	5,748	660	-	-3	6,405
of which depreciation and amortisation	1,814	199	178	197	117	2,505	124	-	-68	2,561
of which other operating expenses	2,003	272	227	820	294	3,616	1,374	-	-1,563	3,427
Results of equity investments ¹⁾	-32	-102	28	-65	-34	-205	6	-	-	-199
of which result of invest- ments accounted for using the equity method	-39	-102	17	-52	-34	-210	-	-	-	-210
Adjusted EBIT ²⁾	-4,674	-703	772	-383	-284	-5,272	-314	-	135	-5,451
Reconciliation items	-1,425	-99	-55	-125	-147	-1,851	-49	-	-2	-1,902
Impairment losses/gains	-1,398	-97	-55	-108	-201	-1,859	-43	-	32	-1,870
Effects from pension provisions	-27	-	-	-3	1	-29	-2	-	-	-31
Results of disposal of assets	-	-2	-	-14	53	37	-4	-	-34	-1
EBIT	-6,099	-802	717	-508	-431	-7,123	-363	-	133	-7,353
Other financial result										-1,278
Profit/loss before income taxes										-8,631
Capital employed ³⁾	12,171	950	2,287	3,467	943	19,818	3,399	-	-131	23,086
of which from investments accounted for using the equity method	17	77	49	173	86	402	2	-	-1	403
Segment capital expenditure ⁴⁾	958	106	222	152	28	1,466	48	-	-202	1,312
of which from investments accounted for using the equity method	-	-	-	31	-	31	-	-	-	31
Number of employees at end of period	57,363	3,088	4,373	22,745	13,227	100,796	9,269	-	-	110,065
Average number of employees	59,191	3,203	4,436	23,519	25,288	115,637	9,570	-	-	125,207

¹⁾ The result from equity investments does not include any impairment losses on investments accounted for using the equity method.

²⁾ For reconciliation from Adjusted EBIT to EBIT → T021, p. 40, in the Group management report.

³⁾ The capital employed results from total assets adjusted for non-operating items (deferred taxes, positive market values, derivatives) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

⁴⁾ Investment in intangible assets and property, plant and equipment, as well as in loans to and shares in companies. Investment is shown without capitalised borrowing costs.

T144 SEGMENT INFORMATION FOR THE 2019 REPORTING SEGMENTS¹⁾

	Network Airlines	Eurowings	Logistics	MRO	Catering	Total reportable operating segments	Additional Businesses and Group Functions	Reconciliation		Group
in €m								Not allocated	Consoli- dation	
External revenue	24,111	2,321	2,437	4,378	2,623	35,870	554	-	-	36,424
of which traffic revenue	23,055	2,307	2,318	-	-	27,680	-	456	-	28,136
Inter-segment revenue	967	-10	41	2,194	737	3,929	239	-	-4,168	-
Total revenue	25,078	2,311	2,478	6,572	3,360	39,799	793	-	-4,168	36,424
Other operating income	1,014	176	103	256	95	1,644	1,939	-	-1,068	2,515
Total operating income	26,092	2,487	2,581	6,828	3,455	41,443	2,732	-	-5,236	38,939
Operating expenses	24,362	2,633	2,621	6,425	3,355	39,396	2,971	-	-5,243	37,124
of which cost of materials and services	13,843	1,861	1,778	3,902	1,441	22,825	287	-	-3,285	19,827
of which staff costs	4,801	259	406	1,448	1,290	8,204	913	-	-6	9,111
of which depreciation and amortisation	1,927	233	160	191	119	2,630	107	-	-45	2,692
of which other operating expenses	3,791	280	277	884	505	5,737	1,664	-	-1,907	5,494
Results of equity investments	46	24	41	60	28	199	12	-	-	211
of which result of investments accounted for using the equity method	31	24	18	34	24	131	-	-	-	131
Adjusted EBIT²⁾	1,776	-122	1	463	128	2,246	-227	-	7	2,026
Reconciliation items	-52	-4	-34	9	-30	-111	6	-	-64	-169
Impairment losses/gains	-26	-	-38	12	-42	-94	7	-	-52	-139
Effects from pension provisions	-3	-	-1	-2	-1	-7	-4	-	1	-10
Results of disposal of assets	-23	-4	5	-1	13	-10	3	-	-13	-20
EBIT	1,724	-126	-33	472	98	2,135	-221	-	-57	1,857
Other financial result										3
Profit/loss before income taxes										1,860
Capital employed ³⁾	12,957	1,480	2,135	5,607	1,513	23,692	1,904	-	-561	25,035
of which from investments accounted for using the equity method	25	178	49	279	139	670	5	-	-3	672
Segment capital expenditure ⁴⁾	2,738	127	286	313	127	3,591	107	-	-32	3,666
of which from investments accounted for using the equity method	-	-	-	49	28	77	-	-	-	77
Number of employees at end of period	60,913	3,432	4,539	23,855	35,636	128,375	9,978	-	-	138,353
Average number of employees	60,594	3,666	4,543	23,077	35,904	127,784	10,000	-	-	137,784

¹⁾ Previous year's figures have been adjusted.

²⁾ For reconciliation from Adjusted EBIT to EBIT → T021, p. 40, in the Group management report.

³⁾ The capital employed results from total assets adjusted for non-operating items (deferred taxes, positive market values, derivatives) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

⁴⁾ Investment in intangible assets and property, plant and equipment, as well as in loans to and shares in companies. Investment is shown without capitalised borrowing costs.

The reconciliation column includes both the effects of consolidation activities and the amounts resulting from different definitions of segment item contents compared with the corresponding Group items.

The amounts in the reconciliation column for Group EBIT include the effects of consolidation procedures on profit or loss in which income and expense do not figure for two companies at the same amount, or in the same period.

Eliminated segment revenue generated with other consolidated segments is shown in the reconciliation column for revenue.

NOTES ON GEOGRAPHICAL REGIONS IN 2020

The allocation of traffic revenue to geographic regions is based on the original location of sale. Non-current assets are allocated according to the location of the relevant asset. The allocation of other revenue to the individual regions is based on the geographical location of the customer.

The regions are defined on a geographical basis. As an exception to this rule, traffic revenue generated in Turkey is attributed to Europe.

The Lufthansa Group controls its air traffic operations on the basis of network results and not on the basis of regional earnings contributions. The same applies to the Catering segment. Consequently, the presentation of regional segment results is of no informational value for the Lufthansa Group.

A presentation of traffic revenue generated in the Network Airlines, Eurowings and Logistics segments by traffic region, rather than by original location of sale, is included in the information on the respective segments in the management report.

External revenue, non-current assets and capital expenditure are as follows:

T145 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY REGION FOR 2020

in €m	Europe	North America	Central and South America	Asia / Pacific	Middle East	Africa	Group
Traffic revenue ¹⁾	5,852	1,113	222	1,541	166	184	9,078
Other revenue	1,875	1,474	160	704	179	119	4,511
Non-current assets ^{2) 3) 4)}	20,123	777	37	201	2	11	21,151
Capital expenditure on non-current assets ³⁾	1,224	33	-	10	-	1	1,268

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

⁴⁾ Including rights of uses in accordance with first-time application of IFRS 16.

The figures for the main countries are as follows:

T146 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY COUNTRIES FOR 2020

in €m	Germany	USA
Traffic revenue ¹⁾	2,691	1,005
Other revenue	677	1,200
Non-current assets ^{2) 3) 4)}	14,183	527
Capital expenditure on non-current assets ³⁾	991	27

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

⁴⁾ Including rights of uses in accordance with first-time application of IFRS 16.

NOTES ON GEOGRAPHICAL REGIONS IN 2019

External revenue, non-current assets and capital expenditure are as follows:

T145 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY REGION FOR 2019

in €m	Europe	North America	Central and South America	Asia / Pacific	Middle East	Africa	Group
Traffic revenue ¹⁾	18,586	4,707	615	3,073	621	534	28,136
Other revenue	3,437	2,557	463	1,394	281	156	8,288
Non-current assets ^{2) 3)}	23,385	564	48	305	4	26	24,332
Capital expenditure on non-current assets ³⁾	3,434	50	6	17	–	3	3,510

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

The figures for the main countries are as follows:

T146 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY COUNTRIES FOR 2019

in €m	Germany	USA
Traffic revenue ¹⁾	8,610	4,233
Other revenue	1,077	2,097
Non-current assets ^{2) 3)}	16,207	533
Capital expenditure on non-current assets ³⁾	2,913	45

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

In 2020 and in the previous year, no more than 10% of Lufthansa Group revenue was generated with any one customer.

NOTES TO THE CONSOLIDATED CASH FLOW STATEMENT

44 Notes to cash flow from operating, investing and financing activities

The cash flow statement shows how cash and cash equivalents have changed over the reporting period at the Lufthansa Group. In accordance with IAS 7, cash flows are divided into cash inflows and outflows from operating activities, from investing activities and from financing activities. The cash and cash equivalents shown in the cash flow statement comprise the statement of financial position items bank balances and cash-in-hand, without fixed-term deposits with terms of four to twelve months, amounting to EUR 2m (previous year: EUR 0m). The amount of liquidity in the broader sense is reached by adding securities that can be liquidated at short notice.

Interest paid and interest income from the corresponding interest rate hedges are netted to avoid overemphasising the items interest income and interest paid in the cash flow statement.

ADDITIONAL INFORMATION ON THE CASH FLOW STATEMENT

Cash flow from operating activities

Cash flow from operating activities is derived from profit/loss before income taxes using the indirect method. It is adjusted for non-cash income and expenses as well as changes in trade working capital and in other assets/liabilities that are not attributable to investing or financing activities.

Lease accounting in line with IFRS 16 means that payments under operating leases are presented as capital repayments or interest payments within cash flow from financing activities. In the financial year this had a positive impact of EUR 432m on cash flow from operating activities (of which EUR 379m capital repayment) and a corresponding negative effect in cash flow from financing activities.

In the current financial year, the Group primarily recognised the following non-cash income and expenses:

T147 SIGNIFICANT NON-CASH INCOME AND EXPENSES

in €m	2020	2019
Result of miscellaneous financial items	306	84
Write-down on receivables	248	120
Reversal of write-downs on receivables	-36	-27
Income from the reversal of provisions and accruals	-338	-321
Adjustments to retirement and transitional benefit systems	32	10
Total	212	-134

Trade working capital consists of changes in the carrying amounts of inventories, trade receivables and payables, contract assets and down payments, other current assets and other current liabilities, contract liabilities and current deferrals and prepaid expenses.

Other assets/liabilities mainly include corrections between pensions expenses and payments, changes in other provisions, accruals/deferrals and corrections for non-cash effects from currency translation.

Special effects from other activities to strengthen the Group's liquidity included the realisation and repeat purchase of existing exchange rate hedges with positive market values (known as restriking). This generated cash inflows of EUR 428m. Emissions certificates were also used in repo transactions to generate cash inflows (EUR 275m). Liquidity was improved by a total of EUR 918m thanks to deferrals of taxes and charges (mainly import VAT and air traffic tax) in Germany and Austria.

Cash flow from investing (and cash management) activities

Cash flows from investing and financing activities are calculated on the basis of payments.

Cash flow from investing activities results mainly from investments and disinvestments in non-current assets. The negative cash effect from disposals of consolidated companies results from the net total of the purchase price received and the decrease in cash and cash equivalents through the sale of the LSG group's European business.

The Lufthansa Group contributed EUR 235m to pension assets in 2020 (previous year: EUR 394m). These payments were categorised as cash flow from investing and cash management activities. By contrast, pension payments from fund assets led to cash inflows from investments (EUR 510m; previous year: EUR 138m). They correspond to cash outflows from operating activities.

Cash flow from financing activities

Cash flow from financing activities now also includes capital repayments and interest payments on lease liabilities.

Borrowing and the instruments used to hedge it changed as follows in the financial year:

T148 FINANCIAL LIABILITIES 2020

	31 Dec 2019	Cash effective	Non-cash effective						31 Dec 2020
			Addition due to changes in consoli- dation	Addition due to lease liabilities	Currency translation differences	Accrued interest	Reclassi- fication/ Reclassi- fication in held for sale	Changes in fair value	
in €m									
Non-current borrowings	8,396	5,547	-	340	-278	25	-2,022	244	12,252
Current borrowings	1,634	-532	-	2	-19	-	2,036	-5	3,116
Other borrowings ¹⁾	17	-1	-	-	-2	-	-	-	14
Total financial liabilities	10,047	5,014	-	342	-299	25	14	239	15,382
Interest rate swaps and currency futures used for hedging – assets	-205	69	-	-	-	-	-	-20	-156
Interest rate swaps and currency futures used for hedging – liabilities	27	-	-	-	-	-	-	17	44

¹⁾ Mainly relate to bank overdrafts.

Changes in borrowing in the previous year were as follows:

T148 FINANCIAL LIABILITIES 2019

	31 Dec 2018	Cash effective	Non-cash effective						31 Dec 2019
			Addition due to changes in consoli- dation	Addition due to lease liabilities ¹⁾	Currency translation differences	Accrued interest	Reclassi- fication/ Reclassi- fication in held for sale	Changes in fair value	
in €m									
Non-current borrowings	5,008	2,688	-	2,410	66	21	-1,867	70	8,396
Current borrowings	1,677	-2,236	-	359	3	-	1,833	-2	1,634
Other borrowings ²⁾	39	-22	-	-	-	-	-	-	17
Total financial liabilities	6,724	430	-	2,769	69	21	-34	68	10,047
Interest rate swaps and currency futures used for hedging – assets	-154	89	-	-	-	-	-	-140	-205
Interest rate swaps and currency futures used for hedging – liabilities	49	-	-	-	-	-	-	-22	27

¹⁾ This includes additions from the first-time application of IFRS 16 (non-current borrowings: EUR 1,599m and current borrowings: EUR 359m).

²⁾ Mainly relate to bank overdrafts.

OTHER DISCLOSURES

45 Additional disclosures on financial instruments

FINANCIAL ASSETS BY MEASUREMENT CATEGORY

As of the current reporting date, the financial assets can be broken down into measurement categories with the following carrying amounts:

T149 FINANCIAL ASSETS IN THE BALANCE SHEET AS OF 31 DEC 2020

in €m	Amortised cost	At fair value through profit or loss	At fair value through other comprehensive income (with recycling)	At fair value through other comprehensive income (without recycling)	Derivative financial instruments which are an effective part of a hedging relationship	Total
Other equity investments	-	34	-	-	-	34
Non-current securities	31	-	-	23	-	54
of which equity instruments	-	-	-	23	-	23
of which debt instruments	31	-	-	-	-	31
Loans	121	-	-	-	-	121
Non-current receivables	318	-	-	-	-	318
Non-current derivative financial instruments	-	-	-	-	363	363
Trade receivables and other current receivables	2,793	-	-	-	-	2,793
Current derivative financial instruments	-	2	-	-	258	260
Current securities	-	3,654	-	-	-	3,654
of which equity instruments	-	3,654	-	-	-	3,654
of which debt instruments	-	-	-	-	-	-
Cash and cash equivalents	1,806	-	-	-	-	1,806
Total	5,069	3,690	-	23	621	9,403

T149 FINANCIAL ASSETS IN THE BALANCE SHEET AS OF 31 DEC 2019

in €m	Amortised cost	At fair value through profit or loss	At fair value through other comprehensive income (with recycling)	At fair value through other comprehensive income (without recycling)	Derivative financial instruments which are an effective part of a hedging relationship	Total
Other equity investments	-	33	-	-	-	33
Non-current securities	31	-	-	22	-	53
of which equity instruments	-	-	-	22	-	22
of which debt instruments	31	-	-	-	-	31
Loans	154	-	-	-	-	154
Non-current receivables	316	-	-	-	-	316
Non-current derivative financial instruments	-	6	-	-	899	905
Trade receivables and other current receivables	5,322	-	-	-	-	5,322
Current derivative financial instruments	-	6	-	-	452	458
Current securities	-	360	1,610	-	-	1,970
of which equity instruments	-	360	-	-	-	360
of which debt instruments	-	-	1,610	-	-	1,610
Cash and cash equivalents	1,415	-	-	-	-	1,415
Total	7,238	405	1,610	22	1,351	10,626

The category “At fair value through other comprehensive income” includes derivatives that do not meet the requirements for applying hedge accounting and so are accounted for as stand-alone derivatives. This category also includes equity instruments, consisting of money market funds and equity investments, for which the instrument-specific option of fair value through other comprehensive income without recycling has not been exercised. As of the reporting date, the fair-value option without effect on profit and loss and without recycling was chosen for one share position, in order to avoid the recognition of changes in market value through profit or loss. The item includes shares in VISA Inc. (market value: EUR 23m, dividend payments: EUR 0.1m in the 2020 financial year) whose market valuation reserve increased by EUR 1m in 2020 (cumulative EUR 15m). The decline in debt instruments designated as at fair value (with recycling) without effect on profit and loss was due to the release of bonds in the financial year. This and other liquid funds are being invested in the current year in money market funds, which are equity instruments at fair value through profit or loss.

FINANCIAL LIABILITIES BY MEASUREMENT CATEGORY

The financial liabilities can be divided into measurement categories with the following carrying amounts, whereby the category “at fair value through profit or loss” includes derivatives that do not meet the requirements for applying hedge accounting and so are accounted for as stand-alone derivatives. In addition, the instrument-specific option to designate the convertible bond issued in the reporting year as “at fair value through profit or loss” was also exercised; its fair value as of the reporting date was EUR 712m. The total change of EUR 118m in the market value of the convertible bond was therefore split into a credit-risk-induced share of EUR 22m, which is recognised as an expense in OCI, and a price-induced share of EUR 96m, which is recognised as an expense in the trading result.

T150 FINANCIAL LIABILITIES IN THE BALANCE SHEET AS OF 31 DEC 2020

	Liabilities at fair value through profit or loss	Derivative financial instruments which are an effective part of a hedging relationship	Other financial liabilities at cost	Total
in €m				
Borrowings (without IFRS 16 lease liabilities)	712	-	12,020	12,732
Derivative financial instruments	85	738	-	823
Trade payables	-	-	2,087	2,087
Other financial liabilities	-	-	1,320	1,320
Total	797	738	15,427	16,962

T150 FINANCIAL LIABILITIES IN THE BALANCE SHEET AS OF 31 DEC 2019

	Liabilities at fair value through profit or loss	Derivative financial instruments which are an effective part of a hedging relationship	Other financial liabilities at cost	Total
in €m				
Borrowings (without IFRS 16 lease liabilities)	-	-	7,158	7,158
Derivative financial instruments	67	199	-	266
Trade payables	-	-	3,722	3,722
Other financial liabilities	-	-	1,705	1,705
Total	67	199	12,585	12,851

The net result of the different categories of financial assets and liabilities is made up as follows:

T151 NET RESULT 2020 FOR FINANCIAL ASSETS AND LIABILITIES BY MEASUREMENT CATEGORY

in €m	Interest expenses	Interest income	Depreciation	Result from valuation and sale	Currency result	Net result
Assets at amortised cost	-	3	-213	-	-11	-221
At fair value through other comprehensive income (with recycling)	-3	4	-	15	-2	14
At fair value through other comprehensive income (without recycling)	-	-	-	1	-	1
Assets at fair value through profit or loss	-	-	-	24	-	24
Liabilities at amortised cost	-169	-	-	-	39	-130
Liabilities at fair value through profit or loss	-1	-	-	-886	-	-887
Total	-173	7	-213	-846	26	-1,199

For the result of measurement and disposal of assets at fair value through profit or loss, see [Note 14, p. 162](#).

T151 NET RESULT 2019 FOR FINANCIAL ASSETS AND LIABILITIES BY MEASUREMENT CATEGORY

in €m	Interest expenses	Interest income	Depreciation	Result from valuation and sale	Currency result	Net result
Assets at amortised cost	-	8	-94	-	8	-78
At fair value through other comprehensive income (with recycling)	-10	21	-	13	5	29
At fair value through other comprehensive income (without recycling)	-	-	-	-8	-	-8
Assets at fair value through profit or loss	-	-	-	411	-	411
Liabilities at amortised cost	-123	-	-	-	-22	-145
Liabilities at fair value through profit or loss	-	-	-	-52	-	-52
Total	-133	29	-94	364	-9	157

The table to the right shows the carrying amounts and market values for individual classes of financial liabilities. The stated market values of bonds reflect their stock market listings (Level 1 of the fair value hierarchy). The market values for other types of financial liability have been calculated using the applicable interest rates for the remaining term to maturity and repayment structures at the balance sheet date based on the available market information (Bloomberg) (Level 2 of the fair value hierarchy). For other assets and liabilities, non-current receivables, trade receivables and cash-in-hand carried at amortised cost, the carrying amount is deemed to be a reasonable approximation of the fair value.

T152 FINANCIAL LIABILITIES

in €m	31 Dec 2020		31 Dec 2019	
	Carrying amount	Market value	Carrying amount	Market value
Bonds	2,707	2,643	995	1,026
Commercial Paper to banks	200	199	98	98
Borrower's note loans	1,900	1,894	2,073	2,129
Credit lines	763	740	-	-
State-guaranteed loans	2,907	3,147	-	-
Aircraft financing	3,603	3,590	3,887	3,682
Other borrowings	651	705	104	124
Leasing liabilities ⁹⁾	2,637	-	2,873	-
Total	15,368	12,918	10,030	7,059

⁹⁾ With the introduction of IFRS 16, declaration of market value no longer applies as of 1 January 2019.

FINANCIAL ASSETS HELD AT FAIR VALUE BY LEVEL OF FAIR VALUE HIERARCHY

The following table shows financial assets and liabilities held at fair value by level of fair value hierarchy. The levels are defined as follows:

Level 1: Financial instruments traded on active markets, the quoted prices for which are taken for measurement unchanged.

Level 2: Measurement is made by means of valuation methods with parameters derived directly or indirectly from observable market data.

Level 3: Measurement is made by means of valuation methods with parameters not based exclusively on observable market data.

In the financial year 2020, the fair value hierarchy for assets and liabilities held at fair value was as follows:

T153 FAIR VALUE HIERARCHY OF ASSETS AS OF 31 DEC 2020

in €m	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit and loss	3,654	2	-	3,656
Financial derivatives classified as held for trading	-	2	-	2
Securities	3,654	-	-	3,654
Derivative financial instruments which are an effective part of a hedging relationship	-	620	-	620
Financial assets at fair value through other comprehensive income	11	12	-	23
Equity instruments	11	12	-	23
Debt instruments	-	-	-	-
Total assets	3,665	634	-	4,299

T154 FAIR VALUE HIERARCHY OF LIABILITIES AS OF 31 DEC 2020

in €m	Level 1	Level 2	Level 3	Total
Financial liabilities at fair value through profit or loss	-	-712	-	-712
Derivative financial instruments at fair value through profit or loss	-	-85	-	-85
Derivative financial instruments which are an effective part of a hedging relationship	-	-738	-	-738
Total liabilities	-	-1,535	-	-1,535

In the previous year, the hierarchy was as follows:

T153 FAIR VALUE HIERARCHY OF ASSETS AS OF 31 DEC 2019

in €m	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit and loss	359	13	-	372
Financial derivatives classified as held for trading	-	12	-	12
Securities	359	1	-	360
Derivative financial instruments which are an effective part of a hedging relationship	-	1,352	-	1,352
Financial assets at fair value through other comprehensive income	-	1,632	-	1,632
Equity instruments	-	22	-	22
Debt instruments	-	1,610	-	1,610
Total assets	359	2,997	-	3,356

T154 FAIR VALUE HIERARCHY OF LIABILITIES AS OF 31 DEC 2019

in €m	Level 1	Level 2	Level 3	Total
Derivative financial instruments at fair value through profit or loss	-	-67	-	-67
Derivative financial instruments which are an effective part of a hedging relationship	-	-199	-	-199
Total liabilities	-	-266	-	-266

NETTING OF FINANCIAL ASSETS AND LIABILITIES

The following financial assets and liabilities are subject to global netting agreements and other agreements.

T155 NETTING OF FINANCIAL ASSETS AS OF 31 DEC 2020

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade receivables and other current receivables	2,837	44	2,793	–	2	2,793
Derivative financial instruments – assets	623	–	623	39	38	546
Cash and cash equivalents	1,822	16	1,806	–	–	1,806
Total assets	5,282	60	5,222	39	40	5,145

T156 NETTING OF FINANCIAL LIABILITIES AS OF 31 DEC 2020

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade payables and other financial liabilities	2,144	60	2,084	–	38	2,084
Derivative financial instruments – liabilities	823	–	823	39	2	782
Total liabilities	2,967	60	2,907	39	40	2,866

In the previous year, the net balances were as follows:

T155 NETTING OF FINANCIAL ASSETS AS OF 31 DEC 2019

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade receivables and other current receivables	5,478	156	5,322	–	–	5,322
Derivative financial instruments – assets	1,364	–	1,364	–37	92	1,309
Cash and cash equivalents	1,445	30	1,415	–	–	1,415
Total assets	8,287	186	8,101	–37	92	8,046

T156 NETTING OF FINANCIAL LIABILITIES AS OF 31 DEC 2019

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade payables and other financial liabilities	3,908	186	3,722	–	92	3,722
Derivative financial instruments – liabilities	266	–	266	–37	–	303
Total liabilities	4,174	186	3,988	–37	92	4,025

PRINCIPLES OF HEDGING POLICY

As an aviation group with worldwide operations, the Lufthansa Group is exposed to exchange rate, interest rate and fuel price movement risks, as well as to credit and liquidity risks. Limiting these risks by means of systematic financial management is part of Company policy.

Market risk

The major market and price risks to which the Lufthansa Group is exposed are exchange rate fluctuations between the euro and other currencies, interest rate fluctuations in international money and capital markets, and price fluctuations in the crude oil and oil products markets. Hedging policy for limiting these risks is laid down by the Executive Board and documented by internal Group guidelines. It also provides for the use of financial derivatives. The corresponding financial transactions are concluded only with first-rate counterparties.

Foreign exchange risk

For US dollars, the Lufthansa Group is in a net payer position as regards currency risks from its operating business, since fuel payments are dollar-denominated. There is always a net surplus for other currencies. The main risks in this respect stem from the Chinese renminbi, the Swiss franc, British pound sterling, the Japanese yen and the Indian rupee. Depending on market liquidity, currency risks from projected operational exposure are hedged gradually over a period of 24 months by means of futures contracts, which are accounted for as cash flow hedges. The target hedging level is defined in the Group's internal guidelines. The volume of hedging was reduced following the outbreak of the coronavirus crisis.

At the end of the financial year 2020, exposure from operations for the next 24 months was as follows:

T157 CURRENCY EXPOSURE, AS OF 2020

in millions	USD	CNY	JPY	GBP	INR
Exposure (currency)	-3,971	4,212	56,644	530	30,819
Exposure (EUR at spot rate)	-3,236	525	448	589	344
Hedges (currency)	706	-1,439	-19,113	-180	-3,690
Hedging level	18%	34%	34%	34%	12%
Hedging rate	1.13	8.26	120.35	0.90	88.81

50% of currency risks from capital expenditure on aircraft are hedged when the contract is signed. The hedging level is reviewed and increased, where necessary, if, over the lifetime of the contract, the exchange rate goes significantly above or below that used to calculate the investment. In the last 24 months before payment, the hedging level is increased in half-yearly steps of 10%, reaching 90% by the end. These investment hedges are therefore also accounted for as cash flow hedges. Capital expenditure on aircraft takes place in US dollars and is hedged in euros or in Swiss francs, depending on the functional currency of the Group company making the purchase. There was no exposure in Swiss francs as of the reporting date.

US dollar exposure for capital expenditure as of year-end 2020 was as follows, broken down by the hedged currency:

T158 USD INVESTMENT EXPOSURE, HEDGED IN EUR

in millions	2021	2022	2023	2024	2025	2026	2027	2028	2029
Exposure from net capital expenditure (USD)	-2,157	-2,717	-3,578	-1,607	-1,325	-947	-269	-213	-168
Exposure from net capital expenditure (EUR at spot rate)	-1,755	-2,211	-2,912	-1,308	-1,078	-770	-219	-173	-137
Hedges (USD)	1,904	2,151	2,303	895	754	496	75	-	-
Hedging level	88%	79%	64%	56%	57%	52%	28%	0%	0%
Hedging rate EUR/USD	1.20	1.20	1.22	1.28	1.30	1.28	1.36	-	-

The following sensitivity analysis shows how net profit and equity would change if the currencies identified as price risk variables had been different from those at the balance sheet date.

T159 SENSITIVITY ANALYSIS BY CURRENCY

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Currency – USD		
+10%	-199	589
-10%	165	-482
Currency – JPY		
+10%	-5	-14
-10%	5	12
Currency – CHF		
+10%	39	-3
-10%	-31	2
Currency – GBP		
+10%	4	-19
-10%	-3	16
Currency – CNY		
+10%	9	-16
-10%	-7	13
Currency – INR		
+10%	2	-4
-10%	-1	3

¹⁾ All amounts after deferred tax effects; +/- signs relate to net profit and/or equity.

Interest rate risk

The Lufthansa Group aims to pay interest on 100% of its financial liabilities in euros at floating rates of interest. To do so, interest rate swaps are arranged for interest-bearing, fixed-rate financial debt and leasing liabilities. Financial liabilities denominated in euros are hedged using “plain vanilla” interest rate swaps, while cross-currency interest rate swaps are used to hedge financial liabilities in foreign currencies. Depending on the interest rate structure of the exposure to be hedged, hedges are either classified as fair value hedges or cash flow hedges. The interest rate risk is monitored constantly; strategic interest rate hedges are used as needed in response to different market situations. Depending on the counterparties and the instruments used, cash collateral for interest rate swaps is either deposited with or received from counterparties (➔ T155, T156, p. 209).

The tables below describe the floating/fixed ratio for non-current borrowing as of financial year-end 2020 after taking into consideration interest rate hedging, as well as the distribution of the nominal volume of interest rate hedges.

T160 INTEREST RATE EXPOSURE AFTER HEDGING

in €m	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Fix	2,613	1,997	1,936	1,875	1,820	1,142	145	41	34	27	20
Variable	6,101	4,657	3,835	3,243	1,956	1,554	1,179	969	697	456	198
Float/Fix-Ratio	70%	70%	66%	63%	52%	58%	89%	96%	95%	94%	91%

T161 NOMINAL VOLUME OF INTEREST RATE HEDGES

in €m	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Fix	-3,105	-2,067	-1,676	-652	-457	-169	-493	34	27	20	12
Variable	3,227	2,196	1,800	771	567	258	558	697	456	198	36

The sensitivity analysis to the right shows how net profit and equity would change if the interest rate identified as a price risk variable had been different from the perspective of the balance sheet date. In view of the current low interest rates, a reduction of more than 50 basis points is not considered likely, which is why the analysis was limited to this figure. Only stand-alone interest rate derivatives have been included, because in fair value hedge accounting, interest rate derivatives offset the movements in the underlying hedged items.

T162 SENSITIVITY ANALYSIS BY INTEREST RATE

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Interest		
+100 basis points	31	-
-50 basis points	-48	-

¹⁾ All amounts after deferred tax effects; +/- signs relate to net profit and/or equity.

Effects of the EU benchmark regulation of global reference interest rates

In terms of the financial instruments used by the Lufthansa Group, the global reform of variable reference interest rates means that the variable reference interest rates for transactions that are available today will no longer be so in future or will be calculated differently. As of 31 December 2020, the Lufthansa Group held EUR 6.8bn in outstanding financial transactions, floating rate liabilities and hedging instruments based on variable interest rates. The following table shows the absolute nominal volume of the transactions as of the reporting date.

T163 NOMINAL VOLUME OF FLOATING RATE FINANCIAL INSTRUMENTS IN HEDGING RELATIONSHIPS

Reference floating interest rate by currency	Nominal volume of floating interest rate derivatives in €m	Nominal volume of floating interest rate financial liabilities (without derivatives) in €m
EUR – EURIBOR	5,001	1,085
USD – LIBOR	331	336
CHF – LIBOR	11	–
Total	5,343	1,421

EURIBOR's calculation method was adjusted in 2019 to ensure that it complies with the rules of the EU benchmark regulation. The change therefore has no effect on the Lufthansa Group for financial instruments based on EURIBOR. Since October 2019, the European Central Bank has also published €STR, a benchmark-compliant overnight interest rate. Clearing houses switched their mark-to-market accounting to the new overnight reference rates €STR (EUR) and SOFR (USD) in the course of 2020. The current status is that USD LIBOR rates will be published for the last time as of 31 December 2021, so another reference interest rate will have to be used from 2022 at the latest but none has yet become established in the market. However, the outstanding changes to USD and CHF LIBOR will not have any significant impact on the Group portfolio due to the small volume of transactions.

The Lufthansa Group has opted for the early application of the “Amendments to IFRS 9, IAS 39 and IFRS 7, Reform of Reference Interest Rates” of Phase 1 published in September 2019 and the pronouncements on Phase 2 of the reform of reference interest rates “Amendments to IFRS 9, IAS 39, IFRS 7, IFRS 4 and IFRS 16” from January 2021, which include transitional provisions and accounting exemptions for interest rate hedges. The outstanding changes to the benchmarks will not have any impact on the USD and CHF hedging relationships designated by the Lufthansa Group. Contracts for

derivatives and for financial liabilities have been analysed by the Lufthansa Group and fallback clauses added if necessary. Systematic adjustments will also be made to the extent required.

Fuel price risk

In 2020, fuel costs accounted for 9.0% of the Lufthansa Group's operating expenses (previous year: 18.1%). Significant changes in fuel prices can therefore have a significant effect on the Lufthansa Group's result.

Fuel price risk is limited by the use of crude oil hedges. The hedging level and the time horizon depend on the risk profile, which is derived from the business model of a Group company. As a rule, up to 5% of exposure is hedged monthly for up to 24 months by spread options and other combinations of hedges. Executive Board approval may be obtained to extend the hedging period and to increase the monthly hedging volume in order to exploit market opportunities. The target hedging level is up to 85%. No new fuel exposure was hedged from March 2020 following the outbreak of the coronavirus crisis.

From a year-end perspective, fuel exposure was as follows:

T164 FUEL EXPOSURE

		2021	2022
Fuel requirement	in 1,000 tonnes	5,687	8,541
Hedges	in 1,000 tonnes	3,901	128
Hedging level	%	69	1
Hedging rate	USD/bbl	59.29	55.65

The following sensitivity analysis shows how equity would have been affected by changes in the market value of hedging instruments held as of the reporting date if the fuel price, the identified risk variable, had been different. Since hedge accounting rules mean that changes in the market value of the instruments are only recognised directly in equity without effect on profit and loss, the change in the fuel price of the hedges alone has no effect on earnings. The impact on earnings results from stand-alone derivative as of the current reporting date:

T165 SENSITIVITY ANALYSIS BY FUEL PRICE

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Fuel price		
+10%	–	66
–10%	–	–73

¹⁾ All amounts after deferred tax effects; +/- signs relate to net profit and/or equity.

Market values of the derivative financial instruments used for hedging

Hedging instruments in designated hedging relationships are used to hedge exchange rate, interest rate and fuel price risks as of the reporting date. They changed as follows in the financial year:

T166 DERIVATIVE FINANCIAL INSTRUMENTS USED FOR HEDGING AS OF 31 DEC 2020

in €m	Positive market value	Negative market value	Change in fair value of hedging instrument – designated risk	Change in fair value of hedging instrument – non-designated risk	Basis adjustment of hedged items	OCI – cash flow hedge reserve	OCI – cost of hedging	Ineffective portion of hedges – designated risk	Ineffective portion of hedges – non-designated risk
Fair value hedge									
Interest rate hedges – interest rate swaps	131	-41	-41	-	60	-	-	19	-
Cash flow hedge									
Fuel hedging – options	6	-42	-61	6	-	-63	7	2	-1
Exchange rate hedging – forward transactions	458	-651	-1,151	129	-	-1,019	140	-132	-10
Interest rate hedges – interest rate swaps	26	-3	-21	-	-	-19	-	-2	-
Total	621	-737	-1,274	135	60	-1,101	147	-113	-11
of which current	258	-354							

T166 DERIVATIVE FINANCIAL INSTRUMENTS USED FOR HEDGING AS OF 31 DEC 2019

in €m	Positive market value	Negative market value	Change in fair value of hedging instrument – designated risk	Change in fair value of hedging instrument – non-designated risk	Basis adjustment of hedged items	OCI – cash flow hedge reserve	OCI – cost of hedging	Ineffective portion of hedges – designated risk	Ineffective portion of hedges – non-designated risk
Fair value hedge									
Interest rate hedges – interest rate swaps	156	-27	89	-	-90	-	-	-1	0
Cash flow hedge									
Fuel hedging – options	185	-39	404	163	-	402	164	2	-1
Exchange rate hedging – forward transactions	963	-133	-386	315	-	-380	319	-6	-4
Interest rate hedges – interest rate swaps	49	-	4	-	-	8	-	-4	0
Total	1,353	-199	111	478	-90	30	483	-9	-5
of which current	452	-135							

In terms of exchange hedging, the change of EUR –309m in the market value of the reserve for cash flow hedges relates to derivatives that are still in the portfolio. Other changes result from the realisation of derivatives, in particular EUR 560m that remain in OCI due to the postponement of hedged transactions. The market values stated for financial derivatives correspond to the price at which an independent third party would assume the rights and/or obligations from the financial instrument. The fair values of interest rate derivatives correspond to their respective market values, which are measured using appropriate mathematical methods, such as discounting expected future cash flows. Discounting takes market standard interest rates and the residual term of the respective instruments into account. Currency forward transactions and interest rate swaps are individually discounted to the balance sheet date based on their respective forward rates and the appropriate interest rate curve. The market prices of options used to hedge fuel prices are determined using acknowledged option pricing models.

Depending on the hedged exposure, the Lufthansa Group designates interest rate hedges as both fair value hedges and cash flow hedges and accounts for them accordingly. Interest rate swaps are designated as part of a hedging relationship and are not broken down into individual components. Ineffectiveness in these hedging relationships mainly occurs as a result of the subsequent designation of cross currency swaps in hedges as of 1 January 2018. Other reasons for ineffectiveness in hedging relationships are different parameters in the hedged item and the hedging instrument and the basis spread in cross currency swaps. Ineffectiveness in fair value hedges and cash flow hedges are recognised and presented as part of the financial result, below the other financial items.

Derivatives used in the context of fuel hedging to hedge future kerosene purchases are designated as cash flow hedges. Since 1 January 2018, the Lufthansa Group has applied the IFRS 9 component approach, using crude oil, based on Brent Crude ICE, as the designated risk component of the hedging instrument. The hedged item is composed of a global mix of crude oil types. The base risk between individual crude oil components in the hedging instrument and the crude oil mix in the hedged item is reduced by rebalancing the volumes that make up the hedged item on a quarterly basis. In 2020, the quarterly rebalancing factors for adjusting the hedged item were as follows: 1.022 (Q1), 1.021 (Q2), 1.02 (Q3) and 1.017 (Q4). The Lufthansa Group generally uses options and combinations of options to hedge fuel prices. The intrinsic value of the option is designated as the hedging instrument, so that effective changes in the intrinsic values are recog-

nised in other comprehensive income in the cash flow hedge reserve. The time value of an option is not designated as a hedging instrument and effective changes in the time value are therefore recognised as a cost of hedging. This rule was applied for the first time retroactively. Ineffectiveness in fuel price hedges results from the base risk between the crude oil component and the crude oil mix in the component approach. Ineffectiveness in hedges is recognised and presented as part of the financial result, below the other financial items.

Since 1 January 2018, the Lufthansa Group has prospectively applied the spot-to-spot method for exchange rate forward transactions designated as cash flow hedges. The spot components of a forward contract are designated as a hedging instrument and effective value changes are recognised in the cash flow hedge reserve. The other effective components of a forward contract, the forward component and the cross-currency basis spread are presented in a separate OCI component in line with the legal requirements for the cost of hedging. Ineffectiveness in hedging relationships results from changes in the timing of the planned aircraft purchases. Ineffectiveness is presented as part of the financial result, below the other financial items (➔ **Note 14, p. 162**).

The decline in flight traffic due to the coronavirus crisis meant that fuel prices and foreign currencies were and still are “overhedged”, meaning hedging relationships previously designated under hedge accounting rules had to be terminated early. The corresponding hedges are therefore accounted for as stand-alone derivatives. As of 31 December 2020, the realised result of expired, stand-alone derivatives was EUR –730m, which was recognised in the financial result. Fuel hedges accounted for expenses of EUR 758m and foreign currency hedges for earnings of EUR 28m. In addition, changes in the market value of outstanding “de-designated” hedges reduced the financial result by EUR –21m, of which fuel hedging accounted for EUR –22m and operating exchange rate hedges for EUR 1m. Overall, de-designated hedging relationships therefore had an earnings impact of EUR –752m. EUR 639m has been paid out in cash to date to settle the underlying transactions.

Long-term foreign currency hedges were also realised early in the financial year, generating cash proceeds of EUR 428m. The amounts will remain in the market valuation reserve until the hedged transactions are added and will then be allocated to the corresponding investments. In general, the same accounting method also applies to hedging transactions in which the hedged items (aircraft purchases) have been deferred due to the crisis.

The Lufthansa Group uses the hypothetical derivative method to calculate changes in the value of hedged items designated as being part of a hedging relationship.

T167 DESIGNATED HEDGED ITEMS IN HEDGING RELATIONSHIPS 2020

in €m	Carrying amount of liabilities	Change in fair value of hedged items – designated risk	Change in fair value of hedged items – non-designated risk	Basis adjustment of hedged items from fair value hedges – cumulative
Fair value hedge				
Interest rate hedges – interest rate swaps	-4,920	60	-	-18
Cash flow hedge				
Fuel hedging – options	-	63	-7	-
Exchange rate hedging – forward transactions	-	770	-148	-
Interest rate hedges – interest rate swaps	-	-25	-	-
Total	-4,920	868	-155	-18

T167 DESIGNATED HEDGED ITEMS IN HEDGING RELATIONSHIPS 2019

in €m	Carrying amount of liabilities	Change in fair value of hedged items – designated risk	Change in fair value of hedged items – non-designated risk	Basis adjustment of hedged items from fair value hedges – cumulative
Fair value hedge				
Interest rate hedges – interest rate swaps	-7,330	-90	-	-99
Cash flow hedge				
Fuel hedging – options	-	-407	-164	-
Exchange rate hedging – forward transactions	-	390	-301	-
Interest rate hedges – interest rate swaps	-	9	-	-
Total	-7,330	-98	-465	-99

T168 STATEMENT OF EQUITY RECONCILIATION FOR CASH FLOW HEDGES 2020

in €m	As of 1 Jan 2020	Gains or losses from effective hedging relationships	Reclassification to profit or loss	Reclassification to acquisition costs of inventories	Reclassification to acquisition costs of aircraft	As of 31 Dec 2020
OCI – cash flow hedge reserve	477	-1,463	-811	-163	41	-54
Fuel hedging – options	43	-1,169	-943	-163	-	-21
Exchange rate hedging – futures	440	-309	132	-	41	-42
Interest rate hedges – interest rate swaps	-6	15	-	-	-	9
OCI – cost of hedging	326	142	-	-	-	468
Fuel hedging – options	-70	2	-	-	-	-68
Exchange rate hedging – futures	396	140	-	-	-	536
Total	803	-1,321	-811	-163	41	414

T168 STATEMENT OF EQUITY RECONCILIATION FOR CASH FLOW HEDGES 2019

in €m	As of 1 Jan 2019	Gains or losses from effective hedging relationships	Reclassification to profit or loss	Reclassification to acquisition costs of inventories	Reclassification to acquisition costs of aircraft	As of 31 Dec 2019
OCI – cash flow hedge reserve	451	803	544	89	146	477
Fuel hedging – options	-360	491	-	89	-	43
Exchange rate hedging – futures	820	310	544	-	146	440
Interest rate hedges – interest rate swaps	-9	2	-	-	-	-6
OCI – cost of hedging	-151	476	-	-	-	326
Fuel hedging – options	-227	157	-	-	-	-70
Exchange rate hedging – futures	76	319	-	-	-	396
Total	300	1,279	544	89	146	803

Derivative financial instruments that do not meet the requirements for applying hedge accounting are measured at fair value through profit or loss. As a rule, these derivatives were originally in an economic hedging relationship with a particular exposure, but the exposure can either not be measured for hedge accounting purposes or no longer exists.

Fair values are all calculated solely on the basis of recognised financial and mathematical methods, using publicly available market information.

Changes in the market values of derivatives that do not qualify as effective hedging transactions under IFRS 9 can be seen in the income statement and in the overview of other financial items, [Note 14, p. 162](#).

Liquidity risk

Complex financial planning systems enable the Lufthansa Group to identify its future liquidity position at an early stage. Based on the results of the Group strategy and planning processes, a monthly rolling liquidity plan differentiated by currency is drawn up with a planning horizon of 24 months. This planning method offers an up-to-date picture of anticipated liquidity developments within the Company and corresponding currency effects.

The Lufthansa Group had no free credit lines as of 31 December 2020 (previous year: EUR 774m). Other funds are nonetheless available as of the reporting date from guaranteed government loans. For details see [Note 2, p. 140ff](#).

The Group has taken specific and general measures to safeguard and manage its liquidity to avoid any future potential liquidity restrictions that could result from the coronavirus pandemic. They include the implementation of monitoring on

the basis of detailed, rolling short-term cash plans in order to manage liquidity effectively and track the current financing requirement closely. Specific liquidity risks resulting from reimbursements for cancelled flights were also analysed and managed. In addition, liquidity management in connection with current orders for goods and services was also optimised, greater transparency was achieved across the Group, including an early warning system and an escalation process for outstanding receivables, additional signatures were required to place orders and monitoring concerning short-time working requirements was introduced.

A maturity analysis for the financial liabilities and the derivative financial instruments based on undiscounted gross cash flows including the relevant interest payments shows the following projected cash inflows and outflows from the perspective of the balance sheet date 31 December 2020. As a result of the hedges used, there are generally direct connections between the cash inflows and outflows for the derivative financial instruments shown.

T169 MATURITY ANALYSIS OF LIABILITIES FROM DERIVATIVE FINANCIAL INSTRUMENTS

in €m	From fuel derivatives	Cash inflow from gross settlement of interest rate and exchange rate derivatives	Cash outflow from gross settlement of interest rate and exchange rate derivatives	Net
1st quarter	12	-1,349	1,424	87
Up to 1 year ¹⁾	11	-3,275	3,508	244
1–5 years	1	-5,470	5,813	344
Later	-	-530	567	37

¹⁾ Without payments in 1st quarter.

**T170 MATURITY ANALYSIS OF LIABILITIES FROM
NON-DERIVATIVE FINANCIAL INSTRUMENTS**

in €m	Outflows
1st quarter	-5,404
Up to 1 year ¹⁾	-3,460
1 – 5 years	-5,468
Later	-5,648

¹⁾ Without payments in 1st quarter.

Credit risk

The sale of passenger travel and freight documents mostly takes place via agencies. These agencies are mostly connected to national clearing systems for billing passenger and freight sales. The credit rating of the agents is reviewed by the responsible clearing systems. Due to the broad diversification, credit risk for the agencies is relatively low worldwide. Nonetheless, credit terms for agents in some markets were tightened significantly in cooperation with the International Air Transport Association (IATA) in order to reduce credit risks even further.

Receivables and liabilities between airlines are offset through bilateral arrangements or via the IATA clearing house, insofar as the contracts underlying services do not explicitly specify otherwise. Systematic settlement of weekly receivables and liability balances significantly reduces the default risk. Fidelity guarantee insurance also covers partial risks within a certain range. Service contracts occasionally require collateral for miscellaneous transactions. All other contractual relationships are subject to credit rules, which, depending on the type and volume of the contract involved, require collateral, credit ratings/references or historical data from prior dealings, particularly payment history, in order to avoid defaults. Credit risks from the MRO business are monitored and managed with a separate credit risk management system. It comprises the calculation, authorisation and monitoring of customer-specific credit limits and the daily monitoring of payments received and receivables past due.

Counterparty risks in connection with credit card companies are monitored closely and incoming payments are reviewed daily. To reduce risks even further, a permanent analysis

process examines whether to further tighten credit terms for some settlement partners. In addition to the monitoring of receivables at the Company or segment level, there is also counterparty monitoring at Group level, with individually assigned limits, in order to identify the accumulation of portfolio risks across the entire Group and take appropriate action where necessary. The maximum credit risk for financial assets from the potential insolvency of customers is their carrying amount.

In addition to individual write-downs on receivables if a default event occurs, IFRS 9 requires risk provisions to be recognised for expected losses. The Lufthansa Group's trade receivables are exposed to external credit risks for which expected losses have already been taken into consideration in accordance with IFRS 9, in addition to individual write-downs. A simplified impairment model based on an impairment matrix is used for the portion of the receivables portfolio that does not consist of credit card receivables but is subject to external credit risks. The portfolio is divided into clusters based on customer groups, regions and days past due date. On the basis of historical default events in the Lufthansa Group's receivables portfolio, a default matrix is calculated and supplemented by forward-looking, publicly available insolvency forecasts. This impairment matrix is applied to trade receivables that are exposed to external credit risk and are not credit card receivables. An impairment matrix is also used for trade and other receivables in the MRO segment. It entails dividing the customer portfolio into four risk classes, with a low, medium, high and very high risk of default. Customers are assigned to each category using the MRO segment's own credit risk management system, which is based on fundamental data, market information and payment history. Probabilities of default are derived from historic default events and current market information. Available collateral is taken into account. In addition, the receivables portfolio includes credit card receivables for which the Lufthansa Group is the credit card issuer. Expected losses for these credit card receivables are calculated in a separate model, based on counterparty-specific external ratings and default probabilities. The Lufthansa Group uses a definition of default of 90 days past due for receivables, which are written off in full if the default event occurs. Exceptions are permitted in justified cases, however.

In the following tables, Levels 1 and 2 describe expected credit losses, whereas Level 3 shows individual impairment losses on the basis of actual default events.

T171 STATEMENT OF RISK PROVISIONS 2020

in €m	Opening balance risk provision as of 1 Jan 2020	Additions through profit or loss	Reversals through profit or loss	Utilisation	Closing balance risk provision as of 31 Dec 2020	Opening balance gross carrying amount as of 1 Jan 2020	Closing balance gross carrying amount as of 31 Dec 2020
Trade receivables and other receivables (simplified approach)	398	217	-23	-41	551	2,457	1,475
of which from expected losses	12	61	-10	0	63	2,066	926
of which from individual loss allowances	386	156	-13	-41	488	391	549
Trade receivables and other receivables (credit card receivables)	17	4	-4	-4	14	1,060	210
of which Level 1	14	0	-3	0	12	1,051	204
of which Level 2	0	0	0	0	0	0	1
of which Level 3	3	4	-1	-4	2	9	5
Total	415	221	-27	-45	565	3,517	1,685

T171 STATEMENT OF RISK PROVISIONS 2019

in €m	Opening balance risk provision as of 1 Jan 2019	Additions through profit or loss	Reversals through profit or loss	Utilisation	Closing balance risk provision as of 31 Dec 2019	Opening balance gross carrying amount as of 1 Jan 2019	Closing balance gross carrying amount as of 31 Dec 2019
Trade receivables and other receivables (simplified approach)	314	102	-10	-8	398	2,540	2,457
of which from expected losses	9	6	-3	0	12	2,234	2,066
of which from individual loss allowances	305	96	-7	-8	386	306	391
Trade receivables and other receivables (credit card receivables)	24	6	-5	-8	17	1,052	1,060
of which Level 1	17	1	-4	0	14	1,041	1,051
of which Level 2	0	0	0	0	0	0	0
of which Level 3	7	5	-1	-8	3	11	9
Total	338	108	-15	-16	415	3,592	3,517

An expected loss of EUR 4m was recognised for contract assets as of the reporting date.

The outbreak of the coronavirus pandemic caused a significant rise in credit risk and in expected and actual losses in the financial year. The customer portfolio in the MRO segment was particularly affected. Although the gross amount of

receivables declined significantly, individual loss allowances in the non-credit card business increased by EUR 100m, mainly due to airline insolvencies. Loss allowances for future credit losses under the simplified approach rose by EUR 52m, particularly due to the special risks in the customer portfolio of Lufthansa Technik AG, which were reflected by business-specific risk clustering.

In the reporting year, the Lufthansa Group used the following default rates for each past due category in the impairment matrix for the simplified approach of the impairment model.

T172 IMPAIRMENT MATRIX FOR TRADE RECEIVABLES 2020

		Not overdue	1 – 30 days overdue	31 – 60 days overdue	61 – 90 days overdue	More than 90 days overdue	Total
Default rate	%	3.0	6.0	6.7	10.0	22.0	–
Carrying amounts for trade receivables	€m	610	117	30	10	159	926
Expected loss	€m	18	7	2	1	35	63

T172 IMPAIRMENT MATRIX FOR TRADE RECEIVABLES 2019

		Not overdue	1 – 30 days overdue	31 – 60 days overdue	61 – 90 days overdue	More than 90 days overdue	Total
Default rate	%	0.3	0.8	1.3	1.6	1.8	–
Carrying amounts for trade receivables	€m	1,419	393	47	41	166	2,066
Expected loss	€m	4	3	1	1	3	12

In order to determine expected losses of credit card receivables according to IFRS 9, the off-balance sheet exposure must be considered, in addition to the on-balance sheet exposure. The off-balance sheet exposure describes the portion of a credit card's unused or free limit. The following overview shows the risk data for the credit card portfolio. Expected losses are calculated at the level of the individual credit card, so that the sizes of the average default probability and the average loss relate to the individual credit card.

T173 CONCENTRATION OF CREDIT RISK FROM CREDIT CARD RECEIVABLES

Internal credit rating	Probability of default according to external credit rating	Average probability of default	Average expected loss per concerned credit card in €	Exposure Level 1 impairment model in €m	Exposure Level 2 impairment model in €m	Exposure Level 3 impairment model in €m
On-balance sheet exposure						
Low risk	≤ 2%	1.4%	16	193	–	–
Medium risk	> 2.0% bis ≤ 6.5%	3.1%	16	10	–	–
High risk	> 6.5%	41.2%	1,516	1	1	5
Total				204	1	5
Off-balance sheet exposure						
Low risk	≤ 2%	1.3%	70	1,259	4	–
Medium risk	> 2.0% bis ≤ 6.5%	3.1%	193	135	–	–
High risk	> 6.5%	42.1%	4,863	143	–	–
Total				1,537	4	–

Securities representing debt instruments are rated as follows (Standard & Poor's):

T174 SECURITIES RATINGS – DEBT INSTRUMENTS

in €m

AAA	–
AA+	–
AA	–
AA–	19
A+	–
A	–
A–	4
BBB+	–
BBB	–
Below BBB or unrated	8
Total	31

The credit risk for derivative financial instruments and securities held at fair value through or without effect on profit and loss is the risk that a counterparty defaults. The maximum credit risk from these instruments is their carrying amount. The counterparty default risk for financial market transactions is limited by defining a maximum risk, taking the credit score given by recognised rating agencies into account.

46 Contingencies and events after the reporting period

T175 CONTINGENT LIABILITIES

in €m	31 Dec 2020	31 Dec 2019
From guarantees, bills of exchange and cheque guarantees ¹⁾	664	935
From warranty contracts	192	378
From providing collateral for third-party liabilities	16	47
	872	1,360

¹⁾ Previous year's figures have been adjusted.

Warranties include an amount of EUR 68m (previous year: EUR 248m) relating to contingent liabilities towards creditors of joint ventures. Liabilities under collateral agreements included contingent liabilities of EUR 4m (previous year: EUR 6m) towards creditors of joint ventures. Of the total, EUR 721m (previous year: EUR 1,035m) relates to joint and several guarantees and warranties taken on. These are matched by compensatory claims against the other co-debtors amounting to EUR 692m (previous year: EUR 974m). Insofar as annual financial statements have yet to be published, these figures are preliminary.

Otherwise, provisions for other risks could not be made because utilisation was not sufficiently probable. The potential financial effect of these provisions on the result would have been EUR 61m (previous year: EUR 55m).

Contracts signed for the sale of aircraft are expected to generate cash inflows of EUR 25m in subsequent years (previous year: EUR 0).

LEGAL RISKS

The Lufthansa Group is exposed to a number of legal risks in the course of its normal business. Based on current knowledge, the assumption is that these will not have any major, lasting effects on the net assets, financial and earnings position, beyond those for which provisions for litigation risks have been made, [Note 36, p. 188ff.](#)

Legal disputes and other claims made against the Group are always subject to uncertainty, however. Management estimates of these risks may also change over time. The actual outcome of these legal disputes may differ from earlier management estimates, which could have significant effects on the net assets, financial and earnings position and the reputation of our Company.

Due to the existing uncertainties and to those described below, we cannot make an assessment of the amount of the respective contingent liabilities or of the group of contingent liabilities. The legal disputes that these statements refer to include:

Risk of successful claims for damages in ongoing antitrust proceedings

Various cargo airlines, including Lufthansa Cargo AG and Swiss International Air Lines AG, were involved in a cargo cartel in the period between December 1999 and February 2006. Deutsche Lufthansa AG, Lufthansa Cargo AG and Swiss International Air Lines AG are at risk of civil claims for damages in Norway, Israel, Korea and the Netherlands. The lawsuits have been brought by both direct and indirect customers and are addressed to the airlines as co-debtors.

All litigation in Germany and the United Kingdom was ended by a settlement in the previous year.

At present, it is not possible to give a concrete assessment of the outcome of the lawsuits still pending or of the number and amount of any other claims. When evaluating the risk, it should nonetheless be borne in mind that the European Commission's decision on the cargo cartel, which the claimants in the civil lawsuits refer to, among others, is still not legally binding. Following the reversal of this 2010 decision by the European Court of Justice (ECJ) in December 2015, the European Commission sent revised penalty notices in March 2017 in which the content was the same but the reasoning had been altered. The airlines concerned, including Lufthansa, again contested them, so the penalty notices are still not effective.

Moreover, an expert economic opinion commissioned by Lufthansa Cargo AG and Swiss International Air Lines AG has come to the conclusion that the cartel did not inflict any actual damage on customers. Even if there were damages (i.e. allegedly higher cartel prices), the court would have to examine whether the claimants passed them on to their own customers (in the case of the freight forwarders) or whether they were passed on to the claimants (in the case of the final customers). Nonetheless, significant effects on the net assets, financial and earnings position of the Group cannot be ruled out if it should lose any of these legal proceedings.

Investigations in connection with work and service contracts

The investigations by the customs authorities in previous years into possible breaches of the German Law on Labour Leasing (Arbeitnehmerüberlassungsgesetz – AÜG) concerning the procurement of services by the Lufthansa Group have been concluded amicably. No penalties were imposed on individual employees of the Lufthansa Group nor on companies in accordance with Section 30 of the Administrative Offences Act (Gesetz über Ordnungswidrigkeiten – OWiG). Administrative proceedings with the German Statutory Pension Insurance Scheme concerning the legal status of two workers have been submitted for judicial review to the Higher Social Court in Berlin.

Legal action by Ryanair against the European Commission's decision on state aid

Ryanair DAC has appealed to the European Court of Justice against the decisions by the European Commission approving stabilisation measures for companies in the Lufthansa Group. Stabilisation measures of around EUR 7.6bn in total are affected for Deutsche Lufthansa AG, Austrian Airlines and Brussels Airlines SA/NV. The Lufthansa Group believes the chances of success are slim. If the appeals are successful, however, the decisions by the European Commission would be declared null and void. If the respective public-sector agencies are prevented from continuing to grant the stabilisation measures, the funds concerned would have to be

cancelled and repaid. The companies will probably make use of the possibility to join the proceedings as interveners alongside the respective member states on the side of the European Commission.

TAX RISKS

Tax risks exist largely because of differences in legal opinions between the German tax authorities and the Company. In tax audits for the financial years 2003 to 2015, the tax authorities came to a number of different conclusions to those on which the Company had based its tax returns, relating, in particular, to partial write-downs on shareholder loans, the treatment of various lease structures and the acquisition of a foreign subsidiary, as well as the recognition and measurement of certain provisions and assets. The Lufthansa Group has appealed against the resulting tax assessments. Without abandoning its legal position, almost all the disputed matters were settled in the past by paying the back taxes demanded by the authorities. The Federal Finance Court adopted a different position on partial write-downs in 2019, which led to a more negative assessment of the current proceedings. However, no final decision on this matter has yet been taken. One previously disputed matter concerning the recognition of a provision for long-term MRO contracts has since been decided in the Company's favour. To the extent that success in the remaining points is considered to be more likely than not, the corresponding receivables against the tax authorities have been recognised in accordance with IFRIC 23. The corresponding provisions were recognised for other potentially disputed matters to the extent that their use was probable. No provisions were recognised for matters which in the Company's opinion are more likely than not to be decided in its favour. They could give rise to back payments of around EUR 200m in total. The assessment of the amount is subject to uncertainty.

EVENTS AFTER THE REPORTING PERIOD

Eurowings and ver.di reach crisis agreement

On 12 January 2021, Eurowings and the trade union ver.di reached a crisis agreement for cabin crew. Under the agreement, employees are protected against dismissal until 31 December 2021. In exchange, reductions in Christmas and holiday bonuses were agreed, among other things.

Lufthansa issues another bond and repays KfW loan

Deutsche Lufthansa AG again successfully issued a bond with a total volume of EUR 1.6bn on 4 February 2021. The first tranche has a volume of EUR 750m and pays interest of 2.875% p.a. over a term of four years. The second tranche of EUR 850m matures in seven years and pays interest at 3.75% p.a.

Together with the funds of approximately EUR 2bn raised in the second half of 2020, the Lufthansa Group has thereby secured the refinancing of the financial liability of EUR 2.6bn due in 2021 and repaid the KfW loan of EUR 1bn ahead of schedule on 11 February 2021.

EU Parliament adopts new slot rules

On 10 February 2021, the European Parliament passed a resolution amending the slot rules for summer 2021 in response to the pandemic. The regulation stipulates that airlines can return up to 50% of their slot series in full before the season begins and that they only have to operate half the remaining slot series in order to secure them for the coming season.

Based on its capacity outlook (🔗 [Forecast, p. 118ff.](#)), the Lufthansa Group expects that it will be able to obtain all of the slots for which it has rights, including beyond the 2021 summer season.

47 Other financial obligations

As of 31 December 2020, there were order commitments of EUR 13.0bn for capital expenditure on property, plant and equipment, including repairable spare parts, and for intangible assets (previous year: EUR 14.6bn). There were also capital and shareholder loan commitments of EUR 228m towards equity investments (previous year: EUR 335m).

48 Auditors' fees

The fees paid to the auditors in the financial year and charged to expenses in accordance with Section 314 Paragraph 1 No. 9 HGB are made up as follows:

T176 AUDITORS' FEES

in €m	2020	2019
Annual audit	3.2	4.4
Other assurance services	0.2	1.0
Tax advisory services	0.1	0.9
Other services	1.6	1.9
Total	5.1	8.2

In the financial year 2020, the audit was carried out for the first time by Ernst & Young GmbH Wirtschaftsprüfungsgesellschaft. In 2019, the fees were paid to PricewaterhouseCoopers GmbH Wirtschaftsprüfungsgesellschaft.

The auditing services mainly consist of fees for auditing the consolidated financial statements, the review of the half-yearly financial statements and the audit of the financial statements of Deutsche Lufthansa AG and its consolidated subsidiaries. Tax advisory services mainly relate to tax advice on tax declarations, transfer pricing, international taxes and inspections by tax authorities. Other services particularly relate to advisory services in connection with the sale of the LSG group's European activities.

The following fees from entities in the worldwide EY network, particularly abroad, were also recognised as expenses (the figures for the previous year relate to the worldwide PwC network).

T177 ADDITIONAL AUDITORS' FEES

in €m	2020	2019
Annual audit	2.3	3.8
Other assurance services	-	0.1
Tax advisory services	0.5	0.3
Other services	0.1	0.2
Total	2.9	4.4

The auditor at Ernst & Young GmbH Wirtschaftsprüfungsgesellschaft responsible for the Lufthansa Group is Siegfried Keller. He held this position for the first time in financial year 2020.

COMPOSITION OF THE GROUP

49 Group of consolidated companies

The consolidated financial statements of Deutsche Lufthansa AG include all major subsidiaries, joint ventures and associated companies.

Subsidiaries are entities over which Deutsche Lufthansa AG has rights that give it the ability to control the entity's relevant activities. Relevant activities are those activities that have a significant influence on the entity's return. Deutsche Lufthansa AG therefore only has control over a company when it is exposed to variable returns from the company and its power over the company's relevant activities enables it to influence these returns. This definition of control also applies to structured entities that are identified as such in the list of significant Group companies. In general, the ability to control subsidiaries arises when Deutsche Lufthansa AG holds a direct or indirect majority of voting shares. In structured entities, the ability to control does not come from holding the majority of voting shares, but rather from contractual agreements. Entities are consolidated from the time that the ability to control begins. They cease to be consolidated when the ability to control ends.

Joint arrangements are classified either as joint ventures or as joint operations. A joint arrangement exists for when the Lufthansa Group carries on joint business activities with third parties on the basis of a contractual agreement. Joint management or control only exists when decisions on activities that have a significant effect on the returns from an agreement require the unanimous approval of the parties sharing control.

Significant interests in companies that are managed jointly with one or more partners (joint ventures) are accounted for using the equity method. Joint operations are defined by the fact that the parties exercising joint control over the arrangement have rights to the assets attributed to the arrangement and are liable for its debts. Assets and liabilities, revenue and expenses from the significant joint operations are recognised in the consolidated financial statements of the Lufthansa Group in proportion to these rights and obligations.

Associated companies are companies in which Deutsche Lufthansa AG has the opportunity to exercise major influence over financial and operating policy based on an interest of between 20% and 50%. Significant associated companies are accounted for in the consolidated financial statements using the equity method.

A list of major subsidiaries, joint arrangements and associated companies can be found in [T184 – T187](#), p. 229 – 235, and the list of shareholdings in [T188](#), p. 236 – 239.

In addition to Deutsche Lufthansa AG as the parent company, the group of consolidated companies includes 50 domestic and 242 foreign companies, including structured entities (previous year: 66 domestic and 267 foreign companies).

One material joint operation was also included in the consolidated financial statements on a pro rata basis in accordance with IFRS 11. It consists of a German cargo airline operated jointly by Deutsche Post AG and Deutsche Lufthansa AG, which each hold 50% of the share capital and voting rights. The two shareholders are also customers of the company and use the capacities of its cargo aircraft. In contrast to its capital and voting rights, the company's assets and liabilities, as well as its income and expenses, are allocated based on the user relationship of the shareholders according to their contracts.

Changes in the group of consolidated companies during the 2020 financial year are shown in the following table. One material change was the sale of the European catering activities to gategroup. These entities accounted for revenue

of EUR 366m in 2020 until their disposal on 2 December 2020. Some 8,000 employees left the Group when the disposal was completed.

T178 CHANGES IN THE GROUP OF CONSOLIDATED COMPANIES FROM 1 JAN 2020 TO 31 DEC 2020

Name, registered office	Additions	Disposals	Reasons
Network Airlines business segment			
Austrian Asset Holding GP S. à r.l., Luxembourg, Luxembourg	19 Jun 2020		Acquisition
Austrian Asset Holding S.C.S., Luxembourg, Luxembourg	25 Jun 2020		Established
Raft Co., Ltd., Tokyo, Japan	18 Dec 2020		Established
Swiss Global Air Lines AG, Basel, Switzerland		11 May 2020	Liquidation
Soir Leasing Co. Ltd., Tokyo, Japan		25 Jun 2020	Liquidation
FI Beauty Leasing Ltd., Tokyo, Japan		29 Jun 2020	Liquidation
Jour Leasing Co. Ltd., Tokyo, Japan		31 Aug 2020	Liquidation
Eurowings business segment			
LeaseAir GmbH & Co. Verkehrsflugzeuge V KG		30 Jun 2020	Fusion
Catering business segment			
MIM IFE Limited, Dublin, Ireland		2 Dec 2019	Fusion
LSG Sky Chefs Danmark A/S, Dragør, Denmark		14 Sep 2020	Liquidation
Evertaste GmbH, Alzey		2 Dec 2020	Disposal
Evertaste Limited, Hounslow, UK		2 Dec 2020	Disposal
Evertaste Oy, Vantaa, Finland		2 Dec 2020	Disposal
Evertaste S.r.l., Fiumicino, Italy		2 Dec 2020	Disposal
LSG France SAS, Paris, France		2 Dec 2020	Disposal
LSG Helvetia SAS, Paris, France		2 Dec 2020	Disposal
LSG Linearis S.A.S., Paris, France		2 Dec 2020	Disposal
LSG Lufthansa Service Catering- und Dienstleistungsgesellschaft mbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs - First Catering Schweiz AG, Bassersdorf, Switzerland		2 Dec 2020	Disposal
LSG Sky Chefs Belgium N.V., Zaventem, Belgium		2 Dec 2020	Disposal
LSG Sky Chefs Berlin GmbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs Czechia spol. s.r.o. i. L., Bor, Czech Republic		2 Dec 2020	Disposal
LSG Sky Chefs Düsseldorf GmbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs Europe GmbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs Frankfurt International GmbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs Frankfurt ZD GmbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs Köln GmbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs Lounge GmbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs Lounge, Inc., Wilmington, USA		2 Dec 2020	Disposal
LSG Sky Chefs München GmbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs Objekt- und Verwaltungsgesellschaft mbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Sky Chefs S.p.A., Fiumicino, Italy		2 Dec 2020	Disposal
LSG Sky Chefs Schweiz AG, Bassersdorf, Switzerland		2 Dec 2020	Disposal
LSG Sky Chefs Spain, S.A., Madrid, Spain		2 Dec 2020	Disposal
LSG Sky Chefs Verwaltungsgesellschaft mbH, Neu-Isenburg		2 Dec 2020	Disposal
LSG Transalpino SAS, Paris, France		2 Dec 2020	Disposal
LSG-Food & Nonfood Handel GmbH, Neu-Isenburg		2 Dec 2020	Disposal
RISTO RAIL PORTUGAL, LDA, Lisbon, Portugal		2 Dec 2020	Disposal
SkylogistiX GmbH, Neu-Isenburg		2 Dec 2020	Disposal
Spiriant Asia Pacific Limited, Hong Kong, China		2 Dec 2020	Disposal
Spiriant Bahrain Limited W.L.L., Manama, Bahrain		2 Dec 2020	Disposal
Spiriant GmbH, Neu-Isenburg		2 Dec 2020	Disposal
SPIRIANT North America, Inc., Wilmington, USA		2 Dec 2020	Disposal
Supply Chain S.à.r.l., Contern, Luxembourg		2 Dec 2020	Disposal
Additional Businesses and Group Functions			
AirPlus Air Travel Card Vertriebsgesellschaft mbH i.L., Vienna, Austria		8 Jan 2020	Liquidation
AirPlus Holding GmbH, Vienna, Austria		4 Feb 2020	Liquidation
Airplus International NV/SA, Brussels, Belgium		26 Nov 2020	Fusion

USE OF EXEMPTION PROVISIONS

The following fully consolidated German Group companies made use of the exemption provisions in Section 264 Paragraph 3 and Section 264b HGB in 2020.

T179 USE OF EXEMPTION PROVISIONS

Company name	Registered office
Eurowings Aviation GmbH	Cologne
Eurowings Digital GmbH	Cologne
Eurowings GmbH	Düsseldorf
Eurowings Technik GmbH	Cologne
Germanwings GmbH	Cologne
Jettainer GmbH	Raunheim
LSG Asia GmbH	Neu-Isenburg
LSG Lufthansa Service Europa/Afrika GmbH	Neu-Isenburg
LSG Lufthansa Service Holding AG	Neu-Isenburg
LSG South America GmbH	Neu-Isenburg
LSY GmbH	Norderstedt
Lufthansa AirPlus Servicekarten GmbH	Neu-Isenburg
Lufthansa Asset Management GmbH	Frankfurt/Main
Lufthansa Asset Management Leasing GmbH	Frankfurt/Main
Lufthansa Aviation Training Berlin GmbH	Berlin
Lufthansa Aviation Training Germany GmbH	Frankfurt/Main
Lufthansa Aviation Training GmbH	Hallbergmoos
Lufthansa Cargo AG	Frankfurt/Main
Lufthansa CityLine GmbH	Munich Airport
Lufthansa Commercial Holding GmbH	Frankfurt/Main
Lufthansa Global Business Services GmbH	Frankfurt/Main
Lufthansa Industry Solutions AS GmbH	Norderstedt
Lufthansa Industry Solutions BS GmbH	Raunheim
Lufthansa Industry Solutions GmbH & Co. KG	Norderstedt
Lufthansa Process Management GmbH	Neu-Isenburg
Lufthansa Seeheim GmbH	Seeheim-Jugenheim
Lufthansa Systems GmbH & Co. KG	Raunheim
Lufthansa Technik AERO Alzey GmbH	Alzey
Lufthansa Technik AG	Hamburg
Lufthansa Technik Immobilien- und Verwaltungsgesellschaft mbH	Hamburg
Lufthansa Technik Logistik GmbH	Hamburg
Lufthansa Technik Logistik Services GmbH	Hamburg
Lufthansa Technik Maintenance International GmbH	Frankfurt/Main
Miles & More GmbH	Frankfurt/Main
Retail in Motion GmbH	Neu-Isenburg
time:matters GmbH	Neu-Isenburg
time:matters Holding GmbH	Neu-Isenburg
time:matters Spare Parts Logistics GmbH	Neu-Isenburg

The companies are affiliated with Deutsche Lufthansa AG by means of direct or indirect profit and loss transfer agreements.

Furthermore, LHBD Holding Limited, London, UK, registration number 06939137, is exempt from the obligation to have its individual financial statements audited pursuant to Section 479a of the 2006 Companies Act, United Kingdom.

The consolidated financial statements include equity investments in 29 joint ventures and 35 associates (previous year: 35 joint ventures and 39 associates), of which ten joint ventures (previous year: eleven) and 14 associates (previous year: 15) were accounted for using the equity method. The other joint ventures and associated companies were valued at amortised cost due to their minor overall significance.

50 Related party disclosures

According to the notification on voting rights from 6 July 2020, the Federal Republic of Germany has held an indirect equity interest of 20.05% in Deutsche Lufthansa AG via the Economic Stabilisation Fund (WSF) since 2 July 2020. The WSF can exercise significant influence and represents a related party of Deutsche Lufthansa AG. The stabilisation measures in 2020 are significant transactions with the WSF.

➤ **Note 2, p. 140ff.**

Balances and transactions between the Company and its fully consolidated subsidiaries, which constitute related parties, have been eliminated in the course of consolidation and are not commented on in this Note. Details of transactions between the Lufthansa Group and other related parties are disclosed below.

The Lufthansa Group segments render numerous services to related parties within the scope of their ordinary business activities. Conversely, the companies in question provide services to the Lufthansa Group as part of their normal business. These extensive supply and service relationships take place on the basis of market prices.

In addition, the Lufthansa Group and certain non-consolidated subsidiaries have concluded numerous billing agreements, partly governing the joint use of services. In these cases, the administrative services provided are charged as cost allocations.

The Lufthansa Group's cash management is centralised, and, in this respect, the Lufthansa Group also performs a "banking function" vis-à-vis the non-consolidated companies of the Group. Non-consolidated Group companies included in the Group's cash management invest their available cash with the Group or borrow funds from the Group and carry out their derivative hedging transactions with the Group. All transactions take place at market conditions.

Due to geographical proximity in many cases, a large number of subletting contracts exists between the Lufthansa Group and related parties. In these cases, the Lufthansa Group usually charges the rental costs and incidental expenses incurred to the companies in question on a pro rata basis.

The following table shows the volume of significant services provided to or by related parties:

T180 VOLUME OF SIGNIFICANT SERVICES PROVIDED TO OR BY RELATED PARTIES

in €m	Volume of services rendered		Volume of services received	
	2020	2019	2020	2019
Non-consolidated subsidiaries				
Albatros Versicherungsdienste GmbH, Germany	0	0	40	50
Austrian Airlines Technik-Bratislava, s.r.o., Slovakia	1	2	15	10
Austrian Airlines Tele Sales Service GmbH, Austria	0	0	3	5
Cargo Future Communications (CFC) GmbH, Germany	0	0	5	4
Delvag Versicherungs-AG, Germany	5	6	4	3
DLH Fuel Company mbH, Germany	0	0	131	508
Global Load Control (PTY) LTD, South Africa	0	0	4	8
handling counts GmbH, Germany	0	0	9	9
LGSP Lufthansa Ground Service Portugal, Unipessoal Lda., Portugal	0	1	5	6
Lufthansa Aviation Training Austria GmbH, Austria	2	1	7	6
Lufthansa Aviation Training Operations Germany GmbH, Germany	2	3	13	16
Lufthansa Aviation Training Pilot Academy GmbH, Germany	1	1	6	3
Lufthansa Aviation Training USA Inc., USA	1	1	11	14
Lufthansa Consulting GmbH, Germany	1	2	3	9
Lufthansa Engineering and Operational Services GmbH, Germany	3	4	26	32
Lufthansa Global Business Services Hamburg GmbH, Germany	6	7	26	29
Lufthansa Global Business Services Sp. z o. o., Poland	2	2	30	33
Lufthansa Global Tele Sales GmbH, Germany	3	7	57	60
Lufthansa Group Business Services New York LLC, USA	0	0	5	0
Lufthansa Group Security Operations GmbH, Germany	1	0	6	0
Lufthansa Industry Solutions SHPK, Albania	0	0	6	0
Lufthansa Industry Solutions TS GmbH, Germany	2	1	13	13
Lufthansa Services (Thailand) Ltd., Thailand	0	0	5	7
Lufthansa Services Philippines, Inc., Philippines	1	0	4	5
Lufthansa Systems FlightNav AG, Switzerland	0	1	22	25
Lufthansa Systems Hungaria Kft, Hungary	0	1	24	25
Lufthansa Systems Poland Sp. z o.o., Poland	1	2	33	32
Lufthansa Technical Training GmbH, Germany	5	6	15	19
Lufthansa Technik Brussels N.V., Belgium	1	5	3	2
Lufthansa Technik Component Services Asia Pacific Limited, China	2	1	6	6
Lufthansa Technik Middle East FZE, United Arab Emirates	3	1	10	8
Lufthansa Technik Milan s.r.l., Italy	1	6	2	3
Lufthansa Technik Miskolc Kft., Hungary	2	0	7	0
Lufthansa Technik Services India Private Limited, India	2	1	4	5
Lufthansa Technik Shenzhen Co. Ltd., China	15	22	26	34
Lufthansa Technik Turbine Shannon Limited, Ireland	3	3	9	19
LZ-Catering GmbH, Germany	3	5	11	12
Reservation Data Maintenance India Private Ltd., India	0	0	2	5
time:matters (Shanghai) International Freight Forwarding Ltd., China	17	0	12	0
time:matters Austria GmbH, Austria	25	1	0	0
time:matters Courier Terminals GmbH, Germany	1	1	9	8
ZeroG GmbH, Germany	1	1	3	7

T180 VOLUME OF SIGNIFICANT SERVICES PROVIDED TO OR BY RELATED PARTIES (continued)

in €m	Volume of services rendered		Volume of services received	
	2020	2019	2020	2019
Joint ventures				
Airfoil Services Sdn. Bhd., Malaysia	0	1	2	7
EME Aero Sp. z o.o., Poland	0	0	18	0
LG-LHT Aircraft Solutions GmbH, Germany	3	6	0	0
N3 Engine Overhaul Services GmbH & Co. KG, Germany	1	5	0	0
Shanghai Pudong International Airport Cargo Terminal Co. Ltd., China	1	1	6	6
Spairliners GmbH, Germany	30	59	36	46
Star Alliance Services GmbH, Germany	1	2	5	8
Terminal 2 Gesellschaft mbH & Co oHG, Germany	0	1	11	13
Terminal One Group Association, L.P., USA	0	7	7	7
XEOS Sp. z o.o., Poland	10	4	16	7
Associated companies				
Aircraft Maintenance and Engineering Corp., China	4	10	6	3
Airmail Center Frankfurt GmbH, Germany	0	1	9	9
AviationPower GmbH, Germany	0	0	7	29
HEICO Aerospace Holdings Corp., USA	0	0	5	13
Other affiliated companies				
Shanghai Pudong International Airport Public Cargo Terminal Co. Ltd. (West), Shanghai, China	0	0	17	97
SunExpress Deutschland GmbH, Germany	7	26	65	93

The following tables show receivables owed by and liabilities to related parties:

T181 RECEIVABLES FROM AFFILIATED COMPANIES

in €m	2020	2019
Trade receivables from non-consolidated subsidiaries	39	30
Trade receivables from joint ventures	11	22
Trade receivables from associated companies	5	5
Trade receivables from other affiliated companies	5	1
Total trade receivables	60	58
Other receivables from non-consolidated subsidiaries	19	30
Other receivables from joint ventures	19	49
Other receivables from associated companies	11	10
Other receivables from other affiliated companies	5	-
Total other receivables	54	89
Loans to non-consolidated subsidiaries	99	99
Loans to joint ventures	5	5
Loans to associated companies	-	-
Total non-current receivables	104	104

T182 LIABILITIES TO AFFILIATED COMPANIES

in €m	2020	2019
Trade payables to non-consolidated subsidiaries	37	29
Trade payables to joint ventures	15	8
Trade payables to associated companies	2	3
Trade payables to other affiliated companies	1	2
Total trade payables	55	42
Other liabilities to non-consolidated subsidiaries	282	260
Other liabilities to joint ventures	26	0
Other liabilities to associated companies	0	0
Other liabilities to other affiliated companies	8	0
Total other liabilities	316	260

No individual shareholders of Deutsche Lufthansa AG exercise significant influence over the Group. For transactions involving members of the Executive Board and Supervisory Board ("directors' dealings"), [Note 51, p. 228](#).

51 Supervisory Board and Executive Board

The disclosure of remuneration for key managers required by IAS 24 includes the remuneration of the active members of the Executive Board and Supervisory Board.

The members of the Executive Board and the Supervisory Board as well as the other offices that they hold are named in the combined management report in the section [Corporate governance, p. 122ff.](#)

The principles of the remuneration system and the amount of remuneration paid to Executive Board and Supervisory Board members are shown and explained in detail in the remuneration report. The remuneration report forms part of the Annual Report 2020.

Total Executive Board remuneration under IFRS was EUR 9.7m (previous year: EUR 14.8m) including current service costs for pensions of EUR 3.2m (previous year: EUR 3.3m).

The active members of the Executive Board in past reporting years were remunerated as follows:

T183 EXECUTIVE BOARD REMUNERATION (IFRS)

in € thousands	2020	2019
Basic salary	5,226	5,934
Other ¹⁾	58	110
One-year variable remuneration	1,058	2,091
Total short-term remuneration	6,342	8,135
Long-term variable remuneration ²⁾	-1,339	1,299
Share-based remuneration	-1,976	-1,945
Current service cost for retirement benefits	3,179	3,291
Total long-term remuneration	-136	2,645
Severance payments	3,500	4,055
Total	9,706	14,835

¹⁾ Other remuneration includes in particular benefits from the use of company cars, discounts in connection with cash outflows from share programmes ([Note 39, p. 192ff.](#)) and concessionary travel in accordance with the relevant IATA regulations.

²⁾ Expenses recognised in the reporting year for long-term variable remuneration for the financial years 2018 to 2020.

Pension provisions for Executive Board members active in the 2020 financial year came to EUR 17.9m (previous year: EUR 16.7m).

In addition to the provisions for the one-year variable remuneration of EUR 1,058k (previous year: EUR 2,091k), the value of the provisions recognised for the future payment of long-term variable remuneration for the Executive Board members active in the financial year 2020 was reduced by a total of EUR -1,339k (previous year: EUR 1,299k additionally recognised). In addition, provisions of EUR 519k were recognised for the future payment of long-term, share-based remuneration for the Executive Board members active as of 31 December 2020 (previous year: EUR 3,706k).

Total remuneration (HGB) paid to the Executive Board of Deutsche Lufthansa AG in the financial year 2020 came to EUR 12,078k (previous year: EUR 13,967k). The above remuneration includes the share-based remuneration component Total Shareholder Return (TSR) of EUR 4,305k as part of the long-term variable remuneration incentive.

Current payments and other benefits for former members of the Executive Board and their surviving dependants came to EUR 6.9m (previous year: EUR 6.4m). This includes payments by subsidiaries as well as benefits in kind and concessionary travel.

Pension obligations towards former Executive Board members and their surviving dependants amount to EUR 71.6m (previous year: EUR 69.1m). They are included in pension provisions ([Note 35, p. 180f.](#)).

Expenses for the fixed remuneration of Supervisory Board members came to EUR 1,887k in 2020 (previous year: EUR 2,170k). Other remuneration, mainly attendance fees, amounted to EUR 23k (previous year: EUR 62k). The Deutsche Lufthansa AG Supervisory Board members were also paid EUR 16k for work on supervisory boards of Group companies (previous year: EUR 2k).

In the reporting year, as in the previous year, no loans or advance payments were made to members of the Executive Board and to members of the Supervisory Board.

In addition to their Supervisory Board remuneration, employee representatives on the Supervisory Board received compensation for their work in the form of wages and salaries including pension entitlements amounting to EUR 1.1m in total in 2020 (previous year: EUR 1.1m).

The following Auditor's Report is a translation of the German language independent auditor's report and refers to the audit of the German language consolidated financial statements and the German language group management report, which is neither included or incorporated by reference, and also includes a "Report on the audit of the electronic reproductions of the financial statements and the management report prepared for the purpose of disclosure in accordance with Section 317 Paragraph 3b HGB" ("ESEF Report"). The subject matter on which the ESEF Report is based (ESEF documents to be audited) is not included. The audited ESEF documents can be viewed in or retrieved from the German Federal Gazette (in German language only).

Independent auditor's report

To Deutsche Lufthansa Aktiengesellschaft

REPORT ON THE AUDIT OF THE CONSOLIDATED FINANCIAL STATEMENTS AND OF THE GROUP MANAGEMENT REPORT

Opinions

We have audited the consolidated financial statements of Deutsche Lufthansa Aktiengesellschaft, Cologne, and its subsidiaries (the "Group" or "Lufthansa Group"), which comprise the consolidated income statement and consolidated statement of comprehensive income for the fiscal year from 1 January to 31 December 2020, and the consolidated statement of financial position as at 31 December 2020, consolidated statement of changes in shareholders' equity and consolidated cash flow statement for the fiscal year from 1 January to 31 December 2020, and notes to the consolidated financial statements, including a summary of significant accounting policies. In addition, we have audited the Group management report of Deutsche Lufthansa Aktiengesellschaft, which is combined with the management report of Deutsche Lufthansa Aktiengesellschaft, for the fiscal year from 1 January to 31 December 2020. In accordance with the German legal requirements, we have not audited the content of the Group corporate governance declaration pursuant to Sec. 315d HGB which is published on the website stated in the "Corporate Governance" section of the combined management report and the Group non-financial declaration pursuant to Sec. 315b HGB included in the "Combined non-financial declaration" section. In addition, we have not audited the content of the other information extending beyond the prior year in the tables with multi-year comparisons of the combined management report (information pertaining to fiscal years 2015, 2016, 2017 and 2018). Other information in the combined management report relates to any information whose disclosure in the management report is not required pursuant to Secs. 315, 315a or Secs. 315b to 315d HGB or GAS 20.

In our opinion, on the basis of the knowledge obtained in the audit,

- the accompanying consolidated financial statements comply, in all material respects, with the IFRSs as adopted by the EU, and the additional requirements of German commercial law pursuant to Sec. 315e (1) HGB and, in compliance with these requirements, give a true and fair view of the assets, liabilities and financial position of the

Group as at 31 December 2020 and of its financial performance for the fiscal year from 1 January to 31 December 2020, and

- the accompanying Group management report as a whole provides an appropriate view of the Group's position. In all material respects, this Group management report is consistent with the consolidated financial statements, complies with German legal requirements and appropriately presents the opportunities and risks of future development. Our opinion on the Group management report does not cover the content of the aforementioned Group corporate governance declaration, the content of the aforementioned Group non-financial declaration or the aforementioned multi-year comparisons included in the Group management report.

Pursuant to Sec. 322 (3) Sentence 1 HGB ("Handelsgesetzbuch": German Commercial Code), we declare that our audit has not led to any reservations relating to the legal compliance of the consolidated financial statements and of the Group management report.

Basis for the opinions

We conducted our audit of the consolidated financial statements and of the Group management report in accordance with Sec. 317 HGB and the EU Audit Regulation (No 537/2014, referred to subsequently as "EU Audit Regulation") and in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the Institut der Wirtschaftsprüfer (Institute of Public Auditors in Germany) (IDW). We performed the audit of the consolidated financial statements in supplementary compliance with the International Standards on Auditing (ISAs). Our responsibilities under those requirements, principles and standards are further described in the "Auditor's responsibilities for the audit of the consolidated financial statements and of the Group management report" section of our auditor's report. We are independent of the Group entities in accordance with the requirements of European law and German commercial and professional law, and we have fulfilled our other German professional responsibilities in accordance with these requirements. In addition, in accordance with Art. 10 (2) f) of the EU Audit Regulation, we declare that we have not provided non-audit services prohibited under Art. 5 (1) of the EU Audit Regulation. We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our opinions on the consolidated financial statements and on the Group management report.

Key audit matters in the audit of the consolidated financial statements

Key audit matters are those matters that, in our professional judgment, were of most significance in our audit of the consolidated financial statements for the fiscal year from 1 January to 31 December 2020. These matters were addressed in the context of our audit of the consolidated financial statements as a whole, and in forming our opinion thereon; we do not provide a separate opinion on these matters.

Below, we describe what we consider to be the key audit matters:

1. Implications of the COVID-19 pandemic for the going concern basis

Reasons why the matter was determined to be a key audit matter

The COVID-19 pandemic has had a particularly strong impact on the airline industry. Since the outbreak of the pandemic in spring 2020, national and international travel restrictions have led to a significant reduction in air travel. The restrictions have had a major impact on the operations and liquidity of the Lufthansa Group.

To safeguard the solvency of the Lufthansa Group, stabilization measures and loans of up to EUR 9b were agreed with the Economic Stabilization Fund in the Federal Republic of Germany as well as with the governments of Switzerland, Austria and Belgium. Lufthansa has also made use of other support measures provided by other governments around the world, such as the US.

The Lufthansa Group has significantly reduced available flight capacity and also initiated measures to reduce costs and capital expenditure. The Executive Board has prepared the Group operational planning for the fiscal years 2021 to 2024, together with a liquidity forecast for the same period, and thereby mapped out the possible effects from national and international travel restrictions in connection with the COVID-19 pandemic on the available flight capacity, taking into account the countermeasures. The result of the planning depends to a large extent on the assumptions that the executive directors have made regarding future business developments and the cash inflows and outflows derived from them. The planning for 2021 and subsequent years is based on the assumption that a vaccine will be widely available or an established testing strategy will be in place. Significant cost savings were also assumed based on initiated restructuring measures.

The review of the implications of the COVID-19 pandemic for the going concern basis of accounting was a key audit matter due to the wide range of political, medical and global economic effects and due to the measures taken by the executive directors in this connection to mitigate risk and their consideration in the business planning.

Auditor's response

We discussed what influence the COVID-19 pandemic, with its national and international travel restrictions, could have on Deutsche Lufthansa Aktiengesellschaft and its subsidiaries with the executive directors, both direct and indirect and short and medium term. We discussed what different assumptions and scenarios were used by the executive directors in their assessment.

We analyzed, with the support of internal specialists from Strategy and Transactions, the group operational planning prepared by the executive directors, including the liquidity forecast, and discussed this with the responsible management level. We assessed the plausibility (transparency, consistency, lack of contradiction) of management's disclosures on the key planning assumptions, strategic objectives, expected developments and operational measures and their inclusion/reflection in the cash flows.

We also checked the clerical accuracy, completeness and consistency of the planning model, the conceptual design and in particular the derivation of the cash flows and their individual components. In this connection, we also reviewed the implementation of the measures adopted under the restructuring programs to reduce the outflow of liquidity on the basis of resolutions, internal and external publications and observed facts for the implementation of capacity adjustments and their impact on the fleet as well as on personnel. We also checked the capital expenditures reflected in the financial planning on the basis of purchase commitments and framework agreements.

In order to assess the cash inflows included in the liquidity planning we, among other things, reviewed the agreements on state stabilization measures and other financing schemes, checked incoming payments and assessed the terms and conditions. The defined overarching objectives of the financing strategy and the suitability of the measures underlying them were also assessed. In addition, we examined the key planning assumptions taking into account historical developments and our understanding of the business and performed plausibility tests on the basis of various external information. Our assessment was based on analyst estimates, both for the Lufthansa Group as well as in relation to comparable companies, along with other external forecasts on the development of the airline industry (market studies) and macroeconomic forecasts.

In addition, the updated assessments of the executive directors concerning the availability of a COVID-19 vaccine, a testing strategy, the current travel restrictions as well as the current booking behavior and the consideration of these factors in the group planning were discussed with the executive directors and management of the Company and examined on the basis of the Group's internal assessments.

We also discussed other potential measures to secure liquidity which are not reflected in the liquidity planning due to a lack of necessity with the management and assessed their risk coverage potential.

In addition, we checked whether the consolidated financial statements and Group management report contain disclosures on the implications of the COVID-19 pandemic for the going concern basis and in particular the measures taken by the executive directors to mitigate risk.

Our audit procedures did not lead to any reservations relating to the going concern basis/the application of the going concern basis of accounting.

Reference to related disclosures

With regard to the assessment of the executive directors on the Company's ability to continue as a going concern and the related recognition and measurement policies applied, we refer to the disclosure in the notes to the consolidated financial statements in section "2 Going concern," and with regard to the related information on judgment by the Executive Board and sources of estimation uncertainty, we refer to the disclosure in the notes to the consolidated financial statements "3 New international accounting standards in accordance with IFRS and interpretations and summary of the significant accounting policies and valuation methods."

Please also refer to the "Opportunities and risk report" as well as the "Forecast" in the Group management report and there, in particular, to the section "Overall statement by the Executive Board on the expected development of the Lufthansa Group."

2. Recoverability of own and leased aircraft

Reasons why the matter was determined to be a key audit matter

The aircraft reported in the consolidated financial statements of Deutsche Lufthansa Aktiengesellschaft represent a significant portion of the assets of Deutsche Lufthansa Aktiengesellschaft and its subsidiaries. The aircraft reported includes aircraft which is legally owned and used by Deutsche Lufthansa Aktiengesellschaft and its subsidiaries as well as leased aircraft.

In response to the impact of the COVID-19 pandemic, the Executive Board resolved to permanently reduce the fleet. The executive directors were thus tasked with determining the fair value, taking into consideration the future use of the aircraft (for example via sale or abandonment) and thus also assessing the need to recognize an impairment charge.

From our perspective, the recoverability of aircraft was a key audit matter in our audit as the measurement and resulting impairment of these items significant in amount are based to a large extent on the estimates and assumptions of the executive directors, particularly in light of the impact of the COVID-19 pandemic.

Auditor's response

With regard to the assessment of the recoverability of the aircraft in the economic ownership of the Lufthansa Group and earmarked for continued use in flight operations, we examined the fleet planning in connection with the Group operational planning for internal consistency and analyzed whether these are in line with industry forecasts. In this connection, we checked the impairment test prepared by the executive directors and the assumptions underlying this test in relation to the various cash-generating units to which these aircraft are allocated for internal and external consistency (e.g., comparison with internal planning documents and industry forecasts) and checked whether impairment is required.

With regard to the assessment of the decisions made by the executive directors on the continued use of aircraft in the economic ownership of the Lufthansa Group and which is not due to be used further in flight operations, we assessed whether the measures included in the resolutions made by the executive directors on the abandonment or deep storage were reflected in full in the financial statements. In addition, in relation to the aircraft earmarked for abandonment or deep storage, we assessed the measurement assumptions with regard to for the underlying sale or scrap value used by the executive directors for the purpose of determining the impairment of the aircraft, taking into account the available information. In this connection, we assessed whether the assumptions made by the executive directors were transparently derived from observable market data on prices, taking into account a market price overview published regularly by Aircraft Value Analysis Company Ltd., comparable purchase agreements from the past or from purchase agreements already concluded.

Our audit procedures did not lead to any reservations relating to the recoverability of own and leased aircraft.

Reference to related disclosures

With regard to the accounting for aircraft, we refer to the disclosure in the notes to the consolidated financial statements in sections “19 Aircraft and reserve engines including right-of-use assets,” “22 Leases” and “10 Depreciation, amortisation and impairment.”

3. Accounting for state aid/stabilization measures

Reasons why the matter was determined to be a key audit matter

To safeguard the ability of the Lufthansa Group to continue as a going concern, stabilization measures and state guaranteed loans of up to EUR 9b were agreed with the Economic Stabilization Fund in the Federal Republic of Germany (WSF) as well as with the governments of Switzerland, Austria and Belgium. The state aid and stabilization measures include various instruments, of both an equity and debt nature. In addition to syndicated loans and subordinated loans as well as silent participations, the stabilization measures include a 20% stake of WSF in Deutsche Lufthansa Aktiengesellschaft (EUR 306m), a further silent participation classified as equity (EUR 4.7b, Silent Participation I) and a loan from COFAG (COVID-19-Finanzierungsagentur des (österreichischen) Bundes GmbH) (EUR 150m). The loan is recognized as a grant in profit or loss after certain requirements are met and is offset against Silent Participation I, which is thus reduced to EUR 4.5b.

The audit of the accounting for state aid/stabilization measures was deemed a key audit matter as there is an elevated risk of misstatement due to the number of individual measures and the wide range of complex contractual conditions. Due to the complexity of accounting regulations and their interpretation, this relates in particular to the classification of Silent Participation I in the WSF framework agreement as equity, the accounting for contractual components as separate transactions and the recognition of the aid in Austria in profit or loss in the correct period.

Auditor's response

In the course of our audit procedures, with the support of our internal specialists from the Financial Instruments Expert Group, among other things, we analyzed the contracts underlying the respective measures; in particular we examined the WSF framework agreement to determine if the accounting applied by the Executive Board for the individual agreements contained therein as separate transactions meets the criteria of IAS 32, Financial Instruments. In addition, on the basis of the equity criteria defined in IAS 32 and using the documents on the Group planning and liquidity forecast along with discussions with the responsible management, we examined whether the arrangements made in relation to the separate transactions meet the requirements for qualification as equity.

In addition to the participation of WSF in Deutsche Lufthansa Aktiengesellschaft, we reviewed the notarized minutes (Sec. 130 AktG (“Aktiengesetz”: German Stock Corporation Act)) certifying the resolution of the Extraordinary General Meeting of Deutsche Lufthansa Aktiengesellschaft on the approval of the stabilization measures, including the capital increase and reviewed the entry of this capital increase in the commercial register as well as checked the related incoming payment.

In addition, we checked whether the loan of EUR 150m from COFAG was recognized in fiscal year 2020 in profit or loss as a government grant in accordance with IAS 20. The waiver of repayment of the loan is subject to the condition that Austrian Airlines AG suffered economic damages of at least EUR 150m in fiscal year 2020 due to COVID-19. In this connection, we obtained an understanding of the method used for calculating the damages determined by the management of Austrian Airlines AG for the period from 9 March 2020 to 14 June 2020 and assessed the recognition of COVID-19 crisis-induced expenses on the basis of comparative values (budget/actual and prior year).

We reconciled all incoming payments from the stabilization measures agreed with the WSF and the governments of Switzerland, Austria and Belgium with the bank statements. We also checked the calculation of the fixed and variable interest expenses and their recognition in profit or loss on the basis of the underlying contracts.

In addition, we assessed the disclosures required by IAS 20 in the notes to the consolidated financial statements on government grants.

Our audit procedures did not lead to any reservations relating to the accounting for state aid/stabilization measures.

Reference to related disclosures

The disclosures on the accounting for state aid/stabilization measures are mainly included in the sections “2 Going concern,” “7 Other operating income,” “19 Aircraft and reserve engines including right-of-use assets,” “33 Issued capital” and “37 Financial liabilities” of the notes to the consolidated financial statements.

Furthermore, we refer to the “Economic report” in the Group management report and there to the information on “Financing” in the “Earnings, assets and financial position” section, subsection “Financial position.”

4. Accounting for short-time working allowances as well as corresponding social security contributions and the reimbursement thereof

Reasons why the matter was determined to be a key audit matter

The COVID-19 pandemic had a significant impact on the global operations of the Lufthansa Group in fiscal year in 2020. The executive directors of various Group companies of the Lufthansa Group introduced short-time working arrangements as one of the measures to limit the financial fallout from the ongoing crisis. In this connection, the executive directors applied to the relevant competent state authorities for a reimbursement of the short-time pay within the legally prescribed scope.

Approval of the reimbursements of short-time pay requested monthly is, depending on the specific state, handled partly by way of formal benefit notices or official decrees before payment is made, partly also by way of direct payment of the reimbursements by the competent state authority if its review of the payroll reports or reimbursement lists does not lead to any reservations. However, as this generally only relates to approvals subject to a subsequent review to assess compliance of the submitted reimbursement claims with the respective legal requirements by the same or other state authority after the end of the period of short-time work, the reimbursement amounts received represent an uncertainty until this time for the relevant group companies of the Lufthansa Group.

In order to reflect this special matter in payroll accounting, the underlying IT systems were adjusted and special measures for monitoring were implemented in the accounting-related internal control system. These modifications of the underlying personnel accounting systems as well as the accounting-related internal control system ("ICS") of the Group companies represent an elevated risk of material misstatement of the reimbursement claims in the consolidated financial statements, as mistakes in payroll accounting as well as weaknesses in the monitoring activities of the ICS could lead to incorrectly calculated amounts of reimbursement claims, which could give rise to negative financial effects for the Group.

With regard to the reimbursable amounts from the state authorities, a legally fundamental distinction must be made between claims asserted by the Group companies of the Lufthansa Group and claims asserted by individual employees against the former. The claims asserted by the companies against the competent state authorities must be, as they are subject to the fulfillment of certain legally defined conditions, classified as income-related government grants and deducted from the related expense. Reimbursements from the competent state authorities to the group companies of the Lufthansa Group resulting from claims of individual employees against the former, such as in the case in Germany for short-time pay, merely constitute a transitory item for the employer which

is recognized outside of profit or loss in the consolidated financial statements of the Lufthansa Group as receivables against the corresponding state authorities. The different accounting for the two types of reimbursement claims underscores in particular the risk of an incorrect valuation due to differing assessment bases. As of the reporting date, receivables from the competent state authorities are estimated based on the amounts already paid or transferred but not yet reimbursed by the competent state authorities, taking into account historical data of the past months.

In light of these differences in accounting and due to the quantitative significance of reimbursements of short-time working allowances and social security contributions for the consolidated financial statements of the Lufthansa Group as well as the complexity of the legal requirements and the process-related uncertainty, the accounting for short-time working allowances as well as the corresponding social security contributions constitute a key audit matter in our audit.

Auditor's response

In the course of our audit procedures, we assessed, partly with the support of our internal specialists from People Advisory Services, the accounting for short-time working allowances and the corresponding social security contributions as well as the reimbursement thereof by the competent state authorities in terms of merit and amount. In the course of this, we evaluated on the basis of evidence in samples whether there was a legal basis for the introduction of short-time working arrangements. This included the advance declarations or applications for the first grant, some of which are regularly repeated, to the competent state authorities by the executive directors of the group companies which contain a notification of the reduction in working hours or the introduction of short-time work in the operation, as well as the documentation demonstrating compliance with the legal conditions for the introduction of short-time work and, consequently, the receipt of short-time working allowances. We also obtained the approval notices or official orders confirming compliance with the relevant legal conditions for the introduction of short-time working and approval for the receipt of short-time working allowances and subsequent reimbursements thereof and the related social security contributions based on merit.

Our audit procedures included the assessment of the accounting for the reimbursements of the short-time working allowances by the competent state authorities as well as the corresponding – in some cases payable solely by the employer – social security contributions by the competent state authorities in accordance with IAS 20 as income-related government grants. Taking account of the approved individual short-time working arrangements and the arrangements differentiated by operating area, by carrying out sample-based checks we obtained an understanding of the process for calculating and accounting for the reimbursement amounts particularly in order to assess the results of the modifications to the underlying payroll accounting system related to the

short-time (work scheme) working arrangements. In this connection, we compared the absence times due to short-time working arrangements displayed in the time logging systems against the individual payslips of the sample-based check and against the reimbursement lists or payroll reports completed by the responsible employees of the companies and which are the basis of the monthly reimbursement claims. Furthermore, by performing sample-based checks we verified the clerical accuracy and methods used for calculating and determining the short-time working allowances stated in these reimbursement claims as well as the corresponding social security contributions. Furthermore, we gained an understanding during our process walkthrough of the measures of the accounting-related internal control system and assessed these in the course of a test of design.

We also tested the plausibility of the amount of receivables recognized as of the reporting date taking into account historical data from retroactively calculated discrepancies between the monthly reimbursement claims and benefit notices or official decrees issued by the competent state authorities as well as the actual reimbursement amounts.

Furthermore, we performed analytical audit procedures regarding the amount of short-time working allowances as well as the corresponding social security amounts. In doing so, we tested the plausibility of correlations, fluctuations and trends on the basis of our expectations based on surveys and the claim applications and notices presented above. This included, among other things, the analysis of the development of the reimbursed short-time working allowances and the corresponding social security contributions as well as the reimbursement amounts as a percentage of total staff costs and the total social security expenses taking into account the number of employees.

Our audit procedures did not lead to any reservations with regard to the accounting for short-time working allowances and the corresponding social security contributions or their reimbursement.

Reference to related disclosures

We refer to the disclosure in the notes to the consolidated financial statements under “3 New international accounting standards in accordance with IFRS and interpretations and summary of the significant accounting policies and valuation methods” and “9 Staff costs” for information on the accounting for short-time working allowances and government grants in the form of a reimbursement of social security contributions by the competent state authorities in the context of the reimbursement of short-time working allowances.

5. Accounting for derivative financial instruments in hedging relationships

Reasons why the matter was determined to be a key audit matter

Deutsche Lufthansa Aktiengesellschaft uses a variety of derivative financial instruments to hedge against currency, fuel price and interest rate risks arising from its ordinary business activities. The Lufthansa Group experienced significant restrictions in terms of flight volumes in fiscal year 2020 as a result of the COVID-19 pandemic, resulting in lower cash flows in foreign currency as well as a significant decline in purchases of kerosene. As a result, forecast transactions and thus hedged items in hedging relationships were not carried out. In our view, these matters constituted one of the key audit matters due to the significant volume in terms of amount, the high complexity and number of hedging transactions as well as the extensive accounting requirements.

Auditor's response

In the course of our audit, we analyzed, with the support of our internal specialists from the Financial Instruments Experts Group, among other things, the contractual and economic basis of the hedging relationship between the forecast transactions as hedged items and the derivative financial instruments as hedging instruments as well as the accounting, including earnings effects. Our focus was on testing the controls that address the existence and completeness of the derivative financial instruments and of the designated hedges in the IT systems used for the accounting for the hedging relationships. We also obtained bank confirmations as audit evidence.

We obtained an understanding of the valuation of the hedged items and hedging transactions (including option premiums) by the executive directors in particular in the areas of fuel and foreign currency by reperforming valuations in our own valuation systems. For interest rate derivatives, we performed audit procedures in relation to the software used for valuation purposes. We reviewed the access restrictions, the approval process for software changes as well as the archiving of software changes and subsequently reperformed the valuation of significant interest rate derivatives on a sample basis.

We assessed, together with our specialists, compliance with the requirements for the accounting for the hedging relationships using the designation documents prepared by the executive directors and the internal risk management guidelines. On the basis of business plans and inquiries of management, we checked whether only such forecast transactions constitute a hedged item of a hedging relationship that

are considered highly probable on the basis of the current business plan. We traced the accounting for hedging relationships and their (potentially premature) terminations in terms of value and reconciled them with the respective items in the statement of financial position and income statement. With regard to fuel price hedges, we also checked the clerical accuracy of the calculation of the recognition of the option premiums that are a component of the designated derivatives.

Our audit procedures did not lead to any reservations regarding the accounting for derivative financial instruments in hedging relationships.

Reference to related disclosures

The disclosures on the recognition and measurement policies applied for derivative financial instruments in hedging relationships are included in the sections “3 New international accounting standards in accordance with IFRS and interpretations and summary of the significant accounting policies and valuation methods,” “14 Other financial items” and “45 Additional disclosures on financial instruments” in the notes to the consolidated financial statements.

Other information

The Supervisory Board is responsible for the report of the Supervisory Board in the “Report of the Supervisory Board” section. The executive directors and the Supervisory Board are responsible for the declaration pursuant to Sec. 161 AktG on the German Corporate Governance Code, which is part of the Group corporate governance declaration. In all other respects, the executive directors are responsible for the other information. The other information comprises the aforementioned Group corporate governance declaration, the aforementioned Group non-financial declaration and the aforementioned other information included in the Group management report. The other information also comprises parts to be included in the annual report, of which we received a version prior to issuing this auditor’s report, in particular:

- The letter from the Executive Board to the shareholders in the “Letter from the Executive Board” section of the annual report
- The explanations on the Deutsche Lufthansa Aktiengesellschaft share in the “Lufthansa share” section of the annual report
- The declaration by the executive directors in “General remarks,” section “1 Company information” of the notes to the consolidated financial statements
- The remuneration report in the “Remuneration report” section of the annual report

- The Ten-year overview in the “Ten-year overview” section of the annual report.

but not the consolidated financial statements, not the management report disclosures whose content is audited and not our auditor’s report thereon.

Our opinions on the consolidated financial statements and on the Group management report do not cover the other information, and consequently we do not express an opinion or any other form of assurance conclusion thereon.

In connection with our audit, our responsibility is to read the other information and, in so doing, to consider whether the other information

- is materially inconsistent with the consolidated financial statements, with the Group management report or our knowledge obtained in the audit, or
- otherwise appears to be materially misstated.

If, based on the work we have performed, we conclude that there is a material misstatement of this other information, we are required to report that fact. We have nothing to report in this regard.

Responsibilities of the executive directors and the Supervisory Board for the consolidated financial statements and the Group management report

The executive directors are responsible for the preparation of the consolidated financial statements that comply, in all material respects, with IFRSs as adopted by the EU and the additional requirements of German commercial law pursuant to Sec. 315e (1) HGB, and that the consolidated financial statements, in compliance with these requirements, give a true and fair view of the assets, liabilities, financial position and financial performance of the Group. In addition, the executive directors are responsible for such internal control as they have determined necessary to enable the preparation of consolidated financial statements that are free from material misstatement, whether due to fraud or error.

In preparing the consolidated financial statements, the executive directors are responsible for assessing the Group’s ability to continue as a going concern. They also have the responsibility for disclosing, as applicable, matters related to going concern. In addition, they are responsible for financial reporting based on the going concern basis of accounting unless there is an intention to liquidate the Group or to cease operations, or there is no realistic alternative but to do so.

Furthermore, the executive directors are responsible for the preparation of the Group management report that, as a whole, provides an appropriate view of the Group's position and is, in all material respects, consistent with the consolidated financial statements, complies with German legal requirements, and appropriately presents the opportunities and risks of future development. In addition, the executive directors are responsible for such arrangements and measures (systems) as they have considered necessary to enable the preparation of a Group management report that is in accordance with the applicable German legal requirements, and to be able to provide sufficient appropriate evidence for the assertions in the Group management report.

The Supervisory Board is responsible for overseeing the Group's financial reporting process for the preparation of the consolidated financial statements and of the Group management report.

Auditor's responsibilities for the audit of the consolidated financial statements and of the Group management report

Our objectives are to obtain reasonable assurance about whether the consolidated financial statements as a whole are free from material misstatement, whether due to fraud or error, and whether the Group management report as a whole provides an appropriate view of the Group's position and, in all material respects, is consistent with the consolidated financial statements and the knowledge obtained in the audit, complies with the German legal requirements and appropriately presents the opportunities and risks of future development, as well as to issue an auditor's report that includes our opinions on the consolidated financial statements and on the Group management report.

Reasonable assurance is a high level of assurance, but is not a guarantee that an audit conducted in accordance with Sec. 317 HGB and the EU Audit Regulation and in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the IDW and supplementary compliance with the ISAs will always detect a material misstatement. Misstatements can arise from fraud or error and are considered material if, individually or in the aggregate, they could reasonably be expected to influence the economic decisions of users taken on the basis of these consolidated financial statements and this Group management report.

We exercise professional judgment and maintain professional skepticism throughout the audit. We also:

- Identify and assess the risks of material misstatement of the consolidated financial statements and of the Group management report, whether due to fraud or error, design and perform audit procedures responsive to those risks,

and obtain audit evidence that is sufficient and appropriate to provide a basis for our opinions. The risk of not detecting a material misstatement resulting from fraud is higher than for one resulting from error, as fraud may involve collusion, forgery, intentional omissions, misrepresentations, or the override of internal control.

- Obtain an understanding of internal control relevant to the audit of the consolidated financial statements and of arrangements and measures (systems) relevant to the audit of the Group management report in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of these systems.
- Evaluate the appropriateness of accounting policies used by the executive directors and the reasonableness of estimates made by the executive directors and related disclosures.
- Conclude on the appropriateness of the executive directors' use of the going concern basis of accounting and, based on the audit evidence obtained, whether a material uncertainty exists related to events or conditions that may cast significant doubt on the Group's ability to continue as a going concern. If we conclude that a material uncertainty exists, we are required to draw attention in the auditor's report to the related disclosures in the consolidated financial statements and in the Group management report or, if such disclosures are inadequate, to modify our respective opinions. Our conclusions are based on the audit evidence obtained up to the date of our auditor's report. However, future events or conditions may cause the Group to cease to be able to continue as a going concern.
- Evaluate the overall presentation, structure and content of the consolidated financial statements, including the disclosures, and whether the consolidated financial statements present the underlying transactions and events in a manner that the consolidated financial statements give a true and fair view of the assets, liabilities, financial position and financial performance of the Group in compliance with IFRSs as adopted by the EU and the additional requirements of German commercial law pursuant to Sec. 315e (1) HGB.
- Obtain sufficient appropriate audit evidence regarding the financial information of the entities or business activities within the Group to express opinions on the consolidated financial statements and on the Group management report. We are responsible for the direction, supervision and performance of the group audit. We remain solely responsible for our audit opinions.

- Evaluate the consistency of the Group management report with the consolidated financial statements, its conformity with (German) law, and the view of the Group's position it provides.
- Perform audit procedures on the prospective information presented by the executive directors in the Group management report. On the basis of sufficient appropriate audit evidence we evaluate, in particular, the significant assumptions used by the executive directors as a basis for the prospective information, and evaluate the proper derivation of the prospective information from these assumptions. We do not express a separate opinion on the prospective information and on the assumptions used as a basis. There is a substantial unavoidable risk that future events will differ materially from the prospective information.

We communicate with those charged with governance regarding, among other matters, the planned scope and timing of the audit and significant audit findings, including any significant deficiencies in internal control that we identify during our audit.

We also provide those charged with governance with a statement that we have complied with the relevant independence requirements, and communicate with them all relationships and other matters that may reasonably be thought to bear on our independence and where applicable, the related safeguards.

From the matters communicated with those charged with governance, we determine those matters that were of most significance in the audit of the consolidated financial statements of the current period and are therefore the key audit matters. We describe these matters in our auditor's report unless law or regulation precludes public disclosure about the matter.

OTHER LEGAL AND REGULATORY REQUIREMENTS

Report on the assurance in accordance with Sec. 317 (3b) HGB on the electronic reproduction of the consolidated financial statements and the Group management report prepared for publication purposes

Opinion

We have performed assurance work in accordance with Sec. 317 (3b) HGB to obtain reasonable assurance about whether the reproduction of the consolidated financial statements and the Group management report (hereinafter the "ESEF documents") contained in the attached electronic file „Deutsche_Lufthansa_AG_KA_LB_ESEF-2020-12-31.zip“ and prepared for publication purposes complies in all material respects with the requirements of Sec. 328 (1) HGB for the electronic reporting format ("ESEF format"). In accordance with German legal requirements, this assurance only extends to the conversion of the information contained in the consol-

idated financial statements and the Group management report into the ESEF format and therefore relates neither to the information contained in this reproduction nor to any other information contained in the abovementioned electronic file.

In our opinion, the reproduction of the consolidated financial statements and the Group management report contained in the abovementioned attached electronic file and prepared for publication purposes complies in all material respects with the requirements of Sec. 328 (1) HGB for the electronic reporting format. We do not express any opinion on the information contained in this reproduction nor on any other information contained in the abovementioned file beyond this reasonable assurance opinion and our audit opinion on the accompanying consolidated financial statements and the accompanying Group management report for the fiscal year from 1 January to 31 December 2020 contained in the "Report on the audit of the consolidated financial statements and of the Group management report" above.

Basis for the opinion

We conducted our assurance work on the reproduction of the consolidated financial statements and the Group management report contained in the abovementioned attached electronic file in accordance with Sec. 317 (3b) HGB and Exposure Draft of IDW Assurance Standard: Assurance in Accordance with Sec. 317 (3b) HGB on the Electronic Reproduction of Financial Statements and Management Reports Prepared for Publication Purposes (ED IDW AsS 410) and the International Standard on Assurance Engagements 3000 (Revised). Our responsibilities under that standard are further described in the "Group auditor's responsibilities for the assurance work on the ESEF documents" section. Our audit firm applied the requirements for quality control systems set forth in IDW Standard on Quality Control: "Requirements for Quality Control in Audit Firms" (IDW QS 1).

Responsibilities of the executive directors and the Supervisory Board for the ESEF documents

The executive directors of the Company are responsible for the preparation of the ESEF documents including the electronic reproduction of the consolidated financial statements and the Group management report in accordance with Sec. 328 (1) Sentence 4 No. 1 HGB and for the tagging of the consolidated financial statements in accordance with Sec. 328 (1) Sentence 4 No. 2 HGB.

In addition, the executive directors of the Company are responsible for such internal control as they have considered necessary to enable the preparation of ESEF documents that are free from material non-compliance with the requirements of Sec. 328 Abs. 1 HGB for the electronic reporting format, whether due to fraud or error.

The executive directors of the Company are also responsible for the submission of the ESEF documents together with the auditor's report and the attached audited consolidated financial statements and the audited Group management report as well as other documents to be published to the operator of the Bundesanzeiger (German Federal Gazette).

The Supervisory Board is responsible for overseeing the preparation of the ESEF documents as part of the financial reporting process.

Group auditor's responsibilities for the assurance work on the ESEF documents

Our objective is to obtain reasonable assurance about whether the ESEF documents are free from material non-compliance with the requirements of Sec. 328 (1) HGB, whether due to fraud or error. We exercise professional judgment and maintain professional skepticism throughout the engagement. We also:

- Identify and assess the risks of material non-compliance with the requirements of Sec. 328 (1) HGB, whether due to fraud or error, design and perform audit procedures responsive to those risks, and obtain assurance evidence that is sufficient and appropriate to provide a basis for our assurance opinion.
- Obtain an understanding of internal control relevant to the assurance work on the ESEF documents in order to design assurance procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of these controls.
- Evaluate the technical validity of the ESEF documents, i.e., whether the electronic file containing the ESEF documents meets the requirements of Delegated Regulation (EU) 2019/815, in the version valid as of the reporting date, on the technical specification for this electronic file.
- Evaluate whether the ESEF documents enable an XHTML reproduction with content equivalent to the audited consolidated financial statements and to the audited Group management report.
- Evaluate whether the tagging of the ESEF documents with Inline XBRL technology (iXBRL) enables an appropriate and complete machine-readable XBRL copy of the XHTML reproduction.
- Review of the interim financial statements of Deutsche Lufthansa AG as of 30 June 2020
- Submission of a comfort letter for Deutsche Lufthansa AG in connection with the Euro Medium Term Note (EMTN) program
- Engagement to obtain limited assurance on the non-financial declaration pursuant to Secs. 289b et seq. and 315b et seq. HGB
- Engagement to obtain limited assurance on the report of Deutsche Lufthansa AG on compliance with the conditions and requirements of the parent company and the affected Group companies pursuant to Art. 8 to 17 of the framework agreement for the granting of stabilization measures
- Audit-related services legally prescribed by Sec. 162 (3) AktG in relation to the remuneration report prepared in accordance with the provisions of the AktG
- Voluntary audits of financial statements as of 31 December 2020
- Performance of agreed-upon procedures for Lufthansa Cargo AG, Frankfurt am Main, in relation to the company's registered office
- Project-related audit of a migration of IT-based accounting-related systems pursuant to IDW AuS 850 for Lufthansa AirPlus Servicekarten GmbH, Neu-Isenburg
- Permissible consulting in connection with a report on the expenditure of funds for Lufthansa Technik AG, Hamburg
- Audit of Albatros Service Center GmbH, Cologne, in accordance with Sec. 24 FinVermV ("Finanzanlagenvermittlungsverordnung": German Ordinance on Financial Investment Mediation)
- Agreed-upon procedures relating to a compliance issue of a foreign subsidiary of Deutsche Lufthansa Aktiengesellschaft

Further information pursuant to Art. 10 of the EU Audit Regulation

We were elected as group auditor by the Annual General Meeting on 5 May 2020. We were engaged by the Supervisory Board on 8 May 2020. We have been the group auditor of Deutsche Lufthansa Aktiengesellschaft since fiscal year 2020.

We declare that the opinions expressed in this auditor's report are consistent with the additional report to the audit committee pursuant to Art. 11 of the EU Audit Regulation (long-form audit report).

In addition to the financial statement audit, we have provided or were engaged to provide to Group entities the following services that are not disclosed in the consolidated financial statements or in the Group management report:

GERMAN PUBLIC AUDITOR RESPONSIBLE FOR THE ENGAGEMENT

The German Public Auditor responsible for the engagement is Siegfried Keller.

Eschborn/Frankfurt am Main, 2 March 2021

Ernst & Young GmbH
Wirtschaftsprüfungsgesellschaft

Prof. Dr. Sven Hayn
Wirtschaftsprüfer
(German Public Auditor)

Siegfried Keller
Wirtschaftsprüfer
(German Public Auditor)

***Audited Consolidated Financial Statements of Deutsche Lufthansa Aktiengesellschaft as of
and for the financial year ended December 31, 2019 (IFRS)***

Consolidated income statement for the financial year 2019

T069 CONSOLIDATED INCOME STATEMENT

in €m	Notes	2019	2018 ¹⁾
Traffic revenue	3	28,136	27,801
Other revenue	4	8,288	7,741
Total revenue		36,424	35,542
Changes in inventories and work performed by entity and capitalised	5	685	531
Other operating income ²⁾	6	1,889	1,818
Cost of materials and services	7	- 19,827	- 18,367
Staff costs	8	- 9,121	- 8,811
Depreciation, amortisation and impairment ³⁾	9	- 2,776	- 2,205
Other operating expenses ⁴⁾	10	- 5,585	- 5,708
Profit/loss from operating activities		1,689	2,800
Result of equity investments accounted for using the equity method	11	88	114
Result of other equity investments	11	80	60
Interest income	12	79	68
Interest expenses	12	- 394	- 212
Other financial items	13	318	- 46
Financial result		171	- 16
Profit/loss before income taxes		1,860	2,784
Income taxes	14	- 615	- 588
Profit/loss after income taxes		1,245	2,196
Profit/loss attributable to non-controlling interests		- 32	- 33
Net profit/loss attributable to shareholders of Deutsche Lufthansa AG		1,213	2,163
Basic/diluted earnings per share in €	15	2.55	4.58

¹⁾ Total revenue and cost of materials and services previous year's figures have been adjusted.

²⁾ This includes EUR 46m (previous year: EUR 61m) from the reversal of write-downs on non-current receivables.

³⁾ This includes EUR 23m (previous year: EUR 13m) for the recognition of write-downs on current receivables.

⁴⁾ This includes EUR 120m (previous year: EUR 100m) for the recognition of loss allowances on receivables.

Statement of comprehensive income for the financial year 2019

T070 STATEMENT OF COMPREHENSIVE INCOME

in €m

	2019	2018
Profit/loss after income taxes	1,245	2,196
Other comprehensive income		
Other comprehensive income with subsequent reclassification to the income statement		
Differences from currency translation	115	124
Subsequent measurement of financial assets at fair value without effect on profit and loss	10	- 10
Subsequent measurement of hedges – cash flow hedge reserve	258	434
Subsequent measurement of hedges – costs of hedging	476	- 90
Other comprehensive income from investments accounted for using the equity method	13	- 2
Other expenses and income recognised directly in equity	15	1
Income taxes on items in other comprehensive income	- 182	- 87
	705	370
Other comprehensive income without subsequent reclassification to the income statement		
Revaluation of defined-benefit pension plans	- 1,108	- 974
Other comprehensive income from investments accounted for using the equity method	-	-
Subsequent measurement of financial assets at fair value	7	5
Other expenses and income recognised directly in equity	- 1	-
Income taxes on items in other comprehensive income	345	117
	- 757	- 852
Other comprehensive income after income taxes	- 52	- 482
Total comprehensive income	1,193	1,714
Comprehensive income attributable to non-controlling interests	- 31	- 34
Comprehensive income attributable to shareholders of Deutsche Lufthansa AG	1,162	1,680

Further details on the statement of comprehensive income can be found in [Note 32, Reserves, p. 177f.](#)

Consolidated statement of financial position as of 31 December 2019

T071 CONSOLIDATED STATEMENT OF FINANCIAL POSITION – ASSETS

in €m	Notes	31 Dec 2019	31 Dec 2018
Intangible assets with an indefinite useful life ¹⁾	16	1,395	1,381
Other intangible assets	17	547	512
Aircraft and reserve engines	18 20	18,349	16,776
Repairable spare parts for aircraft		2,270	2,133
Property, plant and other equipment	19 20	4,041	2,221
Investments accounted for using the equity method	21	672	650
Other equity investments	22 43	256	246
Non-current securities	22 43	53	41
Loans and receivables	23 43	469	512
Derivative financial instruments	43	906	828
Deferred charges and prepaid expenses	27	116	118
Effective income tax receivables		32	10
Deferred tax assets	14	2,268	2,131
Non-current assets		31,374	27,559
Inventories	24	980	968
Contract assets	25	277	234
Trade receivables and other receivables	26 43	5,417	5,576
Derivative financial instruments	43	459	357
Deferred charges and prepaid expenses	27	245	217
Effective income tax receivables		153	58
Securities	28 43	1,970	1,735
Cash and cash equivalents	29 43	1,415	1,500
Assets held for sale	30	369	9
Current assets		11,285	10,654
Total assets		42,659	38,213

¹⁾ Including Goodwill.

T072 CONSOLIDATED STATEMENT OF FINANCIAL POSITION – SHAREHOLDERS' EQUITY AND LIABILITIES

in €m	Notes	31 Dec 2019	31 Dec 2018 ¹⁾
Issued capital	31	1,224	1,217
Capital reserve	32	378	343
Retained earnings	32	5,617	4,555
Other neutral reserves	32	1,715	1,185
Net profit/loss		1,213	2,163
Equity attributable to shareholders of Deutsche Lufthansa AG		10,147	9,463
Non-controlling interests		109	110
Shareholders' equity		10,256	9,573
Pension provisions	33	6,659	5,865
Other provisions	34	490	537
Borrowings	35 43	8,396	5,008
Contract liabilities	36	25	22
Other financial liabilities		76	137
Advance payments received, deferred income and other non-financial liabilities	37	32	51
Derivative financial instruments	43	128	222
Deferred tax liabilities	14	611	583
Non-current provisions and liabilities		16,417	12,425
Other provisions	34	794	894
Borrowings	35 43	1,634	1,677
Trade payables and other financial liabilities	39 43	5,351	5,720
Liabilities from unused flight documents	38	4,071	3,969
Other contract liabilities	38	2,675	2,391
Advance payments received, deferred income and other non-financial liabilities	40	382	388
Derivative financial instruments	43	137	393
Effective income tax obligations		402	783
Liabilities in connection with assets held for sale	30	540	-
Current provisions and liabilities		15,986	16,215
Total shareholders' equity and liabilities		42,659	38,213

¹⁾ Other provisions, trade payables and other contract liabilities previous year's figures have been adjusted.

Consolidated statement of changes in shareholders' equity as of 31 December 2019

T073 CONSOLIDATED STATEMENT OF CHANGES IN SHAREHOLDERS' EQUITY

	Issued capital	Capital reserve	Fair value measurement of financial instruments	Currency differences	Revaluation reserve (due to business combinations)	Other neutral reserves	Total other neutral reserves	Retained earnings	Net profit/loss	Equity attributable to shareholders of Deutsche Lufthansa AG	Minority interests	Total shareholders' equity
in €m												
As of 31 Dec 2017	1,206	263	605	264	236	326	1,431	3,449	2,340	8,689	103	8,792
Capital increases/reductions	11	80	-	-	-	-	-	-	-	91	1	92
Reclassifications	-	-	-	-	-	-	-	1,963	-1,963	-	-	-
Dividends	-	-	-	-	-	-	-	-	-377	-377	-28	-405
Transactions with non-controlling interests	-	-	-	-	-	-	-	-	-	-	-	-
Income after taxes	-	-	-	-	-	-	-	-	2,163	2,163	33	2,196
Other expenses and income recognised directly in equity	-	-	252	124	-	-2	374	-857	-	-483	1	-482
Hedging results reclassified from non-financial assets to acquisition costs	-	-	-620	-	-	-	-620	-	-	-620	-	-620
As of 31 Dec 2018	1,217	343	237	388	236	324	1,185	4,555	2,163	9,463	110	9,573
Restatement IFRIC 23	-	-	-	-	-	-	-	33	-	33	-	33
Adjusted as of 31 Dec 2018	1,217	343	237	388	236	324	1,185	4,588	2,163	9,496	110	9,606
Capital increases/reductions	7	35	-	-	-	-	-	-	-	42	1	43
Reclassifications	-	-	-9	-	-	-	-9	1,792	-1,783	-	-	-
Dividends	-	-	-	-	-	-	-	-	-380	-380	-33	-413
Transactions with non-controlling interests	-	-	-	-	-	-	-	-	-	-	-	-
Income after taxes	-	-	-	-	-	-	-	-	1,213	1,213	32	1,245
Other expenses and income recognised directly in equity	-	-	569	115	-	28	712	-763	-	-51	-1	-52
Hedging results reclassified from non-financial assets to acquisition costs	-	-	-173	-	-	-	-173	-	-	-173	-	-173
As of 31 Dec 2019	1,224	378	624	503	236	352	1,715	5,617	1,213	10,147	109	10,256

Consolidated cash flow statement for the financial year 2019

T074 CONSOLIDATED CASH FLOW STATEMENT

in €m	Notes	2019	2018
Cash and cash equivalents 1 Jan		1,434	1,218
Net profit/loss before income taxes		1,860	2,784
Depreciation, amortisation and impairment losses on non-current assets (net of reversals)		2,738	2,178
Depreciation, amortisation and impairment losses on current assets (net of reversals)		99	23
Net proceeds on disposal of non-current assets	6 10	20	-34
Result of equity investments	11	-168	-174
Net interest	12	315	144
Income tax payments/reimbursements		-1,009	-670
Significant non-cash-relevant expenses/income		-134	-276
Change in trade working capital		490	410
Change in other assets/shareholders' equity and liabilities		-181	-276
Cash flow from operating activities		4,030	4,109
Capital expenditure for property, plant and equipment and intangible assets	16 17 18 19 20	-3,486	-3,709
Capital expenditure for financial investments	22 23	-73	-48
Additions/loss to repairable spare parts for aircraft		-231	-388
Proceeds from disposal of non-consolidated equity investments		1	2
Proceeds from disposal of consolidated equity investments		3	4
Cash outflows for acquisitions/capital increases of/at non-consolidated equity investments	21 22 43	-107	-48
Cash outflows for acquisitions of consolidated equity investments	47	-	-12
Proceeds from disposal of intangible assets, property, plant and equipment and other financial investments		132	146
Interest income		70	51
Dividends received		243	143
Net cash from/used in investing activities		-3,448	-3,859
Purchase of securities/fund investments		-4,989	-3,289
Disposal of securities/fund investments		4,570	3,879
Net cash from/used in investing and cash management activities		-3,867	-3,269
Capital increase	31 32	-	-
Transactions by non-controlling interests		1	1
Non-current borrowing		3,843	987
Repayment of non-current borrowing		-3,413	-1,196
Dividends paid		-414	-349
Interest paid		-178	-69
Net cash from/used in financing activities		-161	-626
Net increase/decrease in cash and cash equivalents		2	214
Changes due to currency translation differences		-5	2
Cash and cash equivalents 31 Dec	29	1,431	1,434
Less cash and cash equivalents of companies held for sale as of 31 Dec		16	-
Cash and cash equivalents of companies not classified as held for sale as of 31 Dec¹⁾		1,415	1,434
Securities	28	1,970	1,735
Liquidity		3,385	3,169
Net increase/decrease in liquidity		216	-600

¹⁾ The difference between the bank balances and cash-in-hand shown in the statement of financial position comes from fixed-term deposits of EUR 0m with terms of four to twelve months (previous year: EUR 66m).

Further details can be found in the section [Notes to the consolidated cash flow statement](#), p. 198f.

Notes to the consolidated financial statements

Deutsche Lufthansa AG 2019

GENERAL REMARKS

1 Company information

The Lufthansa Group is a global aviation group whose subsidiaries and equity investments were organised into five operating segments in the financial year 2019: Network Airlines, Eurowings, Logistics, MRO and Catering.

Deutsche Lufthansa AG has its headquarters in Cologne, Germany, and is filed in the Commercial Register of Cologne District Court under HRB 2168.

The declaration on the German Corporate Governance Code required by Section 161 of the German Stock Corporation Act (AktG) was issued and made available to shareholders on the internet at www.lufthansagroup.com/declaration-of-compliance.

The consolidated financial statements of Deutsche Lufthansa AG, Cologne, and its subsidiaries have been prepared in accordance with the International Financial Reporting Standards (IFRS) issued by the International Accounting Standards Board (IASB), taking account of interpretations by the IFRS Interpretations Committee as applicable in the European Union (EU).

The commercial law provisions of Section 315e Paragraph 1 of the German Commercial Code (HGB) have also been applied. All IFRSs issued by the IASB, in effect at the time that these financial statements were prepared and applied by Deutsche Lufthansa AG have been adopted by the European Commission for application in the EU. The consolidated financial statements of Deutsche Lufthansa AG are prepared in millions of euros. Its financial year is the calendar year.

The accounting policies applied in the previous year have been retained, with the exception of changes required by new or amended standards, interpretations and the IFRIC agenda decision on recognising obligations to compensate customers for delayed or cancelled flights.

The Executive Board of Deutsche Lufthansa AG prepared the consolidated financial statements for 2019 on 13 March 2020. These consolidated financial statements were examined and approved for publication by the Supervisory Board of Deutsche Lufthansa AG in its meeting on 18 March 2020.

2 New international accounting standards in accordance with IFRS and interpretations and summary of the significant accounting policies and valuation methods

INTERNATIONAL FINANCIAL REPORTING STANDARDS (IFRS) AND INTERPRETATIONS (IFRIC) TO BE APPLIED FOR THE FIRST TIME IN THE FINANCIAL YEAR AND AMENDMENTS TO STANDARDS AND INTERPRETATIONS

T075 IFRS-PRONOUNCEMENT (APPLICABLE FROM FINANCIAL YEAR 2019)

IFRS 16, Leases
IFRIC 23, Uncertainty over Income Tax Treatments
Amendments to IAS 28, Non-current equity investments in associated companies and joint ventures
Amendments to IFRS 9, Prepayment Features with Negative Compensation
Annual Improvements to IFRS, 2015 – 2017, Cycle for amendments to IFRS 3, IFRS 11, IAS 12 und IAS 23
Amendments to IAS 19, Plan amendments, curtailments and settlements

The standards, clarifications and interpretations mandatory from 1 January 2019 onwards, particularly IFRS 16, Leases, and IFRIC 23, Uncertainty over Income Tax Treatments, had the following effects on the Group's net assets, financial and earnings position. The first-time application of all of the other amended accounting standards listed below had no or no material effect on the presentation of the net assets, financial and earnings position or on earnings per share.

IFRS 16, Leases. The new provisions of IFRS 16 require lessees to recognise a lease liability and a right-of-use asset for the payment obligations resulting from their lease agreements. IFRS 16 was initially applied using the modified retrospective approach, in accordance with the transitional provisions of IFRS 16. The comparative figures for the financial year 2018 were therefore not adjusted. The Lufthansa Group decided not to reassess whether a contract is or contains a lease. Accordingly, the definition of a lease in accordance with IAS 17 and IFRIC 4 will continue to be applied to those contracts entered or modified before 1 January 2019.

As of 1 January 2019, payment obligations from contracts previously classified as operating leases are discounted using the incremental borrowing rate and recognised as lease liabilities. The discount rate is generally calculated using incremental borrowing rates for the specific lease terms

and currencies, unless the implicit interest rate on which the lease payments are based is available. All lease payments are divided into redemption payments and interest expenses. The interest expense is recognised in profit or loss over the term of the lease. The right-of-use asset is depreciated on a straight-line basis over the shorter of the lease term or the useful life of the leased item.

The right-of-use asset corresponds at initial application to the lease liability, adjusted for any prepaid lease instalments. Initial direct costs are not included in the measurement of the right-of-use asset when the standard is applied for the first time. For the initial application of IFRS 16 hindsight was used. The Lufthansa Group has decided not to apply IFRS 16 to intangible assets and to account for leases ending in 2019 in accordance with the practical expedients for short-term leases. Payments under leases with a term of no more than twelve months beginning after 31 December 2018, and leases in which the leased asset is of low value, will be recognised in profit or loss at the payment date in line with this option. For contracts that include non-lease components alongside lease components, these components are separated. At the time of the transition, the Lufthansa Group had no provisions for onerous leases.

At the transition date of IFRS 16, right-of-use assets of EUR 2.0bn and lease liabilities of the same amount were recognised on 1 January 2019. The operating leases as of 31 December 2018 were reconciled with the opening amount of the lease liabilities in the statement of financial position as of 1 January 2019 as follows:

T076 RECONCILIATION LEASE LIABILITIES

in €m	2019
Obligations from operating leases as of 31 December 2018 ¹⁾	2,739
Short-term leases	10
Leases on assets of low value	338
Concluded contracts with right-of-use assets not yet acquired	126
Other	18
Discounting with incremental borrowing rate at the first application of IFRS 16	289
Lease liabilities newly accounted due to IFRS 16 as of 1 January 2019	1,958
Existing finance lease liabilities as of 31 December 2018	596
Total leasing liabilities	2,554

¹⁾ Adjusted figure.

The weighted average incremental borrowing rate used to calculate the lease liabilities as of 1 January 2019 was 1.95%. Reference interest rates based on congruent, risk-free rates in major countries and currencies were used to calculate the incremental borrowing rate. A credit risk premium was added to the respective reference rates.

The right-of-use asset is presented under the same item of property, plant and equipment as would have been used if the underlying asset had been purchased. The right-of-use assets recognised relate to the following types of assets:

T077 RIGHT-OF-USE ASSETS AND LEASE LIABILITIES

in €m	31 Dec 2019	1 Jan 2019
Aircraft and reserve engines		
Right-of-use assets – aircraft and reserve engines	471	401
Right-of-use assets – from former lease liabilities according to IAS 17	466	578
Property, plant and other equipment		
Right-of-use assets – land and property	1,873	1,531
Right-of-use assets – technical equipment	-	-
Right-of-use assets – other equipment, operating and office equipment	20	19
Right-of-use assets – from former lease liabilities according to IAS 17	80	93
Total right-of-use assets	2,910	2,622
of which first-time application due to IFRS 16	2,364	1,951
Non-current financial liabilities		
Lease liabilities newly accounted due to IFRS 16	1,983	1,599
Existing lease liabilities of finance lease liabilities	400	497
Current financial liabilities		
Lease liabilities newly accounted due to IFRS 16	403	359
Existing lease liabilities of finance lease liabilities	86	99
Total lease liabilities	2,872	2,554
of which first-time application due to IFRS 16	2,386	1,958

In terms of property, the Lufthansa Group mainly leases airport infrastructure, including hangars, parking and handling spaces, lounges, and offices. Other office buildings, production and warehouse space are also leased. In addition, the Group uses aircraft and other operating and office equipment on the basis of leases.

The additional right-of-use assets recognised in line with IFRS 16 led to additional depreciation of EUR 401m and additional interest expenses of EUR 53m, due to the accrued interest on the newly recognised lease liabilities. Foreign currency measurement for the lease liabilities resulted in expenses of EUR 6m in the financial result. The first-time application of IFRS 16 and the discontinuation of lease expenses caused the cost of materials and services to fall by EUR 165m and other operating expenses by EUR 267m.

In addition, the change in the presentation of the expenses related to leases resulted in a shift of EUR 432m between cash flow from financing activities and cash flow from operating activities as the lease payments no longer affect the operating cash flow and are instead recognised as interest and redemption payments within cash flow from financing activities, to the extent that they are not payments under current or low-value leases.

The definition of free cash flow was adjusted following the application of IFRS 16. The new figure Adjusted free cash flow consists of free cash flow adjusted for the effects of IFRS 16. This adjustment accounted for EUR 379m in the reporting year.

First-time application of IFRS 16 meant that earnings per share for the period from 1 January 2019 to 31 December 2019 fell by EUR 0.04 per share.

The regulations on lease accounting on the part of the lessors remain almost unchanged. The Lufthansa Group rents engines and buildings on a minor basis.

Detailed information can be found in the corresponding chapters of the Notes. ➤ **Note 20, p. 170ff.**

IFRIC 23, Uncertainty over Income Tax Treatments, was published by the IASB in June 2017. IFRIC 23 is applicable for financial years beginning on or after 1 January 2019. The interpretation supplements the provisions of IAS 12 on accounting for effective and deferred taxes with regard to uncertainties over the treatment of particular circumstances and transactions by the tax authorities and courts pertaining to income tax.

In the past, the Lufthansa Group has only recognised claims against tax authorities when a cash inflow was considered to be virtually certain. Following the transition to IFRIC 23, the claims will be recognised as soon as the cash inflow is deemed to be probable. IFRIC 23 was applied using the modified retrospective approach without adjusting the figures for prior year periods. The transition resulted in an increase in effective income tax receivables of EUR 33m, recognised in retained earnings.

On 17 September 2019, the IFRS Interpretations Committee published an **agenda decision on IFRS 15 concerning compensation for flight delays or cancellations**. Obligations to compensate customers for delayed or cancelled flights (as required by law) are to be recognised as variable compensation components within the meaning of IFRS 15, thus reducing the amount of revenue. The Lufthansa Group had previously recognised these payments in the income statement along with cost of care (e.g. for accommodation and meals in the case of delays, or damage to luggage). The consolidated financial statements were retrospectively changed as of 31 December 2019 as a result of the agenda decision. The change in compensation payments resulted in an equivalent reduction in revenue and the cost of materials of EUR 239m (2018: EUR 302m). At the same time, the other provisions (EUR 31m) and trade payables (EUR 44m) previously recognised for these items were transferred to other contract liabilities. As of 31 December 2019, this led to a reclassification of EUR 58m (2018: EUR 75m).

PUBLISHED INTERNATIONAL FINANCIAL REPORTING STANDARDS (IFRS) AND INTERPRETATIONS (IFRIC) NOT YET APPLIED/APPLICABLE AND AMENDMENTS TO STANDARDS AND INTERPRETATIONS

The following standards and amendments have already been adopted by the European Union but are only mandatory for financial statements after 31 December 2019:

T078 IFRS-PRONOUNCEMENT (ADOPTED BY THE EU)

	Mandatory application for financial years beginning on or after
Changes to conceptual framework	1 Jan 2020
Amendments to IAS 1 and IAS 8: Definition of materiality	1 Jan 2020
Amendments to IFRS 9, IAS 39 and IFRS 7, reform of reference interest rates	1 Jan 2020

Amendments to IFRS 9, IAS 39 and IFRS 7 aim to reduce the potential impact of reforms to the interbank offered rates (IBORs) on companies' financial reporting. The Lufthansa Group is applying this pronouncement in advance. First-time adoption has no effect on the presentation of the net assets, the financial and earnings position or on earnings per share, but does require additional disclosures in the notes.

➤ **Note 43, p. 207.**

The IASB and the IFRS Interpretations Committee have adopted other standards and interpretations whose application is not mandatory for the financial year 2019:

**T079 IFRS-PRONOUNCEMENT
(NOT YET ENDORSED BY THE EU)**

	Mandatory application for financial years beginning on or after
Amendments to IFRS 3, Business combinations	1 Jan 2020
IFRS 17, Insurance Contracts	1 Jan 2021
Amendments to IAS 1, Classification of financial liabilities	1 Jan 2022

Currently, the new or amended IFRS pronouncements listed in the table are not considered to have a material effect on the presentation of the net assets, financial and earnings position.

The Lufthansa Group has not voluntarily applied any of the new or amended regulations mentioned above before their binding date of application. If the effective dates of the standards and interpretations mentioned above fall within the year, they are applied as of 1 January of the following financial year. This is subject to the endorsement of the standards by the EU.

**SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES
AND VALUATION METHODS**

The companies included in the consolidated financial statements use uniform accounting policies to prepare their financial statements.

The application of the accounting policies prescribed by IFRS and IFRIC requires making a large number of estimates and assumptions with regard to the future that may, naturally, not coincide with actual future conditions. All of these estimates and assumptions are, however, reviewed continuously and are based either on past experience and/or expectations of future events that seem reasonable in the circumstances on the basis of sound business judgement. Estimates and assumptions that are of material importance in determining the carrying amounts for assets and liabilities are explained in the following description of the accounting policies applied to material items in the statement of financial position.

The fundamental valuation method applied in the consolidated financial statements is the acquisition cost principle. Where IFRSs stipulate that other methods of measurement should be applied, these are used instead, and are referred to specifically in the following comments on measuring assets and liabilities.

Amendments to accounting policies as a result of revised and new standards are applied retrospectively unless provided otherwise for a specific standard. The income statement for the previous year and the opening statement of financial position for the comparable period are adjusted as if the new accounting policies had always been applied.

Recognition of income and expenses

Revenue and other operating income are recognised when the service has been provided.

Passenger transport and ancillary services

The Lufthansa Group sells flight tickets and related ancillary services primarily via agents, its own websites or other airlines in the course of interlining. The payments are received by the Lufthansa Group via credit card billing companies, agents or other airlines, generally before the corresponding service is provided. Receivables from the sale of flight tickets and related ancillary services are only amounts payable by credit card billing companies, agents or other airlines.

The Lufthansa Group initially recognises all ticket sales as liabilities from unused flight documents. These are presented as contract liabilities in accordance with IFRS 15. Depending on the terms of the selected fare, the contract liabilities reflect a range of possibilities for refunding services that have not yet been provided. Liabilities include both the deferred income for future flights and ancillary services that are recognised as revenue when the flight documents are used, and the liabilities for award miles credited to the passenger when the flight documents are used. The Lufthansa Group allocates the transaction price to all of the performance obligations identified on the flight ticket on the basis of their individual transaction prices. The individual transaction prices for flight segments are determined using the IATA procedure. The total price payable is allocated to individual flight segments using what is known as a prorate calculation, which meets the IFRS 15 definition of a relative individual transaction price. The individual transaction prices for ancillary services that are not included in the fare are directly observable prices within the meaning of IFRS 15. On average, it takes 2.5 months for a flight coupon to be realised.

The Lufthansa Group reduces liabilities from unused flight documents and recognises revenue for each flight segment (including the related ancillary revenue) when the respective document is used. For tickets that cover more than one flight segment, the Lufthansa Group identifies each flight segment as a distinct performance obligation, since each flight segment is independent and can be distinguished in the context of the contract.

Interlining means that the passenger is carried by another airline for one (or more) flight segments. Only the commission paid by the other carrier is recognised as revenue for these flight segments, since the Lufthansa Group acts solely as an agent in terms of these performance obligations. If passengers with tickets sold by other airlines are carried partly or fully by the Lufthansa Group, the Group shows the pro rata ticket income received from the other airlines less the commission retained by the ticketing airline as revenue.

Generally speaking, the Lufthansa Group does not expect to receive any amount if a flight document is not used (or does not expect the amount to be material) and so for this reason does not anticipate the possibility that documents for a flight segment will not be used. The expected amount if flight documents are not used is only recognised as revenue if the probability that the passengers (in accordance with the portfolios) exercise their remaining rights is low, and no later than when the expiry of flight documents is certain and known.

Logistics

Lufthansa Cargo markets the freight capacities of passenger aircraft at Lufthansa German Airlines, Austrian Airlines, Eurowings and Brussels Airlines and operates a fleet of cargo aircraft. In addition to income from standard cargo services, Lufthansa Cargo generates part of its revenue from ancillary services that are closely connected to the freight service.

In its cargo business, the Lufthansa Group has identified the entire freight service as a distinct performance obligation. The end customer receives the benefit of the transport service and uses the service at the same time as this performance obligation is fulfilled with each transport segment. In this case, the customer takes control of the company's output while the carrier provides its service. The customer receives the benefit of the service as each transport segment is fulfilled. The corresponding cargo revenue is therefore recognised at the prorata value when the documents for each individual freight segment are used.

Lufthansa Cargo typically receives the consideration for performing its service once the transport has been carried out.

MRO

The main distinct performance obligations in the MRO segment are the provision of maintenance and aircraft and engine overhaul services, which are recognised over time since the condition of IFRS 15.35 (b) is generally met. These performance obligations involve estimating the proportion

of the total contract already completed and the profit on the whole contract, so that an input-orientated measurement of the percentage of completion can be made. Contract assets and contract liabilities are therefore both recognised.

Access to Lufthansa Technik's pool of spare parts and components is another key performance obligation, which is satisfied either over time or at a point in time, depending on the contract model agreed.

In some cases, the contracts in the MRO segment make it necessary not to recognise distinct services as individual performance obligations but rather as a series, as described in IFRS 15.22 (b). Furthermore, some of the contracts include standby obligations that require the recognition of revenue over time. This is particularly the case when remuneration is paid in the form of a fixed rate per hour of flying time. For such contracts the percentage of completion is primarily measured on the basis of the hours invoiced monthly to the customer. Revenue from component supply contracts is realised taking into consideration the margin shown in the business plans, which are updated annually.

A significant portion of the contracts in the MRO business segment run for several years and so have price adjustment clauses, but which are only considered in the transaction price when the event that triggers a price adjustment (a wage increase, for example) has occurred.

Catering

The LSG group offers products and services related to in-flight service. These include catering, in-flight sales and entertainment, in-flight service equipment and the associated logistics as well as consultancy services and the operation of lounges.

Airline catering is the main business of the LSG group as far as revenue is concerned. Taking the business model and the value chain for airline catering into account, the preparation of meals and the logistics related to this catering have been identified as distinct performance obligations. The performance obligation to prepare meals is generally fulfilled when the meals are delivered to the customers. The catering logistics performance obligation is fulfilled over time between the transport of the meals to the airport and the disposal of the waste, depending on the services ordered by the customer. For performance obligations over time, the percentage of completion is measured on an output basis in accordance with IFRS 15.B15 in conjunction with IFRS 15.B16.

Billing and payment in the Catering segment generally take place one to two months after the performance obligation has been fulfilled. This gives rise to trade receivables, but no significant contract liabilities or contract assets from catering contracts.

Variable consideration (e.g. volume discounts) must be taken into account when determining the transaction price in the catering business. The majority of the variable consideration is estimated using the expected value method on the basis of historic data and current developments. The LSG group updates the estimated transaction price at the end of each reporting period and accounts for the resulting changes in accordance with IFRS 15.87–90.

Operating expenses are recognised when the product or service is used or the expense arises. Provisions for warranties are generally accounted for when the corresponding revenue is recognised, while provisions for onerous contracts are generally set up when they are identified.

Further disclosures on the Lufthansa Group's revenue from contracts with customers can be found in [Note 3 and 4, p. 157ff.](#)

Interest income and expenses are accrued in the appropriate period. Dividends from shareholdings not accounted for using the equity method are recognised when a legal claim to them arises.

Initial consolidation and goodwill

The initial consolidation of Group companies takes place using the purchase method. This involves measuring the fair value of the assets, liabilities and contingent liabilities identified, in accordance with the provisions of IFRS 3, of the company acquired at the acquisition date, and allocating the acquisition costs to them. The proportion of fair value of assets and liabilities not acquired is shown under non-controlling interests. The ancillary acquisition costs are recognised as expenses in the periods in which they occur.

Any excess of cost over the value of equity acquired is capitalised as goodwill. If the value of the acquirer's interest in the shareholders' equity exceeds the purchase price paid by the acquiring company, the difference is recognised immediately in profit or loss.

Differences from non-controlling interests acquired after control has been gained are set off directly against equity.

Goodwill is not amortised, but is tested annually for impairment. The impairment tests applied to goodwill are carried out using established discounted cash flow methods. This is done on the basis of expected future cash flows from the latest business plan, which are extrapolated on the basis of long-term revenue growth rates and assumptions with regard to margin development and are discounted for the capital costs of the business unit. Tests are performed at the cash generating unit (CGU) level. For the individual premises on which impairment tests were based in the financial year 2019, see [Note 16, p. 164ff.](#)

Additional impairment tests are also applied during the course of the year if events give reason to believe that goodwill could be permanently impaired.

Once an impairment loss has been recognised on goodwill, it is not reversed in subsequent periods.

Notwithstanding the principles described above, Group companies that have no material impact on the Lufthansa Group's net assets, financial and earnings position are not consolidated, but rather recognised in the consolidated financial statements at cost less any impairments.

Currency translation and consolidation methods

The financial statements of the foreign Group companies are prepared in the relevant functional currency and translated into euros before consolidation. The functional currency is mainly the currency of the country in which the company concerned is located. Occasionally, the functional currency differs from the national currency. Assets and liabilities are translated at the middle rates on the balance sheet date. Income statements are translated at the average exchange rates for the year. Any translation differences are recognised directly in equity without effect on profit and loss and are only recognised in profit or loss when control is lost or the equity investment is disposed of.

Goodwill from capital consolidation of foreign subsidiaries prior to 2005 is carried at historical cost net of amortisation accumulated by the end of 2004. Goodwill acquired after 2005 is held in the functional currency of the purchased company and translated at the middle rates on the reporting date.

Transaction differences, however, are recognised in profit or loss. These differences arise in the financial statements of consolidated companies from the measurement of assets and liabilities denominated in a currency other than the company's functional currency. Any resulting exchange rate differences are included in other operating income as exchange rate gains, or in other operating expenses as exchange rate losses.

Translation differences relating to items whose fair value changes are recognised in equity are also recognised in equity without effect on profit and loss.

The most important exchange rates used in the consolidated financial statements have developed in relation to the euro as follows:

T080 EXCHANGE RATES

	2019		2018	
	Balance sheet exchange rate	Income statement average rate	Balance sheet exchange rate	Income statement average rate
AUD	0.62443	0.62094	0.61536	0.63223
CAD	0.68299	0.67379	0.64094	0.65241
CHF	0.92137	0.89864	0.88776	0.86633
CNY	0.12767	0.12955	0.12708	0.12787
GBP	1.17019	1.14121	1.11000	1.13057
HKD	0.11456	0.11410	0.11163	0.10771
INR	0.01251	0.01271	0.01250	0.01235
JPY	0.00819	0.00820	0.00792	0.00767
KRW	0.00077	0.00077	0.00078	0.00077
NOK	0.10137	0.10171	0.10014	0.10426
PLN	0.23505	0.23209	0.23234	0.23463
SEK	0.09564	0.09444	0.09747	0.09727
USD	0.89198	0.89432	0.87409	0.84415

The provisions of IAS 29, Financial Accounting in Hyper-inflationary Economies, were applied to one consolidated company in Argentina. Gains and losses from adjusting for inflation on the carrying amounts of non-monetary assets and liabilities and the income statement were immaterial and were recognised in other operating income.

The effects of intra-Group transactions are completely eliminated in the course of consolidation. Receivables and liabilities between consolidated companies are offset against one another and intra-Group provisions are reversed through profit or loss. Intra-Group profits and losses in non-current assets and inventories are eliminated – mostly in connection with the internal resale of aircraft and maintenance events. Intra-Group income is set off against the corresponding expenses. Tax accruals and deferrals are made as required by IAS 12 for temporary differences arising from consolidation.

Other intangible assets (except goodwill)

Acquired intangible assets are shown at cost, while internally generated intangible assets from which the Lufthansa Group expects to derive future benefit and that can be measured reliably are capitalised at cost of production and amortised regularly using the straight-line method over an estimated useful life. The cost of production includes all costs directly attributable to the production process, including borrowing costs as required under IAS 23, as well as appropriate portions of production-related overhead.

Intangible assets with indefinite useful lives (mainly brands and purchased, resellable take-off and landing rights) are not amortised, but rather subjected to a regular annual impairment test, as is goodwill.

Property, plant and equipment

Tangible assets used in business operations for longer than one year are valued at cost less regular straight-line depreciation. The cost of production includes all costs directly attributable to the manufacturing process as well as appropriate portions of production-related overhead. Borrowing costs in close connection with the financing of the purchase or production of a qualifying asset are also capitalised.

Key components of property, plant and equipment that have different useful lives are recognised and depreciated separately. Seats and in-flight entertainment systems installed in commercial aircraft are recognised separately. If costs are incurred in connection with regular extensive maintenance work (e.g. overhauling aircraft and major engine overhauls), these costs are recognised as a separate component insofar as they meet the criteria for recognition.

The following useful lives are applied throughout the Group:

T081 USEFUL LIVES OF PROPERTY, PLANT AND EQUIPMENT

Property, plant and equipment	Useful life
Buildings	45 years
New commercial aircraft and reserve engines	20 years to a residual value of 5%
Separable aircraft components	4 to 6 years
Technical equipment and machinery	8 to 20 years
Other equipment, operating and office equipment	3 to 20 years

Buildings, fixtures and fittings on rented premises are depreciated according to the terms of the leases or over a shorter useful life.

Assets acquired second-hand are depreciated over their expected remaining useful life.

When assets are sold or scrapped, the difference between the net proceeds and the net carrying amount of the assets is recognised as a gain or loss in the other operating income or expenses, respectively.

In addition to the impairment tests for goodwill, slots and brands, individual items of property, plant and equipment and intangible assets are also tested for impairment if they are no longer intended for future use, either because they are damaged, retired or due to be sold. In this case, the assets are measured individually in line with the applicable standard (full write-down to scrap value, or disposal proceeds less costs to sell). The lowest level at which assets can form a CGU is a production facility, to the extent that separate product lines or customer (groups) can be assigned to it. When aircraft are held for service in the Lufthansa Group fleet and there is no immediate intention to sell them, they are combined with the assets of the respective operating unit for the purposes of impairment testing.

Impairment losses on intangible assets and property, plant and equipment

In addition to depreciation and amortisation on property, plant and equipment and intangible assets, impairment losses are also recognised on the balance sheet date if the asset's recoverable amount has fallen below its carrying amount. The recoverable amount is determined as the higher of an asset's fair value less costs to sell and the present value of the estimated net future cash flows from continued use of the asset (value in use).

Fair value less costs to sell is derived from recent market transactions, if available.

If it is impossible to forecast expected cash flows for an individual asset, the cash flows for the next larger asset unit are estimated, discounted at a rate reflecting the risk involved, and the recoverable amount allocated to the individual assets in proportion to their respective carrying amounts.

If the reason for an impairment loss recognised in previous years should cease to exist in whole or in part in subsequent periods, the impairment loss is reversed up to the amount of the asset's amortised cost.

Repairable spare parts for aircraft

Initial supply of spare parts for aircraft that can be reused after repair are classified as non-current assets.

Repairable spare parts for aircraft are held at continually adjusted prices based on average acquisition costs. For measurement purposes, spare parts to be allocated to a maintenance pool are assigned to individual aircraft models and depreciated on a straight-line basis depending on the life phase of the fleet models for which they can be used. Other spare parts, mainly intended for replacement, are recognised in the statement of financial position at a discount to their acquisition costs, depending on how common they are.

Leases

The Lufthansa Group is a lessee for certain assets, particularly property and aircraft. In accordance with IFRS 16, the Lufthansa Group assesses at contract inception whether a contract is, or contains, a lease. That is, if the contract conveys the right to control the use of an identified asset for a period of time in exchange for consideration.

Right-of-use assets are measured at cost less accumulated depreciation and impairment losses and adjusted for any change in the measurement of the lease liability. The cost of right-of-use assets includes the amount of lease liabilities recognised, initial direct costs incurred, and lease payments made at or before the commencement date less any lease incentives received.

Right-of-use assets are depreciated on a straight-line basis over the shorter of the lease term or the estimated useful life of the leased item.

If ownership of the leased asset passes to the Lufthansa Group at the end of the lease term or is included in the costs of exercising a purchase option, depreciation is calculated for the expected useful life of the leased asset.

Right-of-use assets are also tested for impairment.

At the commencement date of the lease, Lufthansa Group recognises **lease liabilities** measured at the present value of the lease payments to be made over the term of the lease. The lease payments include fixed payments less any lease incentives owed, variable lease payments that depend on an index or a rate, and any amounts that are expected to be paid in the context of residual value guarantees. Lease payments also include the exercise price of a purchase option or penalties for early termination if the exercise of the purchase or termination option by the lessee is reasonably certain.

The Lufthansa Group has several leases that include renewal and termination options, particularly for properties. Judgment is used when determining the probability that the option to renew or terminate the lease will be exercised. When determining lease terms, all the facts and circumstances are considered that offer an economic incentive to exercise renewal options or not to exercise termination options. After the commencement date of the lease, the Lufthansa Group reestimates the term of the lease if a significant event occurs or if circumstances change.

Variable lease payments that do not depend on an index or a reference rate are recognised as expenses in the period in which the event or condition triggering the payment occurs.

Lease payments are discounted at the interest rate implicit in the lease, if it can be determined. Otherwise they are discounted at the incremental borrowing rate.

As the lessor in an operating lease, the Lufthansa Group presents the leased item as an asset at amortised cost in property, plant and equipment. Lease payments received in the period are shown as other operating income. The Lufthansa Group leases some of its properties and engines to other entities. There are currently no finance leases at the Lufthansa Group.

Further information about accounting for leases can be found in [🔗 Note 20, p. 170ff.](#)

Equity investments accounted for using the equity method

Equity investments accounted for using the equity method are capitalised at cost at the time of acquisition.

In subsequent periods, the carrying amounts are either increased or reduced annually by changes in the shareholders' equity of the associated company or joint venture that is held by the Lufthansa Group. The principles of purchase price allocation that apply to full consolidation are applied accordingly to the initial measurement of any difference between the acquisition cost of the investment and the pro rata share of shareholders' equity of the company in question. An impairment test is only carried out in subsequent periods if there are indications of a potential impairment in the entire investment valuation.

Financial instruments

Financial assets are classified within the Lufthansa Group in accordance with IFRS 9 as “at amortised cost”, “at fair value through profit or loss”, “at fair value through other comprehensive income (with and without recycling)” and “derivative financial instruments as an effective part of a hedging relationship”.

The category “**at amortised cost**” consists of financial assets that are debt instruments and are intended to be held to maturity on the basis of the company's business model. Furthermore, these instruments have fixed payment terms and meet the criteria for cash flow characteristics, i.e. contractual payments of principal and interest. For the Lufthansa Group, this item particularly includes loans and receivables, cash-in-hand and bank balances. They are classified as non-current or current assets according to their remaining maturity.

The category “**at fair value through profit or loss**” comprises debt instruments for which the business model is neither to hold nor to sell them, or which do not pass the cash flow characteristics test. This is generally not the case for the Lufthansa Group. Equity instruments are also allocated to this category as a rule, so the Lufthansa Group generally recognises shares and equity investments that are financial instruments in this category. Derivatives are also classified in this category if they do not meet the criteria for hedge accounting.

Debt instruments are classified as “**at fair value through other comprehensive income (with recycling)**” when the business model is to both hold and sell these instruments and they pass the cash flow characteristics test. For the Lufthansa Group, this applies to securities representing debt instruments.

An option can be exercised to classify specific equity instruments as “**at fair value through other comprehensive income (without recycling)**”. The Lufthansa Group exercises this option for individual share positions.

The Lufthansa Group uses derivatives for hedging, which are classified as “**derivative financial instruments as an effective part of a hedging relationship**” if all the requirements for hedge accounting are satisfied.

Financial instruments are recognised on the settlement date, i.e. on the date that they are created or transferred. Financial assets are capitalised at fair value plus transaction costs. Unrealised gains and losses are recognised directly in equity, taking deferred taxes into account. Long-term low or non-interest-bearing loans are recognised at net present value

using the effective interest method. Subsequent measurement of the financial instrument depends on the classification, either at amortised cost using the effective interest method, or at fair value, through profit or loss or in equity without effect on profit and loss.

Receivables denominated in foreign currencies are measured at the balance sheet date rate.

The fair value of securities is determined by the price quoted on an active market. For unlisted fixed-interest securities, the fair value is determined from the difference between effective and market interest rate at the valuation date.

If there are doubts as to the recoverability of receivables, then impairment losses are recognised and these receivables are recognised at the lower recoverable amount. Subsequent reversals (write-backs) are recognised in profit or loss. IFRS 9 requires that when a receivable is recognised for the first time, an expected loss is provided for that reflects the credit risk of the receivable before a default event occurs. An external credit risk exists for the Lufthansa Group, especially in its portfolio of trade receivables, for which an expected credit loss is recognised.

Derivative financial instruments are measured at fair value on the basis of published market prices. If there is no quoted price on an active market, other appropriate valuation methods are applied. Appropriate valuation methods take all factors into account that independent, knowledgeable market participants would consider in arriving at a price and that constitute recognised, established economic models for calculating the price of financial instruments.

In accordance with its internal guidelines, the Lufthansa Group uses derivative financial instruments to hedge interest rate and exchange rate risks and to hedge fuel price risks. This is based on the hedging policy defined by the Executive Board and monitored by a committee.

Interest rate swaps and interest rate/currency swaps are used to manage interest rate risks. Interest rate/currency swaps also hedge exchange rate risks arising from borrowing in foreign currencies.

Fuel price hedging takes the form of spread options and other hedging combinations, primarily for crude oil. To a limited extent, hedging may also be undertaken for other products, such as jet fuel or gas oil.

Hedging transactions are used to secure either fair values (fair value hedge) or future cash flows (cash flow hedge).

To the extent that the financial instruments used qualify as effective cash flow hedging instruments within the scope of a hedging relationship, in accordance with the provisions of IFRS 9, the fluctuations in market value will not affect the result for the period during the term of the derivative. They are recognised without effect on profit or loss in the corresponding reserves. If the hedged cash flow is an investment, the result of the hedging transaction that has previously been recognised in equity is set off against the cost of the capital expenditure at the time the underlying transaction matures. In all other cases, the cumulative gain or loss previously stated in equity is included in net profit or loss for the period on maturity of the hedged cash flow.

In the case of effective hedging of fair values that are designated as a fair value hedge, the changes in the market value of the hedged asset or the hedged debt and those of the financial instrument will balance out in the income statement.

Derivatives that do not meet the criteria for hedge accounting are presented in the category “at fair value through profit or loss”. Changes in fair value are then recognised directly in profit or loss. For the Lufthansa Group, this generally occurs when the exposure or item being hedged cannot be measured reliably or the exposure ceases to exist prematurely over the course of the hedge.

Embedded derivatives – to the extent that they should, but cannot, be separated from the financial host contract – are also considered with these as trading transactions for measurement purposes. Changes in market value are also recognised directly as profit or loss in the income statement. Both types must be classified as financial assets stated at fair value through profit or loss.

It is the Lufthansa Group’s hedging policy (🔗 **Note 43, p. 200ff.**) only to acquire effective derivatives for the purpose of hedging interest rate, exchange rate and fuel price risks.

Initial recognition of **financial guarantees** given to third parties is at fair value. Thereafter, financial guarantees are either measured in the category “at fair value through profit or loss” or at the higher of the originally recognised amount, less any cumulative amortisation through profit or loss in line with IFRS 15, and the value of the contractual obligation measured in line with IAS 37.

Emissions certificates

CO₂ emissions certificates are recognised as intangible assets and presented under other receivables. Rights, both those purchased and those allocated free of charge, are measured at cost and not amortised.

Contract assets and receivables

If the customer has fulfilled their contractual obligations, a contract asset or receivable is recognised. Receivables are recognised if the right to receive consideration is no longer subject to conditions. This is generally the case when the Group is contractually entitled to send the customer an invoice. Contract assets mainly relate to construction or service contracts for MRO and IT services.

Inventories

The item "Inventories" comprises non-repairable spare parts and assets used in production or the provision of services (raw materials, consumables and supplies), purchased merchandise, finished and unfinished goods and advance payments for them. They are measured at cost, determined on the basis of average prices, or at production costs. The cost of production includes all costs directly attributable to the production process, including borrowing costs as required under IAS 23, as well as appropriate portions of production-related overheads. Average capacity utilisation of 96% is assumed in determining the costs of production (previous year: 100%). Measurement on the balance sheet date is at the lower of cost and net realisable value. Net realisable value is defined as the estimated selling price less the estimated cost of completion and the estimated costs necessary to make the sale.

Assets held for sale

Individual, formerly non-current assets or groups of assets that are expected to be sold within the next twelve months are measured at the lower of their carrying amount at the time they are reclassified and fair value less costs to sell. Fair value less costs to sell is derived from recent market transactions, if available.

Property, plant and equipment and intangible assets are no longer depreciated or amortised and affiliated companies accounted for using the equity method are no longer accounted for in this way once they are classified as held for sale or distribution. While the impairment charge from the

last measurement before reclassification is recognised as an impairment loss, all subsequent changes in the measurement of current assets held for sale, e.g. due to exchange rate movements, are shown in other operating expenses or income.

Cash and cash equivalents

Cash and cash equivalents comprise cash-in-hand, cheques received and credit balances at banks. Cash equivalents are financial investments that can be liquidated at short notice. At the time of purchase or investment, they have a maturity of three months or less.

Pension provisions

The pension provisions for defined benefit plans correspond to the present value of the defined benefit obligations (DBO) on the reporting date less the fair value of plan assets, if necessary taking the rules on the maximum surplus of plan assets over the obligation (asset ceiling) into account.

The DBO is calculated annually by independent actuaries using the projected unit credit method prescribed in IAS 19 for defined benefit pension plans. The measurement of pension provisions within the statement of financial position is based on a number of actuarial assumptions.

Capital account plans are measured using the market value of the assets assigned to the individual capital accounts as of the reporting date, whereby the present value of the minimum benefit payable when the beneficiary becomes entitled to the benefit is compared with the amount of contributions already paid in, measured using the assumptions for the benefit plans. Additional risk premiums that the employer contributes to insure against early entitlements are included in current service expense.

They include, in particular, assumptions about long-term salary and pension trends as well as average life expectancy. The assumptions about salary and pension trends are based on developments observed in the past and take into account national interest and inflation rates and labour market trends. Estimates of average life expectancy are based on recognised biometric calculation formulas.

The interest rate used to discount the individual future payment obligations is based on the return from investment grade corporate bonds in the same currency and with a similar term to maturity. The discount rate is determined by reference to high-quality corporate bonds with an issue volume of at least EUR 100m and an AA rating from at least one of the rating agencies Moody's Investor Service, Fitch Ratings or Standard & Poor's Rating Services.

Actuarial gains and losses arising from the regular adjustment of actuarial assumptions are recognised directly in equity in the period in which they arise, taking deferred taxes into account. Also presented without effect on profit and loss are differences between the interest income at the beginning of the period calculated on plan assets based on the interest rate used to discount the pension obligations and the earnings from plan assets actually recorded at the end of the period. The actuarial gains and losses and any difference between the forecast result and the actual result from plan assets form part of the remeasurement.

Past service costs are recognised immediately in profit or loss.

Payments to pension providers for defined contribution retirement benefit commitments for which the pension provider or the beneficiary assumes the financial risks are recognised in staff costs as they fall due.

Other provisions

Other provisions are recognised for present legal and constructive obligations to third parties arising from past events that will probably give rise to a future outflow of resources provided that a reliable estimate can be made of the amount of the obligations as of the reporting date.

The amount of the provision is determined by the best estimate of the amount required to settle the present obligation. Past experience, current cost and price information as well as estimates from internal and external experts are used to determine the amount of provisions.

The management regularly analyses the current information on legal risks and makes provisions for probable obligations. These provisions cover estimated payments to the claimant, court and procedural costs, the costs of lawyers and of any out-of-court settlement. Internal and external lawyers assist with the estimate. When deciding on the necessity of a provision for litigation, the management takes into account the probability of an unfavourable outcome and the chance of making a sufficiently accurate estimate of the amount of the obligation.

The commencement of legal proceedings, the formal assertion of a claim against the Group or the disclosure of certain litigation in the Notes does not automatically mean that a provision was made for the risk concerned. A ruling in court proceedings, a decision by a public authority or an out-of-court settlement may cause the Group to incur expenses for which no provision was made because the amount could not be reliably determined or for which the provision made and the insurance coverage is not sufficient.

Provisions for obligations that are not expected to lead to an outflow of resources in the following year are recognised to the amount of the present value of the expected outflow, taking foreseeable price rises into account.

The assigned value of provisions is reviewed on each balance sheet date. Provisions in foreign currencies are translated at reporting date rates.

If no provision could be recognised because one of the stated criteria was not fulfilled, the corresponding obligations are shown as contingent liabilities and discussed in the relevant section.

Liabilities

Trade and other payables are initially recognised at fair value. Fair value is approximately equivalent to the carrying amount.

Measurement in subsequent periods is at amortised cost using the effective interest rate method.

Liabilities in foreign currencies are measured at the balance sheet date.

Obligations from share programmes were measured at fair value as cash-settled share-based payment transactions in accordance with IFRS 2. Fair value was measured using a Monte Carlo simulation.

The liability is recognised on the basis of the resulting fair value, taking the remaining term of the programme into account. Changes are recognised as staff costs in profit or loss.

Details of the assumptions used for the model and the structure of the share programmes can be found in [🔗 Note 37, p. 189ff.](#)

Contract liabilities

A contract liability is an obligation on the part of the Group towards a customer to provide goods or services for which the customer has already performed an obligation, e.g. by making an advance payment.

The Group's contract liabilities consist of liabilities from unused flight documents, unredeemed miles from customer loyalty programmes, construction contracts and other contract liabilities.

Until they are used, sold flight documents are recognised as an **obligation from unused flight documents**. Coupons that are unlikely to be used any more are recognised pro rata temporis as traffic revenue in the income statement at their estimated value. The estimate is based on historical statistical data.

The Lufthansa Group uses various bonus miles programmes with the aim of ensuring long-term **customer loyalty**. Participants in the Miles & More programme, which is the biggest bonus miles programme in the Lufthansa Group, can collect and redeem bonus miles for flights with the airlines in the Lufthansa Group as well as with numerous partners (including other airlines, hotels, global car hire companies, financial and insurance providers, telecommunications companies, retailers, automobile clubs, etc.). Miles expire three years after they are collected, in accordance with the terms of membership, unless they are protected by frequent flyer status or credit card use.

Revenue for award miles is recognised at the point in time or over the time at which the goods and services purchased with the award miles are transferred.

Observable past redemption patterns are used to measure the premium claims that are collected on flights with the airlines in the Lufthansa Group. Miles that are expected to be used for flights with airlines in the Lufthansa Group are measured with the average price of the premium flight or upgrade for the average number of miles used. The price is calculated on the basis of past redemption patterns, weighted for the various geographic regions and booking classes. This is then corrected to allow for the reduced flexibility of premium flights and the award miles granted for normal flights. Miles that are expected to be redeemed for other bonuses are measured at the average price for these bonuses and the average number of miles redeemed. The prices for additional miles are recalculated every year and applied to all additions in that year. Consumption of miles is measured using the average rate for total miles at the beginning of the year (same as previous year).

Premium points collected from other partners are measured at the amounts paid by these partners in relation to the average number of miles collected and redeemed.

The calculation method for the legal and economic expiry rate entails calculating the expiry rate from the rates observed in prior years, increased or decreased as necessary by reference to past trends or future enhancements to the programme. IFRS 15 requires that income from the expiry of miles is recognised in parallel with revenue from the performance obligations that do not expire. A period of three years is therefore assumed for revenue recognition, and the revenue from miles expected to expire is recognised on a straight-line basis over this time.

Tax liabilities

Obligations towards tax authorities that are uncertain with regard to their occurrence, probability and amount are recorded as tax liabilities on the basis of best estimates or expectations. Any contingent liabilities existing in this context are addressed separately as needed.

Deferred tax items

In accordance with IAS 12, deferred taxes are recognised for all temporary differences between the statements of financial position with regard to tax of individual companies and the consolidated financial statements. Tax loss carry-forwards are recognised to the extent that the deferred tax assets are likely to be used in the future. Company earnings forecasts and specific, realisable tax strategies are used to determine whether deferred tax assets from tax losses carried forward are usable or not, i.e. whether they have a value that can be realised. The planning period used to assess this probability is determined by the individual Group company according to the specific circumstances and lies generally between three and five years.

Effective income taxes

The Lufthansa Group is liable for income taxes in various countries. Material assumptions are necessary to calculate the income tax liabilities. For certain transactions and calculations, the final taxation cannot be assessed definitively in the course of normal business. The amount of the liability for future tax inspections is based on estimates of whether additional income taxes will be owed, and if so, at which amount. Estimates will be corrected as necessary in the period in which taxation is definitively assessed.

NOTES TO THE CONSOLIDATED INCOME STATEMENT

3 Traffic revenue

In the consolidated income statement, the Lufthansa Group attributes revenue to the segments Network Airlines, Eurowings, Logistics, MRO, Catering and Additional Businesses and Group Functions.

The IFRIC agenda decision on 17 September 2019 means that compensation payments for flight cancellations and delays are no longer recognised in the income statement but rather reduce traffic revenue. The Lufthansa Group has applied this change retrospectively. The adjustment came to EUR 239m for 2019, with traffic revenue for the previous year being reduced by EUR 302m.

The following table provides a breakdown of traffic revenue according to the different business models for the financial year that ended on 31 December 2019; the table following shows the adjusted previous year's figures:

T082 TRAFFIC REVENUE BY SECTOR 2019

	Total	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia / Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
Network Airlines	21,826	13,867	4,238	509	2,324	567	321
Lufthansa German Airlines	14,875						
SWISS ²⁾	5,010						
Austrian Airlines	1,941						
Eurowings ²⁾	3,992	3,566	204	12	46	23	141
Logistics	2,318	1,153	265	94	703	31	72
Total	28,136						

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Disclosure of traffic revenue, including belly revenue; this is reported in the segment reporting in the reconciliation column.

T082 TRAFFIC REVENUE BY SECTOR 2018

	Total ³⁾	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia / Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
Network Airlines	21,201	13,846	3,731	554	2,257	538	275
Lufthansa German Airlines	14,496						
SWISS ²⁾	4,749						
Austrian Airlines	1,956						
Eurowings ²⁾	4,050	3,626	173	12	68	26	145
Logistics	2,550	1,313	256	103	798	26	54
Total	27,801						

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Disclosure of traffic revenue, including belly revenue; this is reported in the segment reporting in the reconciliation column.

³⁾ Previous year's figures have been adjusted.

Traffic revenue of EUR 28,136m (previous year: EUR 27,801m) includes freight and mail revenue of EUR 2,774m (previous year: EUR 3,106m). Of the total, EUR 2,318m (previous year: EUR 2,550m) relate to the Logistics segment. Other freight and mail revenue of EUR 456m (previous year: EUR 557m) mainly comes from marketing belly capacities on passenger flights by SWISS and Brussels Airlines.

4 Other revenue

The following table provides a breakdown of other revenue by category (type of service) and geography for 2019 and the previous year:

T083 OTHER OPERATING REVENUE BY AREA OF OPERATIONS 2019

	Total	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia / Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
MRO	4,378	1,973	1,054	282	793	173	103
MRO services	3,765						
Other operating revenue	613						
Catering	2,623	502	1,377	164	468	75	37
Catering services	2,210						
Revenue from in-flight sales	169						
Other services	244						
Network Airlines	591	486	41	1	47	8	8
Eurowings	23	20	1	-	-	-	2
Logistics	119	68	41	-	3	7	-
Additional Businesses and Group Functions	554	388	43	16	83	18	6
IT services	188						
Travel management	279						
Other	87						
Total	8,288						

¹⁾ Traffic revenue is allocated according to the original location of sale.

T083 OTHER OPERATING REVENUE BY AREA OF OPERATIONS 2018

	Total	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia / Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾
in €m							
MRO	3,927	1,821	863	196	759	129	159
MRO services	3,402						
Other operating revenue	525						
Catering	2,499	483	1,270	147	495	62	42
Catering services	2,119						
Revenue from in-flight sales	142						
Other services	238						
Network Airlines	621	495	51	7	50	8	10
Eurowings	30	25	2	-	1	-	2
Logistics	131	73	47	-	4	7	-
Additional Businesses and Group Functions	533	386	39	11	74	17	6
IT services ²⁾	175						
Travel management	276						
Other	82						
Total	7,741						

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Adjustment due to change in allocation of three Lufthansa Systems companies.

MRO services make up the majority of external revenue in the MRO segment. Other revenue in the MRO segment from the sale of material and hiring out material and engines, as well as logistics services, are classified as other services.

The revenue listed under catering services originates exclusively in the Catering segment. The Catering segment also generates revenue from other services, particularly in the areas of flight security concepts, in-flight service equipment, transport/warehouse logistics and lounge operations.

Other revenue also includes revenue from customer contracts that are fulfilled over a given period. These are mainly MRO and IT services.

5 Changes in inventories and work performed by entity and capitalised

T084 CHANGES IN INVENTORIES AND WORK PERFORMED BY ENTITY AND CAPITALISED

in €m	2019	2018
Increase/decrease in finished goods and work in progress	- 9	2
Other internally produced and capitalised assets	694	529
	685	531

Other own work capitalised relates almost exclusively to aircraft and engine overhauls.

6 Other operating income

T085 OTHER OPERATING INCOME

in €m	2019	2018
Foreign exchange gains	761	794
Income from the reversal of provisions and accruals	321	260
Income from operating-leasing aircraft	49	48
Rental income	43	40
Compensation received for damages	42	55
Income from the reversal of impairment losses on fixed assets	38	15
Reversal of write-downs on receivables	27	48
Services provided by the Group	26	27
Income from staff secondment	20	19
Income from the disposal of non-current assets	18	50
Commission income	14	14
Income from the disposal of non-current available-for-sale financial assets	2	1
Miscellaneous other operating income	528	447
	1,889	1,818

Foreign exchange gains (excluding financial liabilities) mainly include gains from differences between the average rate for the month on the transaction date and on the payment date, along with foreign exchange gains from measurement at the closing date rate. Income from exchange rate hedging is also recognised here. Foreign exchange losses from these transactions are reported under other operating expenses.

➤ **Note 10, p. 161.** The foreign currency effects of borrowing are recognised in other financial items, in the context of the net results of exchange rate hedging relationships for borrowing.

Income from the reversal of provisions and accruals relates to a number of provisions and accruals recognised in previous years that have not been fully used. In contrast, expenses from insufficient provisions recognised in previous years are recognised together with the primary expense item to which they relate.

First-time application of IFRS 16 had no impact on the accounting for operating leases in which the Lufthansa Group is a lessor. No contracts were classified as finance leases. The Lufthansa Group recognised lease income of EUR 92m in 2019 (previous year: EUR 88m).

The following table shows the contractual lease payments:

T086 CONTRACTUAL LEASE PAYMENTS (LESSOR)

in €m	31 Dec 2019
to 1 year	79
more than 1 year to 2 years	50
more than 2 years to 3 years	35
more than 3 years to 4 years	28
more than 4 years to 5 years	8
more than 5 years	22

Income of EUR 38m was recognised from reversals on assets in 2019 (previous year: EUR 15m). EUR 19m of the total concerns an Airbus A340-600 from the Network Airlines segment. A further EUR 18m relates to write-backs on financial investments.

Income from the disposal of property, plant and equipment also includes EUR 6m from aircraft sold (previous year: EUR 13m).

Other operating income includes items not attributable to any of the aforementioned categories.

7 Cost of materials and services

T087 COST OF MATERIALS AND SERVICES

in €m	2019	2018 ¹⁾
Aircraft fuel and lubricants	6,715	6,087
Other raw materials, consumables and supplies	3,589	3,107
Purchased goods	512	489
Total cost of raw materials, consumables and supplies and of purchased goods	10,816	9,683
Fees and charges	4,523	4,457
External MRO services	1,911	1,848
Charter expenses ²⁾	814	717
External IT services	452	416
In-flight services	409	383
Flight irregularities	199	216
Other services	703	647
Total cost of purchased services	9,011	8,684
	19,827	18,367

¹⁾ Previous year's figures have been adjusted.

²⁾ In 2018 also including operating lease costs according to IAS 17.

The cost of materials has decreased due to the implementation of IFRS 16 by EUR 165 million in the financial year 2019.

Retrospectively changing the recognition of payments to customers in connection with flight irregularities resulted in an adjustment to other purchased services reported in the previous year. Expenses for flight irregularities are also shown separately for the first time. This includes accommodation and meals in the case of delays, for instance, or payments for damaged luggage. ➤ Note 2, p. 144ff.

8 Staff costs

T088 STAFF COSTS

in €m	2019	2018
Wages and salaries	7,452	7,286
Social security contributions	1,024	977
Expenses for pension plans and other employee benefits	645	548
	9,121	8,811

Expenses for pension plans principally consist of additions to the pension provisions. ➤ Note 33, p. 178ff.

In the previous year, the expenses for pension plans and other employee benefits were reduced by EUR 113m because of negative past service expenses resulting essentially from the changes to the retirement benefits system in Switzerland.

T089 EMPLOYEES

	Average for the year 2019	Average for the year 2018	As of 31 Dec 2019	As of 31 Dec 2018
Ground staff	91,241	88,478	91,642	89,278
Flight staff	45,305	44,858	45,281	45,169
Trainees	1,238	994	1,430	1,087
	137,784	134,330	138,353	135,534

The annual average is calculated pro rata temporis from the time companies are consolidated or deconsolidated.

9 Depreciation, amortisation and impairment

The notes to the individual items show the breakdown of depreciation, amortisation and impairment charges between intangible assets, aircraft and other property, plant and equipment. Total depreciation, amortisation and impairment came to EUR 2,776m (previous year: EUR 2,205m).

T090 DEPRECIATION, AMORTISATION AND IMPAIRMENT

in €m	2019	2018
Amortisation of other intangible assets	102	98
Depreciation of aircraft	2,084	1,833
Depreciation of other tangible assets	506	250
Total amortisation/depreciation	2,692	2,181
Impairment of goodwill	–	–
Impairment of other intangible assets	24	2
Impairment of aircraft	36	4
Impairment of other tangible assets	1	5
Impairment of financial assets	23	13
Total impairment	84	24
Total depreciation, amortisation and impairment	2,776	2,205

The increase in depreciation and amortisation resulted mainly from the first-time application of IFRS 16. Additional right-of-use assets under IFRS 16 resulted in an increase in depreciation and amortisation of EUR 401m.

Impairment losses of EUR 84m were recognised in the financial year 2019. EUR 36m of the total was for four Boeing MD-11Fs from the Logistics segment held for sale. Other impairment losses related to write-downs of other receivables (EUR 22m) and discontinued IT projects (EUR 24m).

Other operating expenses included further impairment losses of EUR 50m on the assets of the European catering operations held for sale and already reclassified as current assets (previous year: no impairment of assets already classified as held for sale). ➤ **Note 30, p. 175.**

A write-down of EUR 43m was recognised in the result from equity investments for the shares in the joint venture Alpha Holding Ltd. (Catering segment), which is accounted for using the equity method. It relates to the insolvency of the Thomas Cook Group plc.

10 Other operating expenses

T091 OTHER OPERATING EXPENSES		
in €m	2019	2018
Staff-related expenses	1,201	1,226
Rental and maintenance expenses	742	923
Foreign exchange losses	648	771
Expenses for computerised distribution systems	506	524
Advertising and sales promotions	384	439
Sales commission paid to agencies	368	330
Auditing, consulting and legal expenses	312	252
Commissions for credit cards	232	222
Other services	144	148
Write-downs on receivables	120	100
Communications costs	85	84
Other taxes	84	84
Insurance premiums for flight operations	61	56
Losses on disposal of non-current assets	39	17
Miscellaneous other operating expenses	609	532
	5,585	5,708

Staff-related expenses also include travel and training costs for Group employees and the costs of outside staff.

Foreign exchange losses (excluding financial liabilities) mainly consist of losses from differences between the monthly average rates on the transaction date and on the payment date, expenses from exchange rate hedges and translation losses from measurement at the exchange rate on the balance sheet date. ➤ **Note 6, p. 159.** The foreign currency effects of borrowing are recognised in other financial items, in the context of the net results of exchange rate hedging relationships for borrowing.

Lease expenses were eliminated from rental expenses and right-of-use assets recognised as a result of the first-time application of IFRS 16 in 2019, to the extent that the corresponding leases meet the conditions for recognition. ➤ **Note 20, p. 170ff.** Rental expenses would have been EUR 267m higher

in 2019 without this change. Rental expenses include property maintenance expenses of EUR 166m (previous year: EUR 150m).

Impairment losses of EUR 50m were recognised in other operating expenses for assets already reclassified as current assets in connection with the sale of the European catering operations.

11 Result from equity investments

T092 RESULT FROM EQUITY INVESTMENTS

in €m	2019	2018
Result of joint ventures accounted for using the equity method	14	75
Result of associated companies accounted for using the equity method	74	39
Result of equity investments accounted for using the equity method	88	114
Dividends from other joint ventures	12	13
Dividends from other associated companies	3	6
Income from profit transfer agreements	35	31
Expenses from loss transfer agreements	-19	-26
Dividends from other equity investments	49	36
Result of other equity investments	80	60
	168	174

Write-downs on shares are also presented in the result of joint ventures accounted for using the equity method. ➤ **Note 9, p. 160f.** The increase in the result of associated companies accounted for using the equity method comes from a one-off disposal gain of EUR 33m for an associate in the MRO segment.

Income and expenses from profit and loss transfer agreements are shown including tax contributions.

12 Net interest

T093 NET INTEREST

in €m	2019	2018
Income from other securities and non-current financial loans	8	5
Other interest and similar income	71	63
Interest income	79	68
Interest expenses on pensions obligations	-119	-97
Interest expenses on other provisions	3	-8
Interest and other similar expenses	-278	-107
Interest expenses	-394	-212
	-315	-144

Net interest comprises interest income and expenses – calculated using the effective interest method in accordance with IFRS 9 – from financial assets and liabilities not classified as at fair value through profit or loss.

Lower net interest is mainly due to interest expenses in connection with a supplementary tax payment due to the non-deductibility of partial write-downs on cross-border loans (EUR 146m) and interest expenses for leases (EUR 53m) recognised for the first time as a result of the adoption of IFRS 16 as of 1 January 2019.

13 Other financial items

T094 OTHER FINANCIAL ITEMS

in €m	2019	2018
Result of fair value hedges – change in time value of hedged transactions	- 90	- 47
Result of fair value hedges – change in time value of hedging instruments	89	60
Ineffective portion of derivatives used as cash flow hedges	- 18	17
Result of derivatives held for trading classified as at fair value through profit or loss	359	- 16
Result of measuring securities classified as at fair value through profit or loss	- 17	- 35
Exchange rates effects from financial liabilities	- 5	- 26
	318	- 46

Exchange rate hedges in cash flow hedges for investments were reversed in October 2019 for total proceeds of EUR 402m after the hedges were terminated early as of 30 September 2019. These hedges covered orders for 14 Boeing 777 aircraft, which were originally held as reconfirmable orders and had been hedged using long-term forward currency transactions in accordance with the hedging strategy. In the third quarter of 2019, the orders for 14 B777s were converted into options which, at present, are not sufficiently certain to be exercised. From a hedge accounting perspective, this means the forecast transactions are no longer highly probable, so the hedge relationships had to be terminated.

14 Income taxes

T095 INCOME TAXES

in €m	2019	2018
Current income taxes	547	614
Deferred income taxes	68	- 26
	615	588

Current income taxes include corporation tax, solidarity surcharge, trade tax and other income taxes paid outside Germany totalling EUR 382m for 2019 (previous year: EUR 538m). Other tax expenses of EUR 165m related to prior years (previous year: EUR 76m). The tax rates used to calculate deferred taxes abroad ranged from 3.5% to 35.0% in 2019 (previous year: 3.5% to 35.0%). For measuring deferred taxes, the relevant taxation rules in force or adopted at the balance sheet date are used.

The following table reconciles expected and effective tax expenses. Expected tax expense is calculated by multiplying profit before income taxes by a tax rate of 25% for the parent company (previous year: 25%). This is made up of a tax rate of 15.825% (previous year: 15.825%) for corporation tax/solidarity surcharge and 9.175% for trade tax (previous year: 9.175%). The portion of trade income related to the foreign air transport operations of the German-based airlines is deducted when calculating the tax rate for trade tax.

T096 TAX RECONCILIATION

	2019		2018	
	Basis of assessment	Tax expenses	Basis of assessment	Tax expenses
in €m				
Expected income tax expenses	1,860	465	2,784	696
Tax free gains/losses	-	- 1	-	-
Non-deductible costs	-	59	-	55
Non-taxable income	-	- 84	-	- 61
Non-taxable income from equity investments	-	- 51	-	- 42
Difference between local taxes and the deferred tax rates of the parent company as well as effects of changes in tax rates	-	- 5	-	- 23
Taxes from other periods ¹⁾	-	193	-	-
Effects from use or deferred tax assets not recognised	-	39	-	- 37
Recognised income tax expenses	-	615	-	588

¹⁾ Deferred tax expenses of EUR 28m for other periods (previous year: EUR 76m tax revenue) and effective tax expenses of EUR 165m for other periods (previous year: EUR 76m).

Deferred tax liabilities of EUR 48m (previous year: EUR 47m) were not recognised on temporary differences in connection with shares in subsidiaries, as the temporary differences are not expected to reverse in the foreseeable future.

Deferred tax assets and liabilities in 2019 and 2018 were allocable to the following items in the statement of financial position:

T097 DEFERRED TAX ASSETS AND LIABILITIES

in €m	31 Dec 2019		31 Dec 2018	
	Assets	Liabilities	Assets	Liabilities
Tax loss carry-forwards and tax credits	45	-	121	-
Pension provisions	2,574	-	2,331	-
Intangible assets, property, plant and equipment	-	1,112	-	923
Non-current financial assets	-	24	-	14
Fair value measurement of financial instruments	-	235	-	126
Provisions for contingent losses	6	-	19	-
Receivables/ liabilities/ other provisions	163	-	-	110
Inventories	242	-	247	-
Assets held for sale	-	-	-	1
Other	-	2	4	-
Offset amounts	- 762	- 762	- 591	- 591
	2,268	611	2,131	583

A deferred tax receivable of EUR 28m (previous year: EUR 47m) was recognised for companies incurring a net tax loss in the reporting year because tax and earnings planning indicates that there is a high probability that the tax receivable will be realised.

In addition to recognised deferred tax assets from tax loss carry-forwards, non-deductible interest carry-forwards and tax credits, further tax loss carry-forwards and temporary differences totalling EUR 3,050m (previous year: EUR 2,423m) exist for which no deferred tax assets could be recognised.

The usage restrictions for non-capitalised tax loss carry-forwards and the resulting deferred taxes are distributed as follows:

T098 LIMITS ON THE USE OF NON-CAPITALISED LOSS CARRY-FORWARDS

in €m	Non-capitalised loss carry-forwards	Deferred taxes
Usable		
until 2023	184	40
until 2024	42	9
until 2025	16	5
until 2026	3	1
until 2027	1	1
until 2028	1	-
2029 and beyond	2,250	569
Total	2,497	625
of which from companies in the disposal group under assets held for sale	67	14

15 Earnings per share

Basic/diluted earnings per share are calculated by dividing consolidated net profit by the weighted average number of shares in circulation during the financial year. To calculate the average number of shares, the shares bought back and reissued for the employee share programmes are included pro rata temporis.

T099 EARNINGS PER SHARE

		2019	2018
Basic/diluted earnings per share	€	2.55	4.58
Consolidated net profit/loss	€m	1,213	2,163
Weighted average number of shares		475,730,992	472,553,216

For the financial year 2019, there was a net profit at Deutsche Lufthansa AG of EUR 595m. Following the transfer of EUR 297m to retained earnings, distributable profit comes to EUR 298m. In view of the current impact of the coronavirus on the course of business and to further strengthen the statement of financial position, the Executive Board and Supervisory Board propose to the Annual General Meeting to transfer the distributable profit of EUR 298m in full to retained earnings.

In 2019, EUR 0.80 per share was distributed as a dividend to shareholders from the net profit for 2018, including a transfer from retained earnings.

NOTES TO THE CONSOLIDATED STATEMENT OF FINANCIAL POSITION

Assets

16 Goodwill and intangible assets with an indefinite useful life

T100 GOODWILL AND INTANGIBLE ASSETS WITH AN INDEFINITE USEFUL LIFE

in €m	Goodwill from consolidation	Intangible assets with an indefinite useful life	Total
Cost as of 1 Jan 2018	1,027	634	1,661
Accumulated impairment losses	- 314	- 4	- 318
Carrying amount 1 Jan 2018	713	630	1,343
Currency translation differences	-	15	15
Additions due to changes in consolidation	-	-	-
Additions	23	-	23
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Reclassifications to assets held for sale	-	-	-
Impairment losses	-	-	-
Reversal of impairment losses	-	-	-
Carrying amount 31 Dec 2018	736	645	1,381
Cost as of 1 Jan 2019	1,050	648	1,698
Accumulated impairment losses	- 314	- 3	- 317
Carrying amount 1 Jan 2019	736	645	1,381
Currency translation differences	-	14	14
Additions due to changes in consolidation	-	-	-
Additions	-	-	-
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Reclassifications to assets held for sale	-	-	-
Impairment losses	-	-	-
Reversal of impairment losses	-	-	-
Carrying amount 31 Dec 2019	736	659	1,395
Cost as of 31 Dec 2019	1,044	663	1,707
Accumulated impairment losses	- 308	- 4	- 312

All goodwill and intangible assets with an indefinite useful life were subjected to a regular impairment test in 2019 as required by IAS 36. Furthermore, there is the obligation to perform an impairment test if there is an indication of impairment. For impairment testing following an indication of impairment ➤ **Note 9, p. 160f.**

Acquired brands and slots have an indefinite useful life due to their lasting legal and economic significance. The tests were performed at the level of the smallest cash generating unit (CGU) on the basis of fair value less costs to sell or value in use. Due to the future organisational and commercial integration of Brussels Airlines with Network Airlines and the re-allocation of Eurowings' long-haul business, the fair value of the Eurowings business segment was divided among the remaining Eurowings group, Lufthansa German Airlines and Brussels Airlines in proportion to their market value.

The following table provides an overview of the goodwill tested and the assumptions made in the respective impairment tests regarding the smallest possible cash-generating unit (CGU) in each case.

T101 IMPAIRMENT TESTS OF GOODWILL 2019

Name of the CGU	Lufthansa German Airlines	Brussels Airlines	Eurowings	LSG Sky Chefs USA group	LSG Sky Chefs Korea	Other ¹⁾
Segment	Network Airlines	Brussels Airlines	Eurowings	Catering	Catering	Catering/Service and financial companies
Carrying amount of goodwill (31 Dec)	€ 252m	€ 44m	€ 57m	€ 277m	€ 60m	€ 45m
Impairment losses	–	–	–	–	–	–
Duration of planning period	4 years	4 years	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	2.0%	2.0%	2.0%	1.9%	2.6%	1.7% to 3.5%
Discount rate	3.9% ²⁾	3.9% ²⁾	3.9% ²⁾	4.6% ³⁾	4.1% ³⁾	3.9% ²⁾ to 7.7% ³⁾

¹⁾ Goodwill of less than EUR 25m carrying amount in any individual instance.

²⁾ After-tax rate.

³⁾ Pre-tax rate.

The assumptions on revenue growth used for the impairment tests are based on approved internal budgets and external sources for the planning period. In some cases, reductions were made for risk to allow for special regional features and market share trends specific to the respective companies. The margins used are based on past experience or were developed on the basis of cost-cutting measures initiated. The investment rates are based on past experience and take account of the replacement of any means of production envisaged during the planning period. Costs of the central functions were charged to the individual units based on their use of these functions. The tests also covered the right-of-use assets recognised in line with IFRS 16. Additional debt resulting from the recognition of lease liabilities in accordance with IFRS 16 was included in the weighted cost of capital.

Assuming sustained revenue growth by the CGUs as described in the table, the recoverable amounts would exceed the carrying amount by a significant figure, even if the growth assumptions were reduced by one percentage point in each case. Worsening the scenarios by one percentage point in each case, in terms of planned margins or the discount rates used for the impairment tests, would also not reduce the recoverable amounts below the respective carrying amounts for all of the other CGUs. The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation.

The following table shows the assumptions used for the previous year's impairment tests.

T101 IMPAIRMENT TESTS OF GOODWILL 2018

Name of the CGU	Lufthansa German Airlines	Eurowings	LSG Sky Chefs USA group	LSG Sky Chefs Korea	Other ¹⁾
Segment	Network Airlines	Eurowings	Catering	Catering	Catering/Service and financial companies
Carrying amount of goodwill (31 Dec)	€ 238m	€ 116m	€ 277m	€ 61m	€ 44m
Impairment losses	–	–	–	–	–
Duration of planning period	4 years	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	2.2%	2.2%	2.0%	2.9%	1.0% to 5.0%
Discount rate	4.7% ²⁾	4.7% ²⁾	5.2% ³⁾	4.9% ³⁾	4.5% ²⁾ to 9.0% ³⁾

¹⁾ Goodwill of less than EUR 25m in any individual instance.

²⁾ After-tax rate.

³⁾ Pre-tax rate.

The earnings situation of the Logistics segment provided an indication for an impairment test, but this did not result in an impairment loss.

The earnings situation of Austrian Airlines would also have triggered an impairment test if the value of the assets attributable to Austrian Airlines had not already been confirmed by the impairment testing of the slots described below.

The intangible assets with indefinite useful lives consist of slots purchased as part of company acquisitions (insofar as they are tradeable) and brand names acquired.

The following table shows the assumptions made for regular impairment testing of the smallest cash-generating unit (CGU) in each case.

T102 IMPAIRMENT TESTS OF SLOTS 2019

Group company	SWISS	Austrian Airlines
Carrying amount for slots (31 Dec)	€ 135m	€ 23m
Impairment losses	-	-
Duration of planning period	4 years	4 years
Revenue growth p.a. after end of planning period	2.0%	2.0%
Discount rate	3.9% ¹⁾	3.9% ¹⁾

¹⁾ After-tax rate.

Based on sustainable revenue growth according to assumptions described in the table, the recoverable amounts significantly exceed the carrying amounts. Even if the assumptions on revenue growth, the discount rate and margins are all

reduced by one percentage point in each case, the recoverable amounts are still higher than the carrying amounts.

The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation.

The following table shows the assumptions used for the previous year's impairment tests.

T102 IMPAIRMENT TESTS OF SLOTS 2018

Group company	SWISS	Austrian Airlines
Carrying amount for slots (31 Dec)	€ 129m	€ 23m
Impairment losses	-	-
Duration of planning period	4 years	4 years
Revenue growth p.a. after end of planning period	2.2%	2.2%
Discount rate	4.8% ¹⁾	4.7% ¹⁾

¹⁾ After-tax rate.

The slots purchased by Deutsche Lufthansa AG were allocated to Lufthansa German Airlines and Eurowings at their historic cost in accordance with their use. A carrying amount of EUR 76m was allocated to Lufthansa German Airlines and a carrying amount of EUR 36m to Eurowings as of 31 December 2019. Both carrying amounts were subjected to impairment testing with identical assumptions to those used for the goodwill of the CGUs Lufthansa German Airlines and Eurowings.

The regular impairment test for the brands acquired was carried out on the basis of the revenue generated from each brand.

The following additional assumptions were used in the impairment test for the acquired brands:

T103 IMPAIRMENT TESTS OF BRANDS 2019

Group company	SWISS	Austrian Airlines	Brussels Airlines	Other
Carrying amount for brand (31 Dec)	€ 238m	€ 107m	€ 37m	€ 7m
Impairment losses	-	-	-	-
Duration of planning period	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	2.0%	2.0%	1.0%	0.0% to 2.0%
Savings in hypothetical leasing payments before taxes (royalty rate)	0.63%	0.35%	0.20%	0.23% to 0.50%
Discount rate	3.9% ¹⁾	3.9% ¹⁾	3.9% ¹⁾	3.9% to 4.4% ¹⁾

¹⁾ After-tax rate.

Assuming sustained brand-related revenue growth at the end of the planning period, as described in the table, the recoverable amounts for the brands exceed their carrying amounts significantly. Even if the assumptions for sustained brand-related revenue growth were to be reduced or the

discount rate were to be increased by one percentage point in each case, the recoverable amounts would exceed the carrying amounts.

There was no impairment charge within the other brands in 2019.

The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation.

The assumptions used for the previous year's impairment tests can be derived from the following table.

T103 IMPAIRMENT TESTS OF BRANDS 2018

Group company	SWISS	Austrian Airlines	Brussels Airlines	Other
Carrying amount for brand (31 Dec)	€ 230m	€ 107m	€ 37m	€ 7m
Impairment losses	-	-	-	-
Duration of planning period	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	2.2%	2.2%	1.0%	0.0% to 2.2%
Savings in hypothetical leasing payments before taxes (royalty rate)	0.63%	0.35%	0.20%	0.23% to 0.50%
Discount rate	4.8% ¹⁾	4.7% ¹⁾	4.7% ¹⁾	4.5% to 5.1% ¹⁾

¹⁾ After-tax rate.

17 Other intangible assets

T104 OTHER INTANGIBLE ASSETS

	Concessions, industrial property rights and similar rights and licences to such rights and assets	Internally developed software	Advance payments	Total
in €m				
Cost as of 1 Jan 2018	1,230	133	112	1,475
Accumulated amortisation	- 861	- 107	- 15	- 983
Carrying amount 1 Jan 2018	369	26	97	492
Currency translation differences	5	2	- 1	6
Additions due to changes in consolidation	-	-	-	-
Additions	29	4	77	110
Reclassifications	30	10	- 35	5
Disposals due to changes in consolidation	-	-	-	-
Disposals	- 1	-	- 1	- 2
Reclassifications to assets held for sale	-	-	-	-
Amortisation	- 92	- 7	-	- 99
Reversal of impairment losses	-	-	-	-
Carrying amount 31 Dec 2018	340	35	137	512
Cost as of 1 Jan 2019	1,289	150	152	1,591
Accumulated amortisation	- 949	- 115	- 15	- 1,079
Carrying amount 1 Jan 2019	340	35	137	512
Currency translation differences	4	- 1	-	3
Additions due to changes in consolidation	-	-	-	-
Additions	52	3	101	156
Reclassifications	43	11	- 41	13
Disposals due to changes in consolidation	-	-	-	-
Disposals	- 7	-	- 8	- 15
Reclassifications to assets held for sale	- 1	-	-	- 1
Amortisation	- 99	- 9	- 13	- 121
Reversal of impairment losses	-	-	-	-
Carrying amount 31 Dec 2019	332	39	176	547
Cost as of 31 Dec 2019	1,324	163	205	1,692
Accumulated amortisation	- 992	- 124	- 29	- 1,145

Non-capitalised research and development expenses for intangible assets of EUR 49m (previous year: EUR 44m) were incurred in the period. Fixed orders have been placed for intangible assets worth EUR 14m (previous year: EUR 10m), but they are not yet at the Lufthansa Group's economic disposal.

18 Aircraft and reserve engines including right-of-use assets

T105 AIRCRAFT AND RESERVE ENGINES INCLUDING RIGHT-OF-USE ASSETS

in €m	Aircraft and reserve engines	Advance payments for aircraft and reserve engines	Total
Cost as of 1 Jan 2018	29,405	1,373	30,778
Accumulated amortisation	- 15,445	-	- 15,445
Carrying amount 1 Jan 2018	13,960	1,373	15,333
Currency translation differences	114	13	127
Additions due to changes in consolidation	-	-	-
Additions	2,542	782	3,324
Reclassifications	475	- 475	-
Disposals due to changes in consolidation	- 5	-	- 5
Disposals	- 158	- 3	- 161
Reclassifications to assets held for sale	- 7	-	- 7
Depreciation	- 1,837	-	- 1,837
Reversal of impairment losses	2	-	2
Carrying amount 31 Dec 2018	15,086	1,690	16,776
Cost as of 1 Jan 2019	31,208	1,690	32,898
Accumulated amortisation	- 16,122	-	- 16,122
Carrying amount 1 Jan 2019	15,086	1,690	16,776
First-time application of IFRS 16	401	-	401
Adjusted carrying amount 1 Jan 2019	15,487	1,690	17,177
Currency translation differences	111	10	121
Additions due to changes in consolidation	-	-	-
Additions	2,149	1,098	3,247
Reclassifications	691	- 691	-
Disposals due to changes in consolidation	-	-	-
Disposals	- 69	- 5	- 74
Reclassifications to assets held for sale	- 20	-	- 20
Depreciation	- 2,121	-	- 2,121
Reversal of impairment losses	19	-	19
Carrying amount 31 Dec 2019	16,247	2,102	18,349
Cost as of 31 Dec 2019	32,945	2,102	35,047
Accumulated amortisation	- 16,698	-	- 16,698

Accounting changes in accordance with IFRS 16 and the resulting capitalisation of right-of-use assets meant that the figures were adjusted as of the beginning of the year. For a detailed description of the effects, we refer to [Note 2, p. 144ff](#). Aircraft previously classified under finance leases in line with IAS 17, the accounting for which has not changed following the introduction of IFRS 16, were carried at EUR 466m as of 31 December 2019 (previous year: EUR 578m). [Note 20, p. 170ff](#).

Additions in 2019 included right-of-use assets of EUR 227m. Other additions were for the procurement of new aircraft.

The item also includes 80 aircraft with a carrying amount of EUR 2,549m (previous year: 72 aircraft with a carrying amount of EUR 2,173m), which have mostly been sold to and leased back from foreign leasing companies with the aim of obtaining favourable financing conditions. The leasing companies were fully consolidated as structured entities. The Lufthansa Group is entitled to buy the aircraft back at a fixed price and at a given point in time.

In the reporting year, borrowing costs of EUR 24m were capitalised (previous year: EUR 22m). The financing rate used was 1.3% (previous year: 1.7%).

Order commitments for aircraft and reserve engines amount to EUR 14.3bn (previous year: EUR 13.5bn).

In the aircraft item, aircraft with a carrying amount of EUR 2,550m (previous year: EUR 2,323m) serve as collateral for current financing arrangements.

19 Property, plant and other equipment

T106 PROPERTY, PLANT AND OTHER EQUIPMENT

	Land and buildings	Technical equipment and machinery	Other equipment, operating and office equipment	Advance payments and plant under construction	Total
in €m					
Cost as of 1 Jan 2018	2,648	1,247	1,409	167	5,471
Accumulated depreciation	-1,388	-906	-979	-12	-3,285
Carrying amount 1 Jan 2018	1,260	341	430	155	2,186
Currency translation differences	6	4	5	1	16
Additions due to changes in consolidation	-	-	-	-	-
Additions	38	44	121	94	297
Reclassifications	43	38	33	-118	-4
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-7	-1	-9	-3	-20
Reclassifications to assets held for sale	-	-	-	-	-
Depreciation	-90	-57	-107	-	-254
Reversal of impairment losses	-	-	-	-	-
Carrying amount 31 Dec 2018	1,250	369	473	129	2,221
Cost as of 1 Jan 2019	2,717	1,315	1,509	142	5,683
Accumulated depreciation	-1,467	-946	-1,036	-13	-3,462
Carrying amount 1 Jan 2019	1,250	369	473	129	2,221
First-time application of IFRS 16	1,531	-	19	-	1,550
Adjusted carrying amount 1 Jan 2019	2,781	369	492	129	3,771
Currency translation differences	21	4	2	3	30
Additions due to changes in consolidation	36	-	-	-	36
Additions	692	42	151	102	987
Reclassifications	50	15	18	-98	-15
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-50	-1	-8	-3	-62
Reclassifications to assets held for sale	-112	-22	-56	-9	-199
Depreciation	-327	-58	-122	-	-507
Reversal of impairment losses	-	-	-	-	-
Carrying amount 31 Dec 2019	3,091	349	477	124	4,041
Cost as of 31 Dec 2019	4,715	1,245	1,470	124	7,554
Accumulated depreciation	-1,624	-896	-993	-	-3,513

The increase in property, plant and other equipment was due to the first-time application of IFRS 16. For further information about right-of-use assets, we refer to [Note 20, p. 170ff.](#) In connection with finance lease arrangements formerly covered by IAS 17, there are also leases for property, plant and equipment with a carrying amount of EUR 80m (previous year: EUR 93m).

As in the previous year, charges of EUR 4m exist over land and property. Pre-emption rights are registered for land held at EUR 185m (previous year: EUR 193m). Other property, plant and equipment carried at EUR 4m (previous year: EUR 4m) serves as collateral for existing financing arrangements.

The following items of property, plant and equipment have been ordered, but are not yet at the Lufthansa Group's economic disposal:

T107 ORDERS OF PROPERTY, PLANT AND EQUIPMENT AS OF THE REPORTING DATE

in €m	31 Dec 2019	31 Dec 2018
Land and buildings	20	61
Technical equipment and vehicles	37	31
Operating and office equipment	61	51
	118	143

20 Leases

Leases for items of property, plant and equipment were categorised as either finance or operating leases up to and including 2018. Since 1 January 2019, leases are accounted for as a right-of-use asset and a corresponding lease liability from the time the Lufthansa Group gains control of the leased asset in accordance with IFRS 16. Further details of the change can be found in [Note 2, p. 144ff.](#)

The following table shows the carrying amounts of the recognised right-of-use assets and the changes during the reporting period.

T108 RIGHT-OF-USE ASSETS

	Aircraft and reserve engines	Land and buildings	Technical equipment and machinery	Other intangible assets and technical equipment	Total
in €m					
Cost as of 1 Jan 2018	862	248	1	10	1,121
Accumulated depreciation	- 407	- 152	- 1	- 4	- 564
Carrying amount 1 Jan 2018	455	96	-	6	557
Currency translation differences	3	-	-	-	3
Additions due to changes in consolidation	-	-	-	-	-
Additions	243	-	-	-	243
Reclassifications	2	-	-	-	2
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-	-	-	- 1	- 1
Reclassifications to assets held for sale	-	-	-	-	-
Depreciation	- 125	- 6	-	- 2	- 133
Reversal of impairment losses	-	-	-	-	-
Carrying amount 31 Dec 2018 ¹⁾	578	90	-	3	671
Cost as of 1 Jan 2019	1,055	218	1	9	1,283
Accumulated depreciation	- 477	- 128	- 1	- 6	- 612
Carrying amount 1 Jan 2019	578	90	-	3	671
First-time application of IFRS 16	401	1,531	-	19	1,951
Adjusted carrying amount 1 Jan 2019	979	1,621	-	22	2,622
Currency translation differences	4	14	-	-	18
Additions due to changes in consolidation	-	-	-	-	-
Additions	227	642	-	13	882
Reclassifications	- 1	- 6	-	-	- 7
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	- 8	- 45	-	- 1	- 54
Reclassifications to assets held for sale	-	- 30	-	- 1	- 31
Depreciation	- 264	- 244	-	- 12	- 520
Reversal of impairment losses	-	-	-	-	-
Carrying amount 31 Dec 2019	937	1,952	-	21	2,910
Cost as of 31 Dec 2019	1,588	2,225	-	38	3,851
Accumulated depreciation	- 651	- 273	-	- 17	- 941

¹⁾ In the previous year these were finance lease assets within the meaning of IAS 17.

The Lufthansa Group mainly leases property, particularly at airports, as well as aircraft and other operating and office equipment. Leases may include renewal and termination options. The terms of the leases are negotiated individually and cover a wide range of different areas. Longer-term leases exist particularly for property. There is a residual lease term of up to 36 years for land and buildings (previous year: up to 29 years) as of the reporting date. EUR 359m of the increase in right-of-use assets for land and buildings in 2019 was due to a long-term land lease at Frankfurt Airport.

As a rule, the aircraft leases that were previously deemed to be in the economic ownership of the Lufthansa Group under IAS 17 cannot be terminated during a fixed basic lease term of at least four years and they run for a maximum of 24 years (previous year: 23 years). New right-of-use assets for aircraft under IFRS 16 have an average residual term of three years.

The leases existing under IFRS 16 affected the consolidated income statement as follows:

T109 LEASE EXPENSES RECOGNISED IN PROFIT OR LOSS

in €m	2019
Amortisation of right-of-use assets	520
Interest expenses for lease liabilities	72
Expenses for short-term leases	193
Expenses for low-value leases	120
Variable lease payments	126

In addition, the Lufthansa Group has signed leases for capacities and unspecified assets; the relevant payments are still recognised in the income statement.

Many of the Lufthansa Group's leases for properties (including airports) and aircraft include renewal options and variable lease payments. These options are used to obtain the greatest possible flexibility in terms of capacities. They have not been taken into account in various cases when measuring the lease liability, because it is not reasonably certain that they will be exercised. Potential future lease payments for periods after the exercise date of the renewal options are summarised in the following table:

T110 DISCLOSURES ON RENEWAL OPTIONS AND VARIABLE LEASE PAYMENTS

in €m	Recognised lease liability (discounted)	Potential future lease payments not included in lease liabilities (undiscounted payments)		Total
	31 Dec 2019	Payable 2021 – 2025	Payable after 2026	
Aircraft	486	71	–	71
Property ¹⁾	1,929	55	292	347
Total	2,415	126	292	418

¹⁾ Of which EUR 27m lease liabilities from held-for-sale entities and EUR 3m from potential future lease payments.

As of 31 December 2019, there were no renewal options for aircraft that had been classified as not reasonably certain.

There are also future payments subject to termination options, but the Lufthansa Group considers these to be immaterial.

Amounts included in the cash flow statement are shown in the following table:

T111 CASH OUTFLOWS FOR LEASES¹⁾

in €m	2019
Lease expenses from short-term and low-value leases and variable lease payments not included in the measurement of lease liabilities	439
Repayment of the redemption portion of the lease liability	499
Interest payments	64
Total	1,002

¹⁾ Disclosures include the former IAS 17 finance lease liabilities and the new IFRS 16 lease liabilities.

Lease payments are shown from 2019 as cash flows from financing activities, unless they relate to short-term or low value leases or are variable lease payments. This positively altered cash flows from operating activities in the reporting year by EUR 563m, of which EUR 432m is due to the new lease liabilities for right-of-use assets in accordance with IFRS 16.

The maturity analysis of lease liabilities is shown under borrowings, [Note 35, p. 188](#).

Information about operating leases in which the Lufthansa Group is lessor can be found in [Note 6, p. 159](#).

21 Equity investments accounted for using the equity method

T112 EQUITY INVESTMENTS ACCOUNTED FOR USING THE EQUITY METHOD

in €m	Investments in joint ventures	Investments in associated companies	Total
Cost as of 1 Jan 2018	350	239	589
Accumulated impairment losses	-	-4	-4
Carrying amount 1 Jan 2018	350	235	585
Currency translation differences	3	1	4
Additions due to changes in consolidation	-	-	-
Additions	32	-	32
Changes with and without an effect on profit and loss	75	37	112
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Dividends paid	-69	-14	-83
Reclassifications to assets held for sale	-	-	-
Impairment losses	-	-	-
Reversal of impairment losses	-	-	-
Carrying amount 31 Dec 2018	391	259	650
Cost as of 1 Jan 2019	391	263	654
Accumulated impairment losses	-	-4	-4
Carrying amount 1 Jan 2019	391	259	650
Currency translation differences	3	9	12
Additions due to changes in consolidation	-	-	-
Additions	77	-	77
Changes with and without an effect on profit and loss	70	73	143
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Dividends paid	-62	-101	-163
Reclassifications to assets held for sale	-	-4	-4
Impairment losses	-43	-	-43
Reversal of impairment losses	-	-	-
Carrying amount 31 Dec 2019	436	236	672
Cost as of 31 Dec 2019	479	241	720
Accumulated impairment losses	-43	-5	-48

For the write-down on shares in joint ventures in 2019

[Note 9, p. 160f](#).

Individual interests in companies accounted for using the equity method

The following tables contain summarised data from the income statements and data from the statement of financial position for the individual material joint ventures accounted for using the equity method.

T113 BALANCE SHEET DATA GÜNES EKSPRES HAVACILIK ANONIM SİRKETİ (SUNEXPRESS), ANTALYA, TURKEY

in €m	31 Dec 2019	31 Dec 2018 ¹⁾
Current assets	594	400
of which cash and cash equivalents	317	196
Non-current assets	1,204	820
Current liabilities	769	404
Non-current liabilities	715	559
Current financial liabilities (except trade and other payables and provisions)	312	114
Non-current financial liabilities (except trade and other payables and provisions)	583	412
Shareholders' equity	314	257
Share of equity	157	129
Other	21	19
Carrying amount	178	148

¹⁾ Previous year's figures have been adjusted.

T114 INCOME STATEMENT DATA GÜNES EKSPRES HAVACILIK ANONIM SİRKETİ (SUNEXPRESS), ANTALYA, TURKEY

in €m	2019	2018
Revenue	1,432	1,261
Depreciation and amortisation	148	53
Interest income	6	6
Interest expenses	25	15
Income tax expense or income	14	34
Profit or loss from continuing operations	48	47
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	9	-6
Total comprehensive income	57	41
Share of profit or loss from continuing operations	24	24
Share of comprehensive income	31	21

The item "Other" in the reconciliation with the carrying amount for SunExpress includes the difference from the first-time consolidation of the company.

T115 BALANCE SHEET DATA
TERMINAL 2 GESELLSCHAFT MBH & CO. OHG,
MUNICH AIRPORT, GERMANY

in €m	31 Dec 2019	31 Dec 2018
Current assets	46	80
of which cash and cash equivalents	–	4
Non-current assets	1,323	1,381
Current liabilities	400	266
Non-current liabilities	908	1,133
Current financial liabilities (except trade and other payables and provisions)	256	122
Non-current financial liabilities (except trade and other payables and provisions)	880	1,103
Shareholders' equity	61	62
Share of equity	25	25
Other	–	–
Carrying amount	25	25

T116 INCOME STATEMENT DATA
TERMINAL 2 GESELLSCHAFT MBH & CO. OHG,
MUNICH AIRPORT, GERMANY

in €m	2019	2018
Revenue	357	357
Depreciation and amortisation	84	84
Interest income	–	–
Interest expenses	36	39
Income tax expense or income	10	10
Profit or loss from continuing operations	78	77
Profit or loss after tax from discontinued operations	–	–
Other comprehensive income	16	8
Total comprehensive income	94	85
Share of profit or loss from continuing operations	31	31
Share of comprehensive income	38	34
Dividends received	38	37

The following table contains summarised aggregated data from the income statements and carrying amounts for the individually immaterial joint ventures accounted for using the equity method.

T117 INCOME STATEMENTS DATA AND CARRYING AMOUNTS
OF JOINT VENTURES ACCOUNTED FOR USING
THE EQUITY METHOD

in €m	2019	2018
Profit or loss from continuing operations	1	23
Profit or loss after tax from discontinued operations	–	–
Other comprehensive income	–	–
Total comprehensive income	1	23
Carrying amount	233	218

The following table contains summarised aggregated data from the income statements and carrying amounts for the individually immaterial associated companies accounted for using the equity method.

T118 INCOME STATEMENTS DATA AND CARRYING AMOUNTS
OF ASSOCIATED COMPANIES ACCOUNTED FOR USING
THE EQUITY METHOD

in €m	2019	2018
Profit or loss from continuing operations	74	39
Profit or loss after tax from discontinued operations	–	–
Other comprehensive income	–1	–2
Total comprehensive income	73	37
Carrying amount	236	259

22 Other equity investments and non-current securities

T119 OTHER EQUITY INVESTMENTS AND NON-CURRENT SECURITIES

in €m	31 Dec 2019	31 Dec 2018
Investments in affiliated companies	223	222
Investments	33	24
Other investments	256	246
Non-current securities	53	41

Shares in related parties include shares in affiliated companies, joint ventures and associates that are not consolidated for reasons of materiality. These shares are carried at amortised cost. Disclosures on the equity investments and non-current securities can be found in [Note 43, p. 200ff.](#)

In the current financial year, other equity investments held at EUR 1m (previous year: EUR 8m) were sold for a loss of EUR 0m (previous year: loss of EUR 1m).

23 Non-current loans and receivables

T120 NON-CURRENT LOANS AND RECEIVABLES

in €m	31 Dec 2019	31 Dec 2018
Loans to and receivables from affiliated companies	104	108
Loans to and receivables from other equity investments	–	–
Other loans and receivables	241	351
Emissions certificates	124	53
	469	512

Non-current loans and receivables are carried at amortised cost.

For the impairment test for emissions certificates, we refer to the disclosures on the cash-generating units (CGU) Lufthansa German Airlines, SWISS, Austrian Airlines, Brussels Airlines and Eurowings in [Note 16, p. 164ff.](#)

Other receivables include expected reimbursements for obligations for which provisions have been made amounting to EUR 2m (previous year: EUR 2m). Of the non-current receivables, EUR 1m (previous year: EUR 32m) serve as collateral for liabilities.

24 Inventories

T121 INVENTORIES

in €m	31 Dec 2019	31 Dec 2018
Raw materials, consumables and supplies	888	813
Finished goods and work in progress	88	150
Advance payments	4	5
	980	968

No inventories have been pledged as collateral for loans. EUR 782m of the inventories (previous year: EUR 695m) consist of non-repairable spare parts for aircraft.

The gross value of written-down inventories as of 31 December 2019 was EUR 998m (previous year: EUR 859m). Inventories with a carrying amount of EUR 716m (previous year: EUR 604m) are held at net realisable value. Write-downs to net realisable value of EUR 259m were made at the beginning of the financial year (previous year: EUR 232m). New impairment losses of EUR 41m were recognised in the reporting year (previous year: EUR 36m). Write-downs of EUR 18m made in the previous year were reversed (previous year: EUR 13m). In addition, accumulated impairment losses of EUR 3m were reclassified to assets held for sale.

25 Contract assets

The Lufthansa Group recognised the following contract assets in 2019:

T122 CONTRACT ASSETS

in €m	31 Dec 2019	31 Dec 2018
Contract assets from MRO and IT services	278	234
Impairment of contract assets	–1	–
Total contract assets	277	234

26 Trade receivables and other receivables

T123 TRADE RECEIVABLES AND OTHER RECEIVABLES

in €m	31 Dec 2019	31 Dec 2018
Trade receivables		
Trade receivables from affiliated companies	58	67
Trade receivables from other equity investments	2	7
Trade receivables from third parties	3,802	3,938
	3,862	4,012
Other receivables		
Receivables from affiliated companies	89	93
Receivables from other equity investments	1	–
Other receivables	1,370	1,408
Emissions certificates	95	63
	1,555	1,564
Total	5,417	5,576

For the impairment test for emissions certificates, we refer to the disclosures on the cash-generating units (CGU) Lufthansa German Airlines, SWISS, Austrian Airlines, Eurowings and Brussels Airlines in [Note 16, p. 164ff.](#)

Collateral received for trade receivables has a fair value of EUR 0m (previous year: EUR 2m). Other receivables include expected reimbursements for obligations for which provisions have been made amounting to EUR 1m (previous year: EUR 26m).

For disclosures on impairment losses, credit risks and term structures, we refer to [Note 43, p. 200ff.](#)

Other receivables include claims of EUR 149m (previous year: EUR 154m) against insurers in connection with the accident involving the Germanwings aircraft on 24 March 2015. As of the reporting date, these receivables are offset by provisions of EUR 135m for outstanding obligations relating to this accident (previous year: EUR 139m).

Other receivables of EUR 0m (previous year: EUR 34m) serve to secure negative market values of derivatives.

27 Deferred charges and prepaid expenses

Deferred charges and prepaid expenses consist of various services paid for in advance for subsequent periods.

28 Current securities

Current securities are fixed income securities, participation certificates, shares and investments in money market funds.

29 Cash and cash equivalents

The item does not include any material fixed-term deposits (previous year: EUR 66m) with terms of four to twelve months.

Bank balances in foreign currencies are translated at the exchange rate on the balance sheet date.

30 Assets held for sale

Assets with a carrying amount of EUR 369m were held for sale as of year-end 2019. These largely consist of the assets in the disposal group "European activities of the LSG group". The related assets and liabilities were reclassified as of 30 September 2019 in view of the high probability of a sale within twelve months. On 6/7 December 2019, a contract was signed with Gategroup Holding AG for shares and some assets attributable to the catering activities of the Lufthansa Group in Europe. The agreement is still subject to conditions precedent, in particular approval by the competition authorities. The transaction is expected to be closed in the first half of 2020.

Impairment testing as of 31 December 2019 resulted in a write-down of EUR 50m regarding the expected sale price to reflect the concrete underlying valuation assumptions following the conclusion of the contract. ➤ **Note 10, p. 161.** The following table shows the now consolidated assets and liabilities in terms of main categories:

T124 ASSETS AND LIABILITIES IN THE DISPOSAL GROUP "EUROPEAN BUSINESS OPERATIONS OF THE LSG GROUP" AFTER CONSOLIDATION AS OF 31 DECEMBER 2019 AND OTHER RECEIVABLES

in €m	31 Dec 2019
Assets	
Intangible assets	1
Property, plant and other equipment	151
Other non-current assets	13
Deferred tax assets	31
Inventories	80
Other current assets	82
Total	358
Shareholders' equity and liabilities	
Pension provisions	289
Other non-current provisions	16
Other non-current liabilities	32
Other current provisions	11
Trade payables and other non-financial liabilities	175
Other current liabilities	17
Total	540

In addition to the assets in the disposal group, the assets held for sale include three freighter aircraft and one building (EUR 11m in total). The carrying amount in the previous year was EUR 9m and related to two aircraft and one building.

Shareholders' equity and liabilities

Issued capital

SHARE CAPITAL

Deutsche Lufthansa AG's issued capital totals EUR 1,224m. Issued capital is divided into 478,194,257 registered shares, with each share representing EUR 2.56 of issued capital.

AUTHORISED CAPITAL

A resolution passed at the Annual General Meeting on 7 May 2019 authorised the Executive Board until 6 May 2024, subject to approval by the Supervisory Board, to increase the Company's issued capital by up to EUR 450,000,000 by issuing new registered shares on one or more occasions for payment in cash or in kind (Authorised Capital A). In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board.

A resolution passed at the Annual General Meeting on 7 May 2019 authorised the Executive Board until 6 May 2024, subject to approval by the Supervisory Board, to increase the issued capital by up to EUR 30,000,000 by issuing new registered shares to employees (Authorised Capital B) for payment in cash. Existing shareholders' subscription rights are excluded. In order to issue new shares to employees of Deutsche Lufthansa AG and its affiliated companies, the Executive Board of Deutsche Lufthansa AG decided, with the approval of the Supervisory Board, to make use of the authorisation voted at the Annual General Meeting on 7 May 2019 (Authorised Capital B) and increase the Company's issued capital by EUR 7,637,831.68 excluding shareholders' subscription rights, by issuing 2,983,528 new registered shares with transfer restrictions and profit entitlement from 1 January 2019 for payment in cash. The capital increase was entered in the Commercial Register of Cologne District Court (HRB 2168) on 25 October 2019. As of 31 December 2019, Authorised Capital B amounts to EUR 22,362,168.32.

CONTINGENT CAPITAL

A resolution passed at the Annual General Meeting on 28 April 2016 authorised the Executive Board until 27 April 2021, subject to approval by the Supervisory Board, to issue bearer or registered convertible bonds, bond/warrant packages, profit sharing rights or participating bonds (or combinations of these instruments), on one or more occasions, for a total nominal value of up to EUR 1.5bn, with or without restrictions on maturity. To do so, contingent capital (Contingent Capital II) was created for a contingent capital increase of up to EUR 237,843,840 by issuing up to 92,907,750 new registered shares. The contingent capital increase will only take place insofar as the holders of convertible bonds or warrants from bond/warrant packages decide to exercise their conversion and/or option rights.

AUTHORISATION TO PURCHASE TREASURY SHARES

A resolution passed at the Annual General Meeting held on 7 May 2019 authorised the Executive Board pursuant to Section 71 Paragraph 1 No. 8 Stock Corporation Act (AktG) to purchase treasury shares until 6 May 2024. The authorisation is limited to 10% of current issued capital, which can be purchased on the stock exchange or by a public purchase offer to all shareholders. The authorisation states that the Executive Board can use the shares, in particular, for the purposes defined in the resolution passed at the Annual General Meeting. According to the resolution of the Annual General Meeting held on 7 May 2019, the Executive Board is also authorised to purchase treasury shares by means of derivatives and to conclude corresponding derivative transactions.

At various points in 2019, Deutsche Lufthansa AG bought back 95,753 of its own shares at an average price of EUR 17.27. This corresponds to 0.02% or EUR 245,127.68 of share capital.

The shares purchased or created by means of the capital increase were used as follows:

- 2,098,795 shares were transferred to the employees of Deutsche Lufthansa AG and to 34 other affiliated companies and equity investments as part of the profit-sharing for 2018, at a share price of EUR 14.18.
- 974,540 shares were transferred as part of performance-related variable remuneration in 2019 to managers and non-payscale employees of Deutsche Lufthansa AG and to 45 further affiliated companies and equity investments at a price of EUR 17.43.
- 377 shares were transferred for previous years' programmes (performance-related variable remuneration for 2017 and 2018 to managers, non-payscale employees and other employees of Deutsche Lufthansa AG and to further affiliated companies and equity investments from profit-sharing for 2017) at a price of EUR 29.58.
- 5,569 shares were resold at a price of EUR 16.54.

On the balance sheet date, treasury shares were no longer held.

CAPITAL MANAGEMENT

The Lufthansa Group continues to aim for a sustainable equity ratio of 25%, in order to ensure long-term financial flexibility and stability as a basis for its growth targets for the Group. As of 31 December 2019 and 2018, equity and total assets were as follows:

T125 EQUITY AND LIABILITIES

in €m	31 Dec 2019	31 Dec 2018
Shareholders' equity	10,256	9,573
In % of total assets	24.0	25.1
Liabilities	32,403	28,640
In % of total assets	76.0	74.9
Total capital	42,659	38,213

In the financial year 2019, the equity ratio declined by 1.1 percentage point compared with the figure for the previous year to 24.0%. The IFRS 16 accounting change reduced the equity ratio by 1.5 percentage points.

Deutsche Lufthansa AG's Articles of Association do not stipulate any capital requirements.

32 Reserves

Capital reserves only include the share premium paid on capital increases and a convertible bond that was redeemed in full in previous years. The legal reserve contained in retained earnings is unchanged at EUR 26m; other reserves consist of other retained earnings.

The following table shows changes in other neutral reserves in 2019:

T126 NOTES ON OTHER COMPREHENSIVE INCOME

in €m	2019	2018
Other comprehensive income after income taxes		
Currency translation differences		
Profit/loss for the period	115	124
Reclassification adjustments recognised in profit or loss	-	-
Subsequent measurement of financial assets at fair value (with recycling)	10	-10
Subsequent measurement of financial assets at fair value (without recycling)	7	5
Profit/loss for the period	-	-3
Reclassification adjustments recognised in profit or loss	-	-2
Subsequent measurement of hedges – cash flow hedge reserve	258	434
Subsequent measurement of hedges – costs of hedging	476	-90
Profit/loss for the period	1,278	401
Reclassification adjustments recognised in profit or loss	-544	-57
Other comprehensive income from investments accounted for using the equity method		
Profit/loss for the period – reclassifiable	13	-2
Profit/loss for the period – non-reclassifiable	-	-
Revaluation of defined-benefit pension plans	-1,108	-974
Other expenses and income recognised directly in equity (with recycling)	15	1
Other expenses and income recognised directly in equity (without recycling)	-1	-
Income taxes on items in other comprehensive income	163	30
Other comprehensive income after income taxes	-52	-482

1127 NOTE ON INCOME TAXES RECOGNISED FOR OTHER COMPREHENSIVE INCOME

in €m	2019			2018		
	Amount before income taxes	Tax expenses/ income	Amount after income taxes	Amount before income taxes	Tax expenses/ income	Amount after income taxes
Currency translation differences	115	-	115	124	-	124
Subsequent measurement of financial assets at fair value (with recycling)	10	-	10	- 10	-	- 10
Subsequent measurement of financial assets at fair value (without recycling)	7	-	7	5	-	5
Subsequent measurement of hedges - cash flow hedge reserve	258	- 64	194	434	- 125	309
Subsequent measurement of hedges - costs of hedging	476	- 118	358	- 90	38	- 52
Other comprehensive income from investments accounted for using the equity method - reclassifiable	13	-	13	- 2	-	- 2
Revaluation of defined-benefit pension plans	- 1,108	345	- 763	- 974	117	- 857
Other expenses and income recognised directly in equity (with recycling)	15	-	15	1	-	1
Other expenses and income recognised directly in equity (without recycling)	- 1	-	- 1	-	-	-
Other comprehensive income	- 215	163	- 52	- 512	30	- 482

The overall change in equity is shown in **T073 Consolidated statement of changes in shareholders' equity**, p. 142.

33 Pension provisions

The Group's pension obligations comprise both defined benefit and defined contribution plans and include both obligations to make current payments and entitlements to future pension payments.

Obligations under defined benefit pension plans for employees of the Lufthansa Group related mostly to pension obligations in Germany, Switzerland, Austria and the USA. Various commitments have been made to different groups of employees.

For the employees in Germany and for staff posted abroad by German companies who joined the Company before 1995, the supplementary pension scheme for state employees (VBL) was retained as the Company's pension scheme after the Lufthansa Group's privatisation. Employees who joined after 1994 received a retirement benefit commitment based on an average salary plan, which provided for pension units to be granted annually, with the size of these components depending on the employee's age and salary. In 2003, the VBL pension scheme was changed to the average salary plan applicable to the employees recruited after 1995. Since 2015, the existing domestic retirement benefit commitments to the individual groups of employees have been converted successively into defined contribution schemes.

The Lufthansa collective agreement on benefits for ground staff established a new Company retirement benefit plan in the form of a defined contribution benefit commitment for the ground staff in Germany, in particular those at Deutsche Lufthansa AG, Lufthansa Cargo AG, the Lufthansa Technik group and the LSG group. For employees recruited before 1 January 2016, the entitlements vested up until 31 December 2015 are maintained. For service periods starting from 1 January 2016, employees can reach the same level of benefits by making contributions from their own pocket. For employees recruited from 1 January 2016, the contributions to the new model will be invested on the capital market. When the employee reaches retirement age, the entire account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum and while guaranteeing the contributions that were originally made.

On 17 March 2017, the "Lufthansa Pension Cabin" wage agreement was signed with the trade union UFO for cabin crew at Deutsche Lufthansa AG (DLH). It replaces the agreements on the "Lufthansa Company Pension" for cabin crew and the "Lufthansa Transitional Benefit for Cabin Crew".

For employees recruited up to 5 July 2016, the pension entitlements vested up until 30 June 2016 are maintained. For service periods from 1 July 2016, these employees receive employer contributions to the Company pension scheme depending on their eligible gross salary. An initial

contribution to the transitional benefit scheme was calculated for the staff concerned as of 30 June 2016 on the basis of parameters and valuation methods defined by the collective bargaining partners. This initial transitional benefit contribution will replace all existing claims by the employees concerned under the collective agreement on “Transitional Benefit for Cabin Crew” and will be switched to a contribution commitment with a minimum guaranteed payment. All employees are free to make their own contributions on a voluntary basis. Contributions from both employer and employee, as well as the initial transitional benefit contribution, are invested on the capital markets with a capital guarantee. When the employee reaches retirement age, the available account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum.

On 21 December 2017, the “Lufthansa Pension Cockpit” wage agreement for cockpit staff was signed with the Vereinigung Cockpit pilots’ union. At the same time, a new “Transitional Benefit Cockpit” wage agreement was signed.

For employees recruited before 1 January 2017, the pension entitlements vested up until 31 December 2016 are maintained. For service periods from 1 January 2017, the employees receive employer contributions to the Company pension scheme depending on their eligible gross salary. All employees are free to make their own contributions on a voluntary basis. The capital is invested on capital markets with a capital guarantee and the guaranteed interest rate offered by the life insurance companies (currently 0.9% per annum) as an additional commitment. When the employee reaches retirement age, the available account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum.

In the new Company retirement benefit scheme for ground, cabin and cockpit staff, the obligations from the capital market-oriented components are recognised at the time value of the corresponding assets, insofar as the assets exceed the minimum guaranteed amount. Plan assets and benefit obligations are presented on a net basis. The employer contributions constitute service expense.

Cockpit staff are still additionally entitled to a transitional pension arrangement covering the period from the end of their active in-flight service until the beginning of their statutory/Company pension plans. Benefits depend on the number of years of service and the final salary before retirement (final salary plans). Pension entitlements continue to accrue while transitional benefits are being received.

In line with the revised version of the wage agreement, the old rules continue to apply to transitional benefits for cockpit staff. The collective retirement age for pilots will go up in stages from 58 to 60 by 2021. The benefits in the latest version have been extended to cover cockpit staff at Germanwings who were recruited before 1 January 2005.

Defined-benefit Company pension schemes and transitional pension arrangements for Germany are funded by plan assets, while amounts that have not yet been transferred are covered by pension provisions.

There are no minimum funding requirements in Germany.

In the course of acquiring Swiss International Air Lines AG, pension obligations, mainly statutory obligations, were taken on in Switzerland. The retirement benefits are funded via pension funds known as collective foundations. In addition to retirement benefits, the plans cover invalidity and dependant persons’ benefits. Beneficiaries can choose between an annuity and a lump sum payment. The retirement age for the plans lies between 58 and 63 years. Contributions to the pension funds are made by employers and employees, whereby the Company contributions must be at least equal to the employee contributions defined in the terms of the plan. Contributions are deducted from the qualifying salary according to a sliding scale. If there is a deficit of plan assets, employer and employee contributions can be increased, a lower return can be determined or other steps permissible by law can be taken. The decision is taken by the trustees of the pension fund concerned. The trustees’ strategies for making good a deficit are based on the report by a pension fund expert and must be presented to the regulatory authority. The approval of the authority is not required, however.

The pension obligations for employees of Austrian Airlines AG are mostly on a defined contribution basis and have been outsourced to a pension fund. They consist of retirement, invalidity and dependant persons’ benefits.

Obligations under defined benefit plans at Austrian Airlines AG relate to former directors and Executive Board members and others already drawing their pensions. Obligations under defined benefit plans for ground staff are now contribution-free and are determined by converting plan assets into an annuity. There are no defined benefit plans but only defined contribution pension obligations for active pilots, flight attendants and members of the top management level.

The defined benefit pension plans at LSG Sky Chefs in the USA are largely closed to new entrants and no further benefits are being granted to beneficiaries still in service. Benefit payments are based on average salary and the years of service acquired before the plan was closed or frozen. The retirement age is 65. Pension payments are funded externally. Retirement benefits have been switched to defined contribution plans.

Other staff abroad are also entitled to minor retirement benefits and in some cases to medical care based mainly on length of service and salary earned. As a rule, benefits are financed by means of external funds.

Contributions for defined contribution retirement benefit commitments came to EUR 499m in 2019 (previous year: EUR 470m).

In the financial year 2004, work began on building up plan assets to fund and safeguard future pension payments. The aim was to fund the pension obligations under existing plans in Germany in full. Contractual trust arrangements (CTAs) in the form of a mutual two-stage trusteeship were set up for this purpose.

The main trustee is Lufthansa Pension Trust e.V., a separate legal entity subject to German regulations. Deutsche Lufthansa AG and the trustees/other trustors agree on contributions and, if such a contribution is determined, make a payment to Lufthansa Pension Trust e.V. Deutsche Lufthansa AG and its subsidiaries Lufthansa Technik AG and Lufthansa Cargo AG are parties to the contractual trust arrangement.

The trust assets have largely been held by a Maltese corporate vehicle since 2007. The Investment Board of Lufthansa Malta Pension Holding decides on the fund's asset allocation. The asset management itself is delegated to fund management companies, who invest the assets in accordance with the general investment principles defined by the Investment Board.

The assets to fund pension obligations in the new Lufthansa Pension Ground, Lufthansa Pension Cabin and Lufthansa Pension Cockpit capital market-based benefits system were transferred to an external trustee, Deutsche Treuinvest Stiftung, as part of a contractual trust arrangement. Capital is invested in what are known as age group funds, whose investment strategy is based on a life cycle model. As employees get older, less and less is invested in asset classes with a higher risk-return profile and a greater percentage in more conservative asset classes. The Company has set up an Investment Committee that is responsible for defining and monitoring the investment strategy, e.g. how the age group funds are composed and how the asset allocation changes over time.

Assets to fund pension obligations for other German subsidiaries have also been invested with Deutsche Treuinvest Stiftung.

EUR 250m was contributed to plan assets for employees in Germany in the reporting year (previous year: EUR 330m).

Amounts shown in the statement of financial position for defined benefit commitments are made up as follows:

T128 DEFINED-BENEFIT RETIREMENT COMMITMENTS

	31 Dec 2019				31 Dec 2018			
	Defined-benefit obligations (DBO)	Fair value of plan assets	Effect of asset ceiling	Net carrying amount for defined-benefit obligations	Defined-benefit obligations (DBO)	Fair value of plan assets	Effect of asset ceiling	Net carrying amount for defined-benefit obligations
in €m								
Retirement benefits Germany	18,227	- 13,135	-	5,092	15,627	- 11,446	-	4,181
Transitional benefits Germany	1,462	- 441	-	1,021	1,391	- 395	-	996
Switzerland	3,840	- 3,434	2	408	3,224	- 2,921	-	303
Austria	450	- 183	-	267	427	- 184	-	243
USA	369	- 327	-	42	339	- 279	-	60
Other countries	517	- 440	-	77	404	- 351	-	53
Carrying amounts	24,865	- 17,960	2	6,907	21,412	- 15,576	-	5,836
of which pension provisions	-	-	-	6,658	-	-	-	5,865
of which other assets	-	-	-	40	-	-	-	29
of which shown under liabilities for disposal	-	-	-	289	-	-	-	-

Reconciliation between the funding status and the amounts shown in the consolidated balance sheet is as follows:

T129 RECONCILIATION FUNDING STATUS

in €m	2019	2018
Present value of funded pension obligations	24,374	20,847
Plan assets	- 17,960	- 15,576
Funding status (net)	6,414	5,271
Present value of unfunded pension obligations	491	565
Adjustment for asset ceiling	2	-
Carrying amounts	6,907	5,836
of which pension provisions	6,658	5,865
of which other assets	40	29
of which shown under liabilities for disposal	289	-

During the reporting period, the present value of defined benefit pension obligations changed as follows:

T130 CHANGE IN PRESENT VALUE OF PENSION OBLIGATIONS

in €m	2019	2018
Balance on 1 Jan	21,412	21,006
Current service costs	514	552
Interest expenses	418	388
Past service cost/effects of curtailments	11	- 113
Effects of settlements	-	-
Revaluations		
Actuarial gains/losses from changes in demographic assumptions	- 64	146
Actuarial gains/losses from changes in financial assumptions	2,461	- 246
Experience adjustments	281	- 119
Currency translation differences	154	137
Changes in the group of consolidated companies	-	-
Plan contributions - employees	116	124
Pension payments	- 449	- 463
Settlement payments	-	-
Other ¹⁾ /reclassifications	11	-
As of 31 Dec	24,865	21,412
of which shown under liabilities for disposal	- 755	-

¹⁾ The amounts are partly for benefit obligations which were measured in accordance with IAS 19 for the first time.

Actuarial gains/losses from changes in financial assumptions include losses due to the fall in the discount rate for Germany, Austria and Switzerland compared with the previous year. Adjustments to obligations regarding capital market-based pension plans, which are due to changes in exchange rates, are shown in adjustments based on past experience.

The following table provides a detailed reconciliation of changes in the fair value of plan assets:

T131 CHANGE IN FAIR VALUE OF PLAN ASSETS

in €m	2019	2018
Balance on 1 Jan	15,576	15,917
Interest income	299	291
Revaluations		
Income from plan assets, without amounts included in interest	1,571	- 1,195
Currency translation differences	139	121
Changes in the group of consolidated companies	-	-
Plan contributions - employers	394	463
Plan contributions - employees	111	124
Pension payments	- 138	- 126
Settlement payments	-	-
Administrative costs related to obligations	- 3	- 3
Other ¹⁾ /reclassifications	11	- 16
As of 31 Dec	17,960	15,576
of which shown under liabilities for disposal	- 466	-

¹⁾ The amounts are partly for benefit obligations which were measured in accordance with IAS 19 for the first time.

In the financial years 2019 and 2018, pension provisions developed as follows:

T132 PENSION PROVISIONS

in €m	2019	2018
Balance on 1 Jan	5,865	5,116
Currency translation differences carried forward	15	16
Changes in the group of consolidated companies	-	-
Pensions payments	- 311	- 337
Current service costs	514	552
Interest expenses	418	388
Interest income	- 299	- 291
Effects of amendments incl. curtailments, settlements and administrative costs	14	- 110
Revaluations		
Actuarial gains/losses and experience adjustments	2,678	- 219
Income from plan assets, without amounts included in interest	- 1,571	1,195
Net effect of adjustments for asset ceiling	2	- 2
Plan contributions/reclassifications	- 378	- 443
As of 31 Dec	6,947	5,865
of which shown under liabilities for disposal	- 289	

Expenses and income for defined-benefit plans are made up as follows:

T133 EXPENSES AND INCOME FOR DEFINED-BENEFIT PENSION PLANS

in €m	2019	2018
Current service costs	514	552
Past service cost/effects of curtailments	11	- 113
Income from settlements	-	-
Accrued interest on projected pension obligations	418	388
Interest income on plan assets	- 299	- 291
Administrative costs related to obligations	3	3
Balance of expenses and income recognised in the income statement	647	539
Income from plan assets, without amounts included in interest	- 1,571	1,195
Actuarial gains and losses	2,678	- 219
Net effect of adjustment for asset ceiling	2	- 2
Other comprehensive income	1,109	974
	1,756	1,513

Interest expenses on pension provisions and interest income on plan assets are shown in the financial result. Current service expense and past service expense are recognised in staff costs.

A profit of EUR 1,870m was generated from plan assets in the financial year 2019. This amount is made up of the interest income recognised in the income statement and the revaluation component for plan assets. Total expenses of EUR 904m were recognised in the previous year.

There were no significant effects from the asset ceiling defined in IAS 19.64.

The past service cost incurred in the reporting year stemmed mainly from the agreement on retirement benefit plans for cockpit and cabin staff at Lufthansa CityLine GmbH.

The main actuarial assumptions used to calculate pension obligations and the corresponding plan assets are shown below:

T134 MAIN ACTUARIAL ASSUMPTIONS FOR GERMAN COMPANIES

in %	31 Dec 2019	31 Dec 2018
Interest rate		
Retirement benefits	1.4	2.0
Transitional benefits	1.4	2.0
Salary increase		
Retirement benefits	2.5	2.5
Transitional benefits	2.5	2.5
Pension increase		
Retirement benefits	1.0	1.0
Transitional benefits	1.0	1.0

The Heubeck Actuarial Tables 2018 G were used in the biometric calculations for the German companies in the Group.

T135 MAIN ACTUARIAL ASSUMPTIONS FOR FOREIGN COMPANIES

in %	31 Dec 2019	31 Dec 2018
Interest rates		
Austria	1.4	2.0
Switzerland	0.3	1.1
USA	3.2	4.3
Salary increase		
Austria	1.8	1.9
Switzerland	1.5	1.5
USA	-	-
Pension increase		
Austria	2.0	2.0
Switzerland	0.0	0.0
USA	-	-

The BVG 2015 generation tables are used for the biometric calculations for Switzerland. Country-specific mortality tables are used in the other countries.

The following table shows how the present value of defined benefit obligations would have been affected by changes in the relevant actuarial assumptions for the main pension plans described above:

T136 CHANGE IN ACTUARIAL ASSUMPTIONS, AS OF 2019

	Effect on the defined-benefit contribution as of 31 Dec 2019 in €m	Change in %
Present value of the obligation ¹⁾	24,865	-
Interest rate		
Increase by 0.5 percentage points	22,862	- 8.1
Decrease by 0.5 percentage points	27,201	+ 9.4
Salary trend		
Increase by 0.5 percentage points	25,022	+ 0.6
Decrease by 0.5 percentage points	24,717	- 0.6
Pension trend		
Increase by 0.5 percentage points	25,140	+ 1.1
Decrease by 0.5 percentage points	24,606	- 1.0

¹⁾ Present value of the obligation using the assumptions shown in the "Actuarial assumptions" tables.

T136 CHANGE IN ACTUARIAL ASSUMPTIONS, AS OF 2018

	Effect on the defined-benefit contribution as of 31 Dec 2018 in €m	Change in %
Present value of the obligation ¹⁾	21,412	-
Interest rate		
Increase by 0.5 percentage points	19,712	- 7.9
Decrease by 0.5 percentage points	23,399	+ 9.3
Salary trend		
Increase by 0.5 percentage points	21,556	+ 0.7
Decrease by 0.5 percentage points	21,283	- 0.6
Pension trend		
Increase by 0.5 percentage points	21,662	+ 1.2
Decrease by 0.5 percentage points	21,187	- 1.1

¹⁾ Present value of the obligation using the assumptions shown in the "Actuarial assumptions" tables.

A reduction of 10% in the mortality rates used to calculate the pension obligations increases the life expectancy of the beneficiaries by a given amount depending on their individual ages. It roughly corresponds to an increase of one year in the life expectancy of a male employee who is 55 years old today. A 10% reduction in the mortality rate would therefore increase the present value of the main benefit obligations in Germany and Switzerland by EUR 176m as of 31 December 2019 (previous year: EUR 151m).

The sensitivity analysis examines changes in one assumption and leaves the other assumptions unchanged compared with the original calculation. The effects of any correlation between the individual assumptions are therefore not taken into account.

Plan assets for funded defined benefit pension obligations consist mainly of fixed-income securities, equities and cash and cash equivalents. They do not include financial instruments issued by companies in the Group nor properties used by Group companies.

Plan assets serve solely to meet the defined benefit obligations. Funding these benefit obligations with assets provides security for future payments. In some countries, this takes place on the basis of statutory regulations, while in others (Germany, for example), this takes place on a voluntary basis.

The Lufthansa Group aims to completely cover its German pension obligations by means of cover assets and positive capital market returns in the medium to long term. Additional capital is transferred to the trustees for this purpose. Investment performance plays a crucial role in meeting this target.

The investment strategy for the capital market-based pension plans is initially defined by the Company and is regularly reviewed in the course of an allocation study. Where necessary, it is adjusted by the Investment Committee to reflect changes in capital market requirements. This may result in changes to the investment strategy for amounts that have already been invested.

The Lufthansa Group manages and monitors the financial risks that arise from outsourcing the defined benefit pension obligations. There was no change in the risk management and monitoring processes compared with the previous year. Derivative financial instruments are used, especially to manage foreign exchange risks.

The allocation of the funds to asset classes (e.g. equities) for the defined benefit plans is carried out on the basis of asset-liability matching studies performed by the Lufthansa Group. The Asset-Liability Matching (ALM) study is conducted

every three years with an external adviser in order to review the funding strategy on a regular basis and to make adjustments as necessary. The results of the study should indicate what combination of investments (annuities, equities, etc.) can be used to cover the long-term pension obligations. Step one of this process is for the actuary to draft a long-term forecast charting how the pension obligations will develop.

In addition to this, target figures are needed for the relative return and relative risk as regards coverage of the obligations. Last but not least, a risk budget must also be defined.

A simulation is used to test all permissible investment allocations for their future compliance with these objectives. Those which do not fulfil the criteria are eliminated. Preference is given to allocations that are return-oriented yet conservative and that have a high probability of achieving the investment target.

The results of the ALM study show whether there will be strategic shifts in the existing allocation. Alternative investments (e.g. property, private equity, infrastructure) are currently being further developed.

Plan assets are made up as follows:

T137 COMPOSITION OF PLAN ASSETS

	31 Dec 2019				31 Dec 2018			
	Listed price in an active market in €m	No listed price in an active market in €m	Total in €m	in %	Listed price in an active market in €m	No listed price in an active market in €m	Total in €m	in %
Equities			5,189	28.9			4,003	25.9
Europe	3,299	–			2,847	–		
Other	1,890	–			1,156	–		
Fixed-income securities			6,111	34.0			6,716	43.6
Government bonds	3,053	–			3,246	–		
Corporate bonds	3,058	–			3,470	–		
Share funds	847	–	847	4.7	525	–	525	3.4
Fixed-income funds	1,482	–	1,482	8.3	422	–	422	2.7
Mixed funds ¹⁾	247	–	247	1.4	148	–	148	1.0
Money market investments	1,039	–	1,039	5.8	1,443	–	1,443	9.4
Property			1,015	5.7			792	5.1
Direct investments	–	7			–	7		
Indirect investments	744	264			638	147		
Insurance contracts	–	133	133	0.7	–	174	174	1.1
Bank balances	334	191	525	2.9	311	203	514	2.4
Other investments ²⁾	288	1,084	1,372	7.6	201	638	839	5.4
Total	16,281	1,679	17,960	100.0	14,407	1,169	15,576	100.0

¹⁾ Includes equities and interest-bearing securities.

²⁾ Other investments include, in particular, alternative investments such as hedge funds, commodities and private equity funds.

In addition to various actuarial risks such as interest rate risk, life expectancy risk and the risk of salary increases, the pension plans expose the Group primarily to financial risks in connection with plan assets.

The return on plan assets is assumed at the beginning of the period to be the discount rate, which is determined on the basis of investment grade corporate bonds. For the old pension plans, if the actual return on plan assets is less than the discount rates applied, the net obligation from the pension plan goes up. With the new capital market-based pension plans, a gross obligation is recognised for the time value of the corresponding plan assets, taking the minimum guaranteed amount into account.

The share price risk that arises from the proportion of plan assets invested in equities is considered to be reasonable. The risk of default by bond issuers is limited, because investments are only made in investment grade bonds.

The amount of the net obligation under the old pension plans depends to a large extent on the rates of interest, whereby the current low-interest environment results in a relatively high net obligation. If yields on corporate bonds continue to decline, this would lead to a further increase in defined benefit obligations, which could probably only be partly offset by positive developments in the market values of the corporate bonds held in plan assets.

Based on current knowledge, an estimated EUR 808m is expected to be transferred to pension plans in 2020 (previous year: EUR 737m). The transfers are made up of planned allocations and benefit payments that are not covered by equivalent reimbursements from plan assets. The weighted duration of pension obligations was 18 years as of 31 December 2019 (previous year: 17 years).

Over the next ten years, the following pension payments are forecast for the defined benefit commitments in existence as of the reporting date:

T138 FORECAST MATURITIES OF UNDISCOUNTED PENSION PAYMENTS, AS OF 2019	
	Forecast pension payments 31 Dec 2019
in €m	
2020	574
2021	596
2022	612
2023	637
2024	656
2025 – 2029	3,828

T138 FORECAST MATURITIES OF UNDISCOUNTED PENSION PAYMENTS, AS OF 2018	
	Forecast pension payments 31 Dec 2018
in €m	
2019	531
2020	547
2021	571
2022	586
2023	615
2024 – 2028	3,533

34 Other provisions

Other provisions disclosed in the statement of financial position as non-current and current other provisions are made up as follows:

T139 NON-CURRENT AND CURRENT OTHER PROVISIONS

in €m	31 Dec 2019			31 Dec 2018 ¹⁾		
	Total	Non-current	Current	Total	Non-current	Current
Obligations under partial retirement contracts	20	18	2	31	14	17
Other staff costs	202	159	43	196	150	46
Obligation to return emissions certificates	95	–	95	63	–	63
Onerous contracts	49	20	29	64	34	30
Environmental restoration	30	27	3	27	24	3
Legal proceedings	77	11	66	85	18	67
Restructuring/severance payments	171	79	92	210	119	91
Maintenance of lease aircraft	293	139	154	281	131	150
Warranties	46	–	46	49	–	49
Other provisions	301	37	264	425	47	378
Total	1,284	490	794	1,431	537	894

¹⁾ Previous year's figures have been adjusted.

Provisions for staff costs mainly relate to staff anniversary bonuses and other current obligations.

A provision for the obligation to submit CO₂ emissions certificates to the relevant authorities is recognised for an amount equivalent to the carrying amount of the capitalised CO₂ certificates. If the obligation is not fully covered by available certificates, the outstanding amount of the provision is measured using the market price of the emissions certificates as of the reporting date.

Expected losses from onerous contracts result from ongoing obligations or other contractual relationships in which performance and consideration are out of balance.

Provisions for environmental restoration are based on surveyors' findings and the assumption that all contamination is removed within ten years without any further legal requirements.

Provisions for ongoing legal proceedings were based on an assessment of the likely outcome of the proceedings.

The provisions for the overhaul of leased aircraft mainly relate to obligations for the maintenance, overhaul and repair of aircraft.

Other provisions of EUR 135m (previous year: EUR 139m) relate to outstanding obligations in connection with the accident involving the Germanwings aircraft on 24 March 2015.

Changes in groups of individual provisions in 2019 were as follows:

T140 CHANGES IN OTHER PROVISIONS 2019

in €m	Obligations under partial retirement contracts	Other staff costs	Obligation to return emissions certificates	Expected losses from onerous contracts	Environmental restoration	Legal proceedings
As of 1 Jan 2019	31	196	63	64	27	85
Changes in the group of consolidated companies	-	-	-	-	-	-
Currency translation differences	-	-	-	-	-	-
Utilisation	-57	-30	-73	-11	-2	-14
Increase/addition	54	42	104	16	5	28
Interest added back	-7	2	-	-	-	-
Reversal	-	-1	-	-19	-	-22
Transfers	-1	-7	1	-1	-	-
As of 31 Dec 2019	20	202	95	49	30	77

T140 CHANGES IN OTHER PROVISIONS 2019 (continued)

in €m	Restructuring/severance payments	Maintenance of lease aircraft	Warranties	Other provisions	Total
As of 1 Jan 2019	210	281	49	425	1,431
Changes in the group of consolidated companies	-	-	-	-	-
Currency translation differences	-	4	-	-	4
Utilisation	-77	-95	-10	-85	-454
Increase/addition	45	127	14	84	519
Interest added back	-	-	-	1	-4
Reversal	-6	-24	-6	-34	-112
Transfers	-1	-	-1	-90	-100
As of 31 Dec 2019	171	293	46	301	1,284

Changes in groups of individual provisions in the previous year were as follows:

T140 CHANGES IN OTHER PROVISIONS 2018

in €m	Obligations under partial retirement contracts	Other staff costs	Obligation to return emissions certificates	Expected losses from onerous contracts	Environmental restoration	Legal proceedings
As of 1 Jan 2018	30	186	31	103	28	119
Changes in the group of consolidated companies	-	2	-	-	-	-
Currency translation differences	-	1	-	-	-	-1
Utilisation	-60	-28	-31	-35	-1	-32
Increase/addition	60	35	63	6	-	28
Interest added back	5	2	-	-	-	-
Reversal	-	-2	-	-10	-	-29
Transfers	-4	-	-	-	-	-
As of 31 Dec 2018	31	196	63	64	27	85

T140 CHANGES IN OTHER PROVISIONS 2018 (continued)

in €m	Restructuring/severance payments	Maintenance of lease aircraft	Warranties	Other provisions ¹⁾	Total ¹⁾
As of 1 Jan 2018	208	292	41	457	1,495
Changes in the group of consolidated companies	-	-31	-	1	-28
Currency translation differences	-	7	-	-	7
Utilisation	-69	-84	-13	-161	-514
Increase/addition	82	110	25	195	604
Interest added back	-	-	-	-	7
Reversal	-10	-13	-4	-33	-101
Transfers	-1	-	-	-34	-39
As of 31 Dec 2018	210	281	49	425	1,431

¹⁾ Previous year's figures have been adjusted.

The funding status for provisions for obligations to employees under partial retirement agreements is as follows:

T141 FUNDING STATUS

in €m	2019	2018
Present value of funded obligations under partial retirement agreements	146	166
External plan assets	-141	-156
	5	10
of which other provisions	20	31
of which other assets	15	21

A total of EUR 121m has been transferred to an external trust fund as insolvency insurance for employer's performance arrears as part of partial retirement agreements under which the employee at first works full-time for less pay and then retires early on the same reduced pay. These assets, which fulfil the requirements for plan assets and therefore reduce the gross amount of obligations accordingly, are measured at market value on the balance sheet date.

Assets held for sale include a trust fund of EUR 23m and a provision of EUR 17m for partial retirement agreements.

Obligations under partial retirement agreements were measured in 2019 using an interest rate of 0.01% (previous year: 0.34%).

The following cash outflows are estimated for the non-current portion of the other groups of provisions:

T142 CASH OUTFLOWS FOR NON-CURRENT PROVISIONS, AS OF 2019

in €m	2021	2022	2023	2024 and thereafter
Expected losses from onerous contracts	11	3	-	7
Environmental restoration	3	3	3	18
Restructuring/severance payments	54	23	9	5
Maintenance of aircraft on leases	87	16	12	29
Other provisions	15	6	4	44

At the end of 2018, the corresponding cash outflows were estimated as follows:

T142 CASH OUTFLOWS FOR NON-CURRENT PROVISIONS, AS OF 2018

in €m	2020	2021	2022	2023 and thereafter
Expected losses from onerous contracts	21	8	2	3
Environmental restoration	3	3	3	16
Restructuring/severance payments	47	47	19	8
Maintenance of aircraft on leases	80	19	11	24
Other provisions	23	16	7	31

35 Financial liabilities

Financial liabilities consist of a non-current portion with a residual term of more than one year and a current portion of less than one year, which is shown under current liabilities. The following table shows the total amount of borrowings:

T143 FINANCIAL LIABILITIES 31 DEC 2019

in €m	Total	Non-current	Current
Bonds	1,094	996	98
Liabilities to banks	2,110	1,453	657
Leasing liabilities	2,872	2,384	488
Other loans	3,954	3,563	391
	10,030	8,396	1,634

T143 FINANCIAL LIABILITIES 31 DEC 2018

in €m	Total	Non-current	Current
Bonds	1,007	508	499
Liabilities to banks	1,957	1,401	556
Leasing liabilities	596	497	99
Other loans	3,125	2,602	523
	6,685	5,008	1,677

Collateral was provided for EUR 2m of the liabilities to banks (previous year: EUR 75m).

There were no delays or defaults on payment obligations under these loan agreements in either 2019 or 2018.

The Lufthansa Group applied IFRS 16 for the first time as of 1 January 2019. This means that the Lufthansa Group recognised lease liabilities for leases that had previously been classified as operating leases for the first time. This increased lease liabilities by EUR 2,386m. [Note 2, p. 144ff.](#)

The Lufthansa Group's lease liabilities have the following term structure: The disclosures are based on contractual, undiscounted payments.

T144 MATURITY ANALYSIS OF LEASE LIABILITIES

in €m	31 Dec 2019
1st quarter	146
Up to 1 year ¹⁾	392
1–5 years	1,350
Later	1,461

¹⁾ Without payments in 1st quarter.

Other loans include in particular aircraft financing transactions that are described in [Note 18, p. 168](#). A total of 19 additional aircraft financing arrangements were concluded in 2019.

36 Non-current contract liabilities

T145 NON-CURRENT CONTRACT LIABILITIES

in €m	31 Dec 2019	31 Dec 2018
Non-current contract liabilities	25	22
	25	22

Non-current contract liabilities consist of long-term deferrals for construction contracts where the payments received exceed the performance to date.

37 Non-current advance payments received, deferred income and other non-financial liabilities

T146 NON-CURRENT ADVANCE PAYMENTS RECEIVED, DEFERRED INCOME AND OTHER NON-FINANCIAL LIABILITIES

in €m	31 Dec 2019	31 Dec 2018
Advance payments received	7	4
Deferred income	13	21
Other non-financial liabilities	12	26
	32	51

Deferred income includes EUR 7m (previous year: EUR 8m) for grants and subsidies received for capital expenditure, which are realised over the useful life of the assets in the following years.

Other non-financial liabilities include obligations under share-based remuneration agreements for Executive Board members, managers and non-payscale employees. As part of the share-based remuneration agreements, the Lufthansa Group and other consolidated and non-consolidated Group companies participating in the programme offer a 50% discount on employee investment in Lufthansa shares to Executive Board members, managers and non-payscale employees. The option packages granted in 2016, 2017, 2018 and 2019 consist of an outperformance option and a performance option. At the end of the programme, the participants receive a cash payment if the conditions are met.

The outperformance option is linked to the performance of the Lufthansa share compared with a notional index of shares in European competitors. When the beneficiary exercises the outperformance option, they receive a cash payment for every percentage point of outperformance, with a hurdle rate of 1%. The cash payment is capped at an outperformance of more than 20%.

The performance option is linked to the absolute performance of the Lufthansa share. The amount of the payment depends on meeting defined performance targets, whereby both a hurdle rate and a cap apply.

The programmes are scheduled to run for four years. The performance and the outperformance in all programmes are calculated on the principle of total shareholder return. The shares invested in personally may not be sold until the option is exercised.

T147 OUTPERFORMANCE OPTION – OVERVIEW

	€ per outperformance level	Maximum per tranche in €
Board member	1,000 per percentage point from 1%	20,000
Managers	400 per percentage point from 1%	8,000
Non-payscale staff	200 per 5 percentage points from 1%	1,000

T148 PERFORMANCE OPTION – PERFORMANCE TARGETS

Performance option per year	Hurdle rate	Cap
2016	27%	41%
2017	23%	35%
2018	22%	33%
2019	22%	33%

T149 PERFORMANCE OPTION – PAYMENT AMOUNTS

	€ per performance level	Maximum per tranche in €
Board member	10,000 + 1,000 per performance unit	20,000
Managers	4,000 + 500 per performance unit	8,000
Non-payscale staff	500 + 100 per performance unit	1,000

A different system of variable remuneration has applied to the Executive Board since 2019. Under this system, one of the financial targets for the share-based remuneration compares the performance of the Lufthansa share, expressed as the total shareholder return (TSR), to other DAX companies. The performance period is still four years. To calculate the TSR in the performance period, the arithmetic mean closing prices in XETRA trading at Deutsche Börse for shares in Deutsche Lufthansa AG and its peer group over the 30 days immediately preceding the beginning of the performance period are compared with the prices over the 30 days immediately preceding the end of the performance period.

The performance of all the companies in the DAX index at the beginning and end of the period is ranked and the relative position of Deutsche Lufthansa AG is determined by its achieved percentile. Performance against the target and the amount of the payment depend on the percentile position of Deutsche Lufthansa AG. Target achievement is adjusted on a linear basis for positions between the 25th and 50th percentile and between the 50th and 75th percentile.

T150 RELATIVE TSR – PERFORMANCE AND PAYMENT

Position of Lufthansa share compared with peer group	Target achievement	Payment in € for Executive Board members	Payment in € for Executive Board Chairman
≤ 25th percentile	0%	–	–
≥ 50th percentile	100%	467,500	888,250
≥ 75th percentile	200%	935,000	1,776,500

Over the financial years 2019 and 2018, the number of options changed as follows:

T151 CHANGE IN NUMBER OF OPTIONS

	2019		2018	
	Number of options/option packages	Cash settlement in € thousands	Number of options/option packages	Cash settlement in € thousands
Outstanding options on 1 Jan	20,546	–	19,167	–
Options issued	7,420	–	6,066	–
Expired or unused options	1,110	–	523	–
Options exercised	4,543	31,726	4,164	36,868
Outstanding options on 31 Dec	22,313	–	20,546	–

The performance and outperformance option of the share programme 2015, which has now expired, resulted in a payment of EUR 32m (previous year: EUR 37m). A total of 974,863 shares were issued in the reporting year at an average price of EUR 17.43 for the share programme 2019, adding up to a total value of EUR 17m. The 50% discount for employees meant the new share issuance resulted in staff costs of EUR 8.5m. Participants in the programme therefore hold 3,594,679 shares as of the reporting date (previous year: 3,561,026 shares).

The fair values of the option rights in the share programmes still running were calculated using Monte Carlo simulations. This involves simulating the future returns of the shares in the comparative index and of Deutsche Lufthansa AG and calculating the value of the option rights as the forecast amount of a dividend.

The following fair values were measured in total:

T152 FAIR VALUE OF OPTION RIGHTS AS OF 31 DEC 2019

	Number of options	Fair value per option in €	Proportional vested benefit	Total fair value in €
Board member				
Options 2016	135	31,582	0.70	2,980,551
Options 2017	181	5,376	0.58	559,576
Options 2018	165	5,770	0.36	342,594
Options 2019	6	281,633	0.25	422,450
Managers				
Options 2016	2,062	12,628	0.79	20,614,158
Options 2017	2,173	2,148	0.54	2,528,286
Options 2018	2,525	2,354	0.29	1,733,623
Options 2019	3,071	5,479	0.04	701,084
Non-payscale staff				
Options 2016	2,470	1,579	0.79	3,087,603
Options 2017	2,301	274	0.54	341,507
Options 2018	3,164	286	0.29	263,930
Options 2019	4,060	659	0.04	111,481
Total	22,313			33,686,841

T152 FAIR VALUE OF OPTION RIGHTS AS OF 31 DEC 2018

	Number of option packages	Fair value per option package in €	Vested benefit as a proportion	Fair value in €
Board member				
Options 2015	135	37,052	0.71	3,543,098
Options 2016	135	33,643	0.46	2,081,661
Options 2017	165	12,029	0.29	578,896
Options 2018	165	11,481	0.04	78,932
Managers				
Options 2015	1,869	14,816	0.79	21,922,124
Options 2016	2,247	13,454	0.54	16,375,200
Options 2017	2,333	4,809	0.29	3,272,324
Options 2018	2,622	4,680	0.04	511,290
Non-payscale staff				
Options 2015	2,539	1,850	0.79	3,718,577
Options 2016	2,631	1,681	0.54	2,395,635
Options 2017	2,426	606	0.29	428,796
Options 2018	3,279	566	0.04	77,330
Total	20,546			54,983,861

Staff fluctuation of 4.9% is again assumed when accounting for the liability resulting from the valuation of option rights, so that the recognised liability is less than their calculated time value. The measurement of option rights therefore resulted in a liability of EUR 32m as of the reporting date (previous year: EUR 53m), of which EUR 7m (previous year: EUR 25m) is shown under non-current liabilities. The payment of EUR 32m in the financial year on the basis of option rights that have now expired reduced the previously recognised liability, so that the change in option rights in the reporting year increased staff costs by EUR 11m.

The weighted average share prices at the calculation date (excluding the Executive Board TSR programme) were used in the Monte Carlo simulation. As stated in the terms of the programme, these are 50-day averages for the shares of Deutsche Lufthansa AG and the competitors included in the comparative index. The volatilities and correlations used are forecasts for a specific date and maturity on the basis of current market estimates.

Swap rates were used as the interest rate for the remaining term of the outperformance option in each case. The maximum term of the programmes was used for measurement purposes.

Time values for the Executive Board TSR programme were also measured using a Monte Carlo simulation based on historical and current market data for the relevant peer group of DAX companies. Forecast volatilities are based on the historic TSR data. The share prices for the past four years were used to calculate historical volatility. A remaining term of 37 months and a risk-free interest rate of – 0.66% were applied.

The parameters used by the external service provider for the notional airline peer group index are shown in the following table:

T153 REFERENCE PRICE

		Options 2016	Options 2017	Options 2018	Options 2019
Lufthansa	EUR	10.55	23.00	21.18	14.84
Air France-KLM	EUR	5.00	13.12	8.56	9.90
IAG	GBP	400.94	612.70	646.28	480.37
Ryanair	EUR	12.40	17.25	12.82	11.00
easyJet	GBP	1,018.21	1,240.72	1,340.31	1,131.63
Air Berlin	EUR	0.68	–	–	–
Norwegian	NOK	–	218.45	236.70	37.49
WIZZair	GBP	–	2,991.52	2,858.46	3,667.66

T154 PROJECTED VOLATILITIES

in % for:	Options 2016 as of 31 Dec 2019	Options 2016 as of 31 Dec 2018	Options 2017 as of 31 Dec 2019	Options 2017 as of 31 Dec 2018	Options 2018 as of 31 Dec 2019	Options 2018 as of 31 Dec 2018	Options 2019 as of 31 Dec 2019	Options 2019 as of 31 Dec 2018
Lufthansa	30.24	29.76	31.05	31.77	29.94	31.46	31.40	–
Air France-KLM	39.30	37.48	38.66	37.22	38.07	36.85	37.73	–
IAG	31.04	24.38	27.42	33.00	26.76	32.51	32.51	–
Ryanair	35.17	29.00	34.01	31.72	31.16	31.41	32.59	–
easyJet	39.22	30.08	34.95	34.97	33.35	33.82	36.06	–
Air Berlin	–	140.85	–	–	–	–	–	–
Norwegian	–	–	81.10	51.97	70.93	50.01	64.75	–
WIZZair	–	–	32.96	35.27	32.14	35.27	33.70	–
Risk-free interest rate	Options 2016: – 0.64% for euro zone (previous year: – 0.64%) 0.67% for UK (previous year: 0.74%)				Options 2018: – 0.66% for euro zone (previous year: – 0.41%) 0.50% for UK (previous year: 0.85%) 1.25% for Norway (previous year: 1.31%)			
	Options 2017: – 0.66% for euro zone (previous year: – 0.54%) 0.55% for UK (previous year: 0.79%) 1.25% for Norway (previous year: 1.20%)				Options 2019: – 0.64% for euro zone 0.48% for UK 1.24% for Norway			
Fluctuation	4.9% (previous year: 5.1%)				4.9% (previous year: 5.1%)			

38 Current contract liabilities

The Lufthansa Group recognised the following contract liabilities:

T155 CONTRACT LIABILITIES

in €m	31 Dec 2019	31 Dec 2018 ¹⁾
Contract liabilities from unused flight documents	4,071	3,969
Liabilities from customer loyalty programmes	2,200	2,186
Liabilities from MRO and IT services	196	78
Miscellaneous contract liabilities	279	127
Other contract liabilities	2,675	2,391
Liabilities from contracts with customers	6,746	6,360
Revenue recognised in the reporting period		
Revenue recognised that was included in the contract liability balance at the beginning of the period		
Revenue from unused flight documents	3,768	3,491
Revenue from customer loyalty programmes	576	578
Revenue from MRO and IT services	76	69
Other	50	56
Total	4,470	4,194

¹⁾ Previous year's figures have been adjusted.

Liabilities under customer loyalty programmes as of 31 December 2019 included 225 billion miles from bonus miles programmes, as in the previous year.

Other contract liabilities include repayment obligations of EUR 58m for compensation payments in connection with passenger rights (previous year: EUR 75m) (➔ **Note 2, p. 144ff.**).

The unsatisfied performance obligation under existing long-term service contracts came to EUR 6.4bn in total, assuming that the services are performed as agreed, of which EUR 1.2bn relate to the next twelve months. These essentially consist of maintenance contracts in the MRO segment for the long-term maintenance and overhaul of airline sub-fleets. To calculate the outstanding performance obligations, the number of maintenance events derived from the respective flight plans and agreed in the contracts are taken into account, along with the expected revenue and fixed prices for certain services (VIP and cabin modifications). Around 62% of performance obligations beyond twelve months are expected to have been fulfilled by 2025.

As in the previous year, no revenue was recognised in 2019 for performance obligations fulfilled in prior financial years.

In line with the simplification rules of IFRS 15, no disclosures are made for the performance obligations as of 31 December 2019 or 31 December 2018 that have a forecast original term of one year or less. Award miles can be redeemed for at least three years, but may be redeemed at short notice.

The Lufthansa Group applies the simplification rule defined in IFRS 15.94, which allows contract initiation costs to be expensed if the amortisation period otherwise to be taken into account would be twelve months or less.

39 Trade payables and other current financial liabilities

T156 TRADE PAYABLES AND OTHER CURRENT FINANCIAL LIABILITIES

in €m	31 Dec 2019	31 Dec 2018 ¹⁾
Trade payables		
Trade payables to affiliated companies	42	43
Trade payables to other equity investments	2	-
Trade payables to third parties	3,678	3,825
	3,722	3,868
Other liabilities		
Liabilities to banks	17	39
Other liabilities to affiliated companies	260	269
Other liabilities to equity investments	-	-
Liabilities from equity investments	356	574
Other financial liabilities	996	970
	1,629	1,852
Total	5,351	5,720

¹⁾ Previous year's figures have been adjusted.

The carrying amount of these liabilities corresponds to their fair value.

40 Current advance payments received, deferred income and other non-financial liabilities

T157 CURRENT ADVANCE PAYMENTS RECEIVED, DEFERRED INCOME AND OTHER NON-FINANCIAL LIABILITIES

in €m	31 Dec 2019	31 Dec 2018
Advance payments received	17	16
Deferred income	59	39
Other non-financial liabilities	306	333
	382	388

Other non-financial liabilities include EUR 279m (previous year: EUR 303m) in deferred amounts for outstanding holiday allowance and overtime. Other non-financial liabilities also include the current portion of obligations under share-based remuneration agreements measured at fair value (➤ **Note 37, p. 189ff.**)

NOTES TO THE SEGMENT REPORTING

41 Notes to the reportable segments and segment data

NOTES TO THE REPORTABLE SEGMENTS

As of 31 December 2019, the Lufthansa Group operates in five reporting segments, which make up its Group activities. The segments are defined in line with the internal reporting and management structure.

Segmentation has been changed compared with the financial statements as of 31 December 2018. Part of the Lufthansa Systems group is managed by the Lufthansa Technik group as of financial year 2019 and so has been allocated to the MRO segment. The figures for the previous year have been adjusted accordingly.

The airline activities were combined in their respective reporting segments based on the similarity between the economic characteristics of the individual airlines, such as network and sales structures, as well as customers and services. The Network Airlines segment comprises Lufthansa German Airlines, SWISS and Austrian Airlines. Further information about the individual airlines can be found in the Group management report ➤ **starting on p. 45.**

The Eurowings segment comprises Eurowings, Germanwings and Brussels Airlines, as well as the equity investment in SunExpress.

The Logistics segment comprises the scheduled airfreight activities of the Lufthansa Cargo group. Lufthansa Cargo is Europe's leading cargo airline.

The MRO segment is a leading global provider of maintenance, repair and overhaul services for civil and commercial aircraft and is represented by the Lufthansa Technik group.

The Catering segment, represented by the LSG Lufthansa Service/Sky Chefs group, is the global market leader in airline catering.

Business activities not allocated to a reportable segment are presented in the "Additional Businesses and Group Functions" column of the segment reporting along with the income and expenses of central Group functions. They include income and expenses of Lufthansa Commercial Holding GmbH, Lufthansa AirPlus group, the Lufthansa Systems group, the Lufthansa Aviation Training group and other Group companies.

NOTES TO SEGMENT DATA AND INTERNAL MANAGEMENT

The accounting policies of the reportable segments are the same as those described in ➤ **Note 2, p. 144ff.**

Due to the IFRIC Agenda decision of 17 September 2019 compensations for flight cancellations and delays can no longer be recognised as an expense, but result in a reduction of traffic revenues. The Lufthansa Group has applied this amendment retrospectively.

The Lufthansa Group measures the performance of its segments using two segment result indicators: EBIT and Adjusted EBIT. EBIT is made up of the IFRS operating result and the result from equity investments. Adjusted EBIT is obtained by correcting EBIT for gains and losses on the disposal of assets and impairment losses and earnings attributable to other periods in connection with pension obligations (plan adjustments and plan settlements).

Sales and revenue between reportable segments are based on arm's length prices. Administrative services are charged as cost allocations.

For information on external traffic revenue ➤ **Note 3, p. 157.**

Capital employed largely comprises segment assets, adjusted for derivative financial instruments, and deferred tax items less non-interest-bearing debt.

The result of the equity valuation for the segment's equity investments is part of its segment result. However, from a Group perspective, it is not attributed to the operating result, but rather to the financial result.

T158 SEGMENT INFORMATION FOR THE 2019 REPORTING SEGMENTS

	Network Airlines	Eurowings	Logistics	MRO	Catering	Total reportable operating segments	Additional Businesses and Group Functions	Reconciliation		Group
in €m								Not allocated	Consoli- dation	
External revenue	22,417	4,015	2,437	4,378	2,623	35,870	554	-	-	36,424
of which traffic revenue	21,375	3,987	2,318	-	-	27,680	-	456	-	28,136
Inter-segment revenue	689	108	41	2,543	737	4,118	239	-	-4,357	-
Total revenue	23,106	4,123	2,478	6,921	3,360	39,988	793	-	-4,357	36,424
Other operating income	785	342	103	260	95	1,585	1,939	-	-1,009	2,515
Total operating income	23,891	4,465	2,581	7,181	3,455	41,573	2,732	-	-5,366	38,939
Operating expenses	22,132	4,655	2,621	6,748	3,355	39,511	2,971	-	-5,358	37,124
of which cost of materials and services	12,799	3,005	1,778	3,927	1,441	22,950	287	-	-3,410	19,827
of which staff costs	4,210	628	406	1,671	1,290	8,205	913	-	-7	9,111
of which depreciation and amortisation	1,678	465	160	207	119	2,629	107	-	-44	2,692
of which other operating expenses	3,445	557	277	943	505	5,727	1,664	-	-1,897	5,494
Results of equity investments¹⁾	46	24	41	60	28	199	12	-	-	211
of which result of investments accounted for using the equity method	31	24	18	34	24	131	-	-	-	131
Adjusted EBIT²⁾	1,805	-166	1	493	128	2,261	-227	-	-8	2,026
Reconciliation items	-48	-8	-34	9	-30	-111	6	-	-64	-169
Impairment losses/gains	-23	-4	-38	12	-42	-95	7	-	-51	-139
Effects from pension provisions	-3	-	-1	-2	-1	-7	-4	-	1	-10
Results of disposal of assets	-22	-4	5	-1	13	-9	3	-	-14	-20
EBIT	1,757	-174	-33	502	98	2,150	-221	-	-72	1,857
Other financial result										3
Profit/loss before income taxes										1,860
Capital employed ³⁾	11,834	2,570	2,135	5,607	1,513	23,659	1,904	-	-528	25,035
of which from investments accounted for using the equity method	25	178	49	279	139	670	5	-	-3	672
Segment capital expenditure ⁴⁾	2,605	260	286	313	127	3,591	107	-	-32	3,666
of which from investments accounted for using the equity method	-	-	-	49	28	77	-	-	-	77
Number of employees at end of period	52,741	8,809	4,539	26,650	35,679	128,418	9,935	-	-	138,353
Average number of employees	52,378	9,087	4,543	25,872	35,954	127,834	9,950	-	-	137,784

¹⁾ The result from equity investments does not include any impairment losses on investments accounted for using the equity method.

²⁾ For reconciliation from Adjusted EBIT to EBIT see T024, p. 36, in the Group management report.

³⁾ The capital employed results from total assets adjusted for non-operating items (deferred taxes, positive market values, derivatives) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

⁴⁾ Investment in intangible assets and property, plant and equipment, as well as in loans to and shares in companies. Investment is shown without capitalised borrowing costs.

T158 SEGMENT INFORMATION FOR THE 2018 REPORTING SEGMENTS

	Network Airlines	Eurowings	Logistics	MRO	Catering	Total reportable operating segments	Additional Businesses and Group Functions	Reconciliation		Group
in €m								Not allocated	Consoli- dation	
External revenue	21,822	4,080	2,681	3,927	2,499	35,009	533	-	-	35,542
of which traffic revenue	20,707	3,986	2,550	-	-	27,243	-	558	-	27,801
Inter-segment revenue	727	18	32	2,178	718	3,673	250	-	-3,923	-
Total revenue	22,549	4,098	2,713	6,105	3,217	38,682	783	-	-3,923	35,542
Other operating income	699	290	57	270	77	1,393	1,879	-	-988	2,284
Total operating income	23,248	4,388	2,770	6,375	3,294	40,075	2,662	-	-4,911	37,826
Operating expenses	20,854	4,643	2,538	5,936	3,208	37,179	2,914	-	-4,929	35,164
of which cost of materials and services	11,714	3,042	1,753	3,376	1,385	21,270	246	-	-3,149	18,367
of which staff costs	4,159	619	420	1,552	1,218	7,968	963	-	-7	8,924
of which depreciation and amortisation	1,497	372	104	126	66	2,165	52	-	-37	2,180
of which other operating expenses	3,484	610	261	882	539	5,776	1,653	-	-1,736	5,693
Results of equity investments	35	24	36	7	29	131	43	-	-	174
of which result of investments accounted for using the equity method	31	24	25	9	25	114	1	-	-1	114
Adjusted EBIT¹⁾	2,429	-231	268	446	115	3,027	-209	-	18	2,836
Reconciliation items	120	-	-5	-1	-5	109	27	-	2	138
Impairment losses/gains	1	-	-4	3	-5	-5	-5	-	1	-9
Effects from pension provisions	110	-	-	-	-	110	2	-	1	113
Results of disposal of assets	9	-	-1	-4	-	4	30	-	-	34
EBIT	2,549	-231	263	445	110	3,136	-182	-	20	2,974
Other financial result										-190
Profit/loss before income taxes										2,784
Capital employed ²⁾	9,635	2,220	1,430	4,836	1,264	19,385	1,727	-	-139	20,973
of which from investments accounted for using the equity method	25	148	52	284	138	647	6	-	-3	650
Segment capital expenditure ³⁾	2,573	515	374	244	79	3,785	56	-	-24	3,817
of which from investments accounted for using the equity method	-	-	-	32	-	32	-	-	-	32
Number of employees at end of period	51,778	9,255	4,505	24,594	35,512	125,644	9,890	-	-	135,534
Average number of employees	51,327	9,296	4,422	23,833	35,548	124,426	9,904	-	-	134,330

¹⁾ For reconciliation from Adjusted EBIT to EBIT ➔ T024, p. 36, in the Group management report.

²⁾ The capital employed results from total assets adjusted for non-operating items (deferred taxes, positive market values, derivatives) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

³⁾ Investment in intangible assets and property, plant and equipment, as well as in loans to and shares in companies. Investment is shown without capitalised borrowing costs.

The reconciliation column includes both the effects of consolidation activities and the amounts resulting from different definitions of segment item contents compared with the corresponding Group items.

The amounts in the reconciliation column for Group EBIT include the effects of consolidation procedures on profit or loss in which income and expense do not figure for two companies at the same amount, or in the same period.

Eliminated segment revenue generated with other consolidated segments is shown in the reconciliation column for revenue.

The change in accounting for compensation payments had the following effects on segment results:

T159 RESTATEMENT EFFECTS OF COMPENSATION PAYMENTS BY REPORTABLE SEGMENTS

in €m	Network Airlines	Eurowings	Group
Traffic revenue	- 146	- 93	- 239
Cost of materials and services	- 146	- 93	- 239
Effect on Adjusted EBIT 2019	-	-	-
Effect on EBIT 2019	-	-	-
Traffic revenue	- 170	- 132	- 302
Cost of materials and services	- 170	- 132	- 302
Effect on Adjusted EBIT 2018	-	-	-
Effect on EBIT 2018	-	-	-

NOTES ON GEOGRAPHICAL REGIONS IN 2019

The allocation of traffic revenue to geographic regions is based on the original location of sale. Non-current assets are allocated according to the location of the relevant asset. The allocation of other revenue to the individual regions is based on the geographical location of the customer.

The regions are defined on a geographical basis. As an exception to this rule, traffic revenue generated in Turkey is attributed to Europe.

The Lufthansa Group controls its air traffic operations on the basis of network results and not on the basis of regional earnings contributions. The same applies to the Catering segment. Consequently, the presentation of regional segment results is of no informational value for the Lufthansa Group.

A presentation of traffic revenue generated in the Network Airlines, Eurowings and Logistics segments by traffic region, rather than by original location of sale, is included in the information on the respective segments in the management report.

External revenue, non-current assets and capital expenditure are as follows:

T160 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY REGION FOR 2019

in €m	Europe	North America	Central and South America	Asia/Pacific	Middle East	Africa	Group
Traffic revenue ¹⁾	18,586	4,707	615	3,073	621	534	28,136
Other revenue	3,437	2,557	463	1,394	281	156	8,288
Non-current assets ^{2) 3) 4)}	23,385	564	48	305	4	26	24,332
Capital expenditure on non-current assets ³⁾	3,434	50	6	17	-	3	3,510

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

⁴⁾ Including rights of uses in accordance with first-time application of IFRS 16.

The figures for the main countries are as follows:

T161 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY COUNTRIES FOR 2019

in €m	Germany	USA
Traffic revenue ¹⁾	8,610	4,233
Other revenue	1,077	2,097
Non-current assets ^{2) 3) 4)}	16,207	533
Capital expenditure on non-current assets ³⁾	2,913	45

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

⁴⁾ Including rights of uses in accordance with first-time application of IFRS 16.

NOTES ON GEOGRAPHICAL REGIONS IN 2018

External revenue, non-current assets and capital expenditure are as follows:

T160 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY REGION FOR 2018

	Europe	North America	Central and South America	Asia/Pacific	Middle East	Africa	Group
in €m							
Traffic revenue ¹⁾	18,785	4,160	669	3,123	590	474	27,801
Other revenue	3,283	2,272	361	1,383	223	219	7,741
Non-current assets ^{2) 3)}	20,342	276	41	206	2	23	20,890
Capital expenditure on non-current assets ³⁾	3,773	39	5	16	-	5	3,838

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

The figures for the main countries are as follows:

T161 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY COUNTRIES FOR 2018

in €m	Germany	USA
Traffic revenue ¹⁾	8,819	3,730
Other revenue	1,066	1,896
Non-current assets ^{2) 3)}	13,984	261
Capital expenditure on non-current assets ³⁾	2,410	36

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

In 2019 and in the previous year, no more than 10% of Lufthansa Group revenue was generated with any one customer.

SEGMENT REPORTING FROM 2020

Brussels Airlines is managed by Network Airlines as of the start of financial year 2020 and has therefore been allocated to the Network Airlines segment. Brussels Airlines generated revenue of EUR 1,471m and Adjusted EBIT of EUR – 26m in the reporting period. The previous year's figures will be adjusted accordingly in the 2020 financial reporting.

The long-haul business of the Eurowings segment will remain in the Lufthansa German Airlines segment in organisational terms, and operational management will mostly take place there too. Segment reporting therefore follows management reporting and presents this unit as part of Network Airlines, adjusting the previous year's figures accordingly. In 2019, the activities concerned accounted for revenue of EUR 340m and Adjusted EBIT of EUR – 60m for the Eurowings segment.

In the medium term, one aim of restructuring the Eurowings segment is to reduce the number of flight operations. In this context, the flight operations of Germanwings are to be allocated to Lufthansa German Airlines for organisational purposes. Until the restructuring is complete, Germanwings will provide services to the other airlines in the Lufthansa Group as purely a wet-lease operator. Segment reporting from 2020 will present Germanwings operations as part of Lufthansa German Airlines. The figures for the previous year will be adjusted accordingly. In 2019, the company had no external revenue and reported positive earnings of EUR 29m.

As of 1 January 2020, the line maintenance business of Lufthansa Technik was transferred to Deutsche Lufthansa AG, which will carry out the work itself from this point on. To facilitate comparison, the previous year's figures for the MRO and Network Airlines segments have been adjusted in line with the situation in place from 2020. In 2019, this concerned MRO segment revenue of EUR 354m and Adjusted EBIT of EUR 30m.

NOTES TO THE CONSOLIDATED CASH FLOW STATEMENT

42 Notes to cash flow from operating, investing and financing activities

The cash flow statement shows how cash and cash equivalents have changed over the reporting period at the Lufthansa Group. In accordance with IAS 7, cash flows are divided into cash inflows and outflows from operating activities, from investing activities and from financing activities. The cash and cash equivalents shown in the cash flow statement comprise the statement of financial position items bank balances and cash-in-hand, without fixed-term deposits with terms of three to twelve months, amounting to EUR 0m (previous year: EUR 66m). The amount of liquidity in the broader sense is reached by adding securities that can be liquidated at short notice.

Interest paid and interest income from the corresponding interest rate hedges are netted to avoid overemphasising the items interest income and interest paid.

ADDITIONAL INFORMATION ON THE CASH FLOW STATEMENT

Cash flow from operating activities

Cash flow from operating activities is derived from profit/loss before income taxes using the indirect method. It is adjusted for non-cash income and expenses as well as changes in trade working capital and in other assets/liabilities that are not attributable to investing or financing activities.

Lease accounting in line with IFRS 16 means that payments under operating leases previously shown in cash flow from operating activities are now presented as capital repayments or interest payments within cash flow from financing activities. In the financial year this had an impact of EUR +432m on cash flow from operating activities and a corresponding negative effect in cash flow from financing activities.

In the current financial year, the Group primarily recognised the following non-cash income and expenses:

T162 SIGNIFICANT NON-CASH INCOME AND EXPENSES

in €m	2019	2018
Result of miscellaneous financial items	84	46
Write-down on receivables	120	99
Reversal of write-downs on receivables	-27	-48
Income from the reversal of provisions and accruals	-321	-260
Adjustments to retirement and transitional benefit systems	10	-113
Total	-134	-276

Trade working capital consists of changes in the carrying amounts of inventories, trade receivables and payables, contract assets and down payments, other current assets and other current liabilities, contract liabilities and current deferrals and prepaid expenses.

Other assets/liabilities mainly include corrections between pension expenses and payments, changes in other provisions, accruals/deferrals and corrections for non-cash effects from currency translation.

Cash flow from investing (and cash management) activities

Cash flows from investing and financing activities are calculated on the basis of payments.

Cash flow from investing activities results mainly from investments and disinvestments in non-current assets.

The Lufthansa Group contributed EUR 394m to pension assets in 2019 (previous year: EUR 463m). These payments were categorised as cash flow from investing and cash management activities. By contrast, pension payments from fund assets led to cash inflows from investments (EUR 138m; previous year: EUR 126m). They correspond to cash outflows from operating activities.

Cash flow from financing activities

Cash flow from financing activities now also includes capital repayments and interest payments on lease liabilities following the first-time application of IFRS 16.

Borrowing and the instruments used to hedge it changed as follows in the financial year:

T163 FINANCIAL LIABILITIES 2019

	31 Dec 2018	Cash effective	Non-cash effective						31 Dec 2019
			Addition due to changes in consoli- dation	Addition due to lease liabilities ¹⁾	Currency translation differences	Accrued interest	Reclassi- fication	Changes in fair value	
in €m									
Non-current borrowings	5,008	2,688	-	2,410	66	91	-1,867	-	8,396
Current borrowings	1,677	-2,236	-	359	3	-2	1,833	-	1,634
Other borrowings	39	-22	-	-	-	-	-	-	17
Interest rate swaps and currency futures used for hedging – assets	-154	89	-	-	-	-	-	-140	-205
Interest rate swaps and currency futures used for hedging – liabilities	49	-	-	-	-	-	-	-22	27

¹⁾ This includes additions from the first-time application of IFRS 16 (non-current borrowings: EUR 1,599m and current borrowings: EUR 359m).

Changes in borrowing in the previous year were as follows:

T163 FINANCIAL LIABILITIES 2018

	31 Dec 2017	Cash effective	Non-cash effective					31 Dec 2018
			Addition due to changes in consolidation	Currency translation differences	Accrued interest	Reclassi- fication	Changes in fair value	
in €m								
Non-current borrowings	6,142	491	-	90	13	-1,728	-	5,008
Current borrowings	672	-722	-	4	-5	1,728	-	1,677
Other borrowings	18	22	-	-1	-	-	-	39
Interest rate swaps and currency futures used for hedging – assets	-83	30	-	-	-	-	-101	-154
Interest rate swaps and currency futures used for hedging – liabilities	2	-	-	-	-	-	47	49

OTHER DISCLOSURES

43 Additional disclosures on financial instruments

FINANCIAL ASSETS BY MEASUREMENT CATEGORY

As of the current reporting date, the financial assets can be broken down into measurement categories with the following carrying amounts:

T164 FINANCIAL ASSETS IN THE BALANCE SHEET AS OF 31 DEC 2019

in €m	Amortised cost	At fair value through profit or loss	At fair value through other comprehensive income (with recycling)	At fair value through other comprehensive income (without recycling)	Derivative financial instru- ments which are an effective part of a hedging relationship
Other equity investments	-	33	-	-	-
Non-current securities	-	-	-	-	-
of which equity instruments	-	-	-	22	-
of which debt instruments	31	-	-	-	-
Loans	154	-	-	-	-
Non-current receivables	316	-	-	-	-
Non-current derivative financial instruments	-	6	-	-	899
Trade receivables and other current receivables	5,322	-	-	-	-
Current derivative financial instruments	-	6	-	-	452
Current securities	-	-	-	-	-
of which equity instruments	-	360	-	-	-
of which debt instruments	-	-	1,610	-	-
Cash and cash equivalents	1,415	-	-	-	-
Total	7,238	405	1,610	22	1,351

T164 FINANCIAL ASSETS IN THE BALANCE SHEET AS OF 31 DEC 2018

in €m	Amortised cost	At fair value through profit or loss	At fair value through other comprehensive income (with recycling)	At fair value through other comprehensive income (without recycling)	Derivative financial instru- ments which are an effective part of a hedging relationship
Other equity investments	-	24	-	-	-
Non-current securities	-	-	-	-	-
of which equity instruments	-	-	-	30	-
of which debt instruments	11	-	-	-	-
Loans	174	-	-	-	-
Non-current receivables	279	-	-	-	-
Non-current derivative financial instruments	-	4	-	-	824
Trade receivables and other current receivables	5,513	-	-	-	-
Current derivative financial instruments	-	23	-	-	334
Current securities	-	-	-	-	-
of which equity instruments	-	280	-	-	-
of which debt instruments	-	-	1,457	-	-
Cash and cash equivalents	1,500	-	-	-	-
Total	7,477	331	1,457	30	1,158

The category “At fair value through other comprehensive income” includes derivatives that do not meet the requirements for applying hedge accounting and so are accounted for as stand-alone derivatives. This category also includes equity instruments, consisting of shares and equity investments, for which the instrument-specific option of fair value through other comprehensive income without recycling has not been exercised. As of the reporting date, the fair-value option without effect on profit and loss and without recycling was chosen for one share position, in order to avoid the recognition of changes in market value through profit or loss. The item includes shares in VISA Inc. (market value: EUR 22m, dividend payments: EUR 0.1m in the 2019 financial year) whose market valuation reserve increased by EUR 7m in 2019 (cumulative EUR 13m).

A share position in Alliance Aviation Services Ltd. held at fair value without effect on profit and loss and without recycling was sold for EUR 15m in the 2019 financial year. The cumulative market valuation reserve at the time of the sale was EUR 12m, which was transferred to retained earnings without effect on profit and loss.

The market valuation reserve for financial assets measured at fair value (with recycling) without effect on profit and loss was EUR 15m as of the reporting date.

FINANCIAL LIABILITIES BY MEASUREMENT CATEGORY

The financial liabilities can be divided into measurement categories with the following carrying amounts, whereby the category “at fair value through profit or loss” includes derivatives that do not meet the requirements for applying hedge accounting and so are accounted for as stand-alone derivatives.

T165 FINANCIAL LIABILITIES IN THE BALANCE SHEET AS OF 31 DEC 2019

in €m	Liabilities at fair value through profit or loss	Derivative financial instruments which are an effective part of a hedging relationship	Other financial liabilities at cost
Borrowings (without IFRS 16 lease liabilities)	-	-	7,158
Derivative financial instruments	67	199	-
Trade payables	-	-	3,722
Other financial liabilities	-	-	1,705
Total	67	199	12,585

T165 FINANCIAL LIABILITIES IN THE BALANCE SHEET AS OF 31 DEC 2018

in €m	Liabilities at fair value through profit or loss	Derivative financial instruments which are an effective part of a hedging relationship	Other financial liabilities at cost
Borrowings (without IFRS 16 lease liabilities)	-	-	6,685
Derivative financial instruments	29	586	-
Trade payables	-	-	3,868
Other financial liabilities	-	-	1,989
Total	29	586	12,542

The net result of the different categories of financial assets and liabilities is made up as follows:

T166 NET RESULT 2019 FOR FINANCIAL ASSETS AND LIABILITIES BY MEASUREMENT CATEGORY

in €m	Interest expenses	Interest income	Depreciation	Result from valuation and sale	Currency result	Net result
Assets at amortised cost	-	8	- 94	-	8	- 78
At fair value through other comprehensive income (with recycling)	- 10	21	-	13	5	29
At fair value through other comprehensive income (without recycling)	-	-	-	- 8	-	- 8
Assets at fair value through profit or loss	-	-	-	411	-	411
Liabilities at amortised cost	- 123	-	-	-	- 22	- 145
Liabilities at fair value through profit or loss	-	-	-	- 52	-	- 52
Total	- 133	29	- 94	364	- 9	157

For the result of measurement and disposal of assets at fair value through profit or loss ➤ Note 13 p. 162.

T166 NET RESULT 2018 FOR FINANCIAL ASSETS AND LIABILITIES BY MEASUREMENT CATEGORY

in €m	Interest expenses	Interest income	Depreciation	Result from valuation and sale	Currency result	Net result
Assets at amortised cost	-	5	- 50	-	29	- 16
At fair value through other comprehensive income (with recycling)	- 12	22	-	- 10	7	7
At fair value through other comprehensive income (without recycling)	-	-	-	5	-	5
Assets at fair value through profit or loss	-	-	-	- 32	-	- 32
Liabilities at amortised cost	- 124	-	-	-	- 39	- 163
Liabilities at fair value through profit or loss	-	-	-	- 18	-	- 18
Total	- 136	27	- 50	- 55	- 3	- 217

The table to the right shows the carrying amounts and market values for individual classes of financial liabilities. The stated market values of bonds reflect their stock market listings (Level 1 of the fair value hierarchy). The market values for other types of financial liability have been calculated using the applicable interest rates for the remaining term to maturity and repayment structures at the balance sheet date based on the available market information (Bloomberg) (Level 2 of the fair value hierarchy). For other assets and liabilities, non-current receivables, trade receivables and cash-in-hand carried at amortised cost, the carrying amount is deemed to be a reasonable approximation of the fair value.

T167 FINANCIAL LIABILITIES

in €m	31 Dec 2019		31 Dec 2018	
	Carrying amount	Market value	Carrying amount	Market value
Bonds	1,094	1,026	1,007	1,026
Liabilities to banks	2,110	2,150	1,957	1,984
Leasing liabilities ¹⁾	2,872	-	596	581
Other borrowings	3,954	3,883	3,125	3,083
Total	10,030	7,059	6,685	6,674

¹⁾ With the introduction of IFRS 16, declaration of market value no longer applies as of 1 January 2019.

FINANCIAL ASSETS HELD AT FAIR VALUE BY LEVEL OF FAIR VALUE HIERARCHY

The following table shows financial assets and liabilities held at fair value by level of fair value hierarchy. The levels are defined as follows:

- Level 1: Financial instruments traded on active markets, the quoted prices for which are taken for measurement unchanged.
- Level 2: Measurement is made by means of valuation methods with parameters derived directly or indirectly from observable market data.
- Level 3: Measurement is made by means of valuation methods with parameters not based exclusively on observable market data.

In the financial year 2019, the fair value hierarchy for assets and liabilities held at fair value was as follows:

T168 FAIR VALUE HIERARCHY OF ASSETS AS OF 31 DEC 2019

in €m	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit and loss	359	13	-	372
Financial derivatives classified as held for trading	-	12	-	12
Securities	359	1	-	360
Derivative financial instruments which are an effective part of a hedging relationship	-	1,352	-	1,352
Financial assets at fair value through other comprehensive income	-	1,632	-	1,632
Equity instruments	-	22	-	22
Debt instruments	-	1,610	-	1,610
Total assets	359	2,997	-	3,356

T169 FAIR VALUE HIERARCHY OF LIABILITIES AS OF 31 DEC 2019

in €m	Level 1	Level 2	Level 3	Total
Derivative financial instruments at fair value through profit or loss	-	- 67	-	- 67
Derivative financial instruments which are an effective part of a hedging relationship	-	- 199	-	- 199
Total liabilities	-	- 266	-	- 266

In the previous year, the hierarchy was as follows:

T168 FAIR VALUE HIERARCHY OF ASSETS AS OF 31 DEC 2018

in €m	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit and loss	278	29	-	307
Financial derivatives classified as held for trading	-	27	-	27
Securities	278	2	-	280
Derivative financial instruments which are an effective part of a hedging relationship	-	1,158	-	1,158
Financial assets at fair value through other comprehensive income	15	1,470	-	1,485
Equity instruments	15	15	-	30
Debt instruments	-	1,455	-	1,455
Total assets	293	2,657	-	2,950

T169 FAIR VALUE HIERARCHY OF LIABILITIES AS OF 31 DEC 2018

in €m	Stufe 1	Stufe 2	Stufe 3	Gesamt
Derivative financial instruments at fair value through profit or loss	-	- 29	-	- 29
Derivative financial instruments which are an effective part of a hedging relationship	-	- 586	-	- 586
Total liabilities	-	- 615	-	- 615

NETTING OF FINANCIAL ASSETS AND LIABILITIES

The following financial assets and liabilities are subject to global netting agreements and other agreements.

T170 NETTING OF FINANCIAL ASSETS AS OF 31 DEC 2019

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade receivables and other current receivables	5,478	156	5,322	-	-	5,322
Derivative financial instruments – assets	1,364	-	1,364	- 37	92	1,309
Cash and cash equivalents	1,445	30	1,415	-	-	1,415
Total assets	8,287	186	8,101	- 37	92	8,046

T171 NETTING OF FINANCIAL LIABILITIES AS OF 31 DEC 2019

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade payables and other financial liabilities	3,908	186	3,722	-	92	3,722
Derivative financial instruments – liabilities	266	-	266	- 37	-	303
Total liabilities	4,174	186	3,988	- 37	92	4,025

In the previous year, the net balances were as follows:

T170 NETTING OF FINANCIAL ASSETS AS OF 31 DEC 2018

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade receivables and other current receivables	5,983	470	5,513	-	33	5,513
Derivative financial instruments – assets	1,185	-	1,185	10	13	1,162
Cash and cash equivalents	1,508	8	1,500	-	-	1,500
Total assets	8,739	478	8,261	10	46	8,238

T171 NETTING OF FINANCIAL LIABILITIES AS OF 31 DEC 2018

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade payables and other financial liabilities	4,390	478	3,912	-	13	3,912
Derivative financial instruments – liabilities	615	-	615	10	33	572
Total liabilities	5,005	478	4,527	10	46	4,484

PRINCIPLES OF HEDGING POLICY

As an aviation group with worldwide operations, the Lufthansa Group is exposed to exchange rate, interest rate and fuel price movement risks, as well as to credit and liquidity risks. Limiting these risks by means of systematic financial management is part of Company policy.

Market risk

The major market and price risks to which the Lufthansa Group is exposed are exchange rate fluctuations between the euro and other currencies, interest rate fluctuations in international money and capital markets, and price fluctuations in the crude oil and oil products markets. Hedging policy for limiting these risks is laid down by the Executive Board and documented by internal Group guidelines. It also provides for the use of financial derivatives. The corresponding financial transactions are concluded only with first-rate counterparties.

Foreign exchange risk

For US dollars, the Lufthansa Group is in a net payer position as regards currency risks from its operating business, since fuel payments are dollar-denominated. There is always a net surplus for other currencies. The main risks in this respect stem from the Chinese renminbi, the Swiss franc, British pound sterling, the Japanese yen and the Indian rupee. Depending on market liquidity, currency risks from projected operational exposure are hedged gradually over a period of 24 months by means of futures contracts, which are accounted for as cash flow hedges. The target hedging level is defined in the

Group's internal guidelines. At the end of 2019, exposure from operations for the next 24 months was as follows:

T172 CURRENCY EXPOSURE, AS OF 2019

in millions	USD	CNY	JPY	GBP	INR
Exposure (currency)	-7,872	9,100	121,313	830	65,650
Exposure (EUR at spot rate)	-7,007	1,164	995	975	819
Hedges (currency)	3,076	-3,744	-49,122	-374	-19,760
Hedging level	39%	41%	40%	45%	20%
Hedging rate	1.18	8.14	124.55	0.90	82.50

50% of currency risks from capital expenditure on aircraft are hedged when the contract is signed. The hedging level is reviewed and increased, where necessary, if, over the life-time of the contract, the exchange rate goes significantly above or below that used to calculate the investment. In the last 24 months before payment, the hedging level is increased in half-yearly steps of 10%, reaching 90% by the end. These investment hedges are therefore also accounted for as cash flow hedges. Capital expenditure on aircraft takes place in US dollars and is hedged in euros or Swiss francs, depending on the functional currency of the Group company making the purchase.

US dollar exposure for capital expenditure as of year-end 2019 was as follows, broken down by the hedged currency:

T173 USD INVESTMENT EXPOSURE, HEDGED IN EUR

in millions	2020	2021	2022	2023	2024	2025	2026	2027
Exposure from net capital expenditure (USD)	-1,687	-2,567	-2,496	-2,779	-1,516	-1,114	-368	-147
Exposure from net capital expenditure (EUR at spot rate)	-1,505	-2,290	-2,226	-2,479	-1,352	-994	-328	-131
Hedges (USD)	1,491	1,875	1,598	1,651	784	582	205	73
Hedging level	88%	73%	64%	59%	52%	52%	56%	56%
Hedging rate EUR/USD	1.26	1.31	1.38	1.35	1.39	1.35	1.35	1.36

T174 USD INVESTMENT EXPOSURE, HEDGED IN CHF

in millions	2020	2021	2022	2023	2024
Exposure from net capital expenditure (USD)	-308	-228	-185	-208	-224
Exposure from net capital expenditure (EUR at spot rate)	-275	-203	-165	-185	-200
Hedges (USD)	255	196	135	104	112
Hedging level	83%	86%	73%	50%	50%
Hedging rate USD/CHF	0.89	0.85	0.85	0.85	0.82

The following sensitivity analysis shows how net profit and equity would change if the currencies identified as price risk variables had been different from those at the balance sheet date.

T175 SENSITIVITY ANALYSIS BY CURRENCY

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Currency – USD		
+10%	- 188	1,060
-10%	156	- 867
Currency – JPY		
+10%	1	- 34
-10%	- 1	27
Currency – CHF		
+10%	11	- 44
-10%	- 9	36
Currency – GBP		
+10%	8	- 36
-10%	- 7	30
Currency – CNY		
+10%	2	- 39
-10%	- 2	32
Currency – INR		
+10%	1	- 13
-10%	- 1	11

¹⁾ All amounts after deferred tax effects; +/- signs relate to net profit and/or equity.

Interest rate risk

The Lufthansa Group aims to pay interest on 100% of its financial liabilities in euros at floating rates of interest. To do so, interest rate swaps are arranged for interest-bearing, fixed-rate financial debt and leasing liabilities. Financial liabilities denominated in euros are hedged using “plain vanilla” interest rate swaps, while cross-currency interest rate swaps are used to hedge financial liabilities in foreign currencies. Depending on the interest rate structure of the exposure to be hedged, hedges are either classified as fair value hedges or cash flow hedges. The interest rate risk is monitored constantly; strategic interest rate hedges are used as needed in response to different market situations. Depending on the counterparties and the instruments used, cash collateral for interest rate swaps is either deposited with or received from counterparties (➤ T170, T171, p. 204).

The tables below describe the floating/fixed ratio for non-current borrowing as of financial year-end 2019 after taking into consideration interest rate hedging, as well as the distribution of the nominal volume of interest rate hedges.

T176 INTEREST RATE EXPOSURE AFTER HEDGING

in €m	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Fix	407	203	172	189	174	127	101	106	6	5	5
Variable	6,235	5,801	4,407	3,609	3,044	1,781	1,484	1,124	919	700	459
Float/Fix-Ratio	94%	97%	96%	95%	95%	93%	94%	91%	99%	99%	99%

T177 NOMINAL VOLUME OF INTEREST RATE HEDGES

in €m	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Fix	- 4,107	- 2,913	- 1,885	- 1,528	- 525	- 321	- 48	- 878	- 664	- 420	- 162
Variable	4,014	2,831	1,846	1,504	512	332	54	880	661	420	162

The sensitivity analysis to the right shows how net profit and equity would change if the interest rate identified as a price risk variable had been different from the perspective of the balance sheet date. In view of the current low interest rates, a reduction of more than 50 basis points is not considered likely, which is why the analysis was limited to this figure.

T178 SENSITIVITY ANALYSIS BY INTEREST RATE

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Interest		
+100 basis points	50	- 35
-50 basis points	- 26	19

¹⁾ All amounts after deferred tax effects; +/- signs relate to net profit and/or equity.

Effects of the EU benchmark regulation of global reference interest rates

In terms of the financial instruments used by the Lufthansa Group, the global reform of variable reference interest rates means that the variable reference interest rates for transactions that are available today will no longer be so in future or will be calculated differently. As of 31 December 2019, the Lufthansa Group held EUR 6.9bn in outstanding financial transactions, floating rate liabilities and hedging instruments based on variable interest rates. The following table shows the absolute nominal volume of the transactions as of the reporting date.

T179 NOMINAL VOLUME OF FLOATING RATE FINANCIAL INSTRUMENTS IN HEDGING RELATIONSHIPS

Reference floating interest rate by currency	Nominal volume of floating interest rate derivatives in €m	Nominal volume of floating interest rate financial liabilities (without derivatives) in €m
EUR – EURIBOR	5,055	1,154
USD – LIBOR	319	325
CHF – LIBOR	21	–
Total	5,395	1,479

EURIBOR's calculation method was adjusted in 2019 to ensure that it complies with the rules of the EU benchmark regulation. The change therefore has no effect on the Lufthansa Group for financial instruments based on EURIBOR. Since October 2019, the European Central Bank has also published €STR, a benchmark-compliant overnight interest rate. Clearing houses are expected to switch their mark-to-market accounting to the new overnight reference rates €STR (EUR) and SOFR (USD) in the course of 2020. No significant impact is expected for the Lufthansa Group from the change in the relevant USD and CHF LIBORs, since the volume of transactions in the portfolio is low.

The Lufthansa Group is applying early the Amendments to IFRS 9, IAS 39 and IFRS 7 Interest Rate Benchmark Reform published in September 2019, which include transitional rules and accounting exemptions for interest rate hedges. The outstanding changes to the benchmarks will not have any impact on the USD and CHF hedging relationships designated by the Lufthansa Group. Contracts for derivatives and financial liabilities will be analysed by the Lufthansa Group and fallback clauses added if necessary. Systematic adjustments will also be made to the extent required.

Fuel price risk

In 2019, fuel costs accounted for 18.1% of the Lufthansa Group's operating expenses (previous year: 17.1%). Significant changes in fuel prices can therefore have a significant effect on the Lufthansa Group's result.

Fuel price risk is limited by the use of crude oil hedges. The hedging level and the time horizon depend on the risk profile, which is derived from the business model of a Group company. As a rule, up to 5% of exposure is hedged monthly for up to 24 months by spread options and other combinations of hedges. Executive Board approval may be obtained to extend the hedging period and to increase the monthly hedging volume in order to exploit market opportunities. The target hedging level is up to 85%.

From a year-end perspective, fuel exposure was as follows:

T180 FUEL EXPOSURE

		2020	2021
Fuel requirement	in 1,000 tonnes	11,040	11,061
Hedges	in 1,000 tonnes	8,106	2,635
Hedging level	%	73	24
Hedging rate	USD/bbl	65.32	60.45

The following sensitivity analysis shows how equity would have been affected by changes in the market value of hedging instruments held as of the reporting date if the fuel price, the identified risk variable, had been different. Since hedge accounting rules mean that changes in the market value of the instruments are only recognised directly in equity without effect on profit and loss, the change in the fuel price alone has no effect on earnings.

T181 SENSITIVITY ANALYSIS BY FUEL PRICE

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Fuel price		
+10%	–	255
–10%	–	–238

¹⁾ All amounts after deferred tax effects; +/– signs relate to net profit and/or equity.

Market values of the derivative financial instruments used for hedging

Hedging instruments in designated hedging relationships are used to hedge exchange rate, interest rate and fuel price risks as of the reporting date. They changed as follows in the financial year:

T182 DERIVATIVE FINANCIAL INSTRUMENTS USED FOR HEDGING AS OF 31 DEC 2019

in €m	Positive market value	Negative market value	Change in fair value of hedging instrument – designated risk	Change in fair value of hedging instrument – non-designated risk	Basis adjustment of hedged items	OCI – cash flow hedge reserve	OCI – cost of hedging	Ineffective portion of hedges – designated risk	Ineffective portion of hedges – non-designated risk
Fair value hedge									
Interest rate hedges – interest rate swaps	156	- 27	89	-	- 90	-	-	- 1	-
Cash flow hedge									
Fuel hedging – options	185	- 39	404	163	-	402	164	2	- 1
Exchange rate hedging – forward transactions	963	- 133	- 386	315	-	- 380	319	- 6	- 4
Interest rate hedges – interest rate swaps	49	-	4	-	-	8	-	- 4	-
Total	1,353	- 199	111	478	- 90	30	483	- 9	- 5
of which current	452	- 135							

T182 DERIVATIVE FINANCIAL INSTRUMENTS USED FOR HEDGING AS OF 31 DEC 2018

in €m	Positive market value	Negative market value	Change in fair value of hedging instrument – designated risk	Change in fair value of hedging instrument – non-designated risk	Basis adjustment of hedged items	OCI – cash flow hedge reserve	OCI – cost of hedging	Ineffective portion of hedges – designated risk	Ineffective portion of hedges – non-designated risk
Fair value hedge									
Interest rate hedges – interest rate swaps	101	- 47	60	-	- 47	-	-	13	-
Cash flow hedge									
Fuel hedging – options	27	- 459	- 833	- 173	-	- 840	- 174	6	1
Exchange rate hedging – forward transactions	977	- 78	463	83	-	462	76	1	6
Interest rate hedges – interest rate swaps	54	- 1	19	-	-	15	-	3	-
Total	1,159	- 585	- 291	- 90	- 47	- 363	- 98	23	7
of which current	334	- 383							

The market values stated for financial derivatives correspond to the price at which an independent third party would assume the rights and/or obligations from the financial instrument. The fair values of interest rate derivatives correspond to their respective market values, which are measured using appropriate mathematical methods, such as discounting expected future cash flows. Discounting takes market standard interest rates and the residual term of the respective instruments into account. Currency forward transactions and interest rate swaps are individually discounted to the balance sheet date based on their respective forward rates and the appropriate interest rate curve. The market prices of options used to hedge fuel prices are determined using acknowledged option pricing models.

Depending on the hedged exposure, the Lufthansa Group designates interest rate hedges as both fair value hedges and cash flow hedges and accounts for them accordingly. Interest rate swaps are designated as part of a hedging relationship and are not broken down into individual components. Ineffectiveness in these hedging relationships mainly occurs as a result of the subsequent designation of cross currency swaps in hedges as of 1 January 2018. Other reasons for ineffectiveness in hedging relationships are different parameters in the hedged item and the hedging instrument and the basis spread in cross currency swaps. Ineffectiveness in fair value hedges and cash flow hedges are recognised and presented as part of the financial result, below the other financial items.

Derivatives used in the context of fuel hedging to hedge future kerosene purchases are designated as cash flow hedges. Since 1 January 2018, the Lufthansa Group has applied the IFRS 9 component approach, using crude oil, based on Brent Crude ICE, as the designated risk component of the hedging instrument. The hedged item is composed of a global mix of crude oil types. The base risk between individual crude oil

components in the hedging instrument and the crude oil mix in the hedged item is reduced by rebalancing the volumes that make up the hedged item on a quarterly basis. In 2019, the quarterly rebalancing factors for adjusting the hedged item were as follows: 1.02 (Q1), 1.02 (Q2), 1.022 (Q3) and 1.021 (Q4). The Lufthansa Group generally uses options and combinations of options to hedge fuel prices. The intrinsic value of the option is designated as the hedging instrument, so that effective changes in the intrinsic values are recognised in other comprehensive income in the cash flow hedge reserve. The time value of an option is not designated as a hedging instrument and effective changes in the time value are therefore recognised as a cost of hedging. This rule was applied for the first time retroactively. Ineffectiveness in fuel price hedges result from the base risk between the crude oil component and the crude oil mix in the component approach. Ineffectiveness is recognised and presented as part of the financial result, below the other financial items.

Since 1 January 2018, the Lufthansa Group has prospectively applied the spot-to-spot method for exchange rate forward transactions designated as cash flow hedges. The spot component of a forward contract is designated as a hedging instrument and effective value changes are recognised in the cash flow hedge reserve. The other effective components of a forward contract, the forward component and the cross-currency basis spread are presented in a separate OCI component in line with the legal requirements for the cost of hedging. Ineffectiveness in hedging relationships results from changes in the timing of the planned aircraft purchases. Ineffectiveness is presented as part of the financial result, below the other financial items (🔗 **Note 13, p. 162**).

The Lufthansa Group uses the hypothetical derivative method to calculate changes in the value of hedged items designated as being part of a hedging relationship.

T183 DESIGNATED HEDGED ITEMS IN HEDGING RELATIONSHIPS 2019

in €m	Carrying amount of liabilities	Change in fair value of hedged items – designated risk	Change in fair value of hedged items – non-designated risk	Basis adjustment of hedged items from fair value hedges – cumulative
Fair value hedge				
Interest rate hedges – interest rate swaps	- 7,330	- 90	-	- 99
Cash flow hedge				
Fuel hedging – options	-	- 407	- 164	-
Exchange rate hedging – forward transactions	-	390	- 301	-
Interest rate hedges – interest rate swaps	-	9	-	-
Total	- 7,330	- 98	- 465	- 99

T183 DESIGNATED HEDGED ITEMS IN HEDGING RELATIONSHIPS 2018

in €m	Carrying amount of liabilities	Change in fair value of hedged items – designated risk	Change in fair value of hedged items – non-designated risk	Basis adjustment of hedged items from fair value hedges – cumulative
Fair value hedge				
Interest rate hedges – interest rate swaps	6,685	– 47	–	– 65
Cash flow hedge				
Fuel hedging – options	–	840	174	–
Exchange rate hedging – forward transactions	–	– 502	– 61	–
Interest rate hedges – interest rate swaps	–	– 16	–	–
Total	6,685	275	113	– 65

T184 STATEMENT OF EQUITY RECONCILIATION FOR CASH FLOW HEDGES 2019

in €m	As of 1 Jan 2019	Gains or losses from effective hedging relationships	Reclassification to profit or loss	Reclassification to acquisition costs of inventories	Reclassification to acquisition costs of aircraft	As of 31 Dec 2019
OCI – cash flow hedge reserve	451	803	544	89	146	477
Fuel hedging – options	– 360	491	–	89	–	43
Exchange rate hedging – futures	820	310	544	–	146	440
Interest rate hedges – interest rate swaps	– 9	2	–	–	–	– 6
OCI – cost of hedging	– 151	476	–	–	–	326
Fuel hedging – options	– 227	157	–	–	–	– 70
Exchange rate hedging – futures	76	319	–	–	–	396
Total	300	1,279	544	89	146	803

T184 STATEMENT OF EQUITY RECONCILIATION FOR CASH FLOW HEDGES 2018

in €m	As of 1 Jan 2018	Gains or losses from effective hedging relationships	Reclassification to profit or loss	Reclassification to acquisition costs of inventories	Reclassification to acquisition costs of aircraft	As of 31 Dec 2018
OCI – cash flow hedge reserve	837	490	57	786	33	451
Fuel hedging – options	479	– 53	–	786	–	– 360
Exchange rate hedging – futures	358	552	57	–	33	820
Interest rate hedges – interest rate swaps	–	– 9	–	–	–	– 9
OCI – cost of hedging	– 61	– 90	–	–	–	– 151
Fuel hedging – options	– 61	– 166	–	–	–	– 227
Exchange rate hedging – futures	–	76	–	–	–	76
Total	776	400	57	786	33	300

Derivative financial instruments that do not meet the requirements for applying hedge accounting are measured at fair value through profit or loss. As a rule, these derivatives were originally in an economic hedging relationship with a particular exposure, but the exposure can either not be measured for hedge accounting purposes or no longer exists.

Fair values are all calculated solely on the basis of recognised financial and mathematical methods, using publicly available market information.

Changes in the market values of derivatives that do not qualify as effective hedging transactions under IFRS 9 can be seen in the income statement and in the overview of other financial items ➤ **Note 13, p. 162.**

Liquidity risk

Complex financial planning systems enable the Lufthansa Group to identify its future liquidity position at an early stage. Based on the results of the Group strategy and planning processes, a monthly rolling liquidity plan differentiated by currency is drawn up with a planning horizon of 24 months. This planning method offers an up-to-date picture of anticipated liquidity developments within the Company and corresponding currency effects.

The Lufthansa Group always holds a liquidity reserve of at least EUR 2,300m that is available at short notice. In addition, the Lufthansa Group held confirmed unused lines of credit as of 31 December 2019 totalling EUR 774m (previous year: EUR 849m).

A maturity analysis for the financial liabilities and the derivative financial instruments based on undiscounted gross cash flows including the relevant interest payments shows the following projected cash inflows and outflows from the perspective of the balance sheet date 31 December 2019. As a result of the hedges used, there are generally direct connections between the cash inflows and outflows for the derivative financial instruments shown.

T185 MATURITY ANALYSIS OF LIABILITIES FROM DERIVATIVE FINANCIAL INSTRUMENTS

in €m	From fuel derivatives	Cash inflow from gross settlement of interest rate and exchange rate derivatives	Cash outflow from gross settlement of interest rate and exchange rate derivatives	Net
1st quarter	5	-1,020	1,048	33
Up to 1 year ¹⁾	12	-2,094	2,169	87
1-5 years	1	-1,823	1,909	87
Later	-	-344	361	17

¹⁾ Without payments in 1st quarter.

T186 MATURITY ANALYSIS FOR NON-DERIVATIVE FINANCIAL INSTRUMENTS

in €m	Outflows
1st quarter	-4,614
Up to 1 year ¹⁾	-2,363
1-5 years	-4,955
Later	-2,274

¹⁾ Without payments in 1st quarter.

Credit risk

The sale of passenger travel and freight documents mostly takes place via agencies. These agencies are mostly connected to national clearing systems for billing passenger and freight sales. The creditworthiness of the agents is reviewed by the responsible clearing systems. Due to the broad diversification, credit risk for the agencies is relatively low worldwide. Nonetheless, credit terms for agents in some markets were tightened significantly in cooperation with the International Air Transport Association (IATA) in order to reduce credit risks even further.

Receivables and liabilities between airlines are offset through bilateral arrangements or via the IATA clearing house, insofar as the contracts underlying services do not explicitly specify otherwise. Systematic settlement of weekly receivables and liability balances significantly reduces the default risk. Fidelity guarantee insurance also covers partial risks within a certain range. Service contracts occasionally require collateral for miscellaneous transactions. All other contractual relationships are subject to credit rules, which, depending on the type and volume of the contract involved, require collateral, credit ratings/references or historical data from prior dealings, particularly payment history, in order to avoid defaults.

Counterparty risks in connection with credit card companies are monitored closely and incoming payments are reviewed daily. To reduce risks even further, a permanent analysis process examines whether to further tighten credit terms for some settlement partners. In addition to the monitoring of receivables at the Company or segment level, there is also counterparty monitoring at Group level, with individually assigned limits, in order to identify the accumulation of portfolio risks across the entire Group and take appropriate action where necessary. The maximum credit risk for financial assets from the potential insolvency of customers is their carrying amount.

In addition to individual write-downs on receivables if a default event occurs, IFRS 9 requires risk provisions to be recognised for expected losses. The Lufthansa Group's trade receivables are exposed to external credit risks for which expected losses have already been taken into consideration in accordance with IFRS 9, in addition to individual write-downs. A simplified impairment model based on an impairment matrix is used for the portion of the receivables portfolio that does not consist of credit card receivables but is subject to external credit risks. The portfolio is divided into clusters based on customer groups, regions and days past due. A default matrix is calculated on the basis of historical

default events in the Lufthansa Group's receivables portfolio, which is adjusted for forward-looking, publicly available insolvency forecasts. This impairment matrix is applied to trade receivables that are exposed to external credit risk and are not credit card receivables. In addition, the receivables portfolio includes credit card receivables for which the Lufthansa Group is the credit card issuer. Expected losses for these credit card receivables are calculated in a separate model, based on counterparty-specific external ratings

and default probabilities. The Lufthansa Group uses a definition of default of 90 days past due for receivables, which are written off in full if the default event occurs. Exceptions are permitted in justified cases, however.

In the following tables, Levels 1 and 2 describe expected credit losses, whereas Level 3 shows individual impairment losses on the basis of actual default events.

T187 STATEMENT OF RISK PROVISIONS 2019

in €m	Opening balance risk provision as of 1 Jan 2019	Additions through profit or loss	Reversals through profit or loss	Utilisation	Closing balance risk provision as of 31 Dec 2019	Opening balance gross carrying amount as of 1 Jan 2019	Closing balance gross carrying amount as of 31 Dec 2019
Trade receivables and other receivables (simplified approach)	314	102	- 10	- 8	398	2,540	2,457
of which from expected losses	9	6	- 3	0	12	2,234	2,066
of which from individual loss allowances	305	96	- 7	- 8	386	306	391
Trade receivables and other receivables (credit card receivables)	24	6	- 5	- 8	17	1,052	1,060
of which Level 1	17	1	- 4	0	14	1,041	1,051
of which Level 2	0	0	0	0	0	0	0
of which Level 3	7	5	- 1	- 8	3	11	9
Total	338	108	- 15	- 16	415	3,592	3,517

T187 STATEMENT OF RISK PROVISIONS 2018

in €m	Opening balance risk provision as of 1 Jan 2018	Additions through profit or loss	Reversals through profit or loss	Utilisation	Closing balance risk provision as of 31 Dec 2018	Opening balance gross carrying amount as of 1 Jan 2018	Closing balance gross carrying amount as of 31 Dec 2018
Trade receivables and other receivables (simplified approach)	276	95	- 31	- 15	325	2,371	2,522
of which from expected losses	15	2	- 8	0	9	2,095	2,216
of which from individual loss allowances	261	93	- 23	- 15	316	276	306
Trade receivables and other receivables (credit card receivables)	12	19	- 2	- 4	24	906	1,048
of which Level 1	12	7	- 2	0	17	899	1,037
of which Level 2	0	0	0	0	0	0	0
of which Level 3	0	12	0	- 4	7	7	11
Total	288	114	- 33	- 19	349	3,277	3,570

An expected loss of EUR 1m was recognised for contract assets as of the reporting date. In addition, a risk provision of EUR 0.4m was recognised as of the reporting date in Level 1 of the general impairment model for securities measured at fair value through other comprehensive income.

In the reporting year, the Lufthansa Group used the following default rates for each past due category in the impairment matrix for the simplified approach of the impairment model.

T188 IMPAIRMENT MATRIX FOR TRADE RECEIVABLES 2019

		Not overdue	1 – 30 days overdue	31 – 60 days overdue	61 – 90 days overdue	More than 90 days overdue	Total
Default rate	%	0.3	0.8	1.3	1.6	1.8	–
Carrying amounts for trade receivables	€m	1,419	393	47	41	166	2,066
Expected loss	€m	4	3	1	1	3	12

T188 IMPAIRMENT MATRIX FOR TRADE RECEIVABLES 2018

		Not overdue	1 – 30 days overdue	31 – 60 days overdue	61 – 90 days overdue	More than 90 days overdue	Total
Default rate	%	0.2	0.5	1.2	1.5	1.7	–
Carrying amounts for trade receivables	€m	1,484	364	163	30	175	2,216
Expected loss	€m	2	2	2	–	3	9

In order to determine expected losses of credit card receivables according to IFRS 9, the off-balance sheet exposure must be considered, in addition to the on-balance sheet exposure. The off-balance sheet exposure describes the portion of a credit card's unused or free limit. The following overview shows the risk data for the credit card portfolio. Expected losses are calculated at the level of the individual credit card, so that the sizes of the average default probability and the average loss relate to the individual credit card.

T189 CONCENTRATION OF CREDIT RISK FROM CREDIT CARD RECEIVABLES

Internal credit rating	Probability of default according to external credit rating	Average probability of default	Average expected loss per concerned credit card in €	Exposure Level 1 impairment model in €m	Exposure Level 2 impairment model in €m	Exposure Level 3 impairment model in €m
On-balance sheet exposure						
Low risk	≤ 2%	0.6%	36	802	2	–
Medium risk	> 2.0% to ≤ 6.5%	3.0%	65	224	–	–
High risk	> 6.5%	48.0%	4,905	21	1	9
Total				1,047	3	9
Off-balance sheet exposure						
Low risk	≤ 2%	0.6%	48	1,161	2	–
Medium risk	> 2.0% to ≤ 6.5%	3.0%	134	249	1	–
High risk	> 6.5%	48.0%	1,020	150	–	–
Total				1,560	3	–

Securities representing debt instruments are rated as follows (Standard & Poor's):

T190 SECURITIES RATINGS – DEBT INSTRUMENTS

in €m	
AAA	344
AA+	36
AA	130
AA-	95
A+	280
A	263
A-	252
BBB+	125
BBB	72
Below BBB or unrated	44
Total	1,641

The credit risk for derivative financial instruments and securities held at fair value through or without effect on profit and loss is the risk that a counterparty defaults. The maximum credit risk from these instruments is their carrying amount. The counterparty default risk for financial market transactions is limited by defining a maximum risk, taking the credit score given by recognised rating agencies into account.

44 Contingencies and events after the reporting period

T191 CONTINGENT LIABILITIES

in €m	31 Dec 2019	31 Dec 2018
From guarantees, bills of exchange and cheque guarantees	1,634	988
From warranty contracts	378	218
From providing collateral for third-party liabilities	47	45
	2,059	1,251

Warranties include an amount of EUR 248m (previous year: EUR 114m) relating to contingent liabilities towards creditors of joint ventures. As in the previous year, liabilities under collateral agreements included contingent liabilities of EUR 6m towards creditors of joint ventures. Of the total, EUR 1,715m (previous year: EUR 1,094m) relates to joint and several guarantees and warranties taken on. These are matched by compensatory claims against the other co-debtors amounting to EUR 1,654m (previous year: EUR 1,042m). Insofar as annual financial statements have yet to be published, these figures are preliminary.

Otherwise, provisions for other risks could not be made because utilisation was not sufficiently probable. The potential financial effect of these provisions on the result would have been EUR 55m (previous year: EUR 55m).

As in the previous year, no profits or cash inflows are expected from contracts for the sale of aircraft.

LEGAL RISKS

The Lufthansa Group is exposed to a number of legal risks in the course of its normal business. Based on current knowledge, the assumption is that these will not have any major, lasting effects on the net assets, financial and earnings position, beyond those for which provisions for litigation risks have been made, [Note 34, p. 185ff.](#)

Legal disputes and other claims made against the Group are always subject to uncertainty, however. Management estimates of these risks may also change over time. The actual outcome of these legal disputes may differ from earlier management estimates, which could have significant effects on the net assets, financial and earnings position and the reputation of our Company.

Due to the existing uncertainties and to those described below, we cannot make an assessment of the amount of the respective contingent liabilities or of the group of contingent liabilities. The legal disputes that these statements refer to include:

Risk of successful claims for damages in ongoing antitrust proceedings

Various cargo airlines, including Lufthansa Cargo AG and Swiss International Air Lines AG, were involved in a cargo cartel in the period between December 1999 and February 2006. Deutsche Lufthansa AG, Lufthansa Cargo AG and Swiss International Air Lines AG are at risk of civil claims for damages in Norway, Israel, Korea and the Netherlands. The lawsuits have been brought by both direct and indirect customers and are addressed to the airlines as co-debtors.

All litigation in Germany and the United Kingdom was ended by a settlement in the course of 2019.

In Germany, a lawsuit against Lufthansa Cargo AG and others for information and damages was filed with Cologne regional court by a subsidiary of Deutsche Bahn AG. Litigation proceedings were started in late 2013 and expanded in late 2014. The lawsuit is addressed to a total of eleven cargo airlines and claims for purported damages of around EUR 3bn in total, including interest. Settlements with some of the other airlines concerned had reduced the claim in the past to EUR 1.6bn. Intense negotiations with Deutsche Bahn AG led to a settlement in August 2019, thus bringing the proceedings against Group companies to an end. It is very unlikely that the other litigants will challenge the settlements.

At present, it is not possible to give a concrete assessment of the outcome of the lawsuits still pending or of the number and amount of any other claims. When evaluating the risk, it should nonetheless be borne in mind that the European Commission's decision on the cargo cartel, which the claimants in the civil lawsuits refer to, among others, is still not legally binding. Following the appeal of this 2010 decision by the European Court of Justice (ECJ) in December 2015, the European Commission sent revised penalty notices in March 2017 in which the content was the same but the reasoning had been altered. The airlines concerned, including Lufthansa, again contested them, so the penalty notices are still not effective.

Moreover, an expert economic opinion commissioned by Lufthansa Cargo AG and Swiss International Air Lines AG has come to the conclusion that the cartel did not inflict any actual damage on customers. Even if there were damages (i.e. allegedly higher cartel prices), the court would have to examine whether the claimants passed them on to their own customers (in the case of the freight forwarders) or whether they were passed on to the claimants (in the case of the final customers). Nonetheless, significant effects on the net assets, financial and earnings position of the Group cannot be ruled out if it should lose any of these legal proceedings.

Investigations in connection with work and service contracts

The investigations by the customs authorities in previous years into possible breaches of the German Law on Labour Leasing (Arbeitnehmerüberlassungsgesetz – AÜG) concerning the procurement of services by the Lufthansa Group have been concluded amicably. No penalties were imposed on individual employees of the Lufthansa Group nor on companies in accordance with Section 30 of the Administrative Offences Act (Gesetz über Ordnungswidrigkeiten – OWiG). Administrative proceedings with the German Statutory Pension Insurance Scheme concerning the legal status of two workers have been submitted for judicial review to the Higher Social Court in Berlin.

TAX RISKS

Tax risks exist largely because of differences in legal opinions between the German tax authorities and the Company. In tax audits for the financial years 2001 to 2012, the tax authorities came to a number of different conclusions to those on which the Company had based its tax returns, relating, in particular, to partial write-downs on shareholder loans, the treatment of various lease structures and the acquisition of a foreign subsidiary, as well as the recognition and measurement of certain provisions and assets. The Lufthansa Group has appealed against the resulting tax assessments. Without abandoning its legal position, almost all the disputed matters were settled in the past by paying the back taxes demanded by the authorities. As far as the partial write-downs are concerned, a change in Federal Finance Court case law in the reporting year resulted in a negative ruling for the Company. The Lufthansa Group still assumes, however, that there is a very strong likelihood of winning in all the matters still being disputed. To the extent that payments have already been made, claims against the tax authorities were recognised in line with IFRIC 23. No provisions were recognised for the remaining points as of the reporting date given the low probability of their use. There is, however, the risk of a possible subsequent payment totalling some EUR 200m for the circumstances mentioned. The assessment of the amount is subject to uncertainty. If the legal position of Deutsche Lufthansa AG should prevail, no negative effects are expected for the net assets, financial and earnings position.

EVENTS AFTER THE REPORTING PERIOD

Spread of the coronavirus significantly affects financial performance of the Lufthansa Group

The ongoing spread of the coronavirus has led to a significant fall in demand for air travel. Some countries, including the USA, imposed a travel ban for passengers from the European Union. This led to a decline in bookings at the Group airlines and to flight cancellations. In response, the Group has decided to cut its flight capacity significantly and to implement a wide range of cost-cutting measures affecting staff costs, operating costs and project budgets. Steps to raise liquidity will also improve the Group's capital structure.

The Lufthansa Group is assuming that Adjusted EBIT will fall significantly in 2020 compared with the previous year as a result of the coronavirus crisis. The precise extent of the fall depends above all on the further spread of the virus, the necessary capacity adjustments, the scope and impact of the cost-cutting measures and the development of fuel costs. ➔ Forecast, p. 106ff.

Agreement with UFO on resolution process for collective bargaining dispute

The Lufthansa Group and the independent flight attendants' union UFO agreed on 31 January 2020 on a multi-stage process to resolve the collective bargaining dispute. The process provides for a separation of wage and non-wage topics in three stages: mediation, arbitration and an out-of-court settlement. In the mediation process, questions of cooperation will be discussed, with a focus on future dealings with one another, and solutions found. At the same time the parties will begin a comprehensive arbitration process covering the wage claims made by UFO and other collective bargaining topics. In a separate, out-of-court settlement process, the specific legal affairs of both parties and individual officials will be presented to a labour law judge, ideally resulting in definitive solutions.

45 Other financial obligations

As of 31 December 2019, there were order commitments of EUR 14.6bn for capital expenditure on property, plant and equipment, including repairable spare parts, and for intangible assets (previous year: EUR 13.9bn). There were also capital and shareholder loan commitments of EUR 335m towards equity investments (previous year: EUR 336m).

46 Auditors' fees

The fees paid to the auditors in the financial year and charged to expenses in accordance with Section 314 Paragraph 1 No. 9 HGB are made up as follows:

T192 AUDITORS' FEES

in €m	2019	2018
Annual audit	4.4	4.2
Other assurance services	1.0	0.4
Tax advisory services	0.9	0.6
Other services	1.9	0.3
Total	8.2	5.5

The auditing services mainly consist of fees for auditing the consolidated financial statements, the review of the half-yearly financial statements and the audit of the financial statements of Deutsche Lufthansa AG and its consolidated subsidiaries, and for assistance in connection with the implementation of new accounting standards. Fees shown under other advisory services mainly relate to the audit of information systems and processes as well as services in connection with statutory and contractual requirements. Tax advisory services mainly relate to intended reorganisations at Group level, tax advice on transfer pricing, international taxes and inspections by tax authorities. Other services relate mainly to IT advisory services.

The following fees paid to the global PricewaterhouseCoopers group, especially abroad, were additionally recognised as expenses:

T193 ADDITIONAL AUDITORS' FEES

in €m	2019	2018
Annual audit	3.8	3.9
Other assurance services	0.1	0.2
Tax advisory services	0.3	0.4
Other services	0.2	0.2
Total	4.4	4.7

The auditor at PricewaterhouseCoopers GmbH Wirtschaftsprüfungsgesellschaft responsible for the Lufthansa Group is Eckhard Sprinkmeier. He held this position for the second time in financial year 2019.

COMPOSITION OF THE GROUP

47 Group of consolidated companies

The consolidated financial statements of Deutsche Lufthansa AG include all major subsidiaries, joint ventures and associated companies.

Subsidiaries are entities over which Deutsche Lufthansa AG has rights that give it the ability to control the entity's relevant activities. Relevant activities are those activities that have a significant influence on the entity's return. Deutsche Lufthansa AG therefore only has control over a company when it is exposed to variable returns from the company and its power over the company's relevant activities enables it to influence these returns. This definition of control also applies to structured entities that are identified as such in the list of significant Group companies. In general, the ability to control subsidiaries arises when Deutsche Lufthansa AG holds a direct or indirect majority of voting shares. In structured entities, the ability to control does not come from holding the majority of voting shares, but rather from contractual agreements. Entities are consolidated from the time that the ability to control begins. They cease to be consolidated when the ability to control ends.

Joint arrangements are classified either as joint ventures or as joint operations. A joint arrangement exists for when the Lufthansa Group carries on joint business activities with third parties on the basis of a contractual agreement. Joint management or control only exists when decisions on activities that have a significant effect on the returns from an agreement require the unanimous approval of the parties sharing control.

Significant interests in companies that are managed jointly with one or more partners (joint ventures) are accounted for using the equity method. Joint operations are defined by the fact that the parties exercising joint control over the arrangement have rights to the assets attributed to the arrangement and are liable for its debts. Assets and liabilities, revenue and expenses from the significant joint operations are recognised in the consolidated financial statements of the Lufthansa Group in proportion to these rights and obligations.

Associated companies are companies in which Deutsche Lufthansa AG has the opportunity to exercise major influence over financial and operating policy based on an interest of between 20% and 50%. Significant associated companies are accounted for in the consolidated financial statements using the equity method.

A list of major subsidiaries, joint arrangements and associated companies can be found in [T201 – T204](#), p. 231 – 238, and the list of shareholdings in [T205](#), p. 239 – 242.

In addition to Deutsche Lufthansa AG as the parent company, the group of consolidated companies includes 66 domestic and 267 foreign companies, including structured entities (previous year: 66 domestic and 256 foreign companies).

One material joint operation was also included in the consolidated financial statements on a pro rata basis in accordance with IFRS 11. It consists of a German cargo airline operated jointly by Deutsche Post AG and Deutsche Lufthansa AG, which each hold 50% of the share capital and voting rights. The two shareholders are also customers of the company and use the capacities of its cargo aircraft. In contrast to its capital and voting rights, the company's assets and liabilities, as well as its income and expenses, are allocated based on the user relationship of the shareholders according to their contracts.

Changes in the group of consolidated companies during the 2019 financial year are shown in the following table:

T194 CHANGES IN THE GROUP OF CONSOLIDATED COMPANIES FROM 1 JAN 2019 TO 31 DEC 2019

Name, registered office	Additions	Disposals	Reasons
Network Airlines business segment			
Yamasa Aircraft LH17 Kumiai, Okayama, Japan	17 Jan 2019		Established
Yamasa Aircraft LH18 Kumiai, Okayama, Japan	17 Jan 2019		Established
Yamasa Aircraft LH19 Kumiai, Okayama, Japan	17 Jan 2019		Established
Yamasa Aircraft LH15 Kumiai, Okayama, Japan	17 Jan 2019		Established
Yamasa Aircraft LH16 Kumiai, Okayama, Japan	17 Jan 2019		Established
Fleur Leasing Co. Ltd., Tokyo, Japan		23 Jan 2019	Liquidation
Yamasa Aircraft LH3 Kumiai, Tokyo, Japan		19 Mar 2019	Liquidation
Yamasa Aircraft LH21 Kumiai, Okayama, Japan	27 Mar 2019		Established
Dia Kranich Ltd., Tokyo, Japan		28 Mar 2019	Liquidation
SMLC Crater Co. Ltd., Tokyo, Japan		28 Mar 2019	Liquidation
Yamasa Aircraft LH4 Kumiai, Okayama, Japan		28 Mar 2019	Liquidation
Yamasa Aircraft LH 20 Kumiai, Okayama, Japan	3 Apr 2019		Established
Dia Orff Ltd., Tokyo, Japan	16 Apr 2019		Established
Dia Wagner Ltd., Tokyo, Japan	16 Apr 2019		Established
Yamasa Aircraft LH22 Kumiai, Okayama, Japan	16 Apr 2019		Established
Yamasa Aircraft LH23 Kumiai, Okayama, Japan	16 Apr 2019		Established
Dia Bach Ltd., Tokyo, Japan	18 Apr 2019		Established
AUA LNR/LNS/LNT/LNU Ltd., George Town, Grand Cayman		13 Jun 2019	Liquidation
Benjamin LH6 Kumiai, Okayama, Japan		23 Jul 2019	Liquidation
Tim LH5 Kumiai Japan, Okayama, Japan		23 Jul 2019	Liquidation
Nicolai LH7 Kumiai, Okayama, Japan		24 Jul 2019	Liquidation
AirUtopia Ltd. Japan, Tokyo, Japan		29 Jul 2019	Liquidation
Air Sylph Ltd., Tokyo, Japan		30 Jul 2019	Liquidation
Lufthansa Leasing Austria GmbH & Co. OG Nr. 40, Salzburg, Austria	17 Aug 2019		Established
Lufthansa Leasing Austria GmbH & Co. OG Nr. 41, Salzburg, Austria	17 Aug 2019		Established
Lufthansa Leasing Austria GmbH & Co. OG Nr. 42, Salzburg, Austria	17 Aug 2019		Established
Lufthansa Leasing Austria GmbH & Co. OG Nr. 43, Salzburg, Austria	17 Aug 2019		Established
Lufthansa Leasing Austria GmbH & Co. OG Nr. 44, Salzburg, Austria	17 Aug 2019		Established
Lufthansa Leasing Austria GmbH & Co. OG Nr. 45, Salzburg, Austria	17 Aug 2019		Established
SJ Frankfurt Co. Ltd., Tokyo, Japan		19 Sep 2019	Liquidation
FK Yocasta Leasing Ltd., Tokyo, Japan	15 Oct 2019		Established
NBB Rothenburg Lease Co., Ltd., Tokyo, Japan	18 Oct 2019		Established
ORIX Aquila Corporation, Tokyo, Japan	23 Oct 2019		Established
ORIX Lysithea Corporation, Tokyo, Japan	23 Oct 2019		Established
ORIX Telesto Corporation, Tokyo, Japan	23 Oct 2019		Established
Dia Hausen Ltd., Tokyo, Japan	29 Oct 2019		Established
CRANE LTD., Tokyo, Japan	17 Dec 2019		Established
Tusker Leasing Co., Ltd., Tokyo, Japan	17 Dec 2019		Established
Eurowings business segment			
Luftfahrtgesellschaft Walter mit beschränkter Haftung, Dortmund		1 Apr 2019	Disposal
Catering business segment			
Inversiones Turísticas Aeropuerto Panama, S.A., Panama City, Panama		15 Mar 2019	Fusion
SC International Services, Inc., Wilmington, USA		29 Mar 2019	Fusion
Sky Chefs Argentine, Inc., Wilmington, USA		29 Mar 2019	Fusion
Servicios Complementarios de Cabina, S.A. de C.V., Mexico City, Mexico	1 May 2019		Beginning of operations
LSG Sky Chefs Brussels International BVBA, Brussels, Belgium		17 Dec 2019	Liquidation
Additional Businesses and Group Functions			
LCH Grundstücksgesellschaft Berlin mbH, Frankfurt/Main	1 Jan 2019		Extension of operations

USE OF EXEMPTION PROVISIONS

The following fully consolidated German Group companies made use of the exemption provisions in Section 264 Paragraph 3 and Section 264b HGB in 2019.

T195 USE OF EXEMPTION PROVISIONS

Company name	Registered office
Eurowings Aviation GmbH	Cologne
Eurowings Digital GmbH	Cologne
Eurowings GmbH	Dusseldorf
Eurowings Technik GmbH	Cologne
Germanwings GmbH	Cologne
Hamburger Gesellschaft für Flughafenanlagen GmbH	Hamburg
Jettainer GmbH	Raunheim
LSG Asia GmbH	Neu-Isenburg
LSG Lufthansa Service Europa/Afrika GmbH	Neu-Isenburg
LSG Lufthansa Service Holding AG	Neu-Isenburg
LSG South America GmbH	Neu-Isenburg
LSY GmbH	Norderstedt
Lufthansa AirPlus Servicekarten GmbH	Neu-Isenburg
Lufthansa Asset Management GmbH	Frankfurt/Main
Lufthansa Asset Management Leasing GmbH	Frankfurt/Main
Lufthansa Aviation Training Berlin GmbH	Berlin
Lufthansa Aviation Training Germany GmbH	Frankfurt/Main
Lufthansa Aviation Training GmbH	Hallbergmoos
Lufthansa Cargo AG	Frankfurt/Main
Lufthansa CityLine GmbH	Munich Airport
Lufthansa Commercial Holding GmbH	Frankfurt/Main
Lufthansa Global Business Services GmbH	Frankfurt/Main
Lufthansa Industry Solutions AS GmbH	Norderstedt
Lufthansa Industry Solutions BS GmbH	Raunheim
Lufthansa Industry Solutions GmbH & Co. KG	Norderstedt
Lufthansa Process Management GmbH	Neu-Isenburg
Lufthansa Seeheim GmbH	Seeheim-Jugenheim
Lufthansa Systems GmbH & Co. KG	Raunheim
Lufthansa Technik AERO Alzey GmbH	Alzey
Lufthansa Technik AG	Hamburg
Lufthansa Technik Immobilien- und Verwaltungsgesellschaft mbH	Hamburg
Lufthansa Technik Logistik GmbH	Hamburg
Lufthansa Technik Logistik Services GmbH	Hamburg
Lufthansa Technik Maintenance International GmbH	Frankfurt/Main
Lufthansa Technik Objekt- und Verwaltungsgesellschaft mbH	Hamburg
Miles & More GmbH	Frankfurt/Main
time:matters GmbH	Neu-Isenburg
time:matters Holding GmbH	Neu-Isenburg
time:matters Spare Parts Logistics GmbH	Neu-Isenburg

The companies are affiliated with Deutsche Lufthansa AG by means of direct or indirect profit and loss transfer agreements respectively.

Furthermore, LHBD Holding Limited, London, UK, registration number 06939137, is exempt from the obligation to have its individual financial statements audited pursuant to Section 479a of the 2006 Companies Act, United Kingdom.

The consolidated financial statements include equity interests in 35 joint ventures and 39 associates (previous year: 35 joint ventures and 37 associates), of which eleven joint ventures (previous year: ten) and 15 associates (previous year: 15) were accounted for using the equity method. The other joint ventures and associated companies were valued at amortised cost due to their minor overall significance.

48 Related party disclosures

Balances and transactions between the Company and its fully consolidated subsidiaries, which constitute related parties, have been eliminated in the course of consolidation and are not commented on in this Note. Details of transactions between the Lufthansa Group and other related parties are disclosed below.

The Lufthansa Group segments render numerous services to related parties within the scope of their ordinary business activities. Conversely, the companies in question provide services to the Lufthansa Group as part of their normal business. These extensive supply and service relationships take place on the basis of market prices.

In addition, the Lufthansa Group and certain non-consolidated subsidiaries have concluded numerous billing agreements, partly governing the joint use of services. In these cases, the administrative services provided are charged as cost allocations.

The Lufthansa Group's cash management is centralised, and, in this respect, the Lufthansa Group also performs a "banking function" vis-à-vis the non-consolidated companies of the Group. Non-consolidated Group companies included in the Group's cash management invest their available cash with the Group or borrow funds from the Group and carry out their derivative hedging transactions with the Group. All transactions take place at market conditions.

Due to geographical proximity in many cases, a large number of subletting contracts exists between the Lufthansa Group and related parties. In these cases, the Lufthansa Group usually charges the rental costs and incidental expenses incurred to the companies in question on a pro rata basis.

The following table shows the volume of significant services provided to or by related parties:

T196 VOLUME OF SIGNIFICANT SERVICES PROVIDED TO OR BY RELATED PARTIES

in €m	Volume of services rendered		Volume of services received	
	2019	2018	2019	2018
Non-consolidated subsidiaries				
Albatros Versicherungsdienste GmbH, Germany	0	1	50	42
Austrian Airlines Technik-Bratislava, s.r.o., Slovakia	2	2	10	6
Austrian Airlines Tele Sales Service GmbH, Austria	0	0	5	5
Delvag Versicherungs-AG, Germany	6	9	3	6
DLH Fuel Company mbH, Germany	0	0	508	476
Global Load Control (PTY) LTD, South Africa	0	0	8	7
handling counts GmbH, Germany	0	1	9	11
LGSP Lufthansa Ground Service Portugal, Unipessoal Lda., Portugal	1	0	6	8
Lufthansa Aviation Training Austria GmbH, Austria	1	2	6	5
Lufthansa Aviation Training Operations Germany GmbH, Germany	3	4	16	13
Lufthansa Aviation Training USA Inc., USA	1	1	14	13
Lufthansa Consulting GmbH, Germany	2	1	9	9
Lufthansa Engineering and Operational Services GmbH, Germany	4	4	32	33
Lufthansa Global Business Services Hamburg GmbH, Germany	7	7	29	30
Lufthansa Global Business Services Sp. z o. o., Poland	2	2	33	30
Lufthansa Global Tele Sales GmbH, Germany	7	1	60	57
Lufthansa Industry Solutions TS GmbH, Germany	1	0	13	10
Lufthansa Services (Thailand) Ltd., Thailand	0	0	7	5
Lufthansa Services Philippines, Inc., Philippines	0	1	5	5
Lufthansa Systems FlightNav AG, Switzerland	1	1	25	23
Lufthansa Systems Hungaria Kft, Hungary	1	1	25	21
Lufthansa Systems Poland Sp. z o.o., Poland	2	2	32	28
Lufthansa Technical Training GmbH, Germany	6	6	19	23
Lufthansa Technik Brussels N.V., Belgium	5	1	2	2
Lufthansa Technik Component Services Asia Pacific Limited, China	1	1	6	6
Lufthansa Technik Middle East FZE, United Arab Emirates	1	0	8	5
Lufthansa Technik Milan s.r.l., Italy	6	2	3	2
Lufthansa Technik Services India Private Limited, India	1	2	5	4
Lufthansa Technik Shenzhen Co. Ltd., China	22	19	34	28
Lufthansa Technik Turbine Shannon Limited, Ireland	3	5	19	15
LZ-Catering GmbH, Germany	5	5	12	13
Reservation Data Maintenance India Private Ltd., India	0	0	5	3
time:matters Courier Terminals GmbH, Germany	1	0	8	1
ZeroG GmbH, Germany	1	0	7	5

T196 VOLUME OF SIGNIFICANT SERVICES PROVIDED TO OR BY RELATED PARTIES (continued)

in €m	Volume of services rendered		Volume of services received	
	2019	2018	2019	2018
Joint ventures				
Airfoil Services Sdn. Bhd., Malaysia	1	5	7	7
LG-LHT Aircraft Solutions GmbH, Germany	6	0	0	0
Lufthansa Bombardier Aviation Services GmbH, Germany	2	2	8	9
N3 Engine Overhaul Services GmbH & Co. KG, Germany	5	13	0	0
Shanghai Pudong International Airport Cargo Terminal Co. Ltd., China	1	1	6	6
Spairliners GmbH, Germany	59	63	46	48
Star Alliance Services GmbH, Germany	2	3	8	7
Terminal 2 Gesellschaft mbH & Co. oHG, Germany	1	1	13	14
Terminal One Group Association, L.P., USA	7	5	7	5
XEOS Sp. z o.o., Poland	4	5	7	0
Associated companies				
Aircraft Maintenance and Engineering Corp., China	10	16	3	5
Airmail Center Frankfurt GmbH, Germany	1	0	9	9
AviationPower GmbH, Germany	0	0	29	27
HEICO Aerospace Holdings Corp., USA	0	0	13	12
Other affiliated companies				
Shanghai Pudong International Airport Public Cargo Terminal Co. Ltd. (West), Shanghai, China	0	0	97	90
SunExpress Deutschland GmbH, Germany	26	25	93	98

The following tables show receivables owed by and liabilities to related parties:

T197 RECEIVABLES FROM AFFILIATED COMPANIES

in €m	2019	2018
Trade receivables from non-consolidated subsidiaries	30	19
Trade receivables from joint ventures	22	22
Trade receivables from associated companies	5	14
Trade receivables from other affiliated companies	1	12
Total trade receivables	58	67
Other receivables from non-consolidated subsidiaries	30	42
Other receivables from joint ventures	49	41
Other receivables from associated companies	10	10
Total other receivables	89	93
Loans to non-consolidated subsidiaries	99	81
Loans to joint ventures	5	27
Loans to associated companies	-	-
Total non-current receivables	104	108

T198 LIABILITIES TO AFFILIATED COMPANIES

in €m	2019	2018
Trade payables to non-consolidated subsidiaries	29	27
Trade payables to joint ventures	8	9
Trade payables to associated companies	3	4
Trade payables to other affiliated companies	2	3
Total trade payables	42	43
Other liabilities to non-consolidated subsidiaries	260	269
Other liabilities to joint ventures	0	0
Other liabilities from associated companies	0	0
Total other liabilities	260	269

No individual shareholders of Deutsche Lufthansa AG exercise significant influence over the Group. For transactions involving members of the Executive Board and Supervisory Board ("directors' dealings") ➔ **Note 49, p. 222.**

49 Supervisory Board and Executive Board

The disclosure of remuneration for key managers required by IAS 24 includes the remuneration of the active members of the Executive Board and Supervisory Board.

The members of the Executive Board and the Supervisory Board as well as the other offices that they hold are named in the combined management report in the section [Corporate governance, p. 110ff.](#)

The principles of the remuneration system and the amount of remuneration paid to Executive Board and Supervisory Board members are shown and explained in detail in the remuneration report. The remuneration report forms part of the combined management report [p. 115 – 130.](#)

Total Executive Board remuneration under IFRS was EUR 14.8m (previous year: EUR 16.2m), including current service costs for pensions of EUR 3.3m (previous year: EUR 3.5m).

The active members of the Executive Board in past reporting years were remunerated as follows:

T199 EXECUTIVE BOARD REMUNERATION (IFRS)

in € thousands	2019	2018
Basic salary	5,934	4,832
Other ¹⁾	110	427
One-year variable remuneration	2,091	4,845
Total short-term remuneration	8,135	10,104
Long-term variable remuneration ²⁾	877	3,570
Share-based remuneration	-1,523	-968
Current service cost for retirement benefits	3,291	3,510
Total long-term remuneration	2,645	6,112
Severance payments	4,055	-
Total	14,835	16,216

¹⁾ Other remuneration includes in particular benefits from the use of company cars, discounts in connection with cash outflows from share programmes ([Notes to the consolidated financial statements, Note 37, p. 189ff.](#)) and concessionary travel in accordance with the relevant IATA regulations.

²⁾ Expenses recognised in the reporting year for long-term variable remuneration for the financial years 2017 to 2019.

Pension provisions for Executive Board members active in the 2019 financial year came to EUR 16.7m (previous year: EUR 12.4m).

In addition to the provision for the one-year variable remuneration of EUR 2,091k (previous year: EUR 4,845k), provisions totalling EUR 877k were recognised for the future payment of long-term variable remuneration for the Executive Board members active in the financial year 2019 (previous year: EUR 3,570k). In addition, provisions of EUR 4,129k were recognised for the future payment of long-term, share-based remuneration for the Executive Board members active as of 31 December 2019 (previous year: EUR 5,651k).

Total remuneration (HGB) paid to the Executive Board of Deutsche Lufthansa AG in the financial year 2019 came to EUR 13,967k (previous year: EUR 13,015k). The above remuneration includes the new sharebased remuneration component Total Shareholder Return (TSR) of EUR 2,988k as part of the long-term variable remuneration incentive. Further comments on the new share-based remuneration component TSR can be found in [Note 37, p. 189ff.](#)

Current payments and other benefits for former members of the Executive Board and their surviving dependants came to EUR 6.4m (previous year: EUR 6.8m). This includes payments by subsidiaries as well as benefits in kind and concessionary travel.

Pension obligations towards former Executive Board members and their surviving dependants amount to EUR 69.1m (previous year: EUR 68.3m). They are included in pension provisions ([Note 33, p. 178ff.](#)).

Expenses for the fixed remuneration of Supervisory Board members came to EUR 2,170k in 2019 (previous year: EUR 2,107k). Other remuneration, mainly attendance fees, amounted to EUR 62k (previous year: EUR 68k). The Deutsche Lufthansa AG Supervisory Board members were also paid EUR 2k for work on supervisory boards of Group companies (previous year: EUR 27k).

In the reporting year, as in the previous year, no loans or advance payments were made to members of the Executive Board and to members of the Supervisory Board.

In addition to their Supervisory Board remuneration, employee representatives on the Supervisory Board received compensation for their work in the form of wages and salaries including pension entitlements amounting to EUR 1.1m in total in 2019 (previous year: EUR 1.1m).

The following auditor's report has been issued in accordance with Section 322 HGB and refers to the audit of the German language consolidated financial statements and the German language group management report, which is neither included nor incorporated by reference.

Independent auditor's report

To Deutsche Lufthansa AG, Cologne

REPORT ON THE AUDIT OF THE CONSOLIDATED FINANCIAL STATEMENTS AND OF THE GROUP MANAGEMENT REPORT

Audit Opinions

We have audited the consolidated financial statements of Deutsche Lufthansa AG, Cologne, and its subsidiaries (the Group), which comprise the consolidated statement of financial position as at 31 December 2019, the consolidated statement of profit or loss, consolidated statement of comprehensive income, consolidated statement of changes in equity and consolidated statement of cash flows for the financial year from 1 January to 31 December 2019, and notes to the consolidated financial statements, including a summary of significant accounting policies. In addition, we have audited the group management report of Deutsche Lufthansa AG, which is combined with the Company's management report, for the financial year from 1 January to 31 December 2019. In accordance with the German legal requirements, we have not audited the content of those parts of the group management report listed in the "Other Information" section of our auditor's report.

In our opinion, on the basis of the knowledge obtained in the audit,

- the accompanying consolidated financial statements comply, in all material respects, with the IFRSs as adopted by the EU, and the additional requirements of German commercial law pursuant to § (Article) 315e Abs. (paragraph) 1 HGB (Handelsgesetzbuch: German Commercial Code) and, in compliance with these requirements, give a true and fair view of the assets, liabilities, and financial position of the Group as at 31 December 2019, and of its financial performance for the financial year from 1 January to 31 December 2019, and
- the accompanying group management report as a whole provides an appropriate view of the Group's position. In all material respects, this group management report is consistent with the consolidated financial statements, complies with German legal requirements and appropriately presents the opportunities and risks of future development. Our audit opinion on the group management report does not cover the content of those parts of the group management report listed in the "Other Information" section of our auditor's report.

Pursuant to § 322 Abs. 3 Satz (sentence) 1 HGB, we declare that our audit has not led to any reservations relating to the legal compliance of the consolidated financial statements and of the group management report.

Basis for the Audit Opinions

We conducted our audit of the consolidated financial statements and of the group management report in accordance with § 317 HGB and the EU Audit Regulation (No. 537/2014, referred to subsequently as "EU Audit Regulation") in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the Institut der Wirtschaftsprüfer (Institute of Public Auditors in Germany) (IDW). We performed the audit of the consolidated financial statements in supplementary compliance with the International Standards on Auditing (ISAs). Our responsibilities under those requirements, principles and standards are further described in the "Auditor's Responsibilities for the Audit of the Consolidated Financial Statements and of the Group Management Report" section of our auditor's report. We are independent of the group entities in accordance with the requirements of European law and German commercial and professional law, and we have fulfilled our other German professional responsibilities in accordance with these requirements. In addition, in accordance with Article 10 (2) point (f) of the EU Audit Regulation, we declare that we have not provided non-audit services prohibited under Article 5 (1) of the EU Audit Regulation. We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our audit opinions on the consolidated financial statements and on the group management report.

Key Audit Matters in the Audit of the Consolidated Financial Statements

Key audit matters are those matters that, in our professional judgment, were of most significance in our audit of the consolidated financial statements for the financial year from 1 January to 31 December 2019. These matters were addressed in the context of our audit of the consolidated financial statements as a whole, and in forming our audit opinion thereon; we do not provide a separate audit opinion on these matters.

In our view, the matters of most significance in our audit were as follows:

1. Recognition of traffic revenue, including contract liabilities in respect of unused flight documents and customer loyalty programs
2. Pension provisions
3. Recoverability of non-current assets, in particular goodwill and intangible assets with indefinite useful lives
4. Accounting treatment of hedging transactions
5. Effects of the first-time application of IFRS 16 on the accounting of leases

Our presentation of these key audit matters has been structured in each case as follows:

- a. Matter and issue
- b. Audit approach and findings
- c. Reference to further information

Hereinafter we present the key audit matters:

1. Recognition of traffic revenue, including contract liabilities in respect of unused flight documents and customer loyalty programs

- a. Until they are used due to departure, sold flight documents are recognized as contract liabilities for unused flight documents in the Company's consolidated financial statements. Once a passenger coupon or an airfreight document has been used due to departure, the corresponding traffic revenue is recognized as revenue in the income statement. First of all, the part of the flight documents that has not yet been used in the financial year and is still valid in the following year is added to the contract liabilities for unused flight documents. In addition, based on historical data, the amount of expired flight documents which are expected to not be used is estimated; this amount is also recognized under contract liabilities for unused flight documents. In the financial year 2019, the Lufthansa Group realized a total of EUR 28.1 billion in traffic revenue, of which EUR 25.4 billion was attributable to passenger airlines. As of 31 December 2019, EUR 4.1 billion was recognized in the consolidated statement of financial position as contract liabilities for unused flight documents.

Until they are redeemed, bonus miles awarded to Miles & More members are recognized as contract liabilities on the basis of the individual selling price per bonus mile. Where bonus miles are collected with external cooperation partners, these are recognized as liabilities until

redeemed and measured at the prices paid to Lufthansa by the external cooperation partners. Bonus miles which are not expected to be used are realized as revenue over a period of three years. The contract liabilities from customer loyalty programs amounted to EUR 2.2 billion as of the balance sheet date.

From our point of view, these matters were of particular significance for our audit, as recognition and measurement of these items, which are specific to the business model and significant in amount, are highly dependent on the estimates and assumptions of the executive directors regarding the use behavior of passengers regarding flight documents, to which calculation processes that are, in some cases, complex are applied.

- b. We also included our specialists in the Risk Assurance Service (RAS) to audit traffic revenue. With their support, we evaluated, among other things, the appropriateness and effectiveness of the established internal control system used to settle and realize traffic revenue, including the IT systems used. To the extent that we were not able to conduct our own evaluation of the internal control system of services relating to IT systems or processes outsourced to third parties, we obtained an assurance report attesting to the appropriateness and effectiveness of the internal control system established by the service provider (ISAE 3402 Type II or SSAE 16), which our specialists assessed. In our audit of the contract liabilities for unused flight documents, we reconstructed among other things the individual steps used in the calculations. Specifically, we examined the open flight documents and their measurement by year of sale and validity. Furthermore, we considered the consistency and continuity of the methods used to calculate the flight prices, fees, taxes and other levies attributable to the flight documents no longer expected to be used. In order to assess the appropriateness of the contract liabilities from customer loyalty programs accounted for as of the balance sheet date, we evaluated among other things the fair value measurement for each category of use and the underlying assumptions and parameters derived therefrom. Furthermore, we assessed the mathematical accuracy of the calculation of the contract liabilities from customer loyalty programs. We were able to satisfy ourselves that the estimates and assumptions made by the executive directors are continuously derived and sufficiently documented.
- c. The disclosures on traffic revenue, contract liabilities in respect of unused flight documents and customer loyalty programs are contained in sections 2, 3 and 38 of the notes to the consolidated financial statements.

2. Pension provisions

- a. In the Company's consolidated financial statements pension provisions amounting to EUR 6.9 billion are reported (thereof EUR 0.3 billion within assets held for sale), comprising the net amount of the obligations under various pension plans amounting to EUR 24.9 billion and the fair values of the plan assets amounting to EUR 18.0 billion. The majority of these provisions relates to old-age and transitional pension commitments in Germany and Switzerland. The obligations from defined benefit pension plans were measured using the projected unit credit method in accordance with IAS 19. This requires in particular that assumptions be made as to long-term salary and pension trends and average life expectancy. The discount rate must be determined by reference to market yields on high-quality corporate bonds with matching currencies and consistent maturities. The plan assets are measured at fair value, which in turn involves making estimates that are subject to uncertainties.

From our point of view, these matters were of particular significance in the context of our audit because the measurement of this significant item in terms of its amount is based to a large extent on estimates and assumptions made by the Company's executive directors.

- b. As part of our audit we evaluated the actuarial expert reports obtained and the professional qualifications of the external experts. Due to the specific features of the actuarial calculations, we were assisted by internal specialists from Pension Consulting. Together with them, we evaluated the numerical data, the actuarial parameters and the valuation methods and assumptions on which the valuations were based for compliance with the standards and appropriateness. On this basis, we then assessed the calculations of the figures presented on the statement of financial position, the accounting entries for the provisions and the disclosures in the notes to the consolidated financial statements based on the expert opinions. Our evaluation of the fair values of plan assets was based on bank confirmations submitted to us. Based on our audit procedures, we were able to satisfy ourselves that the estimates and assumptions made by the executive directors are substantiated and sufficiently documented.
- c. The disclosures relating to pension provisions are contained in sections 2 and 33 of the notes to the consolidated financial statements.

3. Recoverability of non-current assets, in particular goodwill and intangible assets with indefinite useful lives

- a. In the Company's consolidated financial statements a total amount of EUR 1.4 billion is reported under the line item "Intangible assets with an indefinite useful life, incl. goodwill" of the consolidated statement of financial position. Goodwill and intangible assets with indefinite useful lives are regularly tested for impairment ("impairment test") once per financial year or if there are indications of an impairment. In addition, items of property, plant and equipment are tested for impairment if there are indications of impairment. The carrying amount of the relevant cash-generating units, including goodwill, is compared with the corresponding recoverable amount in the context of the impairment test. These measurements are generally based on the present value of future cash flows of the cash-generating unit/group of cash-generating units to which the respective asset is to be allocated. The present values are calculated using discounted cash flow models whose parameters were adjusted due to the first-time application of IFRS 16. The measurements are based on projections that were also used to prepare the four-year budget for the Lufthansa Group prepared by the executive directors and acknowledged by the Supervisory Board. The discount rate used is the weighted average cost of capital for the relevant cash-generating unit.

The outcome of this valuation exercise is dependent to a large extent on the estimates made by the executive directors with respect to the future cash inflows, the discount rate used, the rate of growth and other assumptions, and is therefore subject to considerable uncertainty. Against this background and due to the complex nature of the valuation, this matter was of particular significance in the context of our audit.

- b. As part of our audit and with the assistance of our internal specialists from Valuation & Strategy, we assessed the methodology used for the purposes of performing the impairment test, among other things. After matching the future cash inflows used for the calculation against the adopted four-year budget of the Group, we assessed the appropriateness of the calculation, in particular by reconciling it with general and sector-specific market expectations. In addition, we assessed the appropriate consideration of the costs of Group functions.

With the knowledge that even relatively small changes in the discount rate applied can have material effects on the values calculated in this way, we also focused our testing on the parameters used to determine the discount rate applied, and evaluated the calculation model. In order to reflect the uncertainty inherent in the projections, we assessed the sensitivity analyses performed by the Company and carried out our own additional sensitivity analyses with respect to those cash-generating units with low headroom (recoverable amount compared with the carrying amount). Taking into account the information available, we determined that the carrying amounts of the cash-generating units, including the allocated goodwill, were adequately covered by the discounted future net cash flows. Overall, the valuation parameters and assumptions used by the executive directors are in line with our expectations and are also within the ranges considered by us to be reasonable.

- c. The Company's disclosures relating to impairment testing are contained in sections 2, 9 and 16 of the notes to the consolidated financial statements.

4. Accounting treatment of hedging transactions

- a. The Lufthansa Group uses a variety of derivative financial instruments to hedge against currency, fuel price and interest rate risks arising from its ordinary business activities. The executive directors' hedging policy is documented in corresponding internal guidelines and serves as the basis for these transactions. Currency risks arise primarily from sale transactions, procurement transactions (in particular fuel and investments in aircraft) and financing denominated in foreign currencies. The risk associated with changes in fuel prices results from future procurement transactions that are subject to market price risks. The risk of changes in interest rates results from changes in the fair values of fixed-rate financing.

Derivative financial instruments are recognized at fair value as of the balance sheet date. The positive fair values of all of the derivative financial instruments used as hedges amounted to EUR 1.4 billion as of the balance sheet date and the negative fair value amounted to EUR 0.3 billion. If the financial instruments used by the Lufthansa Group are effective hedges of future cash flows in the context of hedging relationships in accordance with the requirements of IFRS 9, changes in fair value are recognized in other comprehensive income over the duration of the hedging relationship (cash flow hedges). As of the balance sheet date, a cumulative

amount of EUR 0.8 billion was recognized in other comprehensive income as effective fair value changes before income taxes (increasing equity) (EUR 0.5 billion in the cash flow hedge reserve and EUR 0.3 billion in the cost of hedging reserve).

In our view, these matters were of particular importance for our audit due to the high complexity and number of hedging transactions, as well as the extensive accounting requirements.

- b. We involved our internal specialists from Corporate Treasury Solutions (CTS) to assist in the audit of the accounting including the effects of the various hedging transactions on equity and profit or loss. Together with these specialists, we assessed, among other things, the internal control system with regard to derivative financial instruments, including the internal activities to monitor compliance with the hedging policy. In our audit of the fair values, we also evaluated the measurement methods based on market data and the underlying data used. With respect to the hedging of expected future cash flows, we mainly carried out a retrospective assessment of past hedge effectiveness and an estimate of expected future hedge effectiveness, and assessed the corresponding effectiveness tests. We obtained bank confirmations in order to assess the completeness of and to examine the fair values of the recorded transactions. In doing so we were able to satisfy ourselves that, overall, the hedging transactions were appropriately accounted for and measured.
- c. The disclosures on the hedging transactions are contained in sections 2 and 43 of the notes to the consolidated financial statements.

5. Effects of the first-time application of IFRS 16 on the accounting of leases

- a. In the Company's consolidated financial statements right of use assets of EUR 2.9 billion and leasing liabilities of EUR 2.9 billion are reported as of the balance sheet date. In the financial year, the first-time application of the new accounting standard relating to leases (IFRS 16) resulted in material effects on the opening balance sheet figures and their updating throughout the financial year. The modified retrospective approach was applied for the conversion to IFRS 16. The comparable figures from the prior year's periods were not adjusted. Due to the large volume of leases and transactions resulting from them, the Company has established group-wide processes and controls for the complete and accurate recording of the leases. Furthermore, the first-time application required a central IT system to be implemented to report these leases.

The new IFRS 16 accounting standard necessitates that executive directors make estimates and take discretionary decisions for certain areas which have been assessed in the context of our audit. This relates, among other things, to assessments regarding exercising options with implications for the term of the leasing arrangement. Against this background and due to the complexity of the new requirements set forth in IFRS 16, the accounting of leases was of particular significance within the course of our audit.

- b. As part of our audit and with the assistance of our internal specialists from Risk Assurance Services (RAS), we assessed, among other things, the appropriateness and operating effectiveness of the processes and controls established by the Group to record its leases. This also applies to the implementation of the central IT system to report the leases.

In addition, as part of our audit and with the assistance of our internal specialists from Capital Markets & Accounting Advisory Services (CMAAS), we assessed the impact of the first-time application of IFRS 16. Together we assessed the implementation work and evaluated the design of the processes set up to report the transactions in accordance with IFRS 16 and of the central system IT system in place to support the implementation of the new requirements. In this context, we inspected, on a sample basis, lease arrangements and assessed whether these were recorded completely and appropriately in the newly implemented central system in place to report the leases. In doing so, we also evaluated those assessments relating to exercising options with implications for the term of the lease by means of inquiring the Company's employees and examining suitable supporting documentation.

We were able to satisfy ourselves that the processes and controls implemented for the complete and correct recording of leasing arrangements are appropriate. Furthermore, we were able to assess that the estimates and assumptions made by the executive directors are sufficiently documented and substantiated to ensure that the leases are properly accounted for under the first-time application of IFRS 16.

- c. The Company's disclosures relating to the accounting of leases and the effects of the first-time application of IFRS 16 are contained in sections 2 and 20 of the notes to the consolidated financial statements.

Other Information

The executive directors are responsible for the other information. The other information comprises the following non-audited parts of the group management report:

- the statement on corporate governance pursuant to § 289f HGB and § 315d HGB included in section "Corporate Governance" of the group management report
- the corporate governance report pursuant to No. 3.10 of the German Corporate Governance Code,
- the non-financial report pursuant to § 289b Abs. 1 HGB and § 315b Abs. 1 HGB included in section "Combined non-financial report" of the group management report.

The other information comprises further the remaining parts of the annual report – excluding cross-references to external information – with the exception of the audited consolidated financial statements, the audited group management report and our auditor's report.

Our audit opinions on the consolidated financial statements and on the group management report do not cover the other information, and consequently we do not express an audit opinion or any other form of assurance conclusion thereon.

In connection with our audit, our responsibility is to read the other information and, in so doing, to consider whether the other information

- is materially inconsistent with the consolidated financial statements, with the group management report or our knowledge obtained in the audit, or
- otherwise appears to be materially misstated.

If, based on the work we have performed, we conclude that there is a material misstatement of this other information, we are required to report that fact. We have nothing to report in this regard.

Responsibilities of the Executive Directors and the Supervisory Board for the Consolidated Financial Statements and the Group Management Report

The executive directors are responsible for the preparation of the consolidated financial statements that comply, in all material respects, with IFRSs as adopted by the EU and the additional requirements of German commercial law pursuant to § 315e Abs. 1 HGB and that the consolidated financial statements, in compliance with these requirements, give a true and fair view of the assets, liabilities, financial position,

and financial performance of the Group. In addition, the executive directors are responsible for such internal control as they have determined necessary to enable the preparation of consolidated financial statements that are free from material misstatement, whether due to fraud or error.

In preparing the consolidated financial statements, the executive directors are responsible for assessing the Group's ability to continue as a going concern. They also have the responsibility for disclosing, as applicable, matters related to going concern. In addition, they are responsible for financial reporting based on the going concern basis of accounting unless there is an intention to liquidate the Group or to cease operations, or there is no realistic alternative but to do so.

Furthermore, the executive directors are responsible for the preparation of the group management report that, as a whole, provides an appropriate view of the Group's position and is, in all material respects, consistent with the consolidated financial statements, complies with German legal requirements, and appropriately presents the opportunities and risks of future development. In addition, the executive directors are responsible for such arrangements and measures (systems) as they have considered necessary to enable the preparation of a group management report that is in accordance with the applicable German legal requirements, and to be able to provide sufficient appropriate evidence for the assertions in the group management report.

The supervisory board is responsible for overseeing the Group's financial reporting process for the preparation of the consolidated financial statements and of the group management report.

Auditor's Responsibilities for the Audit of the Consolidated Financial Statements and of the Group Management Report

Our objectives are to obtain reasonable assurance about whether the consolidated financial statements as a whole are free from material misstatement, whether due to fraud or error, and whether the group management report as a whole provides an appropriate view of the Group's position and, in all material respects, is consistent with the consolidated financial statements and the knowledge obtained in the audit, complies with the German legal requirements and appropriately presents the opportunities and risks of future development, as well as to issue an auditor's report that includes our audit opinions on the consolidated financial statements and on the group management report.

Reasonable assurance is a high level of assurance, but is not a guarantee that an audit conducted in accordance with § 317 HGB and the EU Audit Regulation and in compliance with German Generally Accepted Standards for Financial

Statement Audits promulgated by the Institut der Wirtschaftsprüfer (IDW) and supplementary compliance with the ISAs will always detect a material misstatement. Misstatements can arise from fraud or error and are considered material if, individually or in the aggregate, they could reasonably be expected to influence the economic decisions of users taken on the basis of these consolidated financial statements and this group management report.

We exercise professional judgment and maintain professional skepticism throughout the audit. We also:

- Identify and assess the risks of material misstatement of the consolidated financial statements and of the group management report, whether due to fraud or error, design and perform audit procedures responsive to those risks, and obtain audit evidence that is sufficient and appropriate to provide a basis for our audit opinions. The risk of not detecting a material misstatement resulting from fraud is higher than for one resulting from error, as fraud may involve collusion, forgery, intentional omissions, misrepresentations, or the override of internal control.
- Obtain an understanding of internal control relevant to the audit of the consolidated financial statements and of arrangements and measures (systems) relevant to the audit of the group management report in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an audit opinion on the effectiveness of these systems.
- Evaluate the appropriateness of accounting policies used by the executive directors and the reasonableness of estimates made by the executive directors and related disclosures.
- Conclude on the appropriateness of the executive directors' use of the going concern basis of accounting and, based on the audit evidence obtained, whether a material uncertainty exists related to events or conditions that may cast significant doubt on the Group's ability to continue as a going concern. If we conclude that a material uncertainty exists, we are required to draw attention in the auditor's report to the related disclosures in the consolidated financial statements and in the group management report or, if such disclosures are inadequate, to modify our respective audit opinions. Our conclusions are based on the audit evidence obtained up to the date of our auditor's report. However, future events or conditions may cause the Group to cease to be able to continue as a going concern.

- Evaluate the overall presentation, structure and content of the consolidated financial statements, including the disclosures, and whether the consolidated financial statements present the underlying transactions and events in a manner that the consolidated financial statements give a true and fair view of the assets, liabilities, financial position and financial performance of the Group in compliance with IFRSs as adopted by the EU and the additional requirements of German commercial law pursuant to § 315e Abs. 1 HGB.
- Obtain sufficient appropriate audit evidence regarding the financial information of the entities or business activities within the Group to express audit opinions on the consolidated financial statements and on the group management report. We are responsible for the direction, supervision and performance of the group audit. We remain solely responsible for our audit opinions.
- Evaluate the consistency of the group management report with the consolidated financial statements, its conformity with German law, and the view of the Group's position it provides.
- Perform audit procedures on the prospective information presented by the executive directors in the group management report. On the basis of sufficient appropriate audit evidence we evaluate, in particular, the significant assumptions used by the executive directors as a basis for the prospective information, and evaluate the proper derivation of the prospective information from these assumptions. We do not express a separate audit opinion on the prospective information and on the assumptions used as a basis. There is a substantial unavoidable risk that future events will differ materially from the prospective information.

We communicate with those charged with governance regarding, among other matters, the planned scope and timing of the audit and significant audit findings, including any significant deficiencies in internal control that we identify during our audit.

We also provide those charged with governance with a statement that we have complied with the relevant independence requirements, and communicate with them all relationships and other matters that may reasonably be thought to bear on our independence, and where applicable, the related safeguards.

From the matters communicated with those charged with governance, we determine those matters that were of most significance in the audit of the consolidated financial statements of the current period and are therefore the key audit matters. We describe these matters in our auditor's report unless law or regulation precludes public disclosure about the matter.

OTHER LEGAL AND REGULATORY REQUIREMENTS

Further Information pursuant to Article 10 of the EU Audit Regulation

We were elected as group auditor by the annual general meeting on 7 May 2019. We were engaged by the supervisory board on 19 July 2019. We have been the group auditor of the Deutsche Lufthansa AG, Cologne, without interruption since the financial year 1955.

We declare that the audit opinions expressed in this auditor's report are consistent with the additional report to the audit committee pursuant to Article 11 of the EU Audit Regulation (long-form audit report).

GERMAN PUBLIC AUDITOR RESPONSIBLE FOR THE ENGAGEMENT

The German Public Auditor responsible for the engagement is Eckhard Sprinkmeier.

Düsseldorf, 13 March 2020

PricewaterhouseCoopers GmbH
Wirtschaftsprüfungsgesellschaft

Petra Justenhoven
Wirtschaftsprüferin
(German Public Auditor)

Eckhard Sprinkmeier
Wirtschaftsprüfer
(German Public Auditor)

***Audited Consolidated Financial Statements of Deutsche Lufthansa Aktiengesellschaft as of
and for the financial year ended December 31, 2018 (IFRS)***

Consolidated income statement for the financial year 2018

T067 CONSOLIDATED INCOME STATEMENT

in €m	Notes	2018	2017 ¹⁾
Traffic revenue	3	28,103	28,399
Other revenue	4	7,741	7,180
Total revenue		35,844	35,579
Changes in inventories and work performed by entity and capitalised	5	531	438
Other operating income ²⁾	6	1,818	2,276
Cost of materials and services	7	-18,669	-19,028
Staff costs	8	-8,811	-8,172
Depreciation, amortisation and impairment ³⁾	9	-2,205	-2,382
Other operating expenses ⁴⁾	10	-5,708	-5,571
Profit/loss from operating activities		2,800	3,140
Result of equity investments accounted for using the equity method	11	114	118
Result of other equity investments	11	60	39
Interest income	12	68	178
Interest expenses	12	-212	-373
Other financial items	13	-46	56
Financial result		-16	18
Profit/loss before income taxes		2,784	3,158
Income taxes	14	-588	-784
Profit/loss after income taxes		2,196	2,374
Profit/loss attributable to minority interests		-33	-34
Net profit/loss attributable to shareholders of Deutsche Lufthansa AG		2,163	2,340
Basic/diluted earnings per share in €	15	4.58	4.98

¹⁾ Previous year's figures have been adjusted.

²⁾ This includes EUR 61m (previous year: EUR 46m) from the reversal of write-downs on receivables.

³⁾ This includes EUR 13m (previous year: EUR 83m) for the recognition of write-downs on receivables.

⁴⁾ This includes EUR 100m (previous year: EUR 98m) for the recognition of loss allowances on receivables.

Statement of comprehensive income

for the financial year 2018

T068 STATEMENT OF COMPREHENSIVE INCOME

in €m	2018	2017 ¹⁾
Profit/loss after income taxes	2,196	2,374
Other comprehensive income		
Other comprehensive income with subsequent reclassification to the income statement		
Differences from currency translation	124	-406
Subsequent measurement of financial assets at fair value without effect on profit and loss	-10	21
Subsequent measurement of hedges – cash flow hedge reserve	434	-546
Subsequent measurement of hedges – costs of hedging	-90	13
Other comprehensive income from investments accounted for using the equity method	-2	3
Other expenses and income recognised directly in equity	1	-3
Income taxes on items in other comprehensive income	-87	137
	370	-781
Other comprehensive income without subsequent reclassification to the income statement		
Revaluation of defined-benefit pension plans	-974	1,186
Other comprehensive income from investments accounted for using the equity method	-	-
Subsequent measurement of financial assets at fair value	5	-
Income taxes on items in other comprehensive income	117	-136
	-852	1,050
Other comprehensive income after income taxes	-482	269
Total comprehensive income	1,714	2,643
Comprehensive income attributable to minority interests	-34	-34
Comprehensive income attributable to shareholders of Deutsche Lufthansa AG	1,680	2,609

¹⁾ Previous year's figures have been adjusted.

Consolidated balance sheet

as of 31 December 2018

T069 CONSOLIDATED BALANCE SHEET – ASSETS

in €m	Notes	31.12.2018	31.12.2017 ¹⁾	1.1.2017 ¹⁾
Intangible assets with an indefinite useful life ²⁾	16	1,381	1,343	1,265
Other intangible assets	17	512	492	472
Aircraft and reserve engines	18 20	16,776	15,333	14,189
Repairable spare parts for aircraft		2,133	1,758	1,604
Property, plant and other equipment	19 20	2,221	2,186	2,199
Investments accounted for using the equity method	21	650	585	516
Other equity investments	22 44	246	221	212
Non-current securities	22 44	41	32	23
Loans and receivables	23 44	512	475	513
Derivative financial instruments	44	828	642	1,474
Deferred charges and prepaid expenses	27	118	9	11
Effective income tax receivables		10	12	4
Deferred tax assets	14	2,131	1,661	1,547
Non-current assets		27,559	24,749	24,029
Inventories	24	968	907	816
Contract assets ³⁾	25	234	-	-
Trade receivables and other receivables	26 44	5,576	5,313	4,570
Derivative financial instruments	44	357	600	534
Deferred charges and prepaid expenses	27	217	197	167
Effective income tax receivables		58	58	37
Securities	28 44	1,735	2,551	2,681
Cash and cash equivalents	29 44	1,500	1,397	1,256
Assets held for sale	30	9	6	132
Current assets		10,654	11,029	10,193
Total assets		38,213	35,778	34,222

¹⁾ Previous year's figures have been adjusted.

²⁾ Including Goodwill.

³⁾ Recognised separately for the first time from the 2018 financial year in accordance with IFRS 15.

T070 CONSOLIDATED BALANCE SHEET – SHAREHOLDERS' EQUITY AND LIABILITIES

in €m	Notes	31.12.2018	31.12.2017 ¹⁾	1.1.2017 ¹⁾
Issued capital	31	1,217	1,206	1,200
Capital reserve	32	343	263	222
Retained earnings	32	4,555	3,724	2,908
Other neutral reserves	32	1,185	1,474	2,255
Net profit/loss		2,163	2,340	-
Equity attributable to shareholders of Deutsche Lufthansa AG		9,463	9,007	6,585
Minority interests		110	103	89
Shareholders' equity		9,573	9,110	6,674
Pension provisions	33	5,865	5,116	8,364
Other provisions	34	537	601	503
Borrowings	35 44	5,008	6,142	5,811
Contract liabilities ²⁾	36	22	-	-
Other financial liabilities	37	137	243	124
Advance payments received, deferred income and other non-financial liabilities	38	51	1,289	1,246
Derivative financial instruments	44	222	190	54
Deferred tax liabilities	14	583	449	437
Non-current provisions and liabilities		12,425	14,030	16,539
Other provisions	34	925	990	1,066
Borrowings	35 44	1,677	672	764
Trade payables and other financial liabilities	40 44	5,764	5,249	4,689
Liabilities from unused flight documents	39	3,969	3,773	3,040
Other contract liabilities ²⁾	39	2,316	-	-
Advance payments received, deferred income and other non-financial liabilities	41	388	992	875
Derivative financial instruments	44	393	124	185
Effective income tax obligations		783	838	390
Current provisions and liabilities		16,215	12,638	11,009
Total shareholders' equity and liabilities		38,213	35,778	34,222

¹⁾ Previous year's figures have been adjusted.

²⁾ Recognised separately for the first time from the 2018 financial year in accordance with IFRS 15.

Consolidated statement of changes in shareholders' equity as of 31 December 2018

T071 CONSOLIDATED STATEMENT OF CHANGES IN SHAREHOLDERS' EQUITY

	Issued capital	Capital reserve	Fair value measurement of financial instruments	Currency differences	Revaluation reserve (due to business combinations)	Other neutral reserves	Total other neutral reserves	Retained earnings	Net profit/loss	Equity attributable to shareholders of Deutsche Lufthansa AG	Minority interests	Total shareholders' equity
in €m												
As of 31.12.2016	1,200	222	1,081	670	236	326	2,313	1,549	1,776	7,060	89	7,149
Amendments to accounting policies and valuation methods	-	-	-	-	-	-	-	-475	-	-475	-	-475
Restatement IFRS 9	-	-	-58	-	-	-	-58	58	-	-	-	-
Adjusted as of 31.12.2016	1,200	222	1,023	670	236	326	2,255	1,132	1,776	6,585	89	6,674
Capital increases/reductions	6	41	-	-	-	-	-	-	-	47	1	48
Reclassifications	-	-	-	-	-	-	-	1,542	-1,542	-	-	-
Dividends to Lufthansa shareholders/minority interests	-	-	-	-	-	-	-	-	-234	-234	-21	-255
Transactions with minority interests	-	-	-	-	-	-	-	-	-	-	-	-
Consolidated net profit/loss attributable to Lufthansa shareholders/minority interests	-	-	-	-	-	-	-	-	2,340	2,340	34	2,374
Other expenses and income recognised directly in equity	-	-	-375	-406	-	-	-781	1,050	-	269	-	269
As of 31.12.2017	1,206	263	648	264	236	326	1,474	3,724	2,340	9,007	103	9,110
Amendments to accounting policies and valuation methods	-	-	-	-	-	-	-	-	-	-	-	-
Restatement IFRS 9	-	-	-43	-	-	-	-43	35	-	-8	-	-8
Restatement IFRS 15	-	-	-	-	-	-	-	-310	-	-310	-	-310
Adjusted as of 31.12.2017	1,206	263	605	264	236	326	1,431	3,449	2,340	8,689	103	8,792
Capital increases/reductions	11	80	-	-	-	-	-	-	-	91	1	92
Reclassifications	-	-	-	-	-	-	-	1,963	-1,963	-	-	-
Dividends to Lufthansa shareholders/minority interests	-	-	-	-	-	-	-	-	-377	-377	-28	-405
Transactions with minority interests	-	-	-	-	-	-	-	-	-	-	-	-
Consolidated net profit/loss attributable to Lufthansa shareholders/minority interests	-	-	-	-	-	-	-	-	2,163	2,163	33	2,196
Other expenses and income recognised directly in equity	-	-	252	124	-	-2	374	-857	-	-483	1	-482
Hedging results reclassified from non-financial assets to acquisition costs	-	-	-620	-	-	-	-620	-	-	-620	1	-620
As of 31.12.2018	1,217	343	237	388	236	324	1,185	4,555	2,163	9,463	110	9,573

Consolidated cash flow statement for the financial year 2018

T072 CONSOLIDATED CASH FLOW STATEMENT

in €m	Notes	2018	2017 ¹⁾
Cash and cash equivalents 1.1.		1,218	1,138
Net profit/loss before income taxes		2,784	3,158
Depreciation, amortisation and impairment losses on non-current assets (net of reversals)		2,178	2,300
Depreciation, amortisation and impairment losses on current assets (net of reversals)		23	39
Net proceeds on disposal of non-current assets	6 10	-34	-37
Result of equity investments	11	-174	-157
Net interest	12	144	195
Income tax payments/reimbursements		-670	-385
Significant non-cash-relevant expenses/income		-276	-705
Change in trade working capital		410	259
Change in other assets/shareholders' equity and liabilities		-276	701
Cash flow from operating activities		4,109	5,368
Capital expenditure for property, plant and equipment and intangible assets	16 17 18 19	-3,709	-3,338
Capital expenditure for financial investments	22 23	-48	-136
Additions/loss to repairable spare parts for aircraft		-388	-210
Proceeds from disposal of non-consolidated equity investments		2	7
Proceeds from disposal of consolidated equity investments		4	-
Cash outflows for acquisitions of non-consolidated equity investments	21 22 44	-48	-55
Cash outflows for acquisitions of consolidated equity investments	48	-12	191
Proceeds from disposal of intangible assets, property, plant and equipment and other financial investments		146	142
Interest income ²⁾		51	61
Dividends received		143	87
Net cash from/used in investing activities		-3,859	-3,251
Purchase of securities/fund investments		-3,289	-2,660
Disposal of securities/fund investments		3,879	905
Net cash from/used in investing and cash management activities		-3,269	-5,006
Capital increase	31 32	-	-
Transactions by minority interests		1	1
Non-current borrowing		987	1,106
Repayment of non-current borrowing		-1,196	-967
Dividends paid		-349	-232
Interest paid		-69	-153
Net cash from/used in financing activities		-626	-245
Net increase/decrease in cash and cash equivalents		214	117
Changes due to currency translation differences		2	-37
Cash and cash equivalents 31.12.³⁾	29	1,434	1,218
Securities	28	1,735	2,551
Liquidity		3,169	3,769
Net increase/decrease in liquidity		-600	-50

¹⁾ Previous year's figures have been adjusted.

²⁾ Cash inflows from interest hedges (EUR 152m) are netted with the corresponding interest payments as of 2018. The figure for the previous year (EUR 136m) was adjusted accordingly.

³⁾ The difference between the bank balances and cash-in-hand shown in the statement of financial position comes from fixed-term deposits of EUR 66m with terms of four to twelve months (previous year: EUR 179m).

Notes to the consolidated financial statements

Deutsche Lufthansa AG 2018

GENERAL REMARKS

1 Company information

The Lufthansa Group is a global aviation group whose subsidiaries and equity investments were organised into five operating segments in the financial year 2018: Network Airlines, Eurowings, Logistics, MRO and Catering.

Deutsche Lufthansa AG has its headquarters in Cologne, Germany, and is filed in the Commercial Register of Cologne District Court under HRB 2168.

The declaration on the German Corporate Governance Code required by Section 161 of the German Stock Corporation Act (AktG) was issued and made available to shareholders on the internet at www.lufthansagroup.com/declaration-of-compliance.

The consolidated financial statements of Deutsche Lufthansa AG, Cologne, and its subsidiaries have been prepared in accordance with the International Financial Reporting Standards (IFRS) issued by the International Accounting Standards Board (IASB), taking account of interpretations by the IFRS Interpretations Committee as applicable in the European Union (EU).

The commercial law provisions of Section 315e Paragraph 1 of the German Commercial Code (HGB) have also been applied. All IFRSs issued by the IASB and in effect at the time that these financial statements were prepared and applied by Deutsche Lufthansa AG have been adopted by the European Commission for application in the EU. The consolidated financial statements of Deutsche Lufthansa AG are prepared in millions of euros. Its financial year is the calendar year.

The accounting policies used in the previous year have been retained, with the exception of changes due to new or amended standards and the capitalisation of engine overhaul expenses, see [Note 2, p. 106ff.](#)

The Executive Board of Deutsche Lufthansa AG prepared the consolidated financial statements for 2018 on 5 March 2019. The consolidated financial statements for 2018 were examined and approved for publication by the Supervisory Board of Deutsche Lufthansa AG in its meeting on 13 March 2019.

2 New international accounting standards in accordance with IFRS and interpretations and summary of the significant accounting policies and valuation methods

INTERNATIONAL FINANCIAL REPORTING STANDARDS (IFRS) AND INTERPRETATIONS (IFRIC) TO BE APPLIED FOR THE FIRST TIME IN THE FINANCIAL YEAR AND AMENDMENTS TO STANDARDS AND INTERPRETATIONS

T073 IFRS-PRONOUNCEMENT (APPLICABLE FROM FINANCIAL YEAR 2018)

IFRS 9, Financial instruments
IFRS 15, Revenue from Contracts with Customers
Clarification of IFRS 15, Revenue from Contracts with Customers
Amendments to IFRS 2, Share-based Payment
Amendments to IFRS 4, Insurance Contracts
Amendments to IAS 40, Transfers from Investment Property
IFRIC 22, Foreign Currency Transactions and Advance Consideration
Annual Improvements to IFRS, 2014 – 2016, Cycle for amendments to IFRS 1, First-time application of the International Financial Reporting Standards, and IAS 28, Investments in associated companies and joint ventures

The standards, clarifications and interpretations mandatory from 1 January 2018 onwards, particularly IFRS 9 (Financial Instruments) and IFRS 15 (Revenue from Contracts with Customers), had the following effects on the Group's net assets, financial and earnings position. The first-time application of all of the other amended accounting standards listed below had no or no material effect on the presentation of the net assets, financial and earnings position or on earnings per share.

IFRS 9

In accordance with the transitional provisions of IFRS 9, Financial Instruments, the Lufthansa Group has not adjusted the figures for the previous year, except for hedging relationships containing options, and recognised the cumulative transitional effects as of 1 January 2018 in retained earnings. The amendments to the accounting for hedging relationships including options were applied retrospectively and the figures for the previous year were adjusted accordingly.

In phase I ("classification"), the transition of share items held as securities from the IAS 39 category "available for sale" (AfS) to the IFRS 9 category "fair value through profit or loss"

(FVTPL) only leads to a transfer within reserves, between the cumulative market value reserve and retained earnings. Share items with a cumulative market value reserve of EUR 12m are also reclassified from AfS to at fair value through other comprehensive income (without recycling).

Debt instruments are still generally classified as at fair value through other comprehensive income. There are no reclassification effects in phase I for loans and receivables either, since they are still held at amortised cost.

As part of phase II ("Impairment standards"), the first-time application of the expected loss model required under IFRS 9 resulted in additional impairment of trade receivables, which is recognised in equity without effect on profit and loss as of 1 January 2018. The effects of this on income as of 31 December 2018 were immaterial.

For fuel hedging transactions, the Group uses the component approach, with crude oil as the designated component and regular rebalancing. This leads to a reduction in volatility in the income statement from changes in the market value of derivatives. The Lufthansa Group uses options as hedging instruments for fuel hedging, with the intrinsic value of the option designated as the hedging instrument in a cash flow hedge. IFRS 9 stipulates that changes in the time values of options are recognised as a cost of hedging in equity without effect on profit and loss. With the retrospective recognition of the time values of options without effect on profit and loss, the changes in time values previously recognised through profit or loss were reclassified within equity to the market valuation reserve as of 1 January 2017. This was also carried out for changes in time values of options in 2017, which were retrospectively reclassified from retained earnings to the market value reserve as a cost of hedging. Comparative figures and results for 2017 have been adjusted accordingly in the financial reporting.

In the area of exchange rate hedging with forward contracts, the Lufthansa Group has been using the spot-to-spot method since 1 January 2018. This involves the spot component of a forward contract being designated as a hedging instrument. The other components of the forward – the scheduling components and the basis spread – are recognised as the forward component and the cross currency basis spread within the market value reserve. This does not have any material impact on the statement of financial position, the income statement or the statement of comprehensive income.

Detailed information can be found in the corresponding chapters of the Notes. ➤ **Note 44, p. 164ff.**

The following table summarises the changes made following the first-time application of IFRS 9.

T074 EFFECTS OF RESTATEMENT FROM IFRS 9 CHANGES

in €m	1.1.2017		
	reported	adjusted	difference
Statement of financial position			
Other comprehensive income – subsequent measurement of cash flow hedges – cost of hedging	-	-76	-76
Income taxes on items in other comprehensive income	-337	-319	18
Retained earnings	1,549	1,607	58

in €m	31.12.2017		
	reported	adjusted	difference
Statement of financial position			
Other comprehensive income – subsequent measurement of cash flow hedges – cost of hedging	-	16	16
Income taxes on items in other comprehensive income	137	134	-3

Income statement			
Profit/loss before income taxes	3,187	3,171	-16
Income taxes	-789	-786	3
Profit/loss after income taxes	2,398	2,385	-13
Earnings per share in €	5.00	5.00	-

Cash flow statement			
Profit/loss before income taxes	3,187	3,171	-16
Significant non-cash-relevant expenses/income	-721	-705	16
Cash flow from operating activities	5,035	5,035	-

in €m	1.1.2018		
	reported	adjusted	difference
Statement of financial position			
Trade receivables and other current receivables	5,283	5,273	-10
Other comprehensive income – subsequent measurement of financial assets at fair value through other comprehensive income (with recycling)	62	19	-43
Retained earnings	4,141	4,174	33
of which reclassification of equity instruments at fair value through other comprehensive income (with recycling)	-	43	43
of which reclassification of equity instruments to at fair value through profit or loss	-	-10	-10
Income taxes on items in other comprehensive income	137	139	2

IFRS 15

Based on the modified retrospective method, the cumulative effects of the changes were recognised in retained earnings as of 1 January 2018. The comparative information for 2017 was therefore not adjusted. Furthermore, the disclosures required by IFRS 15 were generally not made for comparative information. The following table summarises the adjustment effects of the first-time adoption of IFRS 15 on retained earnings as of 1 January 2018:

T075 RETAINED EARNINGS

in €m	Effect of adopting IFRS 15 at 1 Jan 2018
Shift in timing of recognition for fees	29
Customer loyalty programmes	385
Related taxes	-104
Effect at 1 Jan 2018	310

IFRS 15 resulted in a shifting of the recognition date for certain items of other revenue (particularly rebooking fees) from the transaction date to the date of use. The shifting of the recognition date also has an impact on the recognition of expired miles in the miles programme. These will no longer be recognised directly through profit or loss in the year of collection, but rather recognised as collected pro rata. The sum continues to include adjustments due to the introduction of a redesigned data model for the total amount of miles outstanding in relation to the introduction of IFRS 15.

Contractual items that have not been performed in full have to be presented in the balance sheet as contract assets or liabilities (current and non-current, in each case).

Obligations in respect of unused flight documents are still presented separately. As of 1 January 2018, liabilities relating to customer loyalty programmes are recognised under other current contract liabilities. They were previously reported under other non-financial liabilities and deferred items (non-current and current). Since the timing of the fulfilment of these obligations is beyond the control of the Company, they are all presented as current, in accordance with IFRS 15. In addition, the short-term component of the customer loyalty programmes, which was previously recognised under received advance payments and deferred income, was reclassified as other contract liabilities. Also included are obligations from works in progress in connection with long-term production and/or service contracts. This was reclassified from advance payments received and other provisions to other contract liabilities.

From 2018 onwards, for ticket revenue, the airport fees received and the corresponding airport invoices will no longer be recognised in the income statement. Applied as of 31 December 2018, this approach reduced revenue and expenses by EUR 2,257m. Otherwise, there were no material differences between revenue recognition under IFRS 15 and revenue recognition under IAS 11 or IAS 18. In connection with IFRS 15, income from training and travel management was reclassified from other operating income to revenue. This had the effect of increasing revenue by EUR 359m as of 31 December 2018. In the prior-year period, EUR 342m was reported under other operating income.

The following tables summarise the effects of applying IFRS 15 to the relevant items of the consolidated statement of financial position as of 31 December 2018 and the consolidated income statement for financial year 2018.

T076 RECONCILIATION FROM IFRS 15 TO IAS 11/18 – STATEMENT OF FINANCIAL POSITION

in €m	31.12.2018 as reported	Adjustments	31.12.2018 IAS 11/18
Contract assets	234	-234	-
Trade receivables and other receivables	5,576	234	5,810
Deferred tax assets	2,131	104	2,235
Retained earnings	4,555	317	4,872
Other provisions (non-current)	537	22	559
Contract liabilities (non-current)	22	-22	-
Advance payments received, deferred income (non-current) and other non-financial liabilities	51	1,238	1,289
Other provisions (current)	925	59	984
Trade payables (current) and other financial liabilities	5,764	20	5,784
Contract liabilities from unused flight documents (current)	3,969	-19	3,950
Other contract liabilities (current)	2,316	-2,316	-
Advance payments received, deferred income and other non-financial liabilities (current)	388	597	985
Total assets	38,213		38,317

**T077 RECONCILIATION FROM IFRS 15 TO IAS 11/18 –
INCOME STATEMENT**

in €m	31.12.2018 as reported	Adjustments	31.12.2018 IAS 11/18
Total revenue	35,844	1,898	37,742
Traffic revenue	28,103	2,257	30,360
Other revenue	7,741	-359	7,382
Other operating income	1,818	359	2,177
Cost of materials and services	-18,669	-2,257	-20,926

Detailed information can be found in the corresponding chapters of the Notes. ➤ **Notes 3 and 4, p. 121f.**

PUBLISHED INTERNATIONAL FINANCIAL REPORTING STANDARDS (IFRS) AND INTERPRETATIONS (IFRIC) NOT YET APPLIED/APPLICABLE AND AMENDMENTS TO STANDARDS AND INTERPRETATIONS

The following standards and amendments have already been adopted by the European Union but are only mandatory for financial statements after 31 December 2018:

T078 IFRS-PRONOUNCEMENT (ADOPTED BY THE EU)

	Mandatory application for financial years beginning on or after
IFRS 16, Leases	1.1.2019
IFRIC 23, Uncertainty over Income Tax Treatments	1.1.2019
Amendments to IAS 28, Non-current equity investments in associated companies and joint ventures	1.1.2019
Amendments to IFRS 9, Prepayment Features with Negative Compensation	1.1.2019

IFRS 16, Leases, must be applied from 1 January 2019.

Currently, the payment obligations arising from operating leases only have to be disclosed in the Notes. In the future, the rights and obligations related to such leases are required to be recognised as assets (right-of-use asset) and liabilities (lease liability) in the statement of financial position. The Lufthansa Group will adopt the modified retrospective approach for the introduction of this standard. Under this approach, the comparable figures for the previous year are not adjusted and all adjustment effects as of 1 January 2019 have therefore to be presented as adjustments to retained earnings. In addition, the Lufthansa Group will recognise the right-of-use assets on the basis of the corresponding lease liabilities upon first-time application and not in the amount of the carrying amounts of the lease liabilities at the start of the contract, such that IFRS 16 has no impact on equity as

of 1 January 2019. Short-term leases with a term of less than twelve months (and containing no purchase options) and leases where the underlying asset has a low value will not be recognised. The same applies to contracts with a remaining term of less than a year upon first-time application.

In view of the contracts concluded as of the reporting date, the Group is anticipating an increase in total assets of approximately EUR 2.0bn. Material contracts are still under negotiation, which will result in additional right of uses and lease liabilities of EUR 0.4bn if they are signed, so that total assets are currently expected to go up by EUR 2.4bn in total. Based on the total assets as of 31 December 2018, the first-time application of IFRS 16 would result in a decrease of approximately 1.5 percentage points in the equity ratio.

The expenses related to operating leases are currently shown in the income statement under cost of materials and services and other operating expenses. The current volume of operating leases is shown in ➤ **Note 20, p. 134ff.**

From 2019 onwards the item is expected to include additional amortisation on the right of uses of around EUR 360m and interest expenses on the lease liability of around EUR 50m. These changes in presentation are expected to impact operating earnings (EBIT) by around EUR 25m.

In addition, the change in the presentation of the expenses related to operating leases will result in a transfer from cash flow from financing activities to cash flow from operating activities as the lease payments no longer affect the operating cash flow and are instead recognised as interest and redemption payments within cash flow from financing activities. This effect will amount to around EUR 385m in 2019.

The Lufthansa Group has decided to recognise the right-of-use assets as property, plant and equipment and the lease liabilities as other liabilities in the statement of financial position. The relevant detailed disclosures will be made separately in the Notes from 2019.

IFRIC 23, Uncertainty over Income Tax Treatments, was published by the IASB in June 2017. IFRIC 23 is applicable to financial years beginning on or after 1 January 2019; early adoption is permitted. The interpretation supplements the provisions of IAS 12 on accounting for effective and deferred taxes with regard to uncertainties over the treatment of particular circumstances and transactions by the tax authorities and courts pertaining to income tax.

In the past, the Lufthansa Group has only recognised claims against tax authorities when a cash inflow was considered to be as good as definite. Following the transition to IFRIC 23, the claims will be recognised as soon as the cash inflow is deemed to be probable. When IFRIC 23 takes effect, the Lufthansa Group expects to transfer a low two-figure million euro amount to retained earnings as a result.

The IASB and the IFRS Interpretations Committee have adopted other standards and interpretations whose application is not mandatory for the financial year 2018.

**T079 IFRS-PRONOUNCEMENT
(NOT YET ENDORSED BY THE EU)**

	Mandatory application for financial years beginning on or after
Annual Improvements to IFRS, 2015–2017 Cycle	1.1.2019
Amendments to IAS 19, Plan amendments, curtailments and settlements	1.1.2019
Changes to conceptual framework	1.1.2020
Amendments to IFRS 3, Business combinations	1.1.2020
Amendments to IAS 1 and IAS 8: Definition of materiality	1.1.2020
IFRS 17, Insurance Contracts	1.1.2021

Currently, the new or amended IFRS pronouncements listed in the table are not considered to have a material effect on the presentation of the net assets, financial and earnings position.

The Group has not voluntarily applied any of the new or amended regulations mentioned above before their binding date of application. If the effective dates of the standards and interpretations mentioned above fall within the year, they are applied as of 1 January of the following financial year. This is subject to the endorsement of the standards by the EU.

SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES AND VALUATION METHODS

The companies included in the consolidated financial statements use uniform accounting policies to prepare their financial statements.

The application of the accounting policies prescribed by IFRS and IFRIC requires making a large number of estimates and assumptions with regard to the future that may, naturally, not coincide with actual future conditions. All of these estimates and assumptions are, however, reviewed continuously and are based either on past experience and/or expectations of future events that seem reasonable in the circumstances on the basis of sound business judgement. Estimates and assumptions that are of material importance in determining the carrying amounts for assets and liabilities are explained in the following description of the accounting policies applied to material items in the statement of financial position.

The fundamental valuation method applied in the consolidated financial statements is the acquisition cost principle. Where IFRSs stipulate that other methods of measurement should be applied, these are used instead, and are referred to specifically in the following comments on measuring assets and liabilities.

Amendments to accounting policies as a result of revised and new standards are applied retrospectively unless provided otherwise for a specific standard. The income statement for the previous year and the opening balance sheet for the comparable period are adjusted as if the new accounting policies had always been applied.

Recognition of income and expenses

Revenue and other operating income are recognised when the service has been provided.

Passenger transport and ancillary services

The Lufthansa Group sells flight tickets and related ancillary services primarily via agents, its own websites or other airlines in the course of interlining. The payments are received by the Lufthansa Group via credit card billing companies, agents or other airlines, generally before the corresponding service is provided. Receivables from the sale of flight tickets and related ancillary services are only amounts payable by credit card billing companies, agents or other airlines.

The Group initially recognises all ticket sales as liabilities from unused flight documents. These are presented as contract liabilities as of the initial application of IFRS 15. Depending on the terms of the selected fare, the contract liabilities will reflect a range of possibilities for refunding services that have not yet been provided. Liabilities include both the deferred income for future flights and ancillary services that are recognised as revenue when the flight documents are used, and the liabilities for award miles credited to the passenger when the flight documents are used. The Lufthansa Group allocates the transaction price to all of the performance obligations identified on the flight ticket on the basis of their individual transaction prices. The individual transaction prices for flight segments are determined using the IATA procedure. The total price payable is allocated to individual flight segments using what is known as a prorate calculation, which meets the IFRS 15 definition of a relative individual transaction price. The individual transaction prices for ancillary services that are not included in the fare are directly observable prices within the meaning of IFRS 15. On average, it takes 2.5 months for a flight coupon to be realised.

The Lufthansa Group reduces liabilities from unused flight documents and recognises revenue for each flight segment (including the related ancillary revenue) when the respective document is used. For tickets that cover more than one flight segment, the Lufthansa Group identifies each flight segment as a distinct performance obligation, since each flight segment is independent and can be distinguished in the context of the contract.

Interlining means that the passenger is carried by another airline for one (or more) flight segments. As soon as a third party is involved in providing the service, a distinction must be made between the principal and the agent. When Lufthansa acts as the principal, the gross amount of revenue is recognised. If Lufthansa acts as an agent, the net amount is recognised instead. The transaction price for this specific service is limited to the amount of the commission received. The performance obligation is fulfilled when the respective flight document for the flight segment is used.

Generally speaking, the Lufthansa Group does not expect to receive any amount if a flight document is not used (or if the amount is significant) and for this reason does not anticipate the possibility that documents for a flight segment

are not used. The expected amount if flight documents are not used is only recognised as revenue if the probability that the passengers (in accordance with the portfolios) exercise their remaining rights is low, and no later than when the expiry of flight documents is certain and known. This accounting practice was not changed by the implementation of IFRS 15.

Logistics

Lufthansa Cargo markets the freight capacities of passenger aircraft at Lufthansa German Airlines, Austrian Airlines, Eurowings and Brussels Airlines and operates a fleet of cargo aircraft. In addition to income from standard cargo services, Lufthansa Cargo generates part of its revenue from ancillary services that are closely connected to the freight service.

In its cargo business, the Lufthansa Group has identified the entire freight service as a distinct performance obligation. The final customer receives the benefit of the transport service and at the same time uses the service as is provided with each transport segment. In this case, the customer takes control of the company's output while the carrier provides its service. The customer receives the benefit of the service as each transport segment is fulfilled. The corresponding cargo revenue is therefore recognised at the prorate value when the documents for each individual segment are used.

Lufthansa Cargo typically receives the consideration for performing its service once the transport has been carried out.

MRO

The performance obligations in the MRO segment are the provision of maintenance and aircraft and engine overhaul services, which are recognised over time since the condition of IFRS 15.35 (b) is generally met. These performance obligations involve estimating the proportion of the total contract already completed and the profit on the whole contract, so that an input-orientated measurement of the percentage of completion can be made. Contract assets and contract liabilities are therefore both recognised.

Access to Lufthansa Technik's pool of spare parts and components is another key performance obligation, which is satisfied either over time or at a point in time, depending on the contract model agreed.

In some cases, the contracts in the MRO segment make it necessary not to recognise distinct services as individual performance obligations but rather as a series, as described in IFRS 15.22 (b). Furthermore, some of the contracts include standby obligations that require the recognition of revenue over time. This is particularly the case when remuneration is paid in the form of a fixed rate per hour of flying time. In this case, the percentage of completion is primarily measured on the basis of the hours invoiced monthly to the customer. Revenue for component supply contracts is realised taking into consideration the margin shown in the business plans, which are updated annually.

A significant portion of the contracts in the MRO business segment run for several years and so have indexation clauses, but which only impact the transaction price when the index event (e.g. a wage increase) has occurred.

Catering

The LSG group offers products and services related to in-flight service. These include catering, in-flight sales and entertainment, in-flight service equipment and the associated logistics as well as consultancy services and the operation of lounges.

Airline catering is the main business of the LSG group as far as revenue is concerned. Taking the business model and the value chain for airline catering into account, the preparation of meals and the logistics related to this catering have been identified as distinct performance obligations. The performance obligation to prepare meals is generally fulfilled when the meals are delivered to the customers. The catering logistics performance obligation is fulfilled over time between the transport of the meals to the airport and the disposal of the waste, depending on the services ordered by the customer. For performance obligations satisfied over time the percentage of completion is measured on an output basis in accordance with IFRS 15.B15 in conjunction with IFRS 15.B16.

Billing and payment in the Catering segment generally takes place one to two months after the performance obligation has been fulfilled. This gives rise to trade receivables, but no significant contract liabilities or assets from catering contracts.

Variable consideration (e.g. volume discounts) has to be taken into account when determining the transaction price in the catering business. The majority of the variable consideration is estimated using the expected value method on the basis of historic data and current developments. The LSG group updates the estimated transaction price at the end of each reporting period and accounts for the resulting changes in accordance with IFRS 15.87-90.

Operating expenses are recognised when the product or service is used or the expense arises. Provisions for warranties are generally accounted for when the corresponding revenue is recognised, while provisions for onerous contracts are generally set up when they are identified.

Further disclosures on the Group's revenue from contracts with customers can be found in [Notes 3 and 4, p. 121f.](#)

Interest income and expenses are accrued in the appropriate period. Dividends from shareholdings not accounted for using the equity method are recognised when a legal claim to them arises.

Initial consolidation and goodwill

The initial consolidation of Group companies takes place using the purchase method. This involves measuring the fair value of the assets, liabilities and contingent liabilities identified, in accordance with the provisions of IFRS 3, of the company acquired at the acquisition date, and allocating the acquisition costs to them. The proportion of fair value of assets and liabilities not acquired is shown under minority interests. The ancillary acquisition costs are recognised as expenses in the periods in which they occur.

Any excess of cost over the value of equity acquired is capitalised as goodwill. If the value of the acquirer's interest in the shareholders' equity exceeds the purchase price paid by the acquiring company, the difference is recognised immediately in profit or loss.

Differences from minority interests acquired after control has been gained are set off directly against equity.

Goodwill is not amortised, but is tested annually for impairment. The impairment tests applied to goodwill are carried out using established discounted cash flow methods. This is done on the basis of expected future cash flows from the latest business plan, which are extrapolated on the basis of long-term revenue growth rates and assumptions with

regard to margin development and are discounted for the capital costs of the business unit. Tests are performed at the cash generating unit (CGU) level. For the individual premises on which impairment tests were based in the financial year 2018, see [Note 16, p. 127ff.](#)

Additional impairment tests are also applied during the course of the year if events give reason to believe that goodwill could be permanently impaired.

Once an impairment loss has been recognised on goodwill, it is not reversed in subsequent periods.

Notwithstanding the principles described above, Group companies that have no material impact on the Group's net assets, financial and earnings position are not consolidated, but rather recognised in the consolidated financial statements at cost less any impairments.

Currency translation and consolidation methods

The financial statements of the foreign Group companies are prepared in the relevant functional currency and translated into euros before consolidation. The functional currency is mainly the currency of the country in which the company concerned is located. Occasionally, the functional currency differs from the national currency. Assets and liabilities are translated at the middle rates on the balance sheet date. Income statements are translated at the average exchange rates for the year. Any translation differences are recognised directly in equity without effect on profit and loss and are only recognised in profit or loss when control is lost or the equity investment is disposed of.

Goodwill from capital consolidation of foreign subsidiaries prior to 2005 is carried at historical cost net of amortisation accumulated by the end of 2004. Goodwill acquired after 2005 is held in the functional currency of the purchased company and translated at the middle rates on the reporting date.

Transaction differences, however, are recognised in profit or loss. These differences arise in the financial statements of consolidated companies from assets and liabilities based on currency other than the company's functional currency. Any resulting exchange rate differences are included in other operating income as exchange rate gains, or in other operating expenses as exchange rate losses.

Translation differences relating to items whose fair value changes are recognised in equity are also recognised in equity without effect on profit and loss.

The most important exchange rates used in the consolidated financial statements have developed in relation to the euro as follows:

T080 EXCHANGE RATES

	2018		2017	
	Balance sheet exchange rate	Income statement average rate	Balance sheet exchange rate	Income statement average rate
AUD	0.61536	0.63223	0.64984	0.67792
CAD	0.64094	0.65241	0.66404	0.68083
CHF	0.88776	0.86633	0.85440	0.90106
CNY	0.12708	0.12787	0.12774	0.13050
GBP	1.11000	1.13057	1.12499	1.13930
HKD	0.11163	0.10771	0.10648	0.11318
INR	0.01250	0.01235	0.01303	0.01356
JPY	0.00792	0.00767	0.00739	0.00788
KRW	0.00078	0.00077	0.00078	0.00078
NOK	0.10014	0.10426	0.10161	0.10740
PLN	0.23234	0.23463	0.23952	0.23505
SEK	0.09747	0.09727	0.10170	0.10382
USD	0.87409	0.84415	0.83184	0.88200

The provisions of IAS 29, Financial Accounting in Hyper-inflationary Economies, were applied to two consolidated companies, in Venezuela and Argentina. Gains and losses from adjusting for inflation the carrying amounts of non-monetary assets and liabilities and the income statement were immaterial and were recognised in other operating income.

The effects of intra-Group transactions are completely eliminated in the course of consolidation. Receivables and liabilities between consolidated companies are offset against one another and intra-Group provisions are reversed through profit or loss. Intra-Group profits and losses in non-current assets and inventories are eliminated – mostly in connection with the internal resale of aircraft and maintenance events. Intra-Group income is set off against the corresponding expenses. Tax accruals and deferrals are made as required by IAS 12 for temporary differences arising from consolidation.

Other intangible assets (except goodwill)

Acquired intangible assets are shown at cost, while internally generated intangible assets from which the Group expects to derive future benefit and that can be measured reliably are capitalised at cost of production and amortised regularly using the straight-line method over an estimated useful life. The cost of production includes all costs directly attributable to the production process, including borrowing costs as required under IAS 23, as well as appropriate portions of production-related overhead.

Intangible assets with indefinite useful lives (mainly brands and purchased, resalable take-off and landing rights) are not amortised, but rather subjected to a regular annual impairment test, as is goodwill.

Property, plant and equipment

Tangible assets used in business operations for longer than one year are valued at cost less regular straight-line depreciation. The cost of production includes all costs directly attributable to the manufacturing process as well as appropriate portions of production-related overhead. Borrowing costs in close connection with the financing of the purchase or production of a qualifying asset are also capitalised.

Key components of property, plant and equipment that have different useful lives are recognised and depreciated separately. Seats and in-flight entertainment systems installed in commercial aircraft are recognised separately. If costs are incurred in connection with regular extensive maintenance work (e.g. overhauling aircraft), these costs are recognised as a separate component insofar as they meet the criteria for recognition.

More pronounced investment cycles for aircraft and the better data analytics now available for engine maintenance processes lead to the change that from the beginning of financial year 2018, major engine overhauls are now also carried as separate components with a typical useful life of six years. Expenses for major engine overhauls incurred during the useful life of the aircraft are capitalised and depreciated accordingly. This change improves the presentation of the assets and earnings position because expenses for maintenance work now cause earnings to fluctuate less over time. The effects of this change in the accounting method were applied retrospectively in line with IAS 8 and so the comparative figures for

prior years were adjusted accordingly. The following table shows the effects of the change on the relevant items of the statement of financial position, the income statement and the cash flow statement.

T081 EFFECTS OF RESTATEMENT CAPITALISATION ENGINE OVERHAUL EVENTS 2017

in €m	1.1.2017		
	reported	adjusted	difference
Statement of financial position			
Aircraft and reserve engines	14,798	14,189	- 609
Deferred tax assets	1,413	1,547	134
Shareholders' equity	7,149	6,674	- 475
in €m	31.12.2017		
	reported	adjusted	difference
Statement of financial position			
Aircraft and reserve engines	15,959	15,333	- 626
Deferred tax assets	1,523	1,661	138
Shareholders' equity	9,598	9,110	- 488
Income statement			
Changes in inventories and work performed by entity and capitalised	106	438	332
Cost of materials and services	- 19,013	- 19,028	- 15
Depreciation, amortisation and impairment	- 2,052	- 2,382	- 330
Profit/loss from operating activities	3,153	3,140	- 13
Profit/loss before income taxes	3,187	3,174	- 13
Income taxes	- 789	- 787	2
Profit/loss after income taxes	2,398	2,387	- 11
Earnings per share in €	5.03	5.01	- 0.02
Cash flow statement			
Net profit/loss before income taxes	3,187	3,174	- 13
Depreciation, amortisation and impairment losses on non-current assets (net of reversals)	1,970	2,300	330
Change in other assets/ equity and liabilities	685	701	16
Cash flow from operating activities	5,035	5,368	333
Net cash from/used in investing activities	- 2,782	- 3,115	- 333

If the accounting change had not taken place, the relevant items would have been affected as follows:

**T081 EFFECTS OF RESTATEMENT CAPITALISATION
ENGINE OVERHAUL EVENTS 2018**

in €m	31.12.2018		
	reported	without accounting change	difference
Statement of financial position			
Aircraft and reserve engines	16,776	17,280	- 504
Deferred tax assets	2,131	2,013	118
Shareholders' equity	9,573	9,959	- 386
Income statement			
Changes in inventories and work performed by entity and capitalised	531	61	470
Cost of materials and services	- 18,669	- 18,659	- 10
Depreciation, amortisation and impairment	- 2,205	- 1,867	- 338
Profit/loss from operating activities	2,800	2,678	122
Profit/loss before income taxes	2,784	2,662	122
Income taxes	- 588	- 568	- 20
Profit/loss after income taxes	2,196	2,094	102
Earnings per share in €	4.58	4.36	0.22
Cash flow statement			
Net profit/loss before income taxes	2,784	2,662	122
Depreciation, amortisation and impairment losses on non-current assets (net of reversals)	2,178	1,840	338
Change in other assets/ shareholders' equity and liabilities	- 276	- 286	10
Cash flow from operating activities	4,109	3,639	470
Net cash from/used in investing activities	- 3,859	- 3,389	- 470

The following useful lives are applied throughout the Group:

T082 USEFUL LIVES OF PROPERTY, PLANT AND EQUIPMENT

Property, plant and equipment	Useful life
Buildings	45 years
New commercial aircraft and reserve engines	20 years to a residual value of 5%
Separable aircraft components	4 to 6 years
Technical equipment and machinery	8 to 20 years
Other equipment, operating and office equipment	3 to 20 years

Buildings, fixtures and fittings on rented premises are depreciated according to the terms of the leases or over a shorter useful life.

Assets acquired second-hand are depreciated over their expected remaining useful life.

When assets are sold or scrapped, the difference between the net proceeds and the net carrying amount of the assets is recognised as a gain or loss in the other operating income or expenses, respectively.

In addition to the impairment tests for goodwill, slots and brands, individual items of property, plant and equipment and intangible assets are also tested for impairment if they are no longer intended for future use, either because they are damaged, retired or due to be sold. In this case, the assets are measured individually in line with the applicable standard (full write-down to scrap value, or disposal proceeds less costs to sell). The lowest level at which assets can form a CGU is a production facility, to the extent that separate product lines or customer (groups) can be assigned to it. When aircraft are held for service in the Group fleet and there is no immediate intention to sell them, they are combined with the assets of the respective operating unit for the purposes of impairment testing.

Impairment losses on intangible assets and property, plant and equipment

In addition to depreciation and amortisation on property, plant and equipment and intangible assets, impairment losses are also recognised on the balance sheet date if the asset's recoverable amount has fallen below its carrying amount. The recoverable amount is determined as the higher of an asset's fair value less costs to sell and the present value of the estimated net future cash flows from continued use of the asset (value in use).

Fair value less costs to sell is derived from recent market transactions, if available.

If it is impossible to forecast expected cash flows for an individual asset, the cash flows for the next larger asset unit are estimated, discounted at a rate reflecting the risk involved, and the recoverable amount allocated to the individual assets in proportion to their respective carrying amounts.

If the reason for an impairment loss recognised in previous years should cease to exist in whole or in part in subsequent periods, the impairment loss is reversed up to the amount of the asset's amortised cost.

Repairable spare parts for aircraft

Initial supply of spare parts for aircraft that can be reused after repair are classified as non-current assets.

Repairable spare parts for aircraft are held at continually adjusted prices based on average acquisition costs. For measurement purposes, spare parts to be allocated to a maintenance pool are assigned to individual aircraft models and depreciated on a straight-line basis depending on the life phase of the fleet models for which they can be used. Other spare parts, mainly intended for replacement, are recognised in the statement of financial position at a discount to their acquisition costs, depending on how common they are.

Finance leases

In accordance with IAS 17, the economic ownership of leased assets is deemed to be transferred to the lessee if the lessee bears all of the risks and rewards associated with ownership of the leased asset. In addition to the duration of the non-terminable initial term of the lease and the present value of the leasing payments as a proportion of the total investment, particular consideration is given to the distribution of risks and rewards relating to the non-amortised residual value of the asset over the term of the lease. Insofar as its economic ownership is deemed to be with the Lufthansa Group, the asset is capitalised at the start of the leasing contract at the lower of the present value of the leasing instalments and the asset's time value, plus any incidental expenses borne by the lessee. Depreciation methods and useful economic lives correspond to those applied to comparable purchased assets.

Operating leases

With an operating lease, the lease payment made by the lessee is recognised as an expense and the payment received by the lessor as income. The leased asset is still recognised in the consolidated statement of financial position as property, plant and equipment.

Equity investments accounted for using the equity method

Equity investments accounted for using the equity method are capitalised at cost at the time of acquisition.

In subsequent periods, the carrying amounts are either increased or reduced annually by changes in the shareholders' equity of the associated company or joint venture that is held by the Lufthansa Group. The principles of purchase price allocation that apply to full consolidation are applied accordingly to the initial measurement of any difference between the acquisition cost of the investment and the pro rata share of shareholders' equity of the company in question. An impairment test is only carried out in subsequent periods if there are indications of a potential impairment in the entire investment valuation.

Financial instruments

Financial assets are classified within the Lufthansa Group in accordance with IFRS 9 as "at amortised cost", "at fair value through profit or loss", "at fair value through other comprehensive income (with and without recycling)" and "derivative financial instruments as an effective part of a hedging relationship".

The category "**at amortised cost**" consists of financial assets that are debt instruments and are intended to be held to maturity on the basis of the company's business model. Furthermore, these instruments have fixed payment terms and meet the criteria for cash flow characteristics, i.e. contractual payments of principal and interest. At Lufthansa this item particularly includes loans and receivables, cash-in-hand and bank balances. They are classified as non-current or current assets according to their remaining maturity.

The category "**at fair value through profit or loss**" comprises debt instruments for which the business model is neither to hold nor to hold and sell them, or which do not pass the cash flow characteristics test. This is generally not the case at Lufthansa. Equity instruments are also allocated to this category as a rule, so Lufthansa generally recognises shares and equity investments that are financial instruments in this category. Derivatives that do not meet the criteria for hedge accounting are also classified in this category.

Debt instruments are classified "**At fair value through other comprehensive income (with recycling)**" when the business model is to hold and sell these instruments and they pass the cash flow characteristics test. At Lufthansa, this applies to securities representing debt instruments.

An option can be exercised to classify specific equity instruments "**At fair value through other comprehensive income (without recycling)**". Lufthansa exercises this option for individual share positions.

Lufthansa uses derivatives for hedging, which are classified as "**derivative financial instruments as an effective part of a hedging relationship**" if all the requirements for hedge accounting are satisfied.

Financial instruments are recognised on the settlement date, i.e. on the date that they are created or transferred. Financial assets are capitalised at fair value plus transaction costs. Unrealised gains and losses are recognised directly in equity, taking deferred taxes into account. Long-term low or non-interest-bearing loans are recognised at net present value

using the effective interest method. Subsequent measurement of the financial instrument depends on the classification, either at amortised cost using the effective interest method, or at fair value, through profit or loss or in equity without effect on profit and loss.

Receivables denominated in foreign currencies are measured at the balance sheet date rate.

The fair value of securities is determined by the price quoted on an active market. For unlisted fixed-interest securities, the fair value is determined from the difference between effective and market interest rate at the valuation date.

If there are doubts as to the recoverability of receivables, then impairment losses are recognised and these receivables are recognised at the lower recoverable amount. Subsequent reversals (write-backs) are recognised in profit or loss. IFRS 9 requires that when a receivable is recognised for the first time, an expected loss is provided for that reflects the credit risk of the receivable before a default event occurs. External credit risk exists for the Lufthansa Group especially in its portfolio of trade receivables, for which an expected credit loss allowance is recognised.

Derivative financial instruments are measured at fair value on the basis of published market prices. If there is no quoted price on an active market, other appropriate valuation methods are applied. Appropriate valuation methods take all factors into account that independent, knowledgeable market participants would consider in arriving at a price and that constitute recognised, established economic models for calculating the price of financial instruments.

In accordance with its internal guidelines, the Lufthansa Group uses derivative financial instruments to hedge interest rate and exchange rate risks and to hedge fuel price risks. This is based on the hedging policy defined by the Executive Board and monitored by a committee.

Interest rate swaps and interest rate/currency swaps are used to manage interest rate risks. Interest rate/currency swaps also hedge exchange rate risks arising from borrowing in foreign currencies.

Fuel price hedging takes the form of spread options and other hedging combinations, primarily for crude oil. To a limited extent, hedging may also be undertaken for other products, such as gas oil.

Hedging transactions are used to secure either fair values (fair value hedge) or future cash flows (cash flow hedge).

To the extent that the financial instruments used qualify as effective cash flow hedging instruments within the scope of a hedging relationship, in accordance with the provisions of IFRS 9, the fluctuations in market value will not affect the result for the period during the term of the derivative. They are recognised without effect on profit or loss in the corresponding reserves. If the hedged cash flow is an investment, the result of the hedging transaction that has previously been recognised in equity is set off against the cost of the capital expenditure at the time the underlying transaction matures. In all other cases, the cumulative gain or loss previously stated in equity is included in net profit or loss for the period on maturity of the hedged cash flow.

In the case of effective hedging of fair values that are designated as a fair value hedge, the changes in the market value of the hedged asset or the hedged debt and those of the financial instrument will balance out in the income statement.

Derivatives that do not meet the criteria for hedge accounting are presented in the category “at fair value through profit or loss”. Changes in fair value are then recognised directly in profit or loss. At Lufthansa, this generally occurs when the exposure or item being hedged cannot be measured reliably or the exposure ceases to exist prematurely over the course of the hedge.

Embedded derivatives – to the extent that they should, but cannot, be separated from the financial host contract – are also considered with these as trading transactions for measurement purposes. Changes in market value are also recognised directly as profit or loss in the income statement. Both types must be classified as financial assets stated at fair value through profit or loss.

It is the Group's hedging policy (➤ **Note 44, p. 164ff.**) only to acquire effective derivatives for the purpose of hedging interest rate, exchange rate and fuel price risks.

Initial recognition of **financial guarantees** given to third parties is at fair value. Thereafter, financial guarantees are either measured in the category “at fair value through profit or loss” or at the higher of the originally recognised amount, less any cumulative amortisation through profit or loss in line with IFRS 15, and the value of the contractual obligation measured in line with IAS 37.

Emissions certificates

CO₂ emissions certificates are recognised as intangible assets and presented under other receivables. Rights, both those purchased and those allocated free of charge, are measured at cost and not amortised.

Contract assets and receivables

If the customer has fulfilled their contractual obligations. Receivables are recognised if the right to receive the consideration is no longer subject to conditions. This is generally the case when the Group is contractually entitled to send the customer an invoice. Contract assets mainly relate to construction or service contracts for MRO and IT services.

Inventories

The item "Inventories" comprises non-repairable spare parts and assets used in production or the provision of services (raw materials, consumables and supplies), purchased merchandise, finished and unfinished goods and advance payments for them. They are measured at cost, determined on the basis of average prices, or at production costs. The cost of production includes all costs directly attributable to the production process, including borrowing costs as required under IAS 23, as well as appropriate portions of production-related overheads. Average capacity utilisation of 100% is assumed in determining the costs of production (previous year: 96%). Measurement on the balance sheet date is at the lower of cost and net realisable value. Net realisable value is defined as the estimated selling price less the estimated cost of completion and the estimated costs necessary to make the sale.

Assets held for sale

Individual, formerly non-current assets or groups of assets that are expected to be sold within the next twelve months are measured at the lower of their carrying amount at the time they are reclassified and fair value less costs to sell. Fair value less costs to sell is derived from recent market transactions, if available.

Property, plant and equipment and intangible assets are no longer depreciated or amortised and affiliated companies accounted for using the equity method are no longer accounted for in this way once they are classified as held

for sale or held for distribution. While the impairment charge from the last measurement before reclassification is recognised as an impairment loss, all subsequent changes in the measurement of current assets held for sale, e.g. due to exchange rate movements, are shown in other operating expenses or income.

Cash and cash equivalents

Cash and cash equivalents comprise cash-in-hand, cheques received and credit balances at banks. Cash equivalents are financial investments that can be liquidated at short notice. At the time of purchase or investment, they have a maturity of three months or less.

Pension provisions

The pension provisions for defined-benefit plans correspond to the present value of the defined-benefit obligations (DBO) on the reporting date less the fair value of plan assets, if necessary taking the rules on the maximum surplus of plan assets over the obligation (asset ceiling) into account.

The DBO is calculated annually by independent actuaries using the projected unit credit method prescribed in IAS 19 for defined-benefit pension plans. The measurement of pension provisions within the statement of financial position is based on a number of actuarial assumptions.

Capital account plans are measured using the market value of the assets assigned to the individual capital accounts as of the reporting date, whereby the present value of the minimum benefit payable when the beneficiary becomes entitled to the benefit is compared with the amount of contributions already paid in, measured using the assumptions for the benefit plans. Additional risk premiums that the employer contributes to insure against early entitlements are included in current service expense.

They include, in particular, assumptions about long-term salary and pension trends as well as average life expectancy. The assumptions about salary and pension trends are based on developments observed in the past and take into account national interest and inflation rates and labour market trends. Estimates of average life expectancy are based on recognised biometric calculation formulas.

The interest rate used to discount the individual future payment obligations is based on the return from investment grade corporate bonds in the same currency and with a similar term to maturity. The discount rate is determined by reference to high-quality corporate bonds with an issue volume of at least EUR 100m and an AA rating from at least one of the rating agencies Moody's Investor Service, Fitch Ratings or Standard & Poor's Rating Services.

Actuarial gains and losses arising from the regular adjustment of actuarial assumptions are recognised directly in equity in the period in which they arise, taking deferred taxes into account. Also presented without effect on profit and loss are differences between the interest income at the beginning of the period calculated on plan assets based on the interest rate used to discount the pension obligations and the earnings from plan assets actually recorded at the end of the period. The actuarial gains and losses and any difference between the forecast result and the actual result from plan assets form part of the remeasurement.

Past service costs are recognised immediately in profit or loss.

Payments to pension providers for defined contribution retirement benefit commitments for which the pension provider or the beneficiary assumes the financial risks are recognised in staff costs as they fall due.

Other provisions and tax provisions

Other provisions are recognised for present legal and constructive obligations to third parties arising from past events that will probably give rise to a future outflow of resources provided that a reliable estimate can be made of the amount of the obligations as of the reporting date.

The amount of the provision is determined by the best estimate of the amount required to settle the present obligation. Past experience, current cost and price information as well as estimates from internal and external experts are used to determine the amount of provisions.

The management regularly analyses the current information on legal risks and makes provisions for probable obligations. These provisions cover estimated payments to the claimant, court and procedural costs, the costs of lawyers and of any out-of-court settlement. Internal and external lawyers assist with the estimate. When deciding on the necessity of a provision for litigation, the management takes into account the

probability of an unfavourable outcome and the chance of making a sufficiently accurate estimate of the amount of the obligation. The commencement of legal proceedings, the formal assertion of a claim against the Group or the disclosure of certain litigation in the Notes does not automatically mean that a provision was made for the risk concerned. A ruling in court proceedings, a decision by a public authority or an out-of-court settlement may cause the Group to incur expenses for which no provision was made because the amount could not be reliably determined or for which the provision made and the insurance coverage is not sufficient.

Provisions for obligations that are not expected to lead to an outflow of resources in the following year are recognised to the amount of the present value of the expected outflow, taking foreseeable price rises into account.

The assigned value of provisions is reviewed on each balance sheet date. Provisions in foreign currencies are translated at reporting date rates.

If no provision could be recognised because one of the stated criteria was not fulfilled, the corresponding obligations are shown as contingent liabilities and discussed in the relevant section.

Obligations towards tax authorities that are uncertain with regard to their occurrence, probability and amount are recorded as **tax provisions** on the basis of reasonable estimates. Existing contingent liabilities in connection with this are addressed separately.

Liabilities

Liabilities arising from finance leases are recognised at the present value of the lease payments at the inception of the lease term. Other financial liabilities are recognised at fair value. Liabilities for which interest is not payable at a market rate are recognised at present values.

Measurement in subsequent periods is at amortised cost using the effective interest rate method.

Liabilities in foreign currencies are measured at the balance sheet date.

Obligations from share programmes were measured at fair value as cash-settled share-based payment transactions in accordance with IFRS 2. Fair value was measured using a Monte Carlo simulation.

The liability is recognised on the basis of the resulting fair value, taking the remaining term of the programme into account. Changes are recognised as staff costs in profit or loss.

Details of the assumptions used for the model and the structure of the share programmes can be found in

➤ **Note 38, p. 153ff.**

Contract liabilities

A contract liability is an obligation on the part of the Group towards a customer to provide goods or services for which the customer has already performed an obligation, e.g. by making an advance payment.

The Group's contract liabilities consist of liabilities from unused flight documents, unredeemed miles from the customer loyalty programmes, construction contracts and other contract liabilities.

Until they are used, sold flight documents are recognised as an **obligation from unused flight documents**. Coupons that are unlikely to be used any more are recognised pro rata temporis as traffic revenue in the income statement at their estimated value. The estimate is based on historical statistical data.

The Lufthansa Group uses various bonus miles programmes with the aim of ensuring long-term **customer loyalty**. Participants in the Miles & More programme, which is the biggest bonus miles programme in the Lufthansa Group, can collect and redeem bonus miles for flights with the airlines in the Lufthansa Group as well as with numerous partners (including other airlines, hotels, global car hire companies, financial and insurance providers, telecommunications companies, retailers, automobile clubs etc.). Miles expire three years after they are collected, in accordance with the terms of membership, unless they are protected by frequent flyer status or credit card use.

Revenue for premium miles is recognised at the point in time or over the time at which the goods and services purchased with the premium miles are transferred.

Experience of passengers' past behaviour is used to measure the premium claims that are collected on flights with the airlines in the Lufthansa Group. Miles that are expected to be used for flights with airlines in the Lufthansa Group are measured with the average price of the premium flight or upgrade for the average number of miles used. The price is calculated on the basis of past redemption patterns, weighted for the various geographic regions and booking classes.

This is then corrected to allow for the reduced flexibility of premium flights and the award miles granted for normal flights. Miles that are expected to be redeemed for other bonuses are measured at the average price for these bonuses and the average number of miles redeemed. The prices for additional miles are recalculated every year and applied to all additions in that year. Consumption of miles is measured using the average rate for total miles at the beginning of the year (same as previous year).

Premium points collected from other partners are measured at the amounts paid by these partners in relation to the average number of miles collected and redeemed.

The previous calculation method for the legal and economic expiry rate can still be applied unchanged under IFRS 15. It entails calculating the expiry rate from the rates observed in prior years, increased or decreased as necessary by reference to past trends or future enhancements to the programme. There are changes to the recognition of income from the expiry of award miles. IFRS 15 requires that income from the expiry of miles is recognised in parallel with revenue from the performance obligations that do not expire. A period of three years is therefore assumed for revenue recognition, and the revenue from miles expected to expire is recognised on a straight-line basis over this time.

Deferred tax items

In accordance with IAS 12, deferred taxes are recognised for all temporary differences between the statements of financial position with regard to tax of individual companies and the consolidated financial statements. Tax loss carry-forwards are recognised to the extent that the deferred tax assets are likely to be used in the future. Company earnings forecasts and specific, realisable tax strategies are used to determine whether deferred tax assets from tax losses carried forward are usable or not, i.e. whether they have a value that can be realised. The planning period used to assess this probability is determined by the individual Group company according to the specific circumstances and lies generally between three and five years.

Effective income taxes

The Lufthansa Group is liable for income taxes in various countries. Material assumptions are necessary to calculate the income tax liabilities. For certain transactions and calculations, the final taxation cannot be assessed definitively in the course of normal business. The amount of the liability for future tax inspections is based on estimates of whether additional income taxes will be owed, and if so, at which amount. Estimates will be corrected as necessary in the period in which taxation is definitively assessed.

NOTES TO THE CONSOLIDATED INCOME STATEMENT

3 Traffic revenue

In the consolidated income statement, the Lufthansa Group attributes revenue to the segments Network Airlines, Eurowings, Logistics, MRO, Catering and Other. The following table provides a breakdown of traffic revenue according to the different business models for the financial year that ended on 31 December 2018:

Traffic revenue of EUR 28,103m (previous year: EUR 28,399m) includes freight and mail revenue of EUR 3,106m (previous year: EUR 2,932m). The Logistics segment accounted for EUR 2,549m (previous year: EUR 2,371m) of this amount. Other freight and mail revenue of EUR 557m (previous year: EUR 561m) stems mainly from marketing belly capacities on passenger flights by SWISS and Brussels Airlines.

T083 TRAFFIC REVENUE BY SECTOR

	2018	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia/Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾	2017 ²⁾
in €m								
Network Airlines	21,371	13,960	3,759	559	2,274	542	277	22,012
Lufthansa German Airlines	14,610							15,242
SWISS ³⁾	4,776							4,620
Austrian Airlines	1,985							2,150
Eurowings ³⁾	4,182	3,749	178	12	70	26	147	4,014
Logistics	2,550	1,313	256	103	798	26	54	2,373
Total	28,103							28,399

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Application of the modified retrospective approach; revenue measured for 2017 according to IAS 11 and IAS 18.

³⁾ Disclosure of traffic revenue, including belly revenue; this is reported in the segment reporting in the reconciliation column.

4 Other revenue

The following table provides a breakdown of other revenue by category (type of service) and geographic distribution for the 2018 financial year:

T084 OTHER OPERATING REVENUE BY AREA OF OPERATIONS

	2018	Europe ¹⁾	North America ¹⁾	Central and South America ¹⁾	Asia/Pacific ¹⁾	Middle East ¹⁾	Africa ¹⁾	2017 ²⁾
in €m								
MRO	3,812	1,718	855	194	757	129	159	3,568
MRO services	3,402							3,042
Other operating revenue	410							526
Catering	2,499	483	1,270	147	495	62	42	2,556
Catering services	2,119							2,193
Revenue from in-flight sales	142							112
Other services	238							251
Network Airlines	621	495	51	7	50	8	10	629
Eurowings	30	25	2	-	1	-	2	33
Logistics	131	73	47	-	4	7	-	125
Additional Businesses and Group Functions	648	489	47	13	76	17	6	269
IT services	291							291
Travel management	276							-
Other	81							-
Total	7,741							7,180

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Application of the modified retrospective approach; revenue measured for 2017 according to IAS 11 and IAS 18.

MRO services make up the majority of external revenue in the MRO segment. Other revenue in the MRO segment from the sale of material and hiring out material and engines, as well as logistics services, are classified as other services.

The revenue listed under catering services originates exclusively in the Catering segment. The Catering segment also generates revenue from other services, particularly in the areas of flight security concepts, in-flight service equipment, transport/warehouse logistics and lounge operations.

Other revenue also includes revenue from customer contracts that are fulfilled over a given period. These are mainly MRO and IT services.

5 Changes in inventories and work performed by entity and capitalised

T085 CHANGES IN INVENTORIES AND WORK PERFORMED BY ENTITY AND CAPITALISED

in €m	2018	2017
Increase/decrease in finished goods and work in progress	2	5
Other internally produced and capitalised assets	529	433
	531	438

The increase in other own work capitalised results mainly from greater engine overhaul activity in the Group (EUR 470m, previous year: EUR 332m).

In the course of changing the accounting method for engine maintenance events, the figures for the previous year were adjusted accordingly. ➤ **Note 2, p. 106ff.**

6 Other operating income

T086 OTHER OPERATING INCOME

in €m	2018	2017
Foreign exchange gains	794	885
Income from the reversal of provisions and accruals	260	151
Compensation received for damages	55	64
Income from the disposal of non-current assets	50	38
Reversal of write-downs on receivables	48	46
Income from operating-leasing aircraft	48	44
Rental income	40	33
Services provided by the Group	27	27
Income from staff secondment	19	26
Income from the reversal of impairment losses on fixed assets	15	83
Commission income	14	272
Income from the disposal of non-current available-for-sale financial assets	1	5
Income from sub-leasing aircraft	1	1
Miscellaneous other operating income	446	601
	1,818	2,276

Foreign exchange gains (excluding financial liabilities) mainly include gains from differences between the average rate for the month on the transaction date and on the payment date, along with foreign exchange gains from measurement at the closing date rate. Income from exchange rate hedging is also recognised here. Foreign exchange losses from these transactions are reported under other operating expenses.

➤ **Note 10, p. 124.** The foreign currency effects of borrowing are recognised in other financial items, in the context of the net results of exchange rate hedging relationships for borrowing.

Income from the release of provisions relates to a number of provisions recognised in previous years that have not been fully used. In contrast, expenses from insufficient provisions recognised in previous years are recognised together with the primary expense item to which they relate.

Income from the disposal of property, plant and equipment also includes EUR 13m from aircraft sold (previous year: EUR 24m).

In the previous year, commission income included revenue from training and travel management of EUR 342m, which as of 2018 is shown in revenue as a result of the IFRS 15 changes.

Other operating income includes items not attributable to any of the aforementioned categories.

7 Cost of materials and services

T087 COST OF MATERIALS AND SERVICES

in €m	2018	2017
Aircraft fuel and lubricants	6,087	5,232
Other raw materials, consumables and supplies	3,107	3,095
Purchased goods	489	460
Total cost of raw materials, consumables and supplies and of purchased goods	9,683	8,787
Fees and charges	4,457	6,357
External MRO services	1,848	1,534
Charter expenses	626	651
External IT services	416	381
In-flight services	383	379
Operating lease payments	91	88
Other services	1,165	851
Total cost of purchased services	8,986	10,241
	18,669	19,028

As of 2018, the airport fees collected and the corresponding airport charges are no longer recognised in the income statement following the IFRS 15 changes. This led to a reduction of fees and charges by EUR 2,257m.

The retrospective change in the accounting method for engine overhauls has resulted in an adjustment of the external MRO services reported in the previous year. For a detailed description of the effects, we refer to [Note 2, p. 106ff.](#)

8 Staff costs

T088 STAFF COSTS

in €m	2018	2017
Wages and salaries	7,286	7,015
Social security contributions	977	938
Expenses for pension plans and other employee benefits	548	219
	8,811	8,172

Expenses for pension plans principally consist of additions to the pension provisions. [Note 33, p. 142ff.](#)

In the previous year, the expenses for pension plans and other employee benefits were reduced by EUR 582m because of negative past service expenses resulting from the changes to the retirement benefits system for cockpit staff within the Group's wage agreement for Germany.

T089 EMPLOYEES

	Average for the year 2018	Average for the year 2017	As of 31.12.2018	As of 31.12.2017
Ground staff	88,478	86,304	89,278	86,044
Flight staff	44,858	41,496	45,169	42,279
Trainees	994	1,056	1,087	1,101
	134,330	128,856	135,534	129,424

The annual average is calculated pro rata temporis from the time companies are consolidated or deconsolidated.

9 Depreciation, amortisation and impairment

The notes to the individual items show the breakdown of depreciation, amortisation and impairment charges between intangible assets, aircraft and other property, plant and equipment. Total depreciation, amortisation and impairment came to EUR 2,205m (previous year: EUR 2,382m).

T090 DEPRECIATION, AMORTISATION AND IMPAIRMENT

in €m	2018	2017
Amortisation of other intangible assets	98	98
Depreciation of aircraft	1,833	1,702
Depreciation of other tangible assets	250	239
Total amortisation/depreciation	2,181	2,039
Impairment of goodwill	-	3
Impairment of other intangible assets	2	24
Impairment of aircraft	4	213
Impairment of other tangible assets	5	20
Impairment of financial assets	13	83
Total impairment	24	343
Total depreciation, amortisation and impairment	2,205	2,382

The retrospective change in the accounting method for engine overhauls resulted in an adjustment of the depreciation reported in the previous year. For a detailed description of the effects, we refer to [Note 2, p. 106ff.](#) EUR 338m of the depreciation of aircraft related to capitalised engine overhaul events (previous year: EUR 330m).

Impairment losses of EUR 24m were recognised in the financial year 2018. EUR 4m of the total was for two Boeing MD-11s from the Logistics segment held for sale. Other impairment losses related to write-downs of loans to trainee pilots (EUR 11m) and LSG Italia (EUR 5m).

In the previous year, impairment losses came to EUR 343m, of which EUR 204m was for aircraft destined to be scrapped. Other significant impairment losses in the previous year related to financial assets written down in the course of the insolvency and the failed takeover of NIKI Luftfahrt GmbH.

As in the previous year, no other impairment losses on assets held for sale were recognised in other operating expenses.

10 Other operating expenses

T091 OTHER OPERATING EXPENSES

in €m	2018	2017
Staff-related expenses	1,226	1,114
Rental and maintenance expenses	923	884
Foreign exchange losses	771	758
Expenses for computerised distribution systems	524	505
Advertising and sales promotions	439	400
Sales commission paid to agencies	330	335
Auditing, consulting and legal expenses	252	272
Commissions for credit cards	222	227
Other services	148	137
Write-downs on receivables	100	98
Communications costs	84	77
Other taxes	84	87
Insurance premiums for flight operations	56	56
Losses on disposal of non-current assets	17	6
Miscellaneous other operating expenses	532	615
	5,708	5,571

Staff-related expenses also include travel and training costs for Group employees and the costs of outside staff.

Foreign exchange losses (excluding financial liabilities) mainly consist of losses from differences between the monthly average rates on the transaction date and on the payment date, expenses from exchange rate hedges and translation losses from measurement at the exchange rate on the balance sheet date. ➤ **Note 6, p. 122.** The foreign currency effects of borrowing are recognised in other financial items in the context of the net results of exchange rate hedging relationships of borrowing.

11 Result from equity investments

T092 RESULT FROM EQUITY INVESTMENTS

in €m	2018	2017
Result of joint ventures accounted for using the equity method	75	79
Result of associated companies accounted for using the equity method	39	39
Result of equity investments accounted for using the equity method	114	118
Dividends from other joint ventures	13	11
Dividends from other associated companies	6	8
Income from profit transfer agreements	31	24
Expenses from loss transfer agreements	-26	-16
Dividends from other equity investments	36	12
Result of other equity investments	60	39
	174	157

Income and expenses from profit and loss transfer agreements are shown including tax contributions.

12 Net interest

T093 NET INTEREST

in €m	2018	2017
Income from other securities and non-current financial loans	5	115
Other interest and similar income	63	63
Interest income	68	178
Interest expenses on pensions obligations	-97	-159
Interest expenses on other provisions	-8	-6
Interest and other similar expenses	-107	-208
Interest expenses	-212	-373
	-144	-195

Net interest comprises interest income and expenses – calculated using the effective interest method in accordance with IFRS 9 – from financial assets and liabilities not classified as at fair value through profit or loss.

13 Other financial items

T094 OTHER FINANCIAL ITEMS

in €m	2018	2017
Result of fair value hedges – change in time value of hedged transactions	- 47	20
Result of fair value hedges – change in time value of hedging instruments	60	- 20
Ineffective portion of derivatives used as cash flow hedges	17	18
Result of derivatives held for trading classified as at fair value through profit or loss	- 16	- 250
Result of measuring securities classified as at fair value through profit or loss	- 35	-
Exchange rates effects from financial liabilities	- 26	288
	- 46	56

Cross currency swaps have been designated as hedging instruments for hedge accounting since 1 January 2018 and so in contrast to the previous year are no longer presented as stand-alone derivatives. Furthermore, changes in the time value of options used as hedging instruments have been retrospectively recognised in equity without effect on profit and loss as a cost of hedging, instead of through profit or loss, since 1 January 2017. In 2017, this led to an adjustment in the net result from the valuation of derivatives, which was accounted for at the fair value of EUR -16m (before taxes)

➤ Note 2, p. 106ff.

14 Income taxes

T095 INCOME TAXES

in €m	2018	2017
Current income taxes	614	813
Deferred income taxes	- 26	- 29
	588	784

Current income taxes include corporation tax, solidarity surcharge, trade tax and other income taxes paid outside Germany totalling EUR 538m for 2018 (previous year: EUR 629m). Tax expenses of EUR 76m were also incurred for prior years (previous year: EUR 184m).

The tax rates used to calculate deferred taxes abroad ranged from 3.5% to 35.0% in 2018 (previous year: 3.5% to 41.4%). For measuring deferred taxes, the relevant taxation rules in force or adopted at the balance sheet date are used.

The following table reconciles expected and effective tax expenses. Expected tax expense is calculated by multiplying profit before income taxes by a tax rate of 25% for the parent company (previous year: 25%). This is made up of a tax rate of 15.825% (previous year: 15.825%) for corporation tax and solidarity surcharge and 9.175% for trade tax (previous year: 9.175%). The portion of trade income related to the foreign air transport operations of the German-based airlines is deducted when calculating the tax rate for trade tax.

T096 TAX RECONCILIATION

	2018		2017	
	Basis of assessment	Tax expenses	Basis of assessment	Tax expenses
in €m				
Expected income tax expenses/refund	2,784	696	3,158	790
Tax free gains/losses	-	-	-	- 2
Non-deductible costs	-	55	-	42
Non-taxable income	-	- 61	-	- 75
Non-taxable income from equity investments	-	- 42	-	- 37
Difference between local taxes and the deferred tax rates of the parent company as well as effects of changes in tax rates	-	- 23	-	- 1
Taxes from other periods ¹⁾	-	-	-	72
Effects from use or deferred tax assets not recognised	-	- 37	-	- 5
Recognised income tax expenses	-	588	-	784

¹⁾ Deferred tax income of EUR 76m for other periods (previous year: EUR 112m) and effective tax expenses of EUR 76m for other periods (previous year: EUR 184m).

Deferred tax liabilities of EUR 47m (previous year: EUR 37m) were not recognised on temporary differences in connection with shares in subsidiaries, as the temporary differences are not expected to reverse in the foreseeable future.

Deferred tax assets and liabilities in 2018 and 2017 were allocable to the following items in the statement of financial position:

T097 DEFERRED TAX ASSETS AND LIABILITIES

in €m	31.12.2018		31.12.2017	
	Assets	Liabilities	Assets	Liabilities
Tax loss carry-forwards and tax credits	121	-	196	-
Pension provisions	2,331	-	1,949	-
Finance leases aircraft	-	36	17	-
Intangible assets, property, plant and equipment	-	887	-	931
Non-current financial assets	-	14	-	4
Fair value measurement of financial instruments	-	126	-	227
Provisions for contingent losses	19	-	34	-
Receivables/ liabilities/ other provisions	-	110	-	7
Inventories	247	-	173	-
Assets held for sale	-	1	-	-
Other	4	-	12	-
Offset amounts	- 591	- 591	- 720	- 720
	2,131	583	1,661	449

The deferred tax assets and liabilities in the category receivables/liabilities/other provisions and inventories are largely expected to reverse within twelve months of the reporting date.

A deferred tax asset of EUR 47m (previous year: EUR 18m) was recognised for companies incurring a net tax loss in the reporting year or in the previous year, because tax and earnings planning indicates that there is a high probability that the tax asset will be realised.

In addition to recognised deferred tax assets from tax loss carry-forwards, non-deductible interest carry-forwards and tax credits, further tax loss carry-forwards and temporary differences totalling EUR 2,423m (previous year: EUR 3,095m) exist for which no deferred tax assets could be recognised. The total amount of deferred tax assets from tax loss carry-forwards that could not be capitalised as of 31 December 2018 was EUR 584m (previous year: EUR 684m).

The usage restrictions for non-capitalised tax loss carry-forwards are distributed as follows:

T098 LIMITS ON THE USE OF NON-CAPITALISED LOSS CARRY-FORWARDS

in €m	Non-capitalised loss carry-forwards	Deferred taxes
Usable		
until 2022	32	7
until 2023	40	8
until 2024	16	4
until 2025	3	1
until 2026	1	-
until 2027	1	-
2028 and beyond	2,221	564
Total	2,314	584

15 Earnings per share

Basic/diluted earnings per share are calculated by dividing consolidated net profit by the weighted average number of shares in circulation during the financial year. To calculate the average number of shares, the shares bought back and reissued for the employee share programmes are included pro rata temporis.

T099 EARNINGS PER SHARE

	2018	2017
Basic/diluted earnings per share €	4.58	4.98
Consolidated net profit/loss €m	2,163	2,340
Weighted average number of shares	472,553,216	469,820,411

For the effects on earnings per share of the new accounting method for engine maintenance events, we refer to

➔ **Note 2, p. 106ff.**

As the parent company of the Group, Deutsche Lufthansa AG reported a distributable profit according to HGB of EUR 380m for the financial year 2018, including a transfer of EUR 41m from retained earnings. The Executive Board and Supervisory Board will table a proposal at the Annual General Meeting to be held on 7 May 2019 to pay a dividend of EUR 0.80 per share. This represents a total dividend of EUR 380m or 12.8% of EBIT for 2018.

In 2018, EUR 0.80 per share was distributed as a dividend to shareholders from the net profit for 2017.

NOTES TO THE CONSOLIDATED STATEMENT OF FINANCIAL POSITION

Assets

16 Goodwill and intangible assets with an indefinite useful life

T100 GOODWILL AND INTANGIBLE ASSETS WITH AN INDEFINITE USEFUL LIFE

in €m	Goodwill from consolidation	Intangible assets with an indefinite useful life	Total
Cost as of 1.1.2017	949	631	1,580
Accumulated impairment losses	- 313	- 2	- 315
Carrying amount 1.1.2017	636	629	1,265
Currency translation differences	- 2	- 34	- 36
Additions due to changes in consolidation	-	37	37
Additions	82	-	82
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	- 1	- 1
Reclassifications to assets held for sale	-	-	-
Impairment losses	- 3	- 1	- 4
Reversal of impairment losses	-	-	-
Carrying amount 31.12.2017	713	630	1,343
Cost as of 1.1.2018	1,027	634	1,661
Accumulated impairment losses	- 314	- 4	- 318
Carrying amount 1.1.2018	713	630	1,343
Currency translation differences	-	15	15
Additions due to changes in consolidation	-	-	-
Additions	23	-	23
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Reclassifications to assets held for sale	-	-	-
Impairment losses	-	-	-
Reversal of impairment losses	-	-	-
Carrying amount 31.12.2018	736	645	1,381
Cost as of 31.12.2018	1,050	648	1,698
Accumulated impairment losses	- 314	- 3	- 317

All goodwill and intangible assets with an indefinite useful life were subjected to a regular impairment test in 2018 as required by IAS 36. Furthermore, there is the obligation to perform an impairment test if there is an indication of impairment. For impairment testing following an indication of impairment ➤ **Note 9, p. 123ff.**

Acquired brands and slots have an indefinite useful life due to their lasting legal and economic significance. The tests were performed at the level of the smallest cash generating

unit (CGU) on the basis of fair value less costs to sell or value in use. Because Brussels Airlines is integrated into the Eurowings group for organisational and commercial purposes, goodwill for Brussels Airlines has been attributed to Eurowings and is tested for impairment there. The goodwill arising from the acquisition of Luftfahrtgesellschaft Walter is also part of the goodwill for Eurowings, since the company only acts as a service platform for flight services within Eurowings.

The following table provides an overview of the goodwill tested and the assumptions made in the respective impairment tests regarding the smallest possible cash-generating unit (CGU) in each case.

T101 IMPAIRMENT TESTS OF GOODWILL 2018

Name of the CGU	Lufthansa German Airlines	Eurowings	LSG Sky Chefs USA group	LSG Sky Chefs Korea	Other ¹⁾
Segment	Network Airlines	Eurowings	Catering	Catering	Catering/Service and financial companies
Carrying amount of goodwill (31.12.)	€ 238m	€ 116m	€ 277m	€ 61m	€ 44m
Impairment losses	–	–	–	–	–
Duration of planning period	4 years	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	2.2%	2.2%	2.0%	2.9%	1.0% to 5.0%
Discount rate	4.7% ²⁾	4.7% ²⁾	5.2% ³⁾	4.9% ³⁾	4.5% to 9.0% ³⁾

¹⁾ Goodwill of less than EUR 25m in any individual instance.

²⁾ After-tax rate.

³⁾ Pre-tax rate.

The assumptions on revenue growth used for the impairment tests are based on approved internal budgets and external sources for the planning period. In some cases, reductions were made for risk to allow for special regional features and market share trends specific to the respective companies. The margins used are based on past experience or were developed on the basis of cost-cutting measures initiated. The investment rates are based on past experience and take account of the replacement of any means of production envisaged during the planning period. Costs of the central functions were charged to the individual units based on their use of these functions.

Assuming sustained revenue growth by the CGUs as described in the table, the recoverable amounts would exceed the carrying amount by a significant figure, even if the growth assumptions were reduced by one percentage point in each

case, with the exception of LSG Sky Chefs Korea. Worsening the scenarios by one percentage point in each case, in terms of planned margins or the discount rates used for the impairment tests, would also not reduce the recoverable amounts below the respective carrying amounts for all of the other CGUs. The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation. At LSG Sky Chefs Korea, the recoverable amount exceeds the carrying amount for the CGU by 50%. An increase in the discount factor of 0.9 percentage points, a reduction of revenue growth by 0.4 percentage points or a reduction of 1.5 percentage points in the planned margin of 11.3% would all result in an impairment charge.

An impairment test was performed for LSG Sky Chefs Italy following the loss of a key customer, which resulted in an impairment loss of EUR 5m on property, plant and equipment.

The following table shows the assumptions used for the previous year's impairment tests.

T101 IMPAIRMENT TESTS OF GOODWILL 2017

Name of the CGU	Lufthansa German Airlines	Eurowings	Brussels Airlines	LSG Sky Chefs USA group	LSG Sky Chefs Korea	Other ¹⁾
Segment	Network Airlines	Eurowings	Eurowings	Catering	Catering	Catering/Service and financial companies
Carrying amount of goodwill (31.12.)	€ 238m	€ 11m	€ 82m	€ 277m	€ 60m	€ 45m
Impairment losses	–	–	–	–	–	€ 3m
Duration of planning period	3 years	3 years	3 years	3 years	3 years	3 years
Revenue growth p.a. after end of planning period	2.2%	2.2%	2.2%	2.0%	3.0%	2.0% to 4.0%
Discount rate	4.7% ²⁾	4.7% ²⁾	5.3% ³⁾	5.4% ³⁾	5.4% ³⁾	4.7% ²⁾ to 9.7% ³⁾

¹⁾ Goodwill of less than EUR 25m in any individual instance.

²⁾ After-tax rate.

³⁾ Pre-tax rate.

The intangible assets with indefinite useful lives consist of slots purchased as part of company acquisitions (insofar as they are tradeable) and brand names acquired.

The following table shows the assumptions made for regular impairment testing of the smallest cash-generating unit (CGU) in each case.

T102 IMPAIRMENT TESTS OF SLOTS 2018

Group company	SWISS	Austrian Airlines
Carrying amount for slots (31.12.)	€ 129m	€ 23m
Impairment losses	–	–
Duration of planning period	4 years	4 years
Revenue growth p.a. after end of planning period	2.2%	2.2%
Discount rate	4.8% ¹⁾	4.7% ¹⁾

¹⁾ After-tax rate.

The slots purchased by Deutsche Lufthansa AG with a carrying amount of EUR 112m as of 31 December 2018, unchanged from the previous year, were subjected to an impairment test on the same assumptions as those used for impairment testing the goodwill of the CGU Lufthansa German Airlines.

Based on sustainable revenue growth according to assumptions described in the table, the recoverable amounts significantly exceed the carrying amounts. Even if the assumptions on revenue growth, the discount rate and margins are all reduced by one percentage point in each case, the recoverable amounts are still higher than the carrying amounts.

The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation.

The following table shows the assumptions used for the previous year's impairment tests.

T102 IMPAIRMENT TESTS OF SLOTS 2017

Group company	SWISS	Austrian Airlines
Carrying amount for slots (31.12.)	€ 124m	€ 23m
Impairment losses	–	–
Duration of planning period	3 years	3 years
Revenue growth p.a. after end of planning period	2.2%	2.2%
Discount rate	4.8% ¹⁾	4.7% ¹⁾

¹⁾ After-tax rate.

The regular impairment test for the brands acquired was carried out on the basis of the revenue generated from each brand.

The following additional assumptions were used in the impairment test for the acquired brands:

T103 IMPAIRMENT TESTS OF BRANDS 2018

Group company	SWISS	Austrian Airlines	Brussels Airlines	Other
Carrying amount for brand (31.12.)	€ 230m	€ 107m	€ 37m	€ 7m
Impairment losses	-	-	-	-
Duration of planning period	4 years	4 years	4 years	4 years
Revenue growth p.a. after end of planning period	2.2%	2.2%	1.0%	0.0% to 2.2%
Savings in hypothetical leasing payments before taxes (royalty rate)	0.63%	0.35%	0.20%	0.23% to 0.50%
Discount rate	4.8% ¹⁾	4.7% ¹⁾	4.7% ¹⁾	4.5% to 5.1% ¹⁾

¹⁾ After-tax rate.

Assuming sustained brand-related revenue growth at the end of the planning period, as described in the table, the recoverable amounts for the brands exceed their carrying amounts significantly. Even if the assumptions for sustained brand-related revenue growth were to be reduced or the discount rate were to be increased by one percentage point in each case, the recoverable amounts would exceed the carrying amounts.

There was no impairment charge within the other brands in 2018. Brand rights for a company in the LSG group valued at EUR 1m were written off in 2017, since they will no longer be used.

The sensitivity analysis takes into account changes in one assumption at a time, whereby the other assumptions remain unchanged from the original calculation.

The assumptions used for the previous year's impairment tests can be derived from the following table.

T103 IMPAIRMENT TESTS OF BRANDS 2017

Group company	SWISS	Austrian Airlines	Brussels Airlines	Other
Carrying amount for brand (31.12.)	€ 221m	€ 107m	€ 37m	€ 6m
Impairment losses	-	-	-	€ 1m
Duration of planning period	3 years	3 years	3 years	3 years
Revenue growth p.a. after end of planning period	2.2%	2.2%	1.0%	0.0% to 2.2%
Savings in hypothetical leasing payments before taxes (royalty rate)	0.63%	0.35%	0.20%	0.23% to 0.50%
Discount rate	4.8% ¹⁾	4.7% ¹⁾	4.5% ¹⁾	4.5% to 5.7% ¹⁾

¹⁾ After-tax rate.

17 Other intangible assets

T104 OTHER INTANGIBLE ASSETS

	Concessions, industrial property rights and similar rights and licences to such rights and assets	Internally developed software	Advance payments	Total
in €m				
Cost as of 1.1.2017	1,135	124	147	1,406
Accumulated amortisation	- 816	- 103	- 15	- 934
Carrying amount 1.1.2017	319	21	132	472
Currency translation differences	- 12	-	- 3	- 15
Additions due to changes in consolidation	56	-	4	60
Additions	31	4	62	97
Reclassifications	81	8	- 86	3
Disposals due to changes in consolidation	-	-	-	-
Disposals	- 4	-	-	- 4
Reclassifications to assets held for sale	-	-	-	-
Amortisation	- 102	- 7	- 12	- 121
Reversal of impairment losses	-	-	-	-
Carrying amount 31.12.2017	369	26	97	492
Cost as of 1.1.2018	1,230	133	112	1,475
Accumulated amortisation	- 861	- 107	- 15	- 983
Carrying amount 1.1.2018	369	26	97	492
Currency translation differences	5	2	- 1	6
Additions due to changes in consolidation	-	-	-	-
Additions	29	4	77	110
Reclassifications	30	10	- 35	5
Disposals due to changes in consolidation	-	-	-	-
Disposals	- 1	-	- 1	- 2
Reclassifications to assets held for sale	-	-	-	-
Amortisation	- 92	- 7	-	- 99
Reversal of impairment losses	-	-	-	-
Carrying amount 31.12.2018	340	35	137	512
Cost as of 31.12.2018	1,289	150	152	1,591
Accumulated amortisation	- 949	- 115	- 15	- 1,079

Non-capitalised research and development expenses for intangible assets of EUR 44m (previous year: EUR 30m) were incurred in the period. Fixed orders have been placed for intangible assets worth EUR 10m (previous year: EUR 9m), but they are not yet at the Group's economic disposal.

18 Aircraft and reserve engines

T105 AIRCRAFT AND RESERVE ENGINES

in €m	Aircraft and reserve engines	Advance payments for aircraft and reserve engines	Total
Cost as of 1.1.2017	26,590	1,348	27,938
Accumulated amortisation	- 13,749	-	- 13,749
Carrying amount 1.1.2017	12,841	1,348	14,189
Currency translation differences	- 240	- 30	- 270
Additions due to changes in consolidation	259	3	262
Additions	2,450	460	2,910
Reclassifications	404	- 404	-
Disposals due to changes in consolidation	-	-	-
Disposals	- 51	- 4	- 55
Reclassifications to assets held for sale	122	-	122
Depreciation	- 1,907	-	- 1,907
Reversal of impairment losses	82	-	82
Carrying amount 31.12.2017	13,960	1,373	15,333
Cost as of 1.1.2018	29,405	1,373	30,778
Accumulated amortisation	- 15,445	-	- 15,445
Carrying amount 1.1.2018	13,960	1,373	15,333
Currency translation differences	114	13	127
Additions due to changes in consolidation	-	-	-
Additions	2,542	782	3,324
Reclassifications	475	- 475	-
Disposals due to changes in consolidation	- 5	-	- 5
Disposals	- 158	- 3	- 161
Reclassifications to assets held for sale	- 7	-	- 7
Depreciation	- 1,837	-	- 1,837
Reversal of impairment losses	2	-	2
Carrying amount 31.12.2018	15,086	1,690	16,776
Cost as of 31.12.2018	31,208	1,690	32,898
Accumulated amortisation	- 16,122	-	- 16,122

The previous year's figures were adjusted in connection with the changes to reporting regarding the capitalisation of engine maintenance events. For a description of the effects, we refer to [Note 2, p. 106ff](#). The additions in 2018 include EUR 470m for engine maintenance events (previous year: EUR 332m).

Other additions were for the procurement of new and used aircraft. In the previous year, the additions and changes in the group of consolidated companies included aircraft and reserve engines from the first-time consolidation of Brussels Airlines, as well as aircraft that had previously been in service at the Air Berlin group.

The item also includes 72 aircraft with a carrying amount of EUR 2,173m (previous year: 77 aircraft with a carrying amount of EUR 2,498m), which have mostly been sold to and leased back from foreign leasing companies with the aim of obtaining favourable financing conditions. The leasing companies were fully consolidated as structured entities. The Group is entitled to buy the aircraft back at a fixed price and at a given point in time.

In the reporting year, borrowing costs of EUR 22m were capitalised (previous year: EUR 20m). The financing rate used was 1.7% (previous year: 1.7%).

Order commitments for aircraft and reserve engines amount to EUR 13.5bn (previous year: EUR 12.8bn).

In the aircraft item, aircraft with a carrying amount of EUR 2,323m (previous year: EUR 2,660m) serve as collateral for current financing arrangements and aircraft with a carrying amount of EUR 565m (previous year: EUR 437m) were also acquired under finance leases; [Note 20, p. 134ff](#).

19 Property, plant and other equipment

T106 PROPERTY, PLANT AND OTHER EQUIPMENT

	Land and buildings	Technical equipment and machinery	Other equipment, operating and office equipment	Advance payments and plant under construction	Total
in €m					
Cost as of 1.1.2017	2,680	1,263	1,384	120	5,447
Accumulated depreciation	-1,366	-917	-965	-	-3,248
Carrying amount 1.1.2017	1,314	346	419	120	2,199
Currency translation differences	-28	-14	-11	-9	-62
Additions due to changes in consolidation	6	1	2	1	10
Additions	25	31	120	147	323
Reclassifications	40	38	8	-90	-4
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-7	-5	-6	-2	-20
Reclassifications to assets held for sale	-	-	-	-	-
Depreciation	-90	-56	-102	-12	-260
Reversal of impairment losses	-	-	-	-	-
Carrying amount 31.12.2017	1,260	341	430	155	2,186
Cost as of 1.1.2018	2,648	1,247	1,409	167	5,471
Accumulated depreciation	-1,388	-906	-979	-12	-3,285
Carrying amount 1.1.2018	1,260	341	430	155	2,186
Currency translation differences	6	4	5	1	16
Additions due to changes in consolidation	-	-	-	-	-
Additions	38	44	121	94	297
Reclassifications	43	38	33	-118	-4
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-7	-1	-9	-3	-20
Reclassifications to assets held for sale	-	-	-	-	-
Depreciation	-90	-57	-107	-	-254
Reversal of impairment losses	-	-	-	-	-
Carrying amount 31.12.2018	1,250	369	473	129	2,221
Cost as of 31.12.2018	2,717	1,315	1,509	142	5,683
Accumulated depreciation	-1,467	-946	-1,036	-13	-3,462

As in the previous year, charges of EUR 4m exist over land and property. Pre-emption rights are registered for land held at EUR 193m (previous year: EUR 205m). Other property, plant and equipment carried at EUR 4m (previous year: EUR 13m) serves as collateral for existing financing arrangements. Other property, plant and equipment carried at EUR 93m was acquired by means of finance leases (previous year: EUR 102m). ➤ Note 20, p. 134ff.

The following items of property, plant and equipment have been ordered, but are not yet at the Group's economic disposal:

T107 ORDERS OF PROPERTY, PLANT AND EQUIPMENT AS OF THE REPORTING DATE

in €m	31.12.2018	31.12.2017
Land and buildings	61	30
Technical equipment and vehicles	31	35
Operating and office equipment	51	44
	143	109

20 Assets for which the Group is lessor or lessee

Property, plant and equipment also includes leased assets that are deemed to be the property of the Group since the underlying contracts are structured as finance leases. The following table shows leased assets for which the Group is either lessor or lessee:

T108 ASSETS FOR WHICH THE GROUP IS LESSOR OR LESSEE

in €m	Lessee of aircraft and reserve engines	Lessee and lessor of aircraft and reserve engines	Lessee of buildings	Lessee of intangible assets and technical equipment	Lessee of other equipment, operating and office equipment
Cost as of 1.1.2017	695	-	299	1	3
Accumulated depreciation	-406	-	-148	-1	-2
Carrying amount 1.1.2017	289	-	151	-	1
Currency translation differences	-8	-	-3	-	-
Additions due to changes in consolidation	227	-	-	-	-
Additions	61	19	5	-	6
Reclassifications	1	-	-41	-	-
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-15	-	-4	-	-
Reclassifications to assets held for sale	-	-	-	-	-
Depreciation	-118	-1	-12	-	-1
Reversal of impairment losses	-	-	-	-	-
Carrying amount 31.12.2017	437	18	96	-	6
Cost as of 1.1.2018	843	19	248	1	10
Accumulated depreciation	-406	-1	-152	-1	-4
Carrying amount 1.1.2018	437	18	96	-	6
Currency translation differences	3	-	-	-	-
Additions due to changes in consolidation	-	-	-	-	-
Additions	243	-	-	-	-
Reclassifications	2	-	-	-	-
Disposals due to changes in consolidation	-	-	-	-	-
Disposals	-	-	-	-	-1
Reclassifications to assets held for sale	-	-	-	-	-
Depreciation	-121	-4	-6	-	-2
Reversal of impairment losses	-	-	-	-	-
Carrying amount 31.12.2018	564	14	90	-	3
Cost as of 31.12.2018	1,036	19	218	1	9
Accumulated depreciation	-472	-5	-128	-1	-6

The carrying amount of leased aircraft attributed to the Group's economic ownership under IAS 17 is EUR 564m (previous year: EUR 437m) and relates to 26 Airbus A319s, 27 A320s, two A321s, eight A330s and four Boeing 777s.

FINANCE LEASES

As a rule, aircraft finance lease agreements cannot be terminated during a fixed basic lease term of at least four years and they run for a maximum of 23 years (previous year: 20 years).

At the end of the lease term, the lessee sometimes has the right to buy the asset or renew the lease. If the lessee does not exercise this option, the lessor will sell the aircraft at the best possible market price.

In addition, the Group has a variety of finance leases for buildings, fixtures and for operating and office equipment. For buildings and fixtures, the leases run for 19 to 29 years (previous year: 15 to 30 years). In some cases, the leases have purchase or renewal options for the lessee at the end of the lease term. The agreements cannot generally be terminated.

The following lease payments are due for finance leases, whereby the lease payments dependent on reference interest rates have been extrapolated on the basis of the most recent interest rate:

T109 LEASE PAYMENTS DUE UNDER FINANCE LEASES, AS OF 2018

in €m	2019	2020 – 2023	from 2024
Lease payments	126	343	169
Discounted amounts	20	43	17
Present values	106	300	152

In the previous year, the following figures were given for finance leases:

T109 LEASE PAYMENTS DUE UNDER FINANCE LEASES, AS OF 2017

in €m	2018	2019 – 2022	from 2023
Lease payments	131	328	98
Discounted amounts	4	28	20
Present values	127	300	78

OPERATING LEASES

In addition to the finance leases, a large number of leases were signed which, on the basis of their economic parameters, qualify as operating leases, i.e. the leased asset is deemed to belong to the lessor. The leases are for property and land, as well as 66 aircraft (previous year: 62 aircraft).

Operating leases for aircraft have remaining terms of up to ten years (previous year: twelve years). These agreements generally end automatically after the term has expired, but there is sometimes an option to extend the agreement.

T110 LEASE PAYMENTS DUE UNDER OPERATING LEASES, AS OF 2018

in €m	2019	2020 – 2023	from 2024
Aircraft and reserve engines	142	338	15
Land and property	279	853	1,103
Other leases	114	245	–
	535	1,436	1,118
Payments from sub-leasing (Sublease)	14	9	2

Annual payments from 2024 onwards amount to EUR 149m, of which land and buildings account for EUR 144m and aircraft and reserve engines account for EUR 5m. In the previous year, the following figures were given for operating leases:

T110 LEASE PAYMENTS DUE UNDER OPERATING LEASES, AS OF 2017

in €m	2018	2019 – 2022	from 2023
Aircraft and reserve engines	119	338	41 p. a.
Land and property	307	1,026	223 p. a.
Other leases	102	322	69 p. a.
	528	1,686	333 p. a.
Payments from sub-leasing (Sublease)	56	183	24 p. a.

Eight aircraft, twelve reserve engines and other non-current assets that were legally and economically the property of the Group at the end of 2018 have been leased to third parties under non-terminable operating leases. These leases give rise to the following forecast payments:

T111 FORECAST PAYMENTS FROM OPERATING LEASES, AS OF 2018

in €m	2019	2020 – 2023	from 2024
Payments received from operating leases	22	23	8

34 aircraft in the economic ownership of Lufthansa, mostly from the acquisition of aircraft that were in service at companies in the Air Berlin group, 13 reserve engines and other non-current assets were used in the previous year on the basis of non-terminable operating leases.

These leases resulted in the following forecast payments:

T111 FORECAST PAYMENTS FROM OPERATING LEASES, AS OF 2017

in €m	2018	2019 - 2022	ab 2023
Payments received from operating leases	24	47	10

21 Equity investments accounted for using the equity method

T112 EQUITY INVESTMENTS ACCOUNTED FOR USING THE EQUITY METHOD

in €m	Investments in joint ventures	Investments in associated companies	Total
Cost as of 1.1.2017	274	247	521
Accumulated impairment losses	-	-5	-5
Carrying amount 1.1.2017	274	242	516
Currency translation differences	-12	-25	-37
Additions due to changes in consolidation	-	-	-
Additions	55	-	55
Changes with and without an effect on profit and loss	81	41	122
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Dividends paid	-48	-23	-71
Reclassifications to assets held for sale	-	-	-
Impairment losses	-	-	-
Reversal of impairment losses	-	-	-
Carrying amount 31.12.2017	350	235	585
Cost as of 1.1.2018	350	239	589
Accumulated impairment losses	-	-4	-4
Carrying amount 1.1.2018	350	235	585
Currency translation differences	3	1	4
Additions due to changes in consolidation	-	-	-
Additions	32	-	32
Changes with and without an effect on profit and loss	75	37	112
Reclassifications	-	-	-
Disposals due to changes in consolidation	-	-	-
Disposals	-	-	-
Dividends paid	-69	-14	-83
Reclassifications to assets held for sale	-	-	-
Impairment losses	-	-	-
Reversal of impairment losses	-	-	-
Carrying amount 31.12.2018	391	259	650
Cost as of 31.12.2018	391	263	654
Accumulated impairment losses	-	-4	-4

Individual interests in companies accounted for using the equity method

The following tables contain summarised data from the income statements and data from the statement of financial position for the individual material joint ventures accounted for using the equity method.

**T113 BALANCE SHEET DATA
GÜNES EKSPRES HAVACILIK ANONİM ŞİRKETİ
(SUNEXPRESS), ANTALYA, TURKEY**

in €m	31.12.2018	31.12.2017
Current assets	400	449
of which cash and cash equivalents	196	234
Non-current assets	816	844
Current liabilities	404	362
Non-current liabilities	559	719
Current financial liabilities (except trade and other payables and provisions)	114	96
Non-current financial liabilities (except trade and other payables and provisions)	412	477
Shareholders' equity	253	212
Share of equity	127	106
Other	21	21
Carrying amount	148	127

**T114 INCOME STATEMENT DATA
GÜNES EKSPRES HAVACILIK ANONİM ŞİRKETİ
(SUNEXPRESS), ANTALYA, TURKEY**

in €m	2018	2017
Revenue	1,261	1,157
Depreciation and amortisation	53	48
Interest income	6	4
Interest expenses	15	21
Income tax expense or income	34	18
Profit or loss from continuing operations	47	65
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	-6	-7
Total comprehensive income	41	58
Share of profit or loss from continuing operations	24	33
Share of comprehensive income	21	29

The item "Other" in the reconciliation with the carrying amount for SunExpress includes the difference from the first-time consolidation of the company.

T115 BALANCE SHEET DATA**TERMINAL 2 GESELLSCHAFT MBH & CO. OHG,
MUNICH AIRPORT, GERMANY**

in €m	31.12.2018	31.12.2017
Current assets	80	84
of which cash and cash equivalents	4	-
Non-current assets	1,381	1,472
Current liabilities	266	277
Non-current liabilities	1,133	1,210
Current financial liabilities (except trade and other payables and provisions)	122	136
Non-current financial liabilities (except trade and other payables and provisions)	1,103	1,182
Shareholders' equity	62	69
Share of equity	25	28
Other	-	-
Carrying amount	25	28

T116 INCOME STATEMENT DATA**TERMINAL 2 GESELLSCHAFT MBH & CO. OHG,
MUNICH AIRPORT, GERMANY**

in €m	2018	2017
Revenue	357	317
Depreciation and amortisation	84	86
Interest income	-	-
Interest expenses	39	42
Income tax expense or income	10	5
Profit or loss from continuing operations	77	37
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	8	17
Total comprehensive income	85	54
Share of profit or loss from continuing operations	31	15
Share of comprehensive income	34	22
Dividends received	92	55

The following table contains summarised aggregated data from the income statements and carrying amounts for the individual immaterial joint ventures accounted for using the equity method.

**T117 INCOME STATEMENTS DATA AND CARRYING AMOUNTS
OF JOINT VENTURES ACCOUNTED FOR USING
THE EQUITY METHOD**

in €m	2018	2017
Profit or loss from continuing operations	21	31
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	-	-
Total comprehensive income	21	31
Carrying amount	218	195

The following table contains summarised aggregated data from the income statements and carrying amounts for the individual immaterial associated companies accounted for using the equity method.

**T118 INCOME STATEMENTS DATA AND CARRYING AMOUNTS
OF ASSOCIATED COMPANIES ACCOUNTED FOR USING
THE EQUITY METHOD**

in €m	2018	2017
Profit or loss from continuing operations	37	39
Profit or loss after tax from discontinued operations	-	-
Other comprehensive income	-	-
Total comprehensive income	37	39
Carrying amount	259	235

22 Other equity investments and non-current securities

T119 OTHER EQUITY INVESTMENTS AND NON-CURRENT SECURITIES

in €m	31.12.2018	31.12.2017
Investments in affiliated companies	222	204
Investments	24	17
Other investments	246	221
Non-current securities	41	32

Shares in related parties include shares in affiliated companies, joint ventures and associates that are not consolidated for reasons of materiality. These shares are carried at amortised cost. Disclosures on the equity investments and non-current securities can be found in [Note 42, p. 157ff.](#)

In the current financial year, other equity investments held at EUR 8m (previous year: EUR 5m) were sold for a loss of EUR 1m (previous year: profit of EUR 2m).

23 Non-current loans and receivables

T120 NON-CURRENT LOANS AND RECEIVABLES

in €m	31.12.2018	31.12.2017
Loans to and receivables from affiliated companies	108	89
Loans to and receivables from other equity investments	–	–
Other loans and receivables	351	358
Emissions certificates	53	28
	512	475

Non-current loans and receivables are carried at amortised cost.

For the impairment test for emissions certificates, we refer to the disclosures on the cash-generating units (CGU) Lufthansa German Airlines, SWISS, Austrian Airlines and Eurowings in [Note 16, p. 127ff.](#)

Other receivables include expected reimbursements for obligations for which provisions have been made amounting to EUR 2m (previous year: EUR 2m). Of the non-current receivables, EUR 32m (previous year: EUR 32m) serve as collateral for liabilities.

24 Inventories

T121 INVENTORIES

in €m	31.12.2018	31.12.2017
Raw materials, consumables and supplies	813	663
Finished goods and work in progress	150	156
Advance payments	5	88
	968	907

No inventories have been pledged as collateral for loans. EUR 695m of the inventories (previous year: EUR 556m) consist of non-repairable spare parts for aircraft.

The gross value of written-down inventories as of 31 December 2018 was EUR 859m (previous year: EUR 855m). Inventories with a carrying amount of EUR 604m (previous year: EUR 611m) are held at net realisable value. Write-downs to net realisable value of EUR 232m were made at the beginning of the financial year (previous year: EUR 211m). New impairment losses of EUR 36m were recognised in the reporting year (previous year: EUR 40m). Write-downs of EUR 13m made in the previous year were reversed (previous year: EUR 8m).

25 Contract assets

The Lufthansa Group recognised the following contract assets in 2018:

T122 CONTRACT ASSETS

in €m	31.12.2018	1.1.2018
Contract assets from MRO and IT services	234	185
Impairment of contract assets	–	–
Total contract assets	234	185

26 Trade receivables and other receivables

T123 TRADE RECEIVABLES AND OTHER RECEIVABLES

in €m	31.12.2018	31.12.2017
Trade receivables		
Trade receivables from affiliated companies	67	80
Trade receivables from other equity investments	7	4
Trade receivables from third parties	3,938	3,901
	4,012	3,985
Other receivables		
Receivables from affiliated companies	93	58
Receivables from other equity investments	-	1
Other receivables	1,408	1,238
Emissions certificates	63	31
	1,564	1,328
Total	5,576	5,313

For the impairment test for emissions certificates, we refer to the disclosures on the cash-generating units (CGU) Lufthansa German Airlines, SWISS, Austrian Airlines, the Eurowings group and Brussels Airlines in [Note 16, p. 127ff.](#)

Collateral received for trade receivables has a fair value of EUR 2m, as in the previous year. Other receivables include expected reimbursements for obligations for which provisions have been made amounting to EUR 26m (previous year: EUR 11m).

Other receivables include claims of EUR 154m (previous year: EUR 171m) against insurers in connection with the accident involving the Germanwings aircraft on 24 March 2015. As of the reporting date, these receivables are offset by provisions of EUR 139m for outstanding obligations relating to this accident (previous year: EUR 149m).

Other receivables of EUR 34m (previous year: EUR 77m) serve to secure negative market values of derivatives.

27 Deferred charges and prepaid expenses

Deferred charges and prepaid expenses consist of various services paid for in advance for subsequent periods.

28 Current securities

Current securities are fixed income securities, participation certificates, shares and investments in money market funds.

29 Cash and cash equivalents

This item includes EUR 66m (previous year: EUR 179m) in fixed-term deposits with terms of four to twelve months.

Bank balances in foreign currencies are translated at the exchange rate on the balance sheet date.

30 Assets held for sale

Assets with a carrying amount of EUR 9m were held for sale as of year-end 2018. They consist of two aircraft and one building. In the previous year, the carrying amount was EUR 6m and related to one aircraft, buildings and spare parts for aircraft (carrying amount of EUR 2m).

Shareholders' equity and liabilities

31 Issued capital

ISSUED CAPITAL

Deutsche Lufthansa AG's issued capital totals EUR 1,217m. Issued capital is divided into 475,210,729 registered shares, with each share representing EUR 2.56 of issued capital.

AUTHORISED CAPITAL

A resolution passed at the Annual General Meeting on 29 April 2015 authorised the Executive Board until 28 April 2020, subject to approval by the Supervisory Board, to increase the Company's issued capital by up to EUR 561,160,092 by issuing new registered shares on one or more occasions for payment in cash or in kind (Authorised Capital A). In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board. In order to issue new shares to settle dividend entitlements, the Executive Board of Deutsche Lufthansa AG decided, with the approval of the Supervisory Board at the Annual General Meeting on 8 May 2018, to make partial use of the authorisation voted at the Annual General Meeting on 29 April 2015 (Authorised Capital A) and increase the Company's issued capital by EUR 6,130,027.52 by issuing 2,394,542 new registered shares with transfer restrictions and profit entitlement from 1 January 2018. The capital increase was entered in the Commercial Register of Cologne District Court (HRB 2168) on 7 June 2018. As of 31 December 2018, Authorised Capital A amounted to EUR 547,180,702.56.

A resolution passed at the Annual General Meeting on 29 April 2014 authorised the Executive Board until 28 April 2019, subject to approval by the Supervisory Board, to increase the issued capital by up to EUR 29m by issuing new registered shares to employees (Authorised Capital B) for payment in cash. Existing shareholders' subscription rights are excluded. In order to issue new shares to employees of Deutsche Lufthansa AG and its affiliated companies, the Executive Board of Deutsche Lufthansa AG decided, with the approval of the Supervisory Board, to make partial use of the authorisation voted at the Annual General Meeting on 29 April 2014 (Authorised Capital B) and increase the Company's issued capital by EUR 3,984,750.08, excluding shareholders' subscription rights, by issuing 1,556,543 new registered shares with transfer restrictions and profit entitlement from 1 January 2018 for payment in cash. The capital increase was entered in the Commercial Register of Cologne District Court (HRB 2168) on 25 October 2018. As of 31 December 2018, Authorised Capital B amounts to EUR 6,791,923.20.

CONTINGENT CAPITAL

A resolution passed at the Annual General Meeting on 28 April 2016 authorised the Executive Board until 27 April 2021, subject to approval by the Supervisory Board, to issue bearer or registered convertible bonds, bond/warrant packages, profit sharing rights or participating bonds (or combinations of these instruments), on one or more occasions, for a total nominal value of up to EUR 1.5bn, with or without restrictions on maturity. To do so, contingent capital (Contingent Capital II) was created for a contingent capital increase of up to EUR 237,843,840 by issuing up to 92,907,750 new registered shares. The contingent capital increase will only take place insofar as the holders of convertible bonds or warrants from bond/warrant packages decide to exercise their conversion and/or option rights.

AUTHORISATION TO PURCHASE TREASURY SHARES

A resolution passed at the Annual General Meeting held on 29 April 2015 authorised the Executive Board pursuant to Section 71 Paragraph 1 No. 8 Stock Corporation Act (AktG) to purchase treasury shares until 28 April 2020. The authorisation is limited to 10% of current issued capital, which can be purchased on the stock exchange or by a public purchase offer to all shareholders. The authorisation states that the Executive Board can use the shares, in particular, for the purposes defined in the resolution passed at the Annual General Meeting. According to the resolution of the Annual General Meeting held on 28 April 2016, the Executive Board is also authorised to purchase treasury shares by means of derivatives and to conclude corresponding derivative transactions.

In 2018, Deutsche Lufthansa AG bought back 249,185 of its own shares at an average price of EUR 19.40. This represents 0.05% of issued capital.

The shares purchased or created by means of the capital increase were used as follows:

- 1,023,955 shares were transferred to the employees of Deutsche Lufthansa AG and to 35 other affiliated companies and equity investments as part of the profit-sharing for 2017, at a share price of EUR 22.04.
- 755,598 shares were transferred as part of performance-related variable remuneration in 2018 to managers and non-payscale employees of Deutsche Lufthansa AG and to 41 further affiliated companies and equity investments at a price of EUR 17.76.

- 16,891 shares were transferred to Executive Board members at a price of EUR 17.76 as part of the share programme for 2018.
- 3,348 shares were transferred for previous years' programmes (performance-related variable remuneration for 2017 to managers, non-payscale employees and other employees of Deutsche Lufthansa AG and to further affiliated companies and equity investments from profit-sharing for 2016) at a price of EUR 27.12.
- 2,394,542 shares were transferred to shareholders to settle dividend entitlements for 2017.

5,936 shares were resold at a price of EUR 19.91.

On the balance sheet date, treasury shares were no longer held.

CAPITAL MANAGEMENT

The Lufthansa Group continues to aim for a sustainable equity ratio of 25%, in order to ensure long-term financial flexibility and stability as a basis for its growth targets. As of 31 December 2018 and 2017, equity and total assets were as follows:

T124 EQUITY AND LIABILITIES

in €m	31.12.2018	31.12.2017
Shareholders' equity	9,573	9,110
In % of total assets	25.1	25.5
Liabilities	28,640	26,668
In % of total assets	74.9	74.5
Total capital	38,213	35,778

In the financial year 2018, the equity ratio declined by 0.4 percentage points compared with the adjusted figure for the previous year to 25.1%.

Lufthansa's Articles of Association do not stipulate any capital requirements.

32 Reserves

Capital reserves only include the share premium paid on capital increases and a convertible bond that was redeemed in full in previous years. The legal reserve contained in retained earnings is unchanged at EUR 26m; other reserves consist of other retained earnings.

The following table shows changes in other neutral reserves in 2018:

T125 NOTES ON OTHER COMPREHENSIVE INCOME

in €m	2018	2017
Other comprehensive income after income taxes		
Currency translation differences		
Profit/loss for the period	124	- 408
Reclassification adjustments recognised in profit or loss	-	2
Subsequent measurement of financial assets at fair value (with recycling)	- 10	21
Subsequent measurement of financial assets at fair value (without recycling)	5	-
Profit/loss for the period	- 3	139
Reclassification adjustments recognised in profit or loss	- 2	- 118
Subsequent measurement of hedges – cash flow hedge reserve	434	- 546
Subsequent measurement of hedges – costs of hedging	- 90	13
Profit/loss for the period	401	- 457
Reclassification adjustments recognised in profit or loss	- 57	- 31
Transfer to cost of hedged items	-	- 58
Other comprehensive income from investments accounted for using the equity method		
Profit/loss for the period – reclassifiable	- 2	3
Profit/loss for the period – non-reclassifiable	-	-
Transfer to cost of hedged items	-	-
Revaluation of defined-benefit pension plans	- 974	1,186
Revaluation of defined-benefit pension plans within disposal groups	-	-
Other expenses and income recognised directly in equity	1	- 3
Income taxes on items in other comprehensive income	30	1
Other comprehensive income after income taxes	- 482	269

T126 NOTE ON INCOME TAXES RECOGNISED FOR OTHER COMPREHENSIVE INCOME

in €m	2018			2017		
	Amount before income taxes	Tax expenses/ income	Amount after income taxes	Amount before income taxes	Tax expenses/ income	Amount after income taxes
Currency translation differences	124	-	124	- 406	-	- 406
Subsequent measurement of financial assets at fair value (with recycling)	- 10	-	- 10	21	- 1	20
Subsequent measurement of financial assets at fair value (without recycling)	5	-	5	-	-	-
Subsequent measurement of hedges - cash flow hedge reserve	434	- 125	309	- 546	142	- 404
Subsequent measurement of hedges - costs of hedging	- 90	38	- 52	13	- 4	9
Other comprehensive income from investments accounted for using the equity method - reclassifiable	- 2	-	- 2	3	-	3
Revaluation of defined-benefit pension plans	- 974	117	- 857	1,186	- 136	1,050
Other expenses and income recognised directly in equity	1	-	1	- 3	-	- 3
Other comprehensive income	- 512	30	- 482	268	1	269

The overall change in equity is shown in the [Consolidated statement of changes in shareholders' equity](#), p. 104.

33 Pension provisions

The Group's pension obligations comprise both defined-benefit and defined-contribution plans and include both obligations to make current payments and entitlements to future pension payments.

Obligations under defined-benefit pension plans for Group employees related mostly to pension obligations in Germany, Switzerland, Austria and the USA. Various commitments have been made to different groups of employees.

For the employees in Germany and for staff posted abroad by German companies who joined the Company before 1995, the supplementary pension scheme for state employees (VBL) was retained as the Company's pension scheme after Lufthansa's privatisation. Employees who joined after 1994 received a retirement benefit commitment based on an average salary plan, which provided for pension units to be granted annually, with the size of these components depending on the employee's age and salary. In 2003, the VBL pension scheme was changed to the average salary plan applicable to the employees recruited after 1995. Since 2015, the existing domestic retirement benefit commitments to the individual groups of employees have been converted successively into defined contribution schemes.

The Lufthansa collective agreement on benefits for ground staff established a new Company retirement benefit plan in the form of a defined contribution benefit commitment for the ground staff in Germany, in particular those at Deutsche Lufthansa AG, Lufthansa Cargo AG, the Lufthansa Technik group and the LSG group. For employees recruited before 1 January 2016, the entitlements vested up until 31 December 2015 are maintained. For service periods starting from 1 January 2016, employees can reach the same level of benefits by making contributions from their own pocket. For employees recruited from 1 January 2016, the contributions to the new model will be invested on the capital market. When the employee reaches retirement age, the entire account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum and while guaranteeing the contributions that were originally made.

On 17 March 2017, the "Lufthansa Pension Cabin" wage agreement was signed with the trade union UFO for cabin crew at Deutsche Lufthansa AG (DLH). It replaces the agreements on the "Lufthansa Company Pension" for cabin crew and the "Lufthansa Transitional Benefit for Cabin Crew".

For employees recruited up to 5 July 2016, the pension entitlements vested up until 30 June 2016 are maintained. For service periods from 1 July 2016, these employees receive employer contributions to the Company pension scheme depending on their eligible gross salary. An initial

contribution to the transitional benefit scheme was calculated for the staff concerned as of 30 June 2016 on the basis of parameters and valuation methods defined by the collective bargaining partners. This initial transitional benefit contribution will replace all existing claims by the employees concerned under the collective agreement on “Transitional Benefit for Cabin Crew” and will be switched to a contribution commitment with a minimum guaranteed payment. All employees are free to make their own contributions on a voluntary basis. Contributions from both employer and employee, as well as the initial transitional benefit contribution, are invested on the capital markets with a capital guarantee. When the employee reaches retirement age, the available account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum.

On 21 December 2017, the “Lufthansa Pension Cockpit” wage agreement for cockpit staff was signed with the Vereinigung Cockpit pilots’ union. At the same time, a new “Transitional Benefit Cockpit” wage agreement was signed.

For employees recruited before 1 January 2017, the pension entitlements vested up until 31 December 2016 are maintained. For service periods from 1 January 2017, the employees receive employer contributions to the Company pension scheme depending on their eligible gross salary. All employees are free to make their own contributions on a voluntary basis. The capital is invested on capital markets with a capital guarantee and the guaranteed interest rate offered by the life insurance companies (currently 0.9% per annum) as an additional commitment. When the employee reaches retirement age, the available account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum.

In the new Company retirement benefit scheme for ground, cabin and cockpit staff, the obligations from the capital market-oriented components are recognised at the time value of the corresponding assets, insofar as the assets exceed the minimum guaranteed amount. Plan assets and benefit obligations are presented on a net basis. The employer contributions constitute service expense.

Cockpit staff are still additionally entitled to a transitional pension arrangement covering the period from the end of their active in-flight service until the beginning of their statutory/Company pension plans. Benefits depend on the number of years of service and the final salary before retirement (final salary plans). Pension entitlements continue to accrue while transitional benefits are being received.

In line with the revised version of the wage agreement, the old rules continue to apply to transitional benefits for cockpit staff. The collective retirement age for pilots will go up in stages from 58 to 60 by 2021. Rules on transitional benefits were extended to cover cockpit staff at Germanwings who were recruited before 1 January 2005.

Defined-benefit Company pension schemes and transitional pension arrangements for Germany are funded by plan assets, while amounts that have not yet been transferred are covered by pension provisions.

There are no minimum funding requirements in Germany.

In the course of acquiring Swiss International Air Lines AG, pension obligations, mainly statutory obligations, were taken on in Switzerland. The retirement benefits are funded via pension funds known as collective foundations. In addition to retirement benefits, the plans cover invalidity and dependant persons’ benefits. Beneficiaries can choose between an annuity and a lump sum payment. The retirement age for the plans lies between 58 and 63 years. Contributions to the pension funds are made by employers and employees, whereby the Company contributions must be at least equal to the employee contributions defined in the terms of the plan. Contributions are deducted from the qualifying salary according to a sliding scale. If there is a deficit of plan assets, employer and employee contributions can be increased, a lower return can be determined or other steps permissible by law can be taken. The decision is taken by the trustees of the pension fund concerned. The trustees’ strategies for making good a deficit are based on the report by a pension fund expert and must be presented to the regulatory authority. The approval of the authority is not required, however.

The pension obligations for employees of Austrian Airlines AG are mostly on a defined contribution basis and have been outsourced to a pension fund. They consist of retirement, invalidity and dependant persons’ benefits.

Obligations under defined-benefit plans at Austrian Airlines AG relate to former directors and Executive Board members and others already drawing their pensions. Obligations under defined-benefit plans for ground staff are now contribution-free and are determined by converting plan assets into an annuity. There are no defined-benefit plans but only defined contribution pension obligations for active pilots, flight attendants and members of the top management level.

The defined-benefit pension plans at LSG Sky Chefs in the USA are largely closed to new entrants and no further benefits are being granted to beneficiaries still in service. Benefit payments are based on average salary and the years of service acquired before the plan was closed or frozen. The retirement age is 65. Pension payments are funded externally. Retirement benefits have been switched to defined contribution plans.

Other staff abroad are also entitled to minor retirement benefits and in some cases to medical care based mainly on length of service and salary earned. As a rule, benefits are financed by means of external funds.

Contributions for defined contribution retirement benefit commitments came to EUR 470m in 2018 (previous year: EUR 426m).

In the financial year 2004, work began on building up plan assets to fund and safeguard future pension payments. The aim was to fund the pension obligations under existing plans in Germany in full. Contractual trust arrangements (CTAs) in the form of a mutual two-stage trusteeship were set up for this purpose.

The main trustee is Lufthansa Pension Trust e.V., a separate legal entity subject to German regulations. Deutsche Lufthansa AG and the trustees/other trustors agree on contributions and, if such a contribution is determined, make a payment to Lufthansa Pension Trust e.V. Deutsche Lufthansa AG and its subsidiaries Lufthansa Technik AG and Lufthansa Cargo AG are parties to the contractual trust arrangement.

The trust assets have largely been held by a Maltese corporate vehicle since 2007. The Investment Board of Lufthansa Malta Pension Holding decides on the fund's asset allocation. The asset management itself is delegated to fund management companies, who invest the assets in accordance with the general investment principles defined by the Investment Board.

The assets to fund pension obligations in the new Lufthansa Pension Ground, Lufthansa Pension Cabin and Lufthansa Pension Cockpit capital market-based benefits system were transferred to an external trustee, Deutsche Treuinvest Stiftung, as part of a contractual trust arrangement. Capital is invested in what are known as age group funds, whose investment strategy is based on a life cycle model. As employees get older, less and less is invested in asset classes with a higher risk-return profile and a greater percentage in more conservative assets classes. The Company has set up an Investment Committee that is responsible for defining and monitoring the investment strategy, e.g. how the age group funds are composed and how the asset allocation changes over time.

Assets to fund pension obligations for other German subsidiaries have also been invested with Deutsche Treuinvest Stiftung.

EUR 330m was contributed to plan assets for employees in Germany in the reporting year (previous year: EUR 1,824m).

Amounts shown in the statement of financial position for defined-benefit commitments are made up as follows:

1127 DEFINED-BENEFIT RETIREMENT BENEFIT COMMITMENTS

	31.12.2018				31.12.2017			
	Defined-benefit obligations (DBO)	Fair value of plan assets	Effect of asset ceiling	Net carrying amount for defined-benefit obligations	Defined-benefit obligations (DBO)	Fair value of plan assets	Effect of asset ceiling	Net carrying amount for defined-benefit obligations
in €m								
Retirement benefits Germany	15,627	- 11,446	-	4,181	13,449	- 10,075	-	3,374
Transitional benefits Germany	1,391	- 395	-	996	3,130	- 2,126	-	1,004
Switzerland	3,224	- 2,921	-	303	3,251	- 2,867	2	386
Austria	427	- 184	-	243	402	- 193	-	209
USA	339	- 279	-	60	350	- 293	-	57
Other countries	404	- 351	-	53	424	- 363	-	61
Carrying amounts	21,412	- 15,576	-	5,836	21,006	- 15,917	2	5,091
of which pension provisions	-	-	-	5,865	-	-	-	5,116
of which other assets	-	-	-	29	-	-	-	25

Reconciliation between the funding status and the amounts shown in the consolidated balance sheet is as follows:

T128 RECONCILIATION FUNDING STATUS

in €m	2018	2017
Present value of funded pension obligations	20,847	20,444
Plan assets	- 15,576	- 15,917
Funding status (net)	5,271	4,527
Present value of unfunded pension obligations	565	562
Adjustment for asset ceiling	-	2
Carrying amounts	5,836	5,091
of which pension provisions	5,865	5,116
of which other assets	29	25

During the reporting period, the present value of defined-benefit pension obligations changed as follows:

T129 CHANGE IN PRESENT VALUE OF PENSION OBLIGATIONS

in €m	2018	2017
Balance on 1.1.	21,006	21,442
Current service costs	552	671
Interest expenses	388	404
Past service cost/effects of curtailments	- 113	- 551
Effects of settlements	-	-
Revaluations		
Actuarial gains/losses from changes in demographic assumptions	146	6
Actuarial gains/losses from changes in financial assumptions	- 246	- 602
Experience adjustments	- 119	317
Currency translation differences	137	- 358
Changes in the group of consolidated companies	-	120
Plan contributions - employees	124	56
Pension payments	- 463	- 431
Settlement payments	-	- 4
Other ¹⁾ /reclassifications	-	- 64
As of 31.12.	21,412	21,006

¹⁾ The amounts are partly for benefit obligations which were measured in accordance with IAS 19 for the first time.

Actuarial gains/losses from changes in financial assumptions include gains due to the increase in the discount rate for Switzerland compared with the previous year. The significant change that results from the demographic assumptions stems almost exclusively from adjustments to the mortality tables for Germany and Austria. Adjustments to obligations regarding capital market-based pension plans, which are due to changes in exchange rates, are shown in adjustments based on past experience.

The following table provides a detailed reconciliation of changes in the fair value of plan assets:

T130 CHANGE IN FAIR VALUE OF PLAN ASSETS

in €m	2018	2017
Balance on 1.1.	15,917	13,092
Interest income	291	245
Revaluations		
Income from plan assets, without amounts included in interest	- 1,195	910
Currency translation differences	121	- 300
Changes in the group of consolidated companies	-	88
Plan contributions - employers	463	1,950
Plan contributions - employees	124	56
Pension payments	- 126	- 123
Settlement payments	-	- 4
Administrative costs related to obligations	- 3	- 3
Other ¹⁾ /reclassifications	- 16	6
As of 31.12.	15,576	15,917

¹⁾ The amounts are partly for benefit obligations which were measured in accordance with IAS 19 for the first time.

In the financial years 2018 and 2017, pension provisions developed as follows:

T131 PENSION PROVISIONS

in €m	2018	2017
Balance on 1.1.	5,116	8,364
Currency translation differences carried forward	16	- 58
Changes in the group of consolidated companies	-	32
Pensions payments	- 337	- 308
Current service costs	552	671
Interest expenses	388	404
Interest income	- 291	- 245
Effects of amendments incl. curtailments, settlements and administrative costs	- 110	- 548
Revaluations		
Actuarial gains/losses and experience adjustments	- 219	- 279
Income from plan assets, without amounts included in interest	1,195	- 910
Net effect of adjustments for asset ceiling	- 2	2
Plan contributions/reclassifications	- 443	- 2,009
As of 31.12.	5,865	5,116

Expenses and income for defined-benefit plans are made up as follows:

T132 EXPENSES AND INCOME FOR DEFINED-BENEFIT PENSION PLANS

in €m	2018	2017
Current service costs	552	671
Past service cost/effects of curtailments	- 113	- 551
Income from settlements	-	-
Accrued interest on projected pension obligations	388	404
Interest income on plan assets	- 291	- 245
Administrative costs related to obligations	3	3
Balance of expenses and income recognised in the income statement	539	282
Income from plan assets, without amounts included in interest	1,195	- 910
Actuarial gains and losses	- 219	- 279
Net effect of adjustment for asset ceiling	- 2	2
Other comprehensive income	974	- 1,187
	1,513	- 905

Interest expenses on pension provisions and interest income on plan assets are shown in the financial result. Current service expense and past service expense are recognised in staff costs.

A loss of EUR 904m was generated from plan assets in the financial year 2018. This amount is made up of the interest income recognised in the income statement and the revaluation component for plan assets. Total income of EUR 1,155m was recognised in the previous year.

There were no significant effects from the asset ceiling defined in IAS 19.64.

Past service expense incurred in the reporting year is partly due to adjustments by the trustees of the pension funds for Swiss International Air Lines AG to the rates for converting pension contributions to pension entitlements in view of higher life expectancy and partly due to the agreement on future core elements of company retirement and transitional benefits for the cockpit staff of Lufthansa CityLine GmbH.

The main actuarial assumptions used to calculate pension obligations and the corresponding plan assets are shown below:

T133 MAIN ACTUARIAL ASSUMPTIONS FOR GERMAN COMPANIES

in %	31.12.2018	31.12.2017
Interest rate		
Retirement benefits	2.0	2.0
Transitional benefits	2.0	2.0
Salary increase		
Retirement benefits	2.5	2.5
Transitional benefits	2.5	2.5
Pension increase		
Retirement benefits	1.0	1.0
Transitional benefits	1.0	1.0

The updated "Heubeck Actuarial Tables 2018 G" were used in the biometric calculations for the German companies in the Group.

T134 MAIN ACTUARIAL ASSUMPTIONS FOR FOREIGN COMPANIES

in %	31.12.2018	31.12.2017
Interest rates		
Austria	2.0	2.0
Switzerland	1.1	0.7
USA	4.3	3.7
Salary increase		
Austria	1.9	1.8
Switzerland	1.5	1.5
USA	-	-
Pension increase		
Austria	2.0	1.7
Switzerland	0.0	0.0
USA	-	-

The BVG 2015 generation tables are used for the biometric calculations for Switzerland. Country-specific mortality tables are used in the other countries.

The following table shows how the present value of defined-benefit obligations would have been affected by changes in the relevant actuarial assumptions for the main pension plans described above:

T135 CHANGE IN ACTUARIAL ASSUMPTIONS, AS OF 2018

	Effect on the defined-benefit contribution as of 31.12.2018 in €m	Change in %
Present value of the obligation ¹⁾	21,412	-
Interest rate		
Increase by 0.5 percentage points	19,712	- 7.9
Decrease by 0.5 percentage points	23,399	+ 9.3
Salary trend		
Increase by 0.5 percentage points	21,556	+ 0.7
Decrease by 0.5 percentage points	21,283	- 0.6
Pension trend		
Increase by 0.5 percentage points	21,662	+ 1.2
Decrease by 0.5 percentage points	21,187	- 1.1

¹⁾ Present value of the obligation using the assumptions shown in the "Actuarial assumptions" tables.

T135 CHANGE IN ACTUARIAL ASSUMPTIONS, AS OF 2017

	Effect on the defined-benefit contribution as of 31.12.2017 in €m	Change in %
Present value of the obligation ¹⁾	21,006	-
Interest rate		
Increase by 0.5 percentage points	19,289	- 8.2
Decrease by 0.5 percentage points	22,998	+ 9.5
Salary trend		
Increase by 0.5 percentage points	21,158	+ 0.7
Decrease by 0.5 percentage points	20,868	- 0.7
Pension trend		
Increase by 0.5 percentage points	21,274	+ 1.3
Decrease by 0.5 percentage points	20,758	- 1.2

¹⁾ Present value of the obligation using the assumptions shown in the "Actuarial assumptions" tables.

A reduction of 10% in the mortality rates used to calculate the pension obligations increases the life expectancy of the beneficiaries by a given amount depending on their individual ages. It roughly corresponds to an increase of one year in the life expectancy of a male employee who is 55 years old today. A 10% reduction in the mortality rate would therefore increase the present value of the main benefit obligations in Germany and Switzerland by EUR 151m as of 31 December 2018 (previous year: EUR 148m).

The sensitivity analysis examines changes in one assumption and leaves the other assumptions unchanged compared with the original calculation. The effects of any correlation between the individual assumptions are therefore not taken into account.

Plan assets for funded defined-benefit pension obligations consist mainly of fixed-income securities, equities and cash and cash equivalents. They do not include financial instruments issued by companies in the Group nor properties used by Group companies.

Plan assets serve solely to meet the defined-benefit obligations. Funding these benefit obligations with assets provides security for future payments. In some countries, this takes place on the basis of statutory regulations, while in others (Germany, for example), this takes place on a voluntary basis.

The Lufthansa Group aims to completely cover its German pension obligations by means of capital contributions and positive capital market returns in the medium term. Regular annual contributions to the trustee have been made for this purpose to date. Investment performance plays a crucial role in meeting this target.

The investment strategy for the capital market-based pension plans is initially defined by the Company and is regularly reviewed in the course of an allocation study. Where necessary, it is adjusted by the Investment Committee to reflect changes in capital market requirements. This may result in changes to the investment strategy for amounts that have already been invested.

The Lufthansa Group manages and monitors the financial risks that arise from outsourcing the defined-benefit pension obligations. There was no change in the risk management and monitoring processes compared with the previous year. Derivative financial instruments are used, especially to manage foreign exchange risks.

The allocation of the funds to asset classes (e.g. equities) for the defined-benefit plans is carried out on the basis of asset-liability matching studies performed by the Lufthansa

Group. The Asset-Liability Matching (ALM) study is conducted every three years with an external adviser in order to review the funding strategy on a regular basis and to make adjustments as necessary. The results of the study should indicate what combination of investments (annuities, equities, etc.) can be used to cover the long-term pension obligations. Step one of this process is for the actuary to draft a long-term forecast charting how the pension obligations will develop.

In addition to this, target figures are needed for the relative return and relative risk as regards coverage of the obligations. Last but not least, a risk budget must also be defined.

A simulation is used to test all permissible investment allocations for their future compliance with these objectives. Those which do not fulfil the criteria are eliminated. Preference is given to allocations that are return-oriented yet conservative and that have a high probability of achieving the investment target.

The results of the ALM study show whether there will be strategic shifts in the existing allocation. Alternative investments (e.g. property, private equity, infrastructure) are currently being further developed.

Plan assets are made up as follows:

T136 COMPOSITION OF PLAN ASSETS

	31.12.2018				31.12.2017			
	Listed price in an active market in €m	No listed price in an active market in €m	Total		Listed price in an active market in €m	No listed price in an active market in €m	Total	
			in €m	in %			in €m	in %
Equities			4,003	25.9			4,358	27.4
Europe	2,847	–			2,993	–		
Other	1,156	–			1,365	–		
Fixed-income securities			6,716	43.6			5,959	37.5
Government bonds	3,246	–			2,542	2		
Corporate bonds	3,470	–			3,415	–		
Share funds	525	–	525	3.4	593	–	593	3.7
Fixed-income funds	422	–	422	2.7	1,016	–	1,016	6.4
Mixed funds ¹⁾	148	–	148	1.0	161	–	161	1.0
Money market investments	1,443	–	1,443	9.4	1,958	–	1,958	12.3
Property			792	5.1			753	4.7
Direct investments	–	7			302	6		
Indirect investments	638	147			319	126		
Insurance contracts	–	174	174	1.1	–	147	147	0.9
Bank balances	311	203	514	2.4	480	–	480	3.0
Other investments ²⁾	201	638	839	5.4	242	250	492	3.1
Total	14,407	1,169	15,576	100.0	15,386	531	15,917	100.0

¹⁾ Includes equities and interest-bearing securities.

²⁾ Other investments include, in particular, alternative investments such as hedge funds, commodities and private equity funds.

In addition to various actuarial risks such as interest rate risk, life expectancy risk and the risk of salary increases, the pension plans expose the Group primarily to financial risks in connection with plan assets.

The return on plan assets is assumed at the beginning of the period to be the discount rate, which is determined on the basis of investment grade corporate bonds. For the old pension plans, if the actual return on plan assets is less than the discount rates applied, the net obligation from the pension plan goes up. With the new capital market-based pension plans, a gross obligation is recognised for the time value of the corresponding plan assets, taking the minimum guaranteed amount into account.

The share price risk that arises from the proportion of plan assets invested in equities is considered to be reasonable. The risk of default by bond issuers is limited, because investments are only made in investment grade bonds.

The amount of the net obligation under the old pension plans depends to a large extent on the rates of interest, whereby the current low-interest environment results in a relatively high net obligation. If yields on corporate bonds continue to decline, this would lead to a further increase in defined-benefit obligations, which could probably only be partly offset by positive developments in the market values of the corporate bonds held in plan assets.

Based on current knowledge, an estimated EUR 737m is expected to be transferred to pension plans in 2019 (previous year: EUR 674m). The transfers are made up of planned allocations and benefit payments that are not covered by equivalent reimbursements from plan assets. The weighted duration of pension obligations was 17 years as of 31 December 2018 (previous year: 18 years).

Over the next ten years, the following pension payments are forecast for the defined-benefit commitments in existence as of the reporting date:

T137 FORECAST MATURITIES OF UNDISCOUNTED PENSION PAYMENTS, AS OF 2018

in €m	Forecast pension payments 31.12.2018
2019	531
2020	547
2021	571
2022	586
2023	615
2024 – 2028	3,533

T137 FORECAST MATURITIES OF UNDISCOUNTED PENSION PAYMENTS, AS OF 2017

in €m	Forecast pension payments 31.12.2017
2018	514
2019	526
2020	545
2021	559
2022	567
2023 – 2027	3,224

34 Other provisions

Other provisions disclosed in the statement of financial position as non-current and current other provisions are made up as follows:

T138 NON-CURRENT AND CURRENT OTHER PROVISIONS

in €m	31.12.2018			31.12.2017		
	Total	Non-current	Current	Total	Non-current	Current
Obligations under partial retirement contracts	31	14	17	30	14	16
Other staff costs	196	150	46	186	145	41
Obligation to return emissions certificates	63	–	63	31	–	31
Onerous contracts	64	34	30	103	54	49
Environmental restoration	27	24	3	28	25	3
Legal proceedings	85	18	67	119	21	98
Restructuring/severance payments	210	119	91	208	94	114
Fixed-price customer maintenance contracts	–	–	–	148	42	106
Maintenance of operating lease aircraft	281	131	150	292	149	143
Warranties	49	–	49	41	–	41
Other provisions	456	47	409	405	57	348
Total	1,462	537	925	1,591	601	990

Provisions for construction contracts in the MRO business segment are shown under contract liabilities following the introduction of IFRS 15.

Provisions for staff costs mainly relate to staff anniversary bonuses and other current obligations.

A provision for the obligation to submit CO₂ emissions certificates to the relevant authorities is recognised for an amount equivalent to the carrying amount of the capitalised CO₂ certificates. If the obligation is not fully covered by available certificates, the outstanding amount of the provision is measured using the market price of the emissions certificates as of the reporting date.

Expected losses from onerous contracts result from ongoing obligations or other contractual relationships in which performance and consideration are out of balance.

Provisions for environmental restoration are based on surveyors' findings and the assumption that all contamination is removed within ten years without any further legal requirements.

Provisions for ongoing legal proceedings were based on an assessment of the likely outcome of the proceedings.

The provisions for the overhaul of leased aircraft mainly relate to obligations for the maintenance, overhaul and repair of aircraft.

Other provisions of EUR 139m (previous year: EUR 149m) relate to outstanding obligations in connection with the accident involving the Germanwings aircraft on 24 March 2015.

Changes in groups of individual provisions in 2018 were as follows:

T139 CHANGES IN OTHER PROVISIONS 2018

in €m	Obligations under partial retirement contracts	Other staff costs	Obligation to return emissions certificates	Onerous contracts	Environmental restoration	Legal proceedings
As of 1.1.2018	30	186	31	103	28	119
Changes in the group of consolidated companies	-	2	-	-	-	-
Currency translation differences	-	1	-	-	-	-1
Utilisation	-60	-28	-31	-35	-1	-32
Increase/addition	60	35	63	6	-	28
Interest added back	5	2	-	-	-	-
Reversal	-	-2	-	-10	-	-29
Transfers	-4	-	-	-	-	-
As of 31.12.2018	31	196	63	64	27	85

T139 CHANGES IN OTHER PROVISIONS 2018 (continued)

in €m	Restructuring/severance payments	Maintenance of lease aircraft	Warranties	Other provisions	Total
As of 1.1.2018	208	292	41	457	1,495
Changes in the group of consolidated companies	-	-31	-	1	-28
Currency translation differences	-	7	-	-	7
Utilisation	-69	-84	-13	-161	-514
Increase/addition	82	110	25	195	604
Interest added back	-	-	-	-	7
Reversal	-10	-13	-4	-33	-101
Transfers	-1	-	-	-3	-8
As of 31.12.2018	210	281	49	456	1,462

Changes in groups of individual provisions in the previous year were as follows:

T139 CHANGES IN OTHER PROVISIONS 2017

in €m	Obligations under partial retirement contracts	Other staff costs	Obligation to return emissions certificates	Onerous contracts	Environmental restoration	Legal proceedings
As of 1.1.2017	21	167	28	115	28	104
Changes in the group of consolidated companies	-	2	8	-	-	15
Currency translation differences	-	-3	-	-	-	6
Utilisation	-58	-26	-30	-41	-2	-31
Increase/addition	62	47	25	37	2	59
Interest added back	4	-	-	-	-	-
Reversal	-	-2	-	-2	-	-16
Transfers	1	1	-	-6	-	-18
As of 31.12.2017	30	186	31	103	28	119

T139 CHANGES IN OTHER PROVISIONS 2017 (continued)

in €m	Restructuring/severance payments	Fixed-price customer maintenance contracts	Maintenance of lease aircraft	Warranties	Other provisions	Total
As of 1.1.2017	219	167	291	50	379	1,569
Changes in the group of consolidated companies	2	-	61	-	13	101
Currency translation differences	-	-2	-20	-1	-3	-23
Utilisation	-72	-97	-194	-17	-110	-678
Increase/addition	63	88	168	16	135	702
Interest added back	-	-	-	-	-	4
Reversal	-4	-7	-14	-7	-11	-63
Transfers	-	-1	-	-	2	-21
As of 31.12.2017	208	148	292	41	405	1,591

The funding status for provisions for obligations to employees under partial retirement agreements is as follows:

T140 FUNDING STATUS

in €m	2018	2017
Present value of funded obligations under partial retirement agreements	166	167
External plan assets	-156	-162
	10	5
of which other provisions	31	30
of which other assets	21	25

A total of EUR 175m has been transferred to an external trust fund as insolvency insurance for employer's performance arrears as part of partial retirement agreements under which the employee at first works full-time for less pay and then retires early on the same reduced pay. These assets, which fulfil the requirements for plan assets and therefore reduce the gross amount of obligations accordingly, are measured at market value on the balance sheet date.

Obligations under partial retirement agreements were measured in 2018 using an interest rate of 0.34% (previous year: 0.03%).

The following cash outflows are estimated for the non-current portion of the other groups of provisions:

T141 CASH OUTFLOWS FOR NON-CURRENT PROVISIONS, AS OF 2018

in €m	2020	2021	2022	2023 and thereafter
Onerous contracts	21	8	2	3
Environmental restoration	3	3	3	16
Restructuring/severance payments	47	47	19	8
Maintenance of aircraft on leases	80	19	11	24
Other provisions	23	16	7	31

At the end of 2017, the corresponding cash outflows were estimated as follows:

T141 CASH OUTFLOWS FOR NON-CURRENT PROVISIONS, AS OF 2017

in €m	2019	2020	2021	2022 and thereafter
Onerous contracts	25	15	4	10
Environmental restoration	3	3	3	17
Restructuring/severance payments	19	19	40	15
Fixed-price customer maintenance contracts ¹⁾	22	21	-	-
Maintenance of aircraft on leases	73	54	21	11
Other provisions	23	20	9	31

¹⁾ Figures for the current year are shown in contract liabilities in accordance with IFRS 15.

35 Borrowings

Financial liabilities consist of a non-current portion with a residual term of more than one year and a current portion of less than one year, which is shown under current liabilities. The following table shows the total amount of borrowings:

T142 BORROWINGS 31.12.2018

in €m	Total	Non-current	Current
Bonds	1,007	508	499
Liabilities to banks	1,957	1,401	556
Leasing liabilities and other loans	3,721	3,099	622
	6,685	5,008	1,677

T142 BORROWINGS 31.12.2017

in €m	Total	Non-current	Current
Bonds	1,005	1,005	-
Liabilities to banks	2,044	1,881	163
Leasing liabilities and other loans	3,765	3,256	509
	6,814	6,142	672

Collateral was provided for EUR 75m of the liabilities to banks (previous year: EUR 101m).

There were no delays or defaults on payment obligations under these loan agreements in either 2018 or 2017.

Leasing liabilities and other loans relate almost exclusively to finance leases described in [Note 20, p. 134ff.](#), and to aircraft financing arrangements described in [Note 18, p. 132](#).

36 Non-current contract liabilities

T143 NON-CURRENT CONTRACT LIABILITIES

in €m	31.12.2018	1.1.2018
Non-current contract liabilities	22	43
	22	43

Following the introduction of IFRS 15, non-current deferred income from construction contracts previously recognised in other provisions is now presented in non-current contract liabilities.

37 Other non-current financial liabilities

T144 OTHER NON-CURRENT FINANCIAL LIABILITIES

in €m	31.12.2018	31.12.2017
Other non-current financial liabilities	137	243
	137	243

38 Non-current advance payments received, deferred income and other non-financial liabilities

T145 NON-CURRENT ADVANCE PAYMENTS RECEIVED, DEFERRED INCOME AND OTHER NON-FINANCIAL LIABILITIES

in €m	31.12.2018	31.12.2017
Advance payments received	4	4
Deferred income	21	752
Other non-financial liabilities	26	533
	51	1,289

As of 31 December 2017, deferred income included EUR 739m and other non-financial liabilities included EUR 499m in obligations under bonus mile programmes, which as of 2018 are shown under current contract liabilities following the introduction of IFRS 15.

In addition, deferred income includes EUR 8m (previous year: EUR 8m) for grants and subsidies received for capital expenditure, which are realised over the useful life of the assets in the following years.

Other non-financial liabilities include obligations under share-based remuneration agreements for Executive Board members, managers and non-payscale employees. As part of the share-based remuneration agreements, Lufthansa and other consolidated and non-consolidated Group companies participating in the programme offer a 50% discount on employee investment in Lufthansa shares to Executive Board members, managers and non-payscale employees. The option packages granted in 2015, 2016, 2017 and 2018 consist of an outperformance option and a performance option. At the end of the programme, the participants receive a cash payment if the conditions are met.

The outperformance option is linked to the performance of the Lufthansa share compared with a fictitious index composed of European competitors' shares, whereas the performance option is linked to the absolute performance of the Lufthansa share. With the outperformance option, the holder receives a cash payment for each percentage point of outperformance on exercising the option. The cash payment is capped at an outperformance of more than 20%.

The performance option for 2015 results in a cash payment if the share price goes up by more than 29%. This is capped at a share price increase of more than 44%. The performance option for 2016 results in a cash payment if the share price goes up by more than 27%. This is capped at a share price increase of more than 41%. The performance option for 2017 results in a cash payment if the share price goes up by more than 23%; the maximum cash payment will be made if there is a share price increase of 35%. The performance option for 2018 results in a cash payment if the share price goes up by more than 22%. This is capped at a share price increase of more than 33%.

T146 2015, 2016, 2017 AND 2018 OUTPERFORMANCE OPTION PROGRAMMES

	€ per outperformance level	Maximum per tranche in €
Board member	1,000 per percentage point from 1%	20,000
Managers	400 per percentage point from 1%	8,000
Non-payscale staff	200 per 5 percentage points from 1%	1,000

T147 2015, 2016, 2017 AND 2018 PERFORMANCE OPTION PROGRAMMES

	€ per performance level	Maximum per tranche in €
Board member	10,000 + 1,000 per performance unit	20,000
Managers	4,000 + 500 per performance unit	8,000
Non-payscale staff	500 + 100 per performance unit	1,000

The programmes are scheduled to run for four years. All options can be exercised at a fixed time in the final year. The performance and outperformance in all programmes are calculated on the principle of total shareholder return. The shares invested in personally may not be sold until the option is exercised.

Over the financial years 2018 and 2017, the number of options changed as follows:

T148 CHANGE IN NUMBER OF OPTIONS

	2018		2017	
	Number of options / option packages	Cash settlement in € thousands	Number of options / option packages	Cash settlement in € thousands
Outstanding options on 1.1.	19,167	-	18,277	-
Options issued	6,066	-	5,178	-
Expired or unused options	523	-	156	-
Options exercised	4,164	36,868	4,132	36,191
Outstanding options on 31.12.	20,546	-	19,167	-

The performance and the outperformance option of the share programme 2014, which has now expired, each resulted in a payment of EUR 37m (previous year: EUR 36m). A total of 772,489 shares were issued in the reporting year at an average price of EUR 17.76 for the share programme 2018, adding up to a total value of EUR 14m. The 50% discount for employees meant the new share issuance resulted in staff costs of EUR 7m (previous year: EUR 5m). Participants in the programme therefore hold 3,561,026 shares as of the reporting date (previous year: 3,907,293 shares).

The fair values of the option rights in the share programmes still running were calculated using Monte Carlo simulations. This involves simulating the future returns of the shares in the comparative index and of Deutsche Lufthansa AG and calculating the value of the option rights as the forecast amount of a dividend.

The following fair values were measured in total:

T149 FAIR VALUE OF OPTION RIGHTS

	Number of option packages as of 31.12.2018	Fair value per option package in € as of 31.12.2018	Vested benefit as a proportion of the programme term	Fair value in € as of 31.12.2018
Board member				
Options 2015	135	37,052	0.71	3,543,097
Options 2016	135	33,643	0.46	2,081,661
Options 2017	165	12,029	0.29	578,896
Options 2018	165	11,481	0.04	78,932
Managers				
Options 2015	1,869	14,816	0.79	21,922,123
Options 2016	2,247	13,454	0.54	16,375,200
Options 2017	2,333	4,809	0.29	3,272,324
Options 2018	2,622	4,680	0.04	511,290
Non-payscale staff				
Options 2015	2,539	1,850	0.79	3,718,577
Options 2016	2,631	1,681	0.54	2,395,635
Options 2017	2,426	606	0.29	428,796
Options 2018	3,279	566	0.04	77,330
Total	20,546			54,983,861

T149 FAIR VALUE OF OPTION RIGHTS

	Number of option packages as of 31.12.2017	Fair value per option package in € as of 31.12.2017	Vested benefit as a proportion of the programme term	Fair value in € as of 31.12.2017
Board member				
Options 2014	135	39,180	0.75	3,991,463
Options 2015	135	37,671	0.50	2,566,337
Options 2016	135	35,697	0.25	1,227,084
Options 2017	165	18,384	0.04	126,390
Managers				
Options 2014	1,824	15,671	0.79	22,628,924
Options 2015	1,917	15,102	0.54	15,681,539
Options 2016	2,303	14,277	0.29	9,589,980
Options 2017	2,453	7,351	0.04	751,333
Non-payscale staff				
Options 2014	2,205	1,958	0.79	3,417,934
Options 2015	2,554	1,887	0.54	2,610,507
Options 2016	2,646	1,784	0.29	1,376,802
Options 2017	2,695	922	0.04	103,533
Total	19,167			64,071,826

Staff fluctuation of 5.1% is again assumed when accounting for the obligation resulting from the valuation of option rights, so that the recognised provision is less than their calculated time value. The measurement of options therefore results in another non-financial liability of EUR 53m as of the reporting date (previous year: EUR 61m), of which EUR 25m (previous year: EUR 32m) is shown under non-current liabilities. The payment of EUR 37m in the financial year on the basis of option rights that have now expired reduced the previously recognised liability, so that the change in option rights in the reporting year resulted in staff costs of EUR 29m (previous year: EUR 82m).

The weighted average share prices at the calculation date were used in the Monte Carlo simulation. As stated in the terms of the programme, these are 50-day averages for the shares of Deutsche Lufthansa AG and the competitors included in the comparative index. The volatilities and correlations used are forecasts for a specific date and maturity on the basis of current market estimates.

Swap rates were used as the interest rate for the remaining term of the outperformance option in each case. The maximum term of the programmes was used for measurement purposes.

The parameters used by the external service provider are shown in the following table:

T150 REFERENCE PRICE

		Options 2015	Options 2016	Options 2017	Options 2018
Lufthansa	EUR	12.34	10.55	23.00	21.18
Air France-KLM	EUR	6.32	5.00	13.12	8.56
IAG	GBP	574.73	400.94	612.70	646.28
Ryanair	EUR	13.05	12.40	17.25	12.82
easyJet	GBP	1,733.22	1,018.21	1,240.72	1,340.31
Air Berlin	EUR	1.00	0.68	-	-
Norwegian	NOK	-	-	218.45	236.70
WIZZair	GBP	-	-	2,991.52	2,858.46

T151 PROJECTED VOLATILITIES

in % for:	Options 2015 as of 31.12.2018	Options 2015 as of 31.12.2017	Options 2016 as of 31.12.2018	Options 2016 as of 31.12.2017	Options 2017 as of 31.12.2018	Options 2017 as of 31.12.2017	Options 2018 as of 31.12.2018	Options 2018 as of 31.12.2017
Lufthansa	31.88	31.66	29.76	31.30	31.77	31.78	31.46	-
Air France-KLM	38.09	36.78	37.48	36.45	37.22	37.58	36.85	-
IAG	23.35	36.90	24.38	35.04	33.00	34.45	32.51	-
Ryanair	32.79	31.14	29.00	30.89	31.72	30.59	31.41	-
easyJet	30.17	37.15	30.08	34.96	34.97	34.15	33.82	-
Air Berlin	147.57	100.16	140.85	84.39	-	-	-	-
Norwegian	-	-	-	-	51.97	42.08	50.01	-
WIZZair	-	-	-	-	35.27	34.44	35.27	-
Risk-free interest rate	Options 2015: -0.68% for euro zone (previous year: -0.72%) 0.75% for UK (previous year: 0.49%)				Options 2017: -0.54% for euro zone (previous year: -0.46%) 0.79% for UK (previous year: 0.69%) 1.20% for Norway (previous year: 0.89%)			
	Options 2016: -0.64% for euro zone (previous year: -0.60%) 0.74% for UK (previous year: 0.58%)				Options 2018: -0.41% for euro zone 0.85% for UK 1.31% for Norway			
Fluctuation	5.1% (previous year: 5.1%)				5.1% (previous year: 5.1%)			

39 Current contract liabilities

The Lufthansa Group recognised the following contract liabilities:

T152 CONTRACT LIABILITIES

in €m	31.12.2018	1.1.2018
Contract liabilities from unused flight documents	3,969	3,773
Liabilities from customer loyalty programmes	2,186	2,151
Liabilities from MRO and IT services	79	110
Miscellaneous contract liabilities	51	60
Other contract liabilities	2,316	2,321
Liabilities from contracts with customers	6,285	6,094
Revenue recognised in the reporting period		
Revenue recognised that was included in the contract liability balance at the beginning of the period		
Revenue from unused flight documents	3,491	
Revenue from customer loyalty programmes	578	
Revenue from MRO and IT services	69	
Other	56	
Total	4,194	
Revenue recognised from performance obligations satisfied in previous periods		
Revenue from MRO and IT services	-	

Liabilities under customer loyalty programmes as of 31 December 2018 included 225 billion miles from bonus miles programmes (previous year: 220 billion miles).

The unsatisfied performance obligation under existing long-term service contracts came to EUR 6.1bn in total, assuming that the services are performed as agreed, of which EUR 1.2bn relate to the next twelve months. These essentially consist of maintenance contracts in the MRO segment for the long-term maintenance and overhaul of airline sub-fleets. To calculate the outstanding performance obligations, the number of maintenance events derived from the respective flight plans and agreed in the contracts is taken into account, along with the expected revenue and fixed prices for certain services (VIP and cabin modifications). Around 79% of performance obligations beyond twelve months are expected to have been fulfilled by 2024.

In line with the practical expedient of IFRS 15, no disclosures are made for the performance obligations as of 31 December 2018 that have a forecast original term of one year or less. Award miles can be redeemed for at least three years, but may be redeemed at short notice.

The Lufthansa Group applies practical expedient defined in IFRS 15.94, which allows contract initiation costs to be expensed if the amortisation period otherwise to be taken into account would be twelve months or less.

40 Trade payables and other current financial liabilities

T153 TRADE PAYABLES AND OTHER CURRENT FINANCIAL LIABILITIES

in €m	31.12.2018	31.12.2017
Trade payables		
Trade payables to affiliated companies	43	74
Trade payables to other equity investments	-	-
Trade payables to third parties	3,869	3,357
	3,912	3,431
Other liabilities		
Liabilities to banks	39	18
Other liabilities to affiliated companies	269	233
Other liabilities to equity investments	-	-
Liabilities from equity investments	574	591
Other financial liabilities	970	976
	1,852	1,818
Total	5,764	5,249

The carrying amount of these liabilities corresponds to their fair value.

41 Current advance payments received, deferred income and other non-financial liabilities

T154 CURRENT ADVANCE PAYMENTS RECEIVED, DEFERRED INCOME AND OTHER NON-FINANCIAL LIABILITIES

in €m	31.12.2018	31.12.2017
Advance payments received	16	51
Deferred income	39	380
Other non-financial liabilities	333	561
	388	992

Obligations of EUR 290m under bonus mile programmes were recognised in the previous year in current deferred income, and obligations of EUR 242m in other current non-financial liabilities. These amounts are presented in contract liabilities in accordance with IFRS 15.

Other non-financial liabilities include EUR 303m (previous year: EUR 286m) in deferred amounts for outstanding holiday allowance and overtime. Other non-financial liabilities also include the current portion of obligations under share-based remuneration agreements measured at fair value (➤ **Note 38, p. 153ff.**).

NOTES TO THE SEGMENT REPORTING

42 Notes to the reportable segments and segment data

NOTES TO THE REPORTABLE SEGMENTS

As of 31 December 2018, the Lufthansa Group operates in five reporting segments, which make up its Group activities. The segments are defined in line with the internal reporting and management structure.

The airline activities were combined in their respective reporting segments based on the similarity between the economic characteristics of the individual airlines, such as network and sales structures, as well as customers and services. The Network Airlines segment comprises Lufthansa German Airlines, SWISS and Austrian Airlines. Further information about the individual airlines can be found in the Group management report ➤ **starting on p. 12.**

The Eurowings segment comprises Eurowings, Germanwings and Brussels Airlines, as well as the equity investment in SunExpress.

The Logistics segment comprises the scheduled airfreight activities of the Lufthansa Cargo group. Lufthansa Cargo is Europe's leading cargo airline.

The MRO segment is a leading global provider of maintenance, repair and overhaul services for civil and commercial aircraft and is represented by the Lufthansa Technik group.

The Catering segment, represented by the LSG Lufthansa Service/Sky Chefs group, is the global market leader in airline catering.

Business activities not allocated to a reportable segment are presented in the "Additional Businesses and Group Functions" column of the segment reporting along with the income and expenses of central Group functions. They include income and expenses of Lufthansa Commercial Holding GmbH, Lufthansa AirPlus group, the Lufthansa Systems group, the Lufthansa Aviation Training group and other Group companies.

NOTES TO SEGMENT DATA AND INTERNAL MANAGEMENT

The accounting policies of the reportable segments are the same as those described in ➤ **Note 2, p. 106ff.**

The Group measures the performance of its segments using two segment result indicators: EBIT and Adjusted EBIT. EBIT is made up of the IFRS operating result and the result from equity investments. Adjusted EBIT is obtained by correcting EBIT for gains and losses on the disposal of assets and impairment losses and earnings attributable to other periods in connection with pension obligations (plan adjustments and plan settlements).

Sales and revenue between reportable segments are based on arm's length prices. Administrative services are charged as cost allocations.

For information on external traffic revenue ➤ **Note 3, p. 121.**

Capital employed largely comprises segment assets, adjusted for derivative financial instruments, and deferred tax items less non-interest-bearing debt.

The result of the equity valuation for the segment's equity investments is part of its segment result. However, from a Group perspective, it is not attributed to the operating result, but rather to the financial result.

T155 SEGMENT INFORMATION FOR THE 2018 REPORTING SEGMENTS

	Network Airlines	Eurowings	Logistics	MRO	Catering	Total reportable operating segments	Additional Businesses and Group Functions	Reconciliation		Group
in €m								Not allocated	Consoli- dation	
External revenue	21,992	4,212	2,681	3,812	2,499	35,196	648	-	-	35,844
of which traffic revenue	20,877	4,118	2,550	-	-	27,545	-	558	-	28,103
Inter-segment revenue	727	18	32	2,106	718	3,601	344	-	-3,945	-
Total revenue	22,719	4,230	2,713	5,918	3,217	38,797	992	-	-3,945	35,844
Other operating income	699	290	57	264	77	1,387	1,884	-	-987	2,284
Total operating income	23,418	4,520	2,770	6,182	3,294	40,184	2,876	-	-4,932	38,128
Operating expenses	21,024	4,775	2,538	5,764	3,208	37,309	3,109	-	-4,952	35,466
of which cost of materials and services	11,884	3,174	1,753	3,359	1,385	21,555	261	-	-3,147	18,669
of which staff costs	4,159	619	420	1,435	1,218	7,851	1,080	-	-7	8,924
of which depreciation and amortisation	1,497	372	104	124	66	2,163	54	-	-37	2,180
of which other operating expenses	3,484	610	261	846	539	5,740	1,714	-	-1,761	5,693
Results of equity investments	35	24	36	7	29	131	44	-	-1	174
of which result of investments accounted for using the equity method	31	24	25	9	25	114	1	-	-1	114
Adjusted EBIT¹⁾	2,429	-231	268	425	115	3,006	-189	-	19	2,836
Reconciliation items	120	-	-5	1	-5	111	27	-	-	138
Impairment losses/gains	1	-	-4	5	-5	-3	-5	-	-1	-9
Effects from pension provisions	110	-	-	-	-	110	2	-	1	113
Results of disposal of assets	9	-	-1	-4	-	4	30	-	-	34
EBIT	2,549	-231	263	426	110	3,117	-162	-	19	2,974
Other financial result										-190
Profit/loss before income taxes										2,784
Capital employed ²⁾	9,635	2,220	1,430	4,825	1,263	19,373	1,890	-	-290	20,973
of which from investments accounted for using the equity method	25	148	52	284	138	647	6	-	-3	650
Segment capital expenditure	2,573	515	374	241	79	3,782	59	-	-24	3,817
of which from investments accounted for using the equity method	-	-	-	32	-	32	-	-	-	32
Number of employees at end of period	51,778	9,255	4,505	23,219	35,512	124,269	11,265	-	-	135,534
Average number of employees	51,327	9,296	4,422	22,537	35,548	123,130	11,200	-	-	134,330

¹⁾ For reconciliation from Adjusted EBIT to EBIT → T023, p. 32, in the Group management report.

²⁾ The capital employed results from total assets adjusted for non-operating items (deferred taxes, positive market values, derivatives) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

T155 SEGMENT INFORMATION FOR THE 2017 REPORTING SEGMENTS

	Network Airlines	Eurowings	Logistics	MRO	Catering	Total reportable operating segments	Additional Businesses and Group Functions	Reconciliation		Group
in €m								Not allocated	Consoli- dation	
External revenue	22,644	4,045	2,497	3,568	2,556	35,310	269	-	-	35,579
of which traffic revenue	21,538	3,927	2,373	-	-	27,838	-	561	-	28,399
Inter-segment revenue	673	-4	27	1,836	663	3,195	177	-	-3,372	-
Total revenue	23,317	4,041	2,524	5,404	3,219	38,505	446	-	-3,372	35,579
Other operating income	801	281	73	326	55	1,536	2,243	-	-1,191	2,588
Total operating income	24,118	4,322	2,597	5,730	3,274	40,041	2,689	-	-4,563	38,167
Operating expenses	21,841	4,294	2,357	5,345	3,240	37,077	2,839	-	-4,561	35,355
of which cost of materials and services	12,617	3,044	1,575	3,039	1,396	21,671	231	-	-2,874	19,028
of which staff costs	4,203	484	435	1,356	1,220	7,698	1,031	-	-6	8,723
of which depreciation and amortisation	1,479	262	100	115	64	2,020	53	-	-33	2,040
of which other operating expenses	3,542	504	247	835	560	5,688	1,524	-	-1,648	5,564
Results of equity investments	19	32	23	30	32	136	20	-	1	157
of which result of investments accounted for using the equity method	15	32	20	25	26	118	1	-	-1	118
Adjusted EBIT¹⁾	2,296	60	263	415	66	3,100	-130	-	-1	2,969
Reconciliation items	453	-93	-1	-4	-21	334	-3	-	-3	328
Impairment losses/gains	-129	-78	-6	-4	-26	-243	-16	-	-1	-260
Effects from pension provisions	559	-15	4	-	-	548	2	-	1	551
Results of disposal of assets	23	-	1	-	5	29	11	-	-3	37
EBIT	2,749	-33	262	411	45	3,434	-133	-	-4	3,297
Other financial result										-139
Profit/loss before income taxes										3,158
Capital employed ²⁾	9,139	1,996	1,317	4,226	1,219	17,897	2,233	-	-143	19,987
of which from investments accounted for using the equity method	48	131	48	247	129	603	6	-	-24	585
Segment capital expenditure	2,051	972	79	233	84	3,419	62	-	-143	3,338
of which from investments accounted for using the equity method	17	-	-	38	-	55	-	-	-	55
Number of employees at end of period	50,190	7,501	4,511	21,502	34,563	118,267	11,157	-	-	129,424
Average number of employees	49,679	7,190	4,504	21,200	35,122	117,695	11,161	-	-	128,856

¹⁾ For reconciliation from Adjusted EBIT to EBIT ➔ T023, p. 32, in the Group management report.

²⁾ The capital employed results from total assets adjusted for non-operating items (deferred taxes, positive market values, derivatives) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).

The reconciliation column includes both the effects of consolidation activities and the amounts resulting from different definitions of segment item contents compared with the corresponding Group items.

Eliminated segment revenue generated with other consolidated segments is shown in the reconciliation column for revenue.

The amounts in the reconciliation column for Group EBIT include the effects of consolidation procedures on profit or loss in which income and expense do not figure for two companies at the same amount, or in the same period.

The change in accounting for engine overhaul events had the following effects on segment results:

T156 RESTATEMENT EFFECTS OF ENGINE MAINTENANCE BY REPORTABLE SEGMENTS

	Network Airlines	Eurowings	Logistics	Consoli- dation	Group
Increase in depreciation and amortisation 2018	-266	-88	-19	35	-338
Reduction in MRO expense 2018	422	35	22	-489	-10
Increase in capitalised internal expenses 2018	-	-	-	470	470
Effect on Adjusted EBIT¹⁾ 2018	156	-53	3	16	122
Impairment losses/results of disposal	-	-	-	-	-
Effect on EBIT 2018	156	-53	3	16	122
Increase in depreciation and amortisation 2017	-268	-66	-17	30	-321
Reduction in MRO expense 2017	301	32	38	-386	-15
Increase in capitalised internal expenses 2017	-	-	-	332	332
Effect on Adjusted EBIT¹⁾ 2017	33	-34	21	-24	-4
Impairment losses/results of disposal	-4	-8	1	2	-9
Effect on EBIT 2017	29	-42	22	-22	-13

¹⁾ For reconciliation from Adjusted EBIT to EBIT ↗ T023, p. 32, in the Group management report.

NOTES ON GEOGRAPHICAL REGIONS IN 2018

The allocation of traffic revenue to geographic regions is based on the original location of sale. Non-current assets are allocated according to the location of the relevant asset. The allocation of other revenue to the individual regions is based on the geographical location of the customer.

The regions are defined on a geographical basis. As an exception to this rule, traffic revenue generated in Turkey is attributed to Europe.

Lufthansa controls its air traffic operations on the basis of network results and not on the basis of regional earnings contributions. The same applies to the Catering segment. Consequently, the presentation of regional segment results is of no informational value for the Lufthansa Group.

A presentation of traffic revenue generated in the Network Airlines, Eurowings and Logistics segments by traffic region, rather than by original location of sale, is included in the information on the respective segments in the management report.

External revenue, non-current assets and capital expenditure are as follows:

T157 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY REGION FOR 2018

	Europe	North America	Central and South America	Asia/ Pacific	Middle East	Africa	Group
in €m							
Traffic revenue ¹⁾	19,022	4,193	674	3,142	594	478	28,103
Other revenue	3,283	2,272	361	1,383	223	219	7,741
Non-current assets ^{2) 3)}	20,342	276	41	206	2	23	20,890
Capital expenditure on non-current assets ³⁾	3,773	39	5	16	-	5	3,838

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

The figures for the main countries are as follows:

**T158 EXTERNAL REVENUE AND NON-CURRENT ASSETS
BY COUNTRIES FOR 2018**

in €m	Germany	USA
Traffic revenue ¹⁾	8,955	3,759
Other revenue	1,066	1,896
Non-current assets ^{2) 3)}	13,984	261
Capital expenditure on non-current assets ³⁾	2,410	36

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

NOTES ON GEOGRAPHICAL REGIONS IN 2017

External revenue, non-current assets and capital expenditure are as follows:

T157 EXTERNAL REVENUE AND NON-CURRENT ASSETS BY REGION FOR 2017

in €m	Europe	North America	Central and South America	Asia/Pacific	Middle East	Africa	Group
Traffic revenue ¹⁾	19,040	4,338	676	3,174	681	490	28,399
Other revenue	2,888	2,103	312	1,375	294	208	7,180
Non-current assets ^{2) 3)}	18,818	261	42	210	3	20	19,354
Capital expenditure on non-current assets ³⁾	3,384	34	2	9	-	2	3,431

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

The figures for the main countries are as follows:

**T158 EXTERNAL REVENUE AND NON-CURRENT ASSETS
BY COUNTRIES FOR 2017**

in €m	Deutschland	USA
Traffic revenue ¹⁾	8,761	3,889
Other revenue	946	1,766
Non-current assets ^{2) 3)}	12,999	247
Capital expenditure on non-current assets ³⁾	2,501	32

¹⁾ Traffic revenue is allocated according to the original location of sale.

²⁾ Non-current assets include property, plant and equipment and intangible assets with the exception of repairable spare parts for aircraft.

³⁾ Aircraft are allocated according to their location of registration.

SEGMENT REPORTING FROM 2019

Part of the Lufthansa Systems group is managed by the Lufthansa Technik group as of financial year 2019 and so has been allocated to the MRO segment. The companies affected by the change generated revenue of EUR 234m in 2018 and Adjusted EBIT of EUR 20m. The previous year's figures will be adjusted accordingly in the 2019 financial reporting.

In 2018 and in the previous year, no more than 10% of Lufthansa Group revenue was generated with any one customer.

NOTES TO THE CONSOLIDATED CASH FLOW STATEMENT

43 Notes to cash flow from operating, investing and financing activities

The cash flow statement shows how cash and cash equivalents have changed over the reporting period at the Lufthansa Group. In accordance with IAS 7, cash flows are divided into cash inflows and outflows from operating activities, from investing activities and from financing activities. The cash and cash equivalents shown in the cash flow statement comprise the statement of financial position items bank balances and cash-in-hand, without fixed-term deposits with terms of three to twelve months, amounting to EUR 66m (previous year: EUR 179m). The amount of liquidity in the broader sense is reached by adding securities that can be liquidated at short notice.

ADDITIONAL INFORMATION ON THE CASH FLOW STATEMENT

Cash flow from operating activities

Cash flow from operating activities is derived from profit/loss before income taxes using the indirect method. It is adjusted for non-cash income and expenses as well as changes in trade working capital and in the other assets/liabilities position that are not attributable to investing or financing activities.

In the current financial year, the Group primarily recognised the following non-cash income and expenses:

T159 SIGNIFICANT NON-CASH INCOME AND EXPENSES

in €m	2018	2017
Result of miscellaneous financial items	46	-56
Write-down on receivables	99	99
Reversal of write-downs on receivables	-48	-46
Income from the reversal of provisions and accruals	-260	-151
Adjustments to retirement and transitional benefit systems	-113	-551
Total	-276	-705

Trade working capital consists of changes in the carrying amounts of inventories, trade receivables and payables, contract assets and down payments, other current assets and other current liabilities, contract liabilities and current deferrals and prepaid expenses.

Other assets/liabilities mainly include corrections between pensions expenses and payments, changes in other provisions, accruals/deferrals and corrections for non-cash effects from currency translation.

For the effects of the new accounting methods applied retroactively, we refer to [Note 2, p. 106ff.](#)

Cash flow from investing (and cash management) activities

Cash flows from investing and financing activities are calculated on the basis of payments.

Cash flow from investing activities results mainly from investments and disinvestments in non-current assets.

Assets capitalised by the Lufthansa Group that meet the criteria for finance leases are categorised as cash flow from investing activities. In the reporting year, they amounted to EUR 243m (previous year: EUR 91m).

The Group contributed EUR 587m to pension assets in 2018 (previous year: EUR 2,006m). These payments were categorised as cash flow from investing and cash management activities. By contrast, pension payments from fund assets lead to cash inflows from investments (EUR 126m; previous year: EUR 123m). They correspond to cash outflows from operating activities.

For the effects of the new accounting methods for engine overhaul events, we refer to [Note 2, p. 106ff.](#)

Cash flow from financing activities

Cash flow from financing activities includes outflows for the repayment of finance lease liabilities. Inflows of EUR 241m from new borrowing (previous year: EUR 69m) are shown corresponding to cash outflows from investing activities for new finance lease transactions.

Borrowing and the instruments used to hedge it changed as follows in the financial year:

T160 FINANCIAL LIABILITIES

	31.12.2017		Non-cash effective					31.12.2018
		Cash effective	Addition due to changes in consolidation	Currency translation differences	Accrued interest	Reclassification	Changes in fair value	
Non-current borrowings	6,142	491	-	90	13	-1,728	-	5,008
Current borrowings	672	-722	-	4	-5	1,728	-	1,677
Other borrowings	18	22	-	-1	-	-	-	39
Interest rate swaps and currency futures used for hedging – assets	-83	30	-	-	-	-	-101	-154
Interest rate swaps and currency futures used for hedging – liabilities	2	-	-	-	-	-	47	49

Changes in borrowing in the previous year were as follows:

T160 FINANCIAL LIABILITIES

	31.12.2016		Non-cash effective					31.12.2017
		Cash effective	Addition due to changes in consolidation	Currency translation differences	Accrued interest	Reclassification	Changes in fair value	
Non-current borrowings	5,811	1,104	199	-290	-	-682	-	6,142
Current borrowings	764	-921	155	-10	2	682	-	672
Other borrowings	63	-44	-	-1	-	-	-	18
Interest rate swaps and currency futures used for hedging – assets	-98	6	-	-	-	-	9	-83
Interest rate swaps and currency futures used for hedging – liabilities	-	-	-	-	-	-	2	2

OTHER DISCLOSURES

44 Additional disclosures on financial instruments

FINANCIAL ASSETS BY MEASUREMENT CATEGORY

The introduction of IFRS 9 as of 1 January 2018 alters both the classification of financial assets and their measurement,

particularly because of the introduction of the new impairment model based on expected losses. The following overview describes the changes in the measurement categories and the measurement of financial assets when IFRS 9 is applied for the first time. IFRS 9 does not imply any changes for Lufthansa in the classification and measurement of financial liabilities.

T161 IFRS 9 CHANGE – CHANGES IN THE MEASUREMENT CATEGORY AND CARRYING AMOUNTS OF FINANCIAL ASSETS

in €m	IAS 39 Carrying amounts by measurement category as of 31.12.2017					
	Loans and receivables	At fair value through profit or loss	Available for sale	Derivative financial instruments which are an effective part of a hedging relationship	Changes in carrying amounts due to revaluation for the expected loss impairment model	Changes in carrying amount due to change of measurement category
Other equity investments	-	-	17	-	-	-
Non-current securities	-	-	32	-	-	-
of which equity instruments	-	-	32	-	-	-
of which debt instruments	-	-	-	-	-	-
Loans	158	-	-	-	-	-
Non-current receivables	289	-	-	-	-	-
Non-current derivative financial instruments	-	77	-	565	-	-
Trade receivables and other current receivables	5,283	-	-	-	-10	-
Current derivative financial instruments	-	56	-	544	-	-
Current securities	-	-	2,551	-	-	-
of which equity instruments	-	-	391	-	-	-
of which debt instruments	-	-	2,160	-	-	-
Cash and cash equivalents	1,397	-	-	-	-	-
Total	7,127	133	2,600	1,109	-10	-

T161 IFRS 9 CHANGE – CHANGES IN THE MEASUREMENT CATEGORY AND CARRYING AMOUNTS OF FINANCIAL ASSETS (continued)

in €m	IFRS 9 Carrying amounts by measurement category as of 1.1.2018				
	Amortised cost	At fair value through profit or loss	At fair value through other comprehensive income (with recycling)	At fair value through other comprehensive income (without recycling)	Derivative financial instruments which are an effective part of a hedging relationship
Other equity investments	-	17	-	-	-
Non-current securities	-	-	-	32	-
of which equity instruments	-	-	-	32	-
of which debt instruments	-	-	-	-	-
Loans	158	-	-	-	-
Non-current receivables	289	-	-	-	-
Non-current derivative financial instruments	-	77	-	-	565
Trade receivables and other current receivables	5,273	-	-	-	-
Current derivative financial instruments	-	56	-	-	544
Current securities	-	391	2,160	-	-
of which equity instruments	-	391	-	-	-
of which debt instruments	-	-	2,160	-	-
Cash and cash equivalents	1,397	-	-	-	-
Total	7,117	541	2,160	32	1,109

As of the current reporting date, the financial assets can be broken down into measurement categories with the following carrying amounts:

T162 FINANCIAL ASSETS IN THE BALANCE SHEET AS OF 31.12.2018

in €m	Amortised cost	At fair value through profit or loss	At fair value through other comprehensive income (with recycling)	At fair value through other comprehensive income (without recycling)	Derivative financial instruments which are an effective part of a hedging relationship
Other equity investments	-	24	-	-	-
Non-current securities	-	-	-	-	-
of which equity instruments	-	-	-	30	-
of which debt instruments	11	-	-	-	-
Loans	174	-	-	-	-
Non-current receivables	279	-	-	-	-
Non-current derivative financial instruments	-	4	-	-	824
Trade receivables and other current receivables	5,513	-	-	-	-
Current derivative financial instruments	-	23	-	-	334
Current securities	-	-	-	-	-
of which equity instruments	-	280	-	-	-
of which debt instruments	-	-	1,457	-	-
Cash and cash equivalents	1,500	-	-	-	-
Total	7,477	331	1,457	30	1,158

The category "At fair value through other comprehensive income" includes derivatives that do not meet the requirements for applying hedge accounting and so are accounted for as stand-alone derivatives. This category also includes equity instruments, consisting of shares and equity investments, for which the instrument-specific option of fair value through other comprehensive income without recycling has not been exercised. As of the reporting date, the fair-value option without effect on profit and loss and without recycling was chosen for two share positions, in order to avoid the recognition of changes in market value through profit or loss. The first position comprises shares in VISA Inc. (market value: EUR 15m, dividend payments: EUR 0.1m), the second position comprises shares in Alliance Aviation Services Ltd. (market value: EUR 15m, dividend payments: EUR 0.6m). As of the reporting date, both share positions had reserves of EUR 18m. The market valuation reserve for financial assets measured at fair value (with recycling) without effect on profit and loss was EUR 3m as of the reporting date.

FINANCIAL LIABILITIES BY MEASUREMENT CATEGORY

The financial liabilities can be divided into measurement categories with the following carrying amounts, whereby the category "at fair value through profit or loss" includes derivatives that do not meet the requirements for applying hedge accounting and so are accounted for as stand-alone derivatives.

T163 FINANCIAL LIABILITIES IN THE BALANCE SHEET AS OF 31.12.2018

in €m	Liabilities at fair value through profit or loss	Derivative financial instruments which are an effective part of a hedging relationship	Other financial liabilities at cost
Financial liabilities	-	-	6,685
Derivative financial instruments	29	586	-
Trade payables	-	-	3,912
Other financial liabilities	-	-	1,989
Total	29	586	12,586

T163 FINANCIAL LIABILITIES IN THE BALANCE SHEET AS OF 31.12.2017

in €m	Liabilities at fair value through profit or loss	Derivative financial instruments which are an effective part of a hedging relationship	Other financial liabilities at cost
Financial liabilities	-	-	6,814
Derivative financial instruments	123	191	-
Trade payables	-	-	3,431
Other financial liabilities	-	-	2,061
Total	123	191	12,306

The net result of the different categories of financial assets and liabilities is made up as follows:

T164 NET RESULT 2018 FOR FINANCIAL ASSETS AND LIABILITIES BY MEASUREMENT CATEGORY

in €m	Interest expenses	Interest income	Depreciation	Result from valuation and sale	Currency result	Net result
Assets at amortised cost	-	59	43	-	29	131
At fair value through other comprehensive income (with recycling)	-12	22	-	-10	7	7
At fair value through other comprehensive income (without recycling)	-	-	-	-	-	-
Assets at fair value through profit or loss	-	-	-	-32	-	-32
Liabilities at amortised cost	-124	-	-	-	-39	-163
Liabilities at fair value through profit or loss	-	-	-	-18	-	-18
Total	-136	81	43	-60	-3	-75

T164 NET RESULT 2017 FOR FINANCIAL ASSETS AND LIABILITIES BY MEASUREMENT CATEGORY

in €m	Interest expenses	Interest income	Depreciation	Result from valuation and sale	Currency result	Net result
Loans and receivables	-	8	135	-	-135	8
Financial assets held for sale	-	130	-	-	-26	104
Financial assets held for trading	-	-	-	-216	-	-216
Liabilities at continued acquisition cost	-127	-	-	-	510	383
Total	-127	138	135	-216	349	279

The following table shows the carrying amounts and market values for individual classes of financial liabilities. The stated market values of bonds reflect their stock market listings (Level 1 of the fair value hierarchy). The market values for other types of financial liability have been calculated using the applicable interest rates for the remaining term to maturity and repayment structures at the balance sheet date based on the available market information (Reuters) (Level 2 of the fair value hierarchy). For other assets, non-current receivables, trade receivables and cash-in-hand carried at amortised cost, the carrying amount is deemed to be a reasonable approximation of the fair value.

T165 FINANCIAL LIABILITIES

in €m	31.12.2018		31.12.2017	
	Carrying amount	Market value	Carrying amount	Market value
Bonds	1,007	1,026	1,005	1,063
Liabilities to banks	1,957	1,984	2,044	2,113
Leasing liabilities and other loans	3,721	3,664	3,765	3,722
Total	6,685	6,674	6,814	6,898

FINANCIAL ASSETS HELD AT FAIR VALUE BY LEVEL OF FAIR VALUE HIERARCHY

The following table shows financial assets and liabilities held at fair value by level of fair value hierarchy. The levels are defined as follows:

- Level 1: Financial instruments traded on active markets, the quoted prices for which are taken unchanged for the measurement.
- Level 2: Measurement is made by means of valuation methods with parameters derived directly or indirectly from observable market data.
- Level 3: Measurement is made by means of valuation methods with parameters not based exclusively on observable market data.

In the financial year 2018, the fair value hierarchy for assets and liabilities held at fair value was as follows:

T166 FAIR VALUE HIERARCHY OF ASSETS AS OF 31.12.2018

in €m	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit and loss				
Financial derivatives classified as held for trading	-	27	-	27
Securities	278	2	-	280
Total financial assets through profit and loss	278	29	-	307
Derivative financial instruments which are an effective part of a hedging relationship	-	1,158	-	1,158
Financial assets at fair value through other comprehensive income				
Equity instruments	15	15	-	30
Debt instruments	-	1,455	-	1,455
	15	1,470	-	1,485
Total assets	293	2,657	-	2,950

T167 FAIR VALUE HIERARCHY OF LIABILITIES AS OF 31.12.2018

in €m	Level 1	Level 2	Level 3	Total
Derivative financial instruments at fair value through profit or loss	-	-29	-	-29
Derivative financial instruments which are an effective part of a hedging relationship	-	-586	-	-586
Total liabilities	-	-615	-	-615

In the previous year, the fair value hierarchy for assets and liabilities held at fair value was as follows:

T166 FAIR VALUE HIERARCHY OF ASSETS AS OF 31.12.2017

in €m	Level 1	Level 2	Level 3	Total
Financial assets at fair value through profit and loss				
Financial derivatives classified as held for trading	-	131	-	131
Securities	-	-	-	-
Total financial assets through profit and loss	-	131	-	131
Derivative financial instruments which are an effective part of a hedging relationship	-	1,110	-	1,110
Financial assets available for sale				
Equity instruments	410	13	-	423
Debt instruments	-	2,160	-	2,160
	410	2,173	-	2,583
Total assets	410	3,414	-	3,824

T167 FAIR VALUE HIERARCHY OF LIABILITIES AS OF 31.12.2017

in €m	Level 1	Level 2	Level 3	Total
Derivative financial instruments at fair value through profit or loss	-	123	-	123
Derivative financial instruments which are an effective part of a hedging relationship	-	191	-	191
Total liabilities	-	314	-	314

NETTING OF FINANCIAL ASSETS AND LIABILITIES

The following financial assets and liabilities are subject to global netting agreements and other agreements.

T168 NETTING OF FINANCIAL ASSETS AS OF 31.12.2018

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade receivables and other current receivables	5,983	470	5,513	-	33	5,513
Derivative financial instruments – assets	1,185	-	1,185	10	13	1,162
Cash and cash equivalents	1,508	8	1,500	-	-	1,500
Total assets	8,676	478	8,198	10	46	8,175

T169 NETTING OF FINANCIAL LIABILITIES AS OF 31.12.2018

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade payables and other financial liabilities	4,390	478	3,912	-	13	3,912
Derivative financial instruments – liabilities	615	-	615	10	33	572
Total liabilities	5,005	478	4,527	10	46	4,484

In the previous year, the net balances were as follows:

T168 NETTING OF FINANCIAL ASSETS AS OF 31.12.2017

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade receivables and other current receivables	5,434	151	5,283	-	77	5,283
Derivative financial instruments – assets	1,242	-	1,242	72	10	1,160
Cash and cash equivalents	1,422	25	1,397	-	-	1,397
Total assets	8,098	176	7,922	72	87	7,840

T169 NETTING OF FINANCIAL LIABILITIES AS OF 31.12.2017

in €m	Gross amount	Netted amounts	Reported net amount	Amounts not netted	Cash collateral	Net amount
Trade payables and other financial liabilities	3,607	176	3,431	-	10	3,431
Derivative financial instruments – liabilities	314	-	314	72	77	165
Total liabilities	3,921	176	3,745	72	87	3,596

PRINCIPLES OF HEDGING POLICY

As an aviation group with worldwide operations, the Lufthansa Group is exposed to exchange rate, interest rate and fuel price movement risks, as well as to credit and liquidity risks. Limiting these risks by means of systematic financial management is part of company policy.

Market risk

The major market and price risks to which the Lufthansa Group is exposed are exchange rate fluctuations between the euro and other currencies, interest rate fluctuations in international money and capital markets, and price fluctuations in the crude oil and oil products markets. Hedging policy for limiting these risks is laid down by the Executive Board and documented by internal Group guidelines. It also provides for the use of financial derivatives. The corresponding financial transactions are concluded only with first-rate counterparties.

Foreign exchange risk

For US dollars, Lufthansa is mainly in a net payer position as regards currency risks from its operating business, since fuel payments are dollar-denominated. There is always a net surplus for other currencies. The main risks in this respect stem from the Chinese renminbi, the Swiss franc, British pound sterling, the Japanese yen and the Indian rupee. Depending on market liquidity, currency risks from projected operational exposure are hedged gradually over a period of 24 months by means of futures contracts, which are accounted for as

cash flow hedges. The target hedging level is defined in the Group's internal guidelines. At the end of 2018, exposure from operations for the next 24 months was as follows:

T170 CURRENCY EXPOSURE, AS OF 2018

in millions	USD	CNY	JPY	GBP	INR
Exposure (currency)	-9,512	9,906	148,644	1,223	61,881
Exposure (EUR at spot rate)	-8,374	1,256	1,152	1,373	783
Hedges (currency)	3,892	-3,825	-60,508	-482	-13,019
Hedging level	41%	39%	41%	39%	21%
Hedging rate	1.22	8.25	130.19	0.90	85.64

50% of currency risks from capital expenditure on aircraft are hedged when the contract is signed. The hedging level is reviewed and increased, where necessary, if, over the life-time of the contract, the exchange rate goes significantly above or below that used to calculate the investment. In the last 24 months before payment, the hedging level is increased in half-yearly steps of 10%, reaching 90% by the end. These investment hedges are therefore also accounted for as cash flow hedges. Capital expenditure on aircraft takes place in US dollars and is hedged in euros or Swiss francs, depending on the functional currency of the Group company making the purchase.

The US dollar exposure for capital expenditure as of year-end 2018 was as follows, broken down by the hedged currency:

T171 USD INVESTMENT EXPOSURE, HEDGED IN EUR

in millions	2019	2020	2021	2022	2023	2024	2025
Exposure from net capital expenditure (USD)	-1,588	-1,356	-1,860	-2,220	-2,125	-1,436	-578
Exposure from net capital expenditure (EUR at spot rate)	-1,388	-1,186	-1,626	-1,941	-1,857	-1,255	-505
Hedges (USD)	1,417	1,052	1,227	1,790	1,696	1,082	396
Hedging level	89%	78%	66%	81%	80%	75%	68%
Hedging rate EUR/USD	1.26	1.36	1.38	1.44	1.45	1.54	1.47

T172 USD INVESTMENT EXPOSURE, HEDGED IN CHF

in millions	2019	2020	2021	2022	2023	2024	2025
Exposure from net capital expenditure (USD)	-265	-227	-228	-185	-208	-224	-
Exposure from net capital expenditure (EUR at spot rate)	-232	-198	-199	-162	-182	-196	-
Hedges (USD)	259	203	192	135	104	112	-
Hedging level	98%	89%	84%	73%	50%	50%	-
Hedging rate USD/CHF	0.92	0.87	0.85	0.84	0.85	0.82	-

The following sensitivity analysis shows how net profit and equity would change if the currencies identified as price risk variables had been different from those at the balance sheet date.

T173 SENSITIVITY ANALYSIS BY CURRENCY

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Currency – USD		
+10%	-140	1,053
-10%	118	-862
Currency – JPY		
+10%	1	-40
-10%	-1	33
Currency – CHF		
+10%	11	-72
-10%	-12	58
Currency – GBP		
+10%	7	-44
-10%	-9	36
Currency – CNY		
+10%	-1	-39
-10%	-	32
Currency – INR		
+10%	1	-13
-10%	-1	11

¹⁾ All amounts after deferred tax effects; +/- signs relate to net profit and/or equity.

Interest rate risk

The Lufthansa Group aims to pay interest on 100% of its financial liabilities in euros at floating rates of interest. To do so, interest rate swaps are arranged for interest-bearing, fixed-rate financial debt and leasing liabilities. Financial liabilities denominated in euros are hedged using “plain vanilla” interest rate swaps, while cross-currency interest rate swaps are used to hedge financial liabilities in foreign currencies. Depending on the interest rate structure of the exposure to be hedged, hedges are either classified as fair value hedges or cash flow hedges. The interest rate risk is monitored constantly; strategic interest rate hedges are used as needed in response to different market situations. Depending on the counterparties and the instruments used, cash collateral for interest rate swaps is either deposited with or received from counterparties (➔ T168, T169, p. 168).

The tables below describe the floating/fixed ratio for non-current borrowing as of financial year-end 2018 after taking into consideration interest rate hedging, as well as the distribution of the nominal volume of interest rate hedges.

T174 INTEREST RATE EXPOSURE AFTER HEDGING

in €m	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
Fix	505	283	197	164	173	155	104	73	78	-	-
Variable	5,786	4,383	3,784	2,015	1,687	1,194	902	679	425	288	139
Float/Fix-Ratio	92%	94%	95%	92%	91%	89%	90%	90%	84%	100%	100%

T175 NOMINAL VOLUME OF INTEREST RATE HEDGES

in €m	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
Fix	-2,458	-2,319	-1,198	-1,051	-770	-627	-505	-340	-242	-101	-46
Variable	2,392	2,257	1,148	1,037	766	633	530	355	250	101	46

The following sensitivity analysis shows how net profit and equity would change if the interest rate identified as a price risk variable had been different from the perspective of the balance sheet date. In view of the current low interest rates, a reduction of more than 50 basis points is not considered likely, which is why the analysis was limited to this figure.

T176 SENSITIVITY ANALYSIS BY INTEREST RATE

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Interest		
+100 basis points	55	-28
-50 basis points	-29	16

¹⁾ All amounts after deferred tax effects; +/- signs relate to net profit and/or equity.

Fuel price risk

In 2018, fuel costs accounted for 17.2% of the Lufthansa Group's operating expenses (previous year: 15.0%). Significant changes in fuel prices can therefore have a significant effect on the Group's result.

Fuel price risk is limited by the use of crude oil hedges. The hedging level and the time horizon depend on the risk profile, which is derived from the business model of a Group company. As a rule, up to 5% of exposure is hedged monthly for up to 24 months by spread options and other combinations of hedges. Executive Board approval may be obtained to extend the hedging period and to increase the monthly hedging volume in order to exploit market opportunities. The target hedging level is up to 85%.

From a year-end perspective, fuel exposure was as follows:

T177 FUEL EXPOSURE

		2019	2020
Fuel requirement	in 1,000 tonnes	11,145	11,320
Hedges	in 1,000 tonnes	8,117	2,736
Hedging level	%	73	24
Hedging rate	USD/bbl	66.41	70.09

The following sensitivity analysis shows how net profit and equity would change if the fuel price identified as a risk variable had been different from the perspective of the balance sheet date.

T178 SENSITIVITY ANALYSIS BY FUEL PRICE

in €m	Effects on earnings after taxes ¹⁾	Effects on equity ¹⁾
Fuel price		
+10%	-	236
-10%	-	-240

¹⁾ All amounts after deferred tax effects; +/- signs relate to net profit and/or equity.

Market values of the derivative financial instruments used for hedging

At the balance sheet date, exchange rate, interest and fuel price risks are hedged by means of the following hedging transactions:

T179 DERIVATIVE FINANCIAL INSTRUMENTS USED FOR HEDGING AS OF 31.12.2018

in €m	Positive market value	Negative market value	Change in fair value of hedging instrument – designated risk	Change in fair value of hedging instrument – non-designated risk	Basis adjustment of hedged items	OCI – cash flow hedge reserve	OCI – cost of hedging	Ineffective portion of hedges – designated risk	Ineffective portion of hedges – non designated risk
Fair value hedge									
Interest rate hedges – interest rate swaps	101	-47	60	-	-47	-	-	13	-
Cash flow hedge									
Fuel hedging – options	27	-459	-833	-173	-	-840	-174	6	1
Exchange rate hedging – forward transactions	977	-78	463	83	-	462	76	1	6
Interest rate hedges – interest rate swaps	54	-1	19	-	-	15	-	3	-
Total	1,159	-585	-291	-90	-47	-363	-98	23	7
of which current	334	-383							

The market values stated for financial derivatives correspond to the price at which an independent third party would assume the rights and/or obligations from the financial instrument. The fair values of interest rate derivatives correspond to their respective market values, which are measured using appropriate mathematical methods, such as discounting expected future cash flows. Discounting takes market standard interest rates and the residual term of the respective instruments into account. Currency forward transactions and interest rate swaps are individually discounted to the balance sheet date based on their respective forward rates and the appropriate interest rate curve. The market prices of options used to hedge fuel prices are determined using acknowledged option pricing models.

Depending on the hedged exposure, Lufthansa designates interest rate hedges as both fair value hedges and cash flow hedges and accounts for them accordingly. Interest rate swaps are designated as part of a hedging relationship and are not separated or broken down into individual components. Ineffectiveness in these hedging relationships mainly occurs as a result of the subsequent designation of cross currency swaps in hedges as of 1 January 2018. Other reasons for ineffectiveness in hedging relationships are different parameters in the hedged item and the hedging instrument and the basis spread in cross currency swaps. Ineffectiveness in fair value hedges and cash flow hedges are recognised and presented as part of the financial result, below the other financial items.

Derivatives used in the context of fuel hedging to hedge future kerosene purchases are designated as cash flow hedges. Since 1 January 2018, Lufthansa has applied the IFRS 9 component approach, using crude oil, based on Brent Crude ICE, as the designated risk component of the hedging instrument. The hedged item is composed of a global mix of crude oil types. The base risk between individual crude oil components

in the hedging instrument and the crude oil mix in the hedged item is reduced by rebalancing the volumes that make up the hedged item on a quarterly basis. In 2018, the quarterly rebalancing factors for adjusting the hedged item were as follows: 1.016 (Q1), 1.017 (Q2), 1.019 (Q3) and 1.020 (Q4). Lufthansa generally uses options and combinations of options to hedge fuel prices. The intrinsic value of the option is designated as the hedging instrument, so that effective changes in the intrinsic values are recognised in other comprehensive income in the cash flow hedge reserve. The time value of an option is not designated as a hedging instrument and effective changes in the time value are therefore recognised as a cost of hedging. This rule was applied for the first time retroactively. Ineffectiveness in fuel price hedges result from the base risk between the crude oil component and the crude oil mix in the component approach. Ineffectiveness is recognised and presented as part of the financial result, below the other financial items.

Since 1 January 2018, Lufthansa has prospectively applied the spot-to-spot method for exchange rate forward transactions designated as cash flow hedges. The spot component of a forward contract is designated as a hedging instrument and effective value changes are recognised in the cash flow hedge reserve. The other effective components of a forward contract, the forward component and the cross currency basis spread are recognised in a separate OCI component in line with the legal requirements for the cost of hedging. Ineffectiveness in hedging relationships result from changes in the timing of the planned aircraft purchases. Ineffectiveness in hedges is recognised and presented as part of the financial result, below the other financial items.

Lufthansa uses the hypothetical derivative method to calculate changes in the value of hedged items designated as being part of a hedging relationship.

T180 DESIGNATED HEDGED ITEMS IN HEDGING RELATIONSHIPS

in €m	Carrying amount of liabilities	Change in fair value of hedged items – designated risk	Change in fair value of hedged items – non-designated risk	Basis adjustment of hedged items from fair value hedges – cumulative
Fair value hedge				
Interest rate hedges – interest rate swaps	6,685	-47	-	-65
Cash flow hedge				
Fuel hedging – options	-	840	174	-
Exchange rate hedging – forward transactions	-	-502	-61	-
Interest rate hedges – interest rate swaps	-	-16	-	-
Total	6,685	275	113	-65

T181 STATEMENT OF EQUITY RECONCILIATION FOR CASH FLOW HEDGES

in €m	As of 1.1.2018	Gains or losses from effective hedging relationships	Reclassification to profit or loss	Reclassification to acquisition costs of inventories	Reclassification to acquisition costs of aircraft	As of 31.12.2018
OCI – cash flow hedge reserve	837	490	57	786	33	451
Fuel hedging – options	479	-53	-	786	-	-360
Exchange rate hedging – futures	358	552	57	-	33	820
Interest rate hedges – interest rate swaps	-	-9	-	-	-	-9
OCI – cost of hedging	-61	-90	-	-	-	-151
Fuel hedging – options	-61	-166	-	-	-	-227
Exchange rate hedging – futures	-	76	-	-	-	76
Total	776	400	57	786	33	300

Derivative financial instruments that do not meet the requirements for applying hedge accounting are measured at fair value through profit or loss. As a rule, these derivatives were originally in an economic hedging relationship with a particular exposure, but the exposure can either not be measured for hedge accounting purposes or no longer exists.

Fair values are all calculated solely on the basis of recognised financial and mathematical methods, using publicly available market information.

Changes in the market values of derivatives that do not qualify as effective hedging transactions under IFRS 9 can be seen in the income statement and in the overview of other financial items ➤ **Note 13, p. 125.**

Liquidity risk

Complex financial planning systems enable Lufthansa to identify its future liquidity position at an early stage. Based on the results of the Group strategy and planning processes, a monthly rolling liquidity plan differentiated by currency is drawn up with a planning horizon of 24 months. This planning method offers an up-to-date picture of anticipated liquidity developments within the Company and corresponding currency effects.

In principle, the Lufthansa Group holds a liquidity reserve of at least EUR 2,300m that is available at short notice. In addition, the Lufthansa Group held confirmed unused lines of credit as of 31 December 2018 totalling EUR 849m (previous year: EUR 855m).

A maturity analysis for the financial liabilities and the derivative financial instruments based on undiscounted gross cash flows including the relevant interest payments shows the following projected cash inflows and outflows from the perspective of the balance sheet date 31 December 2018. As a result of the hedges used, there are generally direct connections between the cash inflows and outflows for the derivative financial instruments shown.

T182 MATURITY ANALYSIS OF LIABILITIES FROM DERIVATIVE FINANCIAL INSTRUMENTS

in €m	From fuel derivatives	Cash inflow from gross settlement of interest rate and exchange rate derivatives	Cash outflow from gross settlement of interest rate and exchange rate derivatives	Net
1st quarter	51	-939	951	63
Up to 1 year ¹⁾	218	-1,405	1,444	257
1-5 years	100	-1,520	1,517	97
Later	-	-734	708	-26

¹⁾ Without payments in 1st quarter.

T183 MATURITY ANALYSIS FOR NON-DERIVATIVE FINANCIAL INSTRUMENTS

in €m	Outflows
1st quarter	-5,304
Up to 1 year ¹⁾	-3,024
1-5 years	-3,889
Later	-1,662

¹⁾ Without payments in 1st quarter.

Credit risk

The sale of passenger travel and freight documents mostly takes place via agencies. These agencies are mostly connected to national clearing systems for billing passenger and freight sales. The creditworthiness of the agents is reviewed by the responsible clearing systems. Due to the broad diversification, credit risk for the agencies is relatively low worldwide. Nonetheless, credit terms for agents in some markets were tightened significantly in cooperation with the International Air Transport Association (IATA) in order to reduce credit risks even further.

Receivables and liabilities between airlines are offset through bilateral arrangements or via the IATA clearing house, insofar as the contracts underlying services do not explicitly specify otherwise. Systematic settlement of weekly receivables and liability balances significantly reduces the default risk. Fidelity guarantee insurance also covers partial risks within a certain range. Service contracts occasionally require collateral for miscellaneous transactions. All other contractual relationships are subject to credit rules, which, depending on the type and volume of the contract involved, require collateral, credit ratings/references or historical data from prior dealings, particularly payment history, in order to avoid defaults.

Counterparty risks in connection with credit card companies are monitored closely and incoming payments are reviewed daily. To reduce risks even further, a permanent analysis process examines whether to further tighten credit terms for some settlement partners. In addition to the monitoring of receivables at the Company or segment level, there is also counterparty monitoring at Group level, with individually assigned limits, in order to identify the accumulation of portfolio risks across the entire Group and take appropriate action where necessary. The maximum credit risk for financial assets from the potential insolvency of customers is their carrying amount.

In addition to individual write-downs on receivables if a default event occurs, IFRS 9 has since 1 January 2018 required risk provisions to be recognised for expected losses. Lufthansa's trade receivables are exposed to external credit risks for which expected losses have already been taken into consideration in accordance with IFRS 9, in addition to individual write-downs. A simplified impairment model based on an impairment matrix is used for the portion of the receivables portfolio that does not consist of credit card receivables but is subject to external credit risks. The portfolio is divided into clusters based on customer groups, regions and days past due. A default matrix is calculated on the basis of historical default events in Lufthansa's receivables portfolio, which is adjusted for forward-looking, publicly available insolvency forecasts. This impairment matrix is applied to trade receivables that are exposed to external credit risk and are not credit card receivables. In addition, the receivables portfolio includes credit card receivables for which Lufthansa is the credit card issuer. Expected losses for these credit card receivables are calculated in a separate model, based on counterparty-specific external ratings and default probabilities. Lufthansa uses a definition of default of 90 days past due for receivables, which are in principle written off in full if the default event occurs.

The following table shows the reconciliation of risk provisions between IAS 39 and IFRS 9 and the first-time application effect of IFRS 9. The statement of risk provisions that follows it shows changes in risk provisions in the financial year and the related carrying amounts. Impairment losses of EUR 17m were recognised as of 31 December 2017 under IAS 39, of which EUR 11m were trade receivables and EUR 6m were credit card receivables, which were carried over to expected

losses as defined in IFRS 9. The additional need to recognise an impairment of EUR 10m (EUR 8m after tax) required as of 1 January 2018 under IFRS 9 was recognised in equity without effect on profit and loss. In the following tables, Levels 1 and 2 describe expected credit losses, whereas Level 3 shows individual impairment losses on the basis of actual default events.

T184 IFRS 9 CHANGE – RECONCILIATION OF RISK PROVISION FOR FINANCIAL ASSETS

in €m	Risk provision IAS 39 as of 31.12.2017	Risk provision IFRS 9 as of 1.1.2018			
		Level 1	Level 2	Level 3	Simplified approach
Trade receivables and other receivables (simplified approach)	272	-	-	-	276
Trade receivables and other receivables (credit card receivables)	6	12	-	-	-
Total	278	12	-	-	276

T185 STATEMENT OF RISK PROVISIONS

in €m	Opening balance risk provision as of 1.1.2018	Additions through profit or loss	Reversals through profit or loss	Utilisation	Closing balance risk provision as of 31.12.2018	Opening balance gross carrying amount as of 1.1.2018	Closing balance gross carrying amount as of 31.12.2018
Trade receivables and other receivables (simplified approach)	276	95	-31	-15	325	2,371	2,522
of which from expected losses	15	2	-8	-	9	2,095	2,216
of which from individual loss allowances	261	93	-23	-15	316	276	306
Trade receivables and other receivables (credit card receivables)	12	19	-2	-4	24	906	1,048
of which Level 1	12	7	-2	-	17	899	1,037
of which Level 2	-	-	-	-	-	-	-
of which Level 3	-	12	-	-4	7	7	11
Total	288	114	-33	-19	349	3,277	3,570

In addition, a risk provision of EUR 0.3m was recognised as of the reporting date in Level 1 of the general impairment model for securities measured fair value through other comprehensive income. In the reporting year, the Lufthansa Group used the following default rates for each past due category in the impairment matrix for the simplified approach of the impairment model.

T186 IMPAIRMENT MATRIX FOR TRADE RECEIVABLES

		Not overdue	1 – 30 days overdue	31 – 60 days overdue	61 – 90 days overdue	More than 90 days overdue	Total
Default rate	%	0.2	0.5	1.2	1.5	1.7	-
Carrying amounts for trade receivables	€m	1,484	364	163	30	175	2,216
Expected loss	€m	2	2	2	-	3	9

In order to determine expected losses of credit card receivables according to IFRS 9, the off-balance sheet exposure must be considered, in addition to the on-balance sheet exposure. The off-balance sheet exposure describes the portion of a credit card's unused or free limit. The following overview shows the risk data for the credit card portfolio. Expected losses are calculated at the level of the individual credit card, so that the sizes of the average default probability and the average loss relate to the individual credit card.

T187 CONCENTRATION OF CREDIT RISK FROM CREDIT CARD RECEIVABLES

Internal credit rating	Probability of default according to external credit rating	Average probability of default	Average expected loss in €m	Exposure in €m Level 1 impairment model	Exposure in €m Level 2 impairment model	Exposure in €m Level 3 impairment model
On-balance sheet exposure						
Low risk	≤ 2.0%	0.7%	13	591	-	-
Medium risk	> 2.0% to ≤ 6.5%	2.5%	41	432	-	-
High risk	> 6.5%	11.9%	154	14	-	11
Total				1,037	-	11
Off-balance sheet exposure						
Low risk	≤ 2.0%	0.7%	18	665	-	-
Medium risk	> 2.0% to ≤ 6.5%	2.5%	64	763	-	-
High risk	> 6.5%	11.9%	274	21	-	-
Total				1,449	-	-

Securities representing debt instruments are rated as follows (Standard & Poor's):

T188 SECURITIES RATINGS – DEBT INSTRUMENTS

in €m	
AAA	306
AA+	37
AA	114
AA-	123
A+	265
A	222
A-	198
BBB+	101
BBB	66
Below BBB or unrated	36
Total	1,468

The credit risk for derivative financial instruments and securities held at fair value through or without effect on profit and loss is the risk that a counterparty defaults. The maximum credit risk from these instruments is their carrying amount. The counterparty default risk for financial market transactions is limited by defining a maximum risk, taking the credit score given by recognised rating agencies into account.

45 Contingencies and events after the balance sheet date

T189 CONTINGENT LIABILITIES

in €m	31.12.2018	31.12.2017
From guarantees, bills of exchange and cheque guarantees	988	881
From warranty contracts	218	354
From providing collateral for third-party liabilities	45	39
	1,251	1,274

Warranties include an amount of EUR 114m (previous year: EUR 267m) relating to contingent liabilities towards creditors of joint ventures. Liabilities under collateral agreements included contingent liabilities of EUR 6m (previous year: EUR 4m) towards creditors of joint ventures. Of the total, EUR 1,094m (previous year: EUR 1,137m) relates to joint and several guarantees and warranties taken on. These are matched by compensatory claims against the other co-debtors amounting to EUR 1,042m (previous year: EUR 1,048m). Insofar as annual financial statements have yet to be published, these figures are preliminary.

Otherwise, provisions for other risks could not be made because utilisation was not sufficiently probable. The potential financial effect of these provisions on the result would have been EUR 55m (previous year: EUR 80m).

As in the previous year, no profits or cash inflows are expected from contracts for the sale of aircraft.

LEGAL RISKS

The Group is exposed to a number of legal risks in the course of its normal business. Based on current knowledge, the assumption is that these will not have any major, lasting effects on the net assets, financial and earnings position, beyond those for which provisions for litigation risks have been made see [➤ Note 34, p. 149ff.](#)

Legal disputes and other claims made against the Group are always subject to uncertainty, however. Management estimates of these risks may also change over time. The actual outcome of these legal disputes may differ from earlier management estimates, which could have significant effects on the net assets, financial and earnings position and the reputation of our Company.

Due to the existing uncertainties and to those described below, we cannot make an assessment of the amount of the respective contingent liabilities or of the group of contingent liabilities. The legal disputes that these statements refer to include:

Risk of successful claims for damages in ongoing antitrust proceedings

Various cargo airlines, including Lufthansa Cargo AG and Swiss International Air Lines AG, were involved in a cargo cartel in the period between December 1999 and February 2006. Deutsche Lufthansa AG, Lufthansa Cargo AG and

Swiss International Air Lines AG are at risk of civil claims for damages in Germany, the United Kingdom, Norway, Israel, South Korea and the Netherlands. The lawsuits have been brought by both direct and indirect customers and are addressed to the airlines as co-debtors.

In Germany, a lawsuit against Lufthansa Cargo AG and others for information and damages was filed with Cologne regional court by a subsidiary of Deutsche Bahn AG. Litigation proceedings were started in late 2013 and expanded in late 2014. The lawsuit is addressed to a total of eleven cargo airlines and claims for purported damages of around EUR 3bn in total, including interest. Most recently, the respondents filed extensive separate replies to the extended suit on 30 November 2016. Many of the respondent airlines have since reached a settlement with the claimant. As a result, the lawsuit was withdrawn for the portion of revenue supposedly attributable to these airlines, so the disputed amount is now approximately EUR 1.6bn including interest.

At present, it is not possible to give a concrete assessment of the outcome of the lawsuits already pending and of the number and amount of any other claims. When evaluating the risk, it should nonetheless be borne in mind that the European Commission's decision on the cargo cartel, which the claimants in the civil lawsuits refer to, among others, is still not legally binding. Following the appeal of this 2010 decision by the European Court of Justice (ECJ) in December 2015, the European Commission sent revised penalty notices in March 2017 in which the content was the same but the reasoning had been altered. The airlines concerned, including Lufthansa, again contested them, so the penalty notices are still not effective.

Moreover, an expert economic opinion commissioned by Lufthansa Cargo AG and Swiss International Air Lines AG has come to the conclusion that the cartel did not inflict any actual damage on customers. Even if there were damages (i.e. allegedly higher cartel prices), the court will have to examine whether the claimants did not pass them on to their own customers (in the case of the freight forwarders) or whether they were indeed passed on to them (in the case of the final customers). Based on current assessments, there is nonetheless a slight probability of losing some of these lawsuits, which could have not insignificant effects on the Group's net assets, financial and earnings position.

Investigations in connection with work and service contracts

The investigations by the customs authorities in the previous year into possible breaches of the German Law on Labour Leasing (Arbeitnehmerüberlassungsgesetz – AÜG) concerning the procurement of services by the Lufthansa Group have been settled amicably. No penalties were imposed on individual Lufthansa employees nor on companies in accordance with Section 30 of the Administrative Offences Act (Gesetz über Ordnungswidrigkeiten – OWiG). Administrative proceedings with the German Statutory Pension Insurance Scheme concerning the legal status of two workers have been submitted for judicial review to the Social Court in Berlin.

TAX RISKS

Tax risks exist largely because of differences in legal opinions between the German tax authorities and the Company. In tax audits for the financial years 2001 to 2012, the tax authorities came to a number of different conclusions to those on which the Company had based its tax returns, relating, in particular, to partial write-downs on shareholder loans, the treatment of various lease structures and the acquisition of a foreign subsidiary, as well as the recognition and measurement of certain provisions and assets. The Lufthansa Group has appealed against the resulting tax assessments. Some of the disputed claims by the tax authorities were paid in the previous year, without prejudicing the appeal. The Lufthansa Group still assumes, however, that there is a very strong likelihood of winning in all matters being disputed. Provisions have therefore not been made for the outstanding points as of the reporting date. There is, however, the risk of a possible subsequent payment totalling some EUR 500m for the circumstances mentioned. The assessment of the amount is subject to uncertainty. If the legal position of Deutsche Lufthansa AG is upheld, no adverse effects on the earnings, assets and financial position are to be expected.

EVENTS AFTER THE BALANCE SHEET DATE

Since 31 December 2018, no events of particular importance have occurred that would be expected to have a significant influence on the net assets, financial and earnings position.

46 Other financial obligations

As of 31 December 2018, there were order commitments of EUR 13.9bn for capital expenditure on property, plant and equipment, including repairable spare parts, and for intangible assets (previous year: EUR 12.9bn). There were also capital and shareholder loan commitments of EUR 336m towards equity investments (previous year: EUR 475m).

47 Auditors' fees

The fees paid to the auditors in the financial year and charged to expenses in accordance with Section 314 Paragraph 1 No. 9 HGB are made up as follows:

T190 AUDITORS' FEES

in €m	2018	2017
Annual audit	4.2	4.1
Other assurance services	0.4	0.3
Tax advisory services	0.6	0.7
Other services	0.3	0.4
Total	5.5	5.5

Auditing services include above all the fees for the statutory audit of the separate and consolidated financial statements of Deutsche Lufthansa AG and its consolidated subsidiaries, fees for the auditors' review of the interim financial statements and for audit advice in connection with the implementation of new accounting standards and the audit of information systems and processes. The fees shown under other advisory services relate in particular to services in connection with statutory and contractual requirements. Tax advisory services mainly comprise tax advice on transfer pricing matters, international taxes and inspections by tax authorities. Other services particularly relate to advice on the use of biometric identification methods and IT consultancy services.

The following fees paid to the global PricewaterhouseCoopers group, especially abroad, were additionally recognised as expenses:

T191 ADDITIONAL AUDITORS' FEES

in €m	2018	2017
Annual audit	3.9	3.6
Other assurance services	0.2	0.2
Tax advisory services	0.4	0.4
Other services	0.2	0.5
Total	4.7	4.7

The auditor at PricewaterhouseCoopers GmbH Wirtschaftsprüfungsgesellschaft responsible for the Lufthansa Group is Eckhard Sprinkmeier. He held this position for the first time in financial year 2018.

COMPOSITION OF THE GROUP

48 Group of consolidated companies

The consolidated financial statements of Deutsche Lufthansa AG include all major subsidiaries, joint ventures and associated companies.

Subsidiaries are entities over which Deutsche Lufthansa AG has rights that give it the ability to control the entity's relevant activities. Relevant activities are those activities that have a significant influence on the entity's return. Deutsche Lufthansa AG therefore only has control over a company when it is exposed to variable returns from the company and its power over the company's relevant activities enables it to influence these returns. This definition of control also applies to structured entities that are identified as such in the list of significant Group companies. In general, the ability to control subsidiaries arises when Deutsche Lufthansa AG holds a direct or indirect majority of voting shares. In structured entities, the ability to control does not come from holding the majority of voting shares, but rather from contractual agreements. Entities are consolidated from the time that the ability to control begins. They cease to be consolidated when the ability to control ends.

Joint arrangements are classified either as joint ventures or as joint operations. A joint arrangement exists for when the Lufthansa Group carries on joint business activities with third parties on the basis of a contractual agreement. Joint management or control only exists when decisions on activities that have a significant effect on the returns from an agreement require the unanimous approval of the parties sharing control.

Significant interests in companies that are managed jointly with one or more partners (joint ventures) are accounted for using the equity method. Joint operations are defined by the fact that the parties exercising joint control over the arrangement have rights to the assets attributed to the arrangement and are liable for its debts. Assets and liabilities, revenue and expenses from the significant joint operations are recognised in the consolidated financial statements of the Lufthansa Group in proportion to these rights and obligations.

Associated companies are companies in which Deutsche Lufthansa AG has the opportunity to exercise major influence over financial and operating policy based on an interest of between 20% and 50%. Significant associated companies are accounted for in the consolidated financial statements using the equity method.

A list of major subsidiaries, joint arrangements and associated companies can be found in [T198–T201](#), p. 194 – 201, and the list of shareholdings in [T202](#), p. 202 – 205.

In addition to Deutsche Lufthansa AG as the parent company, the group of consolidated companies includes 66 domestic and 256 foreign companies, including structured entities (previous year: 63 domestic and 268 foreign companies).

One material joint operation was also included in the consolidated financial statements on a pro rata basis in accordance with IFRS 11. It consists of a German cargo airline operated jointly by Deutsche Post AG and Deutsche Lufthansa AG, which each hold 50% of the share capital and voting rights. The two shareholders are also customers of the company and use the capacities of its cargo aircraft. In contrast to its capital and voting rights, the company's assets and liabilities, as well as its income and expenses, are allocated based on the user relationship of the shareholders according to their contracts.

Changes in the group of consolidated companies during the 2018 financial year are shown in the following table:

T192 CHANGES IN THE GROUP OF CONSOLIDATED COMPANIES FROM 1.1.2018 TO 31.12.2018

Name, registered office	Additions	Disposals	Reasons
Network Airlines			
Lufthansa Malta Holding Ltd., St. Julians, Malta		1.1.2018	Merger
All Nippon Airways Trading Sky Leasing Ltd., Tokyo, Japan		13.4.2018	Liquidation
Canary Ltd., Tokyo, Japan		13.4.2018	Liquidation
TLC Saffron Ltd., Tokyo, Japan		13.4.2018	Liquidation
Lark Ltd., Tokyo, Japan		20.4.2018	Liquidation
Matterhorn Leasing Co. Ltd., Tokyo, Japan		20.4.2018	Liquidation
SL Opal Ltd., Tokyo, Japan		20.4.2018	Liquidation
Warbler Leasing Ltd., Tokyo, Japan		20.4.2018	Liquidation
Lufthansa Leasing Austria GmbH & Co. OG Nr. 23, Salzburg, Austria		18.7.2018	Merger
Tyrolean Airways Luftfahrzeuge Technik GmbH, Innsbruck, Austria		31.8.2018	Merger
First Valley Highway Kumiai, Tokyo, Japan		28.9.2018	Liquidation
Second Valley Highway Kumiai, Tokyo, Japan		28.9.2018	Liquidation
Third Valley Highway Kumiai, Tokyo, Japan		28.9.2018	Liquidation
Lily Port Leasing Co. Ltd., Tokyo, Japan		17.12.2018	Liquidation
SL Crane Ltd., Tokyo, Japan		17.12.2018	Liquidation
Stork Ltd., Tokyo, Japan		17.12.2018	Liquidation
Celine Leasing Co., Ltd., Tokyo, Japan	18.12.2018		Established
Yamasa Aircraft LH14 Kumiai, Okoyama, Japan	20.12.2018		Established
A319 LDD-LDE-LDF Ltd., George Town, Grand Cayman, Caiman islands		27.12.2018	Liquidation
Eurowings			
Luftfahrtgesellschaft Walter mit beschränkter Haftung, Dortmund	10.1.2018		Acquisition
Eurowings Digital GmbH, Cologne	1.10.2018		Beginning of operations
MRO segment			
Lufthansa Technik Airmotive Ireland Ltd., Dublin, Ireland		14.8.2018	Disposal
Catering segment			
LSG Transalpino SAS, Paris, France	1.1.2018		Beginning of operations
Retail In Motion Mexico S. de R.L. de C.V., Mexico City, Mexico	1.2.2018		Beginning of operations
Retail Inmotion Middle East L.L.C., Abu Dhabi, United Arab Emirates	25.10.2018		Established
LSG Sky Chefs Lounge, Inc., Wilmington, USA	1.11.2018		Established
LSG Sky Chefs RPC West GmbH, Neu-Isenburg	1.11.2018		Established
ALMEIDA & CADIMA, LDA, Lisboa, Portugal	21.11.2018		Acquisition
Additional Businesses and Group Functions			
Lufthansa SICAV-FIS-Fonds, Saint-Josse-ten-Noode, Belgium		16.3.2018	Liquidation

ACQUISITION OF SUBSIDIARIES

With effect from 9 January 2018, Lufthansa Commercial Holding GmbH acquired all of the shares of Luftfahrtgesellschaft Walter mbH. The acquisition is based on the purchase agreement signed by the Lufthansa Group and the Air Berlin group on 13 October 2017. The purchase price is EUR 24m. Within the Eurowings segment, the company acts as a platform with its own air operator certificate and provides services to Eurowings on the basis of wet-lease agreements for 20 Bombardier DH-8 Q400s and ten Airbus A320/A319s at present. The company operates without its own fleet and solely within Eurowings. At the time of initial consolidation, it had gross assets of EUR 19m and net assets of EUR 1m. The difference of EUR 23m resulting from the purchase price

allocation was classified in full as goodwill, given the peculiarities of the acquired business operations, and assigned to Eurowings. In the 2018 financial year, the company generated revenue of EUR 117m and a net profit for the year of EUR 5m. Since the company owns no aircraft and only provides services to Group companies, the effects of its consolidation on the Group's net assets, financial and earnings position are immaterial.

Another acquisition was made in the Catering segment, whose consolidation, taken individually and in aggregate, had no material effect on the net assets, financial and earnings position of the Group and the segment.

USE OF EXEMPTION PROVISIONS

The following fully consolidated German Group companies made use of the exemption provisions of Section 264 Paragraph 3 and Section 264b HGB in 2018.

T193 USE OF EXEMPTION PROVISIONS

Company name	Registered office
Eurowings Aviation GmbH	Cologne
Eurowings Digital GmbH	Cologne
Eurowings GmbH	Düsseldorf
Eurowings Technik GmbH	Cologne
Evertaste GmbH	Alzey
Germanwings GmbH	Cologne
Hamburger Gesellschaft für Flughafenanlagen mbH	Hamburg
Jettainer GmbH	Raunheim
LSG Asia GmbH	Neu-Isenburg
LSG Lufthansa Service Catering- und Dienstleistungsgesellschaft mbH	Neu-Isenburg
LSG Lufthansa Service Europa/Afrika GmbH	Neu-Isenburg
LSG Lufthansa Service Holding AG	Neu-Isenburg
LSG Sky Chefs Düsseldorf GmbH	Neu-Isenburg
LSG Sky Chefs Europe GmbH	Neu-Isenburg
LSG Sky Chefs Frankfurt International GmbH	Neu-Isenburg
LSG Sky Chefs Frankfurt ZD GmbH	Neu-Isenburg
LSG Sky Chefs Lounge GmbH	Neu-Isenburg
LSG Sky Chefs Objekt- und Verwaltungsgesellschaft mbH	Neu-Isenburg
LSG Sky Chefs Verwaltungsgesellschaft mbH	Neu-Isenburg
LSG South America GmbH	Neu-Isenburg
LSG-Food & Nonfood Handel GmbH	Neu-Isenburg
LSY GmbH	Norderstedt
Lufthansa AirPlus Servicekarten GmbH	Neu-Isenburg
Lufthansa Asset Management GmbH	Frankfurt/Main
Lufthansa Asset Management Leasing GmbH	Frankfurt/Main
Lufthansa Aviation Training Berlin GmbH	Berlin
Lufthansa Aviation Training Germany GmbH	Frankfurt/Main
Lufthansa Aviation Training GmbH	Hallbergmoos
Lufthansa Cargo AG	Frankfurt/Main
Lufthansa CityLine GmbH	Munich
Lufthansa Commercial Holding GmbH	Cologne
Lufthansa Global Business Services GmbH	Frankfurt/Main
Lufthansa Industry Solutions AS GmbH	Norderstedt
Lufthansa Industry Solutions BS GmbH	Raunheim
Lufthansa Industry Solutions GmbH & Co. KG	Norderstedt
Lufthansa Process Management GmbH	Neu-Isenburg
Lufthansa Seeheim GmbH	Seeheim-Jugenheim
Lufthansa Systems GmbH & Co. KG	Raunheim
Lufthansa Technik AERO Alzey GmbH	Alzey
Lufthansa Technik AG	Hamburg
Lufthansa Technik Immobilien- und Verwaltungsgesellschaft mbH	Hamburg
Lufthansa Technik Logistik GmbH	Hamburg
Lufthansa Technik Logistik Services GmbH	Hamburg

T193 USE OF EXEMPTION PROVISIONS (continued)

Company name	Registered office
Lufthansa Technik Maintenance International GmbH	Frankfurt/Main
Lufthansa Technik Objekt- und Verwaltungsgesellschaft mbH	Hamburg
Miles & More GmbH	Neu-Isenburg
Spiriant GmbH	Neu-Isenburg
time:matters GmbH	Neu-Isenburg
time:matters Holding GmbH	Neu-Isenburg
time:matters Spare Parts Logistics GmbH	Neu-Isenburg

The companies are affiliated with Deutsche Lufthansa AG by means of direct or indirect profit and loss transfer agreements respectively.

Furthermore, LHBD Holding Limited, London, UK, registration number 06939137, is exempt from the obligation to have its individual financial statements audited pursuant to Section 479A of the UK Companies Act 2006.

The consolidated financial statements include investments in 35 joint ventures and 37 associated companies (previous year: 38 joint ventures and 35 associated companies), of which ten joint ventures (previous year: ten) and 15 associates (previous year: 15) were accounted for using the equity method. The other joint ventures and associated companies were valued at amortised cost due to their minor overall significance.

49 Related party disclosures

Balances and transactions between the Company and its fully consolidated subsidiaries, which constitute related parties, have been eliminated in the course of consolidation and are not commented on in this Note. Details of transactions between the Group and other related parties are disclosed below.

The Lufthansa Group segments render numerous services to related parties within the scope of their ordinary business activities. Conversely, the companies in question provide services to the Lufthansa Group as part of their normal business. These extensive supply and service relationships take place on the basis of market prices.

In addition, the Group and certain non-consolidated subsidiaries have concluded numerous billing agreements, partly governing the joint use of services. In these cases, the administrative services provided are charged as cost allocations.

The Lufthansa Group's cash management is centralised, and, in this respect, the Lufthansa Group also performs a “banking function” vis-à-vis the non-consolidated companies of the Group. Non-consolidated Group companies included in the Group's cash management invest their available cash with the Group or borrow funds from the Group and carry out their derivative hedging transactions with the Group. All transactions take place at market conditions.

Due to geographical proximity in many cases, a large number of subletting contracts exists between the Lufthansa Group and related parties. In these cases, the Group usually charges the rental costs and incidental expenses incurred to the companies in question on a pro rata basis.

The following table shows the volume of significant services provided to or by related parties:

T194 VOLUME OF SIGNIFICANT SERVICES PROVIDED TO OR BY RELATED PARTIES

in €m	Volume of services rendered		Volume of services received	
	2018	2017	2018	2017
Non-consolidated subsidiaries				
Airport Services Dresden GmbH, Germany	0	0	4	5
Albatros Versicherungsdienste GmbH, Germany	1	1	42	53
Austrian Airlines Technik-Bratislava, s.r.o., Slovakia	2	3	6	6
Austrian Airlines Tele Sales Service GmbH, Austria	0	0	5	5
Delvag Versicherungs-AG, Germany	9	8	6	5
DLH Fuel Company mbH, Germany	0	0	476	429
Global Load Control (PTY) LTD, South Africa	0	1	7	7
handling counts GmbH, Germany	1	1	11	9
LGSP Lufthansa Ground Service Portugal, Unipessoal Lda., Portugal	0	0	8	9
Lufthansa Aviation Training Austria GmbH, Austria	2	1	5	4
Lufthansa Aviation Training Operations Germany GmbH, Germany	4	3	13	12
Lufthansa Aviation Training USA Inc., USA	1	0	13	11
Lufthansa Consulting GmbH, Germany	1	1	9	7
Lufthansa Engineering and Operational Services GmbH, Germany	4	3	33	25
Lufthansa Global Business Services Hamburg GmbH, Germany	7	7	30	26
Lufthansa Global Business Services S.A. de C.V., Mexico	1	1	4	5
Lufthansa Global Business Services Sp. z o.o., Poland	2	2	30	27
Lufthansa Global Tele Sales GmbH, Germany	1	4	57	60
Lufthansa Industry Solutions TS GmbH, Germany	0	0	10	9
Lufthansa Services Philippines, Inc., Philippines	1	2	5	5
Lufthansa Services (Thailand) Ltd., Thailand	0	0	5	5
Lufthansa Super Star gemeinnützige Gesellschaft mit beschränkter Haftung, Germany	0	13	4	11
Lufthansa Systems FlightNav AG, Switzerland	1	1	23	23
Lufthansa Systems Hungaria Kft, Hungary	1	0	21	17
Lufthansa Systems Poland Sp. z o.o., Poland	2	1	28	21
Lufthansa Technical Training GmbH, Germany	6	6	23	17
Lufthansa Technik Component Services Asia Pacific Limited, China	1	1	6	4
Lufthansa Technik Logistik of America LLC, USA	7	7	41	33
Lufthansa Technik Middle East FZE, United Arab Emirates	0	0	5	4
Lufthansa Technik Shenzhen Co., Ltd., China	19	23	28	28
Lufthansa Technik Turbine Shannon Limited, Ireland	5	6	15	18
LZ-Catering GmbH, Germany	5	6	13	13
ZeroG GmbH, Germany	0	0	5	4

T194 VOLUME OF SIGNIFICANT SERVICES PROVIDED TO OR BY RELATED PARTIES (continued)

in €m	Volume of services rendered		Volume of services received	
	2018	2017	2018	2017
Joint ventures				
Airfoil Services Sdn. Bhd., Malaysia	5	0	7	8
FraCareServices GmbH, Germany	2	6	1	1
Lufthansa Bombardier Aviation Services GmbH, Germany	2	2	9	7
N3 Engine Overhaul Services GmbH & Co. KG, Germany	13	4	0	0
Shanghai Pudong International Airport Cargo Terminal Co. Ltd., China	1	1	6	7
Spairliners GmbH, Germany	63	52	48	45
Star Alliance Services GmbH, Germany	3	3	7	0
Terminal 2 Gesellschaft mbH & Co. oHG, Germany	1	1	14	13
Terminal One Group Association, L.P., USA	5	5	5	5
XEOS Sp. z o.o., Poland	5	6	0	0
Associated companies				
Aircraft Maintenance and Engineering Corp., China	16	13	5	4
Airmail Center Frankfurt GmbH, Germany	0	0	9	8
AviationPower GmbH, Germany	0	1	27	35
HEICO Aerospace Holdings Corp., USA	0	0	12	11
Other affiliated companies				
Shanghai Pudong International Airport Public Cargo Terminal Co. Ltd. (West), Shanghai, China	0	0	90	77
SunExpress Germany GmbH, Germany	25	28	98	103

The following tables show receivables owed by and liabilities to related parties:

T195 RECEIVABLES FROM AFFILIATED COMPANIES

in €m	2018	2017
Trade receivables from non-consolidated subsidiaries	19	52
Trade receivables from joint ventures	22	16
Trade receivables from associated companies	14	6
Trade receivables from other affiliated companies	12	6
Total trade receivables	67	80
Other receivables from non-consolidated subsidiaries	42	12
Other receivables from joint ventures	41	36
Other receivables from associated companies	10	10
Total other receivables	93	58
Loans to non-consolidated subsidiaries	81	66
Loans to joint ventures	27	23
Loans to associated companies	-	-
Total non-current receivables	108	89

T196 LIABILITIES TO AFFILIATED COMPANIES

in €m	2018	2017
Trade payables to non-consolidated subsidiaries	27	53
Trade payables to joint ventures	9	14
Trade payables to associated companies	4	4
Trade payables to other affiliated companies	3	3
Total trade payables	43	74
Other liabilities to non-consolidated subsidiaries	269	233
Other liabilities to joint ventures	0	0
Other liabilities from associated companies	0	0
Total other liabilities	269	233

No individual shareholders of Deutsche Lufthansa AG exercise significant influence over the Group. For transactions involving members of the Executive Board and Supervisory Board ("directors' dealings"), see [Note 50, p. 184](#).

50 Supervisory Board and Executive Board

The disclosure of remuneration for key managers required by IAS 24 includes the remuneration of the active members of the Executive Board and Supervisory Board.

The members of the Executive Board and the Supervisory Board as well as the other offices that they hold are named in the combined management report in the section [Corporate governance, p. 79ff.](#)

The principles of the remuneration system and the amount of remuneration paid to Executive Board and Supervisory Board members are shown and explained in detail in the remuneration report. The **remuneration report** forms part of the combined management report [p. 84 – 91.](#)

Total Executive Board remuneration under IFRS was EUR 16.2m (previous year: EUR 23.0m), including current service costs for pensions of EUR 3.5m (previous year: EUR 3.2m).

The active members of the Executive Board in past reporting years were remunerated as follows:

T197 EXECUTIVE BOARD REMUNERATION (IFRS)

in € thousands	2018	2017
Basic salary	4,832	4,831
Other ¹⁾	427	449
One-year variable remuneration	4,845	5,430
Total short-term remuneration	10,104	10,710
Long-term variable remuneration ²⁾	3,570	3,014
Share-based remuneration	-968	6,090
Current service cost for retirement benefits	3,510	3,166
Total long-term remuneration	6,112	12,270
Severance payments	-	-
Total	16,216	22,980

¹⁾ Other remuneration includes in particular benefits from the use of company cars, discounts in connection with cash outflows from share programmes ([Notes to the consolidated financial statements, Note 38, p. 153ff.](#)) and concessionary travel in accordance with the relevant IATA regulations.

²⁾ Expenses recognised in the reporting year for long-term variable remuneration for the financial years 2016 to 2018.

Pension provisions for Executive Board members active in the 2018 financial year came to EUR 12.4m (previous year: EUR 10.8m).

In addition to the provision for the one-year variable remuneration of EUR 4,845k (previous year: EUR 5,430k), provisions totalling EUR 3,570k were recognised for the future payment of long-term variable remuneration for the Executive Board members active in the financial year 2018 (previous year: EUR 3,014k). In addition, provisions of EUR 5,651k were recognised for the future payment of long-term, share-based remuneration for the Executive Board members active as of 31 December 2018 (previous year: EUR 7,911k).

Total remuneration (HGB) paid to the Executive Board of Deutsche Lufthansa AG in the financial year 2018 came to EUR 13,015k (previous year: EUR 14,731k). This includes EUR 1,893k for the new share programme, in which the Executive Board acquired a total of 37,125 shares.

Current payments and other benefits for former members of the Executive Board and their surviving dependants came to EUR 6.8m (previous year: EUR 5.2m). This includes payments by subsidiaries as well as benefits in kind and concessionary travel.

Pension obligations towards former Executive Board members and their surviving dependants amount to EUR 68.3m (previous year: EUR 71.0m). They are included in pension provisions ([Note 33, p. 142ff.](#)).

Expenses for the fixed remuneration of Supervisory Board members came to EUR 2,107k in 2018 (previous year: EUR 2,164k). Other remuneration, mainly attendance fees, amounted to EUR 68k (previous year: EUR 89k). The Deutsche Lufthansa AG Supervisory Board members were also paid EUR 27k for work on supervisory boards of Group companies (previous year: EUR 17k).

In the reporting year, as in the previous year, no loans or advance payments were made to members of the Executive Board and to members of the Supervisory Board.

In addition to their Supervisory Board remuneration, employee representatives on the Supervisory Board received compensation for their work in the form of wages and salaries including pension entitlements amounting to EUR 1.1m in total in 2018 (previous year: EUR 1.0m).

The following auditor's report has been issued in accordance with Section 322 HGB and refers to the audit of the German language consolidated financial statements and the German language group management report, which is neither included nor incorporated by reference.

Independent auditor's report

To Deutsche Lufthansa AG, Cologne

REPORT ON THE AUDIT OF THE CONSOLIDATED FINANCIAL STATEMENTS AND OF THE GROUP MANAGEMENT REPORT

Audit Opinions

We have audited the consolidated financial statements of Deutsche Lufthansa AG, Cologne, and its subsidiaries (the Group), which comprise the consolidated statement of financial position as at 31 December 2018, the consolidated statement of profit or loss, consolidated statement of comprehensive income, consolidated statement of changes in equity and consolidated statement of cash flows for the financial year from 1 January to 31 December 2018, and notes to the consolidated financial statements, including a summary of significant accounting policies. In addition, we have audited the group management report of Deutsche Lufthansa AG, which is combined with the Company's management report, for the financial year from 1 January to 31 December 2018. In accordance with the German legal requirements, we have not audited the content of those parts of the group management report listed in the "Other Information" section of our auditor's report.

In our opinion, on the basis of the knowledge obtained in the audit,

- the accompanying consolidated financial statements comply, in all material respects, with the IFRSs as adopted by the EU, and the additional requirements of German commercial law pursuant to § (Article) 315e Abs. (paragraph) 1 HGB (Handelsgesetzbuch: German Commercial Code) and, in compliance with these requirements, give a true and fair view of the assets, liabilities, and financial position of the Group as at 31 December 2018, and of its financial performance for the financial year from 1 January to 31 December 2018, and
- the accompanying group management report as a whole provides an appropriate view of the Group's position. In all material respects, this group management report is consistent with the consolidated financial statements, complies with German legal requirements and appropriately presents the opportunities and risks of future development. Our audit opinion on the group management report does not cover the content of those parts of the group management report listed in the "Other Information" section of our auditor's report.

Pursuant to § 322 Abs. 3 Satz (sentence) 1 HGB, we declare that our audit has not led to any reservations relating to the legal compliance of the consolidated financial statements and of the group management report.

Basis for the Audit Opinions

We conducted our audit of the consolidated financial statements and of the group management report in accordance with § 317 HGB and the EU Audit Regulation (No. 537/2014, referred to subsequently as "EU Audit Regulation") in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the Institut der Wirtschaftsprüfer (Institute of Public Auditors in Germany) (IDW). We performed the audit of the consolidated financial statements in supplementary compliance with the International Standards on Auditing (ISAs). Our responsibilities under those requirements, principles and standards are further described in the "Auditor's Responsibilities for the Audit of the Consolidated Financial Statements and of the Group Management Report" section of our auditor's report. We are independent of the group entities in accordance with the requirements of European law and German commercial and professional law, and we have fulfilled our other German professional responsibilities in accordance with these requirements. In addition, in accordance with Article 10 (2) point (f) of the EU Audit Regulation, we declare that we have not provided non-audit services prohibited under Article 5 (1) of the EU Audit Regulation. We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our audit opinions on the consolidated financial statements and on the group management report.

Key Audit Matters in the Audit of the Consolidated Financial Statements

Key audit matters are those matters that, in our professional judgment, were of most significance in our audit of the consolidated financial statements for the financial year from 1 January to 31 December 2018. These matters were addressed in the context of our audit of the consolidated financial statements as a whole, and in forming our audit opinion thereon; we do not provide a separate audit opinion on these matters.

In our view, the matters of most significance in our audit were as follows:

1. Recognition of traffic revenue, including contract liabilities in respect of unused flight documents and customer loyalty programs, as well as the impact of the initial application of IFRS 15
2. Pension provisions
3. Recoverability of non-current assets, in particular goodwill and intangible assets with indefinite useful lives
4. Accounting treatment of hedging transactions, including the impact of the initial application of IFRS 9
5. Changes in the accounting treatment of engine overhauls

Our presentation of these key audit matters has been structured in each case as follows:

- a. Matter and issue
- b. Audit approach and findings
- c. Reference to further information

Hereinafter we present the key audit matters:

1. Recognition of traffic revenue, including contract liabilities in respect of unused flight documents and customer loyalty programs, as well as the impact of the initial application of IFRS 15

- a. Until they are used due to departure, sold flight documents are recognized as contract liabilities for unused flight documents in the Company's consolidated financial statements. Once a passenger coupon or an airfreight document has been used due to departure, the corresponding traffic revenue is recognized as revenue in the income statement. First of all, the part of the flight documents that has not yet been used in the financial year and is still valid in the following year is added to the contract liabilities for unused flight documents. In addition, based on historical data, the amount of expired flight documents which are expected to not be used is estimated; this amount is also recognized under contract liabilities for unused flight documents. In the financial year 2018, the Lufthansa Group realized a total of EUR 28.1 billion in traffic revenue, of which EUR 25.0 billion was attributable to passenger airlines. As of 31 December 2018, EUR 4.0 billion was recognized in the consolidated statement of financial position as contract liabilities for unused flight documents.

Until they are redeemed, bonus miles awarded to Miles & More members are recognized as contract liabilities on the basis of the individual selling price per bonus mile. Where bonus miles are collected with external cooperation partners, these are recognized as liabilities until redeemed and measured at the prices paid to Lufthansa by the external cooperation partners. Bonus miles which are no

longer expected to be used are realized as revenue over a period of three years. The contract liabilities from customer loyalty programs amounted to EUR 2.2 billion as of the balance sheet date.

In this financial year, the initial application of the new accounting standard on revenue recognition (IFRS 15) resulted in impacts on the recognition and deferral of revenues, as well as on the presentation of revenue. The initial application of IFRS 15 resulted – based on the amount that would have arisen for the financial year 2018 based on the previous accounting standards – in a decrease in traffic revenues amounting to EUR 2.3 billion, which was primarily attributable to the netting of traffic revenues previously reported on a gross basis with airport charges previously recognized under cost of materials and services. In addition, IFRS 15 has shifted the point in time at which certain other traffic revenue (in particular rebooking fees) and revenue resulting from the expected expiry of bonus miles is recognized. The Lufthansa Group has applied IFRS 15 using the modified retrospective approach pursuant to IFRS 15.C3(b), based on which the cumulative effect of initial application of IFRS 15 was presented as an adjustment to opening retained earnings as of 1 January 2018. This resulted in a reduction in retained earnings of EUR 310 million as of 1 January 2018. The initial application of IFRS 15 necessitated a Group-wide analysis of the various business models and contract types within the Group companies.

From our point of view, these matters were of particular significance for our audit, as recognition and measurement of these items, which are specific to the business model and significant in amount, are highly dependent on the estimates and assumptions of the executive directors regarding the use behavior of flight documents, to which calculation processes that are, in some cases, complex are applied. Recognizing and deferring revenue correctly in accordance with the Group-wide application of the new accounting standard IFRS 15 was moreover considered to be complex.

- b. We also included our specialists in the Risk Assurance Service (RAS) to audit traffic revenue. With their support, we evaluated, among other things, the appropriateness and effectiveness of the established internal control system used to settle and realize traffic revenue, including the IT systems used. To the extent that we were not able to conduct our own evaluation of the internal control system of services relating to IT systems or processes outsourced to third parties, we obtained an assurance report attesting to the appropriateness and effectiveness of the internal control system established by the service provider (ISAE 3402 Type II or SSAE 16), which our specialists assessed. In our audit of the contract liabilities for unused

flight documents, we reconstructed among other things the individual steps used in the calculations. Specifically, we examined the open flight documents and their measurement by year of sale and validity. Furthermore, we considered the consistency and continuity of the methods used to calculate the flight prices, fees, taxes and other levies attributable to the flight documents no longer expected to be used. In order to assess the appropriateness of the contract liabilities from customer loyalty programs accounted for as of the balance sheet date, we evaluated among other things the fair value measurement for each category of use and the underlying assumptions and parameters derived therefrom. Furthermore, we assessed the mathematical accuracy of the calculation of the contract liabilities from customer loyalty programs.

In addition, we assessed together with the support of our internal specialists from Capital Markets & Accounting Advisory Services (CMAAS) the impact of the initial application of IFRS 15. We assessed among other things the appropriateness of the procedure used including the impact analyses conducted within the Group and assessed the estimates and judgments made by the executive directors with respect to the recognition and deferral of revenue for the various business models of the Group companies.

We were able to satisfy ourselves that the estimates and assumptions made by the executive directors are sufficiently substantiated and documented to ensure that revenue is appropriately recognized in accordance with IFRS 15 as applied for the first time.

- c. The disclosures on traffic revenue, contract liabilities in respect of unused flight documents and customer loyalty programs, as well as the impact of the initial application of IFRS 15 are contained in sections 2, 3 and 39 of the notes to the consolidated financial statements.

2. Pension provisions

- a. In the Company's consolidated financial statements pension provisions amounting to EUR 5.9 billion are reported, comprising the net amount of the obligations under various pension plans amounting to EUR 21.4 billion and the fair values of the plan assets amounting to EUR 15.5 billion. The majority of these provisions relates to old-age and transitional pension commitments in Germany and Switzerland. The obligations from defined benefit pension plans were measured using the projected unit credit method in accordance with IAS 19. This requires in particular that assumptions be made as to long-term salary and pension trends and average life expectancy. The discount rate must be determined by reference to market yields on high-quality corporate bonds with matching currencies and consistent maturities. The plan assets are measured at fair value, which in turn involves making estimates that are subject to uncertainties.

From our point of view, these matters were of particular significance in the context of our audit because the measurement of this significant item in terms of its amount is based to a large extent on estimates and assumptions made by the Company's executive directors.

- b. As part of our audit we evaluated the actuarial expert reports obtained and the professional qualifications of the external experts. Due to the specific features of the actuarial calculations, we were assisted by internal specialists from Pension Consulting. Together with them, we evaluated the numerical data, the actuarial parameters and the valuation methods and assumptions on which the valuations were based for compliance with the standards and appropriateness. On this basis, we then assessed the calculations of the figures presented on the statement of financial position, the accounting entries for the provisions and the disclosures in the notes to the consolidated financial statements based on the expert opinions. Our evaluation of the fair values of plan assets was based on bank confirmations submitted to us. Based on our audit procedures, we were able to satisfy ourselves that the estimates and assumptions made by the executive directors are substantiated and sufficiently documented.
- c. The disclosures relating to pension provisions are contained in sections 2 and 33 of the notes to the consolidated financial statements.

3. Recoverability of non-current assets, in particular goodwill and intangible assets with indefinite useful lives

- a. In the Company's consolidated financial statements a total amount of EUR 1.4 billion is reported under the line item "Intangible assets with an indefinite useful life, incl. goodwill" of the consolidated statement of financial position. Goodwill and intangible assets with indefinite useful lives are regularly tested for impairment ("impairment test") once per financial year or if there are indications of an impairment. In addition, items of property, plant and equipment are tested for impairment if there are indications of impairment. The carrying amount of the relevant cash-generating units, including goodwill, is compared with the corresponding recoverable amount in the context of the impairment test. These measurements are generally based on the present value of future cash flows of the cash-generating unit/group of cash-generating units to which the respective asset is to be allocated. The present values are calculated using discounted cash flow models. The measurements are based on projections that were also used to prepare the four-year budget for the Lufthansa Group prepared by the executive directors and acknowledged by the Supervisory Board. The discount rate used is the weighted average cost of capital for the relevant cash-generating unit.

The outcome of this valuation exercise is dependent to a large extent on the estimates made by the executive directors with respect to the future cash inflows, the discount rate used, the rate of growth and other assumptions, and is therefore subject to considerable uncertainty. Against this background and due to the complex nature of the valuation, this matter was of particular significance in the context of our audit.

- b. As part of our audit, we assessed the methodology used for the purposes of performing the impairment test, among other things. After matching the future cash inflows used for the calculation against the adopted four-year budget of the Group, we assessed the appropriateness of the calculation, in particular by reconciling it with general and sector-specific market expectations. In addition, we assessed the appropriate consideration of the costs of Group functions.

With the knowledge that even relatively small changes in the discount rate applied can have material effects on the values calculated in this way, we also focused our testing on the parameters used to determine the discount rate applied, and evaluated the calculation model. In order to

reflect the uncertainty inherent in the projections, we assessed the sensitivity analyses performed by the Company and carried out our own additional sensitivity analyses with respect to those cash-generating units with low headroom (recoverable amount compared with the carrying amount). Taking into account the information available, we determined that the carrying amounts of the cash-generating units, including the allocated goodwill, were adequately covered by the discounted future net cash flows. Overall, the valuation parameters and assumptions used by the executive directors are in line with our expectations and are also within the ranges considered by us to be reasonable.

- c. The Company's disclosures relating to impairment testing are contained in sections 2, 9 and 16 of the notes to the consolidated financial statements.

4. Accounting treatment of hedging transactions, including the impact of the initial application of IFRS 9

- a. The Lufthansa Group uses a variety of derivative financial instruments to hedge against currency, fuel price and interest rate risks arising from its ordinary business activities. The executive directors' hedging policy is documented in corresponding internal guidelines and serves as the basis for these transactions. Currency risks arise primarily from sale transactions, procurement transactions (in particular fuel and investments in aircraft) and financing denominated in foreign currencies. The risk associated with changes in fuel prices results from future procurement transactions that are subject to market price risks. The risk of changes in interest rates results from changes in the fair values of fixed-rate financing.

Derivative financial instruments are recognized at fair value as of the balance sheet date. The positive fair values of all of the derivative financial instruments used as hedges amounted to EUR 1.2 billion as of the balance sheet date and the negative fair value amounted to EUR 0.6 billion. If the financial instruments used by the Lufthansa Group are effective hedges of future cash flows in the context of hedging relationships in accordance with the requirements of IFRS 9, changes in fair value are recognized in other comprehensive income over the duration of the hedging relationship (cash flow hedges). As of the balance sheet date, a cumulative amount of EUR 0.3 billion was recognized in other comprehensive income as effective fair value changes before income taxes (increasing equity) (EUR 0.5 billion in the cash flow hedge reserve and EUR -0.2 billion in the cost of hedging reserve).

At the time of transitioning from hedge accounting under IAS 39 to IFRS 9 at the beginning of the financial year, the Lufthansa Group exercised the option, where possible, of implementing the transition prospectively, without restating prior-period figures. With regard to the accounting treatment of hedges with options, the transition was implemented retrospectively, restating prior-period figures, as required by the standard. As a result, fair value changes relating to options that had been recognized through profit or loss in the previous period were retrospectively transferred to the cost of hedging reserve outside profit or loss as cost of hedging.

In our view, these matters were of particular importance for our audit due to the high complexity and number of hedging transactions as well as the extensive accounting and disclosure requirements of IFRS 9 and IFRS 7.

- b. We involved our internal specialists from Corporate Treasury Solutions (CTS) to assist in the audit of the accounting including the effects of the various hedging transactions on equity and profit or loss. Together with these specialists, we assessed, among other things, the internal control system with regard to derivative financial instruments, including the internal activities to monitor compliance with the hedging policy. We also assessed how the effects from transition were determined and how the prior-period figures were adjusted in relation to the initial application of IFRS 9. In our audit of the fair values, we also evaluated the measurement methods based on market data and the underlying data used. With respect to the hedging of expected future cash flows, we mainly carried out a retrospective assessment of past hedge effectiveness and an estimate of expected future hedge effectiveness, and assessed the corresponding effectiveness tests. We obtained bank confirmations in order to assess the completeness of and to examine the fair values of the recorded transactions. In doing so we were able to satisfy ourselves that, overall, the hedging transactions were appropriately accounted for and measured.
- c. The disclosures on the hedging transactions as well as the impact of the initial application of IFRS 9 are contained in sections 2 and 44 of the notes to the consolidated financial statements.

5. Changes in the accounting treatment of engine overhauls

- a. The Company's executive directors made changes to the accounting treatment of engine overhauls in the financial year 2018. In the Company's consolidated financial statements, material components of items of property, plant and equipment that have different useful lives are recognized and depreciated separately pursuant to IAS 16 (component approach). In the past, the component approach was applied for physical aircraft components such as seats and inflight entertainment systems, as well as for general overhauls. Due to more pronounced investment cycles for aircraft and based on the better data analyses that are now available for engine maintenance processes, large-scale engine overhauls have now also been capitalized as a separate component and depreciated over a useful life of six years since the beginning of the financial year 2018. This change improves the presentation of the Company's assets, liabilities and financial performance by reducing the extent to which the impact of maintenance work expenses on earnings fluctuates over time. In line with the provisions set out in IAS 8, the impact of this change has been applied retrospectively and the prior-period figures have been adjusted. This accounting change has reduced the "Aircraft and reserve engines" line item in the statement of financial position by EUR 609 million as of 1 January 2017 and by EUR 626 million as of 31 December 2017. Equity was reduced by EUR 475 million as of 1 January 2017 and by EUR 488 million as of 31 December 2017. The profit after income taxes deteriorated by EUR 11 million in 2017 and improved by EUR 102 million in 2018. The executive directors made various assumptions and used simplifications for the purposes of calculating, with retroactive effect, the historical cost and cumulative depreciation of engine overhaul components and aircrafts, as well as for the purposes of eliminating intercompany profits.

In our view, this matter was of particular importance for our audit due to the complexity involved in calculating, with retroactive effect, the historical cost and cumulative depreciation of engine overhaul components and aircrafts. In addition, the retrospective adjustments to the "Aircraft and reserve engines" line item, which is significant in terms of amount, are based to a large extent on estimates and assumptions made by the Company's executive directors.

- b. As part of our audit, we firstly assessed whether the requirements for changing the accounting treatment of engine overhauls had been met. This also involved assessing whether the changes in accounting treatment were consistent with the requirements set out in IAS 16.

With regard to the retrospective calculation of the historical cost and cumulative depreciation of engine overhaul components and aircrafts, we evaluated the methods used and the appropriateness of the underlying estimates and assumptions. On this basis, we assessed the calculation of the impact, in terms of amount, of the retrospective change on the consolidated statement of financial position as of 1 January 2017 and as of 31 December 2017, as well as on the consolidated income statement for the financial years 2017 and 2018 and compared it, among other things, with financial information from the Group's accounts. We also assessed the elimination of intracompany profits. In addition, we evaluated the presentation of the change in accounting in the consolidated statement of financial position and the consolidated income statement, as well as the corresponding disclosures in the notes to the consolidated financial statements.

On the basis of our audit procedures, we were able to satisfy ourselves that the methods used are appropriate and that the estimates and assumptions made by the executive directors are sufficiently documented and substantiated to ensure that the retrospective change in the accounting treatment of engine overhauls is appropriate.

- c. The Company's disclosures on the reasons for, and the impact of, the changes in the accounting treatment of engine overhauls are contained in section 2 of the notes to the consolidated financial statements.

Other Information

The executive directors are responsible for the other information. The other information comprises the following non-audited parts of the group management report:

- the statement on corporate governance pursuant to § 289f HGB and § 315d HGB included in section "Corporate Governance" of the group management report
- the corporate governance report pursuant to No. 3.10 of the German Corporate Governance Code

The other information comprises further the remaining parts of the annual report – excluding cross-references to external information – with the exception of the audited consolidated financial statements, the audited group management report and our auditor's report.

Our audit opinions on the consolidated financial statements and on the group management report do not cover the other information, and consequently we do not express an audit opinion or any other form of assurance conclusion thereon.

In connection with our audit, our responsibility is to read the other information and, in so doing, to consider whether the other information

- is materially inconsistent with the consolidated financial statements, with the group management report or our knowledge obtained in the audit, or
- otherwise appears to be materially misstated.

If, based on the work we have performed, we conclude that there is a material misstatement of this other information, we are required to report that fact. We have nothing to report in this regard.

Responsibilities of the Executive Directors and the Supervisory Board for the Consolidated Financial Statements and the Group Management Report

The executive directors are responsible for the preparation of the consolidated financial statements that comply, in all material respects, with IFRSs as adopted by the EU and the additional requirements of German commercial law pursuant to § 315e Abs. 1 HGB and that the consolidated financial statements, in compliance with these requirements, give a true and fair view of the assets, liabilities, financial position, and financial performance of the Group. In addition, the executive directors are responsible for such internal control as they have determined necessary to enable the preparation of consolidated financial statements that are free from material misstatement, whether due to fraud or error.

In preparing the consolidated financial statements, the executive directors are responsible for assessing the Group's ability to continue as a going concern. They also have the responsibility for disclosing, as applicable, matters related to going concern. In addition, they are responsible for financial reporting based on the going concern basis of accounting unless there is an intention to liquidate the Group or to cease operations, or there is no realistic alternative but to do so.

Furthermore, the executive directors are responsible for the preparation of the group management report that, as a whole, provides an appropriate view of the Group's position and is, in all material respects, consistent with the consolidated financial statements, complies with German legal requirements, and appropriately presents the opportunities and risks of future development. In addition, the executive directors are responsible for such arrangements and measures (systems) as they have considered necessary to enable the preparation of a group management re-port that is in accordance with the applicable German legal requirements, and to be able to provide sufficient appropriate evidence for the assertions in the group management report.

The supervisory board is responsible for overseeing the Group's financial reporting process for the preparation of the consolidated financial statements and of the group management report.

Auditor's Responsibilities for the Audit of the Consolidated Financial Statements and of the Group Management Report

Our objectives are to obtain reasonable assurance about whether the consolidated financial statements as a whole are free from material misstatement, whether due to fraud or error, and whether the group management report as a whole provides an appropriate view of the Group's position and, in all material respects, is consistent with the consolidated financial statements and the knowledge obtained in the audit, complies with the German legal requirements and appropriately presents the opportunities and risks of future development, as well as to issue an auditor's report that includes our audit opinions on the consolidated financial statements and on the group management report.

Reasonable assurance is a high level of assurance, but is not a guarantee that an audit conducted in accordance with § 317 HGB and the EU Audit Regulation and in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the Institut der Wirtschaftsprüfer (IDW) and supplementary compliance with the ISAs will always detect a material misstatement. Misstatements can arise from fraud or error and are considered material if, individually or in the aggregate, they could reasonably be expected to influence the economic decisions of users taken on the basis of these consolidated financial statements and this group management report.

We exercise professional judgment and maintain professional skepticism throughout the audit. We also:

- Identify and assess the risks of material misstatement of the consolidated financial statements and of the group management report, whether due to fraud or error, design and perform audit procedures responsive to those risks, and obtain audit evidence that is sufficient and appropriate to provide a basis for our audit opinions. The risk of not detecting a material misstatement resulting from fraud is higher than for one resulting from error, as fraud may involve collusion, forgery, intentional omissions, misrepresentations, or the override of internal control.
- Obtain an understanding of internal control relevant to the audit of the consolidated financial statements and of arrangements and measures (systems) relevant to the audit of the group management report in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an audit opinion on the effectiveness of these systems.
- Evaluate the appropriateness of accounting policies used by the executive directors and the reasonableness of estimates made by the executive directors and related disclosures.
- Conclude on the appropriateness of the executive directors' use of the going concern basis of accounting and, based on the audit evidence obtained, whether a material uncertainty exists related to events or conditions that may cast significant doubt on the Group's ability to continue as a going concern. If we conclude that a material uncertainty exists, we are required to draw attention in the auditor's report to the related disclosures in the consolidated financial statements and in the group management report or, if such disclosures are inadequate, to modify our respective audit opinions. Our conclusions are based on the audit evidence obtained up to the date of our auditor's report. However, future events or conditions may cause the Group to cease to be able to continue as a going concern.
- Evaluate the overall presentation, structure and content of the consolidated financial statements, including the disclosures, and whether the consolidated financial statements present the underlying transactions and events in a manner that the consolidated financial statements give a true and fair view of the assets, liabilities, financial position and financial performance of the Group in compliance with IFRSs as adopted by the EU and the additional requirements of German commercial law pursuant to § 315e Abs. 1 HGB.
- Obtain sufficient appropriate audit evidence regarding the financial information of the entities or business activities within the Group to express audit opinions on the consolidated financial statements and on the group management report. We are responsible for the direction, supervision and performance of the group audit. We remain solely responsible for our audit opinions.
- Evaluate the consistency of the group management report with the consolidated financial statements, its conformity with German law, and the view of the Group's position it provides.

- Perform audit procedures on the prospective information presented by the executive directors in the group management report. On the basis of sufficient appropriate audit evidence we evaluate, in particular, the significant assumptions used by the executive directors as a basis for the prospective information, and evaluate the proper derivation of the prospective information from these assumptions. We do not express a separate audit opinion on the prospective information and on the assumptions used as a basis. There is a substantial unavoidable risk that future events will differ materially from the prospective information.

We communicate with those charged with governance regarding, among other matters, the planned scope and timing of the audit and significant audit findings, including any significant deficiencies in internal control that we identify during our audit.

We also provide those charged with governance with a statement that we have complied with the relevant independence requirements, and communicate with them all relationships and other matters that may reasonably be thought to bear on our independence, and where applicable, the related safeguards.

From the matters communicated with those charged with governance, we determine those matters that were of most significance in the audit of the consolidated financial statements of the current period and are therefore the key audit matters. We describe these matters in our auditor's report unless law or regulation precludes public disclosure about the matter.

OTHER LEGAL AND REGULATORY REQUIREMENTS

Further Information pursuant to Article 10 of the EU Audit Regulation

We were elected as group auditor by the annual general meeting on 8 May 2018. We were engaged by the supervisory board on 18 June 2018. We have been the group auditor of the Deutsche Lufthansa AG, Cologne, without interruption since the financial year 1955.

We declare that the audit opinions expressed in this auditor's report are consistent with the additional report to the audit committee pursuant to Article 11 of the EU Audit Regulation (long-form audit report).

GERMAN PUBLIC AUDITOR RESPONSIBLE FOR THE ENGAGEMENT

The German Public Auditor responsible for the engagement is Eckhard Sprinkmeier.

Düsseldorf, 5 March 2019

PricewaterhouseCoopers GmbH
Wirtschaftsprüfungsgesellschaft

Petra Justenhoven	Eckhard Sprinkmeier
Wirtschaftsprüferin	Wirtschaftsprüfer
(German Public Auditor)	(German Public Auditor)

Audited Annual Financial Statements of Deutsche Lufthansa Aktiengesellschaft as of and for the financial year ended December 31, 2020 (HGB)

Balance Sheet

as of 31 December 2020

T01 BALANCE SHEET - ASSETS

in €m	Notes	31.12.2020	31.12.2019
Intangible assets		385	309
Aircraft	3	5,783	6,679
Property, plant and other equipment		69	82
Financial investments	4	16,800	16,440
Non-current assets	3	23,037	23,510
Inventories	5	111	108
Trade receivables	6	102	642
Other receivables and other assets	6	1,299	1,725
Securities	7	3,655	41
Cash and cash equivalents	7	788	897
Current assets		5,955	3,413
Prepaid expenses	8	73	60
Deferred tax assets	9	3,207	2,142
Excess of plan assets over provisions for pensions	10	-	7
Total assets		32,272	29,132

Balance Sheet

as of 31 December 2020

T02 BALANCE SHEET - SHAREHOLDERS' EQUITY AND LIABILITIES

in €m	Notes	31.12.2020	31.12.2019
Issued capital*	11	1,530	1,224
Capital reserve	12	378	378
Retained earnings	12	6,498	6,200
Distributable earnings	30	-780	298
Shareholders' equity		7,626	8,100
Provisions for pensions and similar obligations		2,230	1,066
Tax provisions		523	287
Other provisions		4,111	5,249
Provisions	13	6,864	6,602
Bonds		2,827	1,098
Liabilities to banks		4,215	2,089
Payables to affiliated companies		4,659	6,094
Other liabilities		6,067	5,135
Liabilities	14	17,768	14,416
Deferred income		14	14
Total shareholders' equity and liabilities		32,272	29,132

*contingent capital as of 31 December 2020 amounts to EUR 1.122m (previous year EUR 238m)

Income statement

For the financial year 2020

T03 INCOME STATEMENT

in €m	Notes	2020	2019
Traffic revenue	18	3,502	14,885
Other revenue	19	778	1,388
Total revenue		4,280	16,273
Other operating income	20	6,825	1,663
Cost of materials and services	21	-5,071	-10,762
Staff costs	22	-2,217	-3,049
Depreciation, amortisation and impairment	23	-1,135	-594
Other operating expenses	24	-2,342	-2,985
Result from operating activities		340	546
Result from equity investments	25	-806	476
Net interest	26	-1,162	-352
Impairment on investments and current securities	27	-147	-45
Financial result		-2,115	79
Current income taxes	28	-51	-293
Deferred income taxes	28	1,065	291
Profit after income taxes		-761	623
Other taxes	28	-19	-28
Net income/Net loss for the year		-780	595
Gewinnvortrag aus dem Vorjahr		298	-
Transfer to retained earnings	12	-298	-297
Distributable earnings	30	-780	298

Statement of changes in non-current assets

As of 31 December 2020

T04 STATEMENT OF CHANGES IN NON-CURRENT ASSETS										
in €m										
	as of 01.01.2020	Acquisitions			as of 31.12.2020	Accumulated depreciation and amortisation			Carrying amounts	
		Additions	Disposals	Reclassifi- cations		as of 01.01.2020	Additions	Disposals	Write-ups	as of 31.12.2020
I. Intangible assets										
1. Purchased concessions, intellectual property and similar rights and assets and licences in such rights and assets	588	6	0	21	615					
2. Geschäfts- oder Firmenwerte	-	103	-	-	103					
3. Advance payments	80	17	15	-21	61					
	668	126	15	-	779					
II. Aircraft										
1. Aircraft and equipment	8,000	233	60	100	8,273					
2. Advance payments and plant under construction	1,573	158	196	-100	1,435					
	9,573	391	256	-	9,708					
III. Property, plant and equipment										
1. Land, leasehold rights and buildings including buildings on third-party land	182	4	7	1	180					
2. Other equipment, operating and office equip- ment	179	9	16	0	172					
3. Advance payments and plant under construction	8	2	6	-1	3					
	369	15	29	-	355					
IV. Investments										
1. Shares in affiliated companies	12,919	6,018	4,568	-	14,369					
2. Loans to affiliated companies	4,195	736	1,702	-11	3,218					
3. Equity investments	169	0	0	-	169					
4. Non-current securities	5	-	0	-	5					
5. Other loans	87	4	26	11	76					
6. Prefinancing of leasehold	6	-	0	-	6					
	17,381	6,758	6,296	-	17,843					
Total	27,991	7,290	6,596	-	28,685					

Notes

Deutsche Lufthansa AG 2020

GENERAL REMARKS

1 Principles

The financial statements of Deutsche Lufthansa AG, Cologne, registered at Cologne District Court under the number HRB 2168, have been prepared in accordance with the German Commercial Code (HGB), the supplementary provisions of the German Stock Corporation Act (AktG) and the Articles of Association, and have been audited by Ernst & Young GmbH Wirtschaftsprüfungsgesellschaft, Eschborn/Frankfurt am Main. In accordance with Section 315e Paragraph 1 HGB, Deutsche Lufthansa AG, the parent company of the Deutsche Lufthansa AG Group, prepares consolidated financial statements on the basis of the International Financial Reporting Standards (IFRS) as adopted by the EU.

Its financial year is the calendar year.

The separate and consolidated financial statements are published in the Federal Gazette. They are permanently available online at <https://investor-relations.lufthansagroup.com/de/publikationen/finanzberichte.html>.

The income statement has been prepared using the total cost method.

To make the presentation clearer, certain items of the balance sheet and the income statement have been grouped together and are shown and explained separately in the notes. For the same reason, disclosures indicating how these items also belong to other items and disclosures marked “of which” have likewise been made at this point. Over and above the statutory classification system, the entry relating to aircraft is listed separately in order to improve the clarity of the financial statements.

2 Summary of significant accounting policies and valuation methods

As before, preparation of the financial statements was essentially based on the following accounting policies.

GOING CONCERN

The business of the Lufthansa Group companies was impacted significantly by the consequences of the coronavirus pandemic. International travel restrictions and quarantine rules that began in winter 2020 (in China) and spring 2020 elsewhere were tightened worldwide over the course of the year, bringing flight traffic at the Lufthansa Group compa-

nies almost to a complete standstill at times. Only the cargo business was able to continue successfully, albeit with a sharp decline in the use of belly capacities. Around 700 Group aircraft were grounded temporarily in the course of the year. The slight recovery in summer 2020, when available capacity was around 25% of the previous year's, was slowed by the end of the year due to the renewed sharp increase in the number of infections. Payments for existing, invariable cost components and the high level of refunds for cancelled flights had an impact on liquidity. Liquidity at Deutsche Lufthansa AG and its domestic and foreign subsidiaries was secured in the middle of the financial year in an initial step by the government support measures as part of the stabilisation packages.

Over the remainder of the year, the companies initiated and partly implemented wide-ranging steps to safeguard short-term liquidity and initiate a medium-term return to a profitable new normal once the pandemic comes to an end. As part of the Group-wide programme ReNew, which is made up of ReStructure, ReFocus, ReOrg and RePay, measures were taken to streamline structures and reduce the number of full-time employees by around 27,000, as well as to downsize the operating fleet permanently by around 150 aircraft. Until further notice, some four-engined wide-bodied models (Airbus A340-600 and A380 and Boeing 747-400) are to be put into deep storage or decommissioned entirely. Previously planned investment spending, mainly in aircraft, was postponed to a later date whenever possible and appropriate. The plan is to take delivery of no more than 80 new aircraft by 2023. In addition to state aid in the form of short-time working pay and the reimbursement of social security contributions, crisis agreements extending into 2022 were reached with the collective bargaining partners to safeguard liquidity. Agreements to protect liquidity were also signed with other system partners. The Lufthansa Group's liquidity management also profited from tax deferrals in the reporting year.

In addition to the government stabilisation packages, Deutsche Lufthansa AG successfully implemented its own financing activities in the financial year, generating cash of EUR 3.8bn for Deutsche Lufthansa AG. Short-term credit lines, commercial paper and money-market securities were supplemented on favourable terms by a euro bond for EUR 1.0bn, a convertible bond for EUR 0.6bn and loans secured by promissory notes and aircraft for around EUR 0.7bn. They serve both to ensure ongoing liquidity and to refinance funding arrangements that expire in 2020 and 2021. At year-end, Deutsche Lufthansa AG had available liquidity of EUR 2.0bn. A further EUR 4.6bn was available to Deutsche Lufthansa AG from the WSF stabilisation package.

The current corporate forecast expect an end to the crisis situation only from 2024 onwards, whereby the Lufthansa Group assumes that even then only 90 % of the pre-crisis level will be reached. This means the planning is in line with sector forecasts and the most recent IATA study from February 2021, which in a positive scenario also predicts a return to pre-crisis levels in 2024 (and in a negative scenario only in 2025). The restructuring measures initiated shall then lead to a profitable growth path again. Risks remain for the ramp-up phase from 2021 until 2024. The duration of the crisis will mainly depend on when travel restrictions, particularly in North America and other important traffic regions, are loosened or lifted. The effects of potential testing options, progress with international vaccination programmes and the risks of new mutations in the COVID-19 virus will be of crucial consequence in the months ahead. On the other hand, how customers behave once travel restrictions are lifted represents a risk for the further course of business, especially for Corporate Business.

Deutsche Lufthansa AG has prepared a rolling liquidity forecast until the end of 2022 on the basis of the uncertainty mentioned above concerning travel restrictions and customers' travel patterns. The main parameters, "available capacity", "load factors" and "yield", were set over the course of the planning process in agreement with the Group companies. Specifically, they assume that average capacity should reach just under 50% of its pre-crisis level in 2021, increasing to around 80% in 2022. For 2021, the assumption is that state aid will still be needed, in the form of short-time working pay and the reimbursement of social security contributions. The parameters were reviewed against publicly available expert opinions (e.g. IATA, Eurocontrol) to ensure they were reasonable. On this basis, the earnings and cost forecasts for the individual units and the Group as a whole were determined using top-down definition of parameters and bottom-up detailed planning. The resulting cash flows were adjusted for balance-sheet-related payments (e.g. ticket refunds), planned capital expenditure and inflows such as the planned financing measures to calculate changes in liquidity over time. The analysis comprises liquidity for the Group as a whole and for Deutsche Lufthansa AG, as well as the other important subsidiaries.

In terms of financing activities, Deutsche Lufthansa AG issued two further bonds with a maturity of four and seven years for a total of EUR 1.6bn in February 2021. They partly secure the refinancing of all liabilities falling due in 2021 and also enable the early repayment of the KfW credit facility of EUR 1.0bn. This means that certain restrictions on the Group's own financing arrangements no longer apply, such as the hypothecation of Group aircraft leasing companies.

In addition to the detailed forecast described above, an additional negative scenario was prepared as a top-down stress test, with more pessimistic assumptions for the level of capacity and changes over time of around 40% of pre-crisis levels, as well as for load factors and pricing. Even in this scenario, liquidity is secured for Deutsche Lufthansa AG for the planning period.

Taking into account the corporate planning, the liquidity planning, the existing and potential funding measures as presented and the existing uncertainties about the future course of business, the Executive Board of the Company considers the liquidity of Deutsche Lufthansa AG to be secure for the eighteen months after the close of the financial year. These financial statements have therefore been prepared on a going concern basis.

MAIN ESTIMATION ASSUMPTIONS

The outbreak of the coronavirus pandemic and the steps taken worldwide to contain the virus have had a massive impact on the business operations of Deutsche Lufthansa AG. The uncertainties resulting from the crisis are vital for the general assessment of the Company's status as a going concern, but also for specific accounting judgements and estimates. It is therefore of fundamental importance how long and at what intensity the COVID-related restrictions to air travel and the constraints for the economy as a whole will continue, and over which period a recovery will take place. At the same time, it is not possible to say what the new normal level will be for air transport in general and Deutsche Lufthansa AG in particular. The main estimated assumptions were therefore based on the liquidity and profit forecasts of Deutsche Lufthansa AG and those of its significant equity investments. Critical accounting areas that are affected most severely by the ongoing uncertainty about the further course of the pandemic are:

- Carrying amount of equity investments, which depend to a large degree on the speed of the recovery and the level of post-crisis business.
- Carrying amount of aircraft, which are affected by uncertainties about future capacities and so the extent to which the fleet can be used.
- Carrying amount of deferred tax assets, especially in relation to the tax loss carry-forwards created in the financial year for which a potential use over a five-year period was used as a basis.
- Financial instruments, for which assumptions were made about the volume of future items to be hedged (particularly kerosene consumption) when judging effective hedging relationships and so the necessity of recognising gains and losses through profit or loss.
- Accounting for unused flight documents, for which there was – and will continue to be – greater uncertainty about how customers would redeem miles or use tickets. Estimating the expected expiration of tickets will largely depend on when and to what extent air travel is possible again.

CURRENCY TRANSLATION

In-house conversion rates for foreign currencies are set monthly in advance according to the exchange rates on international markets. These serve as the basis for convert-

ing foreign currency items into euros in the month in which entries are made.

Receivables / liabilities in foreign currencies, cash and provisions are translated at the mean spot rate on the reporting date in accordance with Section 256a HGB. For other non-current receivables / liabilities in foreign currency, the lower / higher-of-cost-or-market principle is observed by comparing the purchase cost with the value on the balance sheet date.

The cost of capital goods purchased in foreign currencies – mainly aircraft invoiced in US dollars – is determined by translation at the exchange rates in effect at the time of payment. Assets for which payments are hedged against exchange rate fluctuations are recognised within the framework of separate valuation units.

Fair value and cash flow hedges of interest rate, exchange rate and fuel price risks are described in Note 17.

INTANGIBLE ASSETS

Intangible assets are measured at cost and generally amortised on a straight-line basis over five years or their contractual useful lives, whichever is longer. Internally developed intangible assets are not capitalised. Purchased take-off and landing rights are not amortised unless permanently impaired.

As a rule, acquired goodwill is amortised over the expected useful life of five to fifteen years. This is based on the expected benefit of the businesses acquired and is primarily determined by economic factors such as future growth and profit forecasts, synergy effects and workforce.

PROPERTY, PLANT AND EQUIPMENT

Straight-line depreciation of property, plant and equipment is based on the purchase and manufacturing costs depreciated over the asset's expected useful life. Interest on liabilities is not recognised as part of the purchase or manufacturing costs.

Movable assets with a finite useful life and acquisition costs of up to EUR 250 are depreciated in full in the year of purchase. Minor capital goods costing between EUR 251 and EUR 1,000 are pooled in an annual tax account and in the commercial balance sheet for reasons of simplicity. They are depreciated on a straight-line basis over five years.

AIRCRAFT

New commercial aircraft have been depreciated on a straight-line basis over a period of 20 years to a residual value of 5%.

Aircraft purchased in used condition are depreciated individually on a straight-line basis depending on their age at the time of acquisition. Aircraft less than 16 years old at the time of acquisition are depreciated up to an age of 20 years to a residual carrying amount of 5%. Aircraft more than 16

years old at the time of acquisition are depreciated in full over four years without any residual value.

Aircraft are either the legal property of the Company or are leased from aircraft holding entities in which the Company holds a direct or indirect equity interest or from external third parties. Leased aircraft are recognised as non-current assets when the Company is deemed to have economic ownership of them. Economic ownership is determined on the basis of general commercial law and the tax provisions concerning leased assets, if applicable.

OTHER PROPERTY, PLANT AND EQUIPMENT

Buildings are assigned a useful life of between 20 and 50 years. Buildings and installations on land belonging to third parties are depreciated on a straight-line basis according to the term of the lease or are assigned a shorter useful life. Operating and office equipment is depreciated over three to fourteen years on a straight-line basis in normal circumstances of usage.

FINANCIAL INVESTMENTS

Financial investments are shown at cost, adjusted by any necessary impairment charges or write-ups. No write-downs are recognised if the impairment is not permanent.

CURRENT ASSETS

Raw materials, consumables and supplies are valued at cost, with stock risks being accounted for by appropriate mark-downs.

Other current securities are recognised at cost or, if applicable, at lower values as per stock exchanges or market prices on the reporting date, in accordance with Section 253 Paragraph 4 HGB.

Emissions certificates issued free of charge are held at a residual amount; those purchased are held at acquisition cost.

Receivables and other assets are recognised at their nominal value.

In addition to individual write-downs necessary for known risks applying to other current assets, adequate provision is made for general credit risk by a write-down of each item by a standard amount. The standardised write-downs on trade receivables reflect previous defaults, days past due, the business model and the region of the customer.

PENSION OBLIGATIONS

To meet retirement benefit obligations, phased early retirement obligations and claims on employees' lifetime working hours accounts, appropriate funds have been invested in insolvency-proof funds and insurance policies, which are not accessible to the Company's other creditors.

Pension assets are measured at fair value using external price information and netted out with the underlying obliga-

tions. If there is an excess of obligations over assets, it is recognised in provisions. If the time value of the relevant pension assets exceeds that of the corresponding obligations, the difference is shown separately as “excess of plan assets over provisions for pensions”. If the fair value of the relevant pension assets is higher than their historical acquisition cost, the resulting income may not be distributed as a dividend (Section 268 Paragraph 8 Sentence 3 HGB).

PROVISIONS

Pension obligations are calculated using actuarial principles based on the projected unit credit method using the Heubeck 2018 G actuarial tables. As well as appropriate projected rates of fluctuation and career progress, a salary trend of 2.5% as well as a basic pension trend of 1% and transitional benefits for cockpit staff of 2.5% are used, as in the previous year.

Discounting took place at the average market interest rate for the past ten years with an assumed term to maturity of 15 years as published by the German Bundesbank. For measurement as of 31 December 2020, the corresponding interest rate is forecast as of 31 December 2020 on the basis of interest rate information published as of 30 November 2020. In the reporting year, the rate was 2.30% (previous year: 2.71%). The effect of this interest rate change is recognised in interest expense. The difference between the amount of provisions calculated using the ten-year and the seven-year average interest rate as of 31 December 2020 may not be distributed as a dividend. As of 31 December 2020, the seven-year average interest rate used to calculate this difference was 1.60% (previous year: 1.97%).

Benefit obligations from retirement benefit commitments that are funded by reinsurance or capital market investments are recognised at the fair value of the underlying securities, insofar as this amount exceeds the present value of the guarantee.

The provision for partial retirement agreements is recognised at the amount needed to settle the obligation. This amount is composed of the salary outstanding as of 31 December 2020, which is paid during the early retirement phase, as well as the superannuation premiums comprising the salary portion and the additional employer contributions to statutory pension insurance. The provision is calculated making reasonable use of biometric probabilities and a salary trend of 0.83%, which was reduced as a result of the crisis. It is discounted on the basis of average terms to maturity at a seven-year average interest rate forecast as of 31 December 2020. In the reporting year, the rate was 0.54% (previous year: 0.72%).

The other provisions are made for the amount considered necessary to settle the obligations using sound commercial judgement, including future cost and price increases. Provisions with a term to maturity of more than one year are discounted at the average market interest rate for the past seven years corresponding to their remaining term.

LIABILITIES

Liabilities are shown at the amount needed to settle them. Advance payments received are recognised at their nominal value.

DEFERRED TAXES

For temporary or quasi-permanent differences between the valuations of assets, liabilities and deferred income in the financial statements for commercial law and tax purposes, or resulting from tax loss carry-forwards, these are measured and recognised using the individual tax rates at the time when the differences are removed. Deferred tax assets and liabilities are recognised as a net amount. Deutsche Lufthansa AG not only recognises differences resulting from items in its own balance sheet, but also for companies in the same income tax group.

Tax loss carry-forwards are recognised to the extent that the deferred tax assets are likely to be used in the future. Company earnings forecasts are used to determine whether deferred tax assets from tax losses carried forward are usable or not, i.e. whether they have a value that can be realised. A five-year period is used to account for deferred taxes on loss carry-forwards.

VALUATION OPTIONS

To improve the presentation of the net assets, financial and earnings position, the option offered by Section 274 Paragraph 1 Sentence 2 HGB of capitalising the net asset of EUR 3,207m resulting from offsetting deferred tax assets and liabilities has been used.

To improve the presentation of the earnings position, instruments to hedge the price of future fuel requirements, foreign currency hedging transactions to hedge exchange rates as well as interest rate hedges for interest-bearing financial liabilities are combined with corresponding hedged items within valuation units in accordance with Section 254 HGB. Possible onerous contracts in the form of a valuation unit are calculated in line with sales markets, so that, according to the principal of loss-free valuation, no impending losses are recognised, insofar as no loss is incurred from future sales business.

Interests which are acquired through a contribution in kind or premium in kind are usually measured as acquisition costs at the time value of the asset contributed. The time value is normally calculated using generally accepted valuation methods (e.g. as the value of future income based on the discounted cash flow method) while applying the principles of IDW S 1.

NOTES TO BALANCE SHEET

Assets

3 Non-current assets

Changes in individual non-current asset items during the financial year 2020 are shown in a separate table.

With effect from 1 January 2020, the continuing airworthiness management organisation ("CAMO") and line maintenance functions of Lufthansa Technik AG were transferred to Deutsche Lufthansa AG as part of an asset deal. Goodwill of EUR 103m was capitalised in connection with this. Due to the presence of estimation uncertainties, the standardised useful life of 10 years was applied for amortisation.

In addition to the Company's own aircraft listed in the statement of changes in non-current assets and in the balance sheet, further aircraft were chartered, in some cases complete with crews. In addition, the following aircraft leased exclusively from Group companies are in service for Deutsche Lufthansa AG:

T05 NUMBER OF LEASED AIRCRAFT

Aircraft type	31.12.2020	31.12.2019
Airbus A319-100	24	25
Airbus A320-200	49	48
Airbus A321-100	20	20
Airbus A321-200	29	28
Airbus A330-300	19	15
Airbus A340-300	17	17
Airbus A340-600	13	13
Airbus A380-800	5	5
Boeing 747-400	10	13
Boeing 747-8	17	17
Bombardier CRJ 900	6	6
Embraer 190	9	9
Embraer 195	17	17
	235	233

4 Financial investments

The main indirect and direct equity investments of Deutsche Lufthansa AG can be found in the annexe to the notes, "List of shareholdings".

Deutsche Lufthansa AG approved and conducted a capital increase at its wholly owned equity investment Lufthansa Commercial Holding GmbH in the financial year. In connection with this, Deutsche Lufthansa AG increased the share capital of Lufthansa Commercial Holding GmbH through a capital increase by issuing a new shareholding for a nominal amount of EUR 100 and acquiring the new shareholding itself. In addition to the obligation to pay the nominal amount of the new shareholding, payment of an additional

premium in kind as further consideration for granting the new shareholding was agreed between Deutsche Lufthansa AG, Lufthansa Commercial Holding GmbH and AirTrust AG. As part of this, Deutsche Lufthansa AG ceded its rights in relation to its wholly owned equity investment AirTrust AG to Lufthansa Commercial Holding GmbH and assigned and transferred the shares of AirTrust AG to Lufthansa Commercial Holding GmbH.

With regard to the wholly owned equity investment in Crane Strategic Investment S.C.S, a capital reduction of EUR 1,779m was conducted in the financial year as a liquidity measure of Deutsche Lufthansa AG.

Lufthansa Commercial Holding GmbH also conducted a capital reduction of EUR 2,550m in the financial year as an additional liquidity measure.

5 Inventories

T06 INVENTORIES

in €m	31.12.2020	31.12.2019
Raw materials, consumables and supplies	11	14
Emission certificates	90	80
Merchandise	10	14
	111	108

6 Receivables and other assets

T07 RECEIVABLES AND OTHER ASSETS

in €m	31.12.2020	thereof due after more than one year	31.12.2019	thereof due after more than one year
Trade receivables	102	–	642	–
Receivables from affiliated companies	488	7	801	–
Receivables from companies held as other equity investment	4	–	4	–
Other assets	807	162	920	204
	1,401	169	2,367	204

Receivables from affiliated companies primarily relate to financial receivables. Receivables from affiliated companies include trade receivables from affiliated companies of EUR 69m (previous year: EUR 323m).

7 Securities and liquid assets

Money market funds valued at EUR 3,655m and primarily managed by Amundi, BNP Paribas and Morgan Stanley were held as of the reporting date. Cash in hand and bank balances consist almost entirely of deposits held with banks.

Foreign currency bank balances of EUR 32m (previous year: EUR 13m) that are not likely to be transferred in the near future and which are discounted appropriately are reported as other assets.

8 Prepaid expenses

This item essentially consists of payments of EUR 32m to several aircraft leasing companies for future maintenance and leasing services (previous year: EUR 18m).

9 Deferred tax assets

This item consists of the net asset balance of EUR 3,207m remaining after deferred tax assets and liabilities on temporary differences between carrying amounts for commercial law and tax purposes have been offset. They are made up as follows:

T08 DEFERRED TAX ASSETS AND LIABILITIES

in €m	31.12.2020	31.12.2019
Non-current assets	92	3
Inventories, receivables and other assets	293	151
Provisions	2,362	1,942
Liabilities	5	46
Loss carried forward	455	–
	3,207	2,142

Deferred tax assets result primarily from different amounts of pension provisions and similar obligations, other provi-

sions and inventories, and, for the first time in the reporting year, tax loss carry-forwards. Deferred tax liabilities, mainly arising from different valuations of aircraft and other equipment, are more than offset by deferred tax assets.

In addition to recognised deferred tax assets from tax loss carry-forwards, further tax loss carry-forwards exist for which no deferred tax assets in the amount of EUR 746m (previous year: EUR 0m) could be recognised.

Deferred taxes are calculated using the individual tax rates for Deutsche Lufthansa AG's tax group, which are between 24% and 31%. The tax rate used in each case comprises corporation tax, trade tax and the solidarity surcharge.

10 Excess of plan assets over provisions for pensions

The excess results from offsetting retirement benefit obligations in accordance with Section 264 Paragraph 2 Sentence 2 HGB against assets that are exclusively used to fund the retirement benefit obligations and are not accessible to the Company's other creditors. The assets in question are securities. No excess of plan assets over provisions for pensions was recognised in the reporting year.

T09 AKTIVER UNTERSCHIEDSBETRAG AUS DER VERMÖGENS-VERRECHNUNG

in Mio. €	31.12.2020	31.12.2019
Erfüllungsbetrag der Verpflichtungen aus Pensionszusagen	–	2,556
beizulegender Zeitwert des Sicherungsvermögens	–	2,563
Überschuss des Vermögens über die Verpflichtung aus Pensionszusagen (aktiver Unterschiedsbetrag)	–	7
Anschaffungskosten des Sicherungsvermögens	–	2,289

Shareholders' equity and liabilities

11 Issued capital

SHARE CAPITAL

By resolution of the extraordinary general meeting on 25 June 2020, the issued capital of Deutsche Lufthansa AG was increased by EUR 306,044,326.40 to EUR 1,530,221,624.32 by issuing 119,548,565 registered shares excluding shareholders' subscription rights. The shares were bought by the Economic Stabilisation Fund at the nominal amount of EUR 2.56 per share. The capital increase took effect on 2 July 2020 when it was entered in the commercial register. Issued capital is divided into 597,742,822 registered shares, with each share representing EUR 2.56 of issued capital.

AUTHORISED CAPITAL

A resolution passed at the Annual General Meeting on 7 May 2019 authorised the Executive Board until 6 May 2024, subject to approval by the Supervisory Board, to increase the Company's issued capital by up to EUR 450,000,000 by issuing new registered shares on one or more occasions for payment in cash or in kind (Authorised Capital A). In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board.

A resolution passed at the Annual General Meeting on 7 May 2019 authorised the Executive Board until 6 May 2024, subject to approval by the Supervisory Board, to increase the issued capital by up to EUR 30,000,000 by issuing new registered shares to employees (Authorised Capital B) for payment in cash. Existing shareholders' subscription rights are excluded. As of 31 December 2020, shares with a nominal amount of EUR 7,637,832.68 were issued under this authorization, so that Authorised Capital B still amounted to EUR 22,362,168.32 as of the reporting date.

CONTINGENT CAPITAL

A resolution of the Annual General Meeting on 5 May 2020 increased the company's contingent capital by up to EUR 122,417,728. The contingent capital increase serves to issue up to 47,819,425 shares to the holders or creditors of conversion and/or option rights from convertible bonds with a total nominal value of up to EUR 1.5bn that may be issued by the Company or its Group companies until 4 May 2025 in accordance with the resolution of the Annual General Meeting on 5 May 2020. In certain cases, the shareholders' subscription rights can be excluded with the approval of the Supervisory Board.

A resolution of the extraordinary general meeting on 25 June 2020 increased the contingent capital of Deutsche Lufthansa AG by up to EUR 102,014,776.32. The contingent capital increase serves to provide up to 39,849,522 shares for the exercise of conversion rights granted to the Economic Stabilisation Fund created by the Stabilisation Fund Act as a silent shareholder for Silent Participation II-A at a strike

rate of EUR 2.56 per share by resolution of the extraordinary general meeting on 25 June 2020. The rights can be exercised if a decision is published to make a takeover offer pursuant to Section 10 Securities Acquisition and Takeover Act (WpÜG) or if control is acquired pursuant to Sections 35 and 29 WpÜG. The buyer can exercise the conversion rights at any time if the Silent Participation II-A is sold to a private purchaser.

A resolution of the extraordinary general meeting on 25 June 2020 increased the contingent capital of Deutsche Lufthansa AG by up to EUR 897,985,223.68. The contingent capital increase serves to provide up to 350,775,478 shares for the exercise of conversion rights granted to the Economic Stabilisation Fund created by the Stabilisation Fund Act as a silent shareholder for antidilution and/or coupon protection for Silent Participation II-B by resolution of the extraordinary general meeting on 25 June 2020. If the conversion right is exercised to protect against dilution, the new shares will be issued at the current market price on the conversion date, less 10%. If the conversion right is exercised to protect the coupon, they are issued at the current market price on the conversion date, less 5.25%. The conversion rights expire if Silent Participation II-B is assigned to a third party.

AUTHORISATION TO PURCHASE TREASURY SHARES

A resolution passed at the Annual General Meeting held on 7 May 2019 authorised the Executive Board pursuant to Section 71 Paragraph 1 No. 8 Stock Corporation Act (AktG) to purchase treasury shares until 6 May 2024. The authorisation is limited to 10% of current issued capital, which can be purchased on the stock exchange or by a public purchase offer to all shareholders. The authorisation states that the Executive Board can use the shares, in particular, for the purposes defined in the resolution passed at the Annual General Meeting. According to the resolution of the Annual General Meeting held on 7 May 2019, the Executive Board is also authorised to purchase treasury shares by means of derivatives and to conclude corresponding derivative transactions.

At various points in 2020, Deutsche Lufthansa AG bought back 1,299,165 of its own shares at an average price of EUR 7.27. This corresponds to 0.22% or EUR 3,325,862.40 of issued capital.

The shares purchased were used as follows:

- 1,296,504 shares were transferred to the employees of Deutsche Lufthansa AG and to 33 other affiliated companies and equity investments as part of the profit-sharing for 2019, at a share price of EUR 7.56.
- 517 shares were transferred for employees of Deutsche Lufthansa AG and another affiliated company as part of the profit-sharing for 2018, at a share price of EUR 14.18.
- 2,144 shares were resold at a price of EUR 11.14.

On the balance sheet date, treasury shares were no longer held.

SHAREHOLDER STRUCTURE

Notifications received in accordance with Section 33 Paragraph 1 Securities Trading Act (WpHG) on changes in voting rights in the share capital held by third parties are disclosed in abbreviated form below. If a notifying party reaches, exceeds or falls below the thresholds defined in Section 33 Paragraph 1 WpHG more than once, only the last notification that the threshold has been reached or exceeded or is no longer reached is disclosed below.

- On 30 May 2016, Deutsche Asset & Wealth Management Investment GmbH, Frankfurt, Germany, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 19 May 2016 and on this date came to 2.90% (13,471,331 voting shares). 2.90% (13,471,331 voting shares) were attributable to Deutsche Asset & Wealth Management Investment GmbH in accordance with Section 34 WpHG (previously Section 22 WpHG).
- On 14 November 2016, Franklin Templeton Investment Funds, Luxembourg, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 8 November 2016 and on this date came to 2.96% (13,861,910 voting shares). 2.96% (13,861,910 voting shares) are attributable to Franklin Templeton Investment Funds directly in accordance with Section 33 WpHG (previously Section 21 WpHG).
- On 14 November 2016, Franklin Templeton International Services S.à r.l., Luxembourg, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 8 November 2016 and on this date came to 2.96% (13,861,910 voting shares). 2.96% (13,861,910 voting shares) are indirectly attributable to Franklin Templeton International Services S.à r.l. in accordance with Section 34 WpHG (previously Section 22 WpHG).
- On 22 March 2017, Templeton Investment Counsel, LLC, Fort Lauderdale, Florida, USA, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 17 March 2017 and on this date came to 2.98% (13,959,454 voting shares). 2.98% (13,959,454 voting shares) were indirectly attributable to Templeton Investment Counsel, LLC, in accordance with Section 34 WpHG (previously Section 22 WpHG).
- Templeton Growth Fund, Inc., Maryland, USA, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 6 April 2017 and on this date came to 2.90% (13,611,171 voting shares) in accordance with Section 33 WpHG (previously Section 21 WpHG).
- Templeton Global Advisors Limited, Nassau, Bahamas, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 13 June 2017 and on this date came to 2.87% (13,508,944 voting shares). 2.87% (13,508,944 voting shares) were indirectly attributable to Templeton Growth Fund, Inc., in accordance with Section 34 WpHG (previously Section 22 WpHG).
- Norges Bank, Oslo, Norway, notified us that its voting rights in Deutsche Lufthansa AG exceeded the threshold of 3% on 24 April 2019 and on this date came to 3.20% (15,199,406 voting shares). 3.20% (15,199,406 voting shares) were indirectly attributable to Norges Bank in accordance with Section 34 WpHG.
- Amundi S.A., Paris, France, notified us that its voting rights in Deutsche Lufthansa AG exceeded the threshold of 3% on 31 July 2019 and on this date came to 3.15% (14,952,168 voting shares). 3.15% (14,952,168 voting shares) were indirectly attributable to Amundi S.A. in accordance with Section 34 WpHG.
- Amundi S.A., Paris, France, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 19 September 2019 and on this date came to 2.69% (12,784,379 voting shares). 2.69% (12,784,379 voting shares) were indirectly attributable to Amundi S.A. in accordance with Section 34 WpHG.
- BlackRock, Inc., Wilmington, Delaware, USA, notified us that its voting rights in Deutsche Lufthansa AG exceeded the threshold of 3% on 2 March 2020 and on this date came to 3.09% (14,780,221 voting shares). 3.09% (14,780,221 voting shares) are attributable to BlackRock, Inc., indirectly in accordance with Section 34 WpHG.
- BlackRock, Inc., Wilmington, Delaware, USA, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 4 March 2020 and on this date came to 2.81% (13,439,503 voting shares). 2.81% (13,439,503 voting shares) are attributable to BlackRock, Inc., indirectly in accordance with Section 34 WpHG.
- Lansdowne Partners International Ltd., George Town, Grand Cayman, Cayman Islands, notified us that its voting rights in Deutsche Lufthansa AG exceeded the threshold of 5% on 17 May 2019 and on this date came to 5.08% (24,124,898 voting shares). 5.08% (17,045,008 voting shares) were attributable to Lansdowne Partners International Ltd. indirectly in accordance with Section 34 WpHG.
- Lansdowne Partners International Ltd., George Town, Grand Cayman, Cayman Islands, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 3% on 6 March 2020 and on this date came to 2.86% (13,665,474 voting shares). 2.86% (13,665,474 voting shares) were attributable to Lansdowne Partners International Ltd. indirectly in accordance with Section 34 WpHG.
- KB Holding GmbH, Grünwald, on behalf of Heinz Hermann Thiele, notified us that its voting rights in Deutsche Lufthansa AG exceeded the threshold of 15% on 15 June 2020 and on this date came to 15.52% (74,230,000 voting shares). 15.52% (74,230,000 voting shares) were indirectly attributable to KB Holding GmbH in accordance with Section 34 WpHG.

- KB Holding GmbH, Grünwald, on behalf of Heinz Hermann Thiele, notified us that its voting rights in Deutsche Lufthansa AG fell below the threshold of 15% on 2 July 2020 and on this date came to 12.42% (74,230,000 voting shares). 12.42% (74,230,000 voting shares) were indirectly attributable to KB Holding GmbH in accordance with Section 34 WpHG.
- The Economic Stabilisation Fund (WSF), Berlin, on behalf of the Federal Republic of Germany, notified us that its voting rights in Deutsche Lufthansa AG exceeded the threshold of 20% on 2 July 2020 and on this date came to 20.05% (119,864,299 voting shares). 20.05% (119,864,299 voting shares) were indirectly attributable to the Economic Stabilisation Fund (WSF) in accordance with Section 34 WpHG.

For further details, we refer to the individual notifications on voting rights published on our website www.lufthansa.com.

12 Reserves

The capital reserve contains the premiums resulting from capital increases and the proceeds from the issue of debt securities for conversion options to acquire Company shares. Since the nominal share value was used for the capital increase conducted in the 2020 financial year as part of the stabilisation measures of the Economic Stabilisation Fund (WSF), the capital reserve was unchanged compared with the previous year.

The legal reserve contained in retained earnings is unchanged at EUR 26m; other reserves consist of other retained earnings. EUR 298m was transferred to retained earnings in the reporting year.

An amount of EUR 5,320m may not be distributed as dividends. This is made up of EUR 3,207m from the recognition of deferred tax assets for temporary differences between the valuations for commercial and tax purposes as well as on loss carry-forwards, EUR 1,418m from the difference between the application of ten-year or seven-year average interest rates to discount the pension obligations and EUR 695m from the amount by which the fair value of plan assets exceeds their cost. There are still free retained earnings of EUR 398m to cover the amount that may not be distributed.

13 Provisions

T10 PROVISIONS

in €m	31.12.2020	31.12.2019
Provisions for pensions and similar obligations	2,230	1,066
Tax provisions	523	287
Obligations in respect of unused flight documents	1,218	2,290
Other provisions	2,893	2,959
	6,864	6,602

A Company pension scheme exists for staff working in Germany and staff seconded abroad. Benefit obligations are mainly funded by means of contributions to an external trust fund to which access is restricted.

There are also obligations from the conversion of salary components that are funded by assets held in insurance policies. The actuarial obligations are netted with the corresponding assets measured at fair value as of 31 December 2020 to obtain the carrying amount for the balance sheet. The acquisition costs of the fund assets were EUR 9,308m as of 31 December 2020. Their fair value as of the same date was EUR 9,921m. The fund assets which can be netted amount to EUR 9,901m. The actuarial amount required to settle the obligation as of 31 December 2020 is EUR 12,131m.

In 2016, a new company retirement benefit plan for ground staff employed in Germany was established in the form of a defined-contribution benefit commitment. For employees recruited before 1 January 2016, the entitlements vested up until 31 December 2015 are maintained. For service periods starting from 1 January 2016, employees can reach the same level of benefits by making contributions from their own pocket. For employees recruited from 1 January 2016, the contributions to the new model will be invested on the capital market. When the employee reaches retirement age, the entire account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum and while guaranteeing the contributions that were originally made.

In 2017, a new pension agreement was concluded for cabin crew. It includes the provision that employees who started work at Lufthansa before 30 June 2016 maintain their pension entitlements vested up to this point. For service periods from 1 July 2016, the employees receive employer contributions to the Company pension scheme depending on their eligible gross salary. In addition, the previous entitlements to transitional benefits were replaced by an initial contribution to the new capital-market-based benefits system, calculated as of the date on which the change was made. All employees are free to make their own contributions on a voluntary basis. Contributions from both employer and employee, as well as the initial transitional benefit contribution, are invested on the capital markets with a capital guarantee. When the employee reaches retirement age, the available account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum.

In 2017, new pension agreements were concluded for cockpit employees for their retirement and transitional benefits. For employees recruited before 1 January 2017, the pension entitlements vested up until 31 December 2016 are maintained. For service periods from 1 January 2017, the employees receive employer contributions to the Company pension scheme depending on their eligible gross salary. All employees are free to make their own contributions on a voluntary basis. The capital is invested on capital markets with a capital guarantee, in addition to the guaranteed interest rate

offered by life insurance companies (currently 0.9% p.a.) for cockpit crew as an extra commitment. When the employee reaches retirement age, the available account balance is converted into an annuity on the basis of the applicable BilMoG interest rate, subject to a pension adjustment of 1% per annum.

The old rules fundamentally continue to apply to transitional benefits. The collective retirement age for pilots increases gradually from 58 to 60 by 2021. In connection with the forecast long-term staff surplus among cockpit crew, the Company has proposed a temporary suspension of the clause relating to the collective retirement age. A corresponding agreement has not yet been reached on a change to the wage settlement as part of an overall agreement. This results in an increase of EUR 41m in the transitional benefit obligation in the reporting year.

In the new Company retirement benefit scheme for ground, cabin and cockpit staff, the obligations from the capital market components are recognised at the fair value of the corresponding plan assets and are offset against these, insofar as they exceed the minimum guaranteed amount. The employer contributions recognised in staff costs constitute service expense.

Under the crisis collective bargaining agreements concluded in the reporting year, reductions in contributions to the Company retirement benefit scheme have been agreed with cabin and cockpit staff.

As part of the transfer of operations from Lufthansa Technik AG to Deutsche Lufthansa AG disclosed under item 3, employee obligations of EUR 348m (essentially pension obligations) and corresponding fund assets of EUR 241m were transferred to Deutsche Lufthansa AG.

Accrued interest expenses on provisions and expenses from measuring the obligation with a lower discount rate than in the previous year of EUR 652m, as well as expenses of EUR 361m from the negative market valuation of plan assets were recognised in net interest.

The change from a seven-year average interest rate to a ten-year average for calculating the settlement amounts resulted in a difference of EUR 1,418m as of 31 December 2020, which is not available for distribution.

Obligations under partial retirement agreements are recognised in other provisions. Obligations of EUR 95m are netted with fund assets with a fair value of EUR 46m. The acquisition costs of the fund assets are EUR 51m. Accrued interest expenses on provisions of EUR 0.4m, as well as expenses of EUR 3.8m from the negative market valuation of pension fund assets were recognised.

Working hours accounts have been managed for cabin crew since 2017 and had a value of EUR 81m as of the reporting date. They are offset by plan assets with an identical amount. The acquisition costs of the fund assets are EUR 81m.

Other provisions essentially comprise amounts for aircraft maintenance (EUR 1,510m) and provisions for impending losses (EUR 461m). This includes a provision for impending losses of EUR 337m in connection with the lease agreements with aircraft leasing companies. The measures to cut flight capacities in the longer term will have an impact on leases insofar as the obligation to make lease payments for the aircraft affected is expected to outweigh their contribution to the Company's success.

T11 RÜCKSTELLUNGEN FÜR PENSIONEN

in Mio. €	31.12.2020	31.12.2019
Erfüllungsbetrag der Verpflichtungen aus Pensionszusagen	12,131	8,721
beizulegender Zeitwert des Sicherungsvermögens	9,901	7,655
Nettowert der Verpflichtungen aus Pensionszusagen	- 2,230	- 1,066
Anschaffungskosten des Sicherungsvermögens	9,308	6,723

14 Liabilities

T12 LIABILITIES

in €m	31.12.2020	thereof due			31.12.2019	thereof due		
	Total	within one year	between one and five years	after more than five years	Total	within one year	between one and five years	after more than five years
Bonds*	2,827	227	1,100	1,500	1,098	98	1,000	-
Liabilities to banks	4,215	2025	2,156	34	2,089	665	1,390	34
Payables to affiliated companies	4,659	4,659	-	-	6,094	6,094	-	-
Other liabilities	6,067	1,104	2,262	2,701	5,135	1,470	1,686	1,979
thereof Advance payments for orders	6	6	-	-	6	6	-	-
thereof Trade payables	316	316	-	-	697	697	-	-
thereof Payables to affiliated companies	2	2	-	-	6	6	-	-
thereof other remaining liabilities	5,743	780	2,262	2,701	4,426	761	1,686	1,979
thereof for taxes	99	99	-	-	92	92	-	-
thereof relating to social security obligations	2	2	-	-	3	3	-	-
	17,768	8,015	5,518	4,235	14,416	8,327	4,076	2,013

*Share of convertible bonds amounts to EUR 600m (previous year: 0)

Liabilities to affiliated companies include trade payables to affiliated companies of EUR 41m (previous year: EUR 30m).

Under the framework agreement to implement stabilisation measures between the Economic Stabilisation Fund on behalf of the Federal Republic of Germany and Deutsche Lufthansa AG, a syndicated credit facility of up to EUR 3bn was granted for a term of three years with the participation of KfW and private banks. Under the stabilisation funding granted as borrowings in Switzerland, Austria and Belgium for other companies in the Group, the loan portion for Deutsche Lufthansa AG was EUR 1bn. The shares in Austrian and Maltese aircraft leasing companies, which operate a total of 323 aircraft in the Group, were pledged as collateral.

In addition, the stabilisation fund granted a silent participation in July 2020 that excludes loss-sharing. It has an amount of EUR 1bn, which is recognised under other liabilities.

In November 2020, Deutsche Lufthansa AG also decided not to exercise the first right to call its EUR 500m hybrid bond, which matures on 12 August 2075 and carries an interest rate of 5.125%. The right of termination can be exercised again as of 12 February 2026. Furthermore, the coupon will be reset on 12 February 2021.

In November 2020, Deutsche Lufthansa AG issued a EUR 1bn bond under its existing EMTN programme. The bond matures on 29 May 2026 and pays interest of 3.0% per annum.

In November 2020, Deutsche Lufthansa AG issued a convertible bond with a total volume of EUR 600m. Unless previously converted, the bonds will be redeemed at their nominal value in November 2025. Investors also have the option of converting the bond into new and/or existing registered shares of Lufthansa AG. The initial conversion price was set at EUR 12.96, which corresponds to a conversion premium of 40.0% above the reference share price of EUR 9.2545.

EUR 252m of the liabilities to banks were secured by aircraft. Of the other liabilities, obligations arising from finance leases totalling EUR 3,362m are secured by the aircraft concerned.

15 Contingent liabilities

T13 CONTINGENT LIABILITIES

in €m	31.12.2020	31.12.2019
Relating to guarantees, bills of exchange and cheque guarantees	775	1,819
thereof to affiliated companies	117	195
Relating to warranty agreements	342	400
thereof to affiliated companies	108	111
thereof to joint ventures	188	244
Relating to the provision of collateral for third-party obligations	0	25

The amounts listed under liabilities from guarantees include EUR 657m in co-debtors' guarantees given in favour of North American fuelling and handling companies. There was no requirement to recognise these guarantee obligations as a liability, because the fuelling and handling companies are expected to be able to meet the underlying liabilities and a claim is therefore unlikely. Furthermore, this amount is matched by compensatory claims against the other co-debtors amounting to EUR 644m. These amounts are in some cases preliminary, since current financial statements from some counterparties are not available yet.

EUR 64m of the liabilities relating to warranties refer to co-debtors' guarantees in favour of the Terminal One Group Association, L.P. joint venture at New York's JFK Airport. No provision was recognised, as a claim is not considered likely. Based on the Company's current forecasts, the assumption is that the Company will be able to meet its obligations itself. The obligations under the warranties are matched by compensatory claims against the other co-debtors amounting to EUR 48m. The liabilities under warranties to joint ventures include bank guarantees from the financing of two B777 freighter aircraft in service at Aerologic GmbH.

16 Other financial obligations

Financial obligations on the basis of order commitments and loan commitments with no long-term ongoing obligations came to EUR 13,323m as of the reporting date. For ongoing obligations, in some cases with terms of up to 30 years, there were expenses of EUR 1,195m in the reporting year.

ORDER COMMITMENTS

Order commitments for capital expenditure on property, plant and equipment came to EUR 11,984m as of 31 December 2020. Of the corresponding payment obligations, EUR 10,875m falls due in the years 2021 to 2025, and EUR 1,110m falls due in the years 2026 and 2027.

Obligations to acquire Company shares and to contribute capital to investee companies totalling EUR 201k existed as of the reporting date. There were also loan commitments of EUR 1,338m towards affiliated companies.

OBLIGATIONS UNDER TENANCY AGREEMENTS

The Company carries on its business almost exclusively in rented premises. Rental agreements generally run for up to ten years. Facilities at Frankfurt and Munich airports are sometimes rented for longer periods, in some cases for up to 30 years, and are partly prefunded by Lufthansa. Annual rental payments amounted to around EUR 181m in the financial year.

To optimise financing costs, aircraft are regularly leased from affiliated companies and external lessors. Expenses for longer-term operating leases pertaining to aircraft with terms up to 2028 came to EUR 910m in the financial year. It was possible to use them as a qualified assumption for amounts payable annually under these ongoing obligations.

Expenses for operating leases were mainly payable to affiliated companies; EUR 69m was paid to several external lessors (previous year: EUR 65m).

OBLIGATIONS UNDER LONG-TERM MAINTENANCE CONTRACTS

Maintenance contracts for aircraft and aircraft components usually have terms of between 5 and 18 years to secure contractual conditions on a long-term basis. Long-term maintenance contracts with external providers signed as of the reporting date with terms up to 2030 gave rise to expenses of EUR 105m in the financial year. There was a significant decrease in maintenance inspections in the reporting year due to the pandemic. Of the future payment obligations resulting from the long-term maintenance contracts, EUR 455m falls due in the years 2021 to 2025 and EUR 339m falls due in the years 2026 to 2030 if the contractual services are requested as scheduled. A not unrealistic reduction in the contractual services requested will lead to a decrease in the payment obligations.

17 Hedging policy and financial derivatives

As an international airline, Deutsche Lufthansa AG is exposed to the risk of changes in exchange rates, interest rates and fuel priced in US dollars.

EXCHANGE RATE HEDGES

For US dollars, Lufthansa is mainly in a net payer position as regards currency risks from its operating business, since fuel payments are dollar-denominated. There is always a net surplus for other currencies. The main risks in this respect stem from the Chinese renminbi, the Swiss franc, the British pound sterling, the Japanese yen and the Indian rupee. Depending on market liquidity, currency risks from projected operational exposure are hedged gradually over a period of 24 months by means of futures contracts. The target hedging level is defined in the Group's internal guidelines. At the end of 2020, exposure to the major foreign currency items from operations for the next 24 months was as follows:

T14 FOREIGN CURRENCY EXPOSURE FROM OPERATIONS

31.12.2020 in €m	USD	CNY	JPY	GBP	INR
Exposure (currency)	-623	2,091	34,685	390	17,510
Exposure (EUR at spot rate)	-507	261	273	431	195
Hedges (currency)	176	-699	-11,727	-137	-2,397
Hedge ratio	28%	33%	34%	35%	14%
Hedge rate	1.31	8.35	119.86	0.90	88.96

Anticipated macro valuation units are formed prospectively for operational currency hedges in accordance with Section 254 HGB and presented using the net hedge method. The hedged items are the net positions of highly probable future cash flows in foreign currencies from the operating business for each foreign currency and hedging month. Since the target hedging level is always less than the total foreign

currency exposure, the hedges are considered to be fundamentally effective, so no provision for impending losses from valuation units has to be recognised.

Forward currency transactions and swaps are valued individually at their respective forward curve and discounted to the reporting date based on the corresponding interest rate curve. The market prices of currency options are calculated using recognised option pricing models.

The following table shows the market values of external hedges for Deutsche Lufthansa AG and its subsidiaries.

T15 FOREIGN EXCHANGE RATE HEDGES FOR EXPOSURE FROM OPERATIONS

31.12.2020 in €m	Nominal volume	Market value	Maturities up to	Carrying amounts of other provisions
External hedges	3,615	-50	2023	-
External hedges for subsidiaries	3,758	49	2023	-

HEDGED CAPITAL EXPENDITURE

Exchange rate hedges in the form of micro hedges are combined with expected aircraft deliveries to form valuation units for the purpose of hedging the risk of price increases due to exchange rate movements and presented in the balance sheet using the net hedge method. Aircraft purchases are now only hedged by means of forward transactions. The exposure for capital expenditure at year-end 2020, the relevant hedging volume and the effects of the hedges on the acquisition costs of the hedged investments are as follows:

T16 HEDGED USD CAPITAL EXPENDITURE

Year	Exposure in USD	Volume hedged in USD	Market values in €m	Hedge ratio
2021	-2,018	1,779	-56	88%
2022	-2,491	1,967	-75	79%
2023	-3,370	2,199	-50	65%
2024	-1,383	783	7	57%
2025	-1,182	642	12	54%
2026	-947	496	-7	52%
2027	-269	75	2	28%
2028	-213	0	0	0%
2029	-168	0	0	0%
	-12,041	7,941	-167	-66%

In the financial year, long-term foreign currency hedges were realised early, generating cash proceeds of EUR 428m. The amounts will be recognised as liabilities until the hedged transactions occur and will then be allocated to the corresponding investments. Hedging transactions in which the hedged items (aircraft purchases) have been deferred due to the crisis are accounted for in the same way.

INTEREST RATE HEDGES

Suitable interest rate swaps and combined interest rate/currency swaps are arranged with external parties to hedge interest rate risks on bonds, loans and lease liabilities recognised in the balance sheet. They are combined in valuation units as micro hedges and presented in the balance sheet using the net hedge method. Hedged items and hedges have identical maturities, up to 2031 at the latest. As the reciprocal cash flows balance each other out, the interest rate swaps are not presented in the balance sheet. As of 31 December 2020, provisions of EUR 91m are recognised for impending losses regarding interest rate hedges with no hedged items.

Furthermore, Deutsche Lufthansa AG and its subsidiaries have arranged combined interest rate / currency swaps that are matched by interest rate / currency swaps of the same type, volume and maturity with external third parties. They are also combined in valuation units as micro hedges. Hedged items and hedges have identical maturities, up to 2026 at the latest. The hedged cash flows balance each other fully, so that the valuation units are fully effective. As of 31 December 2020, provisions of EUR 5m are recognised for impending losses regarding other internal interest rate hedges.

T17 INTEREST RATE HEDGES

31.12.2020 in €m	Volume hedged	Market value	Maturities up to	Carrying amount of other provision
External hedges with hedged items	5,331	43	2031	-
External hedges without hedged items	940	-90	2028	-91
External hedges for subsidiaries	461	-	2026	-
Internal hedges with subsidiaries	58	-4	2027	-5

The fair values of interest rate derivatives correspond to their respective market values, which are measured using appropriate mathematical methods, such as discounting future cash flows. Discounting takes market standard interest rates and the residual term of the respective instruments into account.

FUEL HEDGING

As of 31 December 2020, exposure to fuel prices was as follows:

T18 FUEL PRICE EXPOSURE

		2021	2022
Fuel exposure	in thousand t	2,558	4,258
Volume hedged	in thousand t	2,455	85
Hedge ratio	in %	96%	2%

Suitable forward transactions, spread options and combinations of hedges are arranged with external counterparties to hedge price risks from future fuel requirements. They have been combined with the hedged items as macro valuation units and presented using the net hedge method to improve presentation of the earnings position.

T19 FUEL PRICE HEDGES

31.12.2020	Volume of hedged items in thousand t	Market value in €m	Maturities up to	Carrying amount of other assets in €m
Range options	2,343	-29	2022	36
Swaps	173	0	2022	-
Hedging combinations	25	-3	2022	0

The collapse of air travel as a result of the pandemic has led to overhedging for some months of 2021 as of the current reporting date. The existing volume of hedging instruments is therefore not fully covered by planned fuel requirements. It is not permitted to recognise valuation units for this share of derivatives. As of 31 December 2020, provisions of EUR 11m are therefore recognised for negative market values in relation to this.

The market prices of options used to hedge fuel prices are determined using acknowledged option pricing models. The market values correspond to the price at which an independent third party would assume the rights and/or obligations from the financial instrument.

FINANCIAL INSTRUMENTS HELD AS FINANCIAL INVESTMENTS

T20 BALANCE SHEET ITEMS - FINANCIAL INVESTMENTS

in €m	Market values 31.12.2020	Carrying amounts 31.12.2020
Shares in affiliated companies	2	2
Loans to affiliated companies	465	465
Other loans	1	1

Shares in affiliated companies measured at market value relate to a wholly owned equity investment in Crane Strategic Investment S.C.S., Belgium. This is an equity investment in a foreign investment vehicle within the meaning of Section 1 Paragraphs 1 and 2 of the German Investment Code (KAGB). As disclosed under Note 3, the liquidity held in the equity investment was almost entirely utilised. The remaining amount is EUR 1.5m and will be utilised in 2021.

Loans to affiliated companies essentially comprise the loan of CHF 500m granted to AirTrust AG in the financial year. As of the reporting date, the market value of this loan was EUR 461m, which is the same as the carrying amount.

NOTES TO THE INCOME STATEMENT

18 Traffic revenue

T21 TRAFFIC REVENUE BY TRAFFIC REGION

in €m	2020	2019
Europe	1,520	5,589
North America	817	4,341
Asia /Pacific	536	2,878
South America	319	959
Africa	201	630
Middle East	109	488
	3,502	14,885

T22 TRAFFIC REVENUE BY SECTOR

in €m	2020	2019
Scheduled	3,117	14,366
Charter	385	519
	3,502	14,885

19 Other revenue

The other revenues made up as follows:

T23 OTHER REVENUE

in €m	2020	2019
Travel services (commissions / fees)	161	513
Aircraft on operating leases	179	195
Matrix allocation	183	252
Ground services / in-flight sales	44	169
Services rendered	126	141
Staff secondment	11	15
Rent for land / buildings	45	55
Other	29	48
	778	1,388

87% of other revenue was generated in Europe (previous year: 81%).

The significant decrease compared with the previous year is due to the sharp decline in the business activities of Deutsche Lufthansa AG in the reporting year as a result of the pandemic.

Revenue from other periods came to EUR 16m in the reporting year and consists largely of income from unused flight documents from prior years.

20 Other operating income

T24 OTHER OPERATING INCOME

in €m	2020	2019
Proceeds on the disposal of non-current assets	5610	13
Exchange rate gains from foreign currency translation	693	1,150
Reversal of provisions	269	178
Compensation received for damages	21	17
Earnings from write-backs on assets	37	58
Other operating income	195	247
	6,825	1,663

Other operating income was primarily due to the recognition of a book gain of EUR 5,608m in connection with the capital contribution at AirTrust AG. The book gain resulted from measuring the additional acquisition costs of the wholly owned equity investment in Lufthansa Commercial Holding GmbH based on the time value of the contributed wholly owned equity investment in AirTrust AG. This led to the release of hidden reserves in the shares of AirTrust AG and therefore to an increase in the shareholders' equity of Deutsche Lufthansa AG. For a detailed description of the capital contribution, we refer to Note 4, page 10.

Income from other periods came to EUR 290m in the reporting year and consisted mainly of write-backs of provisions.

21 Cost of materials and services

T25 COST OF MATERIALS AND SERVICES

in €m	2020	2019
Aircraft fuel and lubricants	959	3,672
Other costs of raw materials, consumables and supplies and goods purchased	62	97
Costs of services purchased	4,050	6,993
	5,071	10,762

The costs of purchased services include a provision for impending losses of EUR 337m in connection with the lease agreements with aircraft leasing companies. The measures to cut flight capacities in the longer term will have an impact on leases insofar as the obligation to make lease payments for the aircraft affected is expected to outweigh their contribution to the Company's success.

The cost of materials and services include expenses from other periods of EUR 46m.

22 Staff costs

T26 STAFF COSTS

in €m	2020	2019
Wages and salaries	1,750	2,405
Social security, pensions and benefit contributions	467	644
thereof for retirement benefits	270	322
	2,217	3,049

T27 AVERAGES NUMBER OF EMPLOYEES

	2020	2019
Flight staff	25,381	25,874
Ground staff	14,070	11,811
Trainees	47	36
	39,498	37,721

The decline in staff costs is due to various effects such as the use of short-time working to preserve jobs, salary waivers and the reduction of variable salary components. The effects with regard to short-time working pertain to reimbursement claims for short-time working pay (EUR 249m) and the social security contributions relating to them (EUR 181m).

Staff costs include expenses from other periods of EUR 5m.

23 Depreciation, amortisation and impairment

Depreciation, amortisation and impairment of intangible assets, aircraft and other property, plant and equipment are detailed in the statement of changes in non-current assets. Impairment losses of EUR 674m were recognised in the financial year. They primarily relate to various mostly four-engine aircraft which have been permanently retired or transferred to long-term parking mode.

Total depreciation and amortisation in 2020 shown in the statement of changes in non-current assets includes amortisation of financial investments.

24 Other operating expenses

Other operating expenses are made up as follows:

T28 OTHER OPERATING EXPENSES

in €m	2020	2019
Exchange rate losses from foreign currency translation	604	713
Expenses for computerised distribution systems	52	277
Travel expenses	76	244
Sales commission paid to agencies	39	241
Advertising and sales promotions	60	211
Rental and maintenance expenses	229	178
Payment system expenses (especially credit card commission payments)	32	160
Matrix allocation	128	159
Auditing, consulting and legal expenses	82	130
Courses / training for flight staff	45	76
Insurance for flight operations	14	25
Impairment charges / depreciation and amortisation for current assets	56	21
Treibstoffpreisübersicherung	467	-
Other operating expenses	458	550
	2,342	2,985

The decline in flight traffic due to the pandemic meant that fuel hedging included overhedging, as a result of which hedging relationships previously designated in measurement units had to be terminated early. The corresponding hedges were therefore accounted for as stand-alone derivatives. For the 2020 financial year, the realised result of expired, stand-alone derivatives was EUR -456m, which is recognised under the special item "fuel overhedging". A further EUR 11m was also recognised in connection with this as a provision for impending losses.

The remaining operating expenses primarily comprise call centre services utilised, various administrative services outsourced to Lufthansa Group Business Services GmbH and external advisory services.

Expenses from other periods came to EUR 38m in the reporting year and are mainly attributable to insufficient provisions in the previous year.

25 Result from equity investments

T29 RESULTS FROM EQUITY INVESTMENTS

in €m	2020	2019
Income from profit transfer agreements	756	493
Expenses from loss transfer agreements	1,903	714
Income from equity investments	341	697
thereof from affiliated companies	334	691
	-806	476

Income/expenses from profit and loss transfer agreements are shown including tax contributions. Income from equity investments consists primarily the accrued dividends from

the Austrian leasing companies for financial year 2020 due to aligning the timing of profit recognition.

26 Net interest

T30 NET INTEREST

in €m	2020	thereof affiliated companies	2018	thereof affiliated companies
Income from other securities and non-current financial loans	37	35	42	40
Other interest and similar income	117	14	96	10
Interest and similar expenses	-506	-1	-1,471	-1
thereof accrued interest	-1,251	-	-734	-
thereof from market valuation of pension fund assets	1,077	-	-568	-
	-352	48	-1,333	49

Changes in interest expenses result primarily from the negative year-on-year change in the market value of pension assets used to fund retirement benefit obligations.

Net interest includes income from other periods of EUR 44m and expenses of EUR 57m. Both resulted from internal audit findings.

27 Impairment of investments and current securities

Impairment losses of EUR 147m were recognised on investments. They related to impairment losses due to the expected lasting impairment of the carrying amounts for Air Dolomiti S.p.A. Linee Aeree Regionali Europee (EUR 117m) and Crane Strategic Investment S.C.S (EUR 10m) as well as a pandemic-related impairment of the remaining portfolio of training loans for novice pilots (EUR 11m). In addition, an impairment loss was recognised for the foreign currency valuation of a loan to Swiss International Air Lines AG denominated in CHF (EUR 9m).

28 Taxes

T31 TAXES

in €m	2020	2019
Income taxes	2	-58
thereof deferred taxes (income)	-291	-421
Other taxes	28	41
	30	-17

The positive effect of taxes on income and earnings is attributable to the earnings from deferred tax assets. The deferred tax assets are primarily due to the higher amount for pension obligations recognised in the financial statements for commercial law purposes compared with the measurement for tax purposes. Furthermore, the prohibition on recognising provisions for impending losses as well as

value-based differences in inventories at Lufthansa Technik AG resulted in deferred tax assets. In addition, a deferred tax receivable on loss carry-forwards was recognised in the financial year for the first time.

Taxes on income and earnings and other taxes also include net items from previous years of EUR 47m.

OTHER DISCLOSURES

29 Supervisory Board and Executive Board

The members of the Supervisory Board and the Executive Board are listed on page 27 f.

The principles of the remuneration system and the amount of remuneration paid to the individual Executive Board and Supervisory Board members are shown and explained in detail in the remuneration report. The reporting of Executive Board remuneration for the 2020 financial year is based on the provisions of the new Section 162 AktG.

EXECUTIVE BOARD

The framework agreement between Deutsche Lufthansa AG and the Economic Stabilisation Fund (WSF) includes significant restrictions on Executive Board remuneration. No bonuses or other variable or similar remuneration components may be awarded to Executive Board members for the duration of the stabilisation measures. The same applies to special payments, gratuities and other forms of compensation in addition to the fixed salary, other discretionary payments by the Company and severance payments not required by law.

Remuneration for the Executive Board members active in the reporting year is as follows:

T32 TOTAL REMUNERATION OF EXECUTIVE BOARD MEMBERS

in €k	2020	2019
Basic salary	5,226	5,934
Other	58	110
One-year variable remuneration	1,058	2,091
Long-term variable remuneration	1,431	2,844
Share programme 1)	4,305	2,988
Total remuneration	12,078	13,967
Severance payment	3,500	4,055
Staff costs of pension commitments	3,331	3,076

1) Fair value at the time the options are granted

The employment contract with Ms Bettina Volken was terminated prematurely as of 31 December 2019 in accordance with a Supervisory Board resolution of 3 December 2019 and the corresponding severance agreement. Ms Volken shall receive a severance payment of EUR 4,055k for the remaining term of her employment contract and keeps her pro rata entitlement for 2019 to the annual and long-term variable remuneration.

The Executive Board's remuneration consists of the following components:

Non-performance-related remuneration

- **Fixed annual salary.** Basic remuneration, paid monthly as a salary.

- **Retirement benefit commitments.** The members of the Executive Board receive retirement benefit commitments based on a defined contribution plan. Since the financial year 2019, every Executive Board member receives, for the duration of their employment, a fixed annual amount, which is credited to their personal pension account.
- **Ancillary benefits.** Ancillary benefits include in-kind benefits from the use of company cars and concessionary travel in accordance with the relevant IATA guidelines.

Performance-related remuneration:

- **One-year variable remuneration (annual bonus).** The annual bonus is based on financial targets (85%) and non-financial targets (15%). For the financial targets, half each of the EBIT margin and the Adjusted EBIT growth compared with the four-year average are compared with a target figure. As a result of the massive negative impact of the global coronavirus pandemic on the Lufthansa Group's business, target achievement for the financial targets in the short-term variable remuneration for 2020 was 0%. For 2020 the Supervisory Board defined the individually assessed targets "customers" and "employees" as well as the collective target "overcoming the crisis situation" as focal points for the sustainability targets in the one-year variable remuneration. The non-financial targets form the basis for payments; however, these have essentially been suspended until further notice.
- **Long-term variable remuneration.** The current long-term variable remuneration includes still ongoing programmes from several financial years, which are partly based on remuneration systems in effect before 1 January 2020. Since 2020, the agreed long-term variable remuneration for Executive Board members has been share-based. At the beginning of the performance period, the Executive Board members receive a number of virtual shares corresponding to the contractually agreed target value. The final number of virtual shares is 85% dependent on achievement of the financial targets and 15% on the non-financial targets. The performance period for the targets is four years. Payment is in cash after the consolidated financial statements have been adopted for the last year of the respective performance period.
- **Share-based remuneration.** Up to and including 2018, the Executive Board members were also obliged to take part in the share programme for Executive Board members and managers.

In addition to the provision for the one-year variable remuneration of EUR 1,058k (previous year: EUR 2,091k), provisions totalling EUR 1,431k were recognised for the future payment of long-term variable remuneration for the Executive Board members active as of 31 December 2020 (previous year: EUR 2,844k).

Current payments to former members of the Executive Board and their surviving dependants came to EUR 6.9m (previous year: EUR 6.4m). This includes payments by subsidiaries as well as benefits in kind and concessionary travel.

Pension obligations towards former Executive Board members and their surviving dependants amount to EUR 64.0m (previous year: EUR 62.6m).

SUPERVISORY BOARD

Fixed remuneration for the Supervisory Board came to EUR 1,887k for the financial year (previous year: EUR 2,107k). Other remuneration, mainly attendance fees, amounted to EUR 23k (previous year: EUR 62k). .

The Deutsche Lufthansa AG Supervisory Board members were also paid EUR 16k for work on supervisory boards of Group companies (previous year: EUR 2k). . In addition to their Supervisory Board remuneration, employee representatives on the Supervisory Board received compensation for their work in the form of wages and salaries including pension entitlements amounting to EUR 1.1m in total (previous year: EUR 1.1m).

In the reporting year, as in the previous year, no loans or advance payments were made to members of the Executive Board or Supervisory Board.

30 Financial result

The net loss for the year of EUR -780m has been carried forward.

31 Events after the reporting period

LUFTHANSA ISSUES ANOTHER BOND AND REPAYS KfW LOAN

Deutsche Lufthansa AG again successfully issued a bond with a total volume of EUR 1.6bn on 4 February 2021. The first tranche has a volume of EUR 750m and pays interest at 2.875% p.a. over a term of four years. The second tranche with a volume of EUR 850m has a term of seven years and pays interest at 3.75% p.a.

Together with the funds of approximately EUR 2bn raised in the second half of 2020, the Lufthansa Group has thereby secured the refinancing of the financial liability of EUR 2.6bn due in 2021 and repaid the KfW loan of EUR 1bn ahead of schedule on 11 February 2021.

EU PARLIAMENT ADOPTS NEW SLOT RULES

On 10 February 2021, the European Parliament passed a resolution amending the slot rules for summer 2021 in response to the pandemic. The regulation stipulates that airlines can return up to 50% of their slot series in full before the season begins and that they only have to operate half the remaining slot series in order to secure them for the coming season.

Based on its capacity outlook, ➤ **Forecast, p. 118** Deutsche Lufthansa AG expects that it will be able to obtain all of the slots for which it has rights, including beyond the 2021 summer season.

32 Declaration of compliance in accordance with Section 161 German Stock Corporation Act (AktG)

The declaration of compliance with the German Corporate Governance Code required by Section 161 of the German Stock Corporation Act (AktG) was issued by the Executive Board and Supervisory Board and made public permanently as part of the declaration on corporate governance in line with Section 289f HGB on the Company's website at **www.lufthansagroup.com/declaration-of-compliance**.

33 Auditors' fees

Total auditors' fees calculated for the financial year in accordance with Section 319 Paragraph 1 HGB are made up as follows:

T33 Auditors' fees		
in €m	2020	2019
Audit services	2.8	3.5
Other certification services	0.1	0.2
Tax advisory services	0.1	0.9
Other services	1.5	1.7
	4.5	6.3

The auditing services mainly consist of fees for auditing the consolidated financial statements, the review of the half-yearly financial statements and the audit of the financial statements of Deutsche Lufthansa AG and its consolidated subsidiaries. The fees recognised under other advisory services pertain to the audit relating to the update of the Euro Debt Issuance Programme of DLH. Tax advisory services mainly consist of tax advice on transfer pricing, international taxes and inspections by tax authorities. The other services primarily relate to the sale of the LSG Europe business.

Corporate Governance

SUPERVISORY BOARD AND EXECUTIVE BOARD

Supervisory Board

Karl-Ludwig Kley

Chairman of the Supervisory Board
E.ON SE
Chairman

Christine Behle

Deputy Chairwoman of the trade union
ver.di
Employee representative¹⁾
Deputy Chairwoman

Alexander Behrens

Flight attendant
Employee representative¹⁾

Jörg Cebulla

Flight captain
Employee representative

Erich Clementi

Deputy Chairman of the Supervisory
Board E.ON SE
(since 5 May 2020)

Thomas Enders

Member in various Supervisory Boards
(since 5 May 2020)

Herbert Hainer

President FC Bayern München e. V.
(until 5 Mai 2020)

Christian Hirsch

Information Management Consultant /
Works Council member on leave of
absence – ver.di section
Employee representative (until 31
December 2020)

Jürgen Jennerke

Chairman of General Works Council
Lufthansa Cargo / Works Council
member on leave in absence – ver.di
section (since 8 December 2020)

Michael Kerkloh

Former Chairman of the Executive
Board Flughafen München GmbH
(since 2 September 2020)²⁾

Carsten Knobel

Chairman of the Executive Board and
CEO
Henkel AG & Co. KGaA

Holger Benjamin Koch

Senior Director Airport /
Industry Charges & Commercial
Provider Management
Employee representative

Martin Koehler

Former head of the Aviation Compe-
tence Center at
Boston Consulting Group
(since 31 August 2020)

Harald Krüger

Former Chairman of the Executive
Board Bayerische Motorenwerke Ak-
tiengesellschaft (BMW AG)
(since 5 May 2020)

Martina Merz

Chairwoman of the Executive Board
thyssenkrupp AG
(until 5 May 2020)

Michael Nilles

Chief Digital &
Information Officer (CDIO)
Henkel AG & Co. KGaA
(until 5 May 2020)

Monika Ribar

President of Board of Directors
Schweizerische Bundesbahnen SBB
AG, Switzerland (until 31 August 2020)

Birgit Rohleder

Teamlead IT Application
Management Airport Services
Employee representative

Miriam Sapiro

Managing Director &
Vice Chairman (Public Affairs),
Sard Verbinen & Co., USA

Ilja Schulz

Flight captain and member of the
Cockpit pilots' union
Employee representative¹⁾

Birgit Spineux

Purser / Employee representative on
leave in absence
Employee representative
(since 1 January 2021)

Astrid Stange

Group Chief Operating Office,
AXA SA
(since 5 May 2020)

Olivia Stelz

Purser / Employee representative on
leave in absence
Employee representative

Stephan Sturm

Chairman of the Executive Board
Fresenius Management SE

Angela Titzrath

Chairwoman of the Executive Board
Hamburger Hafen und Logistik AG
(since 2 September 2020)²⁾

Christina Weber

Commercial Employee
Employee representative
(until 2 December 2020)

Klaus Winkler

Engine mechanic
Employee representative

Matthias Wissmann

Senior International Counsel
WilmerHale (until 5 May 2020)

Honory Chairman**Dipl.-Ing. Jürgen Weber**

Former Chairman of the
Supervisory Board
Deutsche Lufthansa AG

1) Trade union representative in accordance with Section 7 Paragraph 2 Co-determination Act (MitbestG).

2) Member designated by the WSF in accordance with framework agreement.

Executive Board

(Structure since 1 January 2021)

Carsten Spohr

Chief Executive Officer

Thorsten Dirks

Member of the Executive Board
IT, Digital & Innovation
(until 30 June 2020)

Christina Foerster

Chief Customer Officer

Harry Hohmeister

Chief Commercial Officer

Detlef Kayser

Chief Operations Officer

Michael Niggemann

Chief HR & Legal Officer

Remco Steenbergen

Chief Financial Officer
(since 1 January 2021)

Ulrik Svensson

Member of the Executive Board
Finances (until 30 April 2020)

MANDATES

Other mandates of the Supervisory Board members of Deutsche Lufthansa AG

(As of 31 December 2020)

Karl-Ludwig Kley

- a) BMW AG³⁾ (Deputy Chairman)
E.ON SE³⁾ (Chairman)

Christine Behle

- a) Bremer Lagerhaus-Gesellschaft
- Aktiengesellschaft von 1877 - ³⁾
(Deputy Chairwoman)

Jörg Cebulla

- a) Sparda-Bank Hessen eG
- b) Albatros Versicherungsdienste GmbH

Erich Clementi

- a) E.ON SE³⁾ (Deputy Chairman)

Thomas Enders

- a) Knorr Bremse AG
- b) Linde plc.

Herbert Hainer

- a) Allianz SE³⁾
FC Bayern München AG
(Chairman)
- b) Accenture plc., Ireland³⁾

Jürgen Jennerke

- a) Lufthansa Cargo AG
(Deputy Chairman)

Martin Koehler

- a) Delton Technology SE (Deputy
Chairman)
- b) American Funds Investment
Fonds, managed by the Capital
Group, USA
FlixBus GmbH

Harald Krüger

- a) Deutsche Telekom AG³⁾

Martina Merz

- b) AB Volvo, Schweden³⁾
Imerys SA, Frankreich³⁾
(until 4 May 2020)
SAF-HOLLAND SA,
Luxembourg³⁾

Michael Nilles

- a) Lufthansa Technik AG
- b) Medela Holding AG, Switzerland
(Board of Directors),
Medela AG, Switzerland (Board
of Directors)

Monika Ribar

- b) Chain IQ Group AG, Switzerland
SBB Schweizerische Bundesbahnen, Switzerland (President of the Board of Directors)
Sika AG, Switzerland³⁾

Miriam Sapiro

- b) Project HOPE, USA

Astrid Stange

- b) AXA Group Operations SAS
(President of the Executive
Committee)
GIE AXA

Stephan Sturm

- a) Fresenius Kabi AG¹⁾ (Chairman)
Fresenius Medical Care
Management AG¹⁾ (Chairman)
- b) VAMED AG, Austria²⁾ (Deputy
Chairman)

Angela Titzrath

- a) Evonik Industries AG³⁾
- b) Talanx AG³⁾

Christina Weber

- a) LSG Lufthansa Service
Holding AG

Matthias Wissmann

- b) ODDO BHF SCA

a) Membership of supervisory boards required by law.

b) Membership of comparable supervisory bodies at companies in Germany and abroad.

1) Group mandate in accordance with Section 100 Paragraph 2 Sentence 2 AktG.

2) Other group mandate

3) Publicly listed company.

Mandates of the Executive Board members of Deutsche Lufthansa AG

(As of 31 December 2020)

Carsten Spohr

- a) Lufthansa Technik AG¹⁾
(Chairman, until 2 April 2020)
Münchener Rückversicherungs-
Gesellschaft Aktiengesellschaft
(since 27 April 2020)

Thorsten Dirks

- a) Eurowings GmbH¹⁾
(Chairman, until 30 June 2020)
Germanwings GmbH¹⁾ (Chair-
man, until 14 January 2020)
- b) Eurowings Europe GmbH
(Chairman, until 29 February
2020)
Günes Ekspres Havacilik A.S.
(SunExpress), Turkey (Deputy
Chairman, until 31 March 2020)
SN Airholding SA/NV,
Belgium²⁾ (Chairman, until 30
June 2020)

Christina Foerster

- a) Eurowings GmbH¹⁾
(since 14 April 2020)
LSG Lufthansa Service Holding
AG¹⁾ (since 17 April 2020)
Lufthansa CityLine GmbH¹⁾
(since 31 March 2020)
- b) Austrian Airlines AG, Austria
(Chairwoman, since 2 April
2020)
Lufthansa AirPlus Servicekarten
GmbH (Chairwoman, since 6
April 2020)
SN Airholding SA/NV
(since 6 April; Chairwoman,
since 3 July 2020)
Swiss International Air Lines AG,
Switzerland (since 4 April 2020)

Harry Hohmeister

- a) Lufthansa Cargo AG¹⁾ (Chairman)
- b) Aircraft Maintenance and Engi-
neering Corporation (AMECO),
China
Austrian Airlines AG, Austria
(Chairman, until 2 April 2020)
Günes Ekspres Havacilik A.S.
(SunExpress), Turkey (since 16
July 2020)
Swiss International Air Lines AG,
Switzerland (until 2 April 2020)

Detlef Kayser

- a) Aerodata AG (until 31 March
2020)
LSG Lufthansa Service Holding
AG¹⁾ (Chairman)
Lufthansa Technik AG
(Chairman, since 2 April 2020)¹⁾
- b) Austrian Airlines AG, Austria
(Deputy Chairman, since 7 April
2020)
Swiss International Air Lines AG,
Switzerland (since 4 April 2020)
Günes Ekspres Havacilik A.S.
(SunExpress), Turkey (until 31
March 2020)

Michael Niggemann

- a) Lufthansa CityLine GmbH¹⁾
(until 31 March 2020)
- b) Austrian Airlines AG,
Austria (Deputy Chairman,
until 7 April 2020)

Ulrik Svensson

- b) Lufthansa AirPlus Servicekarten
GmbH (Chairman, until 3 April
2020)
Swiss International Air Lines AG,
Switzerland
(until 31 März 2020)

SUPERVISORY BOARD COMMITTEES

as of 31 Dec 2020

Steering Committee	Audit Committee	Nomination Committee	Arbitration Committee in accordance with Section 27 Paragraph 3 Co-determination Act (MitbestG)
Karl-Ludwig Kley, Chairman Christine Behle, Deputy Chairwoman Thomas Enders Ilja Schulz	Stephan Sturm, Chairman Alexander Behrens Jörg Cebulla Michael Kerkloh Carsten Knobel Klaus Winkler	Karl-Ludwig Kley, Chairman Thomas Enders Harald Krüger	Karl-Ludwig Kley, Chairman Christine Behle, Deputy Chairwoman Thomas Enders Ilja Schulz
16 meetings in 2020	Six meetings in 2020	One meeting in 2020	No meetings in 2020

a) Membership of supervisory boards required by law.

b) Membership of comparable supervisory bodies at companies in Germany and abroad.

1) Group mandate in accordance with Section 100 Paragraph 2 Sentence 2 AktG.

2) Other group mandate.

3) Publicly listed company.

LIST OF SHAREHOLDINGS – SIGNIFICANT EQUITY INVESTMENTS

T34 SIGNIFICANT EQUITY INVESTMENTS

Name, registered office	Stake in %	Net profit of the last business year in €m *	Shareholders' equity in €m *
Aerococina S.A. de C.V., Ciudad de Mérida, Mexiko	51.98%	1	7
Aerologic GmbH, Schkeuditz	50.00%	8 ⁹⁾	36
AFS Aviation Fuel Services GmbH, Hamburg, Deutschland	33.33%	2 ⁸⁾ ⁹⁾	3
Air Dolomiti S.p.A. Linee Aeree Regionali Europee, Dossobuono di Villafranca (Verona), Italien	100.00%	-20	42
Aircraft Maintenance and Engineering Corp., Peking, China	25.00%	-58 ⁹⁾ ¹⁰⁾	5
AIRO Catering Services - Ukraine, Boryspil, Ukraine	100.00%	0 ¹³⁾	0 ¹¹⁾
AIRO Catering Services Eesti OÜ, Tallinn, Estland	100.00%	0 ¹¹⁾	0 ¹¹⁾
Airo Catering Services Latvija SIA, Marupe, Lettland	100.00%	-2	-2
AIRO Catering Services Sweden AB, Upplands Väsby, Schweden	100.00%	0 ¹¹⁾	12
AirPlus International AG, Kloten, Schweiz	100.00%	-2	17
AirPlus International Limited, London, Großbritannien	100.00%	0 ¹¹⁾	10
AirPlus International S.r.l., Bologna, Italien	100.00%	0 ¹¹⁾	16
AirPlus International, Inc., Alexandria, USA	100.00%	-2	16
AirPlus Payment Management Co. Ltd., Shanghai, China	100.00%	-1	45
AirTrust AG, Zug, Schweiz	100.00%	0 ¹¹⁾	98
Arlington Services Mexico, S.A. de C.V., Mexiko-Stadt, Mexiko	100.00%	4	17
Arlington Services Panama S.A., Panama-Stadt, Panama	100.00%	0 ¹¹⁾	10
AUA Beteiligungen Gesellschaft m.b.H., Wien-Flughafen, Österreich	100.00%	0 ¹¹⁾	10
Austrian Airlines AG, Wien-Flughafen, Österreich	100.00%	-454	94
Austrian Airlines Lease and Finance Company Ltd. i. L., Guernsey, Großbritannien	100.00%	0 ¹²⁾	0 ¹¹⁾ ¹²⁾
Austrian Asset Holding GP S.à r.l., Luxemburg, Luxemburg	100.00%	0 ¹¹⁾	0 ¹¹⁾
Austrian Asset Holding S.C.S., Luxemburg, Luxemburg	100.00%	10	257
AVIAPIT-SOCHI OOO, Sotschi, Russland	100.00%	0 ¹¹⁾	0 ¹¹⁾
AviationPower GmbH, Hamburg, Deutschland	49.00%	3 ⁷⁾ ⁹⁾	12
Bahia Catering Ltda., Sao Cristovao (Salvador), Brasilien	100.00%	0 ¹¹⁾	1
Belém Serviços de Bordo Ltda., Maracangalha, Belém, Brasilien	70.00%	0 ¹¹⁾	0 ¹¹⁾
BizJet International Sales & Support, Inc., Tulsa, USA	100.00%	2	20
Brussels Airlines SA/NV, Brüssel, Belgien	100.00%	-377	-258
Capital Gain International (1986) Ltd., Hong Kong, China	100.00%	0 ¹¹⁾	4
Cater Suprimento de Refeicoes, Ltda., Rio de Janeiro, Brasilien	100.00%	0 ¹¹⁾	-5
Caterair Servicos de Bordo e Hotelaria Ltda., Ilha do Governador, Brasilien	100.00%	-3	2
Charm Food Service Co. Ltd., Incheon, Süd-Korea	80.00%	0 ¹¹⁾	3
CLS Catering Services Ltd., Vancouver, British Columbia, Kanada	70.00%	-4	23
Comercializadora de Servicios Limitada, ENEA, Pudahuel, Santiago, Chile	100.00%	0 ¹³⁾	0 ¹¹⁾
Comisariato de Baja California, S.A. de C.V., Tijuana, Mexiko	51.00%	0 ¹¹⁾	3
Comisariatos Gotre, S.A. de C.V., Torreon, Mexiko	51.00%	0 ¹¹⁾	1
Constance Food Group, Inc., New York, USA	100.00%	0 ¹¹⁾	16
Cosmo Enterprise Co. Ltd., Narita City, Japan	20.00%	1 ⁸⁾ ⁹⁾ ¹⁰⁾	0
Crane Strategic Investment S.C.S., Grevenmacher, Luxemburg	100.00%	0 ¹¹⁾	2
Delvag Versicherungs-AG, Köln	100.00%	-6 ⁹⁾	60
Diners Club Spain S.A., Madrid, Spanien	25.00%	2 ⁸⁾ ⁹⁾	12
DLH Fuel Company mbH, Hamburg	100.00%	3 ⁸⁾ ⁹⁾	7

Edelweiss Air AG, Zürich, Schweiz	100.00%	-78	4
EME Aero Sp.z.o.o, Jasionka, Polen	50.00%	-25 ⁹⁾	12
Eurowings Aviation GmbH, Köln	100.00%	3	6
Eurowings Digital GmbH, Köln	100.00%	0 ¹¹⁾	0 ¹¹⁾
Eurowings Europe GmbH, Wien-Flughafen, Österreich	100.00%	1	6
Eurowings GmbH, Düsseldorf	100.00%	-516	381
Eurowings Technik GmbH, Köln	100.00%	0 ¹¹⁾	0 ¹¹⁾
Fortaleza Serviços de Bordo Ltda., Fortaleza, Brasilien	70.00%	0 ¹¹⁾	0 ¹¹⁾
Gansu HNA LSG Sky Chefs Co., Ltd, Lanzhou, China	49.00%	0 ^{8) 9)}	4
Germanwings GmbH, Köln	100.00%	-76	2
Global Brand Management AG, Basel, Schweiz	100.00%	10	476
Günes Ekspres Havacilik Anonim Sirketi (Sun Express), Antalya, Türkei	50.00%	56 ^{8) 9)}	298
Hamburger Gesellschaft für Flughafenanlagen mbH, Hamburg	100.00%	24	175
Hawker Pacific Aerospace, Sun Valley, USA	100.00%	0 ¹¹⁾	12
HEICO Aerospace Holdings Corp., Florida, USA	20.00%	-26 ⁹⁾	-3
Hongkong Beijing Air Catering Ltd., Hong Kong, China	45.00%	1 ^{8) 9)}	3
Hongkong Shanghai Air Catering Ltd., Hong Kong, China	45.00%	2 ^{8) 9)}	9
Inflight Catering (Pty) Ltd., Johannesburg, Südafrika	100.00%	0 ¹²⁾	0 ¹²⁾
Inflight Catering Services Limited, Dar es Salaam, Tansania	61.99%	-2	1
Inflite Holdings (Cayman) Ltd., Grand Cayman, Kaimaninseln	49.00%	3 ^{8) 9) 10)}	20
Inflite Holdings (St. Lucia) Ltd., Castries, St. Lucia	49.00%	2 ^{8) 9) 10)}	11
International Food Services Ltd., Hong Kong, China	100.00%	0 ¹¹⁾	4
JASEN Grundstücksgesellschaft mbH & Co. oHG, Grünwald	100.00%	1 ¹⁾	23
Jettainer Americas, Inc., East Meadow, USA	100.00%	0 ¹¹⁾	7
Jettainer GmbH, Raunheim	100.00%	0 ¹¹⁾	6
LCH Grundstücksgesellschaft Berlin mbH, Frankfurt am Main	100.00%	-11	20
LG-LHT Aircraft Solutions GmbH, Hamburg	51.00%	-2 ⁹⁾	4
LG-LHT Passenger Solutions GmbH, Hamburg	51.00%	-2 ⁹⁾	4
LHAMI LEASING LIMITED, Dublin, Irland	100.00%	86	2,838
LHAMIH LIMITED, Dublin, Irland	100.00%	575	3,535
LHBD Holding Limited, London, Großbritannien	100.00%	0 ^{11) 2)}	0 ¹¹⁾
LSG Asia GmbH, Neu-Isenburg	100.00%	-1	0 ¹¹⁾
LSG Catering (Thailand) Ltd., Bangkok, Thailand	100.00%	0 ¹¹⁾	2
LSG Catering China Ltd., Hong Kong, China	100.00%	0 ¹¹⁾	-5
LSG Catering Guam, Inc., Guam, USA	100.00%	0 ¹¹⁾	0 ¹¹⁾
LSG Catering Hong Kong Ltd., Hong Kong, China	100.00%	-20	70
LSG Catering Saipan, Inc., Saipan, Mikronesien	100.00%	0 ¹³⁾	0 ¹¹⁾
LSG Holding Asia Ltd., Hong Kong, China	86.88%	0 ¹¹⁾	18
LSG Lufthansa Service - Sky Chefs do Brasil Catering, Refeições Ltda., Guarulhos, Brasilien	100.00%	-7	-19
LSG Lufthansa Service Asia Ltd., Hong Kong, China	100.00%	0 ¹¹⁾	33
LSG Lufthansa Service Cape Town (Pty) Ltd., Boksburg, Südafrika	100.00%	0 ¹³⁾	0 ¹¹⁾
LSG Lufthansa Service Enterprises Ltd., Hong Kong, China	100.00%	0 ¹¹⁾	1
LSG Lufthansa Service Europa/Afrika GmbH, Neu-Isenburg	100.00%	-62	226
LSG Lufthansa Service Guam, Inc., Tamuning, Guam, USA	100.00%	0 ¹¹⁾	8
LSG Lufthansa Service Holding AG, Neu-Isenburg	100.00%	-293	707
LSG Lufthansa Service Hong Kong Ltd., Hong Kong, China	47.90%	-10 ⁴⁾	14
LSG Lufthansa Service Saipan, Inc., Saipan, Mikronesien	100.00%	0 ¹¹⁾	7
LSG Sky Chefs (India) Private Ltd., Bangalore, Indien	100.00%	0 ¹¹⁾	6

LSG Sky Chefs (Qingdao) Co., Ltd., Laixi City, China	100.00%	0	¹¹⁾	2
LSG Sky Chefs (Thailand) Ltd., Bangkok, Thailand	64.30%	-4		9
LSG Sky Chefs Argentina S.A., Ezeiza, Argentinien	100.00%	2		10
LSG Sky Chefs Bremen GmbH, Neu-Isenburg	100.00%	0	¹¹⁾	-2
LSG Sky Chefs de Venezuela C.A., Caracas, Venezuela	99.99%	0	¹¹⁾	1
LSG Sky Chefs Hamburg GmbH, Neu-Isenburg	100.00%	-1		-9
LSG Sky Chefs Havacilik Hizmetleri A.S., Bakırköy/Istanbul, Türkei	100.00%	-3		-2
LSG Sky Chefs Istanbul Catering Hizmetleri A.S., Istanbul, Türkei	100.00%	0	¹¹⁾ ⁶⁾	9
LSG Sky Chefs Kenya Limited, Nairobi, Embakasi District, Kenia	50.20%	-8		-4
LSG Sky Chefs Korea Co Ltd., Incheon, Südkorea	80.00%	-59		26
LSG Sky Chefs Leipzig GmbH, Neu-Isenburg	100.00%	0	¹¹⁾	-1
LSG Sky Chefs Malmö AB, Kungsör, Schweden	100.00%	0	¹¹⁾	1
LSG Sky Chefs New Zealand Limited, Auckland, Neuseeland	100.00%	-2		26
LSG Sky Chefs Norge AS, Gardermoen, Norwegen	100.00%	0	¹¹⁾	2
LSG Sky Chefs North America Solutions, Inc., Wilmington, USA	100.00%	2		26
LSG Sky Chefs RPC West GmbH, Neu-Isenburg	100.00%	0	¹¹⁾	0 ¹¹⁾
LSG Sky Chefs Rus, Moskau, Russland	100.00%	-1		1
LSG Sky Chefs South Africa (Proprietary) Ltd., Johannesburg, Südafrika	100.00%	-5		-11
LSG Sky Chefs Supply Chain Solutions, Inc., Wilmington, USA	100.00%	4		30
LSG Sky Chefs Sverige AB, Kungsör, Schweden	100.00%	0	¹¹⁾	2
LSG Sky Chefs TAAG Angola S.A., Luanda, Angola	40.00%	-1	⁴⁾	4
LSG Sky Chefs UK Ltd. i.L., Sidcup, Großbritannien	100.00%	0	¹¹⁾	0 ¹¹⁾
LSG Sky Chefs USA, Inc., Wilmington, USA	100.00%	0	¹¹⁾	703
LSG South America GmbH, Neu-Isenburg	100.00%	-12		12
LSG/Sky Chefs Europe Holdings Ltd., Hounslow, Großbritannien	100.00%	0	¹¹⁾	7
LSY GmbH, Norderstedt	100.00%	13		709
Lufthansa AirPlus Servicekarten GmbH, Neu-Isenburg	100.00%	-97		244
Lufthansa Asset Management GmbH, Frankfurt am Main	100.00%	390		3,530
Lufthansa Asset Management Leasing GmbH, Frankfurt am Main	100.00%	7		437
Lufthansa Aviation Training Berlin GmbH, Berlin	100.00%	-2		6
Lufthansa Aviation Training Germany GmbH, Frankfurt am Main	100.00%	-13		27
Lufthansa Aviation Training GmbH, Hallbergmoos	100.00%	-55		122
Lufthansa Aviation Training Switzerland AG, Opfikon, Schweiz	100.00%	4		60
Lufthansa Cargo AG, Frankfurt am Main	100.00%	508		250
Lufthansa CityLine GmbH, München-Flughafen	100.00%	-57		140
Lufthansa Commercial Holding GmbH, Frankfurt am Main	100.00%	3		3,182
Lufthansa Engineering and Operational Services GmbH, Frankfurt am Main	100.00%	-9	⁹⁾	6
Lufthansa Global Business Services GmbH, Frankfurt am Main	100.00%	0	¹¹⁾	-3
Lufthansa Global Tele Sales GmbH, Berlin	100.00%	5	⁹⁾	7
Lufthansa Industry Solutions AS GmbH, Norderstedt	100.00%	9		12
Lufthansa Industry Solutions BS GmbH, Raunheim	100.00%	2		5
Lufthansa Industry Solutions GmbH & Co. KG, Norderstedt	100.00%	10		40
Lufthansa Leasing Austria 1. Beteiligungs GmbH, Salzburg, Österreich	100.00%	4		4
Lufthansa Leasing Austria GmbH & Co. OG Nr. 10, Salzburg, Österreich	100.00%	73		84
Lufthansa Leasing Austria GmbH & Co. OG Nr. 12, Salzburg, Österreich	100.00%	26		35
Lufthansa Leasing Austria GmbH & Co. OG Nr. 14, Salzburg, Österreich	100.00%	39		53
Lufthansa Leasing Austria GmbH & Co. OG Nr. 15, Salzburg, Österreich	100.00%	23		27
Lufthansa Leasing Austria GmbH & Co. OG Nr. 16, Salzburg, Österreich	100.00%	20		26
Lufthansa Leasing Austria GmbH & Co. OG Nr. 17, Salzburg, Österreich	100.00%	7		10
Lufthansa Leasing Austria GmbH & Co. OG Nr. 18, Salzburg, Österreich	100.00%	5		8
Lufthansa Leasing Austria GmbH & Co. OG Nr. 19, Salzburg, Österreich	100.00%	10		11

Lufthansa Leasing Austria GmbH & Co. OG Nr. 20, Salzburg, Österreich	100.00%	10	17
Lufthansa Leasing Austria GmbH & Co. OG Nr. 21, Salzburg, Österreich	100.00%	19	26
Lufthansa Leasing Austria GmbH & Co. OG Nr. 22, Salzburg, Österreich	100.00%	17	77
Lufthansa Leasing Austria GmbH & Co. OG Nr. 24, Salzburg, Österreich	100.00%	3	58
Lufthansa Leasing Austria GmbH & Co. OG Nr. 25, Salzburg, Österreich	100.00%	2	50
Lufthansa Leasing Austria GmbH & Co. OG Nr. 26, Salzburg, Österreich	100.00%	5	165
Lufthansa Leasing Austria GmbH & Co. OG Nr. 27, Salzburg, Österreich	100.00%	6	28
Lufthansa Leasing Austria GmbH & Co. OG Nr. 28, Salzburg, Österreich	100.00%	28	206
Lufthansa Leasing Austria GmbH & Co. OG Nr. 29, Salzburg, Österreich	100.00%	16	24
Lufthansa Leasing Austria GmbH & Co. OG Nr. 30, Salzburg, Österreich	100.00%	10	20
Lufthansa Leasing Austria GmbH & Co. OG Nr. 31, Salzburg, Österreich	100.00%	2	22
Lufthansa Leasing Austria GmbH & Co. OG Nr. 32, Salzburg, Österreich	100.00%	11	106
Lufthansa Leasing Austria GmbH & Co. OG Nr. 33, Salzburg, Österreich	100.00%	0 ¹¹⁾	52
Lufthansa Leasing Austria GmbH & Co. OG Nr. 34, Salzburg, Österreich	100.00%	8	105
Lufthansa Leasing Austria GmbH & Co. OG Nr. 35, Salzburg, Österreich	100.00%	4	68
Lufthansa Leasing Austria GmbH & Co. OG Nr. 36, Salzburg, Österreich	100.00%	0 ¹¹⁾	18
Lufthansa Leasing Austria GmbH & Co. OG Nr. 37, Salzburg, Österreich	100.00%	18	282
Lufthansa Leasing Austria GmbH & Co. OG Nr. 38, Salzburg, Österreich	100.00%	-1	65
Lufthansa Leasing Austria GmbH & Co. OG Nr. 39, Salzburg, Österreich	100.00%	-3	46
Lufthansa Leasing Austria GmbH & Co. OG Nr. 42, Salzburg, Österreich	100.00%	8	156
Lufthansa Leasing Austria GmbH & Co. OG Nr. 43, Salzburg, Österreich	100.00%	3	48
Lufthansa Leasing Austria GmbH & Co. OG Nr. 50, Salzburg, Österreich	100.00%	35	44
Lufthansa Malta Aircraft-Leasing Ltd., St. Julians, Malta	100.00%	94	743
Lufthansa Malta Blues LP, St. Julians, Malta	99.99%	0 ¹¹⁾	-562
Lufthansa Malta Corporate Finance Limited, St. Julians, Malta	100.00%	0 ¹¹⁾	235
Lufthansa Malta Finance Holding Limited, St. Julians, Malta	100.00%	7	249
Lufthansa Malta Treasury Services Limited, St. Julians, Malta	100.00%	0 ¹¹⁾	0 ¹¹⁾
Lufthansa Process Management GmbH, Neu-Isenburg	100.00%	0 ¹¹⁾	1
Lufthansa Seeheim GmbH, Seeheim-Jugenheim	100.00%	-6	4
Lufthansa Systems Americas, Inc., Miami Lakes, USA	100.00%	1	3
Lufthansa Systems GmbH & Co. KG, Raunheim	100.00%	0 ¹¹⁾	-53
Lufthansa Technik AERO Alzey GmbH, Alzey	100.00%	2	42
Lufthansa Technik AG, Hamburg	100.00%	-536	994
Lufthansa Technik Airmotive Ireland Holdings Ltd., Dublin, Irland	100.00%	0 ¹¹⁾	157
Lufthansa Technik Airmotive Ireland Leasing Ltd., Dublin, Irland	100.00%	40	266
Lufthansa Technik Budapest Repülőgépi Nagyjavító Kft., Budapest, Ungarn	100.00%	0 ¹¹⁾	9
Lufthansa Technik Component Services LLC, Tulsa, USA	100.00%	-2	21
Lufthansa Technik Immobilien- und Verwaltungsgesellschaft mbH, Hamburg	100.00%	-3	35
Lufthansa Technik Landing Gear Services UK Ltd., Kestrel Way, Hayes, Großbritannien	100.00%	-9	-22
Lufthansa Technik Logistik GmbH, Hamburg	100.00%	-4	3
Lufthansa Technik Logistik Services GmbH, Hamburg	100.00%	-9	1
Lufthansa Technik Maintenance International GmbH, Frankfurt am Main	100.00%	0 ¹¹⁾	6
Lufthansa Technik Malta Limited, Luqa, Malta	92.00%	0 ¹¹⁾	6
Lufthansa Technik North America Holding Corp., Tulsa, USA	100.00%	0 ¹¹⁾	243
Lufthansa Technik Objekt- und Verwaltungsgesellschaft mbH, Hamburg	100.00%	8	94
Lufthansa Technik Philippines, Inc., Manila, Philippinen	51.00%	-22	23
Lufthansa Technik Puerto Rico LLC, San Juan, Puerto Rico	100.00%	0 ¹¹⁾	10
Lufthansa Technik Shannon Limited, Claire, Irland	100.00%	0 ¹¹⁾	37
Lufthansa Technik Sofia OOD, Sofia, Bulgarien	75.10%	0 ¹¹⁾	18
MARDU Grundstücks-Verwaltungsgesellschaft mbH & Co. oHG, Grünwald	100.00%	18 ¹⁾	152
Miles & More GmbH, Frankfurt am Main	100.00%	76	56
MUSA Grundstücks-Verwaltungsgesellschaft mbH & Co. oHG, Grünwald	100.00%	4 ¹⁾	29

Myanmar LSG Lufthansa Service Ltd., Yangon, Myanmar	100.00%	0	¹¹⁾	8
N3 Engine Overhaul Services GmbH & Co. KG, Arnstadt	50.00%	-59	⁹⁾	-6
Nanjing Lukou International Airport LSG Catering Co. Ltd., Nanjing City, China	40.00%	2	⁸⁾ ⁹⁾	12
Natal Catering Ltda., Aeroporto São Gonçalo do Amarante, Brasilien	70.00%	0	¹¹⁾	0 ¹¹⁾
Oakfield Farms Solutions, L.L.C., Wilmington, USA	100.00%	0	¹¹⁾ ⁶⁾	9
ÖLB Österreichische Luftverkehrs-Beteiligungs GmbH, Wien-Flughafen, Österreich	100.00%	0	¹¹⁾	502
ÖLH Österreichische Luftverkehrs-Holding GmbH, Wien-Flughafen, Österreich	100.00%	8	³⁾	176
ÖLP Österreichische Luftverkehrs-Privatstiftung, Wien Flughafen, Österreich	0.00%	0	¹¹⁾ ⁴⁾	0 ¹¹⁾
Quinto Grundstücksgesellschaft mbH & Co. oHG, Grünwald	100.00%	4	¹⁾	74
Retail In Motion Asia Limited, Hong Kong, China	100.00%	-4		-3
Retail in Motion GmbH, Neu-Isenburg	100.00%	2		26
Retail in Motion Latin America SpA, ENEA, Pudahuel, Santiago, Chile	100.00%	0	¹¹⁾	0 ¹¹⁾
Retail in Motion Limited, Dublin, Irland	100.00%	-10		36
Retail In Motion Mexico S. de R.L. de C.V., Mexiko-Stadt, Mexiko	51.00%	0	¹¹⁾	0 ¹¹⁾
Retail inmotion Middle East L.L.C., Abu Dhabi, Vereinigte Arabische Emirate	100.00%	0	¹¹⁾	-1
SCIS Air Security Corporation, Wilmington, USA	100.00%	1		7
ServCater Internacional Ltda., Guarulhos, Brasilien	90.00%	-12		-27
Servicios Complementarios de Cabina, S.A. de C.V., Mexiko-Stadt, Mexiko	51.88%	0	¹¹⁾	1
Shanghai Pudong International Airport Cargo Terminal Co. Ltd., Shanghai, China	29.00%	60	⁹⁾	171
Siam Flight Services Ltd., Bangkok, Thailand	49.00%	0	¹¹⁾	4
Silver Wings Bulgaria OOD, Sofia, Bulgarien	28.75%	0	¹¹⁾ ⁵⁾	5
Sky Chefs Chile SpA, ENEA, Pudahuel, Santiago, Chile	100.00%	-5		15
Sky Chefs De Mexico, S.A. de C.V., Mexiko-Stadt, Mexiko	51.00%	5		9
Sky Chefs de Panama, S.A., Panama-Stadt, Panama	100.00%	-3		22
Sky Chefs Things Remembered Services FZE, Lagos, Nigeria	51.00%	-5		-7
Sky Chefs Things Remembered Services Limited, Lagos, Nigeria	51.00%	0	¹¹⁾	0 ¹¹⁾
Sky Chefs, Inc., Wilmington, USA	100.00%	-35		112
SN Airholding SA/NV, Brüssel, Belgien	100.00%	1		366
Spairliners GmbH, Hamburg	50.00%	0	⁹⁾ ¹¹⁾	20
Swiss Aviation Software AG, Basel, Schweiz	100.00%	1		19
Swiss International Air Lines AG, Basel, Schweiz	100.00%	-647		1,047
Terminal 2 Gesellschaft mbH & Co oHG, München-Flughafen	40.00%	-97	⁹⁾	41
TGV DLH, Düsseldorf	100.00%	-4		1
time:matters GmbH, Neu-Isenburg	100.00%	9		0 ¹¹⁾
time:matters Holding GmbH, Neu-Isenburg	100.00%	10		69
time:matters Spare Parts Logistics GmbH, Neu-Isenburg	100.00%	2		0 ¹¹⁾
Tolmachevo Catering OOO, Novosibirsk, Russland	26.00%	1	⁸⁾ ⁹⁾	1
Wenzhou Longwan International Airport LSG Sky Chefs Co. Ltd., Wenzhou City, China	40.00%	1	⁸⁾ ⁹⁾	7
Western Aire Chef, Inc., Wilmington, USA	100.00%	5		10
XEOS Sp.z.o.o., Środa Śląska, Polen	51.00%	0	⁹⁾ ¹¹⁾	20
Xian Eastern Air Catering Co. Ltd., Xian, China	30.00%	4	⁸⁾ ⁹⁾	7
Yunnan Eastern Air Catering Co. Ltd., Kunming, China	24.90%	4	⁸⁾ ⁹⁾	12
ZAO Aeromar, Moskau Region, Russland	49.00%	12	⁸⁾ ⁹⁾	56
ZAO AeroMEAL, Yemelyanovo, Russland	100.00%	0	¹¹⁾	0 ¹¹⁾

* IFRS Angaben

1) Gemäß IFRS 10 vollkonsolidiertes strukturiertes Unternehmen

- 2) Die Registrierungsnummer im Companies House lautet: 06939137
- 3) 50,20% der Kapitalanteile bzw. Stimmrechte sind über die ÖLP zuzurechnen
- 4) Die unternehmerische Führung der Gesellschaft liegt im Konzern
- 5) 28,75% der Kapitalanteile und Stimmrechte sind über eine Call-Option zuzurechnen.
- 6) 33,34% der Kapitalanteile bzw. 50,01% der Stimmrechte sind über eine Call-Option zuzurechnen.
- 7) Abschluss 2017
- 8) Abschluss 2019
- 9) Lokaler Abschluss
- 10) Abweichendes Geschäftsjahr
- 11) Wert kleiner 0,5 Mio. EUR.
- 12) in Liquidation
- 13) keine Daten vorhanden

The following Auditor's Report is a translation of the German language independent auditor's report and refers to the audit of the German language annual financial statements and the German language management report, which is neither included or incorporated by reference.

Independent Auditor's Report

To Deutsche Lufthansa AG, Cologne

REPORT ON THE AUDIT OF THE ANNUAL FINANCIAL STATEMENTS AND OF THE MANAGEMENT REPORT

Audit Opinions

We have audited the annual financial statements of Deutsche Lufthansa Aktiengesellschaft, Cologne, which comprise the balance sheet as at 31 December 2020, and the income statement for the fiscal year from 1 January to 31 December 2020, and notes to the financial statements, including the recognition and measurement policies presented therein. In addition, we have audited the combined management report of Deutsche Lufthansa Aktiengesellschaft for the fiscal year from 1 January to 31 December 2020. In accordance with the German legal requirements, we have not audited the content of the corporate governance declaration pursuant to Sec. 289f HGB ("Handelsgesetzbuch": German Commercial Code) which is published on the website stated in the "Corporate Governance" section of the combined management report and the non-financial declaration pursuant to Sec. 289b HGB included in the "Combined non-financial declaration" section. In addition, we have not audited the content of the other information extending beyond the prior year in the tables with multi-year comparisons of the combined management report (information pertaining to fiscal years 2015, 2016, 2017 and 2018). This relates to any information whose disclosure in the combined management report is not required pursuant to Secs. 289, 289a HGB or Secs. 289b to 289f HGB.

In our opinion, on the basis of the knowledge obtained in the audit,

- the accompanying annual financial statements comply, in all material respects, with the requirements of German commercial law applicable to business corporations and give a true and fair view of the assets, liabilities and financial position of the Company as at 31 December 2020 and of its financial performance for the fiscal year from 1 January to 31 December 2020 in compliance with German legally required accounting principles, and
- the accompanying combined management report as a whole provides an appropriate view of the Company's position. In all material respects, this combined management report is consistent with the annual financial statements, complies with German legal requirements and appropriately presents the opportunities and risks of future development. Our opinion on the combined management report does not cover the content of aforementioned corporate governance declaration, the content of the non-financial declaration or the aforementioned multi-year comparisons of the combined management report.

Pursuant to Sec. 322 (3) Sentence 1 HGB, we declare that our audit has not led to any reservations relating to the legal compliance of the annual financial statements and of the combined management report.

Basis for the Audit Opinions

We conducted our audit of the annual financial statements and of the management report in accordance with Sec. 317 HGB and the EU Audit Regulation (No 537/2014, referred to subsequently as "EU Audit Regulation") and in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the Institut der Wirtschaftsprüfer (Institute of Public Auditors in Germany) (IDW). Our responsibilities under those requirements and principles are further described in the "Auditor's responsibilities for the audit of the annual financial statements and of the combined management report" section of our auditor's report. We are independent of the Company in accordance with the requirements of European law and German commercial and professional law, and we have fulfilled our other German professional responsibilities in accordance with these requirements. In addition, in accordance with Art. 10 (2) f) of the EU Audit Regulation, we declare that we have not provided non-audit services prohibited under Art. 5 (1) of the EU Audit Regulation. We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our opinions on the annual financial statements and on the combined management report.

Key Audit Matters in the Audit of the Annual Financial Statements

Key audit matters are those matters that, in our professional judgment, were of most significance in our audit of the annual financial statements for the fiscal year from 1 January to 31 December 2020. These matters were addressed in the context of our audit of the annual financial statements as a whole, and in forming our opinion thereon; we do not provide a separate opinion on these matters.

Below, we describe what we consider to be the key audit matters:

1. Recognition of hidden reserves in connection with the AirTrust transaction

Reasons why the matter was determined to be a key audit matter

Deutsche Lufthansa Aktiengesellschaft is the direct sole shareholder of Lufthansa Commercial Holding GmbH, Frankfurt am Main ("Lufthansa Commercial Holding"), and AirTrust Aktiengesellschaft, Zug, Switzerland ("AirTrust"). As the holding company, AirTrust is the parent of the operating subsidiaries SWISS International Air Lines Aktiengesellschaft, Basel, Switzerland, and Edelweiss Air Aktiengesellschaft, Kloten, Switzerland. As the sole shareholder, Deutsche Lufthansa Aktiengesellschaft resolved to increase the capital stock of Lufthansa Commercial Holding by EUR 100. The capital increase was carried out by way of a cash contribution with a premium in kind. In return for the issuance of a share, Deutsche Lufthansa Aktiengesellschaft made a contribution of EUR 100. Additionally, it contributed

all shares in AirTrust to Lufthansa Commercial Holding at fair value as a premium in kind. The calculation of acquisition costs as part of this cash contribution with a premium in kind was a key audit matter in our audit due to the significant level of hidden reserves and the underlying complexity of the valuation performed.

Auditor's response

For the valuation of the shares in AirTrust at fair value, the Company obtained an appraisal in accordance with IDW S1 ("Principles for the Performance of Business Valuations") by an independent expert as of the cut-off date of 31 October 2020. We consulted with our internal Strategy and Transactions specialists for the purpose of assessing the suitability of the expert's work as audit evidence for the valuation of the shares in AirTrust. We discussed material plausibility tests carried out by the expert, in particular in relation to the business planning prepared by the executive directors and the significant assumptions and developments, on the basis of explanations by the expert, discussions with management and analysis of the information provided by the expert and management (including the report prepared by management, model excerpts made available and working papers as well as business plans developed by management) and assessed these with regard to their transparency, consistency and lack of contradiction.

In addition, we tested the methodical conformity of the valuation model used by the expert against the requirements set out under IDW S1 by analyzing the logical design, the applied valuation process and the conceptual procedure in deriving the individual components of the cash flows relevant to the valuation. Furthermore, we gained an understanding of the full recognition of the valuation subject and the relevant cash flows by comparing and reconciling the planning data used in the valuation model for Group operational planning. Additionally, we gained an understanding of the arithmetical accuracy of the derivation of the estimated future cash flows on the basis of the calculations performed by the expert as well as assessed the consistent presentation of key matters particularly with regard to the delimitation of the valuation subject, the inclusion of restructuring measures, sale-and-leaseback transactions and the consideration of tax loss carryforwards.

Furthermore, we reviewed the capital costs derived by the expert for methodological and arithmetic accuracy by tracing the individual parameters and their derivation and compared these to our own analyses. We also checked the clerical accuracy of the method used for discounting the future cash flows with the capital costs to the valuation date.

We discussed and gained an understanding of the considerations made by the expert concerning the plausibility testing of the derived valuation result on the basis of the working papers provided by the expert and in conversations. Furthermore, we checked a reconciliation performed by management of the value of AirTrust calculated by the expert as of 31 October 2020 to the reporting date of 31 December 2020 using documents provided and on the basis of discussions with management.

To assess the recoverability of the carrying amount of Lufthansa Commercial Holding GmbH as of the reporting date, we checked in particular whether there were any significant changes in the key parameters used for the carrying amount of AirTrust AG as of 31 October 2020 using a reconciliation prepared by the executive directors to the reporting date.

Our audit procedures did not lead to any reservations relating to the recognition of hidden reserves in connection with the AirTrust transaction.

Reference to related disclosures

The disclosures concerning the AirTrust transaction and the valuation methods as well as the related disclosures on the exercise of judgment are contained in sections "2 Summary of significant accounting policies and valuation methods," "4 Financial investments" and "20 Other operating income" of the notes to the financial statements.

2. Implications of the COVID-19 pandemic for the going concern basis

Reasons why the matter was determined to be a key audit matter

The COVID-19 pandemic has had a particularly strong impact on the airline industry. Since the outbreak of the pandemic in spring 2020, national and international travel restrictions have led to a significant reduction in air travel. The restrictions have had a major impact on the operations and liquidity of the Company.

To safeguard the solvency of Deutsche Lufthansa AG, stabilization measures and loans of up to EUR 9b were agreed with the Economic Stabilization Fund in the Federal Republic of Germany as well as with the governments of Switzerland, Austria and Belgium. Lufthansa has also made use of other support measures provided by other governments around the world, such as the US. Deutsche Lufthansa AG has significantly reduced available flight capacity and also initiated measures to reduce costs and capital expenditure. The Executive Board has prepared the Group operational planning for the fiscal years 2021 to 2024, together with a liquidity forecast for the same period, and thereby mapped out the possible effects from national and international travel restrictions in connection with the COVID-19 pandemic on the available flight capacity, taking into account the countermeasures. The result of the planning depends to a large extent on the assumptions that the executive directors have made regarding future business developments and the cash inflows and outflows derived from them. The planning for 2021 and subsequent years is based on the assumption that a vaccine will be widely available or an established testing strategy will be in place. Significant cost savings were also assumed based on initiated restructuring measures.

The review of the implications of the COVID-19 pandemic for the going concern basis of accounting was a key audit matter due to the wide range of political, medical and global economic effects and due to the measures taken by the

executive directors in this connection to mitigate risk and their consideration in the business planning.

Auditor's response

We discussed what influence the COVID-19 pandemic, with its national and international travel restrictions, could have on Deutsche Lufthansa Aktiengesellschaft with the executive directors, both direct and indirect and short and medium term. We discussed what different assumptions and scenarios were used by the executive directors in their assessment.

We analyzed, with the support of our internal specialists from Strategy and Transactions, the Group operational planning prepared by the executive directors, including the liquidity forecast, and discussed this with the responsible management level. We assessed the plausibility (transparency, consistency, lack of contradiction) of management's disclosures on the key planning assumptions, strategic objectives, expected developments and operational measures and their inclusion/reflection in the cash flows.

We also checked the clerical accuracy, completeness and consistency of the planning model, the conceptual design and in particular the derivation of the cash flows and their individual components. In this connection, we also reviewed the implementation of the measures adopted under the restructuring programs to reduce the outflow of liquidity on the basis of resolutions, internal and external publications and observed facts for the implementation of capacity adjustments and their impact on the fleet as well as on personnel. We also checked the capital expenditures reflected in the financial planning on the basis of purchase commitments and framework agreements.

In order to assess the cash inflows included in the liquidity planning we, among other things, reviewed the agreements on state stabilization measures and other financing schemes, checked incoming payments and assessed the terms and conditions. The defined overarching objectives of the financing strategy and the suitability of the measures underlying them were also assessed. In addition, we examined the key planning assumptions taking into account historical developments and our understanding of the business and performed plausibility tests on the basis of various external information. Our assessment was based on analyst estimates, both for Deutsche Lufthansa Aktiengesellschaft and its subsidiaries (the "Lufthansa Group") as well as in relation to comparable companies, along with other external forecasts on the development of the airline industry (market studies) and macroeconomic forecasts.

In addition, the updated estimates of the executive directors concerning the availability of a COVID-19 vaccine, a testing strategy, the current travel restrictions as well as the current booking behavior and the consideration of these factors in the Group planning were discussed with the executive directors and management of the Company and examined on the basis of the Company's internal assessments. We also discussed other potential measures to secure liquidity which are not reflected in the liquidity planning due to a lack of

necessity with the management and assessed their risk coverage potential. In addition, we checked whether the annual financial statements and combined management report contain disclosures on the implications of the COVID-19 pandemic for the going concern basis and in particular the measures taken by the executive directors to mitigate risk.

Our audit procedures did not lead to any reservations relating to the going concern basis / the application of the going concern basis of accounting.

Reference to related disclosures

With regard to the assessment of the executive directors on the Company's ability to continue as a going concern and the related recognition and measurement policies applied, we refer to the disclosure in the notes to the financial statements "2 Summary of significant accounting policies and valuation methods (going concern)."

Please also refer to the "Forecast" as well as the "Opportunities and risk report" in the combined management report and there to the section "Outlook for the Lufthansa Group - Overall statement by the Executive Board on the expected development of the Lufthansa Group."

3. Recoverability of aircraft including investments in aircraft owning companies

Reasons why the matter was determined to be a key audit matter

In its financial statements, the Company discloses aircraft that it legally owns or of which it has economic ownership in the balance sheet item aircraft. The Company also leases aircraft from aircraft owning companies in which the Company has a direct or indirect equity investment and which are disclosed under the item financial investments, as well as from external lessors. Due to the coronavirus crisis, the Company's Executive Board resolved to permanently reduce the fleet. The executive directors were thus tasked with determining the fair value, taking into consideration the future use of the aircraft (for example via sale, abandonment or deep storage) and thus also assessing the need to recognize an impairment charge. In addition to the assessment of recoverability of the Company's own aircraft, management also assessed the recoverability of equity investments in aircraft owning companies which are also affected by the fleet reduction. The fair value of the equity investment was primarily determined on the basis of the fair value of the assets held by the aircraft owning company (mainly aircraft).

From our perspective, the recoverability of aircraft including the equity investments in aircraft owning companies was a key audit matter in our audit as the valuation of these items significant in amount is based to a large extent on the estimates and assumptions of the executive directors, particularly in light of the impact of the COVID-19 pandemic and the corresponding impairment recognized in the fiscal year.

Auditor's response

With regard to the assessment of aircraft in the legal or economic ownership of the Company and earmarked for continued use in flight operations, we examined the fleet planning in connection with the Group operational planning for internal consistency and analyzed whether these are in line with industry forecasts. In this context, we performed a methodological assessment of the impairment test conducted by the executive directors in accordance with the item-by-item principle.

For the purpose of assessing the executive directors' decision to continue to use the aircraft in the Company's legal or economic ownership but not set for continued use in flight operations, we obtained an understanding of whether the measures included in the resolutions adopted by the executive directors for the abandonment or deep storage of aircraft were fully accounted for in the balance sheet. For the purpose of assessing the executive directors' decision to continue to use the aircraft in the legal ownership of the aircraft owning companies but not set for continued use in flight operations, we obtained an understanding of whether the resulting effects on the impairment of the carrying values of the equity investments were accounted for in the balance sheet by the executive directors. Furthermore, for aircraft in the legal ownership of the aircraft owning companies and earmarked for continued use in flight operations, we gained an understanding of the method used in the impairment test of equity investments performed by the executive directors.

In relation to the aircraft earmarked for abandonment or deep storage that are in the legal ownership of the aircraft owning companies as well as in the legal or economic ownership of the Company, we assessed the measurement assumptions used by the executive directors for the underlying sale or scrap value for the purpose of determining the impairment of the aircraft, taking into account the available information. In this connection, we assessed whether the assumptions made by the executive directors were transparently derived from observable market data on prices, taking into account a market price overview published regularly by Aircraft Value Analysis Company Ltd., comparable purchase agreements from the past or from purchase agreements already concluded.

Our audit procedures did not lead to any reservations relating to the recoverability of aircraft or aircraft owning companies.

Reference to related disclosures

The Company's information on the impairment of financial investments is contained in the section "2 Summary of significant accounting policies and valuation methods," "3 Fixed assets" and "23 Depreciation, amortisation and impairment" in the notes to the financial statements.

4. Accounting for short-time working allowances as well as corresponding social security contributions and the reimbursement thereof

Reasons why the matter was determined to be a key audit matter

The COVID-19 pandemic had a significant impact on the global operations of Deutsche Lufthansa Aktiengesellschaft in fiscal year 2020. The executive directors of Deutsche Lufthansa Aktiengesellschaft introduced short-time working arrangements as one of the measures to limit the financial fallout from the ongoing crisis. In this connection, the executive directors also applied to the Agentur für Arbeit (Federal Employment Agency) for a reimbursement of short-time pay within the legally prescribed scope.

As the benefit notices from the Federal Employment Agency confirming the granting of short-time working allowances are issued subject to review, the receipt of reimbursement amounts constitutes an uncertainty for the Company until a final review is performed by the Federal Employment Agency to confirm compliance with the general requirements set out under the German Social Security Code III (Sozialgesetzbuch III) and the specific requirements for submitted reimbursement claims.

In order to reflect this special matter in payroll accounting, the underlying IT systems were adjusted, in particular for the provision of relevant information for the completion of the reimbursement claims, and special measures for monitoring were implemented in the accounting-related internal control system. These modifications of the underlying personnel accounting system as well as the accounting-related internal control system ("ICS") of Deutsche Lufthansa Aktiengesellschaft represent an elevated risk of material misstatement of the reimbursement claims in the annual financial statements, as mistakes in payroll accounting as well as weaknesses in the monitoring activities of the ICS could lead to erroneously calculated amounts of reimbursement claims, which could give rise to negative financial effects for the Company. With regard to the reimbursable amounts from the Federal Employment Agency, a legally fundamental distinction must be made between claims by the Company and claims by individual employees against the former. The claims by the Company against the Federal Employment Agency in the form of contributions to the social security scheme constitute, as they are subject to the fulfillment of certain legally defined conditions, non-refundable government grants that are deducted from the related expense. However, as the employee has a claim to payment of short-time pay against the Federal Employment Agency it merely constitutes a transitory item for the employer and is recognized outside of profit or loss in the annual financial statements of Deutsche Lufthansa Aktiengesellschaft as receivables from Federal Employment Agency. The different accounting for the two types of reimbursement claims underscores in particular the risk of an incorrect valuation due to differing assessment bases. As of the reporting date, the receivables from the Federal Employment Agency are estimated based on the amounts already paid or transferred but

not yet reimbursed by the Federal Employment Agency, taking into account historical data of the past months.

In light of these differences in accounting and due to the quantitative significance of reimbursements of short-time working allowances and social security contributions for the annual financial statements as well as the complexity of the legal requirements and the process-related uncertainty, the accounting for short-time working allowances as well as the corresponding social security contributions constitutes a key audit matter in our audit.

Auditor's response

In the course of our audit procedures, we assessed the accounting for the short-time working allowances and the corresponding social security contributions as well as the reimbursement thereof by the Federal Employment Agency in terms of merit and amount. In the course of this, we evaluated on the basis of evidence whether there was a legal basis for the introduction of short-time working arrangements. This included, in particular, the notification to the Federal Employment Agency of the reduction in working hours by the executive directors for the application of short-time working allowances as well as the corresponding approval notices confirming compliance with the relevant requirements pursuant to Sec. 95 to 99 German Social Security Code III (Sozialgesetzbuch III) for the receipt of short-time working allowances based on merit.

Our audit procedures included the assessment of the accounting for the reimbursements of short-time working allowances by the Federal Employment Agency and the corresponding social security contributions payable solely by the employer in accordance with IDW HFA 1/1984 as amended in 1998 as non-refundable government grants in the form of expense subsidies. Taking account of the approved individual short-time working arrangements and the arrangements differentiated by employee group, by carrying out sample-based checks we obtained an understanding of the process for calculating and accounting for the reimbursement amounts particularly in order to assess the results of the modifications to the underlying payroll accounting system related to the short-time working arrangements. In this connection, we compared the absences due to short-time working arrangements displayed in the time logging systems against the individual payslips of the sample-based check and against the reimbursement lists completed by the responsible employees of the Company and which are the basis of the monthly reimbursement requests. Furthermore, by performing sample-based checks we verified the clerical accuracy and methods used for calculating and determining the short-time working allowances stated in these reimbursement claims as well as the corresponding social security contributions. Furthermore, we gained an understanding during our process walkthrough of the measures of the accounting-related internal control system and assessed these in the course of a test of design.

We also tested the plausibility of the amount of receivables recognized as of the reporting date taking into account historical data from retroactively calculated discrepancies

between the monthly reimbursement claims and benefit notices or official decrees issued by the Federal Employment Agency as well as the actual reimbursement amounts.

Furthermore, we performed analytical audit procedures regarding the amount of short-time working allowances as well as the corresponding social security expenses. In doing so, we tested the plausibility of correlations, fluctuations and trends on the basis of our expectations based on surveys and the claim applications and notices presented above. This included, among other things, the analysis of the development of the reimbursed short-time working allowances and the corresponding social security contributions as well as the reimbursement amounts as a percentage of total staff costs and the total social security expenses taking into account the number of employees.

Our audit procedures did not lead to any reservations with regard to the accounting for short-time working allowances and the corresponding social security contributions or their reimbursement.

Reference to related disclosures

We refer to the disclosure in the notes to the financial statements in "22 Staff costs" for information on the accounting for short-time working allowances and government grants in the form of a reimbursement of social security contributions by the Federal Employment Agency.

5. Accounting for derivative financial instruments in hedges

Reasons why the matter was determined to be a key audit matter

Deutsche Lufthansa Aktiengesellschaft uses a variety of derivative financial instruments to hedge against currency, fuel price and interest rate risks arising from its ordinary business activities. Deutsche Lufthansa Aktiengesellschaft experienced significant restrictions in terms of flight volumes in fiscal year 2020 as a result of the COVID-19 pandemic, resulting in lower cash flows in foreign currency as well as a significant decline in purchases of kerosene. As a result, forecast transactions and thus hedged items in hedging relationships were not carried out. In our view, these matters constituted one of the key audit matters due to the significant volume in terms of amount, the high complexity and number of hedging transactions as well as the extensive accounting requirements.

Auditor's response

In the course of our audit, we analyzed, with the support of our internal specialists from the Financial Instruments Experts Group, among other things, the contractual and economic basis of the hedging relationship between the forecast transactions as hedged items and the derivative financial instruments as hedging instruments as well as the accounting, including earnings effects. Our focus was on testing the controls that address the existence and completeness of the derivative financial instruments and of the designated hedges in the IT systems used for the accounting

for the hedging relationships. We also obtained bank confirmations as audit evidence.

We obtained an understanding of the valuation of the hedged items and hedging transactions (including option premiums) by the executive directors in particular in the areas of fuel and foreign currency by reperforming valuations in our own valuation systems. For interest rate derivatives, we performed audit procedures in relation to the software used for valuation purposes. We reviewed the access restrictions, the approval process for software changes as well as the archiving of software changes and subsequently reperformed the valuation of significant interest rate derivatives on a sample basis.

We assessed, together with our specialists, compliance with the requirements for the accounting for the hedging relationships using the designation documents prepared by the executive directors and the internal risk management guidelines. On the basis of business plans and inquiries of management, we checked whether only such forecast transactions constitute a hedged item of a hedging relationship that are considered highly probable on the basis of the current business plan. We traced the accounting for hedging relationships and their (potentially premature) terminations in terms of value and reconciled them with the respective items in the balance sheet and income statement. With regard to fuel price hedges, we also checked the clerical accuracy of the calculation of the recognition of the option premiums that are a component of the designated derivatives.

Our audit procedures did not lead to any reservations regarding the accounting for derivative financial instruments in hedges.

Reference to related disclosures

The disclosures of the recognition and measurement policies applied for hedging transactions are included in sections “2 Summary of significant accounting policies and valuation methods” and “17 Hedging policy and financial derivatives” of the notes to the financial statements.

Other Information

The executive directors and the Supervisory Board are responsible for the declaration pursuant to Sec. 161 AktG (“Aktiengesetz”: German Stock Corporation Act) on the German Corporate Governance Code, which is part of the corporate governance declaration. In all other respects, the executive directors are responsible for the other information. The other information comprises the aforementioned corporate governance declaration, the aforementioned non-financial declaration and the aforementioned other information included in the combined management report.

A further component of the annual report is the declaration by the executive directors.

Our opinions on the annual financial statements and on the combined management report do not cover the other infor-

mation, and consequently we do not express an opinion or any other form of assurance conclusion thereon.

In connection with our audit, our responsibility is to read the other information and, in so doing, to consider whether the other information

- is materially inconsistent with the annual financial statements, with the combined management report or our knowledge obtained in the audit, or
- otherwise appears to be materially misstated.

If, based on the work we have performed, we conclude that there is a material misstatement of this other information, we are required to report that fact. We have nothing to report in this regard.

Responsibilities of the executive directors and the Supervisory Board for the annual financial statements and the combined management report

The executive directors are responsible for the preparation of the annual financial statements that comply, in all material respects, with the requirements of German commercial law applicable to business corporations, and that the annual financial statements give a true and fair view of the assets, liabilities, financial position and financial performance of the Company in compliance with German legally required accounting principles. In addition, the executive directors are responsible for such internal control as they, in accordance with German legally required accounting principles, have determined necessary to enable the preparation of annual financial statements that are free from material misstatement, whether due to fraud or error.

In preparing the annual financial statements, the executive directors are responsible for assessing the Company’s ability to continue as a going concern. They also have the responsibility for disclosing, as applicable, matters related to going concern. In addition, they are responsible for financial reporting based on the going concern basis of accounting, provided no actual or legal circumstances conflict therewith.

Furthermore, the executive directors are responsible for the preparation of the combined management report that, as a whole, provides an appropriate view of the Company’s position and is, in all material respects, consistent with the annual financial statements, complies with German legal requirements and appropriately presents the opportunities and risks of future development. In addition, the executive directors are responsible for such arrangements and measures (systems) as they have considered necessary to enable the preparation of a combined management report that is in accordance with the applicable German legal requirements, and to be able to provide sufficient appropriate evidence for the assertions in the combined management report.

The Supervisory Board is responsible for overseeing the Company’s financial reporting process for the preparation of the annual financial statements and of the combined management report.

Auditor's responsibilities for the audit of the annual financial statements and of the combined management report

Our objectives are to obtain reasonable assurance about whether the annual financial statements as a whole are free from material misstatement, whether due to fraud or error, and whether the management report as a whole provides an appropriate view of the Company's position and, in all material respects, is consistent with the annual financial statements and the knowledge obtained in the audit, complies with the German legal requirements and appropriately presents the opportunities and risks of future development, as well as to issue an auditor's report that includes our opinions on the annual financial statements and on the combined management report.

Reasonable assurance is a high level of assurance, but is not a guarantee that an audit conducted in accordance with Sec. 317 HGB and the EU Audit Regulation and in compliance with German Generally Accepted Standards for Financial Statement Audits promulgated by the Institut der Wirtschaftsprüfer (IDW) will always detect a material misstatement. Misstatements can arise from fraud or error and are considered material if, individually or in the aggregate, they could reasonably be expected to influence the economic decisions of users taken on the basis of these annual financial statements and this combined management report.

We exercise professional judgment and maintain professional skepticism throughout the audit. We also:

- Identify and assess the risks of material misstatement of the annual financial statements and of the combined management report, whether due to fraud or error, design and perform audit procedures responsive to those risks, and obtain audit evidence that is sufficient and appropriate to provide a basis for our opinions. The risk of not detecting a material misstatement resulting from fraud is higher than for one resulting from error, as fraud may involve collusion, forgery, intentional omissions, misrepresentations, or the override of internal control.
- Obtain an understanding of internal control relevant to the audit of the annual financial statements and of arrangements and measures (systems) relevant to the audit of the combined management report in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of these systems of the Company.
- Evaluate the appropriateness of accounting policies used by the executive directors and the reasonableness of estimates made by the executive directors and related disclosures.
- Conclude on the appropriateness of the executive directors' use of the going concern basis of accounting and, based on the audit evidence obtained, whether a material uncertainty exists relat-

ed to events or conditions that may cast significant doubt on the Company's ability to continue as a going concern. If we conclude that a material uncertainty exists, we are required to draw attention in the auditor's report to the related disclosures in the annual financial statements and in the combined management report or, if such disclosures are inadequate, to modify our respective opinions. Our conclusions are based on the audit evidence obtained up to the date of our auditor's report. However, future events or conditions may cause the Company to cease to be able to continue as a going concern.

- Evaluate the overall presentation, structure and content of the annual financial statements, including the disclosures, and whether the annual financial statements present the underlying transactions and events in a manner that the annual financial statements give a true and fair view of the assets, liabilities, financial position and financial performance of the Company in compliance with German legally required accounting principles.
- Evaluate the consistency of the combined management report with the annual financial statements, its conformity with (German) law, and the view of the Company's position it provides.
- Perform audit procedures on the prospective information presented by the executive directors in the combined management report. On the basis of sufficient appropriate audit evidence we evaluate, in particular, the significant assumptions used by the executive directors as a basis for the prospective information, and evaluate the proper derivation of the prospective information from these assumptions. We do not express a separate opinion on the prospective information and on the assumptions used as a basis. There is a substantial unavoidable risk that future events will differ materially from the prospective information.

We communicate with those charged with governance regarding, among other matters, the planned scope and timing of the audit and significant audit findings, including any significant deficiencies in internal control that we identify during our audit.

We also provide those charged with governance with a statement that we have complied with the relevant independence requirements, and communicate with them all relationships and other matters that may reasonably be thought to bear on our independence and where applicable, the related safeguards.

From the matters communicated with those charged with governance, we determine those matters that were of most significance in the audit of the annual financial statements of the current period and are therefore the key audit matters. We describe these matters in our auditor's report un-

less law or regulation precludes public disclosure about the matter.

Other legal and regulatory requirements

Report on the assurance in accordance with Sec. 317 (3b) HGB on the electronic reproduction of the annual financial statements and the management report prepared for publication purposes

Opinion

We have performed assurance work in accordance with Sec. 317 (3b) HGB to obtain reasonable assurance about whether the reproduction of the annual financial statements and the combined management report (hereinafter the “ESEF documents”) contained in the attached electronic file “Deutsche_Lufthansa_AG_JA_LB_ESEF-2020-12-31.zip” and prepared for publication purposes complies in all material respects with the requirements of Sec. 328 (1) HGB for the electronic reporting format (“ESEF format”). In accordance with German legal requirements, this assurance only extends to the conversion of the information contained in the annual financial statements and the combined management report into the ESEF format and therefore relates neither to the information contained in this reproduction nor to any other information contained in the abovementioned electronic file.

In our opinion, the reproduction of the annual financial statements and the combined management report contained in the abovementioned attached electronic file and prepared for publication purposes complies in all material respects with the requirements of Sec. 328 (1) HGB for the electronic reporting format. We do not express any opinion on the information contained in this reproduction nor on any other information contained in the abovementioned file beyond this reasonable assurance opinion and our audit opinion on the accompanying annual financial statements and the accompanying combined management report for the fiscal year from 1 January to 31 December 2020 contained in the “Report on the audit of the annual financial statements and of the combined management report” above.

Basis for the opinion

We conducted our assurance work on the reproduction of the annual financial statements and the combined management report contained in the abovementioned attached electronic file in accordance with Sec. 317 (3b) HGB and Exposure Draft of IDW Assurance Standard: Assurance in Accordance with Sec. 317 (3b) HGB on the Electronic Reproduction of Financial Statements and Management Reports Prepared for Publication Purposes (ED IDW AsS 410) and the International Standard on Assurance Engagements 3000 (Revised)]. Our responsibilities under that standard are further described in the “Auditor’s responsibilities for the assurance work on the ESEF documents” section. Our audit firm applied the requirements for quality control systems set forth in IDW Standard on Quality Control: “Requirements for Quality Control in Audit Firms” (IDW QS 1).

Responsibilities of the executive directors and the Supervisory Board for the ESEF documents

The executive directors of the Company are responsible for the preparation of the ESEF documents including the electronic reproduction of the annual financial statements and the combined management report in accordance with Sec. 328 (1) Sentence 4 No. 1 HGB.

In addition, the executive directors of the Company are responsible for such internal control as they have considered necessary to enable the preparation of ESEF documents that are free from material non-compliance with the requirements of Sec. 328 (1) HGB for the electronic reporting format, whether due to fraud or error.

The executive directors of the Company are also responsible for the submission of the ESEF documents together with the auditor’s report and the attached audited annual financial statements and the audited combined management report as well as other documents to be published to the operator of the Bundesanzeiger (German Federal Gazette).

The Supervisory Board is responsible for overseeing the preparation of the ESEF documents as part of the financial reporting process.

Auditor’s responsibilities for the assurance work on the ESEF documents

Our objective is to obtain reasonable assurance about whether the ESEF documents are free from material non-compliance with the requirements of Sec. 328 (1) HGB, whether due to fraud or error. We exercise professional judgment and maintain professional skepticism throughout the engagement. We also: □

- Identify and assess the risks of material non-compliance with the requirements of Sec. 328 (1) HGB, whether due to fraud or error, design and perform audit procedures responsive to those risks, and obtain assurance evidence that is sufficient and appropriate to provide a basis for our assurance opinion.
- Obtain an understanding of internal control relevant to the assurance work on the ESEF documents in order to design assurance procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of these controls.
- Evaluate the technical validity of the ESEF documents, i.e., whether the electronic file containing the ESEF documents meets the requirements of Delegated Regulation (EU) 2019/815, in the version valid as of the reporting date, on the technical specification for this electronic file.
- Evaluate whether the ESEF documents enable an XHTML reproduction with content equivalent to

the audited annual financial statements and to the audited management report.

Further information pursuant to Art. 10 of the EU Audit Regulation

We were elected as auditor by the Annual General Meeting on 5 May 2020. We were engaged by the Supervisory Board on 8 May 2020. We have been the auditor of Deutsche Lufthansa AG since fiscal year 2020.

We declare that the opinions expressed in this auditor's report are consistent with the additional report to the audit committee pursuant to Art. 11 of the EU Audit Regulation (long-form audit report).

In addition to the financial statement audit, we have provided to the Company or entities controlled by it the following services that are not disclosed in the annual financial statements or in the combined management report or have been engaged to provide them.

- Review of the interim financial statements of Deutsche Lufthansa AG as of 30 June 2020
- Submission of a comfort letter for Deutsche Lufthansa AG in connection with the Euro Medium Term Note (EMTN) program
- Engagement to obtain limited assurance on the non-financial declaration pursuant to Secs. 289b et seq. and 315b et seq. HGB
- Engagement to obtain limited assurance on the report of Deutsche Lufthansa AG on compliance with the conditions and requirements of the parent company and the affected Group companies pursuant to Art. 8 to 17 of the framework agreement for the granting of stabilization measures
- Audit-related services legally prescribed by Sec. 162 (3) AktG in relation to the remuneration report prepared in accordance with the provisions of the AktG
- Voluntary audits of financial statements as of 31 December 2020
- Performance of agreed-upon procedures for Lufthansa Cargo AG, Frankfurt am Main, in relation to the company's registered office
- Project-related audit of a migration of IT-based accounting-related systems pursuant to IDW AuS 850 for Lufthansa AirPlus Servicekarten GmbH, Neu-Isenburg
- Permissible consulting in connection with a report on the expenditure of funds for Lufthansa Technik AG, Hamburg
- Audit of Albatros Service Center GmbH, Cologne, in accordance with Sec. 24 FinVermV ("Finanzan-

lagenvermittlungsverordnung": German Ordinance on Financial Investment Mediation)

- Agreed-upon procedures relating to a compliance issue of a foreign subsidiary of Deutsche Lufthansa Aktiengesellschaft German Public Auditor responsible for the engagement

German Public Auditor responsible for the engagement

The German Public Auditor responsible for the engagement is Siegfried Keller.

Eschborn/Frankfurt am Main, 2 March 2021

Ernst & Young GmbH

Wirtschaftsprüfungsgesellschaft

Prof. Dr. Sven Hayn

Wirtschaftsprüfer

(German Public Auditor)

Siegfried Keller

Wirtschaftsprüfer

(German Public Auditor)

22. GLOSSARY

€ or EUR	The single European currency adopted by certain participating member states of the European Union, including Germany.
2015 Hybrid Bond	€500,000,000 4.382% hybrid bond due 2075 and issued on August 12, 2015.
80/20 Rule	Air carriers must use at least 80% of a slot series allocated to them, or lose historical precedence for these slots in the next equivalent scheduling period (summer / winter season).
ACTKs	Capacity in terms of industry-wide available cargo ton-kilometers.
ADRs	American Depositary Receipts.
AeroLogic	Aerologic GmbH.
Air Operator Certificate	Prerequisite for the award of an Operating License issued in accordance with Regulation (EU) No 2018/1139 (Article 4 lit. b of the Air Services Regulation).
Air Passengers Regulation	Regulation (EC) No 261/2004 of the European Parliament and of the Council of February 11, 2004.
AirPlus	Lufthansa AirPlus Servicekarten GmbH.
Air Services Regulation	Regulation (EC) No 1008/2008 on common rules for the operation of air services.
Additional Businesses and Group Functions	Mainly comprises the service and financial companies, particularly Lufthansa AirPlus Servicekarten GmbH (“ AirPlus ”), Lufthansa Aviation Training (“ LAT ”) and Lufthansa Systems GmbH & Co. KG (“ Lufthansa Systems ”), as well as the group functions of the Lufthansa Group.
Adjusted EBIT	EBIT adjusted for impairment losses/gains, effects from pension provisions and results of disposal of assets.
Adjusted EBITDA	Adjusted EBIT plus depreciation and amortization. Depreciation and amortization excludes impairments of tangible and intangible assets and of non-current financial assets, as well as impairments of investments accounted for using the equity method and of assets held for sale.
Adjusted EBIT Margin	Adjusted EBIT divided by total revenue.
Adjusted ROCE	Adjusted EBIT plus interest on liquidity less 25% taxes divided by the average capital employed. The capital employed results from total assets adjusted for non-operating items (deferred taxes, derivative financial instruments) less non-interest bearing liabilities (including trade payables and liabilities from unused flight documents).
Alternative Performance Measures or APM	Financial information and operating data presented by the Company that is not prepared in accordance with IFRS, or any other accepted accounting principles within the meaning of the ESMA Guidelines.
Articles of Association	The articles of association of the Company.

Austrian Airlines	Austrian Airlines AG.
Available seat-kilometer or ASK	One seat offered flown for one kilometer.
BADV	Ordinance on groundhandling services at airports (<i>Verordnung über Bodenabfertigungsdienste auf Flugplätzen</i>).
BaFin	The German Federal Financial Supervisory Authority (<i>Bundesanstalt für Finanzdienstleistungsaufsicht</i>).
Barclays	Barclays Bank Ireland PLC, One Molesworth Street, Dublin 2, D02 RF29, Ireland.
BDL	Federal Association of the German Aviation Industry.
BNP PARIBAS	BNP PARIBAS SA, 16, Boulevard des Italiens, 75009 Paris, France.
BofA Securities	BofA Securities Europe SA, 51 rue La Boétie, 75008 Paris, France.
Brexit	The exit of the United Kingdom from the European Union.
Brussels Airlines	Brussels Airlines S.A.
CAEP	Committee on Aviation Environmental Protection.
CAGR	Compound annual growth rate.
Catering	Mainly comprises the LSG Lufthansa Service Holding AG and its subsidiaries (together, the “ LSG Group ”) and, <i>inter alia</i> , the brand ‘LSG Sky Chefs.’
Cargo load factor	The ratio of capacity sold to available capacity. The cargo load factor refers to freight transport or total traffic.
Chicago Convention	Chicago Convention on International Civil Aviation of 1944.
Clearstream	Clearstream Banking Aktiengesellschaft, Mergenthalerallee 61, 65760 Eschborn, Germany.
COMMERZBANK	COMMERZBANK Aktiengesellschaft, Kaiserstraße 16 (Kaiserplatz), 60311 Frankfurt am Main, Germany.
Company	Deutsche Lufthansa Aktiengesellschaft.
CORSIA	Carbon Offsetting and Reduction Scheme for International Aviation.
COVID-19	The outbreak of SARS-CoV-2 first identified in December 2019 and its associated disease.
COVID-19-Act	German Act on Reducing the Effects of the COVID-19 Pandemic in Civil, Insolvency and Criminal Procedure Law (<i>Gesetz zur Abmilderung der Folgen der COVID-19-Pandemie im Zivil-, Insolvenz- und Strafverfahrensrecht</i>) dated March 27, 2020.
Crédit Agricole Corporate and Investment Bank	Crédit Agricole Corporate and Investment Bank, 12 Place des Etats-Unis, CS 70052, 92547 Montrouge Cedex, France.
Deutsche Bank	Deutsche Bank Aktiengesellschaft, Taunusanlage 12, 60325 Frankfurt am Main, Germany.

DHL	DHL International GmbH.
Dry Lease	A lease where the lessor provides the aircraft (without the crew).
DZ BANK	DZ BANK AG Deutsche Zentral-Genossenschaftsbank, Frankfurt am Main, Platz der Republik, 60325 Frankfurt am Main, Germany.
EASA	EU Aviation Safety Agency.
EBIT	The profit/loss from operating activities plus result of equity investments accounted for using the equity method and result of other equity investments.
EBITDA	Earnings before interest, taxes, depreciation and amortization.
ECJ	European Court of Justice.
EEA	European Economic Area.
EFTA	European Free Trade Association.
Equity Ratio	Shareholders' equity divided by total assets.
ESMA	European Securities and Markets Authority.
ESMA Guidelines	Guidelines on alternative performance measures of October 5, 2015.
EU-U.S. ATA	EU-US-Air Transport Agreement entered into force on June 29, 2020, between the United States, and the European Union with Decision 2007/339/EC and Decision (EU) 2020/1110.
Eurowings	Mainly comprises Eurowings and Eurowings Europe, providing point-to-point short-haul flights in Europe to price-sensitive and service-oriented customers.
EU ETS	EU Emission Trading System.
EU ETS Directive	Directive 2003/87/EC of the European Parliament and of the Council of October 13, 2003 establishing a scheme for greenhouse gas emission allowance trading within the Community and amending Council Directive 96/61/EC.
EU Parent Subsidiary Directive	Council Directive 2011/96/EU of November 30, 2011 on the common system of taxation applicable in the case of parent companies and subsidiaries of different member states.
EU Short Selling Regulation	Regulation (EU) No 236/2012 of the European Parliament and of the Council of March 14, 2012 on short selling and certain aspects of credit default swaps, as amended.
EY	Ernst & Young GmbH Wirtschaftsprüfungsgesellschaft, with its registered seat in Stuttgart, Germany, acting through its Frankfurt office at Mergenthalerallee 3-5, 65760 Eschborn, Germany.
Executive Board	The executive board (<i>Vorstand</i>) of the Company.
Framework Agreement	The framework agreement regarding granting stabilization measures (<i>Rahmenvertrag zur Gewährung von Stabilisierungsmassnahmen</i>) and certain ancillary agreements entered into between the WSF and the Company on June 29, 2020,

as amended.

FSMA	United Kingdom Financial Services and Markets Act 2000.
FTE	Full-time equivalent.
GDP	Gross domestic product.
GDPR	General Data Protection Regulation (EU) 2016/679 of the European Parliament and of the Council of April 27, 2016.
General Meeting	The Company's shareholder meeting.
Germany	The Federal Republic of Germany.
Goldman Sachs	Goldman Sachs Bank Europe SE, Marienturm, Taunusanlage 9-10, 60329 Frankfurt am Main, Germany.
HGB	The German Commercial Code (<i>Handelsgesetzbuch</i>).
HSBC	HSBC Trinkaus & Burkhardt AG, Königsallee 21/23, 40212 Düsseldorf, Germany.
IATA	International Air Transport Association.
ICAO	International Civil Aviation Organization.
IDA	New York City Industrial Development Agency.
IDW	Institute of Public Auditors in Germany (<i>Institut der Wirtschaftsprüfer</i>).
IFRS	International Financial Reporting Standards, as adopted by the European Union.
J.P. Morgan	J.P. Morgan AG, Taunustor 1, 60310 Frankfurt am Main, Germany.
Joint Bookrunners	Barclays, BNP PARIBAS, COMMERZBANK, HSBC, UniCredit Bank AG, Crédit Agricole Corporate and Investment Bank, DZ BANK, Landesbank Baden-Württemberg, SMBC Nikko and Société Générale.
Joint Global Coordinators	BofA Securities, Deutsche Bank, Goldman Sachs and J.P. Morgan.
LAT	Lufthansa Aviation Training.
Landesbank Baden-Württemberg	Landesbank Baden-Württemberg, Am Hauptbahnhof 2, 70173 Stuttgart, Germany.
LEI	Legal entity identifier.
LGW	Luftfahrtgesellschaft Walter.
Logistics	Mainly comprises Lufthansa Cargo and the Jettainer group, the time:matters group and the equity investment of 50% in the cargo airline AeroLogic.
LSG Group	LSG Lufthansa Service Holding AG and its subsidiaries.
LTi	Long-term variable cash payment incentive.

Lufthansa Cargo	Lufthansa Cargo Aktiengesellschaft.
Lufthansa Group	The Company together with its consolidated subsidiaries.
Lufthansa Systems	Lufthansa Systems GmbH & Co. KG.
Lufthansa Technik.....	Lufthansa Technik Aktiengesellschaft.
LuftVG.....	German Aviation Act (<i>Luftverkehrsgesetz</i>).
LuftVStG	German Air Traffic Tax Act (<i>Luftverkehrsteuergesetz</i>).
Market Abuse Regulation or MAR.....	Regulation (EU) No 596/2014 of April 16, 2014, as amended.
MiFID II	Markets in Financial Instruments Directive 2014/65/EU of the European Parliament and of the Council of May 15, 2014 on markets in financial instruments, as amended.
MiFID II Product Governance Requirements	The product governance requirements contained within (i) MiFID II, (ii) Articles 9 and 10 of Commission Delegated Directive (EU) 2017/593 supplementing MiFID II; and (iii) local implementing measures.
MMG.....	Miles & More GmbH.
Montreal Convention.....	Convention for the Unification of Certain Rules Relating to International Air Carriage by Air signed at Montreal on May 28, 1999.
Moody's	Moody's Deutschland GmbH.
MRO.....	Maintenance, repair and overhaul.
Network Airlines	Mainly comprises Lufthansa German Airlines (including regional partners and Miles & More), SWISS, Austrian Airlines, Brussels Airlines and Eurowings Discover.
New Shares	Up to 597,742,822 newly issued ordinary registered no-par value shares with restricted transferability (<i>vinkulierte Namensaktien</i>) of the Company from the capital increase against contributions in cash from authorized capital resolved by the Executive Board on September 19, 2021, with approval of the Supervisory Board (<i>Aufsichtsrat</i>) on the same date.
OEMs	Original equipment manufacturers.
Operating License	EU operating license to carry by air passengers, mail and/or cargo for remuneration and/or hire.
Passenger load factor	Measure of capacity utilization in per cent. The passenger load factor refers to passenger transportation.
Prospectus Regulation	Regulation (EU) 2017/1129 of the European Parliament and of the Council of June 14, 2017 on the prospectus to be published when securities are offered to the public or admitted to trading on a regulated market and repealing Directive 2003/71/EC, as amended.
PwC	PricewaterhouseCoopers GmbH Wirtschaftsprüfungsgesellschaft with its registered seat in Frankfurt am Main, Germany, acting through its Düsseldorf office at Moskau Str. 19, 40227 Düsseldorf, Germany.

QIBs	Qualified institutional buyers as defined in Rule 144A.
Regulation S	Regulation S under the Securities Act.
Revenue passenger-kilometer or RPK	One paying passenger transported for one kilometer.
Revenue ton-kilometer or RTK	One ton of load (passengers and/or cargo) transported one kilometer.
ReNew	The restructuring program implemented by the Company.
Rule 144A	Rule 144A under the Securities Act.
SARPs	Standards and Recommended Practices developed by the ICAO.
Scope	Scope Ratings GmbH.
Securities Act	The United States Securities Act of 1933, as amended.
SES-I	The first “Single European Sky” legislation package, consisting of Regulation (EC) No 549/2004 of the European Parliament and of the Council of March 10, and (Regulation (EC) No 550/2004, No 551/2004 and No 552/2004).
SES-II	Regulation (EC) No 1070/2009 revising and extending SES-I.
Shareholder	The Company’s shareholders.
Shares	The existing shares of the Company admitted to trading under the symbol “LHA” on the regulated market segment (<i>regulierter Markt</i>) of the Frankfurt Stock Exchange.
Silent Participation I	The silent participation I (Stille Einlage I) issued by the Company to the WSF in the maximum amount of €4.5 billion.
Silent Participation II	The second acquired silent participation by the WSF consisting of two tranches (the silent participation II-A (<i>Stille Einlage II-A</i>) (the “ Silent Participation II-A ”) and the silent participation II-B (<i>Stille Einlage II-B</i>) (the “ Silent Participation II-B ”) with conversion features.
Slot Allocation Regulation	Council Regulation (EEC) No 95/93 of January 18, 1993 on common rules for the allocation of slots at Community airports.
SMBC Nikko	SMBC Nikko Capital Markets Europe GmbH, Neue Mainzer Straße 52-58, 60311 Frankfurt am Main, Germany.
SOG	Share Ownership Guidelines.
Société Générale S.A.	Société Générale S.A., 29, boulevard Haussmann, 75009 Paris, France.
Standard & Poor’s	S&P Global Ratings Europe Limited.
Subscription Agent	Deutsche Bank Aktiengesellschaft.
SunExpress	Günes Ekspres Havacilik A.S.
Supervisory Board	The supervisory board (<i>Aufsichtsrat</i>) of the Company.
SWISS	SWISS International Air Lines AG.

T2	Terminal 2 Gesellschaft mbH & Co oHG.
Target Market Assessment	The product approval process by the Underwriters, whereby the Shares are (i) compatible with an end target market of retail investors and investors who meet the criteria of professional clients and eligible counterparties, each as defined in MiFID II; and (ii) eligible for distribution through all distribution channels as are permitted by MiFID II.
Temporary Framework or TF	The temporary framework adopted by the European Commission for state aid measures to support the economy in the Covid-19 outbreak.
TOGA	Terminal One Group Association, L.P.
Ton-kilometer offered or TKO	Offered capacity equivalent of one ton of load (passengers and/or cargo) for one kilometer.
TFEU	Treaty on the Functioning of the European Union (<i>Vertrag über die Arbeitsweise der Europäischen Union</i>).
TSR	Total Shareholder Return.
UmwG	The German Transformation Act (<i>Umwandlungsgesetz</i>).
Underwriters	The Joint Global Coordinators and the Joint Bookrunners.
Underwriting Agreement	The underwriting agreement between the Company and the Underwriters dated September 19, 2021.
UNFCCC	United Nations Framework Convention on Climate Change.
UniCredit Bank	UniCredit Bank AG, Arabellastraße 12, 81925 Munich, Germany.
United States	United States of America.
Unit costs or CASK	Operating expenses divided by offered seat-kilometers.
Unit revenue or RASK	Revenue divided by offered seat-kilometers.
Wet Lease	A lease where the lessor provides the aircraft and crew.
WSF	Economic Stabilization Fund (<i>Wirtschaftsstabilisierungsfonds</i>) by the German government.
WSF Stabilization Measures	The Silent Participation I, the Silent Participation II and the WSF Capital Increase.

23. RECENT DEVELOPMENTS AND OUTLOOK

23.1 Recent Developments

On July 1, 2021, the Company entered into an amended and restated dealer agreement with certain financial institutions as dealers to establish a debt issuance program. Under such debt issuance program, the Company may, from time to time, issue notes with an aggregate principal amount of up to €10,000,000,000. Furthermore, the Company issued a dual-tranche corporate bond in the amount of €1.0 billion under its debt issuance program and that was settled on July 14, 2021, consisting of notes in the nominal amount of €500 million due 2024 and a coupon of 2.00% p.a. and notes in the nominal amount of €500 million due 2029 and a coupon of 3.50% p.a.

In July 2021, Eurowings Discover was launched as the new leisure airline of the Lufthansa Group. Eurowings Discover is operating out of Frankfurt, Germany, and is focusing on short-, medium- and long-haul touristic destinations and is part of the Network Airlines.

To further streamline its fleet, the Lufthansa Group sold nine aircraft (two Airbus A321, two Airbus A320, one Airbus A319, one Bombardier CRJ900, two Bombardier Q Series and one Boeing MD-11F aircraft) and terminated leases for two aircraft (one Airbus A319 and one Bombardier Q Series aircraft) in July, August and September 2021. Additionally, the Lufthansa Group took delivery of one new Airbus A321neo in July and one leased Boeing B777F freighter aircraft at the beginning of September 2021. As of the date of this Prospectus, the Lufthansa Group is in the process of delivering a number of aircraft of the A320 family and Bombardier Q Series aircraft to their respective buyers and lessors under aircraft sales agreements and expiring lease agreements as well as in advanced negotiation for the leasing of four Airbus A350 aircraft to be delivered in the fourth quarter of 2021 and the first half of 2022.

Except as described above, there have been no significant changes in the Lufthansa Group's financial and trading position or financial performance between June 30, 2021, and the date of this Prospectus.

23.2 Outlook

The economic conditions in which the Lufthansa Group operates have significantly deteriorated due to the impact of COVID-19 and the related containment measures, such as travel restrictions, restricted movement guidelines and lockdowns. The global spread of COVID-19 and the ensuing travel restrictions had a significant and unprecedented negative impact on the demand for air travel. Generally, the airline industry is expected to remain more severely affected by COVID-19 for a longer period than other sectors of the economy as passenger numbers are not expected to return to pre-COVID-19 levels before 2024.

Despite the uncertainty in relation to the sector outlook, the Company believes that the increased availability of effective vaccination programs will result in the gradual lifting of travel restrictions and drive customer demand for air travel in the Fiscal Year 2021. The recovery of air travel started to slowly gain momentum in the course of the summer 2021, mainly driven by touristic travel to key summer destinations, but the Company does not expect that leisure air travel will reach pre-crisis levels in the Fiscal Year 2021. The recovery on long-haul routes and in the business travel segment is expected to be slower. Correspondingly, the Company seeks to increase its relative share of premium economy seats compared to the share of business and first class seats to offset the slower recovery of the business travel segment. Leisure travel is expected to return to 80% compared to the Fiscal Year 2019 by 2022 and pre-COVID-19 levels and business travel to return to 60% compared to the Fiscal Year 2019 and pre-COVID-19 levels by 2022. However, the Company expects that the lifting of travel restrictions will occur later than previously expected, which will generally delay the recovery of demand for air travel. As a result, leisure travel is expected to fully recover by 2025, whereas the full recovery of business travel may take longer.

Furthermore, customers are expected to continue to book at very short notice. Accordingly, the Company expects that the capacity of the Network Airlines and Eurowings (measured by ASK) will increase in the Fiscal Year 2021 (YTD) to around 40% of the Fiscal Year 2019 and pre-COVID-19 levels and to 90-95% of the Fiscal Year 2019 and pre-COVID-19 levels by 2024. Furthermore, the Company expects such capacity to increase to around 60% in the fourth quarter of the Fiscal Year 2021 compared to the fourth quarter of the Fiscal Year 2019. In order to respond to shifting demands and ensure a high degree of operational agility,

Lufthansa Group is temporarily parking a certain number of aircraft and maintain short-time working arrangements to be able to adjust our flight capacities swiftly.

For the Fiscal Year 2021, the Company expects that total revenue of the Lufthansa Group will increase and that operating loss as measured by Adjusted EBIT of the Lufthansa Group will decrease compared to the Fiscal Year 2020. In this regard, the Company expects a positive Adjusted EBIT (excluding restructuring expenses) for the third quarter of 2021. Similar to July and August 2021, the Company expects passenger numbers to reach around half of the Fiscal Year 2019 and pre-COVID-19 levels in the coming months. The extent of the increase of total revenue will primarily be driven by the recovery of demand of air travel resulting in increased capacity use at the Network Airlines. As a result, the Company expects Adjusted EBIT of the Lufthansa Group for the Fiscal Year 2021 to be less negative than Adjusted EBIT in the Fiscal Year 2020, which amounted to a loss of €5,451 million. Furthermore, the Company expects the Adjusted EBIT margin of the Lufthansa Group for the Fiscal Year 2021 to be less negative than in the Fiscal Year 2020, which amounted to negative 40.1%. This development is expected to occur in respect of the operating segments Network Airlines, Eurowings, MRO, and Catering. More specifically, revenue is expected to gradually recover in the MRO business segment due to the gradual improvement in flying hours across the industry and in the Catering business segment due to the gradual recovery of demand for air travel. Regarding the Lufthansa Group's operating segment Logistics, the Company is expecting to increase revenue and Adjusted EBIT in the Fiscal Year 2021 compared to the Fiscal Year 2020 due to limited capacity in the logistics sector that further drives margins. For additional information on the assumptions on which the Profit Forecast is based, see also *"10 Profit Forecast."*

The Company aims to achieve operating cash flow breakeven in 2022 if capacity use at the Network Airlines increases to at least 50% of the Fiscal Year 2019 and total annual cost savings of €3.5 billion by 2024. To enable the Lufthansa Group to return to profitability and value creation, the implementation of the "ReNew" restructuring program is being further refined. ReNew combines the restructuring programs at our airlines and services companies and includes four modules: (i) ReStructure, (ii) ReOrg, (iii) ReFocus and (iv) RePay. As of June 30, 2021, the number of total employees (by headcount) decreased by 22,000, or 17%, of the total workforce, including 5,000 employees (by headcount) in Germany compared to December 31, 2019 (excluding a workforce reduction by approximately 8,600 employees (headcount) in connection with the divestment of LSG Group's European business). The Lufthansa Group implemented short-term measures in 2020 to reduce staff costs and to counteract the effects of the pandemic. In addition, crisis agreements were concluded for the majority of the other groups of our employees. On August 5, 2021, Lufthansa Group has communicated that it already implemented measures to reduce personnel costs by €1.1 billion. Taking into account additional measures such as, in particular, savings of still to be negotiated amended tariff agreements as well as further staff reduction, the Company expects to achieve a total of €1.8 billion structural annual personnel cost savings from 2024 onwards. As part of the "ReNew" restructuring program, the Lufthansa Group also intends to significantly reduce its fleet. More specifically, the Company aims to reduce its fleet to 650 aircraft by 2023 and to moderately increase the number of aircraft from 2024 onwards. For additional information on the "ReNew" restructuring program, see also *"9.2.7 'ReNew' Restructuring Program."*

Additionally, the Company is targeting an adjusted free cash flow of €2 billion from 2023 onwards. The Lufthansa Group expects to take delivery of around 80 new aircraft until 2023. The Company intends to lease more than half of the future fleet additions. Capital expenditures are expected to slightly increase in the Fiscal Year 2021 to around €1.5 billion compared to the Fiscal Year 2020 and are expected to reach €2.5 billion by 2023. Adjusted ROCE is expected to improve, but remain to be negative in the Fiscal Year 2021. Additionally, we are targeting to achieve an adjusted return on capital employed excluding cash (defined as Adjusted EBIT less 25% taxes on Adjusted EBIT divided by the average capital employed less cash) of at least 10% by 2024. Furthermore, the Company is targeting to achieve an Adjusted EBIT margin of at least 8% by 2024.

The Lufthansa Group further aims to continue its progress in reducing its environmental impact in 2021. The expected improvement in the passenger load factor, effects from the fleet renewal, including early retirement of older and less efficient aircraft as well as the addition of efficient aircraft are, among other factors, expected to have a positive impact on CO₂ emissions per passenger-kilometer. The Lufthansa Group therefore expects carbon emissions to decline year-on-year and aims to achieve a net-reduction of CO₂ emissions per passenger-kilometer by 50% by 2030.

Following the repayment of drawn and repayable amounts of the Silent Participation I in connection with the Offering, the Company intends to repay the Silent Participation II in full and to cancel any undrawn amounts of the Silent Participation I by the end of 2021. Following repayment of the Silent Participation II and cancellation of the Silent Participation I, the Company intends to propose to the General Meeting to cancel the Authorized Capital C.